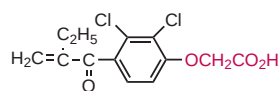


Overview Further modification of furosemide-like structures has led to the development of torsemide. Instead of the sulfonamide group found in furosemide and bumetanide, torsemide contains a sulfonylurea moiety.

Therapeutic Applications Similar to other high-ceiling diuretics, torsemide inhibits the luminal $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ symporter in the ascending limb of the loop of Henle to promote the excretion of sodium, potassium, chloride, calcium, and magnesium ions and water. An additional effect on the peritubular side at chloride channels may enhance the luminal effects of torsemide. In contrast to furosemide and bumetanide, however, torsemide does not act at the proximal tubule and, therefore, does not increase phosphate or bicarbonate excretion. The oral bioavailability of torsemide is very good (~80%), and absorption is not affected by the presence of food in the gastrointestinal tract (18). Peak diuresis is observed 1 to 2 hours following oral or intravenous administration, with a duration of action of approximately 6 hours (Table 22.2). Torsemide is indicated for the treatment of edema resulting from congestive heart failure and for the treatment of hypertension. In patients with cirrhosis and ascites, torsemide should be used with caution. Adverse effects are similar to those induced by furosemide.

ETHACRYNIC ACID



Ethacrynic acid

Mechanism of Action Another major class of high-ceiling diuretics is the phenoxyacetic acid derivatives, of which ethacrynic acid is the prototypical agent. These compounds were developed at about the same time as furosemide but were designed to act mechanistically similar to the organomercurials (i.e., via inhibition of sulfhydryl-containing enzymes involved in solute reabsorption). The mechanism of action of ethacrynic acid appears to be more complex than the simple Michael addition of the α,β -unsaturated ketone of the drug to enzyme sulfhydryl groups. When the double bond of ethacrynic acid is reduced, the resulted compound is still active, although the diuretic activity is diminished. The sulfhydryl groups of the enzyme would not be expected to add to the drug molecule in the absence of the α,β -unsaturated ketone. Ethacrynic acid is the only loop diuretic that is not a sulfonamide derivative and may be useful in patients who are allergic to sulfonamides. The pharmacokinetic properties for ethacrynic acid are listed in Table 22.2.

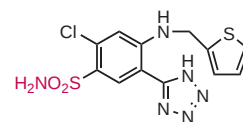
Similar to the other high-ceiling diuretics, ethacrynic acid inhibits the $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ symporter in the ascending limb of the loop of Henle to promote a marked diuresis. Sodium, chloride, potassium, and calcium excretion are

increased following oral or intravenous administration of ethacrynic acid.

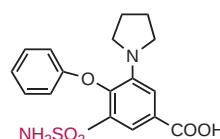
Structure-Activity Relationships Optimal diuretic activity was obtained when an oxyacetic acid group was positioned para to an α,β -unsaturated carbonyl (or other sulfhydryl-reactive group) and chloro or methyl groups were placed at the 2- or 3-position of the phenyl ring (19). In addition, hydrogen atoms on the terminal alkene carbon also provided maximum reactivity. Thus, a molecule with a weakly acidic group to direct the drug to the kidney and an alkylating moiety to react with sulfhydryl groups and lipophilic groups seemed to provide the best combination for a diuretic in this class. These features led to the development of ethacrynic acid as the prototypic agent in this class.

Therapeutic Applications Oral administration of this drug results in diuresis within 1 hour and a duration of action of 6 to 8 hours. Toxicity induced by ethacrynic acid is similar to that induced by furosemide and bumetanide. Ethacrynic acid is not widely used, however, because it induces a greater incidence of ototoxicity and more serious gastrointestinal effects than furosemide or bumetanide.

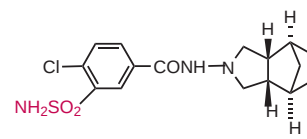
AZOSEMIDE, PIRETANIDE, AND TRIPAMIDE



Azosemide



Piretanide



Tripamide

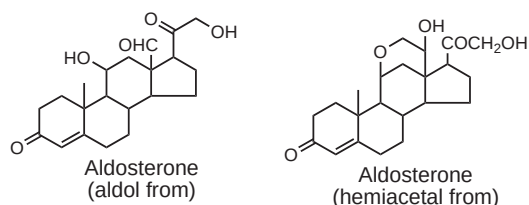
Three additional high-ceiling diuretics are azosemide, piretanide, and tripamide. Azosemide exerts a comparable diuretic effect to furosemide with oral dosing but is 5.5 to 8 times more potent following intravenous administration (20). Its low oral bioavailability (~10% to 15%) could be a consequence of high first-pass metabolism in the liver. Like bumetanide, piretanide is a sulfamoylbenzoic acid derivative. Piretanide exhibits diuretic potency greater than furosemide but lower than bumetanide (21). Acute side effects (pronounced diuresis, nausea, and thirst) with a conventional oral formulation appear to be more common than with furosemide and bumetanide. A sustained-release formulation of piretanide is better tolerated in patients. Similar to furosemide and azosemide, tripamide has a chloro group ortho to the sulfonamide functionality. At lower doses, tripamide exerts an antihypertensive effect

without diuresis, whereas it exhibits both activities at higher doses, suggesting that it may affect other sites in the nephron (22). Azosemide, piretanide, and triparamide are not available in the United States but are used in other countries.

Mineralocorticoid Receptor Antagonists

Mechanism of Action

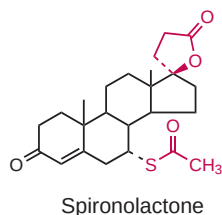
The adrenal cortex secretes a potent mineralocorticoid called aldosterone, which promotes salt and water retention and potassium and hydrogen ion excretion. Aldosterone exerts its biologic effects through binding to the mineralocorticoid receptor (MR), a nuclear transcription factor (23).



Other mineralocorticoids have an effect on the electrolytic balance of the body, but aldosterone is the most potent. Its ability to cause increased reabsorption of sodium and chloride ion and increased potassium ion excretion is approximately 3,000-fold that of hydrocortisone. A substance that antagonizes the effects of aldosterone could conceivably be a good diuretic drug. Spironolactone and eplerenone are examples of such antagonists. These drugs are also classified as potassium-sparing diuretics.

Specific Drugs

SPIRONOLACTONE



Overview Spironolactone competitively inhibits aldosterone binding to the MR, thereby interfering with reabsorption of sodium and chloride ions and the associated water. MR antagonist activity is dependent on the presence of a γ -lactone ring on C-17 and a substituent on C-7 in spironolactone and structurally related compounds (24,25). Interaction of C-7-unsubstituted agonists, such as aldosterone, with a methionine residue in the MR ligand binding domain is important for receptor activation and subsequent transcription. However, this interaction is sterically hindered by C-7 substituents on aldosterone antagonists, thereby leaving MR in

an inactive conformation (24,25). The most important renal site of these receptors, and hence the primary site of action of spironolactone, is in the late distal convoluted tubule and collecting system (collecting duct).

Pharmacokinetics On oral administration, approximately 90% of the dose of spironolactone is absorbed and is significantly metabolized during its first passage through the liver to its major active metabolite, canrenone (see Table 22.2 for their pharmacokinetic properties), which is interconvertible with its canrenoate anion (Fig. 22.3) (26,27). Canrenone is an antagonist to aldosterone and exists in equilibrium with its ring-opened form, canrenoate.

The canrenoate anion is not therapeutically active but acts as an aldosterone antagonist because of its conversion back to canrenone, which exists in the lactone form. Canrenone has been suggested to be the active form of spironolactone as an aldosterone antagonist. The formation of canrenone, however, cannot fully account for the total activity of spironolactone (26,27). Both canrenone and potassium canrenoate are used as diuretics in other countries, but they are not available in the United States.

Therapeutic Applications Spironolactone is useful in treating edema resulting from primary hyperaldosteronism and refractory edema associated with secondary hyperaldosteronism. Spironolactone is considered to be the drug of choice for treating edema resulting from cirrhosis of the liver. The dose of spironolactone is 100 mg/day given in single or divided doses. Another use of spironolactone is coadministration with a potassium-depleting diuretic (e.g., a thiazide or loop diuretic) to prevent or treat diuretic-induced hypokalemia. However, it should not be combined with potassium-sparing diuretics (e.g., triamterene or amiloride). Spironolactone can be administered in a fixed-dose combination with

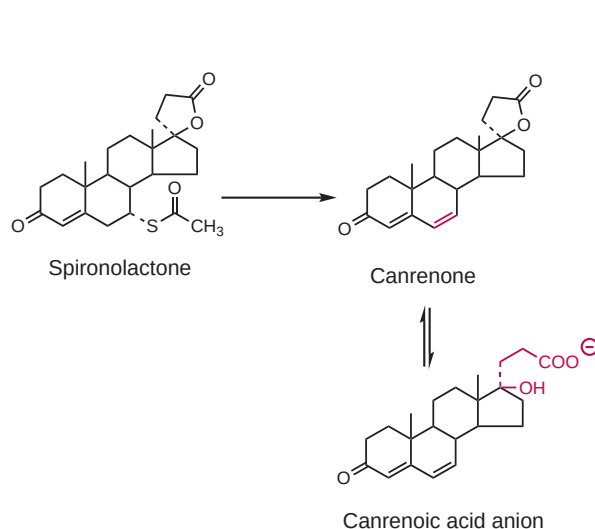
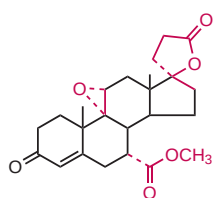


FIGURE 22.3 Metabolic conversion of spironolactone to canrenone.

hydrochlorothiazide for this purpose, but optimal individualization of the dose of each drug is recommended.

Adverse Effects The primary concern with the use of spironolactone is the development of hyperkalemia, which can be fatal. Spironolactone may cause hypersensitivity reactions, gastrointestinal disturbances and peptic ulcer. Sexual side effects (i.e., gynecomastia, decreased libido, and impotence) can also occur and are due to nonselective binding of spironolactone to the androgen receptor (AR), glucocorticoid receptor (GR), or progesterone receptor (PR). It also has been implicated in tumor production during chronic toxicity studies in rats, but human risk has not been documented.

EPLERENONE



Overview Eplerenone is a newer drug that came out of efforts to develop spironolactone analogs with reduced adverse effects (28). In addition to the lactone ring and C-7 substituent (in this case, an acetyl group), which are important for MR antagonism, eplerenone has a 9 α ,11 α -epoxy group as part of its structure. Like spironolactone, it binds to the MR and is an aldosterone antagonist. However, it has a 20- to 40-fold lower affinity for the MR than spironolactone (28). This reduced binding is believed to be due to the epoxy group (29,30). Nevertheless, eplerenone is an effective diuretic and has certain therapeutic advantages over spironolactone.

Pharmacokinetics Eplerenone has good (~70%) oral bioavailability and, unlike spironolactone, only undergoes limited first-pass metabolism (Table 22.2). Absorption is not affected by the presence of food in the gastrointestinal tract. In plasma, it is about 50% bound to plasma α_1 -acid glycoprotein. Eplerenone has a half-life of approximately 5 hours and undergoes extensive metabolism by hepatic CYP3A4 to inactive metabolites (Fig. 22.4) (31). Combination with potent inhibitors of CYP3A4 (i.e., ketoconazole or erythromycin) can alter eplerenone pharmacokinetics (32). Elimination occurs in the urine and feces.

Therapeutic Applications Similar to spironolactone, eplerenone is used alone or with other diuretics for the treatment of hypertension or left ventricular systolic dysfunction and congestive heart failure after myocardial infarction. Single daily oral doses are 25 to 50 mg.

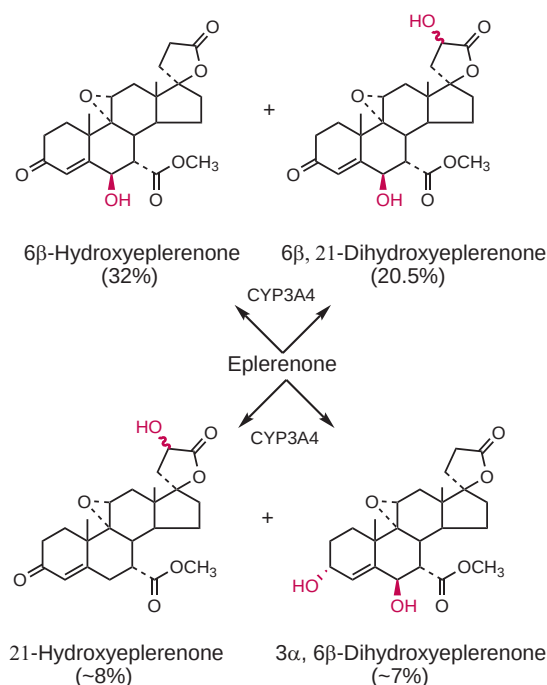


FIGURE 22.4 Major metabolic products formed from eplerenone.

Adverse Effects Hyperkalemia is a serious and potentially fatal side effect. In contrast to spironolactone, eplerenone has limited or no inhibitory effects on AR, GR, and PR and is, therefore, a more selective aldosterone antagonist (28). As a consequence, it has fewer sexual side effects.

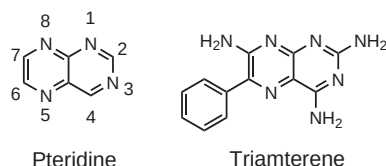
Potassium-Sparing Diuretics

Mechanism of Action

Two drugs in this class of diuretics are triamterene and amiloride (Table 22.2). Individually, amiloride and triamterene exert a mild diuretic effect and are usually used in combination with thiazides or loop diuretics. In vitro experiments have shown that they exert a diuretic effect by blocking an epithelial sodium channel (ENaC) in principal cells of the late distal convoluted tubule and collecting duct (33,34). Both drugs are weak organic bases and inhibit ENaC in a voltage- and pH-dependent manner. Inhibition occurs because amiloride and triamterene bind to negatively charged regions of the sodium channel in the ENaC. The greater potency (approximately 100-fold in vitro) of amiloride is probably due to the fact that it is a stronger base ($pK_a = 8.7$) and is therefore more extensively protonated at physiologic pH than triamterene ($pK_a = 6.2$). Sodium channel inhibitors block the reabsorption of sodium ion and inhibit the secretion of potassium ion. The net result is increased sodium and chloride ion excretion in the urine and almost no potassium excretion. As a consequence, amiloride and triamterene can be used to offset the effect of other diuretics that result in loss of potassium.

Specific Drugs

PTERIDINES: TRIAMTERENE Pteridines have a marked potential for influencing biologic processes. Early screening of pteridine derivatives revealed that 2,4-diamino-6,7-dimethylpteridine was a fairly potent diuretic.



Overview Further structural modification of the pteridine nucleus led to the development of triamterene. Alterations of the triamterene structure are not usually beneficial in terms of diuretic activity. Activity is retained if an amine group is replaced with a lower alkylamine group. Introduction of a para-methyl group on the phenyl ring decreases the activity by approximately half.

Therapeutic Applications Triamterene is more than 70% absorbed on oral administration (see Table 22.2 for its other pharmacokinetic properties). The diuretic effect occurs rapidly (~30 minutes) and reaches a peak plasma concentration in 2 to 4 hours, with a duration of action of more than 24 hours. Triamterene is extensively metabolized to 4'-hydroxytriamterene and its sulfate conjugate, both of which (major metabolite) are still active as diuretics (Fig. 22.5) (35). Both the drug and its metabolites are excreted in the urine. Triamterene is useful in combination with a thiazide or loop diuretic in the treatment of edema or hypertension. Liddle syndrome, which is due to an inherited mutation in ENaC leading to increased activity of the receptor, also may be treated with a sodium channel-blocking drug, such as triamterene or amiloride. Triamterene is administered initially in doses of 100 mg twice a day. A maintenance dose for each patient should be individually determined. This dose may vary from 100 mg a day to as low as 100 mg every other day.

Adverse Effects The most serious side effect associated with the use of triamterene is hyperkalemia. For this reason, potassium supplements are contraindicated, and serum potassium levels should be checked

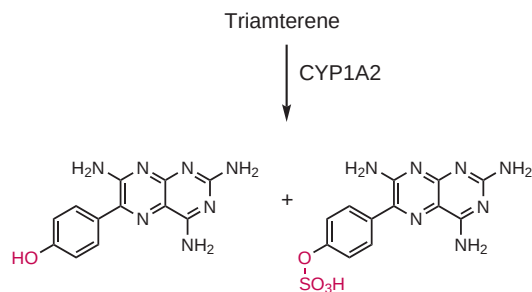
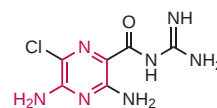


FIGURE 22.5 Metabolism of triamterene.

regularly. Triamterene also is used in combination with hydrochlorothiazide. Here, the hypokalemic effect of the hydrochlorothiazide counters the hyperkalemic effect of the triamterene. Other side effects that are seen with the use of triamterene are nausea, vomiting, and headache.

AMINOPYRAZINES: AMILORIDE



Amiloride is an aminopyrazine structurally related to triamterene as an open-chain analog. It has no effect on the action of aldosterone. Oral amiloride is approximately 50% absorbed (see Table 22.2 for its other pharmacokinetic properties), with a duration of action of 10 to 12 hours, which is slightly longer than that for triamterene. Although triamterene is extensively metabolized, approximately 50% of amiloride is excreted unchanged. Renal impairment can increase its elimination half-life. Like triamterene, amiloride combined with a thiazide or loop diuretic is used to treat edema or hypertension. Aerosolized amiloride has shown some benefit in improving mucociliary clearance in patients with cystic fibrosis. As with triamterene, the most serious side effect associated with amiloride is hyperkalemia, and it also has the other side effects associated with triamterene. The dose of amiloride is 5 to 10 mg/day. Amiloride also is combined with hydrochlorothiazide in a fixed-dose combination.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Kim K. Birtcher, MS, PharmD, BCPS

TB returns to the clinic after 12 weeks. He is taking lisinopril 20 mg daily and HCTZ 25 mg daily in the morning. His blood pressure is at goal (<130/80 mm Hg), and he has lost 10 lb. He reports eating a heart-healthy diet, and he walks at least 45 minutes/day. His K⁺ level is 4.8 mEq/L, his blood urea nitrogen level is 12 mg/dL, and his serum creatinine level is 1.0 ml/min. Because he has been stabilized on an appropriate dose of blood pressure medications, he is given a prescription for the combination product, lisinopril/HCTZ 20/25.

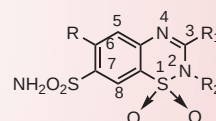
Chemical Analysis

Kim K. Birtcher, S. William Zito, and Victoria Roche

Thiazide diuretics are commonly used as initial or add-on therapy to treat hypertension. The thiazide diuretics decrease blood pressure by several mechanisms. They inhibit the Na⁺/Cl⁻ symporter in the distal convoluted tubule and inhibit the reabsorption of sodium and chloride ions. They also promote water excretion, which decreases plasma volume and the workload of the heart. With chronic therapy, thiazides decrease peripheral vascular resistance. High amounts of sodium in the diet can counter the blood pressure-lowering effects of thiazides. To a lesser extent, thiazide diuretics also inhibit the reabsorption of potassium and

bicarbonate ions. The clinician should periodically monitor K⁺ levels and clinical and laboratory parameters for dehydration.

Chemically, thiazides are 1,2,4-benzothiadiazine-1,1-dioxide derivatives. Changes to their basic structure determines their potency and duration of action. For optimal activity, the sulfonamide group at position 7 is necessary (unsubstituted preferred), and there should be an electron withdrawing group at position 6. Saturation of the 3,4 double bond increases the diuretic activity (~10-fold) and alkyl substitutions at positions 2 and/or 3 increases activity as well as the duration of action.



1,2,4-Benzothiadiazine-1,1-dioxide
(thiazide diuretic pharmacophore)

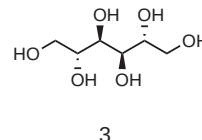
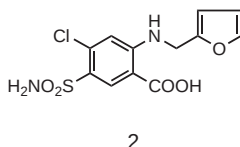
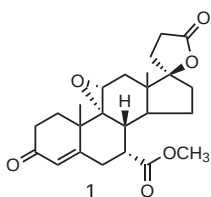
In the clinical scenario, the pharmacist chose to use a combination of an ACE inhibitor (see Chapter 23) and a thiazide diuretic to treat the patient's blood pressure. By doing this, the patient's blood pressure is lowered by different mechanisms of action. In addition, the K⁺ lowering property of the hydrochlorothiazide may be countered with the K⁺ raising property of the lisinopril.

CASE STUDY

S. William Zito and Victoria Roche

SY is a 63-year-old man who presents to the emergency room complaining of breathlessness for the past 3 days. His medical history shows that he had a myocardial infarction (MI) 3 years ago, which was followed by successful bypass surgery. SY has been asymptomatic since surgery with no complaints of chest pain. SY runs a small neighborhood grocery store and over the last 3 months has experienced shortness of breath while unloading groceries and climbing stairs. Two weeks ago he was unable to complete his daily 1-mile walk at the high school track, and 4 days ago he woke at 2 AM short of breath

and had to sleep in his recliner the rest of the night. Yesterday, he became breathless walking from one room to another. He presents today with extreme shortness of breath and swelling in his feet and ankles, and he denies chest pain. A diagnosis of congestive heart failure (CHF) is made based on SY's symptoms, history of MI, and chest X-ray revealing cardiomegaly. The physician wants to initiate treatment with a diuretic to reduce edema before beginning treatment with digoxin for CHF. Evaluate the following three choices for appropriate use in this case.



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 23

Agents Affecting the Renin-Angiotensin Pathway and Calcium Blockers

MARC HARROLD

Drugs Covered in This Chapter*

ANGIOTENSIN-CONVERTING ENZYME INHIBITORS

- Captopril
- Enalapril
- Fosinopril
- Lisinopril
- Moexipril
- Perindopril
- Quinapril
- Ramipril
- *Spirapril*
- Trandolapril

ANGIOTENSIN II RECEPTOR BLOCKERS

- Azilsartan
- Candesartan
- Eprosartan
- Irbesartan
- Losartan
- Olmesartan
- Telmisartan
- Valsartan

CALCIUM CHANNEL BLOCKERS

- Amlodipine
- *Bepridil*

- Clevidipine
- Diltiazem
- Felodipine
- Nicardipine
- Nifedipine
- Nimodipine
- Nisoldipine
- Verapamil
- Ziconotide

RENIN INHIBITOR

- Aliskiren

Abbreviations

ACE, angiotensin-converting enzyme

ARB, angiotensin receptor blocker

AT₁, angiotensin II subtype 1 receptor

AT₂, angiotensin II subtype 2 receptor

AV, atrioventricular

BPF, bradykinin-potentiating factor

CNS, central nervous system

1,4-DHPs, 1,4-dihydropyridines

HVA, high-voltage activated

IC₅₀, half maximal inhibitory concentration

IV, intravenous

LVA, low-voltage activated

MI, myocardial infarction

PSVT, paroxysmal supraventricular tachycardia

SAR, structure–activity relationship

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

SCENARIO

Thomas L. Rihn, PharmD

LB is a 63-year-old man with a 4-year history of hypertension, which has not been well controlled. He has been primarily treated with attempts at lifestyle modification but has not followed up with his appointments. He has also been an insulin-dependent diabetic for 17 years (with proteinuria) and has experienced a few anginal attacks, which have been relieved by nitroglycerin. He presents at the clinic today with a blood pressure of 162/101 mm Hg (this has been his average, measured on three separate occasions over the past week). Considering LB's blood pressure level, what therapeutic approach should be used?

The following questions or considerations are important in deciding on an initial therapeutic approach:

- Begin treatment with a thiazide only?
- Consider initiating therapy with two agents, one of which should be a thiazide diuretic, the other ACEI?
- Begin only with an ARB?
- Begin a triple drug regimen to bring patient under control immediately?
- Refer to a hypertension specialist?

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

THE RENIN-ANGIOTENSIN PATHWAY

The renin-angiotensin system is a complex, highly regulated pathway that is integral in the regulation of blood volume, electrolyte balance, and arterial blood pressure. It consists of two main enzymes, renin and angiotensin-converting enzyme (ACE), the primary purpose of which is to release angiotensin II from its endogenous precursor, angiotensinogen (Fig. 23.1). Angiotensin II is a potent vasoconstrictor that affects peripheral resistance, renal function, and cardiovascular structure (1).

History and Overview of Pathway

Historically, the renin-angiotensin system dates back to 1898, when Tiegerstedt and Bergman demonstrated the existence of a pressor substance in crude kidney extracts. A little more than 40 years later, two independent research groups discovered that this pressor substance, which had previously been named renin, actually was an enzyme and that the true pressor substance was a peptide formed by the catalytic action of renin. This peptide pressor substance initially was assigned two different names, angiotonin and hypertensin; however, these names eventually were combined to produce the current designation, angiotensin. In the 1950s, it was discovered that angiotensin exists as both an inactive decapeptide, angiotensin I, and an active octapeptide, angiotensin II, and that the conversion of angiotensin I to angiotensin II is catalyzed by an enzyme distinct from renin (2).

Angiotensinogen is an α_2 -globulin with a molecular weight of 58,000 to 61,000 daltons. It contains 452 amino acids, is abundant in the plasma, and is continually synthesized and secreted by the liver. A number of hormones, including glucocorticoids, thyroid hormone, and angiotensin II, stimulate its synthesis. The most important portion of this compound is the N-terminus, specifically the Leu¹⁰-Val¹¹ bond. This bond is cleaved by renin and

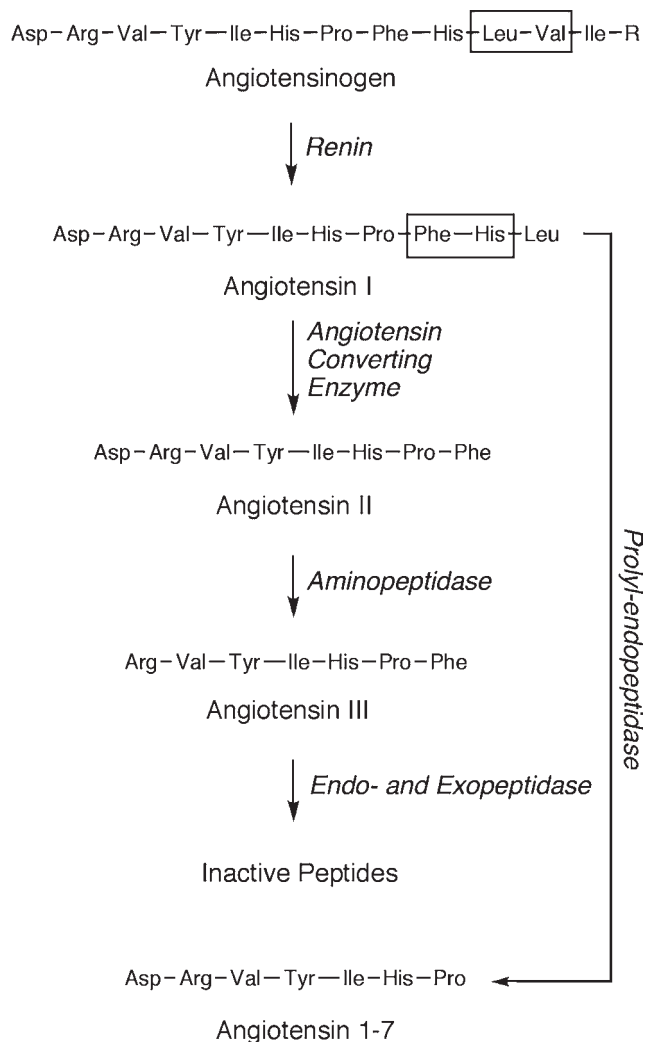


FIGURE 23.1 Schematic representation of the renin-angiotensin pathway. The labile peptide bonds of angiotensinogen and angiotensin I are highlighted.

CLINICAL SIGNIFICANCE



The treatment of hypertension and congestive heart failure (CHF) has improved significantly with the introduction of angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), and calcium channel blockers. The SARs and structural modifications of these agents have produced major therapeutic advances. These drugs have become cornerstones of therapy today. For example, more than 30 years ago, captopril was the first ACE inhibitor to be developed. Subsequent molecular modifications led to the development of newer agents, such as lisinopril. Although lisinopril exerts comparable ACE inhibition, it possesses a superior pharmacokinetic profile. Instead of having to administer captopril three times daily, lisinopril can be administered once daily.

Medication compliance is notoriously poor in cardiovascular patients. Administering an ACE inhibitor such as lisinopril once

daily results in greatly enhanced medication compliance. Clinical trials have clearly demonstrated that the therapeutic outcomes of patients with hypertension and CHF have improved immensely as a result. Similar molecular enhancements have been made with the ARBs and calcium channel blockers. Their therapeutic use today is also evidence-based.

The application of basic science in modifying the chemical structure of these agents has ultimately resulted in patients living longer and suffering fewer cardiovascular events, such as myocardial infarction or worsening CHF. Importantly, their day-to-day quality of life has improved as well.

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produces the decapeptide angiotensin I. The Phe⁸-His⁹ peptide bond of angiotensin I is then cleaved by ACE to produce the octapeptide angiotensin II. Aminopeptidase can further convert angiotensin II to the active heptapeptide angiotensin III by removing the N-terminal arginine residue. Further actions of carboxypeptidases, aminopeptidases, and endopeptidases result in the formation of inactive peptide fragments. An additional compound can be formed by the action of a prolyl-endopeptidase on angiotensin I. Cleavage of the Pro⁷-Phe⁸ bond of angiotensin I produces a heptapeptide known as angiotensin 1-7. The actions of all of these compounds are discussed below.

Actions and Properties of Renin-Angiotensin Pathway Components

Renin is an aspartyl protease that determines the rate of angiotensin II production. It is a much more specific enzyme than ACE. Its primary function is to cleave the leucine-valine bond at residues 10 and 11 of angiotensinogen. The stimulation of renin release is controlled very closely by hemodynamic, neurogenic, and humoral signals (Fig. 23.2). Hemodynamic signals involve the renal juxtaglomerular cells. These cells are sensitive to the hemodynamic stretch of the afferent glomerular arteriole. An increase in the stretch implies a raised blood pressure and results in a reduced release of renin, whereas a decrease in the stretch increases renin secretion. Additionally, these cells also are sensitive to NaCl flux across the adjacent macula densa. Increases in NaCl flux across the macula densa inhibit renin release, but decreases in the flux stimulate release. Furthermore, neurogenic enhancement of renin release occurs via activation of β_1 receptors. Finally, a variety of hormonal signals influence the release of renin. Somatostatin,

atrial natriuretic factor, and angiotensin II inhibit renin release, whereas vasoactive intestinal peptide, parathyroid hormone, and glucagon stimulate renin release (3).

In contrast, ACE, also known as kininase II, is a zinc protease that is under minimal physiologic control. It is not a rate-limiting step in the generation of angiotensin II and is a relatively nonspecific dipeptidyl carboxypeptidase that requires only a tripeptide sequence as a substrate. The only structural feature required by ACE is that the penultimate amino acid in the peptide substrate cannot be proline. For this reason, angiotensin II, which

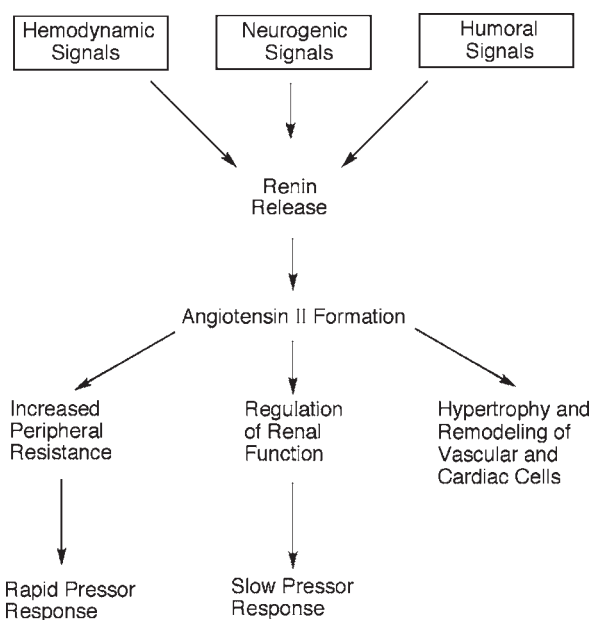


FIGURE 23.2 Summary of the factors involved in renin release and the effects mediated by angiotensin II.

contains a proline in the penultimate position, is not further metabolized by ACE. The lack of specificity and control exhibited by ACE results in its involvement in the bradykinin pathway (Fig. 23.3). Bradykinin is a nonapeptide that acts locally to produce pain, cause vasodilation, increase vascular permeability, stimulate prostaglandin synthesis, and cause bronchoconstriction. Similar to angiotensin II, bradykinin is produced by proteolytic cleavage of a precursor peptide. Cleavage of kininogens by the protease kallikrein produces a decapeptide known as either kallidin or lysyl-bradykinin. Subsequent cleavage of the N-terminal lysine by aminopeptidase produces bradykinin. The degradation of bradykinin to inactive peptides occurs through the actions of ACE. Thus, ACE not only produces a potent vasoconstrictor but also inactivates a potent vasodilator (1,3,4).

Angiotensin II is the dominant peptide produced by the renin-angiotensin pathway (Fig. 23.2). It is a potent vasoconstrictor that increases total peripheral resistance through a variety of mechanisms: direct vasoconstriction, enhancement of both catecholamine release and neurotransmission within the peripheral nervous system, and increased sympathetic discharge. The result of all these actions is a rapid pressor response. Additionally, angiotensin II causes a slow pressor response, resulting in long-term stabilization of arterial blood pressure. This long-term effect is accomplished by the regulation of renal function. Angiotensin II directly increases sodium reabsorption in the proximal tubule. It also alters renal hemodynamics and causes the release of aldosterone from the adrenal cortex. Finally, angiotensin II causes the hypertrophy and remodeling of both vascular and cardiac cells through a variety of hemodynamic and nonhemodynamic effects (1).

The secondary peptides, angiotensin III and angiotensin 1-7, can contribute to the overall effects of the renin-angiotensin pathway. Angiotensin III is equipotent with angiotensin II in stimulating aldosterone secretion; however, it is only 10% to 25% as potent in increasing blood pressure. In contrast, angiotensin 1-7 does not cause either aldosterone secretion or vasoconstriction, but it does have potent effects that are distinct from those of angiotensin II. Similar to angiotensin II, angiotensin 1-7 causes neuronal excitation and vasopressin release.

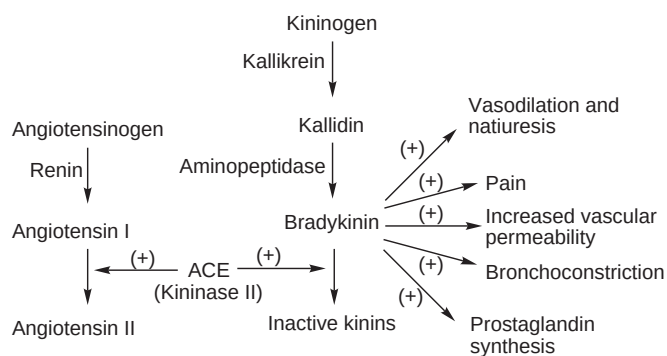


FIGURE 23.3 Schematic representation of the bradykinin pathway and its relationship to ACE and the renin-angiotensin pathway.

Additionally, it enhances the production of prostaglandins via a receptor-mediated process that does not involve an increase in intracellular calcium levels. It has been proposed to be important in the modulation of cell-to-cell interactions in cardiovascular and neural tissues (5).

ROLE OF THE RENIN-ANGIOTENSIN PATHWAY IN CARDIOVASCULAR DISORDERS

Because the renin-angiotensin pathway is central to the maintenance of blood volume, arterial blood pressure, and electrolyte balance, abnormalities in this pathway (e.g., excessive release of renin and overproduction of angiotensin II) can contribute to a variety of cardiovascular disorders. Specifically, overactivity of this pathway can result in hypertension or heart failure via the mechanisms previously described. Abnormally high levels of angiotensin II can contribute to hypertension through both rapid and slow pressor responses. In addition, high levels of angiotensin II can cause cellular hypertrophy and increase both afterload and wall tension. All of these events can cause or exacerbate heart failure.

High blood pressure is a relatively common disorder, affecting more than 50 million Americans. It is more prevalent in males than in females and in blacks than in Caucasians. Onset usually begins during the third, fourth, and fifth decades of life, and the incidence of the disorder increases with age. Hypertension is classified as either primary or secondary. Primary hypertension, also known as essential hypertension, is the most prevalent form of the disorder and is defined as high blood pressure of an unknown etiology. Most cases of primary hypertension are thought to result from a variety of underlying pathophysiologic mechanisms and not from a single, specific cause. Additionally, genetic factors appear to be important in the development of primary hypertension. Secondary hypertension is associated with a specific disorder (e.g., chronic renal disease, pheochromocytoma, and Cushing syndrome), is present in approximately 5% of individuals with high blood pressure, and, in some instances, is potentially curable. Secondary hypertension is much more common in children than in adults (6).

Heart failure (previously designated as congestive heart failure) affects approximately 5 million Americans and is the most common hospital discharge diagnosis in patients older than 65 years. The overall 5-year survival rate is approximately 50% for all patients, with women having an overall lower mortality rate than men. The disease results from conditions in which the heart is unable to supply blood at a rate sufficient to meet the demands of the body. Similar to hypertension, this pathophysiologic state can occur via a variety of mechanisms. Any pathophysiologic event that causes either systolic or diastolic dysfunction will result in heart failure. Systolic dysfunction, or decreased contractility, can be caused by dilated cardiomyopathies, ventricular hypertrophy, or a reduction in muscle mass. Diastolic dysfunction, or restriction in ventricular filling, can be

caused by increased ventricular stiffness, mitral or tricuspid valve stenosis, or pericardial disease. Both ventricular hypertrophy and myocardial ischemia can contribute to increased ventricular stiffness. Angiotensin II causes and/or exacerbates heart failure by increasing systemic vascular resistance, promoting sodium retention, stimulating aldosterone release, and stimulating ventricular hypertrophy and remodeling (7).

OVERVIEW OF DRUG THERAPY AFFECTING THE RENIN-ANGIOTENSIN PATHWAY

Because angiotensin II produces the majority of the effects attributed to the renin-angiotensin pathway, compounds that can block either the synthesis of angiotensin II or the binding of angiotensin II to its receptor should attenuate the actions of this pathway. Indeed, enzyme inhibitors of both renin and ACE, as well as receptor antagonists of angiotensin II, have all been shown to produce beneficial effects in decreasing the actions of angiotensin II. Inhibitors of ACE were the first class of compounds to be marketed. This occurred in 1981 with the approval by the U.S. Food and Drug Administration of captopril. Fourteen years later, losartan was approved as the first angiotensin II receptor blocker (previous referred to as an angiotensin II receptor antagonist), and in 2007, aliskiren was approved as the first orally active renin inhibitor. The development, structure–activity relationships (SARs), physicochemical properties, interactions, and indications of these classes of drugs are discussed later in this chapter.

ANGIOTENSIN-CONVERTING ENZYME INHIBITORS

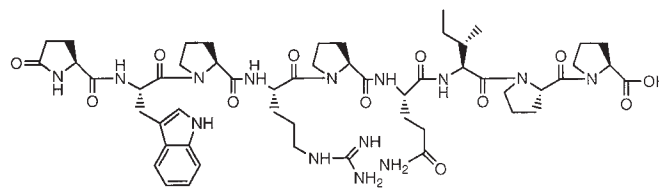
Currently, there are 11 ACE inhibitors approved for therapeutic use in the United States. These compounds can be subclassified into three groups based on their chemical composition: sulfhydryl-containing inhibitors (exemplified by captopril), dicarboxylate-containing inhibitors (exemplified by enalapril), and phosphonate-containing inhibitors (exemplified by fosinopril). Captopril and fosinopril are the lone representatives of their respective chemical subclassifications, whereas the majority of the inhibitors contain the dicarboxylate functionality. All of these compounds effectively block the conversion of angiotensin I to angiotensin II and have similar therapeutic and physiologic effects. The compounds differ primarily in their potency and pharmacokinetic profiles (1). Additionally, the sulfhydryl group in captopril is responsible for certain effects not seen with the other agents. Detailed descriptions of the rationale for the development of captopril, enalapril, and fosinopril are provided in the following sections.

Sulfhydryl-Containing Inhibitors: Development of Captopril

In 1965, Ferreira et al. (8) reported that the venom of the South American pit viper (*Bothrops jararaca*) contained

factors that potentiated the action of bradykinin. These factors, originally designated as bradykinin-potentiating factors (BPFs), were isolated and found to be a family of peptides containing 5- to 13-amino acid residues. Their actions in potentiating bradykinin were subsequently linked to their ability to inhibit the enzymatic degradation of bradykinin. Soon thereafter, Bakhle et al. (9) reported that these same peptides also inhibited the enzymatic conversion of angiotensin I to angiotensin II. This latter enzyme, ACE, is now known to be identical with the former bradykininase enzyme (kininase II). Even at the time of these initial discoveries, however, BPFs were seen as lead compounds for the development of new antihypertensive agents, because they possessed dual activities—inhibition of the degradation of bradykinin, a potent vasodilator, and inhibition of the biosynthesis of angiotensin II, a potent vasoconstrictor (10).

A nonapeptide, SQ 20,881 (teprotide), isolated from the original BPFs, had the greatest in vivo potency in inhibiting ACE and was shown to consistently lower blood pressure in patients with essential hypertension. It also exerted beneficial effects in patients with heart failure; however, because of its peptide nature and lack of oral activity, teprotide had limited activity in the therapeutic treatment of these diseases (10,11).



SQ 20,881

Cushman, Ondetti, and coworkers (12–14) used SQ 20,881 and other peptide analogs to provide an enhanced understanding of the enzymatic properties of ACE. Using knowledge of substrate-binding specificities and the fact that ACE has properties similar to those of pancreatic carboxypeptidases, these researchers developed a hypothetical model of the enzyme active site. Carboxypeptidase A, like ACE, is a zinc-containing exopeptidase. The binding of a substrate to carboxypeptidase A involves three major interactions (Fig. 23.4A). First, the negatively charged carboxylate terminus of the amino acid substrate binds to the positively charged Arg-145 on the enzyme. Second, a hydrophobic pocket in the enzyme provides specificity for a C-terminal aromatic or nonpolar residue. Third, the zinc atom is located close to the labile peptide bond and serves to stabilize the negatively charged tetrahedral intermediate, which results when a molecule of water attacks the carbonyl bond between the C-terminal and penultimate amino acid residues (15). Similarly, the binding of substrates to ACE was proposed to involve three or four major interactions (Fig. 23.4B). First, the negatively charged carboxylate terminus of angiotensin I and other substrates was assumed

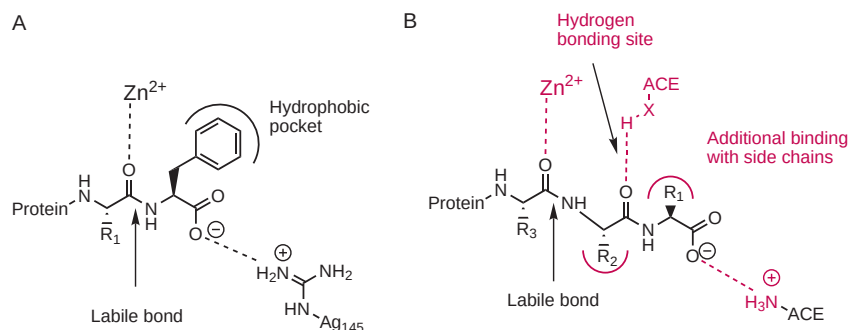


FIGURE 23.4 A model of substrate binding to carboxypeptidase A (A) and ACE (B). ACE substrate binding sites are highlighted in red.

to occur via an ionic bond with a positively charged amine on ACE. Second, the role of the zinc atom in the mechanism of ACE hydrolysis was assumed to be similar to that of carboxypeptidase A. Because ACE cleaves dipeptides instead of single amino acids, the position of the zinc atom was assumed to be located two amino acids away from the cationic center for it to be adjacent to the labile peptide bond. Third, the side chains R_1 and R_2 could contribute to the overall binding affinity; however, ACE, unlike carboxypeptidase A, does not show specificity for C-terminal hydrophobic amino acids and was not expected to have a hydrophobic binding pocket. Finally, the terminal peptide bond is nonlabile and was assumed to provide hydrogen bonding between the substrate and ACE.

The development of captopril and other orally active ACE inhibitors began with the observation that D-2-benzylsuccinic acid was an extremely potent inhibitor of carboxypeptidase A (12–14). The binding of this compound to carboxypeptidase A (Fig. 23.5A) is very similar to that seen for substrates with the exception that the zinc ion binds to a carboxylate group instead of the labile peptide bond. Byers and Wolfenden (16) proposed that this compound is a by-product analog that contains structural features of both products of peptide hydrolysis. Most of the structural features of the compound are identical to the terminal amino acid of the substrate

(Fig. 23.4A), whereas the additional carboxylate group is able to mimic the carboxylate group that would be produced during peptide hydrolysis (16). Applying this concept to the hypothetical model of ACE described earlier resulted in the synthesis and evaluation of a series of succinic acid derivatives (Fig. 23.5B). Because proline was present as the C-terminal amino acid in SQ 20,881 as well as in other potent, inhibitory snake venom peptides, it was included in the structure of newly designed inhibitors. The first inhibitor to be synthesized and tested was succinyl-L-proline (Fig. 23.6). This compound proved to be somewhat disappointing. Although it provided reasonable specificity for ACE, it was only approximately 1/500 as potent as SQ 20,881.

Substitution of other amino acids in place of proline produced compounds that were even less potent; hence, all subsequent SAR studies were conducted using analogs of L-proline (Fig. 23.6). The addition of a methyl group to the 2-position of succinyl-L-proline to mimic the amino acid side chain, R_2 , of the substrate enhanced activity, but only marginally. D-2-Methylsuccinyl-L-proline had effects similar to SQ 20,881 but was still only 1/300 as potent. The D-isomer, rather than the L-isomer normally seen for amino acids, was necessary because of the isosteric replacement of an NH_2 with a CH_2 present in succinyl-L-proline. A comparison of the R_2 group of the substrate (Fig. 23.4B) with the methyl group of D-2-methylsuccinyl-L-proline

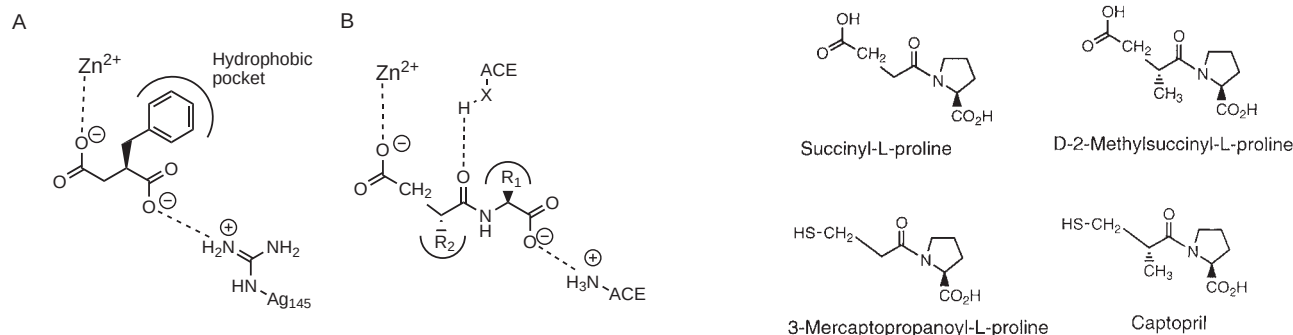


FIGURE 23.5 Inhibitor binding models of (A) D-2-benzylsuccinic acid to carboxypeptidase A and (B) succinic acid derivatives to ACE.

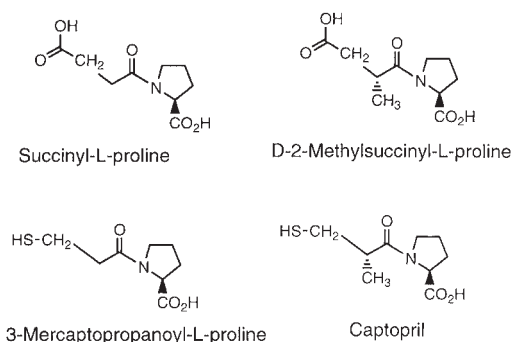


FIGURE 23.6 Compounds prepared in the development of captopril.

illustrates that this methyl group occupies the same binding site as the side chain of an L-amino group.

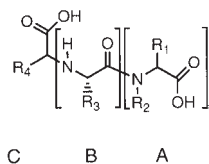
One of the most important alterations to succinyl-L-proline was the replacement of the succinyl carboxylate with other groups having enhanced affinity for the zinc atom bound to ACE. Replacement of this carboxylate with a sulfhydryl group produced 3-mercaptopropanoyl-L-proline. This compound has a half maximal inhibitory concentration (IC_{50}) value of 200 nmol/L and is greater than 1,000-fold more potent than succinyl-L-proline (Fig. 23.6). Additionally, it is 10- to 20-fold more potent than SQ 20,881 in inhibiting contractile and vasopressor responses to angiotensin I. Addition of a 2-D-methyl group further enhanced activity. The resulting compound, captopril (Fig. 23.6), is a competitive inhibitor of ACE with a K_i value of 1.7 nmol/L and was the first ACE inhibitor to be marketed.

The sulfhydryl group of captopril proved to be responsible not only for the excellent inhibitory activity of the compound but also for the two most common side effects, skin rashes and taste disturbances (e.g., metallic taste and loss of taste). These side effects usually subsided on dosage reduction or discontinuation of captopril. They were attributed to the presence of the sulfhydryl group, because similar effects had been observed with penicillamine, a sulfhydryl-containing agent used to treat Wilson disease and rheumatoid arthritis (17,18).

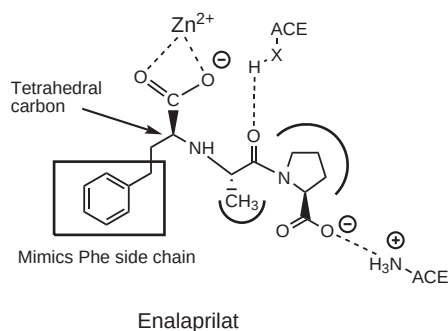
Dicarboxylate-Containing Inhibitors

Development of Enalapril

Researchers at Merck (19) sought to develop compounds that lacked the sulfhydryl group of captopril yet maintained some ability to chelate zinc. Compounds having the general structure shown below were designed to meet this objective.



These compounds are tripeptide substrate analogs in which the C-terminal (A) and penultimate (B) amino



acids are retained but the third amino acid is isosterically replaced by a substituted N-carboxymethyl group (C). Similar to the results seen in the development of captopril, C-terminal proline analogs provided optimum activity. The use of a methyl group at R_3 (i.e., B = Ala) and a phenylethyl group at R_4 resulted in enalaprilat (Fig. 23.7). In comparing the activity of captopril and enalaprilat, it was found that enalaprilat, with a K_i of 0.2 nmol/L, was approximately 10-fold more potent than captopril. Studies investigating the binding of enalaprilat revealed that its ability to chelate the enzyme-bound zinc atom was significantly less than that of captopril. The enhanced binding was proposed to be caused by the ability to mimic the transition state of angiotensin I hydrolysis. As shown in Figure 23.7, enalaprilat possess a tetrahedral carbon in place of the labile peptide bond. The secondary amine, the carboxylic acid, and phenylethyl groups all contribute to the overall binding of the compound to ACE. The secondary amine is located at the same position as the labile amide nitrogen, the ionized carboxylic acid can form an ionic bond with the zinc atom, and the phenylethyl group mimics the hydrophobic side chain of the Phe amino acid, which is present in angiotensin I.

Despite excellent intravenous (IV) activity, enalaprilat has very poor oral bioavailability. Esterification of enalaprilat produced enalapril (Fig. 23.8), a compound with superior oral bioavailability. The combination of structural features in enalaprilat, especially the two carboxylate groups and the secondary amine, is responsible for its overall low lipophilicity and poor oral bioavailability. Zwitterion formation also has been suggested to contribute to the low oral activity (20), and a comparison of the pK_a values for the secondary amine of enalaprilat and enalapril supports this explanation. Ionization of the adjacent carboxylate in enalaprilat greatly enhances the basicity of the secondary amine such that the pK_a of the amine in this compound is 8.02, whereas in enalapril, it is only 5.49. Thus, in the small intestine, the amine in enalaprilat will be primarily ionized and form a zwitterion with the adjacent carboxylate, but the amine in enalapril will be primarily un-ionized (21).

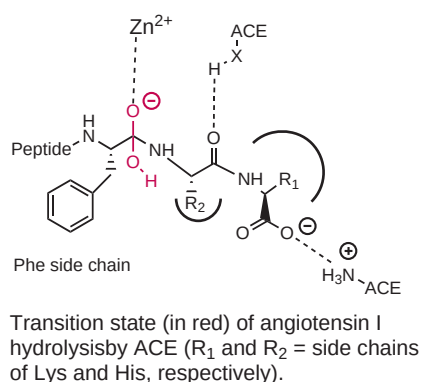


FIGURE 23.7 A comparison of enalaprilat and the transition state of angiotensin I hydrolysis by ACE.

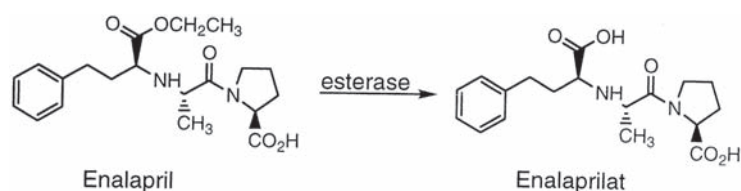


FIGURE 23.8 Bioactivation of enalapril.

IV administration of either enalapril or enalaprilat produced similar effects on angiotensin II production despite the fact that enalapril showed a 1,000-fold decrease in *in vitro* activity. Subsequent studies showed that enalapril undergoes bioactivation and, thus, is a pro-drug of enalaprilat. Because human plasma was reported to lack enalapril esterolytic activity, bioactivation by hepatic esterases (Fig. 23.8) has been suggested as the most probable mechanism for enalaprilat formation (22,23).

Additional Dicarboxylate Inhibitors

Eight other dicarboxylate inhibitors (Table 23.1) have been approved for various therapeutic indications; however, spirapril has never been marketed. Lisinopril is chemically unique in two respects. First, it contains the basic amino acid lysine ($R_1 = \text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$) instead of the standard nonpolar alanine ($R = \text{CH}_3$) residue. Second, it does not require bioactivation, because neither of the carboxylic acid groups are esterified (i.e., $R_2 = \text{H}$). Lisinopril was developed at the same time as

TABLE 23.1 Additional Dicarboxylate-Containing Angiotensin-Converting Enzyme Inhibitors

General structure:		Benazepril		
Compounds	Ring	R_1	R_2	R_3
Lisinopril		$(\text{CH}_2)_4\text{NH}_2$	H	
Moexipril		CH_3	CH_2CH_3	
Perindopril		CH_3	CH_2CH_3	CH_3
Quinapril		CH_3	CH_2CH_3	
Ramipril		CH_3	CH_2CH_3	
Spirapril		CH_3	CH_2CH_3	
Trandolapril		CH_3	CH_2CH_3	

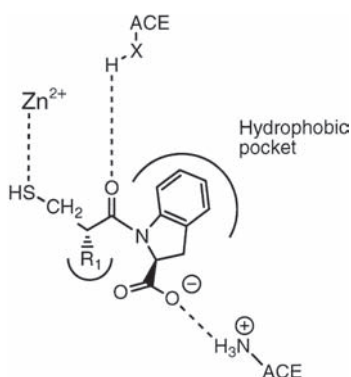


FIGURE 23.9 A modified model of ACE inhibitor binding.

enalapril. Despite the addition of another ionizable group, the oral absorption of lisinopril was found to be superior to that of enalaprilat but less than that of enalapril. In vitro studies of enalaprilat and lisinopril showed lisinopril to be slightly more potent than enalaprilat (22,23). Currently, lisinopril and captopril are the only two ACE inhibitors that are not prodrugs.

The major structural difference among the remaining ACE inhibitors is in the ring of the C-terminal amino acid. Lisinopril, like enalapril and captopril, contains the pyrrolidine ring of proline, whereas all the other compounds contain larger bicyclic or spiro ring systems. Studies of indoline analogs of captopril indicated that a hydrophobic pocket similar to that seen in carboxypeptidase A also was present in ACE. This led to a modification (Fig. 23.9) of Ondetti and Cushman's original model and the development of inhibitors that contained larger hydrophobic ring systems (24). Although this modified model was proposed for captopril analogs, it is readily adaptable to include enalaprilat analogs. In general, the varied ring systems seen in benazepril, moexipril, perindopril, quinapril, ramipril, spirapril, and trandolapril provide enhanced binding and potency. They also lead to differences in absorption, plasma protein binding, elimination, onset of action, duration of action, and dosing among the drugs. These differences are discussed in more detail in a later section in this chapter, "Pharmacokinetic Properties."

Phosphonate-Containing Inhibitors: The Development of Fosinopril

The search for ACE inhibitors that lacked the sulfhydryl group also led to the investigation of phosphorous-containing compounds (25). The phosphinic acid shown in Figure 23.10 is capable of binding to ACE in a manner similar to enalapril. The interaction of the zinc atom with the phosphinic acid is similar to that seen with sulfhydryl and carboxylate groups. Additionally, this compound is capable of forming the ionic, hydrogen, and hydrophobic bonds similar to those seen with enalapril and other dicarboxylate analogs. A feature unique to this compound is the ability of the phosphinic acid to more truly

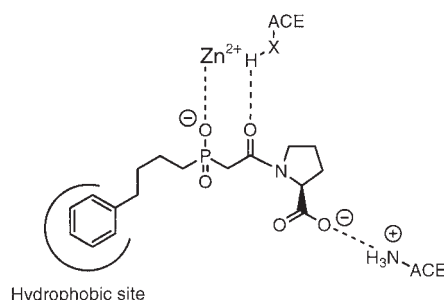


FIGURE 23.10 The binding of phosphinate analogs to ACE.

mimic the ionized, tetrahedral intermediate of peptide hydrolysis. Unlike enalapril and other dicarboxylate analogs, however, the spacing of this tetrahedral species is shorter, being only two atoms removed from the proline nitrogen. Additionally, the spacing between the proline nitrogen and the hydrophobic phenyl ring is one atom longer than that seen in the dicarboxylates.

Structural modification to investigate more hydrophobic, C-terminal ring systems, similar to that described earlier for the dicarboxylate compounds, led to a 4-cyclohexylproline analog of the original phosphinic acid. This compound, fosinoprilat (Fig. 23.11), was more potent than captopril but less potent than enalaprilat. The previously mentioned differences in the spacing of the phosphinic acid and phenyl groups may be responsible for this latter difference in potency. Similar to the dicarboxylates, fosinoprilat was too hydrophilic and exhibited poor oral activity. The prodrug fosinopril contains an (acyloxy)alkyl group that allows better lipid solubility and improved bioavailability (25). Bioactivation via esterase activity in the intestinal wall and liver produces fosinopril (Fig. 23.11).

Mechanism of Action

The ACE inhibitors attenuate the effects of the renin-angiotensin system by inhibiting the conversion of angiotensin I to angiotensin II (Fig. 23.1). They also inhibit the conversion of [des-Asp¹]angiotensin I to angiotensin III; however, this action has only a minor role in the overall cardiovascular effects of these drugs. They are selective in that they do not directly interfere with any other components of the renin-angiotensin system; however, they do cause other effects that are unrelated to the decrease in angiotensin II concentration. Inhibitors of ACE increase bradykinin levels that, in turn, stimulate prostaglandin biosynthesis (Fig. 23.3). Both of these compounds have been proposed to contribute to the overall action of ACE inhibitors. Additionally, decreased angiotensin II levels increase the release of renin and the production of angiotensin I. Because ACE is inhibited, angiotensin I is shunted toward the production of angiotensin 1-7 and other peptides. The contribution of these peptides to the overall effect of ACE inhibitors is unknown (1).

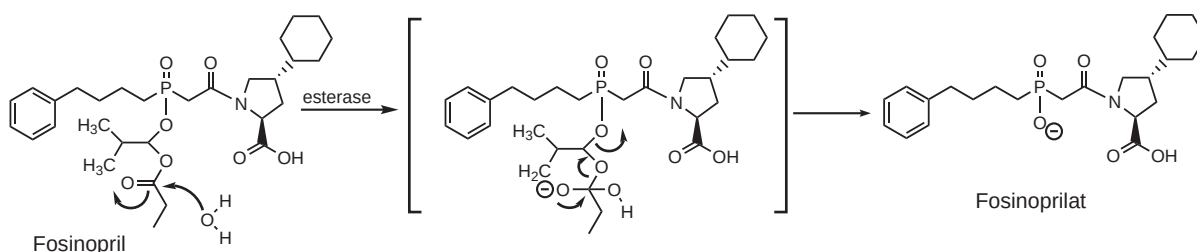


FIGURE 23.11 Bioactivation of fosinopril.

Structure–Activity Relationships

The structural characteristics for ACE inhibitory activity are given in Table 23.2. ACE is a stereoselective drug target. Because currently approved ACE inhibitors act as either di- or tripeptide substrate analogs, they must contain a stereochemistry that is consistent with the L-amino acids present in the natural substrates. This was established very early in the development of ACE inhibitors when compounds with carboxyl-terminal D-amino acids were discovered to be very poor inhibitors (26). Later work by Patchett et al. (19) reinforced this idea. They reported a 100- to 1,000-fold loss in inhibitor activity whenever the configuration of either the carboxylate or the R_1 substituent (Table 23.1) was altered. The *S,S,S*-configuration seen in enalapril and other dicarboxylate inhibitors meets the previously stated criteria and provides for optimum enzyme inhibition.

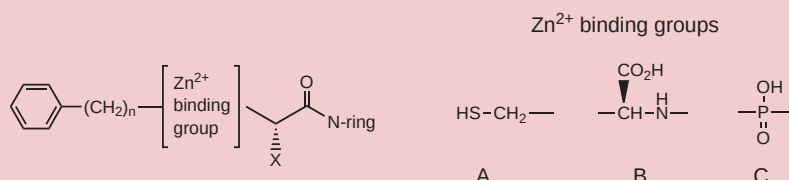
Physicochemical Properties

Captopril and fosinopril are acidic drugs, but all other ACE inhibitors are amphoteric. The carboxylic acid

attached to the N-ring is a common structural feature in all ACE inhibitors. It has a pK_a in the range of 2.5 to 3.5 and will be primarily ionized at physiologic pH. As discussed earlier with enalapril, the pK_a and ionization of the secondary amine in the dicarboxylate series depend on whether the adjacent functional group is in the prodrug or active form. In the prodrug form, the amine is adjacent to an ester, is less basic, and is primarily un-ionized at physiologic pH. Following bioactivation, the amine is adjacent to an ionized carboxylic acid that enhances both the basicity and ionization of the amine. Similarly, the basic nitrogen enhances the acidity of the adjacent carboxylic acid such that it usually has a lower pK_a than the carboxylic acid attached to the N-ring. As an example, the pK_a values of enalapril are 3.39 and 2.30. These values correspond to the carboxylic acid on the N-ring and the carboxylic acid adjacent to the amine, respectively. The analogous values for these functional groups in lisinopril are 3.3 and 1.7 (21).

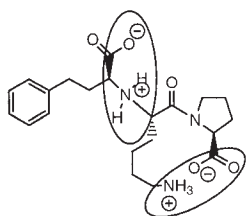
The calculated logP values (21) along with other pharmacokinetic parameters for the ACE inhibitors are shown

TABLE 23.2 Structure–Activity Relationship of ACE Inhibitors



- The N-ring must contain a carboxylic acid to mimic the C-terminal carboxylate of ACE substrates.
- Large hydrophobic heterocyclic rings (i.e., the N-ring) increase potency and alter pharmacokinetic parameters.
- The zinc binding groups can be either sulfhydryl (A), a carboxylic acid (B), or a phosphinic acid (C).
- The sulfhydryl group shows superior binding to zinc (the side chain mimicking the Phe in carboxylate and phosphinic acid compounds partially compensates for the lack of a sulfhydryl group).
- Sulfhydryl-containing compounds produce high incidence of skin rash and taste disturbances.
- Sulfhydryl-containing compounds can form dimers and disulfides, which may shorten duration of action.
- Compounds that bind to zinc through either a carboxylate or phosphinate mimic the peptide hydrolysis transition state and enhance binding.
- Esterification of the carboxylate or phosphinate produces an orally bioavailable prodrug.
- X is usually methyl to mimic the side chain of alanine. Within the dicarboxylate series, when X equals n-butylamine (lysine side chain), this produces a compound that does not require prodrug for oral activity.
- Optimum activity occurs when stereochemistry of inhibitor is consistent with L-amino acid stereochemistry present in normal substrates.

in Table 23.3. With three notable exceptions, captopril, enalaprilat, and lisinopril, all of the compounds possess good lipid solubility. Compounds that contain hydrophobic bicyclic ring systems are more lipid soluble than those that contain proline. A comparison of the logP values of benazepril, fosinopril, moexipril, perindopril, quinapril, ramipril, spirapril, andtrandolapril to those for captopril and enalapril illustrates this fact. As previously discussed, enalaprilat is much more hydrophilic than its ester prodrug and is currently the only ACE inhibitor marked for IV administration. In terms of solubility, lisinopril probably is the most interesting compound in that it is the most hydrophilic inhibitor, yet unlike enalaprilat, it is orally active. One possible explanation for this phenomenon is that in the duodenum, lisinopril will exist as a di-zwitterion in which the ionized groups can internally bind to one another. In this manner, lisinopril may be able to pass through the lipid bilayer with an overall net neutral charge.



Metabolism

Lisinopril and enalaprilat are excreted unchanged, whereas all other ACE inhibitors undergo some degree of metabolic transformation (1,27–29). As previously discussed and illustrated (Figs. 23.8 and 23.11), all dicarboxylate and phosphonate prodrugs must undergo bioactivation via hepatic esterases. Additionally, based on their structural features, specific compounds can undergo metabolic inactivation via various pathways (Fig. 23.12). Because of its sulfhydryl group, captopril is subject to oxidative dimerization or conjugation. Approximately 40% to 50% of a dose of captopril is excreted unchanged, whereas the remainder is excreted as either a disulfide dimer or a captopril-cysteine disulfide. Glucuronide conjugation has been reported for benazepril, fosinopril, quinapril, and ramipril. This conjugation can occur either with the parent prodrug or with the activated drug. Benazepril, with the *N*-substituted glycine, is especially susceptible to this reaction because of a difference in steric hindrance. For all ACE inhibitors, except benazepril, the carbon atom directly adjacent to the carboxylic acid is part of a ring system and provides some steric hindrance to conjugation. The unsubstituted methylene group (i.e., $-\text{CH}_2-$) of benazepril provides less steric hindrance and, thus, facilitates conjugation. Moexipril, perindopril, and ramipril can undergo cyclization to produce diketopiperazines. This

TABLE 23.3 Pharmacokinetic Parameters of ACE Inhibitors

Drug	Calculated LogP	Oral Bioavailability (%)	Effect of Food on Absorption	Active Metabolite	Protein Binding (%)	Onset of Action (hours)	Duration of Action (hours)	Major Route(s) of Elimination
Benazepril	5.50	37	Slows absorption	Benazeprilat	>95	1	24	Renal (primary) Biliary (secondary)
Captopril	0.27	60–75	Reduced	NA	25–30	0.25–0.50	6–12	Renal
Enalapril	2.43	60	None	Enalaprilat	50–60	1	24	Renal/fecal
Enalaprilat	1.54	NA	NA	NA	—	0.25	6	Renal
Fosinopril	6.09	36	Slows absorption	Fosinoprilat	95	1	24	Renal (50%) Hepatic (50%)
Lisinopril	1.19	25–30	None	NA	25	1	24	Renal
Moexipril	4.06	13	Reduced	Moexiprilat	50	1	24	Fecal (primary) Renal (secondary)
Perindopril	3.36	65–95	Reduced	Perindoprilat	60–80	1	24	Renal
Quinapril	4.32	60	Reduced	Quinaprilat	97	1	24	Renal
Ramipril	3.41	50–60	Slows absorption	Ramiprilat	73	1–2	24	Renal (60%) Fecal (40%)
Spirapril	3.16	50	—	Spiraprilat	—	1	24	Renal (50%) Hepatic (50%)
Trandolapril	3.97	70	Slows absorption	Trandolaprilat	80	0.5–1.0	24	Fecal (primary) Renal (secondary)

NA, not applicable; —, data not available.

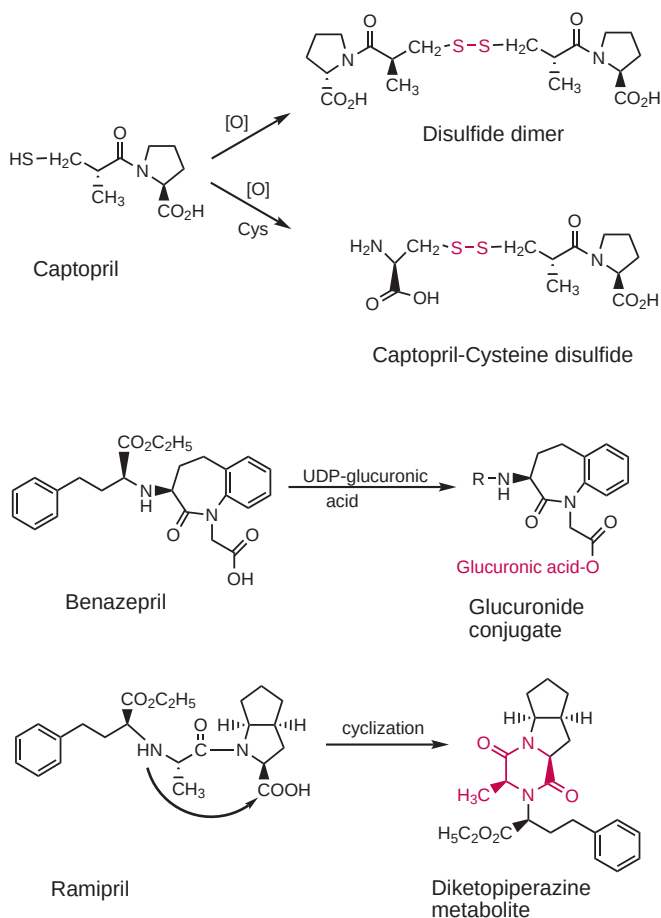


FIGURE 23.12 Metabolic routes of ACE inhibitors.

cyclization can occur with either the parent or active forms of the drugs.

A comparative study of the metabolism and biliary excretion of lisinopril, enalapril, perindopril, and ramipril revealed that whereas neither lisinopril nor enalapril underwent any appreciable metabolism beyond bioactivation of enalapril to enalaprilat, both perindopril and ramipril were extensively metabolized beyond the initial bioactivation. It was proposed that these differences in hepatic metabolism could be explained, in part, by the larger, more hydrophobic rings present on perindopril and ramipril (30).

Pharmacokinetic Parameters

The pharmacokinetic parameters and dosing information for ACE inhibitors are summarized in Tables 23.3 and 23.4, respectively (1,27–29). The oral bioavailability of this class of drugs ranges from 13% to 95%. Differences in both lipid solubility and first-pass metabolism are most likely responsible for this wide variation. Both parameters should be considered when comparing any two or more compounds. With the exceptions of enalapril and lisinopril, the concurrent administration of food adversely affects the oral absorption of ACE inhibitors. Product literature specifically instructs that captopril should be

taken 1 hour before meals and that moexipril should be taken in the fasting state. Although not specifically stated, similar instructions also should benefit patients taking an ACE inhibitor whose absorption is affected by food.

The extent of protein binding also exhibits wide variability among the different compounds. The data suggest that this variation has some correlation with the calculated logP values for the compounds (Table 23.3). Three of the more lipophilic compounds—fosinopril, quinapril, and benazepril—exhibit protein binding of greater than 90%, whereas three of the least lipophilic compounds—lisinopril, enalapril, and captopril—exhibit much lower protein binding. The lack of a protein binding value for spirapril prevents a more definitive statement on this correlation.

Renal elimination is the primary route of elimination for most ACE inhibitors. With the exceptions of fosinopril and spirapril, altered renal function significantly diminishes the plasma clearance of ACE inhibitors, including those that are eliminated primarily by the feces. Therefore, the dosage of most ACE inhibitors should be reduced in patients with renal impairment (1). Studies of fosinopril in patients with heart failure demonstrated that it is eliminated by both renal and hepatic pathways and does not require a dosage reduction in patients with renal dysfunction (31). Spirapril also exhibits similar properties; however, it is not currently available for use. It should be noted that the literature data for routes of elimination are not always consistent. The designation of renal elimination is quite clear, but it is difficult to correlate what some sources call renal/hepatic elimination with what others call renal/fecal elimination. Additionally, it is uncertain whether the designation of fecal elimination also includes unabsorbed drug. As a result, there is some variability for major routes of elimination listed in Table 23.3.

With one exception, all ACE inhibitors have a similar onset of action, duration of action, and dosing interval. Captopril has a more rapid onset of action; however, it also has a shorter duration and requires a more frequent dosing interval than any of the other compounds. When oral dosing is inappropriate, enalaprilat can be used IV. The normal dose administered to hypertensive patients is 0.625 to 1.25 mg every 6 hours. The dose usually is administered over 5 minutes and may be titrated up to 5 mg IV every 6 hours.

Therapeutic Applications

The ACE inhibitors have been approved for the treatment of hypertension, heart failure, left ventricular dysfunction (either after myocardial infarction [MI] or

CHEMICAL/PHARMACOLOGIC CLASSES USED TO TREAT HYPERTENSION

Diuretics (see Chapter 22), ACE inhibitors, angiotensin II blockers, renin inhibitors, calcium channel blockers (present chapter), central α_2 -agonists, peripheral α_1 -antagonists, β -blockers, ganglionic blockers, and vasodilators (see Chapter 24).

TABLE 23.4 Dosing Information for Orally Available ACE Inhibitors

Generic Name	Trade Name(s)	Approved Indications	Dosing Range (Treatment of Hypertension)	Maximum Daily Dose	Dose Reduction with Renal Dysfunction	Available Tablet Strengths (mg)
Benazepril	Lotensin	Hypertension	40 mg once daily or b.i.d.	80 mg	Yes	5, 10, 20, 40
Captopril	Capoten	Hypertension, heart failure, left ventricular dysfunction (post-MI), diabetic nephropathy	25–150 mg b.i.d. or t.i.d.	450 mg	Yes	12.5, 25, 50, 100
Enalapril	Vasotec	Hypertension, heart failure, left ventricular dysfunction (asymptomatic)	2.5–40 mg once daily or b.i.d.	40 mg	Yes	2.5, 5, 10, 20
Fosinopril	Monopril	Hypertension, heart failure	10–40 mg once daily	80 mg	No	10, 20, 40
Lisinopril	Prinivil, Zestril	Hypertension, heart failure, improve survival post-MI	10–40 mg once daily	80 mg	Yes	2.5, 5, 10, 20, 30, 40
Moexipril	Univasc	Hypertension	7.5–30 mg once daily or b.i.d.	30 mg	Yes	7.5, 15
Perindopril	Aceon	Hypertension, stable CAD	4–8 mg once daily or b.i.d.	16 mg	Yes	2, 4, 8
Quinapril	Accupril	Hypertension, heart failure	10–80 mg once daily or b.i.d.	80 mg	Yes	5, 10, 20, 40
Ramipril	Altace	Hypertension, heart failure (post-MI), reduce risk of MI, stroke, and death from cardiovascular causes	2.5–20 mg once daily or b.i.d.	20 mg	Yes	1.25, 2.5, 5, 10
Trandolapril	Mavik	Hypertension, heart failure (post-MI), left ventricular dysfunction (post-MI)	1–4 mg once daily	8 mg	Yes	1, 2, 4

b.i.d., twice a day; CAD, coronary artery disease; MI, myocardial infarction; t.i.d., three times a day.

asymptomatic), improved survival after MI, stable coronary heart disease, diabetic nephropathy, and reduction of the risk of MI, stroke, and death from cardiovascular causes. Although all ACE inhibitors possess the same physiologic actions and, thus, should produce similar therapeutic effects, the approved indications differ among the currently available agents (Table 23.4). Inhibitors of ACE have been designated as first-line agents for the treatment of hypertension (32) and are effective for a variety of cardiovascular disorders. They can be used either individually or with other classes of compounds. They are especially useful in treating patients with hypertension who also suffer from heart failure, left ventricular dysfunction, or diabetes. Arterial and venous dilation seen with ACE inhibitors not only lowers blood pressure but also has favorable effects on both preload and afterload in patients with heart failure. Additionally, the ability of ACE inhibitors to cause regression of left ventricular hypertrophy has been demonstrated to reduce the incidence of further heart disease in patients with hypertension. The use of ACE inhibitors in patients with MI is similarly based on the ability of ACE inhibitors to decrease mortality by preventing postinfarction left ventricular hypertrophy

and heart failure. Current recommendations to give ACE inhibitors to all patients with impaired left ventricular systolic impairment regardless of the presence of observable symptoms also are based on the ability of these inhibitors to block the vascular and cardiac hypertrophy and remodeling caused by angiotensin II. Inhibitors of ACE also have been reported to slow the progression of diabetic nephropathy and, thus, are preferred agents in the treatment of hypertension in patients with diabetes. It also has been suggested that ACE inhibitors be used in

COMBINATION PRODUCTS THAT INCLUDE AN ACE INHIBITOR

ACE Inhibitor/Diuretic: benazepril/hydrochlorothiazide, captopril/hydrochlorothiazide, enalapril/hydrochlorothiazide, fosinopril/hydrochlorothiazide, lisinopril/hydrochlorothiazide, moexipril/hydrochlorothiazide, and quinapril/hydrochlorothiazide

ACE Inhibitor/Calcium Channel Blocker: benazepril/amlodipine, enalapril/diltiazem, enalapril/felodipine, trandolapril/verapamil

UNLABELED USES

Hypertensive crises, renovascular hypertension, neonatal and childhood hypertension, stroke prevention, migraine prophylaxis, nondiabetic nephropathy, chronic kidney disease, scleroderma renal crisis, Raynaud phenomenon, and Bartter syndrome (27,28).

patients with diabetic nephropathy regardless of the presence or absence of hypertension (1,6,7,28).

Adverse Effects and Drug Interactions

The most prevalent or significant side effects of ACE inhibitors are listed below, whereas drug interactions for ACE inhibitors are listed in Table 23.5 (1,27,28). Some adverse effects can be attributed to specific functional groups within individual agents, whereas others can be directly related to the mechanism of action of this class

ADVERSE EFFECTS OF ACE INHIBITORS

Hypotension, hyperkalemia, cough, rash, taste disturbances, headache, dizziness, fatigue, nausea, vomiting, diarrhea, acute renal failure, neutropenia, proteinuria, and angioedema

of compounds. The higher incidence of maculopapular rashes and taste disturbances observed among those using captopril have been linked to the presence of the sulfhydryl group in this compound. All ACE inhibitors can cause hypotension, hyperkalemia, and a dry cough. Hypotension results from an extension of the desired physiologic effect, whereas hyperkalemia results from a decrease in aldosterone secretion secondary to a decrease in angiotensin II production. Cough is by far the most prevalent and bothersome side effect seen with the use of ACE inhibitors. It is seen in 5% to 20% of patients, usually

PEPTIDE MIMETICS: DESIGN OF AGONISTS/ANTAGONISTS

Peptide mimetics have been defined as molecules that mimic the action of peptides, have no peptide bonds (i.e., no amide bonds between amino acids), and have a molecular weight of less than 700 daltons. In comparison with peptide drugs, peptide mimetics have numerous pharmaceutical advantages. Foremost among these are increased bioavailability and duration of action. The majority of known peptide mimetics have been discovered by random screening techniques; however, this process is costly, labor intensive, and unpredictable.

A more logical and rational approach is *de novo* peptide mimetic design (33), and an example of this approach is illustrated in Figure 23.13. In this example, the overall process is divided into three steps (A–C). Initially, the amino acids that comprise the pharmacophore of the peptide must be identified. Thus, a knowledge of the SARs for the peptide under consideration is essential. In Figure 23.13A, the side chains present on amino acid residues 1, 3, and 5 of a hypothetical heptapeptide are assumed to comprise the pharmacophore, and the remainder of the peptide is assumed to provide the proper structural support for these key groups. In the second step of this *de novo* design process, the proper spatial arrangement of the pharmacophoric groups must be elucidated. Nuclear magnetic resonance spectroscopy, x-ray diffraction studies, and molecular modeling programs that allow energy-minimization procedures and molecular dynamics simulation can be used to construct a model of the biologically active conformation. Returning to the example, the side chains representing the pharmacophore are assumed to be located on the inside of the peptide, whereas the remaining residues are assumed to be located on the outside of the peptide (Fig. 23.13B). In the final step of the process, the pharmacophoric groups must be mounted on a nonpeptide template in such a manner that they retain the proper spatial arrangement found in the original peptide. This is shown in Figure 23.13C, where side chains 1, 3, and 5 of the original peptide are connected to a rigid template

(represented by the polygon). A variety of aromatic ring systems (e.g., benzene, biphenyl, phenanthrene, and benzodiazepine) can be used to provide the rigid template, and appropriately placed alkyl groups can be used to enhance spacing and increase flexibility. Additionally, isosteres of the original pharmacophoric groups may be used to circumvent specific synthetic problems (34).

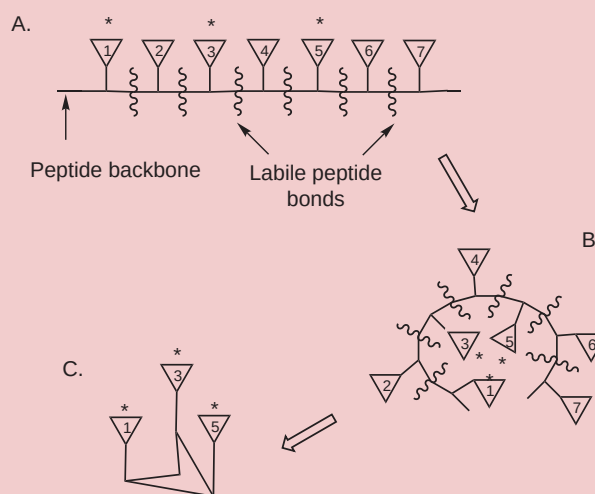


FIGURE 23.13 A general process for the rational design of peptide mimetics: (A) identification of crucial pharmacophoric groups, (B) determination of the spatial arrangement of these groups, and (C) use of a template to mount the key functional groups in their proper conformation. Groups highlighted with an asterisk comprise the pharmacophore of the heptapeptide. (From Harrold MW. Preparing students for future therapies: the development of novel agents to control the renin-angiotensin system. *Am J Pharm Educ* 1997;61:173–178, with permission.)

TABLE 23.5 Drug Interactions for ACE Inhibitors

Drug	ACE Inhibitor	Result of Interaction
Allopurinol	Captopril	Increased risk of hypersensitivity
Antacids	All	Decreased bioavailability of ACE inhibitor (more likely with captopril and fosinopril)
Capsaicin	All	Exacerbation of cough
Digoxin	All	Either increased or decreased plasma digoxin levels
Diuretics	All	Potential excessive reduction in blood pressure; the effects of loop diuretics may be reduced
Iron salts	Captopril	Reduction of captopril levels unless administration is separated by at least 2 hours
K ⁺ preparations or K ⁺ -sparing diuretics	All	Elevated serum potassium levels
Lithium	All	Increased serum lithium levels
NSAIDs	All	Decreased hypotensive effects
Phenothiazides	All	Increased pharmacologic effects of ACE inhibitor
Probenecid	Captopril	Decreased clearance and increased blood levels of captopril
Rifampin	Enalapril	Decreased pharmacologic effects of enalapril
Tetracycline	Quinapril	Decreased absorption of tetracycline (may result from high magnesium content of quinapril tablets)

NSAIDs, nonsteroidal anti-inflammatory drugs.

is not dose related, and apparently results from the lack of selectivity of this class of drugs. As previously discussed, ACE inhibitors also prevent the breakdown of bradykinin (Fig. 23.3), and because bradykinin stimulates prostaglandin synthesis, prostaglandin levels also increase. The increased levels of both bradykinin and prostaglandin have been proposed to be responsible for the cough (35).

The use of ACE inhibitors during pregnancy is contraindicated. This class of compounds is not teratogenic during the first trimester, but administration during the second and third trimester is associated with an increased incidence of fetal morbidity and mortality. Inhibitors of ACE can be used in women of childbearing age; however, they should be discontinued as soon as pregnancy is confirmed.

ANGIOTENSIN II RECEPTOR BLOCKERS

From an historical perspective, the angiotensin II receptor was the initial target for developing compounds that could inhibit the renin-angiotensin pathway. Efforts to

develop angiotensin II receptor antagonists began in the early 1970s and focused on peptide-based analogs of the natural agonist. The prototypical compound that resulted from these studies was saralasin, an octapeptide in which the Asp¹ and Phe⁸ residues of angiotensin II were replaced with Sar (sarcosine, *N*-methylglycine) and Ile, respectively. Saralasin as well as other peptide analogs demonstrated the ability to reduce blood pressure; however, these compounds lacked oral bioavailability and expressed unwanted partial agonist activity. More recent efforts have used peptide mimetics to circumvent these inherent problems with peptide-based antagonists. The culmination of these efforts was the 1995 approval of losartan, a nonpeptide angiotensin II receptor blocker (ARB) (1,36).

Development of Losartan

The development of losartan can be traced back to two 1982 patent publications (37), which described the anti-hypertensive effects of a series of imidazole-5-acetic acid analogs. These compounds are exemplified by S-8308 (Fig. 23.14) and were later found to block the angiotensin II receptor specifically. Although these compounds were relatively weak antagonists, they did not possess the unwanted agonist activity previously seen in peptide analogs. A computerized molecular modeling overlap of angiotensin II with the structure of S-8308 revealed three common structural features: The ionized carboxylate of S-8308 correlated with the C-terminal carboxylate of angiotensin II, the imidazole ring of S-8308 correlated with the imidazole side chain of the His⁶ residue, and the *n*-butyl group of S-8308 correlated with the hydrocarbon side chain of the Ile⁵ residue (Fig. 23.14). The benzyl group of S-8308 was proposed to lie in the direction of the N-terminus of angiotensin II; however, it was not believed to have any significant receptor interactions.

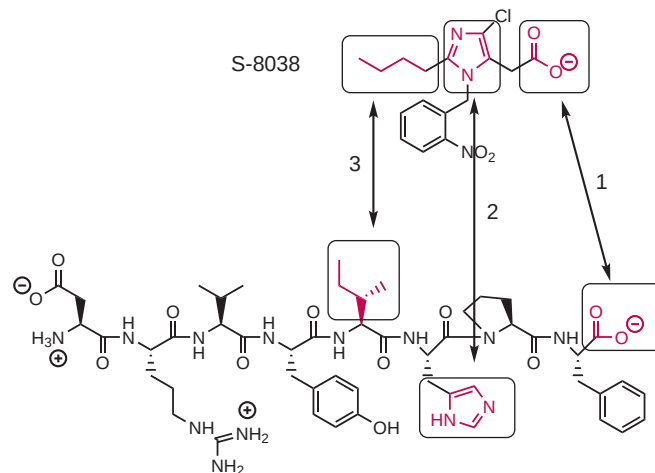


FIGURE 23.14 Structural comparison of S-8308, an imidazole-5-acetic acid analog, with angiotensin II. (Adapted from Timmermans PB, Wong PC, Chiu AT, et al. Angiotensin II receptors and angiotensin II receptor antagonists. *Pharmacol Rev* 1993;45:205–213, with permission.)

From S-8308, a number of molecular modifications were carried out in an attempt to improve receptor binding and lipid solubility, with the latter being important to assure adequate oral absorption. These changes resulted in the preparation of losartan, a compound with high receptor affinity ($IC_{50} = 0.019 \mu\text{mol/L}$) and oral activity (Fig. 23.15).

Additional Angiotensin II Receptor Blockers

Valsartan, irbesartan, telmisartan, candesartan, olmesartan, and azilsartan are biphenyl analogs of losartan (Fig. 23.16). These compounds possess structural features that are similar to those seen in losartan. Valsartan, named for the valine portion of the compound, is the first nonimidazole-containing ARB and is slightly more potent ($IC_{50} = 0.0089 \mu\text{mol/L}$) than losartan. The amide carbonyl of valsartan is isosteric with the imidazole nitrogen of losartan and can serve as a hydrogen bond acceptor similar to the imidazole nitrogen. Irbesartan is a spiro-compound that lacks the primary alcohol of losartan but that has a 10-fold greater binding affinity ($IC_{50} = 0.0013 \mu\text{mol/L}$) for the angiotensin II receptor. Hydrogen bonding, or ion–dipole binding, of the carbonyl group can mimic the interaction of the primary alcohol of losartan, whereas the spirocyclopentane can provide enhanced hydrophobic binding. Azilsartan medoxomil, candesartan cilexetil, and telmisartan contain benzimidazole rings that provide some enhanced hydrophobic binding, similar to that seen with the spirocyclopentane ring of irbesartan. Azilsartan medoxomil, candesartan cilexetil, and olmesartan medoxomil are prodrugs that are rapidly and completely hydrolyzed during absorption from the gastrointestinal tract to their active carboxylic acid metabolites, azilsartan, candesartan, and olmesartan, respectively. These carboxylic acids lie in exactly the same locations as the hydroxyl group of losartan, the carboxylic acid of valsartan, and the ketone of irbesartan and can participate in both ionic and dipole interactions.

Eprosartan was developed using a different hypothesis than that for losartan (Fig. 23.17). Similar to the rationale for losartan, the carboxylic acid of S-8308 was thought to mimic the Phe⁸ (i.e., C-terminal) carboxylate of angiotensin II. The benzyl group of S-8308 was proposed to be an important structural feature that mimicked the aromatic side chain of Tyr⁴ present in the agonist. Thus, the major

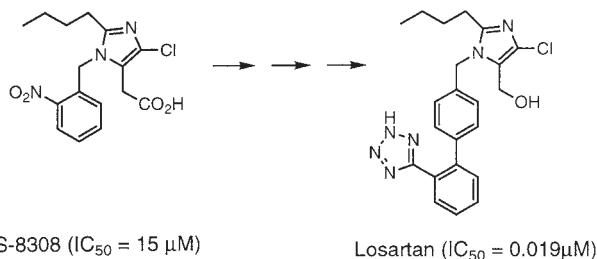


FIGURE 23.15 The development of losartan from S-8308.

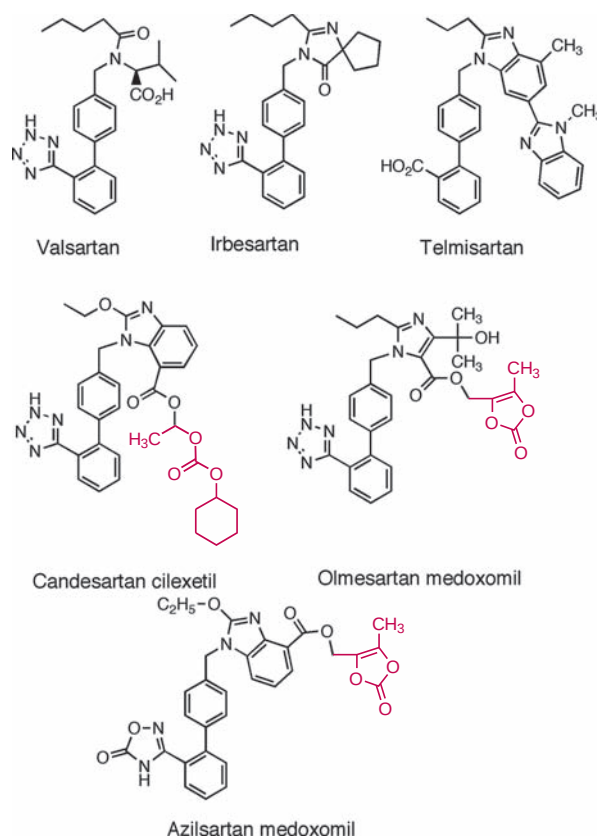


FIGURE 23.16 Structures of losartan analogs. The highlighted portions of candesartan cilexetil and olmesartan medoxomil are hydrolyzed via esterases to produce their respective active, carboxylate metabolites.

structural change was not an extension of the *N*-benzyl group but, rather, an enhancement of the compound's ability to mimic the C-terminal end of angiotensin II. This was accomplished by substituting the 5-acetic acid group with an α -thienylacrylic acid. In addition, a para-carboxylate (a functional group investigated during the development of losartan) also was added. The thienyl ring isosterically mimics the Phe⁸ phenyl ring of angiotensin II and, along with the para-carboxylate, is responsible for the excellent potency ($IC_{50} = 0.0015 \mu\text{mol/L}$) of this compound (36).

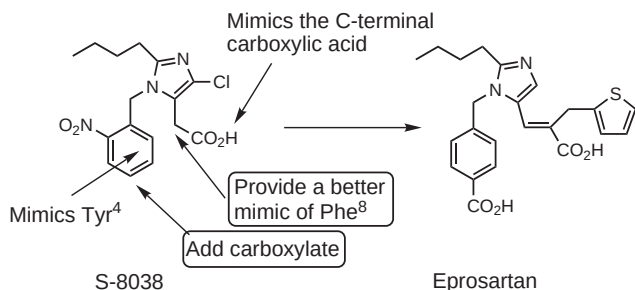


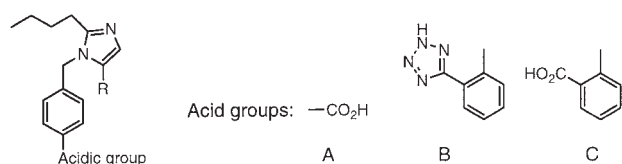
FIGURE 23.17 The development of eprosartan from S-8308. The Phe⁸ residue of angiotensin II contains the C-terminal carboxylic acid.

Mechanism of Action

The angiotensin II receptor exists in at least two subtypes: type 1 (AT_1) and type 2 (AT_2). The AT_1 receptors are located in brain, neuronal, vascular, renal, hepatic, adrenal, and myocardial tissues and mediate the cardiovascular, renal, and central nervous system (CNS) effects of angiotensin II. All currently available ARBs are 10,000-fold more selective for the AT_1 receptor subtype and act as competitive antagonists at this site. In terms of relative affinity for the AT_1 receptor, azilsartan, candesartan, and olmesartan have the greatest affinity; irbesartan and eprosartan have a somewhat lower affinity; and telmisartan, valsartan, and losartan have the lowest affinity. All ARBs prevent and reverse all of the known effects of angiotensin II, including rapid and slow pressor responses, stimulatory effects on the peripheral sympathetic nervous system, CNS effects, release of catecholamines, secretion of aldosterone, direct and indirect renal effects, and all growth-promoting effects. The function of the AT_2 receptors is not as well characterized; however, they have been proposed to mediate a variety of growth, development, and differentiation processes. Some concern has arisen that unopposed stimulation of the AT_2 receptor in conjunction with AT_1 receptor antagonism may cause long-term adverse effects. As a result, compounds that exhibit balanced antagonism at both receptor subtypes are currently being sought (1,38).

Structure–Activity Relationships

All commercially available ARBs are analogs of the following general structure:



1. The “acidic group” is thought to mimic either the Tyr⁴ phenol or the Asp¹ carboxylate of angiotensin II. Groups capable of such a role include the carboxylic acid (A), a phenyl tetrazole or isostere (B), or a phenyl carboxylate (C).
2. In the biphenyl series, the tetrazole and carboxylate groups must be in the ortho position for optimal activity (the tetrazole group is superior in terms of metabolic stability, lipophilicity, and oral bioavailability).
3. The *n*-butyl group of the model compound provides hydrophobic binding and, most likely, mimics the side chain of Ile⁵ of angiotensin II. As seen with azilsartan, candesartan, telmisartan, and olmesartan, this *n*-butyl group can be replaced with either an ethyl ether or an *n*-propyl group.
4. The imidazole ring or an isosteric equivalent is required to mimic the His⁶ side chain of angiotensin II.

5. Substitution can vary at the “R” position. A variety of R groups, including a carboxylic acid, a hydroxymethyl group, a ketone, or a benzimidazole ring, are present in currently available ARBs and are thought to interact with the AT_1 receptor through either ionic, ion–dipole, or dipole–dipole bonds.

Physicochemical Properties

All ARBs are acidic drugs. The tetrazole ring found in losartan, valsartan, irbesartan, candesartan, and olmesartan has a pK_a of approximately 6 and will be at least 90% ionized at physiologic pH. The carboxylic acids found on valsartan, candesartan, olmesartan, telmisartan, and eprosartan have pK_a values in the range of 3 to 4 and also will be primarily ionized. Currently, available agents have adequate, but not excellent, lipid solubility. As previously mentioned, the tetrazole group is more lipophilic than a carboxylic acid. Additionally, the four nitrogen atoms present in the tetrazole ring can create a greater charge distribution than that available for a carboxylic acid. These properties have been proposed to be responsible for the enhanced binding and bioavailability of the tetrazole-containing compounds (39). Similar to ACE inhibitors, the stereochemistry of valsartan is consistent with the L-amino acids in the natural agonist.

Metabolism

Approximately 14% of a dose of losartan is oxidized by the isozymes CYP2C9 and CYP3A4 to produce EXP-3174, a noncompetitive AT_1 receptor antagonist that is 10- to 40-fold more potent than losartan (Fig. 23.18). The overall cardiovascular effects seen with losartan result from the combined actions of the parent drug and the active metabolite; thus, losartan should not be considered to be a pro-drug (1). As previously mentioned, azilsartan medoxomil, candesartan cilexetil, and olmesartan medoxomil are rapidly and completely hydrolyzed to azilsartan, candesartan, and olmesartan, respectively, in the intestinal wall.

None of the other compounds are converted to active metabolites. All of these compounds are primarily (80%) excreted unchanged. Approximately 20% of valsartan is metabolized to inactive compounds via mechanisms that do not appear to involve the CYP450 system. The primary circulating metabolites for irbesartan, telmisartan

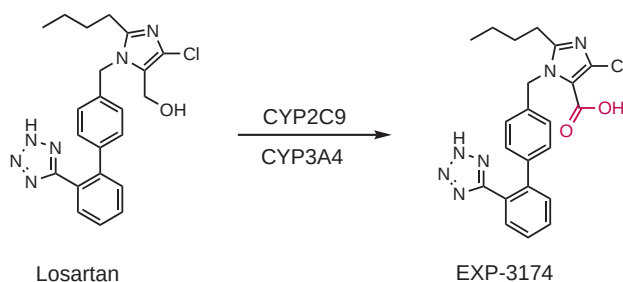


FIGURE 23.18 The metabolic conversion of losartan to EXP-3174 by cytochrome P450 isozymes.

and eprosartan, are inactive glucuronide conjugates. A small amount of irbesartan is oxidized by CYP2CP; however, irbesartan does not substantially induce or inhibit the CYP450 enzymes normally involved in drug metabolism (1,27–29,64). Azilsartan is primarily metabolized by CYP2C9 to an inactive O-dealkylated metabolite.

Pharmacokinetic Parameters

The pharmacokinetic parameters and dosing information for angiotensin receptor antagonists are summarized in Tables 23.6 and 23.7, respectively (27–29,64). With the exception of azilsartan medoxomil (60%), irbesartan (60% to 80%) and, possibly, telmisartan (42% to 58%), all of the compounds have low, but adequate, oral bioavailability (15% to 33%). Given the fact that most of the compounds are excreted unchanged, the most probable reasons for the low bioavailability are poor lipid solubility and incomplete absorption. The effect of food on the absorption of losartan, eprosartan, valsartan, and eprosartan is to reduce absorption; however, this effect has been deemed to be clinically insignificant; thus, the compounds can be taken either with or without food. All of the compounds have similar onsets, are highly protein bound, have elimination half-lives that allow once- or twice-daily dosing, and with the exception of olmesartan, are primarily eliminated via the fecal route. Candesartan and telmisartan appear to require a slightly longer time to reach peak plasma concentrations. As with ACE inhibitors, literature designation of fecal elimination is unclear regarding whether it includes unabsorbed drug.

Azilsartan medoxomil, candesartan cilexetil, losartan, and olmesartan medoxomil differ from the other compounds in several respects. They are the only

compounds with active metabolites, and they have the highest renal elimination of all of the agents. Product labeling indicates that renal impairment does not require a dosage reduction for losartan, but area under the curve values are increased by 50% in patients with a creatinine clearance of less than 30 mL/min and are doubled in hemodialysis patients. These increases are not seen for the other agents. Losartan and telmisartan are the only two agents that require initial dose reductions in patients with hepatic impairment. Because of significantly increased plasma concentration, patients with impaired hepatic function or biliary obstructive disorders should avoid the use of telmisartan.

Therapeutic Applications

All ARBs are currently approved for the treatment of hypertension and, along with ACE inhibitors, diuretics, β -blockers, and calcium channel blockers, have been designated as first-line agents either alone or in combination with other antihypertensive agents (32). A number of other indications have also been approved. Irbesartan and losartan have been approved for the treatment of nephropathy in type 2 diabetes. Losartan has been approved for stroke prevention in hypertensive patients with left ventricular hypertrophy. Candesartan and valsartan have been approved for the treatment of heart failure. Telmisartan has been approved to reduce the risk of MI and stroke, while valsartan has been approved to reduce cardiovascular mortality in clinically stable patients with left ventricular failure or left ventricular dysfunction following MI. Based on their ability to attenuate the renin-angiotensin system, one should expect a gradual increase in the number of uses and approved indications for this class of agents.

TABLE 23.6 Pharmacokinetic Parameters of Angiotensin II Receptor Blockers

Drug	Oral Bioavailability (%)	Active Metabolite	Protein Binding (%)	Time to Peak Plasma Concentration (hours)	Elimination Half-Life (hours)	Major Route(s) of Elimination
Azilsartan Medoxomil	60	Azilsartan	99	1.5–3.0	11	Fecal (55%) Renal (42%)
Candesartan Cilexetil	15	Candesartan	99	3–4	9	Fecal (67%) Renal (33%)
Eprosartan	15	None	98	1–2	5–9	Fecal (90%) Renal (10%)
Irbesartan	60–80	None	90	1.5–2.0	11–15	Fecal (80%) Renal (20%)
Losartan	33	EXP-3174	98.7 (losartan)	1 (losartan)	1.5–2.0 (losartan)	Fecal (60%)
			99.8 (EXP-3174)	3–4 (EXP-3174)	6–9 (EXP-3174)	Renal (35%)
Olmesartan Medoxomil	26	Olmesartan	99	1.5–3.0	10–15	Fecal (35%–50%) Renal (50%–65%)
Telmisartan	42–58	None	100	5	24	Fecal (97%)
Valsartan	25	None	95	2–4	6	Fecal (83%) Renal (13%)

TABLE 23.7 Dose Information for Angiotensin II Receptor Blockers

Generic Name	Trade Name(s)	Approved Indications	Dosing Range (Treatment of Hypertension)	Maximum Daily Dose	Initial Dose Reduction with Hepatic Dysfunction	Dose Reduction with Renal Dysfunction	Available Tablet Strengths (mg)
Azilsartan Medoxomil	Edarbi	Hypertension	40–80 mg once daily	80 mg	No	No	40, 80
Candesartan Cilexetil	Atacand	Hypertension, heart failure	8–32 mg once daily	32 mg	No	Only with severe impairment	4, 8, 16, 32
Eprosartan	Teveten	Hypertension	400–800 mg once daily or b.i.d.	900 mg	No	Decrease maximum daily dose to 600 mg	400, 600
Irbesartan	Avapro	Hypertension, nephropathy in type 2 diabetics	150–300 mg once daily	300 mg	No	No	75, 150, 300
Losartan	Cozaar	Hypertension, nephropathy in type 2 diabetics, hypertension, stroke prevention in hypertensive patients with left ventricular hypertrophy	25–100 mg once daily or b.i.d.	100 mg	Yes (reduce to 25 mg once daily)	Adults, no Children, yes	25, 50, 100
Olmesartan Medoxomil	Benicar	Hypertension	20–40 mg once daily	40 mg	No	No	5, 20, 40
Telmisartan	Micardis	Hypertension, cardiovascular risk reduction of MI and stroke	40–80 mg once daily	80 mg	Yes (reduce to 20 mg once daily)	No	20, 40, 80
Valsartan	Diovan	Hypertension, heart failure, post-MI in patients with left ventricular failure or dysfunction	80–320 mg once daily	320 mg	No	No	40, 80, 160, 320

b.i.d., twice a day; MI, myocardial infarction.

Adverse Effects

The most prevalent side effects of ARBs are listed below (1,27–29). Overall, this class of agents is well tolerated, with CNS effects being the most commonly reported complaint. Similar to ACE inhibitors, some of the adverse effects are directly related to attenuation of the renin-angiotensin

pathway. Notably absent are the dry cough and angioedema seen with ACE inhibitors. Because ARBs are specific in their actions, this class of drugs does not affect the levels of bradykinin or prostaglandins and, thus, does not cause these bothersome side effects. Like ACE inhibitors, the use of ARBs during pregnancy is contraindicated, especially during the second and third trimesters. The use of ARBs should be discontinued as soon as pregnancy is confirmed unless the benefits outweigh the potential risks.

COMBINATION PRODUCTS THAT INCLUDE AN ARB

ARB/Diuretic: candesartan/hydrochlorothiazide, eprosartan/hydrochlorothiazide, irbesartan/hydrochlorothiazide, losartan/hydrochlorothiazide, olmesartan/hydrochlorothiazide, telmisartan/hydrochlorothiazide, valsartan/hydrochlorothiazide

ARB/Calcium Channel Blocker: olmesartan/amlodipine, telmisartan/amlodipine, valsartan/amlodipine

ARB/Diuretic/Calcium Channel Blocker: olmesartan/hydrochlorothiazide/amlodipine, valsartan/hydrochlorothiazide/amlodipine

ARB/Renin Inhibitor: valsartan/aliskiren

Drug Interactions

Coadministration of ARBs with potassium salts, potassium-sparing diuretics, or drospirenone may cause

ADVERSE EFFECTS OF ANGIOTENSIN II RECEPTOR ANTAGONISTS

Headache, dizziness, fatigue, hypotension, hyperkalemia, dyspepsia, diarrhea, abdominal pain, upper respiratory tract infection, myalgia, back pain, pharyngitis, and rhinitis

hyperkalemia. Nonsteroidal anti-inflammatory drugs may alter the response to ARBs and other antihypertensive agents (including ACE inhibitors and calcium channel blockers) because of inhibition of vasodilatory prostaglandins. Studies have shown that indomethacin, naproxen, and piroxicam have a greater propensity for causing this interaction. Telmisartan has been reported to increase digoxin levels and to slightly decrease warfarin levels; however, the reduced warfarin levels were not sufficient to alter the international normalized ratio. Rifampin, because of its ability to induce CYP3A4, can decrease the plasma levels of losartan and its active metabolite, EXP-3174. The clinical significance of drug interactions between ARBs and compounds that can inhibit either CYP3A4 or CYP2C9 has yet to be established.

ALISKIREN, AN ORALLY ACTIVE RENIN INHIBITOR

Historical Overview

In 1957, Skeggs et al. (40) reported the activities of a purified amino acid sequence that could function as a renin substrate and proposed that the inhibition of renin would be beneficial in the treatment of hypertension. As previously mentioned in this chapter, renin is a very specific enzyme that recognizes the Pro⁷-Phe⁸-His⁹-Leu¹⁰-Val¹¹-Ile¹²-His¹³-Asn¹⁴ octapeptide sequence of angiotensinogen. Initial attempts to design renin inhibitors focused upon peptide analogs designed to mimic portions or all of this sequence. Many of these analogs, exemplified by compound A in Figure 23.19, showed effective inhibition of renin; however, susceptibility to proteolytic cleavage, poor or absent oral bioavailability, and short durations of action limited the therapeutic utility of these peptidic agents. Sequential structural modifications produced peptide-based compounds with the ability to mimic the transition state of angiotensinogen cleavage. Compounds such as zankiren (Fig. 23.19) possessed improved potency, favorable lipid solubility, and better oral bioavailability

as compared to their predecessors. Unfortunately, the cost of synthetic preparation coupled with the success of newly approved ARBs temporarily curtailed commercial interest in this area. Success was finally obtained when the peptide backbone was abandoned and replaced with a nonpeptidic template. This template was used in the development of aliskiren (Fig 23.19), the first nonpeptide, low molecular weight, orally active, transition-state renin inhibitor (41–43).

Mechanism of Action

Aliskiren directly inhibits renin, thereby preventing the formation of angiotensin I and angiotensin II. As previously mentioned, this is the rate-limiting step in this pathway and is highly regulated by hemodynamic, neurogenic, and humoral signals. Studies have shown that there are two potential advantages of inhibiting this enzyme as compared to inhibiting ACE or using an ARB (41,44). Inhibition of the renin-angiotensin pathway through any of these mechanisms has been shown to cause a compensatory increase in renin concentrations; however, unlike ACE inhibitors and ARBs, the ability of renin inhibitors to directly bind to the enzyme blocks the increases in plasma renin activity seen with ACE inhibitors and ARBs. Additionally, alternate pathways, such as the chymostatin-sensitive pathway present in the heart, can convert angiotensin I to angiotensin II. While this alternate pathway could affect the efficacy of ACE inhibitors, it would not alter the effects of direct renin inhibition.

Physicochemical Properties

Aliskiren is a basic compound and is marketed as a hemifumarate salt. The calculated logP value for the unionized form of aliskiren is 4.32 (21); however, its salt form is highly water soluble. Aliskiren contains four chiral centers and is marketed as the pure 2S, 4S, 5S, 7S enantiomer (29).

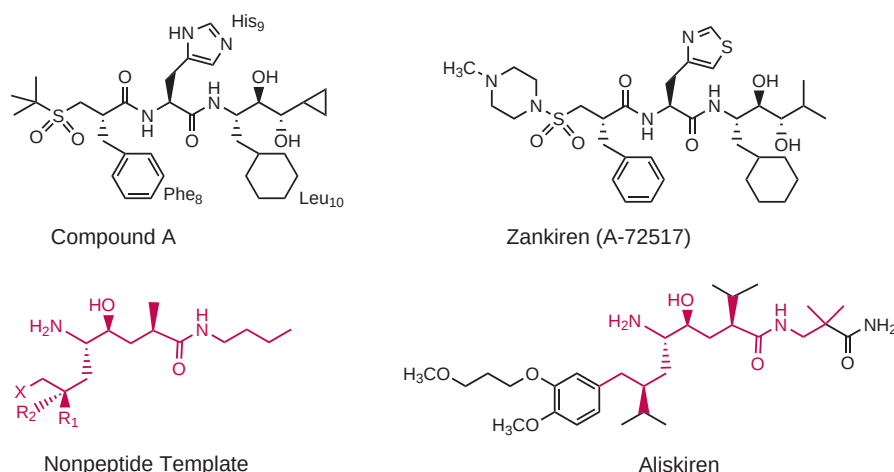


FIGURE 23.19 Aliskiren and predecessor compounds.

Metabolism

Approximately 90% of aliskiren is eliminated unchanged in the feces. Due to its poor bioavailability, the extent of systemic metabolism is unclear; however, clearance through the hepatobiliary tract appears to be the primary route of elimination. While metabolism is a minor pathway, the two major metabolites of aliskiren are an *O*-demethylated alcohol derivative and a carboxylic acid derivative. The primary metabolizing enzyme is CYP3A4. Aliskiren has not been shown to either inhibit or induce any of the CYP450 isoenzymes (27–29).

Pharmacokinetic Parameters

Aliskiren is a poorly absorbed and has a bioavailability of approximately 2.5%. Following oral administration, peak plasma concentrations are reached within 1 to 3 hours, and steady-state blood concentration is reached within 7 to 8 days.

The normal, initial dose of aliskiren is 150 mg once daily. This may be increased to 300 mg once daily in patients whose blood pressure is not adequately controlled. The antihypertensive effects are usually attained within 2 weeks. High-fat meals have been shown to significantly decrease absorption, and patients are encouraged to establish a fixed/routine time to take aliskiren. No adjustment in dose is required for elderly patients or those with renal impairment or hepatic insufficiency. Caution, however, should be used in dosing patients with severe renal impairment (27,28,44).

Therapeutic Applications

Aliskiren is approved for the treatment of hypertension, either as monotherapy or in combination with other antihypertensive agents. Aliskiren is available alone or in combination with hydrochlorothiazide (a diuretic), amlodipine (a calcium channel blocker), or valsartan (an ARB) (27).

Adverse Effects

Similar to ARBs, aliskiren does not affect the levels of bradykinin and prostaglandins and does not cause the dry cough or angioedema seen with ACE inhibitors. Pregnancy warnings are identical to those mentioned for ACE inhibitors and ARBs. Additionally and similar to ACE inhibitors and ARBs, aliskiren can cause hyperkalemia due to its ability to indirectly decrease aldosterone release. The most common adverse effects are mild and include diarrhea, abdominal pain, dyspepsia, gastroesophageal reflux, and rash (28,44).

Drug Interactions

Coadministration of aliskiren with irbesartan has been shown to decrease both the plasma levels and the efficacy of aliskiren. If these are used in combination, an adjustment of aliskiren may be required. Potent P-glycoprotein inhibitors, such as atorvastatin, cyclosporine, and

ketoconazole, may increase the plasma levels of aliskiren. Coadministration of aliskiren and cyclosporine is not recommended. Aliskiren has been shown to decrease the area under the curve and maximum plasma concentration of furosemide (28,44).

ROLE OF CALCIUM AND CALCIUM CHANNELS IN VASCULAR SMOOTH MUSCLE CONTRACTION

Calcium is a key component of the excitation-contraction coupling process that occurs within the cardiovascular system. It acts as a cellular messenger to link internal or external excitation with cellular response. Increased cytosolic concentrations of Ca^{2+} result in the binding of Ca^{2+} to a regulatory protein, either troponin C in cardiac and skeletal muscle or calmodulin in vascular smooth muscle. This initial binding of Ca^{2+} uncovers myosin binding sites on the actin molecule, and subsequent interactions between actin and myosin result in muscle contraction. All of these events are reversed once the cytosolic concentration of Ca^{2+} decreases. In this situation, Ca^{2+} binding to troponin C or calmodulin is diminished or removed, myosin binding sites are concealed, actin and myosin can no longer interact, and muscle contraction ceases (45,46).

Mechanisms of Calcium Movement and Storage

The regulation of cytosolic calcium levels occurs via specific influx, efflux, and sequestering mechanisms (Fig. 23.20). The influx of calcium can occur through receptor-operated channels (site 1), the $\text{Na}^+/\text{Ca}^{2+}$ exchange process (site 2), “leak” pathways (site 3), and potential-dependent channels (site 4). Influx via either receptor-operated or voltage-dependent channels has

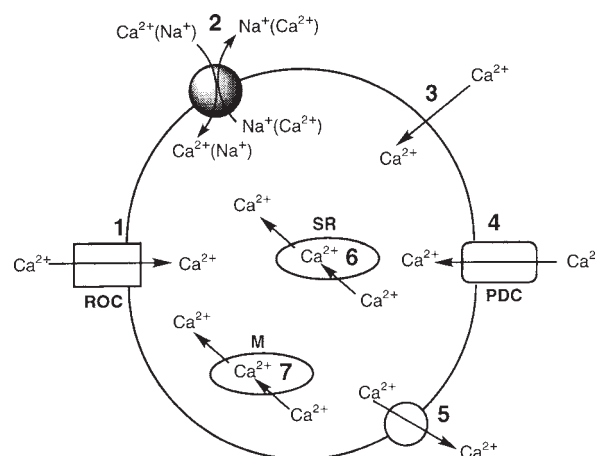


FIGURE 23.20 Cellular mechanisms for the influx, efflux, and sequestering of Ca^{2+} . M, mitochondria; PDC, potential-dependent Ca^{2+} channels; ROC, receptor-operated Ca^{2+} channels; SR, sarcoplasmic reticulum.

been proposed to be the major entry pathway for Ca^{2+} . Receptor-operated channels have been defined as those associated with cellular membrane receptors and activated by specific agonist–receptor interactions. In contrast, potential-dependent channels, also known as voltage-dependent or voltage-gated calcium channels, have been defined as those activated by membrane depolarization. The $\text{Na}^+/\text{Ca}^{2+}$ exchange process can promote either influx or efflux, because the direction of Ca^{2+} movement depends on the relative intracellular and extracellular ratios of Na^+ and Ca^{2+} . The “leak” pathways, which include unstimulated Ca^{2+} entry as well as entry during the fast inward Na^+ phase of an action potential, play only a minor role in calcium influx.

Efflux can occur through either an adenosine triphosphate–driven membrane pump (site 5) or via the $\text{Na}^+/\text{Ca}^{2+}$ exchange process previously mentioned (site 2). In addition to these influx and efflux mechanisms, the sarcoplasmic reticulum (site 6) and the mitochondria (site 7) function as internal storage/release sites. These storage sites work in concert with the influx and efflux processes to assure that cytosolic calcium levels are appropriate for cellular needs. Although influx and release processes are essential for excitation–contraction coupling, efflux and sequestering processes are equally important for terminating the contractile process and for protecting the cell from the deleterious effects of Ca^{2+} overload (47,48).

Potential-Dependent Calcium Channels

The pharmacologic class of agents known as calcium channel blockers produces their effects through interaction with potential-dependent channels. To date, six functional subclasses, or types, of potential-dependent Ca^{2+} channels have been identified: T, L, N, P, Q, and R. These types differ in location and function and can be divided into two major groups: low-voltage activated (LVA) channels and high-voltage activated (HVA) channels. Of the six types, only the T (transient, tiny) channel can be rapidly activated and inactivated with small changes in the cell membrane potential. It is thus designated as an LVA channel. All of the other types of channels require a larger depolarization and are thus designated as HVA channels. The L (long-lasting, large) channel is the site of action for currently available calcium channel blockers and, therefore, has been extensively studied. It is located in skeletal, cardiac, and smooth muscle and, thus, is highly involved in organ and vessel contraction within the cardiovascular system. The N channel is found in neuronal tissue and exhibits kinetics and inhibitory sensitivity distinct from both L and T channels. The functions, sensitivities, and properties of the other three types of channels are not as well known. The P channel has been named for its presence in the Purkinje cells, whereas the Q and R channels have been characterized by their abilities to bind to certain polypeptide toxins (49–51).

The L channel is a pentameric complex consisting of α_1 , α_2 , β , γ , and δ polypeptides (Fig. 23.21). The α_1 subunit is a transmembrane-spanning protein that consists of four domains and that functions as the pore-forming subunit. The α_1 subunit also contains binding sites for all the currently available calcium channel blockers. The other four subunits surround the α_1 portion of the channel and contribute to the overall hydrophobicity of the pentamer. This hydrophobicity is important in that it allows the channel to be embedded in the cell membrane. Additionally, the α_2 , δ , and β subunits modulate the α_1 subunit. Other types of potential-dependent channels are similar to the L channel. They all have a central α_1 subunit; however, molecular cloning studies have revealed that there are at least six α_1 genes: α_{1S} , α_{1A} , α_{1B} , α_{1C} , α_{1D} , and α_{1E} . Three of these genes, α_{1S} , α_{1C} , and α_{1D} , have been associated with L channels. The L channels found in skeletal muscle result from the α_{1S} gene; those in the heart, aorta, lung, and fibroblast result from the α_{1C} gene; and those in endocrine tissue result from the α_{1D} gene. Both α_{1C} and α_{1D} are used for L channels in the brain. Thus, there are some differences among the L channels located in different organs and tissues. Additionally, differences in α_1 genes and differences among the other subunits are responsible for the variations seen among the other five types of potential-dependent channels. As an example, the N channel lacks the γ subunit and contains an α_1 subunit derived from the α_{1B} gene (49,51).

CARDIOVASCULAR DISORDERS ASSOCIATED WITH POTENTIAL-DEPENDENT CALCIUM CHANNELS

As described earlier, the movement of calcium underlies the basic excitation–contraction coupling process. Thus, vascular tone and contraction primarily are determined by the availability of calcium from extracellular or intracellular sources. Potential-dependent Ca^{2+} channels are important in regulating the influx of Ca^{2+} ; therefore, inhibition of Ca^{2+} flow through these channels results in both vasodilation and decreased cellular response to contractile stimuli. Arterial smooth muscle is more sensitive to this action than venous smooth muscle. Additionally, coronary and cerebral arterial vessels are more sensitive than other arterial beds (48,51). As a result of these actions, calcium channel blockers are useful in the treatment of hypertension and ischemic heart disease. A brief overview of hypertension is provided in the renin-angiotensin section of this chapter.

The term “ischemic heart disease,” or coronary heart disease, encompasses a variety of syndromes. These include angina pectoris, silent myocardial ischemia, acute coronary insufficiency, and MI. The overall incidence of coronary heart disease is higher in men than in women

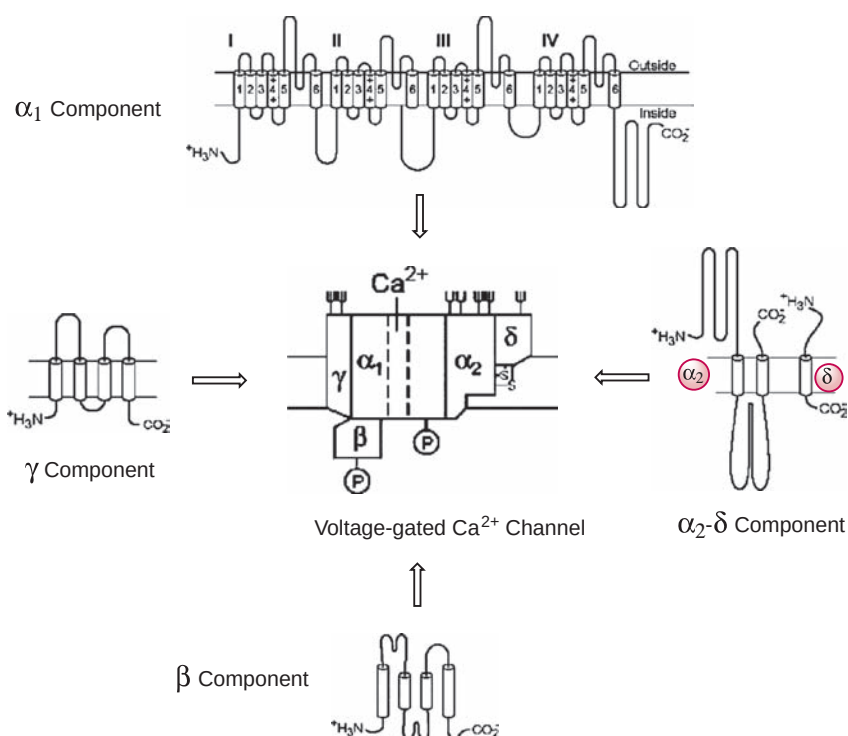


FIGURE 23.21 Representation of the structure of the voltage-gated Ca^{2+} channel (L channel) composed of several subunits— α_1 , α_2 - δ , β , γ —organized as depicted in the central area.

and increases with age. MI is the primary initial event in men, whereas angina is the most common initial presentation in women. Approximately 1.25 million individuals suffer new or recurrent episodes annually. Statistics published in 2006 from the American Heart Association estimate the prevalence of angina at approximately 10.2 million individuals (52).

Angina pectoris is a clinical manifestation that results from coronary atherosclerotic heart disease. It is characterized by a severe constricting pain in the chest that often radiates to the left shoulder, the left arm, or the back. Clinically, angina pectoris can be classified as either exertional, variant, or unstable. Exertional angina, otherwise known as stable angina or exercise-induced angina, is the most common form and results from an imbalance between myocardial oxygen supply and demand. Variant angina, otherwise known as Prinzmetal angina, results from the vasospasm of large, surface coronary vessels or branches. Unstable angina is the most difficult to treat and may occur as a result of advanced atherosclerosis and coronary vasospasm (53).

Excitation-contraction coupling in the heart is different from that in vascular smooth muscle in that a portion of the inward current is carried by Na^+ through the fast channel. In the sinoatrial and atrioventricular (AV) nodes, however, depolarization depends primarily on the movement of Ca^{2+} through the slow channel. Attenuation of this Ca^{2+} movement produces a negative

inotropic effect and decreased conduction through the AV node. This latter effect is especially useful in treating paroxysmal supraventricular tachycardia (PSVT), an arrhythmia primarily caused by AV nodal reentry and AV reentry (51).

CALCIUM CHANNEL BLOCKERS

Historical Overview

Identification of compounds that could block the inward movement of Ca^{2+} through slow cardiac channels occurred in the early 1960s. Verapamil and other phenylalkylamines were shown to possess negative inotropic and chronotropic effects that were distinct from other coronary vasodilators. Further investigations revealed that these agents mimicked the cardiac effects of Ca^{2+} withdrawal: They reduced contractile force without affecting the action potential. The effects of these compounds could be reversed by the addition of Ca^{2+} , thus suggesting that the negative inotropic effect was linked to an inhibition of excitation-contraction coupling. Subsequently, derivatives of verapamil, as well as other chemical classes of compounds, were shown to competitively block Ca^{2+} movement through the slow channel and, thus, alter the cardiac action potential. Therefore, calcium channel blockers also are known as slow channel blockers, calcium entry blockers, and calcium antagonists (47,51).

Chemical Classifications

Overview

Currently, 10 calcium channel blockers are available for therapeutic use. These compounds have diverse chemical structures and can be grouped into one of four chemical classifications (Fig. 23.22), each of which produces a distinct pharmacologic profile: 1,4-dihydropyridines (1,4-DHPs; e.g., nifedipine), phenylalkylamines (e.g., verapamil), benzothiazepines (e.g., diltiazem), and diaminopropanol ethers (e.g., bepridil). The majority of calcium channel blockers are 1,4-DHPs, and a detailed description of the SAR for this chemical class is provided below. In contrast, verapamil, diltiazem, and bepridil are the lone representatives of their respective chemical classes and, thus, are discussed as individual agents. Verapamil and diltiazem are discussed along with the 1,4-DHPs. Bepridil is a nonselective agent that is no longer available in the United States.

1,4-Dihydropyridines

HISTORY AND DEVELOPMENT The chemistry of dihydropyridines can be traced back to an 1882 paper in which Hantzsch described their utility as intermediates in the synthesis of substituted pyridines. Fifty years later, interest in this chemical class of compounds increased when it was discovered that a 1,4-DHP ring was responsible for the “hydrogen-transfer” properties of the coenzyme NADH. Numerous biochemical studies followed this discovery; however, it was not until the early 1970s that the pharmacologic properties of 1,4-DHPs were fully investigated. Loev and coworkers at Smith, Klein & French laboratories investigated the activities of “Hantzsch-type” compounds. As shown in Figure 23.23, the Hantzsch reaction produced a symmetrical compound in which both the esters (i.e., CO_2R_2) and

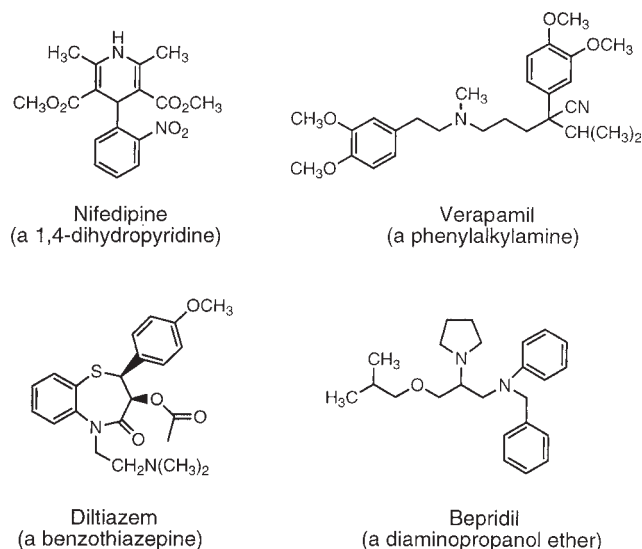


FIGURE 23.22 Chemical classes of calcium channel blockers.

BEPRIDIL

Bepridil is unique among all the calcium channel blockers in that its actions are not based solely on its ability to block potential-dependent L-type (i.e., slow) Ca^{2+} channels (27,51). Unlike other calcium channel blockers, bepridil also blocks fast Na^{+} channels as well as receptor-operated calcium channels. These additional actions are responsible for bepridil's ability to inhibit cardiac conduction, to slow AV nodal conduction, to increase the refractory period, to slow the heart rate, and to prolong the QT interval.

Bepridil was indicated for the oral treatment of chronic stable angina pectoris; however, its manufacturer voluntarily removed it from the US market, primarily because of its ability to cause torsades de pointes. It also should be noted that bepridil was never highly prescribed, most likely because of the significant number of cardiovascular warnings and contraindications associated with its use.

the C_2 and C_6 substituents (i.e., CH_3) are identical with each other. Structural requirements necessary for activity were identified by sequentially modifying the C_4 substituent (i.e., the R_1 group), the C_3 - and C_5 -esters (i.e., the R_2 groups), the C_2 - and C_6 -alkyl groups, and the N_1 -H substituent (54–57).

STRUCTURE–ACTIVITY RELATIONSHIPS The SARs for 1,4-DHP derivatives (see *General Structure* in Table 23.8) indicate that the following structural features are important for activity:

1. A substituted phenyl ring at the C_4 position optimizes activity (heteroaromatic rings, such as pyridine, produce similar therapeutic effects but are not used because of observed animal toxicity), and C_4 substitution with a small nonplanar alkyl or cycloalkyl group decreases activity.
2. Phenyl ring substitution (X) is important for size and position rather than for electronic nature. Compounds with ortho or meta substitutions possess optimal activity, whereas those that are unsubstituted or that contain a para substitution show a significant decrease in activity. Despite the fact that all commercially available 1,4-DHPs have electron-withdrawing ortho and/or meta substituents, this

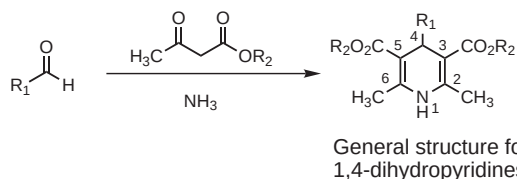
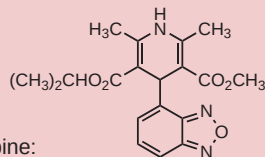
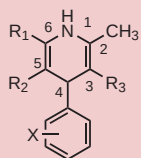


FIGURE 23.23 Synthesis of 1,4-DHPs using the Hantzsch reaction.

TABLE 23.8 Structure of the Dihydropyridine Ca^{2+} Channel Blockers

General structure:



Isradipine:

Compounds	R ₁	R ₂	R ₃	X
Amlodipine	$\text{CH}_2\text{O}(\text{CH}_2)_2\text{NH}_2$	$\text{CO}_2\text{CH}_2\text{CH}_3$	CO_2CH_3	2-Cl
Clevidipine	CH_3	$\text{CO}_2\text{CH}_2\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{nC}_3\text{H}_7$	CO_2CH_3	2,3-Cl ₂
Felodipine	CH_3	$\text{CO}_2\text{CH}_2\text{CH}_3$	CO_2CH_3	2,3-Cl ₂
Nicardipine	CH_3	$\text{CO}_2(\text{CH}_2)_2-\text{N}(\text{CH}_3)-\text{CH}_2-\text{C}_6\text{H}_5$	CO_2CH_3	3-NO ₂
Nifedipine	CH_3	CO_2CH_3	CO_2CH_3	2-NO ₂
Nimodipine	CH_3	$\text{CO}_2\text{CH}_2\text{CH}_2\text{OCH}_3$	$\overset{\text{CH}_3}{\text{CO}_2\text{CHCH}_3}$	3-NO ₂
Nisoldipine	CH_3	$\text{CO}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$	CO_2CH_3	2-NO ₂

is not an absolute requirement. Compounds with electron-donating groups at these same positions also have demonstrated good activity. The importance of the ortho and meta substituents is to provide sufficient bulk to “lock” the conformation of

the 1,4-DHP such that the C₄ aromatic ring is perpendicular to the 1,4-DHP ring (Fig. 23.24). This perpendicular conformation has been proposed to be essential for the activity of the 1,4-DHPs.

3. The 1,4-DHP ring is essential for activity. Substitution at the N₁ position or the use of oxidized (piperidine) or reduced (pyridine) ring systems greatly decreases or abolishes activity.
4. Ester groups at the C₃ and C₅ positions optimize activity. Other electron-withdrawing groups show decreased antagonist activity and may even show agonist activity. For example, the replacement of the C₃ ester of isradipine with a NO₂ group produces a calcium channel activator, or agonist (Fig. 23.25). Thus, the term “calcium channel modulators” is a more appropriate classification for the 1,4-DHPs.
5. When the esters at C₃ and C₅ are nonidentical, the C₄ carbon becomes chiral, and stereoselectivity between the enantiomers is observed. Additionally, evidence suggests that the C₃ and C₅ positions of the dihydropyridine ring are not equivalent positions. Crystal structures of nifedipine, a symmetrical 1,4-DHP, have shown that the C₃ carbonyl is synplanar to the C₂-C₃ bond but that the C₅ carbonyl is antiperiplanar to the C₅-C₆ bond (Fig. 23.26). Asymmetrical compounds have shown enhanced selectivity for specific blood vessels and are being preferentially developed. Nifedipine, the first 1,4-DHP to be marketed, is the only symmetrical compound in this chemical class.
6. With the exception of amlodipine, all 1,4-DHPs have C₂ and C₆ methyl groups. The enhanced

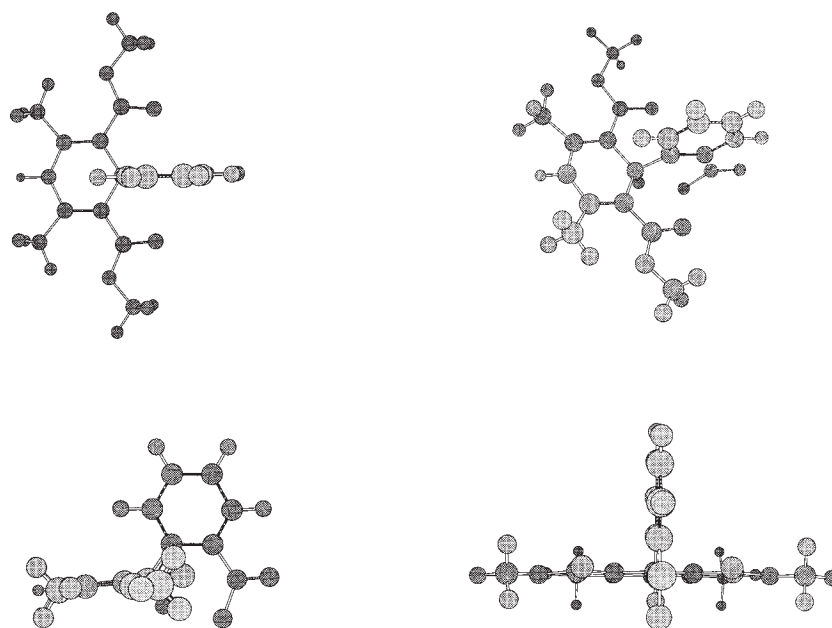


FIGURE 23.24 Molecular models of nifedipine. The ortho-nitro group of nifedipine provides steric bulk and ensures that the required perpendicular nature of the phenyl and dihydropyridine rings is maintained.

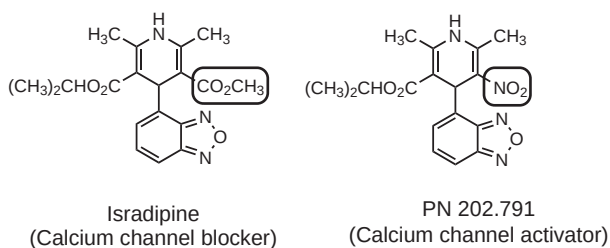


FIGURE 23.25 Structures of isradipine and its analogous 3-nitro derivative.

potency of amlodipine (vs. nifedipine) suggests that the 1,4-DHP receptor can tolerate larger substituents at this position and that enhanced activity can be obtained by altering these groups.

Mechanism of Action

Despite the name, calcium channel blockers do not simply “plug the hole” and physically block the Ca^{2+} channel. Instead, they exert their effects by binding to specific receptor sites located within the central α_1 subunit of L-type, potential-dependent channels. Three distinct, but allosterically interacting, receptors have been identified for verapamil, diltiazem, and the 1,4-DHPs. As shown in Table 23.9, the binding of verapamil to its receptor inhibits the binding of both diltiazem and the 1,4-DHPs to their respective receptors. Likewise, the binding of either diltiazem or the 1,4-DHPs inhibits the binding of verapamil. In contrast, diltiazem and the 1,4-DHPs mutually enhance the binding of each other (54).

Potential-dependent channels can exist in one of three conformations: a resting state, which can be stimulated by membrane depolarization; an open state, which allows the Ca^{2+} to enter; and an inactive state, which is refractory to further depolarization. Calcium

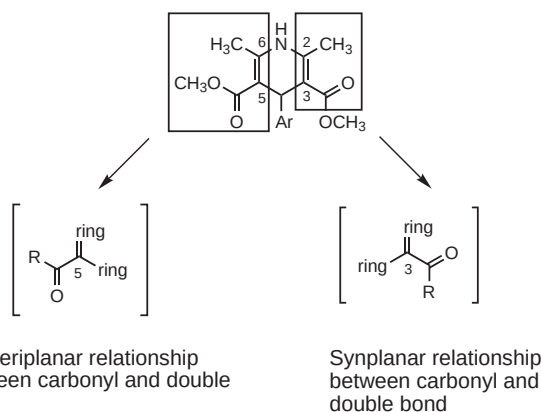
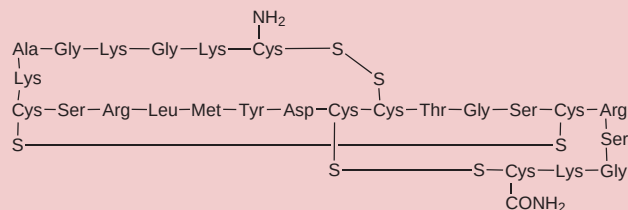


FIGURE 23.26 Conformation of the C_3 and C_5 esters of nifedipine ($\text{Ar} = 2$ -nitrophenyl). The C_3 carbonyl is synplanar to the C_2 - C_3 bond, and the C_5 carbonyl antiperiplanar to the C_5 - C_6 bond.

channel blockers have been shown to be more effective when membrane depolarization is either longer, more intense, or more frequent. This use dependency suggests that these compounds preferentially interact with their receptors when the Ca^{2+} channel is in either the open or inactive state. This state dependence is not identical for all classes of Ca^{2+} blockers and, in combination with the different binding sites, allosteric interactions, acidity, and solubility, may be responsible for the pharmacologic differences among verapamil, diltiazem, and the 1,4-DHPs. A summary of these differences is listed in Table 23.10. The 1,4-DHPs are primarily vasodilators,

ZICONOTIDE: AN N-TYPE CALCIUM CHANNEL BLOCKER

Ziconotide (Prialt) is a synthetic analog of a naturally occurring conopeptide found in the piscivorous marine snail (*Conus magus*). It is structurally, mechanistically, and therapeutically different from the other calcium channel blockers discussed in this chapter. Structurally, it is a polybasic peptide containing 25 amino acids and three disulfide bridges.



It selectively binds to N-type potential-dependent calcium channels located on the primary nociceptive (A-d and C) afferent nerves in the superficial layers (Rexed laminae I and II) of the dorsal horn in the spinal cord. Analgesic effects result from a blockade of neurotransmitter release and a disruption of normal nociceptive signal transmission. Ziconotide is indicated for the management of severe chronic pain in patients who are intolerant of or refractory to other systemic analgesics, adjunct therapies, or intrathecal morphine. Because of its peptide structure, ziconotide is not effective orally. It is administered by continuous intrathecal infusion. The initial recommended dose is 2.4 $\mu\text{g/day}$ (0.1 $\mu\text{g/hour}$) and can be titrated up to a maximum daily dose of 19.2 mg (0.8 $\mu\text{g/hour}$). Ziconotide has a half-life of approximately 4.5 hours. It is metabolized by endopeptidases and exopeptidases in numerous organs and tissues. Common adverse effects of ziconotide include dizziness, nausea, confusion, headache, somnolence, nystagmus, asthenia, and pain. The most serious adverse effects appear to be the development of severe psychiatric symptoms and neurologic impairment. As a result, ziconotide should not be used in patients with a preexisting history of psychosis. Frequent monitoring for evidence of cognitive impairment, hallucinations, or changes in mood or consciousness is essential. These effects are potentially additive with those of other CNS-depressant drugs (27,28,58).

TABLE 23.9 Actions of Calcium Channel Blockers and Interactions among Their Receptor Sites

Calcium Channel Blocker	Effect on Ca ²⁺ Channel	Allosteric Effect on the Binding of:		
		Verapamil	Diltiazem	1,4-Dihydropyridines
Verapamil	Antagonist; blocks channel	NA	Inhibits	Inhibits
Diltiazem	Antagonist; blocks channel	Inhibits	NA	Enhances
1,4-Dihydropyridines	Antagonist/agonist; can either block or open channel	Inhibits	Enhances	NA

NA, not applicable.

whereas verapamil and diltiazem have both vasodilator and cardiodepressant actions. The increased heart rate seen with the 1,4-DHPs results from a reflex mechanism that tries to overcome the vasodilation and subsequent drop in blood pressure caused by these drugs. In contrast, the compensatory mechanism does not occur to the same extent with either verapamil or diltiazem. This difference is the result of the ability of verapamil and diltiazem to block AV nodal conductance and the increased ability of 1,4-DHPs to activate the baroreceptor reflex. Ultimately, these pharmacologic differences are reflected in the clinical use of these agents (48,53,54).

Physicochemical Properties

A comparison of the acid-base properties of verapamil, diltiazem, and the 1,4-DHPs reveals that whereas all of the compounds are basic, the 1,4-DHPs are considerably less basic than verapamil and diltiazem. Verapamil and diltiazem both contain tertiary amines with pK_a values of 8.9 and 7.7, respectively (21). In contrast, the nitrogen of the 1,4-DHPs is part of a conjugated carbamate. Its electrons are involved in resonance delocalization and are much less available for protonation. Thus, at physiologic pH, verapamil and diltiazem are primarily

ionized, whereas 1,4-DHPs are primarily un-ionized. There are two exceptions to this. Amlodipine and nifedipine contain basic amine groups as part of the side chains connected to the 1,4-DHP ring. Although the 1,4-DHP ring of these compounds is un-ionized, the side-chain amines will be primarily ionized at physiologic pH. Because ionic attraction often is the initial interaction between a drug and its receptor, the differences in basicity between the 1,4-DHP ring and the tertiary amines of verapamil and diltiazem are consistent with the previously noted fact that the binding site for the 1,4-DHPs is distinct from those for verapamil and diltiazem.

The calculated logP values for the calcium channel blockers are listed in Table 23.11 (21,59). As evidenced by the data, all of these compounds possess good lipid solubility and, hence, excellent oral absorption (not shown in Table 23.11). Within the 1,4-DHP class, enhanced lipid solubility occurs in compounds that contain either larger ester groups or disubstituted phenyl rings. A comparison of the logP values of nifedipine and nisoldipine illustrates this fact. It should be noted that the calculated logP values listed in Table 23.11 are for the un-ionized compounds. These values significantly decrease for the ionized forms of amlodipine, nifedipine, verapamil, and diltiazem such that the latter three agents possess sufficient water solubility to be used both orally and parenterally.

All calcium channel blockers, with the exception of nifedipine, contain at least one chiral center; however, they are all marketed as their racemic mixtures. As previously noted, 1,4-DHPs with asymmetrically substituted esters exhibit stereoselectivity between the enantiomers. Additionally, the *S*(-)-enantiomers of verapamil and other phenylalkylamines are more potent than the *R*(+)-enantiomers. Very few SAR studies are available for diltiazem; however, the *cis* arrangement of the acetyl ester and the substituted phenyl ring is required for activity (54).

Metabolism

With the exception of clevidipine, all calcium channel blockers undergo extensive first-pass metabolism in the

TABLE 23.10 Similarities and Differences among 1,4-Dihydropyridines (DHPs), Verapamil, and Diltiazem

Cardiovascular Effect	1,4-DHPs	Verapamil and Diltiazem
Peripheral vasodilation	Increase	Increase
Blood pressure	Decrease	Decrease
Heart rate	Increase	Decrease or no effect
Atrioventricular node conduction	No effect	Decrease
Contractility	No effect or moderate increase	Decrease

TABLE 23.11 Pharmacokinetic Parameters of Calcium Channel Blockers

Drug	Calculated LogP	Oral Bioavailability (%)	Effect of Food on Absorption	Active Metabolite	Protein Binding (%)	T _{max} (hours)	Elimination Half-Life (hours)	Major Route(s) of Elimination
1,4-Dihydropyridines								
Amlodipine	2.76	64–90	None	None	93–97	6–12	35–50	Renal (60%) Fecal (20%–25%)
Clevidipine	2.96	NA	NA	None	>99	2–4 (min)	0.15	Renal (63%–74%) Fecal (7%–22%)
Felodipine	4.69	10–25	Increase	None	>99	2.5–5.0	11–16	Renal (70%) Fecal (10%)
Isradipine	3.19	15–24	Reduced rate, same extent	None	95	7–18 (CR)	8	Renal (60%–65%) Fecal (25%–30%)
Nicardipine	4.27	35	Reduced	None	>95	0.5–2.0 (IR) 1–4 (SR)	2–4	Renal (60%) Fecal (35%)
Nifedipine	2.40	45–70 86 (SR)	None	None	92–98	0.5 (IR) 6 (SR)	2–5 (IR) 7 (SR)	Renal (60%–80%) Biliary/fecal (15%)
Nimodipine	3.14	13	Reduced	None	>95	1	8–9	Renal
Nisoldipine	3.86	5	High-fat meal increases immediate release but lowers overall amount	Hydroxylated analog	>99	6–12	7–12	Renal (70%–75%) Fecal (6%–12%)
Phenalkylamines								
Verapamil	3.53	20–35	Reduced (SR form only)	Norverapamil	90	1–2 (IR) 7–11 (SR) 0.1–0.2 (IV)	3–7 (IR) 12 (SR)	Renal (70%) Fecal (16%)
Benzothiazepines								
Diltiazem	3.55	40–60	None	Deacetyl-diltiazem	70–80	2–4 (IR) 6–14 (SR)	3.0–4.5 (IR) 4.0–9.5 (SR) 3.4 (IV)	Renal (35%) Fecal (60%–65%)

CR, controlled-release product; IR, immediate-release product; IV, intravenous administration; NA, not applicable; SR, sustained-release product; T_{max}, time to maximum blood concentration.

liver and are substrates for the CYP3A4 isozyme (27,29). Additionally, several of these compounds can inhibit CYP3A4. Clevidipine was designed to have an ultra-short duration of action. Upon IV infusion, clevidipine exerts its antihypertensive actions within 2 to 4 minutes. Rapid ester hydrolysis (Fig. 23.27) inactivates the compound

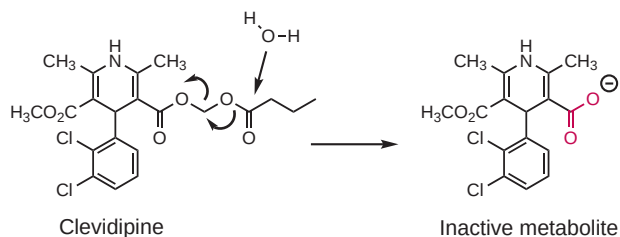


FIGURE 23.27 In vivo metabolic hydrolysis of clevidipine to an inactive metabolite.

and allows it to be used in patients with either renal or hepatic dysfunction without any dosage adjustments (27). All other 1,4-DHPs are oxidatively metabolized to a variety of inactive compounds. In many cases, the dihydropyridine ring is initially oxidized to an inactive pyridine analog (Fig. 23.28). These initial metabolites are then further transformed by hydrolysis, conjugation, and additional oxidation pathways. Nisoldipine also is subject to these processes; however, hydroxylation of its isobutyl ester produces a metabolite that retains 10% of the activity of the parent compound. In addition to the drug–drug interactions listed below (see Table 23.13), an interesting drug–food interaction occurs with the 1,4-DHPs and grapefruit juice (60). Coadministration of 1,4-DHPs with grapefruit juice produces an increase systemic concentration of the 1,4-DHPs. The mechanism of this interaction appears to result from inhibition of intestinal CYP450 by flavonoids and furanocoumarins

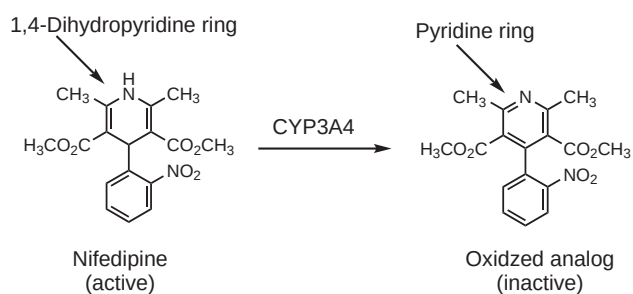


FIGURE 23.28 Oxidation of the 1,4-dihydropyridine ring of nifedipine.

specifically found in grapefruit juice. It has been proposed that limiting daily intake to either an 8-oz. glass of grapefruit juice or half of a fresh grapefruit would likely avoid significant drug interactions with most CYP3A4-metabolized drugs (61).

Verapamil is primarily metabolized to the *N*-demethylated compound, norverapamil, which retains approximately 20% of the pharmacologic activity of verapamil and can reach or exceed the steady-state plasma levels of verapamil. Interestingly, the more active *S*(-)-isomer undergoes more extensive first-pass hepatic metabolism than does the less active *R*(+)-isomer. This is important to note, because when given IV, verapamil prolongs the PR interval of the electrocardiogram to a greater extent than when it is given orally (62). This is because the preferential metabolism of the more active enantiomer does not occur with parenteral administration. Diltiazem is primarily hydrolyzed to deacetyldiltiazem. This metabolite retains 25% to 50% of the coronary vasodilatory effects of diltiazem and is present in the plasma at levels of 10% to 45% of the parent compound.

Pharmacokinetic Parameters

The pharmacokinetic parameters and oral dosing information for calcium channel blockers are summarized in Tables 23.11 and 23.12, respectively (27–29). Some doses (specifically, those for diltiazem and verapamil) may vary for either specific indications (i.e., hypertension vs. angina) or different brand names, and the reader should consult the product literature for additional information. The primary differences among the compounds are onset of action, half-life, and oral bioavailability. All calcium channel blockers have excellent oral absorption; however, because they also are subject to rapid first-pass metabolism in the liver, the actual oral bioavailability of these compounds varies considerably depending on the extent of metabolism. All compounds are highly plasma protein bound and primarily eliminated as inactive metabolites in the urine. Because of extensive hepatic transformation, calcium channel blockers should be used cautiously in patients with hepatic disease. Recommendations for these patients include dosage reductions, careful titrations, and close therapeutic monitoring. Diltiazem and verapamil also require dosage adjustments in patients with renal dysfunction, because

renal impairment can significantly increase the concentrations of the active metabolites of these compounds. Dosage adjustments usually are not required for the other eight compounds, because seven of them produce inactive metabolites and nisoldipine produces active metabolites with significantly lower activity.

Felodipine and nisoldipine are only available as sustained-release (or extended-release) formulations, and amlodipine and nimodipine are only available as immediate-release formulations. Nifedipine, isradipine, nicardipine, verapamil, and diltiazem are available as both immediate-release and sustained-release formulations. The latter three compounds also are available as parenteral preparations, as is clevipidine. Unlike regular tablets and capsules, sustained-release (or extended-release) formulations cannot be chewed or crushed, because this may lead to an immediate, rather than a sustained, release of the compound. This effect not only will decrease the duration of the dose but also could produce an overdose and subsequent toxicities in the patient. Parenteral preparations of nicardipine and verapamil are incompatible with IV solutions containing sodium bicarbonate. In each case, sodium bicarbonate increases the pH of the solution, resulting in the precipitation of the calcium channel blocker. Although this interaction is not listed for diltiazem, it is reasonable to assume that a similar interaction may occur. Additionally, nicardipine is incompatible with lactated Ringer's solution, and verapamil will precipitate in solutions having a pH greater than or equal to 6 (27,28).

Therapeutic Applications

As illustrated in Table 23.12, calcium channel blockers have been approved for the treatment of hypertension, angina pectoris, subarachnoid hemorrhage, and specific types of arrhythmias (27,28). All calcium channel blockers cause vasodilation and decrease peripheral resistance. With the exceptions of nimodipine, all are approved to treat hypertension. Recent studies have indicated that immediate-release formulations of short-acting calcium channel blockers, especially nifedipine, can cause an abrupt vasodilation that can result in MI. As a result, only the sustained-release formulations of nifedipine and diltiazem should be used in the treatment of essential hypertension (63). Five of the 10 agents are approved for the treatment of angina pectoris. Verapamil is the most versatile agent in that it is indicated for all three types of angina: vasospastic, chronic stable, and unstable. Amlodipine, nifedipine, and diltiazem are indicated for both vasospastic and

CHEMICAL/PHARMACOLOGIC CLASSES USED TO TREAT ANGINA PECTORIS

Organic nitrates, β -blockers (see Chapter 21), and calcium channel blockers

TABLE 23.12 Oral Dosing Information for Calcium Channel Blockers

Generic Name	Brand Name(s)	Approved Indications	Normal Dosing Range	Maximum Daily Dose	Precautions with Hepatic Dysfunction	Available Tablet or Capsule Strengths (mg)
1,4-Dihydropyridines						
Amlodipine	Norvasc	Angina (V, CS), hypertension	5–10 mg once daily	10 mg	Reduce dosage	2.5, 5, 10
Clevidipine	Cleviprex	Hypertension	4–6 mg/hr IV	32 mg/hr IV	None	25 mg/50 mL emulsified suspension
Felodipine	Plendil	Hypertension	2.5–10.0 mg once daily	10 mg	Reduce dosage	ER: 2.5, 5, 10
Isradipine	DynaCirc CR	Hypertension	5–20 mg once daily	20 mg	Titrate dosage	2.5, 5 (CR: 5, 10)
Nicardipine	Cardene, Cardene IV	Angina (CS), hypertension	20–40 mg t.i.d. (SR: 30–60 mg b.i.d.) (IV: 5–15 mg/hour)	120 mg	Titrate dosage	20, 30 (ER: 30, 45, 60) (IV: 2.5 mg/mL)
Nifedipine	Procardia, Adalat	Angina (V, CS), hypertension	10–20 mg t.i.d. (SR: 30–60 mg once a day)	180 mg (SR: 90 mg)	Reduce dosage	10, 20 (ER: 30, 60, 90)
Nimodipine	Nimotop	Subarachnoid hemorrhage	60 mg every 4 hours for 21 days	360 mg	Reduce dosage	30
Nisoldipine	Sular	Hypertension	17–34 mg once daily	34 mg	Closely monitor blood pressure	ER: 8.5, 17, 20, 25.5, 30, 34, 40
Phenylalkylamines						
Verapamil	Calan, Covera, Isoptin, Verelan	Angina (V, CS, U), hypertension, atrial fibrillation/flutter, PSVT	80–120 mg t.i.d. or q.i.d. (SR: 240–480 mg once daily or b.i.d.)	480 mg (540 mg for Covera HS only)	Reduce dosage	40, 80, 120 (SR: 100, 120, 180, 200, 240, 300, 360) (IV: 2.5 mg/mL)
Benzothiazepines						
Diltiazem	Cardizem, Cartia, Dilacor, Taztia, Tiazac	Angina (V, CS), hypertension, atrial fibrillation/flutter, PSVT	30–120 mg t.i.d. or q.i.d. (SR: 120–480 mg once daily)	480 mg (SR: 540 mg)	Reduce dosage	30, 60, 90, 120 (SR: 60, 90, 120, 180, 240, 300, 360, 420) (IV: 5 mg/mL)

b.i.d., twice a day; CR, controlled release; CS, chronic stable angina; ER, extended release; IV, intravenous; PSVT, paroxysmal supraventricular tachycardia; q.i.d., four times a day; SR, sustained release; t.i.d., three times a day; U, unstable angina; V, vasospastic angina.

chronic stable angina, whereas nicardipine is indicated only for chronic stable angina. Nimodipine is unique in that it has a greater effect on cerebral arteries than on other arteries. As a result, nimodipine is indicated for the improvement of neurologic deficits because of spasm following subarachnoid hemorrhage from ruptured congenital intracranial aneurysms in patients otherwise in good neurologic condition after the episode. Verapamil and diltiazem are pharmacologically different from the 1,4-DHPs in that they block sinus and AV nodal conduction. As a result, IV formulations of verapamil and diltiazem are indicated for the treatment of atrial fibrillation, atrial flutter, and PSVT. Verapamil also can be used orally, either alone (for prophylaxis of repetitive PSVT) or in combination with digoxin (for atrial flutter or atrial fibrillation).

Adverse Effects and Drug Interactions

The most prevalent or significant side effects of calcium channel blockers are listed below (27,28,48,51,53). Drug interactions for calcium channel blockers are listed in Table 23.13. In most instances, these side effects do not cause long-term complications, and they often resolve with time or dosage adjustments. Many of these effects are simply extensions of the pharmacologic effects of

UNLABELED USES

Migraine headache, Raynaud syndrome, pulmonary hypertension, preterm labor, and hypertrophic cardiomyopathy

this class of compounds. Excessive vasodilation results in edema, flushing, hypotension, nasal congestion, headache, and dizziness. Additionally, the palpitations, chest pain, and tachycardia seen with 1,4-DHPs are a result of sympathetic responses to the vasodilatory effects of this chemical class. The use of a β -blocker in combination with a 1,4-DHP can minimize these compensatory effects and can be very useful in treating hypertension. Verapamil and diltiazem can cause bradycardia and AV block because of their ability to depress AV nodal conduction. Because of risks associated with additive cardiodepressive effects, they should not be used in combination with β -blockers. Clevidipine is formulated as an oil-in-water emulsion that contains soybean oil, glycerin,

and purified egg yolk phospholipids. As a result, this compound is contraindicated in patients with egg hypersensitivity or soya lecithin hypersensitivity. Additionally, the emulsion used in this formulation can aggravate pre-existing disorders of lipid metabolism.

ADVERSE EFFECTS OF CALCIUM CHANNEL BLOCKERS

Edema, flushing, hypotension, nasal congestion, palpitations, chest pain, tachycardia, headache, fatigue, dizziness, rash, nausea, abdominal pain, constipation, diarrhea, vomiting, shortness of breath, weakness, bradycardia, and AV block

TABLE 23.13 Drug Interactions for Calcium Channel Blockers

Drug	Calcium Blocker(s)	Result of Interaction
α_1 -Blockers (prazosin, terazosin)	Verapamil	Increased prazosin and terazosin levels
Amiodarone	Diltiazem, verapamil	Increased bradycardia and cardiotoxicity; decreased cardiac output
Aspirin	Verapamil	Increased incidence of bruising
Azole antifungals	Felodipine, isradipine, nifedipine, nisoldipine	Increased serum concentrations of the calcium channel blockers
Barbiturates	Felodipine, nifedipine, verapamil	Decreased pharmacologic effects of the calcium channel blockers
β -Blockers	All	Coadministration may cause additive or synergistic effects; increased cardiodepressant effects (more extensive with verapamil and diltiazem); inhibition of β -blocker metabolism by diltiazem, isradipine, nicardipine, nifedipine, and verapamil
Buspirone	Diltiazem, verapamil	Increase buspirone levels
Carbamazepine, oxcarbazepine	Felodipine, diltiazem, verapamil	Carbamazepine and oxcarbazepine decrease felodipine levels; verapamil and diltiazem increase carbamazepine levels
Cimetidine	All	Increased 1,4-DHP levels
Cyclosporine	Felodipine, nicardipine, nifedipine, diltiazem, verapamil	Increased cyclosporine levels when used with all of these except for nifedipine; cyclosporine increases felodipine and nifedipine levels
CYP3A4 Inhibitors	All	Potentially can increase the plasma levels of calcium channel blockers
Digoxin	Nifedipine, diltiazem, verapamil	Increased digoxin levels
Disopyramide, flecainide	Verapamil	Additive cardiodepressant effects
Dofetilide	Verapamil	Increased dofetilide levels
Doxorubicin	Verapamil	Increased doxorubicin levels
Erythromycin, clarithromycin	All	Increased 1,4-DHP levels and increased toxicity
Fentanyl	All	Severe hypotension and/or bradycardia
General anesthetics	All	Potential of the cardiac effects and vascular dilation associated with anesthetics
HMG-CoA reductase inhibitors	Diltiazem, verapamil	Increase levels of HMG-CoA reductase inhibitor
Imipramine	Diltiazem, verapamil	Increased imipramine levels
Lithium	Diltiazem, verapamil	Decreased lithium levels with verapamil; neurotoxicity with diltiazem

TABLE 23.13 Drug Interactions for Calcium Channel Blockers (*Continued*)

Drug	Calcium Blocker(s)	Result of Interaction
Lovastatin	Isradipine	Decreased effects of lovastatin
Melatonin	All	Decreased therapeutic effects of calcium channel blockers
Methylprednisolone	Diltiazem, verapamil	Increased methylprednisolone levels
Midazolam, triazolam	Diltiazem, verapamil	Increased effects of these benzodiazepines
Moricizine	Diltiazem	Increased moricizine levels; decreased diltiazem levels
Phenobarbital	All	Decreased bioavailability of calcium channel blocker
Phenytoin	All	Decreased effectiveness of calcium channel blocker due to induction of metabolism
Quinidine	Diltiazem, nifedipine, nisoldipine, verapamil	Variable responses: quinidine decreases AUC of nisoldipine, but increases actions of nifedipine; diltiazem and verapamil increase the effects of quinidine; nifedipine decreases quinidine levels and actions
Rifampin	Diltiazem, isradipine, nicardipine, nifedipine, verapamil	Decreased levels of calcium channel blocker
Sirolimus, tacrolimus	Diltiazem, nifedipine, verapamil	Increased sirolimus and tacrolimus levels
St. John's wort	Nifedipine	Decreased nifedipine levels (St. John's wort most likely increases the metabolism of all calcium channel blockers)
Theophylline	Diltiazem, verapamil	Increased theophylline levels and toxicity
Valproic acid	Nimodipine	Increased nimodipine levels
Vecuronium	Verapamil	Increased vecuronium levels
Vincristine	Nifedipine	Increased vincristine levels

AUC, area under the curve; HMG-CoA, 3-hydroxy-3-methyl-glutaryl-coenzyme A.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Thomas L. Rihn, PharmD

According to the National Institutes of Health Joint National Committee VII Practice Guidelines for High Blood Pressure, LB has stage 2 hypertension (systolic blood pressure >160 mm Hg and diastolic blood pressure >100 mm Hg). A thiazide diuretic is indicated for LB; however, the vast majority of patients with this level of high blood pressure will require a two-drug regimen. Considering LB's longstanding diabetes and proteinuria, an ACE inhibitor should be administered in addition to the diuretic. Lisinopril would be a drug of choice because of its pharmacokinetic profile (once daily therapy) and relatively low cost to the patient. It will allow for better blood pressure control and will likely favorably lower the level of proteinuria and perhaps slow the progression of diabetic nephropathy. Although an ARB could be given, the preferred approach generally would be to add an ACE inhibitor before an ARB. If the patient is intolerant to the ACEI (e.g., cough), an ARB could be substituted. It is not neces-

sary to begin with a three-drug regimen. This would be reserved for patients for sustained severe elevations in blood pressure or in whom end organ damage has occurred. Finally, there would be no need to refer LB to a hypertension specialist at this point in time. Only if he developed refractory hypertension would he be preferred to a specialist.

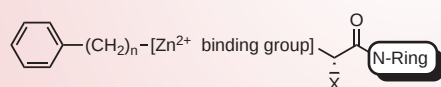
Chemical Analysis

S. William Zito and Victoria Roche

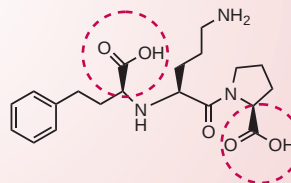
A two-drug regime is recommended for this patient. A thiazide diuretic (see Chapter 22) and an ACE inhibitor. The added ACE inhibitor will contribute to lowering LB's blood pressure as well as aid in controlling his diabetic nephropathy. The natural substrate for ACE is the decapeptide, angiotensin I (Asp-Arg-Val-Tyr-Ile-His-Pro-Phe-His-Leu). The catalytic site of ACE contains an arginine (Arg) residue that binds the carboxyl group of the terminal Leu, and a Zn²⁺ that coordinates with a carbonyl (Phe) to orient angiotensin I so that it can be hydrolyzed to angiotensin

II by attack of the carboxylate anion from Glu. Therefore an ACE inhibitor must have affinity for the ACE active site, have an anionic group to bind to the cationic Arg residue, be capable of binding to the Zn^{2+} and stable to the hydrolytic attack of the Glu carboxylate anion. These structural features are incorporated into a general pharmacophore which represents a di- or tripeptide and contains a stereochemical center with an L-amino acid representation.

Lisinopril contains those structural features that make it an ACE inhibitor. The carboxyl group on the proline can bind to the Arg, a second carboxyl group that can bind to the Zn^{2+} , and a chiral center that can represent an L-amino acid configuration. In addition, lisinopril contains an phenylethyl group and an amino butyl moiety that enhance pharmacokinetics making it have good oral absorption due to the possible formation of zwitterions, no metabolic transformation, and quick onset and long duration of action resulting in once-a-day dosing.



ACEI pharmacophore



Lisinopril

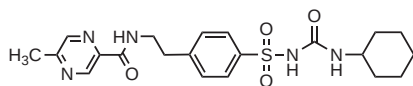
CASE STUDY

S. William Zito and Victoria Roche

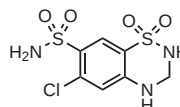
JB is a 46-year-old white man who presents to his local physician for a routine check-up. He is a professional opera singer. He has no complaints except for mild shortness of breath when singing tenor arias on stage. He also says that he experiences shortness of breath when walking long distances or up a flight of stairs. JB's medical history reveals that he has had type 2 diabetes for the past 6 years, which has been successfully treated with glipizide 10 mg po bid. His current blood pressure is 146/101 mm Hg, which is consistent with blood pressure measured during a previous visit. JB has a family history of hypertension and dyslipidemia. His laboratory tests reveal proteinuria and elevated cholesterol. His physician has diagnosed

JB's hypertension as stage 2 according to JNC VII guideline and needs to prescribe a two-drug therapeutic regimen. The drug of choice for hypertension is a thiazide diuretic, and JB's physician prescribes hydrochlorothiazide 25 mg daily but is undecided as to which one of the following three antihypertensives to use as the second drug that would best fit JB's needs.

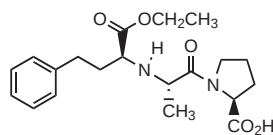
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.



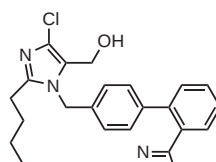
Glipizide



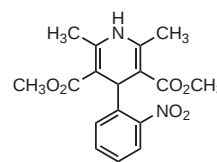
Hydrochlorothiazide



1



2



3

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Chapter 24

Central and Peripheral Sympatholytics and Vasodilators

DAVID A. WILLIAMS

Drugs Covered in This Chapter*

β -NONSELECTIVE BLOCKERS

- Bucindolol
- Carteolol
- Levobunolol
- Nadolol
- Penbutolol
- Pindolol
- Propranolol
- Timolol

β_1 -SELECTIVE BLOCKERS

- Acebutolol
- Atenolol
- Betaxolol
- Bisoprolol
- Esmolol
- Metoprolol
- Nebivolol

α_1 -BLOCKERS

- Prazosin
- Doxazosin

- Terazosin

MIXED α/β -BLOCKERS

- Carvedilol
- Labetalol

CENTRALLY ACTING SYMPATHOLYTICS

- Methyldopa
- Clonidine
- *Moxonidine*
- *Rilmenidine*
- Guanabenz
- Guanfacine
- Metyrosine

ADRENERGIC NEURON BLOCKING AGENTS

- Reserpine
- Guanethidine
- Guanadrel

VASODILATORS

- Hydralazine
- Minoxidil

- Diazoxide

PHOSPHODIESTERASE INHIBITORS

- Inamrinone
- Milrinone
- Sildenafil

NITRODILATOR

- Sodium nitroprusside

GANGLIONIC BLOCKERS

- Mecamylamine
- Trimethaphan

ENDOTHELIAL ANTAGONISTS

- Bosentan
- Ambrisentan
- *Sitaxsentan*

PROSTANOIDS

- Epoprostenol
- Treprostinil
- *Beraprost*
- *Iloprost*

Abbreviations

ACE, angiotensin-converting enzyme
ACEI, angiotensin-converting enzyme inhibitor
ADHD, attention-deficit hyperactivity disorder
ARB, angiotensin receptor blocker
AT-II, angiotensin II

ATP, adenosine triphosphate
BB, β -blocker
CaM, calmodulin
cAMP, cyclic adenosine monophosphate
cGMP, cyclic guanosine monophosphate

CCB, calcium channel blocker
CNS, central nervous system
COMT, catechol-O-methyltransferase
COX-2, cyclooxygenase-2
CSF, cerebrospinal fluid
DAG, diacylglycerol
eNOS, endothelial nitric oxide synthase

* Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in *italics*.

Abbreviations (Continued)

ET , endothelin	MAO , monoamine oxidase	PIP₂ , phosphatidylinositol
FDA , U.S. Food and Drug Administration	MLCK , myosin light-chain kinase	PKC , protein kinase C
GI , gastrointestinal	NMDA , <i>N</i> -methyl-D-aspartate	PLC , phospholipase C
iNOS , inducible nitric oxide synthase	nNOS , neuronal nitric oxide synthase	PVD , peripheral vascular disease
IP₃ , inositol triphosphate	NO , nitric oxide	SR , sarcoplasmic reticulum
ISA , intrinsic sympathomimetic activity	NOS , nitric oxide synthase	VSM , vascular smooth muscle
iv , intravenous	NYHA , New York Heart Association	
L-DOPA , L-dihydroxyphenylalanine	PAH , pulmonary arterial hypertension	

SCENARIO**Judy Cheng, PharmD**

AJ is a 62-year-old white male who presented to his primary care doctor today for a routine annual physical examination. His past medical history includes hyperlipidemia, hypertension, and chronic stable angina. He is currently receiving aspirin 325 mg daily, hydrochlorothiazide 25 mg daily, isosorbide mononitrate 30 mg daily, and simvastatin 40 mg daily. His significant physical

examination findings at the clinic today include blood pressure 165/98 mm Hg and heart rate 80 beats per minute. Others are within normal limit.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

CARDIOVASCULAR HYPERTENSION

Hypertension is the most common cardiovascular disease and is the major risk factor for coronary artery disease, heart failure, stroke, and renal failure. Approximately 50 million Americans have a systolic or diastolic blood pressure $\geq 140/90$ mm Hg. The onset of hypertension is defined as having a blood pressure of 140/90 mm Hg or greater and most commonly appears during the fourth, fifth, and sixth decades of life (1).

The importance of controlling blood pressure is well documented (1), although the rates of awareness, treatment, and control of hypertension have not risen as expected in the National Health and Nutrition Examination Survey (2). This survey showed that 68% of Americans are aware that they have high blood pressure but that only 53% are receiving treatment and only 27% have their blood pressure under control. Since 1976, there has been a significant improvement in the rates of awareness, treatment, and control of hypertension; however, since 1990, whatever progress had been achieved has now reached a plateau (2). Although the age-adjusted death rates from stroke and coronary heart disease during this period have fallen by 59% and 53%, respectively, these rates of decline also appear to have reached a plateau (2). These troubling trends should awaken clinicians to be more aggressive in the treatment of patients with hypertension.

When the decision to initiate hypertensive therapy is made, physicians often are presented with the dilemma of which of more than 80 antihypertensive products, representing more than 8 different drug classes, to use for their patients (Table 24.1) (1,3). Those factors that can affect

the outcomes from the treatment of hypertension, including potential adverse effects, clinically significant drug-drug interactions (especially when so many different drug classes are involved), patient compliance, affordability, risk/benefit ratios, and dosing frequency, must be considered (3). Having considered these factors, the health care provider (clinician or pharmacist) arrives at an appropriate choice of antihypertensive drug (3). Once the patient is stabilized with an antihypertensive medication, some of these issues need to be reevaluated. Patients should be continually asked about side effects, because many of the antihypertensive drugs possess side effects that the patient cannot tolerate (1). This problem and the cost of drug therapy can affect compliance to drug therapy especially for the elderly and those on fixed incomes (4).

Drug therapy in the management of hypertension must be individualized and adjusted based on coexisting risk factors, including the degree of blood pressure elevation, severity of the disease (e.g., presence of target organ damage), presence of underlying cardiovascular or other risk factors, response to therapy (single or multiple drugs), and tolerance to drug-induced adverse effects (1,3). Antihypertensive therapy is generally reserved for patients who fail to respond to nondrug therapies along with lifestyle modifications, such as diet including sodium restriction and adequate potassium intake, regular aerobic physical activity, moderation of alcohol consumption, and weight reduction (3).

It is not surprising that compliance with antihypertensive therapy can be as low as 40% when one considers that the patient, if he or she has other chronic diseases, can be taking as many as 10 different drugs and up to 40 tablets

CLINICAL SIGNIFICANCE



Cardiovascular diseases continue to be the number one cause of death in many industrialized nations, as a result of risk factors such as hypertension, coronary artery diseases, and heart failure. Sympatholytic agents play an important role in managing these diseases, not only by reducing blood pressure, angina, and other clinical symptoms, but also by reducing mortality. β -Blockers are now considered part of the standard of care in patients with coronary artery disease and heart failure and one of the drugs of first choice in hypertension management.

However, not all sympatholytic agents have the same pharmacologic and chemical properties or equal amounts of clinical evidence to support their use in every clinical indication. For instance, carvedilol, metoprolol, and bisoprolol have the most clinical evidence related to improving heart failure mortality, and nonselective β -blockers are probably more detrimental than others in patients with reactive airway disease, peripheral arterial diseases, and cocaine overdose, due to their β_2 -antagonistic

effect (which can be predicted by their chemical structure). Newer generation β -blockers such as nebivolol, which is the only β -blocker that structurally differs fundamentally from that derived from propranolol, also possess additional effects in relaxing smooth muscle by the nitric oxide pathway and may be less harmful in these situations.

The sympathetic nervous system of our body plays important roles in many disease processes. New medications continue to be discovered that augment our sympathetic nervous system activities. Newer agents are designed with a hope to reduce the side effect profiles of older agents and to enhance their therapeutic effects. Understanding the medicinal chemistry of these agents will allow us to select the specific agent that is most optimal for a particular patient situation.

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or capsules per day (4). To achieve better compliance requires educating the patient and simplifying the drug regimen by reducing the number of drugs being taken.

Hypertension in pregnancy presents a formidable therapeutic challenge and requires comprehensive management with close monitoring for both maternal and fetal welfare (5). Mechanisms involved with pregnancy-related hypertension include a hyperadrenergic state, plasma volume reduction, reduction in uteroplacental perfusion, hormonal control of vascular reactivity, and prostacyclin deficiency and can result from or activate the mechanisms that elevate blood pressure. Effective blood pressure control for pregnancy-related hypertension can often be achieved with methyldopa (recommended), β -blockers (BBs), or

mixed α/β -blockers (dual combination of α - and β -blocker activity). The vasodilating agent hydralazine is used to treat hypertensive emergencies associated with eclampsia (1,6). The presence or development of proteinuria (preeclampsia) in a hypertensive pregnant woman implies a major increase in risk to the fetus and warrants immediate admission to a hospital for specialist management (5).

Patients with diabetes mellitus have a much higher rate of hypertension than would be expected in the general population. Regardless of the antihypertensive agent used, a reduction in blood pressure helps to prevent or reduce diabetic microvascular and macrovascular complications, such as blindness and kidney failure. Angiotensin-converting enzyme (ACE) inhibitors

TABLE 24.1 Classification of Antihypertensive Activity According to Mechanism of Action

Drug Class	Drug Subclass
I Diuretics (Chapter 22)	
II Sympatholytic drugs	1. Centrally acting drugs (methyldopa, clonidine, guanabenz, guanfacine) (Chapter 24) 2. Ganglionic blocker drugs (Chapter 24) 3. Adrenergic neuron blocking drugs (Chapter 24) 4. β -Adrenergic blocking drugs (Chapters 10 and 24) 5. α -Adrenergic blocking drugs (Chapters 10 and 24) 6. Mixed α/β -adrenergic blocking drugs (Chapters 10 and 24)
III Vasodilator (Chapter 24)	Arterial (hydralazine, minoxidil, diazoxide) Arterial and venous (sodium nitroprusside)
IV Calcium channel blockers (Chapter 23)	
V Angiotensin-converting enzyme inhibitors (Chapter 23)	
VI Angiotensin receptor antagonists (Chapter 23)	

(ACEIs) and angiotensin receptor blockers (ARBs) are considered first-line therapy in patients with diabetes and hypertension because of their well-established renal protective effects. Most diabetic patients with hypertension require combination therapy with low-dose diuretics and BBs to achieve optimal blood pressure goals.

Combination Antihypertensive Therapy

It is well-documented that monotherapy adequately controls hypertension only in approximately 50% of patients (7,8). Therefore, a large percentage of patients will require at least a combination of two drugs to control their blood pressure and symptoms of hypertension. By combining different antihypertensive drug classes in low doses, their different mechanisms of action result in synergistic blood pressure lowering as well as in minimizing the adverse effects and improving compliance issues (1,8). For example, the addition of a low-dose thiazide diuretic dramatically increases the response rates to methyldopa, ACEIs, and BBs without producing the undesirable side effects. In the latest guidelines for treatment of hypertension, the Joint National Committee for Prevention, Detection, Evaluation and Treatment of High Blood Pressure (JNC-7), clinicians are encouraged to begin initially as stage 1 with thiazide-type diuretics and can also consider adding ACEIs, ARBs, BBs, or calcium channel blockers (CCBs) or combinations of these drugs in hypertensive patients without compelling risk factors (Table 24.2). These drug classes have been shown to decrease morbidity and mortality in long-term clinical trials (2). In patients with compelling risk factors such as heart disease, clinical manifestations of cardiovascular diseases or diabetes, other antihypertensive classes are considered as well as including the stage 1 agents (Table 24.2). In these patients with compelling risk factors for cardiovascular disease, treatment should be more aggressive, with the goal of reducing blood pressure to less than 130/80 mm Hg (Table 24.3). These recommendations reflect the current awareness of

the importance of addressing other cardiovascular conditions aside from just lowering the blood pressure.

Arterial pressure is the product of cardiac output and peripheral vascular resistance and, therefore, can be lowered by decreasing or inhibiting either or both of these physiologic responses with the drug classes represented in Table 24.1 and summarized in Figure 24.1. This chapter will discuss those antihypertensives that are classified as either sympatholytics (i.e., having a central or peripheral mechanism of action) or vasodilators. These classes of drugs are less commonly used today because of the higher incidence of side effects associated with inhibition of the sympathetic nervous system (sympathoinhibition) or vasodilation. In many instances, they have been replaced because of availability of newer and more effective antihypertensive drugs with fewer side effects, such as ACEIs and ARBs.

Overview of Vascular Tone (1)

Before beginning the discussion of the sympatholytics and vasodilators, it is important to review the nature of vascular tone. The term “vascular tone” refers to the degree of constriction experienced by a blood vessel relative to its maximally dilated state. All resistance (arteries) and capacitance (venous) vessels under basal conditions exhibit some degree of smooth muscle contraction, which determines the diameter and, hence, the tone of the vessel.

Basal vascular tone varies among organs. Those organs having a large vasodilatory capacity (e.g., myocardium, skeletal muscle, skin, and splanchnic circulation) have high vascular tone, whereas organs having relatively low vasodilatory capacity (e.g., cerebral and renal circulations) have low vascular tone. Vascular tone is determined by many different competing vasoconstrictor and vasodilator influences acting on the blood vessel. Influences such as sympathetic nerves and circulating angiotensin II regulate arterial blood pressure by increasing vascular tone (i.e., vasoconstriction). On the other hand, mechanisms for local blood flow regulation within an organ include

TABLE 24.2 Initial Drug Choices (JNC-7)

Without Compelling Risk Factors		With Compelling Risk Factors
Blood pressure goal in age ≥ 50 years, $<140/90$ mm Hg		Blood pressure goal, $<130/80$ mm Hg
Stage 1 Hypertension	Stage 2 Hypertension	Compelling risk factors include:
SBP 140–159 or DBP 90–99 mm Hg	SBP ≥ 160 or DBP ≥ 100 mm Hg	Heart failure After myocardial infarction High coronary disease risk Diabetes Chronic kidney disease Recurrent stroke prevention
Thiazide-type diuretics + may consider adding ACEIs, ARBs, BBs, CCBs, or combinations	Two drug combinations + may consider adding thiazide-type diuretics, ACEIs, ARBs, BBs, CCBs	Drugs for compelling indications: Stage 1 drugs and other antihypertensive drugs
Not at Blood Pressure Goal		
Optimize stage 2 drug treatment, or add additional classes of antihypertensive drugs until blood pressure goal is achieved		

TABLE 24.3 Risk Factors for Cardiovascular Disease

Correctable	Noncorrectable
Cigarette smoking	Age >60 years
Hypertension	Sex (men and postmenopausal women)
Elevated cholesterol	Family history of cardiovascular disease or stroke (women <65 years, men <55 years)
Reduced HDL cholesterol	
Diabetes mellitus	
Obesity	Target organ damage
HDL, high-density lipoprotein.	

endothelial factors (e.g., nitric oxide [NO] and endothelin [ET]) or local hormones/chemical substances (e.g., prostanoids, thromboxanes, histamine, and bradykinin) that can either increase or decrease tone. The mechanisms by which the above influences either constrict or relax blood vessels involve a variety of signal transduction mechanisms that, ultimately, influence the interaction between actin and myosin in the smooth muscle.

Overview of the Regulation of Vascular Smooth Muscle Contraction and Relaxation (1)

The contractile characteristics and the mechanisms that cause contraction of vascular smooth muscle (VSM) are very different from those of cardiac muscle. The VSM undergoes slow, sustained, tonic contractions, whereas

cardiac muscle contractions are rapid and of relatively short duration (a few hundred milliseconds). Whereas VSM contains actin and myosin, it does not have the regulatory protein troponin, as is found in the heart. Furthermore, the arrangement of actin and myosin in VSM is not organized into distinct bands, as it is in cardiac muscle. This is not to imply that the contractile proteins of VSM are disorganized and not well developed. Actually, they are highly organized and well suited for their role in maintaining tonic contractions and reducing lumen diameter.

Contraction of the VSM can be initiated by mechanical, electrical, and chemical stimuli. Passive stretching of VSM can cause contraction that originates from the smooth muscle itself and, therefore, is termed a “myogenic response.” Electrical depolarization of the VSM cell membrane also elicits contraction, most likely by opening voltage-dependent calcium channels (L-type calcium channels) and causing an influx (increase) in the intracellular concentration of calcium ion. Finally, a number of chemical stimuli, such as norepinephrine, angiotensin II, vasopressin, endothelin-1, and thromboxane A_2 , can cause contraction. Each of these substances bind to specific receptors on the VSM cell (or to receptors on the endothelium adjacent to the VSM), which then leads to VSM contraction. The mechanism of contraction involves different signal transduction pathways, all of which converge to increase intracellular Ca^{+2} .

The mechanism by which an increase in intracellular Ca^{+2} stimulates VSM contraction is illustrated in the left panel of Figure 24.1. An increase in free intracellular Ca^{+2} results from either increased flux of Ca^{+2} into the VSM cell through calcium channels or by release of Ca^{+2} from intracellular stores of the sarcoplasmic reticulum (SR). The

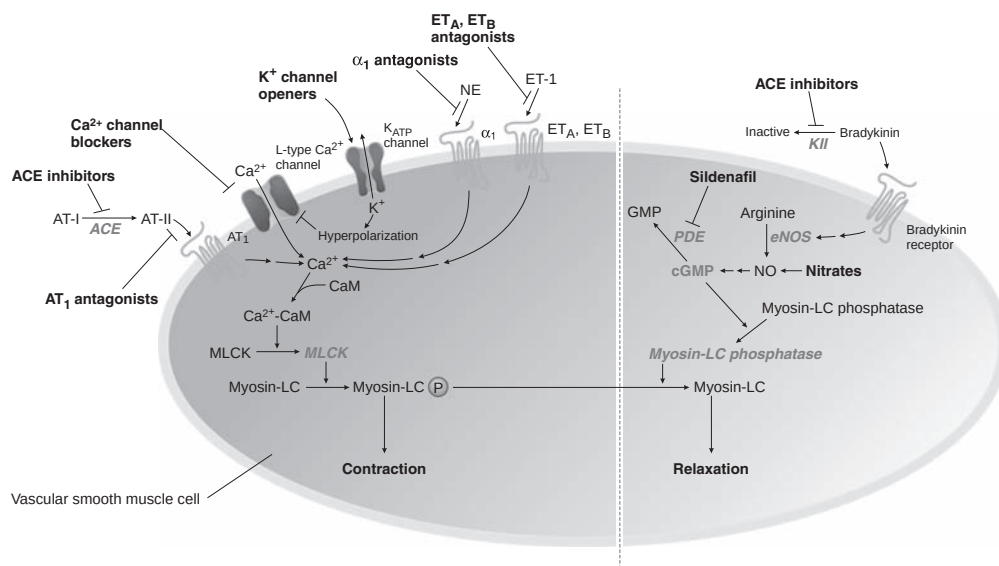


FIGURE 24.1 Calcium second messenger mechanism of vascular smooth muscle contraction and relaxation and sites of action of the peripheral and centrally acting sympatholytics and vasodilators. ACE, angiotensin-converting enzyme; AT, angiotensin; CaM, calmodulin; ET-1, endothelin peptide; ET_A and ET_B , endothelin receptors; eNOS, endothelial nitric oxide synthase; KII, kininase II or ACE; MLCK, myosin light chain kinase; PDE, phosphodiesterase. (From Yeh DC, Michel T. Pharmacology of vascular tone. In: Golan DE, Tashjian A, Armstrong E, et al., eds. Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy, 2nd Ed. Baltimore, MD: Wolters Kluwer/Lippincott Williams & Wilkins, 2008:367–385, with permission.)

SR is an internal membrane system within the VSM that functions as the major regulator of Ca^{+2} for managing VSM contractility and relaxation. The SR releases Ca^{+2} during contraction, and the released free intracellular Ca^{+2} binds to a special calcium binding protein called calmodulin (CaM), which in turn activates myosin light-chain kinase (MLCK), an enzyme that phosphorylates the myosin light chains by means of adenosine triphosphate (ATP). Phosphorylation of the myosin light chain leads to actin–myosin cross-bridge formation between the myosin heads and the actin filaments and, hence, VSM contraction. Dephosphorylation of the phosphorylated myosin light chain by myosin light-chain phosphatase yields myosin light chain, which results in relaxation of the VSM. The concentration of intracellular Ca^{+2} depends on the balance between the Ca^{+2} that enters the VSM cells, the Ca^{+2} released by the SR, and the removal of Ca^{+2} either transported by an ATP-dependent calcium pump back into SR where the Ca^{+2} is resequestered or removed from the VSM cell to the external environment by an ATP-dependent calcium pump or by the sodium–calcium exchanger.

The activation of the calcium second messenger system by hormones, neurotransmitters, local mediators, and sensory stimuli is very important in regulating VSM contraction. Several signal transduction mechanisms modulate intracellular calcium concentration and, therefore, the state of vascular tone. These calcium second messenger systems are the phosphatidylinositol (PIP_2)/ G_q protein-coupled pathway, the cyclic adenosine monophosphate (cAMP)/ G_s protein-coupled pathway, and the NO/cyclic guanosine monophosphate (cGMP) pathway.

The PIP_2 pathway in VSM is similar to that found in the heart (Fig. 24.2). The VSM membrane is lined with specific receptors for norepinephrine (α_1 -adrenoceptors), angiotensin II (AT-II), or endothelin-1 (ET-1, which binds one of two receptors ET_A or ET_B), that stimulate G_q protein, activating phospholipase C (PLC) and resulting in the formation of inositol triphosphate (IP_3) from PIP_2 in the membrane. Then, IP_3 stimulates the SR to release calcium, which in turn activates the phosphorylation of myosin light chain, causing contraction. The formation of diacylglycerol (DAG) activates protein kinase C (PKC), which also contributes to VSM contraction via protein phosphorylation.

The cAMP/ G_s protein-coupled pathway stimulates adenylate (also known as adenylyl) cyclase, which catalyzes the formation of cAMP (Fig. 24.2). In VSM, unlike the heart, an increase in intracellular cAMP concentrations stimulated by a β_2 -adrenoceptor agonist, such as epinephrine or isoproterenol, binding to the β -receptor inhibits myosin light-chain phosphorylation, causing VSM relaxation. Therefore, drugs that increase cAMP (e.g., β_2 -adrenoceptor agonists, PDE3 phosphodiesterase inhibitors) cause vasodilation. On the other hand, stimulation of G_i protein inhibits adenylyl cyclase.

A third mechanism that is also very important in regulating VSM tone is the NO/cGMP pathway (Fig. 24.1, right panel). The formation of NO in the endothelium activates guanylate (also known as guanylyl) cyclase, which causes increased formation of cGMP and

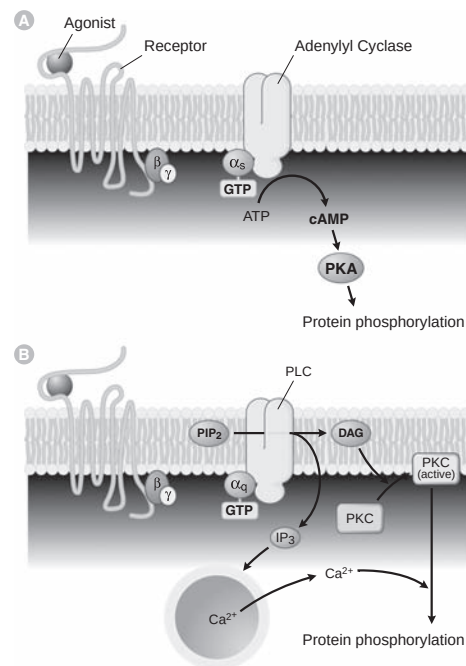


FIGURE 24.2 The mechanism of activation of the cyclic adenosine monophosphate (cAMP)/ G_s protein-coupled pathway and the phospholipase C/phosphatidylinositol (PIP_2) pathway in vascular smooth muscle. PKA, protein kinase A; DAG, diacylglycerol; IP_3 , inositol triphosphate. (From Yeh DC, Michel T. Pharmacology of vascular tone. In: Golan DE, Tashjian A, Armstrong E, et al., eds. *Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy*. Baltimore, MD: Wolters Kluwer/Lippincott Williams & Wilkins, 2008:367–385, with permission.)

vasodilation. The precise mechanisms by which cGMP relaxes VSM is unclear; however, cGMP can activate a cGMP-dependent PKC, inhibit calcium entry into the VSM, activate K^+ channels, and decrease IP_3 .

Drug Therapy of Hypertension

Peripherally Acting Sympatholytics

β -ADRENERGIC RECEPTOR BLOCKERS BBs are drugs that bind to β -adrenoceptors and, thereby, block the binding of norepinephrine and epinephrine to these receptors, causing inhibition of normal sympathetic effects. Therefore, BBs are sympatholytic drugs. Some BBs, when they bind to the β -adrenoceptor, partially activate the receptor while preventing norepinephrine from binding to the receptor. These partial agonists therefore provide some “background” of sympathetic activity while preventing normal and enhanced sympathetic activity. These particular BBs (partial agonists) are said to possess intrinsic sympathomimetic activity (ISA). Some BBs also possess what is referred to as membrane-stabilizing activity. This effect is similar to the membrane-stabilizing activity of sodium channel blockers that represent class I antiarrhythmics.

The first generation of BBs were nonselective, meaning that they blocked both β_1 - and β_2 -adrenoceptors (Fig. 24.3). Second-generation BBs are more cardioselective, because they are relatively selective for

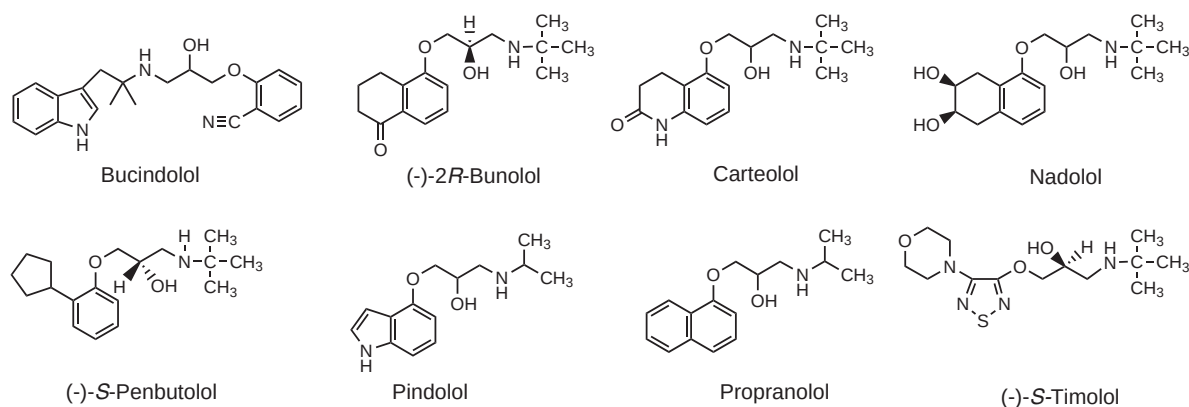


FIGURE 24.3 Nonselective β -adrenergic blockers.

β_1 -adrenoceptors (Fig. 24.4). Note that this relative selectivity can be lost at higher drug doses. Finally, the third-generation BBs are drugs that also possess vasodilator actions through blockade of vascular α -adrenoceptors (mixed α_1/β_1 -adrenergic blockers) (see Fig. 24.6). Their structure–activity relationship, pharmacokinetics, and metabolism are discussed in Chapter 10 and are also presented in Table 24.4. In addition to uncomplicated hypertension, they can also be used as monotherapy in the treatment of angina, arrhythmias, mitral valve prolapse, myocardial infarction, migraine headaches, performance anxiety, excessive sympathetic tone, or “thyroid storm” in hyperthyroidism (6).

Mechanism of Action The VSMs are lined with β_2 -adrenoceptors that normally are activated by norepinephrine released from sympathetic adrenergic nerves or by circulating epinephrine. These receptors, like those in the heart, are coupled to a G_s protein, which stimulates the formation of cAMP. Although increased cAMP enhances cardiac contraction, with VSM an increase in cAMP leads to smooth muscle relaxation (Fig. 24.2). Therefore, increases in intracellular cAMP caused by β_2 -agonists inhibit MLCK, thereby producing less contractile force

(i.e., promoting relaxation). Inhibition of cardiac β_1 - and β_2 -adrenoceptors reduces the contractility of the myocardium (negative inotropic), decreasing heart rate (negative chronotropic), blocking sympathetic outflow from the central nervous system (CNS), and suppressing renin release (9). The antianginal and antiarrhythmic effects of the BBs are discussed in Chapter 21.

Therapeutic Applications (1,6,10) BBs decrease arterial blood pressure by reducing cardiac output. Many forms of hypertension are associated with an increase in blood volume and cardiac output. Therefore, reducing cardiac output by β -blockade can be an effective treatment for hypertension, especially when used in conjunction with a diuretic. Hypertension in some patients is caused by emotional stress, which causes enhanced sympathetic activity. BBs are very effective in these patients and are especially useful in treating hypertension caused by a pheochromocytoma, which results in elevated circulating catecholamines. BBs have an additional benefit as a treatment for hypertension in that they inhibit the release of renin by the kidneys (the release of which is partly regulated by a β_1 -adrenoceptors in the kidney). Decreasing circulating plasma renin leads to a decrease

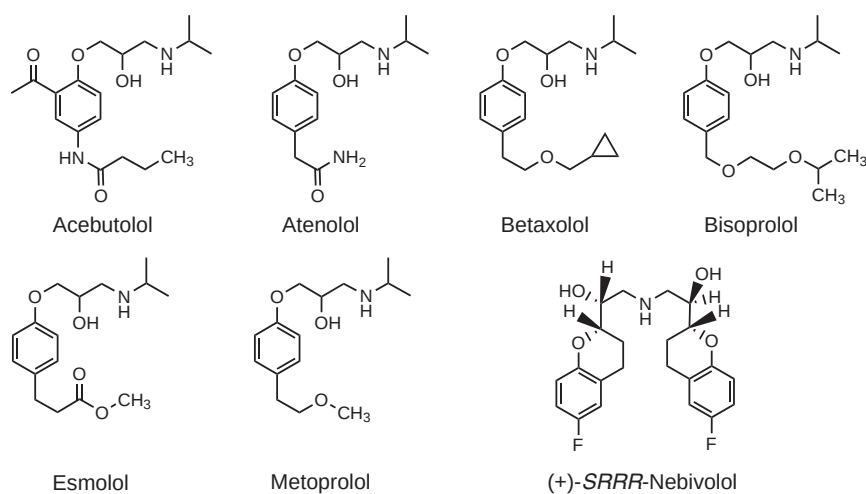


FIGURE 24.4 β_1 -Selective adrenergic blockers.

TABLE 24.4 Pharmacologic/Pharmacokinetic Properties of Antihypertensive β -Adrenergic Blocking Agents

Drug	Adrenergic Receptor Blocking Activity	Membrane-Stabilizing Activity	Intrinsic Sympatho-mimetic activity	Lipophilicity ^a ($\log D_{pH 7.4}$)	Extent of Absorption (%)	Absolute Oral Bioavailability (%)	Half-Life (h)	Protein Binding (%)	Metabolism/Excretion
Acebutolol (Sectral)	β_1^a	+	+	-0.09	90	20-60	3-4	26	Hepatic; renal excretion 30%-40%, nonrenal excretion 50%-60% (bile)
Atenolol (Tenormin)	β_1^a	o	o	-2.01	50	50-60	6-9	5-16	~50% excreted unchanged in feces
Betaxolol (Kerlone)	β_1^a	o	o	2.4	~100	89	14-22	50	Hepatic; >80% recovered in urine, 15% unchanged
Bisoprolol (Zebeta)	β_1^a	o	o	2.2	≥0	80	9-12	30	~50% excreted unchanged in urine, remainder as inactive metabolites; <2% excreted in feces
Esmolol (Brevibloc)	β_1^a	o	o	1.7	NA	NA	0.15	55	Rapid metabolism by esterases in cytosol of red blood cells
Metoprolol (Lopressor)	β_1^a	o ^b	o	-0.24	95	40-50	3-7	12	Hepatic; renal excretion, <5% unchanged
Metoprolol, LA		—	—	—	77	—			
Nebivolol (Bystolic)	β_1^a	o	o	2.34	NA	12-fast metabolizers 96-poor metabolizers	12-19	98	Hepatic; glucuronidation, N-dealkylation and oxidation by CYP2D6 Renal: 38%-67%; <1% renal unchanged; fecal 13%-44%
Carteolol (Cartrol, Ocupress)	$\beta_1 \beta_2$	o	++	-1.56	80	85	6	23-30	50%-70% excreted unchanged in urine
Levobunolol (Betagan)	$\beta_1 \beta_2$	o	o	0.19	NA	NA	NA	NA	Ophthalmic
Nadolol (Corgard)	$\beta_1 \beta_2$	o	o	-1.43	30	30-50	20-24	30	Urine, unchanged
Penbutolol (Levatol)	$\beta_1 \beta_2$	o	+	2.06	~100	>90	5	80-98	Hepatic (conjugation, oxidation); renal excretion of metabolites (17% as conjugate)
Pindolol (Visken)	$\beta_1 \beta_2$	+	+++	-0.36	95	>90	3-4 ^c	40	Urinary excretion of metabolites (60%-65%) and unchanged drug (35%-40%)
Propranolol (Inderal)	$\beta_1 \beta_2$	++	o	1.41	90	30	3-5	90	Hepatic; <1% excreted unchanged in urine
Propranolol, LA						9-18	8-11		
Timolol (Blocadren, Timoptic)	$\beta_1 \beta_2$	o	o	0.03	90	75	4	10	Hepatic; urinary excretion of metabolites and unchanged drug
Labetalol ^d (Normodyne)	$\beta_1 \beta_2 \alpha_1$	o	o	1.08	100	30-40	5.5-8.0	50	55%-60% excreted in urine as conjugates or unchanged drug
Carvedilol (Coreg)	$\beta_1 \beta_2 \alpha_1$	o	o	3.53	>90	25-35	7-10	98	

^aInhibits β_2 -receptors (bronchial and vascular) at higher doses.^bDetectable only at doses much greater than required for β -blockade.^cIn elderly hypertensive patients with normal renal function; half-life variable, 7-15 h.^dNot labetalol monograph.^eMeasured logD at pH 7.4 from Avdeef A. Absorption and Drug Development. Hoboken, NJ: Wiley-Interscience, 2003:59-66.

Adapted from Drug Facts and Comparison 2000; with permission.

NA, not applicable (available as intravenous only); o, none; +, low; ++, moderate; +++, high.

in angiotensin II and aldosterone, which enhances renal loss of sodium and water and further diminishes arterial pressure. Acute treatment with a BB is not very effective in reducing arterial pressure because of a compensatory increase in systemic vascular resistance. This can occur because of baroreceptor reflexes working in conjunction with the removal of β_2 vasodilatory influences that normally offset, to a small degree, α -adrenergic-mediated vascular tone. Chronic treatment with BB lowers arterial pressure more than acute treatment, possibly because of reduced renin release and effects of β -blockade on central and peripheral nervous systems.

Several of the nonselective BBs are also used to reduce intraocular pressure in the treatment of glaucoma. These include carteolol (Occupress), levobunolol (Betagan), and timolol (Timoptic).

The selection of oral BBs as monotherapy for stage 1 or 2 hypertension without compelling risk factors (Table 24.2) is based on several factors, including their cardioselectivity and preexisting conditions, ISA, lipophilicity, metabolism, and adverse effects (exception is esmolol) (Table 24.4). Esmolol is a very short-acting cardioselective β_1 -blocker administered by infusion because of its rapid hydrolysis by plasma esterases to a rapidly excreted zwitterionic metabolite (plasma half-life, 9 minutes). After the discontinuation of esmolol infusion, blood pressure returns to preexisting conditions in approximately 30 minutes. The elderly hypertensive patient (age, ≥ 65 years) cannot tolerate or respond to these drugs because of their mechanism of lowering cardiac output and increasing systemic vascular resistance (11).

Adverse Effects Common adverse effects for the BBs include decreased exercise tolerance, cold extremities, depression, sleep disturbance, and impotence, although these side effects can be less severe with the β_1 -selective blockers, such as metoprolol, atenolol, or bisoprolol (12). The use of lipid-soluble BBs, such as propranolol (Table 24.4), has been associated with more CNS side effects, such as dizziness, confusion, or depression (1,6). These side effects can be avoided, however, with the use of hydrophilic drugs, such as nadolol or atenolol. The use of β_1 -selective drugs also helps to minimize adverse effects associated with β_2 -blockade, including suppression of insulin release and increasing the chances for bronchospasms (asthma) (1,6). It is important to emphasize that none of the BBs, including the cardioselective ones, are cardiospecific. At high doses, these cardioselective BBs can still adversely affect asthma, peripheral vascular disease, and diabetes (1,6). Nonselective BB are contraindicated in patients with bronchospastic disease (asthma), and β_1 -selective blockers should be used with caution in these patients. BBs with ISA, such as acebutolol, pindolol, carteolol, or penbutolol (Table 24.4), partially stimulate the β -receptor while also blocking it (13). The proposed advantages of BBs with ISA over those without ISA include less cardiodepression and resting bradycardia as well as neutral effects on lipid and glucose metabolism.

Neither cardioselectivity nor ISA, however, influences the efficacy of BBs in lowering blood pressure (6).

Unlike the conventional cardioselective β_1 -receptor blocker, nebivolol also exhibits NO-potentiating vasodilatory effect for the treatment of hypertension. Chemically, it is a mixture of stereoisomers (+)-nebivolol [(+)-*SRRR* nebivolol] and (–)-nebivolol [(–)-*RSSS* nebivolol] that differs chemically and pharmacologically from other BBs. The selective β_1 -blocking effect is determined almost exclusively by the (+)-stereoisomer. The combination of (+)-nebivolol and (–)-nebivolol acts synergistically to produce a cardiovascular profile that differs noticeably from that of conventional BBs with respect to enhanced blood pressure reduction at lower doses. The drug is highly cardioselective at low doses, but at higher doses, it loses its cardioselectivity and blocks both β_1 - and β_2 -receptors. The (–)-stereoisomer has minimal effect on systolic and diastolic blood pressure. Nebivolol is unique among the conventional BBs by stimulating endothelial NO synthesis, thereby producing a sustained vasodilation in VSM, which results in decreased peripheral resistance and blood pressure. The (–)-stereoisomer indirectly increases NO availability by the inhibition of endothelial NO synthase, thereby reducing NO inactivation. Neither nebivolol nor its stereoisomers show any intrinsic sympathomimetic activity, without the undesirable BB effects, such as a decrease in cardiac output. Nebivolol is also not cardioselective when taken by patients who are poor CYP2D6 metabolizers of nebivolol (and other drugs). As many as 1 in 10 whites and even more blacks are poor CYP2D6 metabolizers and therefore would not be likely to benefit from nebivolol's cardioselectivity.

α_1 -ADRENERGIC BLOCKERS The structures for the available α_1 -receptor blockers are shown in Figure 24.5. These include prazosin ($\log D_{7.4} = 1.70$), doxazosin ($\log D_{7.4} = 1.97$), terazosin ($\log D_{7.4} = 2.13$), and alfuzosin ($\log D_{7.4} = 0.87$). The structure–activity relationships of these drugs were previously discussed in Chapter 10, along with their pharmacokinetics and metabolism.

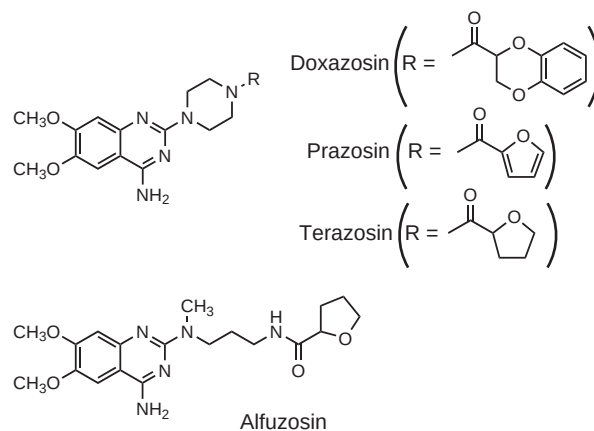


FIGURE 24.5 α_1 -Selective adrenergic blockers.

Mechanism of Action These drugs block the effect of sympathetic nerves on blood vessels by selectively binding to α_1 -adrenoceptors located on the VSM (Fig. 24.1), which then stimulate the G_q protein, activating smooth muscle contraction through the IP_3 signal transduction pathway. Most of these drugs act as competitive antagonists by competing with the binding of norepinephrine to α_1 -adrenergic receptors on VSM. Some α -blockers are noncompetitive (e.g., phenoxybenzamine) (see Chapter 10), which greatly prolongs their action. Prejunctional α_2 -adrenoceptors located on the sympathetic nerve terminals serve as a negative feedback mechanism for norepinephrine release.

α -Blockers dilate both arteries and veins, because both vessel types are innervated by sympathetic adrenergic nerves. The vasodilator effect is more pronounced, however, in the arterial resistance vessels. Because most blood vessels have some degree of sympathetic tone under basal conditions, these drugs are effective dilators. They are even more effective under conditions of elevated sympathetic activity (e.g., during stress) or during pathologic increases in circulating catecholamines caused by an adrenal gland tumor (pheochromocytoma) (9). α_2 -Adrenoceptors are also abundant in the smooth muscle of the bladder neck and prostate and, when inhibited, cause relaxation of the bladder muscle, increasing urinary flow rates and the relief.

Therapeutic Applications (6,14) α_1 -Blockers are effective agents for the initial management of hypertension and are especially advantageous for older men who also suffer from symptomatic benign prostatic hyperplasia. They have been shown to be as effective as other major classes of antihypertensives in lowering blood pressure in equivalent doses. α_1 -Blockers possess a characteristic “first-dose” effect, which means that orthostatic hypotension frequently occurs with the first few doses of the drug. This side effect can be minimized by slowly increasing the dose and by administering the first few doses at bedtime.

Side Effects and Contraindications The most common side effects are related directly to α_1 -adrenoceptor blockade. These side effects include dizziness, orthostatic hypotension (because of loss of reflex vasoconstriction on standing), nasal congestion (because of dilation of nasal mucosal arterioles), headache, and reflex tachycardia (especially with nonselective α -blockers). Fluid retention is also a problem that can be rectified by use of a diuretic in conjunction with the α_1 -blocker. α -Blockers have not been shown to be beneficial in heart failure or angina and should not be used in these conditions.

MIXED α/β -BLOCKERS The two available mixed α/β -receptor blockers are carvedilol (15) and labetalol (16) (Fig. 24.6), and their structure–activity relationships were previously discussed in Chapter 10 along with their

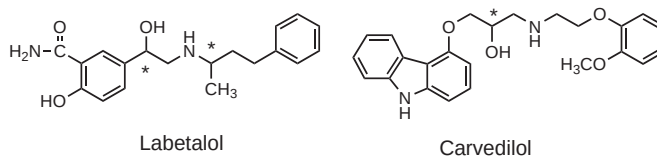


FIGURE 24.6 Mixed α/β -selective adrenergic blockers.

pharmacokinetics and metabolism (Table 24.4). The α -methyl substituent attached to the N -aryalkyl group appears to be responsible for the α -adrenergic blocking effect. Carvedilol is administered as its racemate; its $S(-)$ -enantiomer is both an α - and nonselective β -blocker, whereas its $R(+)$ -enantiomer is an α_1 -blocker. Labetalol possesses two chiral centers and, therefore, is administered as a mixture of four stereoisomers, of which $R(CH_3), R(OH)$ is the active β -blocker diastereomer with minimal α_1 -blocking activity and the $S(CH_3), R(OH)$ diastereomer is predominantly an α_1 -blocker. The R,R diastereomer is also known as dilevalol, which was not approved by the FDA because of hepatotoxicity. The $S(CH_3), S(OH)$ and $R(CH_3), S(OH)$ diastereomers are both inactive. The comparative potency for labetalol reflects the fact that 25% of the diastereomeric mixture is the active R,R -diastereomer.

Mechanism of Action The mixed α/β -receptor blocking properties in the same molecule confer some advantages in the lowering of blood pressure. Vasodilation via α_1 -blockade lowers peripheral vascular resistance to maintain cardiac output, thus preventing bradycardia more effectively when compared to BBs (17). β -Blockade helps to avoid the reflex tachycardia sometimes observed with the other vasodilators listed later in this chapter.

Therapeutic Applications Monotherapy with these mixed-acting antihypertensive drugs reduces blood pressure as effectively as other major antihypertensives and their combinations (15–17). Selection of mixed α/β -blockers is recommended for management of hypertension when the stage 2 family of drugs cannot be used alone or when a compelling indication (Table 24.2) is present that requires the use of a specific drug. Both drugs effectively lower blood pressure in essential and renal hypertension. Carvedilol is also effective in ischemic heart disease.

Adverse Effects Any adverse effects are usually related to β_1 - or α_1 -blockade. The β effects are usually less bothersome, because the α_1 -blockade reduces the effects of β -blockade.

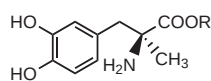
Centrally Acting Sympatholytics

The sympathetic adrenergic nervous system plays a major role in the regulation of arterial pressure. Activation of these nerves to the heart increases the heart rate (positive chronotropy), contractility (positive inotropy), and velocity of electrical impulse conduction

(positive chronotropy). Within the medulla are located preganglionic sympathetic excitatory neurons, which travel from the spinal cord to the ganglia. They have significant basal activity, which generates a level of sympathetic tone to the heart and vasculature even under basal conditions. The sympathetic neurons within the medulla receive input from other neurons within the medulla, and together, these neuronal systems regulate sympathetic (and parasympathetic) outflow to the heart and vasculature. Sympatholytic drugs can block this sympathetic adrenergic system on three different levels. First, peripheral sympatholytic drugs, such as α -adrenoceptor and β -adrenoceptor antagonists, block the influence of norepinephrine at the effector organ (heart or blood vessel). Second, there are ganglionic blockers that block impulse transmission at the sympathetic ganglia. Third, centrally acting sympatholytic drugs block sympathetic activity within the brain. Centrally acting sympatholytics block sympathetic activity by binding to and activating α_2 -adrenoceptors, which reduces sympathetic outflow to the heart, thereby decreasing cardiac output by decreasing heart rate and contractility. Reduced sympathetic output to the vasculature decreases sympathetic vascular tone, which causes vasodilation and reduced systemic vascular resistance, which in turn decreases arterial pressure.

Specific Drugs

METHYLDOPA AND METHYLDOPATE ESTER HYDROCHLORIDE



L-Methyldopa R = H

Methyldopate ethyl ester hydrochloride R = C₂H₅

Physicochemical Properties Methyldopa is structurally and chemically related to L-dihydroxyphenylalanine (L-DOPA) and the catecholamines. To increase its water solubility for parenteral administration, the zwitterion methyldopa is esterified and converted to its hydrochloride salt, methyldopate ethyl ester hydrochloride (referred to as methyldopate). Methyldopate ester hydrochloride is used to prepare parenteral solutions of methyldopa, having a pH in the range of 3.5 to 6.0. Methyldopa is unstable in the presence of oxidizing agents (i.e., air), alkaline pH, and light. Being related to the catecholamines, which are subject to air oxidation, metabisulfite/sulfite can be added to dosage formulations to prevent oxidation. Some patients, especially those with asthma, can exhibit sulfite-related hypersensitivity reactions. Methyldopate hydrochloride injection has been reported to be physically incompatible with drugs that are poorly soluble in an acidic medium (e.g., sodium salts of barbiturates and sulfonamides) and with drugs that are acid labile. Incompatibility depends on several factors (e.g., concentrations of the drugs, specific diluents used, resulting pH, and temperature).

Mechanism of Action As discussed in Chapter 10, the central mechanism for the antihypertensive activity of the prodrug methyldopa is not caused by its inhibition of norepinephrine biosynthesis but, rather, by its metabolism in the CNS to α -methylnorepinephrine, an α_2 -adrenergic agonist (Fig. 24.7) (9). Other more powerful inhibitors of aromatic L-amino acid decarboxylase (e.g., carbidopa) have proven to be clinically useful, but not as antihypertensives. Rather, these agents are used to inhibit the metabolism of exogenous L-DOPA administered in the treatment of Parkinson disease (see Chapter 13).

The mechanism of the central hypotensive action for methyldopa is attributed to its transport into the CNS via an aromatic amino acid transport mechanism, where it is decarboxylated and hydroxylated into α -methylnorepinephrine (Fig. 24.7) (9). This active metabolite of methyldopa decreases total peripheral resistance, with little change in cardiac output and heart rate, through its stimulation of central inhibitory α_2 -adrenoceptors. A reduction of plasma renin activity can also contribute to the hypotensive action of methyldopa. Postural hypotension and sodium and water retention are also effects related to a reduction in blood pressure. If a diuretic is not administered concurrently with methyldopa, tolerance to the antihypertensive effect of the methyldopa during prolonged therapy can result.

Pharmacokinetics (18) The oral bioavailability of methyldopa ranges from 20% to 50% and varies among individuals. Optimum blood pressure response occurs in 12 to 24 hours in most patients. After withdrawal of the drug, blood pressure returns to pretreatment levels within 24 to 48 hours. Methyldopa and its metabolites are weakly bound to plasma proteins. Although 95% of a dose of methyldopa is eliminated in hypertensive patients with normal

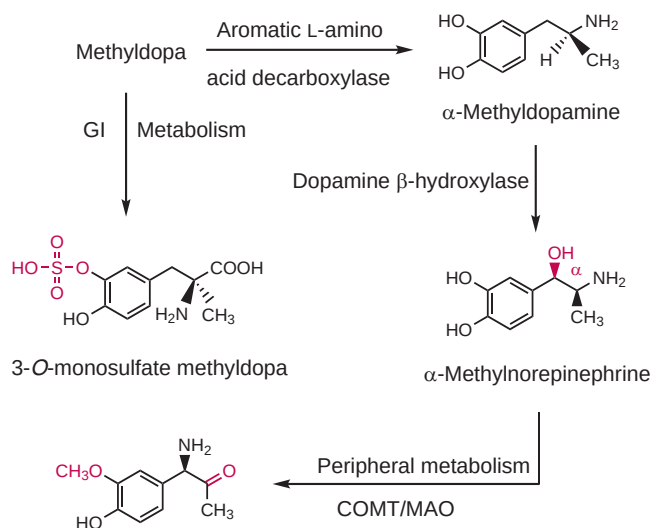


FIGURE 24.7 Metabolism of methyldopa.

renal function, with a plasma half-life of approximately 2 hours, in patients with impaired renal function, the half-life is doubled to approximately 3 to 4 hours, with about 50% of it excreted. Orally administered methyldopa undergoes presystemic first-pass metabolism in the gastrointestinal (GI) tract to its 3-*O*-monosulfate metabolite (Fig. 24.7). Sulfate conjugation occurs to a greater extent when the drug is given orally than when it is given intravenously (IV). Its rate of sulfate conjugation is decreased in patients with renal insufficiency. Methyldopa is excreted in urine as its mono-*O*-sulfate conjugate. Any peripherally decarboxylated α -methylnorepinephrine is metabolized by catechol-*O*-methyltransferase (COMT) and monoamine oxidase (MAO) (Fig. 24.7).

Methyldopate is slowly hydrolyzed in the body to form methyldopa. The hypotensive effect of IV methyldopate begins in 4 to 6 hours and lasts 10 to 16 hours.

Therapeutic Applications (1,6) Methyldopa is used in the management of moderate to severe hypertension and is reserved for patients who fail to achieve blood pressure goals with stage 2 drugs. Methyldopa is also coadministered with diuretics and other classes of antihypertensive drugs, permitting a reduction in the dosage of each drug and minimizing adverse effects while maintaining blood pressure control. Methyldopa has been used in the management of hypertension during pregnancy without apparent substantial adverse effects on the fetus and also for the management of pregnancy-induced hypertension (i.e., preeclampsia) (5).

Intravenous methyldopate can be used for the management of hypertension when parenteral hypotensive therapy is necessary. Because of its slow onset of action, however, other agents, such as sodium nitroprusside, are preferred when a parenteral hypotensive agent is employed for hypertensive emergencies.

Adverse Effects (6) The most common adverse effect for methyldopa is drowsiness, which occurs within the first 48 to 72 hours of therapy and can disappear with continued administration of the drug. Sedation commonly recurs when its dosage is increased. A decrease in mental acuity, including impaired ability to concentrate, lapses of memory, and difficulty in performing simple calculations, can occur and usually necessitates withdrawal of the drug. Patients should be warned that methyldopa can impair their ability to perform activities requiring mental alertness or physical coordination (e.g., operating machinery or driving a motor vehicle). Nightmares, mental depression, orthostatic hypotension, and symptoms of cerebrovascular insufficiency can occur during methyldopa therapy and are indications for dosage reduction. Orthostatic hypotension can be less pronounced with methyldopa than with guanethidine but can be more severe than with reserpine, clonidine, hydralazine, propranolol, or thiazide diuretics. Nasal congestion commonly occurs in patients receiving methyldopa. Positive direct antiglobulin (Coombs') test results have

been reported in approximately 10% to 20% of patients receiving methyldopa, usually after 6 to 12 months of therapy. This phenomenon is dose related. Methyldopa should be used with caution in patients with a history of previous liver disease or dysfunction, and it should be stopped if unexplained drug-induced fever and jaundice occur. These effects commonly occur within 3 weeks after initiation of treatment.

Dosage forms of methyldopa and methyldopate can contain sulfites, which can cause allergic-type reactions, including anaphylaxis and life-threatening or less severe asthmatic episodes. These allergic reactions are observed more frequently in asthmatic than in nonasthmatic individuals. Methyldopa is contraindicated in patients receiving MAO inhibitors.

α_2 -ADRENERGIC AGONISTS The mechanism of action, therapeutic applications, and adverse effects common to the α_2 -adrenergic agonists clonidine, guanabenz, and guanfacine (Fig. 24.8) will be discussed together, but any significant differences between these specific agents will be included in the discussions of the individual drugs.

Agmatine (Fig 24.8), an aminoguanidine, is an endogenous neurotransmitter found in the brain with high affinity for the G protein-coupled imidazoline receptors, I_1 and I_2 , α_2 , and glutaminergic *N*-methyl-D-aspartate (NMDA) receptor channels, where it binds as an antagonist. It is biosynthesized by the decarboxylation of the amino acid, arginine, metabolized by agmatase to the polyamine, putrescine, which binds to NMDA receptor channels (Fig. 24.9). As a neurotransmitter, agmatine is stored in vesicles in neurons and released upon neuronal depolarization. It also irreversibly inhibits neuronal nitric oxide synthase. Agmatine is released from storage vesicles in response to stress (e.g., hypoxic-ischemia) and/or inflammation. The sympathoinhibition of agmatine involves α_2 -adrenoceptors and imidazoline I_1 receptors. Despite binding like clonidine to α_2 -adrenergic receptors and imidazoline I_1 and I_2 receptors, agmatine does not replicate the central

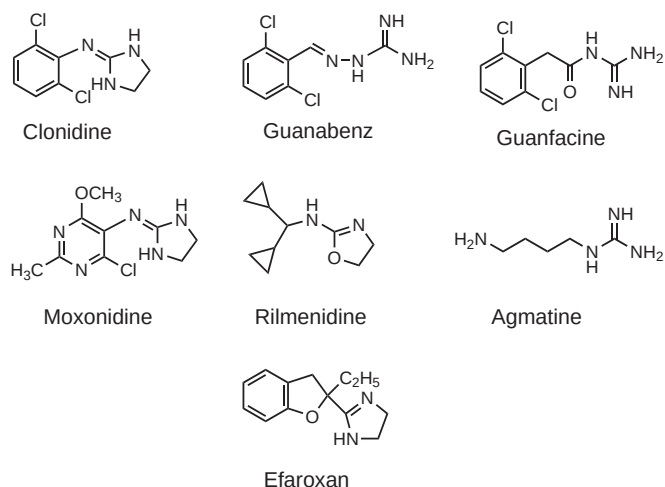


FIGURE 24.8 Centrally acting sympatholytics.

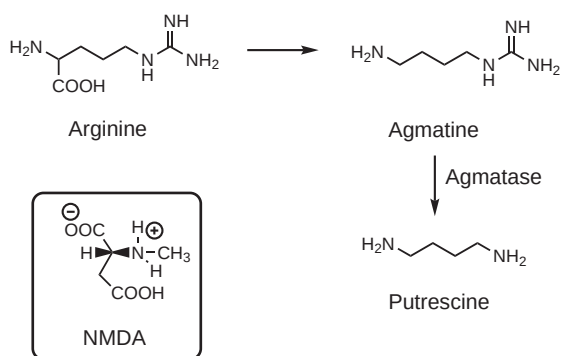


FIGURE 24.9 Biosynthesis and metabolism of agmatine.

or peripheral actions of clonidine. Agmatine appears to be a CNS endogenous neurotransmitter or neuromodulator (not reabsorbed by the presynaptic neuron or metabolized into inactive products) in stress-related disorders, such as depression, anxiety, and posttraumatic stress disorder. Although the physiologic role of agmatine in normal brain function is still unknown, it also blocks the development of morphine tolerance and exhibits antidepressant-like effects in depression models. Agmatine, administered intracerebroventricular or intraperitoneal in animal models, has not been shown to lower arterial pressure, thus ruling out the possibility of α_2 -adrenergic receptor activation in this action of agmatine.

Mechanism of Action The overall mechanism of action for the centrally active sympatholytics, clonidine, guanabenz, and guanfacine, appears to be stimulation of α_2 -adrenoceptors and specific binding to nonadrenergic imidazoline I_1 receptors in the CNS (mainly in the medulla oblongata), causing inhibition of sympathetic output (sympathoinhibition) (19,20). This effect results in reduced peripheral and renovascular resistance and leads to a decrease in systolic and diastolic blood pressure. Through the use of imidazoline and α_2 -adrenergic antagonists, specific I_1 receptors have recently been characterized in CNS control of blood pressure (21). The I_1 receptors are pharmacologically distinct from α_2 -receptors, because they are not activated by catecholamines but characterized by their high affinity for 2-iminoimidazolines (or 2-aminoimidazolines) and low affinity for guanidines (21). Thus, the central hypotensive action for clonidine, other 2-aminoimidazolines, and structurally related compounds needs both the I_1 and α_2 -adrenoceptors to produce their central sympatholytic response (20). As a result of this discovery, a new generation of centrally acting antihypertensive agents selective for the I_1 receptor has been developed and includes moxonidine (a pyrimidinyl aminoimidazoline) and rilmenidine (an alkylaminoimidazoline) (Fig. 24.8). Rilmenidine and moxonidine are both highly selective for the I_1 receptor while having low affinity for α_2 -adrenoceptors, and both control blood pressure effectively without the adverse effects associated with binding to α_2 -receptors (e.g., sedation, bradycardia,

and mental depression) (20). Clonidine appears to be more selective for α_2 -adrenoceptors than for I_1 receptors. Another antihypertensive is efaroxan, which exhibits good affinity for I_1 receptor but is an antagonist at α_2 -receptors.

Pharmacokinetics The effective oral dose range for rilmenidine is 1 to 3 mg, with a dose-dependent duration of action of 10 to 20 hours. Moxonidine is administered once a day at a dose range of 0.2 to 0.4 mg. The oral bioavailability of moxonidine in humans is greater than 90%, with approximately 40% to 50% of the oral dose excreted unmetabolized in the urine (22,23). The principal route of metabolism for moxonidine is oxidation of the 2-methyl group in the pyrimidine ring to 2-hydroxymethyl and 2-carboxylic acid derivative as well as the formation of corresponding glucuronides. After an oral dose of moxonidine, peak hypotensive effects occur within 2 hours, with an elimination half-life of greater than 8 hours (23,24). Rilmenidine is readily absorbed from the GI tract, with an oral bioavailability greater than 95%. It is poorly metabolized and is excreted unchanged in the urine, with an elimination half-life of 8 hours (25,26).

After IV or oral administration of these drugs in normotensive patients, an initial hypertensive response to the drug occurs that is caused by activation of the peripheral α_2 -adrenoceptors and the resulting vasoconstriction. This response is not observed, however, in patients with hypertension.

Therapeutic Applications (1,6) The selection of these drugs for monotherapy or in the combination therapy is based on several factors, including their similar mechanism of action, preexisting or compelling conditions, pharmacokinetics, distribution, and metabolism. The α_2 -adrenergic antagonists show a similarity in adverse effects.

Clonidine, guanabenz, and guanfacine are used in the management of mild to moderate hypertension or when stage 2 drugs have been ineffective in achieving blood pressure goals (1,6). They have been used as monotherapy or to achieve lower dosages in combination with other classes of antihypertensive agents. The centrally acting sympatholytics are generally reserved for patients who fail to respond to therapy with a stage 1 drug (e.g., diuretics, β -adrenergic blocking agents, ACEIs, or ARBs). Clonidine, guanabenz, and guanfacine can be used in combination with diuretics and other stage 1 hypotensive agents, permitting a reduction in the dosage of each drug, which minimizes adverse effects while maintaining blood pressure control. Geriatric patients, however, cannot tolerate the adverse cognitive effects of these sympatholytics. All three drugs reduce blood pressure to essentially the same extent in both supine and standing patients; thus, orthostatic effects are mild and infrequently encountered. Exercise does not appear to affect the blood pressure response to guanabenz and guanfacine in patients with hypertension. Plasma renin activity

can be unchanged or reduced during long-term therapy with these drugs.

Adverse Effects (6) Overall, the frequency of adverse effects produced by clonidine, guanabenz, and guanfacine are similar and appear to be dose related. Drowsiness, tiredness, dizziness, weakness, bradycardia, headache, and dry mouth are common adverse effects for patients receiving clonidine, guanabenz, and guanfacine. The sedative effect for these centrally acting sympatholytics can result from their central α_2 -agonist activity. The dry mouth induced by these drugs can result from a combination of central and peripheral α_2 -adrenoceptor mechanisms, and the decreased salivation can involve inhibition of cholinergic transmission via stimulation of peripheral α_2 -adrenoceptors. Orthostatic hypotension does not appear to be a significant problem with these drugs, because there appears to be little difference between supine and standing systolic and diastolic blood pressures in most patients. Other adverse effects include increased urinary frequency (nocturia), urinary retention, sexual dysfunction (e.g., decreased libido, erectile dysfunction, and impotence), nasal congestion, tinnitus, blurred vision, and dry eyes. These symptoms most often occur within the first few weeks of therapy and tend to diminish with continued therapy, or they can be relieved by a reduction in dosage. Although adverse effects of the drug are generally not severe, discontinuance of therapy has been necessary in some patients because of intolerable sedation or dry mouth. Sodium and fluid retention can be avoided or relieved by administration of a diuretic.

Drug Interactions (6) The hypotensive actions for clonidine, guanabenz, and guanfacine can be additive with, or can potentiate the action of, other CNS depressants, such as opiates or other analgesics, barbiturates or other sedatives, anesthetics, or alcohol. Coadministration of opiate analgesics with clonidine can also potentiate the hypotensive effects of clonidine. Tricyclic antidepressants (i.e., imipramine and desipramine) have reportedly inhibited the hypotensive effect of clonidine, guanabenz, and guanfacine, and the increase in blood pressure usually occurs during the second week of tricyclic antidepressant therapy. Dosage should be increased to adequately control hypertension if necessary. Sudden withdrawal of clonidine, guanabenz, and guanfacine can result in an excess of circulating catecholamines; therefore, caution should be exercised in concomitant use of drugs that affect the metabolism or tissue uptake of these amines (MAO inhibitors or tricyclic antidepressants, respectively). Because clonidine, guanabenz, and guanfacine can produce bradycardia, the possibility of additive effects should be considered if these drugs are given concomitantly with other drugs, such as hypotensive drugs or cardiac glycosides.

Specific Drugs

Clonidine. Clonidine is an aryl-2-aminoimidazoline that is more selective for α_2 -adrenoceptors than for I_1

receptors (Fig. 24.8) in producing its hypotensive effect. It is available as oral tablets, injection, or a transdermal system.

Mechanism of Action. In addition to its central stimulation of I_1 receptors and α_2 -adrenoceptors (20,21), clonidine (as well as other α_2 -adrenergic agonists), when administered epidurally, produces analgesia by stimulation of spinal α_2 -adrenoceptors, inhibiting sympathetically mediated pain pathways that are activated by nociceptive stimuli, thus preventing transmission of pain signals to the brain (9) (also see Chapter 20). The administration of clonidine, while blocking opiate withdrawal, causes sympathetic inhibition and reduction in arterial pressure. The actions of clonidine on morphine pain response and withdrawal are most likely mediated by its ability to block NMDA receptor channels. Analgesia resulting from clonidine therapy is not antagonized by opiate antagonists. Activation of α_2 -adrenoceptors also apparently stimulates acetylcholine release and inhibits the release of substance P, an inflammatory neuropeptide.

Pharmacokinetics. Clonidine has an oral bioavailability of more than 90%, with a $\log D_{7.4}$ of 1.82. It is well absorbed when applied topically to the eye and is well absorbed percutaneously following topical application of a transdermal system to the arm or chest (27–29). Following application of a clonidine transdermal patch, therapeutic plasma concentrations are attained within 2 to 3 days. Studies have indicated that release of clonidine from the patch averages from 50% to 70% after 7 days of wear. Plasma clonidine concentrations attained with the transdermal systems are generally similar to twice-daily oral dosing regimens of the drug. Percutaneous absorption of the drug from the upper arm or chest is similar, but less drug is absorbed from the thigh (29). Replacement of the transdermal system at a different site at weekly intervals continuously maintains therapeutic plasma clonidine concentrations. After discontinuance of transdermal therapy, therapeutic plasma drug concentrations persist for approximately 8 hours and then decline slowly over several days; over this time period, blood pressure returns gradually to pretreatment levels.

Blood pressure begins to decrease within 30 to 60 minutes after an oral dose of clonidine, with the maximum decrease in approximately 2 to 4 hours (6). The hypotensive effect lasts up to 8 hours. Following epidural administration of a single bolus dose of clonidine, it is rapidly absorbed into the systemic circulation and into cerebrospinal fluid (CSF), with maximal analgesia within 30 to 60 minutes. Although the CSF is not the presumed site of action of clonidine-mediated analgesia, the drug appears to diffuse rapidly from the CSF to the dorsal horn of the spinal cord. After oral administration, clonidine appears to be well distributed throughout the body, with the lowest concentration in the brain. Clonidine is approximately 20% to 40% bound to plasma proteins, and it crosses the placenta. The plasma half-life of clonidine is 6 to 20 hours in patients with normal renal

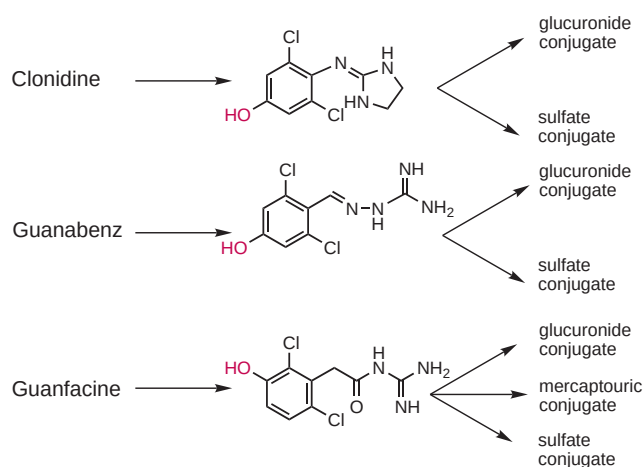


FIGURE 24.10 Metabolites formed from clonidine, guanabenz, and guanfacine.

function and 18 to 41 hours in patients with impaired renal function. Clonidine is metabolized in the liver to its inactive major metabolite 4-hydroxyclo-nidine and its glucuronide and sulfate conjugates (10% to 20%) (Fig. 24.10). In humans, 40% to 60% of an oral or IV dose of clonidine is excreted in urine as unchanged drug within 24 hours. Approximately 85% of a single dose is excreted within 72 hours, with 20% of the dose excreted in feces, probably via enterohepatic circulation.

Therapeutic Applications (6). Clonidine is administered twice a day for the management of mild to moderate hypertension in those patients not achieving the blood pressure goal with stage 2 drugs (6). Transdermal clonidine has also been successfully substituted for oral clonidine in some patients with mild to moderate hypertension whose compliance with a daily dosing regimen can be a problem (28).

When administered by epidural infusion, clonidine is used as adjunct therapy in combination with opiates for the management of severe cancer pain not relieved by opiate analgesics alone. Other nonhypertensive uses for clonidine include the prophylaxis of migraine headaches, the treatment of severe dysmenorrhea, menopausal flushing, rapid detoxification in the management of opiate withdrawal in opiate-dependent individuals, in conjunction with benzodiazepines for the management of alcohol withdrawal, and for the treatment of tremors associated with the adverse effects of methylphenidate in patients with attention-deficit hyperactivity disorder (ADHD). Clonidine has been used to reduce intraocular pressure in the treatment of open-angle and secondary glaucoma.

Clonidine, a nonstimulant drug, is also used in reducing the symptoms of ADHD and bipolar disorders, whose mechanism of action is thought to regulate norepinephrine release from the locus ceruleus, part of the brain stem involved with physiologic responses to stress and panic. Clonidine can also reduce symptoms of aggression as with patients with bipolar disorders and reduce

the insomnia associated with CNS stimulants such as methylphenidate. Once stabilized, children on larger doses can be switched to the transdermal clonidine patch.

Adverse Effects (6). Adverse effects occurring with transdermal clonidine generally appear to be similar to those occurring with oral therapy (28,29). They have been mild and have tended to diminish with continued treatment. Hypotension has occurred in patients receiving clonidine by epidural infusion as adjunct therapy with epidural morphine for the treatment of cancer pain. With the transdermal system, localized skin reactions, such as erythema and pruritus, have occurred in some patients. The use of clonidine and methylphenidate in combination for managing ADHD can adversely affect cardiac conduction and increase risk for arrhythmias. ADHD patients need to be closely monitored and screened for a patient or family history of rhythm disturbances, and periodic monitoring of blood pressure, heart rate, and rhythm is recommended.

Within 2 to 3 hours following the abrupt withdrawal of oral clonidine therapy, a rapid increase in systolic and diastolic blood pressures occurs, and blood pressures can exceed pretreatment levels. Associated with the clonidine withdrawal syndrome, the symptoms observed include nervousness, agitation, restlessness, anxiety, insomnia, headache, sweating, palpitations, increased heart rate, tremor, and increased salivation. The exact mechanism of the withdrawal syndrome following discontinuance of α_2 -adrenergic agonists has not been determined but can involve increased concentrations of circulating catecholamines, increased sensitivity of adrenoceptors, enhanced renin-angiotensin system activity, decreased vagal function, failure of autoregulation of cerebral blood flow, and failure of central α_2 -adrenoceptor mechanisms to regulate sympathetic outflow from the CNS (6). The clonidine withdrawal syndrome is more pronounced after abrupt cessation of long-term therapy and with administration of high oral dosages (>1.2 mg daily). Withdrawal symptoms have been reported following discontinuance of transdermal therapy or when absorption of the drug is impaired because of dermatologic changes (e.g., contact dermatitis) under the transdermal system. Epidural clonidine can prolong the duration of the pharmacologic effects, including both sensory and motor blockade, of epidural local anesthetics.

Guanabenz Acetate. Guanabenz, a centrally active hypotensive agent, is pharmacologically related to clonidine but differs structurally from clonidine by the presence of an aminoguanidine side chain rather than an aminoimidazoline ring (Fig. 24.8). Guanabenz ($pK_a = 8.1$; $\log D_{7.4} = 2.42$) is largely (~80%) in the nonionized, lipid-soluble base form. Guanabenz can be given as a single daily dose administered at bedtime to minimize adverse effects.

Pharmacokinetics (30). The oral bioavailability of guanabenz is 70% to 80%. After an oral dose, the hypotensive effect of guanabenz begins within 1 hour, peaks within

2 to 7 h, and is diminished within 6 to 8 hours. It has an elimination half-life averaging 4 to 14 hours. The blood pressure response can persist for at least 12 hours. After IV dosing, guanabenz is distributed into the CNS, with brain concentrations 3 to 70 times higher than concurrent plasma concentrations. Guanabenz is approximately 90% bound to plasma proteins. In patients with hepatic or renal impairment, its elimination half-life can be prolonged.

Guanabenz is metabolized principally by hydroxylation to its inactive metabolite, 4-hydroxyguanabenz, which is eliminated in the urine as its glucuronide (major) and sulfate conjugates (Fig. 24.10). Guanabenz and its inactive metabolites are excreted principally in urine, with approximately 70% to 80% of its oral dose excreted in urine within 24 hours and approximately 10% to 30% excreted in feces via enterohepatic cycling. Approximately 40% of an oral dose of guanabenz is excreted in urine as 4-hydroxyguanabenz and its glucuronide, and less than 5% is excreted unchanged. The remainder is excreted as unidentified metabolites and their conjugates.

Therapeutic Applications (6,30). The therapeutic applications for guanabenz are similar to those of clonidine and other α_2 -adrenergic agonists. One advantage for guanabenz is its once-a-day dosing schedule. Guanabenz has been used in diabetic patients with hypertension without adverse effect on the control of or therapy for diabetes, and it has been effective in hypertensive patients with chronic obstructive pulmonary disease, including asthma, chronic bronchitis, or emphysema. Guanabenz has been used alone or in combination with naltrexone in the management of opiate withdrawal in patients physically dependent on opiates and undergoing detoxification. Guanabenz has also been used as an analgesic in a limited number of patients with chronic pain.

Adverse Effects (6,30). Overall, the frequency of adverse effects produced by guanabenz is similar to that produced by clonidine and the other α_2 -adrenergic agonists, but the incidence is lower. As with the other centrally acting sympatholytics (e.g., clonidine), abrupt withdrawal of guanabenz can result in rebound hypertension, but the withdrawal syndrome symptoms appear to be less severe.

Guanfacine Hydrochloride. Guanfacine, a phenylacetyl guanidine derivative ($pK_a = 7.1$; $\log D_{7.4} = 1.52$) (Fig. 24.8), is a centrally acting sympatholytic that is more selective for α_2 -adrenoceptors than is clonidine. Its mechanism of action is similar to clonidine and is an effective alternative to that of the other centrally acting antihypertensive drugs. Although guanfacine is 5- to 20-fold less potent than clonidine on a weight basis, comparable blood pressure-lowering effects have been achieved when the two drugs were given in equipotent dosages. Its relatively long elimination half-life permits a once-a-day dosing schedule. Guanfacine activates peripheral α_2 -adrenoceptors

because a transient increase in blood pressure is observed in normotensive, but not in hypertensive, patients.

Pharmacokinetics (31–33). The pharmacokinetic properties for guanfacine differ from those of clonidine, guanabenz, and α -methyldopa. At pH 7.4, guanfacine is predominately (67%) in the nonionized, lipid-soluble base form, which accounts for its high oral bioavailability (>80%). Following an oral dose, peak plasma concentrations occur in 1 to 4 hours, with a relatively long elimination half-life of 14 to 23 hours. The maximum blood pressure response occurs in 8 to 12 hours after oral administration and is maintained up to 36 hours after its discontinuation. After IV dosing, guanfacine achieves the highest concentrations in liver and kidney, with low concentrations in the brain. Guanfacine is 64% bound to plasma proteins. In patients with hepatic or renal impairment, its elimination half-life can be prolonged.

Guanfacine is metabolized principally by hepatic hydroxylation to its inactive metabolite, 3-hydroxyguanfacine (20%), which is eliminated in the urine as its glucuronide (30%), sulfate (8%), or mercapturic acid conjugate (10%), and 24% to 37% is excreted as unchanged guanfacine (Fig. 24.10). Its nearly complete bioavailability suggests no evidence of any first-pass effect. Guanfacine and its inactive metabolites are excreted principally in urine, with approximately 80% of its oral dose excreted in urine within 48 hours.

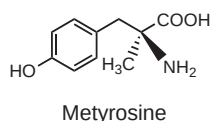
Therapeutic Applications (6,32). The therapeutic applications for guanfacine are similar to those of the other centrally acting α_2 -adrenergic agonists and methyldopa. It has been effective as monotherapy in the treatment of patients with mild to moderate hypertension. One advantage for guanfacine is its once-a-day dosing schedule. The use of diuretics to prevent accumulation of fluid can allow a reduction in the dosage for guanfacine. Somnolence and sedation were commonly reported adverse events in clinical trials. The most common adverse events associated with guanfacine treatment include somnolence/sedation, abdominal pain, dizziness, hypotension/decreased blood pressure, dry mouth, and constipation.

Guanfacine extended-release tablets (Intuniv) have been FDA approved for the non-stimulant treatment of ADHD in children and adolescents age 6 to 17 years. Although the mechanism of action for treatment of ADHD is unknown, guanfacine is thought to directly bind to postsynaptic α_2 -adrenoceptors in the prefrontal cortex of the brain, an area of the brain that has been linked to ADHD. Stimulation of these adrenoceptors is theorized to strengthen working memory, reduce susceptibility to distraction, improve attention regulation, improve behavioral inhibition, and enhance impulse control. Guanfacine treatment can cause decreases in blood pressure and heart rate, which can lead to syncope. This drug should be used with caution in ADHD patients with a history of hypotension, heart block, bradycardia, cardiovascular disease, syncope, orthostatic hypotension, or dehydration and should be used with caution in patients

being treated concomitantly with antihypertensive agents or other drugs that reduce blood pressure or heart rate or increase the risk of syncope. Guanfacine extended-release tablets should not be crushed, chewed, or broken before they are swallowed and not substituted for immediate-release tablets. ADHD patients should be reevaluated periodically for the long-term usefulness of this drug. When ADHD therapy with guanfacine is discontinued, the dose should be tapered over a period of 7 days.

Adverse Effects (6,32). Overall, although the frequency of troublesome adverse effects produced by guanfacine is similar to that produced by clonidine and the other centrally acting sympatholytics, their incidence and severity are lower with guanfacine. Unlike clonidine, abrupt discontinuation of guanfacine rarely results in rebound hypertension. When a withdrawal syndrome has occurred, its onset was slower and its symptoms less severe than the syndrome observed with clonidine.

Metyrosine.



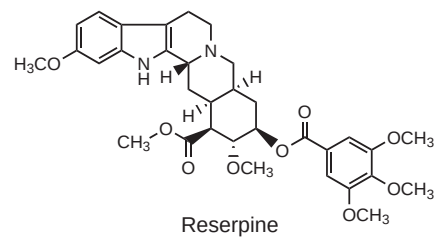
Hypothetically, inhibitors of any of the three enzymes involved in the conversion of L-tyrosine to norepinephrine (see Fig. 10.1) could be used as drugs to moderate adrenergic transmission. Inhibitors of the rate-limiting enzyme tyrosine hydroxylase would be the most logical choice. One inhibitor of tyrosine hydroxylase, metyrosine or α -methyl-L-tyrosine, a competitive inhibitor of tyrosine hydroxylase, is in limited clinical use to help control hypertensive episodes caused by excess catecholamine biosynthesis. The drug can also control other symptoms of catecholamine overproduction in patients with the rare adrenal tumor pheochromocytoma. Although metyrosine is useful in treating hypertension associated with pheochromocytoma, it is not useful for treating essential hypertension. The drug metyrosine is the *S*-enantiomer of α -methyltyrosine. The *R*-enantiomer does not bind to the active site of tyrosine hydroxylase and, thus, has no useful pharmacologic activity.

Adrenergic Neuron Blocking Agents

Reserpine, bretylium, guanethidine and guanadrel are four drugs with similar mechanisms of action involving norepinephrine storage granules. These drugs are transported into the adrenergic neurons by uptake-1, where they bind to the storage vesicles and prevent release of neurotransmitter in response to a neuronal impulse. Reserpine, guanethidine, and guanadrel are orally active antihypertensives that actually replace norepinephrine in the storage vesicles, resulting in a slow release in the amount of norepinephrine that is present. At usual doses, guanethidine and guanadrel act as “false neurotransmitters” in that

they are released into the synapse but do not effectively stimulate the receptors. At higher acute doses, their principal mechanism is a poorly understood inhibition of neurotransmitter release. Bretylium is a quaternary ammonium salt and must be given IV, because it has poor oral absorption. Initially, it can cause a release of norepinephrine and a transient rise in blood pressure, but its clinical utility is limited to cardiac arrhythmias and so will not be discussed in this chapter (see Chapter 21).

RESERPINE



An old and historically important drug that affects the storage and release of norepinephrine is reserpine. Reserpine ($pK_a = 6.6$; $\log D_{7.4} = 3.14$) is one of several indole alkaloids isolated from the roots of *Rauwolfia serpentina*; these roots were used in India for centuries both as a remedy for snake bites and as a sedative. The antihypertensive effects of the root extracts were first reported in India in 1918 and in the West in 1949. Shortly thereafter, reserpine was isolated and identified as the principal active agent. Reserpine was the first effective antihypertensive drug introduced into Western medicine, but it has largely been replaced in clinical use by agents with fewer side effects.

Mechanism of Action (9) Reserpine acts to replace and deplete the adrenergic neurons of their stores of norepinephrine by inhibiting the active transport Mg-ATPase responsible for sequestering norepinephrine and dopamine within the storage vesicles. The norepinephrine and dopamine that are not sequestered in vesicles are destroyed by MAO. As a result, the storage vesicles contain little neurotransmitter, adrenergic transmission is dramatically inhibited, and sympathetic tone is decreased, leading to vasodilation. Reserpine has the same effect on epinephrine storage in the adrenal medulla. Reserpine readily enters the CNS, where it also depletes the stores of norepinephrine and serotonin. The CNS neurotransmitter depletion led to the use of reserpine in treating certain mental illnesses.

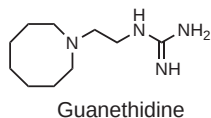
Pharmacokinetics (6,9) Limited information is available regarding the pharmacokinetics of reserpine. Peak blood concentrations for reserpine occur within 2 hours following oral administration, and the full effects for reserpine are usually delayed for at least 2 to 3 weeks. Both CNS and cardiovascular effects can persist for several days to several weeks after chronic oral therapy is discontinued.

Reserpine appears to be widely distributed in body tissues, especially adipose tissue; crosses the blood–brain barrier and the placenta; and is distributed into milk. The elimination of reserpine appears to be biphasic, with a plasma half-life averaging 4.5 hours during the first phase and approximately 11.3 days during the second phase. Reserpine is metabolized to unidentified inactive compounds. Unchanged reserpine and its metabolites are excreted slowly in urine and feces, with an average of 60% reserpine recovered in feces within 96 hours after oral administration of 0.25 mg of radiolabeled reserpine.

Therapeutic Application (6) Reserpine has been used in the management of mild to moderate hypertension, but because of very significant CNS adverse effects and its cumulative action in the adrenergic neurons, reserpine is rarely used. Reserpine and related *Rauwolfia* alkaloids have been used in the symptomatic treatment of agitated psychotic states, such as schizophrenic disorders, although other antipsychotic agents have generally replaced reserpine and the alkaloids.

Adverse Effects (6) The common adverse CNS effects for reserpine include drowsiness, fatigue, and lethargy. Mental depression is one of the most serious potential adverse effects for reserpine, which can be severe enough to require hospitalization or result in suicide attempts. Reserpine-induced depression can persist for several months after the drug is discontinued.

GUANETHIDINE MONOSULFATE



Guanethidine contains two basic nitrogen atoms with pK_a values of 9.0 and 12.0 and, therefore, can form guanethidine monosulfate ($C_{10}H_{22}N_4 \cdot H_2SO_4$) or guanethidine sulfate [$(C_{10}H_{22}N_4)_2 \cdot H_2SO_4$] salts. Caution should be exercised when interchanging between these sulfate forms, because the potency of guanethidine can be expressed in terms of guanethidine sulfate or guanethidine monosulfate, a significant difference in molecular weight.

Mechanism of Action Guanethidine is an adrenergic neuronal blocking agent that produces a selective block of peripheral sympathetic pathways by replacing and depleting norepinephrine stores from adrenergic nerve endings, but not from the adrenal medulla (6,9). It prevents the release of norepinephrine from adrenergic nerve endings in response to sympathetic nerve stimulation. The chronic administration of guanethidine results in an increased sensitivity of these effector cells to catecholamines. Following the oral administration of usual doses of guanethidine, depletion of the catecholamine

stores from adrenergic nerve endings occurs at a very slow rate, producing a more gradual and prolonged fall in systolic blood pressure than in diastolic pressure. Associated with the decrease in blood pressure is an increase in sodium and water retention and expansion of plasma volume (edema). If a diuretic is not administered concurrently with guanethidine, tolerance to the antihypertensive effect of the guanethidine during prolonged therapy can result.

Pharmacokinetics (6) Guanethidine is incompletely absorbed ($\log D_{7.4} = -0.24$) from the GI tract and is metabolized in the liver to several metabolites, including guanethidine *N*-oxide from flavin mononucleotide. These metabolites of guanethidine are excreted in the urine and have less than 10% of its hypotensive activity. The amount of drug that reaches the systemic circulation after oral administration is highly variable from patient to patient and can range from 3% to 50% of a dose. Guanethidine accumulates in the neurons with an elimination half-life of 5 days.

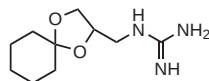
Therapeutic Applications (6) Guanethidine is used in the management of moderate to severe hypertension and in the management of renal hypertension. Guanethidine is for patients with compelling indications or who fail to respond adequately to an antihypertensive regimen that includes stage 2 drugs. Its coadministration with other hypotensive agents permits a reduction in the dosage of each drug and a minimization of adverse effects while maintaining blood pressure control. It has been administered as ophthalmic drops in the treatment of chronic open-angle glaucoma and for endocrine ophthalmopathy, ophthalmoplegia, lid lag, and lid retraction.

Adverse Effects (6) Adverse effects of guanethidine frequently are dose related, including dizziness, weakness, lassitude, and syncope resulting from postural or postexercise hypotension. A hot environment (i.e., a hot bath) can aggravate postural hypotension. Patients should be warned about possible orthostatic hypotension and about the effect of rapid postural changes on blood pressure (e.g., arising in the morning) that can cause fainting, especially during the initial period of dosage adjustment. Sodium retention (edema) is usually controlled by the coadministration of a diuretic.

Drug Interactions (6) Diuretics and other hypotensive drugs can potentiate the hypotensive effects of guanethidine. Reportedly, MAO inhibitors antagonize the hypotensive effect of guanethidine. Oral sympathomimetics, nasal decongestants, and other vasopressor agents should be used cautiously in patients receiving guanethidine, because guanethidine can potentiate their pressor effects. The mydriatic response to ophthalmic administration of phenylephrine is markedly increased in patients receiving guanethidine either ophthalmically or orally. Tricyclic antidepressants and some phenothiazines block

the uptake of guanethidine into adrenergic neurons and, thus, prevent the hypotensive activity of guanethidine. Orthostatic hypotension can be increased by concomitant administration of alcohol with guanethidine, and patients receiving guanethidine should be cautioned to limit alcohol intake.

GUANADREL SULFATE



Guanadrel

Guanadrel sulfate is an adrenergic neuronal blocking agent that is structurally and pharmacologically related to guanethidine: Both are guanidine derivatives. Guanadrel ($\log D_{7.4} = -4.80$) differs structurally from guanethidine by the presence of a dioxaspirodecyl ring system linked to guanidine by a methyl group rather than a hexahydroazocinyl ring linked by an ethyl group.

Mechanism of Action Guanadrel, like guanethidine, produces a selective block of efferent, peripheral sympathetic pathways by replacing and depleting norepinephrine stores from adrenergic nerve endings, thus preventing the release of norepinephrine from adrenergic nerve endings in response to sympathetic nerve stimulation (9,34). Unlike guanethidine, it does not release norepinephrine from the adrenal medulla and reportedly depletes norepinephrine stores in the GI tract to a lesser extent than guanethidine does. Guanadrel decreases systolic blood pressure more than diastolic blood pressure.

Pharmacokinetics (34) Guanadrel, unlike guanethidine, is rapidly and almost completely absorbed following oral administration. Following oral administration, its peak plasma concentrations are usually achieved in approximately 2 hours, and its hypotensive effect usually has an onset of 0.5 to 2.0 hours, with peak activity at 4 to 6 hours and a duration of action of 4 to 14 hours. Approximately 20% of guanadrel is bound to plasma proteins, and little, if any, of the drug crosses the blood-brain barrier or distributes into the eye. Guanadrel has a plasma half-life of approximately 2 hours and an elimination half-life of approximately 10 to 12 hours in patients with normal renal function. Approximately 40% to 50% of guanadrel is metabolized in the liver to 2,3-dihydroxypropylguanidine and several unidentified metabolites, which are excreted principally in the urine (Fig. 24.11). Unlike guanethidine, approximately 85% of an oral dose of the drug is excreted in the urine within 24 hours, with 40% to 50% of the dose excreted in the urine unchanged. In patients with impaired renal function, the half-life of guanadrel is prolonged, and apparent total-body clearance and renal clearances are decreased.

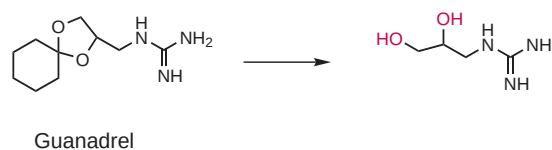


FIGURE 24.11 Metabolism of guanadrel.

Therapeutic Application (6,34) Guanadrel is used in the management of hypertension, and its efficacy is similar to that of guanethidine. Guanadrel is reserved for patients with compelling indications who fail to respond to therapy with stage 2 drugs or for cases requiring more prompt or aggressive therapy. Postural and postexercise hypotension is common in patients receiving guanadrel, and it is also likely that heat-induced vasodilation will augment its hypotensive effect. There is a possibility that geriatric patients cannot tolerate the postural hypotensive effects of guanadrel. Being a peripheral adrenergic neuron blocking drug, guanadrel shares the toxic potentials of guanethidine, and the usual precautions of this drug should be observed.

Adverse Effects (6) Overall, the frequency of adverse effects produced by guanadrel is similar or less than those produced by guanethidine and by methyldopa. In patients with impaired renal function, the elimination half-life of unmetabolized guanadrel is prolonged and its clearance decreased, thus increasing the incidence of adverse effects if the usual dosage is maintained in these patients.

Drug Interactions Being a peripheral adrenergic neuron blocking drug, guanadrel shares the same potential for drug interactions as guanethidine, and the usual precautions of this drug should be observed.

Vasodilators

Vasodilator drugs relax the smooth muscle in blood vessels, which causes the vessels to dilate. Dilation of arterial vessels leads to a reduction in systemic vascular resistance, which leads to a fall in arterial blood pressure. Dilation of venous vessels decreases venous blood pressure.

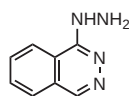
Arterial dilator drugs are used to treat systemic and pulmonary hypertension, heart failure, and angina. They reduce arterial pressure by decreasing systemic vascular resistance, thereby reducing the afterload on the left ventricle and enhancing stroke volume and cardiac output. They also decrease the oxygen demand of the heart and, thereby, improve the oxygen supply/demand ratio. The primary functions of venous dilators in treating cardiovascular hypertension include reduction in venous pressure, thus reducing preload on the heart and decreasing cardiac output and capillary fluid filtration and edema formation (a decrease in capillary hydrostatic pressure). Therefore, venous dilators sometimes are used in the treatment of heart failure along with other drugs, because they help to reduce pulmonary and/or systemic edema that results from heart failure.

There are three potential drawbacks in the use of vasodilators: First, vasodilators can lead to a baroreceptor-mediated reflex stimulation of the heart (increased heart rate and inotropy) from systemic vasodilation and arterial pressure reduction. Second, they can impair the normal baroreceptor-mediated reflex vasoconstriction when a person stands up, which can lead to orthostatic hypotension and syncope on standing. Third, they can lead to renal retention of sodium and water, increasing blood volume and cardiac output.

Vasodilator drugs are classified either based on their site of action (arterial vs. venous) or, more commonly, by their primary mechanism of action.

DIRECT-ACTING VASODILATORS

Hydralazine Hydrochloride



Hydralazine

Mechanism of Action. The only drug in this group, hydralazine, does not fit neatly into the other mechanistic classes, in part because its mechanism of action is not entirely clear. It seems to have multiple, direct effects on the VSM. Hydralazine, a phthalazine-substituted hydrazine antihypertensive drug with a pK_a of 7.3 and $\log D_{7.4}$ of 0.52, is highly specific for arterial vessels, producing its vasodilation by a couple of different mechanisms. First, it causes smooth muscle hyperpolarization, quite likely through the opening of K^+ channels. Activation therefore increases the efflux of potassium ions from the cells, causing hyperpolarization of VSM cells and, thus, prolonging the opening of the potassium channel and sustaining a greater vasodilation on arterioles than on veins (9). It can also inhibit the second messenger, IP_3 -induced release of calcium from the smooth muscle SR (the PIP_2 signal transduction pathway) (Fig. 24.2). Finally, hydralazine stimulates the formation of NO by the vascular endothelium, leading to cGMP-mediated vasodilation (Fig. 24.1). The arterial vasodilator action of hydralazine reduces systemic vascular resistance and arterial pressure. Diastolic blood pressure is usually decreased more than systolic pressure is. The hydralazine-induced decrease in blood pressure and peripheral resistance causes a reflex response, which is accompanied by increased heart rate, cardiac output, and stroke volume and an increase in plasma renin activity. It has no direct effect on the heart (6). This reflex response could offset the hypotensive effect of arteriolar dilation, limiting its antihypertensive effectiveness. Hydralazine also causes sodium and water retention and expansion of plasma volume, which could develop tolerance to its antihypertensive effect during prolonged therapy. Thus, coadministration of a diuretic improves the therapeutic outcome.

Pharmacokinetics (6,9). Hydralazine is well absorbed from the GI tract and is metabolized in the GI mucosa (prehepatic systemic metabolism) and in the liver by acetylation, hydroxylation, and conjugation with glucuronic acid (Fig. 24.12; see Table 4.17). Little of the hydralazine dose is excreted unchanged while most of the dose appears as metabolites in the urine which are without significant therapeutic activity. A small amount of hydralazine is reportedly converted to a hydrazone, most likely with vitamin B_6 (pyridoxine), which can be responsible for some its neurotoxic effects. Following the oral administration of hydralazine, its antihypertensive effect begins in 20 to 30 minutes and lasts 2 to 4 hours. The plasma half-life of hydralazine is generally 2 to 4 hours but, in some patients, can be up to 8 hours (i.e., slow acetylators). In slow acetylator patients or those with impaired renal function, the plasma concentrations for hydralazine are increased and, possibly, prolonged. Approximately 85% of hydralazine in the blood is bound to plasma proteins following administration of usual doses.

First-pass acetylation in the GI mucosa and liver is related to genetic acetylator phenotype (8). Acetylation phenotype is an important determinant of the plasma concentrations of hydralazine when the same dose of hydralazine is administered orally. Slow acetylators have an autosomal recessive trait that results in a relative deficiency of the hepatic enzyme *N*-acetyl transferase, thus prolonging the elimination half-life of hydralazine (see Chapter 4). This population of hypertensive patients will require an adjustment in dose to reduce the increased overactive response. Approximately 50% of African Americans and whites and the majority of American Indians, Eskimos, and Orientals are rapid acetylators of hydralazine. This population of patients will have subtherapeutic plasma concentrations of hydralazine because of its rapid metabolism to inactive metabolites and shorter elimination times. Patients with hydralazine-induced systemic lupus erythematosus frequently are slow acetylators.

Therapeutic Applications (6). Hydralazine is used in the management of moderate to severe hypertension. Hydralazine is reserved for patients with compelling indications and who fail to respond adequately to a stage 2 antihypertensive regimen. This drug is recommended for use in conjunction with cardiac glycosides and other

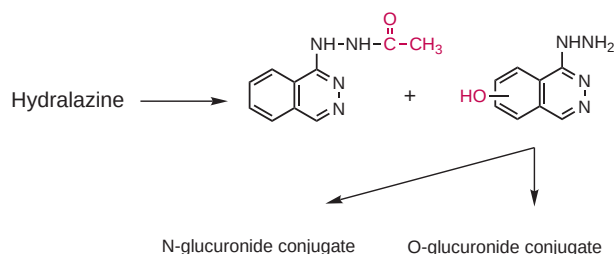


FIGURE 24.12 Metabolism of hydralazine.

vasodilators for the short-term treatment of severe congestive heart failure. Patients who engage in potentially hazardous activities, such as operating machinery or driving motor vehicles, should be warned about possible faintness, dizziness, or weakness. Hydralazine should be used with caution in patients with cerebrovascular accidents or with severe renal damage.

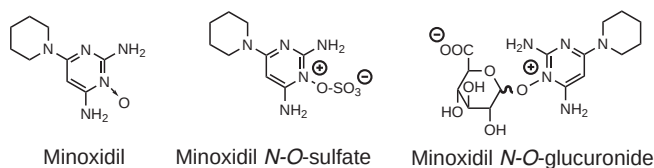
Parenteral hydralazine can be used for the management of severe hypertension when the drug cannot be given orally or when blood pressure must be lowered immediately. Other agents (e.g., sodium nitroprusside) are preferred for the management of severe hypertension or hypersensitive emergencies when a parenteral hypotensive agent is employed.

Drug Interactions. The coadministration of diuretics and other hypotensive drugs can have a synergistic effect, resulting in a marked decrease in blood pressure.

Potassium Channel Openers

SPECIFIC DRUGS

Minoxidil Although several potassium channel openers have been used in research for many years, only one, minoxidil, is approved for use in humans for treating hypertension. Minoxidil is the *N*-oxide of a piperidino-pyrimidine hypotensive agent, has a pK_a 4.6 and $\log D_{7.4}$ of 0.62, and is not an active hypotensive drug until it is metabolized by hepatic thermolabile phenol sulfotransferase (SULT1A1) to minoxidil *N*-*O*-sulfate (9).



Mechanism of Action. Potassium channel openers are drugs that activate (i.e., open) ATP-sensitive K⁺ channels in the VSM (Fig. 24.1) (35). By opening these potassium channels, there is increased efflux of potassium ions from the cells, causing hyperpolarization of VSM, which closes the voltage-gated calcium channels and, thereby, decreases intracellular calcium. With less calcium available to combine with calmodulin, there is less activation of MLCK and phosphorylation of myosin light chains. This leads to relaxation and vasodilation. Because small arteries and arterioles normally have a high degree of smooth muscle tone, these drugs are particularly effective in dilating these resistance vessels, decreasing systemic vascular resistance, and lowering arterial pressure. The fall in arterial pressure leads to reflex cardiac stimulation (baroreceptor-mediated tachycardia).

Minoxidil, as its active metabolite minoxidil *O*-sulfate, prolongs the opening of the potassium channel, sustaining greater vasodilation on arterioles than on veins. The drug decreases blood pressure in both the supine and standing positions, and there is no orthostatic

hypotension. Associated with the decrease in peripheral resistance and blood pressure is a reflex response that is accompanied by increased heart rate, cardiac output, and stroke volume, which can be attenuated by the coadministration of a BB (6). Along with this decrease in peripheral resistance is increased plasma renin activity and sodium and water retention, which can result in expansion of fluid volume, edema, and congestive heart failure. The sodium- and water-retaining effects of minoxidil can be reversed by coadministration of a diuretic. When minoxidil is used in conjunction with a β -adrenergic blocker, pulmonary artery pressure remains essentially unchanged.

Pharmacokinetics (36). Minoxidil is absorbed from the GI tract and is metabolized to its active sulfate metabolite. Plasma concentrations for minoxidil sulfate peak within 1 hour and then decline rapidly. Following an oral dose of minoxidil, its hypotensive effect begins in 30 minutes, is maximal in 2 to 8 hours, and persists for approximately 2 to 5 days. The delayed onset of the hypotensive effect for minoxidil is attributed to its metabolism to its active metabolite. The drug is not bound to plasma proteins. The major metabolite for minoxidil is its *N*-*O*-glucuronide, which unlike the sulfate metabolite, is inactive as a hypotensive agent. Approximately 10% to 20% of an oral dose of minoxidil is metabolized to its active metabolite, minoxidil *O*-sulfate, and approximately 20% of minoxidil is excreted unchanged.

Therapeutic Applications.

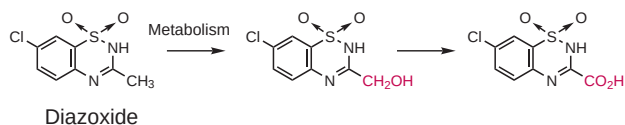
Hypertension (6,36). Being effective arterial dilators, potassium-channel openers are used in the treatment of hypertension. These drugs are not first-line therapy for hypertension because of their side effects; therefore, they are relegated to treating refractory, severe hypertension. They are generally used in conjunction with a BB and a diuretic to attenuate the reflex tachycardia and retention of sodium and fluid, respectively.

Minoxidil is used in the management of severe hypertension. It is reserved for resistant cases of hypertension that have not been managed with maximal therapeutic dosages of a diuretic and two other hypotensive drugs or for patients who have failed to respond adequately with hydralazine. To minimize sodium retention and increased plasma volume, minoxidil must be used in conjunction with a diuretic. A BB must be given before beginning minoxidil therapy and should be continued during minoxidil therapy to minimize minoxidil-induced tachycardia and increased myocardial workload.

Androgenetic Alopecia (6,37). Minoxidil is used topically to stimulate regrowth of hair in patients with androgenetic alopecia (male pattern alopecia, hereditary alopecia, or common male baldness) or alopecia areata. Commercially available topical minoxidil preparations should be used rather than the extemporaneous topical formulations from tablets to reduce the potential of minoxidil being absorbed systemically.

Drug Interactions. When minoxidil is administered with diuretics or other hypotensive drugs, the hypotensive effect of minoxidil increases, and concurrent use can cause profound orthostatic hypotensive effects.

Diazoxide



Diazoxide is a nondiuretic hypotensive and hyperglycemic agent that is structurally related to the thiazide diuretics. Being a sulfonamide with a pK_a of 8.5 and $\log D_{7.4}$ of 1.28, it can be solubilized in alkaline solutions (pH of injection is 11.6). Solutions or oral suspension of diazoxide are unstable to light and will darken when exposed to light. Such dosage forms should be protected from light, heat, and freezing. Darkened solutions can be subpotent and should not be used.

Mechanism of Action. Diazoxide reduces peripheral vascular resistance and blood pressure by a direct vasodilating effect on the VSM with a mechanism similar to that described for minoxidil by activating (opening) the ATP-modulated potassium channel (36). Thus, diazoxide prolongs the opening of the potassium channel, sustaining greater vasodilation on arterioles than on veins (9). The greatest hypotensive effect is observed in patients with malignant hypertension. Although oral or slow IV administration of diazoxide can produce a sustained fall in blood pressure, rapid IV administration is required for maximum hypotensive effects, especially in patients with malignant hypertension (6). Diazoxide-induced decreases in blood pressure and peripheral vascular resistance are accompanied by a reflex response, resulting in an increased heart rate, cardiac output, and left ventricular ejection rate. In contrast to the thiazide diuretics, diazoxide causes sodium and water retention and decreased urinary output, which can result in expansion of plasma and extracellular fluid volume, edema, and congestive heart failure, especially during prolonged administration.

Diazoxide increases blood glucose concentration (diazoxide-induced hyperglycemia) by several different mechanisms: by inhibiting pancreatic insulin secretion, by stimulating release of catecholamines, or by increasing hepatic release of glucose (6,9). The precise mechanism of inhibition of insulin release has not been elucidated but, possibly, can result from an effect of diazoxide on cell-membrane potassium channels and calcium flux.

Pharmacokinetics (6). After rapid IV administration, diazoxide produces a prompt reduction in blood pressure, with maximum hypotensive effects occurring within 5 minutes. The duration of its hypotensive effect varies from 3 to 12 hours, but ranges from 30 minutes to 72 hours have been observed. The elimination half-life of

diazoxide following a single oral or IV dose has been reported to range from 21 to 45 hours in adults with normal renal function. In patients with renal impairment, the half-life is prolonged. Approximately 90% of the diazoxide in the blood is bound to plasma proteins. Approximately 20% to 50% of diazoxide is eliminated unchanged in the urine, along with its major metabolites, resulting from the oxidation of the 3-methyl group to its 3-hydroxymethyl- and 3-carboxyl-metabolites.

Therapeutic Applications.

Severe Hypertension (6). Intravenous diazoxide has been used in hypertensive crises for emergency lowering of blood pressure when a prompt and urgent decrease in diastolic pressure is required in adults with severe non-malignant and malignant hypertension and in children with acute severe hypertension. However, other IV hypotensive agents are preferred for the management of hypertensive crises. Diazoxide is intended for short-term use in hospitalized patients only. Although diazoxide has also been administered orally for the management of hypertension, its hyperglycemic and sodium-retaining effects make it unsuitable for chronic therapy.

Hypoglycemia (6). Diazoxide is administered orally in the management of hypoglycemia caused by hyperinsulinism associated with inoperable islet cell adenoma or carcinoma or extrapancreatic malignancy in adults.

Phosphodiesterase Inhibitors

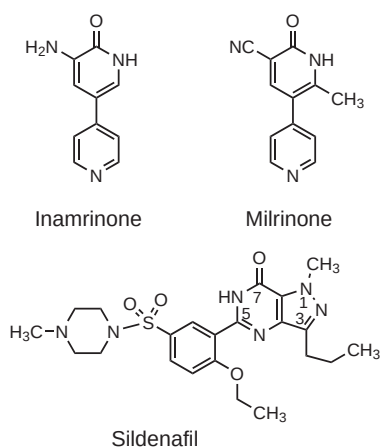
MECHANISM OF ACTION OF CAMP-DEPENDENT PHOSPHODIESTERASE INHIBITORS (PDE3) PDE3 is one of the isoforms of phosphodiesterase found in the heart and VSM. The mechanism by which cAMP and cGMP relaxes VSM has been described previously in the section on VSM contraction and relaxation (Figs. 24.1 and 24.2). The cAMP released is broken down by a cAMP-dependent phosphodiesterase (PDE3). Therefore, inhibition of PDE3 increases intracellular cAMP, which further inhibits MLCK, thereby producing less contractile force (i.e., promoting relaxation). The overall cardiac and vascular effects of cAMP-dependent phosphodiesterase inhibitors cause cardiac stimulation, increasing cardiac output and reducing systemic vascular resistance, thereby lowering arterial pressure. Because cardiac output increases and systemic vascular resistance decreases, the change in arterial pressure depends on the relative effects of the phosphodiesterase inhibitor on the heart versus the VSM. At normal therapeutic doses, PDE3 inhibitors, such as milrinone, have a greater effect on VSM than cardiac muscle so that arterial pressure is lowered in the presence of augmented cardiac output. Because of the dual cardiac and vascular effects of these compounds, they sometimes are referred to as inodilators.

MECHANISM OF ACTION OF cGMP-DEPENDENT PHOSPHODIESTERASE INHIBITORS (PDE5) A second isoenzyme form of phosphodiesterase found in VSM is PDE5, a cGMP-dependent phosphodiesterase, which is also found in the corpus

cavernosum of the penis (erectile dysfunction). This enzyme is responsible for breaking down cGMP that forms in response to increased NO (Fig. 24.1). Increased cGMP leads to smooth muscle relaxation primarily by reducing calcium entry into the cell. Inhibitors of cGMP-dependent phosphodiesterase increase intracellular cGMP, thereby enhancing VSM relaxation and vasodilation.

Therapeutic Indications The cardiostimulatory and vasodilatory actions of PDE3 inhibitors make them suitable for the treatment of heart failure, because VSM relaxation reduces ventricular wall stress and the oxygen demands placed on the failing heart. The cardiostimulatory effects of the PDE3 increase inotropy, which further enhances stroke volume and ejection fraction. A baroreceptor reflex, which occurs in response to hypotension, can contribute to the tachycardia. Clinical trials have shown that long-term therapy with PDE3 inhibitors increases mortality in patients with heart failure; therefore, these drugs are not used for long-term, chronic therapy. They are very useful, however, in treating acute, decompensated heart failure or temporary bouts of decompensated chronic failure. They are not used as a monotherapy. Instead, they are used in conjunction with other treatment modalities, such as diuretics, ACEIs, BBs, or digitalis. The somewhat selective vasodilatory actions of PDE5 inhibitors have made these compounds very useful in the treatment of male erectile dysfunction and as a combination therapy for pulmonary hypertension. The PDE3 inhibitors are used for treating heart failure, whereas the PDE5 inhibitors are used for treating male erectile dysfunction. Note that the generic names for PDE3 inhibitors end in “one” and those for the PDE5 inhibitors end in “fil.”

SPECIFIC DRUGS



Milrinone (Primacor as Lactate Salt) and Inamrinone (Inocor as Lactate Salt) Inamrinone and milrinone ($\log D_{7.4} = -1.89$) are positive inotropes and vasodilators indicated for the short-term IV management of congestive heart failure in patients who have not responded adequately to digitalis, diuretics, and/or vasodilators.

For IV infusion, inamrinone lactate and milrinone lactate injection solutions can be diluted in sodium chloride solution for injection. Inamrinone lactate for injection is preserved with sodium metabisulfite and needs protection from light. It should not be diluted with solutions containing dextrose, because a chemical interaction occurs over 24 hours. For milrinone, an immediate chemical interaction with furosemide with the formation of a precipitate is observed when furosemide is injected into an infusion of milrinone. Patients who are sensitive to bisulfites can also be sensitive to inamrinone lactate injection, which contains sodium metabisulfite.

The pharmacokinetics for inamrinone shows protein binding from 10% to 49%. Its half-life in healthy volunteers is approximately 3.6 hours, whereas in patients with congestive heart failure, the plasma half-life increases to approximately 5.0 to 8.3 hours. For infants younger than 4 weeks, the half-life is 12.7 to 22.2 hours, and for infants older than 4 weeks, the half-life is 3.8 to 6.8 hours. Time to peak effect is less than 10 minutes. Its duration of action is dose related, ranging from 30 minutes at low dose to 2 hours at the higher dosages. Approximately 63% of the administered dose is eliminated via the urine as unchanged drug, and 18% is eliminated in the feces. Elderly patients are more likely to have age-related renal function impairment, which can require adjustment of dosage in patients receiving inamrinone.

The pharmacokinetics for milrinone following IV injections to patients with congestive heart failure showed a volume of distribution of 0.38 to 0.45 L/kg, a mean terminal elimination half-life of 2.3 hours, and a clearance of 0.13 L/kg/hour. These pharmacokinetic parameters were not dose dependent, and the area under the plasma concentration versus time curve following injections was significantly dose dependent. Milrinone is approximately 70% bound to human plasma protein. The primary route of excretion for orally administered milrinone is via the urine, with unchanged milrinone (83%) and its *O*-glucuronide metabolite (12%) being present. Elimination in normal subjects via the urine is rapid, with approximately 60% recovered within the first 2 hours following dosing and approximately 90% within the first 8 hours following dosing. In patients with renal function impairment, elimination of unchanged milrinone is reduced, suggesting that a dosage adjustment can be necessary.

The selective PDE5 inhibitor most commonly used is sildenafil (Viagra as citrate) ($\log D_{7.4} = 1.85$). Studies in vitro have shown that sildenafil is selective for PDE5. Its effect is more potent on PDE5 than on other known phosphodiesterases (10 times for PDE6, >80 times for PDE1, and >700 times for PDE2, PDE3, PDE4, PDE7, PDE8, PDE9, PDE10, and PDE11). The approximately 4,000 times selectivity for PDE5 versus PDE3 is important because PDE3 is involved in control of cardiac contractility. Sildenafil is only approximately 10 times as potent for PDE5 compared to PDE6, an enzyme found in the retina that is involved in the phototransduction pathway of the retina. This lower selectivity

is thought to be the basis for abnormalities related to color vision observed with higher doses or plasma levels. (For a complete discussion of the pharmacokinetics including drug metabolism, see Chapter 40.)

Side Effects and Contraindications The most common and severe side effect for PDE3 inhibitors is ventricular arrhythmias, some of which can be life-threatening. Other side effects include headaches and hypotension, which are not uncommon for drugs that increase cAMP in cardiac and vascular tissues (e.g., β -agonists).

In addition to human corpus cavernosum smooth muscle, PDE5 is also found in lower concentrations in other tissues, including platelets, vascular and visceral smooth muscle, and skeletal muscle. The inhibition of PDE5 in these tissues by sildenafil can be the basis for the enhanced platelet antiaggregatory activity of NO observed *in vitro*, an inhibition of platelet thrombus formation *in vivo*, and peripheral arterial-venous dilatation *in vivo*. The most common side effects for PDE5 inhibitors include headache and cutaneous flushing, both of which are related to vascular dilation caused by increased vascular cGMP. Clinical evidence suggests that nitrodilators can interact adversely with PDE5 inhibitors. The reason for this adverse reaction is that nitrodilators stimulate cGMP production, whereas PDE5 inhibitors inhibit cGMP degradation. When combined, these two drug classes greatly potentiate cGMP levels, which can lead to hypotension and impaired coronary perfusion.

Nitrodilators

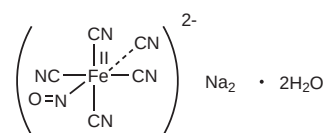
MECHANISM OF ACTION NO, a molecule produced by many cells in the body, has several important actions. NO is a highly reactive gas that participates in many chemical reactions. It is one of the nitrogen oxides ("NO_x") in automobile exhaust and plays a major role in the formation of photochemical smog, but NO has also many physiologic functions. It is synthesized within cells by an enzyme NO synthase (NOS). There are three isoenzymes, neuronal NOS (nNOS or NOS-1), inducible NOS (iNOS or NOS-2) found in macrophages, and endothelial NOS (eNOS or NOS-3) found in the endothelial cells that line the lumen of blood vessels. Whereas the levels of nNOS and eNOS are relatively steady, expression of iNOS genes awaits an appropriate stimulus. All types of NOS produce NO from arginine with the aid of molecular oxygen and NADPH. Because NO diffuses freely across cell membranes, there are many other molecules with which it can interact, and NO is quickly consumed close to where it is synthesized. Thus, NO affects only cells adjacent to its point of synthesis. NO relaxes the smooth muscle in the walls of the arterioles. At each systole, the endothelial cells that line the blood vessels release a puff of NO, which diffuses into the underlying smooth muscle cells, causing them to relax and, thus, to permit the surge of blood to pass through easily. The signaling functions of NO begin with its binding to protein receptors on or in the

cell, triggering the formation of cGMP from soluble guanylyl cyclase (Fig. 24.1). Mice in which the genes for the NOS found in endothelial cells (eNOS) have been "knocked out" suffer from hypertension. Nitroglycerin, which is often prescribed to reduce the pain of angina, does so by generating NO, which relaxes venous walls and arterioles, improving the oxygen supply/demand ratio (see Chapter 21). NO also inhibits the aggregation of platelets and, thus, keeps inappropriate clotting from interfering with blood flow. Other actions on smooth muscle include penile erection and peristalsis aided by the relaxing effect of NO on the smooth muscle in intestinal walls. NO also inhibits the contractility of the smooth muscle wall of the uterus, but at birth, the production of NO decreases, allowing contractions to occur. Nitroglycerin has helped some women who were at risk of giving birth prematurely to carry their baby to full term. The NO from iNOS inhibits inflammation in blood vessels by blocking the release of mediators of inflammation from the endothelial cells, macrophages, and T lymphocytes. The NO from iNOS has been shown to S-nitrosylate cyclooxygenase-2 (COX-2), increasing its activity, and drugs that prevent this interaction could work synergistically with the nonsteroidal anti-inflammatory drugs inhibiting COX-2. NO affects hormonal secretion from several endocrine glands. Hemoglobin transports NO at the same time that it carries oxygen, and when it unloads oxygen in the tissues, it also unloads NO. Fireflies use NO to turn on their flashers.

Since the dawn of recorded human history, nitrates have been used to preserve meat from bacterial spoilage. Harmless bacteria in our throat convert nitrates in our food into nitrites. When the nitrites reach the stomach, the acidic gastric juice (pH ~1.4) generates NO from these nitrites, killing almost all the bacteria that have been swallowed in our food.

In the cardiovascular system, NO is produced primarily by vascular endothelial cells. This endothelial-derived NO has several important functions, including relaxing VSM (vasodilation), inhibiting platelet aggregation (antithrombotic), and inhibiting leukocyte-endothelial interactions (anti-inflammatory). These actions involve NO-stimulated formation of cGMP (Fig. 24.1). Nitrodilators are drugs that mimic the actions of endogenous NO by releasing NO or forming NO within tissues. These drugs act directly on the VSM to cause relaxation and, therefore, serve as endothelial-independent vasodilators.

SODIUM NITROPRUSSIDE (SODIUM NITROFERRICYANIDE; NITROPRESS; NIPRIDE)



Sodium nitroprusside

There are two basic types of nitrodilators: those that release NO spontaneously (e.g., sodium nitroprusside) and those that require an enzyme activation to form NO (organic nitrates). Sodium nitroprusside is a direct-acting vasodilator on VSM, producing its vasodilation by the release of NO. Since 1929, it has been known as a rapidly acting hypotensive agent when administered as an infusion. It is chemically and structurally unrelated to other available hypotensive agents. As a reminder in preparing extemporaneous infusions, the potency of sodium nitroprusside is expressed in terms of the dihydrated drug. When reconstituted with 5% dextrose injection, sodium nitroprusside solutions are reddish-brown in color, with a pH of 3.5 to 6.0. Its crystals and solutions are sensitive and unstable to light and should be protected from extremes of light and heat. The exposure of sodium nitroprusside solutions to light causes deterioration, which can be evidenced by a change from a reddish-brown to a green to a blue color, indicating a rearrangement of the nitroso to the inactive isonitro form. Sodium nitroprusside solutions in glass bottles undergo approximately 20% degradation within 4 hours when exposed to fluorescent light and even more rapid degradation in plastic bags. Sodium nitroprusside solutions should be protected from light by wrapping the container with aluminum foil or other opaque material. When adequately protected from light, reconstituted solutions are stable for 24 hours. Trace metals, such as iron and copper, can catalyze the degradation of nitroprusside solutions, releasing cyanide. Any change in color for the nitroprusside solutions is an indication of degradation, and the solution should be discarded. No other drug or preservative should be added to stabilize sodium nitroprusside infusions.

Mechanism of Action Sodium nitroprusside is not an active hypotensive drug until metabolized to its active metabolite, NO, the mechanism of action of which has been previously described (Fig. 24.1). Studies with sodium nitroprusside suggest that it releases NO by its interaction with glutathione or with sulfhydryl groups in the erythrocytes and tissues to form a *S*-nitrosothiol intermediate, which spontaneously produces NO, which in turn freely diffuses into the VSM, thereby increasing intracellular cGMP concentration (6,9). NO also activates K^+ channels, which leads to hyperpolarization and relaxation.

The hypotensive effect of sodium nitroprusside is augmented by concomitant use of other hypotensive agents and is not blocked by adrenergic blocking agents. It has no direct effect on the myocardium, but it can exert a direct coronary vasodilator effect on VSM. When sodium nitroprusside is administered to hypertensive patients, a slight increase in heart rate commonly occurs, and cardiac output is usually decreased slightly. Moderate doses of sodium nitroprusside in patients with hypertension produce renal vasodilation without an appreciable

increase in renal blood flow or decrease in glomerular filtration (6).

Intravenous infusion of sodium nitroprusside produces an almost immediate reduction in blood pressure. Blood pressure begins to rise immediately when the infusion is slowed or stopped and returns to pretreatment levels within 1 to 10 minutes.

Pharmacokinetics Sodium nitroprusside undergoes a redox reaction that releases cyanide (6,9). The cyanide that is produced is rapidly converted into thiocyanate in the liver by the enzyme thiosulfate sulfotransferase (rhodanase) and is excreted in the urine (6,9). The rate-limiting step in the conversion of cyanide to thiocyanate is the availability of sulfur donors, especially thiosulfate. Toxic symptoms of thiocyanate begin to appear at plasma thiocyanate concentrations of 50 to 100 mg/mL. The elimination half-life of thiocyanate is 2.7 to 7.0 days when renal function is normal but longer in patients with impaired renal function.

Therapeutic Applications (6) Intravenous sodium nitroprusside is used as an infusion for hypertensive crises and emergencies. The drug is consistently effective in the management of hypertensive emergencies, irrespective of etiology, and can be useful even when other drugs have failed. It can be used in the management of acute congestive heart failure.

Adverse Effects (6) The most clinically important adverse effects of sodium nitroprusside are profound hypotension and the accumulation of cyanide and thiocyanate. Thiocyanate can accumulate in the blood of patients receiving sodium nitroprusside therapy, especially in those with impaired renal function. Thiocyanate is mildly neurotoxic at serum concentrations of 60 $\mu\text{g/mL}$ and can be life-threatening at concentrations of 200 $\mu\text{g/mL}$. Other adverse effects of thiocyanate include inhibition of both the uptake and binding of iodine, producing symptoms of hypothyroidism.

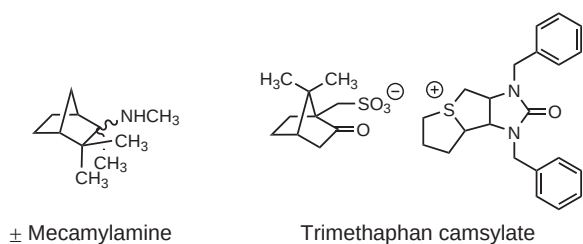
Sodium nitroprusside can bind to vitamin B_{12} , interfering with its distribution and metabolism, and it should be used with caution in patients having low plasma vitamin B_{12} concentrations. Excess cyanide can also bind to hemoglobin, producing methemoglobinemia.

Ganglionic Blockers

MECHANISM OF ACTION Ganglionic blockers block impulse transmission at the sympathetic ganglia. Neurotransmission within the sympathetic and parasympathetic ganglia involves the release of acetylcholine from preganglionic efferent nerves, which binds to nicotinic receptors on the postganglionic efferent nerves. Ganglionic blockers inhibit autonomic activity by interfering with neurotransmission within autonomic ganglia. This reduces sympathetic outflow to the heart, thereby decreasing cardiac output by decreasing

heart rate and contractility. Reduced sympathetic output to the vasculature decreases sympathetic vascular tone, which causes vasodilation and reduced systemic vascular resistance, which decreases arterial pressure. Parasympathetic outflow is also reduced by ganglionic blockers.

THERAPEUTIC INDICATIONS Ganglionic blockers are not commonly used in the treatment of chronic hypertension or in patients with compelling indications largely because of their side effects and because numerous more effective and safer antihypertensive drugs can be used. They are, however, occasionally used for hypertensive emergencies. The ganglionic blockers available for clinical use include trimethaphan camsylate, and mecamlamine, which are competitive antagonists at $\alpha_4\beta_2$ -nicotinic acetylcholine receptor.



Trimethaphan is a tertiary sulfonium ion and cannot cross lipid cell membranes, whereas mecamlamine is a secondary amine with pK_a of 11.2 and $\log D_{7.4}$ of -1.1 . Therefore, trimethaphan is a short-acting peripheral direct vasodilator that must be given as an IV infusion, whereas mecamlamine can be given orally. *S*-(+)-Mecamlamine is the active enantiomer for the $\alpha_4\beta_2$ -nicotinic acetylcholine receptor. Mecamlamine rapidly disappears from the blood with a plasma half-life of 1 hour and crosses the blood-brain barrier into the CNS. Trimethaphan is rapidly excreted in unchanged form by the kidney. Mecamlamine is excreted by the kidney much more slowly. Trimethaphan is the drug of choice for managing acute aortic dissection and for hypertensive emergencies. Both drugs are of limited use, because of the availability of more specific acting vasodilators. Mecamlamine has been used for labeling CNS nicotinic receptors and crosses into the CNS, where it can block neuronal $\alpha_4\beta_2$ -nicotinic acetylcholine receptors. Mecamlamine has been studied for use with nicotine for smoking cessation and as an antidepressant.

SIDE EFFECTS AND CONTRAINDICATIONS Side effects of trimethaphan include prolonged neuromuscular blockade and potentiation of neuromuscular blocking agents. It can produce excessive hypotension and impotence (because of its sympatholytic effect) as well as constipation, urinary retention, and dry mouth (because of its parasympatholytic effect).

PULMONARY ARTERIAL HYPERTENSION

Pulmonary hypertension, which was once a rare life-threatening disease, reportedly affects about 160,000 people today. Pulmonary arterial hypertension (PAH) is defined as a group of diseases characterized by a progressive increase of pulmonary vascular resistance, leading to right ventricular failure. It includes a variety of pulmonary hypertensive diseases with different etiologies but similar clinical presentation (36). Primary pulmonary hypertension can occur without any apparent cause (idiopathic), or it can be inherited. PAH is a disease of the small pulmonary arteries, characterized by progressive narrowing of the pulmonary vascular bed. Vasoconstriction and scarring (or fibrosis) cause the pulmonary wall to become stiffer and thicker, contributing to an increased pulmonary vascular resistance. This extra stress causes the heart to enlarge and become less flexible. Less blood flows from the heart, through the lungs and into the body, resulting in additional symptoms. One direct effect of these abnormally elevated pressures is blood leakage from the pulmonary vessels. A blood-producing cough often is an indicator of leakage from the pulmonary vessels. The pulmonary arterial walls produce a substance called endothelin, which causes these blood vessels to constrict. The reasons for this overproduction are unknown. In some cases, the cause is genetically programmed or the person is predisposed genetically to primary pulmonary hypertension after being exposed to a certain drug (e.g., the diet drugs Fen-Phen [fenfluramine/phen-termine], Redux [dexfenfluramine], or Pondimin [fenfluramine]).

Often, primary pulmonary hypertension is not diagnosed in a timely manner, because its early symptoms can be confused with those of many other conditions (Table 24.5). To establish a diagnosis of pulmonary hypertension, a series of tests are performed that show how well a person's heart and lungs are working. These tests can include assessment of daily living tasks, such as a 6-minute walk test, a computed tomography scan to rule out a pulmonary embolism or lung disease, a pulmonary function test to rule out obstructive lung disease, a formal sleep study to rule out sleep apnea, and laboratory tests to rule out hepatitis, collagen disease, HIV, or other conditions.

Rationale for Pharmacologic Treatment

If PAH has an identifiable cause, then measures can be taken to correct the underlying problem. If the diagnosis is primary PAH, then pharmacologic intervention is required to reduce the pressure. This is done using vasodilator drugs to decrease pulmonary vascular resistance and, thereby, to lower the pressure. Adjunct therapy can include diuretics to reduce blood volume, which will reduce central venous pressure and right

TABLE 24.5 Symptoms of Primary Pulmonary Hypertension

Breathlessness or shortness of breath
Feeling tired all the time
Dizziness, especially when climbing stairs or standing up
Fainting (often the symptom that brings people to their doctors)
Swollen ankles and legs
Chest pain, especially during physical activity

ventricular stroke volume, as well as to reduce some of the signs and symptoms of edema and shortness of breath associated with PAH. Anticoagulants are administered to prevent the formation of pulmonary thrombi. Patients with cardiovascular hypertension are generally treated with antihypertensive drugs that reduce blood volume (which reduces central venous pressure and cardiac output), reduce systemic vascular resistance, or reduce cardiac output by depressing heart rate and stroke volume.

Drugs Used to Treat Pulmonary Hypertension

Classes of drugs used in the treatment of PAH include thiazide diuretics, loop diuretics, vasodilators, calcium channel blockers, prostaglandins, endothelin receptor antagonists, NO, and PDE5 inhibitors (38,39). During the last decade, substantial improvements in the therapeutic options for PAH have emerged that target the mechanisms involved in the pathogenesis of this devastating disease. Intravenous epoprostenol was the first drug to improve symptoms and survival of patients with PAH. Novel prostanoids, including subcutaneous treprostinil and inhaled iloprost, have beneficial effects in many patients, although their long-term efficacy is less well known. Among the newer treatments for PAH, endothelin receptor antagonists and PDE5 inhibitors have reshaped clinical practice for PAH. The endothelin receptor antagonist bosentan is recommended as first-line treatment for patients with PAH and New York Heart Association (NYHA) functional class III heart failure (i.e., patients with marked limitation of activity; they are comfortable only at rest). Other endothelin receptor antagonists include sitaxsentan sodium and ambrisentan. The combination of the PDE5 inhibitor sildenafil and iloprost, a prostacyclin analog, is being studied in patients with pulmonary hypertension. Targeting a single pathway cannot be expected to be uniformly successful, because PAH is a complex disorder. Thus, combining drugs with different modes of action is expected to improve symptoms, hemodynamics, and survival in PAH patients, although combination therapy has yet to undergo the scrutiny of large randomized clinical trials.

Specific Drugs

Phosphodiesterase Inhibitors

SILDENAFIL (REVATIO) Sildenafil has been approved for treatment of PAH through its inhibition of cGMP and smooth muscle relaxation of the pulmonary vasculature.

Endothelin Receptor Antagonists

MECHANISM OF ACTION ET-1 is a 21-amino acid peptide that is produced by the vascular endothelium. It is a very potent vasoconstrictor that binds to VSM endothelin receptors ET_A and ET_B (Fig. 24.13) (40). The ET-1 receptors are linked to the G_q protein and IP_3 signal transduction pathway (Fig. 24.1). Therefore, ET-1 causes SR release of calcium, increasing the VSM contractility. Vascular endothelial cells secrete the majority of ET-1, which binds to two receptor subtypes: ET_A and ET_B . In vascular tissue, ET_A is located predominantly on smooth muscle cells, whereas ET_B is found on both endothelial and smooth muscle cells. Activation of ET_A by ET-1 leads to potent vasoconstriction resulting from an increase in cytosolic calcium levels via influx of extracellular calcium and release from

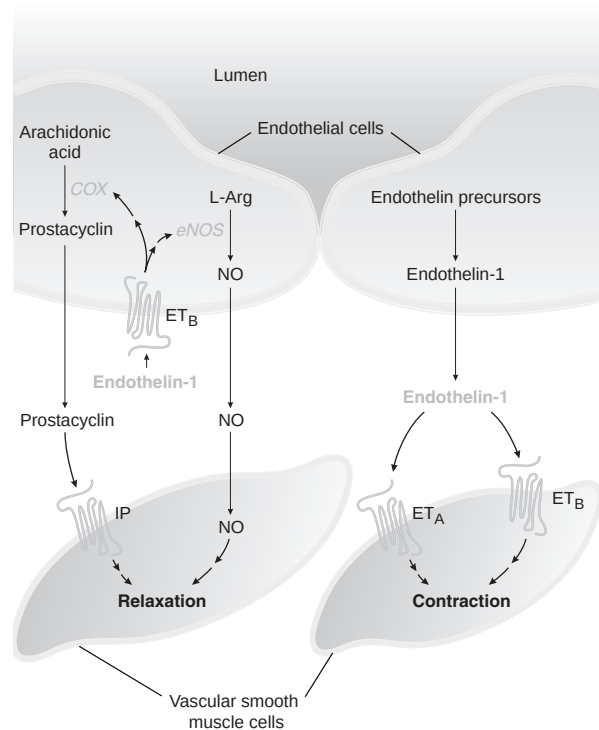


FIGURE 24.13 The effects of endothelin-1, prostacyclin, and nitric oxide on the contraction and relaxation (vasodilation) of vascular smooth muscle cells. ET, endothelin receptor; IP, iso-prostenoid receptor; L-Arg, L-arginine; NO, nitric oxide. (From Yeh DC, Michel T. Pharmacology of vascular tone. In: Golan DE, Tashjian AH, Armstrong E, et al., eds. Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy. Baltimore: Lippincott Williams & Wilkins, 2008, with permission)

intracellular SR stores (Fig. 24.1). The actions of ET_B are more complicated. Like ET_A , ET_1 activation of ET_B on VSM cells leads to vasoconstriction. Furthermore, some studies suggest that in the pulmonary hypertensive state, blockade of both ET_A and ET_B is necessary to achieve maximal vasodilation. Activation of ET_B by ET_1 stimulates COX, which catalyzes the formation of prostacyclin from arachidonic acid. Prostacyclin then binds to and activates the isoprostanoid receptor on VSM. ET_1 also activates ET_B , which stimulates eNOS (eNOS) to produce NO from L-arginine. Both prostacyclin and NO are potent vasodilators of VSM (relaxation). Additionally, ET_1 binds to the ET_B receptors on the endothelium of pulmonary smooth muscle to stimulate the formation of NO, which produces vasodilation in the absence of smooth muscle ET_A and ET_B receptor activation. This receptor distribution helps to explain the phenomenon that ET_1 administration causes transient vasodilation (initial endothelial ET_B activation) and hypotension, followed by prolonged vasoconstriction (smooth muscle ET_A and ET_B activation) and hypertension.

THERAPEUTIC INDICATIONS Because of its powerful vasoconstrictor properties and its effects on intracellular calcium, ET_1 has been implicated in the pathogenesis of hypertension, coronary vasospasm, and heart failure (40). A number of studies suggest a role for ET_1 in pulmonary hypertension as well as in systemic hypertension. Additionally, ET_1 has been shown to be released by the failing myocardium, where it can contribute to cardiac calcium overload and hypertrophy.

Endothelin receptor antagonists, by blocking the vasoconstrictor and cardiotoxic effects of ET_1 , produce vasodilation and cardiac inhibition. Endothelin receptor antagonists have been shown to decrease mortality and to improve hemodynamics in experimental models of heart failure. At present, the only approved indication for endothelin antagonists is pulmonary hypertension.

SPECIFIC DRUGS

Endothelin Receptor Antagonists

Bosentan (Tracleer). Bosentan is an orally administered, nonselective ET_1 receptor sulfonamide-class antagonist blocking both ET_A and ET_B receptors and is approved for the treatment of patients with PAH (Fig. 24.14) (41). Following oral administration, bosentan attains peak plasma concentrations in approximately 3 hours, with an absolute bioavailability of approximately 50%. Food has no clinically relevant effect on its absorption recommended doses. Bosentan is approximately 98% bound to albumin, with a volume of distribution of 30 L. Its terminal half-life after oral administration is 5.4 hours and is unchanged at steady-state. Steady-state concentrations are achieved within 3 to 5 days after multiple-dose

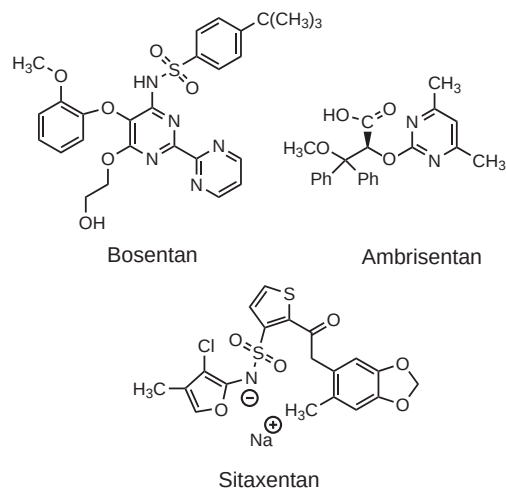


FIGURE 24.14 Endothelin receptor antagonists.

administration. Bosentan is mainly eliminated from the body by hepatic metabolism and subsequent biliary excretion of the metabolites. Three metabolites have been identified, formed by CYP2C9 and CYP3A4 (Fig. 24.15). The pharmacokinetics of bosentan are dose proportional up to 500 mg/d (multiple doses). The pharmacokinetics of bosentan in pediatric patients with PAH are comparable to those in healthy subjects, whereas adult patients with PAH show a two-fold increase in clearance. Severe renal impairment and mild hepatic impairment do not have a clinically relevant influence on its pharmacokinetics. Bosentan should generally be avoided in patients with moderate or severe hepatic impairment and/or elevated liver

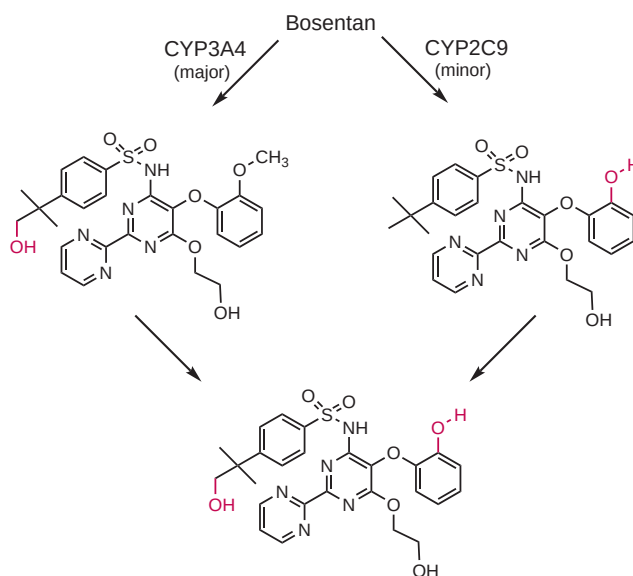


FIGURE 24.15 Metabolism of bosentan.

aminotransferases. Inhibitors of CYP3A4 increase the plasma concentration of bosentan and cause an increase in the clearance of drugs metabolized by CYP3A4 and CYP2C9. The drug induces CYP2C9 expression. The possibility of reduced efficacy of CYP2C9 and CYP3A4 substrates coadministered with bosentan is increased. No clinically relevant interaction was detected for P-glycoprotein. Bosentan can increase plasma levels of ET-1. Adverse effects include hypotension, headache, flushing, increased liver aminotransferases, leg edema, and anemia. Bosentan is teratogenic in animal models and therefore can cause birth defects and is contraindicated in pregnancy. Bosentan can also cause liver injury. Patients with hypersensitivity to sulfonamides should be excluded.

When prescribing any endothelin receptor antagonist, it is important to be aware of potential interactions with concomitant medications. Patients with PAH frequently require anticoagulant therapy, and bosentan affects warfarin metabolism. Because bosentan induces CYP2C9, the principal hepatic enzyme involved in the metabolism of warfarin, bosentan reduces prothrombin clotting time (as measured by international normalized ratio) in patients receiving warfarin. Increases in warfarin dose may be necessary to maintain a desirable clotting time. Thus, monitoring of coagulation status is warranted when bosentan is prescribed.

Ambrisentan (Letairis). Ambrisentan has been FDA approved for the treatment of PAH (Fig. 24.14). The drug is a selective antagonist of the ET_A receptor with a 77-fold selectivity for the ET_A receptor as compared to the ET_B receptor (42,43). The drug is significantly more selective for the ET_A receptor than bosentan (20:1). Ambrisentan exhibits two major adverse warnings. First, it is classified as a category X drug that must not be used in pregnant women because of its known teratogenicity in animal models, and pregnancy must be excluded before the start of treatment and prevented thereafter by the use of a reliable contraceptive method. Second, ambrisentan shows hepatotoxicity, and liver function tests for alanine aminotransferase and aspartate aminotransferase levels should be monitored before and during therapy.

Ambrisentan is administered orally and shows an 80% oral bioavailability. The drug is highly bound to plasma protein (99%). The drug is a substrate for P-glycoprotein and primarily undergoes glucuronidation of the carboxyl group catalyzed by UGT1A9, UGT2B7, and UGT1A3 to form an acyl glucuronide. Ambrisentan does not appear to undergo significant metabolism via CYP3A4/5 and CYP2C9 isoforms and is therefore unlikely to affect the pharmacokinetics of P450 metabolism of the drug. The FDA new drug application identifies two additional metabolites, which include oxidation of the 4-methyl group to the

4-hydroxymethyl group and the formation of a diglucuronide metabolite of the 4-hydroxymethyl and carboxyl groups. Biliary excretion is the primary route of elimination for ambrisentan (~36%) and metabolites (~30%), with another 20% of the metabolites appearing in the urine. Elimination half-life is 15 hours, thus allowing for dose accumulation.

Sitaxsentan Sodium (Thelin). Sitaxsentan is an orphan drug approved for marketing in the European Union (August 2006), Canada, and Australia (March 2007) and undergoing phase III clinical trials in the United States for the treatment of PAH. If FDA approved, sitaxsentan will join bosentan and ambrisentan, which are presently available in the United States for treatment of PAH (Fig. 24.14).

Sitaxsentan is a potent sulfonamide-class inhibitor of the ET_A receptor, exhibiting 6,500-fold selectivity for the ET_A receptor as compared to the ET_B receptor, blocking the action of endothelin (44,45). This selective antagonist of the ET_A receptor increases vascular blood flow and reverses vasoconstriction in human clinical studies. Clinical trials have shown that the efficacy of sitaxsentan is much the same as bosentan, a nonselective ET antagonist, but with reduced liver toxicity, although it carries a black box warning in the Canadian drug-approved labeling. Alanine aminotransferase and aspartate aminotransferase levels should be monitored using liver function tests before and during therapy.

When administered orally, sitaxsentan was rapidly and extensively absorbed, with an oral bioavailability of approximately 55%. The drug can be taken with or without food. Sitaxsentan has a volume of distribution of greater than 60 L and a half-life of 6 to 7 hours and is greater than 98% protein bound. Sitaxsentan or its metabolites did not effectively cross the blood-brain barrier, but it was found in rat breast milk. In vitro tests have shown that sitaxsentan is primarily metabolized by CYP2C9 and, to a lesser extent, CYP3A4/5. It is a strong inhibitor of CYP2C9 and a lesser inhibitor of CYP2C19 and CYP3A4/5. The two main circulating metabolites were identified as 1,2-diketositaxsentan and 1-keto-2-hydroxy-sitaxsentan, which account for approximately 3% and 9% of the parent drug, respectively (Fig. 24.16) (45). These two metabolites were 20 to 30 times less active than sitaxsentan on ET_A receptors and inactive on ET_B receptors. The main routes of excretion were renal (49% to 62%) and fecal (34% to 48%). Renal clearance was decreased in PAH patients compared to healthy individuals.

The coadministration of sitaxsentan did not alter the pharmacokinetic disposition of cyclosporin, which is extensively metabolized by CYP3A4/5 (45). Although this coadministration resulted in a sixfold increase in the plasma concentrations of sitaxsentan, the mechanism for

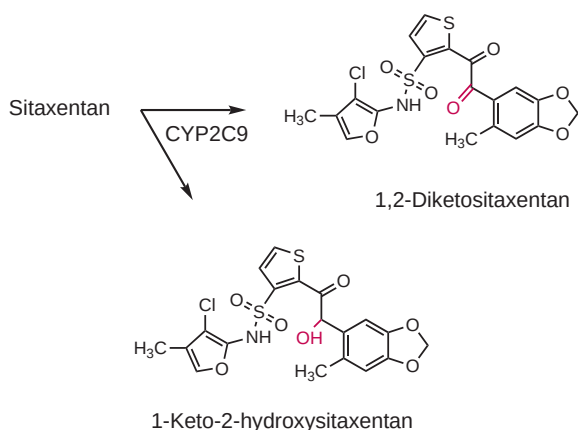


FIGURE 24.16 Metabolism of sitaxsentan.

this interaction is not known but may be due to cyclosporin inhibition of CYP3A4/5. Because of this increase in sitaxsentan exposure, its use in patients receiving cyclosporin is contraindicated.

When prescribing any endothelin receptor antagonist, patients with PAH frequently require anticoagulant therapy, and sitaxsentan affects warfarin metabolism or similar anticoagulants. Sitaxsentan is a potent inhibitor of CYP2C9 and therefore increases the area under the curve and maximum plasma concentration of drugs metabolized by CYP2C9 (45). When sitaxsentan is coadministered with warfarin, the area under the curve of warfarin was increased by approximately 96%, and clearance was decreased by approximately 63%. An enhanced effect on prothrombin clotting times was observed, consistent with the increase in plasma levels of warfarin. Therefore, because sitaxsentan inhibits the metabolism of warfarin, a warfarin dose adjustment is needed when coadministered with sitaxsentan. The dose of warfarin should be decreased by 80% when starting sitaxsentan, and then increased in increments of no greater than 0.5 mg/d while titrating to the desired clotting time. Thus, monitoring of coagulation status is warranted when sitaxsentan is prescribed.

Like other endothelin receptor antagonists, sitaxsentan is also teratogenic, and pregnancy must be excluded before the start of treatment and prevented thereafter by using a reliable contraceptive method. The drug has been associated with a reversible, dose-related increase in the liver enzymes aspartate aminotransferase and alanine aminotransferase, accompanied in some cases by elevated bilirubin (hepatotoxicity). Therefore, testing of liver aminotransferases prior to starting treatment and monthly thereafter is recommended. Patients with hypersensitivity to sulfonamides should be excluded.

The most commonly reported adverse events were headache, peripheral edema, nasal congestion, and nausea.

Prostanoids (46).

Epoprostenol (Flolan). Prostacyclin and its analogs (prostanoids) (Fig. 24.17) are potent vasodilators and possess antithrombotic and antiproliferative properties. Prostacyclin is derived from the endothelium of VSM, and its synthesis is reduced in patients with PAH. Its physiologic antagonist, thromboxane A_2 , is increased, however, causing vasoconstriction. Prostacyclin produces its vasodilation via activation of the PIP_2 signal transduction pathway, increasing concentrations of cAMP (Fig. 24.2). Epoprostenol is the sodium salt of prostacyclin and is administered as an implanted continuous IV infusion because of its very short duration of action (2 to 3 minutes). It must be reconstituted with a special glycine buffer diluent, giving a reconstituted solution with pH 10 to 11 that is stable for 15 minutes at 4°C and for less than 10 minutes at 37°C. Its injection solution is unstable at a lower pH because of acid-catalyzed hydrolysis of the vinyl-ether structure to 6-oxo-PGF_{1α} (Fig. 24.17). It is short acting because of rapid metabolism at the 15-hydroxy group to the inactive 15-oxo metabolite.

Treprostinil (Remodulin). Treprostinil (Fig. 24.17) is a synthetic, stable form of prostacyclin for the treatment of advanced pulmonary hypertension with NYHA class III or IV symptoms, as well as for late-stage peripheral vascular disease (PVD). Its sodium salt injectable form is administered either as a continuous subcutaneous infusion directly into the skin or, if the subcutaneous infusion is not tolerated, as a continuous IV infusion without an implanted catheter. Treprostinil is rapidly absorbed from the subcutaneous site of infusion, with an almost 100% bioavailability and a mean half-life of 85 minutes (34 minutes for the IV infusion). The IV solution must be diluted with normal saline or sterile water before starting the infusion. Unlike epoprostenol, treprostinil is stable at room temperature for up to 5 years, with vasodilation action lasting from 4 to 6 hours, compared with the short, 2- to 3-minute action for epoprostenol. Because of its long life in the body, it can be administered under the skin with a microinfusion subcutaneous infusion pump rather than into the bloodstream and, thus, without hospitalization, as contrasted with the central IV infusion of epoprostenol.

Side effects include jaw pain, headaches, nausea, diarrhea, flushing, and localized pain at the delivery site under the skin. This pain has been reported as slight to severe irritation. Patients using the drug seem to experience improvement in their condition, including decreased fatigue, decreased shortness of breath, and decreased pulmonary artery pressures, as well as overall improvement in quality of life.

Beraprost. Beraprost is an oral formulation of a prostacyclin analog for the treatment of early-stage pulmonary hypertension as well as early-stage PVD (Fig. 24.17).

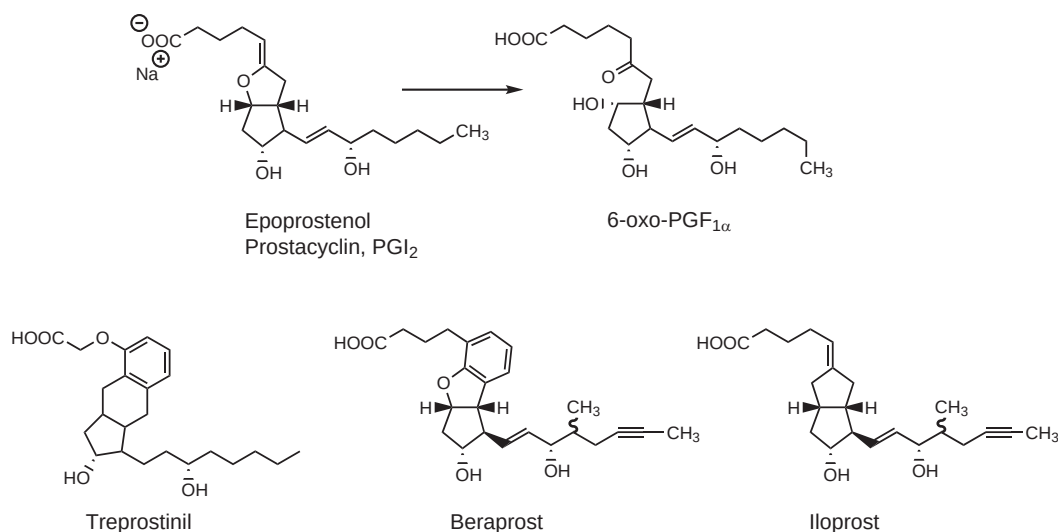


FIGURE 24.17 Prostacyclin analogs.

Beraprost is a chemically stable, oral form of prostacyclin that is readily absorbed from GI tract. Like natural prostacyclin, beraprost dilates blood vessels, prevents platelet aggregation, and prevents proliferation of smooth muscle cells surrounding blood vessels. It can be an important treatment for early-stage PVD and early-stage pulmonary hypertension. Intermittent oral doses of beraprost, however, do not seem to provide the consistent blood levels necessary to treat the advanced stages of pulmonary hypertension. Beraprost has proven to be safe and effective for the treatment of PVD in the clinical studies conducted and has been approved for the treatment of PVD in Japan since 1994. It will soon be available for use in patients with pulmonary hypertension in the United States. Adverse effects include headache, flushing, jaw pain, and diarrhea.

Iloprost (Ventavis). Iloprost is administered as an inhalation solution of a prostacyclin analog for the treatment of NYHA class III and class IV PAH (Fig. 24.17). The drug can also be administered as an IV infusion. It is stable at room temperature and to light, with a body half-life of 30 minutes. Iloprost has approximately 10 times greater potency than prostacyclin as a vasodilator of the pulmonary blood vessels; this greater potency of inhaled iloprost results from coating of the drug on the alveoli of the lungs. It relieves pulmonary vascular resistance. Patients inhale six to eight puffs every 2 to 3 hours. Each puff lasts approximately 15 minutes. This therapy is used mainly in Europe and is not available in United States. Studies have reported minor side effects, such as coughing, headaches, and jaw pain.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

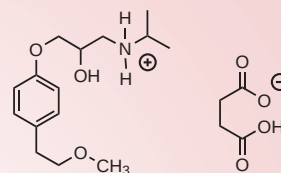
Judy Cheng, PharmD

An additional antihypertensive agent is necessary to better control blood pressure. Because AJ has also history of chronic stable angina, adding a β -blocker will not only help control the patient's blood pressure (his goal needs to be at least $<140/90$ mm Hg) and his heart rate (his goal needs to be 50 to 60 beats per minute to minimize risk of chest pain), but it is also a good antianginal agent for AJ. It is recommended that metoprolol succinate 25 mg once daily be added to the patient's regimen.

Chemical Analysis

Victoria Roche and S. William Zito

Metoprolol succinate is a β_1 -selective adrenergic antagonist commonly used to decrease heart rate and force of myocardial contraction. The aryloxypropanolamine pharmacophore contains all of the functional groups required by adrenoceptors of their endogenous neurotransmitter norepinephrine, but separates the critical binding moieties (aromatic ring, carbinol OH, cationic amino nitrogen) by distances that are longer than what is observed in the agonist. The flexibility of the side chain permits access to key adrenoceptor binding residues but disallows receptor stimulation. Strong competitive antagonism is the pharmacologic result.



Metoprolol succinate

Metoprolol targets β -adrenoceptors rather than α -receptors because the *N*-isopropyl moiety binds with high affinity to the hydrophobic binding site at β sites but encounters affinity-destroying steric hindrance at α sites. The selectivity is further defined by the *p*-substituent on the aromatic ring, which directs the molecule to β_1 -receptors in the heart over β_2 -receptors in the lung. The lack of problematic bronchoconstriction in therapeutic doses enhances the safety of this agent, particularly in patients with compromised lung function.

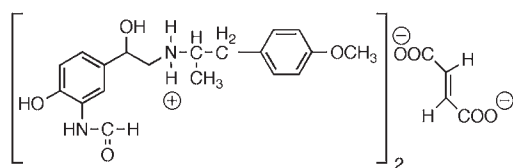
Metoprolol and other β_1 -selective agents can attenuate the symptoms of angina because, with the decrease in cardiac workload, the coronary vessels have more time to fill with blood during diastole, leading to improved coronary artery perfusion. Of importance to AJ is that β_1 -selective blockers do not negatively impact serum lipid levels to the extent that nonselective β -blockers do. They are commonly coadministered with diuretics like hydrochlorothiazide in the treatment of hypertension.

CASE STUDY

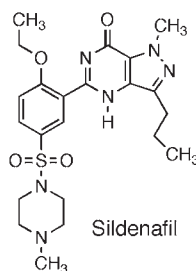
Victoria Roche and S. William Zito

HP is a 72-year-old retiree who lives with his wife in a beachside community in south Florida. HP was born and raised in Iowa, and he was a very successful pig farmer for all of his adult life. As a result of his exposure to organic dusts and gases associated with his agricultural occupation, he has compromised lung function that manifests as chronic bronchitis and emphysema. Like many men his age, he suffers from benign prostatic hypertrophy, which he manages through the use of the nutraceutical saw palmetto. HP and his wife love hitting the open road on day trips to see the south Florida sights. They still enjoy a healthy sexual relationship, although he depends on the PDE5 inhibitor sildenafil (Viagra) to overcome age-related erectile dysfunction.

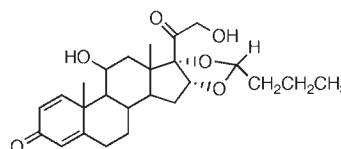
Always borderline hypertensive, HP had been able to keep his blood pressure in check with a proper diet and the active lifestyle that was a natural part of his farm-related occupation. However, now that he's taking life easy (and eating out more often), his pressure is elevated to the point where pharmacotherapy is appropriate. When advising HP's physician about a good starting therapy, you make note that, in addition to Viagra, his patient profile includes Symbicort, a combination product containing budesonide (an inhaled corticosteroid), and formoterol fumarate (a long-acting β_2 -adrenoceptor agonist). Which of the three blood pressure-lowering agents below would you recommend for this "young-at-heart" retiree?



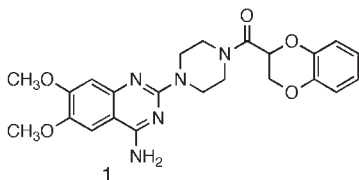
Formoterol fumarate



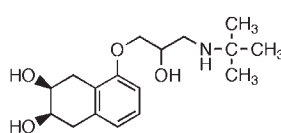
Sildenafil



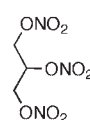
Budesonide



1



2



3

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 25

Antihyperlipoproteinemics and Inhibitors of Cholesterol Biosynthesis

MARC HARROLD

Drugs Covered in This Chapter*

- Atorvastatin
- Bezafibrate
- Cholestyramine
- Ciprofibrate
- Clofibric acid
- Colesevelam
- Colestipol
- Ezetimibe
- Fenofibric acid
- Fluvastatin
- Gemfibrozil
- Lovastatin
- Mevastatin
- Nicotinic acid/niacin
- Pitavastatin
- Pravastatin
- Rosuvastatin
- Simvastatin

Abbreviations

ACAT, acyl CoA–cholesterol acyltransferase
ALT, alanine transaminase
apo, apolipoprotein
AST, aspartate transaminase
CETP, cholesterol ester transfer protein
CHD, coronary heart disease
CoA, coenzyme A

FDA, U.S. Food and Drug Administration
HDL, high-density lipoprotein
HMG, 3-hydroxy-3-methylglutaryl
HMGRIs, HMG-CoA reductase inhibitors
IDL, intermediate-density lipoprotein
LDL, low-density lipoprotein
NAD⁺, nicotinamide adenine dinucleotide

NADP⁺, nicotinamide adenine dinucleotide phosphate
PPAR, peroxisome proliferator-activated receptor
SAR, structure–activity relationship
RCT, reverse cholesterol transport
VLDL, very-low-density lipoprotein

SCENARIO

Thomas L. Rihn, PharmD

TN is a 66-year-old man with a 2-year history of angina and an 8-year history of non–insulin-dependent diabetes mellitus. He also has fairly severe peripheral vascular disease. TN returned to his primary care physician for follow-up after experiencing several recent bouts of chest pain upon exertion. Blood work during that visit revealed the following lipid profile:

- Total Chol. 272 mg/dL

- LDL-C 197 mg/dL
- HDL-C 46 mg/dL
- Normal triglycerides

TN has insurance for prescription coverage.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

THE CHEMISTRY AND BIOCHEMISTRY OF PLASMA LIPIDS

The major lipids found in the bloodstream are cholesterol, cholesterol esters, triglycerides, and phospholipids. An excess plasma concentration of one or more of these compounds is known as hyperlipidemia. Because all lipids require the presence of soluble lipoproteins to be transported in the blood, hyperlipidemia ultimately results in an increased concentration of these transport molecules, a condition known as hyperlipoproteinemia. Hyperlipoproteinemia has been strongly associated with atherosclerotic lesions and coronary heart disease (CHD) (1,2). Before discussing lipoproteins, their role in cardiovascular disease, and agents to decrease their concentrations, it is essential to examine the biochemistry of cholesterol, triglycerides, and phospholipids.

Synthesis and Degradation of Cholesterol

Cholesterol is a C_{27} steroid that serves as an important component of all cell membranes and as the precursor for androgens, estrogens, progesterone, and adrenocorticoids (Fig. 25.1). It is synthesized from acetyl coenzyme A (CoA), as shown in Figure 25.2 (3). The first stage of the biosynthesis is the formation of isopentenyl pyrophosphate from three acetyl CoA molecules. The conversion of 3-hydroxy-3-methylglutaryl (HMG)-CoA to mevalonic acid is especially important, because it is a primary control site for cholesterol biosynthesis. This reaction is catalyzed by HMG-CoA reductase and reduces the thioester

of HMG-CoA to a primary hydroxyl group. The second stage involves the coupling of six isopentenyl pyrophosphate molecules to form squalene. Initially, three isopentenyl pyrophosphate molecules are condensed to form farnesyl pyrophosphate, a C_{15} intermediate. Two farnesyl pyrophosphate molecules are then combined using a similar type of reaction. The next stage involves the cyclization of squalene to lanosterol. This process involves an initial epoxidation of squalene, followed by a subsequent cyclization requiring a concerted flow of four pairs of electrons and the migration of two methyl groups. The final stage involves the conversion of lanosterol to cholesterol. This process removes three methyl groups from lanosterol, reduces the side-chain double bond, moves the other double bond within the ring structure, and requires approximately 20 steps.

Cholesterol is enzymatically transformed by two different pathways. As illustrated in Figure 25.1, cholesterol can be oxidatively cleaved by the enzyme desmolase (side chain–cleaving enzyme). The resulting compound, pregnenolone, serves as the common intermediate in the biosynthesis of all other endogenous steroids. As illustrated in Figure 25.3, cholesterol also can be converted to bile acids and bile salts. This pathway represents the most important mechanism for cholesterol catabolism. The enzyme 7 α -hydroxylase catalyzes the initial, rate-limiting step in this metabolic pathway and, thus, is the key control enzyme for this pathway. Cholic acid and its derivatives are primarily (99%) conjugated with either glycine (75%) or taurine (24%). Bile salts, such as

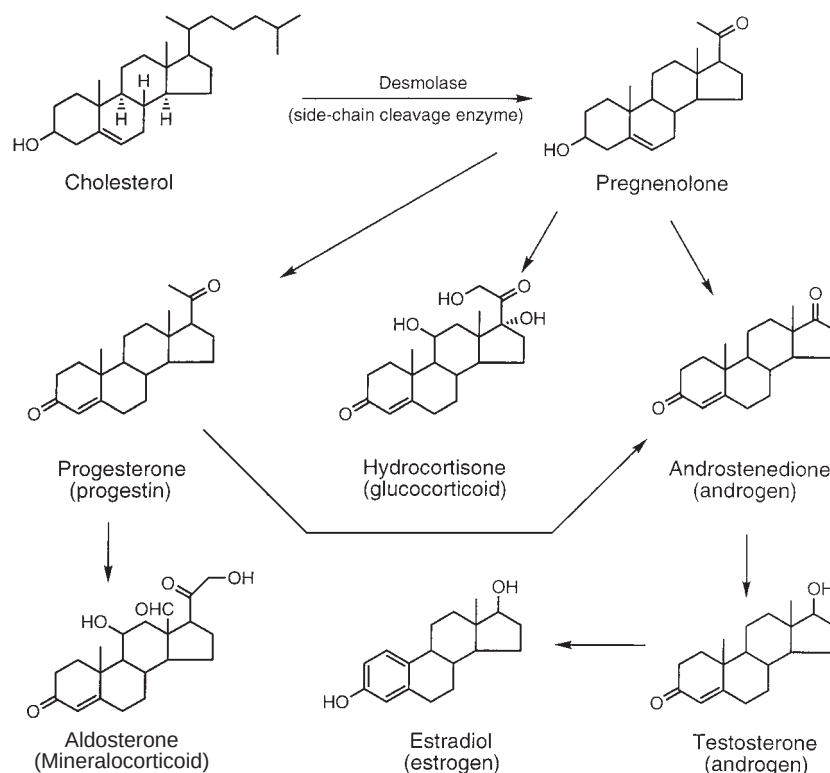
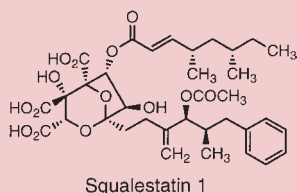


FIGURE 25.1 Cholesterol's role as a key intermediate in the biosynthesis of endogenous steroids.

SQUALENE SYNTHASE: A POTENTIAL DRUG TARGET

Inhibitors of squalene synthase, the enzyme responsible for catalyzing the two-step conversion of two molecules of farnesyl pyrophosphate to squalene, are currently being investigated as antihyperlipidemic agents. Squalene synthase catalyzes the first committed step in sterol biosynthesis and offers some potential advantages over HMG-CoA reductase as a drug target. The latter group of compounds inhibits cholesterol synthesis at an early stage of the pathway and, thus, lacks specificity. Mevalonic acid, the immediate product of HMG-CoA reductase, is a common intermediate in the biosynthesis of other isoprenoids, such as ubiquinone (an electron carrier in oxidative phosphorylation), dolichol (a compound involved in oligosaccharide synthesis), and farnesylated proteins (the farnesyl portion targets the protein to cell membrane as opposed to the cytosol). Inhibitors of squalene synthase target an enzyme involved in a later stage of cholesterol biosynthesis and could potentially accomplish the same desired outcomes as currently available agents without interfering with the biosynthesis of other essential, nonsteroidal compounds. One class of compounds currently under investigation is the squalestatsins. These compounds were originally isolated as fermentation products produced by a species of *Phoma*. Squalestatins are potent inhibitors of squalene synthetase and have been shown to produce a marked decrease in serum cholesterol. Additional analogs and other structural classes continue to be investigated and, ultimately, may produce alternatives to currently available therapy (4,5).



glycocholate, are surface-active agents that act as anionic detergents.

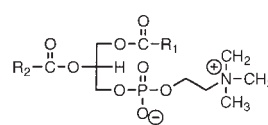
The bile salts are synthesized in the liver, stored in the gallbladder, and released into the small intestine, where they emulsify dietary lipids and fat-soluble vitamins. This solubilization promotes the absorption of these dietary compounds through the intestinal mucosa. Bile salts are predominantly reabsorbed through the enterohepatic circulation and returned to the liver, where they exert a negative feedback control on 7 α -hydroxylase and, thus, regulate any subsequent conversion of cholesterol (3,6).

The terms “bile acid” and “bile salt” refer to the unionized and ionized forms, respectively, of these compounds. For illustrative purposes only, Figure 25.3 shows cholic acid as a unionized bile acid and glycocholate as an ionized bile salt (as the sodium salt). At physiologic and intestinal pH values, both compounds would exist almost exclusively in their ionized forms.

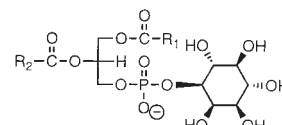
Overview of Triglycerides and Phospholipids

Triglycerides (or, more appropriately, triacylglycerols) are highly concentrated stores of metabolic energy. They are formed from glycerol 3-phosphate and acylated CoA (Fig. 25.4) and accumulate primarily in the cytosol of adipose cells. When required for energy production, triglycerides are hydrolyzed by lipase enzymes to liberate free fatty acids that are then subjected to β -oxidation, the citric acid cycle, and oxidative phosphorylation.

Phospholipids, or phosphoglycerides, are amphipathic compounds that are used to make cell membranes, generate second messengers, and store fatty acids for use in the generation of prostaglandins. They can be synthesized from phosphatidate, an intermediate in triglyceride synthesis. Two common phospholipids, phosphatidyl choline and phosphatidyl inositol, are shown below (3).



Phosphatidyl choline



Phosphatidyl inositol

Lipoproteins and Transport of Cholesterol and Triglycerides

Cholesterol, triglycerides, and phospholipids are freely soluble in organic solvents, such as isopropanol, chloroform, and diethyl ether, but are relatively insoluble in aqueous, physiologic fluids. To be transported within the blood, these lipids are solubilized through association with macromolecular aggregates known as lipoproteins. Each lipoprotein is associated with additional proteins, known as apolipoproteins, on their outer surface. These apolipoproteins provide structural support and stability, bind to cellular receptors, and act as cofactors for enzymes involved in lipoprotein metabolism. The compositions and primary functions of the six major lipoproteins are listed in Table 25.1 (6,7).

Lipoprotein nomenclature is based on mode of separation. When preparative ultracentrifugation is used, lipoproteins are separated according to their density and identified as very-low-density lipoproteins (VLDLs), intermediate-density lipoproteins (IDLs), low-density lipoproteins (LDLs), and high-density lipoproteins (HDLs). When electrophoresis is used in the separation, lipoproteins are designated as pre- β , β , and α . The IDLs are mainly found in the pre- β fraction as a second electrophoretic band and are currently believed to be an intermediate lipoprotein in the catabolism of VLDL to LDL. Chylomicron remnants and IDLs may show similar electrophoretic and ultracentrifugation separation characteristics. In general, VLDL, LDL, and HDL correspond to pre- β , β , and α lipoprotein, respectively.

The interrelationship among the lipoproteins is shown in Figure 25.5 (7,8). As illustrated, the pathway can be divided into both exogenous (dietary intake) and endogenous (synthetic) components. The exogenous

CLINICAL SIGNIFICANCE

The development and availability of cholesterol and triglyceride-lowering agents has evolved significantly over the past 10 to 15 years. The structure–activity relationships and structural modifications of bile acid sequestrants, fibrates, nicotinic acid, and, in particular, 3-hydroxy-3-methylglutaryl-coenzyme A reductase inhibitors form the basis for this evolution and the therapeutic advances that have resulted.

The most widely used cholesterol-lowering agents in cardiovascular medicine today are the “statins” (vastatins). Lovastatin was the first statin to be used clinically on a large scale. Structural modifications of the early statin molecules have produced superior agents in terms of their pharmacokinetic profile, potency, drug interactions, and, perhaps, selective adverse effects. This has resulted in the widespread clinical use of superior agents, such as atorvastatin. In randomized clinical trials, such agents

have been demonstrated to exert a potent effect in lowering low-density lipoprotein cholesterol as well as an important anti-inflammatory action.

The net effect of applying basic science in modifying the chemical structure of cholesterol-lowering drugs is that negative patient outcomes such as death, myocardial infarction, and other cardiovascular events have been greatly reduced. This is clearly a situation in which the application of basic and clinical science has produced a profound effect on tens of millions of patients.

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pathway begins after the ingestion of a fat-containing meal or snack. Dietary lipids are absorbed in the form of cholesterol and fatty acids. The fatty acids are then reesterified within the intestinal mucosal cells and, along with the cholesterol, are incorporated into chylomicrons, the largest lipoprotein. During circulation, chylomicrons are degraded into remnants by the action of lipoprotein

lipase, a plasma membrane enzyme located on capillary endothelial cells in adipose and muscle tissue. The interaction of chylomicrons with lipoprotein lipase requires apolipoprotein (apo) C-II, and the absence of either the enzyme or the apolipoprotein can lead to hypertriglyceridemia and pancreatitis. The liberated free acids are then available for either storage or energy

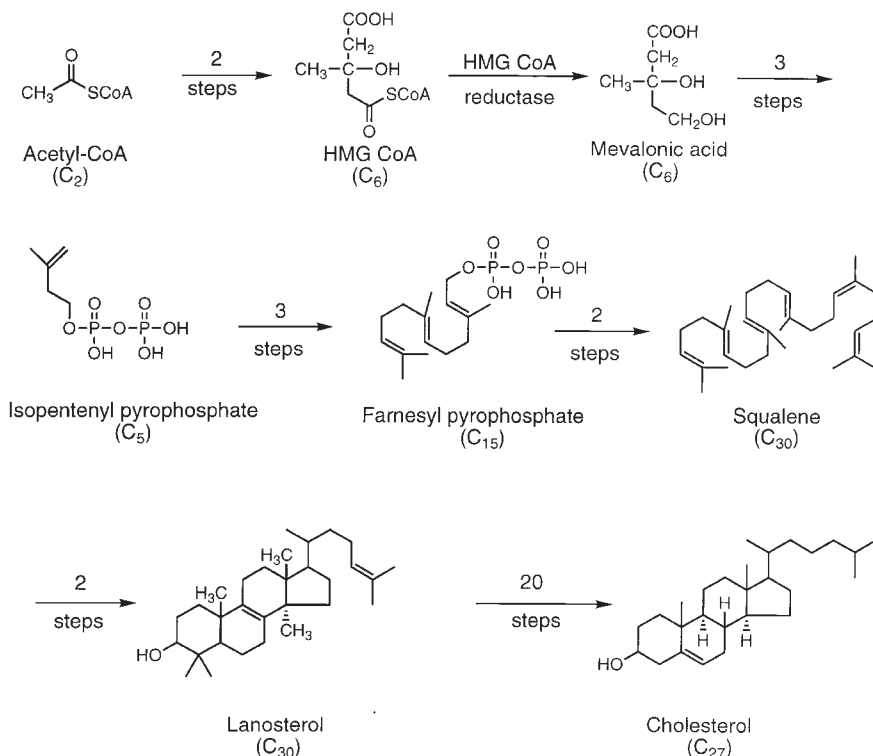


FIGURE 25.2 The biosynthesis of cholesterol.

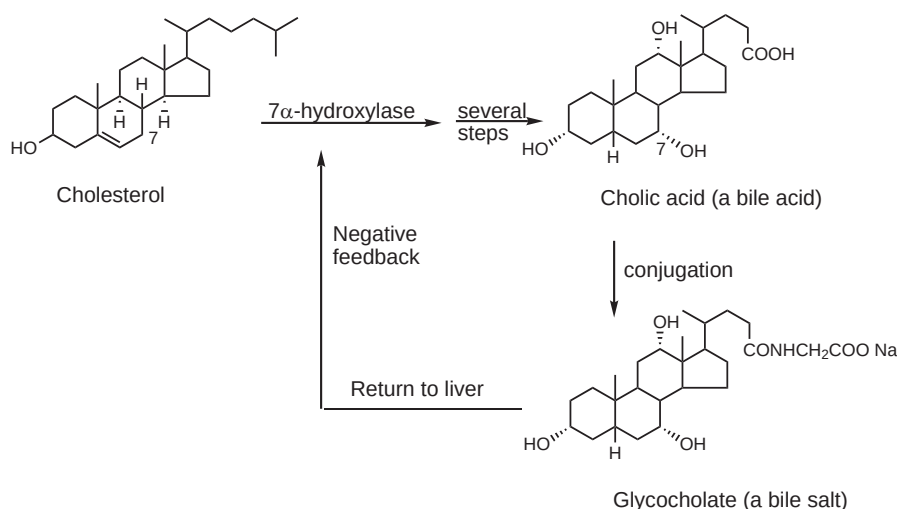


FIGURE 25.3 The conversion of cholesterol to bile acids and bile salts.

generation by these tissues. The remnants are predominantly cleared from the plasma by liver parenchymal cells via recognition of the apoE portion of the carrier.

The endogenous pathway begins in the liver with the formation of VLDL. Similar to chylomicrons, triglycerides are present in a higher concentration than either cholesterol or cholesterol esters; however, the concentration difference between these lipids is much less than that seen in chylomicrons. The metabolism of VLDL also is similar to chylomicrons in that lipoprotein lipase

reduces the triglyceride content of VLDL and increases the availability of free fatty acids to the muscle and adipose tissue. The resulting lipoprotein, IDL, either can be further metabolized to LDL or can be transported to the liver for receptor-mediated endocytosis. This latter effect involves an interaction of the LDL receptor with the apolipoproteins, apoB-100 and apoE, on IDL. The amount of IDL delivered to the liver is approximately the same as that converted to LDL. The half-life of IDL is relatively short as compared to that of LDL and, thus, accounts for only a small portion of total plasma cholesterol. In contrast, LDL accounts for approximately two-thirds of total plasma cholesterol and serves as the primary source of cholesterol for both hepatic and extrahepatic cells. As with IDL, the uptake of LDL by these cells is mediated by a receptor interaction with the apoB-100 on LDL. The number of LDL receptors on the cell surface mediates regulation of cellular LDL uptake. Cells requiring increased amounts of cholesterol will increase the biosynthesis of LDL receptors. Conversely, it has been demonstrated that increased hepatic concentrations of cholesterol will inhibit both HMG-CoA reductase as well as the production of LDL receptors. As previously discussed, hepatic cholesterol can be converted to bile acids and bile salts and reenter the endogenous pathway through the bile and enterohepatic circulation.

Synthesized in the liver and intestine, HDL initially exists as a dense, phospholipid disk composed primarily of apoA-I. The primary function of HDL is to act as a scavenger to remove cholesterol from extrahepatic cells and to facilitate its transport back to the liver. Nascent HDL accepts free, unesterified cholesterol. A plasma enzyme, lecithin-cholesterol acyltransferase, then esterifies the cholesterol. This process allows the resulting cholesterol esters to move from the surface to the core and results in the production of spherical HDL₃ particles. As cholesterol content is added, HDL₃ is converted to HDL₂, which is larger and less dense than HDL₃. The ultimate return of cholesterol from HDL₂ to the liver is known as

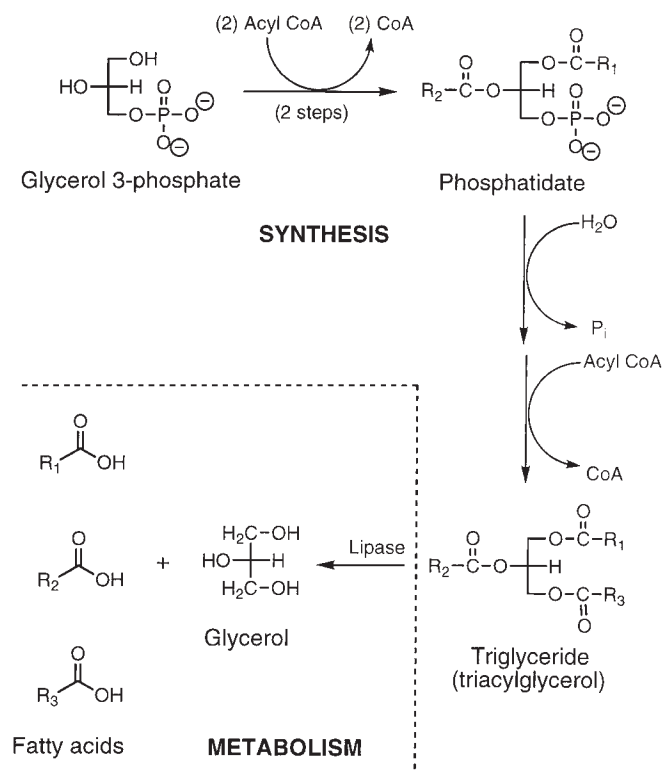


FIGURE 25.4 The biosynthesis and metabolism of triglycerides.

TABLE 25.1 Classification and Characteristics of Major Plasma Lipoproteins

Classification	Composition	Major Apolipoproteins	Primary Function(s)
Chylomicrons	Triglycerides 80%–95%, free cholesterol 1%–3%, cholesterol esters 2%–4%, phospholipids 3%–9%, apoproteins 1%–2%	apoA-I, apoA-IV, apoB-48, apoC-I, apoC-II, apoC-III	Transport dietary triglycerides to adipose tissue and muscle for hydrolysis by lipoprotein lipase
Chylomicron remnants	Primarily composed of dietary cholesterol esters	apoB-48, apoE	Transport dietary cholesterol to liver for receptor-mediated endocytosis
VLDL	Triglycerides 50%–65%, free cholesterol 4%–8%, cholesterol esters 16%–22%, phospholipids 15%–20%, apoproteins 6%–10%	apoB-100, apoE, apoC-I, apoC-II, apoC-III	Transport endogenous triglycerides to adipose tissue and muscle for hydrolysis by lipoprotein lipase
IDL	Intermediate between VLDL and LDL	apoB-100, apoE, apoC-II, apoC-III	Transport endogenous cholesterol for either conversion to LDL or receptor-mediated endocytosis by the liver
LDL	Triglycerides 4%–8%, free cholesterol 6%–8%, cholesterol esters 45%–50%, phospholipids 18%–24%, apoproteins 18%–22%	apoB-100	Transport endogenous cholesterol for receptor-mediated endocytosis by either the liver or extrahepatic tissues
HDL	Triglycerides 2%–7%, free cholesterol 3%–5%, cholesterol esters 15%–20%, phospholipids 26%–32%, apoproteins 45%–55%	apoA-I, apoA-II, apoE, apoC-I, apoC-II, apoC-III	Removal of cholesterol from extrahepatic tissues via transfer of cholesterol esters to IDL and LDL

apo, apolipoprotein; HDL, high-density lipoprotein; IDL, intermediate-density lipoprotein; LDL, low-density lipoprotein; VLDL, very-low-density lipoprotein.

reverse cholesterol transport and is accomplished via an intermediate transfer of cholesterol esters from HDL₂ to either VLDL or IDL. This process regenerates spherical HDL₃ molecules that can recirculate and acquire excess cholesterol from other tissues. In this manner, HDL

serves to prevent the accumulation of cholesterol in arterial cell walls and other tissue and may serve as the basis for its cardioprotective properties (6,7).

Classification of Hyperlipoproteinemias

Hyperlipoproteinemia can be divided into primary and secondary disorders. Primary disorders are the result of genetic deficiencies or mutations, whereas secondary disorders are the result of other conditions or diseases. Secondary hyperlipoproteinemia has been associated with diabetes mellitus, hypothyroidism, renal disease, liver disease, alcoholism, and certain drugs (1,6,7).

In 1967, Fredrickson et al. (8) classified primary hyperlipoproteinemias into six phenotypes (I, IIa, IIb, III, IV, and V) based on which lipoproteins and lipids were elevated. Current literature and practice, however, appear to favor the more descriptive classifications and subclassifications listed in Table 25.2. Primary disorders are currently classified as those that primarily cause hypercholesterolemia, those that primarily cause hypertriglyceridemia, and those that cause a mixed elevation of both cholesterol and triglycerides. Subclassifications are based on the specific biochemical defect responsible for the disorder. Classifications developed by Fredrickson have been included in Table 25.2 under the heading *Previous Classification* for comparative and reference purposes.

As shown in Table 25.2, some disorders are well characterized, whereas others are not (1,6,7). Familial hypercholesterolemia is caused by a deficiency of LDL receptors. This results in a decreased uptake of IDL and LDL by hepatic and extrahepatic tissues and an elevation in plasma LDL levels. The homozygous form of this disorder is rare but results in extremely high LDL levels

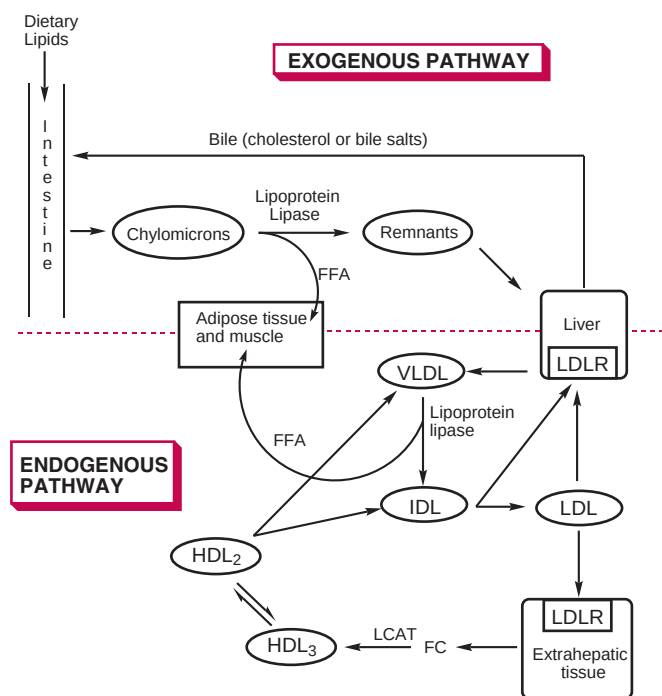


FIGURE 25.5 Endogenous and exogenous pathways for lipid transport and metabolism. FC, free unesterified cholesterol; FFA, free fatty acids; LCAT, lecithin-cholesterol acyltransferase; LDLR, low-density lipoprotein receptor.

TABLE 25.2 Characteristics of the Major Primary Hyperlipoproteinemias

Current Classification	Biochemical Defect	Elevated Lipoproteins	Previous Classification
Hypercholesterolemias			
Familial hypercholesterolemia	Deficiency of LDL receptors	LDL	IIa
Familial defective apoB-100	Mutant apoB-100	LDL	IIa
Polygenic hypercholesterolemia	Unknown	LDL	IIa
Hypertriglyceridemias			
Familial hypertriglyceridemia	Unknown	VLDL	IV
Familial lipoprotein lipase deficiency	Deficiency of lipoprotein lipase	Chylomicrons, VLDL	I (chylomicron elevation only), V
Familial apoC-II deficiency	Deficiency of apoC-II	Chylomicrons, VLDL	I (chylomicron elevation only), V
Mixed hypercholesterolemia and hypertriglyceridemia			
Familial combined hyperlipidemia	Unknown	VLDL, LDL	IIb
Dysbetalipoproteinemia	Presence of apoE ₂ isoforms	VLDL, IDL	III

apo, apolipoprotein; IDL, intermediate-density lipoprotein; LDL, low-density lipoprotein; VLDL, very-low-density lipoprotein.

and early morbidity and mortality because of the total lack of LDL receptors. A related disorder, familial defective apoB-100, also results in elevated LDL levels but is caused by a genetic mutation rather than a deficiency. Alteration of apoB-100 decreases the affinity of LDL for the LDL receptor and thus hinders normal uptake and metabolism. Elevations in chylomicron levels can result from a deficiency of either lipoprotein lipase or apoC-II. These deficiencies cause decreased or impaired triglyceride hydrolysis and result in a massive accumulation of chylomicrons in the plasma. Dysbetalipoproteinemia results from the presence of an altered form of apoE and is the only mixed hyperlipoproteinemia with a known cause. Proper catabolism of chylomicron and VLDL remnants requires apoE. The presence of a binding-defective form of apoE, known as apoE₂, results in elevated levels of VLDL and IDL triglyceride and cholesterol levels.

DISEASES AND DISORDERS CAUSED BY HYPERLIPIDEMIAS

CHD, which includes acute myocardial infarction, ischemic heart disease, and angina pectoris, is the leading cause of mortality in the United States. In 2006, CHD was responsible for 425,00 deaths. In addition, mortality from CHD often occurs rapidly, either in an emergency room or before hospitalization. The highest mortality is seen in patients older than 65 years; however, the vast majority of deaths in patients younger than 65 years occur during an initial attack. Risk factors associated with CHD include hypertension, cigarette smoking, elevated plasma cholesterol levels, physical inactivity, diabetes, and obesity (9).

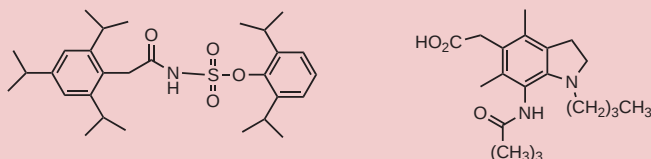
Atherosclerosis, which is named from the Greek terms for “gruel” (*athere*) and “hardening” (*sclerosis*), is the underlying cause of CHD. It is a gradual process in which an initial accumulation of lipids in the arterial intima leads to thickening of the arterial wall, plaque formation, thrombosis, and occlusion (10–12). The involvement of LDL cholesterol in this process is shown in Figure 25.6. Within the extracellular space of the intima, LDL is more

susceptible to oxidative metabolism, because it is no longer protected by plasma antioxidants. This metabolism alters the properties of LDL such that it is readily scavenged by macrophages. Unlike normal LDL, the uptake of oxidized LDL is not regulated; thus, macrophage cells can readily become engorged with oxidized LDL. Subsequent metabolism produces free cholesterol, which either can be released into the plasma or reesterified by the enzyme acyl CoA–cholesterol acyltransferase (ACAT). Cholesterol released into plasma can be scavenged by HDL₃ and returned to the liver, thus preventing any accumulation or damage. In this manner, HDL acts as a cardioprotective agent, because high concentrations of reesterified cholesterol can morphologically change macrophages into foam cells. Accumulation of lipid-engorged foam cells in the arterial intima results in the formation of fatty streaks, the initial lesion of atherosclerosis. Later, the deposition of lipoproteins, cholesterol, and phospholipids causes the formation of softer, larger plaques. Associated with this lipid deposition is the proliferation of arterial smooth muscle cells into the intima and the laying down of collagen, elastin, and glycosaminoglycans, leading to fibrous plaques. Ultimately, the surface of the plaque deteriorates, and an atheromatous ulcer is formed with a fibrous matrix, accumulation of necrotic tissue, and appearance of cholesterol and cholesterol ester crystals. A complicated lesion also shows calcification and hemorrhage with the formation of organized mural thrombi. Thrombosis results from changes in the arterial walls and in the blood-clotting mechanism.

Obviously, individuals with higher cholesterol and LDL levels are more susceptible to these detrimental effects than those with normal cholesterol and LDL levels. Total plasma cholesterol levels less than 200 mg/dL are considered desirable. Levels above 240 mg/dL are considered high, and levels between 200 and 239 mg/dL are considered borderline. For LDL, plasma levels of less than 100 mg/dL are considered optimal, plasma levels equal to or greater than 160 are considered high, plasma levels between 130 and 159 mg/dL are considered

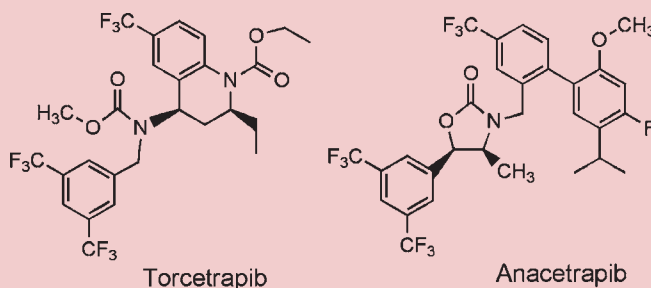
POTENTIAL DRUG TARGETS: ACAT AND CETP

Inhibitors of ACAT are currently being investigated as cholesterol-lowering or antiatherosclerotic agents. In addition to its role in foam cell formation, ACAT also is required for esterification of cholesterol in intestinal mucosal cells and for synthesis of cholesterol esters in hepatic VLDL formation. Thus, ACAT inhibitors have the potential of providing three beneficial effects in patients with hypercholesterolemia: decreased cholesterol absorption, decreased hepatic VLDL synthesis, and decreased foam cell formation. Initial successes at inhibiting ACAT were dampened by the discovery of accompanying adrenal toxicity. Subsequent structural modifications have led to the development of potent, orally active ACAT inhibitors that lack this toxicity. Two promising compounds are shown below. These and other analogs and efforts have given new hope that inhibitors of this enzyme may provide an alternative treatment of atherosclerotic disorders (11,13,14).



Cholesterol ester transfer protein (CETP) interacts with HDL, LDL, and VLDL to mediate the transfer of cholesterol esters from HDL (primarily HDL₂) to LDL and VLDL while balancing this exchange in a 1:1 fashion with a transfer of triglycerides from LDL and VLDL to HDL. The end result of this process is a higher concentration of LDL cholesterol and a lower concentration of HDL cholesterol. It has been postulated that CETP inhibitors

would produce a therapeutic benefit due to their ability to prevent these transfers and favorably increase HDL cholesterol and lower LDL cholesterol. The initial development of potent CETP inhibitors proved to be challenging due to the nonenzymatic nature of a transport protein, the lack of a transition state and/or tightly bound intermediates, and the lack of potent natural and synthetic inhibitors. A number of chemical classes of compounds have been investigated, including thiol-based irreversible inhibitors and tetrahydroquinoline-based reversible inhibitors. Drug development at Pfizer produced torcetrapib, a potent oral CETP inhibitor that was evaluated in a phase III clinical outcomes trial. Unfortunately, torcetrapib caused increased blood pressure as well as increased circulating aldosterone levels and altered serum electrolyte levels. Further studies verified that these effects were not related to the inhibition of CETP and thus would not necessarily be seen with other agents. Another structurally related compound, anacetrapib (Merck), is currently in phase III clinical trials and has been shown to reduce LDL by up to 40% and increase HDL by up to 138% with a more acceptable side effect profile than originally seen with torcetrapib. Since CETP is also involved in reverse cholesterol transport (RCT), the process by which HDL promotes the efflux of cholesterol from macrophages, there has been some debate as to whether inhibition of CETP would cause the production of cholesterol-laden HDL molecules that would be less efficient in the RCT atheroprotective process. Clinical trials involving both torcetrapib and anacetrapib have shown that isolated HDL particles from patients involved in these studies are still capable of promoting this efflux. Additional studies are needed prior to the approval of anacetrapib or another CETP inhibitor (15,16).



borderline, and plasma levels between 100 and 129 mg/dL are considered above optimal. Current guidelines (8,13) also recommend an LDL level below 70 mg/dL as a goal for very high-risk patients (i.e., those with multiple risk factors and known cardiovascular disease).

Elevated plasma triglyceride levels can contribute to atherosclerosis and CHD in mixed hyperlipoproteinemias, whereas pure hypertriglyceridemias are primarily associated with pancreatitis and show little to no relationship to CHD (6,7).

OVERVIEW OF DRUG THERAPY AFFECTING LIPOPROTEIN METABOLISM

Bile acid sequestrants, HMG-CoA reductase inhibitors (HMGRIs), ezetimibe, fibrates, and niacin are all used in the treatment of hyperlipoproteinemia. In general, successful use of these compounds depends on proper identification and classification of the hyperlipoproteinemia affecting the patient. With the possible exceptions of niacin, atorvastatin, and rosuvastatin, currently

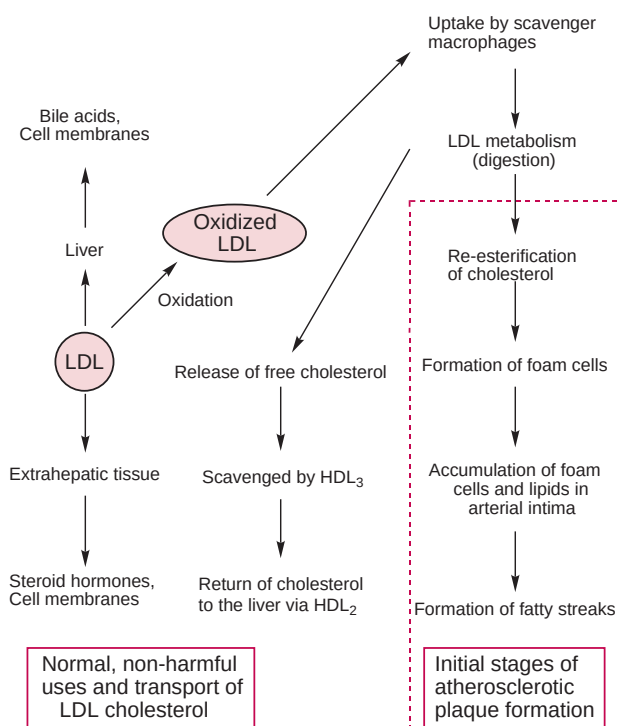


FIGURE 25.6 The role of LDL cholesterol in the development of atherosclerotic plaques. The mechanism by which HDL provides a cardioprotective action also is shown.

available compounds do not have equal efficacy in reducing both hypercholesterolemia and hypertriglyceridemia and, thus, are used primarily for their ability to decrease either cholesterol or triglyceride levels. Bile acid sequestrants, inhibitors of HMG-CoA reductase, and ezetimibe are effective in decreasing plasma cholesterol and LDL levels. The fibrates also have some actions on plasma cholesterol; however, their main effect is to stimulate lipoprotein lipase and to increase the clearance of triglycerides. Niacin, through its ability to decrease VLDL formation, has been shown to decrease the plasma levels of both triglycerides and cholesterol. All of these compounds are discussed in the following sections. References for these sections have been limited to current reviews and texts, selected papers, and product literature, both traditional and electronic. Readers requiring additional references for any of these compounds should consult either the previous edition of this text or the references contained within the cited reviews and texts.

BILE ACID SEQUESTRANTS

Historical Overview

Cholestyramine was originally developed in the 1960s to treat pruritus secondary to elevated plasma concentrations of bile acids in patients with cholestasis. Its ability to bind (i.e., to hold or to sequester) bile acids and to increase their fecal elimination was subsequently shown to produce beneficial effects in lowering serum cholesterol levels. In 1973, cholestyramine was approved for

the treatment of hypercholesterolemia in patients who do not respond to dietary modifications. Colestipol and colestevam, which retain the key structural features required to bind bile acids, were approved in 1977 and 2000, respectively (6,17).

Cholestyramine, colestipol, and colestevam are chemically classified as anion-exchange resins. This term arises from their ability to selectively bind and exchange negatively charged atoms or molecules with one another. The selectivity comes from the fact that these positively charged resins do not bind equally to all anions. For example, the chloride ion of cholestyramine can be displaced by, or exchanged with, other anions (e.g., bile acids) that have a greater affinity for the positively charged functional groups on the resin.

Mechanism of Action

Cholestyramine, colestipol, and colestevam lower plasma LDL levels by indirectly increasing the rate at which LDL is cleared from the bloodstream. Under normal circumstances, approximately 97% of bile acids are reabsorbed into the enterohepatic circulation. As previously discussed, these compounds are returned to the liver where they regulate their own production. Bile acid sequestrants are not orally absorbed but, rather, act locally within the gastrointestinal tract to interrupt this process. They bind the two major bile acids, glycocholic acid and taurocholic acid, and greatly increase their fecal excretion. As a result, decreased concentrations of these compounds are returned to the liver. This removes the feedback inhibition of 7 α -hydroxylase and increases the hepatic conversion of cholesterol to bile acids (Fig. 25.3). The decrease in hepatic cholesterol concentrations leads to several compensatory effects: increased expression of LDL receptors, increased hepatic uptake of plasma LDL, induction of HMG-CoA reductase, and increased biosynthesis of cholesterol. The latter two effects are insufficient to counteract the increases in cholesterol clearance and catabolism; however, concurrent use of an HMGRI can provide an additive effect in lowering LDL cholesterol. Bile acid sequestrants do not alter the removal of plasma LDL by nonreceptor-mediated mechanisms and, thus, are ineffective in treating homozygous familial hypercholesterolemia (6,17–19).

The decreased return of bile acids to the liver also will produce an increase in triglyceride synthesis and a transient rise in VLDL levels. Subsequent compensatory mechanisms will increase VLDL removal, most likely through the increased LDL receptors, and return VLDL levels to predrug levels. For those patients with preexisting hypertriglyceridemia, the compensatory mechanisms are inadequate, and a persistent rise in VLDL levels occurs (6).

Structure–Activity Relationships

Cholestyramine (Fig. 25.7) is a copolymer consisting primarily of polystyrene, with a small amount of divinylbenzene as the cross-linking agent. In addition, it contains

approximately 4 mEq of fixed quaternary ammonium groups per gram of dry resin. These positively charged groups function as binding sites for anions. Virtually all of these sites are accessible to bile acids. Increasing the amount of divinylbenzene from 2% to 4% to 8% increases the cross-linkage and reduces the porosity of the resin. This prevents binding of bile acids to interior sites and decreases the efficacy of the compound.

Colestipol (Fig. 25.7) is a copolymer of tetraethylenepentamine and epichlorhydrin and is commercially marked as its hydrochloride salt. The key functional groups on colestipol are the basic secondary and tertiary amines. Although the total nitrogen content of colestipol is greater than that of cholestyramine, the functional anion-exchange capacity of the resin depends on intestinal pH and may be less than cholestyramine. Comparative *in vitro* studies indicate that cholestyramine has a higher adsorption capacity than colestipol for bile salts (20). Quaternization of colestipol with methyl iodide increases the capacity *in vitro* for glycocholate (21).

Colesevelam is a more diverse polymer; however, its initial formation is similar to colestipol. In the case of colesevelam, poly(allylamine) is initially cross-linked with epichlorhydrin and then alkylated with 1-bromodecane and (6-bromohexyl)-trimethylammonium bromide. The resulting polymer contains four basic and/or quaternary functional groups, as shown in Figure 25.7. Fragment

A is a primary amine, fragment B contains a secondary amine and a quaternary ammonium group, fragment C contains a pair of secondary amines, and fragment D is a decylated amine. The overall polymer is a hydrophilic gel and is insoluble in water (6).

Physicochemical Properties

All bile acid sequestrants are large, hygroscopic, water-insoluble resins. The molecular weight of cholestyramine is reported to be greater than 1,000,000 daltons; however, no specific molecular weight has been assigned to either colestipol or colesevelam. Cholestyramine contains a large number of quaternary ammonium groups and has multiple permanent positive charges. Colestipol contains a large number of secondary and tertiary amines, whereas colesevelam contains quaternary ammonium groups as well as primary and secondary amines. Normal pK_a values for the amines range from 9.0 to 10.5; thus, all of these groups should be primarily ionized at intestinal pH.

Pharmacokinetic Parameters, Metabolism, and Dosing

Cholestyramine, colestipol, and colesevelam are not orally absorbed and are not metabolized by gastrointestinal enzymes. They are excreted in the feces as an insoluble complex with bile acids. Their onset of action occurs

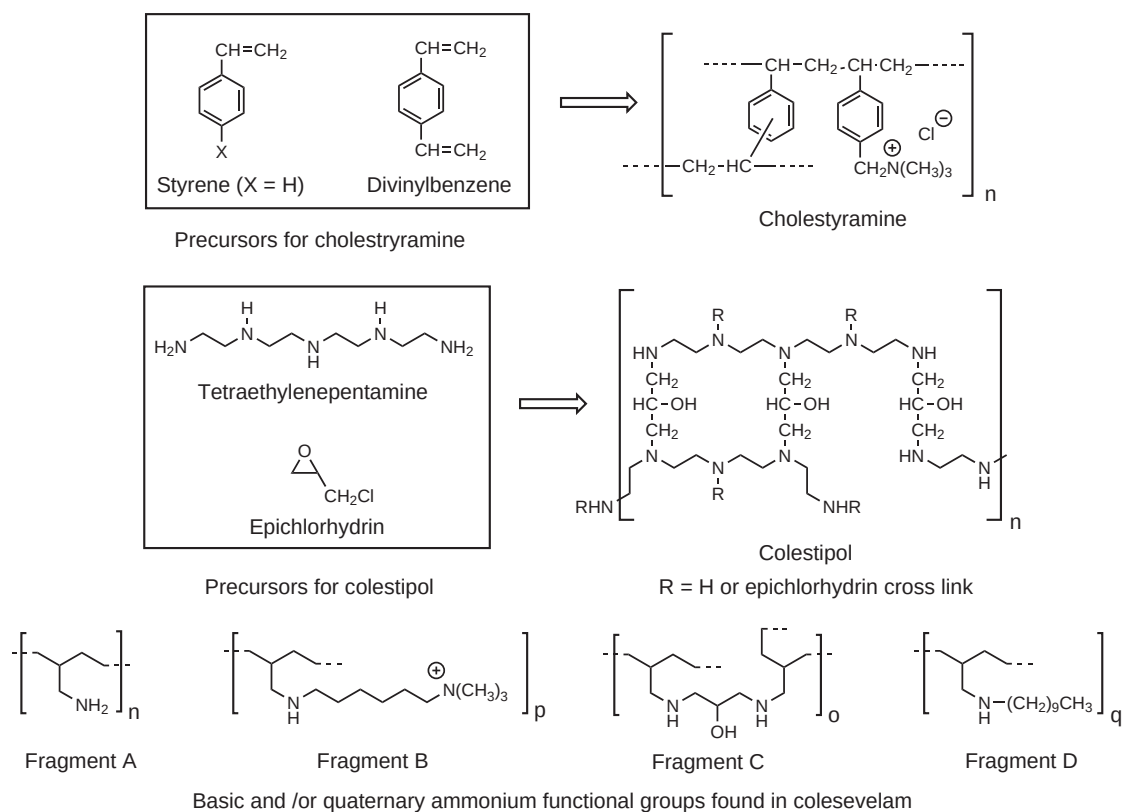


FIGURE 25.7 Structures of cholestyramine, colestipol, and precursors for these polymeric resins. Also included are the basic and quaternary functional groups found on colesevelam. Note that cholestyramine will contain a fraction of unsubstituted aromatic rings (i.e., those that are neither cross-linked nor contain a quaternary ammonium group).

UNLABELED USES

Diarrhea, digitalis toxicity, adjunctive treatment for hyperthyroidism, and pseudomembranous colitis

within 24 to 48 hours; however, it may take up to 1 month to achieve peak response (6,17,22).

Cholestyramine is available as a powder that is mixed with water, juice, or other noncarbonated beverages to create a slurry to drink. Patients should experiment with various liquids to find the most palatable combination; however, patient acceptance and compliance with this dosage formulation can limit its use. Each packet or scoop of cholestyramine is equivalent to 4 g of cholestyramine. The recommended daily dose for the treatment of hypercholesterolemia is 8 to 16 g (two to four packets or scoops) per day divided into two doses and taken with meals. The maximum daily dose for hypercholesterolemia is 24 g. Colestipol is available as either granules or 1-g tablets. The granules should be taken in a manner similar to that described for the cholestyramine powder. The starting dose for the granules is 5 g once or twice daily. This dose can be increased in 5-g increments every 1 to 2 months until therapeutic goals or a maximum of 30 g/d has been reached. The starting dose for the tablets is 2 g once or twice daily. This dose can then be increased in 2-g segments up to a maximum daily dose of 16 g. Patients must be advised that colestipol tablets should not be chewed, crushed, or cut and should be taken with plenty of water (17,23). Colesevelam is available as a 625-mg tablet and a 3.75-g powder for suspension. Both formulations should be taken with a meal. The starting dose is three tablets (1.875 g) twice a day, six tablets (3.75 g) once a day, or one 3.75-g powder packet once a day. The dose may be increased to a maximum of seven tablets per day (4.375 g).

Therapeutic Applications

Bile acid sequestrants are indicated for the treatment of hypercholesterolemia in patients who do not adequately respond to dietary modifications. They may be used either alone or in combination with HMGRIs or niacin. These combinations often can achieve a 50% reduction in plasma LDL levels. Cholestyramine is also approved for the relief of pruritus associated with partial biliary obstruction, whereas colestipol and colesevelam are approved for use as adjunct agents to improve glycemic control in patients with type 2 diabetes mellitus. Bile acid sequestrants should not be used to treat hypertriglyceridemias or mixed hyperlipoproteinemias in which hypertriglyceridemia is the primary concern. These compounds also are contraindicated in patients with cholelithiasis or complete biliary obstruction because of the impaired secretion of bile acids caused by these conditions. Finally, cholestyramine and colestipol are contraindicated in

patients with primary biliary cirrhosis, because this can further raise serum cholesterol (6,17,23).

Adverse Effects

Because bile acid sequestrants are not orally absorbed, they produce minimal systemic side effects and, thus, are one of the safest drugs to use for hypercholesterolemia. Constipation is by far the most frequent patient complaint. Increasing dietary fiber or using bulk-producing laxatives, such as psyllium, often can minimize this adverse effect. Other gastrointestinal symptoms, such as bloating and abdominal discomfort, usually disappear with continued use; however, the possibility of fecal impaction requires that extreme caution be used in patients with preexisting constipation. All three of these compounds release chloride ions as part of their exchange mechanism and can cause hyperchloremic acidosis. This is not a common occurrence, but it may limit the use of bile acid sequestrants in patients with renal disease. Hypoprothrombinemia and bleeding are caused by the ability of bile acid sequestrants to bind with and impair the absorption of dietary vitamin K. These effects also are rare, but they may limit the use of these agents in patients with preexisting clotting disorder and in those being concurrently treated with anticoagulants (2,6,17,23).

Drug Interactions

Because of their mechanism of action, bile acid sequestrants can potentially bind with and decrease the oral absorption of almost any other drug. Because these anion-exchange resins contain numerous positive charges, they are much more likely to bind to acidic compounds than to basic compounds or nonelectrolytes. This is not an absolute, however, because cholestyramine and colestipol have been reported to decrease the oral absorption of propranolol (a base) and the lipid-soluble vitamins A, D, E, and K (nonelectrolytes). As a result, the current recommendation is that all other oral medication should be administered at least 1 hour before or 4 hours after cholestyramine and colestipol. Interestingly, this drug interaction has been used in a beneficial manner to treat digitalis overdose and toxicity.

Colesevelam seems to be less likely to interfere with the absorption of concurrently administered drugs. No significant decreases in absorption were seen when colesevelam was coadministered with digoxin, lovastatin, warfarin, metoprolol, quinidine, or valproic acid. Because of potential interactions, especially for drugs with a low therapeutic index, the administration of drugs that have not been directly studied in combination with colesevelam should be spaced accordingly, as described earlier for cholestyramine and colestipol (6,17).

HMGC_oA REDUCTASE INHIBITORS

Currently, seven HMGRIs are approved for therapeutic use in the United States. Chemically, they can be divided

ADVERSE EFFECTS OF BILE ACID SEQUESTRANTS

Bloating, abdominal discomfort, constipation, bowel obstruction, steatorrhea, anorexia, cholelithiasis, pancreatitis, hyperchloremic acidosis, hypoprothrombinemia, and bleeding

into two groups, natural products and synthetic agents. All of these compounds effectively block the conversion of HMG-CoA to mevalonic acid and have similar effects on plasma cholesterol levels. The compounds differ somewhat in their indications, potencies, and pharmacokinetic profiles. They often are referred to as statins or vastatins. Because of the potential confusion of the terms “statin” and “statine” (i.e., a stable dipeptide mimic [see Chapter 23]), it is suggested here that classifying an HMGRi as a vastatin is preferable to classifying it as a statin.

Historical Overview and Development

The development and use of HMGRIs began in 1976 with the discovery of mevastatin. Originally named compactin, this fungal metabolite was isolated from two different species of *Penicillium* and demonstrated potent, competitive inhibition of HMG-CoA reductase. Its affinity for the enzyme was shown to be 10,000-fold greater than that of the substrate HMG-CoA (24). Several years later, a structurally similar compound was isolated from *Monascus ruber* and *Aspergillus terreus*. This compound was originally known as mevinolin, was later renamed lovastatin, and was more than twofold more potent than mevastatin (Fig. 25.8). Structurally, it differed from mevastatin only by the presence of a methyl group at the 6' position of the bicyclic ring. As illustrated in Figure 25.8, mevastatin and lovastatin can bind very tightly with HMG-CoA reductase, because their hydrolyzed lactones mimic the tetrahedral intermediate produced by the reductase enzyme (6). Studies published in 1985 confirmed this theory and also established that the bicyclic portions of these compounds bind to the CoA site of the enzyme (25). Clinical trials of mevastatin were halted after reports of altered intestinal morphology in dogs (26); however, lovastatin received approval by the U.S. Food and Drug Administration (FDA) in 1987, representing the first HMGRi to be available in the United States for therapeutic use.

Structure–Activity Relationship

Mevastatin and lovastatin served as lead compounds in the development of additional HMGRIs. Initial research published by Merck Pharmaceuticals examined alterations of the lactone and bicyclic rings as well as the ethylene bridge between them. The results demonstrated that the activity of HMGRIs is sensitive to the stereochemistry of the lactone ring, the ability of the lactone ring to be hydrolyzed, and the length of bridge connecting the two ring systems. Additionally, it was found that the bicyclic

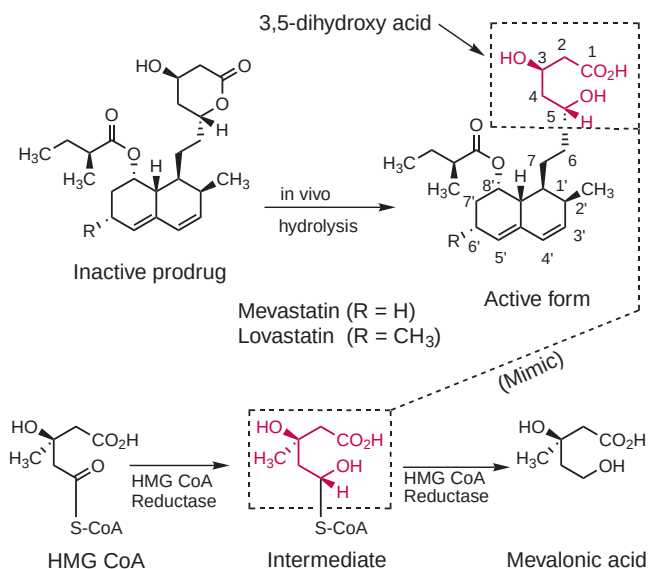


FIGURE 25.8 Mechanism of action of mevastatin and lovastatin. Hydrolysis of these prodrugs produces a 3,5-dihydroxy acid that mimics the tetrahedral intermediate produced by HMG-CoA reductase.

ring could be replaced with other lipophilic rings and that the size and shape of these other ring systems were important to the overall activity of the compounds (27).

Minor modifications of the bicyclic ring and side-chain ester of lovastatin produced simvastatin and pravastatin (Fig. 25.9). Pravastatin, a ring-opened dihydroxy acid with a 6'α-hydroxyl group, is much more hydrophilic than either lovastatin or simvastatin. Proposed advantages of this enhanced hydrophilicity are minimal penetration into the lipophilic membranes of peripheral cells, better selectivity for hepatic tissues, and a reduction in the incidence of side effects seen with lovastatin and simvastatin (28,29).

The replacement of the bicyclic ring with various substituted, aromatic ring systems led to the development of atorvastatin, fluvastatin, pitavastatin, and rosuvastatin (Fig. 25.9). The initial rationale centered on a desire to simplify the structures of mevastatin and lovastatin. The 2,4-dichlorophenyl analog (compound A) was one of the first compounds to demonstrate that this type of substitution was possible; however, compound A was considerably less potent than mevastatin. Subsequent research investigated a variety of aromatic substitutions and heterocyclic ring systems to optimize HMGRi activity. The substituted pyrrole (compound B) retained 30% of the activity of mevastatin (30) and was a key intermediate in the development of atorvastatin. The 4-fluorophenyl and isopropyl (or cyclopropane) substitutions found in compound B also are seen in the indole, quinoline, and pyrimidine ring systems of fluvastatin, pitavastatin, and rosuvastatin, respectively, and most likely represent the optimum substitutions at their respective positions. The design of all four of these compounds included the ring-opened dihydroxy acid functionality first seen in pravastatin.

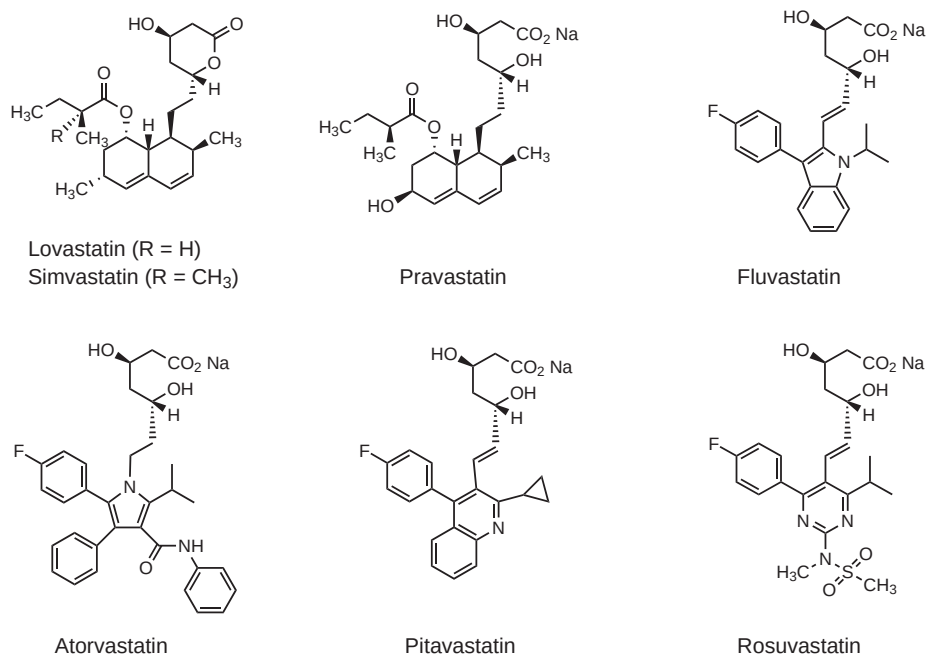
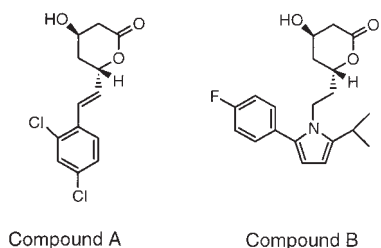


FIGURE 25.9 Commercially available HMG-CoA reductase inhibitors.



All HMGRIs can be chemically classified as 7-substituted-3,5-dihydroxyheptanoic acids, the general structure of which is shown in Table 25.3. Additionally, these compounds can be subclassified based on their lower ring. Compounds structurally related to the natural products mevastatin and lovastatin have structural features common to ring A, whereas those that are completely synthetic have structural features common to ring B (27,29–37).

Mechanism of Action

Inhibitors of HMG-CoA reductase lower plasma cholesterol levels by three related mechanisms: inhibition of cholesterol biosynthesis, enhancement of receptor-mediated LDL uptake, and reduction of VLDL precursors (17,23). As previously discussed, HMG-CoA reductase is the rate-limiting step in cholesterol biosynthesis. Inhibition of this enzyme causes an initial decrease in hepatic cholesterol. Compensatory mechanisms result in an enhanced expression of both HMG-CoA reductase and LDL receptors. The net result of all these effects is a slight to modest decrease in cholesterol synthesis, a significant increase in receptor-mediated LDL uptake, and an overall lowering of plasma LDL levels. Evidence to support the theory that enhanced LDL receptor

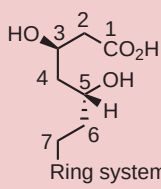
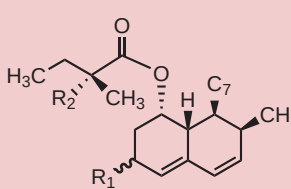
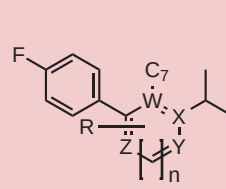
expression is the primary mechanism for lowering LDL levels comes from the fact that most statins do not lower LDL levels in patients who are unable to produce LDL receptors (i.e., homozygous familial hypercholesterolemia). The increased number of LDL receptors also may increase the direct removal of VLDL and IDL. Because these lipoproteins are precursors to LDL, this action may contribute to the overall lowering of plasma LDL cholesterol. Finally, all HMGRIs can produce a modest (8% to 12%) increase in HDL (17).

Atorvastatin, rosuvastatin, and simvastatin appear to have some effects beyond those seen with the other HMGRIs. These compounds have been shown to decrease plasma LDL levels in patients with homozygous familial hypercholesterolemia, an effect that is proposed to result from their ability to produce a more significant decrease in the hepatic production of LDL cholesterol. Additionally, atorvastatin and rosuvastatin can produce a significant lowering in plasma triglycerides. In the case of atorvastatin, this effect has been attributed to its ability to produce an enhanced removal of triglyceride-rich VLDL (17,38,39).

Physicochemical Properties

In their active forms, all HMGRIs contain a carboxylic acid. This functional group is required for inhibitory activity, has a pK_a in the range of 2.5 to 3.5, and will be primarily ionized at physiologic pH. Lovastatin and simvastatin are neutral, lactone prodrugs and should be classified as non-electrolytes. Pravastatin, fluvastatin, and atorvastatin can be classified as acidic drugs. The nitrogen atoms in the indole and pyrrole rings of fluvastatin and atorvastatin, respectively, are aromatic nitrogens that are not ionizable. This is because the lone pair electrons of these atoms are

TABLE 25.3 Structure–Activity Relationship of HMG-CoA Reductase Inhibitors

	 7-substituted-3,5-dihydroxyheptanoic acid	 Ring A	 Ring B
Common for all HGMIs	1. The 3,5-dihydroxycarboxylate is essential for inhibitory activity. Compounds containing a lactone are prodrugs requiring in vivo hydrolysis. 2. The absolute stereochemistry of the 3- and 5-hydroxyl groups must be the same as that found in mevastatin and lovastatin. 3. Altering the two-carbon distance between C5 and the ring system diminishes or fails to improve activity. 4. A double bond between C6 and C7 can either increase or decrease activity. The ethyl group provides optimal activity for compounds containing ring A and some heterocyclic rings (e.g., pyrrole ring of atorvastatin). The ethenyl group is optimal for compounds with other ring systems, including the indole, quinoline, and pyrimidine rings seen in fluvastatin, pitavastatin, and rosuvastatin, respectively.		
Ring A subclass	1. The decalin ring is essential for anchoring the compound to the enzyme active site. Replacement with a cyclohexane ring resulted in a 10,000-fold decrease in activity. 2. Stereochemistry of the ester side chain is not important for activity; however, conversion of this ester to an ether results in a decrease in activity. 3. Methyl substitution at the R₂ position increases activity (i.e., simvastatin is more potent than lovastatin). 4. β-Hydroxyl group substitution at the R₁ position enhances hydrophilicity and may provide some cellular specificity.		
Ring B subclass	1. Substituents W, X, Y, and Z can be either carbon or nitrogen; n is equal to either zero or one (i.e., five- or six-member heterocyclic). 2. The para-fluorophenyl cannot be coplanar with the central aromatic ring. (Structural restraints to cause coplanarity have resulted in a loss of activity). 3. R substitution with aryl groups, hydrocarbon chains, amides, or sulfonamides enhances lipophilicity and inhibitory activity.		

involved in maintaining the aromaticity of their respective rings and are not available to bind protons. Pitavastatin and rosuvastatin are amphoteric compounds; however, their quinoline and pyrimidine rings are weakly basic and will primarily be un-ionized at physiologic pH.

The calculated logP values for the HMGRIs are shown in Table 25.4. Although some variation exists among the values for lovastatin, pravastatin, and simvastatin, the general trends are the same regardless of what program was used to calculate the values. Atorvastatin, fluvastatin, pitavastatin, and the prodrugs lovastatin and simvastatin have a much higher lipid solubility than either pravastatin and rosuvastatin. Hydrolysis of the lactone ring for the two prodrugs produces a 3,5-dihydroxycarboxylate with significantly improves water solubility.

The HMG-CoA reductase enzyme is stereoselective. The 3*R*,5*R* stereochemistry seen in the active forms of mevastatin and lovastatin (Fig. 25.8) is required for inhibitory activity and is present in all other HMGRIs. Stereochemistry of the substituents on the bicyclic rings of lovastatin, simvastatin, and pravastatin is less crucial to activity, as indicated in the summary of the structure–activity relationships (SARs).

Metabolism

As previously mentioned, lovastatin and simvastatin are inactive prodrugs that must undergo in vivo hydrolysis

to produce their effects (Fig. 25.8). The active forms of these two compounds as well as most HMGRIs undergo extensive first-pass metabolism (15,20,36,37). The CYP3A4 isozyme is responsible for the oxidative metabolism of atorvastatin, lovastatin, and simvastatin. In the case of atorvastatin, the *ortho*- and *para*-hydroxylated metabolites are equiactive with the parent compound and contribute significantly to the overall activity of the drug (Table 25.4). Rosuvastatin is metabolized to a limited extent by CYP2C9 to form an *N*-desmethyl metabolite that can contribute to activity, but is seven-fold less potent. In contrast, the activity of lovastatin and simvastatin resides primarily in the initial hydrolysis product (i.e., further oxidation decreases activity). Fluvastatin is metabolized by the CYP2C9 and CYP3A4 isozymes to active hydroxylated metabolites; however, these metabolites do not circulate systemically and do not contribute to the overall activity. Pravastatin also undergoes oxidative metabolism, but the resulting compounds retain only minimal activity and are not significant. Neither pitavastatin, pravastatin, nor rosuvastatin is metabolized by CYP3A4; therefore, these drugs are potentially advantageous for patients who must take concurrent medication that alters the activity of this isozyme. Pitavastatin is primarily metabolized by CYP2C9 and undergoes lactonization to an inactive compound.

Pharmacokinetic Parameters

The pharmacokinetic parameters and dosing information for HMGRI are summarized in Tables 25.4 and 25.5, respectively (2,6,17,21,22,38,40,41). With a few exceptions, all of these compounds have similar onsets of action, durations of action, dosing intervals, and plasma protein binding. Despite the ability to attain a peak plasma concentration in 1 to 4 hours, HMGRI require approximately 2 weeks to demonstrate an initial lowering of plasma cholesterol. Peak reductions of plasma cholesterol occur after 4 to 6 weeks of therapy for most compounds. Studies with atorvastatin, however, indicate that it may only need 2 weeks to produce its peak reduction. Atorvastatin and rosuvastatin also are unique in that they have much longer durations of action than the other compounds. With the exception of pravastatin, which is one of the more hydrophilic compounds in this class, most HMGRI bind extensively to plasma proteins.

Because of first-pass metabolism, the oral bioavailability of this class of drugs generally is low and does not reflect the actual absorption of the individual drugs. For example, 60% to 80% of a dose of simvastatin is orally absorbed, but only 5% is actually available to produce an effect. The same is true with fluvastatin, pravastatin, and lovastatin, which have oral absorptions of 90%, 34%, and 35%, respectively, but much lower bioavailabilities. Pitavastatin has the highest oral bioavailability in this class of compounds (Table 25.4). With the exception of lovastatin, the concurrent administration of food does not affect the overall therapeutic effects of HMGRI. Lovastatin should always be administered with food to maximize oral bioavailability. Failure to do this results in a 33% decrease in plasma concentrations. In general, HMGRI should be

administered in the evening or at bedtime to counteract the peak cholesterol synthesis, which occurs in the early morning hours. Exceptions to this are atorvastatin and rosuvastatin, which because of their long half-lives are equally effective regardless of when they are administered.

The primary route of elimination of these compounds is through the feces. Because of extensive hepatic transformation and the ability to elevate hepatic enzymes, HMGRI are contraindicated in patients with active hepatic disease or unexplained persistent elevations in serum aminotransferase concentrations. Dosage reductions in patients with renal dysfunction depend on the individual agent. Atorvastatin, which has minimal renal excretion, requires no dosage reduction and may be the best agent for patients with renal disorders. Fluvastatin, rosuvastatin, and simvastatin require dosage reductions only in cases of severe renal impairment and are better choices than lovastatin, pitavastatin, and pravastatin, which require dosage reductions in mild or moderate impairment.

Therapeutic Applications

All HMGRI are approved for the treatment of primary hypercholesterolemia and familial combined hyperlipidemia, or mixed dyslipidemia (Fredrickson type IIa and IIb) (Table 25.2) in patients who have not responded to diet, exercise, and other nonpharmacologic methods (Table 25.6) (17,23). They may be used either alone or in combination with bile acid sequestrants, ezetimibe, or niacin. As previously mentioned, they should be administered at least 1 hour before or 4 to 6 hours after bile acid sequestrants when this combination is desired. Atorvastatin, fluvastatin, lovastatin, pravastatin, and simvastatin have been specifically indicated to reduce the mortality of CHD and stroke. By

TABLE 25.4 Pharmacokinetic Parameters of HMG-CoA Reductase Inhibitors

Drug	Calculated LogP ^a	Oral Bioavailability (%)	Active Metabolite(s)	Protein Binding (%)	Time to Peak Concentration (h)	Elimination Half-Life (h)	Major Route(s) of Elimination
Atorvastatin	4.13	12–14	<i>ortho</i> - and <i>para</i> -hydroxylated	98	1–2	14–19	Biliary/fecal (>90%) Renal (<2%)
Fluvastatin	3.62	20–30	None	98	0.5–1.0	1	Biliary/fecal (95%) Renal (5%)
Lovastatin	4.07 (4.04) ^b	5	3,5-Dihydroxy acid	>95	2	3–4	Fecal (83%) Renal (10%)
Pravastatin	1.44 (0.5) ^b	17	None	43–55	1.0–1.5	2–3	Fecal (70%) Renal (20%)
Pitavastatin	3.45	51	None	>99	1	12	Fecal (79%) Renal (15%)
Rosuvastatin	0.42	20	<i>N</i> -Desmethyl	88	3–5	19–20	Fecal (90%) Renal (10%)
Simvastatin	4.42 (4.2) ^b	5	3,5-Dihydroxy acid	95	4	3	Fecal (60%) Renal (13%)

^aA commercial program was used for calculated values (47).

^bCalculated using the CLOG program (40).

TABLE 25.5 Dosing Information for HMG-CoA Reductase Inhibitors

Generic Name	Brand Name(s)	Dosing Range	Maximum Daily Dose	Dose Reduction with Renal Dysfunction	Tablet Strengths (mg)
Atorvastatin	Lipitor	10–80 mg once daily	80 mg	No	10, 20, 40, 80
Fluvastatin	Lescol Lescol XR	20–80 mg once daily or b.i.d.	80 mg	Caution in severe impairment	20, 40, 80 (XR)
Lovastatin	Mevacor Altoprev (XR)	10–80 mg once daily or b.i.d.	80 mg (60 mg if XR) (20 mg with fibrate)	Yes	10, 20, 40, 20 (XR), 40 (XR), 60 (XR)
Pitavastatin	Livalo	1–4 mg once daily	40 mg	Yes	1, 2, 4
Pravastatin	Pravachol	10–80 mg once daily	80 mg	Yes	10, 20, 40, 80
Rosuvastatin	Crestor	5–40 mg once daily	40 mg (10 mg with fibrate)	Only with severe impairment	5, 10, 20, 40
Simvastatin	Zocor	5–40 mg once daily	80 mg (10 mg with fibrate)	Only with severe impairment	5, 10, 20, 40, 80

b.i.d., twice a day; XR, extended release.

reducing plasma LDL levels, these compounds slow the progression of atherosclerosis and reduce the risk of myocardial infarction and other ramifications of vascular occlusion. Atorvastatin, rosuvastatin, and simvastatin have been shown to be effective in homozygous familial hyperlipidemia and are indicated for this use. Additionally, atorvastatin, pravastatin, rosuvastatin, and simvastatin are indicated for primary dysbetalipoproteinemia (Fredrickson type III) (Table 25.2). Finally, with the exception of lovastatin, all HMGRIs are indicated for the treatment of hypertriglyceridemia.

Inhibitors of HMG-CoA reductase are contraindicated in pregnancy. Fetal development requires cholesterol as a precursor for the synthesis of steroids and cell membranes; thus, inhibition of its synthesis may cause fetal harm. Additionally, HMGRIs are excreted in breast milk and should not be used by nursing mothers.

Adverse Effects

The most prevalent or significant side effects of HMGRIs are listed below (6,17,23). In general, this class of drugs is well tolerated. Gastrointestinal disturbances are the most common complaint; however, these and other adverse reactions tend to be mild and transient. Elevations in hepatic transaminase levels can occur with all HMGRIs. These increases usually occur shortly after the initiation of therapy and resolve after the discontinuation of medication. In a small percentage of patients, these levels can increase to more than threefold the upper limit of normal. Therefore, liver function tests should be done at the initiation of therapy, at 6 and 12 weeks after the initiation of therapy, and at periodic intervals (e.g., every 6 months) thereafter. Similar testing should be done with dosage increases. Approximately 5% to 10% of patients will experience mild increases in creatine phosphokinase levels; however, less than 1% will develop symptoms

of myalgia and myopathy (e.g., fever, muscle aches or cramps, and unusual tiredness or weakness). Tests for creatine phosphokinase levels should be performed in patients reporting muscle complaints. Rhabdomyolysis (i.e., massive muscle necrosis with secondary acute renal failure) has occurred, but this is rare. The risk of this very serious adverse effect increases when an HMGRIs is taken with certain other medications, such as cyclosporine, erythromycin, niacin, or fibrates (Table 25.7). Specific dosage reductions have been suggested for the combination use of fibrates with lovastatin, rosuvastatin, or simvastatin (17).

EZETIMIBE, A CHOLESTEROL ABSORPTION INHIBITOR

Historical Overview

The discovery of ezetimibe and its mechanism of action began with a desire to develop novel ACAT inhibitors, a potential target for hypercholesterolemia described earlier in this chapter (43). Compound C was one of the initial azetidinones tested for the ability to inhibit ACAT and to

COMBINATION PRODUCTS THAT INCLUDE AN HMGR

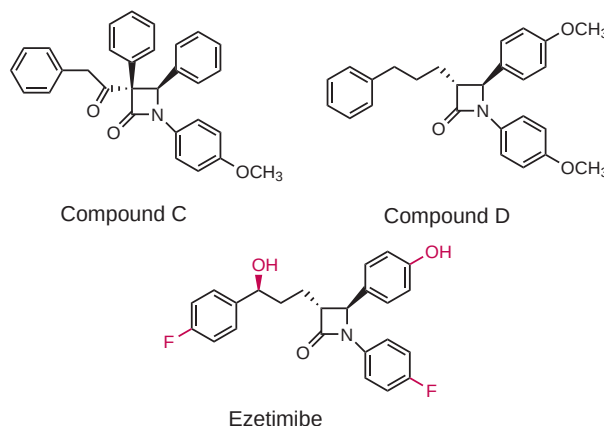
- HMGRIs and antithrombotic
 - Pravastatin/aspirin (Pravigard Pac)
- HMGRIs and calcium channel blocker
 - Atorvastatin/amlodipine (Caduet)
- HMGRIs and additional antihypercholesterolemic agent
 - Lovastatin/niacin (Advicor)
 - Simvastatin/niacin (Simcor)
 - Simvastatin/ezetimibe (Vytorin)

TABLE 25.6 Approved Therapeutics Conditions for HMG-CoA Reductase Inhibitors

Therapeutics Condition	Atorvastatin	Fluvastatin	Lovastatin	Pitavastatin	Pravastatin	Rosuvastatin	Simvastatin
Primary hypercholesterolemia	✓	✓	✓	✓	✓	✓	✓
Primary dysbetalipoproteinemia	✓				✓	✓	✓
Mixed dyslipidemia	✓	✓	✓	✓	✓	✓	✓
Hypertriglyceridemia	✓	✓		✓	✓	✓	✓
Homozygous familial hyperlipidemia	✓					✓	✓
Primary prevention of cardiovascular events	✓		✓		✓	✓	✓
Secondary prevention of cardiovascular events	✓	✓	✓		✓		✓

lower plasma cholesterol. Interestingly, this compound's ability to decrease plasma cholesterol exceeded its ability to inhibit ACAT. Further SAR studies resulted in the development of compound D and confirmed that the cholesterol-lowering activity of this class of compounds was independent of its ability to inhibit ACAT. Using compound D as a lead, as well as in vivo data suggesting that metabolic transformations produced the active compound ultimately responsible for the cholesterol-lowering effect, structural modifications were made that, eventually, led to the development of ezetimibe. The most important changes involved the introduction of hydroxyl groups to help localize the compound in

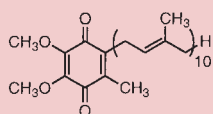
the intestine and the introduction of *p*-fluoro groups to block undesirable metabolism.



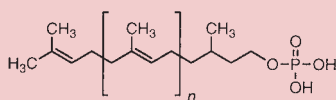
POTENTIAL NON-LIPID-LOWERING USES OF HMGRIS

Cellular metabolites derived from mevalonic acid are required for cell proliferation. Cholesterol is an essential component of cell membranes, farnesyl pyrophosphate is required to covalently bind to intracellular proteins and modify their function, ubiquinone is required for mitochondrial electron transport, and dolichol phosphates are required for glycoprotein synthesis.

Farnesyl pyrophosphate (Fig. 25.2) is an intermediate in the biosynthesis of cholesterol, ubiquinone, and dolichol phosphates. Based on their site of action, HMGRIs will decrease the availability of all four of these compounds and, thus, decrease cell proliferation. Potential applications of this antiproliferative effect include the prevention of restenosis following angioplasty, prevention of glomerular injury in renal disease, treatment of malignant disease, and prevention of organ transplantation rejection (42).



Ubiquinone

Dolichol phosphate
(*n* = 15-19)

Mechanism of Action

Ezetimibe lowers plasma cholesterol levels by inhibiting the absorption of cholesterol at the brush border of the small intestine (17,23). Specifically, ezetimibe has been proposed to bind to a specific transport protein located in the wall of the small intestine, resulting in a reduction of cholesterol transport and absorption (44). Ezetimibe appears to be selective in its actions in that it does not interfere with the absorption of triglycerides, lipid-soluble vitamins, or other nutrients. The decreased absorption

ADVERSE EFFECTS OF HMGRIS

Constipation, flatulence, dyspepsia, abdominal pain, diarrhea, nausea, vomiting, headache, rhinitis, sinusitis, elevated hepatic enzymes, arthralgia, myalgia, myopathy, muscle cramps, rhabdomyolysis, and chest pain

of cholesterol eventually leads to enhanced receptor-mediated LDL uptake similar to that seen with bile acid sequestrants and HMGRI. When used as monotherapy, the decreased absorption of cholesterol causes a compensatory increase in cholesterol biosynthesis. This is similar to that described for bile acid sequestrants and is insufficient to override the overall LDL-lowering effects of ezetimibe.

Physicochemical Properties

Ezetimibe is a crystalline powder that is practically insoluble in water but is freely soluble in ethanol and other organic solvents. Its calculated logP value is 3.50 (40). The phenol present in ezetimibe allows this compound to be classified as an acidic compound; however, the phenol has a pK_a of 9.72 and is predominantly un-ionized at physiologic pH.

Metabolism

After oral administration, ezetimibe is rapidly and extensively metabolized in the intestinal wall and the liver to its active metabolite, a corresponding phenol glucuronide (Fig. 25.10). This glucuronide is reexcreted in the bile back to its active site. A small amount (<5%) of ezetimibe undergoes oxidation to convert the benzylic hydroxyl group to a ketone; however, ezetimibe does not appear to exert any significant effect on the activity of CYP450 enzymes (17,22,44).

Pharmacokinetic Parameters

Ezetimibe is administered orally; however, its absolute bioavailability cannot be determined because of its aqueous insolubility and the lack of an injectable formulation. Based on area under the curve values, the oral absorption ranges from 35% to 60%. Mean peak concentrations of the active glucuronidated metabolite are reached within 1 to 2 hours. Both ezetimibe and its glucuronide conjugate are extensively bound (>90%) to plasma proteins. The relative plasma concentrations of ezetimibe and its glucuronide conjugate range from 10% to 20% and from 80% to 90%, respectively. Both compounds have a long half-life of approximately 22 hours. The coadministration of food with ezetimibe has no effect on the extent of absorption.

The normal dose of ezetimibe is 10 mg once daily. Dosage reduction for patients with renal impairment, intermittent hemodialysis, or mild hepatic impairment is not necessary. Because of insufficient data, the use of ezetimibe is not recommended in patients with moderate to severe hepatic impairment (17,22,23).

Therapeutic Applications

Ezetimibe is indicated as monotherapy or in combination with an HMGRI for the reduction of elevated total cholesterol, LDL cholesterol, and apoB in patients with

ADVERSE EFFECTS OF EZETIMIBE

Abdominal pain, diarrhea, arthralgia, back pain, cough, pharyngitis, sinusitis, fatigue, and viral infection

primary (heterozygous familial and nonfamilial) hypercholesterolemia, homozygous familial sitosterolemia, or homozygous familial hypercholesterolemia. When used as monotherapy, ezetimibe reduces LDL cholesterol by approximately 18%. When used in combination therapy with an HMGRI, LDL levels are reduced by 25% to 65% depending on the dose of the HMGRI inhibitor. Ezetimibe is also indicated for combination use with fenofibrate to treat hypercholesterolemia in patients with mixed hyperlipoproteinemia. All indications are for patients who have not responded to diet, exercise, and other nonpharmacologic methods.

Adverse Effects

Ezetimibe generally is well tolerated. The most common adverse effects are listed above. Whenever ezetimibe is used in combination with an HMGRI, the incidence of myopathy or rhabdomyolysis does not increase above that seen with HMGRI monotherapy (17,23). Drug interactions for ezetimibe are listed in Table 25.8.

FIBRATES

Historical Overview and Development

The use of this class of drugs to treat hyperlipoproteinemias can be traced back to 1962 and thus predates the use of bile acid sequestrants and HMGRI. A random screening test on a series of aryloxyisobutyric acids demonstrated that these compounds could lower both plasma cholesterol and total lipid levels (45). The compound that produced the best balance between activity and toxicity was ethyl *p*-chlorophenoxyisobutyrate (Fig. 25.11). Later renamed clofibrate, this compound was subsequently shown to be a prodrug for *p*-chlorophenoxyisobutyric acid (clofibric acid). It was approved for therapeutic use in 1967, and for a time, it was a very popular and widely prescribed drug. Results from a 1978 World Health Organization trial changed the acceptance of clofibrate and dramatically decreased its use. These trials indicated that despite a 9% lowering of cholesterol, patients taking clofibrate showed no reduction of cardiovascular events and actually had an increase in overall mortality (6). Although clofibrate is no longer available in the United States, it has served as the prototype for the design of safer and more effective fibrates. Structural modifications, focused primarily on ring substitutions and the addition of spacer groups, have produced a number of active compounds (Fig. 25.11). Gemfibrozil and fenofibrate became available for therapy in 1981 and 1998, respectively. Fenofibrate was actually approved in 1993; however,

DRUG INTERACTIONS

Drug interactions for HMGRI are listed in Table 25.7.

TABLE 25.7 Drug Interactions for HMG-CoA Reductase Inhibitors (HMGRIs)

Drug	HMGRIs	Result of Interaction
Amiodarone	Lovastatin, simvastatin	Increased risk of myopathy
Antacids	Atorvastatin, rosuvastatin	Decreased levels of atorvastatin, rosuvastatin; no change in plasma LDL reduction; administer rosuvastatin at least 2 h after antacid
Azole antifungal agents	All	Increased risk of severe myopathy or rhabdomyolysis; increased plasma levels of atorvastatin, lovastatin, and simvastatin because of inhibition of CYP3A4; additive decreases in concentrations or activity of endogenous steroid hormones
Bile acid sequestrants	All	Decreased bioavailability of HMGRIs if administration is not adequately spaced
Cimetidine	Fluvastatin	Increase in plasma fluvastatin levels
Cyclosporine	All	Increased risk of severe myopathy or rhabdomyolysis
Danazol	Lovastatin	Increased risk of severe myopathy or rhabdomyolysis
Digoxin	Atorvastatin, fluvastatin, simvastatin	Slight elevation in plasma concentrations of digoxin
Diltiazem	Atorvastatin, lovastatin, simvastatin	Increased risk of severe myopathy
Erythromycin, clarithromycin	All	Increased risk of severe myopathy or rhabdomyolysis; increased plasma levels of atorvastatin, lovastatin, and simvastatin because of inhibition of CYP3A4
Ethanol	Fluvastatin, lovastatin	Increased risk of hepatotoxicity
Fibrates	All	Increased risk of severe myopathy or rhabdomyolysis
Grapefruit juice (>1 quart/d)	Atorvastatin, lovastatin, simvastatin	Elevated plasma levels of the HMGRIs and increased risk of myopathy
HIV protease inhibitors	Atorvastatin, lovastatin, pitavastatin, simvastatin	Elevated plasma levels of the HMGRIs and increased risk of myopathy
Isradipine	Lovastatin	Increased clearance of lovastatin and its metabolites
Itraconazole	Pitavastatin	Decreased plasma levels of pitavastatin
Niacin	All	Increased risk of severe myopathy or rhabdomyolysis
Omeprazole	Fluvastatin	Increase in plasma fluvastatin levels
Oral contraceptives	Atorvastatin, rosuvastatin	Increased plasma concentrations of norethindrone and ethinyl estradiol
Phenytoin	Fluvastatin	Increased plasma concentrations of both compounds because of CYP2C9 interaction
Ranitidine	Fluvastatin	Increase in plasma fluvastatin levels
Rifampin	Fluvastatin, pitavastatin	Increased plasma clearance of fluvastatin
Ritonavir, saquinavir	Pravastatin	Decreased plasma levels of pravastatin
Spironolactone	All	Additive decreases in concentrations or activity of endogenous steroid hormones
St. John's wort	Lovastatin, simvastatin	Decreased HMGRIs plasma levels
Warfarin	Fluvastatin, lovastatin, rosuvastatin, simvastatin	Anticoagulant effect of warfarin may be increased

its marketing was voluntarily delayed until a more bioavailable, micronized formulation of the drug was available (46). Both of these compounds are safer and more effective than clofibrate in lowering plasma triglyceride levels and increasing plasma HDL levels. Additional compounds, such as ciprofibrate and bezafibrate, are not currently available in the United States but have been used in other countries.

Mechanism of Action

Overall, fibrates decrease plasma triglyceride levels much more dramatically than they decrease plasma cholesterol levels. They significantly decrease VLDL levels, cause a

moderate increase in HDL levels, and have variable effects on LDL concentrations. As an example of this latter point, gemfibrozil will raise LDL levels in patients with hypertriglyceridemia but lower LDL levels in patients with normal triglyceride levels. The exact mechanisms for these actions have not been fully elucidated; however, studies have shown that this class of compounds can produce a variety of beneficial effects on lipoprotein metabolism. Many of these effects have been proposed to be mediated through the activation of peroxisome proliferator-activated receptors (PPARs) and an alteration of gene expression. Specifically, fibrates bind to PPAR α (6,17,29,46).

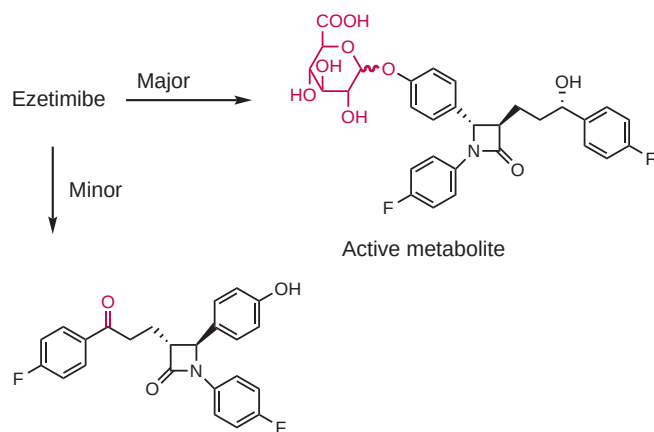


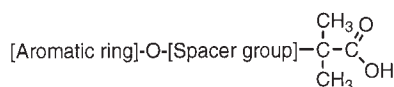
FIGURE 25.10 Metabolism of ezetimibe to its active and inactive metabolites.

Decreases in plasma VLDL primarily result from the ability of these compounds to stimulate the activity of lipoprotein lipase, the enzyme responsible for removing triglycerides from plasma VLDL (Fig. 25.5). Additionally, fibrates can lower VLDL levels through PPAR α -mediated stimulation of fatty acid oxidation, inhibition of triglyceride synthesis, and reduced expression of apoC-III. This latter effect enhances the action of lipoprotein lipase because apoC-III normally serves as an inhibitor of this enzyme. Favorable effects on HDL levels appear to be related to increased transcription of apoA-I and apoA-II as well as a decreased activity of cholesteryl ester transfer protein.

All fibrates accelerate the turnover and removal of cholesterol from the liver. This increases the biliary secretion of cholesterol, enhances its fecal excretion, and may cause cholelithiasis (i.e., gallstone formation).

Structure–Activity Relationships

Fibrates can be chemically classified as analogs of phenoxyisobutyric acid. Literature references to the SARs for this class of drugs are sparse; however, all compounds are analogs of the following general structure.



The isobutyric acid group is essential for activity. Fenofibrate, which contains an ester, is a prodrug and requires *in vivo* hydrolysis. Substitution at the para position of the aromatic ring with a chloro group or a chlorine-containing cyclopropyl ring produces compounds with significantly longer half-lives. Although most compounds contain a phenoxyisobutyric acid, the addition of an *n*-propyl spacer, as seen in gemfibrozil, results in an active drug.

Physicochemical Properties

Similar to HMGRIs, the active forms of all fibrates contain a carboxylic acid. The pK_a of this functional group

TABLE 25.8 Drug Interactions for Ezetimibe

Drug	Result of Interaction
Antacid	Aluminum and magnesium-containing antacids decrease the maximum plasma concentration of ezetimibe
Bile acid sequestrants	Decreased bioavailability of ezetimibe if administration is not adequately spaced
Cyclosporine	Increased ezetimibe concentration
Fibrates	Increased ezetimibe concentration and possible increased risk of cholelithiasis. With the exception of fenofibrate, concomitant use is not recommended.

on clofibric acid is reported to be 3.5 (47), and thus, it will be primarily ionized at physiologic pH. Although not reported, the pK_a and ionization values of gemfibrozil and fenofibric acid can reasonably be assumed to be similar. Both clofibrate and fenofibrate are neutral, ester prodrugs and should be classified as nonelectrolytes. Gemfibrozil can be classified as an acidic drug. The calculated logP values for fenofibrate and gemfibrozil are shown in Table 25.9 (47). Both compounds are highly lipid soluble despite the fact that gemfibrozil contains a water-soluble carboxylic acid. This can be partially explained by examining the π values for the substituents on fenofibrate and gemfibrozil (48). The 2,5-dimethyl ring in gemfibrozil is predicted to be much more hydrophobic than the 4-chloro ring of fenofibrate. Additionally, the propyl bridge seen in gemfibrozil, but not in fenofibrate, significantly adds to its hydrophobicity. The isopropyl ester as well as the additional aromatic ring account for the enhanced lipid solubility of fenofibrate. All currently available fibrates are achiral molecules and not subject to stereochemical concerns.

Metabolism

The prodrug fenofibrate undergoes rapid hydrolysis to produce fenofibric acid. This active metabolite can then be further metabolized by oxidative or conjugative pathways. Gemfibrozil is slightly different in that it does not require initial bioactivation; however, similar to fenofibric acid, it can be oxidized or conjugated. Oxidation of the aromatic methyl groups produces inactive hydroxymethyl and carboxylic acid analogs. As a drug class, fibrates and their oxidized analogs are primarily excreted as glucuronide conjugates in the urine. Oxidation requires the CYP3A4 isozyme; however, because of the ability of these compounds to be conjugated and eliminated either with or without oxidation, drug interactions with other compounds affecting the CYP3A4 system are less important here than with other drug classes.

Pharmacokinetic Parameters

The pharmacokinetic parameters and dosing information for the fibrates are summarized in Tables 25.9 and 25.10, respectively (6,17,22,23,47). The prodrug, fenofibrate,

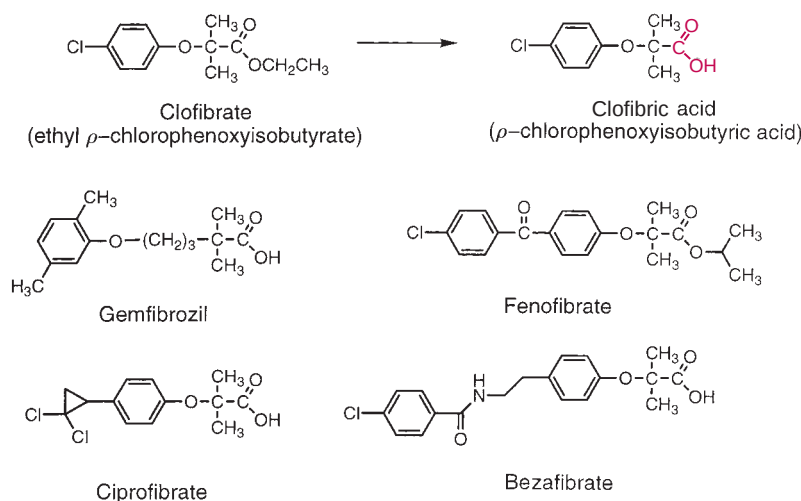


FIGURE 25.11 Bioactivation of clofibrate and chemical structures of other fibrates.

requires a longer time to reach peak concentrations compared with gemfibrozil. Because of differences in aromatic substitution, fenofibrate also has a much longer half-life than gemfibrozil. As previously mentioned, the 2,5-dimethyl substitution in gemfibrozil is much more susceptible to oxidative metabolism than the para-chloro group present in fenofibrate. Similar to HMGRIs, changes in lipid levels are not seen immediately, and up to 2 months may be required to reach maximal clinical effects and to determine the overall clinical efficacy.

Fibrates have excellent bioavailability and are extensively bound to plasma proteins. Because food can significantly enhance their oral absorption, these compounds should be taken either with or just before meals. Fenofibrate was available in Europe and elsewhere as standard tablet and capsule formulations for many years before its approval and marketing in the United States, where it was introduced only after the development of a micronized formulation that allowed better oral absorption, a lower daily dose, and once-daily administration. A variety of formulations and doses of fenofibrate have been developed. Specific dosing ranges, maximum daily doses, and available strengths are noted in Table 25.10.

Renal elimination is the primary route through which these compounds are excreted from the body. Patients with mild renal dysfunction often can be managed with minor dosage adjustments, whereas those with severe impairment or renal failure may have to discontinue its use.

Therapeutic Applications

Fibrates are approved to treat hypertriglyceridemia and familial combined hyperlipidemia (Fredrickson types IIa, IIb, IV, and V) (Table 25.2) in patients who are at risk of pancreatitis and have not responded to dietary adjustments or in patients who are at risk of CHD and have not responded to weight loss, dietary adjustments, and other pharmacologic treatment. They can be used either alone or in combination with niacin, bile acid sequestrants, or

HMGRIs. If used with bile acid sequestrants, fibrates must be taken either 1 hour before or 4 to 6 hours after the sequestrant. As discussed previously and reemphasized below, caution should be used if fibrates are combined with HMGRIs. Fibrates are not effective in the treatment of hypertriglyceridemia associated solely to elevated chylomicron levels (Fredrickson type I).

Adverse Effects

The most prevalent or significant side effects caused by the fibrates are listed below (6,17,23). Despite the potential to cause serious side effects, fibrates usually are well tolerated. Gastrointestinal complaints are the most common but do not usually cause discontinuation of therapy. In general, gemfibrozil and fenofibrate appear to be less problematic than the original compound, clofibrate. In fact, many of the concerns regarding fibrate therapy are based on the effects of clofibrate and the results of a 1978 clinical trial in which patients taking clofibrate had a significantly higher morbidity and mortality from causes other than CHD. These causes included malignancy, gallbladder disease, pancreatitis, and postcholecystectomy complications. Studies with gemfibrozil and fenofibrate have not shown similar increases; however, because all fibrates have similar pharmacologic actions, cautions and contraindications generally are applied to the entire drug class. As an example, even though gemfibrozil and fenofibrate have not demonstrated a significant increase in gallbladder disease, as seen with clofibrate, all three of these compounds are contraindicated in patients with preexisting gallbladder disease or cholelithiasis. Similar to HMGRIs, fibrates can cause myopathy, myositis, and rhabdomyolysis. Although rare, the risk of these serious effects increases when these two classes of agents are used together. Fibrates also cause increases in plasma aspartate transaminase (AST), alanine transaminase (ALT), and creatine phosphokinase levels.

Drug interactions for fibrates are listed in Table 25.11.

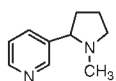
TABLE 25.9 Pharmacokinetic Parameters of Fibrates

Drug	Calculated LogP	Oral Bioavailability (%)	Active Metabolite	Protein Binding (%)	Time to Peak Concentration (h)	Elimination Half-Life (h)	Major Route(s) of Elimination
Fenofibrate	5.24	60–90	Fenofibric acid	99	4–8	20–22	Renal (60%–90%) Fecal (5%–25%)
Gemfibrozil	3.9	>90	None	99	1–2	1.5	Renal (70%) Fecal (6%)

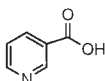
NICOTINIC ACID

Historical Overview

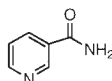
The history of nicotinic acid (niacin) began in 1867, when it was first synthesized by oxidation of nicotine. The name niacin was derived later from the words *nicotinic acid* and *vitamin* in an effort to avoid confusing nicotinic acid and nicotinamide with nicotine. Although the terms “niacin” and “nicotinic acid” are today used interchangeably, only the more chemically descriptive term, “nicotinic acid,” will be used in the following discussions.



Nicotine



Nicotinic acid



Nicotinamide

Discovery of the biochemical and pharmacologic actions of nicotinic acid began in the early 1900s, when brewer's yeast was demonstrated to prevent pellagra in humans. The subsequent isolation of nicotinic acid from brewer's yeast established its role as an essential dietary requirement. In the 1930s, its amide metabolite, nicotinamide, was isolated from liver extracts and found to be a required structural feature of nicotinamide adenine dinucleotide phosphate (NADP⁺), a cofactor involved in electron transport and intermediary metabolism (49). In 1955, Altschul et al. (50) observed that high doses of nicotinic acid lowered cholesterol levels in humans, an activity unrelated to its properties as a vitamin. Subsequent studies have shown that nicotinic

acid also lowers serum triglyceride levels and is effective against a variety of hyperlipoproteinemias. None of these antihyperlipidemic effects are seen with nicotinamide.

Mechanism of Action

Nicotinic acid exerts a variety of effects on lipoprotein metabolism (6,18,51). One of its most important actions is the inhibition of lipolysis in adipose tissue. This initial inhibition, like those of previously discussed antihyperlipidemic agents, produces a sequence of events that ultimately result in the lowering of plasma triglycerides and cholesterol. Impaired lipolysis decreases the mobilization of free fatty acids, thus reducing their plasma levels and their delivery to the liver. In turn, this decreases hepatic triglyceride synthesis and results in a decreased production of VLDL. Enhanced clearance of VLDL through stimulation of lipoprotein lipase also has been proposed to contribute to the reduction of plasma VLDL levels. Because LDL is derived from VLDL (Fig. 25.5), the decreased production of VLDL ultimately leads to a decrease in LDL levels. The sequential nature of this process has been clinically demonstrated. The reduction in triglyceride levels occurs within several hours after initiation of nicotinic acid therapy, whereas the reduction in cholesterol does not occur until after several days of therapy. Nicotinic acid also increases HDL levels because of a reduction in the clearance of apoA-I, an essential component of HDL. Unlike bile acid sequestrants and HMGRI, nicotinic acid does not have any effects on cholesterol catabolism or biosynthesis.

TABLE 25.10 Dosage Information for Fibrates

Generic Name	Brand Name	Dosing Range	Maximum Daily Dose	Dose Reduction with Renal Dysfunction	Tablet/Capsule Strengths (mg)
Fenofibrate	TriCor	48–145 mg once daily	145 mg	Only with severe impairment	48, 145
	Lofibra	54–160 mg once daily	160 mg		54, 160
	Lipofen	50–150 mg once daily	150 mg		50, 150
	Triglide	50–160 mg once daily	160 mg		50, 160
	Antara (micronized)	43–130 mg once daily	130 mg		67, 134, 200
	Lofibra (micronized)	67–200 mg once daily	200 mg		43, 130
Gemfibrozil	Lopid	600 mg b.i.d.	1200 mg	Yes	600

b.i.d., twice a day.

UNLABELED USES

Polymetabolic syndrome X (fenofibrate)

ADVERSE EFFECTS

Abdominal pain, dyspepsia, nausea, vomiting, diarrhea, constipation, cholestasis, jaundice, cholelithiasis, pancreatitis, headache, dizziness, drowsiness, blurred vision, mental depression, impotence, decreased libido, myopathy, myositis, rhabdomyolysis, anemia, leukopenia, eosinophilia, pruritus, and rash

Physicochemical Properties

Nicotinic acid (niacin) is a stable, nonhygroscopic, white, crystalline powder. Its carboxylic acid has a pK_a of 4.76 and, thus, is predominantly ionized at physiologic pH. The pyridine nitrogen is a very weak base ($pK_a = 2.0$) and, therefore, primarily exists in the un-ionized form. Nicotinic acid is freely soluble in alkaline solutions and has a measured logP of -0.20 at pH 6.0 (47).

Metabolism

Nicotinic acid is a B-complex vitamin that is converted to nicotinamide, nicotinamide adenine dinucleotide (NAD^+), and $NADP^+$. The latter two compounds are coenzymes and are required for oxidation/reduction reactions in a variety of biochemical pathways. Additionally, nicotinic acid is metabolized to a number of inactive compounds, including nicotinuric acid and *N*-methylated derivatives. Normal biochemical regulation and feedback prevent large doses of nicotinic acid from producing excess quantities of NAD^+ and $NADP^+$. Thus, small doses of nicotinic acid, such as those used for dietary supplementation, will be primarily excreted as metabolites, whereas large doses, such as those used for the treatment of hyperlipoproteinemia, will be primarily excreted unchanged by the kidney (17).

Pharmacokinetic Parameters

Nicotinic acid is readily absorbed. Peripheral vasodilation is seen within 20 minutes, and peak plasma concentrations occur within 45 minutes. The half-life of the compound is approximately 1 hour, thus necessitating frequent dosing or an extended-release formulation. Extended-release

tablets produce peripheral vasodilation within 1 hour, reach peak plasma concentrations within 4 to 5 hours, and have a duration of 8 to 10 hours.

Dosing of nicotinic acid should be titrated to minimize adverse effects. An initial dose of 50 to 100 mg three times a day often is used with immediate-release tablets. The dose then is gradually increased by 50 to 100 mg every 3 to 14 days, up to a maximum of 6 g/day, as tolerated. Therapeutic monitoring to assess efficacy and prevent toxicity is essential until a stable and effective dose is reached. Similar dosing escalations are available for extended-release products, with doses normally starting at 500 mg once daily at bedtime. (6,17,23).

Therapeutic Applications

Nicotinic acid is approved for the treatment of hypercholesterolemia, hypertriglyceridemia, and familial combined hyperlipidemia (Fredrickson types IIa, IIb, IV, and V) (Table 25.2) in patients who have not responded to diet, exercise, and other nonpharmacologic methods. It also is approved for nutritional supplementation, for the prevention of pellagra, and as adjunct therapy for peripheral vascular disease and circulatory disorders. It is contraindicated in patients with hepatic disease and peptic ulcer disease. Additionally, because of its ability to elevate glucose and uric acid levels, especially when taken in large doses, nicotinic acid should be used with caution in patients who have or are predisposed to diabetes mellitus and gout (17,22,23).

TABLE 25.11 Drug Interactions for Fibrates

Drug	Fibrate	Result of Interaction
Antidiabetic agents	All	Increased hypoglycemic effect through increased sensitivity and decreased glucagon secretion
Bexarotene	Gemfibrozil	Increased bexarotene plasma concentrations
Bile acid sequestrants	All	Decreased bioavailability of fibrate if administration is not adequately spaced
Cyclosporine	Fenofibrate	Increased potential for nephrotoxicity
Ezetimibe	All	Increased ezetimibe concentration and possible increased risk of cholelithiasis; concomitant use is not recommended
HMG-CoA reductase inhibitors	All	Increased risk of severe myopathy or rhabdomyolysis
Oral anticoagulants	All	Increased hypoprothrombinemic effect
Repaglinide	Gemfibrozil	Increased repaglinide plasma concentrations
Ursodiol	All	Increased hepatic cholesterol secretion, which may increase the possibility of gallstone formation and counteract the effectiveness of ursodiol

ADVERSE EFFECTS OF NIAIN

Flushing, pruritus, headache, nausea, vomiting, diarrhea, flatulence, hepatic dysfunction, jaundice, hyperglycemia, hyperuricemia, blurred vision, and tachycardia

Adverse Effects

The most common (and, often, dose-limiting) side effects of nicotinic acid treatment are cutaneous vasodilation (flushing and pruritus) and gastrointestinal intolerance, which may occur in 20% to 50% of treated patients. Flushing and pruritus are prostaglandin-mediated effects and may be prevented by taking aspirin or indomethacin before nicotinic acid. Gastrointestinal side effects, such as flatulence, nausea, vomiting, and diarrhea, can be minimized if nicotinic acid is taken either with or immediately after meals. As previously mentioned, all of these effects can be minimized by slowly titrating the dose of nicotinic acid. Hepatic dysfunction is one of the more

TABLE 25.12 Drug Interactions for Nicotinic Acid

Drug	Result of Interaction
Adrenergic blocking agents	Enhanced vasodilation and postural hypotension
Bile acid sequestrants	Decreased bioavailability of nicotinic acid if administration is not adequately spaced
Ethanol	Potential enhanced hepatotoxicity and excessive peripheral or cutaneous vasodilation
HMG-CoA reductase inhibitors	Increased risk of myopathy and rhabdomyolysis
Vasodilating agents (calcium channel blockers, epoprostenol, nitrates)	Enhanced cutaneous vasodilation

serious complications of high-dose nicotinic acid. Plasma AST, ALT, lactate dehydrogenase, and alkaline phosphatase levels often are elevated but usually return to normal when therapy is either adjusted or discontinued (6,17,23).

Drug interactions for nicotinic acid are listed in Table 25.12.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Thomas L. Rihn, PharmD

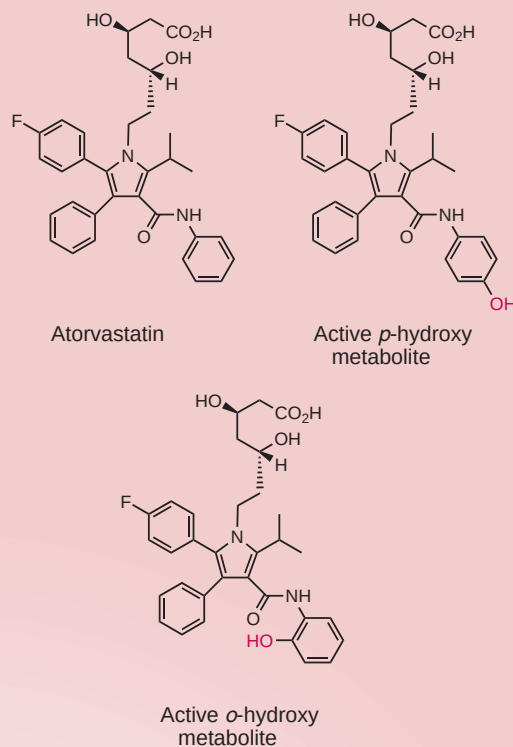
The therapeutic goal in this case is to achieve low-density lipoprotein cholesterol (LDL-C) level of at least <100 mg/dL. The most recent guidelines for a patient like TN—one with extensive vascular disease and diabetes—would suggest lowering the LDL-C even further to <70 mg/dL. Given TN's current lipid profile, a potent statin such as atorvastatin would be the preferred choice. TN does have a prescription plan to cover the cost of the medication. Beginning the patient on atorvastatin 10–20 mg daily and titrating the dose over time to achieve the therapeutic goal would be the treatment of choice. This is a case where the application of medicinal chemistry in the development of newer, more potent statins and their clinical use in a patient would result in a much more rapid and predictable control of the lipid abnormality. The net result is a better patient health outcome and avoidance of adverse cardiovascular events.

Chemical Analysis

Victoria F. Roche and S. William Zito

Atorvastatin is a good antihyperlipidemic candidate for this patient. Its unesterified carboxylic acid moiety confirms its ability to immediately initiate binding at the target enzyme, HMG CoA reductase (HMGR), through an ion-ion bond with an anchoring Lys residue. The C₃ and C₅ hydroxyls, *p*-fluorine and isopropyl moieties make important contributions to high affinity binding, and these groups are common to many (and, in the case of the hydroxyl groups, all) statin structures. A particularly strong hydrogen bond is formed between the carbonyl oxygen of atorvastatin's unique amide moiety and an HMGR Ser residue. Collectively, these high-affinity interactions make atorvastatin one of the most potent of the marketed statins. Once it is bound to HMGR, atorvastatin inhibits the biosynthesis of cholesterol and

stimulates the receptor-mediated uptake of LDL, which results in a lowering of serum LDL levels. The different statins have a variable impact on high-density lipoprotein cholesterol elevation, but all are able to lower triglyceride levels to some extent. The ability of statins to inhibit C-reactive protein, a major biomarker of inflammation, only adds to their value in the treatment of inflammation-based diseases, including coronary artery disease.

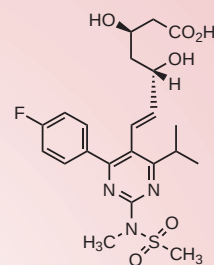


SCENARIO: OUTCOME AND ANALYSIS (Continued)

In addition to high affinity HMGR binding and LDL-lowering potency, atorvastatin has a long duration of action due to CYP3A4-mediated hydroxylation to two equally active metabolites. The 14-hour half-life permits convenient once-daily dosing without regard to time of day.

If TN is taking other medications that compete for or inhibit the 3A4 isoform, he may be at increased risk for statin-induced myopathy. Depending on the level of risk and/or TN's susceptibility to muscle toxicity, a highly potent, long-acting statin alternative that is not metabolized by CYP3A4 (e.g., rosuvastatin) could be considered. Rosuvastatin is vulnerable to CYP2C9-mediated N-demethylation, but only about 10% of a dose undergoes this sluggish reaction. Through hydrogen bonding with the aforementioned Ser and an Arg HMGR residue, the sulfonamide oxygen

atoms of rosuvastatin make this drug the most potent of all currently marketed statin antihyperlipidemic agents.



Rosuvastatin

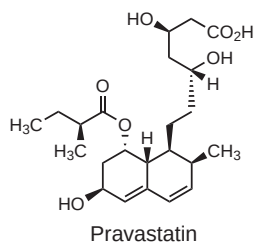
CASE STUDY

Victoria Roche and S. William Zito

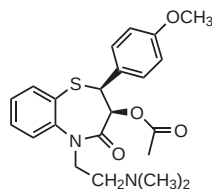
JC is a 60-year-old waitress at "The Southern Comfort," a restaurant famous for its fried chicken and homemade biscuits (always served with lots of butter and honey). JC lives alone and, after a hard day on her feet, just wants to "take a load off." She has a difficult time resisting this comfort food and, with the blessing of the restaurant manager, often takes her dinner home in a box to eat in front of the TV. JC has been blessed throughout her life with general good health, although lately she's been experiencing significant gastrointestinal upset that she's self-treating with over-the-counter omeprazole (a proton pump inhibitor). She's also taking low-dose, extended-release

diltiazem for mildly elevated blood pressure and pravastatin for hypercholesterolemia.

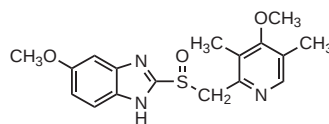
During her most recent checkup, JC's physician noted a rise in her LDL and total serum cholesterol levels. Her serum HDLs are low to normal and her serum triglycerides levels are in the acceptable range. A decision is made to alter her therapy. You remind the physician of the other drugs JC is taking and that diltiazem is an inhibitor of CYP3A4. The physician asks for your opinion about which of the therapeutic choices drawn below would be the best way to go. She wants to either replace pravastatin with compound 1 or add compound 2 or 3 to her pravastatin regimen.



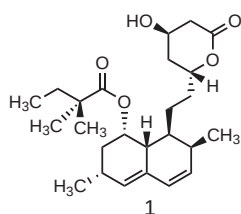
Pravastatin



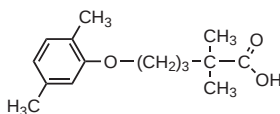
Diltiazem



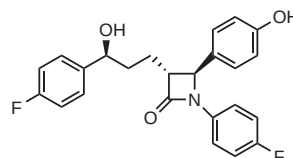
Omeprazole



1



2



3

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 26

Antithrombotics, Thrombolytics, Antiplatelets, and Coagulants

MATTHIAS C. LU

Drugs Covered in This Chapter*

WARFARIN

HEPARINS

- High molecular weight heparin
- Low molecular weight heparin
 - Ardeparin
 - Dalteparin
 - Enoxaparin
 - Tinzaparin
- Fondaparinux

DIRECT THROMBIN INHIBITORS

- Argatroban
- Bivalirudin
- Desirudin
- Lepirudin

- Dabigatran etexilate
- Rivaroxaban

ANTIPLATELET DRUGS

- Abciximab
- Aspirin
- Cilostazol
- Clopidogrel
- Dipyridamole
- Eptifibatide
- *Indobufen*
- Prasugrel
- Ticagrelor
- Ticlopidine
- Tirofiban
- *Triflusal*

THROMBOLYTIC DRUGS

- Alteplase
- Prourokinase
- Reteplase
- Streptokinase
- Tenecteplase
- Urokinase

COAGULANTS

- Protamine
- Vitamin K

THROMBOPOIETIN RECEPTOR AGONISTS

- Eltrombopag
- Romiplostim

ANTIFIBRINOLYTIC AGENTS

- Aprotinin

Abbreviations

ADP, adenosine diphosphate

aPTT, activated partial thromboplastin time

ATP, adenosine triphosphate

a-Xa U, antifactor Xa units

cAMP, cyclic adenosine monophosphate

COX, cyclooxygenase

CYP, cytochrome P450

DTI, direct thrombin inhibitor

DVT, deep vein thrombosis

FDA, U.S. Food and Drug Administration

FXa, activated factor X

GP, glycoprotein

HIT, heparin-induced thrombocytopenia

5-HT, serotonin

HTB, 2-hydroxy-4-trifluoromethylbenzoic acid

IC₅₀, half maximal inhibitory concentration

INR, international normalized ratio

ITP, idiopathic thrombocytopenia purpura

KH₂, vitamin K hydroquinone

LMWH, low molecular weight heparin

NSAID, nonsteroidal anti-inflammatory drug

PAI-1, plasminogen activator inhibitor-1

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

Abbreviations (Continued)

PAR-1, protease-activated receptor-1
PDE, phosphodiesterase
PE, pulmonary embolism
PF₄, platelet factor 4
PGI₂, prostacyclin
PT, prothrombin time

TF, tissue factor
TFPI, tissue factor pathway inhibitor
tPA, tissue-type plasminogen activator
TXA₂, thromboxane A₂
USP, U.S. Pharmacopeia
V_d, volume of distribution

VKORC1, vitamin K 2,3-epoxide reductase complex 1
VTE, venous thromboembolism
vWF, von Willebrand factor

SCENARIO

Kim K. Birtcher, PharmD

JF, a 46-year-old white man, came to the emergency department by ambulance complaining of crushing chest pain, which radiated down his left arm and started 1 hour ago while he was watching television. The patient's wife called 911, and the paramedics gave him an aspirin, sublingual nitroglycerin, and oxygen en route to the hospital. Clinical and laboratory parameters showed that the patient had a non-ST-segment elevation myocardial infarction.

The patient was started on a heparin infusion and intravenous eptifibatide and was sent for cardiac catheterization. The patient received a loading dose of clopidogrel, and a drug-eluting stent was placed in the patient's left anterior descending artery.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

Venous thromboembolism (VTE) is a complicating condition responsible for high morbidity and mortality worldwide (1). This disease is commonly linked to advanced age but has both hereditary and acquired risk factors, such as surgery, any form of trauma, and childbirth, associated with it. It encompasses an avoidable condition of deep vein thrombosis (DVT) that may result in pulmonary embolism (PE). It is estimated that 2 million people in the United States develop DVT each year with approximately one-third of these patients progressing to PE, leading to 200,000 deaths annually. Preventative therapy consists of the use of two different classes of antithrombotic agents, namely anticoagulants and antiplatelet drugs (2,3).

Heparin was discovered serendipitously in 1916 by a young coal miner turned medical student, Jay McLean, while working in the research laboratory of Professor W.H. Howell (4). Its beneficial effects were not realized until the early 1970s, when Kakkar et al. (5) established in a prospective, double-blind trial that low doses of heparin can prevent DVT after major surgery. In 1933, Dr. K.P. Link discovered that hydroxycoumarins are contained in sweet clover after finding cattle hemorrhaging to death (4). This discovery led to the development of orally active anticoagulants for prophylaxis in VTE and other thrombotic disorders. Optimal therapy with warfarin and other coumarin derivatives is not easy, however, because of their narrow therapeutic indexes. The need to carefully assess the benefit and risks for anticoagulation with frequent drug monitoring and dose adjustment also is troublesome (1,6). Thus, further development of novel antithrombotic agents with greater specificity on the coagulation cascade, with more predictable pharmacodynamic and pharmacokinetic profiles, and with fewer or no laboratory monitoring requirements is still needed for optimal treatment of thrombotic disorders (7–10).

Thrombolytics, on the other hand, are drugs needed to dissolve the newly formed thrombi in conditions such as DVT, acute PE, or myocardial infarction. Because of their lack of specificity, however, these agents actually may cause internal bleeding and, thus, are contraindicated with the use of many other therapeutic agents.

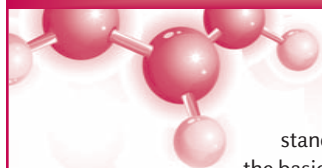
A variety of pathologic and toxicologic conditions can result in excessive bleeding from inadequate coagulation. Depending on the etiology and severity of the hemorrhagic episode, select coagulants that induce blood coagulation are therapeutically used to prevent excessive bleeding in these conditions.

DISEASE STATES REQUIRING ANTITHROMBOTIC THERAPY

A number of serious medical conditions are thrombotic in nature. In fact, in Western society, thrombotic conditions are the major cause of morbidity and mortality, and it is speculated that these disorders will be the leading cause of death worldwide within 20 years (2,8). As would be expected from the gravity of thrombotic disorders, many of the conditions involve the major vasculature, heart, brain, and lungs.

In the heart, a thrombotic condition may be involved in the disease state of acute myocardial infarction, valvular heart disease, unstable angina, and atrial fibrillation as well as surgical procedures, such as percutaneous transluminal coronary angioplasty and prosthetic heart valve replacement. Thrombotic conditions involving the vasculature include VTE, primary and secondary prevention of arterial thromboembolism, and peripheral vascular disease. The most significant such condition involving the lungs is PE and, in the brain, cerebrovascular accidents. Anticoagulation therapy is indicated for all of these conditions.

CLINICAL SIGNIFICANCE



The treatment, management, and prevention of thrombogenesis are complex. Clinicians must understand the steps of the clotting cascade, the basics of clot formation and lysis, and the differences between the individual medications that affect these processes. The clinician's knowledge of the mechanisms of action and pharmacokinetics of the anticoagulants will facilitate safe and effective treatment for individual patients. For example, it takes several days for warfarin to deplete the vitamin K-dependent clotting factors, so warfarin should be given in combination with unfractionated heparin or low molecular weight heparin when initiating treatment of VTE. Using low molecular weight heparin for this indication may allow the patient to receive outpatient therapy, whereas the use of unfractionated heparin requires continuous intravenous infusion in the hospital. In addition, the onset, duration of action, and predictability of anticoagulant effects of the low molecular weight heparins make them safer alternatives to warfarin during the perioperative period. The combined actions of aspirin; a thienopyridine; heparin, low molecular weight heparin, fondaparinux, or bivalirudin; a glycoprotein II_b/III_a inhibitor; and/or a fibrinolytic agent are necessary to manage thrombosis formation during an acute myocardial infarction.

Knowledge of structure–activity relationships helps predict site of action, duration of activity, potency, and side effect profile. As new medications are developed, the clinician will rely on this knowledge to tailor therapy to the individual patient. For example, the low molecular weight heparins' short pentasaccharide-containing heparin chains bind to antithrombin, so they have a greater relative affinity against factor Xa than the large pentasaccharide-containing heparin chains, which bind to both antithrombin and thrombin. Unfractionated heparin and bivalirudin are both antithrombotic agents, so they should not be used at the same time. Fondaparinux, a synthetic pentasaccharide, is not long enough to bind with thrombin, so it must be combined with heparin to prevent thrombus formation during percutaneous coronary intervention procedures. Dabigatran, a direct thrombin inhibitor, may be a safer and more convenient alternative to warfarin for some patients, because it does not require the same frequent laboratory monitoring as warfarin.

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Venous Thromboembolism and Pulmonary Embolism

VTE occurs when red blood cells, fibrin, and, to a lesser extent, platelets and leukocytes coagulate to form a thrombus within an intact cardiovascular system (11). A patient undergoing orthopedic surgery incurs the greatest risk for VTE (3,12). PE occurs when a segment of a thrombus within the deep venous system detaches itself from the blood vessel, travels to the lungs, and lodges within the pulmonary arteries. Both of these conditions, if not properly detected and treated, can have serious consequences, such as sudden death or recurrent VTE and postthrombotic syndrome characterized by persistent pain, swelling, and skin discoloration or necrosis and ulceration of the affected limb (7,8).

Atrial Fibrillation

Atrial fibrillation is the most common cardiac arrhythmia, being found in more than 2 million Americans. It is characterized as a storm of electrical energy that travels in spinning wavelets throughout the atria, causing the upper chambers to quiver or fibrillate. It also is one of the leading risk factors for ischemic stroke for those individuals older than 50 years (13–15). Atrial fibrillation is more prevalent in men than in women, and the median age of patients with atrial fibrillation is approximately 72 years. Anticoagulation with warfarin and lifestyle modification

are the most effective treatment modalities for patients with atrial fibrillation.

Pathophysiology of Thrombogenesis

Arterial thrombosis usually is initiated by the exposure of the thrombogenic material as a result of spontaneous or mechanical rupture of an atherosclerotic plaque (16,17). Arterial thrombi usually form in medium-sized vessels as a result of surface lesions on endothelial cells roughened by atherosclerosis. In most cases, circulating platelets adhere to the areas of abnormal vascular endothelium. More platelets then aggregate with those stuck to the vascular wall, forming a clot known as an occlusive thrombus (16). The formation of the occlusive thrombus at the site of the lesion is the major cause of complications of stroke and myocardial infarction. Venous thrombosis results from either an excessive activation of the coagulation cascade (hypercoagulability) or from some disease process or venous pooling (stasis) of the blood. Venous thrombus is initiated in the same fashion as the arterial thrombus formation, except that the bulk of the clot is formed of long fibrin tails that enmesh red blood cells. Venous thrombosis also can occur from vascular trauma caused by damage to the vessel wall, especially after major orthopedic surgery (11). Hypercoagulability, venous stasis, and vascular injury are known as Virchow's triad. As a general rule, arterial thrombi cause serious

conditions through localized occlusive ischemia, whereas venous thrombi fragments give rise to pleural embolic complications.

Biochemical Mechanism of Blood Coagulation: The Coagulation Cascade

The formation of a blood clot is the result of an intricate and elegant cascade of biochemical events (Fig. 26.1).

The coagulation in arteries or veins is triggered by tissue factor (TF), a small molecular weight glycoprotein that initiates the TF pathway (previous known as the extrinsic clotting cascade) (1,17). The TF glycoprotein is expressed on the surface of macrophages, and TF is a major initiating factor of arterial thrombogenesis. TF binds and activates factor VII to form TF/VIIa complex,

which in turn activates factor IX and factor X from either the intrinsic or extrinsic pathways shown in Figure 26.1.

Initiation of the contact activation pathway (also known as the intrinsic pathway) involves the sequential activation of factors XII, XI, and IX. (The activated form of a coagulation factor is indicated by a lowercase “a.”) In the presence of calcium, factor IXa binds to an activated factor VIII on the surface of activated platelets to form an intrinsic Xase complex, which initiates the activation of factor X to Xa. In addition, initiation of the tissue factor pathway involves activation of factor VII by TF to form TF/VIIa, which, like factor IXa, also catalyzes the conversion of factor X to Xa. The underlying purpose of the contact activation pathway is maintenance of homeostasis, whereas the tissue factor pathway is activated by trauma. The TF and contact activation pathways come together with the conversion of

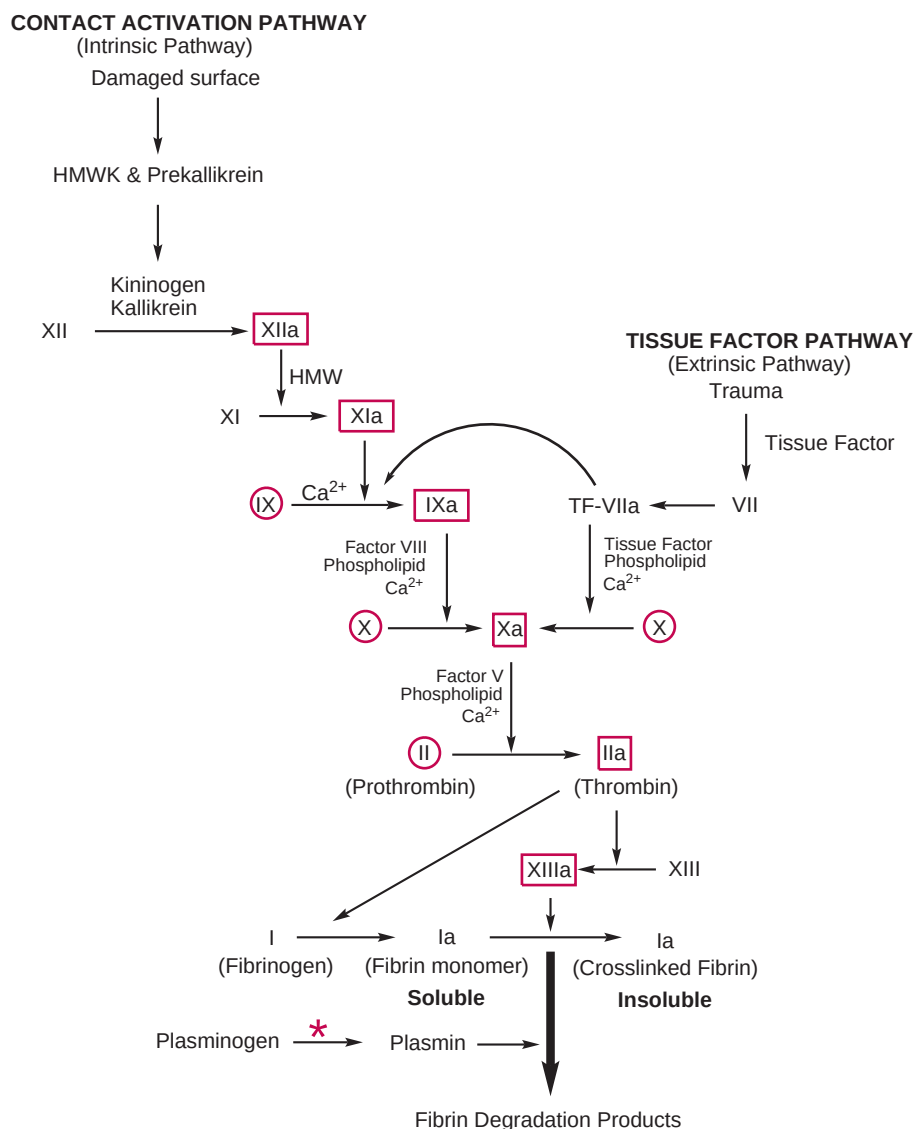


FIGURE 26.1 The coagulation cascade. *Circled factors* are those inhibited by warfarin-like drugs; *boxed factors* are those affected by heparin. The *star* indicates the site of action of thrombolytic drugs, such as streptokinase and urokinase.

factor X to its activated form, factor Xa. The coagulation cascade is unique in that the product of a given reaction (i.e., activated form of a specific factor) catalyzes the activation of the next factor in the cascade. The final steps in the coagulation cascade involve the conversion of prothrombin (factor II) to thrombin (factor IIa) by prothrombinase complex, consisting of factor Xa and activated factor V (Fig. 26.2). In turn, thrombin catalyzes the conversion of fibrinogen to soluble fibrin, which then becomes insoluble fibrin through the action of factor XIIIa. In its activated state, factor XIIIa actually is a transamidase enzyme. This enzyme catalyzes the formation of isopeptide bonds between lysine and glutamine side chains of distinct fibrin molecules, resulting in cross-linked (insoluble) fibrin aggregates (Fig. 26.3) (18).

Strategies for Regulating Coagulation

Coagulation can be regulated at several levels (1,17). These include TF pathway inhibitor (TFPI), antithrombin, and the protein C pathways. Because the designs of many new anticoagulant drugs are aimed at enhancing endogenous anticoagulant or fibrinolytic mechanisms, a brief review of these pathways is warranted here.

TFPI is now recognized as a major physiologic inhibitor of TF-initiated coagulation (19). Its main role is to modulate factor VIIa/TF catalytic activity by a two-step process. First, TFPI binds and inactivates factor Xa by forming a TFPI/factor Xa complex. Then, the inactivated factor Xa/TFPI complex binds factor VIIa within the TF/VIIa complex to modulate its catalytic activity. Additionally, TFPI potentiates the effect of heparins (i.e., TFPI is released from the vascular endothelium after injection of either unfractionated heparin or low molecular weight heparins [LMWHs]), which may then provide high concentrations of TFPI at sites of tissue damage and ongoing thrombosis (20). The propagation of coagulation occurs when TFPI concentrations are low (17).

Antithrombin is a potent inhibitor of thrombin and factors IXa and Xa. It also has inhibitory actions on other activated clotting factors, including the TF/VIIa complex. In the absence of heparin, however, the action of antithrombin is slow. The action of antithrombin is enhanced more than 2,000-fold in the presence of heparin (1). It should be pointed out that even though heparin is not normally found in the blood, the vascular

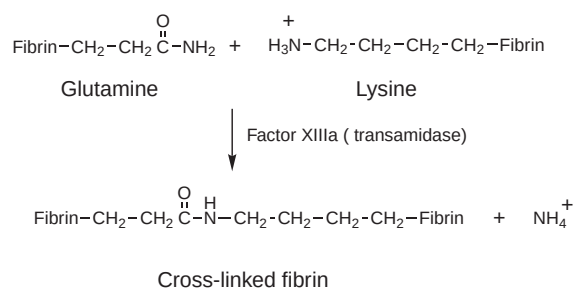


FIGURE 26.3 Cross-linking of soluble fibrin monomers (factor Ia) through the activity of factor XIIIa, a transamidase enzyme.

endothelium is rich in heparin sulfate, which contains the antithrombin-binding pentasaccharide sequence of the heparin. Drugs such as fondaparinux and idraparinix block the propagation of coagulation by inactivating factor Xa and, thereby, inhibiting the thrombin formation (1,17). Direct thrombin inhibitors, such as hirudin and argatroban, work by inhibiting both the clot-bound and free thrombin and, thus, preventing fibrin formation, the final step in the coagulation cascade (1,17).

In the protein C pathway, thrombin also is inhibited by binding to thrombomodulin, a thrombin receptor found in the endothelium. On binding to thrombomodulin, thrombin changes conformation, converting it from a procoagulant into a potent activator of protein C, a vitamin K-dependent protein, which degrades and inactivates factors Va and VIIa, thereby attenuating thrombogenesis (21,22).

GENERAL APPROACHES TO ANTICOAGULANT THERAPY

Overview

Because current clinically available antithrombotic drugs target only a few specific areas within the coagulation cascade, the selection of an appropriate anticoagulant for a given patient should be based on the patient's medical and drug history, age and location of the clot, underlying diagnosis of the disease state, and ultimate goal of the therapeutic intervention. If dissolving of an existing clot is needed, activation of plasminogen with the thrombolytic agents, which degrades insoluble fibrin, is the

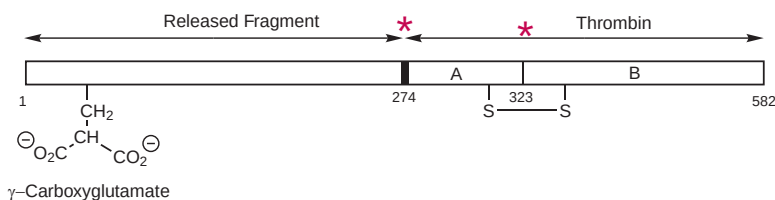


FIGURE 26.2 Structure of prothrombin. Thrombin is liberated through the cleavage of the Arg 274–Thr 275 and Arg 323–Ile 324 peptide bonds (indicated by the red stars). The γ-carboxyglutamate residues are in the released N-terminal portion of prothrombin and are not part of thrombin. The A and B chains of thrombin are joined by a disulfide bond.

typical approach. If the therapeutic goal is prevention of thrombus formation or extension, however, inhibition of factor activation higher in the cascade with heparin and other oral anticoagulants is the most appropriate.

An ideal anticoagulant should have reproducible pharmacodynamic and pharmacokinetic properties such that no coagulation monitoring is necessary (9). It also should have a wide therapeutic window, a rapid onset and offset of action, and minimal adverse effects, particularly with minimal interactions with food and other drugs (9).

Laboratory Assessment and Monitoring

Anticoagulant drug dosing represents a fine balance between reducing the morbidity and mortality associated with the thrombotic condition and minimizing the risk of serious hemorrhage from excessive therapeutic anticoagulation. Because of the potentially life-threatening consequences of either inadequate or excessive anticoagulation, patients receiving antithrombotic medications, such as warfarin and unfractionated heparin, often are closely monitored with specific clinical laboratory assays. A baseline assessment of the patient's coagulation features is performed before the initiation of anticoagulant therapy. This allows detection of congenital coagulation factor deficiencies, thrombocytopenia, hepatorenal insufficiency, and vascular abnormalities, which could prove to be catastrophic if anticoagulant therapy was instituted empirically.

For monitoring oral anticoagulant therapy (i.e., vitamin K antagonists), the prothrombin time (PT) is measured (23–25). This test is used to assess the activity of the vitamin K–dependent clotting factors (II, VII, IX, and X). The PT is particularly sensitive to factor VII, which is not of great clinical significance by itself but serves as a rough estimate for the ability of the liver to synthesize proteins or the extent of vitamin K depletion from warfarin therapy. The PT assay measures the time that it takes for a clot to form in citrated plasma after the addition of tissue thromboplastin and calcium. In normal (i.e., warfarin-free) plasma, this clot formation takes 10 to 13 seconds (23–25). Because of variances in commercially available thromboplastins, most clinical laboratories now report PT results in terms of international normalized ratios (INRs). Patients on warfarin therapy are optimally maintained with an INR of 2.0 to 3.0. In cases of patients who have had mechanical prosthetic heart valves placed, an INR of 2.5 to 3.5 often is recommended. At the initiation of warfarin therapy, daily PTs are performed. As the drug dosage is adjusted appropriately based on these results, the length of time between PT assessments can be extended to weekly. Finally, after warfarin therapy has been optimized and the patient's PT results have stabilized within an acceptable range, monthly or bimonthly PT checks are reasonable.

Heparin directly deactivates clotting factors II and X. Therapy with this drug is monitored based on the activated partial thromboplastin time (aPTT) assay (23–25). This assay monitors factors II and X as well as several

others. Deficiencies of clotting factors that affect the aPTT result can be of little clinical significance (e.g., prekallikrein and factor XII), of potential clinical significance (e.g., factor XI), or of great clinical significance (e.g., factors VIII and IX and the hemophilic factors). In the aPTT assay, a surface activator, such as eleagic acid, kaolin, or silica, is used to activate the intrinsic pathway. When this activator comes in contact with citrated plasma in the presence of calcium and phospholipid, clot formation begins. As with the PT, the time taken for this clot to form is measured. In normal (nonheparinized) plasma, the average aPTT result is 25 to 45 seconds. A therapeutic aPTT in a patient receiving heparin typically is 70 to 140 seconds. In vivo, the platelet membrane rather than the phospholipid is the source of several clotting factors and the site of many of the coagulation reactions in the intrinsic pathway. The phospholipid used in the aPTT assay does not completely substitute for the in vivo actions of the platelets. Although this phospholipid does potentiate the intrinsic pathway, it does so without activating factor VII. This “partial activation” of the intrinsic pathway is the genesis of the name of the assay (aPTT).

Several other laboratory assays are used to assess function at various points within the clotting cascade. Quantitative levels of fibrinogen and fibrin degradation products are used to assess the extent of the effects of conditions such as acute inflammation, disseminated intravascular coagulation, and severe liver disease. The specific clotting factors in which a given patient may be deficient also can be determined using various mixing studies (23–25). These assessments are far more specialized and performed much less frequently than the PT and aPTT.

ORAL ANTICOAGULANTS

There are two different chemical classes of orally active anticoagulants, namely coumarin derivatives and 1,3-indandiones. It has been known since 1921 that cattle eating spoiled sweet clover hay often would die from uncontrollable bleeding after suffering a very minor injury. This discovery and other subsequent findings eventually led to the isolation of bishydroxycoumarin (i.e., dicoumarol) in 1934 by Link and Campbell and its use in humans in 1954 as the first orally active anticoagulant drug (26).

Coumarin Derivatives

Warfarin and other vitamin K antagonists have been the mainstay of oral anticoagulant therapy for more than 50 years (1). Although their effectiveness in the prophylaxis of thrombotic disorders has been established through many well-designed clinical trials, their use in clinical practice is challenging because of their narrow therapeutic index, potential for drug–drug/drug–food interactions, and patient variability including genetic polymorphisms (CYP2C9, VKORC1) that require close assessment and drug monitoring (24,25,27).

Mechanism of Action

Vitamin K antagonists, such as warfarin, produce their effect on blood coagulation by interfering with the cyclic interconversion of vitamin K and vitamin K 2,3-epoxide (Fig. 26.4) (28). Vitamin K is an essential cofactor necessary for the posttranslational carboxylation of the glutamic acid residues on the N-terminal portions of the specific clotting factors (II, VII, IX, and X) and anticoagulant proteins, such as protein C (21). This γ -glutamyl carboxylation results in a new amino acid, γ -carboxyglutamate, which through chelation of calcium ions causes the proteins to undergo a conformational change. This change in tertiary structure allows the four vitamin K-dependent clotting factors to become activated and bind to the negatively charged phospholipid membranes during clotting cascade activation.

The specific enzyme that carboxylates vitamin K-dependent coagulation factors requires a reduced form of vitamin K (vitamin K hydroquinone [KH_2]), molecular oxygen, and carbon dioxide as cofactors. In the process of this reaction, KH_2 is oxidized to vitamin K 2,3-epoxide. The return of the epoxide to the active KH_2 form is the result of a two-step reduction. First, the epoxide is reduced to vitamin

K quinone, the rate-limiting step, by vitamin K 2,3-epoxide reductase complex 1 (VKORC1), in the presence of NADH (29). This quinone intermediate is then further reduced back to KH_2 by vitamin K quinone reductase. The warfarin-like anticoagulants (i.e., vitamin K antagonists) exert their anticoagulant activity through the inhibition of VKORC1 and, possibly, through inhibition of vitamin K quinone reductase, which in turn inhibits activation of the four affected coagulation factors. Unlike heparin, and as a direct result of their mechanism of action, the vitamin K antagonists only inhibit blood coagulation *in vivo*.

Structure–Activity Relationship of Coumarin Derivatives

All of the coumarin derivatives (Fig. 26.5) are water-insoluble lactones. Structure–activity relationship requirements typically are based on substitution of the lactone ring, specifically in positions 3 and 4. Although coumarin is a neutral compound, the clinically available derivatives are weakly acidic because of the presence of a 4-hydroxy substitution. The acidity of the proton on the 4-hydroxy group allows formation of water-soluble sodium salts for commercial preparations. Furthermore, warfarin (and, possibly, acenocoumarol) also can exist in

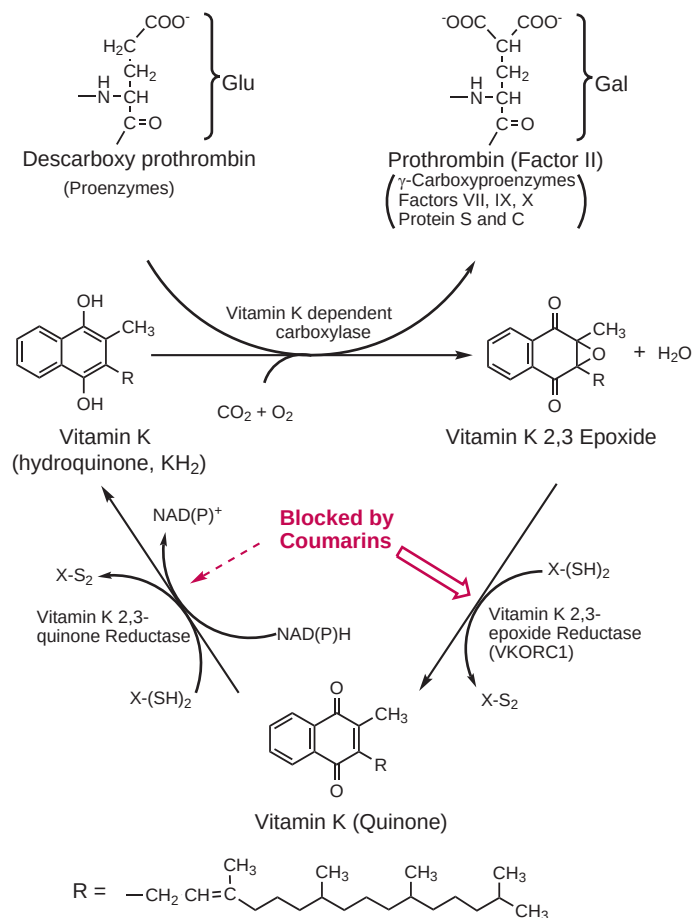


FIGURE 26.4 Redox cycling of vitamin K in the activation of blood clotting, which involves conversion of glutamate residues to γ -carboxyglutamates.

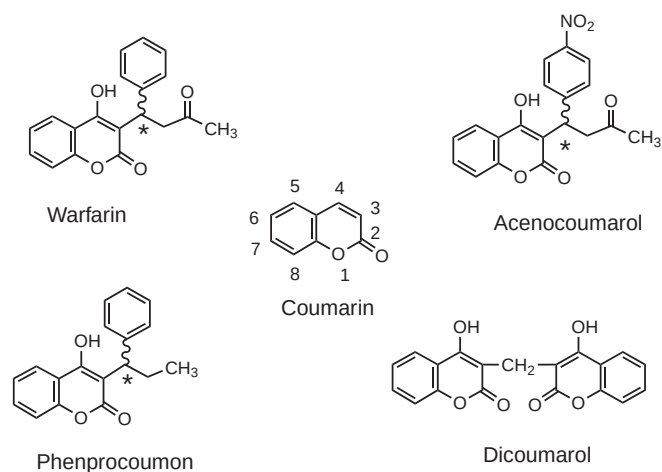


FIGURE 26.5 Chemical structures of coumarin and coumarin-derived drugs.

solution as two diastereomeric cyclic hemiketal conformers in addition to its open-chain conformer (Fig. 26.6). Because it has been suggested that vitamin K forms an active hemiketal in vivo, the cyclic hemiacetals of the vitamin K antagonists, such as warfarin, also may be the active conformers in vivo (30).

Pharmacokinetics

The substituents at position 3 greatly affect the pharmacokinetic and toxicologic properties of warfarin and its derivatives (Table 26.1) (31). Dicoumarol is not completely absorbed in the gastrointestinal tract, often is associated with gastrointestinal discomfort, and is very rarely used clinically. Today, the only coumarin used in the United States is warfarin, but phenprocoumon and acenocoumarol are used in Europe.

Warfarin

Warfarin sodium is rapidly and completely absorbed (~100% bioavailability) following oral, intramuscular, intravenous, or rectal administration. Peak plasma concentrations occur at approximately 3 hours. Its anticoagulant effect is not immediately present, however, following initiation of therapy. Instead, a delay in onset of anticoagulation occurs while the clotting factors with normal activity are cleared and those that have not been carboxylated because of the actions of warfarin reach physiologically significant levels. On average, this delay is approximately

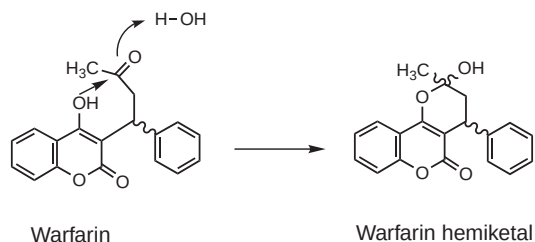


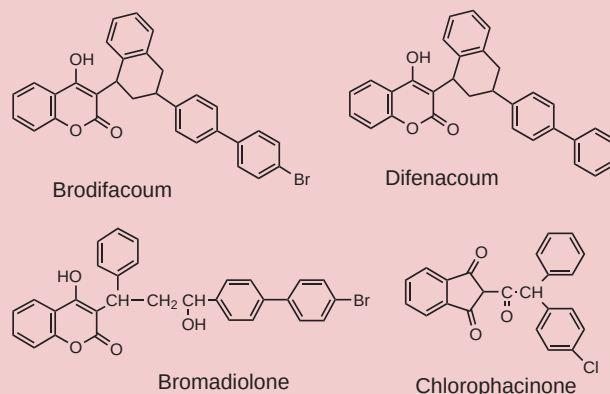
FIGURE 26.6 Formation of a cyclic hemiketal of warfarin.

5 hours for factor V turnover and 2 to 3 days for factor II (thrombin). Consequently, because of the rapid decline in protein C levels, the anticoagulated state frequently is preceded by a period of hypercoagulability (32).

Warfarin also is highly protein bound (95% to 99%) and, as a result, has numerous interactions with other drugs. The free drug (i.e., that not bound to plasma proteins) is the active constituent. Therefore, any other substance that displaces bound drug from protein binding sites increases the levels of free drug and, as a result, can cause warfarin toxicity, which usually is manifested by hemorrhage. The volume of distribution is quite small (0.1 to 0.2 L/kg), and the plasma half-life is quite long, both of which presumably result from the high degree of plasma protein binding (24,33).

“SUPERWARFARIN” RODENTICIDES

Brodifacoum is a member of the second-generation anticoagulant rodenticides known as “superwarfarins” (34). This compound and others like it (e.g., bromadiolone and difenacoum) were developed to combat rodent resistance to warfarin (34).



Human ingestion of brodifacoum typically is accidental in children but usually is intentional in adults (i.e., to commit suicide) (35). In cases when large quantities are ingested, severe and potentially fatal hemorrhaging can result. Brodifacoum is readily available over the counter in hardware stores and supermarkets, and it is marketed under numerous trade names in North America, Europe, Australia, and New Zealand. Brodifacoum, like warfarin, is thought to exhibit its anticoagulant effects through inhibition of vitamin K epoxide reductase. Despite the similarity in mechanism of action between brodifacoum and warfarin, brodifacoum is at least fivefold more potent as an anticoagulant rodenticide (34,36). The half-life of brodifacoum in humans is approximately 24.2 days, which is roughly ninefold longer than that of warfarin (34,37). Brodifacoum also has a volume of distribution roughly six times that of warfarin (37). For these reasons, vitamin K therapy may be needed for weeks to months after ingestion of a superwarfarin rodenticide (38).

TABLE 26.1 Pharmacokinetic Properties of the Coumarins and Anisindione

Drug	Trade Name	Onset (hours)	Duration (days)	Half-Life (days)	Time to Peak Plasma Concentration (hours)
Warfarin	Coumadin	1.5–3.0	2–5	0.62–2.5	4
Dicoumarol		1–5	2–10	1–4	1–9
Phenprocoumon	Marcumar ^a		7–14	5–6	
Acenocoumarol	Sinthrom ^a		2	0.3–1.0	1–3
Anisindione	Miradon	~6	1–3	3–5	2–3

^aNot available in the United States but available in Europe.

The clinically used preparation of warfarin is racemic, but the (*S*)- and (*R*)-enantiomers are not equipotent. In fact, (*S*)-warfarin is at least fourfold more potent as an anticoagulant than (*R*)-warfarin. The difference in the activities and metabolism of the enantiomers is the key to understanding several stereoselective drug interactions. Similar stereochemical properties are noted for the other asymmetric coumarins (Fig. 26.5). In the case of acenocoumarol, the (*R*)-isomer is responsible for the majority of its activity.

Hepatic CYP2C9 isozyme is responsible for metabolizing (*S*)-warfarin and other coumarin derivatives to give 6- and 7-hydroxywarfarins as the major inactive metabolites, whereas hepatic CYP3A4, CYP1A2, and CYP2C19 isozymes inactivate (*R*)-warfarin, the less active enantiomer, to give 4'-, 6-, and 8-hydroxywarfarins, respectively (39). Warfarin also undergoes, to a lesser extent, reductive metabolism of the ketone on the C-3 side chain to a pair of pharmacologically active, diastereomeric 2'-hydroxywarfarins (Fig. 26.7). Almost no unchanged drug is excreted in the urine. As expected, those individuals with compromised hepatic function are at greater risk for warfarin toxicity secondary to diminished clearance. Furthermore, individuals with VKORC1 and CYP2C9 variants may require a much lower warfarin maintenance dose to avoid serious bleeding complications (40). Many of the drug–drug interactions are associated with enhanced or inhibited metabolism of warfarin via CYP2C9 induction or inhibition. Many additional drugs and conditions have profound effects on warfarin therapy. A partial list of these factors is shown in Table 26.2 (31).

Indandiones

Indane-1,3-dione derivatives (Fig. 26.8), such as phenindione and anisindione, are orally active anticoagulants having a similar mechanism of action to the coumarins but are rarely used clinically today because of their significant renal and hepatic toxicities. Furthermore, because of the structural similarity, patients who are allergic to warfarin will experience cross-sensitivity with anisindione and other indandione anticoagulants (41). Although anisindione reportedly has fewer significant side effects, most clinicians still prefer warfarin for oral

anticoagulation. The pharmacokinetic properties also are similar to those of the coumarins (Table 26.1). Most of the newer indandione drugs, such as chlorophacinone, are marketed as potent rodenticides.

HEPARIN-BASED ANTICOAGULANTS

Chemistry

The heparin anticoagulants are represented by a variety of structures, including the natural heparan sulfate that lines the vascular endothelium, unfractionated heparin, LWMHs, and most recently, the synthetic pentasaccharide

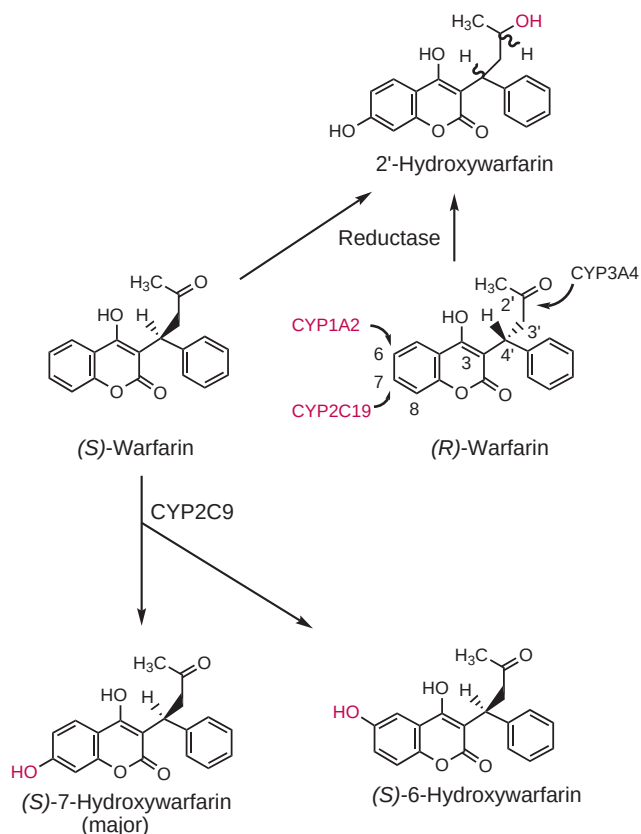
**FIGURE 26.7** Metabolism of warfarin.

TABLE 26.2 Factors Affecting Warfarin Therapy

Potentiate Anticoagulation		Antagonize Anticoagulation
Drugs:		Drugs:
Acetaminophen	Miconazole	Alcohol (chronic abuse)
Alcohol/ethanol (acute intoxication)	Nalidixic acid	Aminoglycosides
Allopurinol	Naproxen	Antacid
Amiodarone	Omeprazole	Antihistamines
Anabolic and androgenic steroids	Oral hypoglycemics	Barbiturates
Aspirin	Pentoxifylline	Carbamazepine
Bromelains	Phenylbutazone	Chlordiazepoxide
Cephalosporins	Phenytoin	Cholestyramine
Chenodiol	Piroxicam	Colestipol
Chloral hydrate	Propafenone	Corticosteroids
Cimetidine	Propranolol	Dextrothyroxine
Clofibrate	Quinidine, quinine	Griseofulvin
Chlorpropamide	Sulfamethoxazole–trimethoprim	Haloperidol
Cotrimoxazole	Sulfinpyrazone	Meprobamate
Dextran	Sulfonylureas	Nafcillin
Diazoxide	Sulindac	Oral contraceptives
Diflunisal	Tamoxifen	Penicillins (large doses)
Disulfiram	Thyroxine	Phenytoin
Erythromycin	Ticlopidine	Primidone
Ethacrynic acid	Tolmetin	Rifampin
Fenoprofen	Tricyclic antidepressants	Sucralfate
Fluconazole		Trazodone
Glucagon	Other Factors:	Vitamin K (large doses)
Heparin	Fever	
Ibuprofen	Stress	Other Factors:
Indomethacin	Congestive heart failure	High–vitamin K diet: spinach, cheddar cheese, cabbage
Inhalation anesthetics	Radioactive compounds	Edema
Isoniazid	Diarrhea	Hypothyroidism
Ketoconazole	Cancer	Nephrotic syndrome
Lovastatin	X-rays	
Mefenamic acid	Hyperthyroidism	
Metronidazole	Hepatic dysfunction	

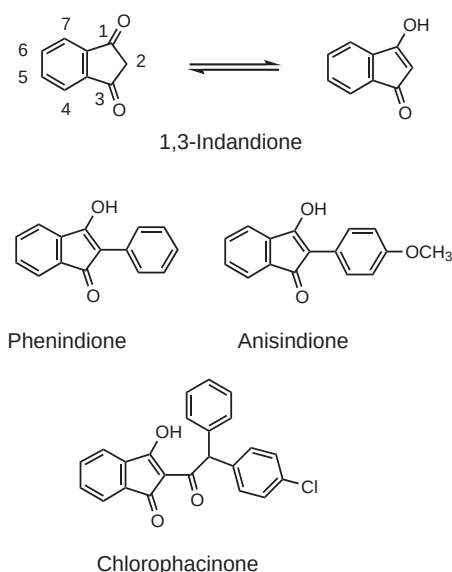


FIGURE 26.8 Chemical structures of 1,3-indandione and 1,3-indandione-derived drugs.

fondaparinux. Heparin is composed of a heterogeneous mixture of straight-chain, sulfated, and negatively charged mucopolysaccharides of a molecular weight range of 5 to 30 kd, isolated from bovine lung or porcine intestinal mucosa. Heparin, also known as heparinic acid, is an acidic molecule similar to chondroitin and hyaluronic acid. The polysaccharide polymer chains are composed of two alternating sugar units, *N*-acetyl-D-glucosamine and uronic acid (either D-glucuronic or L-iduronic), linked by $\alpha, 1 \rightarrow 3$ bonds (Fig. 26.9) (42).

These chains are called glycosaminoglycans and typically are composed of 200 to 300 monosaccharide units. In mast cells, approximately 10 to 15 of these chains are bound to a core protein to yield a proteoglycan (i.e., a protein/sugar conglomerate molecule) with a molecular weight of 750 to 1,000 kd. Before the molecule is capable of binding to antithrombin, the proteoglycan must undergo a series of structural modifications (42). These modifications include *O*-sulfation and *N*-sulfation of the D-glucosamine residues at carbons 6 and 2, respectively; *O*-sulfation of the D-glucuronic acid at carbon 2; epimerization of the D-glucuronic acid at carbon 5 to form L-iduronic acid; *O*-sulfation at carbon 2 of the L-iduronic acid; and *N*-deacetylation of the glucosamine and *O*-sulfation of the glucosamine at position 3. None of these reactions goes

to completion, so the resulting polysaccharide chains are structurally quite diverse (43,44). The heparin proteoglycan then undergoes degradation by an endo- β -glucuronidase in mast cell granules to release the active 5- to 30-kd polysaccharide chains.

Under physiologic pH conditions, heparin exists primarily as polysulfate anions and, therefore, usually is administered as a salt. In clinical use, standard heparin most often is the sodium salt, but calcium heparin also is effective. Lithium heparin is used in blood sample collection tubes to prevent clotting of the blood samples in vitro but not in vivo. The use of heparin salts also is important to maintain aqueous solubility, which is necessary for injection. Heparin can be administered intravenously or subcutaneously but not orally, because the polysaccharide chains are broken down by gastric acid. Intramuscular injection of heparin is associated with a high risk of hematoma formation and is not recommended.

Mechanism of Action

Heparin (unfractionated heparin) was the first parenteral anticoagulant to show efficacy in the treatment of VTE and has been in use since 1937. Heparin acts at multiple sites in the coagulation cascade (42). Anticoagulation occurs when heparin binds, via a distinct pentasaccharide sequence in its molecule, to the circulating antithrombin III (a serine protease inhibitor) and potentiates the antithrombin III-mediated inhibition of thrombin (factor IIa) and factor Xa, two of the key proteases in the blood coagulation cascade (Fig. 26.10) (45,46). The binding between heparin and antithrombin III consists of ionic bonding between sulfate and carboxylate anions in the pentasaccharide chain of heparin and arginine and lysine cations in the antithrombin III (46,47). Antithrombin III works by forming a stable 1:1 complex with both thrombin and factor Xa. Although the rates of these reactions (with IIa and Xa) are slow in the absence of heparin, binding is accelerated more than 2,000-fold when heparin is added (1,46). The reason for this enhancement is that when heparin binds to the antithrombin III, it induces a conformational change, resulting in increased accessibility of its active site and more rapid interaction with its protease substrates (i.e., thrombin and factor Xa). It should be noted that the ability of heparin to expose the active sites of antithrombin III is related to the large molecular size of heparin. With smaller molecules, such as LMWH and fondaparinux, the binding of antithrombin to

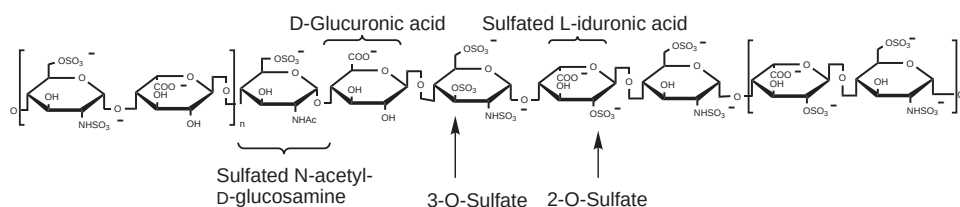


FIGURE 26.9 Chemical structure of heparin polymer.

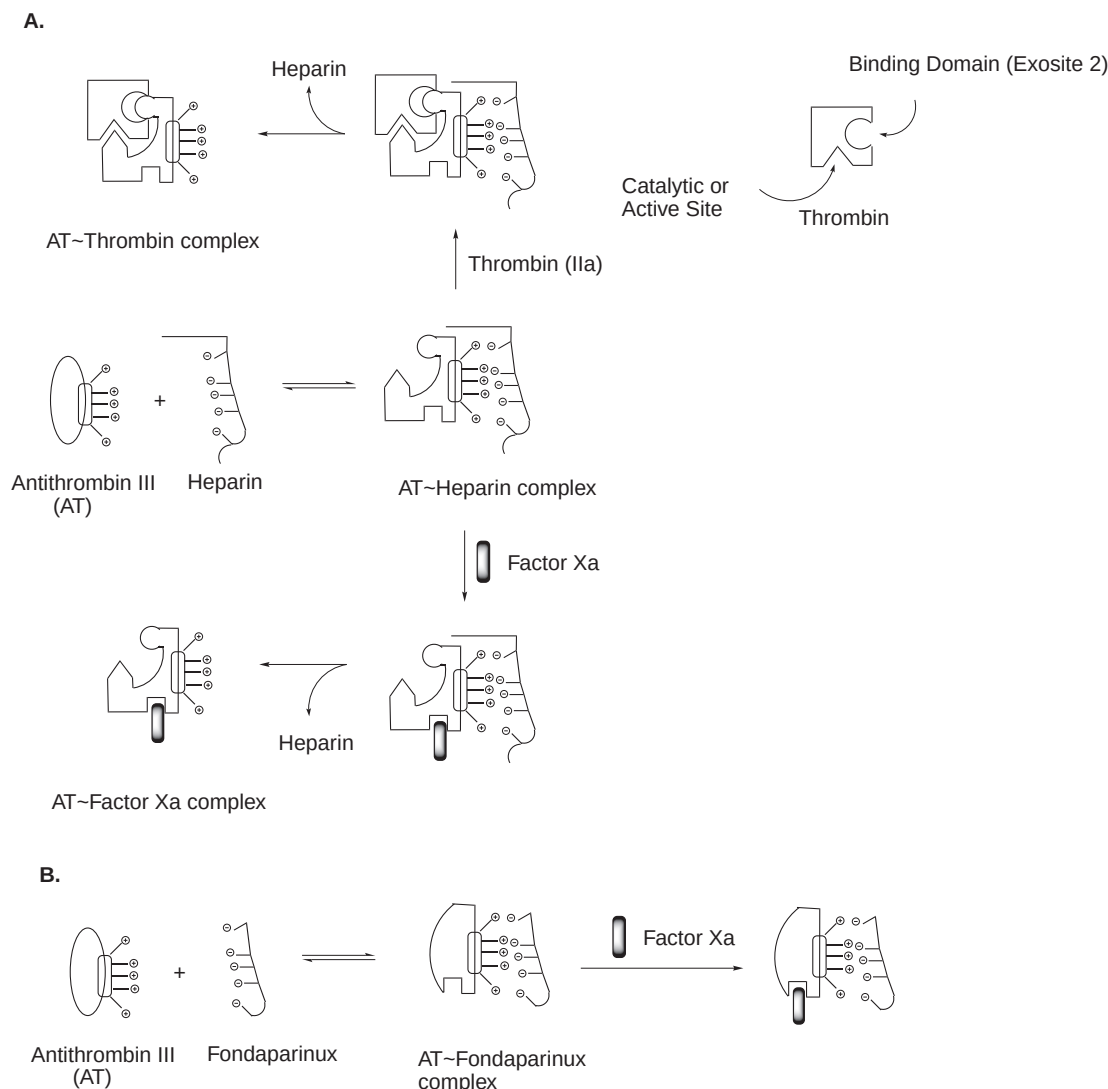


FIGURE 26.10 Schematic representation of catalytic role of the heparin drugs in promoting (A) antithrombin-factor Xa and antithrombin-thrombin complexes and (B) antithrombin-fondaparinux complex.

thrombin is diminished, and the drugs become more selective (see discussions of LMWH and fondaparinux). Interestingly, the role of heparin in this process is only catalytic in nature (i.e., it is not consumed, inactivated, or degraded by the reaction). In fact, once the complex of antithrombin and protease is formed, the heparin is released, with no loss of activity, to catalyze formation of more antithrombin/protease complexes (Fig. 26.10) (45). Additional effects of heparin on the coagulation of blood are a result of heparin's effects on plasminogen activator inhibitor, protein C inhibitor, and TFPI (20,21).

Pharmacokinetics

The pharmacokinetic profiles of heparin and LMWHs are quite different. Whereas heparin is only 30% absorbed following subcutaneous injection, 90% of LMWH is systemically absorbed (40,48). The binding affinity of heparin to various protein receptors, such as those on plasma

proteins, endothelial cells, platelets, platelet factor 4 (PF4), and macrophages, is very high and is related to the high negative-charged density of heparin (44). This high nonspecific binding decreases bioavailability and patient variability. Additionally, heparin's nonspecific binding to PF4 may account for heparin's narrow therapeutic window and heparin-induced thrombocytopenia (HIT), a major limitation of heparin (46,49). These same affinities are quite low, however, in the case of LMWHs. These parameters explain several of the benefits of the LMWHs. The favorable absorption kinetics and low protein binding affinity of the LMWHs result in a greater bioavailability compared with heparin. The lowered affinity of LMWHs for PF4 seems to correlate with a reduced incidence of HIT (44). Heparin is subject to fast zero-order metabolism in the liver, followed by slower first-order clearance from the kidneys (50,51). The LMWHs are renally cleared and follow first-order kinetics. This

makes the clearance of LMWHs more predictable as well as resulting in a prolonged half-life. Finally, the incidence of heparin-mediated osteoporosis is significantly diminished with use of LMWHs as opposed to heparin.

Metabolism

Independent of molecular weight, the metabolic fate of the heparins is essentially the same. The distribution of the compounds is limited primarily to the circulation, but heparins also are taken up by the reticuloendothelial system (33). Once this uptake occurs, rapid depolymerization of the polysaccharide chains ensues, resulting in products that are inactive as anticoagulants. Desulfation also occurs in mononuclear phagocytes, which also produces inactive metabolites. These metabolites, as well as some of the parent compound, are then excreted in the urine (33). Because of the depolymerization of heparin in the liver and ultimate renal elimination of both metabolites and parent drug, half-life is prolonged in patients with hepatic or renal dysfunction. Another heparin-like medication is danaparoid sodium (52). The drug is composed of 84% heparan sulfate, 12% dermatan sulfate, and 4% chondroitin sulfate. The average molecular weight is 5.5 kd, and like the LMWHs, danaparoid is dosed in terms of antifactor Xa activity (with target anti-Xa levels of 0.5 to 0.7 U/mL). Danaparoid is completely bioavailable intravenously or subcutaneously and attains maximal antifactor Xa activity 2 to 12 hours after administration, without bleeding or thromboembolic events during treatment in patients with suspected HIT (52). The elimination half-life is approximately 24 hours, and clearance is through the kidneys. Coagulation assays (e.g., PT and aPTT) are not routinely monitored in patients who are receiving danaparoid therapy.

Specific Heparin Drugs

High Molecular Weight Heparin (Unfractionated Heparin)

Standard heparin is unfractionated and contains mucopolysaccharides ranging in molecular weight from

5 to 30 kd (mean, ~15 kd) and is referred to as high molecular weight heparin (Table 26.3). This group of compounds has a very high affinity for antithrombin III and causes significant *in vivo* anticoagulant effects. Because heparin is a heterogeneous mixture of polysaccharides with different affinities for the target receptor, dosing based on milligrams of drug is inappropriate (i.e., there frequently is a limited correlation between the concentration of heparin given and the anticoagulant effect produced). Therefore, heparin is dosed in terms of standardized activity units that must be established by bioassay. One U.S. Pharmacopeia (USP) unit for heparin is the quantity of heparin required to prevent 1.0 mL of citrated sheep blood from clotting for 1 hour after the addition of 0.2 mL of 1% calcium chloride (23). Commercially available heparin sodium USP must contain at least 120 USP units per milligram. Heparin therapy typically is monitored by the aPTT. A therapeutic aPTT is represented by a clotting time in the assay that is 1.5- to 2.5-fold the normal mean aPTT (23). Monitoring therapy with laboratory testing is critical.

Low Molecular Weight Heparins

In the past two decades, an increased interest has surfaced in a group of compounds known as LMWHs (50). The LMWHs typically are in the 4- to 6-kd molecular weight range and are isolated as fractions from heparin using gel filtration chromatography or differential precipitation with ethanol (46). The LMWHs have more favorable pharmacokinetic and pharmacodynamic profiles relative to standard heparins (Table 26.3) (53,54). The mechanism of action of LMWHs is similar to conventional heparin, but the binding of LMWHs is more selective (i.e., LMWHs have more targeted activity against activated factor Xa and less against activated factor IIa [thrombin]) (Fig. 26.10). The reason for this difference is that even though LMWHs still possess the exact same specific pentasaccharide sequence as that of heparin needed for binding and potentiating the antithrombin

TABLE 26.3 Properties of Heparin Derivatives

Drug	Trade Name	Dosing	Molecular Weight (daltons)	Binding Ratio
Unfractionated heparin				
Heparin	Calciparine, Calcilean	bid, tid	5–30	1:1
Low molecular weight heparins				
Ardeparin	Normiflo	bid	5.6–6.4	2
Dalteparin	Fragmin	qd	3–8	2.2:1
Enoxaparin	Lovenox	qd, bid	3.5–5.5	2.7:3.9:1
Tinzaparin	Innohep	qd	5.5–7.5	2.8:1
Pentasaccharide				
Fondaparinux	Arixtra	qd	1.728	Xa only

bid, twice a day; qd, every day; tid, three times a day.

III-mediated inhibition of activated factor Xa, most of the LMWH/antithrombin III complex is of insufficient length to bind and inhibit factor Xa and thrombin at the same time (53). Thus, although all LMWHs inactivate factor Xa, only 25% to 50% of these molecules also inactivate thrombin (50). This factor selectivity typically is defined as a higher factor Xa:thrombin (anti-Xa:anti-IIa) activity ratio. In fact, although standard (unfractionated) heparin has an anti-Xa:anti-IIa ratio of 1:1, the same ratio in the LMWHs varies from 2:1 to 4:1 (Table 26.3) (50).

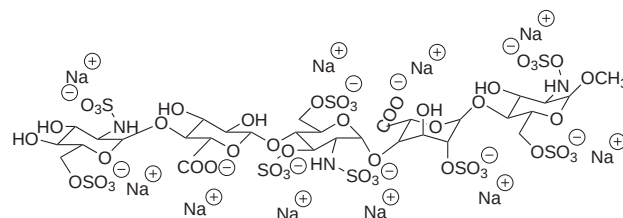
Four LMWHs are commercially available in the United States. The LMWHs are shown in Table 26.3. These drugs differ slightly in their medical indications for use as well as in their molecular weight ranges. All four compounds are indicated for perioperative thromboembolism prevention for specific abdominal and orthopedic surgeries. Enoxaparin and dalteparin are approved for use, in combination with aspirin, for the prophylaxis of ischemic complications of unstable angina and non-Q-wave myocardial infarction (55). Enoxaparin also is used in therapy for DVT with or without concomitant PE (55,56). Because of the increased homogeneity of enoxaparin compared to heparin, dosing of this drug is based on drug weight rather than on USP unitage. A typical dosing scheme for enoxaparin is the administration of 1 mg/kg once or twice daily. In the cases of dalteparin and ardeparin, dosage is based on antifactor Xa units (a-Xa U). Dalteparin is given as a once-daily subcutaneous injection at a dose of 2,500 to 5,000 a-Xa U. Typical dosing for tinzaparin is 175 a-Xa U/kg once daily, but the U.S. Food and Drug Administration (FDA) has issued a warning for using this product in elderly patients with impaired renal function. The LMWHs have a limited anticoagulant effect on *in vitro* clotting assays, such as the aPTT. In contrast to heparin, coagulation parameters, such as aPTT, usually are not monitored in patients receiving LMWHs; in addition, monitoring these assays is not really necessary because the LMWHs have a highly predictable dose-response relationship (50,53).

Newer Heparin Developments

Many recent studies involving heparin have been directed toward either increasing oral bioavailability or decreasing unwanted side effects (43,57). The poor bioavailability of heparin results from its high molecular weight and high anionic charge density. These properties, combined with the instability of the polysaccharides to gastric acid, make penetration of biologic membranes, such as the gut wall, extremely difficult for heparin. Various approaches to increase heparin absorption following oral administration have been investigated (58). Combinations of heparin with assorted calcium-binding substances and non- α -amino acids (*N*-acylated aminoalkanoic acids) for simultaneous oral administration also have been studied (58). Attempts to use structural modifications to heparin to attenuate undesirable side effects also have been investigated (57,59). HIT is caused by the interaction of

heparin with PF4. The PF4 binding domain appears to be distinct from the thrombin binding domain. Therefore, it should be possible to use shorter oligosaccharides that bind specifically to the thrombin inhibitory sites without binding to PF4 (57). These observations were used to develop and synthetically produce a series of oligosaccharides with good thrombin binding profiles that have limited interaction with PF4. This preliminary work is quite promising, but these compounds are not yet clinically available.

Fondaparinux



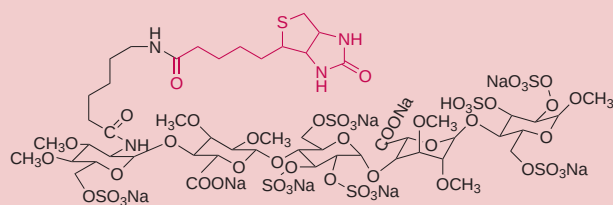
Fondaparinux is a prototype of a novel class of anticoagulants with significant advantages compared to their structurally related heparin (60). Based on the active site of the heparins, fondaparinux is a synthetic, highly sulfonated pentasaccharide. The immediate advantage of fondaparinux is that as a synthetic drug, its composition will not change, which results in improved pharmacokinetics and a more selective anticoagulant action.

MECHANISM OF ACTION The development of fondaparinux, a synthetically derived pentasaccharide that binds specifically to and activates antithrombin III, is a further refinement on the mechanism of action of heparin (61). Fondaparinux and a related analog, idraparinux, are specific, indirect inhibitors of activated factor Xa via their activation of antithrombin (Fig. 26.10). Fondaparinux has strategically located sulfonates that bind to antithrombin. Fondaparinux is structurally related to the antithrombotic binding site of heparin (62,63). Unlike heparin or LMWHs, however, these inhibitors have no effect on thrombin, because they lack the longer saccharide chains required for binding to thrombin. The highly sulfated heparins exhibit nonselective binding to a number of additional proteins, resulting in decreased bioavailability and significant variation in activity.

THERAPEUTIC APPLICATION Fondaparinux is the first selective factor Xa inhibitor that is approved for the prophylaxis of DVT, which may occur in patients undergoing hip fracture surgery or hip or knee replacement surgery (64). The most common side effect is major and minor bleeding, and the patient must be carefully monitored because its anticoagulant effect cannot be neutralized by protamine sulfate (8). The drug is not to be used when spinal anesthesia or spinal puncture is employed because of the potential for developing a blood clot in the spine.

IDRABIOTAPARINUX

Idrabiotaparinux is a biotinylated and polymethylated derivative of fondaparinux that is long-acting yet its anticoagulant effect is easily reversible with intravenous administration of avidin, a tetrameric protein derived from egg whites (7,67,68). Avidin binds with high affinity to the biotin moiety of idrabiotaparinux to form a 1:1 stoichiometric complex that is rapidly cleared via the kidneys (68). Like fondaparinux, this drug is an indirect inhibitor of factor Xa, which binds to antithrombin with very high affinity for antithrombin ($K_d < 1$ nmol/L) (67). The drug can be given subcutaneously on a once-weekly regimen. Idrabiotaparinux, like fondaparinux, is 100% bioavailable and does not bind to other plasma proteins.



Idrabiotaparinux

Fondaparinux has not been reported to cause thrombocytopenia, a condition seen with heparin (61,64–66). It is 100% bioavailable, with little or no protein binding.

PHARMACOKINETICS Fondaparinux is administered via subcutaneous injection with a single daily dose and shows complete absorption. The drug is highly bound to antithrombin III (~94%), with no significant binding to other plasma proteins. Because of the predictable anticoagulant effect, the drug does not require routine coagulation monitoring (66). The drug is not metabolized and is excreted in the urine unchanged within 72 hours in patients with normal renal function. Fondaparinux has an elimination half-life of 17 hours. Presently, no clinically significant drug interactions have been reported.

DIRECT THROMBIN INHIBITORS

In the last 10 years, with a better understanding of the molecular mechanisms of blood coagulation, the availability of molecular modeling and recombinant technologies, and the structure-based drug design strategies, many new anticoagulants that target almost every step in the coagulation pathway have been developed (17,69). Among these, five direct thrombin inhibitors (DTIs; hirudin, lepirudin, desirudin, bivalirudin, and argatroban) have been approved for clinical use in recent years (Table 26.4). Many other anticoagulants that act on the endogenous anticoagulation mechanisms, such as activated protein C pathway, TFPI, and additional orally active DTIs, currently are undergoing clinical trials (15,69,70).

Discovery and Design of Direct Thrombin Inhibitors

Hirudin, the lead compound for the design of DTIs, is a small protein (65 amino acids) that was originally isolated from the salivary glands of the medicinal leech, *Hirudo medicinalis* (71). This protein has potent and specific inhibitory effects on thrombin through the formation of a 1:1 complex with the clotting factor. The anticoagulant activity of hirudin seems to be contained within its highly anionic C-terminus. Several clinical studies have compared hirudin and a small peptidomimetic analog, hirulog, with heparin in the treatment of several thrombotic disorders. In many cases, hirudin seems to be more efficacious and the responses to it more predictable. Furthermore, some of the studies also indicate a lower incidence of bleeding complications with hirudin compared with heparin. Hirudin is now produced by recombinant technology, and many hirulogs continue to be screened (72,73). Significant progress in the design and development of direct thrombin inhibition has been achieved. The recent emergence of orally active DTIs may simplify the prevention and treatment of various thrombotic disorders (74–76).

Mechanism of Action

The DTIs bind and inactivate both free thrombin and thrombin bound to fibrin. Unlike heparin, DTIs, such as lepirudin, bivalirudin, and argatroban, bind directly

TABLE 26.4 Direct Thrombin Inhibitors (DTIs)

Drugs	Trade Name	Route of Administration	Site of Binding	Reversibility	Half-Life (minutes)	Route of Excretion	Protein Binding
Hirudin		IV	CS, Exosite I	Irreversible			
Lepirudin	Refludan	IV	CS, Exosite I	Irreversible	80		
Desirudin	Revasc	SC	CS, Exosite I	Irreversible	120–180	Kidney	0
Bivalirudin	Angiomax	IV	CS, Exosite I	Reversible	25	Kidney	0
Argatroban	Novastan	IV	CS	Reversible	39–51	Hepatobiliary	54%

CS, catalytic site; IV, intravenous; PO, oral; SC, subcutaneous.

and reversibly to the active site of thrombin. Unlike the heparins, these inhibitors do not require an activated antithrombin III as a cofactor for their anticoagulant activity. Furthermore, contrary to the heparins, these agents inhibit only the activity of thrombin, whereas heparin indirectly inhibits factors IIa (thrombin), IXa, Xa, XIa, and XIIa (77). There are three different domains where DTIs bind to and block the action of thrombin: the active site (or catalytic site, CS) as well as two additional exosites. Exosite-1 acts as a dock for substrates such as fibrin and, thereby, orients the appropriate peptide bonds in the active site for its biotransformation. Exosite-2 is the heparin binding domain. Bivalent DTIs, such as lepirudin and bivalirudin, block thrombin at the active site and exosite-1, whereas argatroban is a univalent DTI and, thus, binds only to the active site of the thrombin.

Specific Drugs

Recombinant Hirudin Derivatives: Lepirudin (Refludan) and Desirudin (Revasc)

Lepirudin and desirudin have been approved for the treatment of HIT and of HIT with thrombotic syndrome (78). Both lepirudin and desirudin are recombinant hirudin derivatives that consists of a 65-amino acid protein (79,80). Lepirudin is related to the recombinant product desirudin differing in the two N-terminal amino acids. The N-terminal amino acids in lepirudin are leucine-1 and threonine-2, whereas in desirudin, the N-terminal amino acids are valine-1 and valine-2. Additionally, desirudin lacks a sulfated tyrosine at amino acid 63. The anti-thrombin activity of the two drugs is slightly different.

Lepirudin and desirudin are both bivalent DTIs that bind to both the active site and the exosite-1 of thrombin. The result of this binding is that they create a nearly irreversible inhibition of thrombin. Lepirudin and desirudin inhibit both free thrombin and thrombin bound to fibrin (81).

Lepirudin is administered via intravenous bolus injection, followed by continuous infusion, whereas desirudin is administered subcutaneously twice daily (Table 26.4) (80,82). The drugs are cleared via the kidneys. Lepirudin is nearly totally degraded before excretion (~90%), whereas desirudin is excreted 50% unchanged. Lepirudin has immunogenic properties, and a significant number of patients develop antithrombin antibodies. In addition, hemorrhages may occur in patients treated with lepirudin. The drug half-life is approximately 1.3 hours (83).

Bivalirudin (Angiomax, Angiox)

Bivalirudin, a 20-amino acid peptide, has been approved for use in patients with unstable angina undergoing percutaneous coronary intervention (Fig. 26.11) (84).

Bivalirudin is a rapid-onset, short-acting DTI that binds to both the active site and the exosite-1 of thrombin. Unlike lepirudin, bivalirudin is a reversible inhibitor of both free thrombin and thrombin bound to fibrin. This reversibility is possible because the bound bivalirudin undergoes cleavage at the second N-terminal proline to release the portion of the drug bound to the active site. The carboxyl-terminal portion of bivalirudin dissociates from thrombin to regenerate thrombin (Fig. 26.11) (83). Bivalirudin does not bind to plasma protein.

Bivalirudin is administered via intravenous bolus injection, followed by continuous infusion (Table 26.4). The drug exhibits a rapid onset and a short duration of action. Bivalirudin is eliminated by renal excretion. It has been suggested that dosage adjustments be made in patients with severe renal impairment and in patients undergoing dialysis. Approximately 30% is eliminated unchanged along with proteolytic cleavage products. Because of the reversible nature of bivalirudin, the drug exhibits less risk of bleeding than other antithrombotics.

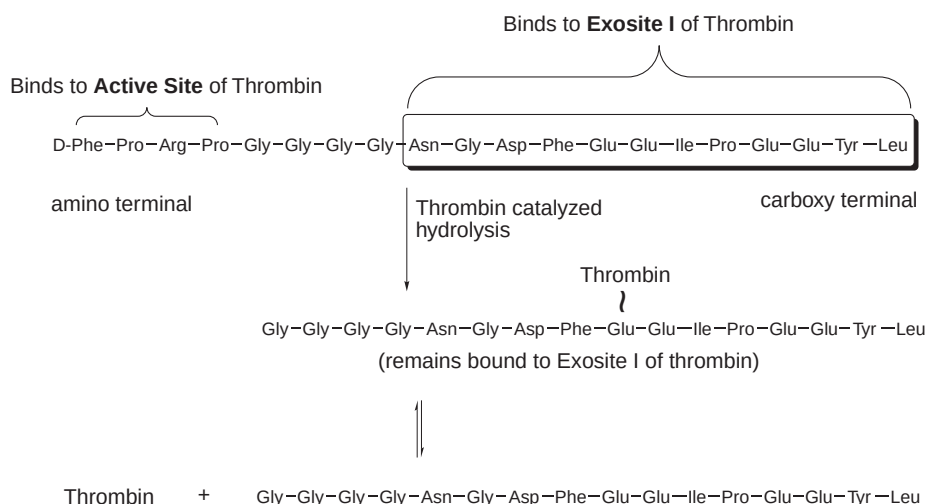
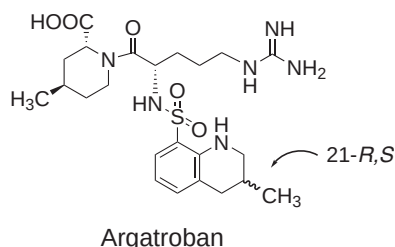


FIGURE 26.11 Chemical structure of bivalirudin, binding sites to thrombin, and release from thrombin.

and there have been no reported cases of antibody formation to bivalirudin (85).

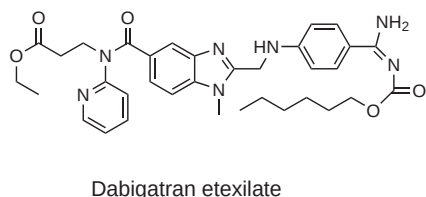
Argatroban



Argatroban has been approved for the prophylaxis and treatment of thrombosis in patients with HIT (86). Argatroban is a peptidomimetic that binds selectively to the catalytic site of thrombin as a univalent competitive DTI (Fig. 26.12). Argatroban is available as a mixture of 21-(*R*) and 21-(*S*) diastereomers (64:36), with the (*S*)-isomer approximately twice as potent as the (*R*)-isomer (87). The drug is a reversible inhibitor of both free thrombin and clot-bound thrombin.

PHARMACOKINETICS (TABLE 26.4) Argatroban is administered subcutaneously because of the low lipophilicity of the drug. The drug is bound to plasma protein and is metabolized via CYP3A4/5 to the aromatized metabolite and the two hydroxylated metabolites (Fig. 26.13). The M-1 metabolite retains 20% to 30% of the antithrombotic activity. Coadministration of argatroban with inhibitors of CYP3A4 does not appear to produce clinically significant effects. Argatroban is eliminated via biliary secretion into the feces (88).

Dabigatran Etxilate (Pradaxa)



Dabigatran etexilate, an orally active DTI, has recently been approved for the prevention of stroke and blood clots in individuals with atrial fibrillation. Dabigatran etexilate is a nonpeptidomimetic prodrug that gives



FIGURE 26.12 Schematic representation of mechanism of action of argatroban.

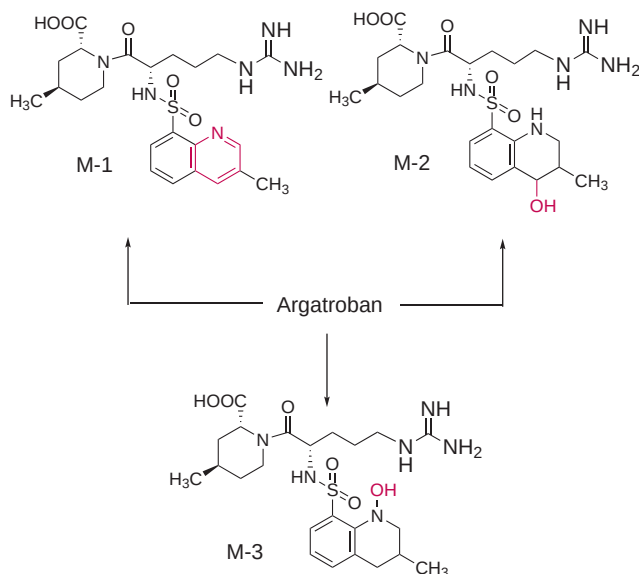


FIGURE 26.13 Metabolism of argatroban.

rise to dabigatran (Fig. 26.14), a reversible, basic benzimidazole-based direct thrombin inhibitor that binds and inactivates both free and fibrin-bound thrombins (89,90). The drug is available in the European Union and Canada, where it is approved for treatment of VTE in patients undergoing hip or knee replacement surgery, and it is being studied for treatment of DVT and PE.

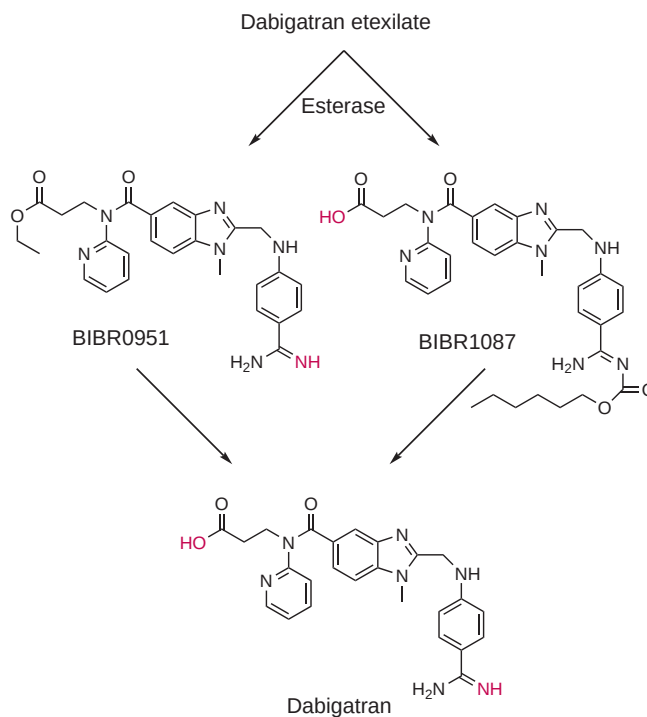


FIGURE 26.14 Metabolic activation of dabigatran etexilate to dabigatran.

PHARMACOKINETICS Dabigatran etexilate has a rapid onset of action, reaching maximum plasma concentrations within 0.5 to 2 hours after oral administration, and is quickly converted to dabigatran by plasma esterases (89,91). Very little dabigatran etexilate or the intermediate metabolites (BIBR0951-1087) are detectable in plasma. Dabigatran itself is quite hydrophilic due to the strongly basic amidine group and, at physiologic pH, exists as a charged species. As a result, dabigatran is not absorbed orally. The less basic etexilate is readily absorbed from the stomach and small intestine, but requires an acid environment. To promote an acidic microenvironment, the dabigatran etexilate is coated on tartaric acid pellets within the drug capsule, which creates an acidic environment thus promoting absorption and reducing variability due to different gastric pH in the individual. Dabigatran etexilate has a low bioavailability (~6.5%), which is offset by the administration of a large dose (75 to 150 mg twice a day). Up to 80% of dabigatran is eliminated unchanged in the kidneys with the remainder excreted via the biliary system as its acylglucuronides (91). Dabigatran etexilate was reported to have no clinically relevant interactions with drugs metabolized by cytochrome P450 (CYP) isozymes; thus, it can be given in fixed doses without coagulation monitoring (89). However, because dabigatran etexilate is excreted predominantly in the kidneys, it is not recommended for patients with severe renal dysfunction (70). Dosage is based on creatinine clearance: For patients with creatinine clearance greater than 30 mL/minute, the recommend dose is 150 mg twice a day, whereas for patients with a creatinine clearance of 15 to 30 mL/min, the recommended dose is 75 mg twice a day. Due to dabigatran's rapid onset and rapid offset, the drug does not require coagulation monitoring, which represents a major advantage over other anticoagulant therapy. However, the drug is reported to produce a significant increase in risk of major gastrointestinal bleeding, and the patient should be made aware of this side effect. It is reported that there is a significant reduction in life-threatening and total bleeding. Other side effects reported include dyspepsia, stomach pain, nausea, heartburn, and bloating.

ANTIPLATELET DRUGS

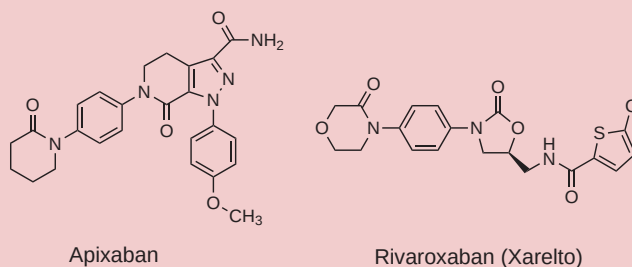
Another site for regulating blood coagulation and subsequent thrombus formation is at the level of the platelets (97). Antiplatelet drugs work by inhibiting platelet activation via a number of different mechanisms (97,98). The major role of antiplatelet drugs is in the prevention of ischemic complications in patients with coronary diseases (99,100). These drugs also are effective in combination with moderate-intensity anticoagulants for patients with atrial fibrillation.

Pathophysiology of Arterial Thrombosis

The pivotal role of platelets in thrombus formation and potential sites for drug interventions is illustrated in Figure 26.15 (98,100). Normal endothelial cells in the vascular wall synthesize and release prostacyclin (PGI_2),

ORALLY ACTIVE DIRECT THROMBIN INHIBITORS IN THE PIPELINE

Several additional orally active small-molecule inhibitors that specifically target thrombin or activated factor X (FXa) are in various stages of their clinical development for VTE, atrial fibrillation, and other thrombotic diseases (1,2,8,15,70,89). Two of these agents that are in the late stages of their clinical development are rivaroxaban and apixaban. The former was approved in the European Union and Canada in 2008 for prevention and treatment of VTE in patients undergoing elective hip or knee surgery.



Orally active direct thrombin inhibitors

Rivaroxaban was the first orally active, direct coagulation FXa inhibitor with the most published clinical trials data showing similar efficacy and safety to enoxaparin in preventing VTE in major orthopedic surgery (92,93). It has received approvals for the prevention of VTE associated with major surgery in several countries and has been recently approved for use in the United States (94). Rivaroxaban is a highly selective and reversible inhibitor of direct FXa ($K_i = 0.4 \text{ nmol/L}$) that binds to the active site of both free and prothrombinase-bound FXa. It is rapidly absorbed, reaching maximum plasma concentrations 3 hours after oral administration, and has a half-life ranging from 5 to 9 hours in young adults and 11 to 13 hours in the elderly.

Apixaban is the second selective and potent oral direct FXa inhibitor ($K_i = 0.8 \text{ nmol/L}$) that exhibits a superior risk-to-benefit ratio with respect to bleeding than enoxaparin (93,95). It is well absorbed after oral administration, reaching peak plasma levels in about 3 hours with a half-life of 12 hours. Metabolic profile studies have revealed that apixaban is the major drug-related component in plasma, urine, and feces in humans, with the *O*-demethylapixaban sulfate as a stable, water-soluble inactive metabolite (96).

which stimulates the conversion of adenosine triphosphate (ATP) to cyclic adenosine monophosphate (cAMP), thus preventing platelet aggregation and degranulation. In the case of an injury to the vascular wall, glycoprotein (GP) receptors (i.e., GPI_a and GPI_b) bind substances such as von Willebrand factor (vWF) and collagen from the exposed subendothelial surface, thereby activating the platelets. The $\text{GPII}_b/\text{III}_a$ receptors (also known as the fibrinogen receptor or integrin $\alpha\text{IIb}\beta 3$ receptor) then mediate the final step of platelet aggregation by binding

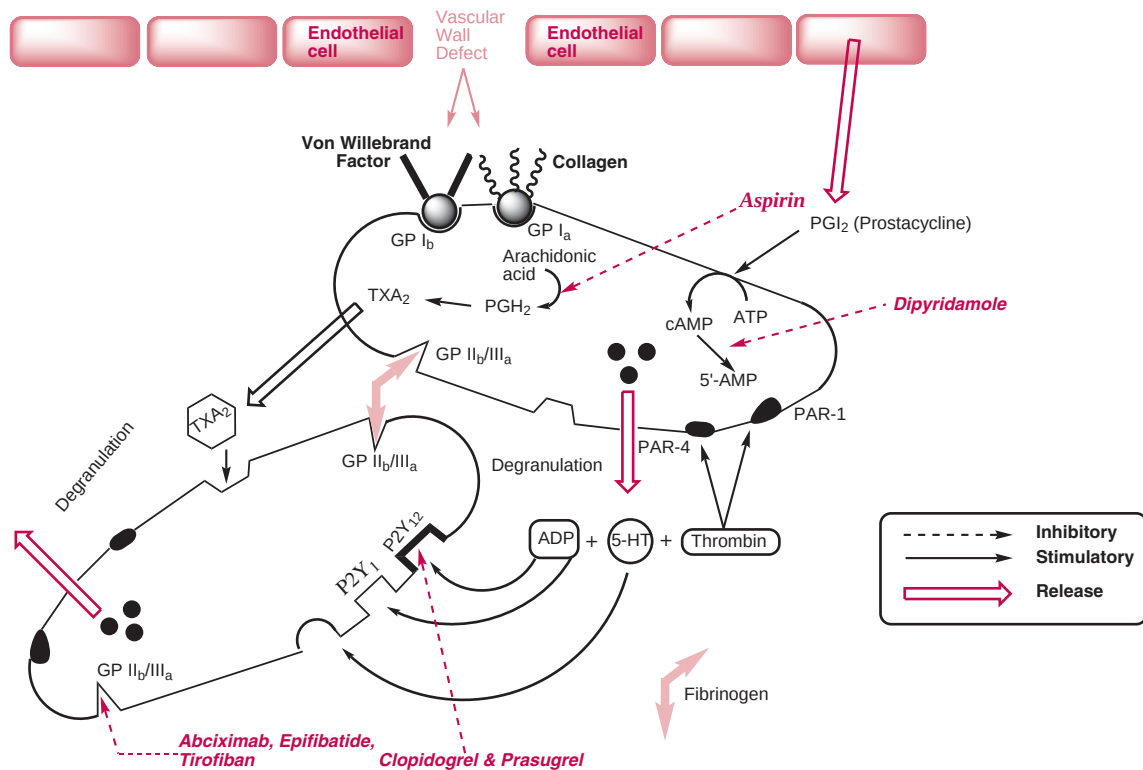


FIGURE 26.15 Scheme describing platelet activation as it relates to blood clot formation. The thrombus is formed at the site of a damaged wall in the vasculature. Normal endothelial cells in the vascular wall provide prostacyclin, which stimulates the conversion of adenosine triphosphate (ATP) to cyclic adenosine monophosphate (cAMP), preventing platelet aggregation. In injury, glycoprotein (GP) receptors bind substances such as von Willebrand factor and collagen, activating the platelet. The GPII_b/III_a receptors cross-link platelets via fibrinogen binding. As the platelet degranulates, additional aggregating substances, including thromboxane A₂ (TXA₂), serotonin (5-HT), and adenosine diphosphate (ADP), are released. ADP, by binding to P2Y₁ and P2Y₁₂, promotes and sustains platelet aggregation, respectively. Also shown (in *italics*) are drugs and sites of inhibition of platelet aggregation.

to fibrinogen or vWF, thus cross-linking platelets to form aggregates (Fig. 26.15).

The adherent platelets degranulate and release additional aggregating substances, such as thromboxane A₂ (TXA₂), serotonin (5-HT), thrombin, and adenosine diphosphate (ADP). These substances serve as secondary chemical messengers to recruit more platelets to the site of vascular injury and, thereby, amplify platelet aggregation (98,100). For example, thrombin production releases ADP, which is a potent inducer of platelet aggregation and stimulates prostaglandin synthesis from arachidonic acid in the platelet. The prostaglandins synthesized, PGI₂ and TXA₂, have opposite effects on thrombogenesis. PGI₂ is synthesized in the walls of the vasculature and inhibits thrombus formation. Conversely, TXA₂, which is synthesized in the platelets, induces vasoconstriction and thrombogenesis. Serotonin, which also is released from the platelets, has similar and additive effects to those of TXA₂.

This rapid platelet aggregation and thrombus formation at the site of vascular injury is the main mechanism of hemostasis (stoppage of bleeding, a normal process of wound healing). When platelets are activated on the ruptured atherosclerotic plaques or in regions of restricted blood flow, however, it can lead to thromboembolic

complications that contribute to common diseases, such as myocardial infarction or ischemic stroke.

Mechanism of Action of Antiplatelet Drugs

Most of the current available antiplatelet drugs, such as aspirin, dipyridamole, and ticlopidine, exert their actions by affecting only the secondary platelet aggregation pathways (97,101). For example, aspirin inhibits the biosynthesis of TXA₂ in the platelets by irreversibly and permanently inactivating cyclooxygenase (COX)-1 through covalent acetylation of a serine residue in close proximity to the active site of the enzyme. A cumulative inactivation effect occurs on platelets with long-term aspirin therapy in doses as low as 30 mg/d (99), because platelets do not synthesize new COX (i.e., platelets are unable to synthesize, via de novo pathway, COX-1, because they are anucleated cells). Therefore, the effects of aspirin last for the lifetime of the platelet (7 to 10 days). Dipyridamole interrupts platelet function through its effect of increasing cellular concentration of cAMP by inhibiting phosphodiesterase, an enzyme needed for degradation of cAMP. Dipyridamole also may stimulate PGI₂ release and inhibits TXA₂ formation. Ticlopidine, clopidogrel,

and prasugrel selectively inhibit ADP-induced platelet aggregation with no direct action on prostaglandin production. New and more selective antiplatelet drugs, such as thrombin receptor (or protease-activated receptor-1 [PAR-1]) antagonists, integrin $\alpha\text{IIb}\beta 3$ receptor antagonists (GPII_b/III_b blockers), thromboxane synthase inhibitor, and TXA₂ receptor antagonists, are currently being developed (7,101). Recent trials with the PAR-1 antagonists have indicated that these agents may provide a better overall platelet inhibition without the liability of increased bleeding when used with aspirin or clopidogrel therapy (100).

COX-1 Inhibitors

Because TXA₂ is a potent vasoconstrictor as well as a labile platelet aggregation inducer, inhibition of production of TXA₂ effectively blocks platelet aggregation. Aspirin and related analogs (Fig. 26.16) exhibit their effectiveness through such a blocking mechanism.

Aspirin

Aspirin is a well-established antiplatelet drug in the treatment of atherothrombotic vascular disease (7,97–99). As stated earlier, aspirin works by its ability to acetylate and irreversibly deactivate platelet COX (COX-1), and its antithrombotic effect remains for the life span of the platelet (7 to 10 days) (Fig. 26.17). Aspirin also has been shown to have other antithrombotic effects that are unrelated to its action on COX-1 (97). These effects include the dose-dependent inhibition of platelet function, the enhancement of fibrinolysis, and the suppression of blood coagulation.

Aspirin is rapidly absorbed in the stomach and quickly degraded by plasma cholinesterases (half-life, 15 to 20 minutes). A once-daily dose of 50 to 100 mg of aspirin, which is much lower than dosages needed for its anti-inflammatory/analgesic actions, is sufficient to completely inactivate platelet COX-1 irreversibly (97,99). Higher doses of aspirin only contribute to its side effects, especially internal bleeding and upper gastrointestinal irritations.

In recent years, the term “aspirin resistance” has been used to denote those situations in which the use of aspirin is unable to protect a patient from thrombotic complications, to cause a prolongation of the bleeding time, or to produce an anticipated effect on one or more *in vitro* tests of platelet function (97,99,102). One possible

explanation for aspirin-resistant TXA₂ biosynthesis is the transient expression of COX-2 in newly formed platelets (103). Many other clinical, pharmacodynamic, biologic, and genetic factors, however, such as tobacco use, drug interaction, alternate pathways for platelet activation, and genetic polymorphism or mutations of the COX-1 gene, may be involved (104). Currently, many questions regarding the biochemical mechanism, diagnosis, prevalence, clinical relevance, and optimal therapeutic intervention for aspirin resistance remain unanswered (97,104).

Triflusal

Triflusal (2-acetoxy-4-trifluoromethyl benzoic acid) is an antiplatelet drug that despite its structural similarity to aspirin (Fig. 26.16) exhibits quite different pharmacologic and pharmacokinetic properties (105). Unlike aspirin, 2-hydroxy-4-trifluoromethylbenzoic acid (HTB), the deacetylated metabolite of triflusal, retains significant antiplatelet activity. Triflusal is rapidly absorbed and metabolized. The area under the concentration–time curve for triflusal is 20.26 mg/L/h after a 900-mg dose, whereas that for HTB is 42.27 mg/L/h. Much of the pharmacokinetic data for triflusal activity are associated with HTB. The inhibition of COX, as measured by reduced production of thromboxane B₂, is 25% after 2 hours and 85% after 7 days with triflusal, whereas the effect of aspirin on thromboxane B₂ is more than 90% reduction after 2 hours and is maintained at this level after 7 days (106). It would appear that the presence of a 4-trifluoromethyl group also greatly enhances triflusal's ability to inhibit the activation of nuclear factor- κ B, which in turn regulates the expression of the mRNA of vascular cell adhesion molecule-1 (107) needed for platelet aggregation. In addition, triflusal increases nitric oxide synthesis in neutrophils, which results in an increased vasodilatory potential (108). Finally, an additional site of action for triflusal/HTB is the inhibition of cAMP phosphodiesterase, leading to increased levels of cAMP. Elevated cAMP levels decrease platelet aggregation through decreased mobilization of calcium. Aspirin and salicylic acid do not significantly increase cAMP levels.

Although recent trials comparing triflusal and aspirin for the prevention of vascular events in patients following a stroke revealed no significant differences between these two antiplatelet drugs, triflusal's use was associated with a significantly lower rate of hemorrhagic complications (105).

Indobufen

Many nonsteroidal anti-inflammatory drugs (NSAIDs) also inhibit TXA₂-dependent platelet function through a competitive, reversible inhibition of COX-1. At a conventional analgesic dosage, these drugs only inhibit COX-1 activity by 70% to 90%, which is inadequate for controlling platelet aggregation. Thus, unlike aspirin, most of

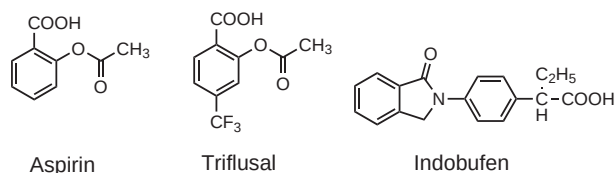


FIGURE 26.16 Cyclooxygenase-1 (COX-1) inhibitors.

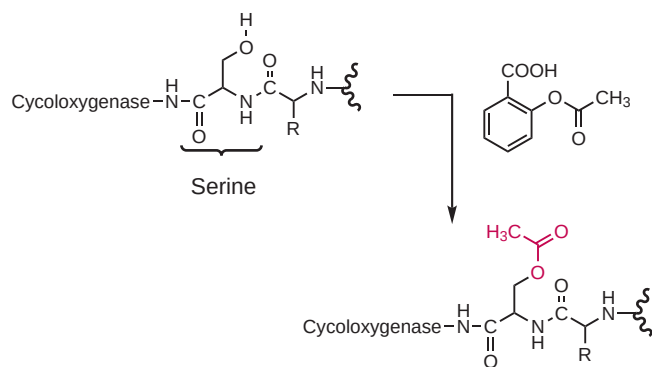


FIGURE 26.17 Irreversible acetylation of serine 530 in cyclooxygenase-1 by aspirin.

the clinically available NSAIDs are not used clinically for their antithrombotic properties.

In contrast, indobufen (Fig. 26.16), a reversible but very potent inhibitor of platelet COX-1 activity, was shown to have comparable clinical efficacy to that of aspirin in prevention of DVT after myocardial infarction and in blocking exercise-induced increase in platelet aggregation (109). In the secondary prevention of thromboembolic events, 100 or 200 mg of indobufen twice daily is as effective as warfarin or aspirin in patients with or without atrial fibrillation (110). Currently, indobufen is only available for routine clinical use in Europe.

Phosphodiesterase Inhibitors

Phosphodiesterase-3 (PDE3) is an enzyme responsible for degradation of cAMP to adenosine monophosphate in platelets and blood vessels. Selective cAMP PDE3 inhibitors, such as dipyridamole and cilostazol (Fig. 26.17), inhibit the degradation of cAMP, thereby increasing cellular concentration of cAMP and leading to inhibition of platelet aggregation and vasodilation (111).

Dipyridamole (Aggrenox)

Dipyridamole is a pyrimidopyrimidine derivative with vasodilatory and antiplatelet properties (Fig. 26.18). Dipyridamole exerts its antiplatelet function by increasing cellular concentrations of cAMP via its inhibition of the degrading enzyme, cyclic nucleotide PDE3. It also

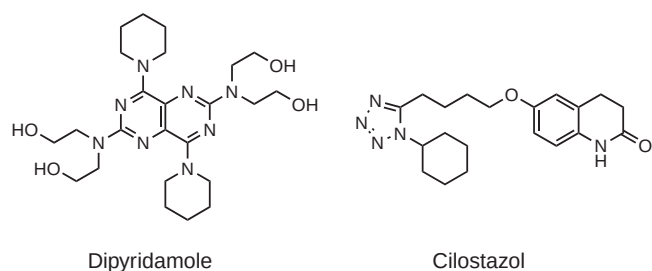


FIGURE 26.18 Chemical structures of phosphodiesterase inhibitors.

blocks adenosine uptake, which acts at A_2 adenosine receptors to stimulate platelet adenylyl cyclase. Less common uses for this drug include inhibition of embolization from prosthetic heart valves when used in combination with warfarin (the only currently recommended use) and reduction of thrombosis in patients with thrombotic disease when used in combination with aspirin. Alone, dipyridamole has little, if any, benefit in the treatment of thrombotic conditions (97,112).

Cilostazol (Pletal)

Cilostazol, a quinolinone derivative, is a potent orally active antiplatelet drug approved for the treatment of intermittent claudication (a peripheral artery disease resulting from blockage of blood vessels in the limbs). Cilostazol exhibits greater selectivity than dipyridamole as an inhibitor of PDE3A (Fig. 26.19) (112). The drug does not affect the other PDEs (PDEs 1, 2, or 4). Cilostazol reversibly inhibit platelet aggregation induced by a number of stimuli, such as thrombin, ADP, collagen, or stress from exercise (113,114). Additionally, cilostazol inhibits adenosine uptake, leading to increased activity of adenosine at A_1 and A_2 receptors. Adenosine's action on A_2 receptors in platelets increases cAMP levels, which, as previously indicated, leads to decreased platelet aggregation (115).

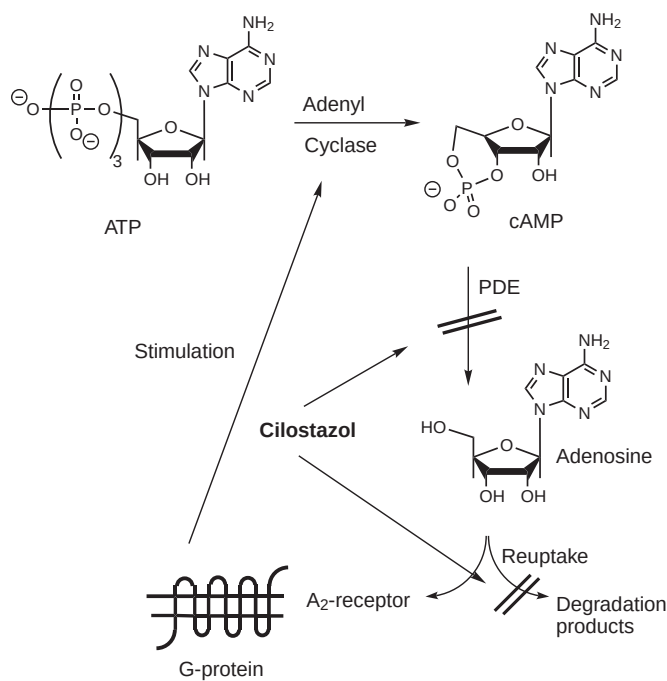


FIGURE 26.19 Sites of action of cilostazol, blocking phosphodiesterase 3A (PDE) and adenosine uptake, leading to increased levels of cyclic adenosine monophosphate (cAMP) directly by inhibition of cAMP breakdown and indirectly through adenosine binding to adenosine 2 receptors (A_2 -receptor), which, through G protein coupling, stimulate adenylyl cyclase.

Cilostazol is rapidly absorbed after oral administration, particularly with a high-fat meal, which greatly increases its bioavailability to approximately 90%. It is extensively metabolized in the liver by various cytochromes. The most important cytochromes appear to be CYP3A4 and, to lesser extent, CYP2C19, with an elimination half-life of approximately 11 to 13 hours. Among the various metabolites produced (11 metabolites are known), the two major metabolites are 3,4-dehydrocilostazol and 4'-trans-hydroxycilostazol (Fig. 26.20). These two metabolites are pharmacologically active. Studies indicate that the concomitant administration of cilostazol with CYP3A inhibitors can greatly increase cilostazol blood concentrations, and a dose reduction may be required (116). Similar results are seen when CYP2C19 is inhibited, leading to decreased formation of 4-trans-hydroxycilostazol and significant increases in cilostazol and 3,4-dehydrocilostazol (117).

Platelet P₂Y Purinergic Receptor

The crucial role that ADP plays in platelet activation and aggregation has been extensively investigated (98,118). The molecular targets of ADP in the platelet are G protein-coupled P₂Y purinergic receptors. There are three nucleotide receptors: P₂X₁, a cation channel receptor activated by ATP, and purinergic receptors P₂Y₁ and P₂Y₁₂, both of which are activated by ADP. Initial binding of ADP to the P₂Y₁ purinergic receptor (373-amino acid protein) induces platelet shape changes, causes intracellular calcium mobilization, and initiates aggregation. Subsequent binding of ADP to the P₂Y₁₂ purinergic receptors (342 amino acids) leads to sustained platelet aggregation by inhibiting adenylate cyclase and, thereby, decreasing cellular cAMP levels (119). The P₂Y₁₂ receptor is coupled to the G α_{12} G protein. The antithrombotic drugs ticlopidine and clopidogrel are irreversible antagonists of this P₂Y₁₂ purinergic receptor (118,120). The clinical relevance of this P₂Y₁₂ receptor as a new target for antiplatelet drug development of novel reversible antagonists has been extensively reviewed (120–122).

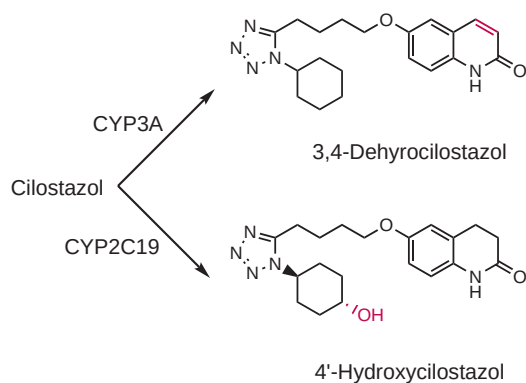
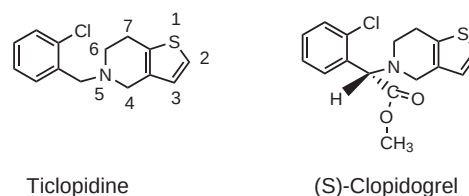


FIGURE 26.20 Metabolism of cilostazol.

Ticlopidine (Ticlid) and Clopidogrel (Plavix)



Ticlopidine and clopidogrel are thienopyridines, which through inhibition of platelet aggregation prolong bleeding time and delay clot retraction (97,122). The thienopyridines are prescribed for reduction of myocardial infarction and stroke, for treatment of peripheral arterial disease, and in combination with aspirin for acute coronary syndromes. This latter utility appears to result from the fact that both aspirin and the thienopyridines block major amplification pathways, leading to platelet aggregation and, thus, producing enhanced effectiveness. Ticlopidine has major safety concerns in that in a small population (1% to 2%), neutropenia occurs that is potentially fatal, may cause thrombotic thrombocytopenic purpura, and is largely replaced with clopidogrel. These same side effects are rare with clopidogrel. Additional side effects include diarrhea, nausea, vomiting, and skin rash (123).

The thienopyridine class exhibits selective inhibition of ADP-induced platelet aggregation. The actions of ticlopidine and clopidogrel appear to be irreversible in that there is still antiplatelet activity for 7 to 10 days after discontinuation of the medications (124). Support of the theory of irreversible inhibition of P₂Y₁₂ is provided by the observation that ticlopidine is not effective in blocking platelet aggregation *in vitro* when compared to the effect of the drug on the platelets of people taking ticlopidine. The thienopyridines function as prodrugs requiring cytochrome P450 activation. The thienopyridines are rapidly absorbed and extensively metabolized in the liver. The most significant metabolites of clopidogrel are shown in Figure 26.21. An inactive carboxylic acid represents the major circulating metabolite, while activation occurs through oxidation by CYP3A4 (with possible involvement of other cytochrome isoforms) giving rise to the 2-oxo derivative, which in turn is hydrolyzed to the thiol. The thiol binds irreversibly to P₂Y₁₂ by forming a disulfide bridge to a cysteine in P₂Y₁₂ (125–127). Specially, clopidogrel is thought to bind to Cys17 or Cys270 and, thus, block the binding of the agonist. In the case of ticlopidine, oxidative activation occurs, but additional metabolites have been identified, dihydrothienopyridinium (M5) and thienodihydropyridinium metabolites (M6). These short-lived metabolites may be responsible for the toxic adverse reactions (128).

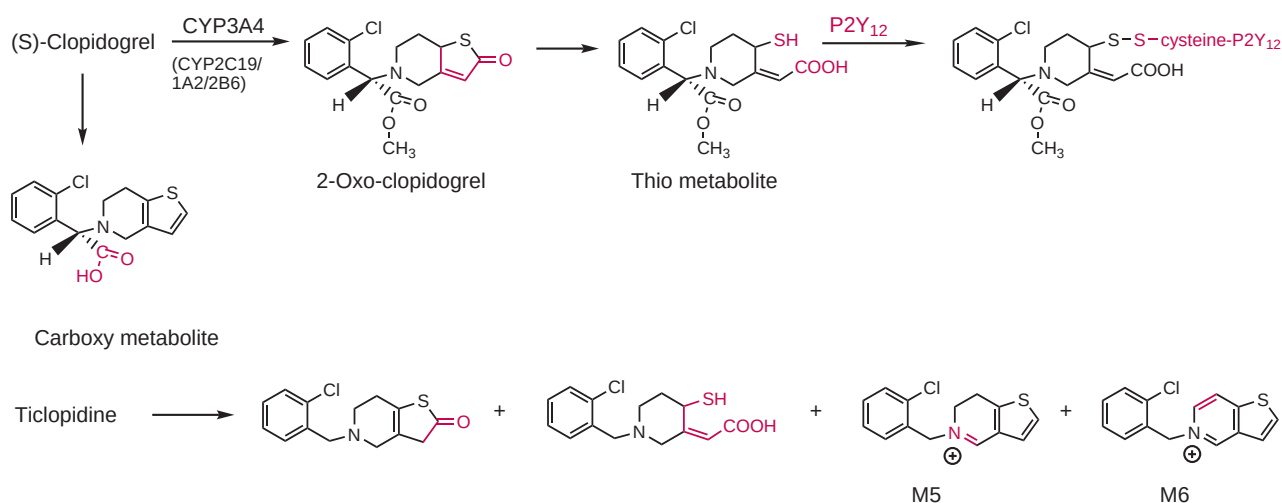
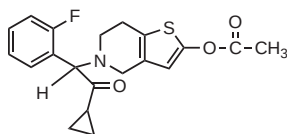


FIGURE 26.21 Metabolism of clopidogrel and ticlopidine. Irreversible reaction of clopidogrel with P2Y₁₂ is shown in red.

Prasugrel (Effient)



Prasugrel

Prasugrel has recently been approved for treatment of acute coronary syndrome when managed with percutaneous coronary intervention (129–131). The drug has also been shown to reduce the rate of thrombotic cardiovascular events in unstable angina and some forms of myocardial infarction. Prasugrel joins clopidogrel and ticlopidine in the prevention of platelet aggregation.

As with ticlopidine and clopidogrel, prasugrel is a prodrug that, upon activation, serves as an irreversible inhibitor of P2Y₁₂ receptors. Unlike the previous agents, prasugrel's activation involves hydrolysis to the thiolactone, which is then hydrolyzed to the thiol intermediate, which is the active metabolite and reacts irreversibly with P2Y₁₂ (Fig. 26.22) (122,132–134).

Several reports in the literature suggest a role for various cytochromes in this activation (CYP3A4 and CYP2B6), as is the case with clopidogrel and ticlopidine, but this does not appear to be true with prasugrel. R-138727, the active metabolite of prasugrel, is metabolically inactivated via S-methylation. Additional minor metabolites have been isolated, and they may be formed through CYP-catalyzed reactions.

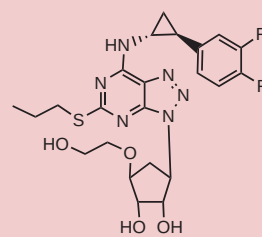
Prasugrel is rapidly absorbed following oral administration, reaching peak plasma concentration of R-138727 within 30 minutes. High-fat diets may slow the time to reach maximum plasma concentration, and it is recommended that the drug be taken on an empty stomach. R-138727 is highly bound to serum albumin (~98%) and

is excreted as the inactive metabolite via both the urine (68%) and fecal route (27%).

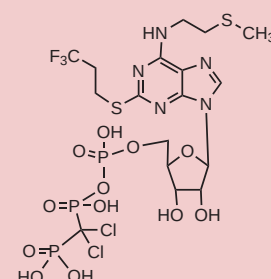
Prasugrel is commonly used in combination with aspirin and is reported to be more effective than clopidogrel in treatment of unstable angina or myocardial infarction. Prasugrel does have a higher rate of major bleeding and life-threatening bleeding than does clopidogrel (135).

REVERSIBLE P2Y PURINERGIC RECEPTOR INHIBITORS

Even though prasugrel has a faster onset and reduced interpatient variability than clopidogrel, newer agents are still needed because prasugrel works via an irreversible inhibition of P2Y₁₂ purinergic receptors and it appears to have a higher rate of bleeding incidences than clopidogrel. Ticagrelor and cangrelor are two reversible inhibitors of the P2Y₁₂ purinergic receptors in their advanced stages of development (122,136,137). Ticagrelor is the first oral reversible P2Y₁₂ antagonist that has improved pharmacokinetic and pharmacodynamic profiles than currently available drugs for treating acute coronary syndromes (138,139). Cangrelor, on the other hand, is being developed as a short-term, intravenous P2Y₁₂ antagonist with a very fast onset and offset, which may provide advantages over current drugs (99,136,140). Recently ticagrelor has been approved by the USFDA for prevention of thrombotic events in patients with acute coronary syndromes.



Ticagrelor (Brilinta)



Cangrelor

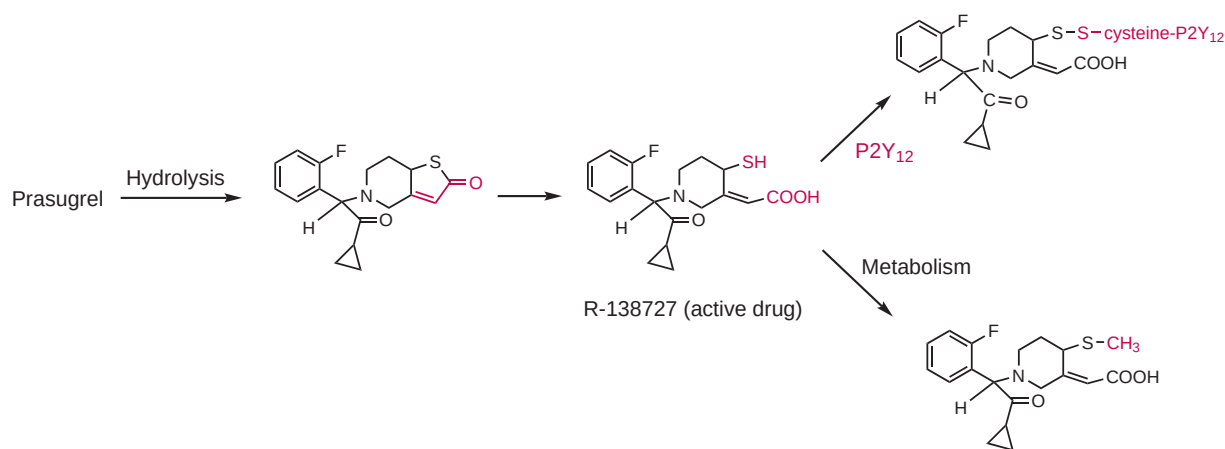


FIGURE 26.22 Activation of prasugrel to R-138727 followed by irreversible binding to P2Y₁₂ or metabolic methylation to inactive metabolite.

Glycoprotein II_b/III_a Receptor Antagonists

One of the newest groups of antithrombotic agents is the platelet receptor GPII_b/III_a antagonists (97,145,146). This novel class of compounds has been shown to provide more comprehensive inhibition of platelet aggregation than the usual combination of aspirin and heparin (140). The final common pathway in platelet aggregation is the expression of functional GPII_b/III_a (integrin α IIb β 3) receptors. These protein receptors are expressed regardless of the origin of the stimulus initiating the clotting cascade. The normal substrate for the GPII_b/III_a receptor is fibrinogen. One fibrinogen molecule acts to cross-link two platelets via binding to the GPII_b/III_a receptors on the platelet surfaces (Fig. 26.15). If the platelet surface receptors are occupied by another substrate that prevents fibrinogen binding and cross-linking, platelet aggregation will not occur. To this end, a number of novel compounds representing diverse structural groups have been prepared as GPII_b/III_a receptor antagonists (Fig. 26.23) (147,148). Included in this list of antagonists are monoclonal antibodies against the natural GPII_b/III_a receptor, naturally occurring peptides isolated from snake venom that contain the Arg-Gly-Asp (RGD) sequence, synthetic peptides containing either the RGD or Lys-Gly-Asp (KGD) sequences, and peptidomimetic and nonpeptide RGD mimetics that compete with fibrinogen and other ligands for occupancy of the receptor (97). The natural binding ligands, such as vWF and fibronectin, contain the natural RGD sequence.

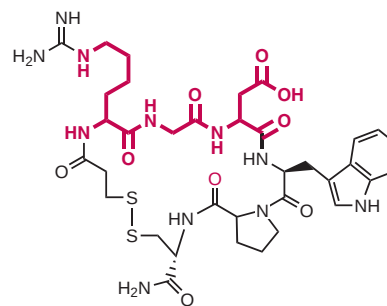
The GPII_b/III_a receptor antagonists are indicated in therapy for unstable angina, non-Q-wave myocardial infarction, and percutaneous coronary procedures. Like other antithrombotic agents, the main concern associated with GPII_b/III_a receptor antagonists is bleeding. Additionally, these drugs have been suggested to possibly increase the risk of thrombocytopenia (97,149). Although a number of orally active GPII_b/III_a receptor antagonists have been prepared and evaluated, their clinical efficacy

in acute treatment of patients with unstable angina and in those undergoing angioplasty has not been fully established (147).

Abciximab (ReoPro)

The initial antibodies against the GPII_b/III_a receptor were murine in origin. Because of concerns about the antigenicity of a pure murine antibody, a chimeric human-mouse 7E3 Fab was developed (150,151). This chimera, marketed as abciximab, is the clinically available form of the antibody that is widely studied in patients undergoing percutaneous coronary intervention (151). For an adult patient, the usual dosing scheme is 0.25 mg/kg as an intravenous bolus given 10 to 60 min before percutaneous coronary intervention, followed by the continuous infusion of 0.125 μ g/kg/minute for 12 hours to a maximum of 10 μ g/kg. Elimination of abciximab is biphasic. The initial phase has a half-life of 10 minutes, whereas the half-life of the second phase is approximately 30 minutes and results from platelet binding. Platelet function returns to normal within 48 hours after infusion, even though abciximab is bound to circulating platelets for approximately 2 weeks (Table 26.5) (152,153).

Eptifibatide (Integrilin)



Eptifibatide (red indicates KGD portion of drug)

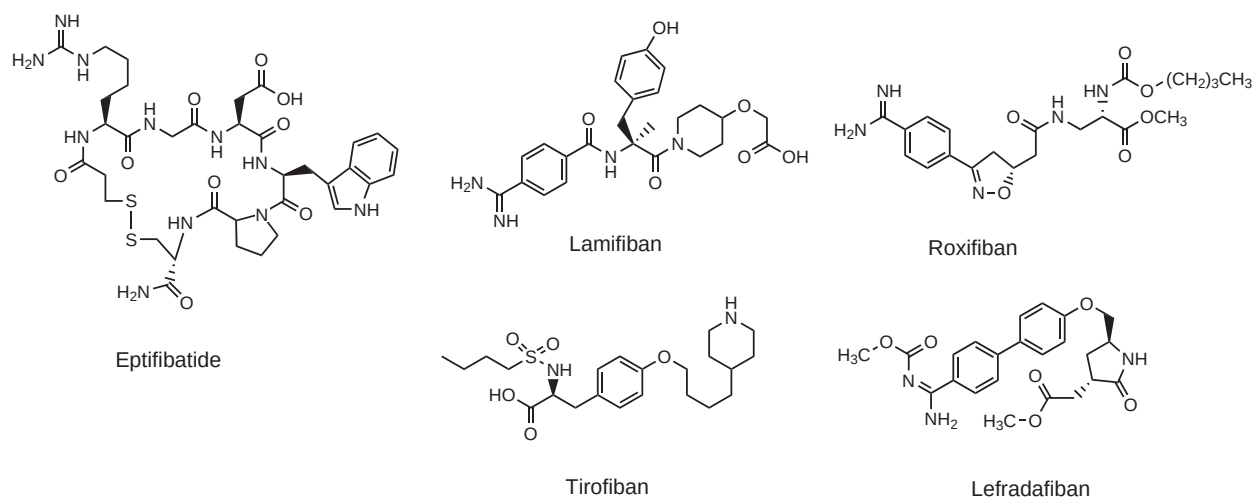


FIGURE 26.23 Structures of selected glycoprotein (GP) II_b/III_a receptor agonists.

Eptifibatide is a cyclic heptapeptide composed of six amino acids and one mercaptopropionyl residue. The cyclization is completed via a disulfide linkage between the cysteine and the mercaptopropionyl moieties. The lysine-glycine-aspartate component of eptifibatide is highly specific for the GPII_b/III_a receptor, with low binding affinity, as indicated by the rapid dissociation constant (Table 26.5). Because of this, eptifibatide is a reversible, parenterally administered antagonist of platelet aggregation.

The drug is eliminated primarily via renal mechanisms as eptifibatide and deaminated eptifibatide. The clinical importance of eptifibatide and its benefits in comparison with other therapeutic agents used in the treatment of acute coronary syndromes and percutaneous coronary intervention have recently been reviewed by Curran and Keating (154).

Tirofiban (Aggrastat)

Tirofiban is a member of a new class of antithrombotic agents known as the “fibans” (Fig. 26.23). These compounds have a structural similarity to disintegrin, which was originally isolated from snake venoms. The location of the $-\text{COO}^-$ and NH_3^+ in the fibans is identical to the distance between the same functional groups of the RGD loop of disintegrin, and as a result, the fibans

are able to effectively block the binding of fibrinogen to the GPII_b/III_a receptor in a reversible manner. Tirofiban, like eptifibatide, has a rapid dissociation constant (Table 26.5). Tirofiban is a peptidomimetic (nonpeptide) that is parenterally administered and exhibits a reduced risk of bleeding because of its shorter biologic half-life than abciximab (155). Additionally, it is less costly than other GPII_b/III_a receptor antagonists. The clinical use of tirofiban in ST elevation for myocardial infarction is effective and has an acceptable safety profile (155,156). The remaining fibans shown in Figure 26.23 are in various stages of development. Lamifiban is administered parenterally, whereas roxifiban and lefradafiban are used orally.

New Developments in Antiplatelet Drugs

Several newer approaches to antiplatelet drug development have been recently discovered (157). These include inhibitions of the vWF/GPI_b interaction, the platelet/collagen interaction, and the thrombin-induced platelet activation. Other approaches to platelet inhibition include the use of serotonin antagonists (because serotonin induces platelet aggregation), nitric oxide-donating antiplatelet agents, phosphodiesterase inhibitors, and inducers of adenylyl cyclase.

TABLE 26.5 Pharmacokinetic Properties of the Glycoprotein IIb/IIIa Receptor Antagonists

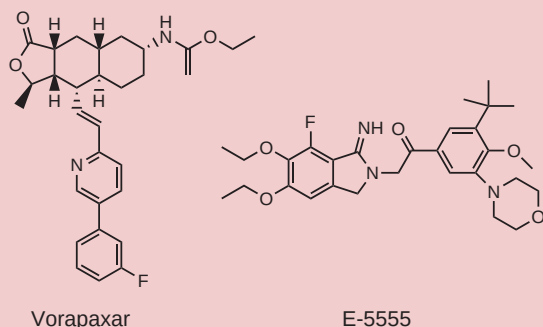
Drug	Trade Name	Molecular Weight (daltons)	Dissociation Constant (nmol/L)	Plasma Half-Life (h)	Protein Binding (%)
Abciximab	ReoPro	47,615	5	72	
Eptifibatide	Integrelin	800	120	4	25
Tirofiban	Aggrastat	495	15	3–4	65

PAR-1 ANTAGONISTS

Despite the effectiveness of aspirin, clopidogrel, and other newer P2Y₁₂ antagonists for the treatment and prevention of atherothrombotic events in patients with acute coronary syndrome, myocardial infarction, and other existing peripheral arterial diseases, the risk of bleeding events, potential drug resistance, and other factors contributing to the interpatient variability remain a therapeutic challenge in these patients (99). Two novel oral antiplatelet agents that inhibit thrombin-mediated platelet activation via G protein-linked PAR-1 antagonists have recently advanced to pivotal phase III clinical trials (100,101,141).

Vorapaxar is a potent, highly selective, himbacined-based PAR-1 antagonist that has greater than 90% oral bioavailability and is well tolerated in patients undergoing nonurgent percutaneous coronary intervention (141,142). It binds competitively with high affinity ($K_i = \sim 8$ nmol/L) yet in a slowly reversible manner to PAR-1 (142). Vorapaxar is primarily metabolized and eliminated via biliary and gastrointestinal routes by CYP3A4 isozymes at a very slow rate (141). Preliminary clinical trial data further indicate that vorapaxar, unlike prasugrel, improves ischemic outcomes without significantly increasing the bleeding liability (99–101).

Another PAR-1 antagonist that has demonstrated similar antiplatelet effects without increasing bleeding times is E-5555 (143). Interestingly, E-5555 also inhibits thrombin-induced release of interleukin-6 (half maximal inhibitory concentration [IC_{50}] = 0.19 nmol/L) and P-selectin expression (IC_{50} = 16 nmol/L) in human coronary artery smooth muscles, thereby preventing vascular spasm in an animal model studies (144).



Specific inhibitors of thromboxane synthase have been tested for their ability to block this enzyme without inhibiting the entire arachidonic acid cascade (99,158). This allows an accumulation of prostaglandin G₂ and prostaglandin H₂, which can then be channeled into an increased production of PGI₂, which also has antithrombotic activity.

THROMBOLYTIC DRUGS

Early application of reperfusion therapy with thrombolytic agents has significantly improved the outcomes of acute myocardial infarction and other conditions, such

as PE, DVT, arterial thrombosis, acute thrombosis of retinal vessel, extensive coronary emboli, and peripheral vascular thromboembolism (159).

Although, the beneficial effect of the thrombolytic drugs, such as streptokinase and urokinase, for dissolving newly formed thrombus in patients with acute myocardial infarction were first reported in 1958, their full acceptance into clinical practice was not realized until the early 1980s as a result of several prospective, randomized, controlled trials. Today, with the approval of many second- and third-generation thrombolytic drugs, thrombolytic therapy has become a standard treatment for patients presenting with acute myocardial infarction or stroke (159,160).

Mechanism of Action

Normally, newly formed blood clots (fibrin) are dissolved by the actions of the fibrinolytic system, the purpose of which is the removal of unwanted clots without damaging the integrity of the vascular system. This system works via a relatively nonspecific protease enzyme called plasmin, the function of which is to digest fibrin (the very last step of the coagulation cascade) (Fig. 26.1). The lack of substrate specificity of plasmin is illustrated by the fact that it degrades fibrin clots as well as some plasma proteins and coagulation factors.

The fibrinolytically active plasmin is produced from the circulating inactive “proenzyme” plasminogen following the cleavage of a single peptide bond by a group of trypsin-like serine proteases known as the plasminogen activators. The principal activator, tissue-type plasminogen activator (tPA), is released from the vascular endothelium. Thrombolytic drugs, such as streptokinase and urokinase, act like a plasminogen activator that converts this proenzyme to the active plasmin. Endogenously, plasmin activity is regulated by two specific inactivators known as tPA inhibitors 1 and 2.

SNAKE VENOM-INDUCED COAGULOPATHY

The venom of snakes of the Crotalinae family, which includes rattlesnakes, produces a state of impaired coagulation. This can lead to both local and systemic hemorrhagic events. Venom consists of many components, including phospholipases and hemolysins, which cause cell lysis by disrupting platelet and red blood cell membranes. The venoms also contain procoagulant components that induce the formation of intravascular clots as well as hemorrhagins that destroy vascular integrity. Typically, both PT and aPTT are elevated with total fibrin levels being lowered and fibrin degradation products being increased. All of the findings are consistent with a consumptive coagulopathy. Because of the multiple mechanisms affecting coagulation, coagulopathies caused by snake venom are best managed using antivenin.

First-Generation Thrombolytic Agents

Streptokinase (Streptase)

Streptokinase, an exocellular protein produced by several strains of β -hemolytic streptococci, is a drug of choice for thrombolytic therapy based on its cost-effectiveness consideration, and it is the only thrombolytic drug approved by the FDA for peripheral vascular disease (161). The drug is approved for treatment of myocardial infarction but rarely is used for this condition today, having been replaced with the fibrin-specific agents discussed below (162).

MECHANISM OF ACTION Streptokinase is a protein purified from culture broths of group C β -hemolytic streptococci bacteria. Streptokinase contains a single polypeptide chain of 414-amino acid residues with a molecular weight of 47 kD (161). Streptokinase by itself has no intrinsic enzymatic activity. To be active, it must bind with plasminogen to form an activator complex (1:1 complex). This complex then acts to convert uncomplexed plasminogen to the active fibrinolytic enzyme, plasmin. The streptokinase/plasminogen complex not only degrades fibrin clots but also catalyzes the breakdown of fibrinogen and factors V and VII (159,161). As a result, streptokinase is considered to be a fibrin-nonspecific drug.

PHARMACOKINETICS Unfortunately, the half-life of the activator complex is less than 30 minutes, which frequently is too short to completely lyse a thrombus. Anistreplase (APSAC; Eminase) is a 1:1 streptokinase/lysine-plasminogen complex that has been acylated with an anisoyl group at the active-site serine within the lysine-plasminogen. Anistreplase is inactive as such, but following complexation with fibrin, the anisoyl group is slowly cleaved, exposing the active site and, thus, leading to degradation of fibrin. The prodrug nature of anistreplase exhibits an improved pharmacokinetic profile, with anistreplase acting as a semiselective lysis agent at the clot site. The inactivity of the circulating anistreplase also allows this drug to be given as a very rapid intravenous infusion (typically, 30 U over 3 to 5 minutes). Tissue reperfusion following anistreplase therapy compares favorably to streptokinase because of the extended half-life (90 minutes).

SIDE EFFECTS Because it is a foreign protein, streptokinase is associated with significant hypersensitivity reactions. Most people have, at some point in their lives, had a streptococcal infection and, therefore, have developed circulating antistreptococcal antibodies. These antibodies frequently are active against streptokinase as well. The response of the streptokinase to these antibodies can vary widely, from inactivation of the fibrinolytic properties of the protein to rash, fever, and rarely, anaphylaxis. Significant allergic reactions to streptokinase occur in approximately 3% of patients.

Urokinase (Abbokinase)

Urokinase is an enzyme with the ability to directly degrade fibrin and fibrinogen. It is now isolated from

cultures of human fetal kidney cells and is composed of two polypeptide chains with molecular weights of 32 and 54 kD. This method of isolation is much more efficient than the original isolation of urokinase from human urine (163). Because of its source, the human body does not see urokinase as a foreign protein. Therefore, it lacks the antigenicity associated with streptokinase and frequently is used for patients with a known hypersensitivity to streptokinase (164). Plasmin cannot be used directly because of the presence of naturally occurring plasmin antagonists in plasma. No such inhibitors of urokinase exist in the plasma, however, allowing this enzyme to have clinical utility. Even so, urokinase is much more expensive (threefold the price of streptokinase) and has an even shorter half-life (15 minutes). Urokinase also has other fibrin-nonspecific actions similar to streptokinase.

Currently, urokinase is only approved for treatment of PE.

Second-Generation Thrombolytic Agents

Alteplase (Activase)

Alteplase (tPA) is a serine protease with a low affinity for free plasminogen but a very high affinity for the plasminogen bound to fibrin in a thrombus (fibrin-specific agent) (Fig. 26.24). Both streptokinase and urokinase lack this specificity (i.e., are nonspecific) and act on free plasminogen, inducing a generalized thrombolytic state. Alteplase also has a greater specificity for older clots compared with newer clots relative to streptokinase and urokinase. Alteplase was originally isolated from cultures of human melanoma cells but is now produced commercially using recombinant DNA technology. Alteplase is unmodified human tPA, whereas reteplase (see below) is human tPA that has had several specific amino acid sequences removed (165). At low doses, alteplase is quite selective for degrading fibrin without concomitant lysis of other proteins, such as fibrinogen. At the higher doses currently used therapeutically, however, alteplase activates free plasminogen to some extent and, therefore, can cause hemorrhage. Many of the therapeutic indications for the other thrombolytic agents also are indications for alteplase (i.e., myocardial infarction, massive PE, and acute ischemic stroke). The half-life of alteplase is very short (~5 minutes), necessitating its administration as a 15-mg intravenous bolus, followed by a 85-mg intravenous infusion over 90 minutes, or as 60 mg infused over the first hour, with the remaining 40 mg given at a rate of 20 mg/h.

Prourokinase (scuPA, r-ProUK)

Prourokinase is a single-chain, urokinase-like plasminogen activator of 411 amino acids that displays clot-lysis activity yet does not interfere with hemostasis (166). It is nonimmunogenic and has a more favorable dose-related safety and efficacy profile than both urokinase and streptokinase. Thus, it is a potentially useful thrombolytic drug in the treatment of peripheral vascular occlusion (167).

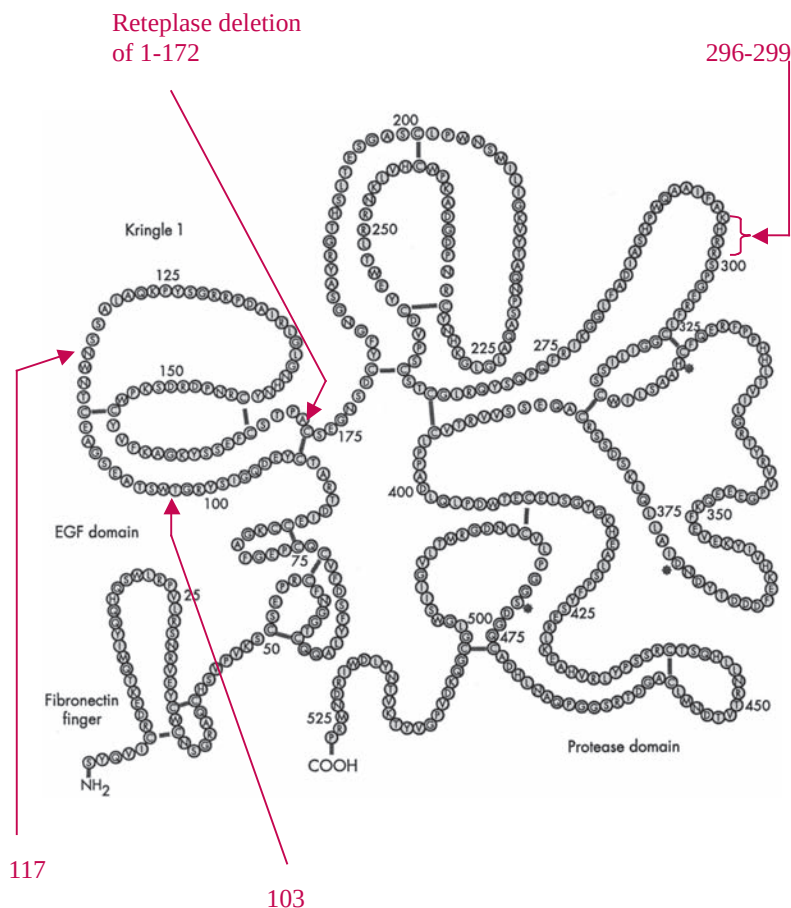


FIGURE 26.24 Schematic diagram of alteplase, reteplase (removal of amino acids 1 to 172), and tenecteplase in which threonine (T) at position 103 is replaced with asparagines, asparagine (N) at position 117 is replaced with glutamine, and lysine (K)-histidine-arginine-asparagine at positions 296 to 299 are replaced with four alanines. (Adapted with permission from Nordt TK, Bode C. Thrombolysis: newer thrombolytic agents and their role in clinical medicine. *Heart* 2003;89:1358–1362.)

Third-Generation Thrombolytic Agents

Many third-generation thrombolytic agents currently are under clinical trials. These agents are derived from structural modifications of the basic plasminogen activators (tPA or other tPA of animal origins) using technologies such as mutations, conjugation with monoclonal antibodies, or hybridization with another thrombolytic agent. Some of these agents are amideplase (hybrid of tPA and prourokinase), lanoteplase (mutant tPA), and staphylokinase (from bacterial tPA) (160).

Reteplase (Retavase)

Reteplase is a recombinant deletion mutant of tPA lacking the finger, epidermal growth factor, kringle 1 domain, and carbohydrate side chain (Fig. 26.24) (168). As a highly fibrin-specific thrombolytic agent, reteplase is missing the first 172 amino acids that are present in alteplase and has 355 amino acids with a molecular weight of 39 kd. Because of the removal of the finger kringle 1 domain, reteplase binding to fibrin is reduced from that of alteplase, and reteplase has reduced fibrin selectivity. In addition, the structural modification reduces hepatic elimination, leading to a

longer half-life (reteplase, 14 to 18 minutes; alteplase, 3 to 4 minutes).

Administered as a double bolus of 10 U every 30 minutes, reteplase is approved for use in acute myocardial infarction.

Tenecteplase (TNKase)

Tenecteplase is composed of 527 amino acids with 17 disulfide bridges. It differs structurally from alteplase by three point mutations (Fig. 26.24). The mutations were bioengineered to occur at amino acid 103, where threonine (T) is replaced by asparagine; at amino acid 117, where asparagine (N) is replaced by glutamine; and at amino acids 296 to 299, where lysine (K)-histidine-arginine-arginine are replaced with four alanines. Thus, the name TNK is derived from the mutations. The replacement of these amino acids along with their attached carbohydrate side chains results in a prolonged half-life (~17 minutes) and allows a single bolus application (168). These point-mutation changes also change the binding of tenecteplase to plasminogen activator inhibitor-1 (PAI-1) by 80-fold, thus improving activity. A physiologic enzyme, PAI-1 inhibits fibrinolysis. Finally,

tenecteplase shows a 15-fold higher fibrin specificity. The drug is still eliminated via hepatic mechanisms.

TOXICITY OF ANTITHROMBOTICS AND THROMBOLYTICS

Antithrombotic Toxicity

Recall that warfarin exhibits its anticoagulation effects by preventing γ -carboxylation of specific glutamate residues necessary for vitamin K–dependent coagulation (Fig. 26.4). However, γ -carboxyglutamate proteins are not unique to coagulation factors. These types of proteins are synthesized in bone as well. As would be expected, warfarin also interferes with the carboxylation of these proteins, resulting in an inhibition of the effects of vitamin K on osteoblast development. It has been suggested that this is the mechanism responsible for bone abnormalities in neonates born to mothers who were treated with warfarin while pregnant (169). No evidence suggests that bone metabolism or development is affected by warfarin when the drug is administered to children or adults. Because of the mechanism of action of the warfarin-like drugs, the management of their toxicity is based largely on vitamin K therapy (see “Coagulants,” later in this chapter).

Unlike warfarin, heparin is safe for anticoagulant therapy during pregnancy (169). Although warfarin is known to cause serious fetal malformations when used in pregnancy, heparin does not cross the placental barrier and has shown no tendency to induce fetal damage. Furthermore, heparin does not increase fetal mortality or prematurity. To minimize the risk of postpartum hemorrhage, it is recommended that heparin therapy be withdrawn 24 hours before delivery.

Despite its safety in pregnancy, several potential problems are associated with heparin therapy. Because heparins (high molecular weight heparins and LMWHs) are isolated from animal sources, the chance of antigenic hypersensitivity exists but is rarely observed. Heparin competitively binds many other plasma proteins (i.e., vitronectin and PF4) in addition to the antithrombin, resulting in inactivation of the heparin as an anticoagulant (44). This may be the reason for heparin resistance and for a serious condition known as HIT. Typically, this condition occurs 7 to 14 days after initiation of heparin therapy, but it may occur earlier in some patients who have had previous exposures to heparin. In these cases, heparin-induced platelet aggregation occurs and may result in the production of antiplatelet antibodies. Development of this condition necessitates termination of heparin therapy and institution of antiplatelet drugs or oral anticoagulants. On withdrawal of heparin, the thrombocytopenia usually is reversible. Mild increases in liver function tests frequently are associated with heparin therapy. Long-term use of full therapeutic doses of heparin (>20,000 U/d for 3 to 6 months) has been associated with osteoporosis, and spontaneous vertebral fractures have been infrequently reported (44).

Hemorrhagic complications of heparin therapy are managed, in part, with the specific antagonist protamine

sulfate (44). This agent is discussed in greater detail in the “Coagulants” section of this chapter.

The structural and mechanistic diversity of the antiplatelet drugs disallows a cohesive description of their toxicities. Hemorrhage is certainly a concern, but other, more drug-specific toxicities may be of greater immediate concern. It is suggested that toxicity information for antiplatelet medications be obtained from appropriate references for the specific agent in question and from the most recent reviews of antiplatelet agents (97,99).

Thrombolytic Toxicities

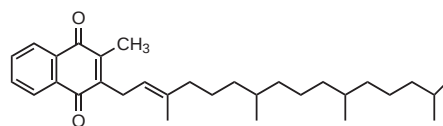
As discussed earlier, plasmin, because of its lack of specificity, not only digests fibrin but also degrades many other plasma proteins, including several coagulation factors and the anticoagulating factor, activated protein C. Thus, as expected, most thrombolytic drugs not only attack pathologic clots but also exert their actions on any other site of compromised vascular integrity. The dissolution of necessary clots results in the principal side effect of thrombolytic therapy, hemorrhage. Its action on the activated protein C also may be responsible for their neurovascular toxicities (170).

Multiple studies have examined the incidence of life-threatening hemorrhage (i.e., intracranial hemorrhage) with the various thrombolytic medications. These studies indicate that the rate of significant hemorrhagic complication is essentially the same (0.1% to 0.7%) regardless of the specific therapeutic agent used. Supportive care is indicated in cases of thrombolytic toxicity. No specific antagonist exists to manage thrombolytic medication-induced hemorrhage, but antifibrinolytic drugs, such as aminocaproic acid and tranexamic acid, often are used. These compounds are described in detail in the following “Coagulants” section of this chapter.

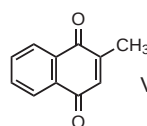
COAGULANTS

A variety of pathologic and toxicologic conditions can result in excessive bleeding from inadequate coagulation. Depending on the etiology and severity of the hemorrhagic episode, several possible blood coagulation inducers can be therapeutically employed.

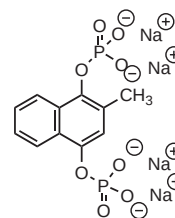
Vitamin K



Vitamin K₁



Vitamin K₃



Vitamin K₄

Because the orally active anticoagulants, such as warfarin and the indandiones, act through interruption of the normal actions of vitamin K, it stands to reason that vitamin K should be effective in the treatment of bleeding induced by these agents (157). Vitamin K₁ (phytonadione, Mephyton) is the form of vitamin K most often used therapeutically. Vitamin K₁ is safe for use in infants, pregnant women, and patients with glucose-6-phosphate deficiency. Furthermore, phytonadione, being more lipid soluble, has a faster onset than other vitamin K preparations and requires smaller doses than vitamin K₃ (menadione) or vitamin K₄ (menadiol sodium diphosphate). Both vitamins K₃ and K₄ may produce hyperbilirubinemia and kernicterus in neonates as well as hemolysis in neonates and glucose-6-phosphate-deficient patients. In fact, the only advantage of vitamins K₃ and K₄ over vitamin K₁ is that whereas absorption of vitamin K₁ requires the presence of bile, absorption of vitamins K₃ and K₄ does not, because they are absorbed via a passive process directly from the intestine (157). This may be a slight advantage for patients with cholestasis or severe pancreatic dysfunction. Only vitamin K₁, however, is appropriate therapy for bleeding associated with warfarin and superwarfarin anticoagulation. Vitamin K₂ is not used therapeutically.

Vitamin K₁ is effective at inducing coagulation when administered orally, subcutaneously, intramuscularly, or intravenously. Although the oral route is preferred, it is not always practical in a patient who is critically hemorrhaging. The other routes of administration, though used clinically, all have significant potential drawbacks. Larger doses (e.g., volume >5 mL) are not appropriate for subcutaneous administration, and intramuscular injection generally is avoided in patients who are at risk for significant hematoma formation (e.g., hemophiliacs). Intravenous dosing of vitamin K has been associated with severe anaphylactoid reactions (including death) presumably secondary to colloidal formulation.

The half-life of vitamin K₁ is quite short—only 1.7 hours via the intravenous route and 3 to 5 hours via the oral route. When given orally, vitamin K₁ is absorbed directly from the proximal small intestine in an energy-dependent and saturable process that requires the presence of bile salts. These kinetic features argue for administration in divided doses rather than larger, single daily doses. The typical starting point for adults with drug-induced hypoprothrombinemia is 2.5 to 10 mg of vitamin K₁ orally, repeating in 12 to 48 hours if needed. In cases of ingestion of long-acting superwarfarin rodenticides (e.g., brodifacoum), therapy may be 125 mg/d for weeks or months. Practically speaking, because vitamin K₁ is dispensed as 5-mg tablets, superwarfarin-poisoned patients may require 10 to 30 tablets every 6 hours.

Because of the short half-life of vitamin K₁, dosing must be repeated two to four times per day for the duration of treatment. Furthermore, regardless of the route of administration, coagulant effects are not evident for up to 24 hours. Because of this delay in onset, severe acute hemorrhage is better managed initially with intravenous infusion of fresh-frozen plasma, followed by vitamin K therapy.

Protamine

Mechanism of Action

Protamine sulfate has been approved in the United States as a specific antagonist to heparin since 1968 (169). Protamines are an arginine-rich, highly basic group of simple proteins derived from salmon sperm. The highly acidic heparin polysaccharides exhibit their anticoagulant activity through binding to antithrombin III. Because of the basicity of protamine, heparin has an increased affinity for protamine relative to antithrombin III. In fact, its binding affinity for protamine is so much greater than that of antithrombin III that protamine actually will induce dissociation of the heparin/antithrombin III complex. If protamine is administered in the absence of heparin, it can have marked effects on coagulation. Protamine is not completely selective for heparin and, *in vivo*, also interacts with fibrinogen, platelets, and other plasma proteins causing anticoagulation. For this reason, the minimal amount of protamine necessary to antagonize heparin-associated bleeding should be used (usually 1 mg of protamine intravenously for every 100 U of heparin remaining in the patient).

Side Effects

Anaphylaxis also has been associated with the use of protamine. Although development of protamine anaphylaxis is not limited to diabetics, those patients with diabetes who have used protamine-containing insulin (NPH or protamine zinc) do have a slightly increased risk of anaphylaxis. Some less common reactions to protamine include pulmonary vasoconstriction, hypotension, and thrombus formation.

Thrombopoietin Receptor Agonists

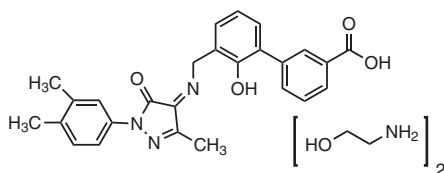
Chronic idiopathic thrombocytopenia purpura (ITP), unlike HIT, is an autoimmune disorder characterized by persistent thrombocytopenia that involves antibody-mediated platelet destruction and decreased platelet production leading to life-threatening bleeding (171,172). The disease is characterized by platelet counts of less than 30,000 platelets/ μ L (normal platelet count, ~150,000 to 450,000 platelets/ μ L). Successful treatment is indicated by platelet counts of 200,000 platelets/ μ L. The diagnosis and management of this autoimmune disorder is a challenge to clinicians because the pathophysiology of ITP is not completely understood (171,173). Recent evidence supports the coexistence of antibody-induced megakaryocyte abnormalities that impair proliferation and differentiation of megakaryocytes, decreased proplatelet formation, and subsequent platelet release in ITP patients (173).

Romiplostim and eltrombopag are two thrombopoietin receptor agonists that have recently been approved for the short-term treatment of thrombocytopenia in patients with chronic ITP (174,175). These drugs work by binding to the transmembrane domain of the human thrombopoietin receptor (c-Mpl) on the platelet surface as well as megakaryocytes and megakaryocyte precursor cells, thereby activating a number of signaling pathways, including tyrosine phosphorylation of c-Mpl, Janus kinase 2, and signal transduction and activation of transcription, which results in an increase in the synthesis of new platelets (174,176).

Romiplostim (Nplate)

Romiplostim is an Fc-peptide fusion protein analog of thrombopoietin with a molecular weight of 59 kd that is administered subcutaneously in a dose range of 3 to 15 $\mu\text{g/kg/wk}$, based on the patient's platelet count (174,177). It was approved in 2003 by the FDA as an orphan drug and subsequently in 2008 for long-term treatment of adult chronic ITP, especially in relapsed postsplenectomy patients (177). Romiplostim is generally well tolerated with many major adverse events that could be attributed to the drug (178). The serum half-life of romiplostim ranges from 1 to 34 days (median, 3.5 days) following subcutaneous injection.

Eltrombopag (Promacta)



Eltrombopag

Eltrombopag has been approved for the treatment of thrombocytopenia in patients with chronic immune ITP. It is the first orally active low molecular weight agent approved in 2008 (175,179).

Eltrombopag is rapidly absorbed, reaching maximum plasma concentrations in 3 to 6 hours, and is highly bound to plasma protein (~99%). Coadministration of eltrombopag with heavy metals found in antacids results in reduced (~70%) systemic exposure, whereas the fat content of food does not affect eltrombopag absorption. The drug should be administered on an empty stomach. Eltrombopag is metabolized in the liver, although only minor CYP-catalyzed oxidation is reported, and some glucuronidation also occurs. The half-life of eltrombopag is approximately 26 to 35 hours, with elimination occurring through the feces (~59%) and urine (~31%). Systemic exposure to eltrombopag is reported to be increased in individuals of East Asian descent, and the initial dose should be decreased in these patients.

Eltrombopag is reported to cause hepatotoxicity, and patients must have liver function tests done every 2 weeks during initial drug therapy and monthly after establishment of a maintenance dose. The drug is only available in pharmacies that are part of the PROMACTA CARES program, and patients must be registered with this program.

Antifibrinolytic Agents

Mechanism of Action

Control of a variety of fibrinolytic states can be achieved using a number of synthetic antifibrinolytic agents, such as tranexamic acid (Cyklokapron) and ϵ -aminocaproic acid (Amicar), that completely inhibit plasminogen activation. Plasmin binds to fibrin through a lysine binding site to activate the final stages of fibrinolysis (Fig. 26.1). Aminocaproic acid, a lysine analog, and tranexamic acid

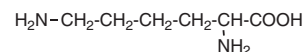
are antifibrinolytic agents with high affinity for the five lysine binding sites of plasminogen, thus effectively competing and preventing the binding of plasmin to fibrin.



Aminocaproic acid



Tranexamic acid



Lysine

Pharmacokinetics

Both ϵ -aminocaproic acid and tranexamic acid are readily absorbed when administered orally. They also can be given intravenously, although significant hypotension can result if the infusion is given too quickly. Elimination of the drugs is primarily renal, with little metabolism taking place. The half-lives of ϵ -aminocaproic acid and tranexamic acid are each approximately 2 hours.

Therapeutic Use

These drugs find clinical utility in settings such as prevention of rebleeding in intracranial hemorrhages, as adjunctive therapy in hemophilia, and of course, in treatment of bleeding associated with fibrinolytic therapy. In most bleeding conditions, however, ϵ -aminocaproic acid therapy has not been shown to be of definitive benefit. In recent trials, tranexamic acid was found to reduce red cell transfusion better than ϵ -aminocaproic acid or placebo in patients undergoing liver transplantation (180).

Side Effects

The major risk associated with ϵ -aminocaproic or tranexamic acid therapy is intravascular thrombosis as a direct result of the inhibition of plasminogen activator. Thrombi that form during therapy are not easily lysed and, therefore, can have additional ischemic consequences. Additional possible complications include hypotension, abdominal discomfort, and rarely, myopathy and muscle necrosis.

Aprotinin (Trasylo)

Operative procedures, such as heart valve replacement, frequently have effects on platelet function and endogenous coagulation factors (181). These effects may result in significant peri- or postoperative bleeding. Aprotinin is a serine protease inhibitor that blocks kallikrein and plasmin and provides some protection to platelets from mechanical injury. The inhibition of fibrinolysis results in profound antihemorrhagic effects (181). Side effects of aprotinin therapy usually are minor, but anaphylaxis has possibly been implicated in a small population (<0.5%). For this reason, it is suggested that a small test dose be given before initiation of the therapeutic infusion.

Plasma Fractions

Spontaneous bleeding can result from dysfunction or deficiencies of specific coagulation factors. A list of coagulation factors and deficiency states is given in Table 26.6.

TABLE 26.6 Clotting Factors

Factor	Common Name	Deficiency State	Source	Half-Life of Infused Factor (days)	Target for Action of
I	Fibrinogen	Afibrinogenemia, defibrination syndrome	Liver	4	
II	Prothrombin	Prothrombin deficiency	Liver (requires vitamin K)	3	Heparin (IIa), warfarin (synthesis)
III	Tissue thromboplastin, thrombokinase, tissue factor		Liver (may require vitamin K)		
IV	Calcium (Ca ²⁺)				
V	Proaccelerin, labile factor	Factor V deficiency	Liver	1	
VI	Deleted factor				
VII	Proconvertin, stable factor	Factor VII deficiency	Liver (requires vitamin K)	0.25	Heparin (VIIa); warfarin (synthesis)
VIII	Antihemophilic A factor (AHF), antihemophilic globulin (AHG)	Hemophilia A (classic) von Willebrand disease	Liver	0.5 Unknown	
IX	Antihemophilic B factor, plasma thromboplastin component (PTC), Christmas factor	Hemophilia B (Christmas disease)	Liver (requires vitamin K)	1	Heparin (IXa); warfarin (synthesis)
X	Stuart or Stuart-Prower factor	Stuart-Prower defect	Liver (requires vitamin K)	1.5	Heparin (IXa); warfarin (synthesis)
XI	Plasma thromboplastin antecedent (PTA)	PTA deficiency	Unknown	3	
XII	Hageman factor, contact factor	Hageman defect	Unknown	Unknown	
XIII	Fibrin-stabilizing factor, fibrinase	Fibrin-stabilizing factor deficiency	Unknown	6	
	Fletcher factor, prekallikrein factor		Liver		
	Fitzgerald factor, high molecular weight kininogen		Liver		
Antithrombin III Proteins C & S Plasminogen		Antithrombin III deficiency		3	Warfarin (synthesis) Thrombolytic enzymes, aminocaproic acid

Spontaneous bleeding usually occurs when the activity of coagulation factors falls below 5% of normal. Typically, these deficiencies are the result of a chronic disease state, such as von Willebrand disease or hemophilia. Management of an acute hemorrhagic event in a coagulation factor-deficient patient includes administration of the appropriate factors in concentrated form. The most common inherited clotting factor deficiencies involve factor VIII (classic hemophilia A) and factor IX (hemophilia B, or Christmas disease).

Two forms of factor VIII concentrate are clinically available, cryoprecipitate and lyophilized factor VIII concentrate. Cryoprecipitate is a factor VIII-rich plasma protein fraction prepared from whole blood that also contains approximately 300 mg of fibrinogen per unit. Immediately before infusion, the required number of cryoprecipitate units are thawed in a sterile saline/citrate solution and pooled. The lyophilized factor VIII concentrates are prepared from large plasma pools and also are rich in fibrinogen. Lyophilized factor VIII concentrates are not useful in therapy for von Willebrand disease, because during the extraction and lyophilization process, the polymeric structure of factor VIII in the von Willebrand protein that supports platelet adhesion is destroyed, rendering the preparation inactive. Because of the pooling of blood from multiple donors in the preparation of lyophilized factor VIII concentrates, it generally is held that cryoprecipitate, which is isolated from a single donor, is safer.

The major concern associated with the use of concentrated clotting factors is the risk of viral transmission (primarily HIV and hepatitis B). This fear has somewhat attenuated the use of concentrated plasma fractions, even in diseases such as hemophilia. Ultrapure factor VIII concentrates produced using recombinant DNA technology have been approved for use. Frequently, however, the expense of these recombinant agents is the reason why the more traditional plasma isolates are used—despite the possibility of viral transmission.

Lyophilized preparations of prothrombin, factor IX, and factor X also are available. The manufacturing process involves plasma extraction with solvents and detergents that renders the preparations virally inactive but still able to activate clotting factors. To prevent excessive thrombus formation in these situations, heparin often is added to the therapeutic regimen.

At times, a hemorrhagic event is possible, but the patient does not require immediate coagulation therapy. For example, if a patient with mild hemophilia A needs to have a dental extraction performed, the potential for hemorrhage exists. In these cases, it is possible to increase the activity of the endogenous factor VIII through pretreatment with desmopressin acetate. This preoperative measure may alleviate the need for clotting factor replacement.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Kim K. Birtcher, MS, PharmD, BCPS

During the hospitalization, the patient received aspirin, clopidogrel, and other evidence-based therapies. He was enrolled in a cardiac rehabilitation program and received instructions for lifestyle modifications to decrease his risk of future cardiovascular events. He was discharged to his home after 3 days. His discharge medications included aspirin, clopidogrel, an angiotensin-converting enzyme inhibitor, a β -blocker, a statin, and fish oil. The pharmacist provided discharge counseling and emphasized the importance of continuous adherence to the dual antiplatelet therapy with aspirin and clopidogrel therapy for at least a year after drug-eluting stent placement to prevent stenosis.

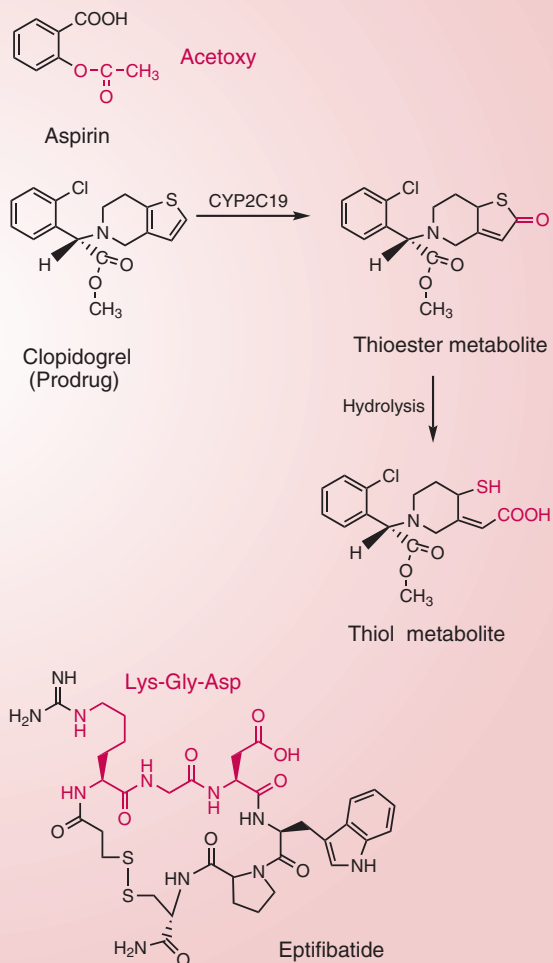
Chemical Analysis

Kim E. Birtcher, Victoria Roche and S. William Zito

Myocardial infarction is commonly caused by the rupture of an unstable atherosclerotic plaque and the subsequent formation of a platelet-mediated thrombus. Shortly after plaque rupture, platelets adhere to collagen fibrils through the GPIa/IIa receptors. Tissue factor and collagen are exposed, which induces adhesion and activation of platelets and the release of ADP and TXA₂. The GPIIb/IIIa receptors change, allowing fibrin bridges to form between the platelets. The extrinsic coagulation pathway is also activated. Thrombin is produced and converts fibrinogen to fibrin. The fibrin stabilizes the thrombus and catches red blood cells.

The use of antiplatelet agents and anticoagulants for the treatment of myocardial infarction in the clinical scenario matches the complexity of clot formation. Aspirin's acetoxy moiety (unique among NSAIDs) irreversibly deactivates platelet COX-1 through acetylation of a COX-1 Ser residue, thus rapidly inhibiting the synthesis of TXA₂ and attenuating platelet aggregation. Clopidogrel, a thienopyridine, is a prodrug that requires activation by cytochrome 2C19 to exert its antiplatelet effect. The active metabolite, the thiol product of the hydrolysis of the CYP2C19-generated thioester, selectively and irreversibly inhibits the binding of ADP to its platelet P2Y₁₂ receptor, which impairs the ADP-mediated activation of the GPIIb/IIIa complex and ultimately inhibits platelet aggregation.

Unfractionated heparin binds to antithrombin and inhibits the activity of clotting factors Xa and IIa (thrombin). The lysine-glycine-aspartate component of eptifibatide is highly specific for the GPIIb/IIIa receptor. Eptifibatide is a reversible inhibitor of the receptor, since it has a low binding affinity for the receptor. The combined actions of these medications provide comprehensive management of the thrombus formed during myocardial infarction.



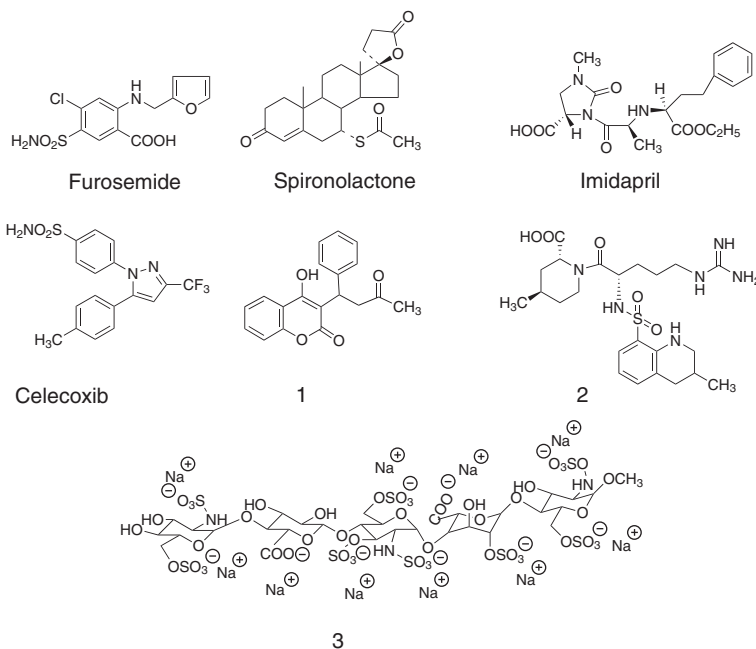
CASE STUDY

Matthis C. Lu

DY is a 78-year-old Asian woman who was admitted to the hospital because she needs to have hip-replacement surgery because she fell in her house recently. Her significant medical history includes chronic renal failure (creatinine clearance, 14 mL/min), osteoarthritis, hypertension, and congestive heart failure. Her current medication list includes furosemide 20 mg po daily, spironolactone 12.5 mg po daily, imidapril 2.5 mg po bid, and celecoxib 200 mg po bid. Unfractionated heparin therapy at a subcutaneous dosage

of 5,000 units tid was initiated after surgery for deep vein thrombosis prophylaxis. Five days after surgery, DY complained of pain and swelling in her left leg. A compression ultrasound reading demonstrated superficial femoral and popliteal vein thrombosis. A laboratory evaluation for heparin-induced thrombocytopenia (serotonin release assay) was ordered and yielded a positive result. Her physician wants to change her unfractionated heparin to a different anticoagulant and asks your opinion of the choices 1–3.

(continued)



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 27

Insulin and Drugs Used to Treat Diabetes

S. WILLIAM ZITO

Drugs Covered in This Chapter*

ACARBOSE

AMYLIN

EXENATIDE

INSULIN

- Aspart
- Detemir
- Glulisine
- Glargine
- Lispro

LIRAGLUTIDE

LINAGLIPTIN

METFORMIN

MIGLITOL

NATEGLINIDE

PIOGLITAZONE

PRAMLINTIDE

REPAGLINIDE

ROSIGLITAZONE

SAXAGLIPTIN

SITAGLIPTIN

SULFONYLUREAS

- Acetohexamide
- Chlorpropamide
- Glimepiride
- Glipizide
- Glyburide
- Tolazamide
- Tolbutamide

TROGLITAZONE

Vildagliptin

VOGLIBOSE

Abbreviations

ADA, American Diabetes Association
ADP, adenosine diphosphate
AGE, advanced glycation end-product
Arg, arginine
Asp, aspartate
ATP, adenosine triphosphate
DAG, diacylglycerol
DPP-IV, dipeptidyl peptidase-IV
FDA, U.S. Food and Drug Administration
FFA, free fatty acid
FPG, fasting plasma glucose

GAPDH, glyceraldehyde-3-phosphate dehydrogenase
GIP, glucose-dependent insulinotropic polypeptide
GLP-1, glucagon-like peptide-1
Glu, glutamate
GLUT, facilitative glucose transporter
Gly, glycine
HbA_{1c}, hemoglobin A_{1c}
HLA, human leukocyte antigen
IRS, insulin receptor substrate
Lys, lysine
NPH, neutral protamine Hagedorn

OGTT, oral glucose tolerance test
PARP, poly(ADP-ribose) polymerase
PKC, phosphokinase C
PPAR, peroxisome proliferator-activated receptor
PTP, protein tyrosine phosphatase
Ser, serine
SGLT, sodium-glucose co-transporters
SUR, sulfonylurea receptor
Thr, threonine
TNF- α , tumor necrosis factor- α
TZD, thiazolidinedione
UPD, uridine diphosphate

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in *italics*.

SCENARIO

JL is a 57-year-old Hispanic man who was initially prescribed metformin 500 mg PO BID, but his doctor increased it to 1000 mg PO BID 4 months ago. He is also taking glipizide 10 mg BID. His body mass index is 35 and hemoglobin A1c value is 7.7%. He is very reluctant to use insulin because he thinks it is too complicated, he fears hypoglycemia, and has heard that insulin use is associated with weight

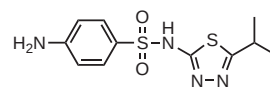
gain. He monitors his blood glucose one to two times daily, including after meals. His fasting blood glucose levels are approximately 140 mg/dL; most postprandial values are greater than 200 mg/dL.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

The history of diabetes dates back to antiquity and is probably a disease that has always afflicted humankind. The recorded history of medicine begins with the Egyptian and Greek civilizations. The discovery of the Ebers Papyrus, written in 1550 BC, contains descriptions of a number of diseases including an account resembling the signs and symptoms of diabetes (1). Aretaeus of Cappadocia (81–138 AD) used the term “diabetes” in connection with this disease, which means “to run through” or “siphon,” and Galen (131–201 AD) described the disease as “diarrhea urinosa” (diarrhea of urine) and “dipsakos” (the disease of thirst). In the 5th and 6th centuries, a number of cultures (Indian, Chinese, and Japanese) described the sweetness of the urine of diabetics, but it was not until the 16th century that the term “mellitus” (Latin for “honey” or “sweet”) was applied to distinguish it from a similar polyuric disease, diabetes insipidus, where the urine was tasteless (insipid) (2). Sugar was positively identified in the urine of diabetics by Dobson in 1776, the cause of which was attributed by him and others as being associated with defects in the blood, kidney, and liver. Two seminal discoveries guided the understanding of the etiology of diabetes. The first was the identification of a polysaccharide in the liver termed “glycogen” by Claude Bernard in 1875. The second was the discovery, in 1889 by Oskar Minkowski, that when the pancreas was removed from a dog, the animal developed all the signs and symptoms of diabetes mellitus (2,3). In 1869, Paul Langerhans noticed small clusters of ductless cells on the pancreas but gave no function to them, and in 1902, the work of Eugene Opie clearly linked these ductless cells, which by now were called the isles of Langerhans, to diabetes mellitus. Although a hypothetical secretion of the islet cells was postulated and called insulin (from insula, which means island), it took the work of Frederick Banting, Charles Best, and John Macleod to isolate insulin and use it as a diabetes medication. In 1923, Banting and Macleod shared the Nobel Prize in medicine, and Banting subsequently shared his prize with Best (2–4). Since the discovery of insulin, a number of types have been developed to improve its pharmacokinetics. The amino acid sequence of insulin was identified by Sanger and Tuppy in 1951 (5,6), and the total synthesis of active insulin was accomplished in 1963 by Katsoyannis et al. (7). The late 1970s

saw recombinant DNA technology applied to the synthesis of human insulin in *Escherichia coli* (8), and the use of small, nonprotein molecules for the treatment of diabetes was developed from the discovery by Janbon et al. in 1942 that the sulfonamide 2-(*p*-aminobenzenesulfonamide)-5-isopropylthiadiazole induced hypoglycemia (9).



IPTD

DIABETES MELLITUS

Diabetes mellitus is a metabolic disorder characterized by hyperglycemia and associated with impaired fat, carbohydrate, and protein metabolism. The disease is the result of defects in insulin secretion and/or insulin action, which progressively leads to chronic microvascular, macrovascular, and neuropathic complications (10). There are two major types of diabetes, type 1 and type 2. In addition diabetes mellitus is associated with certain other hyperglycemic conditions, including gestational diabetes, maturity-onset diabetes of youth, diabetes secondary to other disease states, and drug- or chemical-induced diabetes (11).

Type 1 was formerly called insulin-dependent diabetes mellitus or juvenile-onset diabetes. It accounts for 5% to 10% of patients with diabetes and is largely recognized as an autoimmune disease whereby the β -cells are destroyed by the body's own antibodies. Because the pancreas can no longer produce insulin, type 1 diabetics have an absolute requirement for exogenous insulin. The symptoms of type 1 diabetes often come on suddenly, can be severe, and include polydipsia (thirst), polyuria (frequent urination), polyphagia (hunger), weight loss, fatigue, and diabetic ketoacidosis (12).

Type 2 diabetes (formerly called non-insulin-dependent diabetes mellitus or adult-onset diabetes) accounts for 90% to 95% of adult cases of diabetes. Type 2 diabetes slowly progresses from a state where the patient develops insulin resistance to a state where the pancreas loses its ability to produce enough insulin to compensate for the insulin resistance of peripheral tissues. Insulin resistance is the state where tissues do not utilize insulin properly. Insulin resistance is associated with a number of

physiologic risk factors (hyperinsulinemia, hypertension, dyslipidemia, hypercoagulation, proinflammatory state, and abdominal obesity) most commonly referred to as “the metabolic syndrome.” Nondiabetic patients with metabolic syndrome (Table 27.1) are at high risk for the development of type 2 diabetes, which then gives these patients a two to four times greater risk of developing coronary heart disease and stroke (13). In addition, type 2 diabetes is usually associated with race, lack of physical activity, and a family history of the disease (14,15). Unlike some traits, diabetes does not seem to be inherited in a simple pattern. Yet clearly, some people inherit a greater predisposition for diabetes than others.

Gestational diabetes is usually diagnosed during pregnancy. It occurs more often in women who are obese, have a family history of diabetes, and are members of a high-risk ethnic group (African American, Hispanic/Latino, Asian/Pacific Islander, and American Indian). Gestational diabetes requires treatment to control the hyperglycemia and to avoid complications to the infant. Most women return to normal postpartum; however, there is increased risk of developing diabetes in the next 10 years (16).

Maturity-onset diabetes of youth is characterized by faulty secretion of insulin, is rare (<5% of type 2 diabetes), and is associated with a number of genetic defects of β -cell function. These defects are inherited and occur at six chromatin loci identified on chromosomes 20q, 7p, 12q, 13q, 17q, and 2q (11). This form of diabetes is the result of impaired secretion of insulin, and there is no evidence that insulin action on tissue targets is decreased.

TABLE 27.1 International Diabetes Federation Worldwide Definition of Metabolic Syndrome (13)

Central obesity	Waist circumference* (race and gender specific) U.S. male: 40 inches U.S. female: 35 inches *If body mass index (BMI) is $>30 \text{ kg/m}^2$, then central obesity is assumed. Plus any two of the following:
Triglycerides	$\geq 150 \text{ mg/dL}$ or if under treatment for this dyslipidemia
HDL cholesterol	Males : $<40 \text{ mg/dL}$ Females: $<50 \text{ mg/dL}$ or if under treatment for this dyslipidemia
Blood pressure	$\geq 130/85 \text{ mm Hg}$ or if under treatment for hypertension
Fasting plasma glucose	$\geq 100 \text{ mg/dL}$ or previously diagnosed with type 2 diabetes

HDL, high-density lipoprotein.

Diagnosis of Diabetes

The American Diabetes Association (ADA) has established four criteria for the diagnosis of diabetes: 1) a fasting plasma glucose (FPG) $\geq 126 \text{ mg/dL}$; 2) a 2-hour oral glucose tolerance test (OGTT) of $\geq 200 \text{ mg/dL}$; 3) in patients with overt symptoms of hyperglycemia (polyphagia, polyuria, polydipsia, weight loss), a random plasma glucose of $\geq 200 \text{ mg/dL}$; or 4) hemoglobin A_{1c} (HbA_{1c}) of $\geq 6.5\%$. In addition, the ADA has recognized a prediabetic state for patients who are

CLINICAL SIGNIFICANCE



Diabetes is a condition wherein the body no longer produces insulin (β -cell dysfunction) or uses insulin efficiently (insulin resistance).

Insulin is a hormone that is needed to convert carbohydrates and other food into energy needed for life. The cause of diabetes remains unknown, although genetics and environmental factors, such as obesity and a sedentary lifestyle, seem to play important roles.

Many oral diabetes medications are available with different mechanisms of action. Combination therapy, using medications from different classes to address the deficiencies seen in diabetes, is standard in diabetes care.

Differences in the structures and structure-activity relationships account for the differences in the secretagogues, such as the sulfonylureas and the meglitinides. Both of these classes increase pancreatic insulin secretion, but their structures allow for different binding capacities and adverse effect profiles. Metformin blocks hepatic output of glucose and is a modification of phenformin, which causes a significant amount of lactic acidosis. Thiazolidinediones enhance insulin sensitivity. The

initial concern about the hepatotoxicity of thiazolidinediones has been relaxed, with few cases being reported with available agents, especially after troglitazone was removed from the market. Differences in the structure-activity of the two remaining agents, pioglitazone and rosiglitazone, may account for the reported difference in adverse events, such as myocardial infarction and impact in lipids.

With the introduction of the incretin mimetics, the first new class of diabetes agents in nearly 20 years is now available. Incretins improve glucose control but also have the potential to improve insulin resistance and to restore β -cell function. Many drugs affecting the incretin system are in development and are all modifications or mimetics of endogenous incretins.

With every new medication being developed and new structure modifications being pursued, there will be more options in our arsenal to treat a disease that is very prevalent and complex.

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at risk of developing diabetes. The criteria include an FPG of 100 to 125 mg/dL, referred to as impaired fasting glucose, and/or a 2-hour OGTT of 140 to 199 mg/dL (impaired glucose tolerance), and/or an HbA_{1c} of 5.7% to 6.4%. The HbA_{1c} is perhaps the most accurate indicator of glucose because it reflects plasma glucose levels over the previous 2 to 3 months and is now accepted as the ideal standard for assessing glycemic control (17,18).

Epidemiology

Diabetes is a global health problem. More than 220 million people worldwide have diabetes. In 2004, an estimated 3.4 million people died from consequences of high blood sugar, and most of the deaths occurred in low- to middle-income countries. The World Health Organization projects that diabetes deaths will double between 2005 and 2030. The most recent data from the ADA report that 23.5 million children and adults have diabetes (19). The total prevalence includes 5.7 million people who are undiagnosed and does not include the 57 million people who are defined as prediabetics. The ADA also predicts that 1.6 million new cases of diabetes will be diagnosed each year. Relatively few diabetics (~0.2%) are under the age of 20, and approximately 1 in 500 of them has type 1 diabetes. Of the diabetics over the age of 20 (23.5 million), more than half are 60 years or older. There is an almost equal distribution between adult men and women with diabetes (12 million vs. 11.5 million, respectively).

There are race and ethnic differences in the prevalence of diabetes in adults. Figure 27.1 shows that

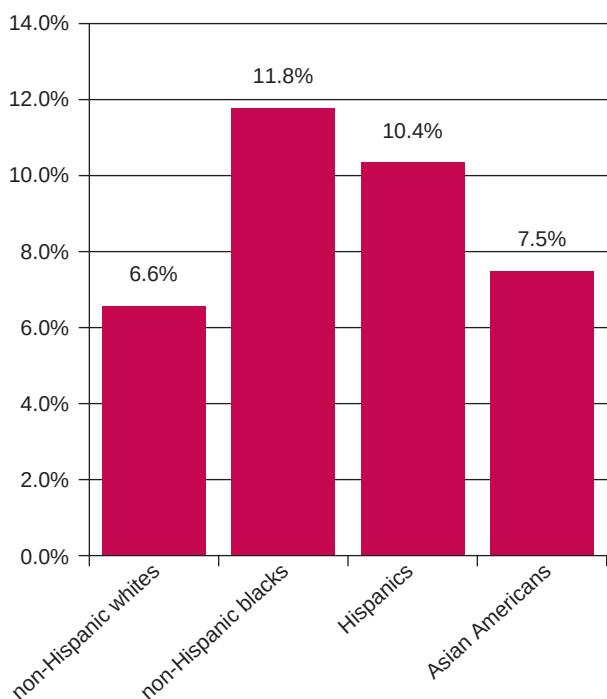


FIGURE 27.1 Prevalence of diabetes in adults by race/ethnicity—United States 2007.

non-Hispanic blacks have the highest prevalence (11.8%), followed closely by Hispanics, which include Cubans, Mexican Americans, and Puerto Ricans (10.4%). The prevalence of diabetes in Asian Americans is 7.5%, whereas non-Hispanic whites have a prevalence of 6.6%.

Diabetes was the seventh leading cause of death in the United States in 2006. Diabetes is not likely to be reported as the cause of death because the cause of death might be listed as being due to one of the many complications associated with diabetes, such as heart disease, hypertension, kidney disease, and nervous system disease. It is most likely that the risk of death for people with diabetes is twice that of people without diabetes (16).

Biochemistry of Diabetes

Regulation of Glucose Homeostasis

Glucose is the primary source of cellular energy, and therefore, plasma glucose concentration is tightly controlled, normally between 65 and 104 mg/dL. Glucose homeostasis is maintained by a number of hormones, the most important being insulin and glucagon. Insulin is secreted by β -cells when blood glucose concentration rises. Insulin reduces blood glucose levels either by inhibition of hepatic glucose production (glycogenolysis and gluconeogenesis) or by increasing glucose uptake into liver, muscle, and fat tissue. Glucagon is secreted by pancreatic α -cells in response to low concentrations of glucose. It acts principally at the liver and antagonizes the effects of insulin by increasing glycogenolysis and gluconeogenesis. In addition to glucagon, cortisol (hydrocortisone) and catecholamines also raise plasma glucose levels.

Other hormones also function in maintaining normal plasma glucose levels. These include amylin (a 37-amino acid peptide), glucagon-like peptide-1 (GLP-1) (a 30-amino acid peptide), and glucose-dependent insulinotropic polypeptide (GIP) (a 42-amino acid peptide). Amylin is actually co-secreted with insulin from β -cells and functions in decreasing gastric emptying, which enhances glucose absorption following a meal. GLP-1 and GIP are incretins, or gut-derived peptides, which have a multitude of effects, the primary of which is to promote the synthesis and secretion of insulin from β -cells (20).

It is surprising that such an essential nutrient as glucose is not freely absorbed from the intestines or by cells that require it for energy. Instead, glucose must be transported across membranes by glucose transporters. The glucose transporters are a family of membrane-bound glycoproteins divided into two main types: sodium-glucose co-transporters (SGLT) and facilitative glucose transporters (GLUT). The SGLT1 type is expressed in the absorptive epithelial cells of the intestines and transports glucose against its concentration gradient. SGLT1 is composed of 664 amino acids arranged into 14-transmembrane helices, and both the N- and C-terminals face the extracellular fluid. The SGLT2 type is expressed in the brush border membrane of the kidney and is the major

transporter involved in the reabsorption of glucose from the glomerular filtrate. There are as many as six SGLTs found in a variety of tissues including liver, brain, lung, and heart (21).

In contrast to the SGLTs, the GLUT family of transporters is sodium independent and composed of 12 membrane-spanning α -helices connected through hydrophilic loops with their N- and C-terminals on the cytoplasmic side of the cell membrane. Helices 7, 8, and 11 are believed to form an aqueous pore providing a channel for substrate passage. Mammalian cells have 12 transporters (GLUT1 to GLUT12). GLUT1, GLUT3, and GLUT4 have D-glucose specificity, whereas GLUT2 and GLUT5 have specificity for D-fructose. The GLUT4 transporter is by far the most abundant type, is expressed in adipose and muscle (heart, smooth, and skeletal), and is responsible for insulin-stimulated transport of glucose (22). GLUT4 has been the only insulin-responsive GLUT transporter characterized to date (Fig. 27.2).

When insulin binds to its receptor on sensitive cells, it sets off a complicated cascade of events involving both the phosphatidylinositol-3-kinase and protein kinase Akt or protein kinase B pathways that result in the release of GLUT4 from storage vesicles and its translocation to the cell membrane (Fig. 27.3). The insulin receptor is a transmembrane glycoprotein composed of two α -subunits and two β -subunits linked by disulfide bonds (23). The α -subunits contain the insulin binding site and are located extracellularly. The β -subunits contain tyrosine kinase activity, which is activated by insulin binding-induced autophosphorylation. When insulin binds to the receptor's two α -subunits, it enables a conformational change that allows for adenosine triphosphate (ATP) binding to the β -subunit's intracellular domain. ATP binding activates receptor autophosphorylation, which, in turn, enables the receptor's tyrosine kinase to phosphorylate insulin receptor substrates (IRS). The IRS family of

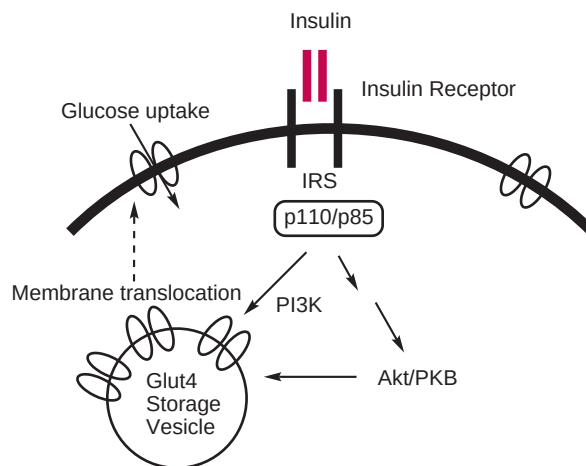


FIGURE 27.3 Overview of insulin signaling pathway leading to translocation of the GLUT4 transporter from storage vesicles. Akt, protein kinase Akt; IRS, insulin receptor substrates; p110, catalytic subunit of PI3K; p85, regulatory subunit of PI3K; PI3K, phosphatidylinositol-3-kinase; PKB, protein kinase B.

proteins consists of four closely related members (IRS-1 to IRS-4) and the related homolog Gab-1. These IRS proteins act as intracellular messengers that begin the cascade of events that result in the translocation of GLUT4 to the cell surface as well as other processes necessary for cell survival (24).

Pathogenesis of Diabetes

Type 1 Diabetes

It is well established that type 1 diabetes results from immunologic destruction of the insulin-producing β -cells of the pancreas. However, abnormal response to external factors can have also a role in disease pathogenesis including damage caused by viruses (e.g., mumps, coxsackie B4, enteroviruses), cytotoxins, and antibodies released from lymphocytes and environmental chemical exposure. It is now understood that type 1 diabetes results from the interplay between genetic susceptibility and certain external triggers such as viruses, environmental toxins (nitrosamines), or foods (e.g., cow's milk proteins, cereals, gluten). Genetic susceptibility to type 1 diabetes is linked to two genes found on chromosome 6. These genes code for the production of human leukocyte antigens (HLAs) DR3 and DR4. Most but not all of patients with type 1 diabetes have both of these antigens. Interestingly, patients who carry HLA-DQA1*0102 or HLA-DQB1*0602 antigens are actually resistant to type 1 diabetes. Type 1 diabetes is characterized by a long preclinical period marked by the presence of immune markers such as circulating antibodies, islet cell antibodies, and insulin autoantibodies. It is not known as to just when the interplay between genetic susceptibility and environmental factors combine to develop the full blown disease with the result in an absolute insulin deficiency (25,26).

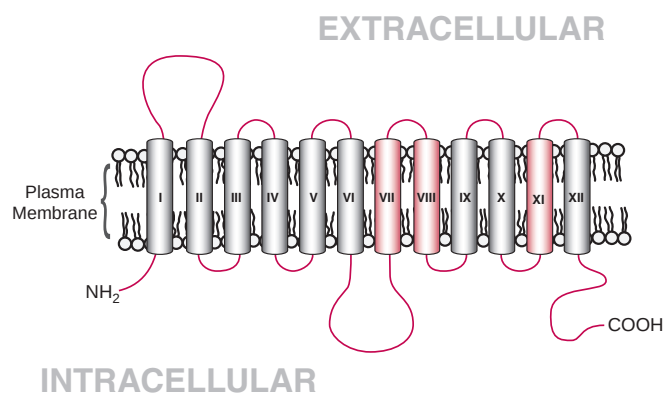


FIGURE 27.2 Schematic representation of the GLUT receptor, showing the membrane-spanning α -helices (1–12). Helices 7, 8, and 11 (red) are believed to form the hydrophilic channel, and the extracellular loop between helices 1 and 2 is the location of glycosylation.

Type 2 Diabetes

The pathogenesis of type 2 diabetes is complex but typically begins with insulin resistance at target organs such as liver, muscle, and adipose tissue. In order to compensate for insulin resistance, there is an initial increase in insulin production. This hyperinsulinemic state is only temporary, and over time, insulin secretion diminishes due to progressive β -cell deterioration. The combined effect of insulin resistance and β -cell dysfunction results in a diminished capacity to limit hepatic glucose production and the ability to uptake and utilize glucose in muscle and adipose tissue.

Insulin resistance is a complex disease that typifies the metabolic syndrome and is likely caused by a number of defects along the insulin signaling cascade (27). Other factors include increased concentrations of free fatty acids (FFAs), tumor necrosis factor- α (TNF- α), and the hormone resistin (27). The increase in plasma FFAs produces insulin resistance by inhibiting glucose uptake and its glycolysis in skeletal muscle and also increases hepatic gluconeogenesis. Both TNF- α and resistin are produced by adipose tissue in greater amounts in obese diabetic individuals. TNF- α impairs insulin action, whereas resistin is known to antagonize the effects of insulin.

Increased hepatic glucose production in type 2 diabetes is attributed to both hepatic insulin resistance and increased glucagon levels (28). β -Cells compensate for this insulin resistance by secreting more insulin. The hyperinsulinemic state is only temporary, because β -cells cannot maintain insulin levels required to maintain normal plasma glucose levels. Impaired insulin secretion and increased glucagon contribute to continued hepatic

glucose output, resulting in elevated fasting glucose levels. When insulin resistance can no longer be overcome, transition to type 2 diabetes occurs.

Hyperglycemia can be caused by a number of other mechanisms. Some patients have abnormal hormone levels including elevated glucagon, somatostatin, growth hormone, cortisol, and epinephrine. Of special importance for the treatment of diabetes is the effect of certain drugs on plasma glucose levels. Table 27.2 lists drugs that both increase and decrease plasma glucose levels.

Diabetic Complications

All types of diabetics constantly battle to control their chronic hyperglycemia. If high levels of plasma glucose are uncontrolled or poorly controlled, it will result in the development of both acute and chronic pathologies. The short-term effects are usually those symptoms usually associated with type 1 diabetics, such as polydipsia, polyuria, polyphagia, blurred vision, urinary tract infections, weight loss, and fatigue. Although these acute effects are relatively minor, they can lead to two rather serious complications: diabetic ketoacidosis and a hyperosmolar hyperglycemic state.

Diabetic ketoacidosis can be life threatening. Insulin blocks the action of lipases that hydrolyze stored fats to free fatty acids. The diabetic with diminished or no insulin, therefore, has increased free fatty acids, which are oxidized to acetone, acetoacetic acid, and β -hydroxybutyric acid, presumably to make up for lack of glucose for oxidative energy (glycolysis). These keto acids can be metabolized, but in prolonged periods of insulin deficiency, the body cannot keep up with their production,

TABLE 27.2 Drugs that Alter Plasma Glucose Levels

Drugs that Increase Plasma Glucose	Drugs that Decrease Plasma Glucose
Acetazolamide	Alcohol (ethanol)
Birth control pills	Anabolic steroids
Atypical antipsychotics: clozapine; olanzapine, risperidone	ACE inhibitors
β -Adrenergic blockers	Chloramphenicol
Caffeine	Clofibrate
Calcium Channel blockers	Coumadin when coadministered with oral hypoglycemic agents
Clonidine	Gatifloxacin
Diuretics: thiazides > loop > K-sparing	MAO inhibitors
Glucocorticoids	Saquinavir
Niacin	
Phenytoin	
Rifampin	
Thyroid hormones	

ACE, angiotensin-converting enzyme; MAO, monoamine oxidase.

and ketoacidosis occurs. Decreased insulin levels also allow for unchecked glucagon activity to increase plasma glucose by gluconeogenesis and glycogenolysis. The lowering of plasma pH by keto acids and hyperglycemia lead to symptoms of ketoacidosis, including vomiting, dehydration, hyperventilation, confusion, and possibly coma and death.

When plasma glucose levels become higher than 600 mg/dL, increased amounts of glucose are excreted in the urine, leading to development of hyperglycemic state characterized by dehydration, hyperosmolarity, and electrolyte imbalance. Symptoms of hyperglycemic state include tachycardia, dry skin, and orthostatic hypotension, which if not treated can lead to death in as many as 30% of those afflicted (20).

The long-term effects of diabetes are serious and often not detected until they become overt. Both microvascular and macrovascular effects occur. The microvascular pathologies involve the retina, renal glomerulus, and peripheral nerves, and as a consequence, diabetes is a leading cause of blindness, end-stage renal disease, and painful neuropathies. The macrovascular effects involve arteries that supply the heart, brain, and lower extremities. Therefore diabetics have a much higher risk of cardiovascular disease, including atherosclerosis, myocardial infarct, stroke, and limb amputation (17).

The essential question now is how does diabetes result in so many diverse microvascular and macrovascular pathologies? The answer centers around four main molecular mechanisms, all related to the diabetic hyperglycemic state and all seemingly related to the overproduction of superoxide by the mitochondrial electron-transport chain. The four biochemical mechanisms are increased polyol pathway flux; increased advanced glycation end-product (AGE) formation; activation of protein kinase C (PKC) isoforms; and increased hexosamine pathway flux (29,30).

The polyol pathway (Fig. 27.4) involves the reduction of aldehydes generated by reactive oxygen species to inactive alcohols, and glucose is reduced to sorbitol by

the enzyme aldose reductase. Aldose reductase is a cytosolic oxidoreductase that uses NADPH. The sorbitol produced is oxidized by sorbitol dehydrogenase to fructose using NAD^+ as cofactor. When glucose levels are high, the activation of the polyol pathway can deplete reduced glutathione, leading to cellular oxidative stress. The effect of polyol pathway flux is implicated in the formation of cataracts in the eye as well as development of peripheral neuropathies (31).

Increased levels of AGEs have been found in both retinal blood vessels and renal glomeruli. They arise from intracellular hyperglycemia by the auto-oxidation of glucose to glyoxal and the subsequent formation of 3-deoxyglucosone and dephosphorylation of glyceraldehydes 3-phosphate and dihydroxyacetone phosphate to methylglyoxal (Fig. 27.5). These products are often referred to as dicarbonyls, and they react with protein amino groups of lysine (Lys), arginine (Arg), and the thiol group of cysteine to form AGEs. AGEs damage vascular cells and also alter several cellular functions, including changes in gene expression in endothelial cells and macrophages (32).

Activation of the family of PKC enzymes is brought about by the secondary messenger, diacylglycerol (DAG). In hyperglycemic cells, DAG is increased by de novo synthesis from reduction of dihydroxyacetone phosphate to glycerol-3-phosphate followed by stepwise acylation. PKCs can also be activated indirectly through ligation of AGE receptors and increased activity of the polyol pathway (30). PKC, when increased in cells, is involved in a significant number of biochemical pathways, leading to blood flow abnormalities, vascular permeability and angiogenesis, capillary and vascular occlusion, proinflammatory gene expression, and reactive oxygen species elevation (Table 27.3). Elevated levels of PKCs, primarily the α - and β -isoforms, have been found in the retina and renal glomeruli of diabetic animals as well as in cultured vascular cells (33).

When intracellular glucose levels are in excess, glucose gets shunted into the hexosamine pathway.

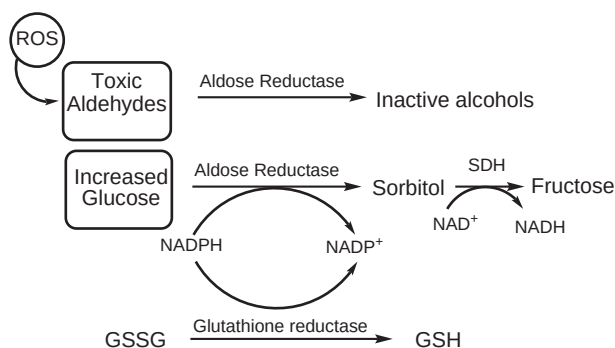


FIGURE 27.4 The polyol pathway showing the role of aldose reductase in reducing toxic aldehydes and glucose to sorbitol. GSH, reduced glutathione; ROS, reactive oxygen species; SDH, sorbitol dehydrogenase.

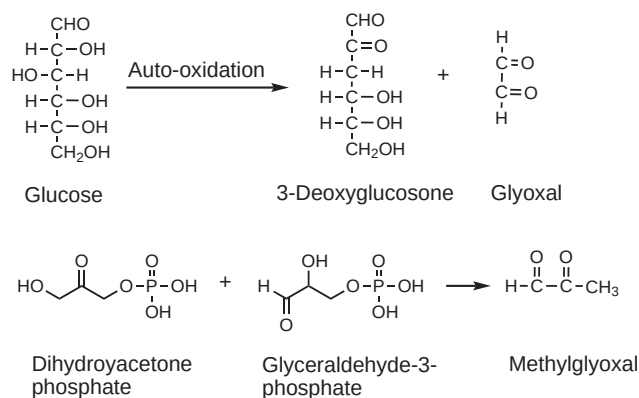


FIGURE 27.5 Formation of advanced glycation end-product precursors: glyoxal, 3-deoxyglucosone, and methylglyoxal.

TABLE 27.3 Pathologic Consequences of Protein Kinase C Activation

Factor Effected	Pathology
↓ Endothelial nitric oxide synthase (eNOS)	Blood flow abnormalities
↑ Endothelin-1 (ET-1)	Blood flow abnormalities
↑ Vascular endothelial growth factor (VEGF)	Vascular permeability and angiogenesis
↑ Collagen	Capillary occlusion
↑ Fibronectin	Capillary occlusion
↑ Plasminogen activator inhibitor-1 (PAI-1)	↓ Fibrinolysis and vascular occlusion
↑ Nuclear factor κ light-chain enhancer of activated B cells (NF- κ B)	Proinflammatory gene expression
↑ NAD(P)H oxidases	Increased reactive oxygen species and multiple effects

In this pathway, fructose-6-phosphate is converted to glucosamine-6-phosphate and then to uridine diphosphate (UDP)-*N*-acetylglucosamine by the rate-limiting enzyme glutamine:fructose-6-phosphate amidotransferase (Fig. 27.6). UDP-*N*-acetylglucosamine is coupled to intracellular protein serine (Ser) and threonine (Thr) residues by *O*-linked *N*-acetylglucosamine transferase. Because both phosphorylation and *O*-linked

N-acetylglucosamine transferase acylation compete for the same substrates, the two processes can compete for sites. The increased linkage of *N*-acetylglucosamine to Ser and Thr residues on transcription factors leads to alteration in both gene expression and protein function, which together contribute to the pathologies of diabetic complications.

Each of the four different pathogenic mechanisms is activated by a single hyperglycemic event, that is, overproduction of superoxide by the mitochondrial electron transport chain (30). It is now believed that the overproduction of superoxide activates the four pathogenic pathways (Fig. 27.6). Excess superoxide inhibits the glycolytic enzyme glyceraldehyde-3-phosphate dehydrogenase (GAPDH) through its effect on poly(adenosine diphosphate [ADP]-ribose) polymerase (PARP). This inhibition causes intermediate metabolites of the glycolysis to accumulate. Thus, the inhibition of the conversion of glyceraldehyde-3-phosphate to 1,3-diphosphoglycerate results in increased amounts of dihydroxyacetone phosphate, which in turn increases DAG, the intracellular activator of PKC. In addition, the increase in DAG also results in the formation of methylglyoxal, the main precursor to AGEs. Earlier in the glycolytic chain, increased levels of fructose-6-phosphate are shunted into the hexosamine pathway, producing UDP-*N*-acetylglucosamine, which in turn forms *O*-linked glycoproteins that affect transcription. Finally, increased amounts of glucose are diverted through the polyol pathway, which consumes NADPH and depletes glutathione. It was originally thought that superoxide itself directly inhibited GAPDH; however, further investigations revealed that GAPDH is actually

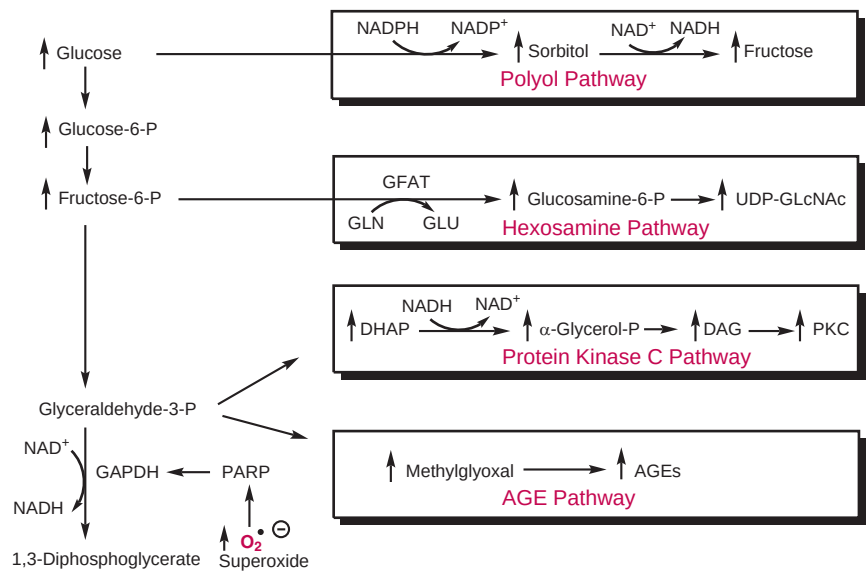


FIGURE 27.6 The role of hyperglycemia-induced mitochondrial excess superoxide in activation of the four pathways involved in diabetic microvascular and macrovascular damage. AGEs, advanced glycation end-products; DAG, diacylglycerol; DHAP, dihydroxyacetone phosphate; GAPDH, glyceraldehyde-3-phosphate dehydrogenase; GFAT, glutamine:fructose-6-phosphate amidotransferase; Gln, glutamine; GLcNAc, *N*-acetylglucosamine; PARP, poly(ADP-ribose)polymerase; PKC, protein kinase C.

inhibited by PARP. PARP is a DNA repair enzyme found in the nucleus. PARP is activated in response to DNA strand breaks caused by superoxide radicals. Once activated, PARP splits NAD^+ into nicotinic acid and ADP-ribose. PARP then makes polymers of ADP-ribose, which accumulate on GAPDH, inhibiting its activity (29,34).

Therapeutic Approaches to the Treatment of Diabetes

Diabetes is a complex chronic disease with no cure. Therefore, therapy is directed at controlling hyperglycemia and reducing the symptoms and morbidities associated with microvascular and macrovascular complications. Early diagnosis and aggressive maintenance of euglycemia (normal blood glucose levels) will go a long way to moderate the microvascular pathologies (35); however, reduction of the risk for macrovascular pathologies requires management of cardiovascular risk factors, such as smoking cessation, treatment for dyslipidemia, control of hypertension, and antiplatelet therapy. The ADA recommends that appropriate medical care for the diabetic requires setting goals for glycemia, as well as for therapeutic lifestyle changes (diet and exercise) along with control of blood pressure, plasma lipids, and the use of appropriate medications (36).

Glycemic control involves two primary techniques, patient self-monitoring and measurement of HbA_{1c} . HbA_{1c} is perhaps the most accurate indicator of glucose because it reflects plasma glucose levels over the previous 2 to 3 months and is now accepted as the gold standard for assessing glycemic control. HbA_{1c} is formed by the glycosylation of hemoglobin's amino terminal valine residue and has a half-life equivalent to that of an erythrocyte. The United Kingdom Prospective Diabetes Study and the Diabetes Control and Complications Trial/Epidemiology of Diabetes Interventions and Complications Study have established that if a patient's HbA_{1c} is maintained below 7%, the development and progression of neuropathy, nephropathy, retinopathy, and cardiovascular disease in type 1 or 2 patients can be significantly decreased (37–39).

Pharmacologic treatment for type 1 diabetics requires intensive insulin therapy. The large number of short- and long-acting insulin analogs allows for the use of multiple doses of basal and prandial (at meals) doses of insulin. Therefore, patients can match their dose of prandial insulin to carbohydrate intake, premeal plasma glucose, and anticipated activity. The most common side effect of the use of insulin is severe hypoglycemia; however, the development of quick-acting and long-acting insulin analogs has moderated this side effect while still maintaining equal HbA_{1c} lowering (36).

Medical treatment of the type 2 diabetic requires management of hyperglycemia as measured by the patient's HbA_{1c} below 7%. Along with lifestyle interventions, the array of different classes of hypoglycemic drugs offers the physician many different treatment options. Obesity and sedentary lifestyle are the major risk factors for diabetes;

therefore, weight loss, dietary changes, and increased activity levels should be the initial approach to treating the type 2 diabetic. In fact, morbidly obese patients who have undergone weight-loss surgery and who have maintained at least a 40-pound weight loss have shown elimination of the disease (40). When diet and exercise are insufficient to maintain the patient's HbA_{1c} below 7%, it then becomes necessary to initiate blood glucose-lowering medications. In addition to insulin, there are several classes of oral hypoglycemic agents available. They include the insulin secretagogues (sulfonylureas, meglitinides), biguanides (metformin), insulin sensitizers (glitazones or thiazolidinediones), α -glucosidase inhibitors, GLP-1 analogs, dipeptidyl peptidase-IV (DPP-IV) inhibitors, amylin antagonists, and a number of new drugs in development.

Currently, ADA recommends beginning medication therapy with metformin along with lifestyle interventions (Fig. 27.7). If this therapy fails to maintain or sustain glucose levels, then another medication should be added, usually a sulfonylurea or insulin. Insulin (intermediate or long acting) is indicated for patients who have trouble decreasing their HbA_{1c} level below 8.5%. If lifestyle, metformin, and sulfonylurea or insulin do not result in achievement of target glycemia, the next step is to intensify the insulin therapy. This usually consists of additional injections of a short- or rapid-acting insulin analog given before meals. At this stage, insulin secretagogues (sulfonylureas or meglitinides) should be terminated because they are not considered to be synergistic. Under certain conditions (e.g., patients who have hazardous jobs, allergic to sulfonylureas, promotion of weight loss), the addition of pioglitazone or exenatide to metformin can be considered. The addition of a third medication to metformin/pioglitazone or metformin/exenatide can be considered if target HbA_{1c} is not reached. The use of amylin agonists, α -glucosidase inhibitors, meglitinides, and DPP-IV inhibitors is usually reserved for patients who cannot tolerate the first-line drugs, because these drugs do not have equivalent glucose-lowering ability and are relatively expensive and there are limited clinical data demonstrating effectiveness (41).

Therapeutic Classes of Drugs Used to Treat Diabetes

Insulin

Insulin was isolated in crystalline form by Abel in 1926, only 5 years after it was isolated from the canine pancreas by Banting, Best, Collip, and Macleod. Banting and Macleod received the 1923 Nobel Prize for their work, and Banting announced that he would share his prize with Best; Macleod did the same with Collip. In 1958, Sanger received the Nobel Prize for the determination of the amino acid sequence of insulin, and Hodgkin received the Nobel Prize in 1964 for determining its three-dimensional structure. The development of an immunoassay for insulin by Berson and Yalow in 1960 earned Yalow the Nobel Prize in 1977, after Berson had passed away.

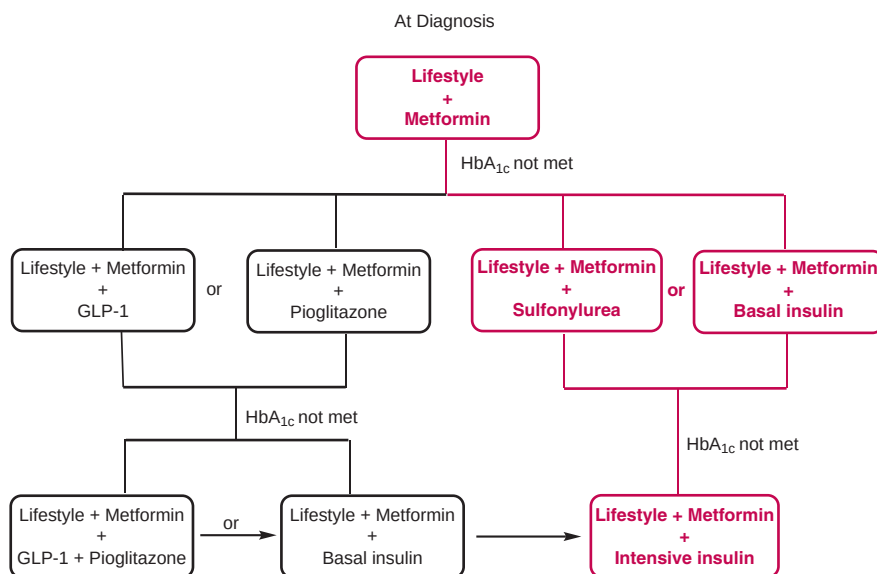


FIGURE 27.7 Decision tree for the management of type 2 diabetes. Lifestyle interventions are reinforced at every step, and HbA_{1c} is recommended to be less than 7%. The path in red represents first-line treatment protocols recommended by the consensus of the American Diabetes Association and the European Associations for the Study of Diabetes (41).

Clearly, the awarding of so many Nobel Prizes points to the importance of insulin-related research throughout the 20th century (42).

The insulin molecule is composed of two polypeptide chains (A and B) linked together by two disulfide bonds. There is an additional disulfide bond in chain A. Chain A contains 21 amino acid residues, and chain B has 30 amino acids, giving a molecular weight of 5,734 Daltons. Insulin is biosynthesized in the β -cells of the pancreas from preproinsulin, a 110-amino acid chain with a molecular weight of 12,000 Daltons. Preproinsulin is cleaved in the endoplasmic reticulum, losing a 24-amino acid unit from the N-terminus. The product is called proinsulin (molecular weight = 9,000 Daltons), which folds to allow the disulfide bonds to form and undergoes further proteolytic modification in the Golgi apparatus, losing four basic amino acids (Arg^{B31}, Arg^{B32}, Lys^{A64}, and Arg^{A65}) and releasing connector C-chain by the action of prohormone convertases PC1 and PC2 (Fig. 27.8) (43,44).

The biologically active form of insulin is the monomer. However, in solution, insulin can exist as a dimer and as a hexamer (six monomeric units attached together). The hexamer is formed by its coordination with two zinc ions, the storage form of insulin in the granules of the β -cells. When released from the granules, the hexamer gets diluted in the plasma (nanomolar) and dissociates into monomers. Secretion of insulin is primarily regulated by glucose, but many other nutrients and hormones also have a role. Amino acids, fatty acids, and ketone bodies promote the secretion of insulin. On the other hand, stimulation of the α_2 -adrenergic receptors on the β -cells inhibits secretion, whereas β_2 -adrenergic receptor stimulation and vagal nerve stimulation enhance the release

of insulin. It follows that any physiologic condition that activates the sympathetic system causes a decrease in the secretion of insulin via stimulation of the α_2 -adrenergic receptors. Such conditions include exercise, hypothermia, surgery, hypoxia, and of course, hypoglycemia. In addition, α_2 -adrenergic receptor antagonists will be expected to increase basal insulin levels and β_2 -adrenergic receptor blockers to decrease basal insulin levels (45).

Glucose enters the β -cell facilitated by GLUT2 transport, is phosphorylated to glucose-6-phosphate by a specific isoform of glucokinase, and enters glycolysis ultimately generating ATP. The increase in ATP changes the ratio of ATP to ADP and prevents an ATP-sensitive K⁺ channel from functioning, which in turn leads to depolarization of the β -cells. This prompts activation of a voltage-gated calcium channel, and calcium flows into the β -cells. The elevated intracellular calcium concentration causes activation of phospholipases A₂ and C and increased levels of inositol triphosphate, an intracellular second messenger. Inositol triphosphate facilitates additional release of calcium into the cytosol, and concentrations of calcium are now sufficiently high to promote insulin secretion from the β -cells. The ATP-sensitive K⁺ channel is an octameric hetero-complex consisting of four pore-forming inwardly rectifying K⁺ channel subunits (Kir6.2) and four regulatory sulfonylurea receptor subunits (SUR1). ATP binding to Kir6.2 closes the channel. Binding of sulfonylureas and the meglitinides to SUR1 also closes the channel, whereas binding of ADP to SUR1 opens the channel (46). Therefore sulfonylurea binding to SUR1 causes the same effect as an increase in the ATP/ADP ratio, which is to close the channel leading to the depolarization of the β -cell membrane and the ultimate secretion of insulin (Fig. 27.9).

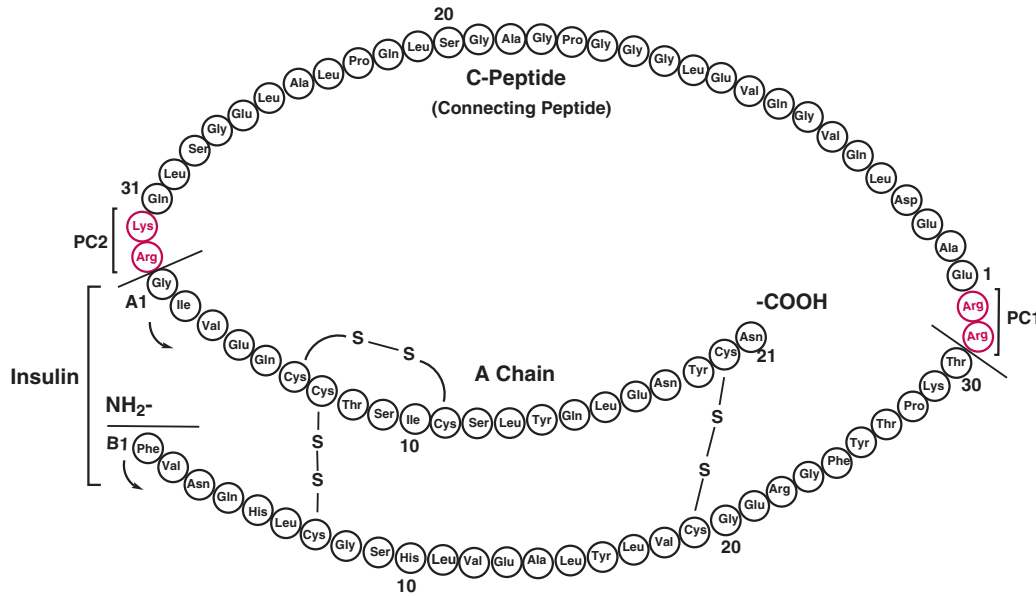


FIGURE 27.8 Schematic structure of proinsulin. Prohormone convertases (PC1 and 2) cleave dipetptides (red) to form insulin.

Fifty percent of the insulin secreted from the pancreas is degraded in the liver and never reaches the general circulation. Hepatic degradation of insulin is the result of the action of a thiol metalloproteinase, which can also be involved in the degradation of glucagon. Insulin is internalized into the hepatocyte by receptor-mediated endocytosis and stored along with its receptor in small vesicles termed endosomes. Insulin is filtered by the renal glomeruli to be reabsorbed by the renal tubules, which also degrade it. Insulin is also degraded at the cell surface of insulin-sensitive tissues (45).

The primary target cells for insulin are liver, muscle, and adipocytes. Insulin interacts with amino acid residues of the α -subunit of its receptor via key amino acid

residues on both the A and B chains. Table 27.4 shows that the N-terminus and C-terminus of the A and B chains are involved in receptor binding (45). Insulin binding to its receptor activates a series of intracellular events that lead to translocation of the GLUT4 transporter to the cell surface. The details of this process have previously been described (Fig. 27.3).

SOURCES AND STABILITY OF INSULIN Historically, patients only had the option of administering either bovine-based or porcine-based insulin, which were alternatives to human insulin because their amino acid sequence homology between species was superb. However, animal sources have become less relevant and have fallen into disuse. Today, the following sources of insulin are available: biosynthetic human, semisynthetic human, and analogs of human insulin. Human insulin is the least antigenic of the available insulins and tends to be more soluble than animal insulin. Human insulin and insulin analogs are formed by recombinant DNA techniques carried out by inserting the human or a modified human gene for proinsulin into *Escherichia coli* or yeast. These genetically altered organisms are fermented and produce proinsulin, which is harvested and enzymatically altered to produce insulin. Human insulin is also produced semisynthetically

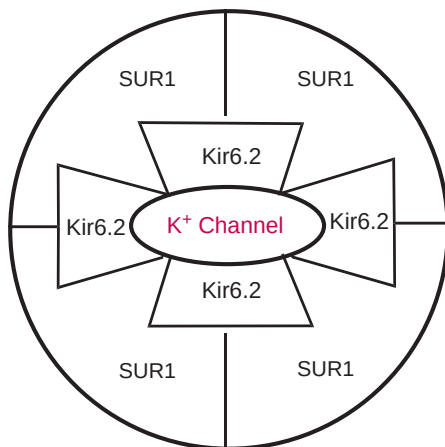


FIGURE 27.9 Idealized topographic view of the ATP-sensitive potassium channel showing four pore-forming inwardly rectifying potassium subunits (Kir6.2) surrounded by four regulatory sulfonylurea receptor subunits (SUR1).

TABLE 27.4 Amino Acid Interactions with Insulin Receptor

Chain	N-Terminus	C-Terminus
A	Gly A1, Glu A4, Gln A5	Tyr A19, Asn A21
B	Val B12	Tyr B16, Gly B23, Phe B24, Phe B25, Tyr B26

by the enzymatic transpeptidation of pork insulin at position 30 of the B chain with the substitution of the amino acid threonine for alanine (20).

Only the insulin monomer is able to interact with insulin receptors, and native insulin exists as a monomer at low, physiologic concentrations ($<0.1 \mu\text{mol/L}$). Insulin dimerizes at the higher concentrations (0.6 mmol/L) found in pharmaceutical preparations. At neutral pH in the presence of zinc ions, hexamers will form, the storage form of insulin in β -cells. At concentrations greater than 0.2 mmol/L , hexamers form even in the absence of zinc ions. Changes in the concentration of insulin can profoundly change absorption after subcutaneous administration. The monomer form of insulin is the only absorbable form, so when insulin is administered subcutaneously at concentrations when dimers and hexamers exist, it is absorbed significantly more slowly (43).

The importance of zinc ions for stabilizing insulin preparations has been known since the first reported crystallization of insulin in the presence of zinc ions in 1934. Presently, all pharmaceutical preparations are either solutions of zinc insulin or suspensions of insoluble forms of zinc insulin. A longer-acting and more stable form of insulin is protamine zinc insulin, which is prepared by precipitating insulin in the presence of zinc ions and protamine, a basic protein. This precipitate is known to contain two zinc ions per insulin hexamer. A somewhat shorter-acting and more useful preparation is neutral protamine Hagedorn (NPH) insulin, which includes *m*-cresol as a preservative. Six *m*-cresol molecules occupy cavities in the hexamer involving the B1-B8 helix formed as a result of the presence of the *m*-cresol. Later, it was reported that when small additional amounts of zinc ions were added to hexameric two-zinc insulin in neutral acetate buffer, insulin could be made to crystallize in several forms with varying rates of dissolution in water. Thus, the slowly soluble lente insulin is a two-zinc insulin, and the crystalline, more slowly soluble ultralente insulin is a four-zinc form.

Fibrils (partially unfolded insulin) are a viscous or insoluble insulin precipitate. Shielding of hydrophobic domains is the principal driving force for the aggregation. Further studies revealed that when the exposed hydrophobic domain (A2, A3, B11, and B15) interacts with the normally buried aliphatic residues (A13, B6, B14, and B18) in the hexameric structure, fibrils form. Insulin fibrils do not resuspend on shaking; thus, they are pharmaceutically inactive. Insulin fibril formation is particularly important with the advent of infusion pumps to deliver insulin. In these devices, insulin is exposed to elevated temperatures, the presence of hydrophobic surfaces, and shear forces—all factors that increase insulin's tendency to aggregate. These problems can be overcome if the insulin is prepared with phosphate buffer or other additives. Another physical stability problem associated with insulin is adsorption to tubing and other surfaces. This normally occurs if the insulin concentration is less than 5 IU/mL (0.03 mmol/L), and it can be prevented

by adding albumin to the dosage form if a dilute insulin solution must be used (43).

There also are chemical instability issues associated with insulin. For many years, the only rapid-acting form of insulin was a solution of zinc insulin, with pH 2 to 3. If this insulin is stored at 4°C , deamidation of the AsnA21 occurs at a rate of 1% to 2% per month. The C-terminal Asn, under acidic conditions, undergoes cyclization to the anhydride, which in turn reacts with water, leading to deamidation to aspartate. The anhydride also can react with the *N*-terminal phenylalanine of another chain to yield a cross-linked molecule. If insulin is stored at 25°C , the inactive deamidated derivative constitutes 90% of the total insulin protein after 6 months (Fig. 27.10).

If a solution of insulin is stored at neutral pH, a different reaction can occur. Deamidation occurs on the AsnB3, and the products, the aspartate (Asp) and isoaspartate-containing insulins, are equiactive with native insulin (Fig. 27.10). More problematic transformations are possible, including chain cleavage between ThrA8 and SerA9 and covalent cross-linking, either with a second insulin chain or with protamine, if added to the solution. These processes are relatively slow compared to the deamidations, but they have the potential of leading to products that can cause allergic reactions (43).

TYPES OF INSULIN The insulin analogs available for treatment of diabetes are classified according to their rate of onset and duration of action. Structure–activity relationship

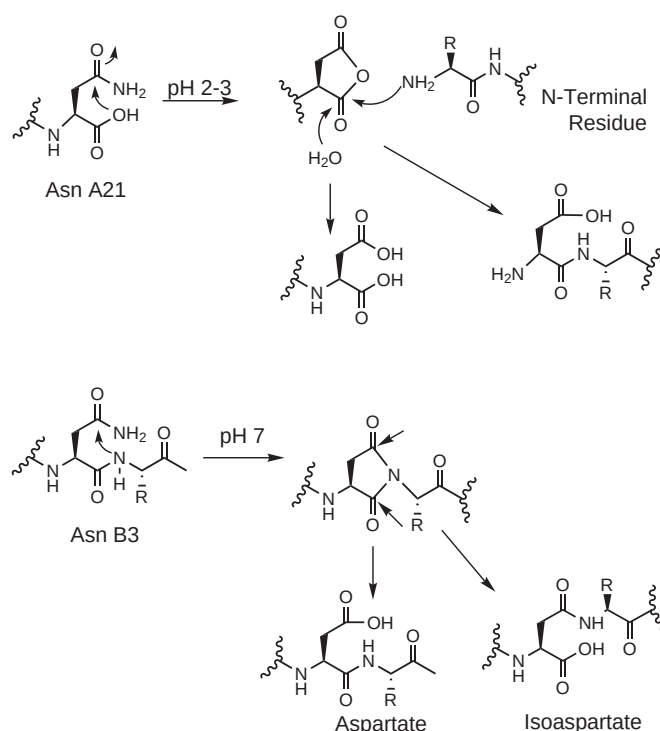


FIGURE 27.10 Chemical degradation of insulin.

studies revealed that variations or removal of amino acid residues from the C-terminus of the B chain could influence the rate of dimer formation while not drastically changing the biologic activity. Inhibiting dimer formation can allow for rapid-acting insulin. Thus, the various insulin analogs that have been developed have substitutions in or additions to the C-terminus of the B chain starting at residue B28. The resulting analogs have either a faster onset or a longer duration of action relative to native insulin. These analogs are all produced by recombinant DNA technology using a modified DNA template. The available insulin preparations are summarized in Table 27.5.

Rapid-acting insulin analogs include insulin lispro, insulin aspart, and insulin glulisine. All have changes made to the amino acid residues in the C-terminus of the B chain (Table 27.6). In insulin lispro, the LysB29 is switched with ProB28, whereas in insulin aspart, the ProB28 has been substituted with an Asp. Glulisine's ValB3 is substituted with a Lys, and LysB29 is changed to a glutamate (Glu). These modifications, as already stated, result in insulin analogs that do not form dimers in solution and that dissociate immediately into monomers, producing a very quick onset of action. Pharmacodynamically, lispro, aspart, and glulisine bind as well to insulin receptors as human insulin and have a low mitogenic potency. Mitogenic activity is the ability of insulin to induce cell division and is believed to be associated with insulin's binding to insulin-like growth factor receptors I and II (47). Lispro, aspart, and glulisine have an onset of action within 15 minutes, peak activity at 30 to 90 minutes, and duration of action of 3 to 4 hours (Table 27.5).

Regular human insulin is the prototype of short-acting insulin. Its onset of action is 30 to 60 minutes, its peak effect 2 to 3 hours after injection, and its duration of action ranges from 4 to 6 hours. Its slow onset requires it to be administered 30 to 60 minutes before meals, which is not convenient, but its slow onset also occurs via intravenous administration, making regular insulin a good choice for intravenous treatment of diabetics.

Intermediate-acting insulin is prepared by adding stoichiometric (equal) amounts of the positively charged polypeptide protamine to regular insulin, forming a poorly soluble insulin-protamine complex. The complex is known as NPH insulin. NPH insulin has an onset of action of 2 to 4 hours, a peak effect of 4 to 12 hours, and a duration of action of 18 to 26 hours after injection. However, most patients receive little effect after 13 to 15 hours (48). Another way to prepare intermediate-acting insulin is to combine regular insulin and zinc in an acetate buffer to form a crystalline complex that dissolves slowly in subcutaneous fluids. This product is termed insulin lente and has a similar onset, peak effect, and duration of action as NPH insulin.

The first long-acting insulin analog to be introduced to the market was insulin glargine. This analog results from the replacement of AsnA21 by glycine (Gly) and the addition of two Arg amino acids to the C-terminus of the B chain (Table 27.6). The resulting analog has an isoelectric point close to 7, but is formulated at an acidic pH 4, where it is completely water soluble. After subcutaneous injection of the acidic solution into tissue at physiologic pH (approximately 7.4), the increase in pH to 7.4 causes the analog to precipitate from solution, forming microcrystals of insulin hexamers, which then slowly dissociate into insulin monomers. The slow dissolution of the hexamer to monomeric insulin from the site of injection results in an onset of 1 to 4 hours, a peak between 5 and 24 hours, and a duration of 20 to 24 hours, which represents a fairly constant release of insulin glargine over 24 hours, giving an almost peakless profile. It has been demonstrated to be comparable or slightly better than NPH insulin at maintaining or reducing HbA_{1c} levels without nocturnal hypoglycemia (43). Insulin ultralente is a long-acting insulin that is a four-zinc acetate crystalline product that has an even slower dissolution rate than the two-zinc acetate insulin lente. Insulin detemir is the newest long-acting analog. This analog results from N-acylation of the LysB29 with the 14-carbon myristic acid (Table 27.6). The fatty acid side chain complexes

TABLE 27.5 Classification, Appearance, and Pharmacokinetics of Insulin Preparations

Type	Appearance	Onset (hours)	Peak (hours)	Duration (hours)
Rapid-acting				
Lispro	Clear	<0.25	0.5–1.5	3–4
Aspart	Clear	<0.25	0.5–1.5	3–4
Glulisine	Clear	<0.25	0.5–1.5	3–4
Short-acting				
Human insulin	Clear	0.5–1	2–3	4–6
Intermediate-acting				
Lente	Cloudy	2–4	6–12	18–26
NPH	Cloudy	2–4	4–12	18–26
Long-acting				
Ultralente	Cloudy	4–8	10–30	>36
Glargine	Clear	1–4	5–24	20–24
Detemir	Cloudy	1–4	5–24	20–24

TABLE 27.6 Insulin Analogs

Generic Name	Trade Name	Change in A Chain	Change in B Chain
Lispro	Humalog	None	B28 Pro → Lys B29 Lys → Pro
Aspart	NovoLog	None	B28 Pro → Asp
Gulisin	Apidra	None	B3 Val → Lys B29 Lys → Glu
Glargine	Lantus	A21 Asn → Gly	Add: B31 Arg and B32 Arg
Detemir	Levemir	None	Remove: B30 Thr Add: C14 fatty acid to B29 Lys

with plasma albumin to produce a depot and longer duration of action. It has approximately the same duration of action as insulin glargine, when administered in appropriate doses.

A new long-acting insulin analog currently in phase III clinical trials is insulin degludec. Insulin degludec is an ultra long-acting insulin that can be administered daily or three times a week and is comparable to glargine and detemir in onset and duration of action. Insulin degludec results from the removal of GluB30 and *N*-acylation of LysB29 with *L*- γ -Glu amidated with hexadecandioic acid. The long duration of action of insulin degludec is believed to be due to the combination of the formation of soluble multihexamer assemblies upon subcutaneous administration, which slowly release monomers, and hexadecandioic acid side chain binding to plasma albumin to produce a depot (49).

An important difference between glargine, ultralente, and detemir is that glargine is a clear solution, whereas the others are cloudy. This can be a problem for patients who rely on the physical appearance of their insulin to distinguish NPH, lente, and ultralente from regular, lispro, aspart, gulisin, or glargine, which are all clear (Table 27.5).

Different types of insulin can be premixed in the same syringe and are usually prescribed for patients needing a simple insulin treatment plan. NPH insulin can be premixed with either aspart or lispro rapid-acting insulins. Mixtures of NPH and lispro can be 50:50 or 75:25, whereas NPH and aspart are available in a 70:30 ratio. A 70:30 NPH/regular insulin premix is also available. The benefit of using premixed insulin is that rapid-acting and long-acting insulin can be administered at the same time and can be given twice a day, usually at breakfast and supper. The drawback of using such a regimen is that to be effective, the amount of carbohydrate to be eaten at each meal is preset. Therefore this works best for patients who can follow strict adherence to a consistent schedule of meals and activity and are able to follow a prescribed diet (49).

Oral Hypoglycemic Agents

The hallmark characteristic of type 2 diabetes is diminished insulin secretion due to impaired β -cell function and/or insulin resistance of peripheral tissues such as liver, adipose, and skeletal muscle, which causes hyperglycemia. The current approach to treatment of type 2 diabetes is to institute lifestyle changes and to initiate therapy with suitable pharmacologic agents (Fig 27.7). The current line of treatment includes: 1) insulin secretagogues, 2) insulin sensitizers, 3) DPP-IV inhibitors, 4) biguanides, 5) α -glucosidase inhibitors, and 6) hormone-related drugs and drugs in development.

INSULIN SECRETAGOGUES The insulin secretagogues include the sulfonylureas and meglitinides, and both increase insulin release from the pancreas by a common mechanism. Sulfonylureas and meglitinides stimulate insulin secretion by binding to the sulfonylurea receptor (SUR) of the ATP-sensitive K^+ channel on the pancreatic β -cells.

SULFONYLUREAS The discovery of the ability of the sulfonylureas to lower plasma glucose resulted from research done in 1942 that noted the hypoglycemic effect of sulfonamides used to treat typhoid fever (50). Subsequent investigations revealed that modifying the sulfonamide antibacterial agents with a urea moiety resulted in sulfonylureas with significant hypoglycemic effects, and consequently, the first-generation oral hypoglycemic drugs were born, which include tolbutamide, chlorpropamide, tolazamide, and acetohexamide (Fig. 27.11).

The first-generation sulfonylureas differ structurally in that they have a small lipophilic substituent on the para position of the phenyl ring (R_1 of the pharmacophore) and an alkyl or cyclic lipophilic substituent on the non-sulfonyl-attached urea nitrogen (R_2 of the pharmacophore) as shown in Figure 27.11. However, these structural modifications do little to increase their binding efficiency to the ATP-sensitive K^+ channel, thus requiring relatively high doses to achieve effectiveness and therefore increasing the potential for adverse events. In addition, the plasma half-life of these first-generation sulfonylureas is fairly long (5 to 36 hours), which also increases their potential for adverse effects.

The second-generation sulfonylureas (Fig. 27.11) developed out of research efforts intended to design hypoglycemic agents with increased potency and that possessed a more rapid onset, shorter plasma half-lives, and longer durations of action. Thus, glyburide, glipizide, and glimepiride are 50 to 100 times more active, with plasma half-lives of 1 to 4 hours and durations of action up to 24 hours. This enhancement of activity is the result of the strong binding affinity to the ATP-sensitive K^+ channel associated with the larger *p*-(β -arylcarboxyamidoethyl) group, which replaces the small lipophilic *p*-substituents found in the first-generation agents (Fig. 27.11).

As mentioned previously, the mechanism of action of all the sulfonylureas is to stimulate the release of insulin from β -cells of the pancreas. These cells metabolize

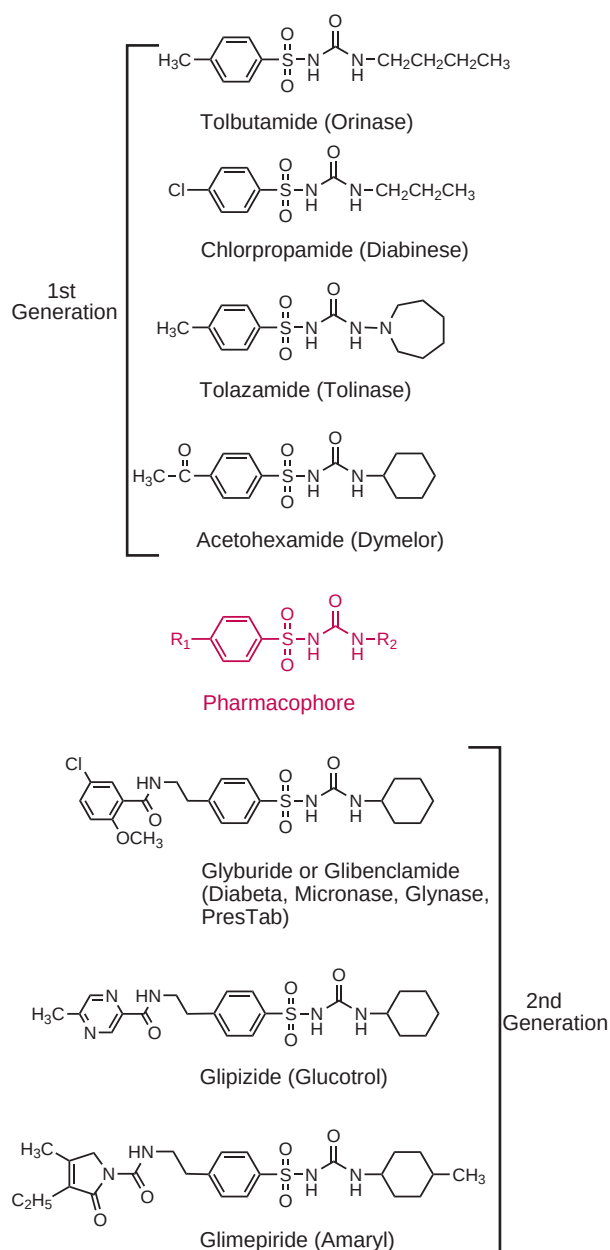


FIGURE 27.11 First- and second-generation sulfonylurea oral hypoglycemics.

glucose in the mitochondria to produce ATP, which increases the intracellular ratio of ATP/ADP, resulting in the closure of the ATP-sensitive K^+ channel on the plasma membrane. Closure of this channel triggers the opening of voltage-sensitive Ca^{2+} channels, leading to rapid influx of Ca^{2+} . Increased intracellular Ca^{2+} causes an alteration in the cytoskeleton and stimulates translocation of insulin-containing granules to the plasma membrane and the exocytotic release of insulin (51).

The ATP-sensitive K^+ channel is an octameric hetero-complex consisting of two units of the binding site for both sulfonylureas and ATP designated as the sulfonylurea receptor type 1 (SUR1) and an inwardly rectifying

K^+ channel (Kir6.2) (46). Sulfonylurea binding to the SUR1 causes the same effect as an increase in the ATP/ADP ratio, which is to close the channel leading to the secretion of insulin. The ATP-sensitive K^+ channels are not only found in the plasma membrane of β -cells of the pancreas but also in the vasculature (SUR2B) and the cardiac myocytes (SUR2A). The binding of sulfonylureas to vascular and cardiac SURs, like the binding to the β -cell receptors, has a number of physiologic consequences. In cardiac myocytes, binding to the SUR2B prevents the opening of these channels, which prevents the reduction of Ca^{2+} influx and results in myocardial cell death. In vascular cells, blocking the SUR2B by sulfonylureas increases muscular tone, resulting in decreased flow. Both of these effects are detrimental to the heart during an ischemic attack (52). Furthermore, inhibition of SUR2A in the heart prevents ischemic preconditioning. Ischemic preconditioning results in reducing the size of an infarct and is a protection resulting from repeated ischemic episodes (53).

Meyer et al. (54) have studied the structure–activity relationships of the sulfonylureas for interaction with SUR subtypes. They concluded that the anionic groups ($pK_a \sim 5$) on these compounds were key to their interaction with the three subtypes of SUR1, SUR2A, and SUR2B. Their selectivity for the SUR1 subtype is due to the lipophilic substituents (cyclohexyl, azepinyl, butyl, propyl) on the non-sulfonyl-attached urea nitrogen. Most interestingly, they found that the *p*-(β -arylcarboxyamidoethyl) groups on the second-generation sulfonylureas, which greatly enhanced binding affinity (54,55), contributed little to their selectivity for the SUR1 subunit.

SULFONYLUREA PHARMACOKINETICS AND METABOLISM Sulfonylureas are highly protein bound, primarily to albumin, which leads to a large volume of distribution (~ 0.2 L/kg) (Table 27.7). Food can delay the absorption of these drugs but does not typically affect bioavailability. Metabolism takes place in the liver, and the metabolites are renal excreted.

Chlorpropamide has a considerably longer half-life than the other sulfonylureas and, as a result, has a greater tendency for adverse effects. One explanation for the long half-life is that its metabolism (ω and ω -1 hydroxylation of the propyl group) is slow. A significant amount of the drug ($\sim 20\%$) is excreted unchanged.

In contrast, tolbutamide and tolazamide undergo a more rapid benzylic oxidation, leading to an inactive benzoic acid derivative (Fig. 27.12). An alternative hydroxylation of the aliphatic ring of tolazamide to an active metabolite results in a prolonged duration of action relative to tolbutamide.

The major metabolite of acetohexamide is reduction of the keto group, forming an alcohol. The hydroxy metabolite exhibits 2.5-fold the hypoglycemic activity of the parent molecule. An additional reported metabolite results from hydroxylation of the cyclohexyl group at the 4'-position, leading to inactivity.

TABLE 27.7 Pharmacokinetic Properties of the Sulfonylureas

Drug (sulfonylureas)	Equivalent Dose (mg)	Serum Protein Binding (%)	Half-Life (hours)	Duration (hours)	Renal Excretion (%)
Tolbutamide	1,000	95–97	4.5–6.5	6–12	100
Chlorpropamide	250	88–96	36	Up to 60	80–90
Tolazamide	250	94	7	12–14	85
Acetohexamide	500	65–88	6–8	12–18	60
Glyburide	5	99	1.5–3.0	Up to 24	50
Glipizide	5	92–97	4	Up to 24	68
Glimepiride	2	99	2–3	Up to 24	40

Glipizide and glyburide are extensively metabolized (Fig. 27.13) to less active or inactive metabolites. Glipizide metabolites are excreted primarily in the urine, whereas glyburide metabolites are excreted equally in the urine and bile.

Glimepiride is metabolized in the liver, primarily by CYP2C9, to the active metabolite M-1 (Fig. 27.14), which is then further metabolized to the inactive metabolite M-2.

MEGLITINIDES Meglitinide is the prototype structure that defines this class of insulin secretagogues. It is the benzoic acid derivative of the non-sulfonylurea moiety of glibenclamide [i.e., the *p*-(β -arylcarboxyamidoethyl)] group (Fig. 27.15). These agents exert their effects by inducing closure of the ATP-sensitive K^+ channel found on the plasma membrane of the pancreatic β -cells (56).

Repaglinide is an analog of meglitinide containing additional substituents (*m*-ethoxy, isobutyl and piperidine ring) added to the basic benzoic acid structure. Repaglinide stimulates insulin secretion by binding to three different receptors on the β -cells; one of them is the SUR1, whereas the other two receptors have been identified via Western blots but have yet to be completely characterized (57). Repaglinide is not tissue specific and

binds well to SUR1, SUR2A, and SUR2B found on cardiac and smooth muscle cells, therefore conferring extrapancreatic effects in addition to the stimulation of insulin in much the same way as glimepiride (58).

Repaglinide has a rapid onset and short duration of action compared to other hypoglycemic drugs. It is not associated with the prolonged hyperinsulinemia seen with the sulfonylureas, and possibly for this reason, it produces fewer side effects, including weight gain and potentially dangerous hypoglycemia. Repaglinide is at least 5-fold more potent than glyburide on intravenous administration and nearly 10-fold more active on oral administration.

Nateglinide is a phenylalanine analog of meglitinide where the phenyl carboxyl group is transposed to the α -carbon of the ethyl side chain, creating the amino acid functionality. Nateglinide binds selectively to the SUR1 on the β -cells and has a much lower affinity for cardiac and skeletal muscle tissue (59). Nateglinide is a rapidly absorbed insulin secretagogue that has a mechanism of action similar to that of repaglinide, with effects appearing within 20 minutes following oral dosing. Bioavailability is

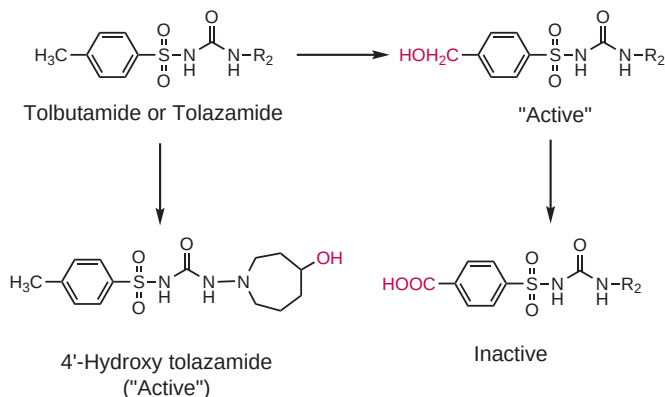


FIGURE 27.12 Metabolism of tolbutamide and tolazamide.

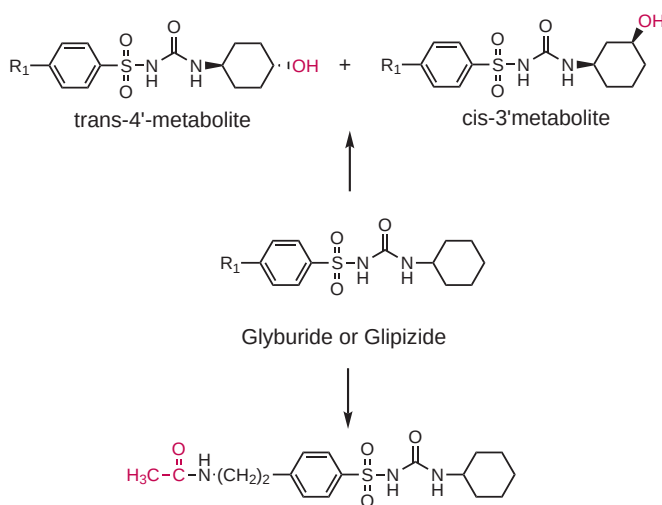


FIGURE 27.13 Metabolism of glyburide and glipizide.

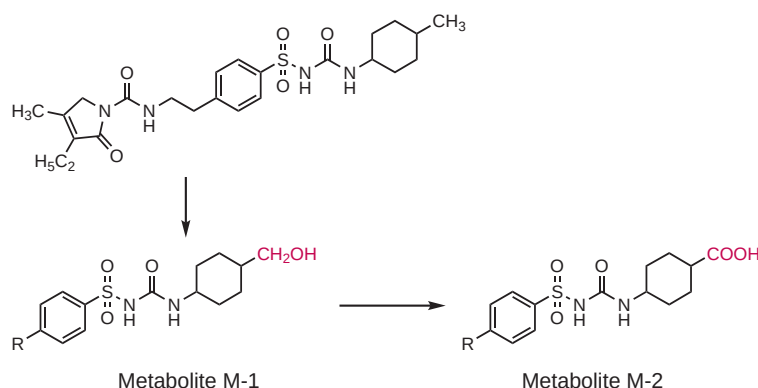


FIGURE 27.14 Metabolism of glimepiride.

73%, and it is 98% protein bound, primarily to albumin. Nateglinide is tissue selective, with low affinity for cardiac and skeletal muscle. It is metabolized in the liver, with 16% excreted in the urine unchanged. The major metabolites are hydroxyl derivatives (CYP2C9, 70%; CYP3A4, 30%) that are further conjugated to glucuronide derivatives (Fig. 27.16). The drug has an elimination half-life of 1.5 hours.

Molecular modeling studies have shown that there is a similar conformation between the sulfonylurea hypoglycemics, glibenclamide and glimepiride, and the meglitinides, which might be important in the interaction of these drugs with the SUR1 on the β -cells. Specifically, these agents display a common U-shape formed by the hydrophobic interaction between the saturated hydrocarbon cyclic moieties, such as cyclohexyl, that are located at the extremity of these molecules and a peptide bond at the bottom of the U. Moreover, the inactive analogs of the meglitinides, including the inactive enantiomer of repaglinide, are unable to adopt this U-shape and thus cannot bind effectively to the SUR1 (60).

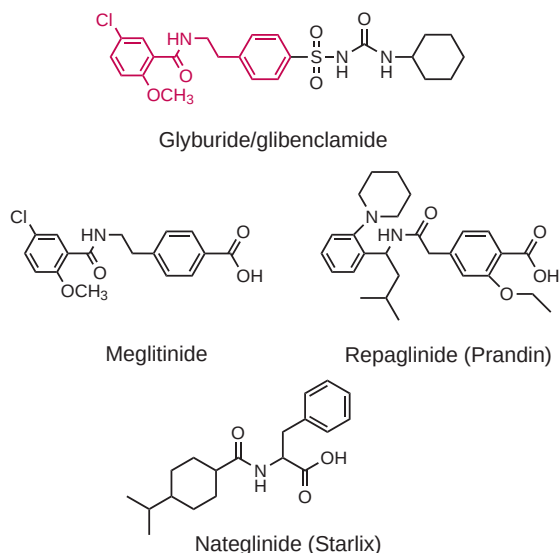


FIGURE 27.15 The meglitinide hypoglycemic agents.

BIGUANIDES Oral hypoglycemic agents from the biguanide chemical class primarily act by reducing hepatic glucose production and by enhancing insulin sensitivity. Although the mechanism of action is not completely understood, biguanides reduce hepatic glucose output by decreasing gluconeogenesis and stimulating glycolysis (61,62). They also increase insulin signaling, which means an increase in insulin receptor activity, as well as metabolic responsiveness by both, the liver and skeletal muscle (61). Unlike the insulin secretagogues, insulin secretion is not directly affected by the biguanides, so they do not cause hypoglycemia. Importantly, biguanides do not induce weight gain, which is very useful for insulin-resistant obese patients (63). They are also considered to have antihypertriglyceridemic effects and to possess vasoprotective properties, which are important actions in treating cardiovascular complications (64). In addition, the biguanides block the breakdown of fatty acids through the activation of AMP-dependent protein kinase.

The use of biguanides can be traced back to medieval times. The plant *Galega officinalis* was not only traditionally used for promoting perspiration during plague epidemics and as a galactagogue (a substance that stimulates lactation) in animals like cows, but it was also prescribed for the relief of intense urination associated with diabetes. *G. officinalis* has several common names, including Goat's rue, French lilac, and Italian fitch. This plant contains active compounds, such as galegine (isoamylene guanidine), that have been shown to have blood glucose-lowering effects (65). The hypoglycemic properties of the plant ultimately led to the synthesis of the biguanide compounds. The biguanides are chemically represented by the linkage of two guanidine groups with different side chains. Despite the toxicity associated with guanidine, the biguanides were shown to exert beneficial effects and quickly became available therapeutic agents for diabetes in the 1950s.

The biguanides (Fig. 27.17) include metformin, phenformin, and buformin. Approved in 1995, metformin is currently the only available biguanide in the United States and is considered to be the first-line treatment for

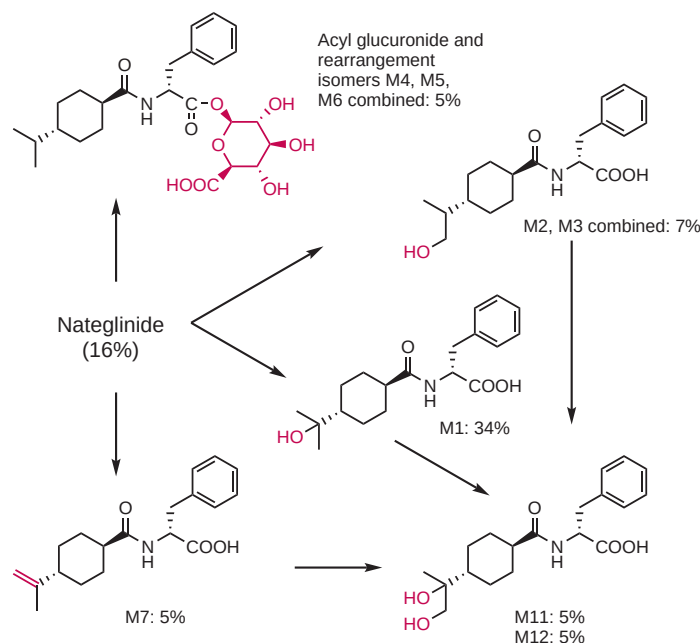


FIGURE 27.16 Urinary metabolites of nateglinide.

type 2 diabetes. It is widely used as monotherapy or in combination with oral agents from the class of sulfonylureas or thiazolidinediones (66).

Metformin makes peripheral muscle tissues more sensitive to insulin for glucose uptake, resulting in the lowering of blood glucose levels. In addition to improving insulin sensitivity in the muscle and liver, it decreases hepatic glucose production by gluconeogenesis, as well as increases oxidative metabolism and even glucose utilization by the intestine through nonoxidative metabolism (64). Metformin can also contribute to the translocation of glucose transporters to the cell surface (65). Although the mechanism of action is not clear, studies provide support showing AMP-dependent protein kinase to be its primary target. Metformin is also useful in treating other types of conditions, such as the insulin resistance experienced by women with polycystic ovary syndrome (67). Side effects include diarrhea, which can be minimized with food intake, mild gastrointestinal distress, and lactic acidosis (68). Metformin is quickly absorbed from the small intestine. Bioavailability is from 50% to 60%, and the drug is not protein bound. Peak plasma concentrations occur at approximately 2 hours. The drug is widely distributed in the body and accumulates in the wall of the small intestine. This depot of drug

serves to maintain plasma concentrations. Metformin is excreted in the urine, via tubular excretion, as unmetabolized drug with a half-life of approximately 2 to 5 hours; therefore, renal impairment and hepatic disease are contraindications for the drug. One key drug–drug interaction of note is the competitive inhibition of renal excretion of metformin by cimetidine, which can lead to increased metformin blood levels (43).

The other two antidiabetic drugs from the biguanide class are phenformin and buformin (Fig. 27.17). Although relatively popular in the 1960s, they were withdrawn from the market in the early 1970s due to toxic lactic acidosis effects and an increased incidence of cardiac mortality. Despite the removal of phenformin from the U.S. market, it has been available illegally as a metformin replacement, requiring the U.S. Food and Drug Administration (FDA) to recall many phenformin-containing medications (69).

Phenformin and buformin are derivatives of metformin. Specifically the addition of a phenyl ring yields phenformin, and the addition of an *N*-butyl side chain yields buformin. Both derivatives have increased lipophilicity and have the same mechanism of action as metformin. However, they both have the major drawback of a high frequency of lactic acidosis in diabetic patients. Lactic acidosis is the result of the accumulation of lactic acid associated with the decreased gluconeogenesis caused by the biguanides. Metformin can also induce such a complication; however, the risk is considered to be minimal when administered at a proper dose. In fact, diabetic subjects taking phenformin are at 10 to 20 times higher risk of developing lactic acidosis than those taking metformin (70). Nevertheless, metformin is not recommended for

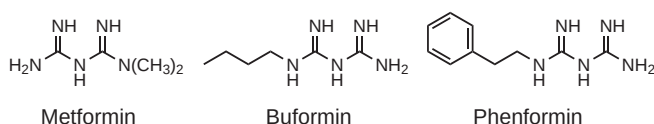


FIGURE 27.17 Chemical structures of the biguanides.

type 2 diabetic patients who are inclined toward metabolic ketoacidosis due to certain conditions including hepatic disease, heart failure, respiratory disease, hypoxemia, severe infection, alcohol abuse, or renal disease (61).

INSULIN SENSITIZERS (PEROXISOME PROLIFERATOR-ACTIVATED RECEPTOR [PPAR] AGONISTS) Activators of PPAR γ in the treatment of insulin resistance and type 2 diabetes mellitus are a much sought after target, because PPARs are central regulators of lipid, carbohydrate, and inflammatory pathways and help maintain homeostasis (71). They belong to the nuclear hormone receptor superfamily of ligand-activated transcription factors and are closely related to steroid, retinoid, and thyroid hormone receptors. This receptor family is comprised of three members: PPAR α , δ , and γ . PPAR δ is ubiquitously present in tissues of adult mammals, whereas the α subtype is abundantly present in tissues catalyzing lipid oxidation, which include the liver, kidney, and heart (72). PPAR γ is primarily expressed in adipose tissue, where it helps control its differentiation.

The thiazolidinediones (TZDs) are classic examples of PPAR γ agonists and are commonly referred to as the “glitazones.” These agents were developed when clofibrate acid analogs were being screened for antihyperglycemic and lipid-lowering activity. Although initially the mechanism of action of the TZDs was unclear, it was soon discovered that they enhanced adipocyte differentiation by activation of the nuclear hormone receptor superfamily, PPAR (73). A ligand, which can be endogenous, upon binding to PPAR, induces a conformational change in the receptor, thus stabilizing the interaction with the retinoid X receptor and, in turn, resulting in the stimulation of transcription of target genes (74). The endogenous ligands for PPAR γ have not been identified; however, studies suggest that certain arachidonic acid metabolites and long-chain unsaturated fatty acids such as linoleic acid are the intrinsic agonists (Fig. 27.18) (75).

PPAR γ agonists, such as the glitazones, act by increasing the sensitivity of cells to insulin. The glitazones also decrease both systemic fatty acid production and fatty acid uptake, which contribute to increased sensitization of cells to insulin. Patients with type 2 diabetes are known to have high triglyceride and low high-density lipoprotein levels. The glitazones increase the lipolysis of triglycerides in very low-density lipoproteins and, as a result, increase high-density lipoprotein levels. However, during the lipolysis of very low-density lipoproteins, the production of low-density lipoproteins could be a major drawback to the use of these drugs (76,77).

PPAR γ activation improves glucose uptake by skeletal muscle and, at the same time, reduces glucose production by slowing down gluconeogenesis. Hence, these drugs improve metabolism of glucose in not only diabetic patients, but also in obese individuals who have impaired glucose tolerance (71). As mentioned earlier, the first PPAR γ agonists to be introduced were the glitazones. The pharmacophore responsible for activity is the thiazolidinedione moiety outlined in red in

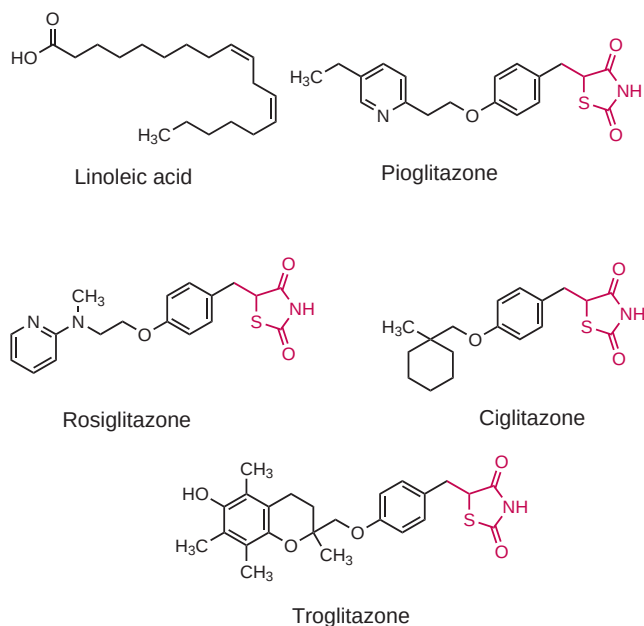


FIGURE 27.18 Linoleic acid and thiazolidinedione (TDZ) PPAR γ agonists (glitazones).

Figure 27.18. A phenyl ring attached to the central nucleus via a methylene group is essential for activity, and in many instances, a saturated linker is found to be more potent than the unsaturated counterpart. The first generation of TZDs includes pioglitazone, rosiglitazone, and ciglitazone. The rationale used for the development of these agents was the fact that the structure of troglitazone (the first drug in this class to be marketed) includes the structure of α -tocopherol, an antioxidant, which retards the oxidation of low-density lipoproteins (78). However, due to severe drug-induced hepatotoxicity and cardiovascular effects, troglitazone and rosiglitazone have been withdrawn, leaving pioglitazone as the only clinically used drug in the TZD family (79). The metabolism of pioglitazone has been studied in rats and dogs and has led to the discovery of up to eight metabolic products. These products result from oxidation at either carbon adjacent to the pyridine ring and are found as various conjugates in the urine and bile (Fig. 27.19). Metabolites M-1, M-2, and M-3 appear to contribute to the biologic activity of pioglitazone (43).

Recently, dual PPAR α / γ agonists have become much sought after targets, and many research groups are actively involved in synthesizing such bioactive compounds as novel antidiabetic agents. Combined activation of PPAR α and PPAR γ is believed to induce complementary and synergistic action on lipid metabolism, insulin sensitivity, and inflammation control, possibly circumventing or reducing the side effects of PPAR γ . Cotreatment was shown to improve overall metabolic parameters including free fatty acids, glycemia, and insulin resistance. In addition, edema associated with rosiglitazone treatment was reduced, and a lowering of

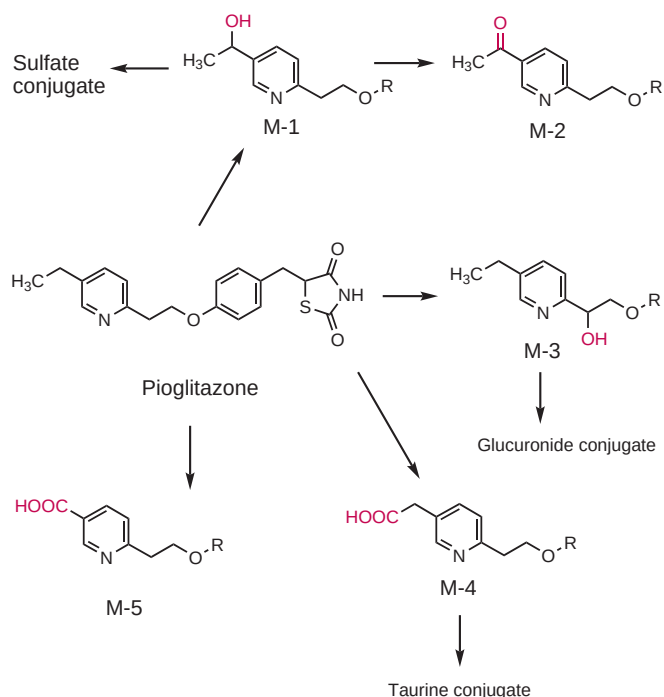


FIGURE 27.19 Metabolic pathway of pioglitazone.

HbA_{1c} levels was more effective than with rosiglitazone used alone. Ragaglitazar and muraglitazar (Fig. 27.20) are examples of dual PPAR α/γ agonists; however, the latter was withdrawn from clinical trials due to cardiovascular toxicity (80,81). Several 2,4-thiazolidinedione derivatives of 1,3-benzoxazinone were synthesized and evaluated in the ob/ob mouse model. DRF-2519, the most active analog, is being further developed, due to its equivalence in activity to the existing TZDs and significant improvement in lipid parameters in fat-fed rats, in which it tested better than the fibric acid antihyperlipidemics (82).

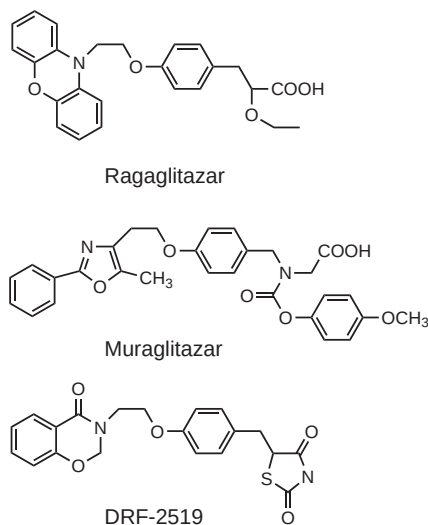


FIGURE 27.20 Examples of dual PPAR α/γ agonists.

α -GLUCOSIDASE INHIBITORS α -Amylase (EC 3.2.1.1) and α -glucosidase (EC 3.2.1.20) are key enzymes responsible for the metabolism of carbohydrates. The salivary and pancreatic α -amylases are responsible for the breakdown of complex polysaccharides into oligo- and disaccharides, preparing them for intestinal absorption. α -Glucosidase, which consists of maltase, sucrase, isomaltase and glucoamylase, is a membrane-bound enzyme present in the brush border of the small intestine in relatively high concentrations in the proximal part of the jejunum. This enzyme catalyzes the conversion of the disaccharides sucrose and maltose into glucose. The resulting monosaccharides are then absorbed by the enterocytes of the jejunum and enter systemic circulation, as well as various biochemical pathways for the production of energy (83). Thus, inhibiting α -glucosidase will delay the process of carbohydrate absorption in the gut by moving these undigested disaccharides into the distal sections of the small intestine and colon. The result is the prevention of glucose production, thereby reducing postprandial hyperglycemia. All known α -glucosidase inhibitors are excreted unchanged in the feces, obviating metabolic drug interactions (84).

The α -glucosidase inhibitors were first introduced in 1996 with the drug acarbose (Fig. 27.21) (85). Acarbose is an oligosaccharide obtained from *Actinomyces utahensis* and is the drug of choice in this category. It is a competitive inhibitor with a high affinity for sucrase and a lesser affinity for glucoamylase and pancreatic α -amylase in humans (86). When used in monotherapy, there is no risk of hypoglycemia and weight gain, as seen with the first- and second-generation sulfonylureas. However, gastrointestinal irritation, bloating, and flatulence caused by fermentation of undigested sugars in the large bowel by intestinal microflora are some drawbacks common to all α -glucosidase inhibitors (83). These side effects can be minimized to a certain extent by gradual dose titration

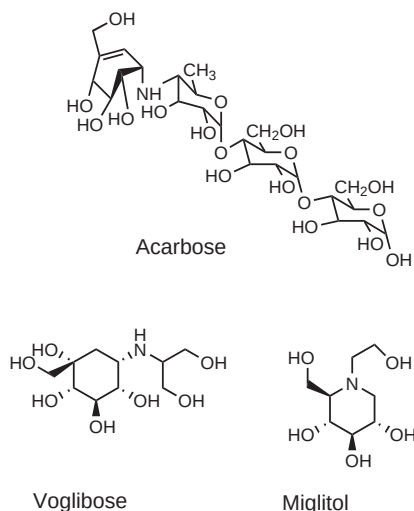


FIGURE 27.21 Clinically useful α -glucosidase inhibitors.

and the right combination therapy with other orally active hypoglycemic drugs.

Voglibose and miglitol are other α -glucosidase inhibitors in clinical use in the management of diabetes. α -Glucosidase inhibitors reduce postprandial hyperglycemia to a lesser extent as compared to other oral antidiabetic agents, and clinical trials with acarbose have shown that the reduction in HbA_{1c} levels is 0.5% to 1% when compared to placebo (85). Therefore, these agents are seldom used as monotherapy and frequently find use in combination therapy especially with sulfonylureas (86). Apart from their use in diabetes, their use can be extended to the treatment of glycosphingolipid lysosomal storage disease, HIV infections, and certain tumors (87,88).

A number of iminosugars isolated from plants were found to have potent α -glucosidase inhibitory activity. The presence of polyhydroxy groups on these compounds is critical for α -glucosidase inhibition activity, because most mimic the natural substrates maltose and sucrose (87). A polyhydroxy piperidine compound, nojirimycin (Fig. 27.22), isolated from *Bacillus* species, was the first compound representative of this family. Research modifying the structure of iminosugars yielded the most potent compound 1-deoxynojirimycin, which is now clinically used in the treatment of Gaucher syndrome (87).

Naturally occurring compounds from different chemical classes have been identified to have α -glucosidase inhibitory activity. Genistein (Fig. 27.22) is a potent

isoflavone isolated from soy fermentation broths of *Streptomyces* species and has been demonstrated to be a reversible, noncompetitive α -glucosidase inhibitor (89).

Some cinnamic acid analogs of particular interest have been isolated from the rhizomes of *Kaempferia galangal*. A 4-methoxy-*trans*-cinnamic acid and its corresponding ethyl ester are inhibitors of maltase ($K_i = 0.05$ mmol/L) (Fig. 27.22). Both of these methoxy derivatives of cinnamic acid showed improved enzyme inhibitory activity, but replacing the electron-rich hydroxy groups on the molecule or introduction of an electron-withdrawing nitro group renders them inactive (90). Closely related to the cinnamic acids is the noncompetitive inhibitor bisdemethoxycurcumin (Fig. 27.22), isolated from rhizomes of *Curcuma longa*, which was found to be twice as active as the clinically used pseudotetrasaccharide acarbose (91).

Green tea, which is an excellent source of polyphenols such as flavonoids, catechins, and theaflavins, has been studied extensively in the control of blood glucose by inhibition of the carbohydrate-metabolizing enzymes. Among the theaflavin derivatives, 3-*O*-gallate and its 3,3'-di-*O*-gallate analog are potent inhibitors of rat and porcine intestinal maltase (92) (Fig. 27.22).

The alkaloid castanospermine (Fig. 27.22), an indolizine that closely resembles 1-deoxynojirimycin, was isolated from the seeds of *Castanospermum australe*. This has been one of the few isolated compounds that, in addition to its α - and β -glucosidase-inhibiting properties, inhibits β -glucocerebrosidase and viral replication of HIV (87).

Although there are few α -glucosidase inhibitors that have been approved by the FDA, current literature shows an increasing awareness in the field of these carbohydrate-metabolizing enzyme inhibitors, which can lead to novel α -glucosidase inhibitors in the near future.

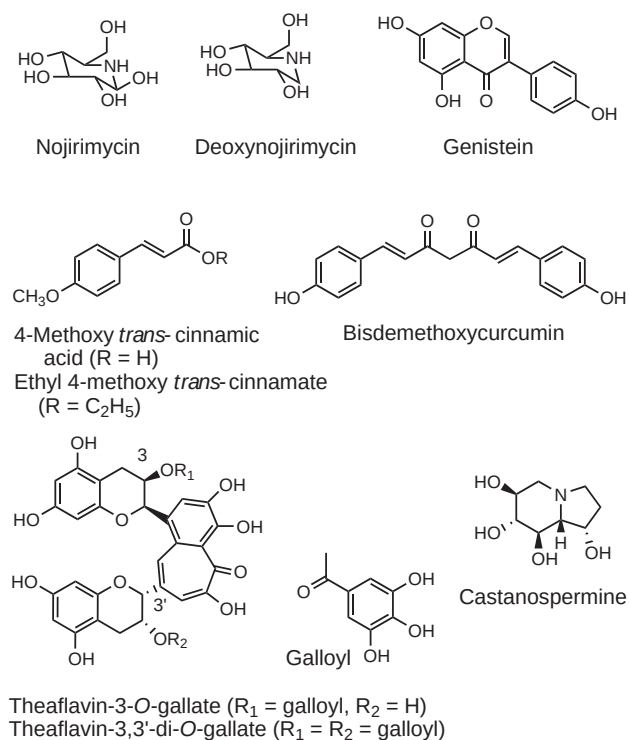
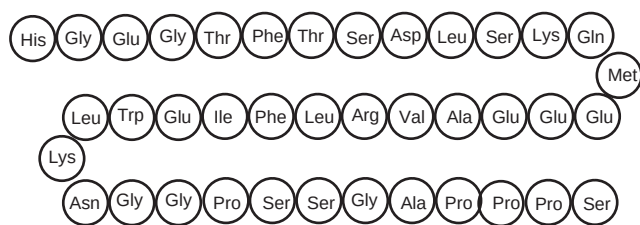


FIGURE 27.22 Naturally occurring α -glucosidase inhibitors.

GLP-1 AGONISTS AND DPP-IV INHIBITORS GLP-1 is a 36-amino acid peptide secreted by L-cells of the gut in response to a meal. It exerts control over glucose levels by promoting insulin secretion in a glucose-dependent manner (93). The role of GLP-1 was first proposed based on the observation that the amount of insulin secreted following an oral glucose dose exceeded that of an equivalent glucose dose administered intravenously in both diabetic and nondiabetic individuals (94). This observation was termed the incretin effect and is the result of two gut hormones, GLP-1 and GIP (95). GLP-1 secretion from L-cells is similar to that of glucose-induced insulin secretion from pancreatic β -cells. Metabolism of glucose in the intestinal L-cells leads to closure of ATP-linked potassium (K^+) channels, resulting in depolarization of the membrane and entry of Ca^{2+} , which leads to the secretion of GLP-1 (96). GLP-1 is rapidly metabolized, with a half-life of 1 to 2 minutes, by an aminopeptidase enzyme, DPP-IV, yielding an inactive peptide that is two amino acids shorter (97,98). It follows, therefore, that GLP-1 agonists or DPP-IV inhibitors would be effective agents to control blood glucose levels in diabetic patients.

GLP-1 Analogs GLP-1 is deactivated by DPP-IV, which removes a dipeptide from the N-terminus. One of the principal reasons why GLP-1 is so susceptible to DPP-IV is because it contains an alanine in the penultimate N-terminal position. Substitution in this position should result in analogs with increased stability. Indeed, substitution of the Ala with Thr (half-life = 197 minutes), Gly (half-life = 159 minutes), Ser (half-life = 174 minutes), or α -aminoisobutyric acid each gives analogs that are more stable in vitro than GLP-1 (half-life = 28 minutes) under the same conditions (99). In fact, the α -aminoisobutyric acid analog exhibited no degradation even after 6 hours. While such analogs are found to be more stable to DPP-IV in vitro, the in vivo half-life is increased only from 1 to 2 to 3 to 4 minutes, and this is attributed to rapid elimination by the kidneys. Interestingly, these analogs still retained binding affinity to the GLP-1 receptor, with the α -aminoisobutyric analog almost twice as potent as GLP-1. More useful in vivo analogs, therefore, require both DPP-IV resistance and decreased renal elimination.

This was achieved with the introduction of the GLP-1 analog exenatide as a parenteral therapy into the market in April 2005. Exenatide is a 39-amino acid peptide analog of GLP-1, isolated from the saliva of the Gila monster (*Heloderma suspectum*), that is resistant to the action of DPP-IV and is a GLP-1 receptor agonist. It has a glycine instead of alanine in the penultimate N-terminal position, is 53% homologous to human GLP-1, and has an in vivo half-life of approximately 3 hours.

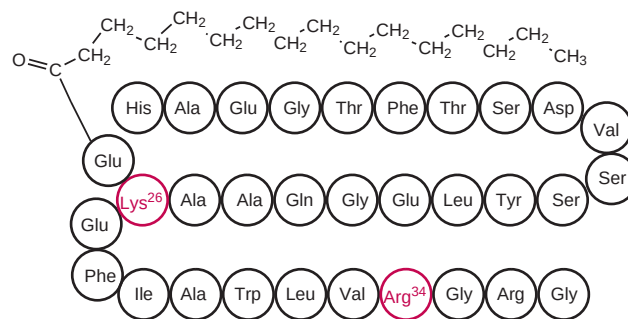


Exenatide (Byetta)

Exenatide has been shown to reduce HbA_{1c} levels in sulfonylurea-treated patients with type 2 diabetes and is associated with weight loss (100). This weight loss is probably due to a decrease in appetite. In *db/db* mice, it increased β -cell mass and delayed the onset of diabetes, so it can even have a prophylactic use (101).

Several other DPP-IV-resistant GLP-1 analogs are currently undergoing human clinical trials. The most promising was liraglutide (102). Liraglutide, which represents amino acid residues 7 to 37 of GLP-1, was developed from a series of acylated GLP-1(7-37) derivatives. Several positions of GLP-1 were substituted with different acyl moieties ranging in length from 12 to 18 carbons. However, the C16 derivative, α -L-glutamoyl-(*N*- α -hexadecanoyl)-Lys²⁶Arg³⁴-GLP-1 (liraglutide) was found to have the best combination of albumin binding to retard renal elimination and resistance to DPP-IV degradation due

to the replacement of Lys³⁴ with Arg (103,104). This analog of GLP-1(7-37) has multiple actions in addition to reduction of hyperglycemia, including, suppression of inappropriate glucagon secretion, slowing of gastric emptying, and enhancement of β -cell function and mass (105). Liraglutide was approved for marketing under the name Victoza by the FDA in 2010; however, due to a possible link to the production of thyroid C-cell tumors, the product label has a Black Box Warning with the recommendation that it be used only in patients for whom the potential benefits outweigh the potential risk (106).



Liraglutide (Victoza)

The major problem with GLP-1 analogs is their need for subcutaneous administration, which can limit patient compliance. Also, GLP-1 analogs, as with any peptide drug, have the potential to be immunogenic. Therefore, this stimulated the search for small-molecule inhibitors of DPP-IV.

DPP-IV Inhibitors DPP-IV (EC 3.4.14.5) is a serine protease that exists as both a membrane-bound and a plasma-soluble form. It is a proline-specific aminopeptidase responsible for the degradation of a number of biologically important peptides other than GLP-1 and GIP. DPP-IV belongs to a family of dipeptidyl peptidase enzymes, including DPP-VI to DPP-IX, DPP-X, fibroblast protein-a, acylaminoacyl carboxypeptidase, prolyl carboxypeptidase, and quiescent cell proline dipeptidase (107). Because the clinical importance of these other prolyl peptidases has not been established, it is important that inhibitors have specificity for DPP-IV.

There are currently four DPP-IV inhibitors commercially available: sitagliptin, vildagliptin, linagliptin, and saxagliptin. The development of many structural variants has led to the knowledge that there is an absolute requirement for all DPP-IV inhibitors to contain a basic amino function in the position equivalent to the penultimate amino acid (alanine) in GLP-1. Most active inhibitors are therefore peptide derivatives of α -aminoacylpyrrolidines, which also takes advantage of the enzyme's high preference for proline binding. The most active inhibitors also have an electrophilic group in the 2-position of the pyrrolidine ring. The most common group is cyano and is present in vildagliptin and saxagliptin (Fig. 27.23). The

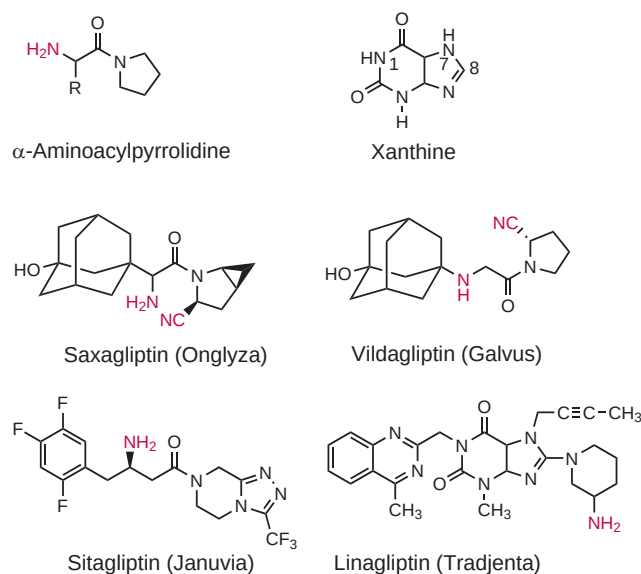


FIGURE 27.23 α -Aminoacylpyrrolidine/thiazolidine pharmacophore and currently available DPP-IV inhibitors.

ciano group forms a covalent imide adduct with the active site Ser⁶³⁰ to irreversibly inactivate it. Sitagliptin has a piperazine ring fused to a pyrazole in place of the pyrrole ring but still contains the α -amino acyl moiety and the amide bond. Recently, linagliptin, a novel xanthine-derived, potent, and selective DPP-IV inhibitor, has been developed (Fig. 27.23). It contains an N-1 quinazoline, a C-8 aminopiperidine, and an N-7 butynyl substituent, all of which play an important role in binding to the enzyme active site. The primary amine of the C-8 aminopiperidine occupies the recognition site for the amino terminus of peptide substrates for DPP-IV and hydrogen bonds to Glu²⁰⁵, Glu²⁰⁶, and Tyr⁶⁶². The butynyl substituent occupies a hydrophobic pocket, and the xanthine scaffold is positioned above Tyr⁵⁴⁷ and held in place via π -stacking aromatic interactions with the phenol of the Tyr⁵⁴⁷. The C-6 xanthine carbonyl hydrogen bonds with the backbone NH of Tyr⁶³¹, and the quinazoline moiety at N-1 interacts by π -stacking with the aromatic ring of Trp⁶²⁹ on a hydrophobic surface area of the protein (108,109).

Sitagliptin was approved by the FDA in 2006 for monotherapy or in combination with metformin, TZDs, or sulfonylureas for treatment of type 2 diabetes. Sitagliptin has a good oral bioavailability of 87% with no effect from food. It is protein bound only to about 37%, and 80% is excreted unchanged in the urine. It is important to have high specificity for DPP-IV, as inhibition of DPP-8 and DPP-9 has been shown to cause severe toxicity in animal studies. The selectivity of sitagliptin for DPP-IV compared to DPP-VIII and DPP-IX is greater than 2,600-fold. Sitagliptin has no effect on glucose levels in healthy patients; however, it reduces HbA_{1c} about 0.6% to 0.9% in type 2 diabetics. Sitagliptin has a number of

side effects including upper respiratory tract infection, nasopharyngitis, and headache. Patients experience nausea when sitagliptin is taken with metformin, leg swelling when taken with TZDs, and hypoglycemia when administered with a sulfonylurea. There have also been reports of anaphylaxis, angioedema, and rashes, although a causal link has not been established (110).

Vildagliptin was approved in 2008 by the European Medicines Agency for use in the European Union, but the manufacturer has withdrawn its intent to submit it for FDA approval because the FDA demanded additional clinical data assuring that skin lesions and kidney impairment seen in animal studies were not seen in humans. In phase III clinical trials, vildagliptin showed good oral bioavailability of 85% and low protein binding (9.3%), and was 21% excreted unchanged in the urine. It has a high specificity for DPP-IV compared to DPP-VIII and DPP-IX of 32- to 250-fold. In clinical trials, no patients discontinued medication due to adverse effects, which are similar to sitagliptin (i.e., upper respiratory tract infections, diarrhea, nausea, and hypoglycemia) (110).

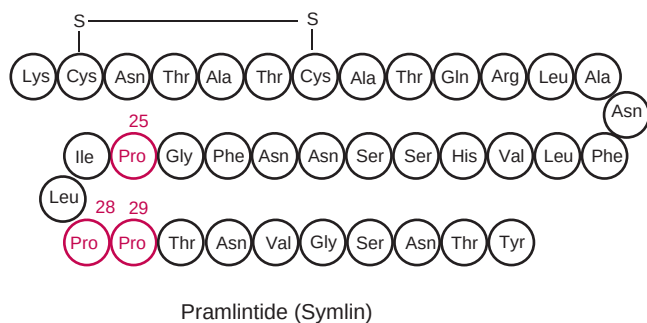
The FDA approved saxagliptin in 2009. It is 10 times more potent an inhibitor of DPP-IV than either vildagliptin or sitagliptin. It shows a higher specificity for DPP-IV compared to DPP-VIII and DPP-IX (400- and 75-fold, respectively). Trials using doses of 2.5 to 10 mg have shown reductions of HbA_{1c} of 0.5% to 0.8%, with no significant weight gain. Saxagliptin was well tolerated, with side effects equivalent to placebo and a very low incidence of hypoglycemia (110).

Linagliptin was approved by the FDA in 2011 as an oral treatment for type 2 diabetes either as a stand-alone medication or in combination with other therapies. Linagliptin should not be prescribed for patients with type 1 diabetes or those who have diabetic ketoacidosis. Its most common adverse effects are upper respiratory tract infection, stuffy or runny nose, sore throat, muscle pain, and headache. Linagliptin has high selectivity for DPP-IV versus the related enzymes DPP-VIII and DPP-IX and inhibits plasma DPP-IV activity by more than 80% over 24 hours. Linagliptin binds extensively to plasma proteins and is not significantly metabolized, with approximately 85% excreted unchanged via the feces (111).

AMYLIN AGONISTS Amylin is a hormone that consists of a single chain of 37 amino acids and is released from pancreatic β -cells, co-secreted with insulin, and primarily involved in controlling postprandial glucose levels. Amylin, like insulin, shows similar fasting and postprandial patterns in healthy individuals by a variety of mechanisms, including delayed gastric emptying and suppression of glucagon secretion (not normalized by insulin alone), which leads to a suppression of endogenous glucose output from the liver (112). Amylin also regulates food intake by modulating the appetite center of the brain. The observation that amylin was deficient in both type 1 and type 2 diabetics stimulated research

and development of amylin analogs that would be able to control postprandial glucose levels by 1) modulation of gastric emptying, 2) prevention of postprandial rise in glucagon, and 3) inhibition of caloric intake and potential weight gain. Amylin itself is unsuitable as a drug because it aggregates and is insoluble in solution, which encouraged development of chemical analogs (113).

Pramlintide is a chemical analog of amylin given enhanced water solubility and reduced aggregation liability by replacing the Ala²⁵, Ser²⁸, and Ser²⁹ of the amylin peptide chain with prolines (114). The subcutaneous administration of 15 mcg of the drug shows optimum peak effect within 20 minutes, thus permitting the drug to be used before meals. Pramlintide has a duration of action of 150 minutes, no significant accumulation liability with repetitive administration, and a renal clearance rate of 1 to 2 L/min. The drug is primarily metabolized in the kidneys, with an elimination half-life of 30 to 50 minutes. Pramlintide is used together with insulin for those who are unable to achieve their target postprandial blood sugar levels on insulin alone (115,116). Pramlintide delays gastric emptying, suppresses glucagon release, and has a central nervous system anorectic effect via unknown mechanisms. The nucleus accumbens and dorsal vagal complex of the brain have been shown to contain the amylin receptor, which can be involved in the central effects of this hormone. Because of the wide differences between the pH of pramlintide and insulin products (4.0 vs. 7.8, respectively), concurrent mixing within the same syringe is not recommended. To avoid severe hypoglycemia with initial drug titration, only short or rapid insulin dosage forms should be used together with pramlintide. In this case, the dose of insulin should be reduced by 50% (117).



Drugs in Development

Considering the medical and economic importance of type 2 diabetes and the need for improvement over existing treatments, new agents for existing targets and the development of new therapeutic targets are being actively pursued in the laboratory and the clinic. Many promising targets are only in the early stages of preclinical development (e.g., small-molecule insulin mimetics or glucagon antagonists, glucokinase activators, and β -cell potassium channel openers). The following are selected new classes of therapeutics, some of which are in clinical trials or earlier in research.

GLUCAGON ANTAGONISTS Glucagon is an important counter-regulator protein to the actions of insulin in glucose regulation. The disruption of the glucagon–insulin bimodal relationship is believed to contribute to diabetes mellitus. Glucagon is synthesized in the α -cells of the pancreatic islet cells and consists of a single 29-amino acid chain (118). The hormone is produced by proteolytic cleavage of a 69-amino acid peptide called glicentin. Glicentin, glucagon, GLP-1, and GLP-2 are all members of the enteroglucagon peptide family (119). Glucagon binds to receptors in the liver, leading to a G_s protein–coupled elevation in adenylyl cyclase activity, resulting in an increase in blood glucose as a result of a breakdown of stored glycogen, increased ketogenesis, and gluconeogenesis. Skeletal muscle glycogen does not appear to be affected by glucagon, probably as a result of the lack of specific receptors at this tissue site. Glucagon can also cause pancreatic β -cell insulin release, catecholamine release from adrenal tumors, and calcium release from medullary tumors (120).

Because diabetic hypoglycemic glucose control is antagonized by glucagon and the diabetic patient exhibits high basal glucagon levels, research has been directed toward the development of glucagon antagonists. Recent research has identified several nonpeptide antagonists (121–128). These antagonists cover a variety of chemical classes (Fig. 27.24). These include, but are not limited to, bi-phenyl and 4-phenyl pyridine derivatives, tri-aryl imidazoles and pyrroles, isoxadiazole thiophenes, alkylidene hydrazides, tri-substituted ureas and their methylene analogs, and a 2-thio benzimidazole derivative. Currently the glucagon antagonists NNC92-1687 and Bay27-9955 are in clinical trials. Bay27-9955 appears to be the most promising agent, with a glucagon-binding half maximal inhibitory concentration of approximately 110 nmol/L and a glucagon-stimulated cyclic adenosine monophosphate accumulation in cells reported as approximately 46 nmol/L (129).

PROTEIN TYROSINE PHOSPHATASE 1B INHIBITORS Protein tyrosine phosphatase (PTB1B) and its potential role in the treatment of type 2 diabetes was first discovered in mice lacking the *PTB1B* gene in 1999. Studies using the PTB1B knockout mice demonstrated enhanced sensitivity to insulin due to the prolonged phosphorylation of the insulin receptor. In addition, a lack of PTB1B activity is associated with an increased ability to clear glucose in both glucose and insulin tolerance tests, as well as to cause a decrease in plasma insulin levels. In addition, mice lacking the *PTB1B* gene were resistant to weight gain when placed on a high-fat diet, in which no increase in total plasma triglycerides and cholesterol levels were detected (130,131). Since then, PTB1B has become an attractive, novel drug target in the development of potent and selective inhibitors for the treatment of both type 2 diabetes and obesity (132).

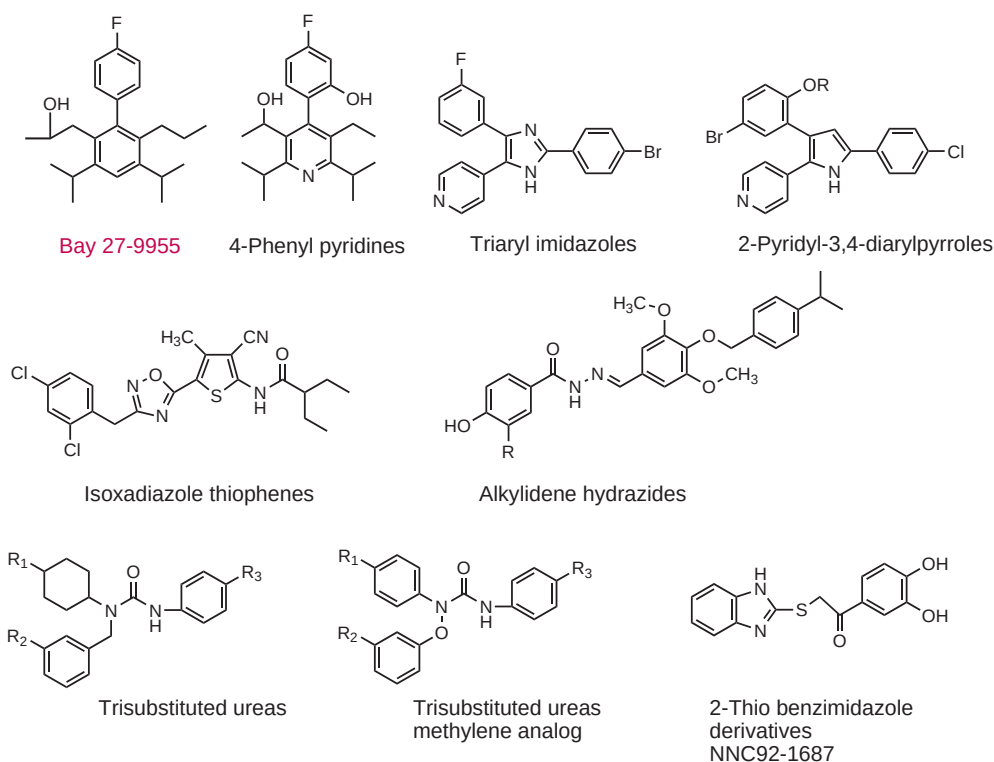


FIGURE 27.24 Structures of representative glucagon antagonists.

The PTPs belong to a superfamily of enzymes that catalyze protein tyrosine dephosphorylation. The importance of this class of enzymes is as modulators in various signal transduction pathways (133). Sequence analysis of the human genome indicates the existence of 107 PTP genes as compared to 90 human protein tyrosine kinase genes (134). The PTP superfamily of enzymes consists of three major subfamilies, which are divided according to their substrate specificity: classical phosphotyrosine-specific PTPs, dual specific PTPs, and low molecular weight PTPs. Currently, there are 38 PTPs belonging to the classical subfamily, and as the name suggests, they target proteins containing phosphotyrosine. The dual specific PTPs recognize proteins that contain phosphothreonine, phosphoserine, and phosphotyrosine residues (134,135).

PTB1B belongs to the subclass of classical phosphotyrosine-specific PTPs. PTB1B was first isolated from human placental tissue and is found on the cytoplasmic face of the endoplasmic reticulum. It is universally expressed and is found in insulin-targeted tissues including the liver, muscle, and fat (136,137). PTB1B plays an important role in the regulation of signal transduction networks. These signaling pathways are regulated by protein tyrosine kinases that catalyze the tyrosine phosphorylation cascade, a major posttranslational modification used by cells. To counterbalance the kinases, the PTPs are responsible for dephosphorylation processes, which control the rate and duration of a particular response. Therefore, both the protein tyrosine kinases and PTPs are equally important in signaling pathways, and an

imbalance or defective operation of these networks can cause abnormal tyrosine phosphorylation, the consequence of which is often seen by the development of human diseases (133–135).

Type 2 diabetics have an inability to propagate the insulin signaling pathway, resulting in insulin resistance (130,138). One possible explanation for insulin resistance would be abnormalities of the insulin signaling events downstream from the receptor (138). In the complex insulin signaling pathway, PTB1B participates by dephosphorylating the tyrosine residues on the activated β subunit of the insulin receptor, as well as dephosphorylating the IRSs (130,132). Therefore, PTB1B acts as a negative regulator in the insulin signal transduction pathway (139). In fact, an overexpression of PTB1B has been shown to cause a decrease in the insulin-stimulated pathway, and thus contributes to diabetes and obesity. Therefore, inhibition of PTB1B activity would prove to be advantageous in augmenting insulin-initiated signaling events. Therefore, there is interest in development of PTB1B inhibitors as a potential treatment for type 2 diabetes.

Numerous PTB1B inhibitors have been synthesized with the hope of developing treatments for type 2 diabetes and obesity. Figure 27.25 shows the variety of chemical structures of some PTB1B inhibitors in development (132). Designing effective PTB1B inhibitors poses multiple challenges to drug development. The first and major challenge is that in order to design selective active site-directed compounds, they must mimic the negatively

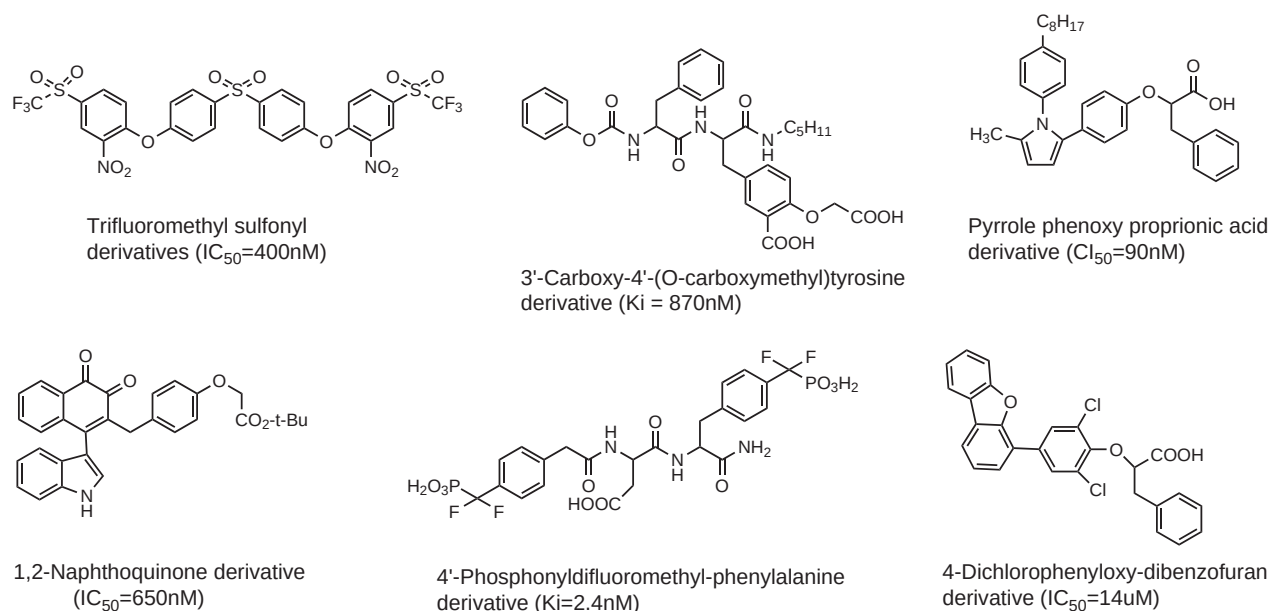


FIGURE 27.25 Structures of PTB1B inhibitors in development. (Adapted from Rotella DP. Novel “second-generation” approaches for the control of type 2 diabetes. *J Med Chem* 2004;47:4111–4112.)

charged phosphotyrosine residues. Highly charged molecules rarely make good drug candidates primarily because they demonstrate poor pharmacokinetics, being poorly absorbed and distributed in the body. Fortunately, research has discovered that the active site of PTB1B contains subpockets (allosteric sites) that can be targeted for inhibitor development. This has led to the concept known as bidentate ligand binding, which aims to design inhibitors that bind to both the active site and a unique

adjacent peripheral site (133). The bioavailability issue has been addressed by charge reduction, increasing the molecule's hydrophobicity, or targeting the subpocket sites only (133).

The development of PTB1B inhibitors continues to be an active area of research, and although as far as we know, there are no compounds in clinical trials, it is most likely that effective PTB1B-selective inhibitors with improved pharmacokinetics will be developed in the near future.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Nathan A. Painter, PharmD

JL has both fasting and postprandial hyperglycemia; however, the postprandial values are primarily responsible for increasing the hemoglobin A_{1c} values. Even though insulin is a good choice for JL, his concerns of hypoglycemia and weight gain are real. Of the other options available, the drug exenatide is most likely to reduce postprandial glucose values. Because exenatide stimulates insulin secretion in a glucose-dependent manner, there is no risk of hypoglycemia when this medication is used alone. When used in combination with a sulfonylurea, however, the risk of hypoglycemia is increased, so the dosage of sulfonylurea may be reduced. Exenatide also promotes weight loss.

Increasing glipizide to 20 mg twice daily offers no benefit; sulfonylureas reach maximal therapeutic benefit at approximately half the maximum recommended dosage. Increasing metformin

to 2550 mg daily offers little added benefit for similar reasons. Adding a thiazolidinedione may lower her hemoglobin A_{1c} value an additional percentage point but may be associated with weight gain, congestive heart failure, and myocardial infarction (rosiglitazone).

Chemical Analysis

S. William Zito and Victoria Roche

Exenatide is the recommended choice in this case to help bring JL's fasting and postprandial hyperglycemia down to normal levels. Exenatide is a glucagon-like peptide 1 (GLP-1) receptor agonist. GLP-1 is a 36 amino acid peptide secreted by L-cells of the gut in response to a meal. It exerts control over glucose levels by promoting insulin secretion in a glucose-dependent manner. Exenatide is a 39-amino acid peptide analog of GLP-1 and is isolated from the saliva of the gila monster (*Heloderma suspectum*). It is resistant

SCENARIO: OUTCOME AND ANALYSIS (*Continued*)

to the proteolytic action of DPP-IV, which removes a dipeptide from the N-terminus because it has a glycine instead of alanine in the penultimate N-terminal position and is 53% homologous to human GLP-1. It has an in vivo half-life of approximately 3 hours.

Exenatide has been shown to reduce HbA_{1c} levels in sulfonyl-urea treated patients with type 2 diabetes and is associated with

weight loss. This weight loss is probably due to a decrease in appetite. In db/db mice it increased β -cell mass and delayed the onset of diabetes, so it may even have a prophylactic use. Exenatide is available as a sterile 250 mcg/ml solution for subcutaneous injection using a pen-injector. Dosing should be initiated at 5 mcg twice daily any time within 60 minutes before the morning and evening meals.



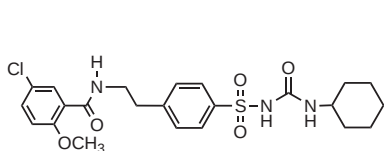
Exenatide (Byetta)

CASE STUDY

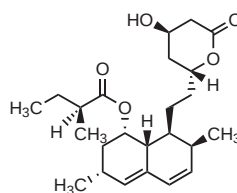
S. William Zito and Victoria Roche

MJ is a 45-year-old Native American male ironworker. Ironworkers are construction workers who build steel structural skeletons for skyscrapers and bridges. MJ is proud of his heritage as one of a long line of Mohawk ironworkers who have been building bridges and buildings all across the United States since the late 1800s. MJ has been a type 2 diabetic since he was 30 years old and has mild hypertension and hyperlipidemia, both of which are under control with enalapril and lovastatin, respectively. He visits his physician complaining that he has been unable to control his blood glucose levels even though he carefully watches what he eats and is conscientious about taking his

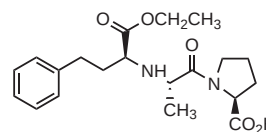
glyburide every day (10 mg twice a day). His last glucose test revealed a level of 195 mg/dL 2 hours after breakfast. In addition, he is very concerned that he has become thirsty, tired, and always wanting to urinate, all of which makes it dangerous for him to “walk the steel.” After eliminating any secondary causes of poor glycemic control (e.g., infection, other medications), and because MJ is currently taking the maximum dose of glyburide, his physician concludes he has most likely developed a “tolerance” to this drug. He would like to add another oral hypoglycemic drug to his therapy. What is your advice as to the following three choices?



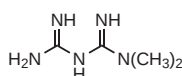
Glyburide



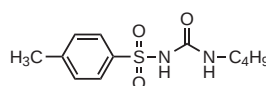
Lovastatin



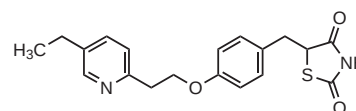
Enalapril



1



2



3

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

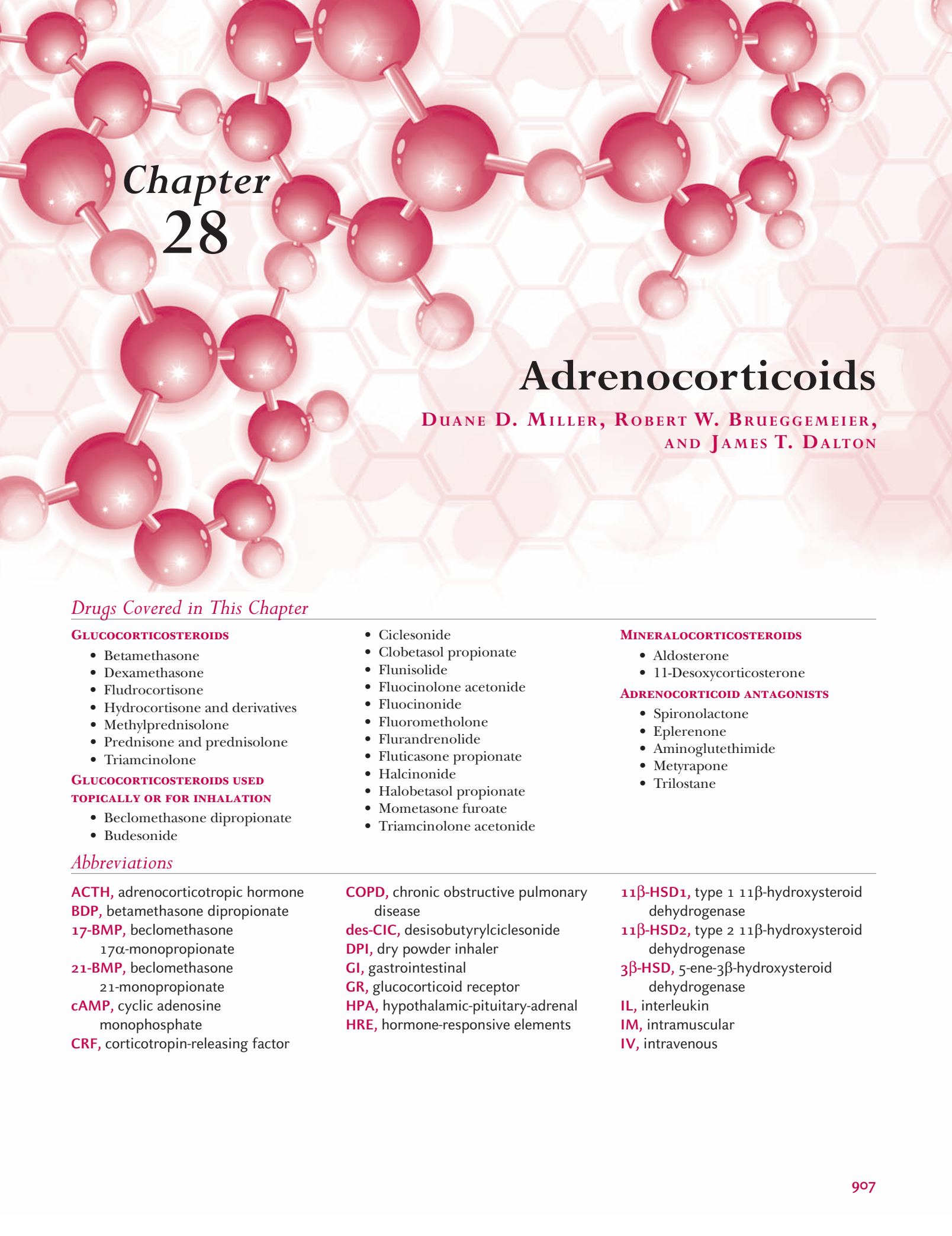
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 28

Adrenocorticoids

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Drugs Covered in This Chapter

GLUCOCORTICOSTEROIDS

- Betamethasone
- Dexamethasone
- Fludrocortisone
- Hydrocortisone and derivatives
- Methylprednisolone
- Prednisone and prednisolone
- Triamcinolone

GLUCOCORTICOSTEROIDS USED TOPICALLY OR FOR INHALATION

- Beclomethasone dipropionate
- Budesonide

- Ciclesonide
- Clobetasol propionate
- Flunisolide
- Fluocinolone acetonide
- Fluocinonide
- Fluorometholone
- Flurandrenolide
- Fluticasone propionate
- Halcinonide
- Halobetasol propionate
- Mometasone furoate
- Triamcinolone acetonide

MINERALOCORTICOSTEROIDS

- Aldosterone
- 11-Desoxycorticosterone

ADRENOCORTICOID ANTAGONISTS

- Spironolactone
- Eplerenone
- Aminoglutethimide
- Metyrapone
- Trilostane

Abbreviations

ACTH, adrenocorticotrophic hormone

BDP, betamethasone dipropionate

17-BMP, beclomethasone
17 α -monopropionate

21-BMP, beclomethasone
21-monopropionate

cAMP, cyclic adenosine
monophosphate

CRF, corticotropin-releasing factor

COPD, chronic obstructive pulmonary
disease

des-CIC, desisobutylciclesonide

DPI, dry powder inhaler

GI, gastrointestinal

GR, glucocorticoid receptor

HPA, hypothalamic-pituitary-adrenal

HRE, hormone-responsive elements

11 β -HSD1, type 1 11 β -hydroxysteroid
dehydrogenase

11 β -HSD2, type 2 11 β -hydroxysteroid
dehydrogenase

3 β -HSD, 5-ene-3 β -hydroxysteroid
dehydrogenase

IL, interleukin

IM, intramuscular

IV, intravenous

SCENARIO

Jeffrey T. Sherer, PharmD, MPH, BCPS, CGP

LT is a 74-year-old woman with chronic obstructive pulmonary disease due to a 55 pack-year history of cigarette smoking. As an outpatient, she is maintained on a regimen including inhaled tiotropium and inhaled salmeterol. Over the past several days, she has experienced an increase in shortness of breath and a change in the color and quantity of her sputum. A chest

radiograph shows hyperinflated lungs but no evidence of pneumonia. She is admitted to the hospital with a diagnosis of acute exacerbation of chronic obstructive pulmonary disease.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

The adrenal glands are flattened, cap-like structures located above the kidneys. The inner core (medulla) of the gland secretes catecholamines, whereas the shell (cortex) of the gland synthesizes steroid hormones known as the adrenocorticoids. The adrenocorticoid steroids include the glucocorticoids, which regulate carbohydrate, lipid, and protein metabolism, and the mineralocorticoids, which influence salt balance and water retention. A third class of steroids produced by the adrenal glands is called the adrenal androgens, which have weak androgenic activity in men and women and can serve as precursors to the sex hormones, estrogens and androgens.

The adrenocorticoids and sex hormones have much in common. All are steroids; consequently, the rules that define their structures, chemistry, and nomenclature are the same. The rings of these biochemically dynamic and physiologically active compounds have a similar stereochemical relationship. Changes in the geometry of the ring junctures usually result in inactive compounds regardless of the biologic category of the steroid. Similar chemical groups are used to render some of these agents water soluble or active when taken orally or to modify their absorption.

In addition, the adrenocorticoids and the sex hormones, which include the estrogens, progestins, and androgens, are mainly biosynthesized from cholesterol, which in turn is synthesized from acetyl-coenzyme A. Cholesterol and steroid hormone catabolism takes place primarily in the liver. Although the products found in the urine and feces depend on the hormone undergoing catabolism, many of the metabolic reactions are similar for these compounds. For example, reduction of double bonds at positions 4 and 5 or 5 and 6, epimerization of 3 α -hydroxyl groups, reduction of 3-keto groups to the 3 α -hydroxyl function, and oxidative removal of side chains are transformations common to these agents.

The adrenocorticoids have many clinical uses. Glucocorticoids and mineralocorticoids may be used for the treatment of adrenal insufficiency (hypoadrenalism), which results from failure of the adrenal glands to synthesize adequate amounts of the hormones. Major pharmacologic uses of glucocorticoids include the treatment

of rheumatoid diseases, symptomatic relief from asthma and allergic conditions, topical application for various dermatologic disorders, and cancer therapy. Toxicities arise when corticosteroids are used for longer than brief periods, and toxic effects can include glucocorticoid-induced adrenocortical insufficiency, glucocorticoid-induced osteoporosis, and generalized protein depletion. Knowledge of the numerous steroid products, structure-activity relationships, and available dosage forms will result in significant benefits for patients with minimal troublesome toxicities.

Despite the similarities in chemical structures and stereochemistry, each class of steroids demonstrates unique and distinctively different biologic activities. Minor structural modifications to the steroid nucleus, such as changes in or insertion of functional groups at different positions, cause marked changes in physiologic activity. The first part of this chapter focuses on the similarities among the steroids and reviews steroid nomenclature, stereochemistry, and general mechanism of action. The second portion of the chapter focuses on the adrenocorticoids and discusses the biosynthesis, metabolism, medicinal chemistry, pharmacology, and pharmacokinetics of endogenous steroid hormones, synthetic agonists, and synthetic antagonists.

STEROID NOMENCLATURE AND STRUCTURE

Steroids consist of four fused rings (A, B, C, and D) (Fig. 28.1). Chemically, these hydrocarbons are cyclopentanoperhydrophenanthrenes; they contain a five-membered cyclopentane (D) ring plus the three rings of phenanthrene. A perhydrophenanthrene (rings A, B, and C) is the completely saturated derivative of phenanthrene. The polycyclic hydrocarbon known as 5 α -cholestane will be used to illustrate the numbering system for a steroid (Fig. 28.1). The term “cholestane” refers to a steroid with 27 carbons that includes a side chain of eight carbons at position 17. Numbering begins in ring A at C1 and proceeds around rings A and B to C10, then into ring C beginning with C11, and snakes around rings C and D to C17. The angular methyl groups are numbered 18 (attached to C13) and 19 (attached to C10). The 17 side chain begins with C20, and the numbering

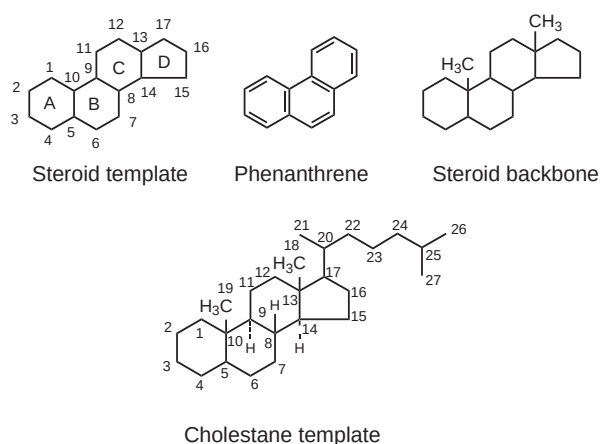


FIGURE 28.1 Basic steroid structure and numbering system.

finishes in sequential order. Using the rigid planar representation for drawing the steroid structure (Fig. 28.2A), the basic steroid structure becomes a plane with two surfaces: The top or β surface is pointing out toward the reader, and the bottom or α surface is pointing away from the reader. Hydrogens or functional groups on the β side of the molecule are denoted by solid lines; those on the α side are designated by dotted lines. The 5α notation is used to denote the configuration of the hydrogen atom at C5, which is opposite from the C19 angular methyl group, making the A/B ring juncture *trans*. The C19 angular methyl group is assigned the β side of the molecule. Similarly, the configuration of the 8β and 9α hydrogens, and the 14α hydrogen and C18 angular methyl group, denote *trans* fusion for rings B/C and C/D. The side chains at position 17 are always β unless indicated by dotted lines or in the nomenclature of the steroid (e.g., 17β or 17α).

Just as cyclohexane can be drawn in a chair conformation, the three-dimensional representation for 5α -cholestane is shown by the following conformational formula (Fig. 28.2B). Although cyclohexane

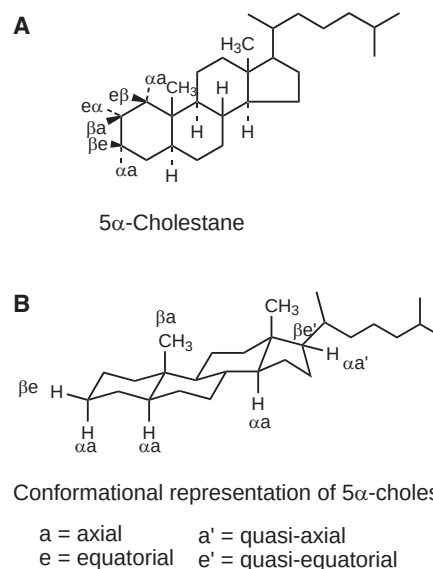


FIGURE 28.2 Planar and conformational structures of 5α -cholestane.

may undergo a flip in conformation, steroids are rigid structures, because they generally have at least one *trans* fused ring system and these rings must be diequatorial to each other.

If one is aware that the angular methyl groups at positions 18 and 19 are β and have an axial orientation (i.e., perpendicular to the plane of the rings), the conformational orientation of the remaining bonds of a steroid can be easily assigned. For example, in 5α -cholestane, the C19 methyl group attached at position 10 is always β -axial (βa); the two bonds at position 3 must be β -equatorial (βe) and α -axial (αa), as indicated. The orientation of the remaining bonds on a steroid may be determined if one recalls that groups on a cyclohexane ring that are positioned on adjacent carbon atoms (vicinal, $-C_1H-C_2H-$) of the ring (i.e., 1,2 to each other) are *trans* if their relationship is 1,2-diaxial or 1,2-diequatorial and are *cis* if their relationship is 1,2-equatorial-axial.

CLINICAL SIGNIFICANCE



Few groups of drugs can rival the adrenocorticoids, which are used for the widest variety of conditions. From asthma to rheumatologic diseases to dermatologic disorders, the adrenocorticoids are commonly prescribed for their beneficial action. Unfortunately, they also exhibit significant toxicity. Only by understanding their complicated mechanisms of action can pharmacists help maximize the therapeutic benefits while minimizing the numerous adverse effects of these agents.

Pharmacists must be familiar with the numerous steroid products and dosage forms that are available as well as with the structure–activity relationships that determine their effects at different receptors. By understanding these factors, the likelihood that patients will derive significant benefits without untoward toxicities is significantly increased.

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Steroid chemists often refer to the series of carbon-carbon bonds shown with heavy lines as the backbone of the steroid (Fig. 28.1). The *cis* or *trans* relationship of the four rings may be expressed in terms of the backbone. The compound 5 α -cholestane (Fig. 28.2) is said to have a *trans-anti-trans-anti-trans* backbone. In this structure, all the fused rings have *trans* (diequatorial) stereochemistry; in other words, the A/B fused ring, the B/C fused ring, and the C/D fused ring are *trans*. The term *anti* is used in backbone notation to define the orientation of rings that are connected to each other and have a *trans*-type relationship. For example, the bond equatorial to ring B, at position 9, which forms part of ring C, is *anti* to the bond equatorial to ring B, at position 10, which forms part of ring A. 5 β -cholestane (Fig. 28.3) has a *cis-anti-trans-anti-trans* backbone in which the A/B rings are fused *cis*. The term *syn* is used in a similar fashion as *anti* to define a *cis*-type relationship. No natural steroids exist with a *syn*-type geometry, although such compounds can be chemically synthesized. Thus, the conventional drawing of the steroid nucleus is the natural configuration and does not show the hydrogens at 8 β , 9 α , or 14 β positions. If the carbon at position 5 is saturated, the hydrogen is always drawn as either 5 α or 5 β . Also, the conventional drawing of a steroid molecule has the C18 and C19 methyl groups shown only as solid lines (no CH₃ drawn).

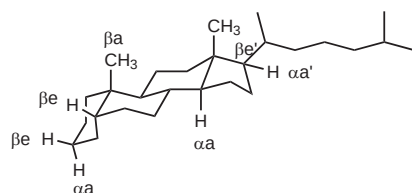
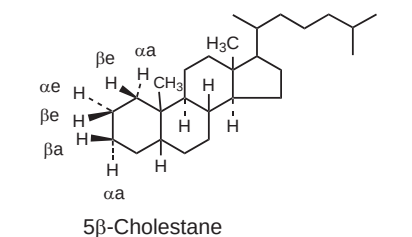
The stereochemistry of the rings markedly affects the biologic activity of a given class of steroids. Nearly all biologically active steroids have the cholestane-type backbone. In most of the important steroids discussed in this chapter, a double bond is present between positions 4 and 5 or 5 and 6; consequently, there is no *cis* or *trans* relationship between rings A and B. The symbol Δ is often used to designate a carbon-carbon

double bond (C=C) in a steroid. If the C=C is between positions 4 and 5, the compound is referred to as a Δ^4 -steroid. If the C=C is between positions 5 and 10, the compound is designated a $\Delta^{5(10)}$ -steroid.

Cholesterol (cholest-5-en-3 α -ol) is a Δ^5 -steroid or, more specifically, a Δ^5 -sterol, because it is an unsaturated alcohol. Biologically active compounds include members of the 5 α -pregnane, 5 α -androstane, and 5 α -estrane steroid classes (Fig. 28.4). Pregnanes are steroids with 21 carbon atoms, androstanes 19 carbon atoms, and estranes 18 carbon atoms, with the C19 angular methyl group at C10 replaced by hydrogen. Numbering is the same as in 5 α -cholestane.

The adrenocorticoids (adrenal cortex hormones) are pregnanes and are exemplified by cortisone, which is a 17 α ,21-dihydroxypregn-4-ene-3,11,20-trione. The acetate ester is named 17 α ,21-dihydroxypregn-4-ene-3,11,20-trione 21-acetate (cortisone acetate) (Fig. 28.4). Progesterone (pregn-4-ene-3,20-dione), a female sex hormone synthesized by the corpus luteum, is also a pregnane analog. The male sex hormones (androgens) are based on the structure of 5 α -androstane. Testosterone, an important and naturally occurring androgen, is named 17 β -hydroxyandrost-4-en-3-one. Dehydroepiandrosterone is the major adrenal androgen and is named 3 β -hydroxyandrost-5-en-17-one (Fig. 28.4). The estrogens, which are female sex hormones synthesized by the graafian follicle of the ovaries, are estrane analogs containing an aromatic A ring. Although the A ring does not contain isolated C=C groups, these analogs are named as if the bonds were in the positions shown in 17 β -estradiol. Hence, 17 β -estradiol, a typical member of this class of drugs, is named *estra-1,3,5,(10)-triene-3,17 β -diol*. Other examples of steroid nomenclature are found throughout this chapter.

Aliphatic side chains at position 17 are always assumed to be β when cholestane or pregnane nomenclature is employed. Hence, the notation 17 β need not be used when naming these compounds. If a pregnane has a 17 α chain, however, this should be indicated in the nomenclature. Finally, the final "e" in the name for the parent steroid hydrocarbon is always dropped when it precedes a vowel, regardless of whether a number appears between the two parts of the word (e.g., note the nomenclature for cholesterol and testosterone versus that for cortisone). For a more extensive discussion of steroid nomenclature, consult the literature (1).



Conformational representation of 5 β -cholestane

a = axial a' = quasi-axial
e = equatorial e' = quasi-equatorial

FIGURE 28.3 Planar and conformational structures of 5 β -cholestane.

MECHANISM OF STEROID HORMONE ACTION

In addition to their structural similarities, adrenocorticoids, estrogens, progestins, and androgens share a common mode of action. They are present in the body only in extremely low concentrations (e.g., 0.1 to 1.0 nmol/L), where they exert potent physiologic effects on sensitive tissues, and they bind with high affinity to intracellular receptors. Extensive research activities directed at elucidation of the general mechanism of

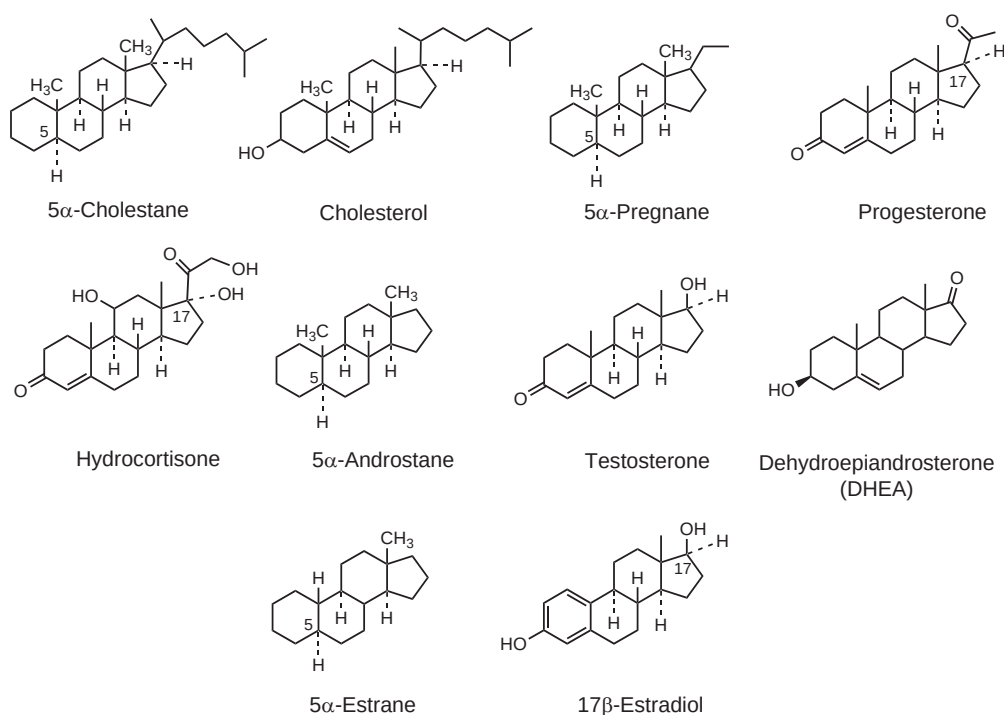


FIGURE 28.4 Steroid classes and corresponding natural hormones.

steroid hormone action have been performed for several decades, and many reviews have appeared (2–7).

Steroid hormones act on target cells to regulate gene expression and protein biosynthesis via the formation of steroid–receptor complexes and subsequent transactivation of gene expression, as outlined in Figure 28.5. The lipophilic steroid hormones are carried in the bloodstream, with the majority of the hormones reversibly

bound to serum carrier proteins. The free steroids can diffuse through the cell membrane and enter cells. Those cells sensitive to the particular steroid hormone (referred to as target cells) contain steroid receptors capable of high-affinity binding with the steroid. These receptors are soluble intracellular proteins that can both bind steroid ligands with high affinity and act as transcriptional factors via interaction with specific DNA sites. Early studies

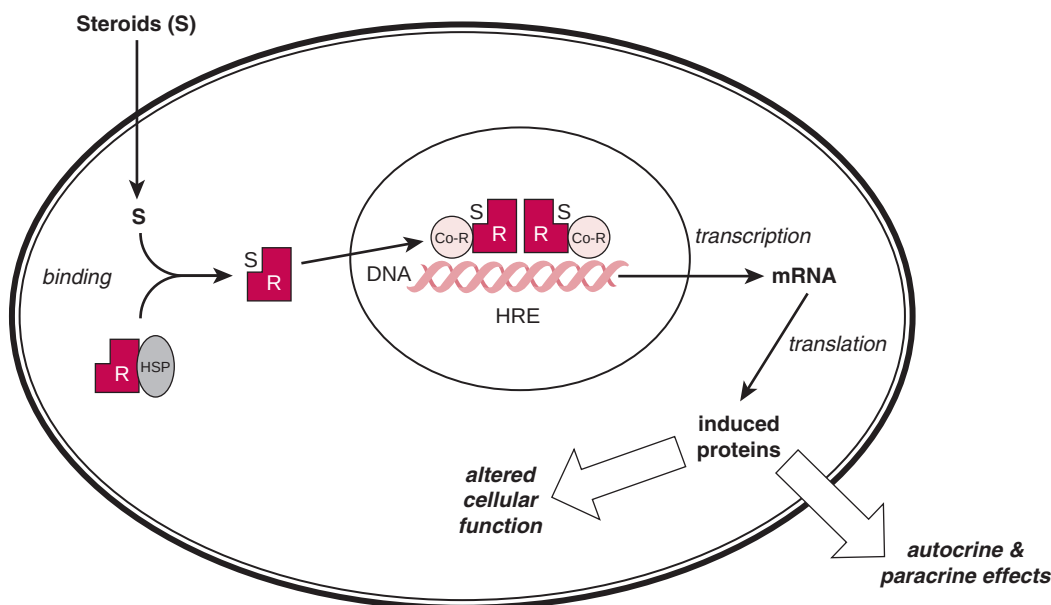


FIGURE 28.5 Mechanism of steroid hormone action.

suggested that the unoccupied steroid receptors were located solely in the cytosol of target cells. Investigations on estrogen, progesterin, and androgen action, however, indicate that active, unoccupied receptors are also present in the nucleus of the cell (2,7). Before the binding of the steroid, the steroid receptor is complexed with heat shock proteins. In the current model, the steroid enters the cell and binds to the steroid receptor in the cytoplasm or nucleus. This binding initiates a conformational change and dissociation of the heat shock protein, allowing steroid receptor dimerization and translocation to the nucleus. The receptor dimer interacts with particular regions of the cellular DNA, referred to as hormone-responsive elements (HRE), and with various coactivators and nuclear transcriptional factors. Binding of the nuclear steroid–receptor complex to DNA initiates transcription of the DNA sequence to produce mRNA. Finally, the elevated levels of mRNA lead to an increase in protein synthesis in the endoplasmic reticulum. These proteins include enzymes, receptors, and secreted factors that subsequently result in the steroid hormonal response regulating cell function, growth, and differentiation and playing central roles in normal physiologic processes as well as in many important diseases.

The primary amino acid sequences of the various steroid hormone receptors have been deduced from cloned cDNA (3,5). The steroid receptor proteins are part of a larger family of nuclear receptor proteins that also include receptors for vitamin D, thyroid hormones, and retinoids. The overall structures of the receptors have strong similarities (Fig. 28.6). A high degree of homology (sequence similarities) in the steroid receptors is found in the DNA binding region that interacts with the HRE. The DNA binding region has critically placed cysteine amino acids that chelate zinc ions, forming finger-like projections, called zinc fingers, that bind to the DNA. Structure–function studies of cloned receptor proteins also identify regions of the molecules that are important for interactions with nuclear transcriptional factors, coactivator or corepressor proteins, activation of gene transcription, and protein-to-protein interactions.

More recently, steroid–receptor complexes have demonstrated negative regulation of gene expression by interacting at negative response elements or by transrepression via direct or indirect protein–protein interactions with other known transcriptional proteins such as

AP-1 and NF κ B (8,9). This can be critical for cross-talk between signaling pathways within the cell and may have an important role in feedback systems. Additional evidence suggests that steroid receptors can activate transcription in the absence of hormone, an effect that appears to depend on the phosphorylation of the receptor via cross-talk with membrane-bound adrenergic and/or growth factor receptors (10). The interactions necessary for formation of the steroid–receptor complexes and subsequent activation or repression of gene transcription are complicated, involve multistage processes, and leave many unanswered questions. Lastly, nongenomic mechanisms involving cytosolic and/or membrane-bound steroid receptors have been identified to explain rapid onset actions of steroid hormones (8).

The basic mechanism of steroid hormone action on target cells is similar for the various classes of agents. Differences in the actions of adrenocorticoids, estrogens, progestins, and androgens arise from the specificity of the particular receptor proteins, the particular genetic processes initiated, and the specific cellular proteins produced.

HISTORY AND DISEASE STATES

The importance of the adrenal glands was recognized long ago. Addison disease, Cushing disease, and Conn syndrome are pathologic conditions related to the adrenal cortex and the hormones produced by the gland.

Addison disease was named after Thomas Addison. In 1855, Addison described a syndrome in which the physiologic significance of the adrenal cortex was emphasized (11). This disease is characterized by extreme weakness, anorexia, anemia, nausea and vomiting, low blood pressure, hyperpigmentation of the skin, and mental depression resulting from decreased secretion of steroid hormones by the adrenal cortex. Addison disease is a rare affliction that affects roughly 1 in 100,000 people and is seen equally in both sexes and in all age groups.

Conditions of this type, usually referred to as hypo-adrenalism, can result from several causes, including destruction of the cortex by tuberculosis or atrophy or decreased secretion of adrenocorticotropin (adrenocorticotrophic hormone [ACTH]) because of diseases of the anterior pituitary (adenohypophysis). Cushing disease, or hyperadrenalism, on the other hand, can result from adrenal cortex tumors or increased production of ACTH caused by pituitary carcinoma. Cushing syndrome is also rare, occurring in only 2 to 5 people for every 1 million people each year. Approximately 10% of newly diagnosed cases are observed in children and teenagers.

Conn syndrome is caused by an inability of the adrenal cortex to carry out 17 α -hydroxylation during the biosynthesis of the hormones from cholesterol. Consequently, the disease is characterized by a high secretory level of aldosterone, which lacks a 17 α -hydroxyl functional group. In addition, hypernatremia, polyuria, alkalosis, and hypertension are observed (12).

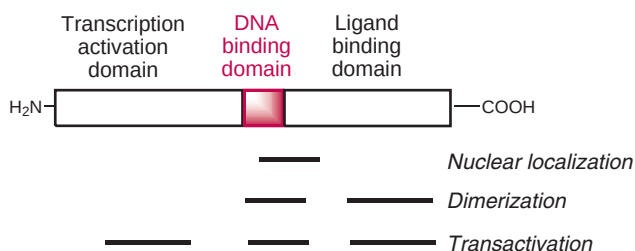
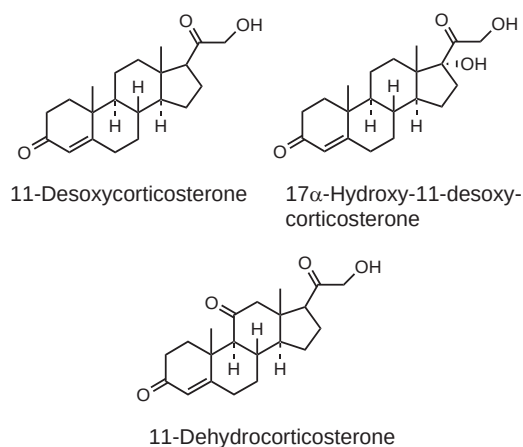


FIGURE 28.6 Structural features of steroid hormone receptors.

The importance of the adrenocorticoids is most dramatically observed in adrenalectomized animals. There is an increase of urea in the blood, muscle weakness (asthenia), decreased liver glycogen, decreased resistance to insulin, lowered resistance to trauma (e.g., cold and mechanical or chemical shock), and electrolyte disturbances. Potassium ions are retained, and excretion of Na^+ , Cl^- , and water is increased. Adrenalectomy in small animals causes death in a few days.

After Addison's observations in 1855, physiologists, pharmacologists, and chemists from many countries contributed to our understanding of adrenocorticoids. It was not until 1927, however, that Rogoff and Stewart found that extracts of adrenal glands, administered by intravenous (IV) injection, kept adrenalectomized dogs alive.

Since that discovery, similar experiments have been repeated many times. Originally, the biologic activity of the extract was thought to result from a single compound. Later, 47 compounds were isolated from such extracts, and some were highly active. Among the biologically active corticoids isolated, hydrocortisone (as called cortisol), corticosterone, aldosterone, cortisone, 11-desoxycorticosterone (17 α -hydroxyprogesterone), 11-dehydrocorticosterone (11-desoxycortisol), and 17 α -hydroxy-11-desoxycorticosterone were found to be most potent (13). The biosynthesis of these steroids is described below.



BIOSYNTHESIS

Pregnenolone Formation

In the adrenal glands, cholesterol is converted by enzymatic cleavage of its side chain to pregnenolone, which serves as the biosynthetic precursor of the adrenocorticoids (Fig. 28.7). This biotransformation is performed by a mitochondrial cytochrome P450 enzyme complex. This enzyme complex found in the mitochondrial membrane consists of three proteins: CYP11A1 (also known as P450_{SCC}), adrenodoxin, and adrenodoxin reductase (14). Defects in CYP11A1 lead to a lack of glucocorticoids, feminization, and hypertension. Three oxidation steps are involved in the conversion, and three moles of NADPH and molecular oxygen are consumed for each

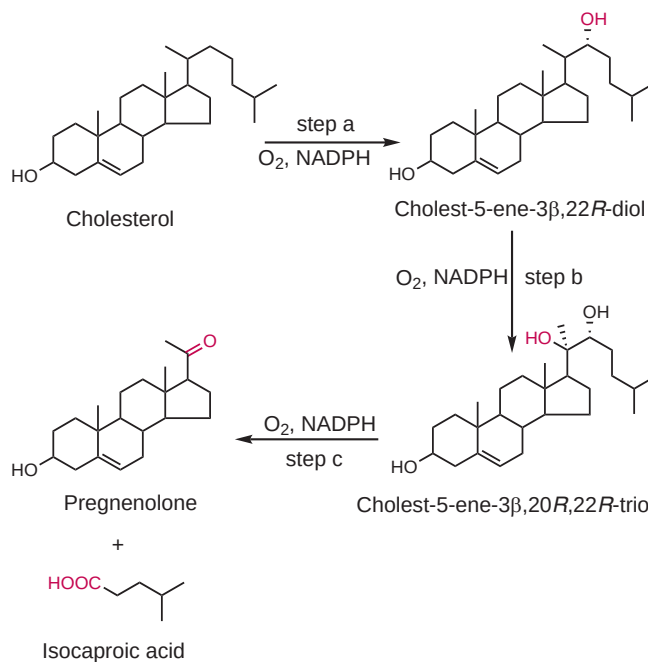


FIGURE 28.7 Biosynthesis of pregnenolone from cholesterol.

mole of cholesterol converted to pregnenolone. The first oxidation results in the formation of cholest-5-ene-3 β ,22 R -diol (step a), followed by the second oxidation yielding cholest-5-ene-3 β ,20 R ,22 R -triol (step b). The third oxidation step catalyzes the cleavage of the C20-C22 bond to release pregnenolone and isocaproic acid (step c).

Pregnenolone serves as the common precursor in the formation of the adrenocorticoids and other steroid hormones. This C21 steroid is converted via enzymatic oxidations and isomerization of the double bond to a number of physiologically active C21 steroids, including the female sex hormone progesterone and the adrenocorticoids hydrocortisone, corticosterone, and aldosterone. Oxidative cleavage of the two-carbon side chain of pregnenolone and subsequent enzymatic oxidations and isomerization lead to C19 steroids, including the androgens testosterone and dihydrotestosterone. The final group of steroids, the C18 female sex hormones, are derived from oxidative aromatization of the A ring of androgens to produce estrogens. More detailed information regarding these biosynthetic pathways is provided in this and the following chapters under the particular class of steroid hormones.

Pregnenolone to Glucocorticoids and Mineralocorticoids

The biosynthesis of the glucocorticoids and mineralocorticoids is regulated by independent mechanisms. The glucocorticoids, such as hydrocortisone (also known as cortisol), are biosynthesized and released under the influence of peptide hormones secreted by the hypothalamus and anterior pituitary (adenohypophysis) to activate the adrenal cortex (the hypothalamic-pituitary-adrenal [HPA] axis). Removal of the pituitary results in atrophy of the adrenal

cortex and a marked decrease in the rate of glucocorticoid formation and secretion. On the other hand, the secretion of the mineralocorticoids, corticosterone and aldosterone, is under the influence of the octapeptide, angiotensin II. Angiotensin II is the active metabolite resulting from the renin-catalyzed proteolytic hydrolysis of plasma angiotensinogen to angiotensin I in the blood. In hypophysectomized animals, the rate of secretion of aldosterone is only slightly decreased or remains unchanged. Consequently, the electrolyte balance remains nearly normal.

The peptide hormone in the anterior pituitary that influences glucocorticoid biosynthesis is ACTH, whereas the peptide hormone in the hypothalamus is corticotropin-releasing factor (CRF). The production of both ACTH and CRF is regulated by the central nervous system and by a negative corticoid feedback mechanism. The CRF is released by the hypothalamus and is transported to the anterior pituitary, where it stimulates the release of ACTH into the bloodstream. Then, ACTH is transported to the adrenal glands, where it stimulates the biosynthesis and secretion of the glucocorticoids. The circulating levels of glucocorticoids act on the hypothalamus and anterior pituitary to regulate the release of both CRF and ACTH. As the levels of glucocorticoids rise, smaller amounts of CRF and ACTH are secreted, and a negative feedback is observed (HPA suppression). Stimuli, such as pain, noise, and emotional reactions, increase the secretion of CRF, ACTH, and consequently, the glucocorticoids. Once the stimulus is alleviated or removed, the negative feedback mechanism inhibits further production and helps to return the body to a normal hormonal balance (15,16).

ACTH acts at the adrenal gland by binding to a receptor protein on the surface of the adrenal cortex cell to stimulate the biosynthesis and secretion of glucocorticoids. The only steroid stored in the adrenal gland is cholesterol, found in the form of cholesterol esters stored in lipid droplets. ACTH stimulates the conversion of cholesterol esters to glucocorticoids by initiating a series of biochemical events through its surface receptor. The ACTH receptor protein is coupled to a G protein and to adenylate cyclase. Binding of ACTH to its receptor leads to activation of adenylate cyclase via the G protein. The result is an increase in intracellular cyclic adenosine monophosphate (cAMP) levels. One of the processes influenced by elevated cAMP levels is the activation of cholesterol esterase, which cleaves cholesterol esters and liberates free cholesterol. Another process is the rapid induction of steroidogenic acute regulatory protein, referred to as StAR, which transfers cholesterol into the mitochondrial membrane.

Free cholesterol is then converted within mitochondria to pregnenolone via the side-chain cleavage reaction described earlier (Fig. 28.7). Pregnenolone is converted to adrenocorticoids by a series of enzymatic oxidations and isomerization of the double bond (Fig. 28.8). The next several enzymatic steps in the biosynthesis of glucocorticoids occur in the endoplasmic reticulum of the adrenal cortex cell. Hydroxylation of pregnenolone at position 17 by the enzyme 17 α -hydroxylase (CYP17) produces

17 α -hydroxypregnenolone (step b). The 17 α -hydroxyl group is important for adrenocorticoid hormone action. In one step, 17 α -hydroxypregnenolone is oxidized to a 3-keto intermediate and isomerized to 17 α -hydroxyprogesterone by the action of a single enzyme, 5-ene-3 β -hydroxysteroid dehydrogenase (3 β -HSD) (step c). Another hydroxylation occurs by the action of 21-hydroxylase (CYP21) to give rise to 11-deoxycortisol, which contains the physiologically important ketol ($-\text{COCH}_2\text{OH}$) side chain at the 17 β position (step d). A lack of CYP21 prevents hydrocortisone biosynthesis, diverting excess 17 α -hydroxypregnenolone and 17 α -hydroxyprogesterone into overproduction of C19 androgens. The final step in the biosynthesis of hydrocortisone is catalyzed by the enzyme 11 β -hydroxylase, a mitochondrial cytochrome P450 enzyme complex (CYP11B2). This last enzymatic step (step e) results in the formation of hydrocortisone (cortisol), the most potent endogenous glucocorticoid secreted by the adrenal cortex. Approximately 15 to 20 mg of hydrocortisone are biosynthesized daily. Several reviews (14,15,17–19) provide more detailed discussions about the enzymology and regulation of adrenal steroidogenesis.

The pathway for the formation of the potent mineralocorticoid molecule, aldosterone, is similar to that for hydrocortisone and uses several of the same enzymes (Fig. 28.8). The preferred pathway involves the conversion of pregnenolone to progesterone by 3 β -HSD (step c). Hydroxylation at position 21 of progesterone by 21-hydroxylase results in 21-hydroxyprogesterone (11-deoxycorticosterone) (step d). Again, these first conversions occur in the endoplasmic reticulum of the cell, whereas the next enzymatic steps occur in the mitochondria. 11 β -Hydroxylase (CYP11B2) catalyzes the conversion of 21-hydroxyprogesterone to corticosterone (step e), which exhibits mineralocorticoid activity. The final two oxidations involve hydroxylations at the C18 methyl group and are catalyzed by 18-hydroxylase (step f). These reactions produce first 18-hydroxycorticosterone (not shown) and then aldosterone, the most powerful endogenous mineralocorticoid secretion of the adrenal cortex. The aldehyde at C18 of aldosterone exists in equilibrium with its hemiacetal form.

METABOLISM

Hydrocortisone (hormonally active) and cortisone (the inactive metabolite of hydrocortisone) are biochemically interconvertible by the enzyme 11 β -hydroxysteroid dehydrogenase (Fig. 28.9). Two isozymes of 11 β -hydroxysteroid dehydrogenase are present, type 1 11 β -hydroxysteroid dehydrogenase (11 β -HSD1), referred to as the “liver” isozyme, and type 2 11 β -hydroxysteroid dehydrogenase (11 β -HSD2), referred to as the “kidney” isozyme (20–22). The 11 β -HSD1 isozyme is a bidirectional enzyme, which readily interconverts hydrocortisone and cortisone, and is found in many tissues in the body. This isozyme has an important role in the regulation of hepatic gluconeogenesis in the liver and in fat production in adipose tissues. By contrast, the 11 β -HSD2 isozyme is only unidirectional,

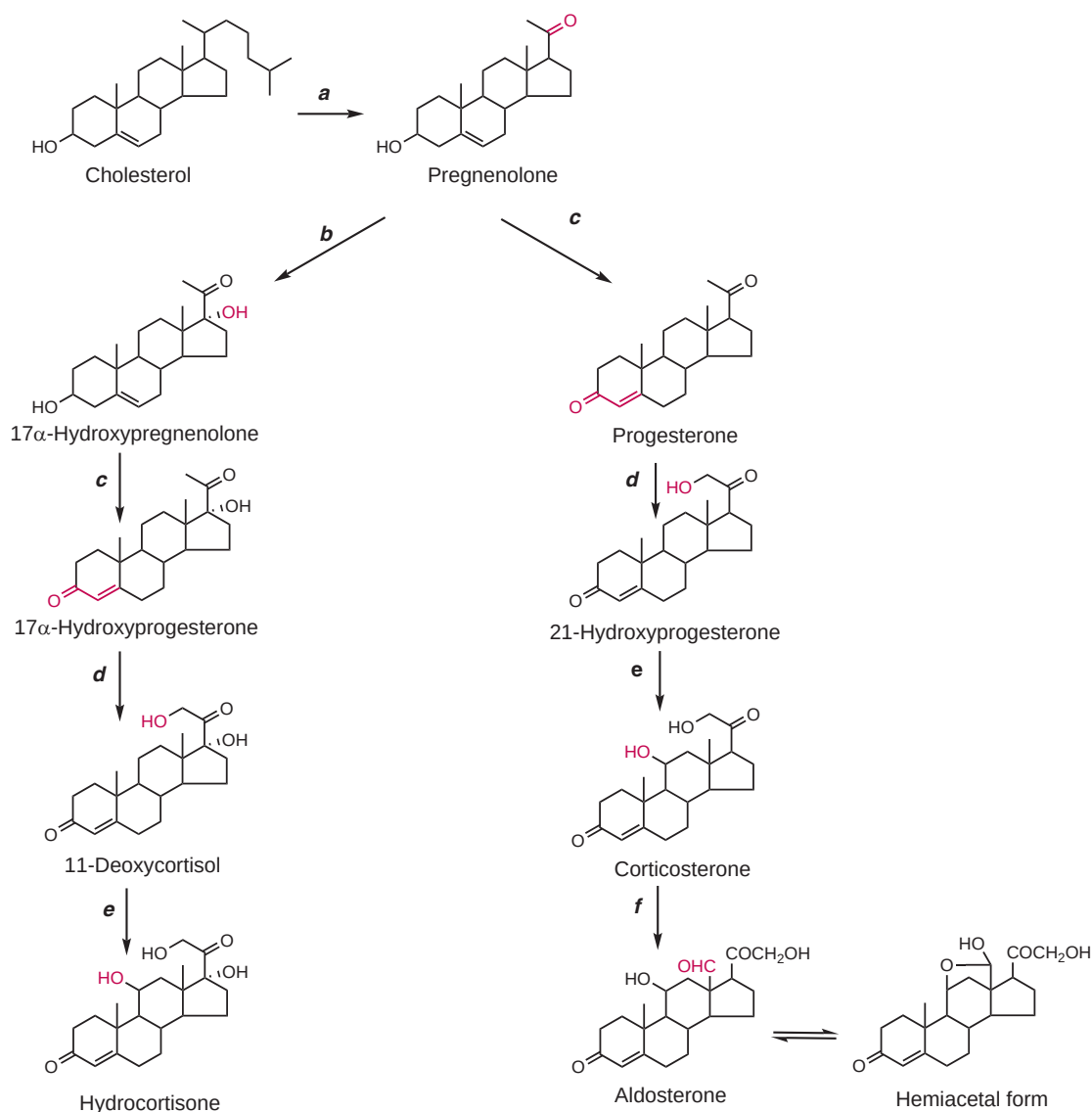


FIGURE 28.8 Biosynthesis of the adrenocorticoids from cholesterol.

catalyzing the 11 β -dehydrogenation of hydrocortisone to give cortisone. The 11 β -HSD2 is present in placenta and in kidney, specifically the distal convoluted tubules and cortical collecting ducts in the kidney. The 11 β -HSD2 isozyme has an important role in the rapid metabolism of hydrocortisone, thus preventing hydrocortisone from binding to the mineralocorticoid receptors present in the same kidney tissues. A deficiency of 11 β -HSD2 is associated with the inherited genetic disease, apparent mineralocorticoid excess, which is characterized by hypertension, excessive salt retention, and hypokalemia caused by the elevated hydrocortisone levels in the kidney.

Hydrocortisone is metabolized by the liver following administration by any route, with a half-life of approximately 1.0 to 1.5 hours (23). Hydrocortisone is mainly excreted in the urine as inactive *O*-glucuronide conjugates and minor *O*-sulfate conjugates of urocortisol, 5 β -dihydrocortisol, and urocortisone (Fig. 28.9). The tetrahydro

metabolite urocortisol is the major metabolite formed and has the 5 β -pregnane geometry and 3 α -hydroxyl function. The 5 β configuration is similar to the ring geometry for the nonhormonal bile acids. Several compounds of this type have been isolated (24,25). All of the biologically active adrenocorticoids contain a ketone at the 3-position and a double bond in the 4,5-position. The formation of inactive 5 β -metabolites from hydrocortisone is characterized by reduction of the 4,5-double bond to a 5 β geometry for rings A and B (a *cis* configuration) by 5 β -reductase or reduction of the 3-ketone to a 3 α -hydroxyl configuration by 3 α -hydroxysteroid dehydrogenase (3 α -hydroxyl configuration) or 3 β -hydroxysteroid dehydrogenase (3 β -hydroxyl configuration). These reactions represent the major pathways of metabolism for the glucocorticoids and their endogenous counterparts for terminating glucocorticoid activity. Urocortisol and urocortisone are named after cortisol (hydrocortisone) and cortisone. Reversible oxidation of the 11 β -hydroxyl

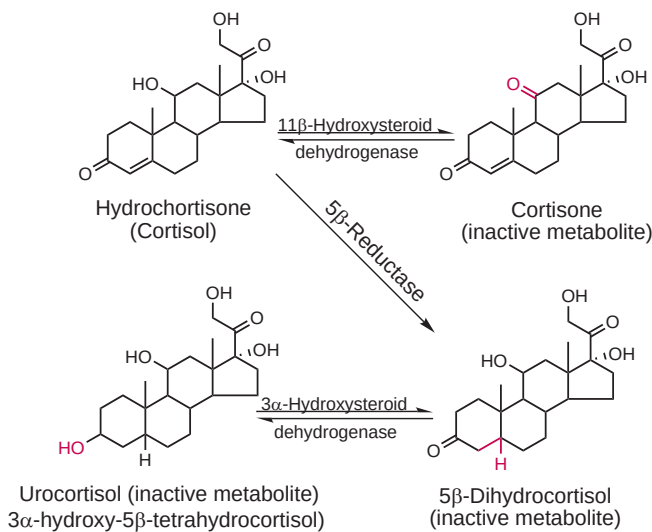


FIGURE 28.9 Major routes of metabolism for hydrocortisone.

group of many glucocorticoids (e.g., hydrocortisone, prednisolone, and methylprednisolone, but not dexamethasone and other 9α-fluorinated glucocorticoids) by 11β-hydroxysteroid dehydrogenase inactivates these drugs and limits their mineralocorticoid activity in the kidneys.

Other routes of metabolism include 6β-hydroxylation (CYP3A4) and reduction of the 20-ketone (e.g., prednisolone) to form 20-hydroxyl analogs as well as oxidation of the 17-ketol side chain to 17β-carboxylic acids and loss of the 17-ketol side chain, resulting in 11β-hydroxy-17-keto-C19 steroids with the geometry of either 5α-androstane or 5β-androstane (15,17–19). In addition, some ring A aromatic adrenocorticoid metabolites that resemble the estrogens have been isolated (26). Biliary and fecal excretion contributes little to the elimination of the adrenocorticoids. The rate of formation of 6β-hydroxyhydrocortisone is a biomarker for determining the level of HPA suppression and adrenal insufficiency.

DEVELOPMENT OF ADRENOCORTICOID DRUGS

Systemic Corticosteroids

The route of administration depends on the disease being treated and the physicochemical, pharmacologic, and pharmacokinetic properties of the drug (Table 28.1). The clinically available adrenocorticoids may be administered by IV injection, oral tablets or solutions, topical formulations, intra-articular administration, and oral or nasal inhalation (Table 28.2). Only a handful of corticosteroids

TABLE 28.1 Pharmacological and Pharmacokinetic Properties for Some Adrenocorticoids

Adrenocorticoid	Oral Glucocorticoid Dose ^a (mg)	Potency Relative to Hydrocortisone		Protein Binding (%) ^d	Half-life (hours)		Duration of Action (days)
		Glucocorticoid Activity ^b	Mineralocorticoid Activity ^c		Plasma	Biologic (tissue)	
Short-acting							
Hydrocortisone	20	1	2+	>90	1.5–2.0	8–12	1.0–1.5
Cortisone	25	0.8	2+	>90	0.5	8–12	1.0–1.5
Intermediate-acting							
Prednisone	5	3.5	1+	>90	3.4–3.8	18–36	1.0–1.5
Prednisolone	5	4	1+	>90	2.1–3.5	18–36	1.0–1.5
Methylprednisolone	5	5	0 ^e	>90	>3.5	18–36	1.0–1.5
Triamcinolone	5	5	0 ^e	>90	2–5	18–36	1.0–1.5
Long-acting							
Dexamethasone	0.75	20–30	0 ^e	>90	3.0–4.5	36–54	2.8–3
Betamethasone	0.6	20–30	0 ^e	>90	3–5	36–54	2.8–3
Fludrocortisone	Not employed	10	10	<90	3.5	18–36	1–2
Fluprednisolone	1.6	10	0 ^e				
Aldosterone	Not employed	0.2	800				
11-Desoxycorticosterone	Not employed	0	40				
Corticosterone	IM	0.5	5				

^aBased on the oral dose of an anti-inflammatory agent in rheumatoid arthritis.

^bAnti-inflammatory, immunosuppressant, and metabolic effects.

^cSodium and water retention and potassium depletion effects.

^dHydrocortisone binds to transcortin (corticosteroid binding globulin [CBG]) and to albumin. Prednisone also binds to CBG, but betamethasone, dexamethasone and triamcinolone do not.

^eAlthough these glucocorticoids are considered not to have significant mineralocorticoid activity, hypokalemia and/or sodium and fluid retention may occur depending on the dosage, duration of use, and patient predisposition.

TABLE 28.2 Adrenocorticoids, Their Trade Names, and Their Routes of Administration

Adrenocorticoid	Trade Name	PO	IV	IM	Inhaled/Intranasal	Topical
Alclometasone dipropionate	Alclovate					•
Amcinonide	Cyclocort					•
Beclomethasone dipropionate	Beclovent, Vanceril, Beconase				•	
Beclomethasone dipropionate monohydrate	Beconase AQ, Vancenase AQ				•	
Betamethasone	Celestone, Diprolene, Luxiq	•				
Betamethasone dipropionate	Diprosone, Maxivate					•
Betamethasone sodium phosphate	Celestone Phosphate			•		
Betamethasone valerate	BetaVal, Valisone,					•
Budesonide	Pulmicort, Rhinocort				•	
Ciclesonide	Alvesco, Omnaris				•	
Clobetasol propionate	Clobex, Cormax, Olux, Temovate					•
Clocortolone pivalate	Cloderm					•
Cortisone acetate		•		•		
Desonide	Desonate					•
Dexamethasone	Decadron	•				• ^a
Dexamethasone acetate	Decadron-LA, Dalalone, Dexasone			•		
Dexamethasone sodium phosphate	Decadron Phosphate		•	•	•	
Desoximetasone	Topicort					•
Diflorasone diacetate	ApexiCon					•
Fludrocortisone acetate		•				
Flumethasone pivalate						•
Flunisolide	Aerobid				•	
Fluocinolone acetonide	Retisert, Drema-Smothe					•
Fluocinonide	Capex					•
Fluorometholone	FML					• ^a
Fluorometholone acetate	Flarex					• ^a
Flurandrenolide	Cordran					•
Fluticasone propionate	Cutivate, Flovent, Flonase				•	•
Halcinonide	Halog					•
Halobetasol propionate	Ultravate					•
Hydrocortisone		•				• ^b
Hydrocortisone acetate				•		• ^c
Hydrocortisone buteprate						•
Hydrocortisone butyrate						•
Hydrocortisone cypionate		•				
Hydrocortisone sodium phosphate			•	•		

(Continued)

TABLE 28.2 Adrenocorticoids, Their Trade Names, and Their Routes of Administration (*Continued*)

Adrenocorticoid	Trade Name	PO	IV	IM	Inhaled/Intranasal	Topical
Hydrocortisone sodium succinate			•	•		
Hydrocortisone valerate						•
Methylprednisolone	Medrol	•				
Methylprednisolone acetate	Depo-Medrol			•		•
Methylprednisolone sodium succinate	Solu-Medrol		•	•		
Mometasone furoate	Elocon					•
Mometasone furoate monohydrate	Nasonex				•	
Prednicarbate	Dermatop					•
Prednisolone	Deltacortef	•				
Prednisolone acetate	Predcor, Predalone	•		•		
Prednisolone acetate	Pred-Mild, Pred-Forte					• ^a
Prednisolone sodium phosphate	Pediapred, Orapred,	•		•		• ^a
Prednisolone tebutate				•		
Prednisone		•				•
Triamcinolone acetonide	Aristocort, Kenacort	•				
Triamcinolone acetonide	Kenalog, Triderm, Delta-Trix, Flutex					•
Triamcinolone acetonide	Azmacort, Nasacort, Nasacort AQ				•	
Triamcinolone diacetate	Aristocort Forte, Triam Forte			•		
Triamcinolone hexacetonide	Aristospan			•		

^aOphthalmic formulations.^bAlso in otic, ophthalmic, and rectal formulations (Cortenema).^cAlso in intrarectal foam (Cortifoam).

IM, intramuscular; IV, intravenous; PO, oral.

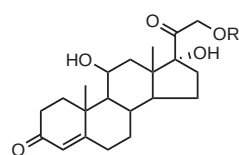
are used clinically by the oral route, including hydrocortisone, prednisone, prednisolone, methylprednisolone, and dexamethasone (Fig. 28.10). These corticosteroids are often described as short-acting, intermediate-acting, or long-acting according to their biologic half-life and duration of action (Table 28.1). They are well-absorbed, undergo little first-pass metabolism in the liver, and demonstrate oral bioavailabilities of 70% to 80%, except for triamcinolone (Table 28.3). The larger volume of distribution for methylprednisolone compared to prednisolone is thought to result from a combination of increased lipophilicity, decrease in metabolism, and better tissue penetration. Glucocorticoids vary in the extent to which they are bound to the plasma proteins, albumin, and corticosteroid-binding globulin (transcortin) (Table 28.3) (27).

Hydrocortisone is extensively bound to the plasma proteins, primarily to transcortin (corticosteroid-binding globulin), with only 5% to 10% of plasma hydrocortisone unbound. Prednisolone and methylprednisolone,

but not the 9 α -fluoro analogs betamethasone, dexamethasone, or triamcinolone, have a high affinity for transcortin and, thus, compete with hydrocortisone for this binding protein. The 9 α -halo analogs bind primarily to albumin. Only the unbound fraction of hydrocortisone and the synthetic corticosteroids are biologically active. As a rule, the amount of transcortin in the plasma determines the distribution of glucocorticoids between free and bound forms, and free glucocorticoid concentrations determine the drug's half-life. Glucocorticoids cross the placenta and can be distributed into milk.

The degree of systemic side effects is dose-dependent, related to the half-life of the drug, frequency of administration, time of day when administered, and route of administration. In other words, the higher the plasma corticosteroid concentration and longer the half-life, the greater will be the systemic side effects.

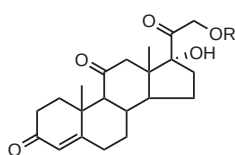
Regardless of the route of administration, all the synthetic adrenocorticoids are excreted from the body in a manner similar to the endogenous adrenocorticoids



Hydrocortisone (cortisol): R = H

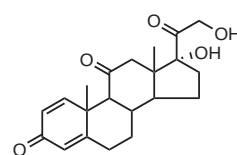
Hydrocortisone acetate: R = CH₃CO-Hydrocortisone butyrate: R = C₃H₇CO-Hydrocortisone butyrate: R = C₂H₅CO-; 17 α -C₃H₇CO-

Hydrocortisone cypionate: R =

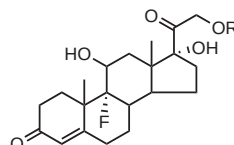
Hydrocortamate sodium succinate: R = Na⁺ OOCCH₂CH₂CO⁻Hydrocortisone sodium phosphate: R = Na₂O₃P-

Cortisone: R = H

Cortisone acetate:

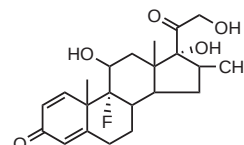
R = CH₃CO-

Prednisone

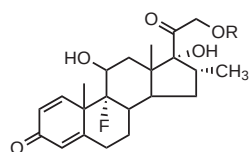


Fludrocortisone: R = H

Fludrocortisone acetate:

R = CH₃CO-

Betamethasone

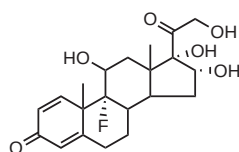


Dexamethasone: R = H

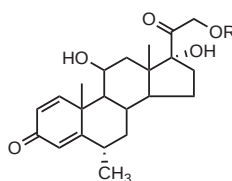
Dexamethasone 21-acetate:

R = CH₃CO-

Dexamethasone sodium phosphate:

R = Na₂O₃P-

Triamcinolone

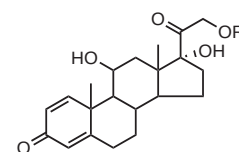


Methylprednisolone: R = H

Methylprednisolone 21-acetate:

R = CH₃CO-

Methylprednisolone sodium succinate:

R = Na OOCCH₂CH₂CO-

Prednisolone: R = H

Prednisolone acetate: R = CH₃CO-

Prednisolone t-butylacetate:

R = (CH₃)₃CCH₂CO-

Prednisolone sodium phosphate:

R = Na₂O₃P-

FIGURE 28.10 Systemic corticosteroids.

(i.e., they are metabolized in the liver and excreted into the urine primarily as glucuronide conjugates but also as sulfate conjugates) (28). In fact, hepatic oxidative metabolism rapidly converts many of the systemic and topical corticosteroids to inactive metabolites and, thus, serves to protect patients from the HPA-suppressive effects of these drugs on endogenous steroid production. The corticosteroids are metabolized in many tissues, including the liver, muscles, and red blood cells (15,18,29); however, the liver metabolizes them most rapidly. The fact

that many of the endogenous corticosteroids are rapidly metabolized by the liver precludes their administration by the oral route. Catabolic products can be isolated from the urine and bile and can be formed in tissue preparations in vitro (30,31).

Specific Drugs

11-Desoxycorticosterone was the first naturally occurring corticoid to be synthesized. It was prepared, before its isolation from the adrenal cortex, by Steiger and Reichstein (32).

TABLE 28.3 Pharmacokinetics of Commonly Used Oral Adrenocorticoids

Adrenocorticoid	Bioavailability (%)	Half-Life (hours)	Protein Binding (%)	Volume Distribution (L/kg)	LogP (experimental)	Clearance (mL/min/70 g)
Dexamethasone	78	3.0	90–95	0.2	1.83	260
Hydrocortisone	96	1.7	90	0.5	1.61	400
Methylprednisolone	90	2.3	90–95	1.5	1.76	430
Prednisone	80	3.6	~90	1.0	1.46	250
Prednisolone	82	2.8	90–95	0.7–1.5	1.62	60
Triamcinolone	23	2.96	~90	1.3	1.16	61

As a result of his synthesis of 11-desoxycorticosterone and other early work with corticoids, Reichstein later shared the Nobel Prize with Kendall, another chemist who was instrumental in carrying out early steroid syntheses, and with Hench, a rheumatologist who in 1929 discovered that cortisone is effective in the treatment of rheumatoid arthritis. Kendall's basic research ultimately led to the synthesis of cortisone from naturally occurring bile acids (33).

CORTISONE, HYDROCORTISONE, AND THEIR DERIVATIVES After the synthesis of 11-desoxycorticosterone in 1937, all the corticoids were synthesized and their structures confirmed. The first synthesis of cortisone from methyl 3 α -hydroxy-11-ketobisnorcholanate was reported by Sarett (34) in 1946. Earlier work of Kendall and coworkers involving its preparation from the methyl ester of desoxycholic acid was used in his research (35). Later, several chemists, including Sarett (36), Kendall, and Tishler, found ways to improve the yields and to decrease the labor involved in the multistep conversion of bile acids to cortisone acetate. In 1949, Merck sold limited quantities of this glucocorticoid to physicians at \$200 per gram for treating rheumatoid arthritis. Subsequent improvements in the methods of synthesis reduced the price to \$10 per gram by 1951. In 1955, Upjohn used an efficient process involving the synthesis of cortisone acetate from progesterone, with the latter steroid being prepared from diosgenin (a steroid sapogenin isolated from the tubers of the *Dioscorea* wild yam). This further reduced the price to \$3.50 per gram. In 2005, the cost was \$8 per gram. Other pharmaceutical companies also began to sell cortisone synthesized from bile acids by a well-developed but lengthy procedure (33).

Cortisone is administered orally or by intramuscular (IM) injection as its 21-acetate (cortisone acetate). Cortisone acetate or hydrocortisone (acetate) are usually the corticosteroid of choice for replacement therapy in patients with adrenocortical insufficiency, because these drugs have both glucocorticoid and mineralocorticoid properties. Following oral administration, cortisone acetate and hydrocortisone acetate are completely and rapidly deacetylated by first-pass esterase metabolism (37). Much of the oral cortisone, however, is inactivated by oxidative metabolism (Fig. 28.9) before it can be converted to hydrocortisone in the liver. The pharmacokinetics for hydrocortisone acetate are indistinguishable from those of orally administered hydrocortisone. Oral hydrocortisone is completely absorbed, with a bioavailability of greater than 95% and a half-life of 1 to 2 hours (23). The metabolism of hydrocortisone (Fig. 28.9) has been previously described. Cortisone acetate is slowly absorbed from IM injection sites over a period of 24 to 48 hours and is reserved for patients who are unable to take the drug orally. The acetate ester derivative demonstrates increased stability and has a longer duration of action when administered by IM injection. Thus, smaller doses can be

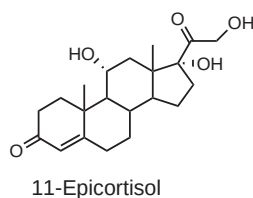
used. Similarly, hydrocortisone may be dispensed as its 21-acetate (hydrocortisone acetate), which is superior to cortisone acetate when injected intra-articularly. Systemic absorption of hydrocortisone acetate from intra-articular injection sites is usually complete within 24 to 48 hours. When administered intrarectally, hydrocortisone is poorly absorbed (38,39).

Other ester derivatives that are available include hydrocortisone cypionate [21-(3-cyclopentylpropionate) ester], hydrocortisone butyrate (17 α -butyrate ester), hydrocortisone buteptrate (17 α -butyrate, 21-propionate esters), hydrocortisone valerate (17 α -valerate ester), hydrocortisone sodium succinate (21-sodium succinate ester), and hydrocortisone sodium phosphate (the 21-sodium phosphate ester) (Fig. 28.10). The water-insoluble hydrocortisone cypionate is used orally in doses expressed in terms of hydrocortisone for slower absorption from the gastrointestinal (GI) tract. The extremely water-soluble 21-sodium succinate and 21-sodium phosphate esters are used for IV or IM injection in the management of emergency conditions that can be treated with anti-inflammatory steroids. The phosphate ester is completely and rapidly metabolized by phosphatases, with a half-life of less than 5 minutes (38). Peak hydrocortisone levels are reached in approximately 10 minutes. The sodium succinate ester is slowly and incompletely hydrolyzed, and peak hydrocortisone levels are attained in 30 to 45 minutes (38). Hydrocortisone butyrate, hydrocortisone buteptrate, and hydrocortisone valerate are used topically.

After the introduction of cortisone (1948) and, later, hydrocortisone (1951) for the treatment of rheumatoid arthritis, many investigators began to search for superior agents having fewer side effects. When these drugs are used in doses necessary to suppress symptoms of rheumatoid arthritis, they also affect other metabolic processes. Side effects, such as excessive sodium retention and potassium excretion, negative nitrogen balance, increased gastric acidity, edema, and psychosis, are exaggerated manifestations of the normal metabolic functions of the hormones.

It was hoped that a compound with high glucocorticoid and low mineralocorticoid activity could be synthesized. Because it was recognized early from structure-activity relationships that a carbonyl group at C3, a double bond between carbons 4 and 5, an oxygen (C=O or β -OH) at carbon 11, and a β -ketol side chain at position 17 are necessary for superior glucocorticoid activity, investigators began to synthesize analogs containing these functions. Additional groups were inserted into other positions of the basic steroid structure, with the expectation that these new substituents might modify the glucocorticoid and mineralocorticoid activities of the parent drugs.

However, the first potent analogs discovered, did not result from a concentrated effort to find a better drug but, rather, from basic chemical research concerned with the preparation of hydrocortisone from 11-epicortisol (11 α -hydrocortisone).



FLUDROCORTISONE A 9 α -bromo analog was prepared that had one-third the glucocorticoid activity of cortisone acetate (40). Other halogens were introduced into the 9 α -position, and it was soon observed that glucocorticoid activity is inversely proportional to the size of the halogen at carbon 9. The 9 α -fluoro analog (fludrocortisone) is approximately 11-fold as potent as cortisone acetate (Fig. 28.10). Fludrocortisone is orally administered as its 21-acetate derivative. When tested clinically in patients with rheumatoid arthritis, it was found to be effective at approximately one-tenth the dose of cortisone acetate. Although glucocorticoid activity is increased 11-fold by insertion of the 9 α -fluoro substituent, mineralocorticoid activity is increased 300- to 800-fold (27). Because of its intense sodium-retaining activity, fludrocortisone is contraindicated in all conditions except those that require a high degree of mineralocorticoid activity, because it leads to edema. Fludrocortisone acetate is used orally for mineralocorticoid replacement therapy in patients with adrenocortical insufficiency, such as Addison disease. This drug, introduced in 1954, helped to provide the impetus for the synthesis and biologic evaluation of newer halogenated analogs.

PREDNISONE, PREDNISOLONE, AND ITS DERIVATIVES One year after the introduction of fludrocortisone, the Δ^1 -corticoids were brought forth into clinical medicine. Investigators at Schering observed that the 1-dehydro derivatives of cortisone and hydrocortisone—namely, prednisone and prednisolone—are more potent antirheumatic and anti-allergenic agents than the parent compounds and produced fewer undesirable side effects at lower dosages. These compounds are known as Δ^1 -corticoids, because they contain an additional double bond between positions 1 and 2 (Fig. 28.10).

The Δ^1 -corticoids, which can be prepared by microbial dehydrogenation of cortisone or hydrocortisone with *Corynebacterium simplex* (41) and by several synthetic methods (33), represent the first chemical innovation leading to the creation of a modified compound that could be prescribed for rheumatoid arthritis. One high-yield route involves oxidation of 5 α - or 5 β -pregnane precursors that have appropriate oxygen substitutions with selenium dioxide (42).

Both prednisone and prednisolone were found to have adrenocortical activity (measured by eosinopenic response, liver glycogen decomposition, and thymus involution in adrenalectomized mice). In these tests, prednisone and prednisolone were found to be three- or

fourfold more potent than cortisone and hydrocortisone. Antiphlogistic (anti-inflammatory) strengths in human subjects were similarly augmented, but their electrolyte activities were not proportionately increased.

The increased potency reflects the effect in the change in geometry of ring A caused by the introduction of C1=C2 function on glucocorticoid receptor (GR) affinity and altered pharmacokinetics (primarily metabolism). Although the remaining portions of the steroid are essentially unchanged (except for less easily visualized molecular perturbations), the conformation of ring A changes from a chair, as in 5 α -pregnan-3-one, to a half-chair (pregn-4-en-3-one) and to a flattened boat (pregna-1,4-dien-3-one) on introduction of unsaturation (Fig. 28.11). The order of GR affinity is dexamethasone (10 $\times$) > triamcinolone (5 $\times$) > methylprednisolone (4 $\times$) > prednisolone (2 $\times$) > hydrocortisone (1 $\times$) (43).

When orally administered, prednisone and prednisolone are almost completely absorbed, with a bioavailability of greater than 80% (Table 28.3) (44,45). As with the relationship between cortisone and hydrocortisone, prednisone and prednisolone are interconvertible by 11 β -hydroxysteroid dehydrogenase in the liver. For practical purposes, prednisone and prednisolone are equally potent and can be used interchangeably. When prednisone or prednisolone is used in the treatment of rheumatoid arthritis, smaller doses are required than with hydrocortisone. The usual dose is 5 mg two to four times a day. Prednisolone is metabolized into a number of hydrophilic and less active metabolites, as shown in Figure 28.12, except there is no reduction of ring A as with hydrocortisone. The major metabolites (6 β - and 16 β -hydroxy) are primarily excreted as glucuronide conjugates in the urine. Prednisolone acetate is available in suspension and ointment forms for use externally. As with hydrocortisone, several other 21-esters of prednisolone are available. Prednisolone tert-butyacetate (3,3-dimethylbutyrate) is used in suspension form and by injection for the same reasons the 21-ester derivatives of hydrocortisone are employed. The butylacetate ester, which is suitable only for use by injection, has a long duration of action because of low water solubility and a slow rate of hydrolysis. The drug is administered in doses of 4 to 20 mg.

Prednisolone sodium phosphate is the water-soluble sodium salt of the 21-phosphate ester, with a half-life of less than 5 minutes because of rapid hydrolysis by

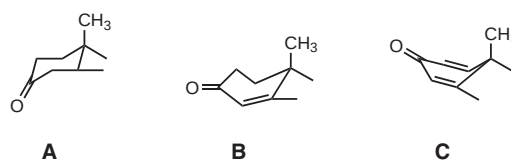


FIGURE 28.11 Ring A conformations for 5 α -pregnan-3-one (A), for pregn-4-en-3-one (B), and for pregena-1,4-dien-3-one (C).

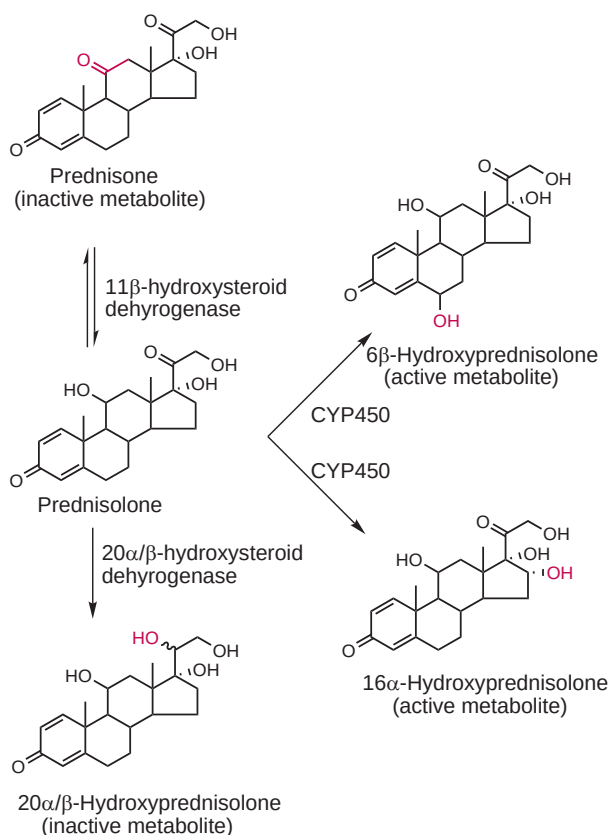


FIGURE 28.12 Major routes of metabolism for prednisolone.

phosphatases (44,46). Peak plasma levels for prednisolone are attained in approximately 10 minutes following its administration by injection (usual dose of 20 mg IV or IM). Topically, one or two drops of a 0.5% solution may be used four to six times daily for its anti-inflammatory action in the eye.

When doses of equivalent antirheumatic potency are given to patients not treated with steroids, the Δ^1 -corticoids promote the same pattern of initial improvement as hydrocortisone. Statistical results of the improvement status during the first few months of therapy have been similar with prednisolone, prednisone, and hydrocortisone. The results of longer-term therapy have been significantly better with the modified compounds. Studies indicate that the Δ^1 -corticoids can be used continuously in patients with rheumatoid arthritis without undue GI hazard. Although the Δ^1 -corticoids are considered not to have significant mineralocorticoid activity, hypokalemia and sodium and fluid retention can occur, depending on the dosage and duration of use.

METHYLPREDNISOLONE Between 1953 and 1962, many derivatives of the Δ^1 -corticoids and the halogen-containing analogs (especially 9 α -fluorinated compounds) were synthesized, and some became useful clinical agents. Studies with methylcorticoids revealed 2 α -methyl derivatives to be inactive, whereas the 2 α -methyl-9 α -fluoro analogs had

potent mineralocorticoid activity. Methylprednisolone (6 α -methyl derivative of prednisolone) was synthesized in 1956 and introduced into clinical medicine (Fig. 28.10). Methylprednisolone is extensively metabolized, with approximately 10% recovered unchanged in urine (47). The metabolic pathways include reduction of the C20 ketone, oxidation of 17 β -ketol group to C21-COOH and C20-COOH, and 6 β -hydroxylation (CYP3A4). These compounds potentiated glucocorticoid activity with negligible salt retention for short-term therapy (Table 28.2) (48).

In human subjects, the metabolic effects did not differ appreciably from those of prednisolone. Its activities with respect to nitrogen excretion, ACTH suppression, and reduction of circulating eosinophils were similar to those of prednisolone. The sodium retention and potassium loss were slightly less than with prednisolone (49).

Methylprednisolone is administered IV as its water-soluble sodium salt of the 21-succinate ester. The succinate ester is slowly and incompletely hydrolyzed. Peak plasma levels for methylprednisolone are attained in approximately 30 to 60 minutes following its IV administration, and approximately 15% of its IV dose is recovered unchanged in urine (45,46). CYP3A4 inhibitors such as the antifungals, ketoconazole, and itraconazole can potentiate the effects of prednisolone.

TRIAMCINOLONE The original interest in 16 α -hydroxycorticosteroids stemmed from their isolation from the urine of a boy with an adrenal tumor. The desire of chemists to synthesize these corticoids, and the hope that such analogs might have potent biologic activity furthered their development (50–52). Therefore, inserting a 16 α -hydroxy group into 9 α -fluoroprednisolone resulted in triamcinolone with glucocorticoid activity equivalent to prednisolone but with decreased mineralocorticoid activity. In fact, 16 α -hydroxy analogs of natural corticoids retain glucocorticoid activity and have a considerably reduced mineralocorticoid activity. Thus, a natural extension of corticoid research involved examination of compounds containing a 9 α -fluoro group, a double bond between positions 1 and 2, and a 16-hydroxy group. Triamcinolone introduced in 1958 combines the structural features of a Δ^1 -corticoid and a 9 α -fluoro corticoid plus a 16 α -hydroxy group (Fig. 28.10). As mentioned previously, the 9 α -fluoro group increases the anti-inflammatory potency, but also markedly increases the mineralocorticoid potency. This is undesirable if the drug is to be used internally for the treatment of rheumatoid arthritis. However, by inserting a 16 α -hydroxy group into the molecule, one decreases the mineralocorticoid activity.

The lower-than-expected oral anti-inflammatory potency for triamcinolone (Table 28.1) has been attributed to its low oral bioavailability (Table 28.3), in part because of increased hydrophilicity from the 16 α -hydroxy group and first-pass metabolism, primarily to its 6 β -hydroxy metabolite. These glucocorticoids actually can cause sodium excretion rather than sodium retention. Triamcinolone

diacetate (17,21-diacetate) and its hexacetonide [16 α , 17 α -methylenedioxy-21-(3,3-dimethylbutyrate)] esters are administered IM or intra-articularly for a prolonged release of triamcinolone. Triamcinolone diacetate has a duration of action depending on its route of administration from 1 to 8 weeks, and triamcinolone hexacetonide has a duration of action of 3 to 4 weeks (53,54).

On a weight-for-weight basis, the antirheumatic potency of triamcinolone is greater than that of prednisolone (~20%) and approximately the same as that of methylprednisolone. Initial improvement following administration of triamcinolone is similar to that noted with other compounds. Reports in the literature, however, indicate that the percentage of patients maintained satisfactorily for long periods has been distinctly smaller than that with prednisolone.

Although triamcinolone has an apparently decreased tendency to cause salt and water retention and edema and can induce sodium and water diuresis, it causes other unwanted side effects, including anorexia, weight loss, muscle weakness, leg cramps, nausea, dizziness, and a general toxic feeling (55). IM triamcinolone is reportedly effective and safe in the treatment of dermatoses, and in combination with folic acid antagonists, it is effective in the treatment of psoriasis (56,57).

DEXAMETHASONE Research with 16-methyl substituted corticoids was initiated in part because investigators hoped to stabilize the 17 β -ketol side chain to metabolism in vivo and improve bioavailability (Fig. 28.10). These studies led to the clinical development of dexamethasone, which combines the structural features of a Δ^1 -corticoid and a 9 α -fluoro corticoid plus a 16 α -methyl group (Fig. 28.10). A 16 α -methyl group decreases the reactivity of the 20-keto group to carbonyl reagents and increases the stability of the drug in human plasma in vitro (58,59). Unlike 16 α -hydroxylation, a methyl group increases the anti-inflammatory activity by increasing lipophilicity and, consequently, receptor affinity. Like the 16 α -hydroxyl group, the methyl group reduces markedly the salt-retaining properties of the corticosteroids (Table 28.2) (60–62). The activity of dexamethasone, as measured by glycogen deposition, is 20-fold greater than that of hydrocortisone. It has fivefold the anti-inflammatory activity of prednisolone. Clinical data indicate that this compound has sevenfold the antirheumatic potency of prednisolone. It is roughly 30-fold more potent than hydrocortisone. Its pharmacokinetics are presented in Table 28.3. Routes of metabolism for dexamethasone are similar to those for prednisolone, with its primary 6 β -hydroxy metabolite being recovered in urine (63). Dexamethasone sodium phosphate is the water-soluble sodium salt of the 21-phosphate ester, with an IV half-life of less than 10 minutes because of rapid hydrolysis by plasma phosphatases (64). Peak plasma levels for dexamethasone usually are attained in approximately 10 to 20 minutes following its IV administered dose. A similar reaction occurs when the phosphate ester is applied topically or by inhalation.

In practical management, 0.75 mg of dexamethasone promotes a therapeutic response equivalent to that from 4 mg of triamcinolone or methylprednisolone, 5 mg of prednisolone, and 20 mg of hydrocortisone. Clinical investigations with small groups of patients indicate that this compound could control patients who did not respond well to prednisolone. Over long periods, the improved status of some patients deteriorated.

In summarizing the biologic properties of this drug, it seems clear that with doses of corresponding antirheumatic strength, this steroid has approximately the same tendency as prednisolone to produce facial mooning, acne, and nervous excitation. Peripheral edema is uncommon (7%) and mild. The more common and most objectionable side effects are excessive appetite and weight gain, abdominal bloating, and distention. The frequency and severity of these symptoms vary with the dose (1 mg maximum for females, and 1.5 mg maximum for males). The longer biologic half-life for dexamethasone significantly increases the potential for glucocorticoid-induced adrenal insufficiency (see *Adverse Effects*).

The striking increase in potency does not confer a general therapeutic index on dexamethasone that is higher than that of prednisolone. Again, this drug probably is best employed as a special-purpose corticoid. It can be useful when other steroids are no longer effective or when increased appetite and weight gain are desirable (64–69). Its efficacy may be increased when it is used in combination with cyproheptadine as an antiallergenic, antipyretic, and anti-inflammatory agent (60).

BETAMETHASONE Shortly after the introduction of dexamethasone, betamethasone, which differs from dexamethasone only in configuration of the 16-methyl group (Fig. 28.10), was made available for the treatment of rheumatic diseases and dermatologic disorders. This analog, which contains a 16 β -methyl group, has received sufficient clinical trial examination to indicate that it is as effective as dexamethasone or, perhaps, even slightly more active. Although this drug has been reported to be less toxic than other steroids, some clinical investigators suggest that it is best used for short-term therapy. Toxic side effects, such as increased appetite, weight gain, and facial mooning, occur with prolonged use. A 0.5-mg tablet of betamethasone is equivalent to a 5.0-mg tablet of prednisolone, which is on par with dexamethasone (70).

Selective Glucocorticoid Receptor Modulators

The long-term use of glucocorticoids is associated with severe adverse effects (see *Adverse Effects*), including osteoporosis, hyperglycemia, muscle wasting, hypertension, and impaired wound healing. Improved glucocorticoids that demonstrate potent anti-inflammatory activity without these serious side effects would provide a significant therapeutic advance and are the focus of intense research efforts by the pharmaceutical industry.

The molecular mechanism of the GR makes it a particularly suitable target for this effort. The majority of unwanted side effects of the glucocorticoids arise from the interaction of the GR with DNA (i.e., GR-mediated transcription), whereas the anti-inflammatory effects are mediated via protein–protein interactions between the GR and proinflammatory transcription factors (i.e., AP-1 and NFκB) that result in repression of the inflammatory response. A variety of selective GR modulators with the ability to repress inflammation but with lesser ability to elicit GR-mediated transcription have been reported (71–75). Early investigations indicate that a variety of different nonsteroidal pharmacophores (e.g., spirocyclic dihydropyridines, triphenylmethanes, pyrazoles, and benzimidazoles) bind with high affinity and selectivity to the GR. Importantly, some of these analogs demonstrate preferential ability to repress proinflammatory genes and a lesser ability to induce GR-mediated transcription. Although these nonsteroidal ligands have yet to be evaluated in clinical studies, these drugs may someday prove to be the first members of an improved class of anti-inflammatory drugs.

Topical Glucocorticoids

Topically applied glucocorticoids are also capable of being systemically absorbed, although to a much smaller extent. The extent of absorption of topical adrenocorticoids is determined by several factors, including the type of cream or ointment, the condition of the skin to which it is being applied, and the use of occlusive dressings. Previous studies with halobetasol propionate showed that approximately 6% of the drug was systemically absorbed after topical application. Although this is a small fraction of the dose, the very high potency of halobetasol propionate contributed to its ability to cause mild adrenal suppression in some patients. The relative potency of the topical glucocorticoids is commonly determined using topical vasoconstriction assays and is dependent on the intrinsic activity of the drug, its concentration in the formulation, and the vehicle in which it is applied (Table 28.4).

Once absorbed through the skin, topical corticosteroids are handled through metabolic pathways similar to the systemically administered corticosteroids. They are metabolized, primarily in the liver, and are then excreted into the urine or in the bile (76). The fact that circulating levels of the topical glucocorticoids are often well below the level of detection does not reduce the risk for potential adverse effects from systemic exposure of topical corticosteroids. The structures for the glucocorticoids applied topically are shown in Figure 28.13 and their relative potencies in Table 28.4. Topical dermatologic products with a low potency ranking have a modest anti-inflammatory effect and are safest for chronic application. Those products with a medium potency ranking are used in moderate inflammatory dermatoses of limited duration. High potency preparations are used in more severe inflammatory dermatoses, but only for a short duration of treatment. Very high potency products

TABLE 28.4 Potency Ranking for Topical Corticosteroids

Potency Ranking	Topical Corticosteroids
I. Very high potency	Augmented betamethasone dipropionate Clobetasol propionate Diflorasone diacetate Halobetasol propionate
II. High potency	Amcinonide Augmented betamethasone dipropionate Betamethasone dipropionate Betamethasone valerate Desoximetasone Diflorasone diacetate Fluocinolone acetonide Fluocinonide Halcinonide Triamcinolone acetonide
III. Medium potency	Betamethasone benzoate Betamethasone dipropionate Betamethasone valerate Clocortolone pivalate Desoximetasone Fluocinolone acetonide Flurandrenolide Fluticasone propionate Hydrocortisone butyrate Hydrocortisone valerate Mometasone furoate Triamcinolone acetonide
IV. Low potency	Alclometasone dipropionate Desonide Dexamethasone Dexamethasone sodium phosphate Fluocinolone acetonide Hydrocortisone Hydrocortisone acetate Prednicarbate

The relative potency is based on the drug concentration, type of vehicle used, and the vasoconstrictor assay as a measure of topical anti-inflammatory activity.

are used primarily as an alternative to systemic corticosteroid therapy when local areas are involved and for only a short duration of therapy and on small surface areas.

Triamcinolone to be used topically is usually dispensed as its more potent and lipophilic acetonide, a 16 α ,17 α -methylenedioxy cyclic ketal or isopropylidene derivative (Fig. 28.14) (see *Inhaled and Intranasal Glucocorticoids*). It is effective in the treatment of psoriasis and other corticoid-sensitive dermatologic conditions. Topically, triamcinolone acetonide is a more potent derivative of triamcinolone and is approximately eightfold more active than prednisolone. The side effects of the drug, however, have occurred with sufficient frequency to discourage its routine use for rheumatoid patients requiring steroid therapy. The drug can be employed advantageously as a special-purpose steroid for instances in which salt and water retention (from other corticoids, hypertension, or

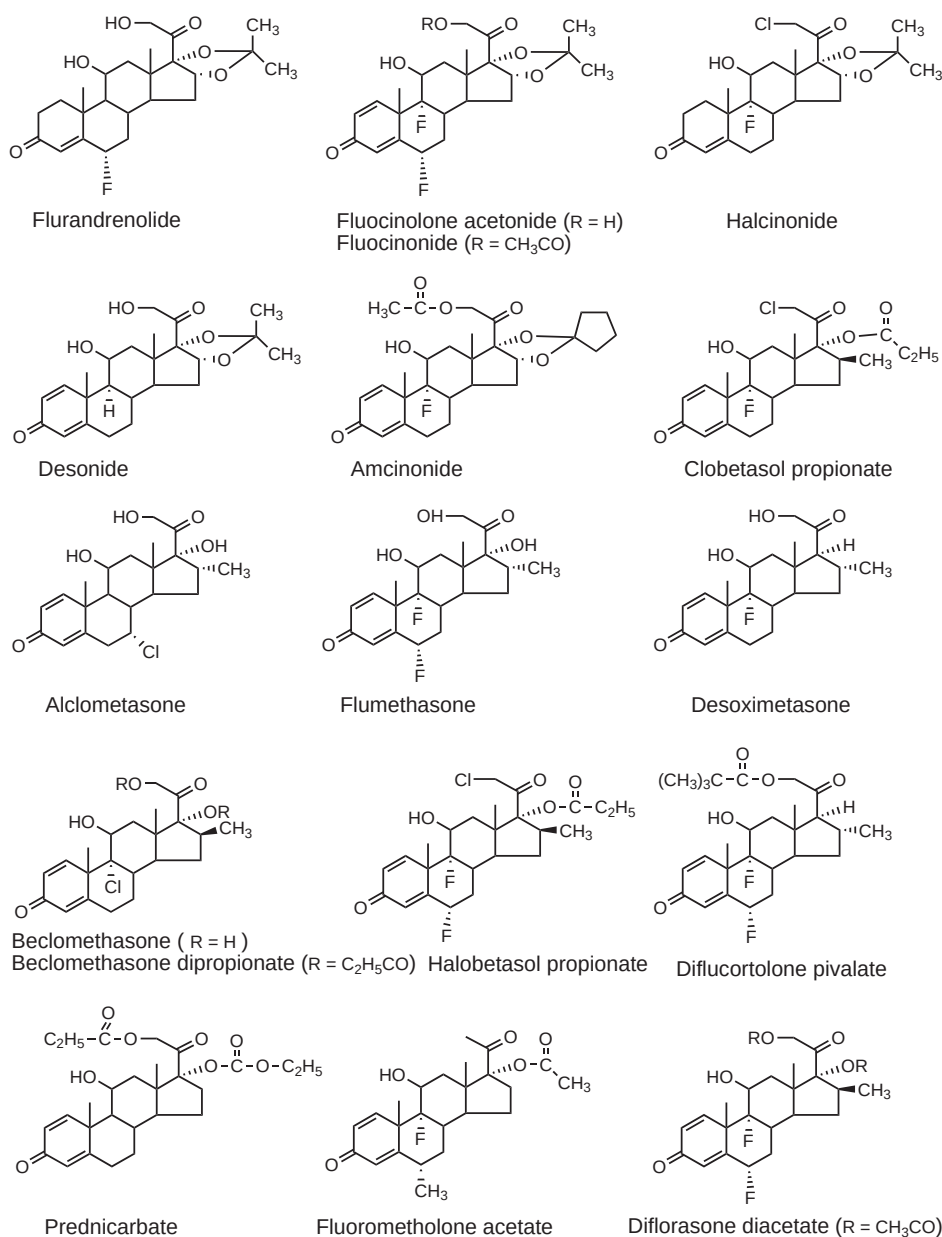


FIGURE 28.13 Topical corticosteroids.

cardiac compensation) or excessive appetite and weight gain are problems in management.

Newer synthetic glucocorticoids have incorporated chlorine atoms onto the steroid molecule as fluorine substitutes. Beclomethasone, a 9 α -chloro analog of betamethasone, is a potent glucocorticoid with approximately half the potency of its fluoro analog. It is used topically as its dipropionate derivative in inhalation aerosol therapy for asthma and rhinitis (see *Inhaled and Intranasal Glucocorticoids*) but not for treatment of steroid-responsive dermatoses (77). The topical anti-inflammatory potency for beclomethasone dipropionate (BDP) is approximately 5,000-fold greater than hydrocortisone, 500-fold greater than betamethasone or dexamethasone, and approximately 5-fold greater than fluocinolone acetonide or

triamcinolone acetonide, as measured by vasoconstrictor assay. Another potent topical glucocorticoid that contains a 7 α -chloro group is alclometasone (78).

Additional mono- and difluorinated analogs for topical application include fluorometholone (a 6 α -methyl-9 α -fluoro; ophthalmic use), flurandrenolide (a 6 α -fluoro-16 α ,17 α -acetonide), fluocinolone acetonide (a 6 α , 9 α -difluoro-16 α ,17 α -acetonide), and fluocinonide (a 21-acetate ester of fluocinolone acetonide) (Fig. 28.13). These compounds are classified as high- to medium-potency anti-inflammatory agents depending on the concentration and vehicle used (Table 28.4). The acetonide (ketal) derivatives at the 16,17-position enhance lipophilicity to provide potent topical anti-inflammatory agents (Table 28.4).

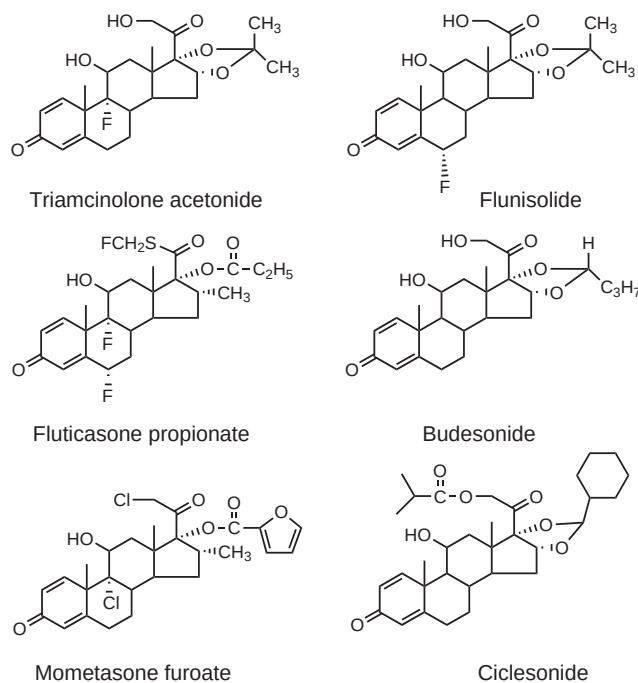


FIGURE 28.14 Inhaled and intranasal corticosteroids.

Psoriasis is one of the few inflammatory dermatoses that has not responded to routine topical steroid therapy, but these more potent steroids appear to work if a special occlusive dressing is used. In this technique, a thin layer of cream or ointment containing flurandrenolide is applied to the individual patch of psoriasis. The area is then covered with plastic food wrap or a similar pliable plastic film.

Clinical investigations show (0.05%) flurandrenolide to be more effective than 1% hydrocortisone acetate and to have approximately the same activity as 0.1% triamcinolone acetonide. Some investigators believe that its greater activity results from an increased biologic half-life. In other words, these analogs are not metabolized as readily. Fluocinolone acetonide has about the same anti-inflammatory activity as fluorometholone.

Clobetasol propionate, halcinonide, halobetasol propionate, and mometasone furoate are examples of 21-chlorocorticoids, in which the 21-chloro group replaces the 21-hydroxyl group (79–82) (Figs. 28.13 and 28.14). Clobetasol propionate, the 21-chloro analog of betamethasone 17-propionate, is approximately eightfold more active as topical anti-inflammatory agent than betamethasone 17 α -valerate, the standard of comparison for topical vasoconstrictor/anti-inflammatory activity. Mometasone furoate, a 9 α ,21-dichloro derivative (see *Inhaled and Intranasal Glucocorticoids*), is also approximately eightfold more active than betamethasone 17 α -valerate as a topical anti-inflammatory agent. Thus, substitution of a chlorine (or a fluorine) atom for the 21-hydroxyl group on the glucocorticoids greatly enhances topical anti-inflammatory activity (80). Clobetasol propionate and halobetasol propionate are classified as very high-potency topical corticosteroid preparations (Table 28.4). The HPA suppression

has occurred following the topical application of 2 g of the 0.05% clobetasol propionate ointment or cream (1 mg of clobetasol propionate total) daily. Because of its high potency and potential for causing adverse systemic effects during topical therapy, the usual dosage for very high-potency topical steroids should not be exceeded. Fluticasone propionate is similar to the 21-chloro steroids, except that it has a 17 α -fluoromethylcarbothioate group instead of the 17-ketol group derivative (see *Inhaled and Intranasal Glucocorticoids*) (Fig. 28.14). Although mometasone furoate and fluticasone propionate are very lipophilic and have the highest binding affinity for the GR (see *Inhaled and Intranasal Glucocorticoids*) when compared to triamcinolone acetonide and dexamethasone, their topical potency is listed as medium, in part because of their insolubility and poor dissolution into inflamed tissue.

Several nonfluorinated analogs of triamcinolone acetonide with the potency-enhancing cyclic ketal moiety are marketed, suggesting that halogens are not always necessary for topical activity. These nonfluorinated cyclic ketals include desonide and amcinonide (Table 28.13) and the cyclic acetal ciclesonide (Fig. 28.14). Amcinonide's potency is greatly enhanced by the more lipophilic cyclopentanone ketal and 21-acetate. A recent addition to the nonhalogenated prednisolone derivatives is prednicarbate, a 17,21-diester (17 α -ethylcarbonate-21-propionate) derivative of prednisolone (Fig. 28.13), which is used for the local treatment of corticoid-sensitive skin diseases (83). Any prednicarbate that is absorbed systemically is readily metabolized by hydrolysis of the 21-ester to its primary and pharmacologically active metabolite, prednisolone-17-ethylcarbonate. This metabolite has a half-life of approximately 1 to 2 hours and is further metabolized by the liver to prednisolone. In vitro binding studies with the GR suggest that the ethyl carbonate metabolite has a receptor binding affinity comparable to that of dexamethasone. The low systemic bioavailability for prednicarbate after dermal application has been attributed to its metabolism to less active prednisolone, which can be a factor for the low systemic side effects of prednicarbate.

Inhaled and Intranasal Glucocorticoids

It is generally accepted that the anti-inflammatory effect of glucocorticosteroids cannot be separated from their adverse effects at the receptor level. Therefore, pulmonary and nasal pharmacokinetics become important determinants for the potential of an inhaled or nasally applied corticosteroid to cause systemic effects, because the lung and nasal tissue provide an enormous surface area from which drug absorption can occur into the systemic circulation (84,85). The main areas of concern with regard to drug-induced systemic effects include HPA axis suppression, change in bone mineral density and growth retardation in children, cataracts, and glaucoma. The degree of systemic side effects is dose-dependent, related to the half-life of the drug, the frequency of administration, the time of day when administered, and the route of administration; in other words, the higher the plasma corticosteroid concentration and longer the half-life, the greater will be the systemic side effects (86).

Thus, the search is to develop inhaled/intranasal corticosteroids with the following desirable pharmacokinetic qualities: They would exhibit fast systemic clearance following GI absorption (high degree of first-pass intestinal/hepatic metabolism), short half-life, lack of active metabolites, and high affinity for the corticosteroid receptor. These qualities determine the proportion of the drug that reaches the target cells as well as the fraction of the dose that reaches the systemic circulation to produce side effects.

Modification of the pharmacokinetics through structural alterations has provided several new steroids with a better GR affinity and therapeutic index and a lower bioavailability than the older drugs (Fig. 28.14). The new inhaled/intranasal glucocorticosteroids like mometasone furoate, budesonide, and fluticasone propionate are more lipophilic than those used in oral and systemic therapy and have greater affinity for the GR than does dexamethasone as a consequence of their greater lipophilicity (43). Several of the topical corticosteroids, such as mometasone furoate, BDP, triamcinolone acetonide, and flunisolide, were reintroduced as inhalation and intranasal dosage forms for treatment of respiratory diseases (e.g., asthma or rhinitis). Inhaled budesonide and flunisolide are readily absorbed from the airway mucosa into the blood and are rapidly biotransformed in the liver into inactive metabolites. Mometasone furoate and fluticasone propionate are very potent anti-inflammatory steroids with an oral bioavailability of less than 1%. Obviously, the risk of systemic side effects for these newer corticosteroids is greatly reduced when compared with the older glucocorticosteroids (e.g., dexamethasone). Beclomethasone dipropionate was discovered to be a prodrug, and this discovery led to the reexamination of other 17α -monoesters as the active form of the corticosteroid esters. The absorption of budesonide, fluticasone propionate, and BDP into the airway tissue was 25- to 130-fold greater than that for dexamethasone and hydrocortisone (86). The GR affinity and the pharmacokinetic properties for the inhaled and intranasal corticosteroids are listed in Table 28.5.

It is generally recognized that when administered by oral inhalation, 10% to 30% of a dose of the corticosteroid is deposited in the respiratory tract depending on type of inhaler (metered dose inhaler or dry powder inhaler [DPI]) and spacer used (87,88). The remainder of the dose is deposited primarily in the mouth and throat to be swallowed into the GI tract, where the drug can be absorbed, metabolized, and eliminated unchanged in the feces. Thus, systemic bioavailability of the inhaled/intranasal steroids is determined by the fraction of the dose absorbed from the lungs/nasal mucosa and the GI tract into systemic circulation and the degree of first-pass metabolism. Although these corticosteroids are very lipid soluble, they display variable degrees of absorption from respiratory and GI tissues, in part because of dissolution problems. When systemically absorbed, they are capable of suppressing the HPA and adrenal function with high and chronic dosing regimens (88,89). Although as much as 40% of the dose for the high-potency adrenocorticoids flunisolide, mometasone furoate, fluticasone propionate, or

ciclesonide is absorbed into airway and nasal tissues during oral inhalation, the remainder of the drug is swallowed to undergo extensive first-pass metabolism in the liver to essentially inactive metabolites, with no apparent suppression effects on adrenal function with long-term therapy.

Lipophilicity can positively or negatively alter the pharmacokinetic and pharmacodynamic actions of the inhaled/intranasal steroids (90). The lipophilic substituents attached to the corticosteroid nucleus can improve receptor affinity (Table 28.5), or they can affect pharmacokinetic properties, such as absorption, protein and tissue binding, distribution, and excretion. The inhaled/intranasal corticosteroids are inhaled as microcrystals and need sufficient water solubility to be dissolved in the nasal or lung epithelial tissue for local anti-inflammatory activity to occur. Lipophilicity, however, can delay their rate of dissolution into these tissues, which can be advantageous, by prolonging their retention within these tissues, affecting their onset and duration of action, or a disadvantage, by facilitating their transport away from these tissues via mucociliary clearance before full dissolution can occur. The systemic steroids prednisolone and hydrocortisone are less effective as inhaled/intranasal steroids because of their higher water solubility and lower lipophilicity. For inhaled/intranasal steroids, there is a sharp drop in water solubility when the lipophilicity is $\log P \geq 4$ ($P = 10,000$), which is the case for BDP, ciclesonide, fluticasone propionate, and mometasone furoate. High lipophilicity correlates well with low oral bioavailability and high protein binding (Table 28.5).

The use of dexamethasone sodium phosphate inhalation aerosol is not recommended because of the potential for extensive systemic absorption and the long metabolic half-life for dexamethasone after absorption, resulting in an increased risk of adverse effects with usual inhalation doses (27). Following the oral inhalation of dexamethasone sodium phosphate, a cumulative dose of 1,200 mcg/day will result in the systemic absorption of 400 to 600 mcg of dexamethasone, which is sufficient to cause HPA suppression. Dexamethasone sodium phosphate nasal aerosol delivers 100 mcg per metered spray. The total daily adult nasal dose is 1,200 mcg.

SPECIFIC DRUGS

Triamcinolone Acetonide Triamcinolone acetonide is frequently used by inhalation for the treatment of lung diseases (e.g., asthma). After inhalation, triamcinolone acetonide can become systemically available when the inhaled formulation is swallowed and absorbed unchanged from the GI tract, causing undesirable systemic effects (91–93). Triamcinolone acetonide that is swallowed is metabolized to 6β -hydroxytriamcinolone acetonide, 21-carboxytriamcinolone acetonide, and 21-carboxy- 6β -hydroxytriamcinolone acetonide, all of which are more hydrophilic than their parent drug. Only approximately 1% of the dose was recovered from the urine as triamcinolone acetonide (94). Triamcinolone is not a major metabolite of triamcinolone acetonide in humans, suggesting that acetonide is resistant to hydrolytic cleavage. Triamcinolone acetonide is approximately eightfold more potent than prednisolone.

TABLE 28.5 Pharmacokinetics of Inhaled and Intranasal Corticosteroids

Parameters	Beclomethasone Dipropionate	Budesonide	Ciclesonide	Flunisolide ^a	Triamcinolone Acetonide	Fluticasone Propionate	Mometasone Furoate
Receptor binding affinity ^b	0.4 BMP 13.5 ^c	9.4 22R, 11.2 22S, 4.2	R-epimer 1,200 R-epimer des-CIC 12,000 ^d	1.8	3.6	18	25–27
Relative Lipophilicity ^e	11,800 3,320 BMP	1580	200,000 14,500 des-CIC	110	270	11,500	40,000
Pulmonary bioavailability	~20% ^f	~39%	~60% des-CIC	40%	25% ^g	~30% (aerosol)	<1% (aerosol)
Nasal bioavailability	~20% ^f	<20%	—	50%	25% ^g	13–16% (powder)	Not detectable (powder)
Oral bioavailability (systemic)	15–20%	~10%, oral	<1% parent and des-CIC	6–10%	23%	<2%	<1%
Protein binding	87% (transcortin & albumin)	85–90% (albumin)	Parent 99% des-CIC <1% (albumin)	Moderate (transcortin & albumin)	68% ^g (albumin)	91% (albumin)	90%
Half-life	30 min IV 10 min inhaled 6.5 hr BMP IV 2–7 hr BMP inhaled	2–3 hr IV	Parent <1 hr des-CIC 6–7 hr inhalation	1–2 hr inhalation	1–2 hr inhalation 1–7 hr nasal 3.1 hr solution	~7.8 hr IV ~14 hr inhaled	4–6 h IV Aerosol (not detectable) Inhaled (not detectable)
Metabolism	Lung and liver esterase, liver (CYP3A4) first pass	Liver (CYP3A) first pass	Liver (CYP3A) Lung esterase	Liver first pass	Liver first pass	Liver (CYP3A4) first pass	Liver
Onset of action	3–7 d	2–3 d	des-CIC 2–4 wks	3–7 d	4–7 d	2–3d	7 hr
Excretion	Feces, urine 12–15%	~60% urine ~30% feces	<20% urine ~60% feces	~50% renal ~40% feces	~40% urine ~60% feces	~5% urine 80–90% feces	6–10% urine 50–90% feces

BMP, beclomethasone; des-CIC, des-ciclesonide; IV, intravenous; d, day; wk, week; hr, hour.

Data from McEvoy GK ed. AHFS 2001 Drug Information. Bethesda, MD: American Society of Health-System Pharmacists, 2001.

^aNasarel and Nasalide are not bioequivalent. Total absorption of Nasarel was 25% less, and the peak plasma concentration was 30% lower, than that of Nasalide. The clinical significance of this is likely to be small, however, because clinical efficacy is dependent on local effects on the nasal mucosa.

^bBinding affinity to human glucocorticoid receptors in vitro relative to dexamethasone. Data from Kelly HW. Establishing a therapeutic index for the inhaled corticosteroids.

Part 1: Pharmacokinetic pharmacodynamic comparison of the inhaled corticosteroids. J Allergy Clin Immunol 1998;102:S36–S51.

^cBeclomethasone dipropionate is hydrolyzed by liver esterases to the more active beclomethasone monopropionate (BMP).

^dCiclesonide is hydrolyzed into des-ciclesonide (des-CIC).

^eCalculated Log P from Advanced Chemical Development, Toronto, Canada. Clog P calculated with ACD/Log P 12v. CLog P were converted to antilogs. *K'* values: water = 1, hydrocortisone = 794, prednisolone = 316, dexamethasone = 400.

^fEstimated for inhaled beclomethasone aerosol.

^gData from oral inhalation administration.

Triamcinolone acetonide inhalation aerosol for pulmonary delivery is a microcrystalline suspension in a chlorofluorocarbon propellant that delivers 200 mcg per metered spray, which is equivalent to 100 mcg delivered at the mouthpiece. The total daily adult nasal dose for triamcinolone acetonide is 1.6 mg. Triamcinolone acetonide nasal aerosol is a microcrystalline suspension in a chlorofluorocarbon propellant that delivers 55 mcg per metered spray. The total daily dose for triamcinolone acetonide is 440 mcg.

Beclomethasone 17,21-dipropionate Beclomethasone dipropionate is used primarily as an inhalation aerosol therapy for asthma and rhinitis (77). A breakthrough in the discovery of new inhalation corticosteroids with reduced risks from systemic absorption was that the 17 α -monopropionate ester of beclomethasone

(17-BMP) was more active than BDP and beclomethasone 21-monopropionate (21-BMP) esters (95). Thus, BDP is a prodrug that is rapidly metabolized by esterases in the lung and other tissues to its more active metabolite, 17-BMP, which has 30-fold greater affinity for the GR than BDP and approximately 14-fold greater affinity than dexamethasone (Table 28.5) (43).

Whether orally administered or swallowed from inhalation, BDP undergoes rapid first-pass metabolism of the unhindered 21-ester via enzymatic hydrolysis in the liver or GI tract primarily to 17-BMP but more slowly to 21-BMP and to beclomethasone and other unidentified metabolites and polar conjugates (96,97). The terminal half-life for 17-BMP is 6.5 hours. The portion of the inhaled dose of BDP that enters the lung is rapidly metabolized to 17-BMP in the respiratory tract before reaching systemic circulation, where it can

be further metabolized by the liver. Following oral administration, BDP and its metabolites are excreted mainly in feces via biliary elimination, and 12% to 15% of a 4-mg dose of BDP is excreted in urine as free and conjugated metabolites. The usual therapeutic dose (<1,200 mcg/day) for BDP oral inhalation does not produce systemic glucocorticoid effects, probably because the drug is rapidly metabolized to less active metabolites. At doses greater than 1,200 mcg/day, HPA suppression has been observed (27).

The BDP monohydrate nasal suspension is available as an aqueous microcrystalline suspension of BDP, which delivers 42 mcg per metered spray. The BDP nasal aerosol or inhalation aerosol consists of microcrystalline suspension of BDP in chlorofluorocarbon propellant, both of which deliver 42, 50, or 84 mcg per metered spray. The total daily adult dose for BDP is 600 mcg from the nasal spray or nasal inhaler and 336 to 1,000 mcg for the aerosol inhaler. Doses exceeding 2,000 mcg/day need to be monitored for HPA suppression.

Flunisolide When administered intranasally or by inhalation, flunisolide (Fig. 28.14) is rapidly absorbed from nasal or lung tissue (Table 28.5) (94). This corticosteroid is efficiently metabolized by the liver to inactive metabolites with no apparent effects on adrenal function with long-term therapy. Flunisolide that is swallowed undergoes extensive first-pass metabolism in the liver, and that which is absorbed directly from the nasopharyngeal mucosa or lung bypasses this initial metabolism (98). It is not known if the drug undergoes metabolism in the GI tract. Flunisolide is rapidly hydroxylated by CYP3A4 at the 6 β position, followed by elimination of the 6 α -fluoro group to its more polar 6 β -hydroxy metabolite, which attains plasma concentrations that usually are greater than those for flunisolide (94,98).

Following IV administration of flunisolide, the 6 β -hydroxy metabolite has 1/100 the potency of flunisolide and a plasma half-life of 3.9 to 4.6 hours. Flunisolide and its 6 β -hydroxy metabolite are conjugated in the liver to inactive glucuronides and sulfates. After intranasal administrations of 100 mcg, the plasma levels for flunisolide were undetectable within 4 hours. The duration of its systemic effects is short because of its short half-life.

Flunisolide nasal solution is available in an aqueous solubilized form, which delivers 25 mcg per spray. The total daily adult dose for flunisolide from the nasal spray is 200 to 400 mcg. Flunisolide inhalation aerosol for pulmonary delivery is a microcrystalline suspension in a chlorofluorocarbon propellant that delivers 250 mcg per metered spray. The total daily adult inhalation dose for flunisolide is 1,000 mcg. Doses exceeding 2,000 mcg/day need to be monitored for HPA suppression.

Budesonide Budesonide is a highly potent, nonhalogenated glucocorticoid intended for the local treatment of lung disease and rhinitis. It was designed to have a high ratio between local and systemic effects. Budesonide is composed of a 1:1 mixture of epimers of the 16,17-butylacetal, creating a chiral center (Fig. 28.14) (99). The 22*R*-epimer binds to the GR with higher affinity than does the 22*S*-epimer

(Table 28.5) (43). The butylacetal chain provided the highest potency for the homologous acetal chains. Its rate of topical uptake into epithelial tissue is more than 100-fold faster than that for hydrocortisone and dexamethasone. Approximately 85% of the orally inhaled dose of budesonide undergoes extensive first-pass hepatic metabolism by CYP3A4 to its primary metabolites, 6 β -hydroxybudesonide and 16 α -hydroxy-prednisolone, which have approximately 1/100 the potency of budesonide (100,101). This is an important inactivation step in limiting budesonide's systemic effect on adrenal suppression. Budesonide was metabolized three- to sixfold more rapidly than triamcinolone acetonide. The pharmacokinetics of budesonide after inhalation, oral, and IV administration displayed a mean plasma half-life of 2.8 hours and a systemic bioavailability of approximately 10% after oral administration (Table 28.5) (101). Pulmonary bioavailability is less than 40% after inhalation (70% to 75% after correction for the amounts of budesonide deposited in the inhalation device and oral cavity). No oxidative metabolism was observed in the lung. When given by inhalation, 32% of the dose is excreted in the urine as metabolites, 15% is excreted in the feces, and 41% remains in the mouthpiece of the inhaler. Following intranasal administration, very little of intranasal budesonide is absorbed from the nasal mucosa. Much of the intranasal dose (~60%) was swallowed, however, and remained in the GI tract to be excreted unchanged in the feces, whereas that fraction of the intranasal dose that was absorbed was extensively metabolized.

Inhaled budesonide, despite its lower lipophilicity, exhibits greater retention within the airways than other inhaled corticosteroids do (100,101). This unusual behavior for inhaled budesonide has been attributed to the subsequent formation of intracellular fatty acid esters of the 21-hydroxy group of budesonide in the airway and lung tissue (102,103). Following inhalation, approximately 70% to 80% of budesonide was reversibly esterified by free fatty acids in the airway tissue. These inactive esters behave like an intracellular depot drug by slowly regenerating free budesonide. Thus, this reversible esterification prolongs the local anti-inflammatory action of budesonide in the airways and may contribute to the high efficacy and safety of budesonide in the treatment of mild asthma when inhaled once daily.

The systemic availability of budesonide in children was estimated to be 6.1% of the nominal dose, and the terminal half-life was 2.3 hours (104,105). Approximately 6% of the nominal dose reached the systemic circulation of young children after inhalation of nebulized budesonide. This is approximately half the systemic availability found in healthy adults using the same nebulizer.

Budesonide nasal aerosol is supplied as a micronized suspension using a chlorofluorocarbon propellant, which delivers 32 mcg from the nasal adapter supplied per metered spray. The total daily adult dose for budesonide is 256 mcg. Budesonide powder for pulmonary inhalation uses micronized dry powder in a turboinhaler (DPI) that delivers 200 mcg per metered spray. The total daily adult dose for budesonide from the DPI is 200 to 800 mcg. Full benefit is attained in approximately 1 to 2 weeks.

Ciclesonide The newest third-generation nonhalogenated prednisolone analog approved in 2006 is ciclesonide (Fig. 28.13). Ciclesonide is the 21-isobutyrate ester, and the 16 α ,17 α -acetal of cyclohexanecarboxaldehyde of prednisolone creates an epimeric center. The *R*-epimer binds to the GR with high affinity (Table 28.5). The 21-isobutyrate ester and 16 α ,17 α -acetal of cyclohexanecarboxaldehyde greatly enhance its lipid solubility. Ciclesonide is a prodrug that is converted locally in airways by carboxylesterases to produce the active metabolite, desisobutyrylciclesonide (des-CIC). Des-CIC has a 100-fold greater relative GR binding affinity than ciclesonide itself (relative glucocorticoid receptor binding affinities are 1,200 and 12, respectively; dexamethasone is 100). If any ciclesonide or des-CIC enters the circulation, it is highly protein bound (99%) and undergoes extensive first-pass hepatic metabolism by CYP3A4 to 6 β -hydroxy metabolites, resulting in very low systemic exposure. Clinical studies demonstrate that ciclesonide is used in inhalation aerosol therapy for asthma and rhinitis (see *Inhaled and Intranasal Glucocorticoids*).

Ciclesonide is supplied as an aqueous inhalation solution pump that dispenses 50 mcg per 70 μ L spray for seasonal allergies or delivered in solution form via a hydrofluoroalkane metered-dose inhaler with a once-daily dosing schedule, which facilitates asthmatic patient compliance. The total daily dose is 200 mcg for seasonal allergies and 80 to 160 mcg for asthma. Following inhalation administration, ciclesonide and des-CIC are not detected in the plasma. Ciclesonide has a half-life of less than 1 hour, and des-CIC has a half-life of 6 to 7 hours and an oral bioavailability of less than 1% due to extensive plasma protein binding (Table 28.5). Ciclesonide produces potent anti-inflammatory effects with an onset of action of approximately 2 to 4 weeks. It is extensively metabolized, with less than 20% of the administered dose recovered in the urine as des-CIC. The majority of the oral inhalation for ciclesonide is deposited in the airway passages and swallowed without absorption in the GI tract until eliminated in the feces (~60% of the administered dose is recovered in the feces). The portion of the inhalation dose that is absorbed is extensively metabolized. These results indicate that inhaled ciclesonide has negligible systemic bioavailability and is extensively metabolized, with reduced risk for causing systemic adrenal suppression effects.

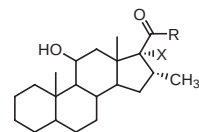
Mometasone Furoate The development of mometasone furoate resulted from reexamination of the effect of 17 α -ester functionalities on topical anti-inflammatory potency relative to the potent 17-benzoate ester of betamethasone. The structure–activity relationship study involved substitution of the 17-benzoate ester with heteroaromatic furoic, thienoic, and pyrrolic esters (81,106). Of the numerous 17 α -heteroaryl esters studied, the 2-furoate ester displayed the greatest increase in potency. Therefore, combining the 17 α -(2-furoate) ester with the potency-enhancing effect of the 21-chloro group, the resultant glucocorticoid (mometasone furoate) (Fig. 28.14) was 5- to 10-fold more potent than the betamethasone benzoate ester, with a more rapid onset of action. Mometasone furoate was originally marketed as a topically applied corticosteroid,

but because of its low systemic bioavailability, it was found to be more useful in the treatment of allergic disorders and lung diseases (107). It has the greatest binding affinity for the GR (Table 28.5), followed by fluticasone propionate, budesonide, triamcinolone acetonide, and dexamethasone (43). Mometasone furoate has strong local anti-inflammatory activity equivalent to that of fluticasone propionate. It has a quick onset of action relative to the other inhaled/intranasal steroids with the least systemic availability and, consequently, the fewest systemic side effects.

After IV suspension or inhalation administration, mometasone furoate was detected in the plasma for up to 8 hours, with a half-life of 4 to 6 hours (Table 28.5) and an oral bioavailability of less than 1%. It is extensively metabolized with less than 10% of the administered dose recovered in the urine unchanged (108). Among the polar metabolites (~80%) and their conjugates (42%) that were recovered were 6 β -hydroxymometasone furoate and its 21-hydroxy metabolite. In contrast, following intranasal administration, its plasma concentrations were below the limit of quantification, and the systemic bioavailability by this route was estimated to be less than 1%. The majority of the intranasal dose for mometasone furoate is deposited in the nasal mucosa and swallowed without absorption in the GI tract until eliminated in the feces (~50% to 90% of the intranasal dose is recovered in the feces). That portion of the intranasal dose that was absorbed was extensively metabolized. These results indicate that inhaled mometasone furoate has negligible systemic bioavailability and is extensively metabolized, with reduced risk for causing systemic adrenal suppression effects.

Mometasone furoate nasal suspension is supplied as an aqueous suspension with an atomizing pump that dispenses 50 mcg per metered spray. The total daily dose for mometasone furoate is 200 mcg.

ANDROSTANE 17 β -CARBOXYLATES AND 17 β -CARBOTHIOATES



Androstane-17 β -carboxylates (R = OCH₂F)
Androstane-17 β -carbothioates (R = SCH₂F)
(X = acetate or propionate)

The androstane 17 β -hydroxyl-17 β -carboxylates and 17 β -carbothioates were designed to be metabolically susceptible to hydrolysis and to have a low systemic bioavailability to minimize systemic glucocorticoid-induced adrenal suppression. The androstane 17 α -hydroxyl-17 β -carboxylates lacked the 17-ketol group found in most of the systemic corticosteroids. When these 17 β -carboxylates were esterified to their 17 α /17 β -diesters, however, they proved to be extremely potent anti-inflammatory corticosteroids, whereas the parent carboxylic acids were inactive (109). Thus, enzymatic hydrolysis of the 17-carboxylate ester function by intestinal or liver esterases would lead to formation of inactive metabolites. The greatest

anti-inflammatory activity was observed with 17 α -acetoxy and 17 α -propionyloxy groups and simple alkyl carboxylate esters, although the fluoromethyl esters showed the highest activity. Superseding the androstane 17 β -carboxylates were the corresponding 17 β -carbothioates (thioesters) derived from flumethasone. The 17 β -fluoromethylcarbothioate when combined with the 17 α -propionyloxy group yielded fluticasone (Fig. 28.14) (110). The androstane 17 β -carbothioates proved not only to be very potent anti-inflammatory agents but also to exhibit weak HPA suppression in the rat. Both the androstane 17 β -carboxylates and the androstane 17 β -carbothioates are very lipophilic and exhibit minimal oral bioavailability and very low systemic activity after inhalation because of intestinal and hepatic enzymatic hydrolysis to inactive metabolites, which have 1/2,000 the activity of the parent molecule (111).

Fluticasone Propionate Fluticasone propionate, a trifluorinated glucocorticoid based on the androstane 17 β -carbothioate nucleus (Fig. 28.14), was designed to be metabolically susceptible to hydrolysis and to have a low systemic bioavailability to minimize the systemic effects on plasma hydrocortisone levels. Its susceptibility to metabolic hydrolysis is doubly enhanced by the combination of a thioester and the high electronegativity of the fluorine group. Fluticasone propionate is approximately as lipophilic as BDP, eightfold more lipophilic than budesonide, and fourfold more lipophilic than triamcinolone acetonide (Table 28.5). It also displays high in vitro selectivity for the GR and a relative receptor affinity 1.5-fold that of 17-BMP and ciclesonide, about equal to mometasone furoate, 2-fold that of budesonide, 18-fold that of dexamethasone, 10-fold that of flunisolide, and 5-fold that of triamcinolone acetonide (43). The rate of association for fluticasone dipropionate with the receptor is faster, and the rate of dissociation is slower, than the other corticosteroids. The half-life of the fluticasone propionate active steroid–receptor complex is greater than 10 hours, compared with approximately 5 hours for budesonide, 7.5 hours for 17-BMP, and 4 hours for triamcinolone acetonide (43).

After topical application to the nasal mucosa or after inhalation, fluticasone propionate produces potent anti-inflammatory effects, with an onset of action of approximately 2 to 3 days. The topical anti-inflammatory potency for fluticasone propionate is approximately equal to that for mometasone furoate, 13-fold greater than that for triamcinolone acetonide, 9-fold greater than that for fluocinolone acetonide, 3-fold greater than that for betamethasone 17-valerate, and 2-fold greater than that for BDP (111). Because of its low systemic bioavailability when administered intranasally or by inhalation and nondetectability in plasma, most pharmacokinetic data for fluticasone propionate are based on IV or oral administration (Table 28.5). Its rate of topical uptake into epithelial tissue is more than 100-fold faster than that for hydrocortisone and dexamethasone but is similar to BDP and budesonide.

As a consequence of its high lipophilicity (Table 28.5), fluticasone propionate is very insoluble and, therefore, is poorly absorbed from the respiratory (10% to 13%) and GI

tracts following nasal inhalation of the drug (112–114). The majority of the intranasal dose for fluticasone propionate is deposited in the nasal mucosa and swallowed into the GI tract until eliminated in the feces (~80% to 90% of the intranasal dose is recovered metabolized and unchanged in the feces). After administration of an IV suspension, fluticasone propionate displayed a systemic bioavailability of less than 2% and underwent extensive hydrolysis and CYP3A4 first-pass metabolism in the liver, with an elimination half-life of approximately 3 hours. Its primary hydrolysis product is 17 β -carboxylate metabolite, which has 1/2,000 the affinity for the GR and can be recovered from the urine along with other unidentified hydroxy metabolites and their conjugates. Following oral administration of 1 to 40 mg, fluticasone propionate is poorly absorbed from the GI tract because of hydrolysis and its insolubility, with an oral bioavailability of less than 1%. Less than 5% of the oral dose is excreted unchanged in the urine. Fluticasone propionate was not detected in plasma for up to 6 hours after IV or oral administration (113). Approximately 80% to 90% of the oral dose is excreted unchanged in the feces, of which approximately 3% to 40% is the hydrolyzed 17 β -carboxylate metabolite. Pulmonary bioavailability ranges between 16% and 30% depending on the inhalation device used, with an elimination half-life of approximately 14 hours, increasing its potential for drug accumulation with repeated dosing (114). The long elimination half-life for fluticasone propionate results, in part, from its very high lipophilicity and very poor water solubility and, consequently, slow dissolution into lung tissue. Some suppression of overnight hydrocortisone levels has been reported with inhaled fluticasone propionate at higher dosages (indicative of HPA axis suppression) (115).

Fluticasone propionate nasal suspension and inhalation aerosol are both available as micronized suspensions in a chlorofluorocarbon propellant that delivers 50 mcg per metered spray. The total daily adult dose for fluticasone propionate is 200 mcg for the nasal suspension and 880 mcg for the inhalation aerosol.

SUMMARY OF STRUCTURE–ACTIVITY RELATIONSHIPS

The structure in Figure 28.15 depicts the ring conformation and the absolute configuration of hydrocortisone and prednisolone. The all-*trans* (B/C and C/D) backbone that is necessary for activity is very evident.

As previously pointed out, the adrenocorticoids are classified as either glucocorticoids, which affect intermediary metabolism and are associated with inhibition of the inflammatory process, or mineralocorticoids. In fact, most

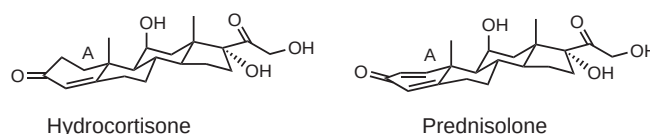


FIGURE 28.15 Conformations of hydrocortisone and prednisolone.

naturally occurring and semisynthetic analogs exhibit both of these actions. The 17 β -ketol ($-\text{COCH}_2\text{OH}$) side chain and the Δ^4 -3-ketone functions are found in clinically used adrenocorticoids, and these groups contribute to the potency of the drugs. Modifications of these groups result in derivatives that retain biologic activity. For example, replacement of the 21-OH group with fluorine increases glucocorticoid and sodium-retaining activities, whereas substitution with chlorine or bromine abolishes activity. Some compounds that do not contain the Δ^4 -3-ketone system have appreciable activity. It has been suggested that this group makes only a minor contribution to the specificity of action by these drugs or to the steroid–receptor association constant (116).

Based on structure–activity studies, the C and D rings, involving positions 11, 12, 13, 16, 17, 18, 20, and 21, are more important for receptor binding than the A and B rings. As a rule, insertion of bulky substituents on the β -side of the molecule abolishes glycogenic activity, whereas insertion on the α -side does not. It has been suggested that association of these steroids with receptors involves β -surfaces of rings C and D and the 17 β -ketol side chain (116). It is possible, however, that association with the α -surface of rings A, C, and D, as well as with the ketol side chain, is essential for sodium-retaining activity. Many functional groups, such as 17 α -OH, 17 α -CH₃, 16 α -CH₃, 16 β -CH₃, 16 α -CH₃O, and 16 α -OH substituents, abolish or reverse this activity in 11-desoxycorticosterone

and 11-oxygenated steroids. Discussions of exceptions of these generalities are found in the literature (116).

Although some steroids cause sodium retention, many have glucocorticoid and either sodium-retaining or sodium-excreting action. Difficulties in correlating the structures of adrenocorticoids with biologic action are compounded because of differences in assay methods, species variation, and mode of drug administration. For example, whereas liver glycogen and anti-inflammatory assays in the rat correlate well, some drugs show high anti-inflammatory action in the rat but little or no antirheumatic activity in humans. The 9 α -F analog, fludrocortisone acetate, is more active than the 9 α -Cl analog in terms of sodium retention in the dog; the reverse is true in the rat. Although 16 α -methylation and 16 β -methylation enhance glucocorticoid activity, anti-inflammatory action is increased disproportionately to glycogenic action in both series.

In humans, eosinopenic and hyperglycemic potencies are essentially the same. There is a close correlation in efficacy ratios derived from these tests and antirheumatic potency (Table 28.6). Because the eosinopenic–hyperglycemic activity and antirheumatic potency show excellent agreement, it has been suggested that these assays afford advantages in the preliminary estimation of anti-inflammatory potency (27).

Structure–activity studies of glucocorticoids have been carried out mainly in animals and are not necessarily applicable to clinical efficacy in man. Relative activity and

TABLE 28.6 Biologic Potencies of Modified Adrenocorticoids in the Rat and Humans

Adrenocorticoid	Thymus Involution	Potency Relative to Hydrocortisone (Cortisol)			
		Rats		Humans	
		Liver Glycogen Deposition	Eosinopenic Potency	Hyperglycemic Potency	Antirheumatic Potency
Corticosterone	—	0.8	0.06	0.06	<0.1
Prednisone	—	3	—	4	4
Prednisolone	2	3.9	4	4	4
Methylprednisolone	10	11	5	5	5
Triamcinolone	4	47	5	5	5
Paramethasone	—	150	12	12	11
Dexamethasone	56	265	28	28	29
Fludrocortisone acetate	6	9	8	8	10
Fluprednisolone	6	81	9	9	10
Triamcinolone acetonide	33	242	3	3	3
Flurandrenolone	4	—	1	—	2
Fluorometholone	25	115	10	10	—
Fluocinolone	19	112	5	6	9

Data from Ringer I. Steroidal activity in experimental animals and man. In: Dorfman RI, ed. *Methods of Hormone Research*, Vol 3, Part A. New York: Academic Press, 1964.

dose correlations for the clinically useful drugs are found in Table 28.1.

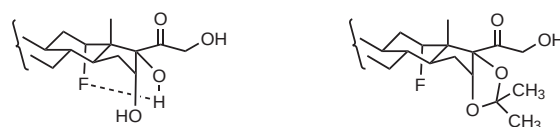
Several other compounds have been studied in animals and used to derive structure–activity relationships. For example, insertion of a double bond between positions 1 and 2 in hydrocortisone increases glucocorticoid activity. The Δ^1 -corticoids have a much longer half-life in the blood than hydrocortisone. Ring A is much more slowly metabolized, but it is oxidatively metabolized at other positions, especially the 6β position and the 17β -ketol (Fig. 28.12). If, however, a double bond is inserted between positions 9 and 11 (no oxygen function at 11), a decrease in glucocorticoid activity is observed. Except for cortisone, which results in an analog with decreased glucocorticoid activity when a double bond is inserted between position 6 and 7, such modification of other glucocorticoids generally produces no change in activity (16).

Insertion of α -CH₃ groups at positions 2 (in 11β -OH analogs), 6, and 16 increases glucocorticoid activity in animals. Again, insertion of a 2α -CH₃ group into the glucocorticoid almost completely prevents reduction of the Δ^4 -3-ketone system in vivo and in vitro. Substitutions at positions 4α , 7α , 9α , 11α , and 21 decrease activity.

Although some analogs, such as $16\alpha,17\alpha$ -isopropylidenedioxy- 6α -methylpregna-1,4-diene-3,20-dione and the 1,2-dihydro derivative, are 11-desoxysteroids and biologically active, the 11β -OH group of hydrocortisone does seem to be involved in the drug–receptor interaction (116). Cortisone, which contains an 11-keto function, is reduced in vivo to hydrocortisone. The drug 2α -methylhydrocortisone exhibits high glucocorticoid activity, probably because of steric hindrance to reduction (i.e., $C=O \rightarrow C-\beta$ -OH) by the methyl group, thus rendering the analog inactive (40,117,118). Insertion of α -OH groups into most other positions (1, 6, 7, 9, 14, and 16) or reduction of the 20-ketone, however, decreases glucocorticoid activity, in part because of increased hydrophilicity.

The 9α -F group increases glucocorticoid activity and nearly prevents metabolic oxidation of the 11β -OH group to a ketone. Redox metabolism of Δ^4 -steroids is mainly restricted to the Δ^4 -3-ketone, 6 and 16 positions, and 17β -ketol side chains, whereas for Δ^1 -steroids, it is only the 6 and 16 positions and 17β -ketol side chains. The 9α -F group increases activity by an inductive effect, which increases the acidic dissociation constant of the 11β -OH group and, thereby, increases the ability of the drug to hydrogen bond to GRs.

A 6α -F group also increases glucocorticoid activity, but it has less effect than the 9α -F function on sodium retention. Insertion of 2α -, 11α - (no OH group at 11), or 21-F groups decreases glucocorticoid activity. Of particular interest is a 12α -F group. When this function is inserted into corticosterone, which has no 17α -OH group, it potentiates activity to the same extent as a 9α -F group. Insertion of a 12α -F group into a $16\alpha,17\alpha$ -dihydroxy steroid, however, renders the compound inactive. A 9α -F group potentiates activity in such analogs.



16 α , 17 α -Dihydroxy steroid 16 α , 17 α -Isopropylidenedioxy steroid

FIGURE 28.16 Hydrogen bonding between 17α -hydroxyl and 12α -fluoro groups.

It has been proposed that hydrogen bonding between the 12α -F and 17α -OH groups renders the analog inactive (Fig. 28.16). Conversion to the $16\alpha,17\alpha$ -isopropylidenedioxy (acetonide) derivative, which cannot hydrogen bond, restores biologic activity (119).

The mineralocorticoid activity of adrenocorticoids is another action of major significance. Many toxic side effects, making it necessary to withdraw steroid therapy in rheumatoid patients, are a result of this action. Highly active, naturally occurring mineralocorticoids have no OH function in positions 11 and 17. In fact, OH groups in any position reduce the sodium-retaining activity of the adrenocorticoid.

9α -F, 9α -Cl, and 9α -Br substitution usually causes increased retention of urinary sodium with an order of activity in which $F > Cl > Br$, but species differences do exist. For these reasons, such compounds are not used internally in the treatment of diseases such as rheumatoid arthritis. Insertion of a 16α -OH group into the molecule affects the sodium retention activity so markedly that it not only negates the effect of the 9α -F atom but also causes sodium excretion.

A double bond between positions 1 and 2 (Δ^1 -corticoids) also reduces the sodium retention activity of the parent drug. However, this functional group contributes to the parent drug approximately one-fifth the sodium-excreting tendency of a 16α -OH group (120).

The 12α -F, 2α -CH₃, and 9α -Cl substitutions contribute equally to sodium retention. A 21-OH group, found in all these drugs, contributes to this action to the same degree. Because 21-OH groups also contribute to glucocorticoid activity, it is easy to understand why it is difficult to develop compounds with only one major action.

A 2α -CH₃ group is approximately threefold, and a 21-F substituent twofold, as effective as unsaturation between positions 1 and 2 in reducing sodium retention. Other substituents reported to inhibit sodium retention include 16α -CH₃, 16β -CH₃, 16α -CH₃O, and 6α -Cl functions. A 17α -OH group, which is present in naturally occurring and semisynthetic analogs, reduces sodium retention to about the same extent as does unsaturation between positions 1 and 2.

Conversion of the 17α -hydroxy to either a 17α -ester or an ether, as with $16\alpha,17\alpha$ -isopropylidenedioxy (acetonide), greatly enhances the anti-inflammatory potency and GR affinity (Table 28.5). As evidenced with BDP, however, esterifying the 21-hydroxy group reduces activity and receptor affinity. On the other hand, 21-halogens

or 21-halomethylene groups greatly increase topical anti-inflammatory activity with no change or a decrease in mineralocorticoid activity. Perhaps a hydrogen bonding group at position 21 enhances or retains mineralocorticoid receptor affinity.

ADRENOCORTICOID ANTAGONISTS

Antagonists of adrenocorticoids include agents that compete for binding to steroid receptors (antiglucocorticoids or antimineralocorticoids) and inhibitors of adrenosteroid biosynthesis. The action of adrenal steroids can be blocked by antagonists that compete with the endogenous steroids for binding sites on their respective cytosolic receptor proteins. The antagonist–receptor complexes are unable to stimulate the production of new mRNA and protein in the target tissues and, thus, are unable to elicit the biologic responses of the hormone agonist. Spironolactone, eplerenone (Fig. 28.17), and related analogs bind to the mineralocorticoid receptor in the kidney and result in the diuretic response of increased Na^+ excretion and K^+ retention. The 3-keto-4-ene A ring is essential for this antagonistic activity, and the opening of the lactone ring dramatically reduces activity. The 7 α -substituent increases both intrinsic activity and oral activity (121,122). Progesterone has also shown antimineralocorticoid activity at 10^{-4} molar concentrations.

Receptor antagonists of glucocorticoids have been described that are derivatives of 19-nortestosterone (123). Mifepristone, also referred to as RU-486 (Fig. 28.17), was originally developed as an antiprogesterin and also exhibits very effective antagonism of glucocorticoids.

Several inhibitors of adrenocorticoid biosynthesis have been described, with the majority of nonsteroidal agents inhibiting one or more of the cytochrome P450 enzyme complexes involved in adrenosteroid biosynthesis (Fig. 28.18). Metyrapone reduces hydrocortisone biosynthesis by primarily inhibiting mitochondrial

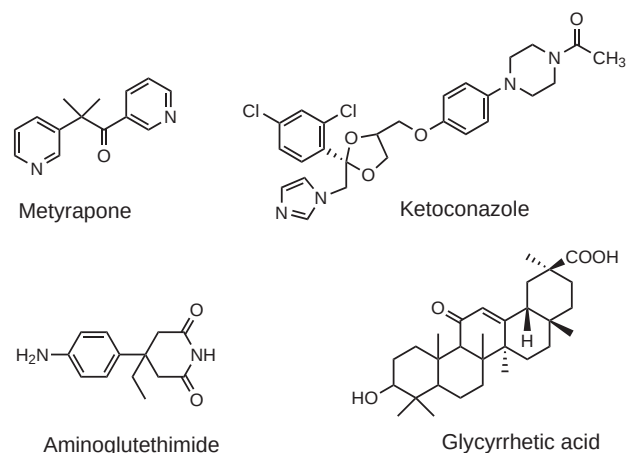


FIGURE 28.18 Inhibitors of adrenocorticoid biosynthesis.

11 β -hydroxylase (124,125). It also inhibits, to a lesser degree, 18-hydroxylase and side-chain cleavage. This agent is used to test pituitary–adrenal function and the ability of the pituitary to secrete ACTH (125). Aminoglutethimide inhibits side-chain cleavage (125) and has been used as a medical adrenalectomy. Several azole antifungal drugs inhibit adrenocorticoid biosynthesis. Ketoconazole is one example of a potent inhibitor of fungal sterol biosynthesis at low concentrations; however, at higher doses, ketoconazole inhibits CYP3A4 in adrenosteroid biosynthesis (126). Trilostane is a steroidal inhibitor of 3 β -hydroxysteroid dehydrogenase (127) and has been used in the treatment of Cushing syndrome.

Inhibition of the two 11 β -hydroxysteroid dehydrogenase isozymes has been described. Excessive ingestion of licorice, an extract of the roots of *Glycyrrhiza glabra*, produces undesirable mineralocorticoid-like intoxication (hypertension, excessive salt retention, and hypokalemia). The active components of licorice, glycyrrhetic acid (Fig. 28.17), and carbenoxolone inhibit both 11 β -HSD isozymes (128), with the undesirable side effects resulting from inhibition of 11 β -HSD2 in the kidney. Bile acids, such as chenodeoxycholic acid and lithocholic acid, have recently shown selective inhibition of 11 β -HSD1 (129,130), and nonsteroidal arylsulfoamidothiazoles also recently have been reported as 11 β -HSD1 inhibitors (131). Such enzyme inhibitors may be useful in the treatment of metabolic syndromes to reduce elevated glucose concentrations and as possible antidiabetic agents.

MECHANISMS OF ADRENOCORTICOID ACTION

Molecular Interaction

Glucocorticoid action is mediated through the GR, which is found primarily in the cytosol of the cell when not bound to glucocorticoids. The GR is stabilized in the cytosol by complexation with phosphorylated proteins, including a 90-kilodalton protein referred to as a heat shock protein 90 (132). The steroid molecule binds to the GR, resulting

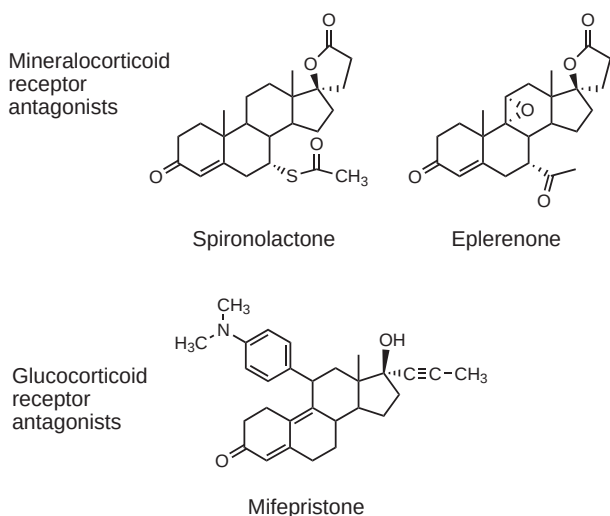


FIGURE 28.17 Adrenocorticoids receptor antagonists.

in a conformational change of the receptor to dissociate the other proteins and initiate translocation of the steroid–receptor complex into the nucleus. The steroid–nuclear GR complex interacts with particular HRE regions of the cellular DNA, referred to as glucocorticoid-responsive elements, and initiates transcription of the DNA sequence to produce mRNA. Finally, the elevated levels of mRNA lead to increased protein synthesis in the endoplasmic reticulum. These proteins then mediate glucocorticoid effects on carbohydrate, lipid, and protein metabolism. An alternative isoform of the GR has been identified. This isoform of the receptor does not bind known glucocorticoids, and its function remains to be determined (133,134). Some of the specific proteins induced by glucocorticoids have been identified and are discussed later. Mineralocorticoid effects are observed in several tissues, and specific mineralocorticoid receptors have been characterized that mediate mineralocorticoid functions (135). Steroid–receptor complexes have also demonstrated negative regulation of gene expression by interacting at negative response elements or by transrepression via direct or indirect protein–protein interactions with other known transcriptional proteins such as AP-1 and NF κ B (8,9). Lastly, nongenomic mechanisms involving cytosolic and/or membrane-bound steroid receptors have been identified to explain rapid onset actions of steroid hormones (8).

Physiologic Effects

Glucocorticoids

Corticosteroids influence all tissues of the body and produce numerous and varying effects in cells (16). These steroids regulate carbohydrate, lipid, and protein biosynthesis and metabolism (glucocorticoid effects), and they influence water and electrolyte balance (mineralocorticoid effects). Hydrocortisone is the most potent glucocorticoid secreted by the adrenal gland, and aldosterone is the most potent endogenous mineralocorticoid. Both naturally occurring glucocorticoids and related, semisynthetic analogs can be evaluated in terms of their ability to sustain life, to stimulate an increase in blood glucose concentrations and a deposition of liver glycogen, to decrease circulating eosinophils (136), and to cause thymus involution in adrenalectomized animals (137,138). In addition, corticosteroids can affect immune system functions, inflammatory responses, and cell growth.

The primary physiologic function of glucocorticoids is to maintain blood glucose levels and, thus, ensure glucose-dependent processes critical to life, particularly brain functions. Hydrocortisone and related steroids accomplish this by stimulating the formation of glucose, by diminishing glucose use by peripheral tissues, and by promoting glycogen synthesis in the liver to increase carbohydrate stores for later release of glucose. For glucose formation, glucocorticoids mobilize amino acids and promote amino acid metabolism and gluconeogenesis. These steroids, acting via the GR mechanism, induce the production of a variety of enzymes important for glucose formation. The

synthesis of tyrosine aminotransferase increases within 30 minutes of glucocorticoid exposure (139–141). This enzyme promotes the transfer of amino groups from tyrosine to α -ketoglutarate to form glutamate and hydroxyphenylpyruvate. Another amino acid–metabolizing enzyme induced rapidly by glucocorticoids is tryptophan oxidase (142). This enzyme oxidizes tryptophan to formylkynurenine, which is subsequently converted to alanine. Alanine transaminase is also induced by glucocorticoids (143). Alanine and, to a lesser extent, glutamate are important for gluconeogenesis in the liver (144).

Several other enzymes important in gluconeogenesis and glycogen formation are elevated for several hours following glucocorticoid administration; these include glycogen synthetase, pyruvate kinase, phosphoenolpyruvate carboxykinase, and glucose-6-phosphate kinase (16,145,146). The delayed increases in these enzymes suggest that their biosyntheses are not regulated directly by glucocorticoids. In peripheral tissues, glucocorticoid-induced inhibition of phosphofructokinase is observed (146). This enzyme catalyzes the formation of D-fructose-1,6-diphosphate from D-fructose-6-phosphate during glycolysis. Inhibition of this enzyme decreases glucose utilization by peripheral tissues and results in maintaining blood glucose levels. Reviews of the multiple effects of glucocorticoids on carbohydrate metabolism have been published (16,146).

Additional effects of glucocorticoids in the body are preventing or minimizing inflammatory reactions and suppressing immune responses. These steroids interfere with both early events in inflammation (e.g., release of mediators, edema, and cellular infiltration) and later stages (e.g., capillary infiltration and collagen formation). Only a few of the mechanisms involved in glucocorticoid suppression of inflammation are known. Hydrocortisone will induce the production of lipocortin and related proteins by increasing gene expression through the GR mechanism (147,148). Lipocortin inhibits the activity of phospholipase A₂, which liberates arachidonic acid and leads to the biosynthesis of eicosanoids (e.g., prostaglandins and leukotrienes) (149). Lipocortin also mediates the decreased production and release of platelet-activating factor (150), and glucocorticoids can suppress the expression of interleukin (IL)-1, tumor necrosis factor, and inducible nitric oxide synthase (151–153). These eicosanoids and peptide factors are important as mediators in the inflammatory response. Some of these factors also have important roles in cellular infiltration and capillary permeability in the inflamed region. Suppression of the immune responses is mediated by inhibition of the synthesis and release of important mediators as well. In macrophages, glucocorticoids inhibit IL-1 synthesis and, thus, interfere with proliferation of B lymphocytes, which are important for antibody production (154). Additionally, IL-1 is important for activation of resting T lymphocytes, which are important for cell-mediated immunity. The activated T cells produce IL-2, the biosynthesis of which is also reduced by glucocorticoids (154).

Mineralocorticoids

The primary physiologic function of mineralocorticoids is to maintain electrolyte balances in the body by enhancing Na^+ reabsorption and increasing K^+ and H^+ secretion in the kidney. Similar effects on cation transport are observed in a variety of secretory tissues, including the salivary glands, sweat glands, and mucosal tissues of the GI tract and the bladder. Aldosterone is the most potent endogenous mineralocorticoid. Deoxycorticosterone is approximately 20-fold less potent than aldosterone. Hydrocortisone exhibits weak mineralocorticoid activity *in vivo* because of rapid metabolism of hydrocortisone by 11β -hydroxysteroid dehydrogenase. The mechanism of action of aldosterone involves binding of the steroid to the mineralocorticoid receptor and initiation of gene transcription, mRNA biosynthesis, and protein production. A protein referred to as aldosterone-induced protein is produced through this mechanism and is thought to aid in Na^+ retention. One possible mode of action of aldosterone-induced protein is to act as a permease to increase the permeability of the cell membrane to Na^+ (155). This results in an accelerated rate of Na^+ influx and elevated activity of Na^+ , K^+ -ATPase to pump Na^+ into extracellular space (156).

PHARMACOLOGIC EFFECTS AND CLINICAL APPLICATIONS

In addition to their natural hormonal actions, the adrenocorticoids have many clinical uses. Glucocorticoids and mineralocorticoids can be used for the treatment of adrenal insufficiency (hypoadrenalism), which results from failure of the adrenal glands to synthesize adequate amounts of the hormones. Adrenocorticoids are also used to maintain patients who have had partial or complete removal of their adrenal glands or adeno-hypophy-sis (adrenalectomy and hypophysectomy, respectively). Glucocorticoids can cross the placenta and can be distributed into milk.

Two major uses of glucocorticoids are in the treatment of rheumatoid diseases and allergic manifestations. Their use in the treatment of severe asthma is well documented, as is the utility of glucocorticoids in sepsis and acute respiratory distress syndrome (157,158). They are effective in the treatment of rheumatoid arthritis, acute rheumatic fever, bursitis, spontaneous hypoglycemia in children, gout, rheumatoid carditis, sprue, allergy (including contact dermatitis), and other conditions. The treatment of chronic rheumatic diseases and allergic conditions with glucocorticoids is symptomatic and continuous. Symptoms return after withdrawal of the drug.

In addition, these drugs are moderately effective in the treatment of ulcerative colitis, dermatomyositis, periarthritis nodosa, idiopathic pulmonary fibrosis, idiopathic thrombocytopenic purpura, regional ileitis, acquired hemolytic anemia, nephrosis, cirrhotic ascites, neurodermatitis, and temporal arteritis. The newer analogs with medium to high potency rankings (Table 28.4), such as

diflorasone diacetate, desoximetasone, flurandrenolide, and fluocinonide (Fig. 28.12), are effective topically in the treatment of psoriasis. Glucocorticoids can be combined with antibiotics to treat pneumonia, peritonitis, typhoid fever, and meningococemia.

When dosages with equivalent antirheumatic potency are given to patients not treated with steroids, the Δ^1 -corticoids (prednisone and prednisolone) promote the same pattern of initial improvement as hydrocortisone. Statistical results of improvement during the first few months of therapy have been similar with prednisone, prednisolone, and hydrocortisone. The results of longer-term therapy have been significantly better with the modified compounds.

Satisfactory rheumatic control, lost after prolonged hydrocortisone therapy, may be regained in an appreciable number of patients by changing to prednisone, prednisolone, or other modified drugs. Of patients whose conditions deteriorate below adequate levels during hydrocortisone administration, nearly half reach their previous level of improvement after Δ^1 -corticoids (in doses slightly larger in terms of antirheumatic strength) are used. With further prolongation of steroid therapy, improvement again wanes in some patients, but in other patients, such management is successful for longer than 2 years. In some instances, the improvement is attributed to increased effectiveness of the drug because of correction of salt and water retention; in other instances, there is no adequate explanation.

When these drugs are administered in doses that have similar antirheumatic strengths, the general incidence of adverse reactions with prednisone and prednisolone is about the same as that with hydrocortisone. The compounds differ, however, in their tendencies to induce individual side effects. The incidence and degree of salt and water retention and blood pressure elevation are less with the Δ^1 -corticoids. Conversely, these analogs are more likely to promote digestive complaints, peptic ulcer, vasomotor symptoms, and cutaneous ecchymosis.

Although these analogs have unwanted side effects, most clinical investigators prefer the Δ^1 -corticoids to hydrocortisone for rheumatoid patients who require steroid therapy. The reasons are that these drugs have less tendency to cause salt and water retention and potassium loss and that they restore improvement in a significant percentage of patients whose therapeutic control has been lost during hydrocortisone therapy.

It seems desirable to administer prednisone and prednisolone in conjunction with nonabsorbable antacids. This affords improvement of long-term therapy. It appears that the therapeutic indices of these two analogs, especially when used in conjunction with nonabsorbable antacids, are higher than those for the naturally occurring glucocorticoids.

Most important, glucocorticoids should not be withdrawn abruptly in cases of acute infections or severe stress, such as surgery or trauma. Myasthenia gravis, peptic ulcer, diabetes mellitus, hyperthyroidism, hypertension,

psychological disturbances, pregnancy (first trimester), and infections may be aggravated by glucocorticoid administration. Hormone therapy is contraindicated in these conditions and should be used only with the utmost precaution.

Semisynthetic analogs exhibiting high mineralocorticoid activity are not employed in the treatment of rheumatic disorders because of toxic side effects resulting from a disturbance of electrolyte and water balance. Some newer synthetic steroids (Table 28.2) are relatively free of sodium-retaining activity. They can show other toxic manifestations, however, and eventually need to be withdrawn.

Glucocorticoids sometimes are used in the treatment of scleroderma, discoid lupus, acute nephritis, osteoarthritis, acute hepatitis, hepatic coma, Hodgkin disease, multiple myeloma, lymphoid tumors, acute leukemia, metastatic carcinoma of the breast, and chronic lymphatic leukemia. Glucocorticoids may be more or less effective in these diseases depending on the clinical condition.

Some modified compounds have been recommended for use when other analogs are no longer effective or when it is desirable to promote increased appetite and weight gain. Triamcinolone can be used advantageously when salt and water retention (from other glucocorticoids, hypertension, or cardiac compensation) or excessive appetite and weight gain are problems in management.

One factor must not be overlooked when applying potent anti-inflammatory agents with high mineralocorticoid activity to the skin. Consideration must be given to percutaneous absorption. Sodium retention and edema occur in patients with dermatitis who apply as much as 75 mg of fludrocortisone acetate (i.e., 30 mL of a 0.25% lotion) to the skin in 24 hours. The relative rate of percutaneous absorption, administered as a cream in rats, was triamcinolone acetate $\geq$ hydrocortisone $>$ dexamethasone, but dexamethasone was deposited in skin longer than the other two drugs. Hydrocortisone disappeared most rapidly (159).

Topical Applications

Topical dermatologic products with a low potency ranking have a modest anti-inflammatory effect and are safest for chronic application (Table 28.4). These products are also the safest products for use on the face with occlusion and in infants and young children. Those products with a medium potency ranking are used in moderate inflammatory dermatoses, such as chronic hand eczema and atopic eczema, and may be used on the face and intertriginous areas (areas where skin comes into contact with itself, such as the armpits, in the groin, and beneath the breasts, which are more prone to infections and rashes because they are hot and often sweaty) for a limited duration. High-potency preparations are used in more severe inflammatory dermatoses, such as severe eczema and psoriasis. They can be used for a limited duration and for

longer periods in areas with thickened skin because of chronic conditions. High-potency preparations may also be used on the face and intertriginous areas, but only for a short treatment duration. Very high-potency products are used primarily as an alternative to systemic corticosteroid therapy when local areas are involved. Examples of conditions for which very high-potency products frequently are used include thick, chronic lesions caused by psoriasis, lichen simplex chronicus, and discoid lupus erythematosus. They may be used for only a short duration of therapy and on small surface areas. Occlusive dressings should not be used with these products. It has been suggested that patients using a lotion or ointment containing these drugs be instructed to apply them sparingly and to spread them lightly over the affected areas. The extent and frequency of applications should be carefully considered. A lotion vehicle is more effective when treating a dermatitis, but a greater degree of percutaneous absorption occurs than when ointments are used.

Intranasal and Inhaled Applications

The pulmonary and nasal bioavailabilities are important determinants for the potential of an inhaled or nasally applied corticosteroid to cause systemic effects, because the lung and nasal tissue provide an enormous surface area from which drug absorption can occur into the systemic circulation. The main areas of concern with regard to systemic effects include HPA axis suppression, change in bone mineral density and growth retardation in children, cataracts, and glaucoma. The degree of systemic side effects is dose-dependent, related to the half-life of the drug, frequency administration, time of day when administered, and route of administration; in other words, the higher the plasma corticosteroid concentration and longer the half-life, the greater will be the systemic side effects (Table 28.5) (27). The amount of an inhaled or nasal corticosteroid reaching the systemic circulation is the sum of the drug concentration available following absorption from the lungs/nasal mucosa, from the GI tract, and the extent of plasma protein binding. The fraction deposited in the mouth will be swallowed, and the systemic availability will be determined by its absorption from the GI tract and the degree of first-pass metabolism and protein binding.

Delivery devices can produce clinically significant differences in activity by altering the dose deposited in the lung (10% to 25%) and, for orally absorbed drugs, the amount deposited in the oropharynx and swallowed (75% to 90%). Clinical studies have shown the following relative potency differences: ciclesonide $>$ mometasone furoate $>$ fluticasone propionate $>$ budesonide = BDP $>$ triamcinolone acetonide = flunisolide. Potency differences can be overcome by giving larger doses of the less potent drug, which increases risks from systemic effects. Adrenal suppression may be associated with high doses of inhaled corticosteroids (>1.5 mg/day, or >0.75 mg/day for fluticasone propionate), although there is a considerable degree of interindividual susceptibility.

All currently used inhaled corticosteroids are rapidly cleared from the body but show varying levels of oral bioavailability, with fluticasone propionate having the lowest (Table 28.5). Following inhalation, there is also considerable variability in the rate of absorption from the lung, and pulmonary residence times are greatest for fluticasone propionate and triamcinolone acetonide and shortest for budesonide and flunisolide. Adrenal suppression has not been observed when intranasal fluticasone propionate was administered in dosages of 200 to 4,000 mcg daily for up to 12 months.

Adverse Effects

Although short-term administration of corticosteroids is unlikely to produce harmful effects, these drugs, when used for longer than brief periods, can produce a variety of devastating effects, including glucocorticoid-induced adrenocortical insufficiency, glucocorticoid-induced osteoporosis, and generalized protein depletion (16,27). The duration of anti-inflammatory activity of glucocorticoids approximately equals the duration of HPA axis suppression. The durations of HPA axis suppression after a single oral dose of glucocorticoids in one study are shown in Table 28.7. When given for prolonged periods, glucocorticoids suppress the HPA axis, thereby decreasing secretion of endogenous corticosteroids and adrenal atrophy. Glucocorticoids inhibit ACTH production by the adenohypophysis, and in turn, this reduces endogenous glucocorticoid production. With time, atrophy of the adrenal glands takes place. The degree and duration of adrenocortical insufficiency produced by the synthetic glucocorticoids is highly variable among patients and depends on the dose, frequency, and time of administration, and duration of glucocorticoid therapy. This effect can be minimized by use of alternate-day therapy.

Patients who develop drug-induced adrenocortical insufficiency may require higher corticosteroid dosage when they are subjected to stress (e.g., infection, surgery,

or trauma). In addition, acute adrenal insufficiency (even death) can occur if the drugs are withdrawn abruptly or if patients are transferred from systemic glucocorticoid therapy to oral inhalation therapy. Therefore, the drugs should be withdrawn very gradually following long-term therapy with pharmacologic dosages. Adrenal suppression can persist up to 12 months in patients who receive large dosages for prolonged periods. Until recovery occurs, patients may show signs and symptoms of adrenal insufficiency when they are subjected to stress, and replacement therapy may be required. Because mineralocorticoid secretion can be impaired, sodium chloride or a mineralocorticoid also should be administered.

Although side effects and toxicities vary with the drug and, sometimes, with the patient, facial mooning, flushing, sweating, acne, thinning of the scalp hair, abdominal distention, and weight gain are observed with most glucocorticoids. Protein depletion (with osteoporosis and spontaneous fractures), myopathy (with weakness of muscles of the thighs, pelvis, and lower back), and aseptic necrosis of the hip and humerus are other side effects. These drugs can cause psychological disturbances, headache, vertigo, and peptic ulcer, and they can suppress growth in children.

Patients with well-controlled diabetes must be closely monitored and their insulin dosage increased if glycosuria or hyperglycemia ensues either during or following glucocorticoid administration. Patients should also be watched for signs of adrenocortical insufficiency after discontinuation of glucocorticoid therapy. Individuals with a history of tuberculosis should receive prophylactic doses of antituberculosis drugs.

Osteoporosis is one of the most serious adverse effects of long-term glucocorticoid therapy. Moderate- to high-dose glucocorticoid therapy is associated with loss of bone and an increased risk of fracture that is most rapid during the initial 6 months of therapy. These adverse effects of glucocorticoids appear to be both dose- and duration-dependent, with oral prednisone dosages of 7.5 mg or more daily for 6 months or longer often resulting in clinically important bone loss and increased fracture risk. Bone loss has even been associated with oral inhalation of glucocorticoids and is of great concern in children. Most patients receiving long-term glucocorticoid therapy will develop some degree of bone loss, and more than 25% will develop osteoporotic fractures. Vertebral fractures have been reported in 11% of patients with asthma who are receiving systemic glucocorticoids for at least 1 year, and glucocorticoid-treated patients with rheumatoid arthritis are at increased risk of fractures of the hip, rib, spine, leg, ankle, and foot. Muscle wasting or weakness and atrophy of the protein matrix of the bone resulting in osteoporosis are manifestations of protein catabolism, which can occur during prolonged therapy with glucocorticoids. These adverse effects can be especially serious in debilitated patients, in geriatric populations, and in postmenopausal women who are especially prone to osteoporosis.

TABLE 28.7 Effect of an Oral Single Dose on the Duration of HPA Suppression

Adrenocorticoid	Duration of Suppression (days)
Hydrocortisone (250 mg)	1.25–1.5
Cortisone (250 mg)	1.25–1.5
Methylprednisolone (40 mg)	1.25–1.5
Prednisone (50 mg)	1.25–1.5
Prednisolone (50 mg)	1.25–1.5
Triamcinolone (40 mg)	2.25
Dexamethasone (5 mg)	2.75
Betamethasone (6 mg)	3.25

To minimize the risk of glucocorticoid-induced bone loss (osteoporosis) and in those with low mineral bone density, the smallest possible effective dosage and duration should be used. Topical and inhaled preparations should be used whenever possible. The immunosuppressive effects of glucocorticoids increase the susceptibility to and mask the symptoms of infections and can result in activation of latent infection or exacerbation of

intercurrent infections. The most common adverse effect of oral inhalation therapy with glucocorticoids is fungal infections of the mouth, pharynx, and occasionally, the larynx. The mineralocorticoid effects are less frequent with synthetic glucocorticoids (except fludrocortisone) than with hydrocortisone but may occur, especially when synthetic glucocorticoids are given in high dosage for prolonged periods.

SCENARIO: OUTCOME AND ANALYSIS

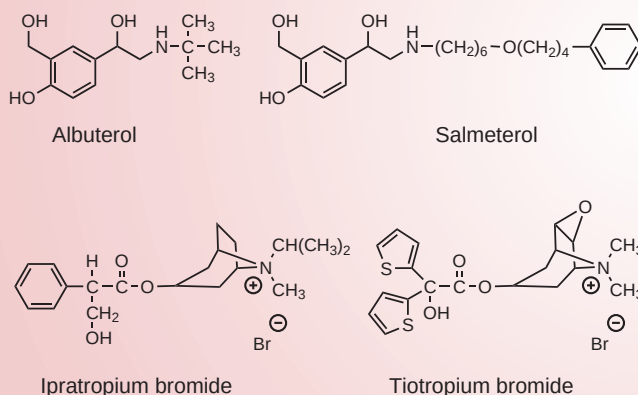
Outcome

Jeffrey T. Sherer, PharmD, MPH, BCPS, CGP

LT is started on intravenous antibiotics and albuterol plus ipratropium administered through a nebulizer. She is also started on intravenous methylprednisolone 25 mg every 12 hours. After 2 days, her symptoms improve dramatically, and she is eating and drinking well. The pharmacist recommends changing her intravenous antibiotic to an oral equivalent and that her intravenous methylprednisolone be switched to oral prednisone 40 mg daily. Her symptoms continue to improve, and she is discharged 2 days later with a short course of oral prednisone plus her usual inhaled medications.

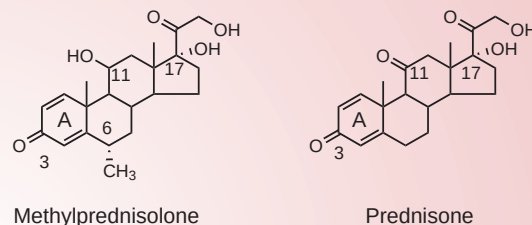
Chemical Analysis

Victoria Roche and S. William Zito



Salmeterol and albuterol are both β_2 -selective adrenergic agonists that, in therapeutic doses, dilate bronchi without unwanted stimulation of the heart (a β_1 -receptor-mediated response). Both structures are phenethylamines with a β_2 -directing salicyl alcohol aromatic ring and a bulky β_2 -directing substituent on the tertiary amino nitrogen. Both contain the carbinol OH group essential to high affinity β -adrenoceptor binding. β_2 -Adrenoceptor agonists are beneficial in the treatment of chronic obstructive pulmonary disease (COPD) due to their ability to relax bronchial smooth muscle and decrease airway resistance.

Tiotropium and ipratropium are quaternary amine-containing antimuscarinic agents based on the structure of naturally occurring Belladonna alkaloids. They are useful in the treatment of COPD due to their antisecretory actions and ability to inhibit acetylcholine-mediated bronchoconstriction. Tiotropium's slower dissociation from muscarinic receptors leads to a longer duration of action compared to ipratropium, but it has also a slower onset of action.



Methylprednisolone and prednisone are gluco-selective adrenocorticoids that act to decrease airway inflammation in patients with asthma and COPD. The 17β -ketol, 17α -OH, and Δ^4 -3-one A ring are essential to corticoid action of all kinds. Both methylprednisolone and prednisone have the Δ^1 -double bond that flattens ring A and provides higher affinity for glucocorticoid receptors compared to mineralocorticoids. The addition of this conjugated double bond also makes the A ring more resistant to inactivating reduction. Methylprednisolone contains a 6α -CH₃ group, which essentially abolishes mineralo side effects (e.g., fluid retention, electrolyte imbalance) in therapeutic doses and provides some steric hindrance to the reduction of the already-resistant A ring. The hydroxyl moiety at 11β (found on methylprednisolone) forms an important affinity-enhancing H-bond with the glucocorticoid receptor, but the 11 -keto (found on prednisone) is rapidly converted to the 11β -OH via the hepatic enzyme 11β -hydroxysteroid dehydrogenase. The equilibrium established between these two forms ensures an equivalent potency of compounds otherwise identical in structure. Although methylprednisolone is available in an oral dosage form, the switch to prednisone (a glucocorticoid with a slightly decreased potency) is a chemically viable alternative for maintenance therapy.

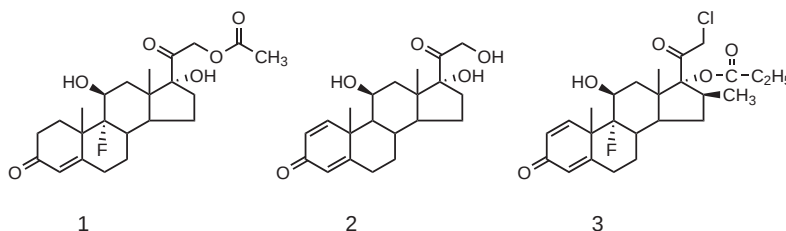
CASE STUDY

Victoria Roche and S. William Zito

You are next door neighbors with LK, a friendly guy who owns a relatively new but increasingly successful lawn service business in your community. LK recently landed a contract with an exclusive real estate company to landscape some model homes in a wooded area being developed on the south side of town. Against his better judgment, LK hired DK, his 22-year-old nephew, for the summer and put him to work on the job. After clearing a large section of one wooded lot of weeds and vines as LK had directed him to do, DK started a controlled burn of the organic debris while his uncle was off inspecting another part of the development property. Fortunately, his uncle returned in time to halt the burning shortly after it began. Unfortunately there was poison ivy in the cleared vegetation and DK was directly exposed

to the urushiol from the plant itself and, briefly, to the smoke from the burning leaves. He is experiencing a significant reaction to the allergen and is thoroughly miserable. Because of the potential for urushiol-induced lung irritation, his uncle knew to seek medical attention, and they are now in the emergency room of the hospital where you practice. While the attending is examining DK, you talk with LK about an anticipated therapeutic plan that will include an oral corticosteroid to provide much-needed anti-inflammatory/antiallergic relief.

Back in the pharmacy, you challenge your ambulatory care resident to employ his understanding of corticosteroid SAR by selecting the most likely drug candidate from three potential adrenocorticoid structures you draw on the whiteboard.



You ask the resident to:

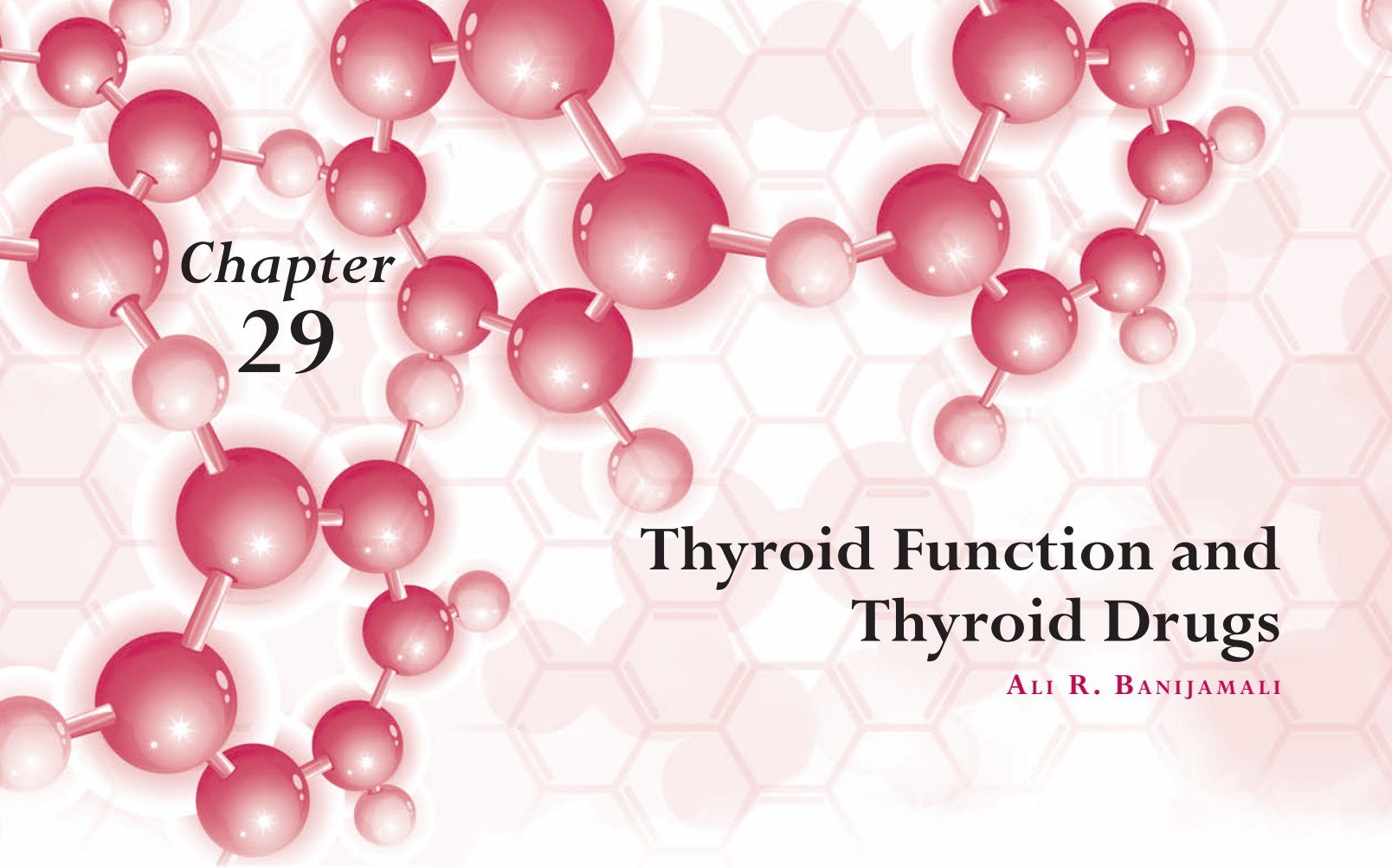
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 29

Thyroid Function and Thyroid Drugs

ALI R. BANIJAMALI

Drugs Covered in This Chapter

DRUGS FOR TREATMENT OF HYPOTHYROIDISM

- Levothyroxine
- Liothyronine
- Liotrix
- Thyroid gland products

DRUGS FOR TREATMENT OF HYPERTHYROIDISM

- Iodide
- Perchlorate
- Radioiodine

- 1-Methyl-2-mercaptoimidazole
- Propylthiouracil

Abbreviations

BMR, basal metabolic rate

cAMP, cyclic adenosine
monophosphate

DIT, diiodo-L-tyrosine

dT₄, dextrothyroxine

FAD, flavin adenine dinucleotide

GRTH, generalized resistance to thyroid
hormone

HSA, human serum albumin

^{99m}TcO₄⁻, radioactive pertechnetate

MIT, monoiodo-L-tyrosine

MMI, 1-methyl-2-mercaptoimidazole

NADPH, reduced form of nicotinamide
adenine dinucleotide phosphate

NIS, sodium/iodide symporter

PTU, propylthiouracil

rT₃, 3,3',5'-triiodo-L-thyronine
(reverse T₃)

SAR, structure–activity relationship

T₂, 3,3'-diiodo-L-thyronine

T₃, Levo or L-triiodothyronine,
Liothyronine

T₄, Levo or L-thyroxine

TBG, thyroxine-binding globulin

Tg, thyroglobulin

TPO, thyroperoxidase

TRH, thyroid-releasing hormone

TSH, thyrotropin
(thyroid-stimulating hormone)

TTR, transthyretin

USP, U.S. Pharmacopeia

SCENARIO

Autumn Stewart, PharmD

PA, a 43-year-old white man, presented to the emergency department with complaints of fever, sleeplessness, and a “racing heart.” PA presented to his physician 1 month prior with symptoms including anxiety, weight loss, and tachycardia. At that time, laboratory and radioactive iodine uptake tests confirmed a diagnosis of Graves’ disease. The patient was initiated on methimazole 5 mg TID and instructed to schedule a follow-up appointment in 1 month with laboratory tests. Upon

admission in the emergency department, PA’s heart rate was 160 beats per minute and irregular; his TSH levels were undetectable and thyroid-stimulating antibodies remained significantly elevated. PA’s white blood cell counts were within normal limits.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

The thyroid gland is a highly vascular, flat structure located at the upper portion of the trachea, just below the larynx. It is composed of two lateral lobes joined by an isthmus across the ventral surface of the trachea. The gland is the source of two fundamentally different types of hormones, thyroxine (T_4) and triiodothyronine (T_3). Both hormones are vital for normal growth and development and control essential functions, such as energy metabolism and protein synthesis.

The word thyroid, meaning shield-shaped, was introduced by Wharton in his description of the gland (1). Like many before him, he attributed a solely cosmetic function to it because of the more frequent presence of enlarged glands in women, giving the throat region a more beautiful roundness. Later, however, it was observed that some characteristic symptoms for diseases always were accompanied by an obvious change in the size of the thyroid. This change was correctly interpreted as evidence that this structure plays a major role in normal body function.

An important step in the understanding of thyroid function was taken by Baumann (2), who discovered that the thyroid gland was the only organ in mammals that had the capability to incorporate iodine into organic substances. That discovery was important in research concerning the phylogeny of the thyroid.

Major clues to the physiologic roles of thyroid hormones were provided when normal and abnormal thyroid functions were related to oxygen uptake (3) and when thyroid hormones were found to induce metamorphosis in tadpoles (4). The first discovery led to investigations regarding the role of thyroid hormone in metabolism and calorogenesis, and the second inspired research concerning specific receptors as points of initiation of thyroid hormone expression. A patient lacking thyroid hormones may be treated with synthetic hormones or natural preparations. Better agents to treat hyperthyroidism are still being sought. Presently available drugs, other compounds affecting thyroid function, and current approaches in the search for new drugs are presented in this chapter within the context of thyroid biochemistry and physiology.

NORMAL BIOCHEMISTRY AND PHYSIOLOGY

Thyroid Follicular Cells

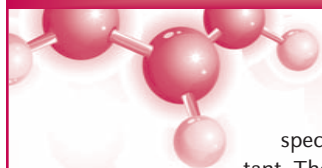
All vertebrates have a thyroid gland consisting of functional units, the follicles. The morphologic and functional characteristics of the follicles are essentially similar in all vertebrate groups.

The follicle is a spherical, cyst-like structure approximately 300 μm in diameter, and it consists of a luminal cavity surrounded by a one-cell-deep layer of cells called follicular or acinar cells. The center of the follicles is filled with a gelatinous colloid, the main component of which is a glycoprotein called thyroglobulin. The follicular cells contain an extensive network of rough endoplasmic reticulum, a well-developed Golgi apparatus, and lysosomes of various sizes (5). Thyroglobulin is synthesized in the rough endoplasmic reticulum of the follicular cells and transported by way of the Golgi complex to the apical membrane and then secreted into the follicle lumen.

The follicular cell contains two major assembly lines operating in opposite directions (6). One line moves in an apical direction (towards the lumen of the cell) and produces thyroglobulin that is delivered to the follicle lumen; the other line begins at the apical cell surface (cell surface facing the lumen of the cell) with endocytosis of thyroglobulin and ends by delivering hormones at the basolateral cell membrane. Therefore, the follicular cell seems to fulfill the functions of secretory and absorptive cells simultaneously. In addition to the functions associated with these two lines, the follicle has the specific ability to metabolize iodine, comprising the accumulation of iodide, iodination of tyrosyl residues in thyroglobulin, and coupling of the iodinated tyrosyls to form thyroid hormones.

Parafollicular cells, also called light cells or C cells, are located individually or in clusters between follicular cells but do not border on the colloid. These cells produce thyrocalcitonin, a peptide hormone involved in calcium homeostasis (see also Chapter 30). The extrafollicular space of the gland is occupied by blood vessels, capillaries, lymphatic vessels, and connective tissue.

CLINICAL SIGNIFICANCE



Differences in the activity profiles of available treatments for thyroid disorders make patient-specific drug selection extremely important. The development of synthetic thyroid

hormones has significantly improved treatment of hypothyroidism by decreasing the variations in levothyroxine and liothyronine blood levels that often resulted from inconsistent bovine and porcine sources. Once-daily or once-weekly dosing, which has been associated with improved patient compliance, is another clinically significant benefit of synthetic T_4 , resulting from its extended half-life. Knowledge of levothyroxine's pharmacokinetic profile has also prompted patient counseling efforts promoting premeal administration because of marked reduction in absorption when combined with meals. The understanding of structure–activity relationships also enabled the development of synthetic T_3 (liothyronine), the metabolically more active form of thyroid hormone with a much shorter duration of action. This

is especially useful for patients with thyroid carcinoma who are to undergo radioiodine imaging and possible treatment. Because of a half-life of only 24 hours, liothyronine substitution enables more timely radioiodine imaging and treatment; as a result, patients experience less symptomatic hypothyroidism.

The study of medicinal chemistry has improved the treatment of thyroid disorders by reducing variability in plasma concentrations of T_3 and T_4 and by increasing the reliability of available monitoring methods. Accurate monitoring and therapy adjustments have resulted in fewer complications and increased quality of life among the millions of patients with thyroid disorders.

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Hormones of the Thyroid Gland

Thyroid hormones are essential for normal development, differentiation, growth, and metabolism of every cell in the body. Thyroid hormones are iodinated amino acids derived from L-tyrosine. They are synthesized in the thyroid gland and stored as amino acid residues of thyroglobulin. The first known biologically active iodine-containing compound of the thyroid gland was isolated from thyroid extracts and named L-thyroxine (T_4). Later, its structure was established as the 3,5,3',5'-tetraiodo-L-thyronine (T_4) (Fig. 29.1), and its synthesis was accomplished. Twenty-five years later, with the availability of chromatographic techniques and radioactive iodine, researchers discovered another major thyroid hormone, 3,5,3'-triiodo-L-thyronine (T_3) (Fig. 29.1), which is derived mainly from T_4 deiodination by deiodinase enzymes outside the thyroid. The body is therefore able to use a dose of T_4 to produce its own T_3 .

T_4 and T_3 play numerous and meaningful roles in regulating metabolism, growth, and development and in maintaining homeostasis. Their reactions and products influence carbohydrate metabolism, protein synthesis and breakdown, and cardiovascular, renal, and brain function. It is usually believed that these actions result from effects of thyroid hormones on protein synthesis.

The thyroid gland also contains two quantitatively important iodinated amino acids, diiodo-L-tyrosine (DIT) and monoiodo-L-tyrosine (MIT). In addition, there are small amounts of other iodothyronines, such as 3,3'-diiodo-L-thyronine (T_2) and

3,3',5'-triiodo-L-thyronine (reverse T_3 [rT_3]). None of the latter compounds possesses any significant hormonal activity. Chemically, MIT is 3-iodo-L-tyrosine, and DIT is 3,5-diiodo-L-tyrosine. The coupling of the two outer rings of DIT or of one outer ring of DIT with that of MIT (each with the net loss of alanine) leads to the formation of the two major thyroid hormones, T_4 and T_3 , respectively.

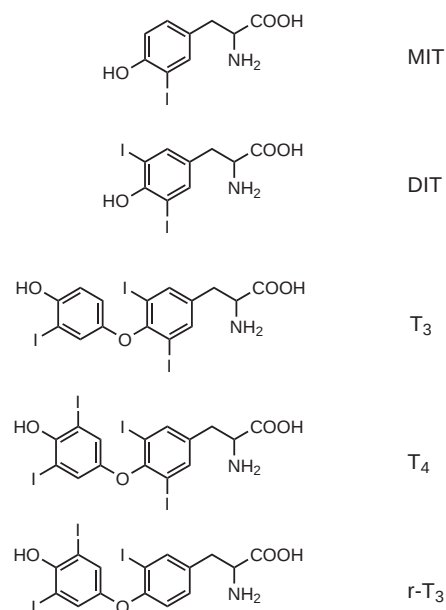


FIGURE 29.1 Structure of the iodinated compounds of the thyroid gland.

Thyroglobulin (Tg) is of special importance, because it serves as the matrix for the synthesis of T_4 and T_3 and as the storage form of the hormones and iodide. Tg, a large glycoprotein with a molecular weight of 660,000 daltons, accounts for about one-third of the weight of the thyroid gland. Tg carries an average of 6 tyrosyl residues as MIT, 5 residues as DIT, 0.3 residues as T_3 , and 1 residue as T_4 . From these values, it can be estimated that a 20-g thyroid stores roughly 10 μ mol (7.8 mg) of T_4 and 3 μ mol (2.0 mg) of T_3 and that the normal human thyroid gland contains enough potential T_4 to maintain a euthyroid state for 2 months without new synthesis. The structures of the iodinated compounds of the thyroid gland are shown in Figure 29.1.

Biosynthesis of Thyroid Hormones

The thyroid contains two hormones, L-thyroxine (T_4) and L-triiodothyronine (T_3). Iodine is an indispensable component of the thyroid hormones, comprising 65% of the weight of T_4 and 58% of the weight of T_3 . The thyroid hormones are the only iodine-containing compounds with established physiologic significance in vertebrates. Ingested iodine is absorbed through the small intestine and transported in the plasma to the thyroid, where it is concentrated, oxidized, and then incorporated into Tg to form MIT and DIT and later T_4 and T_3 . After a variable period of storage in thyroid follicles, Tg is subjected to proteolysis, and the released hormones are secreted into the circulation, where specific binding proteins carry them to target tissues.

The synthesis of the thyroid hormones, T_3 and T_4 , is regulated by thyrotropin (also known as thyroid-stimulating hormone [TSH]), which stimulates the synthesis of Tg, thyroperoxidase (TPO), and hydrogen peroxide. The formation of the thyroid hormones depends on an exogenous supply of iodide. The thyroid gland is unique in that it is the only tissue of the body able to accumulate iodine in large quantities and incorporate it into hormones. Approximately 25% of the body's supply of iodide is located in the thyroid gland. The iodine atoms play a unique role in the conformational preferences for T_3 and T_4 because of their large steric bulkiness. The metabolism of iodine is so closely related to thyroid function that the two must be considered together. The formation of thyroid hormones involves the following complex sequence of events: 1) active uptake of iodide by the follicular cells, 2) oxidation of iodide and formation of iodotyrosyl residues of Tg, 3) formation of iodothyronines from iodotyrosines, 4) proteolysis of Tg and release of T_4 and T_3 into blood, and 5) conversion of T_4 to T_3 . These processes are summarized in Figure 29.2 (7).

Active Uptake of Iodide by Follicular Cells

The first step in the synthesis of the thyroid hormones is the uptake of iodide from the blood by the thyroid gland. An adequate intake of iodide is essential for the synthesis of sufficient thyroid hormone. Ingested iodine

is absorbed through the small intestine and transported in the plasma to the thyroid, where it is concentrated, oxidized, and then incorporated into Tg. Blood iodine is present in a steady state in which dietary iodide, iodide "leaked" from the thyroid gland, and reclaimed hormonal iodide provide the iodide input. Thyroid gland iodide uptake, renal elimination, and a small biliary excretion provide iodide loss. The thyroid gland regulates both the fraction of circulating iodide that it takes up and the amount of iodide that it leaks back into the circulation. A simplified scheme of iodide metabolism is shown in Figure 29.3.

The mechanism enabling the thyroid gland to concentrate blood iodide against a gradient into the follicular cell is the iodide pump (NIS, sodium/iodide symporter), which is regulated by TSH. Decreased stores of thyroid iodine enhance iodide uptake; conversely, dietary iodide can reverse this process. The iodide pump maintains a ratio of thyroid iodide to serum iodide (T:S ratio) of about 20:1 under basal conditions but of more than 100:1 in hyperactive gland. Iodide uptake may be blocked by several inorganic ions, such as thiocyanate (SCN^-) and perchlorate. Because iodide uptake involves concurrent uptake of potassium, it can also be blocked by cardiac glycosides that inhibit potassium accumulation.

Oxidation of Iodide and Formation of Iodotyrosines

To serve as an iodinating agent, iodide must be oxidized to a higher oxidation state, a step that is hydrogen peroxide dependent and is catalyzed by TPO, a membrane-bound heme-enzyme that utilizes hydrogen peroxide as the oxidant. In addition to catalyzing the oxidation of iodide, TPO is essential for the incorporation of iodide into tyrosine residues in Tg (aromatic iodination) and coupling of the iodotyrosyl residues from DIT to form T_4 and T_3 . The activity of TPO is increased by TSH from increased synthesis of TPO.

The second step in the synthesis of the thyroid hormones is a concerted reaction at the apical membrane in which the iodide in the follicle lumen is oxidized by TPO in the presence of hydrogen peroxide to an active iodine species that, in turn, iodates selected tyrosyl residues of Tg. Consistent with the conditions necessary for aromatic halogenation, the iodination of the tyrosyl residues requires the iodinating species to be in a higher oxidation state compared with the iodide anion. The iodinating species is thought to be hypoiodate (OI^-) (7). The two-electron oxidation of iodide to its hypoiodate reactive species is accomplished by TPO. Although the diiodotyrosyl residues constitute the major products, some MIT peptides are also produced. TSH stimulates the generation of hydrogen peroxide and, thus, the process of iodination.

Hydrogen peroxide is an essential and limiting factor in the oxidation of iodide, aromatic iodination of tyrosyl residues, and the coupling reaction. The hydrogen peroxide-generating system is localized at the apical membrane, and its generation involves the oxidation of

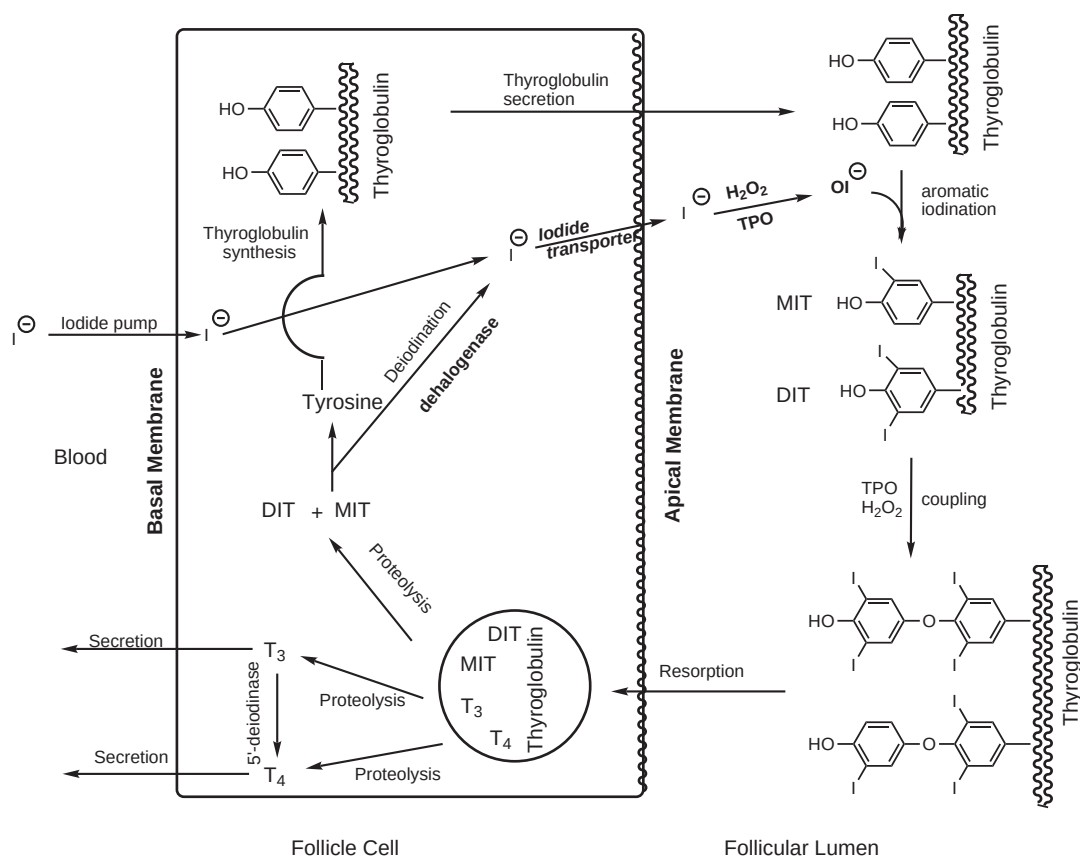


FIGURE 29.2 Summary of the major pathways for the biosynthesis and secretion of the thyroid hormones. When thyrotropin (TSH) binds to the TSH receptor at the basal membrane of the follicular cell, the biosynthesis of thyroglobulin (TG) is stimulated, as is that of thyroperoxidase (TPO) and the production of hydrogen peroxide. Noniodinated TG is synthesized by the rough endoplasmic reticulum of the follicular cell and secreted through the apical membrane of the follicular cell into the follicular lumen. Iodide enters the follicular cell by the iodide pump (NIS, sodium/iodide symporter) and is then transported into the follicular lumen. In the lumen, the iodide is oxidized by TPO-O (a π -cation radical intermediate formed from TPO and hydrogen peroxide) at the apical membrane to form hypoiodate anion (OI^-), followed by aromatic iodination of selected tyrosyl residues on TG to form diiodotyrosyl (DIT) and monoiodotyrosyl (MIT) residues. The tyrosyl ring of DIT couples with adjacent DIT and MIT residues with an ether linkage to form the outer ring of thyroxine (T_4) and of triiodothyronine (T_3), both of which remain attached to TG. Although shown as a sequential reaction, the iodination and coupling reactions occur simultaneously via TPO and hydrogen peroxide. Hydrogen peroxide is generated by a NADPH/FAD thyroid oxidase (THOX) at the apical membrane. Low plasma levels for T_4 cause the iodinated TG to be resorbed into the follicular cell, where complete proteolysis occurs by lysosomal protease to T_4 , T_3 , DIT, MIT, and noniodinated amino acids. Both T_4 and T_3 are secreted by the cell into the blood; T_4 is deiodinated to active T_3 . Both DIT and MIT are recycled by a dehalogenase (or deiodinase) to free tyrosine and iodide, both of which are recycled back into iodinated thyroglobulin.

the reduced form of nicotinamide adenine dinucleotide phosphate (NADPH) by an NADPH/flavin adenine dinucleotide (FAD) oxidase called thyroid oxidase. The reaction product of TPO with hydrogen peroxide is described as TPO-P, a π -cation radical, which generates hypoiodate. The generation of hydrogen peroxide is concentration controlled by iodide, which permits efficient hormone synthesis when iodide is scarce while avoiding excessive hormone synthesis when iodide is abundant. Deficient generation of hydrogen peroxide has been proposed as one explanation for goiter and decreased aromatic iodination in euthyroid patients. The *in vitro* addition of a hydrogen peroxide-generating system to thyroid homogenates or slices from these patients restored normal organification.

In the thyroid follicular cell, intracellular iodide taken up from blood is bound in organic form in a few minutes, so less than 1% of the total iodine of the gland is found as iodide. Therefore, inhibition of the iodide transport system requires blockade of organic binding. This can be achieved by the use of antithyroid drugs, of which n-propyl-6-thiouracil and 1-methyl-2-mercaptoimidazole (MMI) are the most potent.

Coupling of Iodotyrosine Residues

Coupling is also catalyzed by TPO at the apical membrane, and although shown in Figure 29.2 as sequential steps, they occur simultaneously. This coupling reaction takes place at Tg and involves the coupling of the two outer rings from DIT residues to become T_4 , whereas the

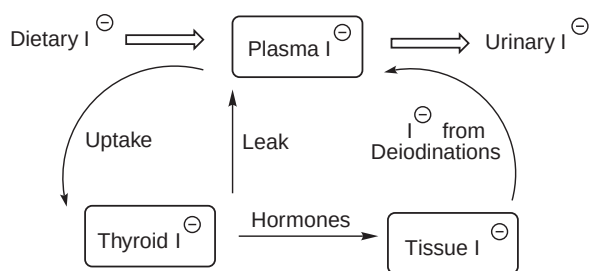


FIGURE 29.3 Simplified scheme of iodide metabolism.

coupling of the outer ring from MIT with DIT results in the formation of T_3 . During the coupling reaction as shown in Figure 29.2, a tyrosyl residue donates its iodinated phenyl group as a DIT radical to become the outer ring of the iodothyronine amino acid at an acceptor site, leaving dehydroalanine at the donor site. The location of the iodothyrosyl residues within Tg creates an optimal spatial alignment, facilitating the coupling reaction. These reactions are catalyzed by TPO and can be blocked by compounds such as thiourea, thiouracils, and sulfonamides.

Proteolysis of Thyroglobulin and Release of Iodothyronines

In response to demand for thyroid hormones, the release of thyroid hormones from Tg begins with the resorption of Tg via endocytosis into the follicular epithelial cells and its subsequent complete proteolysis by the lysosomal digestive enzymes of the follicular cells. Tg proteolysis yields MIT, DIT, T_3 , and T_4 . Although MIT and DIT are formed, they do not leave the thyroid but, instead, are selectively deiodinated to tyrosine and recycled into new Tg. The iodide is recycled into hypoiodate for subsequent iodination, conserving the essential nutrients for the thyroid gland. Both T_3 and T_4 are secreted by the cell into the circulation. A defect in the recyclization of MIT and DIT can lead to hypothyroidism and goiter by increasing their elimination in the urine.

Conversion of Thyroxine to Triiodothyronine

Although T_4 is by far the major thyroid hormone secreted by the thyroid (~8 to 10 times the rate of T_3), it is usually considered to be a prohormone. Because T_4 has a longer half-life, much higher levels of T_4 than of T_3 are in the circulation. The enzymatic conversion of T_4 to T_3 is an obligate step in the physiologic action of thyroid hormones in most extrathyroidal tissues. In the peripheral tissues, approximately 33% of the T_4 secreted undergoes 5'-deiodination to give T_3 , and another 40% undergoes deiodination of the inner ring to yield the inactive material rT_3 (8). The 5'-deiodination of T_4 is a reductive process catalyzed by a group of enzymes named iodothyronine 5'-deiodinases, referred to as 5'-deiodinases and symbolized by 5'-D, which are found in a variety of cells. Approximately 80% of the T_3 is derived from circulating T_4 .

Three types of 5'-deiodinases are currently known, and these are distinguished from each other primarily based on their location, substrate preference, and susceptibility to inhibitors. Type I 5'-deiodinase is found in liver and kidney and catalyzes both inner ring and outer ring deiodination (i.e., T_4 to T_3 and rT_3 to $3,3'$ - T_2). Type II 5'-deiodinase catalyzes mainly outer ring deiodination (i.e., T_4 to T_3 and T_3 to $3,3'$ - T_2) and is found in brain and the pituitary. Type III 5'-deiodinase is the principal source of rT_3 and is present in brain, skin, and placenta (9).

Transport of Thyroid Hormones

Transport in the Blood

More than 99% of the circulating thyroid hormone is bound to plasma proteins but can be liberated rapidly for entry into cells. The thyroid hormone-binding proteins are comprised of thyroxine-binding globulin (TBG), transthyretin (TTR or thyroxine-binding prealbumin), human serum albumin (HSA), and lipoproteins. Their functions are most probably to ensure a constant supply of thyroid hormones to the cells and tissues by preventing urinary loss, protect the organism against abrupt changes in thyroid hormone production and degradation, protect against iodine deficiency, and target the amount of thyroid hormone delivery by ensuring a site-specific, enzymatic alteration of TBG.

Thyroxine-Binding Globulin

TBG carries the major part of both circulating T_4 and T_3 (as well as rT_3), and therefore, quantitative or qualitative changes in TBG concentration have a high impact on total serum T_4 and T_3 . The protein is encoded by a single gene on the X chromosome and is produced and cleared by the liver. It has a single iodothyronine-binding site with higher affinity for T_4 compared to T_3 . When TBG is fully saturated, it carries approximately 20 $\mu\text{g } T_4/\text{L}$. The TBG concentration in serum is between 11 and 21 mg/L (180 to 350 nM), present from the 12th week of fetal life and 1.5 times higher in newborns and children until 2 to 3 years of age. Estrogen has a marked effect on TBG by prolonging the biologic half-life from the normal 5 days, thus resulting in increased plasma concentrations of TBG and total thyroid hormones, whereas testosterone has the opposite effect (10). In children and adolescents, this may have an implication in diseases with a severe sex hormone overproduction related to age, as well as oral contraceptive use and pregnancy in adolescent girls.

Transthyretin

TTR (previously called thyroxine-binding prealbumin) binds only about 15% to 20% of the circulating thyroid hormones and has a lower affinity for the hormones, thus dissociating from them more rapidly, and is responsible for much of the immediate delivery of T_4 and T_3 . TTR is the major thyroid hormone-binding protein in cerebrospinal fluid. It is synthesized in the liver and the choroid

plexus and secreted into the blood and cerebrospinal fluid, respectively. Only 0.5% of the circulating TTR is occupied by T_4 , and it has a rapid turnover of 2 days in plasma. Acquired abnormalities in TTR include major illness, nephrotic syndrome, liver disease, cystic fibrosis, protein fasting, and hyperthyroidism. However, changes in TTR concentrations have little effect on the serum concentrations of thyroid hormones (10).

Albumin

HSA binds about 5% of the circulating T_4 and T_3 . Its affinity for the hormones is even lower, and because HSA associates with a wide variety of substances, including a number of different hormones and drugs, the association between thyroid hormones and HSA can hardly be regarded as specific. Even marked fluctuations in serum HSA concentrations have no effect on thyroid hormone levels.

Lipoproteins

Lipoproteins transport a minor fraction of circulating T_4 and to some extent T_3 . The binding site for thyroid hormones on apolipoprotein A1 is distinct from that which binds to cellular protein receptors.

Consequences of Abnormal Binding Protein Concentrations

Abnormalities of the thyroid hormone-binding proteins do not cause alterations in the metabolic state of the individual and do not result in thyroid disease. Thus, abnormal concentrations of these binding proteins, due to changed synthesis, degradation, or stability, result in maintaining normal free thyroid hormone concentrations.

Metabolism and Excretion

As discussed earlier, T_4 is considered to be a prohormone, and its peripheral metabolism occurs in two ways: outer ring deiodination by the enzyme 5'-D, which yields T_3 , and inner ring deiodination by the enzyme 5-D, which yields rT_3 , for which there is no known biologic function (Fig. 29.4). In humans, deiodination is the most important metabolic pathway of the hormone, not only because of its dual role in the activation and inactivation of T_4 but also in quantitative terms.

A second pathway of thyroid hormone metabolism involves the conjugation of the phenolic hydroxyl group of the outer phenolic ring with sulfonate or glucuronic acid. These conjugation reactions occur primarily in the liver and to a lesser degree in the kidney and result in biotransformation of T_4 and T_3 . The iodothyronine glucuronides are rapidly excreted in bile, but after hydrolysis in intestine by bacterial β -glucuronidases, at least part of the liberated iodothyronine is reabsorbed, constituting an enterohepatic cycle. In contrast, little iodothyronine sulfate appears as a rule in the bile or in the serum because the sulfate esters are rapidly deiodinated

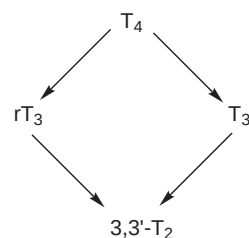


FIGURE 29.4 Deiodination of thyroxine (T_4).

in the liver. Sulfonate conjugation is the primary step in irreversible deactivation of thyroid hormones. Thyroid hormones can also undergo deamination and decarboxylase reactions in the liver, resulting in the formation of so-called acetic acid analogs. These reactions occur at the alanine side chain of the inner tyrosyl ring. Although these analogs are thought to be metabolically active, little is known about the quantities produced or their contribution to hormone activity in humans. The reactions through which thyroid hormone is metabolized are summarized in Figure 29.5.

Physiologic Actions of Thyroid Hormones—Oxygen Consumption and Calorigenesis

The two most important actions of thyroid hormone are those related to oxygen consumption and those related to protein synthesis. A respiratory component of the action of thyroid hormones was first observed almost a century ago. Respiratory exchange was depressed in patients diagnosed as hypothyroid and increased in patients diagnosed as hyperthyroid (3). The increase in respiration that follows the administration of thyroid hormone reflects an increase in metabolic rate, and thyroid function has, indeed, long been assessed by measuring the basal or resting metabolic rate (BMR), a test in which the oxygen consumed, as measured in an individual at rest, is used to calculate total-body energy production. The BMR of a hyperthyroid individual is above the normal range, or positive, and that of a hypothyroid individual is below the normal range, or negative.

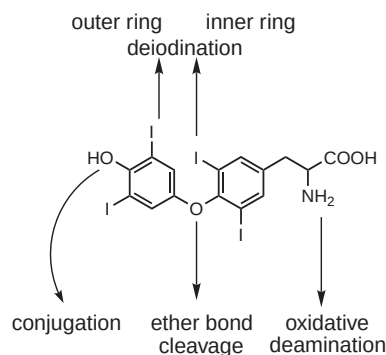


FIGURE 29.5 Metabolic pathways for thyroxine.

Because most of the energy produced by cellular respiration eventually appears as heat, an increase in cellular respiration necessarily leads to an increase in heat production (i.e., to a thermogenic or calorogenic effect). Thus, to the degree that thyroid hormones control BMR, they also control thermogenesis (11).

Clinically, the inability to adjust to environmental temperature is symptomatic of departure from the euthyroid status. Patients with myxedema frequently have subnormal body temperature, have cold and dry skin, and tolerate cold poorly; the thyrotoxic patient, who compensates for excess heat production by sweating (warm, moist hands), does not easily tolerate a warm environment.

Thyroid hormones regulate the turnover of carbohydrates, lipids, and proteins. They promote glucose absorption, hepatic and renal gluconeogenesis, hepatic glycogenolysis, and glucose utilization in muscle and adipose tissue (12). Thyroid hormones are anabolic when present at normal concentrations; they then stimulate the expression of many key enzymes of metabolism. Thyroid hormones at the levels present in hyperthyroidism are catabolic; they lead to the mobilization of tissue protein and, especially, of muscle tissue protein for gluconeogenic processes (13). Thus, the depletion of liver glycogen, the increased breakdown of lipids, and the negative nitrogen balance observed in hyperthyroidism represent toxic effects.

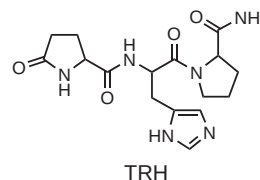
Differentiation and Protein Synthesis

In young mammals, thyroid hormone is necessary not only for general growth but also for proper differentiation of the central nervous system. A deficiency of thyroid hormone during the critical period when the developing human brain is sensitive to thyroid hormone results in an irreversible clinical entity termed cretinism, which is characterized by stunted growth and mental retardation. A great deal of attention has been devoted to the events taking place in the roughly 48-hour interval between the administration of thyroid hormone and the manifestation of certain effects caused by that administration. After Tata (14) had observed that the anabolic effects observed in rats given T_4 could be blocked by inhibitors of protein synthesis, further investigations led Tata and Widnell (15) to the conclusion that thyroid hormones were activating protein synthesis at the ribosomal level.

Control of Thyroid Hormone Biosynthesis

The primary role of thyroid is to produce thyroid hormones. The primary regulator of thyroid function and growth is the pituitary hormone TSH, a glycoprotein with a molecular weight of approximately 28,000 daltons. The most important controller of TSH secretion is thyroid-releasing hormone (TRH). TRH is secreted by hypothalamic neurons into hypothalamic-hypophyseal portal blood and finds its receptors in the anterior pituitary, stimulating the secretion of TSH. TRH is three

amino acids long (a tripeptide). Its basic sequence is *pyro*-glutamyl-histidine-proline amide. Secretion of TRH and, hence, TSH, is inhibited by high blood levels of thyroid hormones in a classical negative feedback loop.



The amount of thyroid hormone circulating in body fluids and present in tissues remains fairly constant. Accounting for this constancy are the relatively long biologic half-life of the thyroid hormones, the regulation of gland activity by the pituitary-hypothalamic system, and the availability of iodide. Thyroid hormone research points toward T_3 as the thyroid hormone; therefore, factors affecting peripheral T_3 formation by the 5'-D enzymes are highly relevant.

T_4 is greater than 99.9% bound to plasma proteins. Thus, T_4 has a biologic half-life of 1 week, as compared to 1 day for the less firmly bound T_3 .

The biosynthesis and secretion of TSH is regulated by TRH and, thus, the quantity of thyroid hormone in circulation through feedback control. The ability of thyroid hormones to prevent the release of TSH is referred to as feedback regulation. The amount of iodide available to the gland for hormone synthesis is also an important regulator of thyroid function. The efficiency of the thyroid pump mechanism and the rate of Tg and TPO synthesis are all TSH dependent. In cases of iodide deficiency, the production of thyroid hormone is lowered, and TSH rises through the pituitary feedback mechanism previously described. The effect of the increased TSH is to produce more thyroid hormone by increasing efficiency of the iodide pump and increasing Tg and TPO synthesis. Thus, in iodide deficiency, there is an increased uptake of iodide (16) and larger MIT-to-DIT ratios, which lead to larger T_3 -to- T_4 ratios. T_3 is the more rapidly acting hormone, mitigating the effect of iodide deficiency. When iodide deficiency is severe, a persistent rise in TSH is observed. This then results in thyroid gland growth.

In the presence of an excess of circulating iodide, the absolute amount of iodide taken up by the thyroid gland remains approximately constant, which is compensated by a decrease in the fraction of the total iodide taken up and an increase in the amount of iodide leaked from the thyroid gland. In addition, there may be a decrease in the formation of iodinated Tg residues and in the release of hormones from the gland. The decrease in the formation of iodinated Tg residues that occurs at excessive physiologic doses of iodide has been called the Wolff-Chaikoff block (17). This block may be caused by an interaction of iodide with NADPH (18), which depletes follicular NADPH and, in turn, depletes the hydrogen

peroxide necessary for TPO activity. The decrease in hormone release occurs when iodide is given in pharmacologic (milligram) quantities; it may last for a few weeks.

The activity of deiodinases reflects thyroid status, general health, and food intake. In hypothyroidism, there is a decrease in hepatic 5'-D-I but an increase in 5'-D-II activity (19). There is a decrease in T_3 and an increase in rT_3 after hepatic disease (20), renal damage (21), chronic illness (22), and starvation (23), indicating a decrease in the activity of the quantitatively more important 5'-D-I. An increase in deiodinase activity has been observed after overfeeding subjects or after administering a high-carbohydrate or high-fat diet (24) to experimental animals. The mechanism of deiodinase control is presumably intricate, but the inhibiting effects of propranolol on 5'-D-I (25) and of prazosin on the 5'-D-II of brown adipose tissue (26) infer the involvement of adrenergic components in the regulation. In addition, the rapid change in activity observed after asphyxia points toward a rapid, possibly cyclic adenosine monophosphate (cAMP)-related control (27). A number of other factors have been shown to influence thyroid hormone secretion. In young children, exposure to a cold environment triggers TRH secretion, leading to enhanced thyroid hormone release. This makes sense considering the known ability of thyroid hormones to spark body heat production.

Diseases Involving the Thyroid Gland

Thyroid disease is associated with both inadequate production and overproduction of thyroid hormones. Both types of disease are relatively common afflictions of man and animals.

Hypothyroidism

Hypothyroidism is the result of any condition that causes thyroid hormone deficiency. Two well-known examples include the following:

- **Iodine deficiency:** Iodide is absolutely necessary for production of thyroid hormones; without adequate iodine intake, thyroid hormones cannot be synthesized. Historically, this problem was seen particularly in areas with iodine-deficient soils, and frank iodine deficiency has been virtually eliminated by iodine supplementation of salt. In the case of iodide deficiency, the thyroid becomes inordinately large and is called a "goiter."
- **Primary thyroid disease:** Inflammatory diseases of the thyroid that destroy parts of the gland are clearly an important cause of hypothyroidism.

Goiter

An enlarged, palpable thyroid gland is referred to as a goiter. When insufficient thyroid hormone is liberated from the thyroid gland, the breakdown of the thyroid-pituitary-hypothalamic feedback mechanism results in the release of excess TSH and in the formation of a

thyroid hypertrophy referred to as a nontoxic goiter. The gland enlarges as it tries to take up more iodine, leading to goiter. Endemic goiters are those that occur in a significant segment of a given population. Goiters are most frequently caused by inadequate intake of dietary iodide in regions not reached by iodide-providing sea mists and, occasionally, by the prolonged intake of goitrogens derived from plant sources or aquifers. This condition can be prevented with iodine supplements, and many industrialized countries now iodize salt.

A characteristic sign of hypothyroidism is a decrease in metabolic rate, with a reduction in calorogenic effect and defective thermoregulation. The elevated serum cholesterol level seen in hypothyroidism is the result of a decrease in cholesterol degradation that exceeds the decrease in cholesterol biosynthesis; it reflects a general slowdown in catabolic processes. Shared symptoms of hypothyroidism and cognitive disorders include fatigue, mental dullness, lethargy, and inattention. Treatment of hypothyroidism includes supplementation of T_4 with the goal of returning TSH levels to normal.

Cretinism

The most severe and devastating form of hypothyroidism is seen in young children with congenital thyroid deficiency. If that condition is not corrected by supplemental therapy soon after birth, the child will suffer from cretinism, a form of irreversible growth and mental retardation.

Myxedema

Common symptoms of hypothyroidism arising after early childhood include lethargy, fatigue, cold intolerance, weakness, hair loss, and reproductive failure. If these signs are severe, the clinical condition is called "myxedema."

Hyperthyroidism

The increased metabolic rate of hyperthyroidism results in symptoms opposite to those seen in hypothyroidism. Common signs of hyperthyroidism include nervousness, insomnia, high heart rate, eye disease, and anxiety. In most species, this condition is less common than hypothyroidism. In humans, the most common form of hyperthyroidism is Graves disease, an autoimmune disease in which autoantibodies bind to and activate the TSH receptor, leading to continual stimulation of thyroid hormone synthesis. Graves disease is commonly treated with antithyroid drugs (e.g., PTU, MMI), which suppress synthesis of thyroid hormones primarily by interfering with iodination of Tg by thyroid peroxidase.

Hashimoto Disease

Hashimoto disease is an autoimmune disease in which the thyroid gland is gradually destroyed by a variety of cell- and antibody-mediated immune processes. It was the first disease to be recognized as an autoimmune disease (28). It was first described by the Japanese

specialist Dr. Hashimoto Hakaru in Germany in 1912. Physiologically, antibodies against thyroid peroxidase and/or Tg cause gradual destruction of follicles in the thyroid gland. Symptoms of Hashimoto thyroiditis include weight gain, depression, mania, sensitivity to heat and cold, paresthesia, fatigue, panic attacks, bradycardia, tachycardia, high cholesterol, reactive hypoglycemia, constipation, migraines, muscle weakness, cramps, memory loss, infertility, and hair loss. Hypothyroidism caused by Hashimoto thyroiditis is treated with thyroid hormone replacement agents such as levothyroxine or desiccated thyroid extract.

Generalized Resistance to Thyroid Hormone

Generalized resistance to thyroid hormone (GRTH) is a rare form of thyroid disorder and may be a heritable disorder characterized by reduced responsiveness of the pituitary and peripheral tissues to the action of thyroid hormones. Behavioral manifestations include hyperactivity, behavioral problems, and cognitive deficits.

Many have evaluated the association of GRTH to attention-deficit hyperactivity disorder and suggested that similar mechanistic pathways are shared between the two conditions (29,30). In GRTH, T_4 fails to block TSH production, leading to excess levels of T_3 and T_4 with either normal or elevated levels of TSH. In conditions of an improperly functioning thyroid, thyroid hormones and TSH ratios are unregulated, producing symptoms that are characteristic of attention-deficit hyperactivity disorder.

Thyroid Cancer

Thyroid cancer is a disease in which malignant cells are found in the tissues of the thyroid gland. This disease is more common in women than in men. Four main types of cancer of the thyroid are differentiated based on how the cancer cells look under a microscope: papillary (75% to 85% of cases) (31), follicular (10% to 20% of cases), medullary (5% to 8% of cases), and anaplastic (<5% of cases). Follicular and papillary carcinomas have the highest incidence in young women but have low mortality; they represent a different disease than anaplastic types or undifferentiated carcinomas, both of which occur among the older patients and have high death rates. Treatment of thyroid cancer is straightforward and effective.

THERAPEUTIC AGENTS

Thyroid Replacement Therapy

Thyroid hormone drugs are natural or synthetic preparations containing sodium salt of T_4 , or the sodium salt of T_3 (liothyronine), or both. Natural hormonal preparations include desiccated thyroid and Tg. Desiccated thyroid is derived from domesticated animals that are used for food by man (either beef or hog thyroid), and Tg is derived from thyroid glands of the hog. The U.S. Pharmacopeia (USP) has standardized the total iodine content of natural preparations. Thyroid USP contains no less than 0.17% and no more than 0.23% iodine, and Tg contains no less than 0.7% of organically bound iodine. Iodine content is only an indirect indicator of

true hormonal biologic activity. Hormone replacement is the established therapy in the treatment of various forms of hypothyroidism, from the complete absence of thyroid function seen in myxedema to simple goiter and cretinism. A large number of organic and some inorganic compounds stimulate or prevent thyroid hormone formation by interfering with iodide uptake into follicular cells, inhibiting TPO, preventing thyroid hormone binding to plasma proteins, or acting as effectors of thyroid deiodinases. Some of the agents described in this section are of therapeutic or diagnostic value, some illustrate potential side effects of drugs, and some are experimental compounds designed to achieve unmet therapeutic goals or to define structural parameters necessary for thyroid hormone actions.

Thyroid hormone preparations belong to two categories: natural hormone preparations derived from animal thyroid and synthetic preparations.

Natural Thyroid Hormone Preparations

Natural preparations include desiccated thyroid and Tg. Desiccated thyroid and Tg are derived from thyroid glands of domesticated animals that are used for food by human. T_4 , T_3 , DIT, and MIT are released from the proteolytic activity of gastrointestinal enzymes. Potency is based on total iodine content or bioassay and is somewhat variable with different preparations.

Desiccated thyroid preparations (Thyroid USP) are essentially acetone powders of bovine or porcine thyroid glands compressed into oral tablets. A diluent is usually present, because the preparations (especially those of porcine origin) commonly exceed the 0.17% to 0.23% iodine content required by the USP. Because the iodine of desiccated thyroid is in the form of iodinated tyrosyl and thyronyl residues of the precipitated Tg, the preparation owes its efficacy to the hormones that eventually are liberated by intestinal proteases. In desiccated preparations, T_3 and T_4 may be present in a ratio of approximately the same as found in humans. Desiccated preparations are less expensive than synthetic hormones but have been shown to produce variable T_4/T_3 blood levels because of inconsistencies both between and within animal sources of the thyroid gland. Most comments regarding desiccated thyroid also apply to partially purified Tg, because the two preparations differ in their total and relative amounts of T_4 and T_3 .

Synthetic Thyroid Hormones

Synthetic, crystalline thyroid hormones are more uniformly absorbed than biologic preparations and contain more precisely measured amounts of active ingredient in their dosage forms. T_4 , T_3 , d T_4 (dextrothyroxine), and T_4 - T_3 mixtures (Liotrix) are commonly used. Table 29.1 describes the pharmacokinetic properties for T_4 and T_3 .

Levothyroxine Sodium

Because of its tight binding to plasma carrier proteins, synthetic crystalline levothyroxine (T_4) sodium salt has a

TABLE 29.1 Some Properties and Pharmacokinetics for Levothyroxine and Liothyronine^a

Parameters	L-Thyroxine	Liothyronine
Trade name	Synthroid, Levothroid, Levoxyl	Cytomel
pK _a (phenolic)	6.7	8.4
Oral bioavailability (%)	50-80 ^b	95
Peak response	Several weeks (hypothyroid)	2-3 days (hypothyroid)
Duration of action	Several weeks (hypothyroid)	Several days (hypothyroid)
Protein binding (%)	99% (weakly)	Weakly bound
Time to peak concentration (hours)	2-4	1-2
Volume of distribution (L/kg)	6-7 euthyroid	1 euthyroid
Elimination half-life (days)	9-10 hypothyroid 3-4 hyperthyroid	1-2 hypothyroid 1 hyperthyroid
Excretion (%)	50 urine 50 feces	urine

^aMicromedex® Healthcare Series, Thompson Healthcare, Inc. Available at: <http://www.thompsonhc.com>. Accessed June 14, 2006.

^bFood decreases bioavailability.

slower onset of action than crystalline T₃ or a desiccated thyroid preparation. Its administration leads to a greater increase in serum T₄ but a lesser increase in serum T₃ than compared with Thyroid USP (32). The availability of 11 different tablet strengths, ranging from 25 to 300 µg, allows individual dosing. T₄ treats hypothyroidism. It is also used to treat or prevent goiter, which can be caused by hormone imbalances, radiation treatment, surgery, or cancer.

Liothyronine Sodium

Liothyronine (T₃) sodium is a synthetic form of T₃. T₃ sodium is the therapy of choice when it is desirable to have a rapid onset or cessation of activity, such as in patients with heart disease.

Liotrix

The medication containing a mixture of the sodium salts of T₄ and T₃ in a 4:1 ratio by weight is called liotrix (Thyrolar). Liotrix is used as replacement or supplemental therapy in patients with hypothyroidism. It works by supplying the thyroid hormones in the ratio they are normally produced by the body.

Dextrothyroxine

Dextrothyroxine (dT₄, CholoXin) is the synthetic (+)-stereoisomer isomer of T₄. It was introduced in hypocholesterolemic-hypolipidemic therapy with the premise that it would be void of calorogenic effects. The possibility of trace contamination with and metabolic conversion to T₄ and congeners has restricted its use, however, especially in patients with coronary heart disease. Dextrothyroxine is used to lower high cholesterol levels in the blood.

Thyroid Imaging Agents

Radioiodine

All isotopes of iodine are rapidly taken up in thyroid follicles. So far, only the isotopes ¹³¹I and ¹²⁵I have been used consistently. Isotope ¹³¹I, which decays to ¹³¹Xe mainly with the emission of a 0.6-meV β-particle and approximately 0.3-meV γ-rays, has a half-life of 8 days. Isotope ¹²⁵I, with a half-life of 60 days, decays to ¹²⁵Te by electron capture. The major component of its decay is 27-keV x-rays, and the minor component is a 35.5-keV γ-ray.

The γ-radiation emitted by ¹³¹I can be detected by a suitably placed scintillation crystal. This is the basis for the diagnostic use of this isotope in iodine uptake and in thyroid-scanning procedures.

The absorption of ¹³¹I β-radiation, which leads to the highly localized destruction of the thyroid follicles in which the isotope is taken up, has promoted radioiodine as a therapeutic alternative to surgical removal of the gland. Advantages of radioiodine therapy over surgery include the simplicity of the procedure, its applicability to patients who are poor surgical risks, and the avoidance of surgical complications, such as hypoparathyroidism. The development of late hypothyroidism (33) and the fear of chromosomal damage are arguments against the use of radioiodine in patients under 20 years of age and during pregnancy.

Radioiodine whole-body scan and serum Tg measurement are commonly used as the follow-up testing in patients with thyroid cancer. Both follow-up tests have limitations and are uncomfortable because they require a hypothyroid state. For thyroid cells to take up the labeled iodine, thyrotropin must be available, and for the pituitary to supply it, the body must be free of T₃ and T₄. This means that patients must completely stop taking medication for several weeks before the scan day. This results in a severe hypothyroidism, and approximately 25% of patients also produce antibodies against Tg, rendering the immunoassay useless. To make the follow-up tests easier, the U.S. Food and Drug Administration recently approved the use of Thyrogen (thyrotropin alfa for injection), a recombinant human thyrotropin produced in Chinese hamster ovary cells. Thyrogen has been shown to significantly enhance the sensitivity of Tg testing in patients maintained on thyroid hormone therapy. It allows patients with thyroid cancer to avoid the debilitating effects of hypothyroidism when undergoing radioiodine imaging scans.

Perchlorate and Pertechnetate

Perchlorates are the salts derived from perchloric acid (HClO₄). They occur both naturally and through manufacturing. They have been used as a medicine for more than 50 years to treat thyroid gland disorders. Perchlorate, which competitively inhibits the uptake of iodide, has been used in both diagnosis and treatment of thyroid disease. In continental Europe, perchlorate has been used for surgical preparation and in the long-term treatment of thyrotoxicosis. In the United States, the use of perchlorate was drastically curtailed after aplastic anemia and severe renal damage were reported following its use. The U.S. Environmental Protection Agency has issued substantial

guidance and analysis concerning the impacts of perchlorate on the environment as well as drinking water (34).

Radioactive pertechnetate ($^{99m}\text{TcO}_4^-$) anion is an important radiopharmaceutical for the study and imaging of the thyroid gland, including its morphology, vascularity, and function. $^{99m}\text{TcO}_4^-$ is conveniently prepared from molybdenum-99, which decays with 87% probability to ^{99m}Tc . The advantages to $^{99m}\text{TcO}_4^-$ include its short half-life of 6 hours and the low radiation exposure to the patient. TcO_4^- and iodide have a comparable charge/radius ratio and therefore are similarly incorporated into the thyroid gland. Pertechnetate anion ion is not incorporated into the Tg. Sodium pertechnetate cannot pass through the blood–brain barrier. $^{99m}\text{TcO}_4^-$ is renally eliminated for the first 3 days after being injected. After a scanning is performed, the patient is recommended to drink large amounts of water in order to expedite elimination of the radionuclide. In contrast to iodide and thiocyanate TcO_4^- and perchlorate do not undergo intrathyroidal metabolism after they are trapped.

Antithyroid Drugs for the Treatment of Hyperthyroidism

Iodide

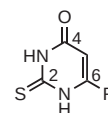
Inhibition of the release of thyroid hormone by iodide is the basis for its use in hyperthyroidism. Iodide decreases the vascularity of the enlarged thyroid gland and also lowers the elevated BMR. It has also been suggested that excess iodide might change the conformation of Tg, making the protein less susceptible to thyroidal proteolysis (35).

With the use of antithyroid drugs, the role of iodide in hyperthyroidism has been relegated to that of preparation for thyroid surgery. Iodide, as Lugol's solution (Strong Iodine Solution USP) or as saturated potassium iodide solution, is administered for approximately 2 weeks to ensure decreased vascularity and firming of the gland. Iodism, a side effect of iodine administration, is apparently an allergic reaction characterized by dermatologic and common cold-like symptoms (36).

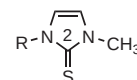
MMI, PTU, and Related Compounds

Thionamides are the major drugs for treatment of thyrotoxicosis and hyperthyroidism. These agents are potent inhibitors of TPO, which is responsible for the iodination of tyrosine residues of Tg and the coupling of iodotyrosine residues to form iodothyronines. These drugs have no effect on the iodide pump or on thyroid hormone release. The most clinically useful thionamides are thioureylenes, which are five- or six-membered heterocyclic derivatives of thiourea and include the thiouracil PTU and the thioimidazole 1-methyl-2-mercaptoimidazole (MMI, Tapazole). The uptake of these drugs into the thyroid gland is stimulated by TSH and inhibited by iodide. PTU is widely used to treat patients with hyperthyroidism. In rare cases, this drug has been found to have severe toxic effects on the liver. The data on hepatotoxicity confirm the findings for agranulocytosis that low-dose MMI is safer than PTU and that MMI toxicity is

more common in people over 40 years old. In contrast, PTU hepatotoxicity often occurs in younger patients. Most cases of hepatic injury occur in the first few months of drug therapy, as with agranulocytosis. The reason that MMI typically causes cholestatic hepatitis whereas PTU causes cytotoxic hepatitis remains unknown.



Thiouracil; R = H
Methylthiouracil; R = CH₃
Propylthiouracil (PTU); R = n-C₃H₇

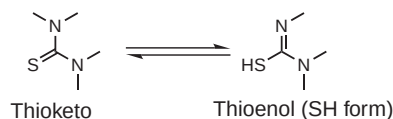


Methimazole (MMI, R = H)
Carbimazole (R = C₂H₅OCO)

Chemically, the grouping R-CS-N- has been referred to as thioamide, thionamide, thiocarbamide, or if R is N, as it is in thiouracil, PTU, and MMI, it is called a thioureylene. This structure may exist in either the thioketo or thioenol tautomeric forms. The study of 6-alkylthiouracil showed maximal antithyroid activity with PTU. 6-Methylthiouracil has less than one-tenth the activity of PTU.



Thioamide



Thioureylene tautomers

The ability of PTU to inhibit the enzyme 5'-D-I (i.e., the peripheral deiodination of T₄ to T₃, in addition to its intrathyroidal inhibition of thyroid hormone formation) has made PTU the drug of choice in the emergency treatment of thyroid storm (37). Single doses of PTU in excess of 300 mg are capable of almost total blockage of peripheral T₃ production (38).

A number of studies have defined the structure–activity relationships (SARs) of the thiouracils and other related compounds as inhibitors of outer ring deiodinase (39). The C₂ thioketo/thioenol group and an unsubstituted N₁ position are essential for activity. The enolic hydroxyl group at C₄ in PTU and the presence of alkyl group at C₅ and C₆ enhance the inhibitory potency.

MMI has more TPO inhibitory activity and is longer acting than PTU but, in contrast to PTU, is not able to inhibit the peripheral deiodination of T₄, presumably because of the presence of the methyl group at N₁

position. The suggested maintenance dosages are 50 to 800 mg daily for PTU and 5 to 30 mg daily for MMI.

PHARMACOKINETICS PTU is rapidly absorbed from the gastrointestinal tract, reaching peak serum levels after 1 hour. The bioavailability of 50% to 80% may be due to incomplete absorption or a large first-pass effect in the liver. The volume of distribution approximates total-body water with accumulation in the thyroid gland. Most of an ingested dose of PTU is excreted by the kidney as the inactive glucuronide within 24 hours. In contrast, MMI is completely absorbed but at variable rates. It is readily accumulated by the thyroid gland and has a volume of distribution similar to that of PTU. Excretion is slower than with PTU; 65% to 70% of a dose is recovered in the urine in 48 hours. The short plasma half-life of these agents (1.5 hours for PTU and 6 hours for MMI) has little influence on the duration of the antithyroid action or the dosing interval because both agents are accumulated by the thyroid gland. For PTU, giving the drug every 6 to 8 hours is reasonable since a single 100-mg dose can inhibit iodine organification by 60% for 7 hours. Since a single 30-mg dose of MMI exerts an antithyroid effect for longer than 24 hours, a single daily dose is effective in the management of mild to moderate hyperthyroidism. The two drugs differ in their binding to serum proteins. MMI is essentially free in serum, whereas 80% to 90% of PTU is bound to albumin.

Efforts to improve the taste and decrease the rate of release of MMI led to the development of 1-carbethoxy-3-methylthioimidazole (carbimazole). Carbimazole, the prodrug derivative of MMI, gives rise to MMI *in vivo* and is used in the same dosage.

The side effects of thioamides include diarrhea, vomiting, jaundice, skin rashes, and at times, sudden onset of agranulocytosis. There does not appear to be a great difference in toxicity among the compounds currently in use.

Antithyroid drugs are associated with a variety of minor side effects, as well as potentially life-threatening or even lethal complications. Side effects of MMI are dose related, whereas those of PTU are less clearly related to dose. This may favor use of low-dose MMI rather than PTU in the average patient with hyperthyroidism. Adverse reactions to the thioamides occur in 3% to 12% of treated patients. Most reactions occur early, especially nausea and gastrointestinal distress. An altered sense of taste or smell may occur with MMI. The most common adverse effect is a maculopapular pruritic rash (4% to 6%), at times accompanied by systemic signs such as fever.

Both PTU and MMI are concentrated several-fold by the thyroid gland and inhibit the iodination and coupling reactions of TPO (7). Taurog (40) described the thioureylenes as potent inhibitors of Tg iodination. He suggested that a thioureyline, such as PTU (PTU-SH), would irreversibly inhibit TPO-catalyzed iodination of Tg when the thioureyline-to-iodide ratio was high and reversibly when the PTU-SH-to-iodide ratio was low. In the course of the iodination reaction, the thioureyline PTU-SH would be oxidized (41), possibly to a disulfide dimer, such as PTU-SS-PTU.

Iodination of tyrosyl residues resumes once oxidation of the drug to disulfide products occurs by either hypoiodate (OI^-) or an enzyme-hypoiodate complex (EOI^-). Under these conditions, the thioureylenes act as competitive inhibitors by competing with tyrosyls for hypoiodate. Conversely, at high drug concentrations, the thioureylenes are only partially oxidized, and the partially oxidized intermediate can presumably inactivate TPO by covalent binding of an oxidized form of the drug to the prosthetic heme group of TPO to prevent formation of the hydrogen peroxide-TPO complex. As a result, iodination is irreversibly blocked. Data obtained from rats fed an iodide-deficient diet are consistent with this *in vitro* model in that intrathyroidal metabolism is radiolabeled PTU and MMI is decreased.

Thioureyline drugs also effectively inhibit the coupling of the DIT/MIT residues on Tg to yield T_4 and T_3 . This effect has been related to an alteration of the conformation of Tg brought on by the binding of the thioureyline to Tg (i.e., by the formation of a compound such as TPO-SS-PTU) (42).

After the observation that PTU inhibited the peripheral deiodination of T_4 (37,43), attempts to relate deiodinase inhibitory activity to structural parameters were undertaken (43). These studies emphasized the need for tautomerization to a thiol form and for the presence of a polar hydrogen on the nitrogen adjacent to the sulfur-bearing carbon. A study of the relation of chemical structure to 5'-D-I inhibitory activity related to similar studies of structural requirements for TPO inhibition could prove fruitful in the design of improved antithyroid drugs.

TOXICITY Agranulocytosis is the most feared side effect of antithyroid drug therapy. Agranulocytosis (an absolute granulocyte count of $<500/\text{mm}^3$) occurred in 0.37% of patients receiving PTU and in 0.35% receiving MMI. Agranulocytosis must be distinguished from the transient, mild granulocytopenia (a granulocyte count of $<1,500/\text{mm}^3$) that occasionally occurs in patients with Graves disease, in some patients of African descent, and occasionally in patients treated with antithyroid drugs. A baseline differential white blood cell count should be obtained before initiation of therapy.

Most cases of agranulocytosis occur within the first 90 days of treatment, but this complication can occur even a year or more after starting therapy. It is important to note that agranulocytosis can develop after a prior uneventful course of drug therapy, a finding that is important since renewed exposure to the drug frequently occurs when patients have a relapse and undergo a second course of antithyroid therapy. Agranulocytosis is thought to be autoimmune mediated, and antigranulocyte antibodies are shown by immunofluorescence and cytotoxicity assays. Agranulocytosis induced by PTU is evidence of a drug-dependent antibody reacting with granulocytes, monocytes, and hematopoietic progenitor cells. All patients should be instructed to discontinue the antithyroid drug and contact a physician immediately if fever or sore throat develops. Hepatotoxicity is another

major side effect of antithyroid drugs. Estimates regarding the frequency of this condition are imprecise, but it probably ranges from 0.1% to 0.2%. The recognition of PTU-related hepatotoxicity may be difficult, since in up to 30% of patients with normal baseline aminotransferase levels who are treated with PTU, transient acute increases in those levels develop, ranging from 1.1 to 6 times the upper limit of normal levels, that resolve while therapy is continued. In addition, asymptomatic elevations in serum aminotransferase levels occur frequently in untreated patients with hyperthyroidism and are not predictive of further increases after the institution of PTU therapy.

Propylthiouracil-Induced Hepatotoxicity

Vasculitis is the third major toxic reaction seen with antithyroid drug treatment, more commonly found in connection with PTU than with MMI. Serologic evidence consistent with lupus erythematosus develops in some patients, fulfilling the criteria for drug-induced lupus. Antineutrophil cytoplasmic antibody-positive vasculitis has also been reported, especially in Asian patients treated with PTU. Most patients have perinuclear antineutrophil cytoplasmic antibodies, with a majority of them having anti-myeloperoxidase antineutrophil cytoplasmic antibodies. It has been hypothesized that antithyroid drugs, especially PTU, can react with myeloperoxidase to form reactive intermediates that promote autoimmune inflammation.

Thyrotoxicosis

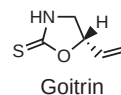
Hyperthyroidism is the term for overactive tissue within the thyroid gland causing an overproduction of thyroid hormones. Hyperthyroidism is thus a cause of thyrotoxicosis (44), the clinical condition of increased thyroid hormones in the blood. It is important to note that hyperthyroidism and thyrotoxicosis are not synonymous. For instance, thyrotoxicosis could instead be caused by ingestion of exogenous thyroid hormone or inflammation of the thyroid gland, causing it to release its stores of thyroid hormones. Because symptoms of thyrotoxicosis resemble those of adrenergic overstimulation, attempts to decrease such symptoms by adrenergic blockade have been undertaken. Reserpine and guanethidine, both of which are depleters of catecholamines, and propranolol, a β -blocking agent, have been used effectively to decrease the tachycardia, tremor, and anxiety of thyrotoxicosis. Because of its less serious side effects, propranolol has become the drug of choice in this adjunctive therapy. Reports of decreased T_3 plasma levels during propranolol treatment suggest that blocking of the peripheral deiodination of T_4 may contribute to the beneficial effects of propranolol. The use of propranolol as a preventive drug in acute thyrotoxicosis has been found to be beneficial by some investigators, but not by others. Hyperthyroidism has become a common and well-recognized disorder of middle-aged to older cats, and treatment with these antithyroid drugs has been equally effective in the long-term control of the feline hyperthyroidism.

Goitrogens and Drugs Affecting Thyroid Function

The presence of environmental goitrogens was suggested by the resistance of endemic goiters to iodine prophylaxis and iodide treatment in Italy and Colombia. In the past, endemic outbreaks of hypothyroidism have pointed toward calcium as a source of waterborne goitrogenicity, and it is presently believed that calcium is a weak goitrogen able to cause latent hypothyroidism to come to the surface.

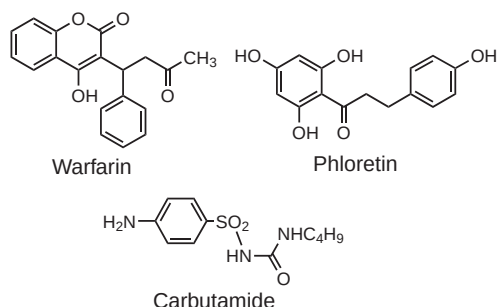
Lithium salts have been used as safe adjuncts in the initial treatment of thyrotoxicosis (45). Lithium is concentrated by the thyroid gland (46), with a thyroid-to-serum ratio of more than 2:1, suggesting active transport. Lithium ion inhibits adenylate cyclase, which forms cAMP. Formed in response to TSH, cAMP is a stimulator of the processes involved in thyroid hormone release from the gland. Inhibition of hormone secretion by lithium has proved to be a useful adjunct in treatment of hyperthyroidism (47).

In view of the role of cysteine residues in the conformation of Tg, the mode of action of TPO, and the deiodination of T_4 , the effect of sulfur-containing compounds on thyroid hormone formation is hardly surprising. Most naturally occurring sulfur compounds are derived from glucosinolates (formerly referred to as thioglucosides) (48), which are present in foods such as cabbage, turnip, mustard seed, salad greens, and radishes (most of these are from the genus *Brassica* or *Cruciferae*) as well as in the milk of cows grazing in areas containing *Brassica* weeds. Chemically, glucosinolates can give rise to many components, primarily to isothiocyanate (SCN^-) and lesser extent to thiocyanate (CNS^-), isothiocyanate (SCN^-), nitriles (RCN), and thiooxazolidones. Thiocyanate is a large anion that competes with iodide for uptake by the thyroid gland; its goitrogenic effect can be reversed by iodide intake. Goitrin, 5-*R*-vinylthiazolidine-2-thione, is a potent thyroid peroxidase inhibitor (49) that is claimed to be more effective than PTU in humans (50) and is held to be the cause of a mild goiter endemic in Finland. In rats, goitrin is actively taken up by the thyroid gland and appears to inhibit the coupling of Tg diiodotyrosyl residues (51). Many workers, however, believe that the goitrogenic effects of *Brassica* result from the additive effects of all goitrogenic components present.

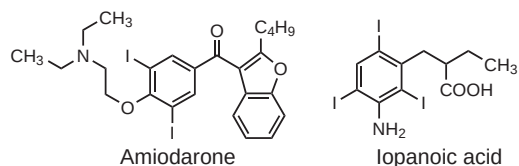


Other compounds affecting thyroid function include sulfonamides, anticoagulants, and oxygenated and iodinated aromatic compounds. The hypoglycemic agent carbutamide and the diuretic acetazolamide (Diamox) are examples of sulfonamides. Of the anticoagulants, heparin appears to interfere with the binding of T_4 to plasma transport proteins (52), but warfarin (Coumadin) and dicoumarol are competitive inhibitors of the substrate T_4 or rT_3 in the 5'-D reaction, with a dissociation constant in the micromolar range (53). Other oxygenated compounds affecting the 5'-D include resorcinol, long known

to be a goitrogen, and phloretin, a dihydrochalcone with a half maximal inhibitory concentration of 4 mol/L.



The ability of oxidation products of 3,4-dihydroxycinnamic acid to prevent the binding of TSH to human thyroid membranes (54) suggests that other oxygenated phenols may interfere with thyroid hormone function in more than one way. Examples of iodinated drugs affecting thyroid function are the antiarrhythmic agent amiodarone and the radiocontrasting agent iopanoic acid. All of these compounds interfere with the peripheral deiodination of T_4 and are being tested as adjuncts in the treatment of hyperthyroidism.



The binding of thyroid hormones to plasma carrier proteins is affected by endogenous agents or by drugs that can change the concentration of these proteins or compete with thyroid hormones for binding sites. Examples of the first group are testosterone and related anabolic agents, which are able to decrease the concentration of T_4 binding globulin, and estrogens and related contraceptive agents, which are able to increase the concentration of T_4 binding globulin. Salicylates, diphenylhydantoin, and heparin are members of the large group competing with thyroid hormones for binding sites. Alterations in the binding of T_3 and T_4 are of no large physiologic consequence, because the steady-state concentrations of free hormone are rapidly restored by homeostatic mechanisms. Knowledge regarding the presence of agents affecting thyroid hormone binding, however, is important for the interpretation of diagnostic tests assessing the presence of free or total hormone in plasma.

STRUCTURE-ACTIVITY RELATIONSHIPS OF THYROID ANALOGS

The synthesis and biologic evaluation of a wide variety of T_4 and T_3 analogs allowed a significant correlation of structural features with their relative importance in the production of hormonal responses. The key findings are

summarized in Table 29.2. In general, only compounds with the appropriately substituted phenyl-X-phenyl nucleus (as depicted in the structure at the top of Table 29.2) have shown significant thyroid hormonal activities. Both single ring compounds such as DIT and a variety of its aliphatic and alicyclic ether derivatives showed no T_4 -like activity in the rat antigoster test (55), the method most often used in determining thyromimetic activity in vivo (56). The SARs are discussed in terms of single structural variations of T_4 in the 1) alanine side chain, 2) 3- and 5-positions of the inner ring, 3) bridging atom, 4) 3'- and 5'-positions of the outer ring, and 5) 4'-phenolic hydroxyl group.

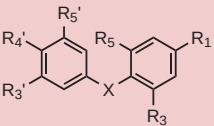
Aliphatic Side Chain

The naturally occurring hormones are biosynthesized from L-tyrosine and possess the L-alanine side chain. The L-isomers of T_4 and T_3 (compounds 1 and 3 in Table 29.2) are more active than the D-isomers (compounds 2 and 4 in Table 29.2). The carboxylate ion and the number of atoms connecting it to the ring are more important for activity than is the intact zwitterionic alanine side chain. In the carboxylate series, the activity is maximum with the two-carbon acetic acid side chain (compounds 7 and 8) but decreases with either the shorter formic acid (compounds 5 and 6) or the longer propionic and butyric acid analogs (compounds 9 to 12). The ethylamine side chain analogs of T_4 and T_3 (compounds 13 and 14) are less active than the corresponding carboxylic acid analogs. In addition, isomers of T_3 in which the alanine side chain is transposed with the 3-iodine or occupies the 2-position were inactive in the rat antigoster test (57), indicating a critical location for the side chain in the 1-position of the inner ring.

Alanine-Bearing Ring

The phenyl ring bearing the alanine side chain, called the inner ring or α -ring, is substituted with iodine in the 3- and 5-positions in T_4 and T_3 . As shown in Table 29.2, removal of both iodine atoms from the inner ring to form 3',5'- T_2 (compound 15) or 3'- T_1 (compound 16) produces analogs devoid of T_4 -like activity, primarily because of the loss of the perpendicular orientation of diphenyl ether conformation. Retention of activity observed on replacement of the 3- and 5-iodine atoms with bromine (compounds 17 and 18) implies that iodine does not play a unique role in thyroid hormone activity. Moreover, a broad range of hormone activity found with halogen-free analogs (compounds 19 and 20) indicates that a halogen atom is not essential for activity. In contrast to T_3 , 3'-isopropyl-3,6-dimethyl-L-thyronine (compound 20) has the capacity to cross the placental membrane and exerts thyromimetic effects in the fetus after administration to the mother. This could prove to be useful in treating fetal thyroid hormone deficiencies or in stimulating lung development (by stimulating lung to synthesize special phospholipids [surfactant], which ensure sufficient functioning of the infant's lungs at birth).

TABLE 29.2 Structure–Activity Relationships of Thyromimetics



Compound	R ₁	R ₃	R ₅	X	R _{3'}	R _{5'}	R _{4'}	Antigoiter Activity ^a
1. L-T ₄	L-Ala	I	I	O	I	I	OH	100
2. D-T ₄	D-Ala	I	I	O	I	I	OH	17
3. L-T ₃	L-Ala	I	I	O	I	H	OH	550
4. D-T ₃	D-Ala	I	I	O	I	H	OH	41
5.	COOH	I	I	O	I	I	OH	0.1
6.	COOH	I	I	O	I	H	OH	0.4
7.	CH ₂ COOH	I	I	O	I	I	OH	50
8.	CH ₂ COOH	I	I	O	I	H	OH	36
9.	(CH ₂) ₂ COOH	I	I	O	I	I	OH	15
10.	(CH ₂) ₂ COOH	I	I	O	I	H	OH	20
11.	(CH ₂) ₃ COOH	I	I	O	I	I	OH	4
12.	(CH ₂) ₃ COOH	I	I	O	I	H	OH	5
13.	(CH ₂) ₂ NH ₂	I	I	O	I	I	OH	0.6
14.	(CH ₂) ₂ NH ₂	I	I	O	I	H	OH	6
15.	L-Ala	H	H	O	I	I	OH	<0.01
16.	L-Ala	H	H	O	I	H	OH	<0.01
17.	DL-Ala	Br	Br	O	I	H	OH	93
18.	L-Ala	Br	Br	O	iPr	H	OH	166
19.	L-Ala	Me	Me	O	Me	H	OH	3
20.	L-Ala	Me	Me	O	iPr	H	OH	20
21.	DL-Ala	iPr	iPr	O	I	H	OH	0
22.	DL-Ala	sBu	sBU	O	I	H	OH	0
23.	DL-Ala	I	I	—	I	I	OH	0
24.	DL-Ala	I	I	S	I	H	OH	132
25.	DL-Ala	I	I	CH ₂	I	H	OH	300
26.	L-Ala	I	I	O	H	H	OH	5
27.	L-Ala	I	I	O	OH	H	OH	1.5
28.	L-Ala	I	I	O	NO ₂	H	OH	<1
29.	DL-Ala	I	I	O	F	H	OH	6
30.	L-Ala	I	I	O	Cl	H	OH	27
31.	DL-Ala	I	I	O	Br	H	OH	132
32.	L-Ala	I	I	O	Me	H	Oh	80
33.	L-Ala	I	I	O	Et	H	OH	517
30.	L-Ala	I	I	O	iPr	H	OH	786
35.	L-Ala	I	I	O	nPr	H	OH	200
36.	DL-Ala	I	I	O	Phe	H	OH	11
37.	DL-Ala	I	I	O	F	F	OH	2.3
38.	L-Ala	I	I	O	Cl	Cl	OH	21
39.	L-Ala	I	I	O	I	H	NH ₂	<1.5
40.	DL-Ala	I	I	O	I	H	H	>150
41.	DL-Ala	I	I	O	CH ₃	H	CH ₃	0
42.	L-Ala	I	I	O	I	H	CH ₃ O	225

^aSee Ekins (68) and Ahmad et al. (69). In vivo activity in rats relative to L-T₄ = 100% or DL-T₄ = 100% for goiter prevention.

immediately before premature birth (58). Substitution in the 3- and 5-positions by alkyl groups significantly larger and less symmetric than methyl groups, such as isopropyl and secondary butyl moieties, produces inactive analogs (compounds 21 and 22). These results show that 3,5-disubstitution by symmetric, lipophilic groups not exceeding the size of iodine is required for activity.

Bridging Atom

Several analogs have been synthesized in which the ether oxygen bridge has been removed or replaced by other atoms. The biphenyl analog of T_4 (compound 23 in Table 29.2), formed by removal of the oxygen bridge, is inactive in the rat antioiter test. The linear biphenyl structure is a drastic change from the normal diphenyl ether conformation found in the naturally occurring hormones. Replacement of the bridging oxygen atom by sulfur (compound 24) or by a methylene group (compound 25) produces highly active analogs. This provides evidence against the Niemann quinoid theory, which postulates that the ability of a compound to form a quinoid structure in the phenolic ring is essential for thyromimetic activity, and emphasizes the importance of the three-dimensional structure and receptor fit of the hormones. Attempts to prepare amino- and carbonyl-bridged analogs of T_3 and T_4 have been unsuccessful (59,60).

Phenolic Ring

The phenolic ring, also called the outer or β -ring, of the thyronine nucleus is required for hormonal activity. Variations in 3'- or 3',5'-substituents on the phenolic ring have dramatic effects on biologic activity and affinity for the nuclear receptor. The unsubstituted parent structure of this series T_2 (compound 26 in Table 29.2) possesses low activity. Substitution at the 3'-position by polar hydroxyl or nitro groups (compounds 27 and 28) causes a decrease in activity as a consequence of both lowered lipophilicity and intramolecular hydrogen bonding with the 4'-hydroxyl (61). Conversely, substitution by nonpolar halogen or alkyl groups results in an increase in activity in direct relation to the bulk and lipophilicity of the substituent—for example, $F < Cl < Br < I$ (compounds 29 to 31) and $CH_3 < CH_2CH_3 < CH(CH_3)_2$ (compounds 32 to 34). Although 3'-isopropylthyronine (compound 34) is the most potent analog known, being approximately 1.4 times as active as T_3 , *n*-propylthyronine (compound 35) is only about one-fourth as active as isopropyl, apparently because of its less compact structure. As the series is further ascended, activity decreases with a further reduction for the more bulky 3'-phenyl substituent (compound 36). Substitution in both 3'- and 5'-positions by the same halogen produces less active hormones (compounds 37 and 38) than the corresponding 3'-monosubstituted analogs (compounds 29 and 30). The decrease in activity has been explained as resulting from the increase in phenolic hydroxyl ionization and the resulting increase in binding to TBG (the primary carrier of thyroid hormones in human plasma) (62). In general,

a second substituent adjacent to the phenolic hydroxyl (5'-position) reduces activity in direct proportion to its size.

Phenolic Hydroxyl Group

A weakly ionized phenolic hydroxyl group at the 4'-position is essential for optimum hormonal activity. Replacement of the 4'-hydroxyl with an amino group (compound 39 in Table 29.2) results in a substantial decrease in activity, presumably as a result of the weak hydrogen bonding ability of the latter group. The retention of activity observed with the 4'-unsubstituted compound (compound 40) provides direct evidence for metabolic 4'-hydroxylation as an activating step. Introduction of a 4'-substituent that cannot mimic the functional role of a phenolic group, such as a methyl group (compound 41), and that is not metabolically converted into a functional residue results in complete loss of hormonal activity. The thyromimetic activity of the 4'-methyl ether (compound 42) was ascribed to the ready metabolic cleavage to form an active 4'-hydroxyl analog. The pK_a of 4'-phenolic hydroxyl group is 6.7 for T_4 (90% ionized at pH 7.4) and 8.5 for T_3 (~10% ionized). The greater acidity for T_4 is reflective of its stronger affinity for plasma proteins and, consequently, of its longer plasma half-life.

Conformational Properties of Thyroid Hormones and Analogs

The importance of the diphenyl ether conformation for biologic activity was first proposed by Zenker and Jorgensen (63,64). Through molecular models, they showed that a perpendicular orientation of the planes of the aromatic

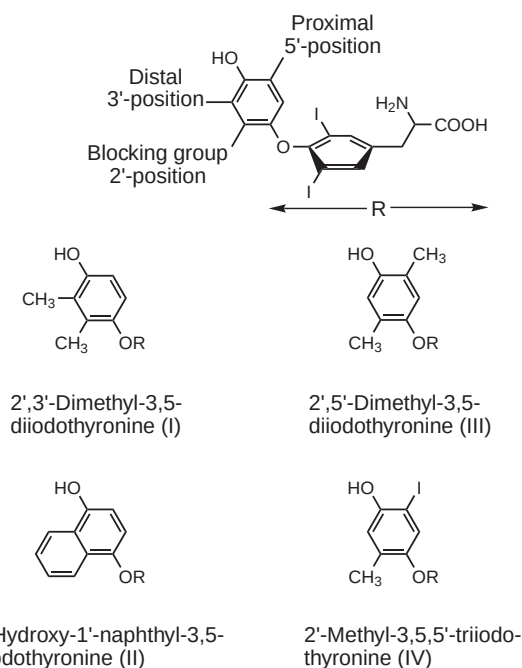


FIGURE 29.6 Structures of representative distal and proximal compounds.

rings of 3,5-diiodothyronines would be favored to minimize interactions between the bulky 3,5-iodines and the 2',6'-hydrogens. In this orientation, the 3'- and 5'-positions of the ring are not conformationally equivalent, and the 3'-iodine of T_3 could be oriented either distal (away from) or 5' proximal (closer) to the side chain-bearing ring (Fig. 29.6). Because the activity of compounds such as 3',5'-dimethyl-3,5-diiodothyronine had demonstrated that alkyl groups could replace the 3'- and 5'-iodine substituents, model compounds bearing alkyl groups in the 3'-position and alkyl or iodine substituents in the 5'-position (in addition to the blocking 2'-methyl group) were synthesized for biologic evaluation (64).

Biologic evaluation of 2',3'- and 2',5'-substituted diiodothyronines (65) revealed that 3'-substitution was favorable for thyromimetic activity but that 5'-substitution was not. The structures of representative distal analogs, 2',3'-dimethyl-3,5-DL-diiodothyronine (compound I) and O-(4'-hydroxy-1'-naphthyl)-3,5-DL-diiodotyrosine (compound II), and of the proximal analogs, 2',5'-dimethyl-3,5-DL-diiodothyronine (compound III) and 2'-methyl-3, 5,5'-DL-triiodothyronine (compound IV), are given in Figure 29.6. The effectiveness of these compounds in rat antigoiter assay (66) is presented in Table 29.3. These results clearly indicate that in 2'-blocked analogs, a distal 3'-substitution is favorable for thyromimetic activity, but a proximal 5'-substitution is not.

In addition to being perpendicular to the inner ring, the outer phenolic ring can adopt conformations relative to the alanine side chain, which would be *cis* or *trans*. In other words, the cisoid and transoid conformations result from the methine group in the alanine side chain being either *cis* or *trans* to the phenolic ring (Fig. 29.7). Although the bioactive conformation of the alanine side chain in thyroid hormone analogs has not yet been defined, these conformations appear to be similar in energy, because both are found in thyroactive structures as determined by x-ray crystallography (67). The synthesis of conformationally fixed cyclic or unsaturated analogs may allow evaluation of the bioactivity of the two conformers.

Transthyretin Receptor Model

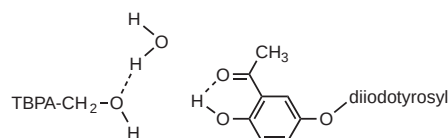
An additional tool in structural analysis and analog design has been TTR. TTR is a serum and cerebrospinal fluid carrier of T_4 and retinol. This is how transthyretin gained its

TABLE 29.3 Effectiveness of Distal and Proximal Compounds Antigoiter Assay

Compound ^a	Dose (mg/kg/day)	% T_4 activity
I	0.025	50
II	0.013	>100
III	2.3	<1
IV	0.5	2

^aSee text for specific descriptions of compounds I to IV.

name—**transports thyroxine and retinol**. TTR was originally called *prealbumin* because it ran faster than albumins on electrophoresis gels. TTR, a plasma protein, binds as much as 27% of plasma T_3 (68). The amino acid sequence of the TTR- T_3 binding site is known, and the protein has therefore served as a model, although admittedly an approximate model, for the T_3 receptor. The TTR model portrays the T_3 molecule as placed in an envelope near the axis of symmetry of the TTR dimer. In this envelope, hydrophobic residues, such as those of leucine, lysine, and alanine, are near pockets accommodating the 3,5,3'- and 5'-positions of T_3 , whereas the hydrophilic groups of serine and threonine (hydrogen bonded to water), are between the 3'-substituent and the 4'-phenolic group. Taking this model into account, Ahmad et al. (69) suggested that 3'-acetyl-3,5-diiodothyronine might be a good analog or a good inhibitor of T_3 , because the carbonyl group of the 3' acetyl substituent would form a strong hydrogen bond with the 4' phenolic hydrogen, thereby preventing its bonding with the hydrated residue of the putative receptor.



This compound, prepared by Benson et al. (70), was found to be indistinguishable from T_3 in oxygen uptake and glycerophosphate activity tests and to be half as active as T_3 in displacing labeled T_3 from rat liver nuclei in specific in vivo conditions.

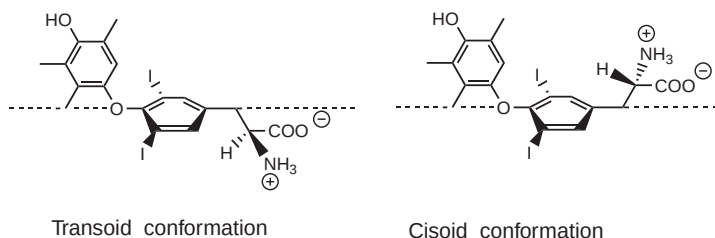


FIGURE 29.7 Side-chain conformations of thyroid hormones: transoid (left) and cisoid (right).

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Autumn Stewart, PharmD

Agranulocytosis ruled out; PA was diagnosed with “thyroid storm” and was stabilized. Upon review of medications at discharge, the clinical pharmacist recommended discontinuation of methimazole and initiation of propylthiouracil 200 mg every 8 hours as an alternative. PA was discharged on propylthiouracil and propranolol 80 mg BID and referred for treatment with radioactive iodine (^{131}I).

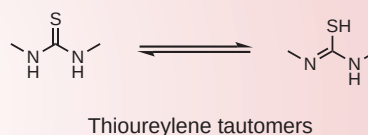
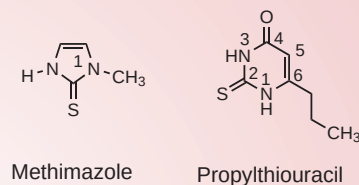
Chemical Analysis

S. William Zito and Victoria Roche

PA’s “thyroid storm” was stabilized in the emergency department, and at discharge he was changed from methimazole to propylthiouracil and placed on propranolol 80 mg BID to control arrhythmia. The change to propylthiouracil was made because of its ability to inhibit the enzyme 5'-iodothyronine deiodinase (i.e., inhibits the peripheral deiodination of T_4 to T_3). Propylthiouracil also inhibits the intrathyroidal formation of thyroid hormones. Taken together, this makes propylthiouracil the drug of choice for the emergency treatment of thyroid storm. Single doses of propylthiouracil in excess of 300 mg are capable of almost total blockage of peripheral T_3 production.

Methimazole and propylthiouracil are classed as thionamides or thioureylenes, which are five- or six-membered heterocyclic derivatives of thiourea. This chemical class contains the major drugs for treatment of thyrotoxicosis and hyperthyroidism. Their mechanism of action is to inhibit thyroperoxidase, which is responsible for the iodination of tyrosine residues of thyroglobulin and the coupling of iodotyrosine residues to form iodothyronines. These drugs have

no effect on the iodide pump or on thyroid hormone release. The uptake of these drugs into the thyroid gland is stimulated by thyroid-stimulating hormone and inhibited by iodide.

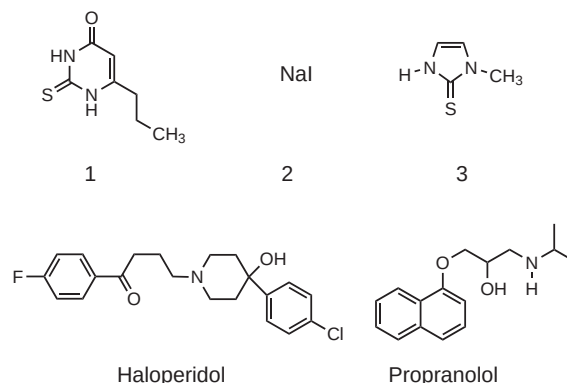


The essential chemical grouping in the thioureylenes is N-CS-N- which can tautomerize to either the thioketo or thioenol forms. Structure-activity relationships of the thioureas indicate that they are inhibitors of outer ring deiodinase. The C_2 thioketo/thioenol group and an unsubstituted N_1 position are essential for activity. The enolic hydroxyl group at C_4 in propylthiouracil and the presence of alkyl group at C_5 and C_6 enhance the inhibitory potency. Methimazole has more thyroperoxidase inhibitory activity and is longer-acting than propylthiouracil but, in contrast to propylthiouracil, is not able to inhibit the peripheral deiodination of T_4 , presumably because of the presence of the methyl group at N_1 position. The suggested maintenance dosages are 50 to 800 mg daily for propylthiouracil and 5 to 30 mg daily for methimazole.

CASE STUDY

S. William Zito and Victoria Roche

ST is brought to the emergency room of the local hospital. He is approximately 35 years old and homeless; he was found in an alley and was unresponsive to verbal questioning. Upon arrival at the emergency room ST was sweating, thrashing, screaming, and displayed combative behavior. His temperature was 102°F, his blood pressure was 144/68, and he had a pulse of 158 beats per minute. He was restrained until his behavior was controlled with intramuscular haloperidol. An electrocardiogram revealed sinus tachycardia and he was treated with propranolol. Based on clinical laboratory values indicating elevated free T_4 and T_3 (5.42 and 458 ng/dL, respectively) and decreased thyroid-stimulating hormone (<0.01 UIU/mL), and after eliminating other possible diagnoses (severe anxiety disorder, neuroleptic malignant syndrome, drug toxicity and meningitis), his condition was diagnosed as thyroid storm. In addition, ST had also elevated thyroid-stimulating immunoglobulin (281 ng/dL), indicating that there was new hormone synthesis, likely from Graves' disease. Evaluate the following three drugs for treatment of ST's thyroid storm.



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Suggested Readings

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Chapter 30

Calcium Homeostasis

ROBIN M. ZAVOD

Drugs Covered in This Chapter*

SELECTIVE ESTROGEN RECEPTOR MODULATORS

- Bazedoxifene acetate
- *Lasofloxifene*
- Raloxifene hydrochloride
- Toremifene citrate

BISPHOSPHONATES

- Alendronate sodium
- *Clodronate tetrasodium*

- *Etidronate disodium*
- Ibandronate sodium
- Pamidronate disodium
- Risedronate sodium
- Tiludronate disodium
- Zoledronic acid

CALCITONIN

CINACALCET HYDROCHLORIDE

TERIPARATIDE

MONOCLONAL ANTIBODY

- Denosumab

INORGANIC SALTS

- Calcium salts
- Gallium nitrate
- Sodium fluoride
- *Strontium ranelate*

Abbreviations

AF-2, activation factor-2

ATP, adenosine triphosphate

BMD, bone mineral density

CaSR, calcium-sensing receptor

DIOP, drug-induced osteoporosis

DNA, deoxyribonucleic acid

ER, estrogen receptor

ERE, estrogen responding element

ERT, estrogen replacement therapy

FDA, U.S. Food and Drug Administration

HMG-CoA, hydroxymethylglutaryl-coenzyme A

IM, intramuscular

IV, intravenous

NaF, sodium fluoride

NHANES III, National Health and Nutrition Examination Survey III

1,25(OH)₂D₃,

1,25-dihydroxycholecalciferol

25(OH)D₃, 25-hydroxycholecalciferol

PPAR γ 2, peroxisome proliferator-activated receptor γ 2

PTH, parathyroid hormone

RANKL, receptor activator of nuclear factor- κ B ligand

RRE, raloxifene responding element

SAR, structure-activity relationship

SC, subcutaneous

SERM, selective estrogen receptor modulator

UGT, uridine diphosphate glucuronosyltransferase

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

SCENARIO

Kathryn Neill, PharmD

PJ is a 70-year-old woman with hypertension, gastroesophageal reflux, and osteoporosis. She was prescribed alendronate and calcium 6 months ago, but the resulting improvement in bone density is less than expected. The pharmacist interviews PJ. Her current medications are alendronate 10 mg daily, calcium carbonate 500 mg three times daily with meals, omeprazole 20 mg daily, and ramipril 5 mg daily.

PJ takes alendronate each morning with only a glass of water. After 30 minutes she takes ramipril, omeprazole, and calcium with coffee. Additional calcium doses are taken with lunch and supper. PJ has lost weight the last few months due to an unsettled stomach.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

Three primary hormones—calcitonin, parathyroid hormone, and vitamin D—control the homeostatic regulation of calcium and its principle counterion, inorganic phosphate. Homeostatic control of these ions is essential not only for the moderation of longitudinal bone growth and bone remodeling but also for blood coagulation, neuromuscular excitability, plasma membrane structure and function, muscle contraction, glycogen and adenosine triphosphate (ATP) metabolism, neurotransmitter/hormone secretion, and enzyme catalysis (1). In an average 70-kg adult, approximately 1 kg of calcium is found, 99% of which is located in the bone. The principle calcium salt contained in the hydroxyapatite crystalline lattice of teeth and bones is $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$. Similarly, approximately 500 to 600 g of phosphate are present, 85% of which is found in the bone. The normal plasma concentration of calcium is approximately 4.5 to 5.7 mEq/L, 50% of which is protein bound. The remainder of the calcium is either complexed to corresponding counterions (46%) or exists in its ionized form (4%). It is only the ionized form of calcium that is tightly hormonally regulated (varies less than 5% to 10%) (1,2). Because serum calcium concentrations fluctuate, so do the plasma levels of the hormones associated with calcium homeostasis. Serum phosphorus levels vary with age, diet, and hormonal status. The most common form of phosphate in the blood (pH 7.4) is HPO_4^{2-} .

The bone is composed of two distinct tissue structures: cortical (compact) bone and trabecular (cancellous) bone (3). Eighty percent of the skeleton is composed of cortical bone (e.g., long bones such as the humerus, radius, and ulna) (4,5), which is a relatively dense tissue (80% to 90% calcified) (4) that provides structure and support (3). Bone marrow cavities, flat bones, and the ends of long bones are all composed of trabecular bone, which is considerably more porous (5% to 20% calcified) (4,5). To maintain healthy, well-mineralized bone, a continuous process of bone resorption (loss of ionic calcium from bone) and bone formation occurs

along the bone surface. Cortical bone is remodeled at the rate of 3% per year, whereas 25% of trabecular bone, which has a considerably higher surface area, is remodeled annually (3). In terms of calcium turnover in bone, approximately 500 mg are removed and replaced on a daily basis.

Both inorganic and organic components are present in the bone. The highly crystalline inorganic component is hydroxyapatite, and the collagen matrix comprises the major portion (90%) of the organic component. The collagen matrix serves as the foundation for hydroxyapatite mineralization. Osteocalcin and osteonectin are minor organic constituents that promote binding of hydroxyapatite and calcium to the collagen matrix and regulate the rate of bone mineralization, respectively (5).

In general, peak bone mass occurs between 30 and 40 years of age (3,6) and is dependent on genetic factors as well as proper intake of calcium, maintenance of quality nutrition, and participation in weight-bearing exercise (6). Thereafter, peak bone mass progressively declines at the rate of 0.3% to 0.5% of cortical bone per year (3). After menopause, bone loss is accelerated (2% per year in the spine) (6) for a period of 5 to 10 years because of the loss of estrogen. This can result in up to a 30% decrease in bone mineral density.

HORMONAL REGULATION OF SERUM CALCIUM LEVELS

Arnaud has developed a “butterfly model” that provides a diagrammatic view of the complex interrelationships among the three hormones (parathyroid, calcitonin, and vitamin D) that control calcium homeostasis (serum concentrations of ionic calcium) and their target organs (bone, kidney, and intestine) (Fig. 30.1) (7). The right side (B loops) of the butterfly model describes the processes that increase the serum calcium concentration in response to hypocalcemia; the left side (A loops) depicts the events that occur in response to hypercalcemia.

CLINICAL SIGNIFICANCE



As our knowledge about the development and risk factors associated with disruptions in calcium homeostasis has increased, so too have the modalities available to prevent and/or treat these disease processes. In general, disorders of calcium homeostasis involve the development of bone disease and/or alterations in serum calcium concentration. In the most basic sense, the development of bone disease is simply an inequity between bone breakdown and bone formation, which also may result in an altered serum calcium concentration. In addition to bone disease, disruptions in normal serum calcium concentrations may be related to an imbalance in calcium intake and renal calcium elimination. These disturbances can result from various factors, including increased activity of cells that cause bone breakdown, decreased activity of cells that form new bone, decreased absorption of calcium, or irregularities in levels of hormones that affect calcium absorption and influence cells involved in bone maintenance. An increased understanding of these physiologic pathways has led to the development of multiple classes of agents targeting the different mechanisms for evolution of these disease processes, including selective estrogen receptor modulators, bisphosphonates, and various calcium salts.

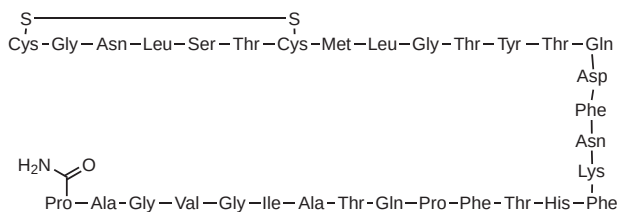
Application of the principles of medicinal chemistry has resulted in the formulation of agents with additional routes of administration, increased potency, and decreased frequency of dosing. These advances have increased the utility of these

agents and improved the quality of life for countless individuals affected by calcium homeostasis disorders. Understanding the development of individual disease processes involved in disorders of calcium homeostasis (e.g., osteoporosis, osteopetrosis, hyperparathyroidism, and Paget's disease) and the pharmacodynamic effects of individual compounds used to treat these disorders is paramount for the practitioner making therapeutic decisions. Incorporation of these factors into the therapeutic plan is necessary to target the valued pharmacodynamic effects of these agents while minimizing unwanted or harmful effects. For example, the selection of raloxifene to treat osteoporosis in a patient with severe gastroesophageal reflux disease, as opposed to an oral bisphosphonate, which could increase the likelihood of developing erosive esophagitis. Finally, it is also important for the clinician to recognize the capacity of certain entities used to treat calcium disorders to be allergenic or more prone to produce adverse effects so that selection of the best agent for an individual patient is facilitated.

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Calcitonin



Human calcitonin

Human calcitonin is a 32-amino acid peptide (molecular weight, 3,527 daltons) biosynthesized in the parafollicular “C” cells found within the thyroid gland. This hormone contains a critical disulfide bridge between residues 1 and 7, with the entire amino acid sequence required for biologic activity. The carboxy-terminal residue is a proline amide. “Procalcitonin,” a precursor peptide, has been identified and proposed to facilitate intracellular transport and secretion. Calcitonin is secreted in response to elevated serum calcium concentrations (>9 mg/100 mL)

and serves to oppose the hormonal effects of parathyroid hormone. In response to a hypercalcemic state (Fig. 30.1, B loops), increased calcitonin secretion drives serum calcium concentrations down via stimulation of urinary excretion of both calcium and phosphate (loop 3B), prevention of calcium resorption from the bone via inhibition of osteoclast activity (loop 1B), and inhibition of intestinal absorption of calcium (loop 2B). When serum calcium concentrations are low (hypocalcemia), the release of calcitonin is slowed, thereby activating loops 1A, 2A, and 3A.

Parathyroid Hormone

Parathyroid hormone (PTH) is biosynthesized as a 115-amino acid preprohormone in the rough endoplasmic reticulum of the parathyroid gland and is cleaved to the prohormone (90 amino acids) in the cisternal space of the reticulum (Fig. 30.2). The active hormone is finally produced (84 amino acids; molecular weight, 9,500 daltons) in the Golgi complex and is stored in secretory granules in the parathyroid gland. This gland is exquisitely sensitive to serum calcium concentrations and is able to monitor these levels via calcium-sensing receptors

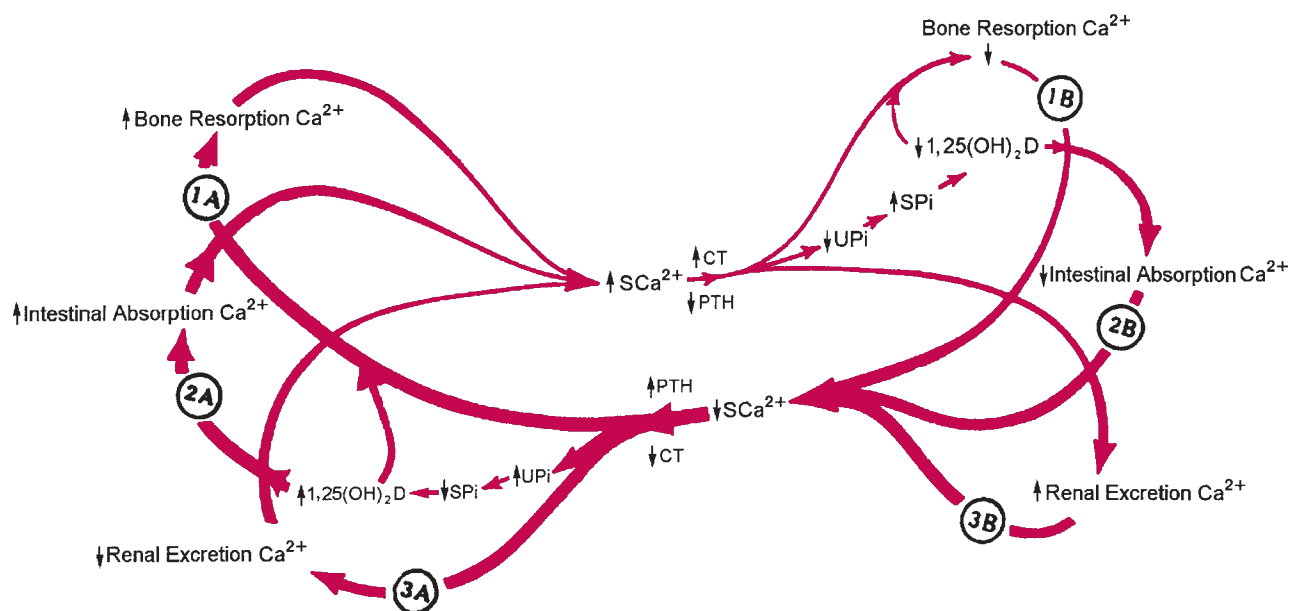


FIGURE 30.1 The Arnaud's butterfly model for regulating calcium homeostasis consists of three overlapping loops that interlock and relate to one another through the serum concentrations of ionic calcium (SCa^{2+}), parathyroid hormone (PTH), and calcitonin (CT). The right side (B loops, where B refers to the effects of CT) of the model describes the physiologic processes that increase the serum calcium concentration in response to hypocalcemia; the left side (A loops, where A refers to the effects of PTH) of the model depicts the events that decrease the serum calcium concentration in response to hypercalcemia. Loop 1 bone resorption; loop 2 intestinal absorption; loop 3 renal excretion; SPI, serum inorganic phosphate; UPI, urinary inorganic phosphate. (Adapted from Arnaud CD. Calcium homeostasis: regulatory elements and their integration. *Fed Proc* 1978;37:2557–2560, with permission).

(CaSR). These cell surface receptors help cells to react to micromolar changes in the concentration of ionized calcium in the serum (8). Binding of calcium to these receptors facilitates activation of phospholipase C and, ultimately, inhibition of PTH secretion. The relatively short-acting PTH is secreted from the parathyroid gland chief cells in response to a hypocalcemic state and serves to oppose the hormonal effects of calcitonin (1). Unlike calcitonin, the biologic activity of PTH resides solely in residues 1 to 34 in the amino terminus.

PTH decreases renal excretion of calcium (Fig. 30.1, loop 3A), indirectly stimulates intestinal absorption of

calcium (Fig. 30.1, loop 2A), and in combination with active vitamin D, promotes bone resorption (Fig. 30.1, loop 1A). PTH stimulates bone resorption by several mechanisms: 1) transformation of osteoprogenitor cells into osteoclasts is stimulated in the presence of PTH, 2) PTH promotes the deep osteocytes to mobilize calcium from perilacunar bone, and 3) surface osteocytes are stimulated by PTH to increase the flow of calcium out of the bone. In addition, the secretion of PTH stimulates the biosynthesis, activation, and release of the third hormone associated with calcium homeostasis, vitamin D. When serum calcium concentrations are high, the release of PTH is inhibited.

Vitamin D

Derived from cholesterol, vitamin D is biosynthesized from its prohormone cholecalciferol (D_3), the product of solar ultraviolet irradiation of 7-dehydrocholesterol in the skin (2). In 1966, it was first recognized that vitamin D must undergo activation via two oxidative metabolic steps (Fig. 30.3). The first oxidation to 25-hydroxycholecalciferol [$25(\text{OH})\text{D}_3$: calcifediol; Calderol] occurs in the endoplasmic reticulum of the liver and is catalyzed by vitamin D 25-hydroxylase. This activation step is not regulated by plasma calcium concentrations. The major circulating form (10 to 80 $\mu\text{g/mL}$) is $25(\text{OH})\text{D}_3$, which also is the primary storage form of vitamin D (2). In response to a hypocalcemic state and the secretion of PTH, a second oxidation step is activated in the mitochondria of the



FIGURE 30.2 Preproparathyroid hormone is the 115-amino acid protein indicated above. Cleavage at site 1 gives rise to parathyroid hormone (89 amino acids), whereas cleavage at site 2 gives rise to parathyroid hormone (PTH, 84 amino acids). The protein shown in red is teriparatide (34 amino acids).

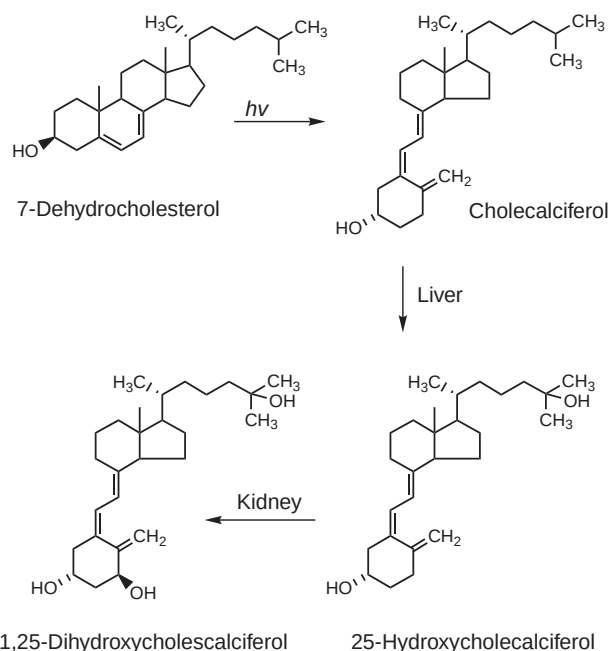


FIGURE 30.3 Bioactivation of vitamin D.

kidney, catalyzed by vitamin D 1α -hydroxylase (2,9). The product of this reaction, 1,25-dihydroxycholecalciferol [$1,25(\text{OH})_2\text{D}_3$: 1,25-calcitriol; Rocaltrol, Calcijex] is the active form of vitamin D. Its concentration in the blood is 1/500 that of its monohydroxylated precursor. The biosynthesis of vitamin D is tightly regulated based on the serum concentrations of calcium, phosphate, PTH, and active vitamin D (2).

Sterol-specific cytoplasmic receptor proteins (vitamin D receptor) mediate the biologic action of vitamin D (9). The active hormone is transported from the cytoplasm to the nucleus via the vitamin D receptor, and as a result of the interaction of the hormone with target genes, a variety of proteins are produced that stimulate the transport of calcium in each of the target tissues. Active vitamin D works in

concert with PTH to enhance active intestinal absorption of calcium, to stimulate bone resorption, and to prohibit renal excretion of calcium (2,9). If serum calcium or 1,25-calcitriol concentrations are elevated, then vitamin D 24-hydroxylase (in renal mitochondria) is activated to oxidize $25(\text{OH})\text{D}_3$ to inactive 24,25-dihydroxy-cholecalciferol and to further oxidize active vitamin D to the inactive 1,24,25-trihydroxylated derivative. Both the 1,24,25-trihydroxylated and the 24,25-dihydroxylated products have been found to suppress PTH secretion as well. Several factors have been identified in the regulation of the biosynthesis of vitamin D, including low phosphate concentrations (stimulatory) as well as pregnancy and lactation (stimulatory).

NORMAL PHYSIOLOGY

During growth periods in childhood and early adulthood, bone formation characteristically exceeds bone loss. In young adulthood, bone formation and bone resorption are nearly equal. After the age of 40 years, however, bone resorption is slightly greater than bone formation, and this results in a gradual decline in skeletal mass. Osteoblasts, osteoclasts, and osteocytes are the three types of cells that make up the bone remodeling unit or bone metabolizing unit and, therefore, are largely responsible for the bone remodeling process (3,4).

The bone remodeling process is comprised of two opposing activities, bone resorption and bone formation. Bone resorption is launched when osteocytes and those cells that line the bone surface release cytokines and growth factors (Fig. 30.4). These endogenous substances signal osteoblasts to release receptor activator of nuclear factor- κB ligand (RANKL), a cytokine (10). This ligand interacts with and activates its receptor (RANK) found on 1) the surface of osteoclast precursor cells, which stimulates osteoclast differentiation, and 2) the surface of mature osteoclasts, which promotes activation (10). RANKL also decreases osteoclast apoptosis.

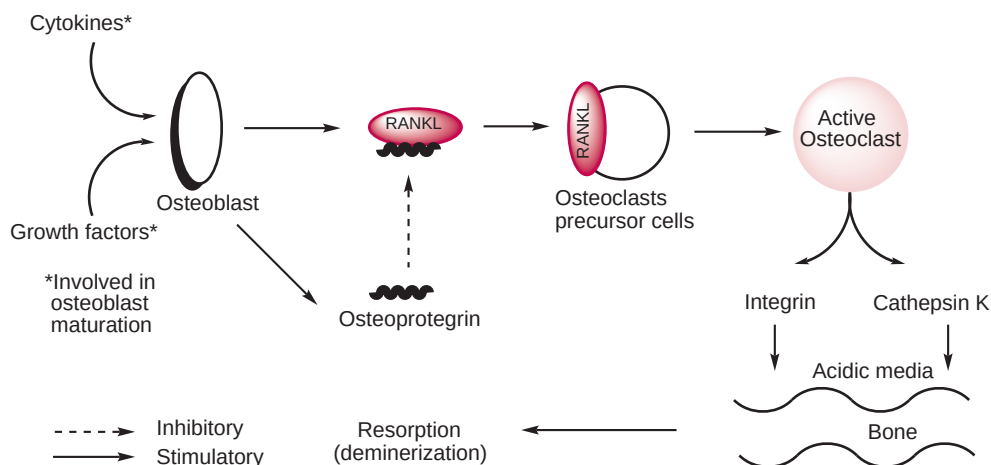


FIGURE 30.4 Bone resorption involving receptor activator of nuclear factor- κB ligand (RANKL).

Osteoclasts are the large multinucleated cells of hemopoietic origin that are responsible for carrying out the bone resorption or destroying process (6). Cytokines (including RANKL), PTH, and the active form of vitamin D are responsible for activation of these cells. Bone-lining flat cells, derived from “retired” osteoclasts and osteoblasts, are located on the bone surface (3). The function of these flat cells is thought to serve to identify areas of the bone that have become weakened or misshapen and to send a signal to the bone remodeling unit to prepare the bone. Lining cells then digest the outer layer of the bone matrix in preparation for bone remodeling. As part of the bone resorption process, the osteoclast membrane comes into contact with the bone surface and, in the presence of integrin, forms an impermeable “sealing zone” of approximately 500 to 1,000 μm in size (2,6). The ruffled border of the osteoclast membrane secretes hydrogen ions, H^+ ATPase, and the cysteine protease cathepsin K (11). As a result, this microenvironment becomes acidified, and ultimately, bone demineralization occurs (6). Several types of lysosomal enzymes have been proposed to digest the collagen matrix, thereby pitting the bone surface to a depth of 50 μm (3–6).

Osteoblasts, which are of mesenchymal origin and are formed in the bone marrow, stimulate bone formation (6). In the maturation process, osteoblasts undergo multiple cell divisions and, in so doing, express the gene products that are needed to form the bone matrix or osteoid, as well as those products responsible for mineralization of that tissue (3,6). It is in the rough endoplasmic reticulum that the biosynthesis of the bone matrix protein occurs (4). Multiple endogenous substances are involved in osteoblast maturation, including many cytokines (interleukins and granulocyte-macrophage colony-stimulating factor), as well as hormones and growth factors.

Not only are osteoblasts involved in bone formation, but they also have a role in limiting bone resorption. Produced by osteoblasts, osteoprotegerin binds to RANKL and therefore prevents its interaction with its RANK receptors on the osteoclast (11). As a result, osteoclast differentiation and bone resorption are inhibited.

Quantification of bone mineral density (BMD) can be measured by noninvasive radiographic tests, such as single-photon or dual-photon absorptiometry (spine, hip, and total body), dual-energy x-ray absorptiometry (spine, hip, and total body), peripheral dual-energy x-ray absorptiometry (wrist, heel, and finger), single energy x-ray absorptiometry (wrist or heel), quantitative computed tomography (spine), peripheral quantitative computed tomography (wrist), and quantitative ultrasound (heel, shin bone, and knee cap) (3,12–14). Dual-energy x-ray absorptiometry is considered to be the gold standard for measuring bone density and has an accuracy that exceeds 95% (4). These techniques measure the attenuation of x-rays or gamma rays as they cross the spine, hip, or radius before they reach the detector (6).

Other methods under development that measure BMD include ultrasound, traditional x-rays, and blood/urine tests (6). Traditional x-rays can identify the site of fracture, but they cannot measure BMD (3). Blood/urine tests can identify if the patient is suffering from a medical condition that is contributing to the loss of BMD and can identify important biochemical markers that can assess the rate of bone resorption and bone turnover. The measurement of serum calcium, phosphorous, and vitamin D levels also may provide insight regarding the cause of decreased BMD (3). Often, patients suffer from multiple vertebral compression fractures without seeking treatment other than an over-the-counter analgesic, and the diagnosis of osteoporosis occurs only after the patient has already lost significant (as much as 30%) bone mass.

DISEASE STATES ASSOCIATED WITH ABNORMAL CALCIUM HOMEOSTASIS

Osteoporosis

Osteoporosis is a skeletal disease that is characterized by loss of bone mass as well as microarchitectural deterioration of the bone tissue. This disease is associated with increased bone fragility and susceptibility to fracture. It is a condition that is characterized not by inadequate bone formation but, rather, by a deficiency in the production of well-mineralized bone mass. Whereas no medical cause typically is evident in primary osteoporosis (3), secondary osteoporosis classically stems from medical illness or medication use. There are two types of primary adult osteoporosis, type I, or postmenopausal, and type II, or senile (Table 30.1) (15,16). In type I osteoporosis, there is an accelerated rate of bone loss via enhanced resorption at the onset of menopause. In this form of the disease, the loss of trabecular bone is threefold greater than the loss of cortical bone. This disproportionate loss of bone mass is the primary cause of the vertebral crush fractures and the wrist and ankle fractures experienced by postmenopausal women. In type II osteoporosis, which is associated with aging, the degree of bone loss is similar in both trabecular and cortical bone and is caused by decreased bone formation by the osteoblasts (5).

Drug- or disease-induced, or type III, osteoporosis (Table 30.2) accounts for up to 30% of the cases of vertebral fractures reported annually. It can be caused by a variety of factors, including long-term suppression of osteoblast function, an inhibition of calcium absorption from the gut, altered vitamin D metabolism, or excessive loss of calcium in the urine (17). Disease states or pharmacologic therapies that result in estrogen deficiency, hyperparathyroidism, hyperthyroidism, or hypogonadism have been correlated with the development of osteoporosis (6,13,17,18). Drug-induced osteoporosis (DIOP) is associated with the use of glucocorticoids, thyroid hormone replacement, lithium,

TABLE 30.1 Classification of Osteoporosis (15)

Etiology	Type I (postmenopausal) Increased Osteoclast Activity and Bone Resorption	Type II (senile) Decreases Osteoblast Activity and Bone Formation; Decreased GI Ca Absorption	Type II(secondary) Drug Therapies; Disease States
Typical age at diagnosis (years)	50–75	>70	Any age
Gender ratio (women:men)	6:1	2:1	1:1
Typical fracture site	Vertebrae, distal radius	Femur, neck, hip	Vertebrae, hip, extremities
Bone morphology	Decreased trabecular bone	Decreased trabecular and normal cortical bone	Decreases cortical bone
Rate of bone loss (per year)	2%–3%	0.3%–0.5%	Variable

antiepileptic agents, selective serotonin reuptake inhibitors, proton pump inhibitors, thiazolidinediones, methotrexate, cyclosporine, aromatase inhibitors, gonadotropin-releasing hormone agonists, and immunosuppressive therapy (6,17–21). See Table 30.3 for a list of the drug-specific mechanisms associated with DIOP (17).

After estrogen deficiency related to menopause, long-term therapy with glucocorticoids represents the most prevalent cause of DIOP. As much as 3% to 27% of total bone loss can occur within the first 6 to 12 months of glucocorticoid therapy (17). From a mechanistic perspective, glucocorticoids cause an initial increase in bone resorption as a result of their ability to increase RANKL and macrophage colony-stimulating factor. This results in an increase in osteoclastogenesis and a decrease in osteoclast apoptosis. With the ability of the glucocorticoids to cause an increase in peroxisome proliferator-activated receptor γ 2 (PPAR γ 2) signaling and a decrease in Wnt signaling protein, a decrease in osteoblast formation and function and an increase in osteoblast apoptosis result, and there is a decrease in bone formation (22).

Vitamin D deficiency, as the cause of pseudohyperparathyroidism, is a common cause of osteoporosis in elder persons who are institutionalized and lack adequate sunlight exposure (2). Many of the older antiepileptic agents induce cytochrome P450 24A1 (CYP24A1) enzymes, which catalyze the conversion of vitamin D to inactive metabolites (17).

According to the National Health and Nutrition Examination Survey III (NHANES III) the National Osteoporosis Foundation estimates that more than 10 million Americans have osteoporosis and an additional 33.6 million have low BMD of the hip (23). It is projected that by 2020, these numbers will increase to 14 and 47 million, respectively (24). With these increases, it is anticipated that the number of hip fractures will double or triple by 2040. Approximately 40% of United States white women and 13% of United States white men will experience at least one fracture related to low BMD in their lifetime. The chance of a 50-year-old woman experiencing a hip, vertebral, or forearm fracture is 17%, 15%, and 16%, respectively. It is predicted that by age 80, 27% of women will be osteopenic and 70% will be osteoporotic. It has been estimated that approximately 20% of hip fracture patients will require long-term nursing home care. A surprising 60% of hip fracture patients do not regain full function, and within 3 to 4 months of hip fracture, as many as 25% die as a result of secondary complications (e.g., pneumonia or infection). Mortality also is increased 17% after both femoral and vertebral fractures. Whereas postmenopausal African Americans have the lowest rate of osteoporosis (4%), Native Americans have the highest (12%) with Hispanic and Asian postmenopausal women falling in between (10% for both) (25). Additional risk factors associated with osteoporosis are presented in Table 30.4. Given these statistics, osteoporosis should be considered a significant health problem that only stands to worsen unless appropriate interventions are pursued.

TABLE 30.2 Causes of Secondary (Type III) Osteoporosis

Gastrointestinal diseases	Anorexia nervosa, chronic liver disease, malabsorption syndromes (e.g., celiac disease, Crohn disease, gastric bypass, short bowel disease)
Nutritional excesses or deficiencies	Alcoholism, calcium, vitamin D, protein deficiency, excess vitamin A, total parenteral nutrition.
Endocrine-based diseases	Acromegaly, diabetes mellitus (types 1 and 2), disease-related elevated hormone levels (hypercortisolism, hyperthyroidism, hyperparathyroidism), disease-related suppressed hormone levels
Other disease states	Chronic obstructive pulmonary disease, hemophilia, myeloma, and some other cancers
Drugs	Corticosteroids, aromatase inhibitors, thiazolidinediones, antiepileptic agents

TABLE 30.3 Mechanisms Related to Drug-Induced Osteoporosis

Drug Therapy	Proposed Mechanism Leading to Altered Bone Mineral Density (BMD)	Citations in Reference 17
Aromatase inhibitors	Inhibit conversion of androgens to estrogen	62
Gonadotropin-releasing hormone agonists	Antiandrogenic effect on pituitary gland suppressing testosterone (results in hypogonadism)	70
Thyroid hormone replacement therapy	Increase in osteoclast activation and RANKL	81
Antipsychotic agents	Stimulation of prolactin secretion lowers estrogen and testosterone levels (results in hypogonadism)	99, 100
Lithium	Hyperparathyroidism	104–106
Thiazolidinediones	Decreases osteoblast differentiation and function via PPAR γ activation	117–119
Loop diuretics	Enhances renal calcium excretion	15

Osteopetrosis

Osteopetrosis, also known as marble bone disease, describes a group of heritable disorders that are centered on a defect in osteoclast-mediated bone resorption. There are four autosomal recessive forms and one autosomal dominant form of osteopetrosis (Table 30.5) (26). It generally is characterized by abnormally dense, brittle bone and increased skeletal mass. Unlike osteoporosis, this disorder results from decreased osteoclast activity, which has an effect on both the shape and structure of the bone. In very extreme cases, the medullary cavity, which houses bone marrow, fills with new bone, and production of hematopoietic cells is hampered. Like osteoporosis, this disease can be detected radiographically and appears as though there is a “bone within a bone.” There is limited evidence that bisphosphonates can induce osteopetrosis via their inhibition of osteoclast activity (27).

Hypocalcemia

Hypocalcemia can be caused by PTH deficiency, vitamin D deficiency, various pharmacologic agents, and miscellaneous disorders (Table 30.6) (28). A state of hypocalcemia will inhibit calcitonin release. This results in an elevation of PTH biosynthesis and release and indirectly causes an

increase in the production of vitamin D. The left wing of Arnaud’s butterfly model (Fig. 30.1) is then activated to increase serum calcium concentrations. In the absence of calcitonin, osteoclast activity is unregulated; therefore, bone resorption is accelerated. In acute cases of hypocalcemia, specifically in the case of hypocalcemic tetany, PTH is administered to correct the hormonal imbalance.

Hypercalcemia

A state of hypercalcemia (Table 30.7) will promote calcitonin biosynthesis and release. As a result, PTH biosynthesis and its secretion are inhibited, as is the production of vitamin D. The right wings of Arnaud’s butterfly model (Fig. 30.1, B loops) are then activated to decrease serum calcium concentrations. In the presence of calcitonin, osteoclast activity is inhibited, so bone resorption is slowed. In acute cases of hypercalcemia, calcitonin is administered to reestablish calcium homeostasis. Hypercalcemia also can be treated with sulfate salts, ethylenediaminetetraacetic acid (EDTA), furosemide, ethacrynic acid, glucocorticoids, and plicamycin.

Hypoparathyroidism

Hypoparathyroidism is caused by decreased serum PTH concentrations. It is characterized by hypocalcemia, hyperphosphatemia, and reduced levels of circulating vitamin D. The right wing of Arnaud’s butterfly model predominates (Fig. 30.1), and serum calcium concentrations precipitously decrease. Administration of intravenous (IV) calcium gluconate and PTH serves to acutely correct plasma calcium levels. Chronic oral administration of active vitamin D as well as calcium supplements has been effective in maintaining appropriate serum calcium concentrations.

Pseudohypoparathyroidism

In this disease state, levels of PTH are normal or even elevated; however, serum calcium concentrations are low. End-organ insensitivity to PTH has been proposed to be the cause of the hypocalcemic state. Treatment of this

TABLE 30.4 Lifestyle and Genetic Risk Factors for Osteoporosis

Lifestyle Factors	Genetic Factors
Smoking	Female
Sedentary lifestyle	Family history
Calcium intake	Small frame
Milk intolerance	Early menopause
Excessive caffeine	
Excessive alcohol	
Nulliparity	

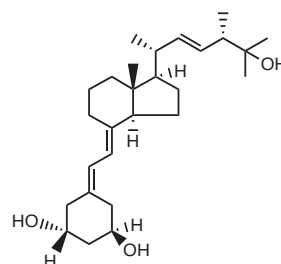
TABLE 30.5 Human Osteopetrosis Genotypes

Gene Involved	Function of Gene	Clinical Symptoms	Patients Affected (%)
Autosomal recessive disorders			
<i>CAII</i>	Carbonic acid and proton production	Less severe: may improve with age, short stature, no hematologic failure	<5
<i>TCIRG1</i>	Proton pump	Severe: apparent in infancy, visual impairment, hypocalcemia, death by 10 years of age	~60
<i>CLCN7</i>	Chloride channel	Severe: apparent in infancy, similar to <i>TCIRG1</i> symptoms	~15
		Intermediate: apparent in infancy but less aggressive, osteomyelitis, fractures	
<i>Gl/gI</i>	Unknown	Extremely severe: death within months	<5
Autosomal dominant disorder			
<i>CLCN7</i>	Chloride channel	Frequent fractures, osteomyelitis, some more severe (like recessive disorders)	

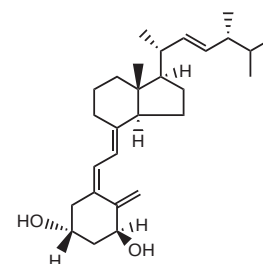
condition with calcium and vitamin D has proven to be successful.

Hyperparathyroidism

Increased levels of PTH lead to moderately to severely elevated serum calcium concentrations and, as a result, a significant loss of calcium from the bone (2). Deposits of calcium salts in soft tissue, as well as formation of renal calculi, also can result from this hormonal imbalance. Treatment of this condition with salmon calcitonin, loop diuretics, or other classical treatments for hypercalcemia has been favorable. The IV vitamin D analog paricalcitol, which is used for both prevention and treatment of hyperparathyroidism secondary to chronic renal failure, has been shown to reduce PTH levels by an average of 30% after 6 weeks of treatment. Whereas paricalcitol is a fully active form of vitamin D, doxercalciferol requires activation by the liver.



Paricalcitol (Zemlar)



Doxercalciferol (Hectorol)

This analog also is indicated for the treatment of secondary hyperparathyroidism. Doxercalciferol capsules should be administered three times weekly at the time of dialysis along with close monitoring of calcium and phosphate levels. Treatment of secondary hyperparathyroidism with vitamin D therapy is problematic, however, because it often leads to hypercalcemia,

TABLE 30.6 Causes of Hypocalcemia

PTH Deficiency	Vitamin D Deficiency	Drugs	Miscellaneous
Hypoparathyroidism	Nutritional deficiency	Chemotherapeutic agents	Osteoblastic metastases
Pseudohypoparathyroidism	Gastrointestinal malabsorption	Diuretics Furosemide	Phosphate infusion
Hypomagnesemia	Renal Failure Tubule disorders Nephrotic syndrome Hepatobiliary disease (decreases synthesis) Pancreatic disease (malabsorption) Anticonvulsant therapy (malabsorption, abnormal metabolism)	Inhibitors of bone resorption	Rapid infusion of citrate buffered plasma or blood or large amounts of albumin

TABLE 30.7 Calcium Homeostasis-Related Disorders

Type of Disorder	Treatment	Examples
Disorders leading to hypercalcemia	Fluids, low-calcium diet, sulfate, loop diuretics, glucocorticoids, calcitonin, EDTA	Hyperparathyroidism Hypervitaminosis D Sarcoidosis Neoplasia Hyperthyroidism Immobilization Paget disease of the bone
Disorders of bone remodeling	Bisphosphonates, calcitonin, estrogen, calcium, fluoride, PTH + vitamin D	Osteoporosis

EDTA, ethylenediaminetetraacetic acid.

hyperphosphatemia, or both because of increased intestinal absorption of both calcium and phosphorous (29). In patients with chronic renal failure, CaSR agonists are able to limit progression of hyperparathyroidism and growth of the parathyroid gland.

Rickets and Osteomalacia

During the Industrial Revolution, there was widespread incidence of rickets in both children and adults, because inadequate exposure to sunlight prevented the biosynthesis of the precursor to active vitamin D in the skin. Both rickets and osteomalacia are metabolic bone diseases that are characterized by poor bone mineralization. Without adequate plasma levels of vitamin D and calcium, deposition of the calcium salts in the bone markedly decreases. Vitamin D supplementation (to improve intestinal absorption of calcium and mineralization of the bone) as well as oral calcium supplementation are required to treat these diseases once established. The incidence of rickets in the United States dropped dramatically through vitamin D-supplemented food programs. The increased use of milk substitutes (e.g., soy) and reduced exposure to sunlight has recently led to a rise in rickets. Rickets is still considered to be a worldwide health problem.

In addition to the classical environmental or nutritional cause of these diseases, both osteomalacia and rickets can have a pharmacologic origin as a result of chronic treatment with anticonvulsants (phenobarbital and phenytoin) or glucocorticoids. These agents interfere with intestinal absorption of calcium and, thereby, cause pseudohyperparathyroidism. As a result, an increase in bone turnover and a decrease in the formation of appropriately mineralized bone are observed. In these patients, treatment with vitamin D improves calcium absorption, ultimately enhancing mineralization of the bone.

Paget Disease of the Bone

Paget disease of the bone (Table 30.7) is characterized by excessive bone resorption, followed by replacement of the normally mineralized bone with soft, poorly mineralized tissue (30). It has been determined that the osteoclasts have an abnormal structure, are hyperactive,

and are present at elevated levels. Patients afflicted with this painful condition often suffer from multiple compression fractures. Administration of calcitonin and oral calcium and phosphate supplements had been the treatment of choice until the bisphosphonate risedronate sodium was approved by the U.S. Food and Drug Administration (FDA). Daily administration of risedronate sodium (see later discussion of bisphosphonates) results in a decreased rate of bone turnover and a decrease in the levels of serum alkaline phosphatase and urinary hydroxyproline, two biochemical markers of bone turnover (4,30). A significant advantage to treatment with the bisphosphonates is long-term suppression of the disease. Calcium supplementation, which often is necessary in these patients, must be dosed separately from risedronate sodium, because calcium- and aluminum- or magnesium-containing antacids interfere with absorption of the bisphosphonates.

DRUG THERAPIES USED TO TREAT OSTEOPOROSIS

Agents used in the treatment and prevention of osteoporosis are categorized as antiresorptive agents or bone-forming agents depending on the primary mechanism of action (31). For most of the effective therapies, bone mass is observed to increase for the first few years of treatment. Eventually, however, all the pits or lacunae will be filled in with new bone, and no additional increase in bone mass will occur. Antiresorptive agents have been shown to increase bone mass by as much as 8% to 9% at the lumbar spine and 3% to 6% in the femoral neck. Once a diagnosis of osteoporosis and the likely cause has been established, it is important to consider both patient fracture history and general medical history when selecting the appropriate treatment for a given patient (Table 30.8) (32).

Antiresorptive Agents

Estrogen Analogs—Estrogen Replacement Therapy

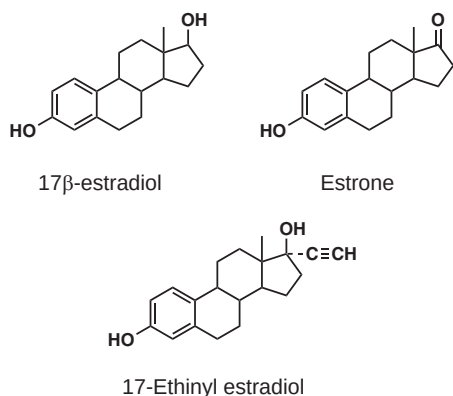
MECHANISM OF ACTION The precise mechanism by which estrogen prevents bone resorption has not been elucidated;

TABLE 30.8 Osteoporosis Treatment Selection Criteria (32)

Patient Data	Alendronate	Risedronate	Raloxifene	Calcitonin	Teriparatide
PM women: (+) osteoporosis/(+) fracture	X	X	X	X	X
PM women: (+) osteoporosis/(-) fracture	X				
Men: (+) osteoporosis	X				X
Corticosteroid-induced osteoporosis	X		X		
(+) Esophageal or upper GI disorder	(-)	(-)	X	X	
(+) Vasomotor symptoms	X	X	(-)	X	
(+) Venous thromboembolic event	X	X	(-)	X	
(+) Vertebral compression fracture pain				X	

X, recommended; (-), not recommended; GI, gastrointestinal; PM, postmenopausal.

however, it has been proposed to be associated with inhibition of osteoclast activity. Limited evidence supports the presence of estrogen-specific receptors (present on osteoclasts) having a biochemical role in the regulation of bone remodeling (15). Estrogen improves calcium absorption, promotes calcitonin biosynthesis, and increases the vitamin D receptors on osteoclasts. Although the primary mechanism of action remains unclear and its use is controversial at best, estrogen replacement therapy (i.e., 17 β -estradiol, estrone sodium sulfate, or 17-ethinyl estradiol) has value in the treatment and prevention of osteoporosis (31,33–35).



In light of the findings of the Women's Health Initiative study, the FDA recommends the use of short-term hormone replacement therapy (estrogen and progestin) in the prevention of osteoporosis only in select cases. The pharmacokinetics of the estrogens are covered in detail in Chapter 41 (33,34).

THERAPEUTIC EFFECTS Fractures of the spine, wrist, and hips decrease by 50% to 70%, and spinal bone density increases by 5%, in those women treated with estrogen within 3 years of the onset of menopause and for 5 to 10 years thereafter (5,13,36,37). The minimum dose required

and that which is considered to be standard therapy is 0.625 mg/d of conjugated estrogens (Premarin); however, a 0.3 mg/d dose of esterified estrogen (e.g., Menest) has been shown to be adequate for the prevention of osteoporosis (5). Estrogen replacement therapy (ERT) is available in several types of formulations, including transdermal patches (e.g., 17 β -estradiol: Climera, Estraderm, Menostar, or Vivelle).

Initiated at the onset of menopause, this therapy also has favorable effects on serum cholesterol levels (reduces low-density lipoprotein and elevates high-density lipoprotein levels). Women taking ERT have found relief from hot flashes, vaginal dryness, and urinary stress incontinence (31). It is recommended that the estrogen be combined with a progestin for those women with an intact uterus so as to decrease the risk of endometrial cancer (15).

Selective Estrogen Receptor Modulators

RALOXIFENE (EVISTA) Tamoxifen citrate, classified chemically as a triarylethylene, was developed as an antiestrogenic agent and as a selective estrogen receptor modulator (SERM). It is indicated as adjuvant therapy in the treatment of axillary node-negative or -positive breast cancer following partial or full mastectomy. Raloxifene hydrochloride, a benzothiophene derivative, also may be considered a semirigid analog of tamoxifen (Fig. 30.5) (36).

The two drugs are similar in that they both possess agonist activity in certain tissues (e.g., bone and cardiovascular) and antagonist activity in others (e.g., breast and uterus) (see Chapter 41) (6,31). Raloxifene hydrochloride, the first SERM approved for the prevention of osteoporosis in postmenopausal women, acts as an estrogen agonist on receptors in osteoblasts and osteoclasts but as an antagonist at breast and uterine estrogen receptors. This selective action means that this agent does not increase the risk of endometrial or breast cancer, as is the case with long-term tamoxifen therapy. Because this

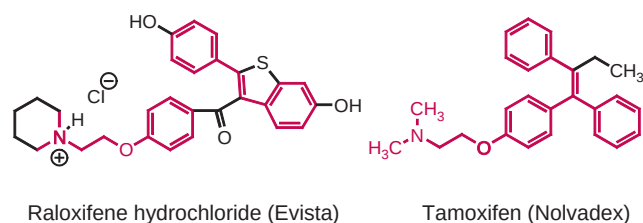


FIGURE 30.5 Structures of raloxifene and tamoxifen highlighting the structural similarity between the two drugs.

agent does not have a stimulatory effect at its receptors on most tissues, it does not prevent the hot flashes and other symptoms of menopause as estrogen does (5).

Therapeutic Action Clinical trials have shown that raloxifene hydrochloride, in combination with oral calcium supplementation, decreases the risk of vertebral fracture and promotes bone formation, albeit to a lesser extent than with estrogen. Raloxifene hydrochloride has been shown to have a beneficial effect on lipid profiles (13). Raloxifene hydrochloride should not be administered in combination with cholestyramine (decreased absorption), warfarin (prothrombin times and international normalized ratios must be monitored more closely), and those drugs that are highly protein bound, such as clofibrate, diazepam, ibuprofen, indomethacin, and naproxen.

Structure–activity Relationship From a structural perspective, the only pure antiestrogens are 7 α -substituted estrogens (31). In the triarylethylene class of agents (e.g., tamoxifen), the A ring phenol is critical for interaction with the portion of the estrogen receptor (ER) protein referred to as the activation factor-2 (AF-2) region, because it mimics the essential 3-phenol group found in estrogen (31). This interaction initiates a change in protein conformation to the form of the receptor able to interact with a specific deoxyribonucleic acid (DNA) sequence known as the estrogen responding element (ERE). As a result, activation of a specific group of genes occurs, and protein biosynthesis ensues. The orientation of the three aryl rings in a propeller type of arrangement also is important for tight receptor binding and biologic activity (31). In raloxifene hydrochloride, the substituted benzothiophene ring mimics the estrogen A ring; however because of the presence of a semirigid, amine-containing side chain, raloxifene is unable to interact with AF-2 (38,39). As a result, interaction with ERE is prevented and antiestrogenic action is observed in reproductive tissues. The raloxifene hydrochloride–ER complex, in concert with specific adapter proteins, is also able to interact with and activate a raloxifene responding element (RRE). Activation of this DNA sequence facilitates activation of another group of genes responsible for the production of proteins that allows for agonist action in nonreproductive tissues (38,39).

Pharmacokinetics Raloxifene hydrochloride is rapidly absorbed following oral administration, with an estimated 60% absorption, but it has a very low bioavailability (2%), associated with extensive phase II metabolism. The metabolites are excreted via the bile, with potential enterohepatic recycling that could account for the interaction with cholestyramine. Supportive of the enterohepatic recycling is the half-life of 28 hours. Metabolism of raloxifene hydrochloride occurs to a great extent in the intestine and consists of glucuronide conjugation catalyzed by uridine diphosphate glucuronosyltransferase (UGT) (40–42). The UGT1A family is responsible for intestinal human metabolism, as shown in Figure 30.6. Efflux by intestinal cells of the resulting glucuronide occurs via P-glycoprotein and multidrug resistance–related protein. The combination of rapid metabolism and efflux can account for the low bioavailability.

LASOFOXIFENE (FABLYN) Lasofoxifene is a very potent second-generation SERM in clinical trials in the United States and approved in the European Union for the treatment and prevention of osteoporosis in postmenopausal women (Fig. 30.7). At low doses, this agent has been shown to prevent bone loss and decrease serum cholesterol (dose range, 0.01 to 20.0 mg/day). In early clinical trials, lasofoxifene improved BMD in the lumbar spine by 3% (after 12 months of therapy), which is twice the improvement observed with similar treatment using raloxifene. Improvement in hip BMD was similar for both agents. Interestingly, a more pronounced reduction in low-density lipoprotein cholesterol was observed with lasofoxifene treatment than with raloxifene treatment. Despite the fact that phase III clinical trial data indicate that there is an increase in deaths from stroke and cancer, lasofoxifene was approved by the European Commission in March 2009 for the treatment of osteoporosis fractures in postmenopausal women.

OSPEMIFENE Ospemifene is a SERM in clinical trials for the treatment of postmenopausal osteoporosis and

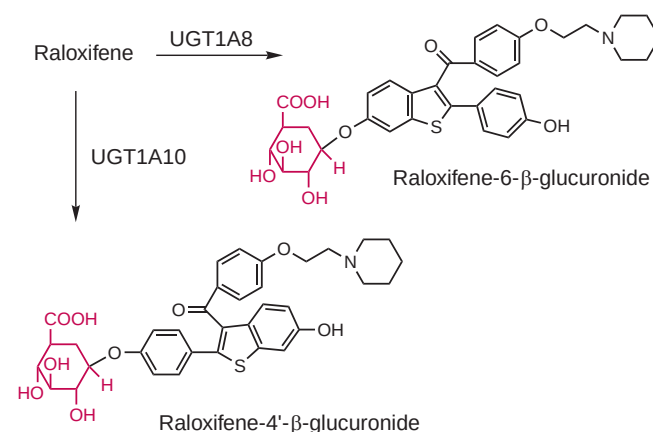


FIGURE 30.6 Metabolism of raloxifene.

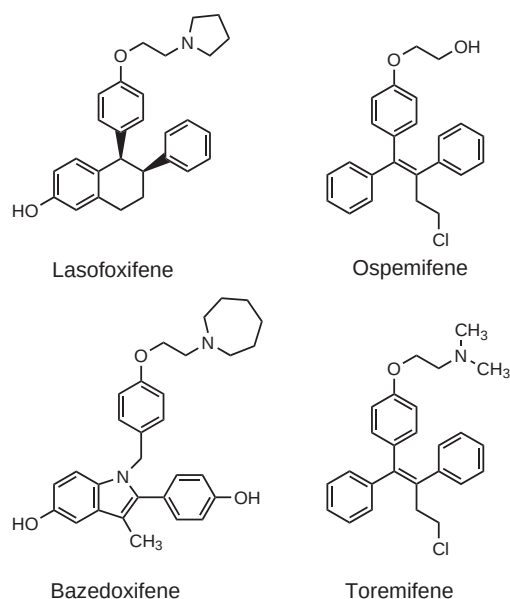


FIGURE 30.7 Investigational second-generation selective estrogen receptor modulators.

postmenopausal vaginal atrophy (Fig. 30.7). It is a known metabolite of toremifene, a triphenylethylene derivative used to treat breast cancer. Ospemifene has been shown to have beneficial effects on the bone without significant estrogen-related side effects. The beneficial effect observed on bone stems from this agent's ability to increase osteoblast proliferation and, as a result, to enhance bone mineralization as well as bone formation. Unlike tamoxifen, ospemifene does not induce osteocyte apoptosis (43). Presently the New Drug Application submitted to the FDA under the trade name of Ophena lists treatment of postmenopausal vaginal atrophy as the only indication.

BAZEDOXIFENE (VIVANT) Bazedoxifene is an indole-based SERM that is under investigation for the treatment and prevention of postmenopausal osteoporosis (Fig. 30.7). It also is being evaluated in combination with Premarin (conjugated estrogens). Bazedoxifene acetate displaces 17β -estradiol from ERs and has excellent binding affinity for the receptor itself. Unlike raloxifene hydrochloride, this agent does not cause hot flashes at the doses required to have a beneficial effect on bone. In addition, it does not cause uterine or mammary gland stimulation (44). Wyeth received approvable letters from the FDA for the use of bazedoxifene acetate in the prevention and treatment of postmenopausal osteoporosis in 2007 and 2008, respectively.

TOREMIFENE Toremifene citrate (Fig. 30.7), a SERM approved for the treatment of breast cancer (Fareston), is under investigation in patients undergoing androgen-deprivation therapy in the treatment of advanced prostate cancer. Preliminary results from phase III clinical trials indicate a 50% reduction in the rate of osteoporotic

fractures, as well as an increase in BMD in the lumbar spine, hip, and femur. Presently, the drug has not been approved for treatment of osteoporosis.

Bisphosphonates

MECHANISM OF ACTION The bisphosphonates are synthetic in origin and are designed to mimic pyrophosphate, where the oxygen in P-O-P is replaced with a carbon atom to create a nonhydrolyzable backbone (Fig. 30.8) (36,37). Because pyrophosphate is a normal constituent of bone, these analogs selectively bind to the hydroxyapatite portion of the bone and can bind to and stabilize calcium phosphate effectively (37,45). The bisphosphonates (Fig. 30.9) effectively inhibit osteoclast proliferation, decrease osteoclast activity, reduce osteoclast life span, and as a result, decrease the number of sites along the bone surface where bone resorption occurs (37). This is largely accomplished via inhibition of the mevalonate pathway within osteoclasts, as well as via inhibition of ATP-dependent enzymes. By these three mechanisms, the bisphosphonates are able to limit bone turnover and allow the osteoblasts to form well-mineralized bone without opposition (3). The precise mechanisms of action of these antiresorptive agents have not been elucidated; it is equally uncertain whether all the bisphosphonates act by a similar mechanism (37). To date, cell surface receptors have not been identified, nor has a second messenger system been detected. There is evidence that bisphosphonates inhibit the mevalonate pathway (specifically farnesyl diphosphate synthase) within osteoclasts, as well as inhibit ATP-dependent enzymes (impairing cellular energetics), both of which are associated with inhibition of osteoclast activity.

STRUCTURE-ACTIVITY RELATIONSHIPS From a structural perspective, the bisphosphonates have been proposed to have specific molecular interactions with their biologic target for drug action, even though precise structure-activity relationships (SARs) have not been elucidated. In fact, the exact molecular target is still under investigation. The central carbon of the geminal phosphonate has been substituted with a variety of functional groups to yield a large family of compounds with differing physicochemical and biologic properties (37). The SAR studies (Fig. 30.8) have concluded that a hydroxyl substituent (R_1) maximizes the affinity of the agent for the hydroxyapatite as

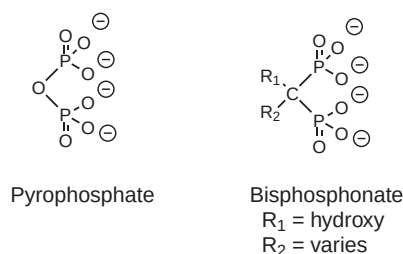
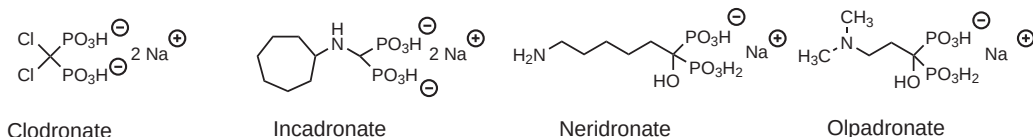


FIGURE 30.8 Bisphosphonate structure-activity relationships.

A. Investigational bisphosphonates:



B. Clinically used bisphosphonates.

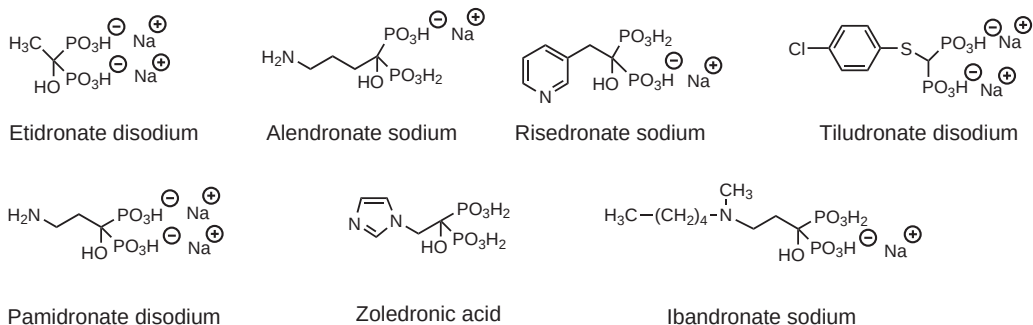


FIGURE 30.9 Bisphosphonates, both investigational and clinically used.

well as improves the antiresorptive character of the agent (31,46). The character of the R_2 substituent varies widely and clearly has a significant influence on the potency of this class of compounds (Fig. 30.9). The R_2 amino-substituted bisphosphonates (pamidronate disodium, alendronate sodium, and neridronate) are more potent than etidronate disodium and clodronate disodium (not available in the United States). The R_2 3-carbon amino linear chain for alendronate sodium is more potent than the R_2 2-carbon derivative pamidronate disodium and the R_2 6-carbon analog neridronate (31). Alkylation of the amine functional group improves potency as is demonstrated by compounds with N substituted amino alkyls at R_2 (e.g., olpadronate and ibandronate sodium) and those that contain rings at R_2 (e.g., risedronate sodium, incadronate, tiludronate disodium, and zoledronic acid). The third-generation analogs contain a basic heterocyclic side chain at R_2 tethered to the central carbon by a variety of linkages (potency: $\text{NH} > \text{CH}_2 > \text{S} > \text{O}$) (31,46). Because structural variation of R_2 has a significant effect on potency, it can be surmised that R_2 interacts at an “active site” and participates in a specific molecular interaction. The bisphosphonate itself as well as the hydroxyl group at R_1 also should be included as critical SAR features (31).

PHARMACOKINETICS To date, four generations of bisphosphonates have been developed for the treatment of osteoporosis (Fig. 30.9). Absorption of these agents from the gut is quite poor (1% to 5%) because of their polar nature, and as a therapeutic class, they have limited cellular penetration (13). Up to 50% of the actual absorbed dose is taken up specifically by the bone within 4 to 6 hours, and the rest is exclusively excreted by the kidney (6,36). Uptake of these agents in the bone is concentrated in areas of the bone that are actively undergoing remodeling (45). Between the selective uptake and

the rapid rate of clearance, the bisphosphonates enjoy a short circulating half-life and very limited drug exposure to nontarget tissues (37). Because the bisphosphonates are only released from the bone when the bone is resorbed, they have a tissue half-life of 1 to 10 years; however, these agents remain pharmacologically active only while they are exposed on bone resorption surfaces (45).

SPECIFIC DRUGS

Etidronate Disodium (Didronel) and Tiludronate Disodium (Skelid) Agents in the first generation of bisphosphonates that were dosed continuously produced poorly mineralized bone, because there was no interval for appropriate bone mineralization to occur (3). Subsequent studies that used a cyclic dosing schedule (400 mg/d for 2 weeks, followed by 2.5 months of calcium supplementation only) showed improvement in bone mineralization (13,43). Both of these bisphosphonates have been approved for treatment of Paget disease of the bone but not for the treatment of osteoporosis (5). Tiludronate disodium is approximately 10-fold more potent than etidronate disodium and, when given orally for 6 months (200, 400, or 800 mg/d), increases BMD by 2% (Fig. 30.9) (6). No further bone loss was detected in patients 6 months after cessation of therapy.

Alendronate Sodium (Fosamax) The second-generation agent alendronate sodium was the first bisphosphonate agent approved by the FDA for the prevention and treatment of osteoporosis and Paget disease of the bone and is 1,000-fold more potent than etidronate disodium (Fig. 30.9) (45,47). Alendronate sodium is also indicated for the treatment of glucocorticoid-induced osteoporosis. This derivative, when dosed continuously (5 to 10 mg/d for osteoporosis and 40 mg/d for Paget disease of the bone) and given with oral calcium supplements (500 mg/d),

produced well-mineralized bone and significantly improved BMD (7% in the spine and 4% in the hip) within 18 months (6). In addition, the vertebral fracture rate was shown to decrease by 47%. A side effect associated with alendronate sodium, chemical esophagitis, has been attributed to inadequate intake of water and lying down after taking the medication (2,3,13,36). Specific patient instructions were developed to limit the incidence of upper gastrointestinal problems and include: 1) taking the medication with 6 to 8 ounces of water on arising in the morning, 2) remaining in an upright position for at least 30 minutes after taking the medication, and 3) delaying drinking other liquids/eating for at least 30 minutes, if not 1 to 2 hours, to allow maximal absorption of the agent. To enhance absorption, calcium supplements and any aluminum- or magnesium-containing antacids should be dosed separately from agents in this class. These agents are not recommended in patients with renal impairment (serum creatinine, <2.5 mg/dL), a history of esophageal disease, gastritis, or peptic ulcer (5). In an attempt to address the inconvenience associated with tablet administration, a once-weekly, 70-mg buffered effervescent formulation of alendronate is under investigation.

Risedronate Sodium (Actonel) The third-generation agent risedronate sodium has been approved for the treatment of osteoporosis (both postmenopausal and in men), Paget disease of the bone, and glucocorticoid-induced osteoporosis (Fig. 30.9). Risedronate sodium is 1,000- to 5,000-fold more potent than etidronate disodium. At the end of an 18-month study, 53% of patients who took risedronate sodium for 2 months remained in remission, as compared to 14% of patients who took etidronate disodium, an earlier-generation bisphosphonate, for 6 months. Oral administration of this agent suffers from the same problems as that of other bisphosphonate agents. Risedronate sodium should not be given to patients with creatinine clearance of less than 30 mL/min. A once-weekly, delayed-release formulation (Atelvia) that can be taken immediately after breakfast with 4 ounces of water was approved by the FDA in October 2010. Other oral formulations of risedronate sodium include tablets to be consumed daily (5 mg), weekly (35 mg), and monthly (150 mg).

Ibandronate Sodium (Boniva) Ibandronate sodium is approved for the treatment and prevention of osteoporosis in postmenopausal women and has a mechanism of action that is identical to the other bisphosphonate agents (Fig. 30.9). Administered daily (2.5 mg), ibandronate sodium has been clinically shown to reduce the risk of vertebral fractures by 62% (48). If administered on an intermittent basis (20 mg), it reduces the risk of vertebral fractures by 50%. Ibandronate sodium (2.5 mg daily), along with 500 mg of supplemental calcium, has been clinically shown to increase BMD in the hip (1.8%), femoral neck (2.0%), and lumbar spine (3.1%). A 150-mg formulation has been approved by the FDA for once-monthly administration as well as a 3-mg IV formulation for quarterly administration.

ADDITIONAL DOSAGE FORMS

A unique formulation of alendronate sodium, FOSAMAX PLUS D, includes 70 mg of alendronate sodium and 2,800 IU or 5,600 IU of vitamin D₃ (i.e., a 7-day supply of both the bisphosphonate and vitamin D). This formulation should not be used in patients with severe kidney disease or low serum calcium levels and should not be the only therapy used to correct a vitamin D deficiency.

Risedronate sodium with calcium carbonate (Actonel with Calcium) represents an additional type of packaging for this class of agents. It addresses the Surgeon General's Report on Bone Health and Osteoporosis, which states that treatments for osteoporosis need to be made simpler and more structured. Sold in units that contain a 1-month supply, each week of therapy includes a total of seven tablets, including one 35-mg tablet of risedronate and six 500-mg tablets of calcium carbonate.

The oral bioavailability of this agent is extremely poor (0.6%) and is adversely affected by the presence of food, beverages other than water, and other medications, including calcium or vitamin D supplements and antacids. Because of the increased calcium content in mineral water, patients should not take this medication with this type of water. Drugs that inhibit gastric acid secretion (e.g., H₂ antagonists and proton pump inhibitors) actually promote ibandronate sodium absorption. Like the other agents in this therapeutic class, ibandronate sodium is not metabolized, and that which is not bound to the bone (40% to 50% of the absorbed dose) is eliminated renally unchanged. It does not inhibit the CYP450 isozymes. This agent does not require any dosage adjustment for patients with hepatic impairment or mild to moderate renal impairment (creatinine clearance, >30 mL/min). Ibandronate sodium should not be prescribed for patients with severe renal impairment (creatinine clearance, <30 mL/min).

Zoledronic Acid (Reclast) Zoledronic acid is approved for the treatment of glucocorticoid-induced osteoporosis and prevention and treatment of postmenopausal osteoporosis, male osteoporosis, and Paget disease of the bone (Fig. 30.9). For the treatment of osteoporosis, zoledronic acid is formulated as a 5-mg, once-yearly IV infusion. The frequency of IV infusion decreases to 5 mg every 2 years for the prevention of osteoporosis. In order to prevent hypocalcemia, concomitant calcium (1,500 mg) and vitamin D (800 to 1,000 IU) intake and/or supplementation is recommended in patients being treated for osteoporosis. On the day of treatment, patients should drink at least two glasses of water and eat normally.

Osteonecrosis of the jaw has been reported in patients receiving IV bisphosphonate therapy (49). The majority of the patients who developed osteonecrosis of the jaw were undergoing chemotherapy (typically for multiple

myeloma, breast, prostate, or lung cancers), taking corticosteroids, and had undergone a dental procedure (e.g., tooth extraction). The FDA recommends that patients receive a thorough dental examination before initiation of IV bisphosphonate therapy and that they avoid invasive dental work during treatment.

In October 2010, the FDA approved safety labeling changes for the bisphosphonates in response to reports of an increase in risk of atypical femur fracture in patients undergoing long-term bisphosphonate therapy (50,51). In addition, an increase in the risk of developing esophageal cancer has surfaced in patients with approximately 5 years of oral bisphosphonate use. Because of these adverse effects, it is recommended that clinicians weigh the benefits against the potential risks.

The remaining bisphosphonates, pamidronate disodium and zoledronic acid, are approved for treatment of hypercalcemia of malignancy as well as other cancer conditions and will be discussed later in the chapter.

Calcitonin (Calcimar [IV, subcutaneous]; Miacalcin and Fortical [nasal spray])

Calcitonin (see earlier discussion in this chapter) has been approved for the treatment of postmenopausal osteoporosis, hypercalcemia of malignancy, and Paget disease of the bone. Several sources are available (e.g., eel, human, salmon, and porcine). The calcitonin isolated from salmon is the preferred source, because it has greater receptor affinity and a longer half-life than the human hormone (3,7). Calcitonin is commercially available as synthetic calcitonin-salmon, which contains the same linear sequence of 32 amino acids, as occurs in natural calcitonin-salmon. Calcitonin-salmon differs structurally from human calcitonin at 16 of 32 amino acids (see Fig. 30.10 for primary structure differences between human and salmon calcitonin). The pharmacologic activity of these calcitonins is the same, but calcitonin-salmon is approximately 50-fold more potent on a weight basis than human calcitonin with a longer duration of action. The duration of action for calcitonin salmon is 8 to 24 hours following intramuscular (IM) or subcutaneous (SC) administration and 0.5 to 12.0 hours following IV administration. The parenteral dose required for the

treatment of osteoporosis is 100 IU/d (46). Initially only available by IM or SC injection, the peptide hormone calcitonin-salmon is available as a nasal spray (Miacalcin) and as a rectal suppository (6). A recombinant DNA form of calcitonin-salmon (Fortical) is available as a nasal spray. The bioavailability of calcitonin-salmon nasal spray shows great variability (range, 0.3% to 30.6% of an IM dose). It is absorbed rapidly from the nasal mucosa, with peak plasma concentrations appearing 30 to 40 minutes after nasal administration, compared with 16 to 25 minutes following parental dosing. Calcitonin-salmon is readily metabolized in the kidney, with an elimination half-life calculated at 43 minutes. As a result, the intranasal dose required is 200 IU/d (3). Once the Miacalcin nasal pump has been activated, the bottle may be kept at room temperature until the medication is finished (2 weeks).

THERAPEUTIC APPLICATION Calcitonin therapy requires the concomitant oral administration of elemental calcium (500 mg/d, see Table 30.9). Clinical studies have shown that the combination of intranasal calcitonin-salmon (200 IU/d), oral calcium supplementation (>1,000 mg/d of elemental calcium), and vitamin D (400 IU/d) has decreased the rate of new fractures by more than 75% and has improved vertebral BMD by as much as 3% annually (3). Calcitonin prevents the abnormal bone turnover characteristic of Paget disease of the bone and has antiresorptive activity. In the presence of calcitonin, the osteoclast brush borders disappear, and the osteoclasts move away from the bone surface undergoing remodeling (52). Side effects are significantly more pronounced when calcitonin-salmon is administered by injection and can include nausea, vomiting, anorexia, and flushing. Because calcitonin-salmon is protein in nature, the possibility of a systemic allergic reaction should be considered, and appropriate measures for treatment of hypersensitivity reaction should be readily available. Although calcitonin-salmon does not cross the placenta, it may pass into breast milk. Calcitonin-salmon is a possible alternative to ERT; however, only limited evidence suggests that it has efficacy in women who already have fractures. Resistance to calcitonin-salmon can result from the development of neutralizing antibodies (53).

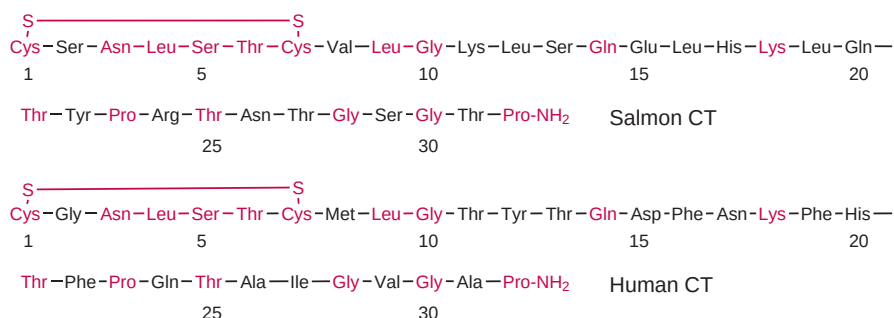


FIGURE 30.10 Primary structures of salmon and human calcitonin (CT). Similarities are highlighted in red.

In addition to its antiresorptive action via suppression of osteoclast activity, calcitonin-salmon exhibits a potent analgesic effect and has provided considerable relief to those patients suffering from the pain associated with Paget disease of the bone and osteoporosis. This analgesic effect is a result of calcitonin-stimulated endogenous opioid release. The potency of this analgesic effect has been demonstrated to be 30- to 50-fold that of morphine in selected patients. Calcitonin is preferred over estrogen and the bisphosphonates when treatment of both osteoporosis and related bone pain is warranted.

Bone-Forming Agents

Teriparatide (Forteo)

In 2002, the FDA approved teriparatide for the treatment of postmenopausal osteoporosis in patients who have a high risk of fracture, for the treatment of glucocorticoid-induced osteoporosis, and to increase bone mass in men with primary or hypogonadal osteoporosis who have a high risk of fracture (54). Teriparatide is recombinant human PTH 1-34 (Fig. 30.2), the biologically active portion of the endogenously produced preprohormone. Unlike the bisphosphonates, which are classified as bone restorative agents, teriparatide is the first approved bone-forming agent. Bone formation is possible because of the ability of this agent to increase the number of osteoblasts. Although teriparatide enhances the function of both osteoclasts and osteoblasts, the exposure incidence dictates its effect on the skeleton. If administered once daily or intermittently, teriparatide preferentially enhances osteoblastic function, and bone formation occurs. Continuous exposure to endogenous PTH may result in poor skeletal composition because of enhanced osteoclast-mediated bone resorption (52). After 18 months of treatment, lumbar BMD increased up to 12% in postmenopausal women. After 10 months of treatment, 53% of men had an increase of 5% or greater in spine BMD. The risk for developing new vertebral fractures was reduced by 65% after 21 months of treatment, and the number of non-vertebral fragility fractures was reduced by 53% (35). There is limited evidence that teriparatide can regrow jaw bone that has been damaged by osteonecrosis and periodontitis.

A black box warning from the FDA advises that treatment with teriparatide should be avoided in patients with an increased baseline risk for osteosarcoma and that treatment should not exceed 2 years in duration.

Administered as a once-daily, 20- μ g SC injection in the thigh or abdominal wall, teriparatide is a clear, colorless liquid that is available as a 750 μ g/3 mL, prefilled, disposable pen that requires refrigeration. Concurrent calcium (1,000 mg) and vitamin D (400 IU) supplementation is recommended. Treatment for longer than 2 years is not recommended. Teriparatide is rapidly absorbed,

demonstrates 95% bioavailability, and is quickly eliminated via both hepatic and extrahepatic routes. The half-life is 1 hour when administered SC. Metabolic studies have not been performed on teriparatide; however, the entire PTH preprohormone has been shown to undergo enzyme-mediated transformations in the liver. Dizziness and leg cramps are the most commonly reported adverse side effects.

Temporary increases in serum calcium levels occur following administration of teriparatide. As a result, this agent is contraindicated in patients who are predisposed to hypercalcemia. Some evidence suggests that these elevations in serum calcium levels may cause a patient who is taking digitalis to experience digitalis toxicity (55). Teriparatide should not be prescribed to patients with Paget disease of the bone, children, young adults, women who are pregnant or nursing, and patients who have received skeletal radiation therapy (52). Because of an increased incidence of osteosarcoma (malignant bone tumors) observed in rats, teriparatide also carries a black box warning.

Inorganic Salts

CALCIUM SALTS Appropriate intake of calcium during childhood, adolescence, and early adulthood increases peak BMD and may reduce the overall risk of developing osteoporosis. For those who are at low risk of developing osteoporosis and have adequate BMD, consumption of the recommended amounts of calcium (1,300 mg/d of elemental calcium for teenagers, 1,000 mg/d for premenopausal women and men, and 1,200 mg/d for postmenopausal women) typically is sufficient to prevent bone loss (56,57). This often can be accomplished by eating a well-balanced diet. For patients with established osteoporosis or areas of poorly mineralized bone, calcium supplementation alone is not sufficient to reverse the bone loss or to significantly improve mineralization of the bone (13).

It should be noted that a study conducted by Bolland et al. (58) reported that calcium supplementation in postmenopausal women may be correlated with significant increases in the rates of vascular events.

The actual amount of elemental calcium that is present in the available calcium salts varies considerably; however, no one particular salt has been identified as an exceptional source of elemental calcium (Table 30.9). Absorption of calcium from the gastrointestinal tract (25% to 40%) improves under acidic conditions; therefore, those medications that change the acidic environment of the stomach (e.g., H_2 antagonists and proton pump inhibitors) have an adverse effect on calcium absorption (3). Total daily doses of elemental calcium that exceed 500 mg should be spaced out over the day to improve absorption (5,56). The more water soluble and, therefore, more easily absorbed salts (e.g., citrate, lactate, and gluconate) are less dependent on the acidic environment for appropriate absorption and would be

TABLE 30.9 Percent of Elemental Calcium Content in Various Salts (3)

Salt	Calcium (%)	Elemental Calcium (mg/tablet)
Calcium carbonate	40	
Tums (500 mg chewable)		200 mg
Titilac (1 g/5 mL suspension)		400 mg/5 mL
Alka-Mints (850 mg chewable)		340 mg
Os-Cal 500 (1,250 mg tablet)		500 mg
Viactive (1,250 mg chewable)		500 mg
Tricalcium phosphate	39	
Calcium chloride	27	
Tribasic calcium phosphate	23	
Posture (1,565.2 mg tablets)		600 mg
Calcium citrate	21	
Citrical (950 mg tablets)		200 mg
Citrical Liquitab (2,376 mg effervescent tablets)		500 mg
Calcium lactate	13	
Generics (325 mg tablets)		42 mg
Generics (650 mg tablets)		84 mg
Calcium gluconate	9	
Neo-Calglucon (1.8 g/5 mL syrup)		115 mg/5 mL

appropriate alternatives for patients who produce low levels of acid. Calcium carbonate is a poorly soluble form of calcium, but it is inexpensive and only requires the patient to take a few tablets per day with acidic food or beverages like citrus juice (56).

SODIUM FLUORIDE Sodium fluoride (NaF) promotes the proliferation and activity of osteoblasts and is classified as a nonhormonal bone-forming agent. Because treatment with NaF induces bone formation, it is essential that this therapy be coupled with oral calcium supplementation (1,000 mg/d). Additionally, NaF exhibits moderate anti-resorptive activity, because it inhibits osteoclastic activity when it is absorbed into the bone matrix. In the treatment of osteoporosis, the therapeutic window for this agent is fairly narrow: Doses less than 45 mg/d are subtherapeutic, and doses in excess of 75 mg/d impair bone mineralization. In addition, the bone that is formed in the presence of NaF is neither as well mineralized nor as strong as normal bone tissue. In fact, some studies have demonstrated that patients taking sodium fluoride have increased bone fragility despite the increase in bone mass and, as a result, have an increased nonvertebral fracture rate as compared to the placebo group (5,31,46). As a result, its use in the treatment of osteoporosis has not been approved and is considered to be somewhat controversial. Several studies have examined the benefits of continuous versus cyclic dosing of NaF in the treatment of osteoporosis. Intermittent dosing (25 mg twice a day for 12 months, followed by 2 months of calcium supplementation alone) of a slow-release formulation of NaF with 400 mg of calcium citrate was shown to effectively

improve bone mass (vertebra, 5% per year; femoral neck, 2% per year) and to decrease the number of vertebral fractures (59,60).

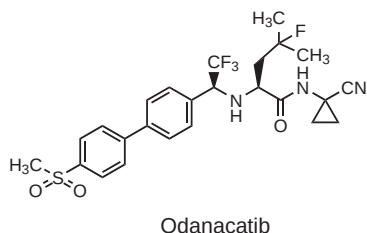
Monoclonal Antibody-Based Therapies

RANKL Inhibitor

DENOSUMAB (PROLIA) Denosumab (Prolia) has been approved by the FDA for the treatment of postmenopausal women with osteoporosis at high risk for fracture. Denosumab is a fully human monoclonal antibody to RANKL, where it functions as a RANKL inhibitor (Fig. 30.4). The RANKL receptor is expressed on the surface of osteoclasts and osteoclast precursors. When bound to its receptor, RANKL promotes the formation and activation of osteoclasts. To balance the effects of RANKL, osteoblasts produce osteoprotegerin, which binds to RANKL and prevents it from binding to and activating its receptor, modulating the production and activation of osteoclasts (61). When an individual develops osteoporosis, this balance is “disrupted,” and RANKL overwhelms osteoprotegerin activity, causing significant bone loss. Denosumab was designed to mimic the biochemical effects of osteoprotegerin. Studies show that denosumab is more effective in improving BMD (4% to 7%) than weekly administration of alendronate (5%) (61). Denosumab is reported to reduce the risk of vertebral fracture (68%), hip fracture (40%), and nonvertebral fracture (20%). This agent is administered subcutaneously once every 6 months and has been associated with adverse events including back pain (35%), serious infections, hypocalcemia, and osteonecrosis of the jaw.

Cathepsin K Inhibitor

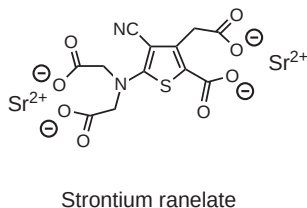
ODANACATIB Odanacatib selectively inhibits cathepsin K, a cytosolic cysteine protease. Produced by osteoclasts, this protease is responsible for degradation of the collagen matrix found in bone tissue.



Phase II clinical studies with odanacatib show that treatment for 2 years resulted in an increase in BMD at the lumbar spine and hip (62). Odanacatib is being evaluated in large-scale phase III clinical trials on vertebral, hip, and nonvertebral fractures.

Miscellaneous Therapies

Strontium ranelate (Protelos) is an orally active agent that can be classified as both an antiresorptive agent and a bone-forming agent (63,64).



It is able not only to stimulate replication of preosteoblastic cells to promote bone formation but also is able to decrease osteoclastic activity to prevent bone resorption. Biochemical markers for bone formation (e.g., bone-specific alkaline phosphatase), which normally decrease in the presence of antiresorptive therapy, are elevated in the presence of strontium ranelate (65). Strontium ranelate is active by virtue of the strontium cation and not the anion portion of the drug. Lumbar spine BMD increased 11.4% in patients treated with this new agent. Although approved as second-line therapy for the treatment of osteoporosis in a number of European nations, Protelos is not FDA approved (66).

The hydroxymethylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors, otherwise known as the statins, have been found to increase bone formation via enhanced activity of the bone morphogenic protein 2 (*BMP-2*) gene. This gene increases osteoblast differentiation. In addition, by inhibiting HMG-CoA reductase, the statins not only prevent the biosynthesis of cholesterol but also prevent the formation of compounds associated with osteoclast activation (67). Unfortunately, clinical data from several large studies conflict, and further study is warranted before the statins can be considered as a viable treatment for osteoporosis (68).

There are a number of investigational agents in the pipeline that act through very novel mechanisms of action (Table 30.10). These include calcilytic inhibitors, inhibitors of protein tyrosine kinase C-src, and Dkk1 monoclonal antibodies (66,69).

DRUG THERAPIES USED TO TREAT HYPERPARATHYROIDISM

Increased levels of PTH lead to moderately to severely elevated serum calcium concentrations and alterations in phosphorous metabolism (2). To modulate the levels of PTH released from the parathyroid gland chief cells, regulation of CaSR sensitivity is required. An agonist at this receptor, a calcimimetic, serves to activate the receptor, whereas an antagonist at this receptor is classified as a calcilytic. There are two types of calcimimetic agents: those that activate the CaSR directly (type I), and those that require the presence of a cation, such as calcium or magnesium (type II), for activation (70). Type I calcimimetics are polycations (e.g., magnesium and neomycin). The first and second generations of type II calcimimetics are phenylalkylamine based. They have an indirect/allosteric action on CaSR mediated by a conformational alteration of these receptors.

Cinacalcet Hydrochloride (Sensipar)

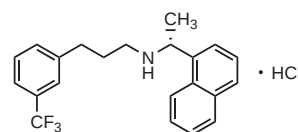


TABLE 30.10 Classes of Experimental Agents for Treatment of Abnormal Calcium Homeostasis

Class	Action
Antiresorptive Agents	
Cathepsin K inhibitors	Prevent degradation of collagen matrix
C-src kinase inhibitors	Prevent formation of osteoclast sealing zone
Bone-Forming Agents	
Calcilytic agents	Inhibit calcium-sensing receptors to promote PTH release
Dkk1 monoclonal antibody	Inhibitor of negative regulators of Wnt signaling

Cinacalcet is the first type II calcimimetic agent approved that improves CaSR sensitivity to calcium (29,71). When calcium is bound to the CaSR, phospholipase C is activated, and the secretion of PTH is inhibited. In the presence of cinacalcet, not only is a decrease in PTH levels observed, but a decrease in serum calcium and phosphorous levels is also observed. This represents a significant therapeutic advantage over vitamin D-based treatments for secondary hyperparathyroidism (29). Cinacalcet hydrochloride is a second-generation calcimimetic approved for the treatment of secondary hyperparathyroidism in patients with chronic kidney disease on dialysis and for the treatment of hypercalcemia in patients with parathyroid cancer. It can be used alone, with vitamin D, and/or with a phosphate binder (72).

DRUG THERAPIES USED TO TREAT HYPERCALCEMIA OF MALIGNANCY

Zoledronic Acid (Zometa)

Zoledronic acid, a bisphosphonate, was approved by the FDA in 2001 for the treatment of hypercalcemia of malignancy, a metabolic complication that can be life-threatening (Fig. 30.9). Hypercalcemia of malignancy can occur in up to 50% of patients diagnosed with advanced breast cancer, multiple myeloma, and non-small-cell lung cancer. This condition arises when chemical moieties produced by the tumor cause overstimulation of osteoclasts. When there is an increase in bone degradation, there is a concomitant release of calcium into the plasma. When serum concentrations of calcium rapidly elevate, the kidneys are unable to handle the overload, and hypercalcemia results. This can lead to dehydration, nausea, vomiting, fatigue, and confusion. Zoledronic acid effectively decreases plasma calcium concentrations via inhibition of bone resorption (inhibition of osteoclastic activity and induction of osteoclast apoptosis). It also prevents the increase in osteoclastic activity caused by tumor-based stimulatory factors. In addition, zoledronic acid has been approved by the FDA for the treatment of multiple myeloma and bone metastases associated with solid tumor-based cancers (e.g., prostate and lung) (73).

Cancer treatment-induced bone loss is a major adverse effect associated with endocrine-based cancer therapies. These therapies may depress ovarian function (e.g., goserelin acetate), decrease ER activation (e.g., ER antagonist), and/or inhibit estrogen biosynthesis (e.g., aromatase inhibition) all of which will lead to significant bone loss. Zoledronic acid has demonstrated efficacy in reducing or delaying these complications.

The maximum recommended dose for the treatment of hypercalcemia of malignancy is 4 mg every 3 to 4 weeks. A clinically significant deterioration in renal

function occurs when single doses of this agent exceed 4 mg and the infusion duration is less than 15 minutes (74). It is recommended that patients be well hydrated before infusion. If serum calcium levels do not fall to normal levels, retreatment is appropriate, but retreatment is not recommended until 7 days have elapsed from the initial treatment. For the treatment of multiple myeloma and metastatic bone lesions, a 4-mg initial dose is recommended, followed by additional doses every 3 to 4 weeks for 9 to 15 months (prostate cancer, 15 months; breast cancer, 12 months; other solid tumors, 9 months).

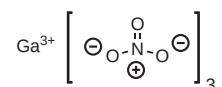
Zoledronic acid is a white, crystalline powder that is available in vials for reconstitution for IV infusion over at least 15 minutes. It does not undergo metabolic transformation and does not inhibit CYP450 enzymes. Clearance of this agent is dependent on the patient's creatinine clearance, not on dose. Serum creatinine levels should be evaluated prior to every treatment. Zoledronic acid is contraindicated in patients with severe renal impairment.

Zoledronic acid should not be mixed with infusion solutions that contain calcium (e.g., lactated Ringer's) and should be administered via IV infusion in its own line. Because of the possibility of a serious deterioration in renal function, the manufacturer requires strict adherence to the infusion duration being no less than 15 minutes.

Pamidronate Disodium (Aredia)

Pamidronate disodium, a second-generation bisphosphonate, is 100-fold more potent than etidronate disodium for the treatment of hypercalcemia of malignancy (Fig. 30.9) (6). It has also been approved for the treatment of Paget disease of the bone and for osteolytic bone metastases of breast cancer and osteolytic lesions of multiple myeloma. When used to treat bone metastases, pamidronate disodium decreases osteoclast recruitment, decreases osteoclast activity, and increases osteoclast apoptosis (75). Administered by IV infusion, a single dose is typically sufficient for the treatment of hypercalcemia and Paget disease of the bone. In the treatment of osteolytic lesions of multiple myeloma, monthly administration is indicated. Pamidronate disodium is cleared renally and therefore is contraindicated in patients with deteriorated renal function.

Gallium Nitrate (Ganite)



Gallium nitrate has been approved for the treatment of hypercalcemia of malignancy in patients who do not

respond to hydration (46). Its effectiveness stems from its ability to inhibit bone resorption despite the presence of tumor-derived factors that promote calcium loss from the bone. Administered by infusion over 24 hours, the typical dose is 200 mg/m²/day for 5 consecutive days. A lower dose is recommended if the symptoms of hypercalcemia are mild (100 mg/m²/d for 5 days). Steady-state is achieved in 24 to 48 hours. Maintenance of patient hydration is essential during treatment. Gallium nitrate is not significantly metabolized and is largely excreted through the kidneys. It is contraindicated in patients with severe renal impairment. Renal function should be closely monitored in all patients receiving this agent.

DRUG THERAPY USED IN THE TREATMENT OF HYPOPARATHYROIDISM

Parathyroid Hormone [rDNA 1-84] (Preos)

In 2007, the FDA Office of Orphan Products Development granted orphan drug designation for Preos for the treatment of hypoparathyroidism (Fig. 30.2). Structurally identical to endogenous PTH, the recombinant human PTH is administered once daily as an SC injection. Although Preos shares a common mechanism of action with teriparatide, it has been suggested that the C-terminal region may possess valuable biologic activity mediated by a novel receptor that specifically interacts with this portion of the peptide hormone.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

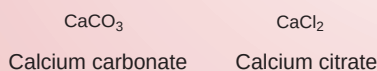
Kathryn Neill, PharmD

The pharmacist recommends changing the calcium carbonate to calcium citrate 200 mg with 200 IU of vitamin D (two tablets three times daily); limiting caffeine intake to no more than two servings daily; and separating the administration of ramipril from calcium by 2 hours.

Chemical Analysis

Victoria F. Roche and S. William Zito

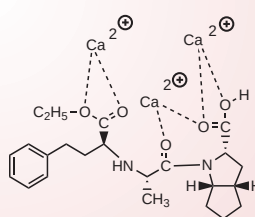
The oral bioavailability of calcium ranges from approximately 5% to 45%. Absorption is dependent on salt form, dose, presence of gastric acid, and presence of activated vitamin D. Calcium absorption is increased when administered with meals as a result of increased secretion of gastric acid in response to a food bolus. The increased gastric acidity releases calcium from water insoluble salt forms (like calcium carbonate) and allows its absorption. For example, calcium carbonate absorption increases as much as 30% when taken with a meal. Absorption of calcium citrate is greater than other salt forms because it is freely water soluble and does not require the action of strong acid to break down the salt to release the calcium. Therefore, it is the most efficacious calcium supplement for patients with achlorhydria.



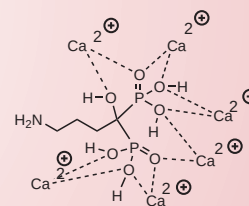
As it dissolves in the gastric acid, calcium carbonate can create gas (CO_2) in the stomach, resulting in stomach upset. Because

it is not a CO_2 generator, this adverse effect is not commonly experienced with calcium citrate. Absorption of calcium in the duodenum is dependent upon active vitamin D. Dietary vitamin D intake is rarely adequate to meet the recommended 800–1,000 units per day (some researchers now recommend 2000 units per day), and supplementation is usually necessary, especially for seniors who may have even higher daily requirements.

Calcium should be separated from doses of other medications by 2 hours to prevent chelation, which can result in decreased absorption and therapeutic effects. As a divalent cation, calcium will be chelated by electron-rich (nucleophilic) atoms like oxygen. Both ramipril and alendronate have many nucleophilic oxygen atoms that could sequester calcium ion, as shown below.



Calcium chelated by ramipril



Calcium chelated by alendronate

It would be important to counsel PJ to take alendronate with a full glass of water and to remain in a sitting or standing position for at least 30 minutes because this bisphosphonate can induce a chemical esophagitis that could exacerbate the distress PJ is experiencing from other GERD-related pathologies.

CASE STUDY

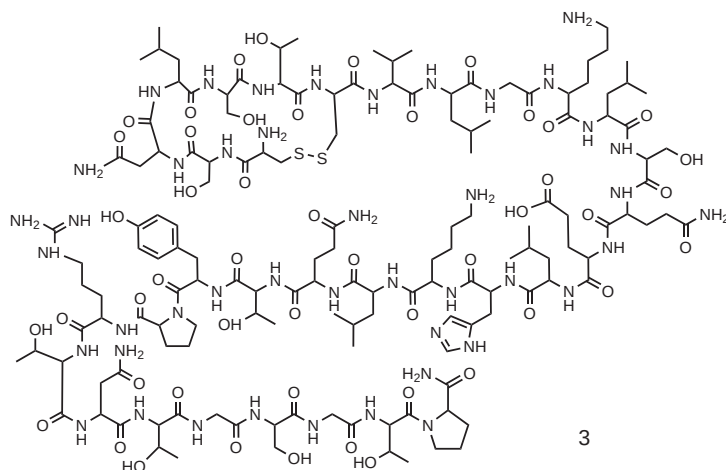
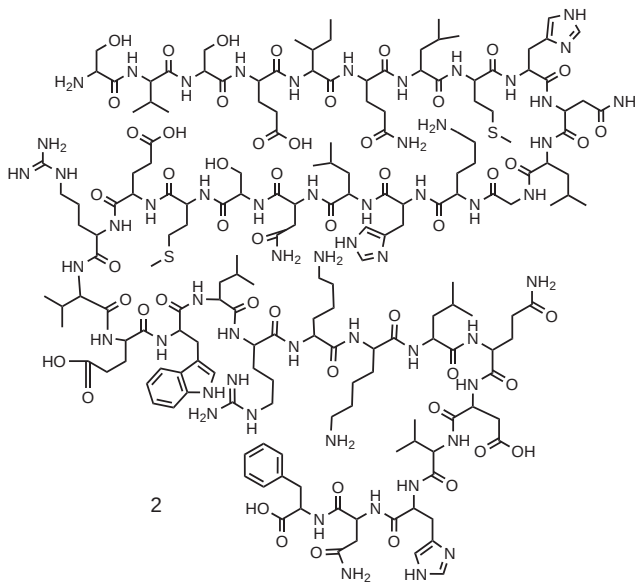
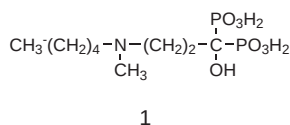
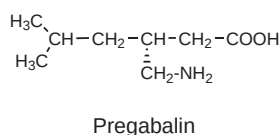
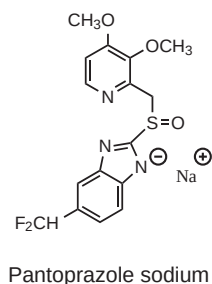
Victoria Roche and S. William Zito

JA is a 73-year-old woman who had polio 58 years ago. Although she had maintained a very active lifestyle throughout her youth and most of her adulthood, postpolio syndrome (PPS) has significantly restricted her mobility and she now uses a wheelchair at home and a power chair to get around outside. JA's relative lack of activity during the past 5 years has caused a troublesome weight gain, and she now carries close to 185 pounds on a 5'5" frame. The extreme fatigue that is a hallmark of PPS is complicated by osteoarthritis in the joints of her upper extremities that came from years of walking in full leg braces with the aid of canes. Some days the fatigue and the joint pain are so disabling that she just stays in bed and sleeps, saying it is the only place she is comfortable.

JA was recently diagnosed with Paget's disease of the bone, a disorder characterized by the deterioration and demineralization of bone. In addition to exacerbating osteoarthritis and leading to loss of

hearing, Paget's disease predisposes patients to the development of osteosarcoma, a rare and therapeutically challenging bone cancer. It is important that JA be treated for this disease because, in addition to these serious complications and the compounding of her pain, she would be at high risk for compression fractures from the stress she must put on her arms when she transfers to and from her wheelchair and power chair.

JA's current medications include the proton pump inhibitor pantoprazole sodium (Protonix) for gastric hyperacidity and pregabalin (Lyrica) for neuropathic pain in her feet. She takes 1,000 mg of elemental calcium daily (2 Viactiv calcium chews) and 2,000 IU of vitamin D. She is depending on you, her ambulatory care pharmacist, to select the right therapy to keep her going in the face of this newest health challenge. You contemplate the structures of three potential drug candidates shown below.



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 31

Nonsteroidal Anti-Inflammatory Drugs

RONALD BORNE, MARK LEVI, AND NORMAN WILSON

Drugs Covered in This Chapter

ANTIPYRETIC ANALGESICS

- Acetaminophen

ANTI-INFLAMMATORY ANALGESICS

- Aspirin and other salicylates
- Bromfenac
- Diclofenac
- Diflunisal
- Etodolac
- Fenoprofen
- Flurbiprofen
- Ibuprofen
- Indomethacin
- Ketoprofen
- Ketorolac
- Mefenamic acid
- Meclofenamic acid
- Meloxicam

- Nabumetone
- Naproxen
- Oxaprozin
- Piroxicam
- Sulindac
- Suprofen
- Tolmetin

COX-2 INHIBITORS

- Celecoxib

DISEASE-MODIFYING DRUGS FOR ARTHRITIS

- Abatacept
- Adalimumab
- Anakinra
- Certolizumab
- Etanercept
- Gold salts

- Golimumab
- Hydroxychloroquine
- Infliximab
- Leflunomide
- Methotrexate
- Rituximab
- Sulfasalazine
- Tocilizumab

DRUGS FOR THE TREATMENT OF GOUT

- Allopurinol
- Colchicine
- Febuxostat
- Pegloticase
- Probenecid
- Sulfinpyrazone

Abbreviations

APC, antigen-presenting cell

5-ASA, 5-aminosalicylic acid

AUC, area under the curve

C_{max}, maximum plasma concentration

C_{min}, minimum plasma concentration

CNS, central nervous system

COX, cyclooxygenase

CTLA-4, cytotoxic T-lymphocyte antigen-4

CYP, cytochrome P450

DMARD, disease-modifying
antirheumatic drug

ED₅₀, median effective dose

FAAH, fatty acid amide hydrolase

FDA, U.S. Food and Drug Administration

GI, gastrointestinal

GPCR, G protein–coupled receptor

HPETE, hydroperoxy-eicosatetraenoic acid

IC₉₀, 90% of maximal inhibitory
concentration

ID₅₀, half maximal inhibitory
concentration

Ig, immunoglobulin

IL, interleukin

IM, intramuscular

IV, intravenous

JAK, Janus activated kinase

LT, leukotriene

6MNA, 6-methoxynaphthalene-2-acetic
acid

Abbreviations (Continued)

μM , micromoles/L	OTC , over the counter	T_{max} , time to maximum serum concentration
mPEG , monomethoxypoly(ethylene glycol)	PEG , polyethylene glycol	TNF , tumor necrosis factor
NF-κB , nuclear factor-κB	PG , prostaglandin	TNFR , tumor necrosis factor receptor
NSAID , nonsteroidal anti-inflammatory drug	SRS-A , slow reacting substance of anaphylaxis	TX , thromboxane
		UMP , uridine monophosphate

SCENARIO**Jill T. Johnson, PharmD, BCPS**

EB is a 53-year-old woman who has had rheumatoid arthritis for the past 20 years. She occasionally wakes with swollen joints but is relatively well controlled with methotrexate and hydroxychloroquine. She does not feel she is severe enough to require anti-tumor necrosis factor drugs at this point. On days she has painful joints, she takes a nonsteroidal drug. Recently she has become fearful of developing a stomach ulcer because

she has to take nonsteroidal drugs quite often. She usually takes naproxen but wonders if there is another effective drug that is safer.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

The classification of drugs covered in this chapter as nonsteroidal anti-inflammatory drugs (NSAIDs) is somewhat misleading, because many of these entities possess antipyretic and analgesic properties in addition to anti-inflammatory properties, which are useful in the treatment of a number of rheumatic disorders. On the other hand, there are drugs that possess analgesic/antipyretic properties but are essentially devoid of anti-inflammatory activity. Additionally, drugs that possess uricosuric properties useful in the treatment of gout will be covered here. The prototype agent of the NSAIDs is acetylsalicylic acid, aspirin, which has therapeutically useful analgesic, antipyretic, and anti-inflammatory actions; other drugs to be covered can possess only one or two of these properties. The corticosteroids that are useful anti-inflammatory drugs are covered separately in Chapter 28.

NSAIDs continue to be one of the more widely used groups of therapeutic drugs. The medicinal drugs covered in this chapter represent a major market in both prescription and nonprescription drugs. Other than caffeine or ethyl alcohol, aspirin remains the most widely used of all drugs (1). As a class, NSAIDs represent one of the most widely used prescription and over-the-counter (OTC) drugs. Rheumatic diseases (Table 31.1) (2) are inflammatory disorders affecting more individuals than any chronic illness and are the most common form of disability in the United States. Based on data obtained from 2003-5 from the National Health Interview Survey, the Centers for Disease Control and Prevention (3,4) estimates that over 46 million adults have self-reported doctor-diagnosed arthritis, whereas 19 million have arthritis and arthritis-attributable limited activity. Women report significantly higher age-related

prevalence than men in nearly all forms of arthritis. Arthritis prevalence increases with age and is higher in women than men in every age group. Arthritis affects all race and ethnic groups with whites being affected approximately nine times more than blacks and Hispanics. Approximately 300,000 children under the age of 18 have some form of arthritis or rheumatoid illness. Individuals who are overweight or obese report more physician-diagnosed arthritis than individuals possessing a lower body mass index. Almost two-thirds of adults with arthritis are overweight, whereas only 16% of those who are underweight or normal weight are affected by the disorder. Osteoarthritis is the most common form of arthritis in the United States, whereas rheumatoid arthritis affects approximately 1.3 million Americans. Because the number of Americans reporting a disability has increased approximately 8% in the past decade, the use of NSAIDs will increase as Americans experience a greater life expectancy. However, the development of new NSAIDs has slowed down because of recent controversies surrounding the safety of selective cyclooxygenase (COX)-2 inhibitors.

The diseases mentioned are considered to be host defense mechanisms. Inflammation is a normal and essential response to any noxious stimulus that threatens the host and can vary from a localized response to a generalized response (5). The resulting inflammation can be summarized as follows: 1) initial injury causing release of inflammatory mediators (e.g., histamine, serotonin, leukokinins, SRS-A, lysosomal enzymes, lymphokinins, and prostaglandins); 2) vasodilation; 3) increased vascular permeability and exudation; 4) leukocyte migration, chemotaxis, and phagocytosis; and 5) proliferation of connective tissue cells. The most common sources of chemical mediators include neutrophils, basophils, mast

cells, platelets, macrophages, and lymphocytes. The etiology of inflammatory and arthritic diseases has received a great deal of recent attention but remains, for the most part, unresolved, hindering the development of new agents that are curative in nature. Currently available drugs relieve the symptoms of the disease but are not considered curative.

Anti-inflammatory drugs can act by interfering with any one of several mechanisms, including immunologic mechanisms such as antibody production or antigen–antibody complexation, activation of complement, cellular activities such as phagocytosis, interference with the formation and release of the chemical mediators of inflammation, or stabilization of lysosomal membranes. The complement system (6–9) is one component of the host defense system that aids in the elimination of various microorganisms and antigens from blood and tissues. Although complement normally has a functional

role in the development of disease states, excessive complement activation, by promoting inflammation locally, is detrimental. Individuals with a deficiency of individual complement proteins, however, either acquired or hereditary, are more susceptible to infections caused by pyrogenic bacteria and diseases resulting from the generation of autoantibodies and immune complexes. Complement proteins are numbered C1 to C9, and their cleavage products are indicated by the suffixes a, b, and so on. The complement system consists of two activating pathways (an antibody-mediated classical pathway and a nonimmunologically activated alternate pathway), a single termination pathway, regulatory proteins, and complement receptors and involves approximately 30 membrane and plasma proteins (7). A major function of complement is to mark antigens and microorganisms with C3 fragments that direct them to cells containing C3 receptors, such as phagocytic cells (7). Complement has been implicated

TABLE 31.1 Classification of Rheumatic Diseases

Classification	Diseases
Acute and chronic polyarthritis and other synovial diseases	Rheumatoid arthritis
Infection-related rheumatic diseases	Septic arthritis Osteomyelitis Lyme disease Rheumatic fever
Spondyloarthropathies	Ankylosing spondylitis Reactive arthritis Psoriatic arthritis Enteropathic spondyloarthropathy (arthritis of inflammatory bowel disease)
Osteoarthritis and related degenerative diseases	
Crystal-induced arthropathies	Gout (monosodium urate) Calcium pyrophosphate dehydrate, apatite, and other calcium crystals deposited in the joint
Metabolic bone disease	Osteoporosis
Connective tissue disease	Systemic lupus erythematosus Scleroderma/polymyositis Mixed connective tissue disease
Musculoskeletal diseases (regional pain syndromes)	Neck, lower back, and lumbar spine stenosis, shoulder, elbow, wrist and hand, hip, and knee
Inflammatory muscle (nonarticular) diseases	Fibromyalgia Bursitis Tendinitis
Vasculitis	Polymyalgia rheumatica
Sjögren syndrome	
Neoplasms	
Drug-induced rheumatologic syndromes	Systemic lupus erythematosus Glucocorticoid-induced arthritis Scleroderma (acrosclerosis) Vasculitis Statin-induced myositis

From Klippel JH, Wayand CM, Wortmann R, et al., eds. *Primer on the Rheumatic Diseases*, 12th Ed. Atlanta, GA: Arthritis Foundation, 2001; with permission.

CLINICAL SIGNIFICANCE



The arachidonic acid pathway that leads to the production of inflammatory mediators provides several targets for intervention on inflammation. The most widely used nonsteroidal anti-inflammatory agents NSAIDs in practice today are drugs such as aspirin, naproxen, ibuprofen, and diclofenac, which inhibit the cyclo-oxygenase enzymes. Although specific NSAIDs have found their niches in particular disease states, the NSAIDs work by the same mechanism with few exceptions. Aspirin, a salicylate, is somewhat different from the other agents; its property of irreversibly inhibiting platelet homeostasis works to its advantage and broadens its use in coronary artery disease prophylaxis, for which it has been shown to reduce mortality, whereas the others have not. Other salicylate varieties lack the ability to irreversibly inhibit platelet function and are therefore not clinically useful in this way.

Variations of the molecular structures have provided improved side effect profiles of agents used. For example, although phenacetin and acetaminophen are not anti-inflammatory agents, they are included in this chapter because they are analgesics and antipyretics and illustrate the point that improvement in molecular structure from phenacetin to acetaminophen helped to reduce the hepatotoxicity and risk for drug-induced hemolytic anemia, with which phenacetin was associated when it was on the market in the past.

Changes to the currently available NSAIDs drugs are being made constantly, hence the abundance of “me-too” drugs. Gastrointestinal (GI) bleeding and complications are the most feared consequence of taking these drugs; therefore, emphasis on minimizing these risks has been one of the motivations to develop less offensive agents. Gastrointestinal side effects are believed to be related to indirect toxic effects (inhibiting COX enzymes) and direct toxic effects (local irritation to the GI mucosa). Molecular changes have been made to the various NSAIDs to produce more potent compounds, reduce the direct irritant effects, and to create prodrugs that lack the direct irritant effect altogether. Other changes have resulted in compounds with no more potency but with lower incidence of GI side effects, which still represents a therapeutic advantage.

As can be seen, a thorough understanding and appreciation of the chemical nature of the NSAIDs is directly linked to clinically significant endpoints and positive therapeutic outcomes.

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in numerous diseases, including allergic, hematologic, dermatologic, infectious, renal, hepatic inflammatory (rheumatoid arthritis and systemic lupus erythematosus), pulmonary, and others (e.g., multiple sclerosis and myasthenia gravis). Complement activation can induce the synthesis or release of inflammatory mediators, such as interleukin-1 (a potent proinflammatory cytokine) and prostaglandins (e.g., PGE₂). Leukotrienes (e.g., LTB₄) and thromboxanes are also released. Complement also aids the immigration of phagocytes associated with inflammation. Thus, inhibition of the complement system by controlling its activation or inhibiting those active fragments that are produced should be beneficial in reducing or eliminating tissue damage associated with inflammatory diseases.

ROLE OF CHEMICAL MEDIATORS IN INFLAMMATION

As indicated previously, a number of chemical mediators have been postulated to have important roles in the inflammatory process. Before 1971, the proposal by Shen (10,11) that the NSAIDs exert their effects by interacting with a hypothetical anti-inflammatory receptor was widely accepted. The topography of the proposed receptor was based on known structure–activity relationships primarily within the series of indole acetic acid derivatives, of which

indomethacin was the prototype. Most NSAIDs, whether they be salicylates, arylalkanoic acids, oxicams, or anthranilic acid derivatives, possess the common structural features of an acidic center, an aromatic or heteroaromatic ring, and an additional center of lipophilicity in the form of either an alkyl chain or an additional aromatic ring. The proposed receptor to which indomethacin was postulated to bind consisted of a cationic site to which the carboxylate anion would bind, a flat area to which the indole ring would bind through van der Waals forces, and an out-of-the-plane trough to which the benzene ring of the *p*-chlorobenzoyl group would bind through hydrophobic or charge-transfer interactions. Additional binding sites for the methoxy and carbonyl groups were also suggested. In 1971, Vane (12) published a classic paper in which he reported that indomethacin, aspirin, and salicylate, in descending order of potency, inhibited the biosynthesis of prostaglandins from arachidonic acid using cell-free preparations of guinea pig lung, and he further suggested that the clinical actions of these drugs resulted from the inhibition of prostaglandins. Vane’s theory has become the most widely accepted mechanism of action of NSAIDs. Gund and Shen (13) subsequently modified Vane’s hypothesis and proposed that the earlier anti-inflammatory receptor model actually described the active site of the key enzyme in prostaglandin biosynthesis (i.e., prostaglandin cyclooxygenase). However,

the complexity of the arachidonic acid pathway and the existence of several isoforms of the cyclooxygenase system have limited the relevance of this model.

Prostaglandins, Thromboxanes, Prostacyclin, and Leukotrienes

Prostaglandins

Prostaglandins are naturally occurring, 20-carbon, cyclopentano-fatty acid derivatives produced in mammalian tissue from polyunsaturated fatty acids. They belong to the class of eicosanoids, a member of the group of autocoids derived from membrane phospholipids. The eicosanoids are derived from unsaturated fatty acids and include the following groups of compounds: prostaglandins, thromboxanes, prostacyclin, and leukotrienes. They have been found in essentially every compartment of the body. In 1931, Kurzrok and Lieb (14) reported that human seminal fluid possessed potent contractile and relaxant effects on uterine smooth muscle. Shortly thereafter, Goldblatt (15) in England and von Euler (16) in Sweden independently reported vasodepressor and smooth muscle-contracting properties in seminal fluid (semen); von Euler identified the active constituent as a lipophilic acidic substance, which he termed “prostaglandin.” These observations attracted little attention during World War II, but shortly thereafter, primarily through the efforts of Bergstrom and Samuelsson (17), it was realized that von Euler’s prostaglandin was actually a mixture of a number of structurally related fatty acids. The first report of the structure of the prostaglandins in 1962 stimulated several studies relating to the chemical and biologic properties of these potent substances.

Prostaglandin Structure

The general structure of the prostaglandins is shown in Figure 31.1. All naturally occurring prostaglandins possess the following substitution pattern: a 15 α -hydroxy group and a *trans* double bond at C-13. Unless a double bond occurs at the C-8, C-12 positions, the two side chains (the carboxyl-bearing chain termed the α -chain and the hydroxyl-bearing chain termed the β -chain) are of the *trans* stereochemistry depicted in Figure 31.1. The prostaglandins are classified by the capital letters A, B, C, D, E, F, G, H, and I (e.g., PGA, PGB, and so on) depending on the nature and stereochemistry of oxygen substituents at the 9- and 11-positions. For example, members of the PGE series possess a keto function at C-9 and an α -hydroxyl group at C-11, whereas members of the PGF series possess α -hydroxyl groups at both of these positions. Members of the PGG and PGH series are cycloendoperoxide

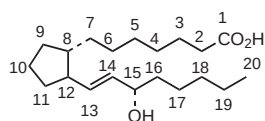


FIGURE 31.1 General structure of prostaglandins.

intermediates in the biosynthesis of prostaglandins, as depicted in Figure 31.2. The number of double bonds in the side chains connected to the cyclopentane ring is designated by subscripts 1, 2, or 3, indicative of the nature of the fatty acid precursor. The subscript 2 indicates an additional *cis* double bond at the C-5, C-6 positions, and the subscript 3 indicates a third double bond of *cis* stereochemistry at the C-17, C-18 positions.

Prostaglandin Biosynthesis

Prostaglandins are derived biosynthetically from unsaturated fatty acid precursors (Fig. 31.2). The number of double bonds contained in the naturally occurring prostaglandins reflects the nature of the biosynthetic precursors. Those containing one double bond are derived from 8,11,14-eicosatrienoic acid, those with two double bonds from arachidonic acid (5,8,11,14-eicosatetraenoic acid), and those with three double bonds from 5,8,11,14,17-eicosapentenoic acid. The most common of these fatty acids in humans is arachidonic acid; hence, prostaglandins of the 2 series have an important biologic role. Arachidonic acid is derived from dietary linoleic acid or is ingested from the diet (18) and esterified to phospholipids (primarily phosphatidylethanolamine or phosphatidylcholine) in cell membranes. Various initiating factors interact with membrane receptors coupled to G proteins (guanine nucleotide-binding regulatory proteins) activating phospholipase A₂, which in turn hydrolyzes membrane phospholipids resulting in the release of arachidonic acid. Other phospholipases (e.g.,

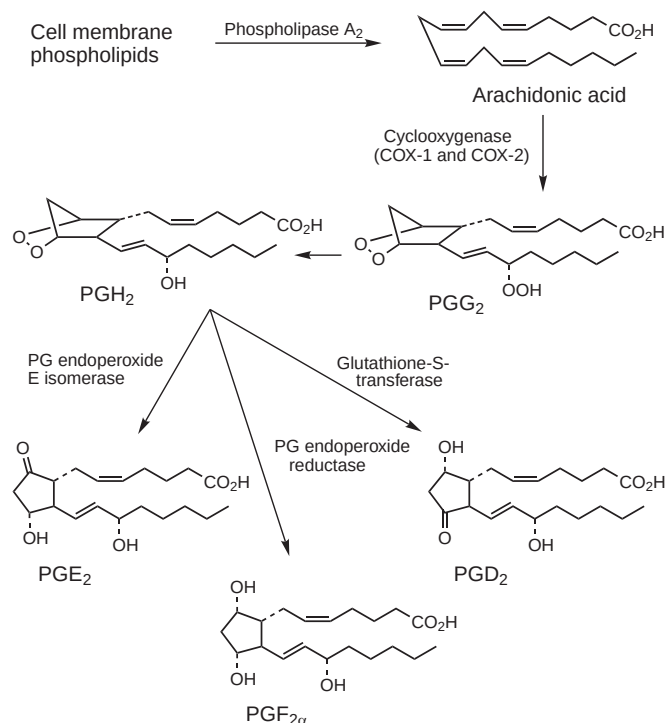


FIGURE 31.2 Biosynthesis of prostaglandins from arachidonic acid.

phospholipase C) are also involved. Phospholipase C differs from phospholipase A₂ by inducing the formation of 1,2-diglycerides from phospholipids with the subsequent release of arachidonic acid by the actions of mono- and diglyceride lipases on the diglyceride (16). A polypeptide produced by leukocytes, interleukin-1, which mediates inflammation, increases phospholipase activity and, thus, prostaglandin biosynthesis. The steroidal anti-inflammatory drugs (corticosteroids) appear to act, in part, by inhibiting these phospholipases, particularly phospholipase A₂. The liberated arachidonic acid can then be acted on by two major enzyme systems: by arachidonic acid cyclooxygenase (prostaglandin endoperoxide synthetase, COX) to produce prostaglandins, thromboxanes, and prostacyclin, or by lipoxygenases to produce leukotrienes.

Cyclooxygenases

Interaction of arachidonic acid with cyclooxygenase (COX) in the presence of oxygen and heme produces, first, the cyclic endoperoxide, PGG₂, and then, through its peroxidase activity, PGH₂, both of which are chemically unstable and decompose rapidly (half-life, 5 minutes). The PGE₂ is formed by the action of PGE isomerase and PGD₂ by the actions of isomerases or glutathione-S-transferase on PGH₂, whereas PGF_{2α} is formed from PGH₂ via an endoperoxide reductase system (Fig. 31.2). It is the COX step at which the NSAIDs inhibit prostaglandin biosynthesis to prevent inflammation. Because PGG₂ and PGH₂ themselves can possess the ability to mediate the pain responses and produce vasoconstriction and because PGG₂ can mediate the inflammatory response, COX inhibition would have a profound effect on the reduction of inflammation.

To date, three isoforms of COX have been identified: COX-1, COX-2, and COX-3. COX was first purified in 1976 (19). Among the more significant advances of the past decade was the isolation of a second isoform of the COX enzyme, COX-2, the expression of which is inducible by cytokines and growth factors (20–24). In 2002, a third distinct COX isoform, COX-3 was reported (25) in addition to two smaller COX-1-derived proteins (partial COX-1, or PCOX-1, proteins, termed PCOX-1α and PCOX-1β). The primary mechanism by which NSAIDs are held to produce their pharmacologic effects is attributed to inhibition of the COX-1 and COX-2 enzymes. Both isoforms carry out the same two reactions in the PG biosynthetic pathway: the double dioxygenation of arachidonic acid to PGG₂ at the cyclooxygenase active site and the subsequent reduction to PGH₂ at the peroxidase site (26).

It has been long known that acetaminophen possesses analgesic and antipyretic activity but little, if any, anti-inflammatory activity. Thus, the identification of COX-3 can be of importance, because COX-3 is selectively inhibited by analgesic/antipyretic drugs and is potently inhibited by some NSAIDs. Inhibition of COX-3 can represent a primary central mechanism by which acetaminophen decreases pain and fever (discussed later).

Both COX-1 and COX-2 are very similar in structure (27,28) and almost identical in length, varying from 599 (human) to 602 (mice) amino acids for COX-1 and from 603 (mice) to 604 (human) amino acids for COX-2. Both isoforms possess molecular masses of 70 to 74 kD and contain just over 600 amino acids, with an approximately 60% homology within the same species (29,30). Cyclooxygenase-2 contains an 18-amino acid insert near the C-terminal end of the enzyme that is not present in COX-1, but all other residues that have been previously identified as being essential to the catalytic activity of COX-1 are present in COX-2. Both isoforms have been cloned from various species, including human, and are heme-containing membrane proteins that exist as dimers (29). The three-dimensional radiographic crystal structure of COX-2 derived from human or murine sources can be superimposed on that of COX-1. Residues that form the substrate binding channel, the catalytic sites, and those residues immediately adjacent are essentially identical with the exception of two minor differences. The isoleucine at positions 434 and 523 in COX-1 is exchanged for valine in COX-2. The smaller size of Val⁵²³ in COX-2 allows inhibitor access to a side pocket off the main substrate channel, whereas the longer side chain of Ile in COX-1 sterically blocks inhibitor access. The available space in the binding pocket of COX-2 is 20% to 25% larger than the COX-1 binding site because of the replacement of the smaller Val⁵²³ in COX-2 for the larger Ile⁵²³ in COX-1. A major difference between COX-1 and COX-2 is that COX-2 lacks a sequence of 17 amino acids from the N-terminus but contains a sequence of 18 amino acids at the C-terminus. The sequence differences cause a disparity in the numbering systems of the two isoforms such that the serine residue acetylated by aspirin in COX-1 is numbered Ser⁵³⁰, whereas in COX-2, the serine residue acetylated is Ser⁵¹⁶. Yet, the amino acid residues that are thought to be responsible for providing the catalytic role are the same, with both isoforms displaying similar ability to convert arachidonic acid to PGH₂. COX-1 appears to be more specific than COX-2 for fatty acid substrates, because COX-2 accepts a wider range of fatty acid substrates than COX-1. COX-1 primarily metabolizes arachidonic acid, and COX-2 metabolizes C-18 and C-20 fatty acid substrates. Selective inhibitors of COX-2 do not bind to Arg-120, which is used by the —COOH of arachidonic acid and the carboxylic acid selective or nonselective COX-1 inhibitors. The active site cavities of COX-1 and COX-2 are illustrated in Figure 31.3 (27).

From a therapeutic standpoint, the major difference between COX-1 and COX-2 lies in physiologic function rather than in structure. Little COX-2 is present in resting cells, but its expression can be induced by cytokines in vascular smooth muscle, fibroblasts, and epithelial cells, leading to the suggestion that COX-1 functions to produce prostaglandins that are involved in normal cellular activity (protection of gastric mucosa, maintenance of kidney function) and COX-2 to produce prostaglandins at inflammatory sites (31). Inducible COX-2 linked

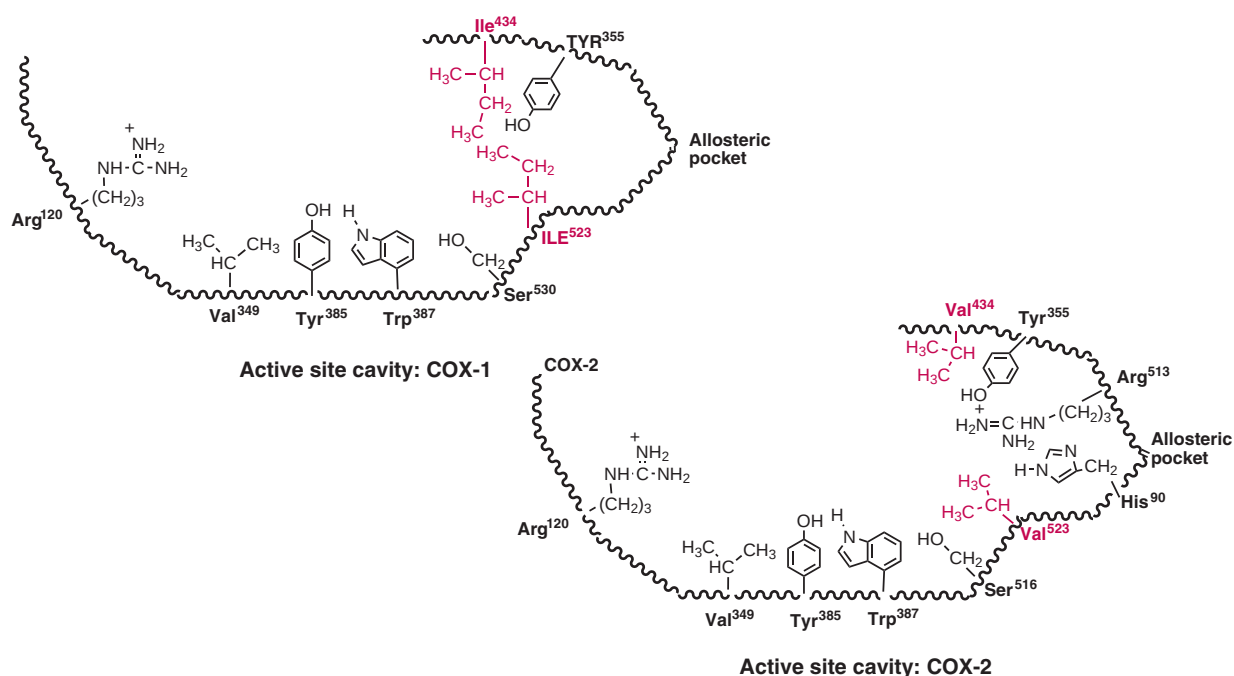


FIGURE 31.3 Representations of COX-1 and COX-2 active sites. (From Roche VF. A receptor-grounded approach to teaching nonsteroidal anti-inflammatory drug chemistry and structure-activity relationships. *Am J Pharm Educ* 2009;78:143.)

to inflammatory cell types and tissues is believed to be the target enzyme in the treatment of inflammatory disorders by NSAIDs. Until recently, most NSAIDs inhibited both COX-1 and COX-2, but with varying degrees of selectivity. Selective COX-2 inhibitors can eliminate side effects associated with NSAIDs because of COX-1 inhibition, such as gastric and renal effects.

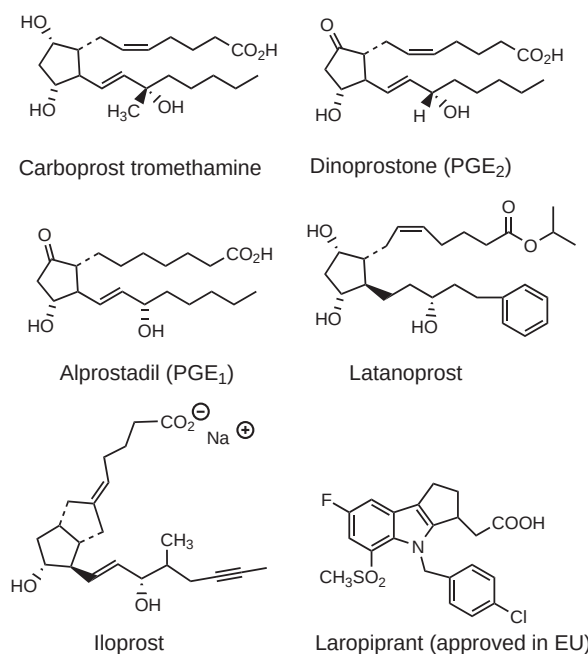
Prostaglandin Metabolism

Prostaglandins are rapidly metabolized and inactivated by various oxidative and reductive pathways. The initial step involves rapid oxidation of the 15 α -OH to the corresponding ketone by the prostaglandin-specific enzyme, prostaglandin 15-OH dehydrogenase, which is followed by reduction of the C-13, C-14 double bond by prostaglandin Δ^{13} -reductase to the corresponding dihydro ketone, which for PGE₂ represents the major metabolite in plasma. Subsequently, enzymes normally involved in β - and ω -oxidation of fatty acids more slowly cleave the α -chain and oxidize the C-20 terminal methyl group to the carboxylic acid derivative, respectively. Hence, dicarboxylic acid derivatives containing only 16 carbon atoms are the major excreted metabolites of PGE₁ and PGE₂.

The effects of prostaglandins on the gastrointestinal (GI) tract deserve special mention. The PGEs and PGI₂ inhibit gastric secretion that can be induced by gastrin or histamine. Prostaglandins appear to have a major cytoprotective role in maintaining the integrity of gastric mucosa. PGE₁ exerts a protective effect on gastroduodenal mucosa by stimulating secretion of an alkaline mucus and bicarbonate ion and by maintaining or

increasing mucosal blood flow. Thus, inhibition of prostaglandin formation in joints produces favorable results, as indicated by a reduction in fever, pain, and swelling. Inhibition of prostaglandin biosynthesis in the GI tract is unfavorable, however, because it can cause disruption of mucosal integrity, resulting in peptic ulcer disease that, as will be discussed later, is commonly associated with the use of NSAIDs and aspirin.

PROSTAGLANDINS AS DRUGS



The promise of prostaglandins as therapeutically effective drugs has been limited because of the pharmacologic diversity displayed by prostaglandins and the metabolic instability observed with their administration. The pharmacologic actions of the various prostaglandins are quite diverse. When administered intravaginally, PGE_2 will stimulate the endometrium of the gravid uterus to contract in a manner similar to uterine contractions observed during labor. Thus, PGE_2 is therapeutically available as dinoprostone (Prostin E2) for use as an abortifacient at 12 to 20 weeks of gestation and for evacuation of uterine content in missed abortion or intrauterine fetal death up to 28 weeks of gestation. Additionally, PGE_2 is a potent stimulator of smooth muscle of the GI tract and can elevate body temperature in addition to possessing potent vasodilating properties in most vascular tissue while possessing constrictor effects at certain sites. The PGEs in general cause pain when administered via the intradermal route. Many of these properties are shared by $\text{PGF}_{2\alpha}$, which is also therapeutically available as an abortifacient at 16 to 20 weeks of gestation and is available as dinoprost tromethamine (Prostin F2 α). The synthetic 15-methyl derivative of $\text{PGF}_{2\alpha}$, carboprost, is also available as the tromethamine salt (Prostin 15/M) as an abortifacient at 13 to 20 weeks gestation. However, $\text{PGF}_{2\alpha}$ differs from PGE_2 in that it does not significantly alter blood pressure in humans. PGD_2 causes both vasodilation and vasoconstriction. The PGEs produce a relaxation of bronchial and tracheal smooth muscle, but the PGFs and PGD_2 cause contraction. PGE_1 is available as alprostadil (Prostin VR Pediatric) to maintain patency of the ductus arteriosus in neonates until surgery can be performed to correct congenital heart defects.

Prostaglandins have been shown to reduce intraocular pressure by increasing the outflow of aqueous humor, and structurally modified prostaglandin analogs have thus found a place in the treatment of glaucoma. The most widely used of these agents is latanoprost (Xalatan), but travaprost (Travatan), bimatoprost (Lumigan), and unoprostone (Rescula) are also available.

Iloprost (Ventavis), a synthetic analog of the metabolically unstable prostacyclin, is indicated for the treatment of pulmonary arterial hypertension, scleroderma, Raynaud phenomenon, and ischemia. It is marketed as the side chain 3*S*, 4*RS*-isomer but the 4*S*-isomer is substantially more potent than the 4*R*-isomer.

Antagonists at prostaglandin receptors also hold therapeutic promise. Laropiprant, an antagonist at the PGD_2 receptor subtype 1 (DP1), is effective in suppressing niacin-induced vasodilation, an adverse effect of niacin that limits its use in dyslipidemia. A combination of nicotinic acid and laropiprant has been approved in Europe as a lipid-modifying therapy for patients with dyslipidemia and primary hypercholesterolemia.

Prostanoid Receptors

The existence of distinct prostaglandin receptors can explain the broad spectrum of action displayed by the

prostaglandins (32,33). Prostaglandins exert their various effects by activating G protein-coupled receptors (GPCRs). Ten members of the prostanoid receptor family have been identified (DP_{1-2} , EP_{1-4} , FP, IP_{1-2} , and TP). The nomenclature of these receptors is based on the affinity displayed by natural prostaglandins, prostacyclin, or thromboxanes at each receptor type. Thus, EP receptors are those receptors for which the PGEs have high affinity, FP receptors are those for PGFs, DP receptors are those for PGDs, IP receptors are those for PGI_2 , and TP receptors are those for thromboxane (TX) A_2 . These receptors are coupled through G proteins to effector mechanisms that include stimulation of adenylate (or adenylyl) cyclase and, hence, increased cyclic adenosine monophosphate levels and phospholipase C, which results in increased levels of inositol-1,4,5-triphosphate. Three distinct receptors for leukotrienes have been identified as well.

Nonprostanoid Products of the Arachidonic Acid Pathway

In addition to forming the various prostaglandins, nonprostanoids can also be formed from PGH_2 , as illustrated in Figure 31.4. Thromboxane synthetase acts on PGH_2 to produce TXA_2 , whereas prostacyclin synthetase converts PGH_2 to prostacyclin (PGI_2), both of which possess short biologic half-lives. A potent vasoconstrictor and inducer of platelet aggregation, TXA_2 has a biologic half-life of approximately 30 seconds, being rapidly nonenzymatically converted to the more stable, but inactive, TXB_2 . Prostacyclin, a potent hypotensive and inhibitor of platelet aggregation, has a half-life of approximately 3 minutes and is nonenzymatically converted to 6-keto- $\text{PGF}_{1\alpha}$. Platelets contain primarily thromboxane synthetase, whereas endothelial cells contain primarily prostacyclin synthetase. Considerable research efforts are being expended in the development of stable prostacyclin analogs and thromboxane antagonists as cardiovascular drugs. The pharmacologic effects of some prostaglandins, TXA_2 , and prostacyclin are summarized in Table 31.2.

LEUKOTRIENES Lipoxygenases are a group of enzymes that oxidize polyunsaturated fatty acids possessing two *cis* double bonds separated by a methylene group to produce lipid hydroperoxides (34). Arachidonic acid is thus metabolized to a number of hydroperoxy-eicosatetraenoic acid (HPETE) derivatives. Lipoxygenases differ in the position at which they peroxidize arachidonic acid and in their tissue specificity. For example, platelets possess only a 12-lipoxygenase, whereas leukocytes possess both a 12-lipoxygenase and a 5-lipoxygenase (35). The HPETE derivatives are not stable, being rapidly converted to a number of metabolites. Leukotrienes are products of the 5-lipoxygenase pathway and are divided into two major classes: hydroxylated eicosatetraenoic acids (LTs), represented by LTB_4 , and peptidoleukotrienes (pLTs), such as LTC_4 , LTD_4 , and LTE_4 . 5-Lipoxygenase will produce

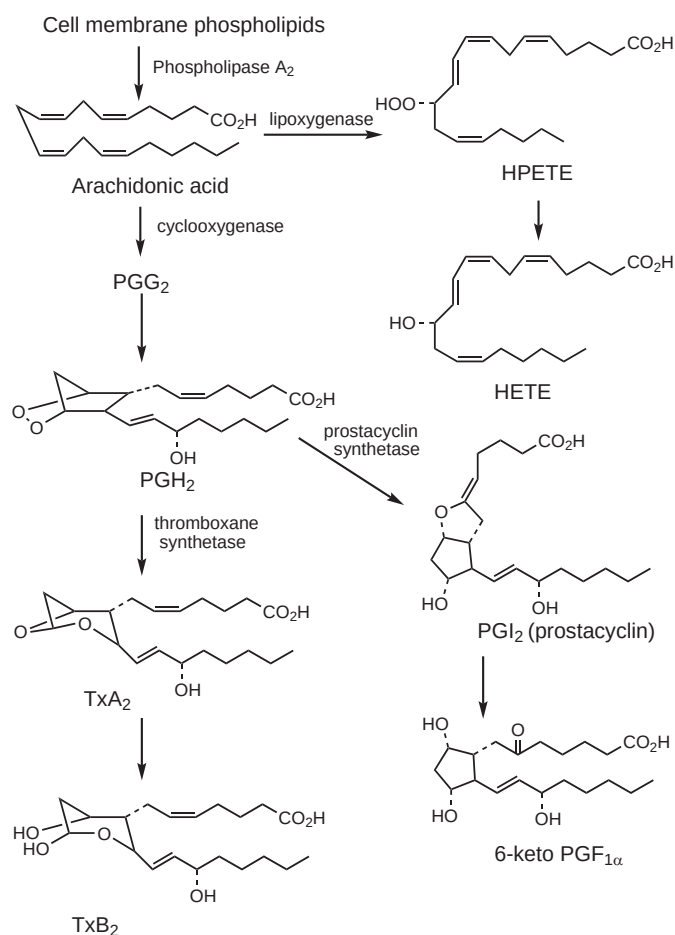


FIGURE 31.4 Biosynthesis of thromboxanes, prostacyclin, and leukotrienes.

leukotrienes from 5-HPETE, as shown in Figure 31.5. LTA synthetase converts 5-HPETE to an unstable epoxide called LTA₄ that can be converted by LTA hydrolase to the leukotriene LTB₄ or by glutathione-S-transferase to LTC₄. Other cysteinyl leukotrienes (e.g., LTD₄, LTE₄, and LTF₄) can then be formed from LTC₄ by the removal of glutamic acid and glycine and then re conjugation with glutamic acid, respectively. The cysteinyl leukotrienes produce airway edema, smooth muscle constriction, and altered cellular activity associated with the inflammatory

process, all of which are associated with the pathophysiology of asthma (see also Chapter 39). Cysteinyl leukotrienes activate at least two receptors, designated as CysLT₁ and CysLT₂. A long-recognized mediator of inflammation, slow-reacting substance of anaphylaxis (SRS-A), is primarily a mixture of two leukotrienes, LTC₄ and LTD₄. The physiologic roles of the various leukotrienes are becoming better understood. LTB₄ is a potent chemotactic agent for polymorphonuclear leukocytes, causes the accumulation of leukocytes at inflammation sites, and leads to the development of symptoms characteristic of inflammatory disorders. Both LTC₄ and LTD₄ are potent hypotensives and bronchoconstrictors. Because of the role played by LTs and pLTs in inflammatory conditions and asthma, it is not surprising that intensive research is being conducted in this area. Zafirlukast (Accolate), the first marketed cysteinyl leukotriene receptor antagonist (termed “lukasts”), was approved in 1996 for the prophylaxis and chronic treatment of asthma (Fig. 31.6). It is a selective and competitive receptor antagonist of the cysteinyl leukotrienes, LTD₄ and LTE₄. In humans, pretreatment with single oral doses of zafirlukast inhibited bronchoconstriction caused by sulfur dioxide and cold air and reduced the both early and late-phase

TABLE 31.2 Pharmacologic Properties of Prostaglandins, Thromboxane, and Prostacyclin

	PGE ₂	PGF _{2α}	PGI ₂	TXA ₂
Uterus	Oxytocic dilation	Oxytocic constriction		
Bronchi	Dilates	Constricts		Constricts
Platelets			Inhibits	Aggregation
Blood vessels	Dilation	Constriction	Dilation	Constriction

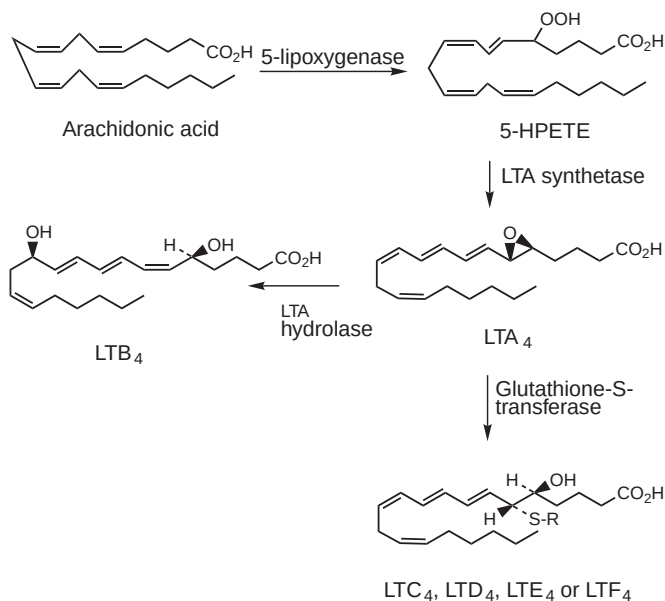


FIGURE 31.5 Biosynthesis of leukotrienes.

reaction in patients with asthma caused by inhalation of various antigens, such as grass, cat dander, and ragweed. Zafirlukast reduced the increase in bronchial hyperresponsiveness to inhaled histamine that followed inhaled allergen challenge. It is rapidly absorbed, but food reduces bioavailability to approximately 40%. Zafirlukast is extensively metabolized via hydroxylation reactions mediated primarily by CYP2C9, and to a minor extent CYP3A4, to essentially inactive metabolites. Excretion is 90% fecal and 10% via urine with an elimination half-life of approximately 10 hours. A second leukotriene antagonist, montelukast (Singulair) (Fig. 31.6), was approved shortly thereafter for prophylaxis and chronic treatment of asthma as an antagonist of LTD₄ at the CysLT₁ receptor. Its oral bioavailability ranges from 60% to 75% depending on the dosage form used. On oral administration, the time to peak concentration ranges from 2 to 4 hours. Food can reduce its bioavailability, prolonging

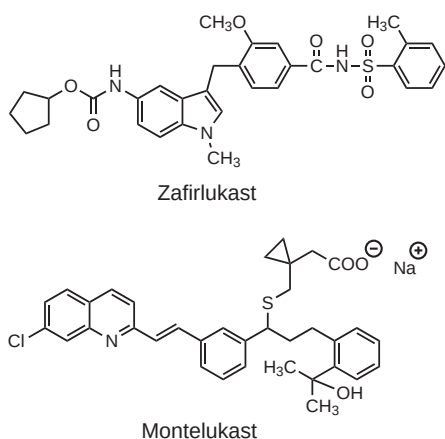
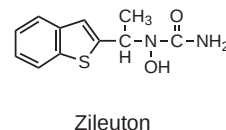


FIGURE 31.6 Leukotriene antagonists.

time to maximum plasma concentration. Protein binding is more than 99%. Like zafirlukast, montelukast is also extensively metabolized via hydroxylation reactions mediated by CYP2C9 and CYP3A4 to essentially inactive metabolites. Montelukast is excreted primarily in feces (86%), with an elimination half-life of 3 to 6 hours. The convenience of once-daily dosing provides montelukast an advantage over zafirlukast in the treatment of asthma.



The first marketed inhibitor of leukotriene biosynthesis, zileuton (Zyflo), was approved shortly after zafirlukast for prophylaxis and chronic treatment of asthma. Zileuton is a specific inhibitor of 5-lipoxygenase and, thus inhibits the formation of LTB₄, LTC₄, LTD₄, and LTE₄. The *R*(+)- and *S*(-)-enantiomers equally inhibit 5-lipoxygenase so the drug is marketed as the racemic mixture. Zileuton is metabolized primarily to two diastomeric glucuronide conjugates and the *N*-dehydroxylated metabolite. The use of zileuton is limited because of reports of liver toxicity, whereas the leukotriene receptor antagonists appear to produce little toxicity.

THERAPEUTIC APPROACH TO ARTHRITIC DISORDERS

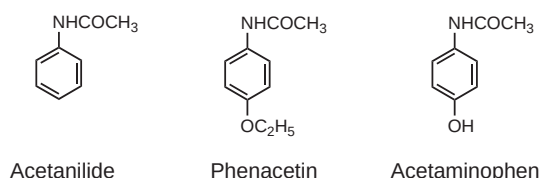
The goal of drug treatment in early rheumatoid arthritis is to induce remission or at least eliminate evidence of disease activity. Rheumatoid arthritis was traditionally treated with a stepwise approach starting with NSAIDs and progressing through more potent drugs such as glucocorticoids, disease-modifying antirheumatic drugs (DMARDs), and biologic response modifiers. The DMARDs were avoided early in the disease because of their potentially serious side effects and were usually reserved for individuals who showed signs of joint damage. Over time, however, this strategy of avoiding DMARDs was recognized as being faulty, because people treated early with DMARDs have better long-term outcomes, with greater preservation of joint function and less work disability.

The current approach, therefore, is to treat rheumatoid arthritis aggressively with DMARDs and biologic response modifiers soon after diagnosis. Treating rheumatoid arthritis early with DMARDs, within 3 to 6 months after symptoms begin, controls inflammation better and is the best way to stop or slow progression of the disease and bring about remission and long-term prevention of joint disease. Methotrexate is the cornerstone of DMARD therapy. Long-term treatment with methotrexate and biologic response modifiers can offer the best control of rheumatoid arthritis for the majority of people, which can eliminate the need for other NSAID medications.

A large number of NSAIDs, inhibitors to varying degrees of COX-1 and COX-2, are therapeutically available for the treatment of pain and inflammation associated with arthritic disorders, differing in efficacy but, perhaps more importantly, differing also in overall toxicity. As a group, NSAIDs can cause GI toxicity, such as dyspepsia, abdominal pain, heartburn, gastric erosion leading to wall perforation, peptic ulcer formation, bleeding, diarrhea, renal disorders (e.g., acute renal failure, tubular necrosis, and analgesic nephropathy), and other effects (e.g., tinnitus and headache). Gastric damage produced by NSAIDs usually involves a dual insult mechanism (Fig. 31.7). Most NSAIDs are acidic substances that produce a primary insult because of direct acid damage, an indirect contact effect, and a back diffusion of hydrogen ions. The secondary insult results from inhibition of prostaglandin biosynthesis in the GI tract, where prostaglandins exert a cytoprotective effect. The dual insult leads to gastric damage. A report from the Arthritis, Rheumatism, and Aging Medical Information System Post-Marketing Surveillance Program, before the introduction of COX-2-selective drugs, ranked the overall toxicity of NSAIDs in the following decreasing order: indomethacin > tolmetin > meclofenamate > ketoprofen > fenoprofen > salsalate > aspirin.

THERAPEUTIC CLASSIFICATIONS

Antipyretic Analgesics



Mechanism of Action

Drugs included in this class possess analgesic and antipyretic actions but lack anti-inflammatory effects. Antipyretics interfere with those processes by which pyrogenic factors produce fever, but they do not appear

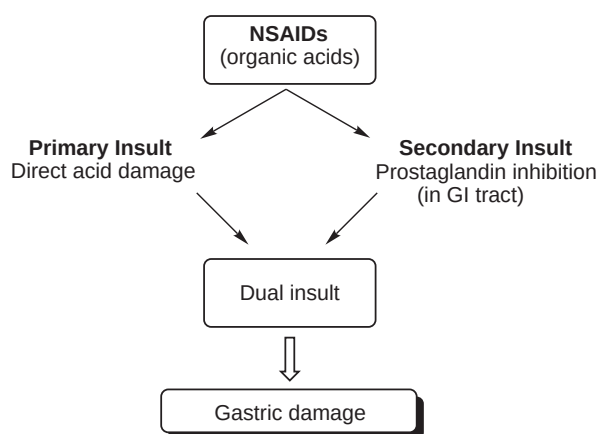


FIGURE 31.7 NSAID-induced production of gastric damage by a dual-insult mechanism.

to lower body temperature in afebrile subjects. It had been historically accepted that the antipyretics exert their actions within the central nervous system (CNS), primarily at the hypothalamic thermoregulatory center, but more recent evidence suggests that peripheral actions can also contribute. Endogenous leukocytic pyrogens can be released from cells that have been activated by various stimuli, and antipyretics can act by inhibiting the activation of these cells by an exogenous pyrogen or by inhibiting the release of endogenous leukocytic pyrogens from the cells once they have been activated by the exogenous pyrogen. Substantial evidence exists suggesting a central antipyretic mechanism, an antagonism that can result from either a direct competition of a pyrogen and the antipyretic agent at CNS receptors, or an inhibition of prostaglandins in the CNS (36). Despite the extensive use of acetaminophen, the mechanism of action has not been fully elucidated. Acetaminophen can inhibit pain impulses by exerting a depressant effect on peripheral receptors; an antagonistic effect on the actions of bradykinin can have a role. The antipyretic effects might not result from inhibition of release of endogenous pyrogen from leukocytes but, rather, from inhibiting the action of released endogenous pyrogen on hypothalamic thermoregulatory centers. The fact that acetaminophen is an effective antipyretic/analgesic but an ineffective anti-inflammatory agent can result from its greater inhibition of prostaglandin biosynthesis via inhibition of the COX-3 isoform in the CNS compared with that in the periphery.

Recent research in dogs has revealed a splice variant of COX-1 that was sensitive to inhibition by acetaminophen. This novel variant of the COX system was named COX-3, and it was hypothesized that inhibition of COX-3 could represent a primary CNS mechanism by which acetaminophen and phenacetin exerted their analgesic and antipyretic effects (25). The ability of selected analgesic/antipyretic drugs to inhibit COX-1, COX-2, and COX-3 is shown in Table 31.3 (25).

Further studies revealed that similar COX variants were present in rodents and humans, but these did not appear to be inhibited by acetaminophen (37,38). These

TABLE 31.3 Binding Constants (IC_{50} , μM) of Selected NSAIDs for the COX Enzymes

Drug	COX-1	COX-2	COX-3
Acetaminophen	>1,000	>1,000	460
Phenacetin	>1,000	>1,000	102
Aspirin	10	>1,000	3.1
Diclofenac	0.035	0.041	0.008
Ibuprofen	2.4	5.7	0.24
Indomethacin	0.010	0.66	0.016

COX variants are proposed to be COX-active, however, and to have a role in the biochemistry of these species (39,40). There is substantial nonhomology between the human, canine, and rodent COX-3 proteins.

Because PGE_2 is crucial to induction of fever, the antipyretic activity of this class is probably due to some degree of inhibition of COX enzymes. The same pertains to the analgesic effects of acetaminophen, where PGE_2 in particular synergizes greatly with other pain-producing mediators such as bradykinin and histamine. However, it appears that not all COX types are inhibited by this class of drugs, and the lack of anti-inflammatory activity suggests that acetaminophen is subtly different from the other NSAIDs. The apparently plausible hypothesis that there is a COX variant (COX-3) sensitive to acetaminophen has been challenged (41). Recent results have shown that acetaminophen increases the levels of endocannabinoids in traumatized tissues by inhibition of the endocannabinoid metabolizing enzyme(s) and that this could be an important mechanism for analgesia. Several, if not all, NSAIDs display the ability to inhibit fatty acid amide hydrolase (FAAH) and raise the levels of endocannabinoids. The body of evidence is such that it is now appropriate to raise the awareness of this mechanism of action (42). The novel proposal that endocannabinoids make a major contribution to the mechanism of action of acetaminophen is based on the observation that ibuprofen and several other NSAIDs inhibited FAAH (43). Further research has augmented this view and can explain the anomalous activity of acetaminophen, since agonists at cannabinoid receptors are known to have analgesic and hypothermic actions (44,45). Further support of the expanding role of endocannabinoids in acetaminophen activity is the observation (46) that the known FAAH inhibitor, AM404, can be formed from the deacylated metabolite of acetaminophen (Fig. 31.8).

Interestingly, esters of acetaminophen were recently found to inhibit FAAH (47).

Additionally, it is known that opioids in combination with cannabinoids, for pain relief, display a large synergistic effect that also decreases the propensity for drug dependence of both drug classes (48). In light of the acetaminophen results, the use of cannabinoid agonists as analgesics should be reconsidered, especially in

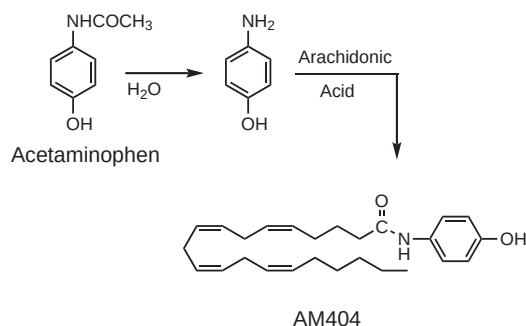


FIGURE 31.8 Formation of AM404 from acetaminophen.

combination with NSAIDs and in situations where opioids are inappropriate because of their more highly addictive character (49). A fuller discussion of the pharmacology of the endocannabinoid system in relation to NSAIDs and other biosystems is available (50).

It is now appreciated that many NSAIDs also reduce the immune response and that patients should not use these drugs after vaccination unless unbearable pain is experienced (51). The reduction in the immune response also fits well with the known agonist effects on the cannabinoid receptors, which modulate the immune system via both CB1 and CB2 receptors. There can be little doubt that the potential medical benefits of actions on this ubiquitous and abundant receptor system in mammals have been much attenuated by traditional abuse of phytocannabinoids. Ajulemic acid, a phytocannabinoid analog, has been studied as a cannabinoid agonist for the possible treatment of arthritis, multiple sclerosis, CNS injury, and analgesia in neuropathic pain (52,53).

Lipoxins, Resolvins, and Protectins

Lipoxins, naturally occurring eicosanoid mediators in mammals, derived from arachidonic acid by specific trihydroxylation, appear to be involved in inflammation resolution by action on specific GPCRs (54). The observation that resolution of inflammation is not a passive action but requires positive signaling by lipoxin-type mediators is an important potential breakthrough.

Similar substances to the lipoxins derived from timnodonic acid (eicosapentaenoic acid) or docosahexaenoic acid have been termed “resolvins” (RvE and RvD, respectively). Dihydroxylated lipids derived from these important polyunsaturated acids have been termed protectins and maserins and add to the spectrum of inflammation-resolving actions (55). These specialized proresolution lipid mediators, or similar substances with anti-inflammatory action, can be generated by COX enzymes that have been only partially inhibited by NSAIDs. Aspirin in particular has been identified as having the ability to increase lipoxin levels by irreversible inhibitory action on the COX system, thereby preventing the normal prostaglandin formation but still allowing oxidation reactions producing lipoxins or lipoxin-related substances (54). This activity can possibly be absent in acetaminophen, and this could explain its low anti-inflammatory effect. It is at present unclear if other NSAIDs have the effect of increasing lipoxins as a major component of their anti-inflammatory mechanism (56,57). However endocannabinoids also increase lipoxin levels, and therefore, typical NSAIDs can also do this via this mechanism.

Therefore, the mode of action of acetaminophen is still in question. Certainly, the analgesic and antipyretic properties parallel the decrease in PGE_2 levels in the CNS caused by acetaminophen. COX-1–deleted, but not the COX-2–deleted, mice showed a decrease in these actions (58). The authors suggest that in this species, it is COX-1, or a variant of it, that is affected by acetaminophen. Unlike other NSAIDs, such as ibuprofen or aspirin,

acetaminophen does not have significant anti-inflammatory, antiplatelet, or gastric ulcerogenic activity. Other authors claim that the mechanism of action of acetaminophen is thought to involve inhibition of COX-2, and this fits with the therapeutic profile of the recently discovered, powerful, and selective COX-2 inhibitors (59). Acetaminophen is only effective, however, when COX-2 activity is at a low level. This view partially explains the lack of anti-inflammatory action, because COX-2 activity will be high in this situation (60).

Graham and Scott (60) have also put forward the interesting hypothesis that acetaminophen acts by depletion of the stores of glutathione, which is a known cofactor for PGE synthase. This would explain the decrease in PGE production and the concomitant analgesic effect. The depletion of glutathione is the main cause of acetaminophen toxicity. The highly reactive benzoquinoneimine, formed by CYP2E1 isoform, must be conjugated with glutathione before it can react with other crucial cell components. In overdose, failure of this molecular mechanism results in serious liver damage. The depletion of the body's supply of glutathione also should affect the biosynthesis of the inflammatory mediator leukotriene C4. Depletion of the body's supply of glutathione does not seem to be significant with therapeutic levels of acetaminophen.

A review article discussing the dichotomies of the acetaminophen mechanism and COX-3 is available (41).

Historical Background

Acetanilide was introduced into therapy in 1886 under the name antifebrin as an antipyretic/analgesic agent but was subsequently found to be too toxic (methemoglobinemia and jaundice), particularly at high doses, to be useful. Phenacetin was introduced the following year and remained in use until the 1960s because of reports of nephrotoxicity. Phenacetin is longer acting than acetaminophen despite the fact that it is metabolized to acetaminophen but is a weaker antipyretic. Shortly thereafter, acetaminophen (paracetamol) was introduced in 1893 but remained unpopular for more than 50 years, until it was observed that it is a metabolite of both acetanilide and phenacetin. It remains the only useful agent of this group and is widely used as a nonprescription antipyretic/analgesic under a variety of trade names (Tylenol, Patrol, and Tempera). The analgesic activity of acetaminophen is comparable to that of aspirin, but acetaminophen lacks useful anti-inflammatory activity. Its advantage over aspirin as an analgesic, however, is that individuals who are hypersensitive to salicylates usually respond well to acetaminophen.

Structure–Activity Relationships

The structure–activity relationships of *p*-aminophenol derivatives have been widely studied. Based on the comparative toxicity of acetanilide and acetaminophen, aminophenols are less toxic than the corresponding aniline derivatives, although *p*-aminophenol itself is too toxic

for therapeutic purposes. Etherification of the phenolic function with methyl or propyl groups produces derivatives with greater side effects than with ethyl groups. Substituents on the nitrogen atom that reduce basicity reduce activity unless that substituent is metabolically labile (e.g., acetyl). Amides derived from aromatic acids (e.g., *N*-phenylbenzamide) are less active or inactive.

ACETAMINOPHEN USP Acetaminophen is weakly acidic (phenolic $pK_a = 9.51$) and synthesized by the acetylation of *p*-aminophenol. It is weakly bound to plasma proteins (18% to 25%). Acetaminophen is indicated for use as an antipyretic/analgesic, particularly in individuals displaying an allergy or sensitivity to aspirin. It does not possess anti-inflammatory activity, but it will produce analgesia in a wide variety of arthritic and musculoskeletal disorders. It is available in various formulations, including suppositories, tablets, capsules, granules, and solutions. The usual adult dose is 325 to 650 mg every 4 to 6 hours. Doses greater than 2.6 g/d are not recommended for long-term therapy because of potential hepatotoxicity issues. Acetaminophen, unlike aspirin, is stable in aqueous solution, making liquid formulations readily available, and particularly advantageous in pediatric cases. Because acetaminophen is water insoluble, liquid formulations are prepared with solvents such as propylene glycol to keep acetaminophen in solution; otherwise, the formulation would be a suspension.

Metabolism and Toxicity The metabolism of acetanilide, acetaminophen, and phenacetin is illustrated in Figure 31.9 (61). As indicated earlier, both acetanilide and phenacetin are metabolized to acetaminophen. Additionally, both undergo hydrolysis to yield aniline derivatives that produce directly, or through their conversion to hydroxylamine derivatives, significant methemoglobinemia and hemolytic anemia, which resulted in their removal from the US market. It has been known for more than 30 years that CYP450-mediated metabolism is responsible for the hepatotoxicity of acetaminophen (62) (see also Chapter 4). However, acetaminophen undergoes rapid first-pass metabolism in the GI tract primarily by conjugation reactions, with the *O*-sulfate being the primary metabolite in children and the *O*-glucuronide being the primary metabolite in adults. A minor, but significant, product of both acetaminophen and phenacetin is the *N*-hydroxyamide produced by CYP2E1 and CYP3A4. CYP2E1 is the rate-limiting enzyme that initiates the cascade of events leading to acetaminophen hepatotoxicity; in the absence of this cytochrome P450 (CYP) enzyme, toxicity will only be apparent at high concentrations. Both CYP2E1 and CYP3A4 are induced by the ingestion of alcohol (63), accounting for the increase in acetaminophen toxicity, especially among alcoholics, observed upon the concomitant consumption of alcoholic beverages with acetaminophen. The *N*-hydroxyamide is then converted to a reactive toxic metabolite, an acetimidoquinone—*N*-acetyl-*p*-benzoquinoneimine—that has been suggested (64)

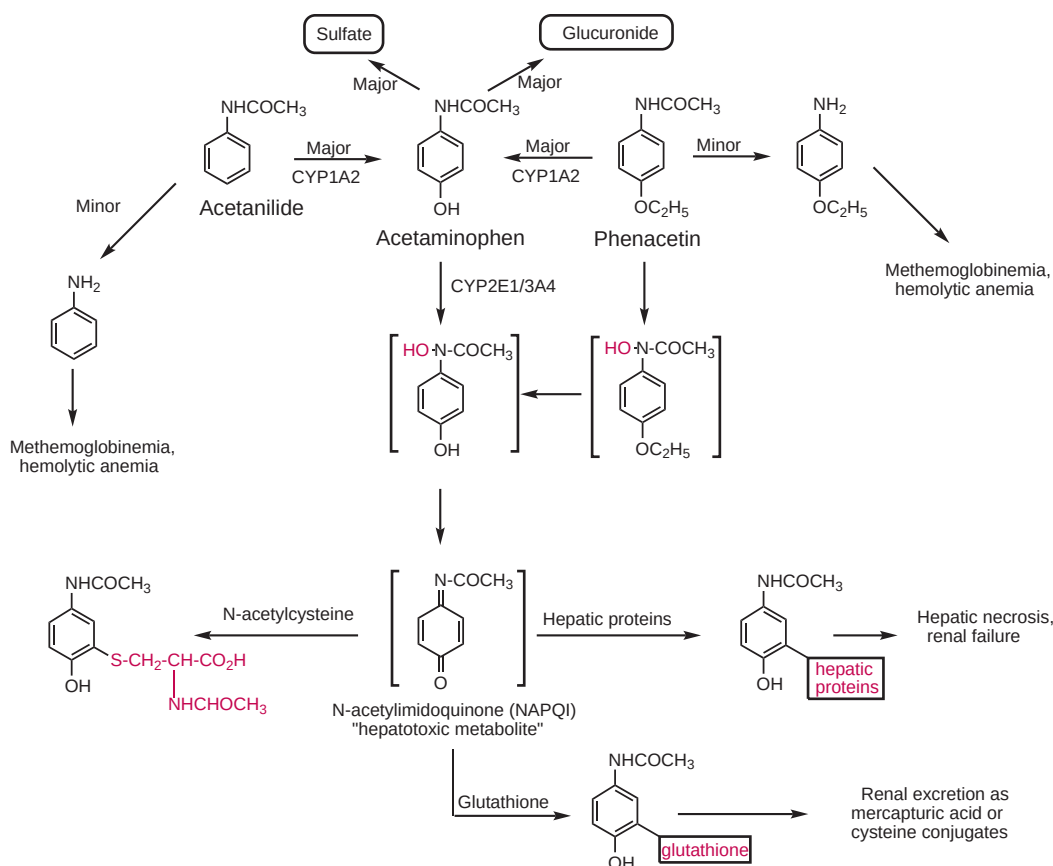


FIGURE 31.9 Metabolism of acetaminophen.

to produce the nephrotoxicity and hepatotoxicity associated with acetaminophen and phenacetin. Normally, this iminoquinone is detoxified by conjugation with hepatic glutathione. In cases of ingestion of large doses or overdoses of acetaminophen, however, hepatic stores of glutathione can be depleted by more than 70%, allowing the reactive quinone to interact with soft nucleophilic functional groups, primarily —SH groups, on hepatic proteins, resulting in the formation of covalent adducts that produce hepatic necrosis. Overdoses of acetaminophen can lead to fulminant acute liver failure, renal tubular necrosis, and hypoglycemic coma. Various sulfhydryl-containing compounds were found to be useful as antidotes to acetaminophen overdoses. The most useful of these antidotes is *N*-acetylcysteine (Mucomyst, Acetadote), which serves as a substitute for the depleted glutathione by enhancing hepatic glutathione stores and/or by enhancing disposition by non-toxic sulfate conjugation (65). *N*-Acetylcysteine can also inhibit the formation of the toxic iminoquinone metabolite (66). In cases of acetaminophen overdoses, *N*-acetylcysteine is administered as a 5% solution in water, soda, or juice or intravenously (IV) at 140 mg/kg followed by 17 maintenance doses of 70 mg/kg every 5 hours.

Drug Interactions Hepatic necrosis develops at much lower doses of acetaminophen in some heavy drinkers

than would be expected (61), perhaps because of the induction of the CYP2E1 system, depletion of glutathione stores, or aberrations in the primary sulfate and glucuronide conjugation pathways. At 4 g/d, acetaminophen has been reported to potentiate the response to oral anticoagulants, increasing prothrombin time (international normalized ratio values) two to three times. Interactions with warfarin (Coumadin), dicumarol, anisindione, and diphenadione have been suggested. The mechanism of these interactions has not been fully elucidated but can be associated with competition for plasma protein binding sites, because acetaminophen is a weak acid and is weakly bound, but can also interfere with the enzymes involved in vitamin K-dependent coagulation factor synthesis. The absorption of acetaminophen is enhanced by polysorbate and sorbitol and is reduced by anticholinergics and narcotic analgesics. Chemical incompatibilities have also been reported based on hydrolysis by strong acids or bases or by phenolic oxidation in the presence of oxidizing agents. Acetaminophen forms "sticky" mixtures with diphenhydramine HCl and discolors under humid conditions in the presence of caffeine or codeine phosphate.

Anti-Inflammatory Drugs

SALICYLATES The use of salicylates dates back to the 19th century. Salicylic acid itself was first obtained in 1838 from

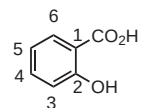
salicin, a glycoside present in most willow and poplar bark. Interestingly, Hippocrates prescribed chewing willow bark for pain relief in the fifth century AD. In 1860, Kolbe synthesized salicylic acid from sodium phenoxide and carbon dioxide, a method that inexpensively produced large quantities. Derivatives of salicylic acid began to receive medical attention shortly thereafter. Sodium salicylate was employed as an antipyretic/antirheumatic agent in 1875, and the phenyl ester was used in 1886. Acetylsalicylic acid was prepared in 1853 but was not used medicinally until 1899. The term “aspirin” was given to acetylsalicylic acid by Dreser, the director of pharmacology at Frederick Bayer and Company in Germany, as a contraction of the letter “a” from acetyl and “spirin,” an older name given to salicylic acid (spiric acid) that was derived from a natural source in spirea plants. Since then, numerous derivatives of salicylic acid have been synthesized and evaluated pharmacologically, yet only a relatively few derivatives have achieved therapeutic utility.

In addition to possessing antipyretic, analgesic, and anti-inflammatory properties, salicylates possess other actions that have been proven to be therapeutically beneficial. Because salicylates promote the excretion of uric acid, they are useful in the treatment of gouty arthritis. More recent attention has been given to the ability of salicylates to inhibit platelet aggregation, which can contribute to heart attacks and stroke. Aspirin appears to inhibit prostaglandin COX in platelet membranes, thus blocking formation of the potent platelet-aggregating factor TXA_2 in a manner that is irreversible. The Physicians Health Study concluded that in a group of 22,071 participants, there was a 44% reduction in the risk of myocardial infarction in the group taking a single 325-mg aspirin tablet taken every other day versus the placebo group (67). The role of aspirin in reducing cardiac mortality has been reviewed (68). Also, aspirin and other NSAIDs might be protective against colon cancer (69). Thus, the therapeutic utility of aspirin continues to increase. Unfortunately, a number of side effects are associated with the use of salicylates, most notably GI disturbances such as dyspepsia, gastroduodenal bleeding, gastric ulcerations, and gastritis.

Mechanism of Action A number of possible mechanisms of action have been proposed for salicylates over the years. Among those that have been suggested are inhibition of the biosynthesis of histamine, antagonism of the actions of various kinins, inhibition of mucopolysaccharide biosynthesis, inhibition of lysosomal enzyme release, and inhibition of leukocyte accumulation. The most widely accepted mechanism of action currently is the ability of these drugs to inhibit the biosynthesis of prostaglandins at the cyclooxygenase stage discussed earlier. Aspirin is the only NSAID that covalently modifies COX by acetylating Ser⁵³⁰ of COX-1 and Ser⁵¹⁶ of COX-2. Aspirin, however, is 10 to 100 times more potent against COX-1 than against COX-2 (70). Aspirin’s actions on COX-1 prevent both endoperoxide and 15-peroxidation

of arachidonic acid, but its action on COX-2 does not prevent formation of 15-OOH arachidonic acid (29).

Structure–Activity Relationships Despite the vast effort that has been expended in the search to find a “better” aspirin—that is, one possessing fewer GI side effects but a greater potency and a longer duration of action yet is inexpensive and an antipyretic, analgesic, and anti-inflammatory agent that is overall superior to aspirin—none has yet to be discovered. The following structure–activity relationships have been established:



Salicylic acid

The active moiety appears to be the salicylate anion. The side effects of aspirin, particularly the GI effects, appear to be associated with the carboxylic acid function. Reducing the acidity of this group (e.g., converting to an amide, salicylamide) maintains the analgesic actions of salicylic acid derivatives but eliminates the anti-inflammatory properties. Substitution on either the carboxyl or phenolic hydroxyl groups can affect potency and toxicity. Benzoic acid itself has only weak anti-inflammatory activity. Placing the phenolic hydroxyl group meta or para to the carboxyl group abolishes this activity. Substitution of halogen atoms on the aromatic ring enhances potency and toxicity. Substitution of aromatic rings at the 5-position of salicylic acid increases anti-inflammatory activity (e.g., diflunisal).

Absorption and Metabolism Most salicylates are rapidly and effectively absorbed on oral administration, with the rate of absorption and bioavailability being dependent on a number of factors, including the dosage formulation, gastric pH, food contents in the stomach, gastric emptying time, the presence of buffering agents or antacids, and particle size. Because salicylates are weak carboxylic acids (acetylsalicylic acid $\text{p}K_a = 3.5$), absorption takes place primarily from the small intestine and, to a lesser extent, from the stomach by the process of passive diffusion of un-ionized molecules across the epithelial membranes of the GI tract. Thus, gastric pH is an important factor in the rate of absorption of salicylates. Any factor that increases gastric pH (e.g., buffering agents, proton pump inhibitors) will slow its rate of absorption, because more of the salicylate will be in the ionized form. The differences in the rates of absorption of aspirin, salicylate salts, and the numerous buffered preparations of salicylates are actually quite small, with absorption half-times in humans ranging from approximately 20 minutes for buffered preparations to 30 minutes for aspirin itself. The presence of food in the stomach also slows the rate of absorption. Formulation factors can contribute to the differences in absorption rates of the various brands of

plain and buffered salicylate preparations. Tablet formulations consisting of small particles are absorbed faster than those of larger particle size. The bioavailability of salicylate from enteric-coated preparations can be inconsistent. Absorption of salicylate from rectal suppositories is slower and incomplete and is not recommended when high salicylate levels are required. Topical preparations of salicylic acid ester (e.g., methyl salicylate) are effective because the rate of salicylate absorption from the skin is rapid. However, 3% to 5% solutions of salicylic acid are also applied topically as a keratolytic agent.

Salicylates are highly bound to plasma protein albumin, with binding being concentration dependent. At low therapeutic concentrations of 100 $\mu\text{g/mL}$, approximately 90% of aspirin is plasma protein bound, whereas at higher concentrations of approximately 400 $\mu\text{g/mL}$, only 76% binding is observed. Plasma protein binding is a major factor in the drug interactions observed for salicylates.

The major metabolic routes of esters and salts of salicylic acid are illustrated in Figure 31.10. The initial route of metabolism of these derivatives is their conversion to salicylic acid, which can be excreted in the urine as the free acid (10%) or undergo conjugation with either glycine, to produce the major metabolite salicyluric acid (75%), or with glucuronic acid, to form the glucuronide ether and ester (15%). In addition, small amounts of metabolites resulting from microsomal aromatic hydroxylation are found. The major hydroxylation metabolite, gentisic acid, was once thought to be responsible for the anti-inflammatory actions of the salicylates, but its presence in trace quantities would rule out a major role for gentisic acid, or the other hydroxylation metabolites, in the pharmacologic action of salicylates. The metabolism of pharmacokinetic properties of salicylates has been extensively reviewed (71).

Side Effects The most commonly observed side effects associated with the use of salicylates relate to disturbances of the GI tract. Nausea, vomiting, epigastric discomfort, intensification of symptoms of peptic ulcer disease (e.g., dyspepsia and heartburn), gastric ulcerations, erosive gastritis, and GI hemorrhage occur in individuals on high doses of aspirin. The incidence of these side effects is rarer at low doses, but a single dose of aspirin can cause GI distress in 5% of individuals. Gastric bleeding induced by salicylates is usually painless but can lead to fecal blood loss and can cause a persistent iron deficiency anemia. At dosages that are useful in anti-inflammatory therapy, aspirin can lead to a loss of 3 to 8 mL/d of blood. The mechanism by which salicylates cause gastric mucosal cell damage can be caused by a number of factors, including gastric acidity, ability of salicylates to damage the normal mucosal barrier that protects against the back diffusion of hydrogen ions, ability of salicylates to inhibit the formation of prostaglandins (particularly those of the PGE series, which normally inhibit gastric acid secretion), and inhibition of platelet aggregation (leading to an increased tendency toward bleeding). Thus, salicylate use before surgery or tooth extraction is contraindicated.

Reye syndrome is an acute condition that can follow influenza and chickenpox infections in children from infancy to their late teens, with the majority of cases occurring between the ages of 4 and 12 years. It is characterized by symptoms including sudden vomiting, violent headaches, and unusual behavior in children who appear to be recovering from an often mild viral illness. Although a rare condition (60 to 120 cases per year, or an incidence of 0.15 per 100,000 population of those ≤ 18 years), it can be fatal, with a death rate of between 20% and 30%. Fortunately, the number of cases

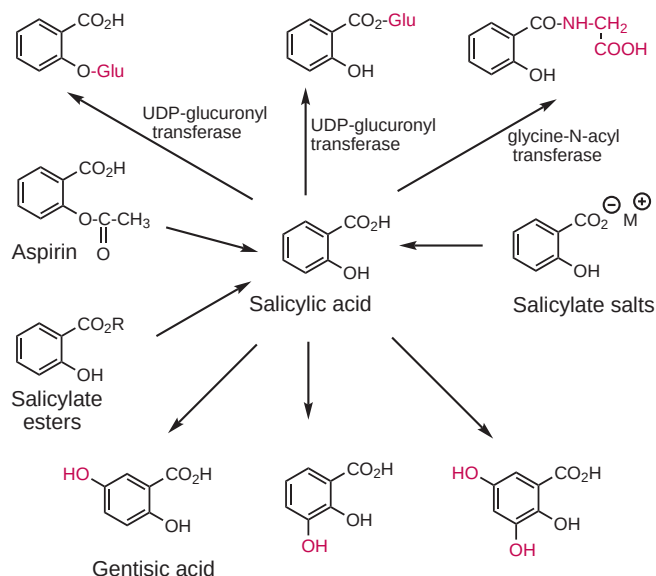


FIGURE 31.10 Metabolism of salicylic acid derivatives. Glu, glucuronide conjugate.

is declining, partly because of the observation that more than 90% of children with Reye syndrome were on salicylate therapy during a recent viral illness. Based on these observations, the FDA has proposed that aspirin and other salicylates be labeled with a warning against their use in children younger than 16 years with influenza, chickenpox, or other flu-like illness. Acetaminophen would appear to be the drug of choice in children with these conditions.

Salicylates account for approximately 25% of all accidental poisonings in the United States.

Drug Interactions Because of the widespread use of salicylates, it is not surprising that interactions with many other drugs used in therapeutic combinations have been observed. Several of these interactions are clinically significant. More data are available for aspirin than for any other specific salicylate product. As mentioned previously, acetylsalicylic acid is a weak acid that is highly bound to plasma proteins (50% to 80%), and it will compete for these plasma protein binding sites with other drugs that are highly bound to these sites. The interaction that results from the combination of salicylates with oral anticoagulants represents one of the most widely documented clinically significant drug interactions reported to date. The plasma concentration of free anticoagulant increases in the presence of salicylates, necessitating a possible decrease in the dosage of anticoagulant required to produce a beneficial therapeutic effect. The ability of salicylates to produce GI ulcerations and bleeding, coupled with the inhibition of the clotting mechanism, results in a clinically significant drug interaction. In addition, salicylates can inhibit the synthesis of prothrombin by antagonizing the actions of vitamin K. Additionally, NSAIDs can produce these interactions. The competition for plasma protein binding sites can also lead to an increase in free methotrexate levels (thus enhancing the toxicity of methotrexate), enhanced toxicity of long-acting sulfonamides, and a hypoglycemic effect (resulting from displacement of oral hypoglycemic drugs). In large doses, salicylates given concomitantly with uricosuric drugs, such as probenecid and sulfinpyrazone, can lead to a retention of uric acid and, thus, antagonize the uricosuric effect, despite the fact that salicylates when used alone increase urinary excretion of uric acid. The diuretic activity of aldosterone antagonists, such as spironolactone, can be antagonized by salicylates. Corticosteroids can decrease blood levels of salicylates because of their ability to increase the glomerular filtration rate. The incidence and severity of GI ulcerations can be increased if corticosteroids, salicylates, and NSAIDs are administered together. The GI bleeding induced by salicylates can be enhanced by the ingestion of ethanol. Numerous other interactions have been reported, but their clinical significance has not been fully established.

Salicylate hypersensitivity, particularly to aspirin, is relatively uncommon but must be recognized, because

severe and potentially fatal reactions can occur. Signs of aspirin hypersensitivity appear soon after administration and include skin rashes, watery secretions, urticaria, vasomotor rhinitis, edema, bronchoconstriction, and anaphylaxis. Less than 1% of the US population can experience aspirin hypersensitivity; this group consists primarily of middle-aged individuals. Females are more likely to experience aspirin intolerance or hypersensitivity. Aspirin-sensitive patients with asthma are especially at high risk. Mild salicylism can occur after repeated administration of large doses. Symptoms include dizziness, tinnitus, nausea, vomiting, diarrhea, and mental confusion. Doses of 10 to 30 g have been known to cause death in adults, but some individuals have ingested up to 130 g without fatality. More than 10,000 cases of serious salicylate toxicity occur in the United States each year.

Available Preparations The structures of the marketed preparations of salicylic acid are presented in Figure 31.11.

Aspirin USP. Acetylsalicylic acid, or aspirin, is a white powder that is stable in a dry environment but that is hydrolyzed to salicylic acid and acetic acid under humid or moist conditions. Hydrolysis can also occur when aspirin is combined with alkaline salts or with salts containing water of hydration. Stable aqueous solutions of aspirin are thus unobtainable despite the addition of modifying drugs that tend to decrease hydrolysis. Aspirin is rapidly absorbed largely intact from the stomach and upper small intestine on oral administration but is rapidly hydrolyzed by plasma esterases. Peak plasma levels are usually achieved within 2 hours after administration. Increasing the pH of the stomach by the addition of buffering agents can affect absorption, because the degree of ionization will be increased.

Aspirin is indicated for the relief of minor aches and mild to moderate pain (325 to 650 mg every 4 hours), for arthritis and related arthritic conditions (3.2 to

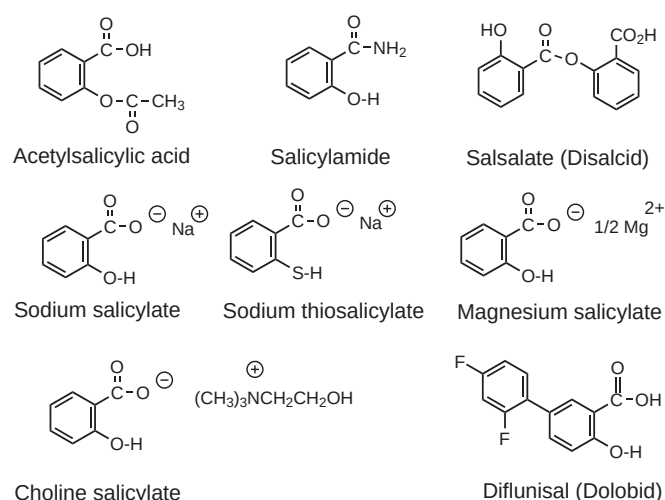


FIGURE 31.11 Structures of marketed derivatives of salicylic acid.

6.0 g/d), to reduce the risk of transient ischemic attacks (1.3 g/d), for myocardial infarction prophylaxis (300 to 325 mg/d), and as a platelet aggregation inhibitor (80 to 325 mg/d). It is available in a large number of dosage forms and strengths as tablets, suppositories, capsules, enteric-coated tablets, and buffered tablets.

Salicylamide. Salicylamide is less acidic ($pK_a = 8.2$) than other salicylic acid derivatives. Although poorly soluble in water, stable solutions can be formed at pH 9 through ionization of the phenolic group. It is absorbed from the GI tract on oral administration and is rapidly metabolized to inactive metabolites by intestinal mucosa, but not by hydrolysis. Activity appears to reside in the intact molecule. Salicylamide is approximately 40% to 55% plasma protein bound, and it competes with other salicylates and acetaminophen for glucuronide conjugation, decreasing the extent of conjugation of these other drugs. Excretion occurs rapidly, primarily in the urine. The major advantages of salicylamide are its general lack of gastric irritation relative to aspirin, and its use in individuals who are hypersensitive to aspirin. Salicylamide enters the CNS more rapidly than other salicylates and will cause sedation and drowsiness when administered in large doses. Whereas salicylamide is reported to be as effective as aspirin as an analgesic/antipyretic and is effective in relieving pain associated with arthritic conditions, it does not appear to possess useful anti-inflammatory activity (72). Thus, indications for the treatment of arthritic disease states are unwarranted, and its use is restricted to the relief of minor aches and pain at a dosage of 325 to 650 mg three or four times per day. Its effects in humans are not reliable, however, and its use is not widely recommended.

Salicylate Salts. Several salts of salicylic acid, sodium salicylate USP, choline salicylate USP, and magnesium salicylate USP, and one salt of thiosalicylic acid, sodium thiosalicylate USP, are available. These salts are used primarily to decrease GI disturbances or because they form stable aqueous solutions. Sodium salicylate is half as potent, on a weight basis, as aspirin as an analgesic/antipyretic, but it produces less GI irritation and equivalent blood levels and is useful in patients exhibiting hypersensitivity to aspirin. It generates salicylic acid in the GI tract, accounting for some GI irritation, and sodium bicarbonate sometimes is given concomitantly to reduce acidity. Sodium salicylate, unlike aspirin, does not affect platelet function, although prothrombin times are increased. It is available as tablets, enteric-coated tablets, and as a solution for injection.

Choline salicylate has a lower incidence of GI side effects compared with aspirin, and it has been shown to be particularly useful in treating juvenile rheumatoid arthritis, in which aspirin was ineffective. It is absorbed more rapidly than aspirin and produces higher salicylate plasma levels. It is available as a mint-flavored liquid.

Magnesium salicylate has a low incidence of GI side effects. Both sodium salicylate and magnesium salicylate

should be used cautiously in individuals in whom excessive amounts of these electrolytes might be detrimental. The possibility of magnesium toxicity in individuals with renal insufficiency exists. It is available as tablets, but its safety in children under 12 years of age has not been fully determined.

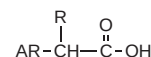
Sodium thiosalicylate (Solate) is administered intramuscularly (IM) for the treatment of rheumatic fever, muscular pain, and acute gout.

Salsalate (Disalcid). Salsalate, or salicylsalicylic acid, ($pK_a = 3.5$ [COOH], 9.8 [AR-OH]) is a dimer of salicylic acid. It is insoluble in gastric juice but is soluble in the small intestine, where it is partially hydrolyzed to two molecules of salicylic acid and absorbed. On a molar basis, it produces 15% less salicylic acid than aspirin. It does not cause GI blood loss and can be given to aspirin-sensitive patients. Salsalate is available as capsules and tablets.

Diflunisal (Dolobid). Diflunisal ($pK_a = 3.3$) was introduced in the United States in 1982 and has gained considerable acceptance as an analgesic and as a treatment of rheumatoid arthritis and osteoarthritis. Diflunisal is metabolized primarily to ether and ester glucuronide conjugates. No metabolism involving changes in ring substituents has been reported. It is more potent than aspirin but produces fewer side effects and has a biologic half-life three to four times greater than that of aspirin. It is rapidly and completely absorbed on oral administration, with peak plasma levels being achieved within 2 to 3 hours of administration. It is highly bound (99%) to plasma proteins after absorption. Its elimination half-life is 8 to 12 hours, and it is excreted into urine primarily as glucuronide conjugates. The most frequently reported side effects include disturbances of the GI system (e.g., nausea, dyspepsia, and diarrhea), dermatologic reactions, and CNS effects (e.g., dizziness and headache).

Diflunisal is a moderately potent inhibitor of prostaglandin biosynthesis, but it differs from the manner in which aspirin inhibits the cyclooxygenase system in that the inhibition is competitive and reversible in nature. Diflunisal does not have an appreciable effect on platelet aggregation, however, and does not significantly produce gastric or intestinal bleeding.

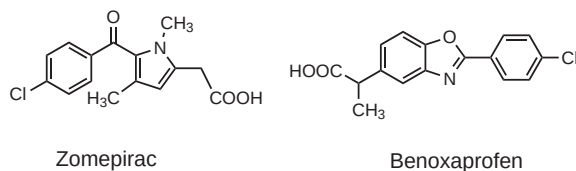
ARYLALKANOIC ACIDS



R = H, CH₃ or alkyl
AR = aryl or heteroaryl

The largest group of NSAIDs is represented by the class of arylalkanoic acids as typified by the general chemical structure, and several factors have caused this to be one of the most active areas of drug development in recent years. The impact that the introduction of

phenylbutazone in the 1950s had on arthritis therapy was more than matched by the interest generated by the introduction of indomethacin in the mid-1960s. As a result of a study designed to investigate the anti-inflammatory activity of 350 indole acetic acid derivatives related structurally to serotonin and metabolites of serotonin, the Merck group led by Shen (73) reported the synthesis and antipyretic and anti-inflammatory activity of the most potent compound in the series, indomethacin. The observation that indomethacin possessed 1,085-fold the anti-inflammatory activity and 20-fold the antipyretic activity of phenylbutazone (and 10-fold the antipyretic activity of aminopyrine) generated considerable interest in the development of other aryl and heteroaryl acetic acid and propionic acid derivatives. The marketplace was ripe for new anti-inflammatory drugs, and most pharmaceutical companies joined in the search for new arylalkanoic acids. The introduction of ibuprofen in the 1970s by Upjohn was quickly followed by the appearance of fenoprofen calcium, naproxen, and tolmetin. Sulindac, an analog of indomethacin, was introduced in the late 1970s. The 1980s produced zomepirac, benoxaprofen, ketoprofen, flurbiprofen, suprofen, and diclofenac sodium. The 1990s produced ketorolac, etodolac, nabumetone, and most significantly, the development of selective COX-2 inhibitors, celecoxib, rofecoxib, and valdecoxib, which reached the market during the period from 1997 to 2000. This rapid development was accompanied by some setbacks, however. Zomepirac, introduced in 1980 as an analgesic, was withdrawn in 1983 because of severe anaphylactoid reactions, particularly in patients sensitive to aspirin.



Benoxaprofen was withdrawn within 6 months of its introduction in 1982 because of several deaths caused by cholestatic jaundice in Europe and the United States. In addition, benoxaprofen produced photosensitivity reactions in patients when they were exposed to sunlight and onycholysis (loosening of the fingernails) in some patients. Suprofen, introduced as an analgesic in 1985, was removed from the market 2 years later because of flank pain and transient renal failure. In 1989, however, it was reintroduced for ophthalmic use. Numerous other arylalkanoic acids currently are being evaluated in various stages of clinical trials.

As discussed earlier, most NSAIDs possess a number of biochemical and pharmacologic actions. As was the case for the salicylates, the arylalkanoic acids share, to various extents, the property of inhibition of prostaglandin biosynthesis by inhibiting COX-1 and COX-2 with varying degrees of selectivity (Table 31.3).

General Structure–Activity Relationships Drugs of this class share a number of common structural features. These general structure–activity relationships will be discussed here as they pertain to the proposed mechanism of action. Specific structure–activity relationships for each drug or drug class will be presented separately, where appropriate.

All nonselective COX inhibitors possess a center of acidity, which can be represented by a carboxylic acid function, an enolic function, a hydroxamic acid function, a sulfonamide, or a tetrazole ring. The relationship of this acid center to the carboxylic acid function of arachidonic acid is obvious. The activity of ester and amide derivatives of carboxylic acids is usually attributed to the metabolic hydrolysis products. One nonacidic drug, nabumetone, has been recently introduced in the United States, but as will be discussed later, its activity is attributed to its bioactivation to an active acid metabolite. The center of acidity is usually located one carbon atom adjacent to a flat surface represented by an aromatic or heteroaromatic ring. The distance between these centers is crucial, because increasing this distance to two or three carbons usually diminishes activity.

$R-COOH$	Carboxylic acid; pKa 4-6
$R-CONHOH$	Hydroxamic acids; pKa (NH) 8-9
$R-SO_2NH_2$	Sulfonamide; pKa (NH) 9-10
	Enolic (phenol); pKa (OH) 8-10
	Tetrazoles; pKa (NH) 4-6

Derivatives of aryl or heteroaryl acetic or propionic acids are most common. This aromatic system appears to correlate with the double bonds at the 5- and 8-positions of arachidonic acid. Substitution of a methyl group on the carbon atom separating the acid center from the aromatic ring tends to increase anti-inflammatory activity. The resulting α -methyl acetic acid, or 2-substituted propionic acid, analogs have been given the class name “profens” by the U.S. Adopted Name Council. Groups larger than methyl decrease activity, but incorporation of this methyl group as part of an alicyclic ring system does not drastically affect activity. Introduction of a methyl group creates a center of chirality. Anti-inflammatory activity in those cases in which the enantiomers have been separated and evaluated, whether determined *in vivo* or *in vitro* by COX assays, is associated with the *S*(+)-enantiomer. Interestingly, in those cases in which the propionic acid is administered as a racemic mixture, *in vivo* conversion of the *R*-enantiomer to the biologically active *S*-enantiomer is observed to varying degrees. A second area of lipophilicity that is noncoplanar with the aromatic or heteroaromatic ring usually enhances activity. This second lipophilic area can correspond to the area of

the double bond in the 11-position of arachidonic acid. This lipophilic function can consist of an additional aromatic ring or alkyl groups either attached to or fused with the aromatic center.

General Metabolism Essentially all the arylalkanoic acid derivatives that are therapeutically available are extensively metabolized. Metabolism occurs primarily through hepatic microsomal enzyme systems and can lead to deactivation or bioactivation of the parent molecules. Metabolism of each drug will be treated separately.

Drug Interactions All the arylalkanoic acids are highly bound to plasma proteins and, thus, can displace other drugs from protein binding sites, resulting in an enhanced activity and toxicity of the displaced drugs. Interestingly, despite the high degree of plasma protein binding, indomethacin does not display this characteristic drug interaction. The most commonly observed interaction is that between the arylalkanoic acid and oral anticoagulants, particularly warfarin (Coumadin). Coadministration can prolong prothrombin time. Potential interactions with other acidic drugs, such as hydantoins, sulfonamides, and sulfonylureas, should be monitored. Concomitant administration of aspirin decreases plasma levels of arylalkanoic acids by as much as 20%. Probenecid, on the other hand, tends to increase these plasma levels. Interactions with drugs that can induce hepatic microsomal enzyme systems, such as phenobarbital, can enhance or diminish anti-inflammatory activity depending on whether the arylalkanoic acid is metabolically bioactivated or inactivated by this enzyme system. Certain diuretics, such as furosemide, inhibit the metabolism of prostaglandins by 15-hydroxy-prostaglandin dehydrogenase, and the resulting increase in PGE_2 levels induces plasma renin activity. Because the arylalkanoic acids block the biosynthesis of prostaglandins, the effects of furosemide can be antagonized, in part, offering a potentially significant drug interaction.

ARYL- AND HETEROARYLACETIC ACIDS The structures of the aryl- and heteroarylacetic acid derivatives and the aryl- and heteroarylpropionic acids ("profens") available are presented in Figure 31.12.

Indomethacin Aqueous solutions of indomethacin are not stable because of the ease of hydrolysis of the *p*-chlorobenzoyl group. The original synthesis of indomethacin by Shen et al. (73) involved the formation of 2-methyl-5-methoxyindole acetic acid and subsequent acylation after protection of the carboxyl group as the *t*-butyl ester. It was introduced in the United States in 1965. It is still one of the most potent NSAIDs in use. It is also a more potent antipyretic than either aspirin or acetaminophen, and it possesses approximately 10 times the analgesic potency of aspirin. The analgesic effect, however, is widely overshadowed by concern over the frequency of side effects.

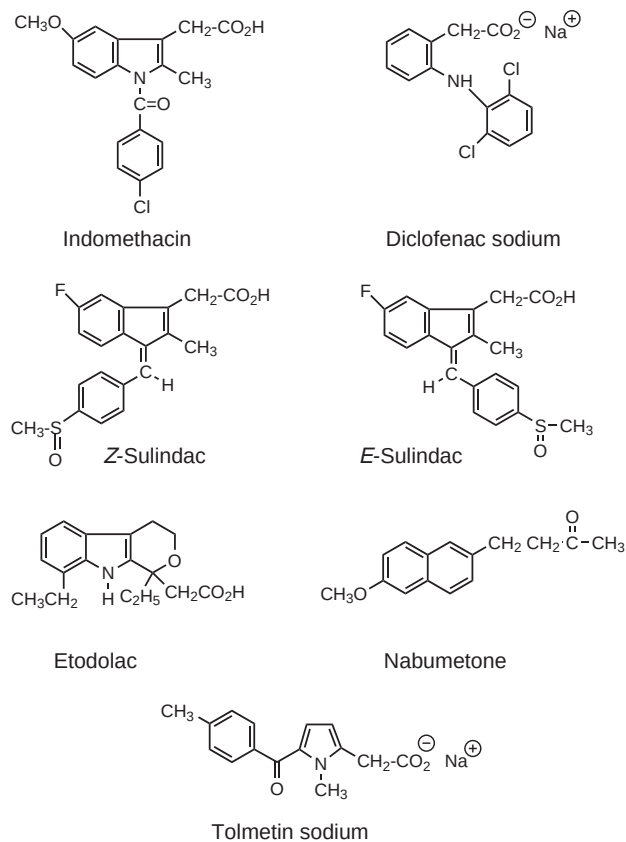
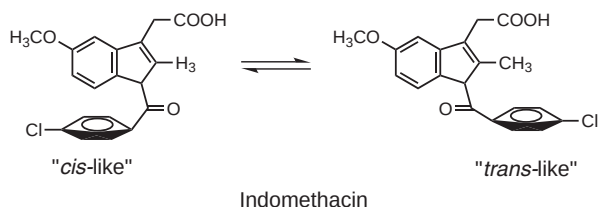


FIGURE 31.12 Structures of aryl- and heteroarylacetic acid derivatives.

Structure–Activity Relationships. Replacement of the carboxyl group with other acidic functionalities decreases activity. Anti-inflammatory activity usually increases as the acidity of the carboxyl group increases and decreases as the acidity is decreased. Amide analogs are inactive. Acylation of the indole nitrogen with aliphatic carboxylic acids or aralkylcarboxylic acids results in amide derivatives that are less active than those derived from benzoic acid. *N*-Benzoyl derivatives substituted in the para-position with fluoro, chloro, trifluoromethyl, or thiomethyl groups are the most active. The 5-position of the indole ring is most flexible with regard to the nature of substituents that enhance activity. Substituents such as methoxy, fluoro, dimethylamino, methyl, allyloxy, and acetyl are more active than the unsubstituted indole ring. The presence of an indole ring nitrogen is not essential for activity, because the corresponding 1-benzylidenylindene analogs (e.g., sulindac) are active. Alkyl groups, especially methyl, at the α -position are more active than aryl substituents. Substitution of a methyl group at the α -position of the acetic acid side chain (to give the corresponding propionic acid derivative) leads to equiactive analogs. The resulting chirality introduced in the molecules is important. Anti-inflammatory activity is displayed only by the *S*(+)-enantiomer. The conformation of indomethacin appears to have a crucial role in

its anti-inflammatory actions. The acetic acid side chain is flexible and can assume a large number of different conformations. The preferred and lower-energy conformation of the *N*-*p*-chlorobenzoyl group is one in which the chlorophenyl ring is oriented away from the 2-methyl group (or *cis* to the methoxyphenyl ring of the indole nucleus) and is noncoplanar with the indole ring because of steric hindrance produced by the 2-methyl group and the hydrogen atom at the 7-position. These conformations are represented as follows:



Absorption and Metabolism. Absorption of indomethacin occurs rapidly on oral administration, and peak plasma levels are obtained within 2 to 3 hours. Being an acidic substance ($pK_a = 4.5$), it is highly bound to plasma proteins (97%). Indomethacin is converted to inactive metabolites; approximately 50% of a single dose is 5-*O*-demethylated by CYP2C9 and 10% is conjugated with glucuronic acid. Nonhepatic enzyme systems hydrolyze indomethacin to *N*-deacylated metabolites. The metabolism of indomethacin is illustrated in Figure 31.13. The ability of indomethacin to potently inhibit prostaglandin

biosynthesis can account for its anti-inflammatory, antipyretic, and analgesic actions.

Adverse Effects. Pronounced side effects are frequently observed at antirheumatic doses. A large number of individuals taking indomethacin, especially those over the age of 70, experience undesirable effects of the GI tract (e.g., nausea, dyspepsia, diarrhea, erosion/ulceration of the stomach walls), CNS (e.g., headache, dizziness, and vertigo), and ears (e.g., tinnitus). Thus, many patients must discontinue its use. As with other arylalkanoic acids, administration of indomethacin with food or milk decreases GI side effects.

Indomethacin is available for the short-term treatment of acute gouty arthritis, acute pain of ankylosing spondylitis, and osteoarthritis. An injectable form to be reconstituted is also available as the sodium trihydrate salt for IV use in premature infants with patent ductus arteriosus. Because of its ability to suppress uterine activity by inhibiting prostaglandin biosynthesis, indomethacin also has an unlabeled use to prevent premature labor.

Sulindac Sulindac was introduced in the United States in 1978 by Merck as a result of chemical studies designed to produce an analog without the side effects commonly associated with the use of indomethacin, particularly GI irritation. It achieved wide popularity and remains a widely used NSAID. Its synthesis was also reported by Shen et al. (74). Sulindac is a prodrug and is converted to a metabolite that appears to inhibit the COX system approximately eightfold as effectively as aspirin. In anti-inflammatory and antipyretic assays, it is only about half as potent as indomethacin but is equipotent in analgesic assays.

Structure–Activity Relationships. The use of classical bioisosteric changes in medicinal chemistry drug design was invoked in the design of sulindac. The isosteric replacement of the indole ring with the indene ring system resulted in a derivative with therapeutically useful anti-inflammatory activity and fewer CNS and GI side effects but with other undesirable effects, particularly poor water solubility and resultant crystalluria. The replacement of the *N*-*p*-chlorobenzoyl substituent with a benzylidene function resulted in active derivatives. However, when the 5-methoxy group of the indene isostere was replaced with a fluorine atom, enhanced analgesic effects were observed. The decreased water solubility of the indene isostere was alleviated by replacing the chlorine atom of the phenyl substituent with a sulfinyl group. The importance of stereochemical features in the action of sulindac, introduced by the benzylidene double bond, is evidenced by the observation that the *Z*-isomer is a much more potent anti-inflammatory agent than the corresponding *E*-isomer (Fig. 31.12). This *cis*-relationship of the phenyl substituent to the aromatic ring bearing the fluoro substituent is similar to the proposed conformation of indomethacin, suggesting that both indomethacin and

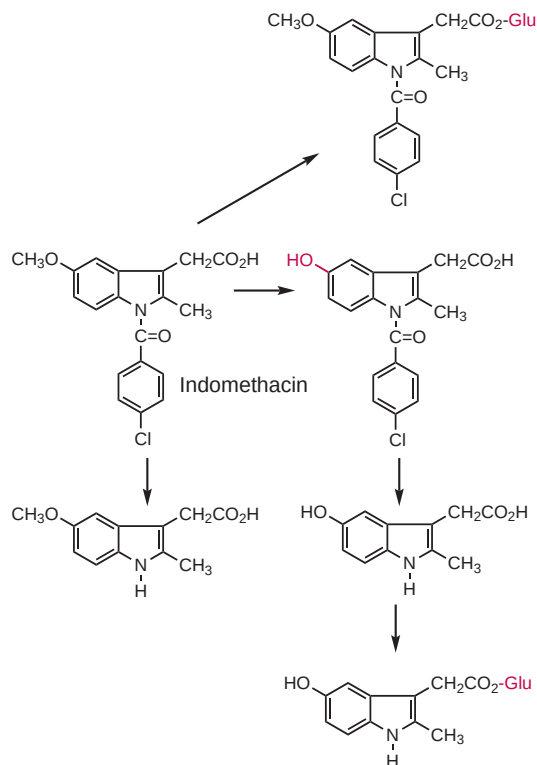


FIGURE 31.13 Metabolism of indomethacin.

sulindac assume similar conformations at the active site of arachidonic acid COX. Although sulindac contains a chiral sulfoxide entity, it is marketed as the racemate because of its reduction to the active, achiral sulfide.

Absorption and Metabolism. Sulindac is well absorbed on oral administration (90%), reaches peak plasma levels within 2 to 4 hours, and being acidic ($pK_a = 4.5$), is highly bound to serum proteins (93%). The metabolism of sulindac has a major role in its actions, because all of the pharmacologic activity is associated with its major metabolite. Sulindac is, in fact, a prodrug, the sulfoxide function being reduced to the active sulfide metabolite. Sulindac is absorbed as the sulfoxide, which is not an inhibitor of prostaglandin biosynthesis in the GI tract. As discussed earlier, prostaglandins exert a protective effect in the GI tract, and inhibition of their synthesis here leads to many of the GI side effects noted for most NSAIDs. Once sulindac enters the circulatory system, it is reduced to the active sulfide, which is an inhibitor of prostaglandin biosynthesis in the joints. Thus, sulindac produces less GI side effects, such as bleeding, ulcerations, and so on, than indomethacin and many other NSAIDs. In addition, the active metabolite has a plasma half-life approximately twice that of the parent compound (~16 hours vs. 8 hours), which favorably affects the dosing schedule. In addition to the sulfide metabolite, sulindac is oxidized to the corresponding sulfone, which is inactive. A minor product results from hydroxylation of the benzylidene function and the methyl group at the 2-position. Glucuronides of several metabolites are also found. Sulindac and the sulfide and the sulfone metabolites are all highly protein bound. Despite the fact that the sulfide metabolite is a major activation product and is found in high concentration in human plasma, it is not found in human urine, perhaps because of its high degree of protein binding. The major excretion product is the sulfone metabolite and its glucuronide conjugate. The metabolism of sulindac is illustrated in Figure 31.14.

Whereas the toxicity of sulindac is lower than that observed for indomethacin and other NSAIDs, the spectrum of adverse reactions is very similar. The most frequent side effects reported are associated with irritation of the GI tract (e.g., nausea, dyspepsia, and diarrhea), although these effects are generally mild. Effects on the CNS (e.g., dizziness and headache) are less common. Dermatologic effects are less frequently encountered.

Sulindac is indicated for long-term use in the treatment of rheumatoid arthritis, osteoarthritis, ankylosing spondylitis, and acute gouty arthritis. The usual maximum dosage is 400 mg/d, with starting doses recommended at 150 mg twice a day. It is recommended that sulindac be administered with food.

Tolmetin Sodium Tolmetin is synthesized straightforwardly from 1-methylpyrrole (75). It was introduced in the United States in 1976, and like other NSAIDs,

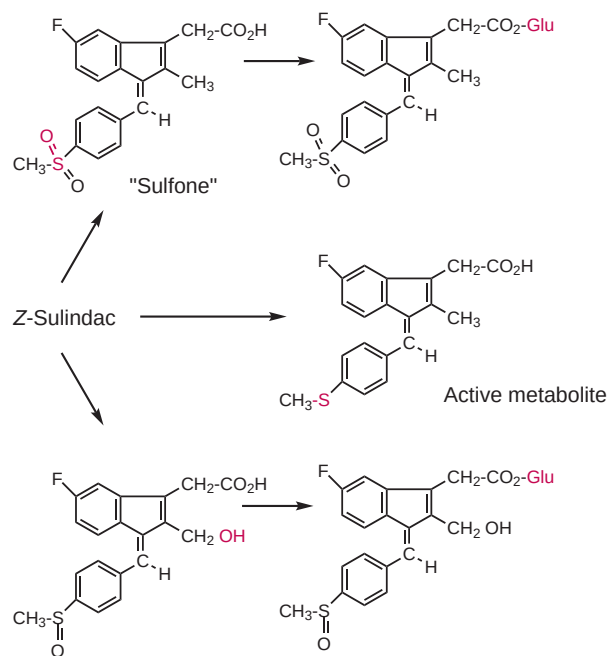


FIGURE 31.14 Metabolism of sulindac. Glu, glucuronide.

inhibits prostaglandin biosynthesis. Tolmetin, however, also inhibits polymorph migration and decreases capillary permeability. Its anti-inflammatory activity, as measured in the carrageenan-induced rat paw edema and cotton pellet granuloma assays, is intermediate between those of phenylbutazone and indomethacin.

Structure–Activity Relationships. The relationship of tolmetin to indomethacin is clear, with each containing a noncoplanar *p*-chlorobenzoyl group and an acetic acid function. Tolmetin possesses a pyrrole ring instead of the indole ring in indomethacin. Replacement of the 5-*p*-toluoyl group with a *p*-chlorobenzoyl moiety produced little effect on activity, whereas introduction of a methyl group in the 4-position of the pyrrole ring produced interesting results. The 4-methyl-5-*p*-chlorobenzoyl analog is approximately four times as potent as tolmetin. Substitution of the *p*-methyl group of tolmetin with a *p*-chloro group blocked oxidative metabolism, increasing duration of action to approximately 24 hours. McNeil marketed this *p*-chloro compound in 1980 as zomepirac, an analgesic that was removed from the market in 1983 because of severe anaphylactic reactions, particularly in patients sensitive to aspirin. Unlike the previous structure–activity relationships discussed for arylalkanoic acids, the propionic acid analog is slightly less potent than tolmetin.

Absorption and Metabolism. Tolmetin sodium is rapidly and almost completely absorbed on oral administration, with peak plasma levels being attained within the first hour of administration. It has a relatively short plasma half-life of approximately 1 hour because of extensive first-pass metabolism, involving hydroxylation of the *p*-methyl group to the primary alcohol, which is

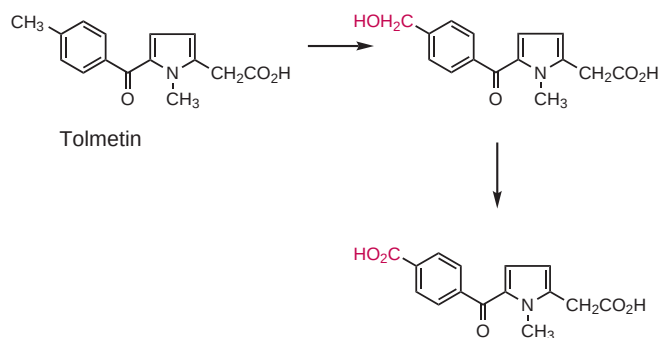


FIGURE 31.15 Metabolism of tolmetin.

subsequently oxidized to the dicarboxylic acid (Fig. 31.15). This metabolite is inactive in standard *in vivo* anti-inflammatory assays. The free acid ($pK_a = 3.5$) is highly bound to plasma proteins (99%), and excretion of tolmetin and its metabolites occurs primarily in the urine. Approximately 15% to 20% of an administered dose is excreted unchanged, and 10% is excreted as the glucuronide conjugate of the parent drug. Conjugates of the dicarboxylic acid metabolite account for the majority of the remaining administered drug.

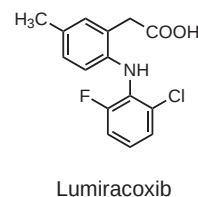
The most frequently adverse reactions are those involving the GI tract (e.g., abdominal pain, discomfort, and nausea) but appear to be less than those observed with aspirin. The CNS effects (e.g., dizziness and drowsiness) are also observed. Few cases of overdosage have been reported, but in such cases, recommended treatment includes elimination of the drug from the GI tract by emesis or gastric lavage and elimination of the acidic drug from the circulatory system by enhancing alkalization of the urine with sodium bicarbonate.

Tolmetin sodium is indicated for the treatment of rheumatoid arthritis, juvenile rheumatoid arthritis, and osteoarthritis.

Diclofenac Sodium Diclofenac is synthesized from *N*-phenyl-2,6-dichloroaniline (76). It is available in 120 different countries and, perhaps, is the most widely used NSAID in the world. It was introduced in the United States in 1989 but was first marketed in Japan in 1974. It ranks among the top prescription drugs in the United States. Diclofenac possesses structural characteristics of both arylalkanoic acid and the anthranilic acid classes of anti-inflammatory drugs, and it displays anti-inflammatory, analgesic, and antipyretic properties. In the carrageenan-induced rat paw edema assay, it is twice as potent as indomethacin and 450 times as potent as aspirin. As an analgesic, it is six times more potent than indomethacin and 40 times as potent as aspirin in the phenyl benzoquinone-induced writhing assay in mice. As an antipyretic, it is twice as potent as indomethacin and more than 350 times as potent as aspirin in the yeast-induced fever assay in rats. Diclofenac is unique among the NSAIDs in that it possesses three possible mechanisms of action: 1)

inhibition of the arachidonic acid COX system (3 to 1,000 time more potent than other NSAIDs on a molar basis), resulting in a decreased production of prostaglandins and thromboxanes; 2) inhibition of the lipoxygenase pathway, resulting in decreased production of leukotrienes, particularly the proinflammatory LKB₄; and 3) inhibition of arachidonic acid release and stimulation of its reuptake, resulting in a reduction of arachidonic acid availability.

Structure–Activity Relationships. Structure–activity relationships in this series have not been extensively studied. It does appear that the function of the two *o*-chloro groups is to force the anilino-phenyl ring out of the plane of the phenylacetic acid portion, this twisting effect being important in the binding of NSAIDs to the active site of the COX, as previously discussed. A selective COX-2 inhibitor, lumiracoxib, which is closely related structurally to diclofenac, was briefly marketed in 2007 but quickly withdrawn from the market because of hepatotoxicity.



Absorption and Metabolism. Diclofenac is rapidly and completely (~100%) absorbed on oral administration, with peak plasma levels being reached within 1.5 to 2.5 hours. The free acid ($pK_a = 4.0$) is highly bound to serum proteins (99.5%), primarily albumin. Only 50% to 60% of an oral dose is bioavailable because of extensive hepatic metabolism. Four major metabolites resulting from aromatic hydroxylation have been identified. The major metabolite via CYP3A4 is the 4'-hydroxy derivative and accounts for 20% to 30% of the dose excreted, whereas the 5-hydroxy, 3'-hydroxy, and 4',5-dihydroxy metabolites via CYP2C9 account for 10% to 20% of the excreted dose. The remaining drug is excreted in the form of sulfate conjugates. Although the major metabolite is much less active than the parent compound, it can exhibit significant biologic activity, because it accounts for 30% to 40% of all of the metabolic products. The metabolism of diclofenac is illustrated in Figure 31.16. Also, diclofenac has been reported to produce reactive benzoquinone imine intermediates (77) similar to that reported in the metabolism of acetaminophen. These reactive intermediates are also normally inactivated via conjugation with glutathione.

Diclofenac sodium is indicated for the treatment of rheumatoid arthritis, osteoarthritis, and ankylosing spondylitis.

Etodolac Etodolac was promoted as the first of a new chemical class of anti-inflammatory drugs, the pyranocarboxylic acids. Although not strictly an arylacetic acid

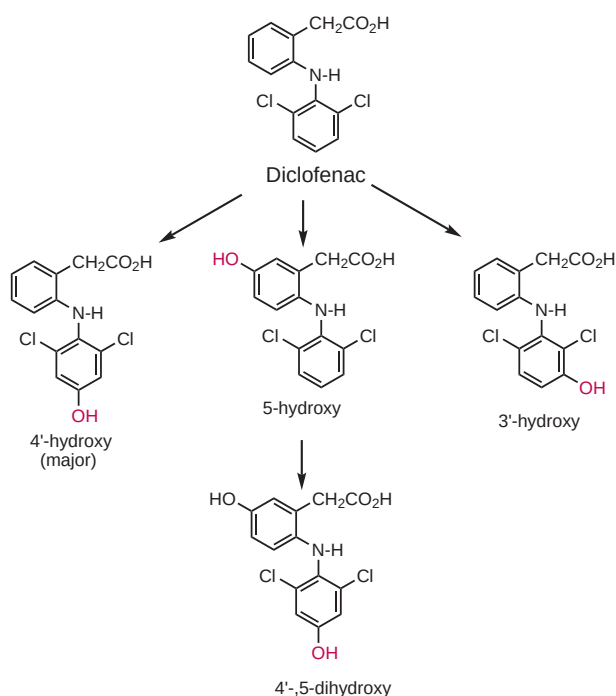
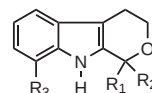


FIGURE 31.16 Metabolism of diclofenac.

derivative (because there is a two-carbon atom separation between the carboxylic acid function and the heteroaromatic ring) (Fig. 31.12), the drug still possesses structural characteristics similar to those of the heteroarylacetic acids and is classified here. It was introduced in the United States in 1991 for acute and long-term use in the management of osteoarthritis and as an analgesic and also possesses antipyretic activity. Etodolac is marketed as a racemic mixture, although only the *S*(+)-enantiomer possesses anti-inflammatory activity in animal models. Etodolac also displays a high degree of enantioselectivity in its inhibitory effects on the arachidonic acid COX system. With regard to its anti-inflammatory actions, etodolac was approximately 50 times more active than aspirin, three times more potent than sulindac, and one-third as active as indomethacin. The ratio of the anti-inflammatory activity (inhibitory dose [ID_{50}]) to the median effective dose (ED_{50}) for gastric ulceration or erosion was more favorable for etodolac ($ID_{50}/ED_{50} = 10$) than for aspirin, naproxen, sulindac, or indomethacin ($ID_{50}/ED_{50} = 4$). At 2.5 to 3.5 times the effective anti-inflammatory dose, etodolac was reported to produce less GI bleeding than indomethacin, ibuprofen, or naproxen. The primary mechanism of action appears to be inhibition of the biosynthesis of prostaglandins at the cyclooxygenase step, with no inhibition of the lipoxygenase system. Interestingly, etodolac is highly selective for COX-2 yet possesses a high safety profile. Etodolac, however, possesses a more favorable ratio of inhibition of prostaglandin biosynthesis in human rheumatoid synoviocytes and chondrocytes than by cultured human gastric mucosal cells compared to ibuprofen, indomethacin, naproxen, diclofenac, and piroxicam.

Thus, although etodolac is no more potent an NSAID than many others, the lower incidence of GI side effects represents a potential therapeutic advantage.

Structure–Activity Relationships. During a search for newer, more effective antiarthritic drugs in the 1970s, the Ayerst group led by Humber investigated a series of pyranocarboxylic acids of the general structure shown below (78,79). Structure–activity relationship studies indicated that alkyl groups at R_1 and an acetic acid function at R_2 enhanced anti-inflammatory activity.



Lengthening the acid chain, or ester or amide derivatives, gave inactive compounds. The corresponding α -methylacetic acid derivatives were also inactive. Increasing the chain length of the R_1 substituent to ethyl or *n*-propyl gave derivatives that were 20 times more potent than methyl. A number of aromatic substituents in the aromatic ring were evaluated, and substituents at the 8-position were most beneficial. Among the most active were the 8-ethyl, 8-*n*-propyl, and 7-fluoro-8-methyl derivatives. Etodolac was found to possess the most favorable anti-inflammatory to gastric distress properties among these analogs.

Absorption and Metabolism. Etodolac is rapidly absorbed following oral administration, with maximum serum levels being achieved within 1 to 2 hours, and it is highly bound to plasma proteins (99%) with a pK_a of 4.7. The penetration of etodolac into synovial fluid is greater than or equal to that of tolmetin, piroxicam, or ibuprofen. Only diclofenac appears to provide greater penetration. Etodolac is metabolized to three inactive hydroxylated metabolites (Fig. 31.17) and to glucuronide conjugates, none of which possesses important pharmacologic activity. Metabolism appears to be the same in the elderly as in the general population, so no dosage adjustment appears necessary.

Etodolac is indicated for the management of the signs and symptoms of osteoarthritis and for the management of pain.

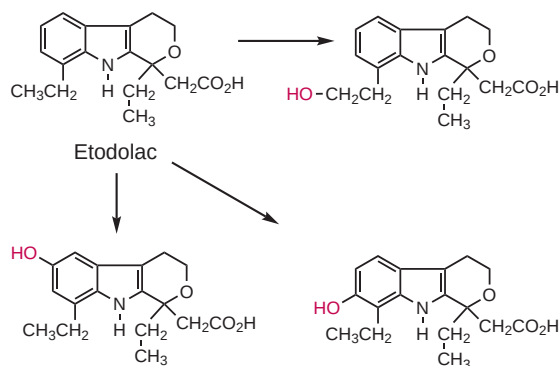


FIGURE 31.17 Metabolism of etodolac.

Nabumetone Nabumetone is unique among the NSAIDs in that it represents a new class of nonacidic prodrugs, being rapidly metabolized after absorption to form a major active metabolite, 6-methoxynaphthaleneacetic acid. It was introduced in the United States in 1992 and is synthesized from 2-acetyl-6-methoxynaphthalene (80). Nabumetone, being nonacidic, does not produce a significant primary insult and is an ineffectual inhibitor of prostaglandin COX in gastric mucosa, thus producing minimum secondary insult. The result is that gastric side effects of nabumetone appear to be minimized. Once the parent drug enters the circulatory system, however, it is metabolized to an active metabolite, 6-methoxynaphthalene-2-acetic acid (6MNA), which is an effective inhibitor of COX in joints. Nabumetone thus represents a classic example of the pro-drug approach in drug design.

In the carrageenan-induced rat paw assay, nabumetone is approximately 13 times more potent than aspirin, one-third as active as indomethacin, and half as active as diclofenac. It is only half as active as aspirin as an analgesic, as measured by the phenylquinone-induced writhing assay in mice. Despite its lower potency, the advantages of nabumetone can reside in its favorable gastric irritancy profile. The ratio of gastric irritancy dose in rats to anti-inflammatory activity in rats (ED_{50}) for nabumetone is 21.25, whereas this ratio is 0.41 for aspirin, 0.55 for indomethacin, 0.72 for diclofenac, 3.00 for tolmetin, and 7.85 for zomepirac.

Structure–Activity Relationships. Introduction of methyl or ethyl groups on the butanone side chain greatly reduced anti-inflammatory activity. The ketone function can be converted to a dioxolane with retention of activity, whereas converting the ketone to an oxime reduced activity. Removal of the methoxy group at the 6-position reduced activity, but replacement of the methoxy with a methyl or chloro group gave active compounds. Replacement of the methoxy with hydroxyl, acetoxy, or *N*-methylcarbamoyl groups, or positional isomers of the methoxy group at the 2- or 4-positions, greatly reduced activity. The active metabolite, 6MNA, is closely related structurally to naproxen, differing only by the lack of an α -methyl group. The ketone precursor, 4-(6-methoxy-2-naphthyl)pentan-2-one, that would be expected to produce naproxen as a metabolite was inactive in chronic models of inflammation.

Absorption and Metabolism. Nabumetone is absorbed primarily from the duodenum. Milk and food increase the rate of absorption and the bioavailability of the active metabolite. Plasma concentrations of unchanged drug are too low to be detected in most subjects after oral administration, so most pharmacokinetic studies have involved the disposition of the active metabolite. Pharmacokinetic properties are altered in elderly patients, with higher plasma levels of the active metabolite being noted. Nabumetone undergoes rapid and extensive metabolism in the liver, with a mean absolute bioavailability of

the active metabolite of 38%. The metabolism of nabumetone is illustrated in Figure 31.18. The major, most active metabolite is 6MNA, but the initial alcohol metabolite, a minor product, and its esters also possess significant anti-inflammatory properties.

Nabumetone is indicated for the acute and chronic treatment of the signs and symptoms of osteoarthritis and rheumatoid arthritis. The recommended starting dosage is 1,000 mg as a single dose with or without food. More symptomatic relief of severe or persistent symptoms can be obtained at doses of 1,500 or 2,000 mg/d.

Bromfenac Bromfenac was recently approved by the FDA for the treatment of postoperative inflammation and reduction of ocular pain in patients who have undergone cataract extraction. It will be available as a 0.09% ophthalmic solution for once-daily dosing. Bromfenac was originally approved by the FDA as a NSAID, but in 1998, 6 months after its approval, the FDA added a “black box” warning that the drug should not be taken for more than 10 days because of the risk of liver failure and hepatitis. Shortly thereafter, bromfenac was removed from the market because of reports of 4 deaths, 8 cases of liver transplantation, and 12 reports of liver damage associated with its use.

ARYL- AND HETEROARYLPROPIONIC ACIDS Members of this class represent the most widely used NSAIDs because several members of the class are available OTC. The introduction of the α -methyl group in the carboxylic acid side chain results in a chiral carbon atom and thus the existence of enantiomers, with oxaprozin being the sole exception. As will be detailed for each drug, pharmacologic activity resides in the *S*-enantiomer, but most drugs of this class are marketed as racemates because of the rapid epimerization of the *R*-enantiomer to the active

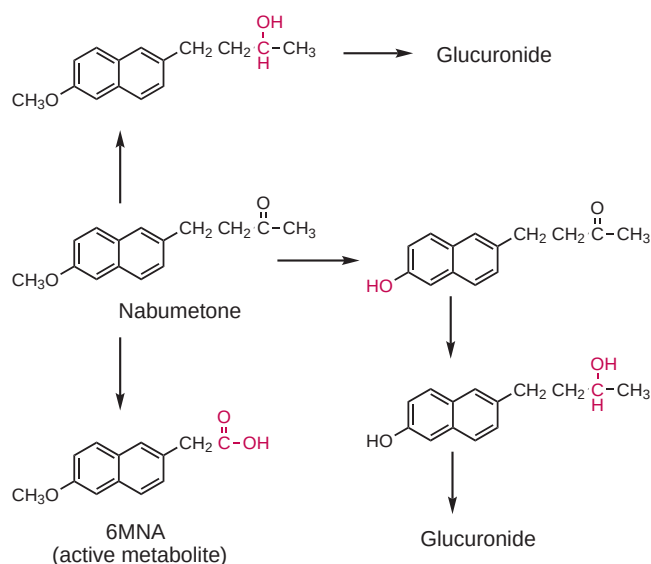


FIGURE 31.18 Metabolism of nabumetone.

Senantiomer. Structures of aryl- and heteroarylpropionic acid derivatives are shown in Figure 31.19.

Ibuprofen The synthesis of ibuprofen was originally reported in 1964 from *p*-isobutylacetophenone (81), but the drug was not marketed in the United States until 1974, despite the fact that it had been available for several years in Europe. It was the first NSAID approved since indomethacin and was immediately accepted into therapy. Its success precipitated the introduction of many new drugs in the 1970s. This chemical class currently comprises the largest group of NSAIDs. Ibuprofen became the first prescription NSAID to become available as a non-prescription (OTC) analgesic in almost 30 years and is available under a number of brand names. It is marketed as the racemic mixture, although biologic activity resides almost exclusively in the *S*(+)-isomer. Ibuprofen is more potent than aspirin but less potent than indomethacin in anti-inflammatory and prostaglandin biosynthesis inhibition assays, and it produces moderate degrees of gastric irritation.

Structure–Activity Relationships. The substitution of an α -methyl group on the alkanolic acid portion of acetic acid derivatives enhances anti-inflammatory actions and reduces many side effects. For example, the acetic acid analog of ibuprofen, ibufenac (*p*-isobutylphenylacetic acid), is less potent and more hepatotoxic than ibuprofen. The stereochemistry associated with the chiral center in the arylpropionic acids, but lacking in the acetic acid derivatives, has an important role in both the in vivo and in vitro activities of these drugs. As indicated earlier, although marketed as a racemic mixture, the

(+)-enantiomer of ibuprofen possess greater activity in vitro than the (–)-isomer. The eudismic (*S/R*) ratio for the inhibition of bovine prostaglandin synthesis is approximately 160, but in vivo, the two enantiomers are equiactive (see next section on absorption and metabolism). The (+)-enantiomer of ibuprofen—and of most of the arylpropionic acids under investigation—has been shown to possess the *S*-absolute configuration.

Absorption and Metabolism. Ibuprofen is rapidly absorbed on oral administration, with peak plasma levels being usually attained within 2 hours and a duration of action of less than 6 hours. As with most of these acidic NSAIDs, ibuprofen ($pK_a = 4.4$) is extensively bound to plasma proteins (99%) and will interact with other acidic drugs that are protein bound. Metabolism occurs rapidly, and the drug is nearly completely excreted in the urine as unchanged drug and oxidative metabolites within 24 hours after administration (Fig. 31.20).

Metabolism by CYP2C9 (90%) and CYP2C19 (10%) involves primarily ω -, ω_1 -, and ω_2 -oxidation of the *p*-isobutyl side chain, followed by alcohol oxidation of the primary alcohol resulting from ω -oxidation to the corresponding carboxylic acid. All metabolites are inactive. When ibuprofen is administered as the individual enantiomers, the major metabolite isolated is the *S*(+)-enantiomer whatever the configuration of the starting enantiomer. Interestingly, the *R*(–)-enantiomer is inverted to the *S*(+)-enantiomer in vivo via an acetyl-coenzyme A intermediate, accounting for the observation that the two enantiomers are bioequivalent in vivo. This is a metabolic phenomenon that has been also observed for other arylpropionic acids, such as ketoprofen, benoxaprofen, fenoprofen, and naproxen (82).

Ibuprofen is indicated for the relief of the signs and symptoms of rheumatoid arthritis and osteoarthritis, the

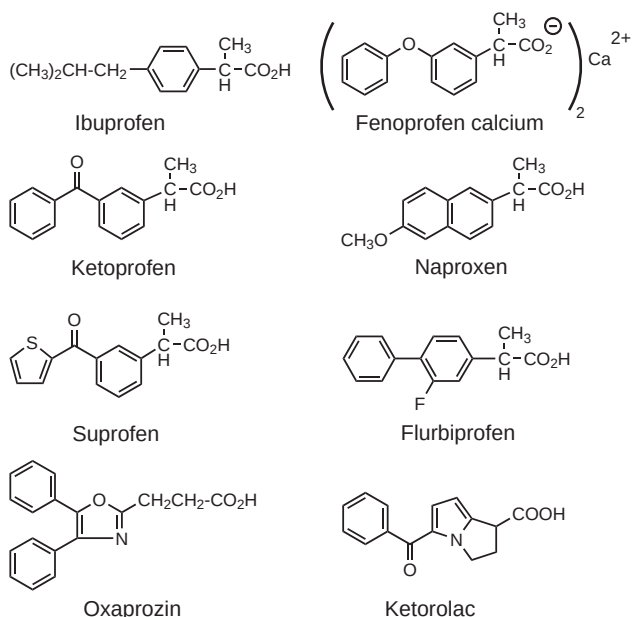


FIGURE 31.19 Structures of aryl- and heteroarylpropionic acid derivatives.

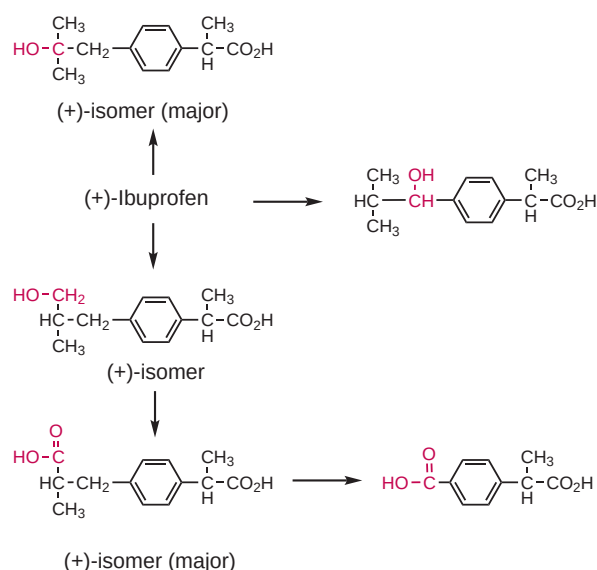


FIGURE 31.20 Metabolism of ibuprofen.

relief of mild to moderate pain, the reduction of fever, and the treatment of dysmenorrhea.

Fenopropfen Calcium The calcium and sodium salts of fenopropfen possess similar bioavailability, distribution, and elimination characteristics. It is the calcium salt that is marketed, however, because it has the advantage of being less hygroscopic. Its original synthesis was reported in 1970 (83), and it was marketed in the United States in 1976. Fenopropfen is less potent in anti-inflammatory assays than ibuprofen, indomethacin, ketoprofen, or naproxen. As an inhibitor of prostaglandin biosynthesis, it is much less potent than indomethacin, more potent than aspirin, and about equipotent with ibuprofen. Fenopropfen also possesses analgesic and antipyretic activity. It possesses other pharmacologic properties, such as inhibition of phagocytic and complement functions and stabilization of lysosomal membranes. Fenopropfen is marketed as a racemic mixture, because no differences have been observed in the *in vivo* anti-inflammatory or analgesic properties of the individual enantiomers. The ability of *R*(-)-arylpropionic acids to undergo inversion to the *S*(+)-enantiomers, however, can be involved. Like other NSAIDs, *in vitro* prostaglandin synthesis assays indicate that the *S*(+)-enantiomer is more potent than the *R*(-)-isomer.

Structure–Activity Relationships. Placing the phenoxy group in the ortho- or para-position of the arylpropionic acid ring markedly decreases activity. Replacement of the oxygen bridge between the two aromatic rings with a carbonyl group results in a marketed analog (ketoprofen).

Absorption and Metabolism. Fenopropfen is readily absorbed (85%) on oral administration and is highly bound (99%) to plasma proteins. Peak plasma levels are attained within 2 hours of administration. The free acid has a pK_a of 4.5, which is within the range of the other arylalkanoic acids. Fenopropfen is rather extensively metabolized, primarily through glucuronide conjugation with the parent drug and the CYP2C9 4'-hydroxy metabolite (84).

Fenopropfen calcium is indicated for treatment of rheumatoid arthritis and osteoarthritis and for the relief of mild to moderate pain.

Ketoprofen Ketoprofen was synthesized from 2-(*p*-aminophenyl)propionic acid via a thiaxanthone intermediate (85) and was introduced in 1986. Ketoprofen, unlike many NSAIDs, inhibits the synthesis of leukotrienes and leukocyte migration into inflamed joints in addition to inhibiting the biosynthesis of prostaglandins. It stabilizes the lysosomal membrane during inflammation, resulting in decreased tissue destruction. Antibradykinin activity has been also observed. Bradykinin is released during inflammation and can activate peripheral pain receptors. In addition to anti-inflammatory activity, ketoprofen also possesses antipyretic and analgesic properties.

Although it is less potent than indomethacin as an anti-inflammatory agent and an analgesic, its ability to produce gastric lesions is about the same (86).

Absorption and Metabolism. Ketoprofen is rapidly and nearly completely absorbed on oral administration, reaching peak plasma levels within 0.5 to 2 hours. It is highly plasma protein bound (99%) despite a lower acidity ($pK_a = 5.9$) than some other NSAIDs. Wide variation in plasma half-lives has been reported. There are no known active metabolites of ketoprofen. It is metabolized by glucuronidation of the carboxylic acid, CYP3A4 and CYP2C9 hydroxylation of the benzoyl ring, and reduction of the keto function.

Ketoprofen is indicated for the long-term management of rheumatoid arthritis and osteoarthritis, for mild to moderate pain, and for primary dysmenorrhea.

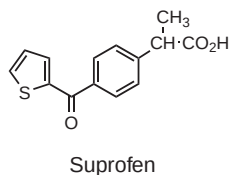
Naproxen Naproxen is synthesized from 2-methoxynaphthalene and the (+)-isomer obtained by resolution with cinchonidine (87). It was introduced in the United States in 1976 and, as a generic drug, has consistently been among the more popular NSAIDs. It is marketed as the *S*(+)-enantiomer, but interestingly, the sodium salt of the (-)-isomer is also on the market as Anaprox. As an inhibitor of prostaglandin biosynthesis, it is 12 times more potent than aspirin, 10 times more potent than phenylbutazone, 3 to 4 times more potent than ibuprofen, and 4 times more potent than fenopropfen, but it is approximately 300 times less potent than indomethacin. *In vivo* anti-inflammatory assays are consistent with this relative order of potency. In the carrageenan-induced rat paw edema assay, it is 11 times more potent than phenylbutazone and 55 times as potent as aspirin, but only 0.7 times as potent as indomethacin. In the phenylquinone writhing assay for analgesia, it is 9 times as potent as phenylbutazone and 7 times as potent as aspirin, but only 10% as potent as indomethacin. In the yeast-induced pyrexia assay for antipyretic activity, it is 7 times as potent as phenylbutazone, 22 times as potent as aspirin, and equipotent as indomethacin. The order of gastric ulcerogenic activity is sulindac < naproxen < aspirin, indomethacin, ketoprofen, and tolmetin.

Structure–Activity Relationships. In a series of substituted 2-naphthylacetic acids, substitution in the 6-position led to maximum anti-inflammatory activity. Small lipophilic groups, such as Cl, CH_3S , and CHF_2O , were active analogs, with CH_3O being the most potent. Larger groups were found to be less active. Derivatives of 2-naphthylpropionic acids are more potent than the corresponding acetic acid analogs. Replacing the carboxyl group with functional groups capable of being metabolized to the carboxyl function (e.g., $-CO_2CH_3$, $-CHO$, or $-CH_2OH$) led to a retention of activity. The *S*(+)-isomer is the more potent enantiomer. Naproxen is the only arylalkanoic acid NSAID currently marketed as optically active isomers.

Absorption and Metabolism. Naproxen is almost completely absorbed following oral administration. Peak plasma levels are achieved within 2 to 4 hours following administration with a duration of action ~12 hours. Like most of the acidic NSAIDs ($pK_a = 4.2$), it is highly bound (99.6%) to plasma proteins. Approximately 70% of an administered dose is eliminated as either unchanged drug (60%) or as conjugates of unchanged drug (10%). The remainder is converted to the 6-*O*-desmethyl metabolite by both CYP3A4 and CYP1A2 and, further, to the glucuronide conjugate of the demethylated metabolite. The 6-*O*-desmethyl metabolite lacks anti-inflammatory activity. Like most of the arylalkanoic acids, the most common side effect associated with the use of naproxen is irritation to the GI tract. The most common other adverse reactions are associated with CNS disturbances (e.g., nausea and dizziness).

Naproxen is indicated for the treatment of rheumatoid arthritis, osteoarthritis, juvenile arthritis, ankylosing spondylitis, tendinitis, bursitis, acute gout, and primary dysmenorrhea and for the relief of mild to moderate pain.

Suprofen Suprofen is a white, microcrystalline powder that is slightly soluble in water.



Suprofen is a thiophene isostere of ketoprofen. It was originally synthesized from thiophene in 1974 (88) and was introduced in the United States in 1985 for the treatment of dysmenorrhea and as an analgesic for mild to moderate pain. Reports of severe flank pain and transient renal failure appeared, however, with the syndrome being noted abruptly within several hours after one or two doses of the drug; therefore, suprofen was removed from the US market in 1987. Obviously, clinical trials are not always sufficient to determine a drug's safety, and postmarketing surveillance becomes most important. Suprofen was reintroduced in the United States in 1990 as a 1% ophthalmic solution for the prevention of surgically induced miosis during cataract extraction. Miosis complicates the removal of lens material and implantation of a posterior chamber intraocular lens that thus increases the risk of ocular trauma. The mechanism of action also involves inhibition of prostaglandin synthesis, because prostaglandins constrict the iris sphincter independent of a cholinergic mechanism. Additionally, prostaglandins also break down the blood-aqueous barrier, allowing the influx of plasma proteins into aqueous humor, resulting in an increase in intraocular pressure.

Flurbiprofen Flurbiprofen synthesis was originally reported in 1974 (89). During a study of the pharmacologic

properties of a large number of substituted phenylalkanoic acids, including ibuprofen and ibufenac, the most potent were found to be substituted 2-(4-biphenyl)propionic acids. Further toxicologic and pharmacologic studies indicated that flurbiprofen possessed the most favorable therapeutic profile, so it was selected for further clinical development. It was not marketed until 1987, when it was introduced as the sodium salt as Ocufen, the first topical NSAID indicated for ophthalmic use in the United States. The indication for Ocufen is the same as that for Profenal—that is, to inhibit intraoperative miosis induced by prostaglandins in cataract surgery. Thus, flurbiprofen is an inhibitor of prostaglandin synthesis. Unlike other members of this class, there is no *R*- to *S*-epimerization observed in rats or humans, although epimerization has been observed in mice (90,91). The oral form was introduced in 1988 as Ansaid (another NSAID) and gained immediate acceptance. In acute inflammation assays in adrenalectomized rats, flurbiprofen was found to be 536-fold more potent than aspirin and 100-fold more potent than phenylbutazone. Orally, it was half as potent as methylprednisolone. As an antipyretic, it was 403 times as potent as aspirin in the yeast-induced fever assay in rats and was 26 times more potent than ibuprofen as an antinociceptive.

Absorption and Metabolism. Flurbiprofen is well absorbed after oral administration, with peak plasma levels being attained within 1.5 hours. Food alters the rate of absorption but not the extent of its bioavailability. It is extensively bound to plasma proteins (99%) and has a plasma half-life of 2 to 4 hours. Metabolism is extensive, with 60% to 70% of flurbiprofen and its metabolites being excreted as sulfate and glucuronide conjugates. Flurbiprofen shows some interesting metabolic patterns, with 40% to 47% as the 4'-hydroxy metabolite, 5% as the 3',4'-dihydroxy metabolite, 20% to 30% as the 3'-hydroxy- 4'-methoxy metabolite, and the remaining 20% to 25% of the drug being excreted unchanged. None of these metabolites demonstrates significant anti-inflammatory activity. The metabolism of flurbiprofen is presented in Figure 31.21.

Flurbiprofen is indicated as an oral formulation for the acute or long-term treatment of rheumatoid arthritis and osteoarthritis and as an ophthalmic solution for the inhibition of intraoperative miosis. The *R*-enantiomer has been proposed to have potential use in the prophylaxis and treatment of colon cancer (92) and Alzheimer disease (93).

Ketorolac Tromethamine Ketorolac represents a cyclized, heteroarylpropionic acid derivative, with the α -methyl group being fused to the pyrrole ring. It was introduced in 1990 and is indicated as a peripheral analgesic for short-term use and for the relief of ocular itching caused by seasonal allergic conjunctivitis, although it exhibits anti-inflammatory and antipyretic activity as well. It was initially introduced only in an injectable form, but recently, an oral formulation has been made available. Its analgesic activity

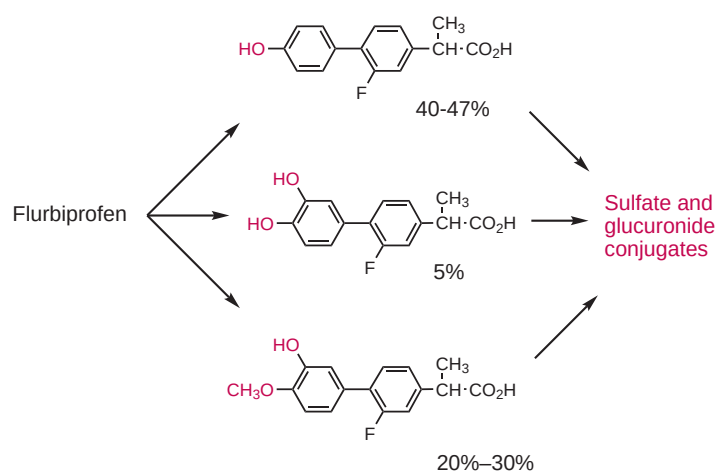


FIGURE 31.21 Metabolism of flurbiprofen.

resembles that of the centrally acting analgesics, with 15 to 30 mg of ketorolac producing analgesia equivalent to a 12-mg dose of morphine, and it has become a widely accepted alternative to narcotic analgesia. Ketorolac inhibits prostaglandin synthesis. Although the analgesic effect is achieved within 10 minutes of injection, peak analgesia lags behind peak plasma levels by 45 to 90 minutes. The free acid has a pK_a of 3.5, and it is not surprising that it is highly plasma protein bound (>99%). Ketorolac is metabolized by CYP2C9 to its *p*-hydroxy derivative and to conjugates that are excreted primarily in the urine.

Oxaprozin Oxaprozin was marketed in 1993 for acute and long-term use in the management of signs and symptoms of osteoarthritis and rheumatoid arthritis. Oxaprozin is synthesized by condensing benzoin with succinic anhydride and cyclizing the resulting benzoin hemisuccinate with ammonium acetate. Although not formally a propionic acid of the α -methylacetic acid type, it appears to be similar to the other propionic acid derivatives considered here. Oxaprozin is well absorbed (100%) after oral administration, but maximum plasma concentrations are not reached until 3 to 5 hours following ingestion. Oxaprozin is an anti-inflammatory agent possessing a rapid onset of action and a prolonged duration of action. In both the carrageenan-induced rat paw edema assay and analgesic tests, it was equipotent with aspirin. Oxaprozin has been associated with the appearance of rash and/or mild photosensitivity. Some patients experience an increased incidence of rash on sun-exposed skin during clinical testing. Oxaprozin, aspirin, ibuprofen, indomethacin, naproxen, and sulindac have comparable efficacy in the treatment of rheumatoid arthritis, whereas oxaprozin, aspirin, naproxen, and piroxicam have comparable efficacy in osteoarthritis. It is highly bound to plasma proteins (99%), is highly lipophilic, and undergoes little first-pass metabolism. Metabolism is via hepatic microsomal oxidation and glucuronidation. Two hydroxylated phenolic metabolites that are produced (<5%)

have been shown to possess anti-inflammatory activity (94) but do not appear to contribute significantly to the overall pharmacologic activity of oxaprozin.

Oxaprozin possesses a relatively long elimination half-life of 59 hours (range, 26 to 92 hours), enabling once-daily dosing. Administration with food appears to delay absorption but not bioavailability.

Several properties of the NSAIDs are summarized in Table 31.4.

***N*-ARYLANTHRANILIC ACIDS (FENAMIC ACIDS)** The anthranilic acid class of NSAIDs is the result of the application of classical medicinal chemistry bioisosteric drug design concepts, because these derivatives are nitrogen isosteres of salicylic acid. Additionally, they could be considered structural analogs of the arylacetic acid derivative, diclofenac. In the early 1960s, the Parke-Davis research group reported the development of a series of *N*-substituted anthranilic acids that have since been given the chemical class name of fenamic acids. The fact that this class of compounds possesses little advantage over the salicylates with respect to their anti-inflammatory and analgesic properties has diminished interest in their large-scale development relative to the arylalkanoic acids. Mefenamic acid was introduced in the United States in 1967 as an analgesic, and this remains the primary indication despite the fact that it possesses modest anti-inflammatory activity. Flufenamic acid has been available in Europe as an antirheumatic agent, but there are no apparent plans to introduce this drug in the United States. With regard to anti-inflammatory activity, mefenamic acid is approximately 1.5 times as potent as phenylbutazone and half as potent as flufenamic acid. Meclofenamic acid was introduced in the United States as its sodium salt in 1980, primarily as an antirheumatic agent and analgesic. The structures of these fenamic acids are shown in Figure 31.22.

The fenamic acids share a number of pharmacologic properties with the other NSAIDs. Because these drugs are potent inhibitors of prostaglandin biosynthesis, it

TABLE 31.4 Some Properties of the NSAIDs

Drug	Year Introduced	Anti-Inflammatory Dose (mg)	Onset (Duration) of Action	Peak Plasma Levels (h)	Protein Binding (%)	Biotransformation	Elimination Half-Life (h)	pK _a
Aspirin	1899	3,200–6,000	ND	2	90	Plasma hydrolysis and hepatic	<30 minutes	3.5
Diclofenac (Voltaren)	1989	100–200	30 min (~8 h)	1.5–2.5	99	Hepatic; first-pass metabolism: 3A4	1–2	4.0
Diflunisal (Dolobid)	1982	500–1,000	1 h (8–12 h)	2–3	99	Hepatic	8–12	3.3
Etodolac (Lodine)	1991	800–1,200	30 min (4–6 h)	1–2	99	Hepatic: 2C9	6–7	4.7
Fenoprofen calcium (Nalfon)	1976	1,200–2,400	NR	2	99	Hepatic: 2C9	3	4.5
Flurbiprofen (Ansaid)	1988	200–300	NR	1.5	99	Hepatic: 2C9	6 (2–12)	4.2
Ibuprofen (Motrin, Advil)	1974	1,200–3,200	30 min (4–6 h)	2	99	Hepatic; first-pass metabolism: 2C9, 2C19	~2	4.4
Indomethacin (Indocin)	1965	75–150	2–4 h (2–3 d)	2–3	97	Hepatic: 2C9	5 (3–11)	4.5
Ketoprofen (Orudis)	1986	150–300	NR	0.5–2	99	Hepatic: 2C9, 3A4	~2	5.9
Meclofenamate sodium (Meclomen)	1980	200–400	1 h (4–6 h)	4.0	99	Hepatic: 2C9	2–3	NR
Mefenamic acid (Ponstel)	1967	1,000	NR	2–4	79	Hepatic: 2C9	2	4.2
Meloxicam (Mobic)	2000	7.5–15	NR	4–5	99	Hepatic 2C9	15–20	1.1, 4.2
Nabumetone ^a (Relafen)	1992	1,500–2,000	NR	2.5 (1–8) 6MNA	99 ^b	Hepatic; first-pass metabolism to 6MNA	6MNA, 23	Neutral
Naproxen (Naprosyn, Anaprox)	1976	500–1,000	NR	2–4	99	Hepatic: 3A4, 1A2	13	4.2
Oxaprozin (Daypro)	1993	1,200	NR	3–5	99	Hepatic: 2C9	25	4.3
Piroxicam (Feldene)	1982	20	2–4 h (24 h)	2	99	Hepatic: 2C9	50	1.8, 5.1
Sulindac ^a (Clinoril)	1978	400	NR	2–4	93	Hepatic; sulfide metabolite active	50	4.5
Tolmetin (Tolectin)	1976	1,200	NR	<1	99	Hepatic	5	3.5

6MNA, 6-methoxynaphthalene-2-acetic acid; ND, not determined; NR, not reported.

^aProdrug.

is tempting to speculate that this represents their primary mechanism of action. Scherrer, like Shen, had proposed a hypothetical receptor for NSAIDs and later modified (95) the receptor to represent the active site

of arachidonic acid COX. Structurally, the fenamic acids fit the proposed active site of arachidonic acid COX proposed by Shen (10), because they possess an acidic function connected to an aromatic ring along

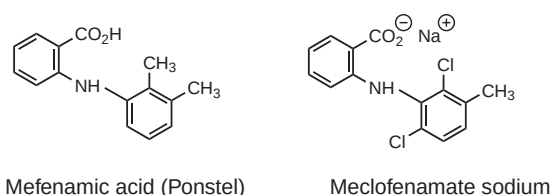
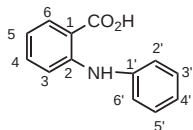


FIGURE 31.22 Structures of *N*-arylanthranilic acids (fenamic acids).

with an additional lipophilic binding site—in this case, the *N*-aryl substituent. The greater anti-inflammatory activity of meclofenamic acid compared to that of mefenamic acid correlates well with its ability to inhibit prostaglandin synthesis. Scherrer (96) compared the *in vivo* anti-inflammatory activities, clinical anti-inflammatory doses in humans, and *in vitro* inhibition of prostaglandin synthesis activities of mefenamic acid, meclofenamic acid, phenylbutazone, indomethacin, and aspirin and suggested an important role of prostaglandin synthesis inhibition in the production of therapeutic effects of the fenamic acids.

Side effects are those primarily associated with GI disturbances (e.g., dyspepsia, discomfort, and especially, diarrhea), some CNS effects (e.g., dizziness, headache, and drowsiness), skin rashes, and transient hepatic and renal abnormalities. Isolated cases of hemolytic anemia have been reported.

General Structure–Activity Relationships Substitution on the anthranilic acid ring reduces activity, whereas substitution of the *N*-aryl ring can lead to conflicting results. In the ultraviolet erythema assay for anti-inflammatory activity, the order of activity is usually $3' > 2' \gg 4'$ for monosubstitution, with the $3'$ -CF₃ derivative (flufenamic acid) being particularly potent.



The opposite order of activity was observed, however, in the rat paw edema assay, with the $2'$ -Cl derivative being more potent than the $3'$ -Cl analog. In disubstituted derivatives, in which the nature of the two substituents is the same, $2',3'$ -disubstitution appears to be the most effective. A plausible explanation can be found in an examination of the proposed topography of the active sites of arachidonic acid COX using either the Shen or Sherrer models. Proposed binding sites include a hydrophobic trough to which a lipophilic group, noncoplanar with the ring bearing the carboxylic acid function, binds. Substituents on the *N*-aryl ring that force this ring to be noncoplanar with the anthranilic acid ring should enhance binding at this site and, thus, activity. This can account for the enhanced anti-inflammatory activity of

meclofenamic acid, which has two ortho-substituents forcing this ring out of the plane of the anthranilic acid ring, over flufenamic acid (no ortho-substituents) and mefenamic acid (one ortho-substituent). Meclofenamic acid possesses 25 times greater anti-inflammatory activity than mefenamic acid. The NH-moiety of anthranilic acid appears to be essential for activity, because replacement of the NH function with O, CH₂, S, SO₂, N-CH₃, or N-COCH₃ functionalities significantly reduces activity. Finally, the position, rather than the nature, of the acidic function is critical for activity. Anthranilic acid derivatives are active, whereas the *m*- and *p*-aminobenzoic acid analogs are not. Replacement of the carboxylic acid function with the isosteric tetrazole moiety has little effect on activity.

Drug Interactions The pK_a values of the *N*-arylanthranilic acids (4.0 to 4.2) resemble those of the arylalkanoic acids; thus, it is not surprising that they are strongly bound to plasma proteins and that interactions with other highly protein bound drugs are very probable. The most common interactions reported are those of mefenamic acid and meclofenamic acid with oral anticoagulants. Concurrent administration of aspirin results in a reduction of plasma levels of meclofenamic acid.

Specific Drugs

Mefenamic Acid. Mefenamic acid is synthesized from *o*-chlorobenzoic acid and 2,3-dimethylaniline under catalytic conditions (97). Mefenamic acid is the only fenamic acid derivative that produces analgesia centrally and peripherally. Mefenamic acid is indicated for the short-term relief of moderate pain and for primary dysmenorrhea.

Mefenamic acid is absorbed rapidly following oral administration, with peak plasma levels being attained within 2 to 4 hours. It is highly bound to plasma proteins (78.5%) and has a plasma half-life of 2 to 4 hours. Metabolism occurs through regioselective oxidation of the $3'$ -methyl group and glucuronidation of mefenamic acid and its metabolites. Urinary excretion accounts for approximately 50% to 55% of an administered dose, with unchanged drug accounting for 6%, the $3'$ -hydroxymethyl metabolite (primarily as the glucuronide) accounting for 25%, and the remaining 20% as the dicarboxylic acid (of which 30% is the glucuronide conjugate) (Fig. 31.23). These metabolites are essentially inactive.

Meclofenamate Sodium. Meclofenamate sodium is rapidly and almost completely absorbed following oral administration, reaching peak plasma levels within 2 hours. It is highly bound to plasma proteins (99%) and has a plasma half-life of 2 to 4 hours. Metabolism involves oxidation of the methyl group, aromatic hydroxylation, monodehalogenation, and conjugation. Urinary excretion accounts for approximately 75% of the administered dose. The major metabolite is the product of $3'$ -methyl oxidation

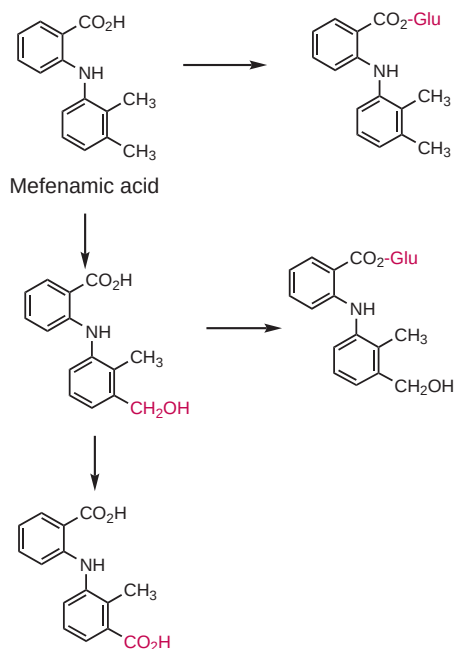


FIGURE 31.23 Metabolism of mefenamic acid.

and has been shown to possess anti-inflammatory activity (Fig. 31.24).

Meclofenamate sodium is indicated for the relief of mild to moderate pain, the acute and chronic treatment of rheumatoid arthritis and osteoarthritis, the treatment of primary dysmenorrhea, and the treatment of idiopathic, heavy menstrual blood loss.

OXICAMS The enolic acid class of NSAIDs has been termed “oxicams” by the U.S. Adopted Name Council to describe the series of 4-hydroxy-1,2-benzothiazine carboxamides that possess anti-inflammatory and analgesic properties. These structurally distinct substances resulted from extensive studies by the Pfizer group in an effort to produce noncarboxylic acid, potent, and well-tolerated anti-inflammatory drugs. Several series were prepared, including 2-aryl-1,3-indanediones, 2-arylbenzothiophen-3-(2*H*)-one 1,1-dioxides, dioxoquinoline-4-carboxamides, and 3-oxa-2*H*-1,2-benzothiazine-4-carboxamide 1,1-dioxides, and evaluated. These results, combined with the previously known activity of 1,3-dicarbonyl derivatives, such as phenylbutazone, led to the development of the oxicams. The first member of this class, piroxicam, was introduced

in the United States in 1982 as Feldene and gained immediate acceptance.

Piroxicam is potent in standard *in vivo* assays, being 200 times more potent than aspirin and at least 10 times as potent as any other standard agent in the ultra-violet erythema assay, as potent as indomethacin and more potent than phenylbutazone or naproxen in the carrageenan-induced rat paw edema assay, and equipotent with indomethacin and 15 times more potent than phenylbutazone in the rat adjuvant arthritis assay. It is less potent than indomethacin, equipotent with aspirin, and more potent than fenoprofen, ibuprofen, naproxen, and phenylbutazone as an analgesic in the phenylquinone writhing assay. Piroxicam inhibits the migration of polymorphonuclear cells into inflammatory sites and inhibits the release of lysosomal enzymes from these cells. It also inhibits collagen-induced platelet aggregation. It is an effective inhibitor of arachidonic acid COX, being almost equipotent with indomethacin and more potent than ibuprofen, tolmetin, naproxen, fenoprofen, phenylbutazone, and aspirin in the inhibition of prostaglandin biosynthesis by methylcholanthrene-transformed mouse fibroblasts (MC-5) assay. A template for designing anti-inflammatory compounds based on CPK space-filling models of the peroxy radical precursor of PGG and inhibitors of COX was proposed (98), and the ability of oxicams, particularly piroxicam, to inhibit this enzyme was subsequently rationalized on the ability of oxicams to assume a conformation resembling that of the peroxy radical precursor (99).

Approximately 20% of individuals on piroxicam report adverse reactions. Not unexpectedly, the greatest incidence of side effects results from GI disturbances. The reported incidence of peptic ulcers, however, is less than 1%.

As will be discussed later, a new oxicam derivative, meloxicam, has been approved as a selective COX-2 inhibitor for the treatment of osteoarthritis.

General Structure–Activity Relationships Within the series of 4-hydroxy-1,2-benzothiazine carboxamides represented by the general structure shown below, optimum activity was observed when R1 was a methyl substituent. The carboxamide substituent, R, is usually an aryl or heteroaryl substituent, because alkyl substituents are less active.

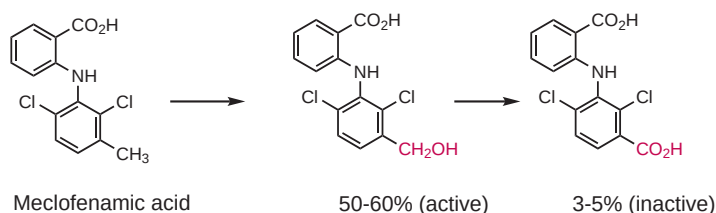
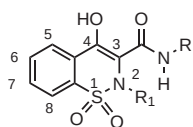
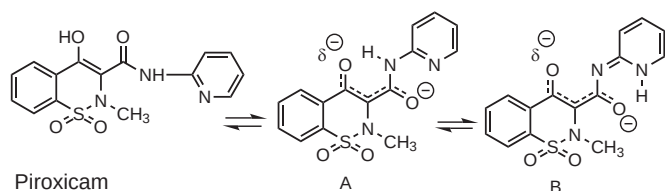


FIGURE 31.24 Metabolism of meclofenamic acid.



Oxicams are acidic compounds, with pK_a values in the range of 4 to 6. *N*-Heterocyclic carboxamides are more acidic than the corresponding *N*-aryl carboxamides, and this enhanced acidity was attributed (100) to stabilization of the enolate anion by the pyridine nitrogen atom, as illustrated in tautomer A and additional stabilization by tautomer B:



This explains the observation that primary carboxamides are more potent than the corresponding secondary derivatives, because no N–H bond would be available to enhance the stabilization of the enolate anion. When the aryl group is *o*-substituted, variable results were obtained, whereas *m*-substituted derivatives are usually more potent than the corresponding *p*-isomers. In the aryl series, maximum activity is observed with an *m*-Cl substituent. No direct correlations were observed between acidity and activity, partition coefficient, and electronic or spatial properties in this series. Two major differences, however, are observed when R = heteroaryl rather than aryl: The pK_a values are usually two to four units lower, and anti-inflammatory activity is increased as much as sevenfold. The greatest activity is associated with the 2-pyridyl (as in piroxicam), 2-thiazolyl, or 3-(5-methyl) isoxazolyl ring systems, with the latter derivative (isoxicam) having been withdrawn from the European market in 1985 following several reports of severe skin reactions. In addition to possessing activity equal to or greater than indomethacin in the carrageenan-induced rat paw edema assay, the heteroaryl carboxamides also possess longer plasma half-lives, providing an improvement in dosing scheduling regimens.

Drug Interactions Few reports of therapeutically significant interactions of oxicams with other drugs have appeared. Concurrent administration of aspirin has been shown to reduce piroxicam plasma levels by approximately 20%, whereas the anticoagulant effect of acenocoumarin is potentiated, presumably as a result of plasma protein displacement.

Specific Drugs

Piroxicam. Piroxicam was synthesized by ring-expansion reactions of saccharin derivatives (101). Piroxicam is readily absorbed on oral administration, reaching peak plasma levels in approximately 2 hours. Peak plasma

levels appear to be lower when given with food at low doses (30 mg), with no differences appearing with a 60-mg dose, but in general, food does not markedly affect bioavailability. Being acidic ($pK_a = 6.3$), it is highly bound to plasma proteins (99%). Piroxicam possesses an extended plasma half-life (38 hours), making single daily dosing possible. Piroxicam is indicated for long-term use in rheumatoid arthritis and osteoarthritis.

Metabolism Although the metabolism of piroxicam varies quantitatively from species to species, qualitative similarities are found in the metabolic pathways of humans, rats, dogs, and rhesus monkeys. It is extensively metabolized in humans, with less than 5% of an administered dose being excreted unchanged. The major metabolites in humans result from CYP2C9 hydroxylation of the pyridine ring and subsequent glucuronidation; other metabolites are of lesser importance (Fig. 31.25). Aromatic hydroxylation at several positions of the aromatic benzothiazine ring also occurs; two hydroxylated metabolites have been extracted from rat urine. Based on nuclear magnetic resonance deuterium-exchange studies, hydroxylation at the 8-position was ruled out, indicating that hydroxylation occurs at two of the remaining positions. Other novel metabolic reactions occur. Cyclodehydration gave a tetracyclic metabolite (the major metabolite in dogs), whereas ring contraction following amide hydrolysis and decarboxylation eventually yields saccharin. All the known metabolites of piroxicam lack anti-inflammatory activity. For example, the major human metabolite is 1,000-fold less effective as an inhibitor of prostaglandin biosynthesis than piroxicam itself. Related oxicams undergo different routes of metabolism. For example, sudoxicam (the *N*-2-thiazolyl analog) undergoes primarily hydroxylation of the thiazole ring, followed by ring

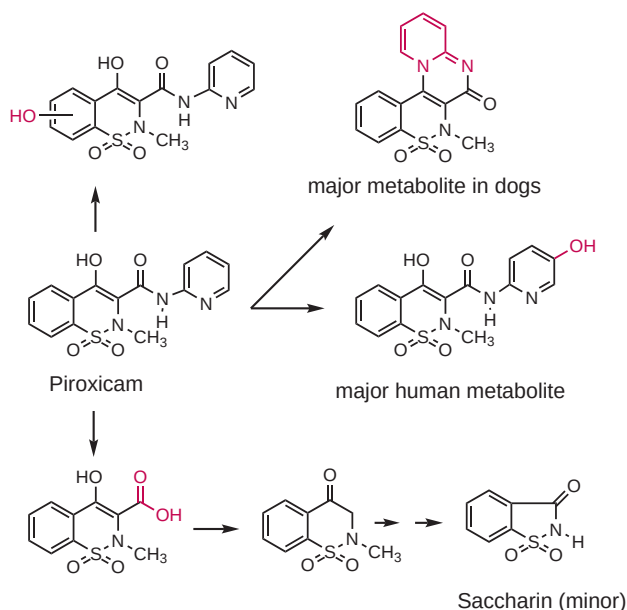
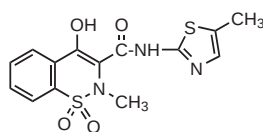


FIGURE 31.25 Metabolism of piroxicam.

opening, whereas isoxicam undergoes primarily cleavage reactions of the benzothiazine ring.

Meloxicam. In April 2000, the FDA approved meloxicam for the treatment of osteoarthritis in the United States. When meloxicam was initially introduced in the United Kingdom, it was promoted as a selective COX-2 inhibitor. Meloxicam, however, is less selective for COX-2 than is celecoxib. Meloxicam is readily absorbed when administered orally and is highly bound to plasma proteins. It is extensively metabolized in the liver, primarily by CYP2C9 and, to a lesser extent, by CYP3A4. The advantages of meloxicam over celecoxib in the treatment of osteoarthritis (or rheumatoid arthritis) are not readily apparent.



Meloxicam

Metabolism. Unlike the metabolism of piroxicam, the metabolism of meloxicam in humans is rather straightforward. Oxidation of the thiazole methyl group mediated by CYP2C9 (CYP3A4 is involved to a much lesser extent), followed by subsequent oxidation to the carboxylic acid, yields metabolites that have not been shown to possess anti-inflammatory activity. The metabolism of meloxicam is shown in Fig. 31.26.

Gastroenteropathy Induced by Nonselective COX NSAIDs

The effectiveness and popularity of the NSAIDs in the United States and Europe make this class one of the most commonly used classes of therapeutic entities. Unfortunately, until the introduction of the selective COX-2 inhibitors, almost all of the current drugs, which are essentially nonselective COX-1 and COX-2 inhibitors, share the undesirable property of producing damaging effects to gastric and intestinal mucosa, resulting in erosion, ulcers, and GI bleeding, and these represent the major adverse reactions to the use of NSAIDs. As many as 20,000 deaths and 100,000 hospitalizations per year have

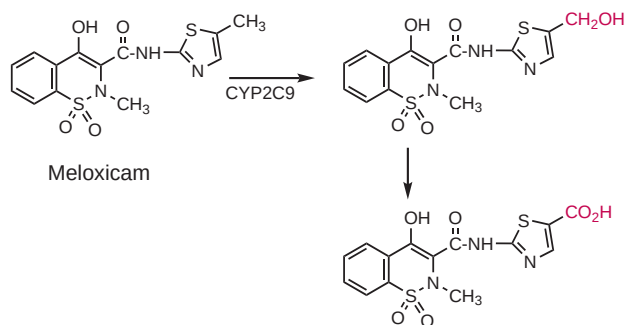
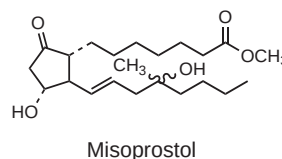


FIGURE 31.26 Metabolism of meloxicam.

been associated with GI complications resulting from the use of NSAIDs. Approximately 30% to 40% of patients taking NSAIDs report some type of gastric injury, and approximately 10% discontinue therapy because of these effects. These acute and chronic injuries to gastric mucosa result in a variety of lesions referred to as NSAID gastropathy, which differs from peptic ulcer disease by localization of lesions more frequently in the stomach rather than in the duodenum. Additionally, NSAID-induced lesions occur more frequently in the elderly than typical peptic ulcers. Normally, the stomach protects itself from the harmful effects of hydrochloric acid and pepsin by a number of protective mechanisms referred to as the gastric mucosal barrier, which consists of epithelial cells, the mucous and bicarbonate layer, and mucosal blood flow. Gastric mucosa is actually a gel consisting of polymers of glycoprotein, which limit the diffusion of hydrogen ions. These polymers reduce the rate at which hydrogen ions (produced in the lumen) and bicarbonate ions (secreted by the mucosa) mix; thus, a pH gradient is created across the mucus layer. Normally, gastric mucosal cells are rapidly repaired when they are damaged by factors such as food, ethanol, or acute ingestion of NSAIDs. Among the cytoprotective mechanisms is the ability of prostaglandins of the PGE series, particularly PGE₁, to increase the secretion of bicarbonate ion and mucus and to maintain mucosal blood flow. The prostaglandins also decrease acid secretion, permitting the gastric mucosal barrier to remain intact. The use of PGE₁ to reduce NSAID-induced gastric damage is limited by the fact that it is ineffective orally and degrades rapidly on parenteral administration, primarily by oxidation of the 15-hydroxy group. To overcome these limitations, misoprostol was synthesized as a prostaglandin prodrug analog in which oral activity was achieved by administering the drug as the methyl ester, allowing the bioactive acid to be liberated after absorption. Oxidation of the 15-hydroxy group was overcome by moving the hydroxy group to the 16-position, thus “fooling” the enzyme prostaglandin 15-OH dehydrogenase. Oxidation was further limited by the introduction of a methyl group at the 16-position, producing a tertiary alcohol that is more difficult to oxidize than the secondary alcohol group of the prostaglandins. Misoprostol was introduced in 1989 as a mixture of stereoisomers at the 16-position as Cytotec for the prevention of NSAID-induced gastric ulcers (but not duodenal ulcers) in patients at high risk of complications from a gastric ulcer, particularly the elderly and patients with concomitant debilitating disease and in individuals with a history of gastric ulcers.



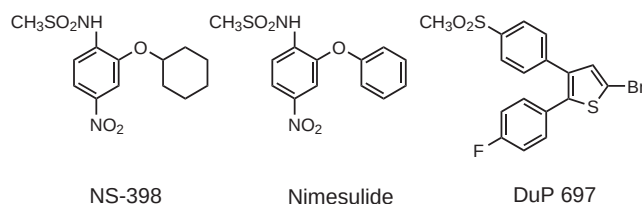
As mentioned earlier, Figure 31.7 illustrates the ability of aspirin and NSAIDs to induce gastric damage by

a dual-insult mechanism. Aspirin and the NSAIDs are acidic substances that can damage the GI tract, even in the absence of hydrochloric acid, by changing the permeability of cell membranes, allowing a back diffusion of hydrogen ions. These weak acids remain un-ionized in the stomach, but the resulting lipophilic nature of these substances allows accumulation or concentration in gastric mucosal cells. Once inside these cells, however, the higher pH of the intracellular environment causes the acids to dissociate and become “trapped” within the cells. The permeability of the mucosal cell membrane is thus altered, and the accumulation of hydrogen ions causes mucosal cell damage. This gastric damage is a result, therefore, of the primary insult of acidic substances. As detailed earlier in this chapter, the primary mechanism of action of the NSAIDs is to inhibit the biosynthesis of prostaglandins at the cyclooxygenase step. The resulting nonselective inhibition of prostaglandin biosynthesis in the GI tract prevents the prostaglandins from exerting their protective mechanism on gastric mucosa; thus, the NSAIDs induce gastric damage through this secondary insult mechanism.

SELECTIVE COX-2 INHIBITORS As previously discussed, the most notable achievement in the development of NSAIDs has been in the area of selective COX-2 inhibitors (“coxibs”). Classic NSAIDs share similar side effect profiles, particularly on the GI tract, many of which have been attributed to the inhibition of COX, the rate-limiting step in prostaglandin biosynthesis. With the discovery of two isoforms of COX, COX-1 and COX-2, and the realization that COX-1 is beneficial in maintaining normal processes in the GI tract by producing cytoprotective prostaglandins, stimulating bicarbonate secretion and mucus, and producing an overall reduction in acid secretion, the search for drugs that selectively inhibit the COX-2 isoform has received much attention. The traditional NSAIDs inhibit COX-1, COX-2, and thromboxane synthetase to varying degrees of selectivity. Decreased gastric mucosal protection (resulting in an enhanced risk of ulceration) and stress-induced decreased renal perfusion result from nonselective inhibition of COX, whereas inhibition of thromboxane synthesis results in increased prostaglandin synthesis and a reduction in platelet aggregation and, thus, an increased bleeding tendency. Those NSAIDs with a greater selectivity for COX-1 as a rule cause greater GI bleeding and renal toxicity than those with greater selectivity for COX-2.

These early studies led to extensive efforts by many laboratories to develop selective inhibitors of the COX-2 isoform with the goal of developing an “ideal” NSAID—that is, one that selectively inhibits COX-2, thus reducing the inflammatory response, but does not interfere with the GI-protective functions of COX-1. Two early lead compounds were developed, NS-398 and DuP 697, which have served as the basis of the development of two widely explored chemical classes. NS-398 and nimesulide are the prototypes of compounds known as “sulides,”

whereas DuP 697 is the prototype of a class of COX-2 inhibitors termed “coxibs.”



These efforts led to the introduction of several selective COX-2 inhibitors in the US market (celecoxib, rofecoxib, lumiracoxib, and valdecoxib), but safety risks and cardiovascular side effects led to the removal of all coxibs except celecoxib, and the development of other selective COX-2 inhibitors such as parecoxib and etoricoxib has been halted (Fig. 31.27). An excellent review of the various chemical classes of selective COX-2 inhibitors has been published (102). Selective COX-2 inhibitors were designed to take advantage of the much larger NSAID binding site on COX-2 compared to the NSAID binding site on COX-1, resulting from the substitution of the smaller amino acid valine in COX-2 for isoleucine at position 523 in COX-1 (see earlier discussion). In COX-1, the larger isoleucine residue near the active site restricts access by larger, relatively rigid side-chain substituents, such as sulfamoyl or sulfonyl side chains usually seen in the selective COX-2 inhibitors.

A comparison of the selectivity of several NSAIDs and selective COX-2 inhibitors has been presented (Fig. 31.28) (103). In this study, rofecoxib displayed greater than 50-fold selectivity for COX-2 versus COX-1,

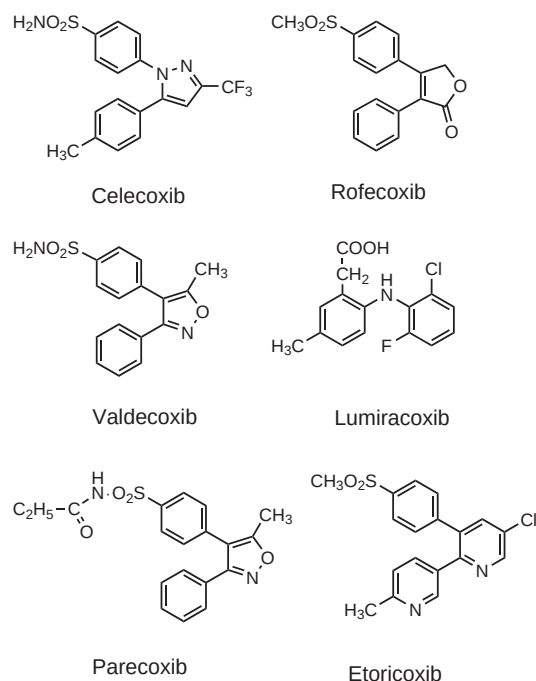


FIGURE 31.27 Structures of selective COX-2 inhibitors.

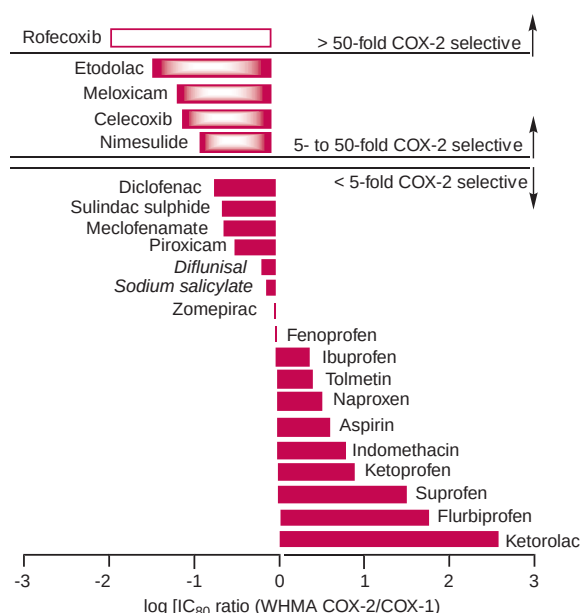


FIGURE 31.28 Selectivity of COX-2 inhibitors and NSAIDs given as log inhibitory concentration (IC_{50}) ratio. The “o” line indicates equipotency. (Adapted from Warner TD, Giuliano F, Vojnovic I, et al. Nonsteroid drug selectivities for cyclooxygenase-1 rather than cyclooxygenase-2 are associated with human gastrointestinal toxicity: a full in vitro analysis. *Proc Natl Acad Sci U S A* 1999;96:7563–7568, with permission.)

whereas celecoxib displayed 5- to 550-fold selectivity. In another study regarding the pharmacologic and biochemical profile of a new investigational selective COX-2 inhibitor, etoricoxib, the comparison of COX-1/COX-2 selectivity for several selective and nonselective inhibitors in human whole blood assays was reported (Table 31.5) (104).

Perhaps as interesting as the role that COX-2 selective inhibitors have in reducing the incidence of GI side effects among NSAIDs are the reports of other potential therapeutic uses for this new class of drugs, including potential use in the treatment of Alzheimer disease and carcinomas of various types. COX-2 appears to be induced in inflammatory plaques that are evident in the CNS in Alzheimer disease. Several reports have appeared indicating that patients taking NSAIDs have a lower incidence and a decreased rate of progression of Alzheimer disease. Epidemiologic studies suggest a significant reduction in the risk for colon cancer in patients regularly taking aspirin. Additionally, NSAIDs have been reported to reduce the growth rate of polyps in the colon in humans as well as the incidence of tumors of the colon in animals. The expression of COX-2 appears to be significantly upregulated in carcinoma of the colon. The effectiveness of NSAIDs in the prevention and treatment of other cancers, such as prostate cancer and mammary carcinoma, has been reported as well. This effectiveness is more noticeable among COX-2 selective drugs.

TABLE 31.5 A Comparison of IC_{50} (μ mole) Binding Constants for Selective Versus Nonselective Cyclooxygenase (COX) Inhibitors

Drug	COX-1	COX-2	COX-1/COX-2 Ratio
Etoricoxib	116	1.1	106
Rofecoxib	18.8	0.53	35
Valdecoxib	26.1	0.87	30
Celecoxib	6.7	0.87	7.6
Nimesulide	4.1	0.56	7.3
Diclofenac	0.15	0.05	3.0
Etodolac	9.0	3.7	2.4
Meloxicam	1.4	0.70	2.0
Indomethacin	0.19	0.44	0.4
Ibuprofen	4.8	24.3	0.2
6MNA	28.9	154	0.2
Piroxicam	0.76	9.0	0.08

IC_{50} , half maximal inhibitory concentration; 6MNA, 6-methoxynaphthalene-2-acetic acid.

Despite the promise of therapeutic effectiveness of the selective COX-2 inhibitors, the potential for severe cardiovascular effects prompted a critical review of these drugs. This concern was initiated through the revelation of long-term clinical trials of rofecoxib that indicated an increased risk of heart attack. One study completed in 2000, named VIGOR (105), showed an increased risk of myocardial infarction. A later retrospective analysis (106) of 1.4 million patients showed that daily rofecoxib doses of greater than 25 mg increased the risk of heart problems by 3.6-fold as compared with older NSAIDs. The coxibs were found (107) to suppress the formation of PGI_2 , which can result in elevated blood pressure and accelerated atherogenesis and predispose patients on coxib therapy to a heightened thrombotic response on the rupture of an atherosclerotic plaque (108). These effects can be related to the different consequences of inhibition of COX-1 and COX-2. Whereas COX-1 mediates the production of prostaglandins and the platelet aggregate stimulator TXA_2 , COX-2 can mediate the production of the platelet aggregate inhibitor prostacyclin. In an excellent review of the adverse cardiovascular effects of selective COX-2 inhibitors, it was suggested that the apparent consequence of selective inhibition of COX-2 is a significant reduction in the production of prostacyclins, whereas the production of TXA_2 by COX-1 is unaffected (109). Of practical interest is a report that resveratrol, an *m*-hydroxyquinone present in red wine that has been suggested to be one agent responsible for the cardioprotective effects observed with the consumption of red wine (i.e., the “French Paradox”), inhibits COX-1 with no apparent effect on COX-2 (110, 111). Thus, the design of highly selective inhibitors of

COX-2 would not be as desirable therapeutically as drugs that preferentially inhibit COX-2 but also inhibit COX-1 to a lesser extent.

In one of the most publicized drug withdrawals, Merck voluntarily withdrew rofecoxib from the US market in September 2004, followed by Pfizer's withdrawal of valdecoxib in April 2005 and the removal of lumiracoxib by Novartis in 2007. These drugs were voluntarily withdrawn because of safety concerns from increased risk of serious cardiovascular thrombotic events, myocardial infarction, and stroke, which can be fatal. This risk can increase with duration of use. Thus, the future of selective COX-2 inhibitors in the treatment of inflammatory disorders remains clouded.

Specific Drugs

Celecoxib (Celebrex). Celecoxib is synthesized by condensing 4-methylacetophenone and ethyltrifluoroacetate with sodium methoxide and the resulting butanedi-one derivative cyclized with 4-hydrazinophenylsulfonamide (111). It was the first NSAID to be marketed as a selective COX-2 inhibitor. Celecoxib is well absorbed from the GI tract, with peak plasma concentrations normally being attained within 3 hours of administration. Peak plasma levels in geriatric patients can be increased, but dosage adjustments in elderly patients are usually not required unless the patient weighs less than 50 kg. Celecoxib is excreted in the urine and feces primarily as inactive metabolites, with less than 3% of an administered dose being excreted as unchanged drug. Metabolism occurs primarily in the liver by CYP2C9 and involves hydroxylation of the 4-methyl group to the primary alcohol, which is subsequently oxidized to its corresponding inactive carboxylic acid, the major metabolite (73% of the administered dose) (Fig. 31.29). The carboxylic acid is conjugated, to a slight extent, with glucuronic acid to form the corresponding glucuronide. None of the isolated metabolites have been shown to exhibit pharmacologic activity as inhibitors of either COX-1 or COX-2.

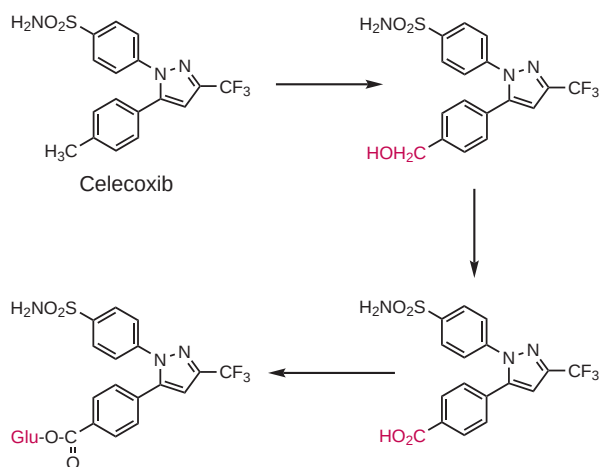


FIGURE 31.29 Metabolism of celecoxib.

ADVERSE EFFECTS

Celecoxib can cause an increased risk of serious cardiovascular thrombotic events, myocardial infarction, and stroke, which can be fatal. All NSAIDs can have a similar risk. This risk can increase with duration of use. Patients with cardiovascular disease or risk factors for cardiovascular disease can be at greater risk. The NSAIDs, including celecoxib, cause an increased risk of serious GI adverse events, including bleeding, ulceration, and perforation of the stomach or intestines, which can be fatal. These events can occur at any time during use and without warning symptoms. Elderly patients are at greater risk for serious GI events.

Celecoxib also inhibits CYP2D6; thus, the potential of celecoxib to alter the pharmacokinetic profiles of other drugs inhibited by this isoenzyme exists. Celecoxib, however, does not appear to inhibit other CYP isoforms, such as CYP2C19 or CYP3A4. Other drug interactions related to the metabolic profile of celecoxib have been noted, particularly with other drugs that inhibit CYP2C. For example, coadministration of celecoxib with fluconazole can significantly increase plasma concentration of celecoxib, because fluconazole inhibits CYP2C9.

Celecoxib is currently indicated for the relief of signs and symptoms of osteoarthritis and rheumatoid arthritis and to reduce the number of adenomatous colorectal polyps in familial adenomatous polyposis as an adjunct to usual care. Celecoxib is at least as effective as naproxen in the symptomatic management of osteoarthritis and at least as effective as naproxen and diclofenac in the symptomatic treatment of rheumatoid arthritis, and it is less likely to cause adverse GI effects. Celecoxib appears to be effective in the management of pain associated with both of these arthritic conditions, but effectiveness in acute or chronic pain has not been fully demonstrated. Unlike aspirin, celecoxib does not exhibit antiplatelet activity. Concomitant administration of aspirin and celecoxib can increase the incidence of GI side effects. Another notable potential drug interaction with celecoxib is its ability, like other NSAIDs, to reduce the blood pressure response to angiotensin-converting enzyme inhibitors. A more detailed discussion of the chemical, pharmacologic, pharmacokinetic, and clinical aspects of celecoxib is available (112).

DISEASE-MODIFYING ANTIRHEUMATIC DRUGS

The drugs previously discussed as NSAIDs, both the nonselective COX and selective COX-2 inhibitors, have proven to be beneficial as analgesics and for the symptomatic treatment of arthritic disorders. Despite their effectiveness and popularity, however, it should be remembered that none of these drugs are effective in preventing or

inhibiting the underlying pathogenic, chronic inflammatory processes. Recent interest has been generated by drugs that are effective in the treatment of arthritic disorders yet fail to demonstrate significant activity in the standard screening assays for antiarthritic drugs. Disease-modifying antirheumatic drugs (DMARDs) differ from the previously discussed drugs in that they are drugs that retard or halt the underlying progression, limiting the amount of joint damage that occurs in rheumatoid arthritis while lacking the anti-inflammatory and analgesic effects observed with NSAIDs. Although both NSAIDs and DMARDs improve symptoms of active rheumatoid arthritis, only DMARDs have been shown to alter the disease course and to improve radiographic outcomes. The DMARDs have an effect on rheumatoid arthritis that is different and more delayed in onset than either NSAIDs or corticosteroids. Once persistent disease activity (chronic synovitis) is established, a DMARD should be considered. The development of erosions or joint space narrowing on radiographs of the involved joints is a clear indication for DMARD therapy; however, one should not wait for radiographic changes to occur. DMARDs are much slower acting, taking as long as 3 months for measurable clinical benefits to be observed. Although these drugs also possess potentially dangerous adverse side effects, which in many cases limit their long-term use, DMARDs are effective in reducing joint destruction and the progression of early rheumatoid arthritis.

A 2004 study reported that taking DMARDs at early stages in the development of rheumatoid arthritis is especially important to slow the disease and to save the joints and other tissues from permanent damage (113). Typically, DMARDs are used with an NSAID or a corticosteroid. The NSAID or corticosteroid handles the immediate symptoms and limits inflammation, and the DMARD goes to work on the disease itself.

The DMARDs can be divided into two general categories: synthetic DMARDs, which can be taken orally, and biologic DMARDs, which are given by IV infusion or subcutaneously, that target and inactivate cell proteins (cytokines) and T lymphocytes (T cells) from causing joint inflammation. The DMARD methotrexate, in combination with the biologic DMARDs, can offer the best control of rheumatoid arthritis for the majority of people, eliminating the need for NSAID medications.

Synthetic Disease-Modifying Antirheumatic Drugs

The synthetic DMARDs include gold salts, hydroxychloroquine, and sulfasalazine. Less common synthetic DMARDs are penicillamine and minocycline. Synthetic immunosuppressants used for the treatment of inflammatory diseases include the antimetabolites methotrexate, leflunomide, and azathioprine.

Gold Compounds

HISTORICAL BACKGROUND At the end of the 19th century, the chemotherapeutic applications of heavy metal derivatives were receiving considerable interest. Among those metals

gaining the greatest attention were gold compounds (or gold salts). The first of these, gold cyanide, was effective in vitro against *Mycobacterium tuberculosis*. This discovery prompted others to extend the use of gold compounds in other disease states that are thought to be tubercular in origin. Early clinical observations had suggested similarities in the symptoms of tuberculosis and rheumatoid arthritis, and some thought rheumatoid arthritis to be an atypical form of tuberculosis. In 1927, aurothioglucose was found to relieve joint pain when used to treat bacterial endocarditis. The area of chrysotherapy had begun. Subsequent investigations led to an extensive study of gold compounds in Great Britain by the Empire Rheumatism Council, which reported in 1961 that sodium aurothiomalate was effective in slowing the development of progressive joint diseases. Both aurothioglucose and sodium aurothiomalate are orally ineffective and are administered by IM injection. In 1985, the first orally effective gold compound for arthritis, auranofin, was introduced in the United States. Several other gold compounds have been evaluated clinically but do not appear to offer advantages in terms of efficacy or toxicity.

MECHANISM OF ACTION The biochemical and pharmacologic properties shared by gold compounds are quite diverse. The mechanism by which they produce their antirheumatic actions has not been totally determined. The earlier observations that gold compounds were effective in preventing arthritis induced by hemolytic streptococci and by pleuropneumonia-like organisms led to the postulation that they acted through an antimicrobial mechanism. The inability of gold compounds to consistently inhibit mycoplasmal growth in vitro while inhibiting the arthritic process independent of microbial origins, however, suggested that they did not directly produce their effects by this mechanism. The involvement of immunologic processes in the pathogenesis of arthritis suggested that a direct suppression of the immunologic response by gold compounds was involved. Available evidence, however, suggests that whereas enzymatic mediators released as a result of the immune response can be inhibited, no direct effect on either immediate or delayed cellular responses is evident to suggest any immunosuppressive mechanism. Suggestions have been made that protein denaturation and macroglobulin formation cause the proteins to become antigenic, thus initiating the immune response and producing biochemical changes in connective tissue, which ultimately leads to rheumatoid arthritis. The possibility that gold compounds inhibit the aggregation of macroglobulins and, in turn, inhibit the formation of immune complexes can account for their ability to slow connective tissue degradation. Interaction with collagen fibrils and, thus, reduction of collagen reactivity that alters the course of the arthritic process has been also postulated. Perhaps the most widely accepted mechanism of action is related to the ability of gold compounds to inhibit lysosomal enzymes, the release of which promotes the inflammatory response. The lysosomal

enzymes glucuronidase, acid phosphatase, collagenase, and acid hydrolases are inhibited, presumably through a reversible interaction of gold with sulfhydryl groups on the enzymes. Gold thiomalate inhibits glucosamine-6-phosphate synthetase, a rate-limiting step in mucopolysaccharide biosynthesis and a property shared, to a lesser extent, by several NSAIDs. Gold sodium thiosulfate is a potent uncoupler of oxidative phosphorylation. Gold sodium thiomalate is also a fairly effective inhibitor of prostaglandin biosynthesis *in vitro*, but the relationship of this effect to the antiarthritic actions of gold compounds has not been clarified.

More recent studies suggest that auranofin suppressed the TLR4-mediated activation of the transcription factors by nuclear factor- κ B (NF- κ B) and IRF3, thus preventing the expression of cytokines and COX-2 (114). Previously it was shown that zinc is a necessary component of NF- κ B for DNA binding and that gold ion can block this binding by oxidizing cysteines associated with zinc (115). Auranofin has also been shown to decrease tumor necrosis factor (TNF)- α -induced NF- κ B activation, suggesting that effective inhibitors of NF- κ B can be useful immunosuppressive and anti-inflammatory agents (116).

SIDE EFFECTS Toxic side effects have been associated with the use of gold compounds, with the incidence of reported adverse reactions in patients on chrysotherapy being as high as 55%. Serious toxicity occurs in 5% to 10% of reported cases. The most common adverse reactions include dermatitis (e.g., erythema, papular, vesicular, and exfoliative dermatitis), mouth lesions (e.g., stomatitis preceded by a metallic taste and gingivitis), pulmonary disorders (e.g., interstitial pneumonitis), nephritis (e.g., albuminuria and glomerulitis), and hematologic disorders (e.g., thrombocytopenic purpura, hypoplastic and/or aplastic anemia, and eosinophilia; blood dyscrasias are rare in incidence but can be severe). Less commonly reported reactions are GI disturbances (e.g., nausea, anorexia, and diarrhea), ocular toxicity (e.g., keratitis with inflammation and ulceration of the cornea and subepithelial deposition of gold in the cornea), and hepatitis. In those cases in which severe toxicity occurs, excretion of gold can be markedly enhanced by the administration of chelating agents, the two most common of which are dimercaprol (British Anti-Lewisite) and penicillamine. Corticoids also suppress the symptoms of gold toxicity and the concomitant administration of dimercaprol, and corticosteroids have been recommended in cases of severe gold intoxication.

GENERAL STRUCTURE-ACTIVITY RELATIONSHIPS Structure-activity relationships of gold compounds have not received a great amount of attention. Two important relationships, however, have been established: 1) Monovalent gold (aurous ion [Au⁺]) is more effective than trivalent gold (auric ion [Au³⁺]) or colloidal gold; and 2)

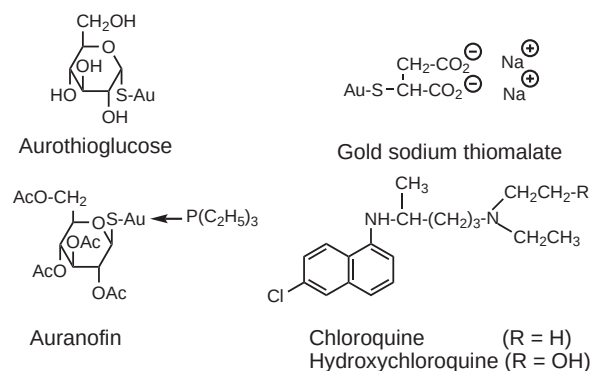


FIGURE 31.30 Structures of disease-modifying antirheumatic drugs (DMARDs).

only those compounds in which aurous ion is attached to a sulfur-containing ligand are active (Fig. 31.30). The nature of the ligands affects tissue distribution and excretion properties and are usually highly polar, water-soluble functions. Aurous ion has only a brief existence in solution and is rapidly converted to metallic gold or auric ion. Aqueous solutions decompose on standing at room temperature, posing a stability problem for the two injectable gold compounds therapeutically available (aurothioglucose and gold sodium thiomalate). Complexation of Au⁺ with phosphine ligands stabilizes the reduced valence state and results in both nonionic complexes that are soluble in organic solvents and an enhancement of oral bioavailability. Other changes also occur. In the phosphine-Au-S compounds, gold has a coordination number of 2, and the molecules are nonconducting monomers in solution. The injectable gold compounds are monocoordinated. Whereas nongold phosphine compounds are ineffective in arthritic assays, the nature of the phosphine ligand in the gold coordination complexes appears to have a greater role in antiarthritic activity than the other groups bound to gold. Within a homologous series, the triethylphosphine gold derivatives provide greatest activity.

The structures of the three therapeutically available gold compounds in the United States are shown in Figure 31.30.

ABSORPTION AND METABOLISM Gold compounds are rapidly absorbed following IM injection, and the gold is widely distributed in body tissues, with the highest concentrations found in the reticuloendothelial system and in adrenal and renal cortices. Binding of gold from orally administered gold to red blood cells is higher than that of injectable gold. Gold accumulates in inflamed joints, where high levels persist for at least 20 days after injection. Although gold is excreted primarily in the urine, the bulk of injected gold is retained. Gold can be found in the urine months later.

DRUG INTERACTIONS The only significant drug interactions reported are the concurrent administration of drugs

that also produce blood dyscrasias (most notably phenylbutazone and the antimalarial and immunosuppressive drugs).

SPECIFIC DRUGS

Gold Sodium Thiomalate Gold sodium thiomalate (actually a mixture of mono- and disodium salts of gold thiomalic acid) is very water soluble. It is available as a light-sensitive, aqueous solution of pH 5.8 to 6.5. The gold content is approximately 50%. It is administered IM, because it is not absorbed on oral administration and is highly bound (95%) to plasma proteins.

Gold sodium thiomalate is indicated in the treatment of active adult and juvenile rheumatoid arthritis as one part of a complete therapy program. It is recommended that injections be given to patients only when they are in a supine position. They must remain so for 10 minutes following injection.

Aurothioglucose Aurothioglucose is highly water soluble, and its aqueous solutions decompose on long standing. Therefore, it is available as a suspension in sesame oil. Gold content is approximately 50%. After IM injection, it is highly protein bound (95%), and peak plasma levels are achieved within 2 to 6 hours. After a single 50-mg dose, the biologic half-life ranges from 3 to 27 days, but following successive weekly doses, the half-life increases to 14 to 40 days after the third dose. The therapeutic effect does not correlate with serum plasma gold levels but appears to depend on total accumulated gold. Aurothioglucose is indicated for the adjunctive treatment of adult and juvenile rheumatoid arthritis.

Auranofin Auranofin contains approximately 29% gold. The carbohydrate portion assumes a chair conformation, with all substituents occupying the equatorial position. It is the first orally effective gold compound used to treat rheumatoid arthritis. On a milligram of gold per kilogram basis, it is reported to be as effective in the rat adjuvant arthritis assay as the parenterally effective drugs. Daily oral doses produce a rapid increase in kidney and blood gold levels for the first 3 days of treatment, with a more gradual increase on subsequent administration. Plasma gold levels are lower than those attained with parenteral gold compounds. The major route of excretion is via the urine. Auranofin can produce fewer adverse reactions than parenteral gold compounds, but its therapeutic efficacy can also be less.

Auranofin is indicated in adults with active rheumatoid arthritis who have not responded sufficiently to one or more NSAIDs.

Aminoquinolines

BACKGROUND The 4-aminoquinoline class of antimalarial drugs has been known to possess pharmacologic actions that are beneficial in the treatment of rheumatoid arthritis. Two of these drugs, chloroquine and hydroxychloroquine (Fig. 31.30), have been used as antirheumatics

since the early 1950s. The corneal and renal toxicity of chloroquine, however, has resulted in its discontinuance for this purpose, although it is still indicated as an antimalarial agent and an amebicide. Whereas hydroxychloroquine is less toxic, it is also less effective than chloroquine as an antirheumatic. The mechanism of action of these drugs as an antirheumatic remains unresolved. Interestingly, most of the data available relate to chloroquine rather than hydroxychloroquine but are assumed to be applicable to the latter. The spectrum of action of the 4-aminoquinolines differs from the NSAIDs in that chloroquine appears to be an antagonist of certain preformed prostaglandins. This effect, however, would indicate an acute, rather than a chronic, antirheumatic effect, whereas chloroquine has been shown to be similar to gold compounds in that it possesses a slow onset of action. Beneficial effects are noted only after 1 to 2 months of administration. Chloroquine inhibits chemotaxis of polymorphonuclear leukocytes *in vitro* but not *in vivo*. Its effects on collagen metabolism in connective tissue are also unclear. The most widely accepted mechanism of action of chloroquine and, presumably, of hydroxychloroquine is related to its ability to accumulate in lysosomes. Although evidence indicating stabilization of lysosomal membranes is not convincing, it can inhibit the activity of certain lysosomal enzymes, such as cartilage chondromucoprotease and cartilage cathepsin B. The possibility that chloroquine can prevent the release of TNF α from phagocytes has been proposed (117). There does not appear to be a correlation of the antirheumatic effects of the 4-aminoquinolines with their antimalarial activity.

HYDROXYCHLOROQUINE SULFATE Hydroxychloroquine sulfate is highly water soluble and exists in two different forms of different melting points. It is readily absorbed on oral administration, reaching peak plasma levels within 1 to 3 hours. It concentrates in organs such as the liver, spleen, kidneys, heart, lung, and brain, thereby prolonging elimination. Hydroxychloroquine is metabolized by N-dealkylation of the tertiary amines, followed by oxidative deamination of the resulting primary amine to the carboxylic acid derivative. In addition to possessing corneal and renal toxicity, hydroxychloroquine can also cause CNS, neuromuscular, GI, and hematologic side effects. Hydroxychloroquine sulfate is indicated for the treatment of rheumatoid arthritis, lupus erythematosus, and malaria.

Immunosuppressants

The discovery of drugs that modify the immune response, whether as immunoregulatory, immunostimulatory, or immunosuppressive agents, has been the focus of much recent research activity. Several substances that suppress the immune system have been explored as antirheumatic drugs, because the etiology of rheumatoid arthritis can involve a destructive immune response. Thus, unlike drugs previously discussed,

immunosuppressive drugs can act at the steps involved in the pathogenesis of the inflammatory disorders. As a group, however, these drugs are cytotoxic, as evidenced by the initial development of these drugs as anticancer agents. Among the more widely employed immunosuppressants for rheumatoid arthritis are leflunomide, methotrexate, and sulfasalazine (Fig. 31.31). Others that have been explored include azathioprine and cyclophosphamide. All of these drugs are quite toxic and, are indicated for rheumatoid arthritis only in patients with severe, active disease who have not responded to full-dose NSAID therapy and at least one DMARD and a corticosteroid. Interestingly, although aspirin and NSAIDs are effective in only one-third of children with juvenile arthritis, methotrexate, when given only once a week at low doses (<20 mg) to minimize side effects, is effective. Cyclosporine (Sandimmune) has been investigated in rheumatoid arthritis and appears to offer short-term benefits, although its toxic effects also limit its long-term use. Cyclosporine appears to inhibit the proliferation of T-helper/inducer lymphocytes, blocking the signaling pathway involved in the etiology of rheumatoid arthritis.

A new class of oral small-molecule immunosuppressants with anti-inflammatory and antiallergic properties that can be administered orally are the Janus activated kinase (JAK) 3 inhibitors, which includes tofacitinib, a pyrrolo[2,3-d]pyrimidine analog. The Janus tyrosine kinases (JAK1, JAK2, JAK3, and Tyk2), named after the Roman god of gates and doors ("Janus"), regulate the signal transduction pathways of numerous cytokines including the interleukins, and therefore have a central role in immunosuppression and immunomodulation. Of the four JAK enzymes, JAK3 controls the signaling pathways of the multiple interleukin cytokines important for various B- and T-cell functions. JAK3 was identified as a target for immunosuppression because its expression is limited to lymphoid cells. Thus inhibition of JAK3 by tofacitinib in clinical trials has shown promising activity in the treatment of rheumatoid arthritis, inflammatory bowel disease, ankylosing spondylitis, psoriasis, and psoriatic arthritis and for the prevention of transplant rejection (118).

SPECIFIC DRUGS

Leflunomide Leflunomide is a DMARD with anti-inflammatory and immunosuppressive activity used for the management of rheumatoid arthritis. It retards structural damage associated with arthritis in adults who have moderate to severe active rheumatoid arthritis. Leflunomide is also being investigated for use in patients with solid tumors and in organ transplant recipients.

Leflunomide is a prodrug that is rapidly and almost completely metabolized (half-life, <60 minutes) after oral administration to teriflunomide, the pharmacologically active α -cyanoenol metabolite (Fig. 31.32). The C₃-H of

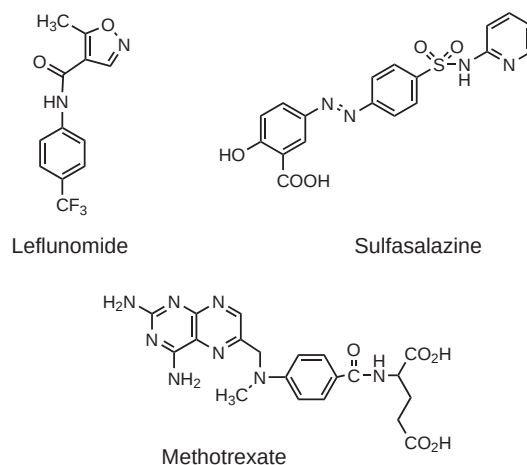


FIGURE 31.31 Structures of DMARD immunosuppressants.

the isoxazole ring is essential for the ring opening to its active metabolite. The reaction is similar to the CYP1A2-catalyzed dehydration of aldoximes. The exact mechanism of action of leflunomide in the management of rheumatoid arthritis has not been fully elucidated but appears to principally involve inhibition of B-lymphocyte (B-cell) proliferation, reducing antibody formation. Activated lymphocytes must proliferate and synthesize large quantities of cytokines, requiring increased de novo synthesis of uridine monophosphate (UMP) and other pyrimidine nucleotides for its cell life cycle. Therefore, any substance that reduces the intracellular concentration of pyrimidine nucleotides will affect the growth of these activated cells.

Leflunomide itself lacks activity, but teriflunomide inhibits pyrimidine de novo synthesis at low therapeutic doses by inhibiting dihydroorotate dehydrogenase (the rate-determining enzyme for the synthesis of UMP), decreasing DNA and RNA synthesis, and arresting the cell proliferation cycle and production of antibodies. The reduction of dihydroorotate to orotate occurs concurrently with the reduction of its cofactor, ubiquinone (coenzyme Q) (Fig. 31.33).

The inhibition of dihydroorotate dehydrogenase by teriflunomide demonstrates noncompetitive and uncompetitive kinetics. Administration of leflunomide in patients with rheumatoid arthritis results in progressive removal of B cells and downregulation of the immune

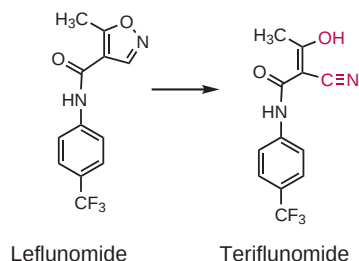


FIGURE 31.32 Metabolism of leflunomide.

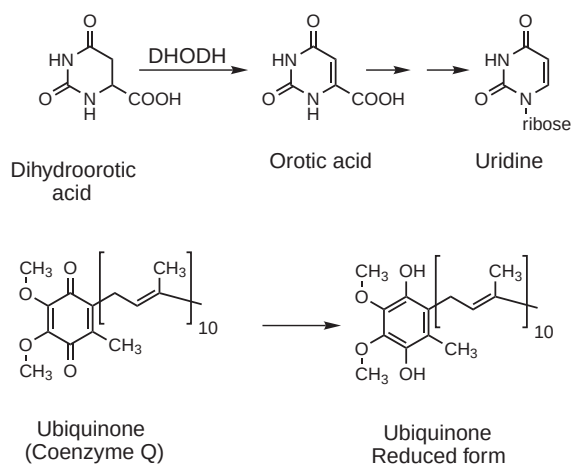


FIGURE 31.33 Dihydroorotatase (DHOH) pathway with cofactor, coenzyme Q.

process. Teriflunomide not only inhibits B-cell proliferation but also T-cell proliferation, blocking the synthesis of immunosuppressive cytokines. At high therapeutic doses, leflunomide inhibits protein tyrosine kinases.

Leflunomide is administered orally as a single daily dose without regard to meals. Therapy can be initiated with a loading dosage given for 3 days, followed by the usual maintenance dose. It undergoes primarily enterohepatic circulation, extending its duration of action. Cholestyramine can be used to enhance its elimination in cases of toxicity.

Methotrexate Methotrexate (Fig. 31.31) is an antifolate drug approved for the treatment of severe active rheumatoid arthritis in adults who are intolerant to or have had an insufficient response to first-line therapy. Although the mechanism of action of methotrexate in rheumatoid arthritis is not known with certainty, several mechanisms have been proposed. Methotrexate blocks pyrimidine biosynthesis by reversibly inhibiting dihydrofolate reductase, blocking the proliferation of B cells by interfering with DNA synthesis, repair, and replication. Other actions of methotrexate that can be involved include inhibition of the production of polyamines such as spermine that can be responsible for inducing tissue damage and the activation of NF- κ B. Oral absorption is dose dependent, being well absorbed at doses of 7.5 to 25 mg once a week. At this dose, oral bioavailability is approximately 60%, and food can delay absorption and reduce peak concentration. The volume of distribution is 0.4 to 0.8 L/kg. Protein binding is approximately 50%. It is metabolized to active metabolites, methotrexate polyglutamates, and 7-hydroxymethotrexate. Some metabolism occurs by intestinal flora after oral administration. Methotrexate is actively transported into the urine (80% to 90% unchanged in the urine within 24 hours) via the folate transporter, an organic anion transporter (hOAT3). Its elimination half-life is 3 to 10 hours.

Life-threatening drug interactions are known to occur between methotrexate and NSAIDs, probenecid, and penicillin G. The NSAIDs (salicylate, ibuprofen, ketoprofen, piroxicam, and indomethacin), probenecid, and penicillin G dose dependently inhibited methotrexate elimination into urine by human organic anion transporters (hOAT1, hOAT3, and hOAT4) (Chapter 5). The inhibitory effects of these drugs on hOAT3 were comparable, with therapeutically relevant plasma concentrations of unbound drugs. Thus, patients with rheumatoid arthritis should not take NSAIDs while taking methotrexate.

Methotrexate therapy requires monitoring of liver enzymes and is contraindicated in those with hepatic disease and in women considering pregnancy.

Sulfasalazine (Azulfidine) Sulfasalazine is a prodrug that is not active in its ingested form; rather, it is hydrolyzed by colonic bacteria into 5-aminosalicylic acid (5-ASA; mesalamine) and sulfapyridine (Fig. 31.34). Some controversy exists regarding which of these two products is responsible for the activity of Azulfidine. 5-Aminosalicylic acid, like acetylsalicylic acid, is an inhibitor of prostaglandin biosynthesis and is known to have therapeutic benefit. However, it is not clear whether sulfapyridine adds any further benefit. It was shown to inhibit purine biosynthesis by inhibiting 5-aminoimidazole-4-carboxamido-ribonucleotide transformylase and to enhance adenosine release, enhancing the ability of adenosine to decrease inflammation by binding to A_2 adenosine receptors on inflammatory cells (119). Sulfasalazine and/or its metabolites have been shown to inhibit the release of inflammatory cytokines and TNF α . In the colon, the products created by the breakdown of sulfasalazine work as anti-inflammatory agents for treating colon inflammation. The beneficial effect of sulfasalazine is believed to result from a local effect on the bowel, although there can also be a beneficial systemic immunosuppressant effect. Sulfasalazine was approved in 1950.

Following oral administration, sulfasalazine is poorly absorbed, with approximately 20% of the ingested sulfasalazine reaching the systemic circulation. The remainder of the ingested dose is metabolized by colonic bacteria into its components, sulfapyridine and mesalamine (5-ASA). Most of the sulfapyridine metabolized

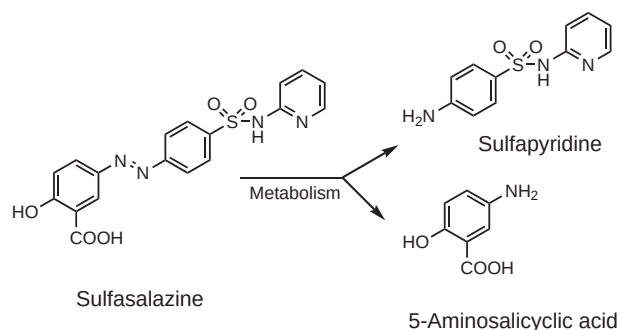


FIGURE 31.34 Metabolism of sulfasalazine.

from sulfasalazine (60% to 80%) is absorbed in the colon following oral administration, and approximately 25% of the 5-ASA metabolized from sulfasalazine is absorbed in the colon. The apparent volume of distribution of sulfasalazine in eight healthy volunteers was 64 L/kg, and that of sulfapyridine was 0.4 to 1.2 L/kg. Protein binding is approximately 99% for sulfasalazine, approximately 50% for sulfapyridine, and approximately 43% for 5-ASA. The absorbed sulfapyridine is acetylated and hydroxylated in the liver, followed by conjugation with glucuronic acid and, for 5-ASA, acetylation in the intestinal mucosal wall and the liver. The elimination half-life is 5 to 10 hours for sulfasalazine and 6 to 14 hours for sulfapyridine, depending on acetylator status of the patient, and 0.6 to 1.4 hours for 5-ASA. Time to peak serum concentration is 1.5 to 6 hours for oral sulfasalazine and 9 to 24 hours for oral sulfapyridine; for enteric-coated tablets, time to peak serum concentration is 3 to 12 hours for sulfasalazine and 12 to 24 hours for sulfapyridine. Approximately 5% of sulfapyridine and approximately 67% of mesalamine are eliminated in the feces, and 75% to 91% of sulfasalazine and sulfapyridine metabolites are excreted in urine within 3 days, depending on the dosage form used. 5-ASA is excreted in urine mostly in acetylated form.

Sulfasalazine is used for the treatment of mild to moderate ulcerative colitis; as adjunct therapy in the treatment of severe ulcerative colitis; for the treatment of Crohn disease; and for the treatment of rheumatoid arthritis or ankylosing spondylitis. Contraindications include hypersensitivity to sulfa drugs, salicylates, intestinal or urinary obstruction, and porphyria.

Tofacitinib Tofacitinib (Orencia) (formerly known as tasocitinib) is an orally active immunosuppressant implicated in the pathogenesis and used for treatment of rheumatoid arthritis, inflammatory bowel disease, ankylosing spondylitis, psoriasis, and psoriatic arthritis and for the prevention of transplant rejection. Tofacitinib specifically inhibits JAK3, blocking proinflammatory cytokine signaling, thus preventing the expression of both B and T cells. Tofacitinib is an investigational drug in FDA phase III trials, which suggest this new drug to be as effective as TNF α inhibitors. Dose-dependent decreases in white blood cell counts and increases in low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, total cholesterol, and serum creatinine levels were observed in patients treated with tofacitinib. The most frequent treatment-related adverse events (<5%) were urinary tract infection, diarrhea, bronchitis, and headache. Some hepatic metabolism of tofacitinib conducted in human liver microsomes and hepatocytes was observed by CYP3A4/5 and CYP2C19. The low clearance of the drug in humans occurred via a combination of renal excretion of unchanged drug and low hepatic metabolic clearance. Volume of distribution in humans was 1.9 L/kg, with a terminal elimination half-life of 7 hours. Pharmacokinetics support daily or twice a day dosing (118).

Biologic Disease-Modifying Antirheumatic Drugs

Cytokine Inhibitors

TUMOR NECROSIS FACTOR BLOCKERS T lymphocytes (T cells), a type of white blood cell, are important cells of the immune system. Patients with rheumatoid arthritis have increased numbers of T cells within the inflamed joints. These T cells are “activated”—that is, they multiply and release chemicals (cytokines) that promote the destruction of tissues surrounding the joints and cause the signs and symptoms of rheumatoid arthritis (120).

As discussed earlier in this chapter (see also Chapter 6 for a detailed discussion of cytokines), there is considerable expression of the cytokines, interleukin (IL)-1, IL-6, and TNF α by the rheumatoid synovium. TNF α is a proinflammatory cytokine (cell protein) that has a major role in the pathologic inflammatory process of rheumatoid arthritis. TNF is an important mediator of local inflammation, and the release of TNF α from T cells produces increased vascular permeability, release of nitric oxide with vasodilation, local activation of vascular endothelium, increased expression of adhesion molecules on endothelial blood vessels, and increased platelet activation and adhesion. As TNF α builds up in the joints, it leads to joint inflammation, which ultimately results in joint destruction. Because of its role in the progression of rheumatoid arthritis, methods for rendering TNF α inactive have become a key focus of therapies for rheumatoid arthritis (119). Patients with rheumatoid arthritis display elevated levels of the cytokines, TNF α , IL-1, and IL-6 in synovial fluid and tissue, and there appears to be a correlation between the amount of these products present and the severity of the disease. A significant advance in the treatment of arthritic diseases was the observation that therapy directed toward diminishing the effects of TNF α appeared to also improve the symptoms of ankylosing spondylitis and psoriatic arthritis.

Two different approaches have been developed to decrease TNF activity that has resulted in marketable drugs: administration of soluble TNF receptors (TNFRs; etanercept), and treatment with anti-TNF α antibodies (e.g., infliximab, adalimumab). These compounds are designed to target and neutralize the effects of TNF, helping to reduce pain, morning stiffness, and tender or swollen joints, usually within 1 or 2 weeks after treatment begins. Evidence suggests that TNF blockers can also halt the progression of the disease. These medications work synergistically with methotrexate and therefore are often taken with methotrexate. TNF blockers approved for treatment of rheumatoid arthritis are etanercept, infliximab, adalimumab, and rituximab. Potential side effects include injection site irritation (adalimumab and etanercept), worsening congestive heart failure (infliximab), blood disorders, lymphoma, demyelinating diseases, and increased risk of infection. These drugs should not be taken if an active infection is present. Effectiveness is lost if the drugs are discontinued.

Because TNF is also important for host defense against infections, the effects of long-term use on toxicity require

further study. Substantial improvements in the course of the disease have been noted with both therapeutic approaches.

Structural illustrations of the anti-TNF agents etanercept, infliximab, adalimumab, and rituximab are provided in Figure 31.35. Two new anti-TNF α antibodies have been introduced recently (certolizumab and golimumab) for rheumatoid arthritis and other forms of inflammatory arthritis.

Etanercept (Enbrel) Etanercept is produced by recombinant DNA technology in a Chinese hamster ovary mammalian cell line and is the first biotechnology-derived drug to be introduced for the reduction of the signs and symptoms of moderately to severely active rheumatoid arthritis in patients who have not adequately responded to one or more of the synthetic DMARDs. It is a dimeric soluble form of the p75 TNFR capable of binding to two TNF molecules in the circulation. It consists of the extracellular ligand binding portion of the 75-kd human TNFR fused to the Fc portion of human immunoglobulin (Ig) G1. The Fc component of etanercept contains the C_H2 domain, the C_H3 domain, and the hinge region, but not

the C_H1 domain of IgG1. It consists of 934 amino acids and has an apparent molecular weight of approximately 150 kd. Two TNFRs have been identified, a 75-kd protein and a 55-kd protein, that occur as monomeric molecules on cell surfaces and soluble forms in the blood. The biologic activity of TNF requires its binding to either of the two cell surface TNFRs. Etanercept can bind specifically to two molecules of TNF α in the circulation, preventing its interaction with cell surface TNFRs. Etanercept also binds TNF β . It can be used as monotherapy or in combination with methotrexate. Some concern exists because of reports that etanercept can, in some cases, cause serious infections and could have contributed to the deaths of several patients using the drug. An excellent review of the properties and use of etanercept is available (121).

Etanercept is available as a powder for injection in single-use vials containing 25 mg of the drug. The reconstituted solution is administered twice a week at 25 mg or once a week at 50 mg as a clear and colorless subcutaneous injection.

Etanercept has been approved for reducing signs and symptoms, inhibiting the progression of structural damage, and improving physical function in patients with moderately to severely active rheumatoid arthritis; for reducing signs and symptoms in patients with active arthritis and patients with psoriatic arthritis; and for reducing signs and symptoms in patients with active ankylosing spondylitis.

Infliximab (Remicade) Infliximab is a chimeric (“humanized”) IgG1 κ monoclonal antibody to human TNF α (see Chapter 6 for detailed discussions of monoclonal antibodies). By combining the Fv domain of the mouse antibody responsible for recognizing TNF α with parts of the human Fc domain of IgG1 (IgG1 κ), the fused protein looks more like normal human IgG1 molecule (“humanized”), so there is a better chance the fused protein will not be destroyed by the patient’s own immune system. Infliximab has an approximate molecular weight of 149,100 daltons and binds specifically, with high affinity, to both the transmembrane and soluble forms of TNF α in the blood, thus neutralizing its biologic activity. It does not bind to TNF β (lymphotoxin A), a related cytokine that uses the same receptors as TNF α . Infliximab is produced by a recombinant cell line cultured by continuous perfusion and is purified by a series of steps that includes measures to inactivate and remove viruses. Cells expressing transmembrane TNF α bound by infliximab can be lysed. The TNF α antibodies decrease synovitis and joint erosions in a murine model of collagen-induced arthritis and, when administered after disease onset, allow eroded joints to heal.

After treatment with infliximab, patients with rheumatoid arthritis or Crohn disease exhibited reduced infiltration of inflammatory cells and TNF α production in inflamed tissues and decreased levels of serum IL-6 and C-reactive protein compared to baseline. In psoriatic arthritis, treatment with infliximab resulted in a

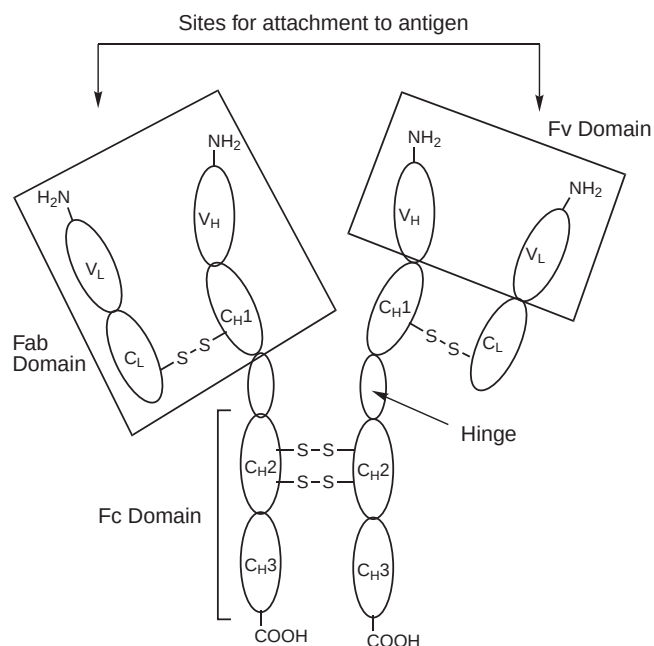


FIGURE 31.35 Anti-tumor necrosis factor (TNF) agents. Etanercept is the extracellular portion of the human TNF receptor fused to the Fc domain of human immunoglobulin (Ig) G1. Infliximab is a partially humanized monoclonal antibody against TNF α . The Fv domains are derived from mouse antihuman sequences, whereas the Fc domain is composed of human IgG1 sequences. Adalimumab is fully humanized antibody to human TNF α (Fv and Fc derived from human sequences). Rituximab is a partially humanized monoclonal antibody against CD40. The Fv domains are derived from mouse antihuman sequences, and the Fc domain is composed of human IgG1 sequences.

reduction in the number of T cells and blood vessels in the synovium and psoriatic skin as well as a reduction of macrophages in the synovium. Single IV infusions showed a linear relationship between the dose administered and the maximum serum concentration. The volume of distribution at steady-state was independent of dose and indicated that infliximab was distributed primarily within the vascular compartment. The terminal half-life of infliximab is 8.0 to 9.5 days. No systemic accumulation of infliximab occurred on continued repeated treatment at 4- or 8-week intervals.

Infliximab is supplied as a sterile, white, lyophilized powder formulated for IV infusion. Following reconstitution with Sterile Water for Injection, the solution should be used immediately after reconstitution, because the vials do not contain antibacterial preservatives. The reconstituted solution should be colorless to light yellow and opalescent.

Infliximab is indicated for the treatment of rheumatoid arthritis in combination with methotrexate and for Crohn disease. Long-term use can be associated with the development of anti-infliximab antibodies, an effect that does not appear when it is used with methotrexate. Warnings associated with the use of infliximab include risks of autoimmunity, infections, and hypersensitivity reactions. An excellent review of the properties and use of infliximab has recently appeared (122). Infliximab is more specific than etanercept, because etanercept binds to both TNF α and TNF β , whereas infliximab is an antibody that binds only to TNF α . Infliximab possesses a longer half-life giving a dosing schedule of approximately every 6 to 8 weeks.

Infliximab, in combination with methotrexate, is indicated for reducing signs and symptoms, inhibiting the progression of structural damage, and improving physical function in patients with moderately to severely active rheumatoid arthritis; for reducing signs and symptoms and maintaining clinical remission in patients with moderately to severely active Crohn disease who have had an inadequate response to conventional therapy; for reducing signs and symptoms in patients with active arthritis and in patients with psoriatic arthritis; for reducing signs and symptoms in patients with active ankylosing spondylitis; and for reducing the signs and symptoms of ulcerative colitis.

Adalimumab (Humira) Adalimumab is a recombinant human IgG1 monoclonal antibody targeted for human TNF α . Adalimumab is produced by recombinant DNA technology in a mammalian cell expression system using a protein-engineering strategy for creating a TNF α antibody with human-derived, heavy- and light-chain variable regions (Fab) and human IgG1 κ constant regions. It consists of 1,330 amino acids and has a molecular weight of approximately 148 kd. Adalimumab, as an antibody, works by targeting and binding TNF α , thus neutralizing the effect of TNF α and, thereby, reducing the symptoms of rheumatoid arthritis and slowing the

progression of structural joint damage caused by the disease. Adalimumab does not bind or inactivate TNF β .

Adalimumab is supplied in single-use, prefilled, glass syringes as a sterile, preservative-free, colorless solution for subcutaneous administration. The pharmacokinetics of adalimumab were linear over the dose range of 0.5 to 10.0 mg/kg following a single IV dose. The mean elimination half-life was approximately 2 weeks.

Rituximab (Rituxan) Rituximab is a genetically engineered, fused mouse/human anti-CD40 monoclonal antibody that targets B lymphocytes by binding specifically to CD20 antigen, a protein found on the surface of B cells at certain stages in their life cycle. Rituximab is composed of two heavy chains of 451 amino acids and two light chains of 213 amino acids with an approximate molecular weight of 145 kd (Fig. 31.30). Its binding affinity for the CD20 antigen is approximately 8.0 nmol/L. The mouse light- and heavy-chain Fab domains of rituximab, which bind to the CD20 antigen on B cells, are linked to the human Fc domains of IgG1 κ . Once the rituximab molecule attaches to the B cells, it initiates B-cell lysis, inducing rapid and profound depletion of peripheral B cells, with patients showing near complete B-cell depletion within 2 weeks after receiving the first dose of rituximab. Because rituximab does not target B cells at the earliest stages of their development, however, these B-cell depletions are usually temporary. In clinical trials, the majority of patients showed peripheral B-cell depletion for at least 6 months, followed by subsequent gradual recovery. A small proportion of patients (4%) had prolonged peripheral B-cell depletion lasting more than 3 years after a single course of treatment.

Rituximab is a sterile, clear, colorless, preservative-free, liquid concentrate formulated for IV administration. It has changed the treatment of rheumatoid arthritis by showing that targeted B-cell therapy in combination with methotrexate can reduce signs and symptoms of rheumatoid arthritis in adult patients with moderately to severely active rheumatoid arthritis who have had an inadequate response to one or more TNF antagonist therapies. Although B cells once were considered to be one of the main contributing factors in the pathogenesis of rheumatoid arthritis, recent evidence has shown that T cells, dendritic cells, and macrophages are also involved. Rituximab has rekindled interest in B cells, highlighting their important role in perpetuating the inflammatory process and showing how they can interact with other cell types and contribute to joint inflammation.

For rheumatoid arthritis, rituximab is given as two 1,000-mg IV infusions separated by 2 weeks. Glucocorticoids are also recommended to reduce the incidence and severity of infusion reactions. Rituximab is given in combination with methotrexate. Its administration has been associated with hypersensitivity reactions (non-IgE-mediated reactions), which can respond to adjustments in the infusion rate and in medical management. People who have not found relief using TNF

blockers might consider using rituximab. Side effects include flu-like signs and symptoms such as fever, chills, and nausea. Some people experience an “infusion-reaction complex,” such as difficulty breathing and heart problems, that can result in death.

Although originally approved for use in people with non-Hodgkin lymphoma, rituximab was approved for rheumatoid arthritis in 2006. A review of the clinical applications of rituximab is available (123).

Certolizumab Pegol (Cimzia) Certolizumab is a monoclonal antibody directed against TNF α (124). It selectively neutralizes TNF α with an IC₉₀ of 4 ng/mL while not neutralizing lymphotoxin A (TNF). Certolizumab is composed of a humanized anti-TNF α antibody Fab' fragment, which is linked to polyethylene glycol (PEG). The Fab' fragment is composed of a light chain with 214 amino acid residues and a heavy chain with 229 amino acids. The antibody lacks an Fc portion and thus binds to only TNF α , without having to bind to the cell surface receptors for antibodies. The drug is produced in higher yields in *Escherichia coli* in only 2 to 3 days; thus, certolizumab offers a lower cost of production with a more reliable supply. The purpose of linking the antibody to PEG (pegolate) is to reduce the immunogenicity of the antibody; increase the circulating half-lives of antibodies, thereby increasing the dosing range to once every 4 weeks, rather than every 2 weeks with several of the other monoclonal antibodies; and enhance TNF α targeting.

Absorption and Metabolism. Subcutaneous and single IV dosing studies of certolizumab demonstrate predictable dose-related plasma concentrations with a linear relationship between the dose administered and maximum plasma concentration ($C_{\max}$) and area under the curve (AUC) versus time. A mean $C_{\max}$ of 43 to 49 mcg/mL occurred at week 5 during the initial dosing period, which led to the development of the dosing regimen of 400 mg subcutaneously at weeks 0, 2, and 4 followed by 200 mg biweekly. Metabolism of the Fab' fragment is unknown, whereas the PEG moiety is excreted in the urine.

Adverse Effects. The major adverse effects are mild and include upper respiratory infections, rash, and urinary tract infections.

Golimumab (Symponi) Golimumab is a human IgG1k monoclonal antibody specific for human TNF α . Its molecular mass is 150 to 151 kD (125). It is an alternative to etanercept, adalimumab, and infliximab for the treatment of rheumatoid arthritis. Golimumab binds to both the soluble and transmembrane bioactive forms of human TNF α , with affinities of 17 pM by surface plasmon resonance and 1.4 pM in solution (126).

Absorption, Metabolism, and Dose. Median time to reach maximum serum concentrations ($T_{\max}$) is 2 to 6 days, whereas the median terminal half-life is 2 weeks.

Metabolism is unknown. Dosing of 50 mg subcutaneously every 4 weeks leads to steady-state serum concentrations by week 12. The advantage golimumab possesses over other anti-TNF α agents is dosing (injected subcutaneously once a month, whereas adalimumab is injected subcutaneously every other week, etanercept is injected subcutaneously weekly, and infliximab is injected by IV infusion every 8 weeks).

Adverse Effects. The most frequent adverse reactions are upper respiratory tract infection, sore throat, and nasopharyngitis. However, serious infections can occur along with lymphomas and other malignancies. Labeling includes a black box warning that alerts patients to the risk of tuberculosis and invasive fungal infections.

INTERLEUKIN-1 RECEPTOR ANTAGONIST

Anakinra (Kineret) Anakinra is a recombinant, nonglycosylated form of the human IL-1 receptor antagonist (IL-1R α) that neutralizes the inflammatory activity of IL-1 by competing with IL-1 for binding to its IL-1 type 1 receptor (IL-1R1). IL-1 is a cytokine for which production is induced in response to inflammatory stimuli and that mediates various physiologic responses, including inflammatory and immunologic responses that promote inflammation. When IL-1 binds to IL-1R1, a signal is produced that increases the formation of nitric oxide, PGE₂, and collagenase in synovial cells, resulting in cartilage degradation and stimulation of bone resorption. Thus, IL-1R α has an important role for regulating synovial proinflammatory IL-1 activity by preventing IL-1 from binding to IL-1R1. Analysis of synovial fluid suggests that the rheumatoid synovium is characterized by an overexpression of IL-1. The resulting imbalance between IL-1 and IL-1R α has been implicated in perpetuating the proinflammatory response and destructive tide of events in rheumatoid arthritis. If IL-1 is prevented from binding to IL-1R1, the inflammatory response decreases. The levels of the naturally occurring IL-1R α in synovium and synovial fluid from rheumatoid arthritis patients are insufficient to compete with the elevated amount of locally produced IL-1.

Therefore, anakinra neutralizes the proinflammatory activity of IL-1 by competitively inhibiting the binding of IL-1 to IL-1R1, similar to the endogenous antagonist, IL-1R α . In vitro studies have shown that anakinra inhibits the induction of the inflammatory mediators, nitric oxide and PGE₂, and collagenase. Anakinra differs from native human IL-1R α in that it has the addition of a single methionine residue at its amino terminus. Anakinra consists of 153 amino acids and has a molecular weight of 17.3 kD. It is produced by recombinant DNA technology using an *E. coli* bacterial expression system.

Anakinra is the first IL-1R α to be approved for use in adults with moderate to severe active rheumatoid arthritis who have not responded adequately to conventional DMARD therapy. It can be used either alone or in combination with methotrexate. Anakinra is supplied

in single-use, prefilled, glass syringes as sterile, clear, preservative-free solution that is administered daily as a self-administered subcutaneous injection under the skin. Some potential side effects include injection site reactions, decreased white blood cell counts, headache, and an increase in upper respiratory infections. There can be a slightly higher rate of respiratory infections in people who have asthma or chronic obstructive pulmonary disease. Persons with an active infection are advised not to use anakinra. Its elimination half-life after subcutaneous administration is 4 to 6 hours.

TOCILIZUMAB (ACTEMRA) Tocilizumab is a recently introduced recombinant humanized antihuman monoclonal antibody that binds to solubilized and membrane-bound IL-6 receptors and inhibits IL-6-mediated signaling (127). It is a member of the IgG1k subclass with a typical H_2L_2 (two heavy and two light chains) polypeptide structure. Each light chain consists of 214 amino acids, whereas the heavy chain contains 448 residues. The four polypeptide chains are linked by disulfide bonds both intra- and intermolecularly. Tocilizumab is the first monoclonal antibody that targets IL-6 and offers an alternative to patients not adequately responding to TNF α blockers. It has a molecular weight of approximately 148 kd.

Absorption, Metabolism, and Dose With a dose of tocilizumab (4 mg/kg IV) given every 4 weeks, the predicted mean steady-state AUC, minimum plasma concentration (C_{min}), and C_{max} of tocilizumab were 13,000 mcg/hour/mL, 1.49 mcg/mL, and 88.3 mcg/mL, respectively. Steady-state was reached following the first administration for C_{max} and AUC, respectively, and after 16 weeks for C_{min} . At 8 mg/kg given every 4 weeks, the predicted mean steady-state AUC, C_{min} , and C_{max} of tocilizumab were 35,000 mcg/h/mL, 9.74 mcg/mL, and 183 mcg/mL, respectively. The half-life is concentration dependent and is 11 days for the 4 mg/kg dose and 13 days for

the 8 mg/kg dose. Metabolism is unknown, and the total dose should not exceed 800 mg per infusion.

Adverse Effects The most frequent adverse effects are upper respiratory tract infection, headache, hypertension, increased alanine transaminase, and nasopharyngitis. All precautions for using an immunosuppressant should be taken.

COSTIMULATION MODULATORS Two signals are required to activate a T-cell response to an antigen, called costimulation. Regulation of costimulatory molecules can be a mechanism whereby the immune system limits the extent of an immune response. If an unactivated antigen-presenting cell (APC; a cell that “presents” an antigen complex that is recognized by the T-cell receptor) presents an antigen to a T cell in the absence of an appropriate costimulatory signal, the T cell does not respond and becomes unreactive and nonresponsive to any further antigenic stimuli (Fig. 31.36). The T-cell costimulatory activation pathway is initiated, however, when an activated APC presents both an antigen and a costimulatory ligand, such as B7 (CD86), that interacts with CD28 on the surface of the T cell to form B7-CD28 complex, initiating T-cell proliferation and differentiation in response to the antigenic stimulus. This stimulus releases cytokines that bind to the T cell, further enhancing its activation. A counterbalance to CD28 is cytotoxic T-lymphocyte antigen-4 (CTLA-4), both of which are expressed on the surface of T cells. The CTLA-4, which is homologous with CD28, becomes expressed on T-cell activation, where it then competes with CD28 for binding to B7 ligands on the surface of APCs. The B7 ligands bind with much greater affinity to CTLA-4 than to CD28, preventing delivery of the costimulatory signal. This built-in limit prevents T-cell activation from spiraling out of control. The CTLA-4 is not expressed constitutively, and its expression is upregulated on T-cell activation. Eventually, however, CD28 is downregulated.

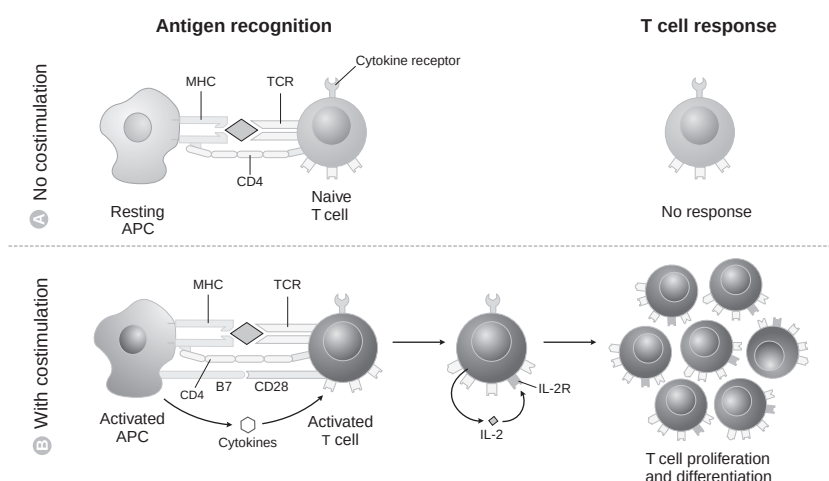


FIGURE 31.36 Costimulation in the T-cell activation pathway.

Because the formation of a B7-CD28 complex between the APC and the T cell results in T-cell proliferation and the release of inflammatory cytokines, whereas the B7-CTLA-4 interaction inhibits the T-cell responses, the design of a pharmacologic agent that prevents costimulation would preferentially inhibit only reactive T cells and be effective in treatment of rheumatoid arthritis—thus, the discovery of abatacept.

Abatacept (Orencia) Abatacept, the first in a new class of immunosuppressant agents, known as costimulation modulators, acts by downregulating T-cell activation for the treatment of rheumatoid arthritis (128). Abatacept is a novel chimeric CTLA-4-IgG1 fused protein created from the fusion of the extracellular domain of the mouse CTLA-4 with the modified heavy-chain constant region of human IgG1.

Abatacept, therefore, acts like an antibody that binds with great affinity to B7 ligands, preventing these ligands from interacting with CD28 on activated T cells. In patients with rheumatoid arthritis, blocking this response by abatacept prevents the generation of positive costimulation signals and stimulation of T-cell activation, suppressing the proliferation of reactive T-cells and the release of more cytokines that destroy tissue, causing the symptoms and signs of arthritis. The extracellular CTLA-4 portion of abatacept is responsible for the affinity of B7. Thus, abatacept slows the damage to bones and cartilage and relieves the symptoms and signs of arthritis.

People with moderate to severe active rheumatoid arthritis who have not been helped by TNF blockers might consider abatacept, which is administered IV monthly. Side effects can include headache, nausea, and mild infections, such as upper respiratory tract infections. Serious infections, such as pneumonia, can occur. There is some concern that blocking of the suppressive signal from B7 to CTLA-4 can have a negative effect on regulatory T cells and, thus, eventually, promote autoimmunity.

Herbs

At least a dozen different herbs are used to ease the symptoms of rheumatoid arthritis; most are considered to be anti-inflammatories. Herbs that have been tried include powdered ginger, borage seed oil, or devil's claw to reduce pain and swelling. Stinging nettles or turmeric can also lessen pain, stiffness, and inflammation. Because these herbs can interact with each other or with prescription comedications, lack of careful studies means that little is known about long-term effects and drug interactions.

Ayurvedic medicine also uses herbal compounds both internally and externally for symptom relief. Topical curcumin can relieve the inflammation of rheumatoid arthritis. When taken in capsule form, it can reduce morning stiffness and boost endurance. A combination of *Withania somnifera*, *Boswellia serrata*, and *Cucurma longa* also caused a significant drop in pain and disability for study participants with osteoarthritis.

Two herbs that have been used for centuries to treat headaches, fever, sore muscles, and rheumatism are white willow bark and meadowsweet, commonly described as “Nature’s aspirin.” White willow bark (*Salix alba*) contains salicin, a glycoside of salicylic acid. Once in the stomach, the salicin hydrolyzes into salicylic acid, which is the active principle for reducing pain and fever. White willow bark has been mentioned in ancient Egyptian, Assyrian, Greek, and Chinese manuscripts, and it was used to treat pain and fever by the ancient physicians Galen, Hippocrates, and Dioscorides. Native Americans used it for headaches, fever, sore muscles, rheumatism, and chills. In the mid-1700s, it was used to treat malaria. Salicin was isolated and identified in the early 1830s, but it was not conclusively shown to reduce the aches and soreness of rheumatism until 1874. White willow bark is recommended for headaches, backache, nerve pain, toothache, and injuries.

Meadowsweet (*Filipendula ulmaria*) is a common wild plant in Britain, Europe, and North America that also contains salicin, but it is not as potent as willow bark, which has a higher salicin content. Its primary medicinal actions are antirheumatic, anti-inflammatory, carminative, antacid, antiemetic, astringent, and diuretic. The flower buds of meadowsweet are the source for salicin and methyl salicylate. Ingestion of the flower buds in a tea results in the breakdown of salicin to salicylic acid. Nicholas Culpeper, a 17th-century English pharmacist, mentioned the use of meadowsweet flower buds to help break fevers and promote sweating during a cold or flu.

A review of the role of traditional Chinese medicine in treating rheumatoid arthritis is available (129).

DRUGS USED TO TREAT GOUT

Pathophysiology

Gout, an ancient and common form inflammatory arthritis, is characterized by elevated levels of uric acid (as urate ion) in the plasma and urine and can take two forms, acute and chronic. It is the most common inflammatory arthritis in men with the incidence occurring in black males at almost twice that of white males. Risk factors include obesity, hypertension, ingestion of alcohol, the use of diuretics, and diets rich in meats and seafood. Gout is associated with an increased risk of kidney stones (130).

Acute gouty arthritis results from the accumulation of needle-like crystals of monosodium urate monohydrate within the joints, synovial fluid, and periarticular tissue and usually appears without warning. Initiating factors can be minor trauma, fatigue, emotional stress, infection, overindulgence in alcohol or food, or drugs, such as penicillin or insulin. Chronic gout symptoms develop as permanent erosive joint deformity appears. The increase in extracellular urate can result from increased uric acid biosynthesis, decreased urinary excretion of uric acid, or perhaps, a combination of both. The formation of uric acid from adenine and guanine is illustrated in Figure 31.37. Uric acid is formed by the oxidation of xanthine by the

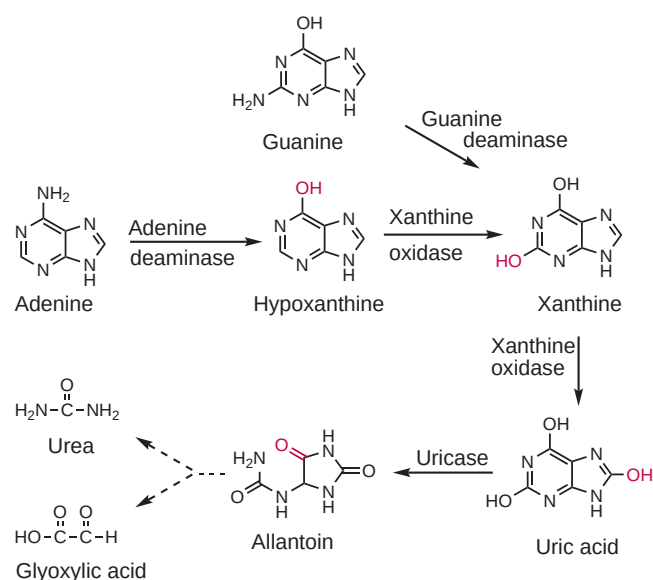


FIGURE 31.37 Formation of uric acid, urea, and glyoxylic acid from purines.

enzyme xanthine oxidase. Xanthine is a metabolic product of adenine (via hypoxanthine) and guanine formed by the enzymes adenine deaminase and guanine deaminase, respectively. Thus, uric acid is the excretory product of purine metabolism in humans as well as the scavenging of potential harmful oxygen free radicals in the body. In other mammals, uric acid is hydrolyzed to allantoin by the enzyme uricase, which is then subsequently hydrolyzed by allantoinase to allantoinic acid. Hydrolysis of allantoinic acid by allantoinase yields the final products, urea and glyoxylic acid.

Normal pool levels of uric acid are approximately 1,000 to 1,200 mg in males and half that in females. In patients suffering from gout, these levels can be as high as two or three times the normal levels. Uric acid is a weak acid with two pK_a values (5.7 and 10.3), with very low water solubility (~ 6 mg/100 mL). At physiologic pH, it exists primarily as the monosodium salt, which is approximately 50 times more soluble in aqueous media than the free acid. Blood levels of urate are maintained by a careful balance between its formation and excretion. The kidney has a dominant role in urate elimination, excreting about 70% of the daily urate production. The excretion of urate has been implicated in the development of hyperuricemia that leads to gout. In humans, the excretion of urate requires the urate anion transporter (URAT1) located in renal proximal tubule cells, which has a central role in urate homeostasis. The URAT1 is targeted by uricosuric and antiuricosuric agents that affect urate excretion.

When levels of uric acid in the body increase, either as a result of decreased excretion or increased formation, the solubility limits of sodium urate are exceeded, and precipitation of the salt from the resulting supersaturated solution causes deposits of urate crystals to form. It is the formation of these urate crystals in joints and connective tissue that initiate attacks of gouty arthritis.

The control of gout has been approached from the following therapeutic strategies: 1) control of acute attacks by drugs that reduce inflammation caused by the deposition of urate crystals (these drugs can possess only an anti-inflammatory component, such as colchicine, or both anti-inflammatory and analgesic actions, such as indomethacin, phenylbutazone, and naproxen), 2) increasing the rate of uric acid excretion (by definition, these drugs are termed “uricosuric drugs” and include probenecid and sulfinpyrazone), and 3) inhibiting the biosynthesis of uric acid by inhibiting the enzyme xanthine oxidase by drugs such as allopurinol.

The structures of the drugs used to treat gout are shown in Figure 31.38.

Treatment of Acute Gout

The management of gout has been approached with the following therapeutic strategies: 1) control of acute attacks with drugs that reduce inflammation caused by the deposition of urate crystals, and 2) control of chronic gout by increasing the rate of uric acid excretion (“uricosuric drugs”) and inhibiting the biosynthesis of uric acid by inhibiting the enzyme xanthine oxidase. Treatment of acute gout includes NSAIDs, such as indomethacin, colchicines, and glucocorticoids. The choice of an NSAID is usually based on the side effect profile.

Colchicine

Colchicine is a pale-yellow powder that is obtained from various species of *Colchicum*, primarily *Colchicum autumnale* L. Its total chemical synthesis has been achieved, but the primary source of colchicine currently remains alcohol extraction

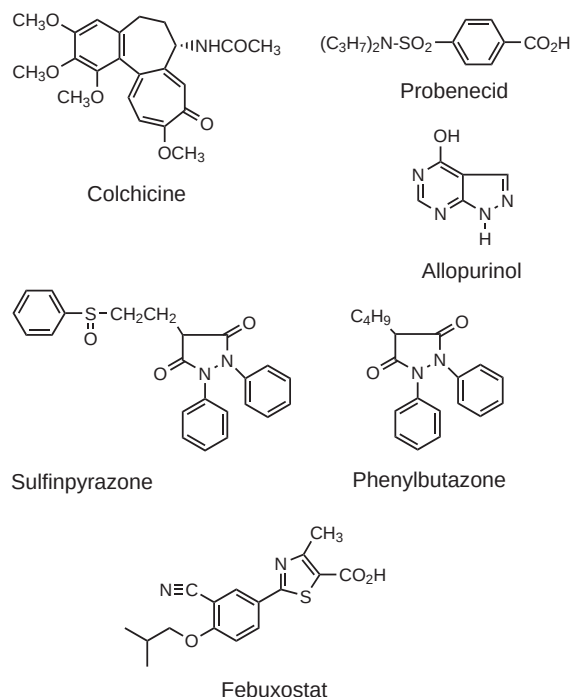


FIGURE 31.38 Structures of agents used to control gout.

of the alkaloid from the corm and seed of *C. autumnale* L. It darkens on exposure to light and possesses moderate water solubility. Colchicine has a pK_a of 12.4. Its use in the treatment of gout dates back to the sixth century AD. Unlike the drugs that will be discussed next, colchicine does not alter serum levels of uric acid. It does, however, appear to retard the inflammation process initiated by the deposition of urate crystals. Acting on polymorphonuclear leukocytes and diminishing phagocytosis, it inhibits the production of lactic acid, causing an increase in the pH of synovial tissue and, thus, a decrease in urate deposition, because uric acid is more soluble at the higher pH. Additionally, colchicine inhibits the release of lysosomal enzymes during phagocytosis that also contributes to the reduction of inflammation. Because colchicine does not lower serum urate levels, it has been found to be beneficial to combine colchicine with a uricosuric agent, particularly probenecid. It is a potent drug, being effective at doses of approximately 1 mg, but doses as small as 7 mg have caused fatalities.

The FDA approved the first single-ingredient oral colchicine formulation, Colcrys, for the treatment of acute gout flares in 2010. The drug is also approved for the treatment of familial Mediterranean fever.

ABSORPTION AND METABOLISM Colchicine is absorbed on oral administration, with peak plasma levels being attained within 0.5 to 2 hours after dosing. Plasma protein binding is only 31%. It concentrates primarily in the intestinal tract, liver, kidney, and spleen and is excreted primarily in the feces, with only 20% of an oral dose being excreted in the urine. It is retained in the body for considerable periods of time, being detected in the urine and leukocytes for 9 to 10 days following a single dose. Metabolism occurs primarily in the liver, with the major metabolite being the amine resulting from amide hydrolysis.

SIDE EFFECTS Colchicine can produce bone marrow depression, with long-term therapy resulting in thrombocytopenia or aplastic anemia. At maximum dose levels, GI disturbances (e.g., nausea, diarrhea, and abdominal pain) can occur. Acute toxicity is characterized by GI distress, including severe diarrhea resulting in excessive fluid loss, respiratory depression, and kidney damage. Treatment normally involves measures that prevent shock as well as morphine and atropine to diminish abdominal pain. A number of drug interactions have been reported. In general, the actions of colchicine are potentiated by alkalinizing substances and are inhibited by acidifying drugs, consistent with its mechanism of action of increasing the pH of synovial fluid. Responses to CNS depressants and to sympathomimetic drugs appear to be enhanced. Clinical tests can be affected; most notably, elevated alkaline phosphatase and serum glutamate oxaloacetate transaminase values and decreased thrombocyte values can be obtained. Because colchicine is a P-gp and CYP3A4 substrate, life-threatening and fatal drug interactions have been reported as a result of significant increases in colchicine plasma levels in patients treated with P-gp and strong CYP3A4 inhibitors.

DOSING Colchicine is indicated for the treatment of acute attacks of gout and is very effective. The usual dose is 1.0 to 1.2 mg, followed by 0.5 to 1.2 mg every 1 to 2 hours until either pain relief is observed or symptoms of GI distress are observed. When a rapid response is required, or if GI reactions warrant discontinuance of oral administration, IV administration (usually 2 mg initially) can be indicated. It is available as 0.5- or 0.6-mg tablets and as an injectable solution of 1 mg in 2-mL ampoules. It often is given in combination with probenecid, and combination products of the two are available in tablets containing 500 mg of probenecid and 0.5 mg of colchicine.

Treatment of Chronic Gout

Drugs That Increase Uric Acid Secretion

PROBENECID Probenecid is insoluble in water and acidic solutions but is soluble in alkaline solutions buffered to pH 7.4. Probenecid initially was synthesized as a result of studies in the 1940s on sulfonamides that indicated the sulfonamides decreased the renal clearance of penicillin, extending the half-life of penicillin as supplies diminished. Probenecid thus was initially used—and is still indicated—for that purpose. Probenecid promotes the excretion of uric acid by inhibiting the urate anion exchange transporter (URAT1), decreasing the reabsorption of uric acid in the proximal tubules. The overall effect is to decrease plasma uric acid concentrations, thereby decreasing the rate and extent of urate crystal deposition in joints and synovial fluids. Within the series of *N*-dialkylsulfamyl benzoates from which probenecid is derived, renal clearance of these compounds is decreased as the length of the *N*-alkyl substituents is increased. Uricosuric activity increases with increasing size of the alkyl group in the series methyl, ethyl, and propyl.

Probenecid is essentially completely absorbed from the GI tract on oral administration, with peak plasma levels observed within 2 to 4 hours. Like most acidic compounds, probenecid ($pK_a = 3.4$) is extensively plasma protein bound (93% to 99%). The primary route of elimination of probenecid and its metabolites is the urine. It is extensively metabolized in humans, with only 5% to 10% being excreted as unchanged drug. The major metabolites detected result from glucuronide conjugation of the carboxylic acid, ω -oxidation of the *n*-propyl side chain and subsequent oxidation of the resulting alcohol to the carboxylic acid derivative, ω -oxidation of the *n*-propyl group, and *N*-dealkylation (Fig. 31.39). Those metabolites possessing a free carboxylic acid functional group usually possess some uricosuric activity. The glycine conjugates possess high affinity to the organic anion transport system thus preventing the reabsorption of uric acid. Probenecid appears to be well tolerated, with few adverse reactions. The major side effect is GI distress (e.g., nausea, vomiting, and anorexia), but these occur in only 2% of patients at low doses. Other effects include headache, dizziness, urinary frequency, hypersensitivity reactions, sore gums, and anemia. Overdosages do not appear to present major

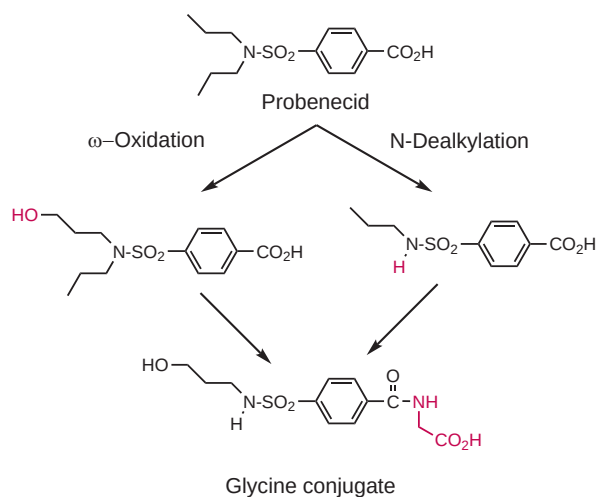


FIGURE 31.39 Metabolism of probenecid.

difficulties; a case of a 49-year-old man who recovered from the ingestion of 47 g in a suicide attempt has been reported. Should overdose occur, treatment consists of emesis or gastric lavage, short-acting barbiturates (if CNS excitation occurs), and epinephrine (for anaphylactic reactions). A number of drug interactions have been reported. Despite the high degree of plasma protein binding, displacement interactions with other drugs bound to plasma proteins do not appear to occur to any significant extent. Salicylates counteract the uricosuric effects of probenecid. Because probenecid inhibits their renal excretion, increased plasma levels of the following drugs can be observed: aminosalicic acid, methotrexate, sulfonamides, dapsone, sulfonyleureas, naproxen, indomethacin, rifampin, and sulfinpyrazone. (The effects on penicillin plasma levels were discussed previously.)

Probenecid is indicated for the treatment of hyperuricemia associated with gout and gouty arthritis and for the elevation and prolongation of plasma levels of penicillins and cephalosporins. In gout, treatment should not begin until an acute attack has subsided. It is not recommended in individuals with known uric acid kidney stones or blood dyscrasias or for children younger than 2 years of age.

SULFINPYRAZONE Sulfinpyrazone is soluble in alkaline solutions. Its structure and synthesis are similar to that of phenylbutazone (131) (Fig. 31.38). It produces its uricosuric effect in a manner similar to that of probenecid. A dose of 35 mg produces a uricosuric effect equivalent to that produced by 100 mg of probenecid, whereas 400 mg/d of sulfinpyrazone produces an effect comparable to that obtained with doses of 1.5 to 2 g of probenecid. It also possesses, not surprisingly, some of the properties of phenylbutazone. It is an inhibitor of human platelet prostaglandin synthesis at the cyclooxygenase step, resulting in a decrease in platelet release and a reduction in platelet aggregation. This antiplatelet effect suggests a role for sulfinpyrazone in reducing the incidence of sudden death, which can occur in the first year following a

myocardial infarction; however, it lacks the analgesic and anti-inflammatory effects of phenylbutazone.

Sulfinpyrazone is a strong acid (enolic OH $pK_a = 2.8$), a factor that is important in the production of the uricosuric effect, because within a series of pyrazolidinedione derivatives, the stronger the acid, the more potent the uricosuric effect. Polar substitution on the side chain also influences uricosuric activity, as discussed previously with regard to the pyrazolidinediones.

Oral administration results in rapid and essentially complete absorption, with peak plasma levels being attained within 1 to 2 hours of administration. It is highly bound (98% to 99%) to plasma proteins, and it is excreted in the urine primarily (50%) as unchanged drug.

The metabolites produced result from sulfoxide reduction, sulfur and aromatic oxidation, and C-glucuronidation of the heterocyclic ring in a manner similar to that for phenylbutazone (Fig. 31.40).

The metabolite resulting from *p*-hydroxylation of the aromatic ring possesses uricosuric effects in humans. The sulfide metabolite, a major metabolic product, can contribute to the antiplatelet effects of sulfinpyrazone but

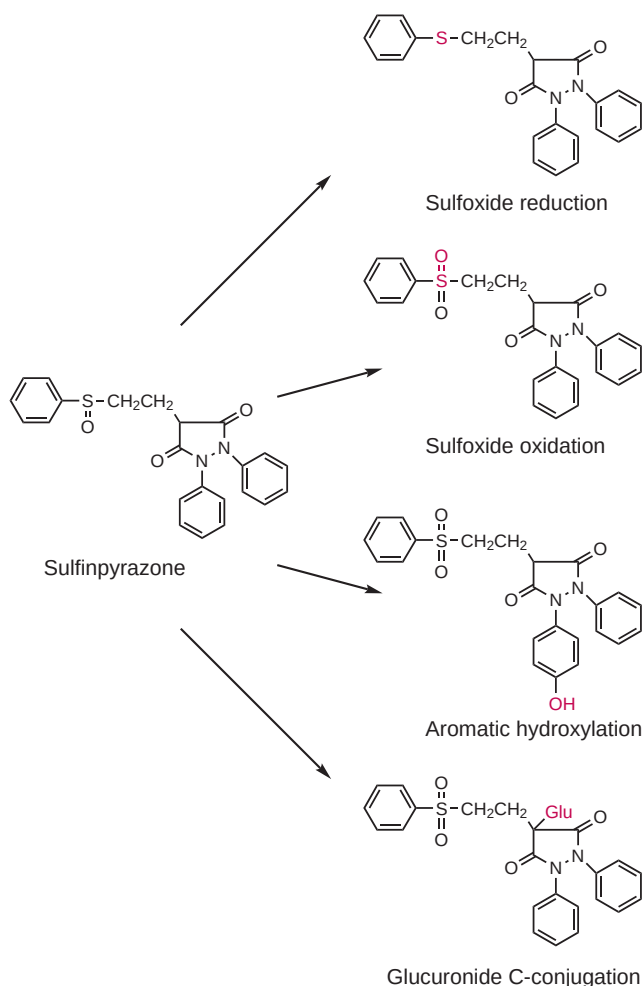


FIGURE 31.40 Metabolism of sulfinpyrazone.

not to the uricosuric effects. The most frequent adverse reactions are GI disturbances; however, the incidence is relatively low. It has been suggested that sulfinpyrazone is a much weaker inhibitor of prostaglandin synthesis in bovine stomach microsomes than either aspirin or indomethacin, a factor that can account for its gastric tolerance. Much rarer have been reports of blood dyscrasias and rash. Overdosage produces symptoms that are primarily GI in nature (e.g., nausea, diarrhea, and vomiting) but can also involve impaired respiration and convulsions. Treatment consists of emesis or gastric lavage and supportive care. Like probenecid, its uricosuric effects are antagonized by salicylates, and probenecid markedly inhibits the renal tubular secretion of sulfinpyrazone. It potentiates the actions of other drugs that are highly plasma protein bound, such as coumarin-type oral anticoagulants, antibacterial sulfonamides, and hypoglycemic sulfonylureas.

Sulfinpyrazone is indicated for the treatment of chronic and intermittent gouty arthritis.

Drugs That Decrease Uric Acid Formation

ALLOPURINOL

Mechanism of Action The biosynthesis of uric acid from the immediate purine precursor xanthine that results from adenine, via the intermediate hypoxanthine, or from guanine is illustrated in Figure 31.37. The enzyme xanthine oxidase (a molybdenum hydroxylase enzyme) is involved in two steps, the conversion of hypoxanthine to xanthine and, the final step, the conversion of xanthine to uric acid. Allopurinol originally was designed as an antineoplastic antimetabolite to antagonize the actions of key purines inasmuch as it differs from normal purines only by the inversion of the nitrogen and carbon atoms at the 7- and 8-positions of the purine ring system but was found to have little or no effect on experimental tissues. It was subsequently found that allopurinol serves as a substrate for xanthine oxidase (15 to 20 times the affinity of xanthine) and reversibly inhibits that enzyme. Normally, uric acid is a major metabolic end product in humans. When allopurinol is administered, however, xanthine and hypoxanthine are elevated in the urine, and uric acid levels decrease. When the synthesis of uric acid is inhibited, plasma urate levels decrease, supersaturated solutions of urate are no longer present, and urate crystal deposits dissolve, eliminating the primary cause of gout. The increased plasma levels of hypoxanthine and xanthine pose no real problem, because they are more soluble than uric acid and are readily excreted.

Absorption and Metabolism Allopurinol was synthesized in 1956 as part of a study of purine antagonists (132). It is well absorbed on oral administration, with peak plasma concentrations appearing within 1 hour. Decreases of uric acid can be observed within 24 to 48 hours. Excretion of allopurinol and its metabolite occurs primarily in the urine,

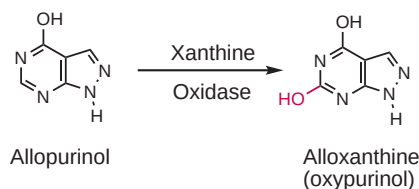
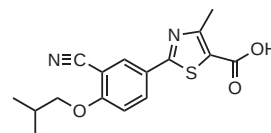


FIGURE 31.41 Metabolism of allopurinol.

with approximately 20% of a dose being excreted in the feces. Allopurinol is rapidly metabolized via oxidation and the formation of numerous ribonucleoside derivatives (Fig. 31.41). The major oxidation metabolite, alloxanthine or oxypurinol, has a much longer half-life (18 to 30 hours vs. 2 to 3 hours) than the parent drug and is an effective, although less potent, inhibitor of xanthine oxidase. The longer plasma half-life of alloxanthine results in an accumulation in the body during chronic administration, thus contributing significantly to the overall therapeutic effects of allopurinol. The major adverse effects are primarily dermatologic in nature (e.g., skin rash and exfoliative lesions). Other effects, such as GI distress (e.g., nausea, vomiting, and diarrhea), hematopoietic effects (e.g., aplastic anemia, bone marrow depression, and transient leukopenia), neurologic disorders (e.g., headache, neuritis, and dizziness), and ophthalmologic effects (e.g., cataracts) are less commonly encountered. Allopurinol can also initiate attacks of acute gouty arthritis during the early stages of therapy and can require the concomitant administration of colchicine. Drug interactions include those drugs that normally are also metabolized by xanthine oxidase. For example, the oxidation of 6-mercaptopurine, a useful antineoplastic agent, is inhibiting, permitting a reduction in the therapeutic dose. Allopurinol also has an inhibitory effect on liver microsomal enzymes, thus prolonging the half-lives of drugs, such as oral anticoagulants, that normally are metabolized and inactivated by these enzymes, although this effect is quite variable. The incidence of ampicillin-related skin rashes increases with the concurrent administration of allopurinol.

Allopurinol is indicated for the treatment of primary and secondary gout, for malignancies such as leukemia and lymphoma, and for the treatment of patients with recurrent calcium oxalate calculi.

FEBUXOSTAT



Febuxostat (Uloric)

Febuxostat, the first new drug to be approved for the treatment of gout in over 40 years, is the first nonpurine, selective xanthine oxidase inhibitor marketed in the

United States (Uloric; Adenuric in European Union). Unlike allopurinol, febuxostat noncompetitively blocks the channel leading to the active site of both the oxidized and reduced forms of xanthine oxidase, thereby inhibiting the synthesis of uric acid (133,134). It was synthesized in seven steps from 4-hydroxy-3-nitrobenzaldehyde (135). Structurally, febuxostat does not contain or mimic a purine or pyrimidine; thus, it does not inhibit enzymes involved in pyrimidine or purine metabolism, resulting in fewer adverse events relative to allopurinol.

Absorption and Metabolism Febuxostat is well absorbed on oral administration, with peak plasma concentrations appearing within 1 to 1.5 hours and a half-life of 5 to 8 hours. Febuxostat is extensively metabolized by conjugation by glucuronosyltransferase enzymes including UGT1A1, UGT1A3, UGT1A9, and UGT2B7, as well as by oxidation by CYP1A1, CYP1A2, CYP2C8, and CYP2C9 and non-CYP enzymes to produce metabolites resulting from ω -oxidation, ω -1 oxidation, *O*-dealkylation, and alcohol oxidation (136,137). Excretion of febuxostat, as various metabolites and their conjugates, occurs primarily in the urine (49%) and feces (45%). From an administered dose, only 3% and 12% are recovered as unchanged drug in urine and feces, respectively. The metabolism of febuxostat is shown in Figure 31.42.

Adverse Effects and Dosing The major adverse effects are primarily liver function abnormalities. Initial dosing with febuxostat can initiate gout flares and can require the concomitant administration of an NSAID or colchicine. Drug interactions include those drugs

that normally are also metabolized by xanthine oxidase, such as azathioprine, mercaptopurine, and theophylline.

Febuxostat is indicated for the chronic management of hyperuricemia in patients with gout and for patients who cannot tolerate allopurinol. Dosing is recommended at 40 or 80 mg once daily. A review of the clinical properties of febuxostat as a new treatment option has recently appeared (138,139).

Drugs Acting by Enhancing the Degradation of Uric Acid

PEGLOTICASE Pegloticase was recently approved in the United States for the treatment of chronic gout in patients for whom traditional small-molecule therapy does not prove effective. Unlike allopurinol or febuxostat, pegloticase does not block the formation of uric acid but enhances the excretion of uric acid by converting it into a more excretable metabolite. The active agent is a recombinant porcine-like uricase (uric oxidase) that catalyzes the oxidation of uric acid to the inert allantoin. Allantoin is more soluble than uric acid and more readily excreted (Fig. 31.37). Pegloticase is a uric acid-specific enzyme that consists of recombinant modified mammalian urate oxidase (uricase) produced by a genetically modified strain of *E. coli* and polyethylene-glycolated (PEGylated). Uricase is covalently conjugated to monomethoxypoly(ethylene glycol) (mPEG) (10 kd molecular weight). The cDNA coding for uricase was derived from mammalian sequences. Each uricase subunit weighs approximately 34 kd (34,000 MW). The average molecular weight of pegloticase (tetrameric enzyme conjugated to mPEG) is approximately 540 kd.

Absorption As would be expected from the administration of an active enzyme, the duration of suppression of plasma uric acid was positively associated with the dose of pegloticase. As for clearance and volume of distribution, the significant covariates were antipegloticase antibodies and body surface area.

Dosing and Adverse Effects The recommended dose of pegloticase is 8 mg IV every 2 weeks. See prescribing information for specific instructions because the drug must not be diluted until it will be administered slowly, over more than 120 minutes. Prior to beginning the infusion, premedication of the patient with antihistamines and corticosteroids is recommended to prevent anaphylaxis and infusion reactions, which are thought to result from cytokine release. During the first 3 months of treatment, gout flares are likely to occur, and thus, NSAIDs and/or colchicine should be started at least 1 week prior to pegloticase therapy.

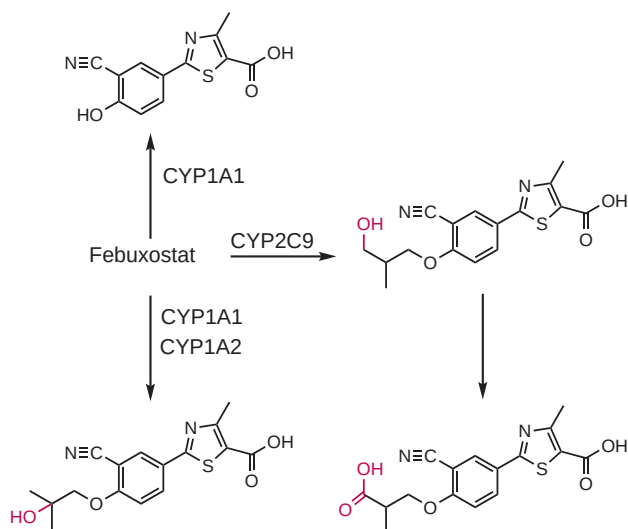


FIGURE 31.42 Metabolism of febuxostat.

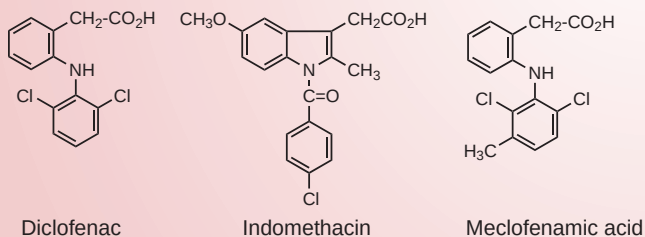
SCENARIO: OUTCOME AND ANALYSIS

Outcome

Jill T. Johnson, PharmD, BCPS

For pain relief, there are no significant short-term (<6 months) differences among oral NSAIDs. For serious harms such as cardiovascular events, celecoxib does not seem to be associated with higher risk compared with nonselective NSAIDs. All nonselective NSAIDs except naproxen are associated with similar increased risk for serious cardiovascular events. Naproxen has a lower cardiovascular risk than nonselective NSAIDs. Compared to nonselective NSAIDs, celecoxib is gastroprotective in the short term (6 months) but not in the long term (at 12 months) in causing clinically significant upper GI events. Nonselective NSAIDs are similar to each other regarding their increased risks of serious GI events, but the partially selective NSAID nabumetone is gastroprotective compared to nonselective NSAIDs.

EB could take nabumetone or celecoxib, which would afford her short-term decreased risk for a stomach ulcer caused by NSAID drug therapy. Because she has a chronic illness necessitating NSAID use, she will require long-term therapy. After 6 months of use, celecoxib failed to be superior to nonselective NSAIDs regarding clinically significant upper GI events; so, in the long term, celecoxib may not provide an advantage over her original therapy. The other alternative is to select an NSAID that relieves her pain and to give her a proton pump inhibitor or a prostaglandin analogue such as misoprostol to protect her stomach from ulcer formation.



Chemical Analysis

Victoria Roche and S. William Zito

With the exception of aspirin and the COX-2 selective NSAIDs (the "coxibs"), NSAIDs have a similar pharmacophoric nucleus because they all bind to the same active site on the COX enzyme. To bind to this site, the NSAID must anchor to a cationic Arg residue (Arg¹²⁰) and, therefore, must contain an acidic functional group with a pKa low enough to generate a significant amount of anionic conjugate base at pH 6.8, the pH of inflamed tissue.

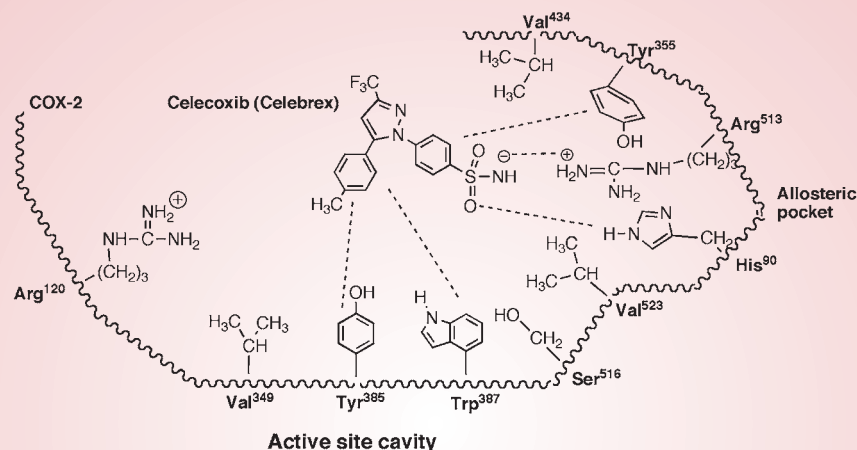
Two conjugated aromatic rings capable of assuming a noncoplanar orientation are ideal for binding to aromatic and aliphatic residues in the active site cavity, although one aromatic system provides therapeutically useful anti-inflammatory activity.

Much of the remaining NSAID SAR relates to increasing potency by enhancing distribution through membranes to reach COX receptors (lipophilic groups) or promoting the ideal noncoplanar conformation between aromatic rings (appropriately placed large groups). For example, a chlorine atom placed at the o-position of one aromatic ring would increase potency significantly over a nonsubstituted analog because it would enhance potency via both mechanisms. This substitution pattern is observed in NSAIDs such as diclofenac and meclofenamate. Another chemical strategy for enhancing NSAID activity is to block degradative metabolism, e.g., through the addition of a *p*-substituent (preferably a lipophilic one) to an aromatic ring. Indomethacin is a good example of this strategy.

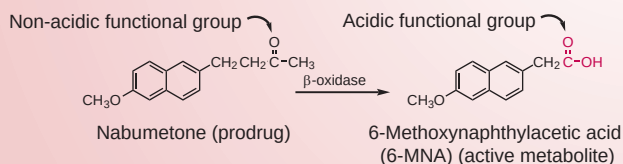
COX-1 is found in the gut, and inhibition of this isoform is responsible for much of the GI distress induced by NSAIDs. When gastric COX-1 is inhibited, the synthesis of the cytoprotective PGE₁ ceases. Coxibs bind selectively to the allosteric site of the COX-2 isoform in inflamed tissue and kidney. The allosteric site of COX-1 is more sterically restricted than COX-2, which denies the coxib drugs access to the binding residues of this isoform. Because they have a lower affinity for gastric COX-1, coxibs claim to have a decreased risk of GI ulceration and hemorrhage, although this holds true mostly in the short term. Celecoxib was the original COX-2 selective NSAID and its anionic sulfonamide group helps it bind securely into the COX-2 allosteric site through an ion-ion interaction with an Arg residue. Many coxibs are diarylheteroaromatic compounds, and the two aromatic rings of these structures enhance affinity for COX-2 through van der Waals bonds with Tyr, His and other aromatic residues in or near the allosteric site. Note the two distinct binding sites on the COX-2 enzyme cartoon below; the active site (which would bind nonselective NSAIDs such as diclofenac, indomethacin, and meclofenamate) and the allosteric site (which binds the coxibs).

Though the degree of inherent COX-2 preference in nonselective (noncoxib) NSAIDs is hard to predict, the lower GI distress of nabumetone can be directly related to its prodrug status. To inhibit COX-1 in the gut, NSAIDs have to bind to the active site Arg¹²⁰ residue, and this requires an anionic drug. Nabumetone has no acidic functional group and can only generate the active anionic form after it has been absorbed from the gut and metabolized via beta-oxidation in the liver. As a nonacidic prodrug, it does not inhibit gastric COX-1, nor can it physically irritate the gastric mucosa as all acidic NSAIDs are capable of doing.

SCENARIO: OUTCOME AND ANALYSIS (Continued)



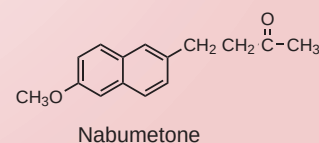
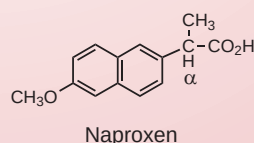
Although it is not as potent as some other NSAIDs because it has only one aromatic moiety, it is often gentler on the stomach. This is an acceptable trade off in mild inflammatory disease.



The original NSAID EB was taking (naproxen) is similar to nabumetone in that it has only one aromatic moiety (a methoxy-substituted naphthylene ring system), but it is significantly different in that it has the carboxylic acid functional group that would permit inhibition of gastric COX-1 and direct gastric mucosal

irritation. The GI distress of naproxen would be anticipated to be greater than its chemical cousin nabumetone. It should be noted that the α -CH₃ group of naproxen is a potency-enhancing feature.

Misoprostol, a PGE₁ analog, protects the gastric mucosa from distress by serving as a replacement for the endogenous gastric PGE₁ inhibited by the use of nonselective NSAIDs. Proton pump inhibitors are cytoprotective in the stomach because they stop all gastric acid secretion by irreversibly inhibiting the adenosine triphosphatase that is responsible for the secretion of acid into the stomach lumen.



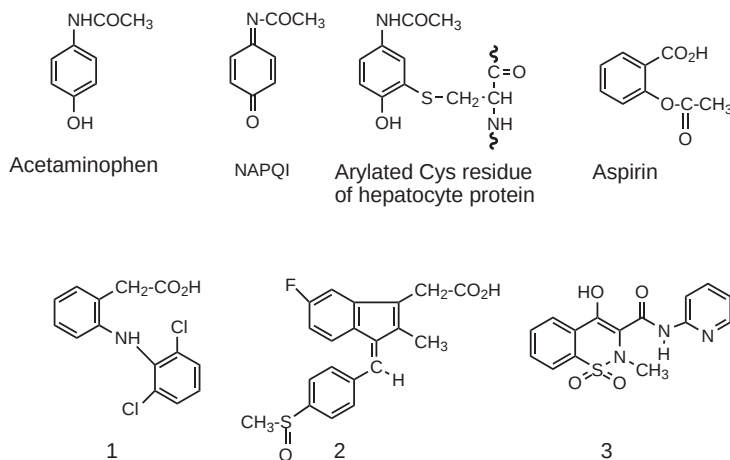
CASE STUDY

Victoria Roche and S. William Zito

This has not been a good week for IG, a 17-year-old figure skater from Minnesota. IG is a rising star in this sport, and has been competing in the 2011 World Championship in Tokyo. While practicing her long program, she took a hard fall on the ice while attempting to execute a quadruple axel and severely strained the adductor (groin) muscle of her right leg. She attributed the mishap, in part, to the fact that she has been feeling so lousy for the past week or so, which she had kept from her coach. Her joints ache, she feels nauseated, and she just does not want to eat much of anything. The team physician sent her to the nearby medical center for a complete workup, and she has been diagnosed with hepatitis A. The assumption is that she acquired this illness when she first arrived in Japan after grabbing a quick bite from a small family-run noodle restaurant a short distance from her downtown hotel.

IG is no stranger to liver dysfunction; she experienced significant hepatotoxicity earlier in her skating career through the chronic overuse of acetaminophen. This common pain reliever can generate a hepatotoxic quinoneimine (known as NAPQI) that can destroy liver cells through the arylation of cysteine residues. IG turned to acetaminophen for pain relief because her GI tract just did not tolerate aspirin. The good news is that her drug-induced hepatic damage has since healed and the physicians expect her to make a full recovery from the hepatitis infection within the next several months.

IG now needs NSAID therapy for her severely strained adductor muscle. Which of the three prescription NSAIDs drawn below would be a logical first choice for this young athlete? She is currently taking no prescription or over-the-counter drugs.



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

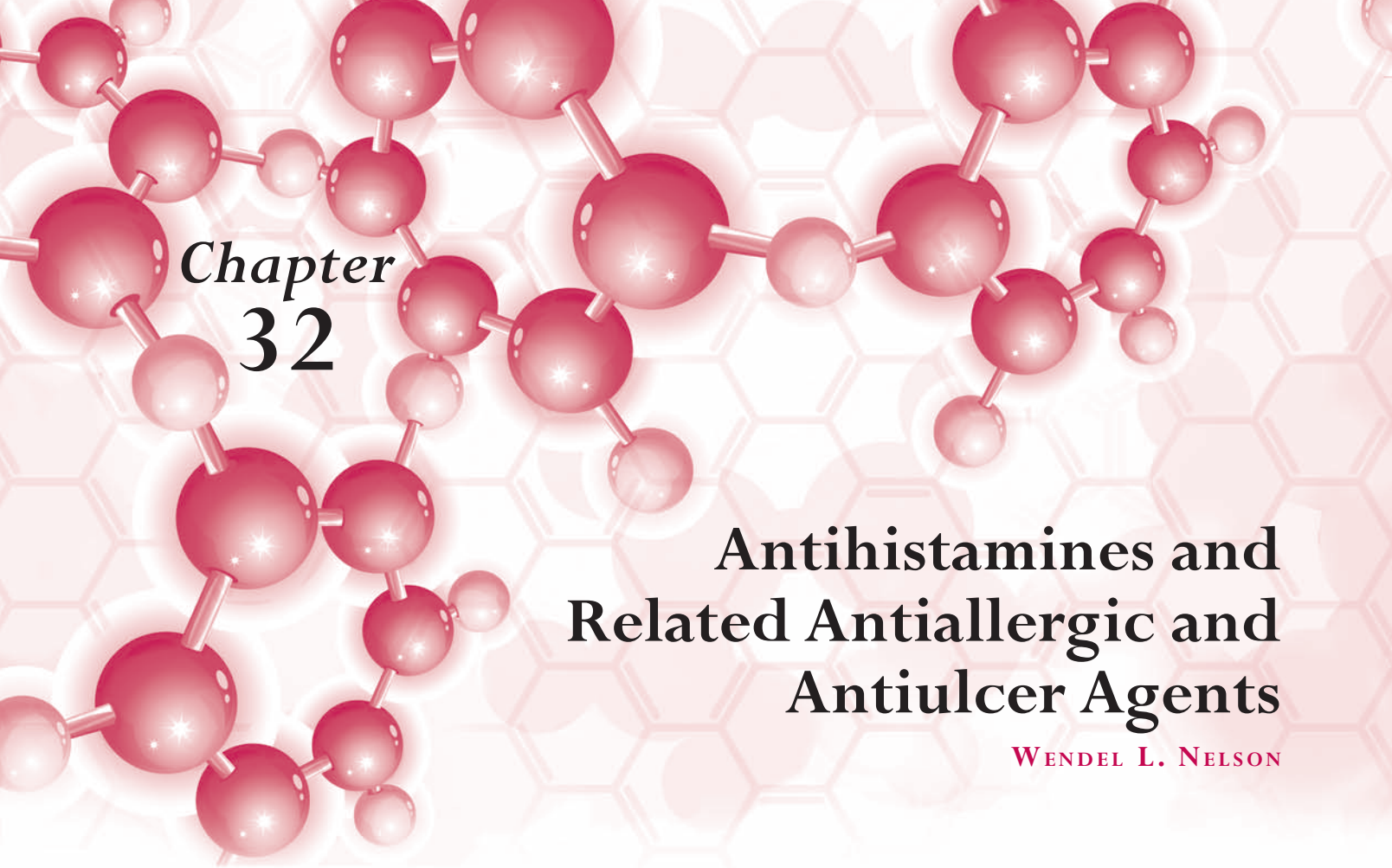
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 32

Antihistamines and Related Antiallergic and Antiulcer Agents

WENDEL L. NELSON

Drugs Covered in This Chapter*

ANTIHISTAMINES

- Acrivastine
- Azelastine
- Bepotastine
- *Bilastine*
- *Carebastine*
- Cetirizine
- Cromolyn
- Desloratadine
- *Ebastine*
- Emedastine
- Epinastine
- Fexofenadine

- Ketotifen
- Levocabastine
- Levocetirizine
- Lodoxamide
- Loratadine
- *Mizolastine*
- Nedocromil
- Olopatadine
- Pemirolast
- *Rupatadine*
- *Terfenadine*

ANTIULCER AGENTS

- Cimetidine

- Dexlansoprazole
- Esomeprazole
- Famotidine
- Lansoprazole
- Metoclopramide
- Misoprostol
- Nizatidine
- Omeprazole
- Pantoprazole
- Rabeprazole
- Ranitidine
- *Revaprazan*
- Sucralfate

Abbreviations

ADO, aldehyde oxidase
ATP, adenosine triphosphate
AUC, area under the curve
cAMP, cyclic adenosine monophosphate
CNS, central nervous system
FDA, U.S. Food and Drug Administration

GERD, gastroesophageal reflux disease
hERG, human ether-a-go-go
Ig, immunoglobulin
IL, interleukin
IP₃, inositol-1,4,5-triphosphate
LTC₄, leukotriene C₄
MAO, monoamine oxidase
NF-κB, nuclear factor-κB

NSAID, nonsteroidal anti-inflammatory drug
OTC, over the counter
PGD₂, prostaglandin D₂
PPI, proton pump inhibitor
T_H1, T-helper type 1
T_H2, T-helper type 2
TNF, tumor necrosis factor

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

SCENARIO

Anne Pace, PharmD

CP, a 70-year-old white man, walked into the pharmacy with a chief complaint of itchy eyes, running nose, and sneezing for the past 2 days. CP states that he experiences these symptoms usually in the spring. His past medical history includes benign prostatic hyperplasia. His current medication is tamsulosin

0.4 mg daily. CP does not have time to visit his physician and wants a recommendation from the pharmacist.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

Histamine (2-[imidazol-4-yl]ethylamine) was synthesized and its effects in model biologic systems were studied before it was found physiologically. Its synthesis occurs in many tissues, including mast cells, parietal cells of the gastric mucosa, and neurons of the central nervous system (CNS) and in the periphery. Early hypotheses about its physiologic function were based on the observed dramatic effects of histamine in guinea pigs. These effects include massive bronchial spasm and effects on smooth muscle and the vasculature resembling anaphylactic shock. Marked species differences in the observed effects occur however, and these dramatic effects are not observed in humans.

Histamine is located in many tissues, and upon release, its effects are principally local ones, as it functions as an autocoid or paracrine (1). Its physiologic function is complex and not completely understood. Histamine is one of the many mediators involved in allergic inflammatory responses, and it has an important role in the regulation of the secretion of gastric acid. These observations have led to development of many important drugs that antagonize its effects and are useful in treatment of allergic inflammatory disorders (H_1 antihistamines) and in the treatment of gastric hypersecretory disorders (H_2 antihistamines).

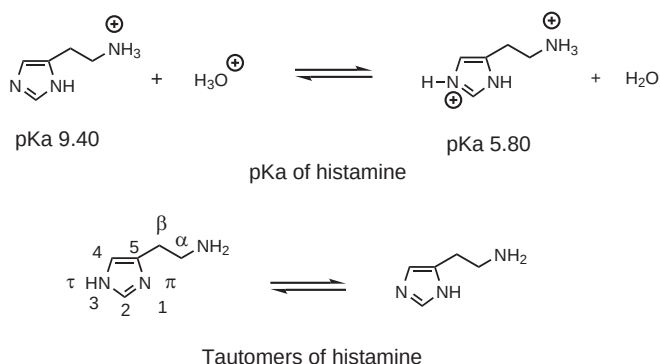
Besides its role in allergic inflammatory processes and gastric acid secretion, a physiologic role at axons in several regions of the CNS has established its role in the regulation of sleeping and waking, in energy and endocrine homeostasis, and in cognition and memory. Histamine modulates the release of neurotransmitters via H_3 auto- and heteroreceptors located at histaminergic and nonhistaminergic neurons centrally and peripherally. A novel H_4 receptor has also been described where histamine facilitates synthesis and release of other proinflammatory mediators and modulates chemotactic responses principally at mast cells and eosinophils.

CHEMISTRY

Histamine has pK_a values of 5.80 (imidazole) and 9.40 (aliphatic primary amine) (2). At physiologic pH, it exists as an equilibrium mixture of tautomeric cations, with the monocation making up more than 96% of the total and the dication about 3%, with only a very small amount of the nonprotonated species. At lower pHs (e.g., the pH

of acidic lipids), a much larger proportion of the dication exists. The two protonated species (monocation and dication) are often considered the biologically active forms. Penetration of membranes by histamine would be expected to occur via the nonprotonated species, and the unprotonated imidazole group would be expected to participate readily in proton transfer processes physiologically. Several aromatic ring congeners of histamine with weakly and very weakly basic heteroaromatic rings, such as 4-chloroimidazole, 1,2,4-triazole, thiazole, and pyridine, exhibit histamine agonist activity (Table 32.1) (2), although they are less potent than histamine. These data suggest that the monocation (protonated aliphatic amine) is sufficient for agonist activity and that protonation of the heterocyclic ring is not an absolute requirement.

In aqueous solutions, the tautomeric equilibrium of the imidazole ring apparently favors the N^H -H tautomer by about 4:1. The free base also prefers the N^H -H tautomer. However, in the crystal of the monohydrochloride salt of histamine, where intermolecular crystal packing forces are important, the N^H -H tautomer is preferred. Changes in tautomeric composition of analogs occur with changes in the 4-substituent (e.g., Me vs. Cl), where the proportion of N^H -H tautomer is decreased in the chlorine-substituted congener to 12% versus 70% for 4-methylhistamine, and decreased agonist potency is observed. An interpretation of these results is that tautomeric composition might be important in the agonist–receptor interaction (3).



Results of conformational studies performed on histamine and its congeners indicate both trans and gauche conformations exist in solution (Fig. 32.1) (3). However,

TABLE 32.1 Histamine-Related Agonists

	Relative H ₁ activity vs. histamine	Relative H ₂ activity vs. histamine	Relative H ₃ activity vs. histamine
	100	100	100
	1.7	12	ND ^b
	0.23	39	<0.008
	0.49	1.0	1550
	12.7	13.7	ND
	26	~0.3	<0.008
	0.01	~0.1	ND
	5.6	2.5	<0.06
	Inactive	Inactive	ND
	~0	~0.4	ND

ND = not determined.

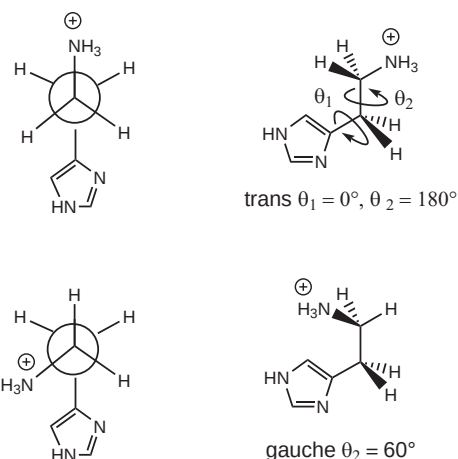
^aActivity expressed relative to histamine = 100, determined in vitro on guinea pig ileum (H₁), guinea pig atrium (H₂) and rat cerebral cortex (H₃)(3).

FIGURE 32.1 Conformers of histamine.

the trans conformation of 4-methylhistamine, which is a selective H₂ agonist, cannot readily adopt the fully extended trans conformation because of interaction of the 4-methyl group with the aliphatic two-carbon chain. Since α - and β -methylhistamine exist predominantly as gauche conformers and both are very weak H₁ and H₂ agonists, it has been suggested that trans conformation of histamine is preferred at both H₁ and H₂ receptors. A gauche conformation has been suggested for histamine at the H₃ receptor because α -methylhistamine and some other more conformationally restricted analogs are potent H₃ agonists.

Addition of other alkyl substituents onto the histamine molecule generally produces compounds with decreased potency at H₁ and H₂ receptors. 2-Methylhistamine is a selective H₁ agonist (vs. 4-methylhistamine, a selective H₂ agonist), but imidazole *N*-substitution (*N*¹ or *N*³) with methyl groups results in nearly inactive agents. Similarly, aliphatic amine nitrogen substitution results in decreasing activity (NH₂ > NHMe > NMe₂ > N⁺Me₃ [quaternary ammonium salt]) at both H₁ and H₂ receptors (3,4).

CLINICAL SIGNIFICANCE



This chapter describes a wide array of medications that have a myriad of different uses. Although many of the compounds presented are structurally similar, their subtle differences in chemical structure produce vastly different pharmacologic effects, potency, and side effect profiles. As a result, the compounds presented are used to treat a variety of conditions, including allergies, motion sickness, and gastroesophageal reflux disease. Studying and understanding the different structures of these compounds allows the reader to distinguish the differences in their pharmacologic effects, potency, and side effects of each of the medications. Many of these drugs

are available over the counter and present an opportunity for pharmacists to be an integral part of the patient's health care. Therefore, only when armed with patient-specific information and a thorough understanding of the drugs, including their chemical structure, can the pharmacist identify the proper medicine and make the best recommendation for their patients.

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PHYSIOLOGIC CHARACTERISTICS OF HISTAMINE

Synthesis and Metabolism of Histamine

Histamine is synthesized in the Golgi apparatus of mast cells and basophils by enzymatic decarboxylation of histidine. This conversion is catalyzed by L-histidine decarboxylase, with pyridoxal phosphate serving as a cofactor for this process (5). The reaction mechanism for this decarboxylation probably involves the formation of an imine intermediate followed by the loss of carbon dioxide, a mechanism demonstrated to occur for decarboxylation of many α -amino acids (Fig. 32.2). Pyridoxal phosphate provides an important catalytic function, and in the final step, the product is released by hydrolysis of the enzyme-bound Schiff base of histamine. Mechanism-based inhibitors of this process, such as α -fluoromethylhistidine, decrease the rate of synthesis of histamine, and thus deplete cells of histamine. As such, α -fluoromethylhistidine is an important pharmacologic tool (4). This approach has not been successfully developed into agents for the treatment of allergic inflammatory disorders, peptic ulcer, or motion of sickness.

Once released, histamine is rapidly metabolized in vivo (based on products from radiolabeled histamine administered intradermally) to nearly inactive metabolites by two

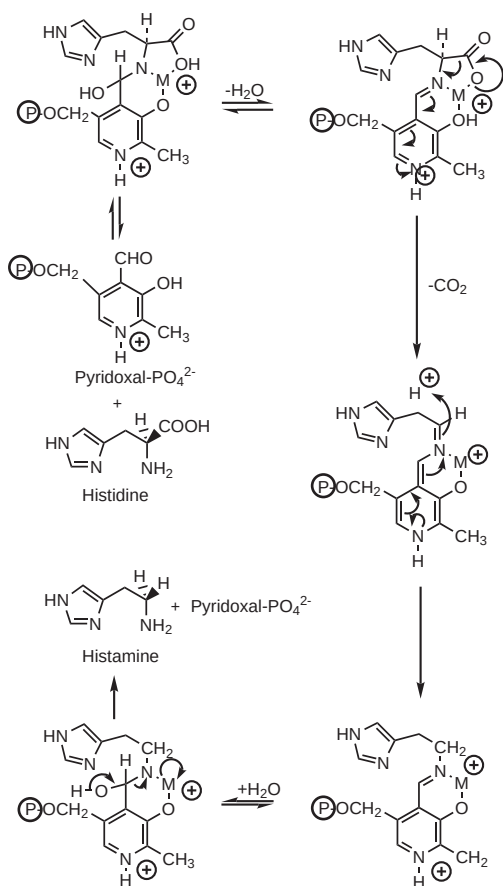


FIGURE 32.2 Formation of histamine by decarboxylation of histidine.

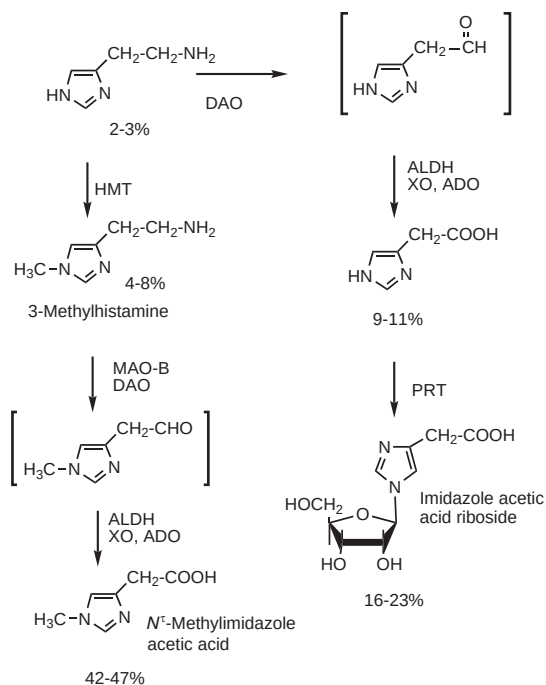


FIGURE 32.3 Major metabolic pathways of histamine, from intradermal histamine as measured in the urine in 12 hours in human men (6). ADO, aldehyde oxidase; ALDH, aldehyde dehydrogenase; DAO, diamine oxidase; HMT, histamine *N*-methyltransferase; MAO-B, monoamine oxidase type B; PRT, phosphoribosyl transferase; XO, xanthine oxidase.

major pathways: *N*-methylation and oxidation (Fig. 32.3). Methylation (*S*-adenosylmethionine), which is catalyzed by the intracellular enzyme *N*-methyltransferase, yields an inactive metabolite. A portion of the *N*-methylated metabolite is oxidized sequentially via monoamine oxidase (MAO) and then via aldehyde oxidase (ADO) to the corresponding *N*-methylimidazole acetic acid. Histamine is also oxidized to imidazole acetic acid by diamine oxidase (histaminase). A small amount of this acid intermediate is converted to the corresponding ribotide, an unusual metabolite (6).

Storage and Release of Histamine

In mast cells, histamine is stored in secretory granules as a complex with acidic residues of the proteoglycan heparin, and in basophils in the blood as a complex with chondroitin sulfate (7,8). Mast cells are distributed in areas of skin and mucous membranes of the respiratory, gastrointestinal, and genitourinary tracts and adjacent to blood and lymph vessels. Although histamine is secreted at low levels from these mast cells and basophils, the primary mechanism of release is associated with cell activation by immunoglobulin (Ig) E-mediated hypersensitivity processes (Fig. 32.4). Immediate hypersensitivity is initiated when allergen molecules crosslink to Fab components of adjacent IgE antibody molecules bound to high-affinity FcεRI receptors on the surface of these cells (FcεRI⁺ cells). Dimerization of occupied IgE-Fc receptors results in several membrane and cytosolic events. These include release

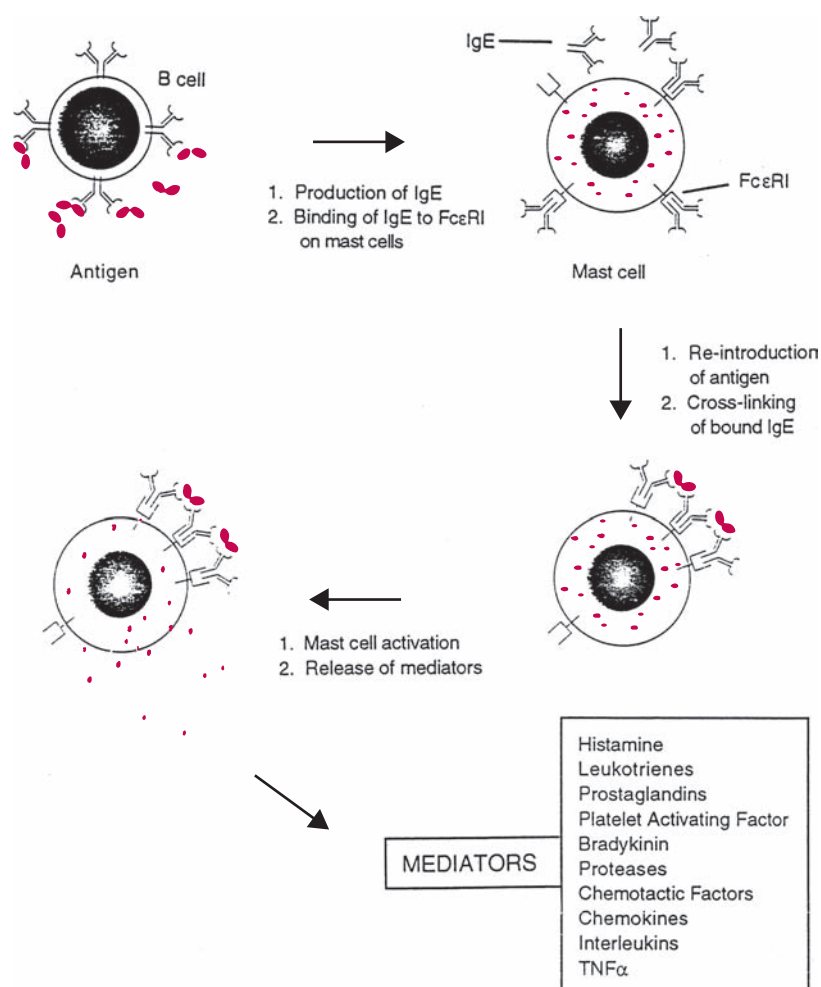


FIGURE 32.4 Sequence of events in immediate hypersensitivity. Initial contact with an antigen leads to specific IgE synthesis by B cells. Secreted IgE binds to mast cells or basophils through high-affinity Fc ϵ receptors (Fc ϵ RI). On subsequent exposure to the antigen, an immediate hypersensitivity reaction is triggered by cross-linking the IgE molecules.

of preformed mediators from secretory granules by exocytosis, synthesis and release of chemotactic mediators and neutral proteases (e.g., tryptase), synthesis and release of lipid mediators (leukotrienes and prostaglandins), and stimulation of synthesis of cytokines and their subsequent release. Other cell activation stimuli for the release of histamine include concanavalin A, substance P, polyamines, opiates, and several lymphokines and cytokines. There are different subpopulations of cells that respond differently to these stimuli. With the exocytotic response, histamine rapidly dissociates from the partially solubilized granule matrix. In basophils, the release process may be slightly different, occurring without degranulation. Other cell types, including lymphocytes, platelets, neutrophils, monocytes, and some macrophages, secrete histamine-releasing factors. These cells also have distinct low-affinity receptors for IgE, which when occupied result in secretion of mediators that selectively recruit and activate secondary effector cells in the inflammatory process. More complete exposition of these processes is available (9,10).

Mast cells play an important role in the early response to allergens, and they provide mediators that lead to initial

and late stages of the process and subsequently to chronic inflammatory reactions (11,12). Early stages appear to be related to degranulation and the release of many mediators, including histamine, prostaglandin D₂ (PGD₂), leukotriene C₄ (LTC₄), platelet activating factor, adenosine triphosphate (ATP), kinins, and some enzymes (e.g., tryptase and chymase). Thus, additional important mediators other than histamine have very significant roles. These mediators include platelet activation factor, substance P, neurokinin A, and others. Besides processes of vasodilation and edema, activation of secondary inflammatory cells, adherence of neutrophils, and migration of eosinophils and T cells to postcapillary venule endothelial cells occur. Specific cell adhesion molecules mediate this process. A further array of mediators and cellular responses follows. A series of interleukins are generated and secreted by subtypes of T cells. Stimulation of many nuclear factor- κ B (NF- κ B)-mediated gene transcription pathways occurs via H₁-receptor activation. Important cytokines are produced and liberated, including tumor necrosis factor (TNF)- α , interleukin (IL)-4, IL-5, IL-1, and IL-6, which are involved in chemokine secretion and

regulation of cell maturation and proliferation processes, changes that occur in late-stage inflammatory processes. Thus, the inflammatory cascade is a complex and intricate one, and the effects of histamine are only a small part of the process. Inhibition of the production of many of the proinflammatory cytokines occurs in the presence of many of the H_1 antihistamines.

As a result of occupation of H_1 receptors by histamine, constriction of bronchial and gastrointestinal smooth muscle occurs. Spasm of the bronchi to inhaled histamine at one time was a test for airway reactivity. Intradermal injection of histamine produces vasodilation of arterioles, as the first step in the "triple response," mediated via H_1 and H_2 receptors. A flare response follows this stimulation, resulting in release of substance P and other neuropeptides. Edema from exudation of plasma fluids follows due to contraction of endothelial cells of the postcapillary venules. The wheal and flare responses are mostly H_1 receptor mediated.

Histamine affects the maturation of the immune system cells altering their activation, chemotactic, and effector functions via H_1 -receptor activation. Histamine regulates functions of monocytes, dendritic cells, and antigen-specific T-helper type 1 (T_H1) and T-helper type 2 (T_H2) cells, as well as regulating antibody responses. These cells synthesize high amounts of histamine, where the synthesis of histamine is modulated by interleukins and other cytokines (13,14).

Histamine Receptors: Molecular and Mechanistic Aspects

Histamine receptors are found in various tissues. Among these are the H_1 receptors in smooth muscle of the bronchi, gut, and uterus. Contraction of the bronchi leads to restriction of air flow in the lungs. Histamine increases the permeability of capillary walls. Plasma constituents flow into extracellular spaces due to contraction of endothelial cells; this process leads to edema. At the level of the CNS, histamine secretion appears to be associated with wakefulness, as H_1 -receptor antagonism centrally is associated with drowsiness. In the stomach, parietal cell stimulation increases production and secretion of acid, mediated through H_2 receptors. The H_2 receptors play a minor role in allergic inflammatory processes. The H_3 receptor is principally an autoreceptor in the CNS at histaminergic neurons, and it is a heteroreceptor on neurons that release other neurotransmitters, also in the brain. The H_4 receptor is expressed primarily on eosinophils and mast cells where it is associated with chemotactic responses.

All of the histamine receptors appear to have constitutive receptor G protein signaling activity independent of the presence of histamine (15). Most of the antihistamines studied are not antagonists, but are inverse agonists. They reduce constitutive G protein signaling activity. Occasionally, some antihistamines are neutral antagonists (i.e., they do not reduce the constitutive G protein-coupled activity of the receptor). A two-state model of inactive and active conformations of the receptor is consistent with this observation.

Histamine H_1 receptors are expressed widely. They are found on nerve cells, airway and vascular smooth muscle, hepatocytes, chondrocytes, endothelial cells, eosinophils, monocytes, macrophages, dendritic cells, and T and B lymphocytes. The human H_1 -receptor gene encodes for a 487-amino acid protein with the signature structural features of G protein-coupled receptors (seven transmembrane domains, N-terminal glycosylation sites, phosphorylation sites for protein kinase A and C, and a large intracellular loop with several serine and threonine residues) (15,16). It is coupled (via $G_{q/11}$ proteins) to phosphatidylinositol turnover as the second messenger system. The H_1 receptor shows 40% homology with the muscarinic M_1 receptor and the M_2 receptor. An aspartic acid in the third transmembrane domain is highly conserved in several species, and it is suggested as a recognition site for the protonated aliphatic amine function of histamine, at both H_1 and H_2 receptors. Based on mutation studies and homologous positions to α -adrenergic receptors, suggestions for sites of binding of the imidazole portion of histamine to amino acids threonine and/or asparagine in the fifth transmembrane have been made.

Signal transduction processes begin with $G_{q/11}$ -coupled hydrolysis of phosphatidylinositide to IP_3 (inositol-1,4,5-triphosphate) and 1,2-diacylglycerol, which occurs via activation of phospholipase C. Elevation of intracellular calcium ion from intracellular stores occurs. Voltage-gated calcium channels may be opened by activation of ion channels permeable to Na^+ and K^+ ions. Calcium channel antagonists block some effects of histamine on intestinal smooth muscle. In addition, the H_1 receptor can activate other signaling pathways, including phospholipase D and phospholipase A_2 , and stimulate NF- κ B-mediated gene transcription.

Histamine H_2 receptors are widely distributed like H_1 receptors. They are found at airway and vascular smooth muscle cells, hepatocytes, chondrocytes, epithelial cells, neutrophils, eosinophils, monocytes, dendritic cells, and T and B lymphocytes. T-cell responses in general are negatively regulated by activation of H_2 receptors.

The H_2 receptor is a 359-amino acid protein in humans. It has some features similar to the H_1 protein (e.g., N-terminal glycosylation sites and phosphorylation sites in the C-terminal). An aspartic acid residue in the third transmembrane loop appears to be critical to agonist and antagonist binding, and threonine/aspartate and tyrosine/aspartate couples in the fifth transmembrane domain appear to be important for interaction of the imidazole part of the histamine molecule. It is positively coupled via G_{as} to activate adenylyl cyclase for synthesis of cyclic adenosine monophosphate (cAMP) as second messenger. In some systems, it is coupled through G_q proteins to stimulate phospholipase C. It appears in some cells that other processes, such as the breakdown of phosphoinositides, control of intracellular calcium ion levels, and phospholipase A_2 activity, can be regulated by other cAMP-independent pathways.

The highest density of H_3 receptors in nerve cells occurs in the brain, principally in the striatum, substantia nigra, and the cortex, and to a much lesser extent at peripheral nerve terminals. The H_3 receptors also occur in eosinophils, dendritic cells, and monocytes. Low expression of these receptors occurs in peripheral tissues. In the brain, the H_3 receptor is a presynaptic auto- and heteroreceptor where activation leads to a decrease in neurotransmitter release. The most widely studied H_3 receptor is 445 amino acids, but many splice variants have been observed (17). It is activated via $G_{ai/o}$ -coupled proteins (negatively to adenylyl cyclase) to the activation of protein kinase A to modulate gene transcription. H_3 receptors, via $G_{ai/o}$ G proteins, may activate phospholipase A_2 , mitogen-activated protein kinase, and phosphatidylinositol-3-kinase. The H_3 receptors have low sequence homology with H_1 and H_2 receptors (~20% each). Activation of histaminergic neurons centrally, which promotes arousal and attention and improves learning in animals, is a potential target for H_3 -receptor antihistamines. Like the H_1 and H_2 antihistamines, the H_3 antihistamines function primarily as inverse agonists.

H_4 receptors are expressed by cells of the hematopoietic system, with expression occurring primarily on eosinophils, neutrophils, dendritic cells, and T cells. The H_4 receptors are also reported at mast cells. Low expression is reported in nerve cells, hepatocytes, and the periphery. Histamine stimulation results in a chemotactic response, cell migration. The presence of H_4 receptors on these cells suggests that this receptor plays a role in the inflammatory response. Evidence suggests that it may be regulation via inflammatory stimulation of TNF- α and cytokine IL-6. The H_4 receptor has a sequence of 390 amino acids; it is coupled via $G_{ai/o}$ to inhibition of adenylyl cyclase. It has highest homology with the H_3 receptor (~35% to 40%), with greater homology (58%) of the transmembrane segments. Thus, antihistamines could be useful agents to antagonize the inflammatory response. Agents studied to date also appear to be inverse agonists.

INHIBITORS OF HISTAMINE RELEASE

Mechanism of Action

The bronchodilatory activity of khellin, a chromone obtained from a plant source (*Ammi visnaga*) used by ancient Egyptians for spasmolytic activity, stimulated the search for related compounds with similar pharmacologic properties (18). From a study of many bischromones, cromolyn sodium was developed and marketed (Fig. 32.5). Although it prevents bronchospasm, cromolyn sodium does not reverse antigen-induced bronchiolar constriction. Thus, it and other agents like it that followed prevent the release of histamine but do not block the effects of histamine at its receptors.

The mechanism by which cromolyn and nedocromil inhibit degranulation of mast cells has been investigated (19). Both agents stimulate phosphorylation of moesin, a 78-kd protein that is phosphorylated by isozymes of protein kinase C. It is suggested that phosphorylation

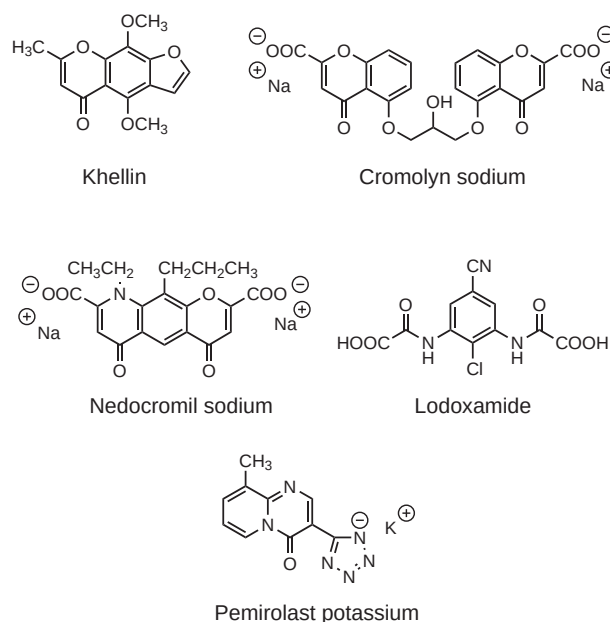


FIGURE 32-5 Mast cell degranulation inhibitors.

at serine hydroxyl groups results in conformational changes that expose domains that promote association with actin and other proteins of the secretory granules. This association results in immobilization of the granules and inhibition of exocytosis. Cromolyn and nedocromil (vide infra) apparently inhibit function of cells other than mast cells; these effects may occur in later stages of inflammatory responses. Cromolyn does not have intrinsic antihistaminic or anti-inflammatory activity.

Therapeutic Applications of Specific Drugs

Cromolyn (Intal, Nasalcrom, Gastrocrom)

Cromolyn is usually used prophylactically for bronchial asthma (as an inhaled powder), for prevention of exercise-induced bronchospasm, and for seasonal and perennial allergic rhinitis (nasal solution). Topically, it is also used as eye drops for allergic conjunctivitis and keratitis. In the management of asthmatic conditions, it is administered using a power-operated nebulizer. The bioavailability is very low on oral administration because of poor absorption. By inhalation, the powder is irritating to some patients. After inhalation, much less than 10% of the dose reaches the systemic circulation. An oral dosage form is used for mastocytosis.

Nedocromil (Alocril)

Nedocromil is a chromone analog used as an ophthalmic solution for the treatment of seasonal and perennial allergic conjunctivitis. Aerosols have previously been used in the prophylaxis of asthma and in reversible obstructive airway disease.

Lodoxamide (Alomide)

Lodoxamide, which shows some structural similarities to cromolyn and nedocromil, is also a mast cell stabilizer

that inhibits the immediate hypersensitivity reaction, preventing increases in vascular permeability associated with antigen-IgE-mediated responses. Its precise mechanism(s) of action is (are) not completely understood. It is used topically in the eye principally for conjunctivitis and keratitis associated with vernal allergens.

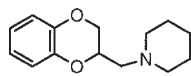
Pemirolast (Alamast)

Pemirolast, with an acidic tetrazole isosteric replacement for a carboxylic acid functionality, is used topically in the eye to prevent itching associated with allergic conjunctivitis. It is an inhibitor of release of histamine and other inflammatory mediators, including leukotrienes.

INHIBITORS OF RELEASED HISTAMINE

Historical Background

The first antihistamine was discovered by Forneau and Bovet, who observed that piperoxan protected guinea pigs against histamine-induced bronchospasm (20). The sensitivity of the guinea pig was initially thought to make it a good model for anaphylaxis. Piperoxan also has important effects related to antagonism of norepinephrine at α -adrenergic receptors.



Piperoxan

Antihistamines, specifically H_1 antihistamines (14,21), are useful in the treatment of allergy and inflammatory disorders, where many effects are mediated via histamine. Compounds that had antagonistic effects in these assays did not antagonize the effects of histamine on the stomach (acid secretion) and heart (positive chronotropic and inotropic effects). These differences led to suggestion of the presence of H_1 and H_2 receptors and, ultimately, to the development of selective H_2 antihistamines to diminish the secretion of gastric acid. A third class of histamine receptors, H_3 receptors, appears primarily to be autoreceptors that control the synthesis and release of histamine presynaptically and heteroreceptors that control the release of other transmitters, primarily in the CNS. The most recently discovered histamine receptors, H_4 receptors, are expressed principally on eosinophils and mast cells where they may play a role in the inflammatory response, especially in late stages. Classical H_1 antihistamines do not bind to this histamine receptor, and H_4 receptors do not modulate degranulation of mast cells.

The first-generation H_1 antihistamines are useful and effective in the treatment of allergic responses (e.g., hay fever, rhinitis, urticaria, and food allergy). These agents also have effects at cholinergic, adrenergic, dopaminergic, and serotonergic receptors. Adverse central effects include sedation, drowsiness, decreased cognitive ability, and somnolence. Peripheral side effects associated

with cholinergic blockade include blurred vision, dry mouth, urinary retention, and constipation. Other effects observed have included appetite stimulation, muscle spasm, anxiety, confusion, and occasionally irritability, tremor, and tachycardia. Of all the side effects, CNS depression is the most common, and it can be so pronounced that some of these agents with short durations of action are used as over-the-counter (OTC) sleep aids. The separation of CNS depressant and anticholinergic effects from peripheral antihistaminic effects in later agents led to the second-generation antihistamines (vide infra).

Structural classes of H_1 antihistamines can be represented by a general structure of two aromatic groups linked through a short chain to a tertiary aliphatic amine (Fig. 32.6). The aromatic groups (Ar_1 , Ar_2) are usually phenyl or substituted phenyl, thienyl, or pyridyl. These aryl substituents are attached to the X group, which is a nitrogen atom in the ethylenediamines, a carbon attached to an ether oxygen atom in the ethanolamine ether series, or only a carbon atom in the alkyl amine series. The spacer is usually two or three carbons in length, and it may be in a ring, branched, and saturated or unsaturated. The R groups attached to the aliphatic amine are usually simple alkyl groups, usually methyl, or occasionally aralkyl groups. One of the early thiophene-containing analogs was carcinogenic in rats. After thiophene versus benzene replacement in other series of compounds demonstrated some toxic effects, thiophene has been relegated to one of the low-priority choices for isosteric replacement of a phenyl group.

Most H_1 antihistamines are inverse agonists rather than neutral antagonists. Like other histamine receptors, H_1 receptors have constitutive G protein-coupled activity in the absence of the agonist histamine that results in activation of second messenger signaling pathways (e.g., phospholipase C activity and NF- κ B-mediated gene transcription) (22). A model of active and inactive conformations of histamine receptors has been advanced to accommodate these biochemical findings. Inverse agonists bind to the inactive conformation of the receptor shifting the equilibrium toward the inactive state. Neutral antagonists interact with both conformations of the receptor.

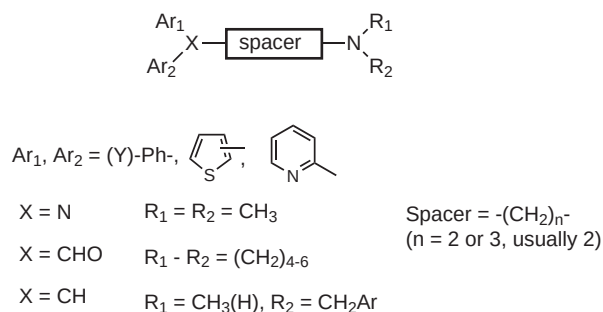


FIGURE 32.6 General structure of first-generation antihistamines.

First-Generation H₁ Antihistamines

Ethylenediamines

The earliest series of H₁ antihistamines is the ethylenediamines. Two two-carbon spacers may appear between the two nitrogen atoms in the piperazine series (vide infra). Examples of agents in the ethylenediamine class, of which phenbenzamine was the first of these agents, appear in Table 32.2. Compounds with several different but closely related aromatic rings are useful and include phenyl, 2-pyridyl, halogen- and methoxy-substituted phenyl, or pyrimidyl. Thiazole-, furanyl-, and thiophene-ring congeners have also been available in the past. The small alkyl substituents on basic nitrogen are usually methyl groups. A number of these agents are still used. With the exception of antazoline, all have the ethylenediamine spacer. In antazoline, the alkyl tertiary amine in phenbenzamine is replaced with an imidazoline group. CNS effects, usually sedation, are very common among agents in this class.

Information on pharmacokinetic and metabolic disposition is limited because this early group of compounds was not studied in great depth. Only later, with the development of second-generation H₁ antihistamines, and/or where issues of potential toxicity have arisen concerning some of the early compounds, have metabolic disposition and pharmacokinetic properties been examined more completely. Thus, the available information on the metabolism of these early antihistamines is sparse. From some of the ethylenediamines, expected products of *N*-demethylation and subsequent deamination have been reported. In addition, some produce quaternary *N*-glucuronides as urinary metabolites, a process that occurs to some extent in many relatively unhindered tertiary aliphatic amines among the antihistamines and also in other lipophilic aliphatic tertiary amine drug classes.

TABLE 32.2 Examples of Ethylenediamine Antihistamines

Drugs	X	Y	Ar
Phenbenzamine	CH	CH	
Tripelennamine	N	CH	
Methapyrilene	N	CH	
Thonzylamine	N	N	
Related compound:			Antazoline

Ethanolamine Ethers

STRUCTURE–ACTIVITY RELATIONSHIPS The prototype of the aminoalkyl ethers is diphenhydramine, a benzhydryl ether still widely used for allergic conditions more than a half-century after its introduction. Structural analogs with various ring substituents (Me, OMe, Cl, Br) in one of the aromatic rings have also been developed, as have compounds with a 2-pyridyl group replacing one of the phenyl groups (Table 32.3).

Significant anticholinergic side effects are observed among members of the group (i.e., dry mouth, blurred vision, tachycardia, urinary retention, constipation). Sedative properties are also very common. Sedation, accompanied with a short half-life and a wide margin of safety, allows some of these compounds to be used as OTC sleep aids. The 8-chlorotheophyllinate salt of diphenhydramine is marketed as dimenhydrinate for use in the treatment of motion sickness. The compound with the *p*-Cl-Ph and 2-pyridyl aryl groups is carbinoxamine, a potent antihistamine. Substitution of a methyl group at the carbon α to the ether function affords the related compound doxylamine, in which the aryl groups are phenyl and 2-pyridyl. Clemastine, a homolog with an additional carbon atom between the oxygen and the basic nitrogen, which is incorporated into a ring, has less sedative properties, as does the drug setastine in which the alkyl amine substituent is incorporated into a seven-membered hexahydroazepine ring.

ANTIHISTAMINIC VERSUS ANTICHOLINERGIC ACTIVITY Besides the structural analogs in Table 32.3 that possess increased selectivity for histamine H₁ receptors over muscarinic receptors, introduction of alkyl substituents at C-2' or C-4' of one aromatic ring results in significant changes

TABLE 32.3 Ethanolamine Ether Antihistamines

Drugs	Trade Name	R ₁	R ₂	X
Diphenhydramine Dimenhydrinate	Benadryl Dramamine	H	H	CH
Bromodiphenhydramine		Br	H	CH
Chlorodiphenhydramine		Cl	H	CH
Carbinoxamine		Cl	H	N
Doxylamine	Decapryn	H	CH ₃	N
	Unisom			
Related compounds:				
Setastine	Clemastine			

in selectivity in tissue-based assays for antihistaminic versus anticholinergic activity. With increasing alkyl group size (Me, Et, *i*Pr, *t*Bu) at C-2', large decreases in antihistaminic activity and increases in anticholinergic activity are observed (23). With larger alkyl groups, the possible spatial orientations of the two aromatic rings with regard to each other are limited due to increasing rotameric restrictions. Introduction of these alkyl substituents at C-4' decreases anticholinergic activity and yields small increases in antihistaminic activity. A chiral center is introduced with these changes, and differences in pharmacologic properties of the enantiomers of each compound are observed; two examples are shown in Table 32.4.

STEREOCHEMICAL AND STRUCTURAL EFFECTS This change in receptor selectivity (histaminergic vs. cholinergic) stimulated the synthesis of a large number of related homologs. However, diphenhydramine itself is also used in the treatment of parkinsonism because of its central anticholinergic properties. Other anticholinergic agents that are structurally related to the benzhydryl ether antihistamines are also used in the treatment of parkinsonism. Compounds with central anticholinergic and antihistaminic effects have been used as one approach to the treatment of Parkinson disease (see Chapter 13).

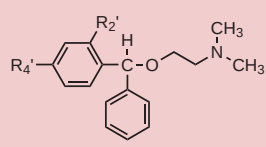
The observed differences in potency of enantiomers in tissue-based assays suggest significant stereoselective interactions of antagonists at the receptor level. Differences in affinity of 60- to 200-fold are noted between enantiomers in several analogs where the chiral center results because of differences in the two aromatic rings (e.g., Ph and 2-pyridyl, or Ph and *p*-Br-Ph) (24). Enantiomers with the *S*-absolute configuration are

usually more potent. Clemastine, a more complex homolog, is marketed as the *R,R*-enantiomer, which is the more potent of the *R,R*- and *S,S*-enantiomeric pair, and more potent than either the *R,S*- and *S,R*-enantiomers of the other diastereomer (25). Consistent with results from related compounds, the chiral center at the benzhydryl carbon has a significant influence on potency, whereas the one in the pyrrolidine ring is of lesser importance (Table 32.5).

Very small changes in the arrangement of aromatic groups in the members of the ethanolamine ether series significantly alter the scope of their pharmacologic properties. Previous work has shown that the two aromatic rings in diphenhydramine can be located slightly differently with respect to each other as in phenyltoloxamine, a potent antihistamine. However, a retro arrangement of carbon and oxygen atoms, prepared in an attempt to investigate structural requirements for antihistamines, afforded significantly different pharmacologic properties and ultimately led to a series of very important selective serotonin reuptake inhibitors, like fluoxetine (Fig. 32.7). Unlike the bioisosteric oxygen to nitrogen atom replacement, conversion of the oxygen atom to a sulfur atom in the diphenhydramine series results in a compound with greatly decreased antihistaminic activity (26).

METABOLISM Only limited information on the metabolic disposition of this group of compounds is available. As expected, sequential *N*-demethylations (formation of the corresponding secondary and the primary amines) followed by subsequent deamination (formation of the carboxylic acid metabolite) is a major pathway for diphenhydramine (Fig. 32.8) and some of its analogs. Although the early experiments are relatively incomplete, it appears that the *N*-demethylation products have shorter half-lives than the corresponding parent drugs, and they probably contribute very little to the observed antihistaminic

TABLE 32.4 Antihistamine and Anticholinergic Activity of Enantiomers of Ring-Substituted Ethanolamine Ethers

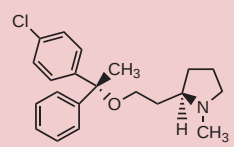


Antihistaminic Activity			Anticholinergic Activity		
$R_{2'}$	$R_{4'}$	pD_2	Ratio of Potency of (+)- to (-)-Isomer	pD_2	Ratio of Potency of (+)- to (-)-Isomer
$(CH_3)_3C$	H	(+) 8.76 ^a (-) 6.87	78	(+) 6.14 (-) 5.89	1.9
H	CH_3	(+) 6.36 (-) 6.00	2.3	(+) 6.03 (-) 8.12 ^a	0.008 ^b

^aMost potent enantiomer in each assay.

^b(-)-Enantiomer is ~125 times as potent as the (+)-enantiomer.

TABLE 32.5 Antihistamine Activity of Stereoisomers of Clemastine



Drug	Diastereomers	pA_2	ED_{50} ^a
Clemastine	(<i>R,R</i>)	9.45	0.04 mg/kg
	(<i>S,S</i>)	7.99	5.0
	(<i>R,S</i>)	9.40	0.28
	(<i>S,R</i>)	8.57	11.0

^a ED_{50} vs. lethal dose of histamine in guinea pigs (22).

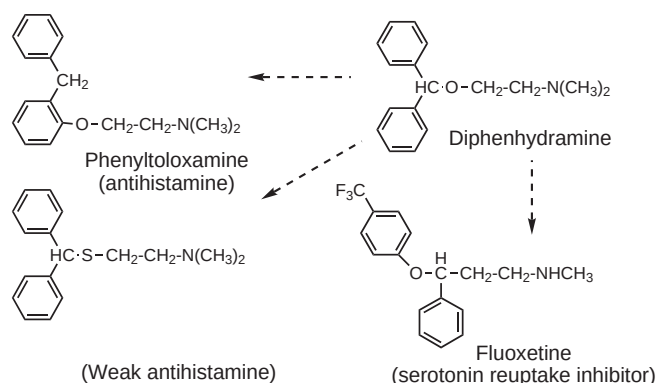


FIGURE 32.7 Structural similarities among diphenhydramine-related structures.

properties. Minor metabolites that are conjugates of the carboxylic acid or of ether cleavage products have been found in some animal species. Compounds with shorter half-lives, like diphenhydramine, when used as antihistamines, require repeated administration, but a very short half-life is an advantage when the same drugs are used as sleep aids (21,27).

Alkyl Amines

A third class of analogs includes those compounds in which a carbon atom replaces the heteroatom spacer in the general structure (Fig. 32.6). Examples are pheniramine, chlorpheniramine, brompheniramine, and the *E*-isomers of olefinic homologs (Fig. 32.9). The ring halogen-substituted compounds are widely used OTC antihistamines for mild seasonal allergies. These agents are characterized by a long duration of antihistaminic action and by a decreased incidence of central sedative side effects, when compared to the ethylenediamines and ethanolamine ether series. This structural change introduces a chiral carbon when the two aromatic rings are different (e.g., Ph and 2-pyridyl). These were the most extensively used antihistamines until the more selective second-generation antihistamines appeared.

STRUCTURAL AND STEREOCHEMICAL EFFECTS *E*- and *Z*-isomers of the alkenes in this series show very large differences in potency in tissue-based assays (e.g., *E*-pyrobutamine is more potent than its *Z*-isomer by 165-fold and *E*-triprolidine

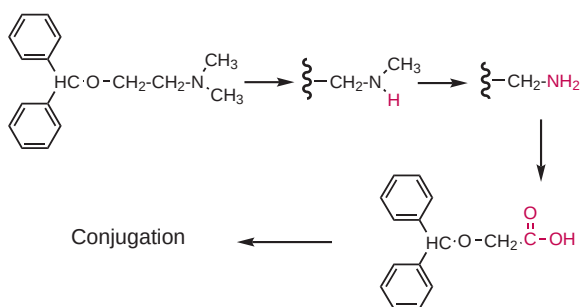


FIGURE 32.8 Metabolism of diphenhydramine.

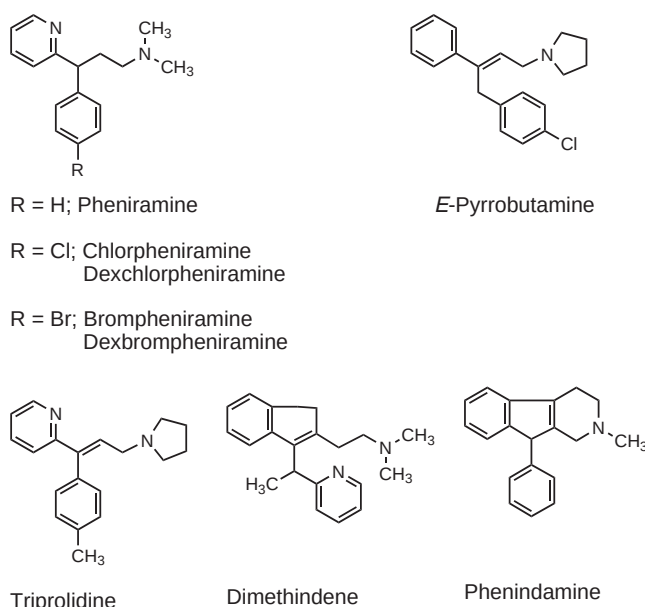


FIGURE 32.9 Examples of alkane and alkene antihistamines.

is more potent than its *Z*-isomer by about 1,000-fold) (Fig. 32.9) (3). Dimethindene has many of the structural features of both of these two agents in a more complex cyclized structure. The observed difference in potency between the *E*- and *Z*-isomers of triprolidine shows that the two aromatic rings probably have quite different binding environments at the receptor (28). Phenindamine, also a related cyclic analog, was also an effective antihistamine. These observations provide evidence suggesting that a 5 to 6 Å distance between the tertiary aliphatic amine and one of the aromatic rings is required at the site of receptor binding (Fig. 32.10) (29).

Differences in potency between the enantiomers of the conformationally mobile amino-alkanes have also been observed. The *S*-enantiomers have greater affinity for H_1 histamine receptors, occasionally by very large amounts [e.g., by 200 to 1,000 fold in radioligand displacement assays and in tissue-based assays for (+)-*S*- vs. (–)-*R*-chlorpheniramine], with the (+)-enantiomer being more potent (Ar = X-Ph, 2-pyridyl). Greater selectivity for H_1 receptors versus muscarinic and adrenergic receptors is also observed. For members of the series, the chiral center of the more potent enantiomer correlates

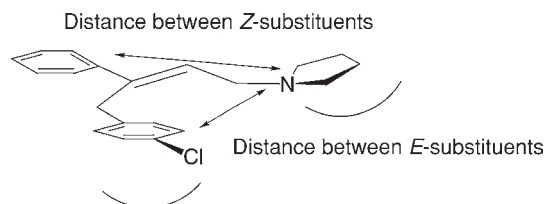


FIGURE 32.10 Potential binding sites based on *E/Z*-configurations.

stereochemically with the more active enantiomer of the oxygen congener carbinoxamine (Ar = Ph, Ar = *p*-Br-Ph) (Table 32.3) (3,30). Single enantiomers of these agents are available in the case of dexchlorpheniramine and dexbrompheniramine.

HALF-LIFE AND METABOLISM The alkyl amines have significantly less CNS depressant effects than benzhydryl ethers of ethanolamines. Additionally, these compounds have long half-lives and extended durations of action. These agents have decreased antiemetic effects and decreased anticholinergic properties compared to ethanolamine ethers. Many are available in OTC preparations for hay fever and other mild allergic conditions, sometimes in combination with decongestants. Most are suitable for once-a-day dosing because of their long half-lives, up to 24 hours, although they are routinely administered more frequently.

Information on the metabolic disposition of several of these agents has been reported. As expected, *N*-dealkylation is a major pathway with the corresponding secondary and primary amines being found in the plasma, as well as the parent drug. In cases where *O*-methyl aryl ethers are present, expected *O*-demethylation products have been reported.

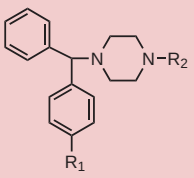
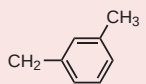
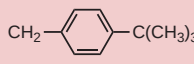
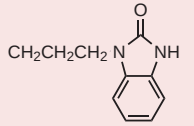
Piperazines

Members of the piperazine class of agents are structurally related to both the ethylenediamines and the benzhydryl ethers of ethanolamines. Their structures include the two-carbon separation between nitrogen atoms, which is incorporated into the piperazine ring (Table 32.6).

Diarylmethylene groups (benzhydryl substituents, like in diphenhydramine) are attached to one of the nitrogen atoms, and an alkyl or aralkyl substituent is attached to the other nitrogen. Early compounds, like cyclizine, chlorcyclizine, meclizine, buclizine, and hydroxyzine, have been widely used as antihistamines and as agents for treatment of motion sickness because they have useful central antiemetic effects.

These agents also have significant anticholinergic and antihistaminic properties. Anticholinergic side effects and drowsiness are common. The primary use of these compounds remains as treatment of motion sickness and vertigo and for suppression of nausea and vomiting. Although teratogenic effects of cyclizine and meclizine have been observed in rodents, large studies have not demonstrated adverse fetal effects in humans. However, these agents are used cautiously in pregnant women and children. Oxatamide is used in Europe principally in allergic rhinitis, urticaria, and in combination with albuterol in asthma. Drowsiness and sedation are noted. Hydroxyzine is used in treatment of pruritus, and at higher dosages, it is used in the management of anxiety and emotional stress. Its acid metabolite, cetirizine, which is formed from oxidation of the terminal primary alcohol to the corresponding

TABLE 32.6 Examples of Piperazine Antihistamines

			
Drugs	Trade name	R ₁	R ₂
Cyclizine	Marezine, Bonine	H	CH ₃
Chlorcyclizine	Mantadil	Cl	CH ₃
Meclizine	Antivert	Cl	
Buclizine	Bucladin-S	Cl	
Oxatamide	Tinset	H	
Hydroxyzine	Vistaril	Cl	CH ₂ CH ₂ OCH ₂ CH ₂ OH
Cetirizine	Zyrtec	Cl	CH ₂ CH ₂ OCH ₂ COOH

carboxylic acid, is usually classified with the second-generation nonsedating antihistamines. The amphoteric nature of cetirizine, having both the tertiary aliphatic amine and carboxylic acid functional groups, appears to be associated with decreased, but not absent, sedative side effects.

Tricyclic H₁ Antihistamines

The two aromatic groups noted in several of the classes of antihistamines can be connected to each other through additional atoms such as the heteroatoms sulfur or oxygen or through a short one- or two-carbon chain. They have a general structure shown in Figure 32.11. The earliest potent tricyclic antihistamines (Table 32.7) were phenothiazines (Y = S, X = N). The phenothiazine antihistamines contain a two- or three-carbon branched alkyl chain between the nonbasic phenothiazine nitrogen and the aliphatic amine. They differ from the antipsychotic phenothiazine derivatives in which the side chain is usually three carbons long and unbranched, and usually without substituents in the aromatic ring. Most had pronounced sedative effects and long durations of action. Other uses included the treatment of nausea and vomiting associated with anesthesia, the treatment of motion sickness, and as antipruritic agents in the treatment of urticaria.

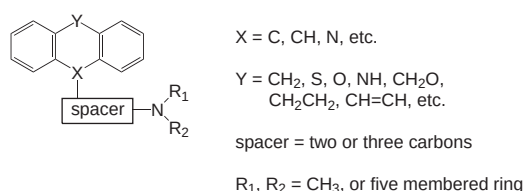


FIGURE 32.11 General structure of tricyclic antihistamines.

CONFORMATIONAL AND STEREOCHEMICAL EFFECTS In the active agents, the steric restrictions and decreased degrees of conformational freedom resulting from the connection of the two aromatic rings together suggests that certain spatial relationships between these two rings are acceptable in the drug receptor interaction of H_1 antihistamines. These ring systems are not flat, but are somewhat puckered with the two aromatic rings not in the same plane. However, the conformations usually undergo rapid intra-conversion. In some closely related systems, where this intraconversion is very slow, conformational enantiomers (atropisomers) have been obtained and studied, including cyproheptadine, doxepin, and hydroxylated metabolites of loratadine (31,32). The enantiomers of 3-methoxycyproheptadine have significantly different pharmacologic potency as antihistamine, antiserotonin, and anticholinergic agents (Fig. 32.12) (33). The (–)-isomer retained antihistaminic, antiserotonin, and appetite-stimulating effects similar to cyproheptadine, whereas the (+)-enantiomer had greater anticholinergic potency. Differences of 9- to 60-fold were observed in the reported assays.

Promethazine has many useful pharmacologic effects other than being an antihistamine. It has significant antiemetic and anticholinergic properties. It has sedative-hypnotic properties and has been used to potentiate the effects of analgesic drugs.

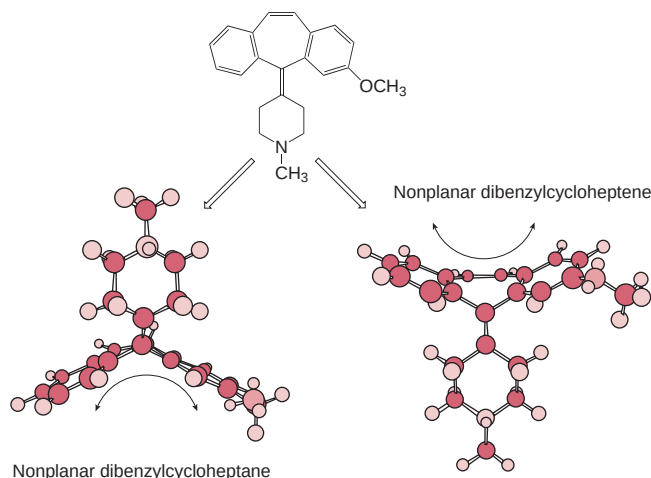


FIGURE 32.12 Enantiomers (atropisomers) of 3-methoxycyproheptadine.

Compounds in which the sulfur atom is replaced with another bridge (e.g., two methylene groups) are also available. Some have a pyridine ring replacing one of the benzenoid systems. Cyproheptadine, with a two-carbon spacer between the aromatic rings, also has anticholinergic, antiserotonergic, and appetite-stimulating properties, which are useful in treatment of anorexia nervosa and cachexia. The pyridine analog apparently lacks most of these qualities. Doxepin, an oxygen-containing congener of cyproheptadine, also has significant affinity for other receptors and has CNS-depressant qualities. It exists as a mixture of *Z*- and *E*-isomers (15:85) in its olefinic nonpiperidine side chain. In tissue-based assays, the *Z*-isomer is more potent than the *E*-isomer by more than threefold (31). Loratadine and desloratadine are widely used, nonsedating, second-generation antihistamines (vide infra).

TABLE 32.7 Examples of Tricyclic Antihistamines

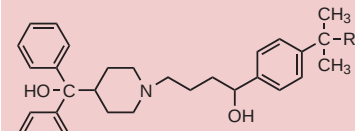
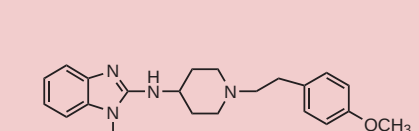
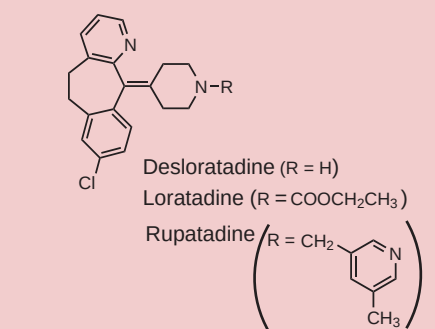
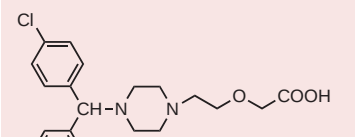
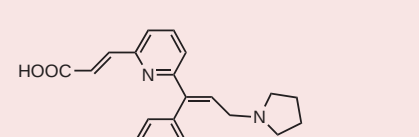
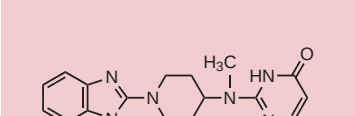
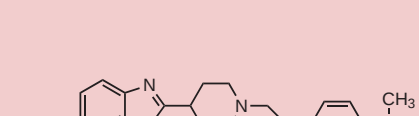
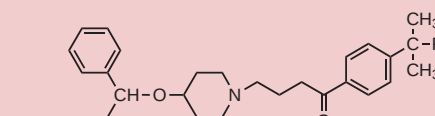
Drugs	Y	Z
Promethazine	S	
Pyrazine	S	
Cyproheptadine		
Azatadine		

METABOLISM Information on the metabolic disposition and pharmacokinetic properties of agents in this group is limited, including incomplete identification of primary metabolic pathways, results of liver microsomal metabolic experiments, and only occasionally pharmacokinetic information. In humans, products from the phenothiazines include products of *N*-demethylation, aromatic hydroxylation, and occasionally sulfoxidation. From tricyclic analogs, metabolites resulting from *N*-demethylation, aromatic hydroxylation, and formation of *N*-quaternary glucuronides have been reported (34).

Second-Generation Nonsedating H_1 Antihistamines Background

The second-generation antihistamines, marketed over the last 25 years, have improved H_1 selectivity, have little or no sedative qualities, and may have antiallergic effects apart from antihistaminic activity (35). They vary widely in structure (Table 32.8), but less so in pharmacologic

TABLE 32.8 Second-Generation Nonsedating Antihistamines

 Terfenadine (R = CH₃) Fexofenadine (R = COOH)	 Astemizole	 Desloratadine (R = H) Loratadine (R = COOCH₂CH₃) Rupatadine (R = CH₂-C₅H₄N-CH₃)
 Cetirizine Levocetirizine	 Acrivastine	
 Mizolastine	 Bilastine	 Ebastine (R = CH₃) Carebastine (R = COOH)

properties, having effects principally in the periphery. Structural resemblance to the first-generation H₁ antagonists is not always obvious as some of these agents were discovered while investigating new molecular structures for other pharmacologic targets. These agents possess selective peripheral H₁ antihistaminic effects, and they usually have less anticholinergic activity. Furthermore, they also have decreased affinity for adrenergic and/or serotonergic receptors and have limited CNS effects. The active agents apparently do not penetrate the blood–brain barrier significantly, perhaps because of their amphoteric nature (most are zwitterionic at physiologic pH) and partitioning characteristics and/or they are substrates for the drug efflux P-glycoprotein transporter or organic anion transporter proteins. Some of these agents have very slow rates of dissociation from H₁ receptors. Several have anti-allergic properties that are separate from their antihistaminic properties, which are not thoroughly understood. In most cases, the parent drug or its important metabolites have half-lives sufficiently long to account for the extended duration of action (36). Most are administered once daily.

Specific Drugs

Terfenadine and Fexofenadine (Allegra)

PHARMACOLOGIC EFFECTS This group of nonsedating antihistamines is usually thought to include fexofenadine and its

parent terfenadine, astemizole, cetirizine, and loratadine and its metabolite desloratadine. Terfenadine was synthesized as an analog of azacyclanol in a search for antipsychotic agents. The initial reports of its antihistaminic properties included the observation of similar effects of its acid metabolite fexofenadine (37). Although terfenadine is no longer available, it was once a very widely used nonsedating antihistamine. Extensive clinical experience resulted in the reports of dangerous cardiac arrhythmias occurring occasionally when certain other drugs were taken concomitantly (38). These cardiac arrhythmias included prolongation of the QT interval and torsades de pointes, a life-threatening ventricular arrhythmia. These cardiac effects are now known to be associated with blockade of the hERG (human ether-a-go-go) gene product, the α -subunit of an inward rectifying cardiac K⁺ channel (I_{Kr}) (39,40). These effects are associated only with the parent molecule. The side effects occur primarily at high concentrations of this lipophilic amine and usually in the presence of other CYP3A4 substrates like ketoconazole or macrolide antibiotics (triacycloleandomycin). In the presence of competing CYP3A4 substrates and inhibitors, high plasma concentrations of the parent agent resulted.

METABOLISM Fexofenadine, the carboxylic acid metabolite of terfenadine, is widely available (Fig. 32.13).

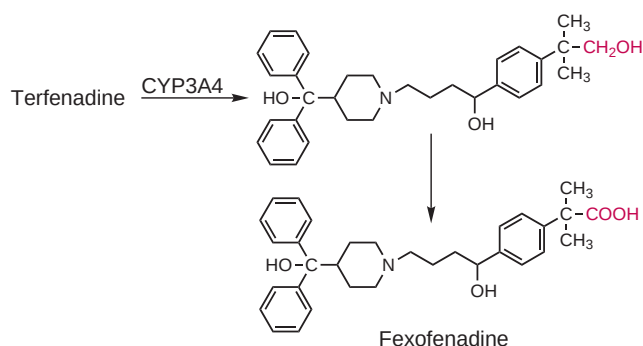


FIGURE 32.13 Metabolism of terfenadine leading to fexofenadine.

It accounted for the antihistaminic properties of terfenadine, which is very rapidly metabolized via CYP3A4-catalyzed processes. Although the enantiomers of terfenadine are reported to have approximately equal antihistaminic activity (41), no data on the activity of the enantiomers of fexofenadine were found. Members of the organic anion transporter protein family and the drug efflux transporter P-glycoprotein are involved in the disposition of fexofenadine. A higher plasma concentration of the *R*-enantiomer is reported, showing some stereoselectivity in this process (42,43). Fexofenadine does not have the antiarrhythmic side effects of terfenadine.

Terfenadine and other lipophilic amines (e.g., astemizole, another second-generation antihistamine that is no longer marketed, grepafloxacin, thioridazine, sertindole, and cisapride) are all lipophilic aliphatic amines containing one or more aromatic groups (40). Each has been associated with life-threatening cardiac arrhythmias associated with binding to the hERG K⁺ ion channel. The disposition of each depends significantly on CYP3A4 for its oxidative metabolism so that in the presence of inhibitors or competitive substrates, significantly elevated plasma levels are observed. Screening of potential drug candidates for hERG K⁺ ion channel affinity is done routinely to aid in attempts to develop safe new drugs. Similarly, proof of safety of agents that are primarily CYP3A4 substrates is required. Studies on the effect of ketoconazole on the plasma levels of the drug in vivo are routinely required by the U.S. Food and Drug Administration (FDA).

Astemizole

Astemizole (Table 32.8), like terfenadine, produced cardiac arrhythmias via interaction with the hERG K⁺ ion channel. Study of its primary metabolites (products of *O*-demethylation and *N*-dealkylation) as possible agents to avoid these effects did not lead to marketing of any of them as drugs.

Loratadine (Claritin) and Desloratadine (Clarinex)

PHARMACOLOGIC EFFECTS Loratadine (Table 32.8) is related to the first-generation tricyclic antihistamines and to

antidepressants. It is nonsedating, and neither it nor its major metabolite, desloratadine (descarboethoxylo-ratadine), is associated with the potentially cardiotoxic effects reported for terfenadine and astemizole. Upon chronic dosing, the area under the curve (AUC; plasma concentration vs. time curve) for the metabolite is greater than for the parent drug, and its half-life is longer. Desloratadine is a more potent H₁ antagonist and a more potent inhibitor of histamine release. This metabolite probably contributes significantly to the antihistaminic effects of administered loratadine. Desloratadine has a slow off-rate from the H₁ receptor in vitro, nearly 100 times slower than first-generation antihistamines (44). One of the hydroxylated metabolites, 3'-hydroxydesloratadine, may contribute to the pharmacologic properties of desloratadine (45).

METABOLISM The metabolic conversion of loratadine to descarboethoxylo-ratadine occurs via an oxidative process and not via direct hydrolysis (Fig. 32.14). CYP2D6 and CYP3A4 appear to be the CYP450 isozymes catalyzing this oxidative metabolic process (46). Apparently the metabolite does not reach the CNS in significant concentrations. Among the nonsedating second-generation antihistamines, this metabolite appears to be the only nonzwitterionic species. Although the failure of zwitterionic molecules to reach CNS sites in significant concentrations can be rationalized readily, a similar explanation is not apparent for loratadine or its metabolite. Competitive substrates for CYP3A4 do not produce a significant drug-drug interaction because the parent molecule lacks effects on hERG K⁺ channel in cardiac tissue.

When desloratadine is administered, its terminal phase half-life is greater than 24 hours, with about half of the administered dose being excreted in the urine and a similar amount in the feces (47,48). The principal hydroxylated metabolite is 3'-hydroxydesloratadine, but a number of other aliphatic and aromatic hydroxylation products have been identified, many appearing as *O*-glucuronides in vivo (49,50).

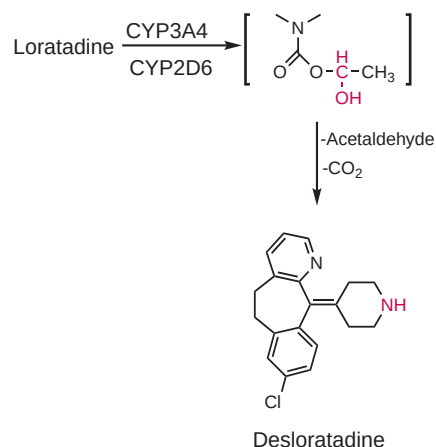


FIGURE 32.14 Metabolism of loratadine.

Rupatadine (Rupafin)

Rupatadine (Table 32.8) is a selective oral H_1 -receptor antagonist available in Europe and in other countries that is also used in seasonal and perennial allergic rhinitis and in chronic idiopathic urticaria. It is dosed once daily. Like loratadine, it is metabolized to desloratadine and to 3'-hydroxydesloratadine. The parent molecule has a terminal phase half-life of 5 to 6 hours. A majority of the metabolites are excreted in the feces. Rupatadine is reported to be more effective than desloratadine in inhibiting release of $TNF-\alpha$ and IL-6 in mast cells and to be an antagonist to platelet-activating factor in vitro, an activity not observed with other second-generation H_1 antagonists. In reported clinical trials, this agent has actions similar to desloratadine, cetirizine, and ebastine (51).

Cetirizine (Zyrtec) and Levocetirizine (Xysal, Xusal)

Cetirizine, the acid metabolite from oxidation of the primary alcohol of the antihistamine hydroxyzine (Fig. 32.15; see also Table 32.6), is a widely used antihistamine (52). It has a long duration of action and is highly selective for H_1 receptors. No cardiotoxicity has been reported, but some drowsiness occurs. The *R*-enantiomer of cetirizine, levocetirizine, is now marketed. Levocetirizine has higher affinity than its *S*-enantiomer for the H_1 receptor (>30-fold) and is more slowly dissociated by more than 20-fold from the receptor (53). Thus the antihistaminic properties of cetirizine are probably accounted for by the *R*-enantiomer. Similar large differences in ratios of dissociation rates and K_i values for closely related enantiomers are also reported. It is excreted primarily unchanged. Only a small amount of levocetirizine is reported as metabolites, including expected products of glucuronidation and aromatic hydroxylation (54).

Acrivastine (Semprex)

Acrivastine (Table 32.8), an acidic congener of triprolidine in which a carboxylic acid-substituted chain has been attached, is also a second-generation nonsedating antihistamine. Penetration of the blood-brain barrier is limited, and it is less sedating than triprolidine. It is used principally in a combination with a decongestant.

Ebastine (Kestine) and Carebastine

Benzhydryl ethers of piperidinols are also useful antihistamines. Those with large *N*-substituents, like those

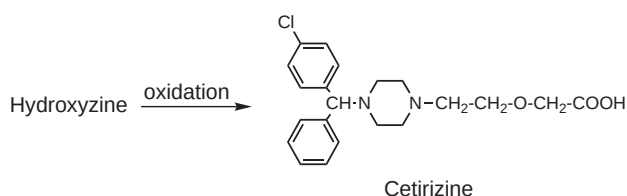


FIGURE 32.15 Metabolism of hydroxyzine to cetirizine.

in terfenadine and other nonsedating antihistamines, are most successful. Ebastine (Table 32.8), structurally similar to terfenadine, is a potent selective H_1 antihistamine as measured in radioligand displacement assays. In these assays, its acid metabolite has significantly higher affinity than the parent molecule. It is nonsedating and apparently free of anticholinergic effects (55). Like some other second-generation antihistamines, ebastine blocks release of PGD_2 and LTC_4/D_4 in cellular assays. Pharmacokinetic data indicate that its acid metabolite carebastine is responsible for its antihistaminic properties, because the parent drug has a very short half-life and the active metabolite a much longer one (Fig. 32.16). In an animal model of torsades de pointes, ebastine, at a high dose, produced significant cardiac conduction abnormalities (e.g., prolongation of the QT interval), whereas the metabolite did not. At lower doses, these effects occurred only in the presence of competitive CYP3A4 substrates. However, some in vitro data suggest that CYP450 isozymes other than CYP3A4 may be important in the initial hydroxylation. The pharmacologically active acid metabolite carebastine is metabolically analogous to fexofenadine (oxidation of a *t*-butyl group), the acid metabolite of terfenadine. Ebastine is marketed in several countries outside the United States.

Mizolastine (Mizollen)

Mizolastine (Table 32.8) is a second-generation nonsedating antihistamine that structurally resembles astemizole (56). Its half-life is long. It is primarily conjugated, and only small amounts of the compound are metabolized oxidatively. The AUC increases about twofold in the presence of ketoconazole, a CYP3A4 substrate and inhibitor. Only weak interaction with the hERG K^+ channel and with muscarinic receptors is reported. It is available in Europe.

Bilastine

Bilastine is structurally related to other second-generation H_1 antihistamines, being an amino acid with structural resemblance to parts of fexofenadine and astemizole (Table 32.8). It is being developed for the treatment of symptoms of seasonal and perennial allergic rhinitis and

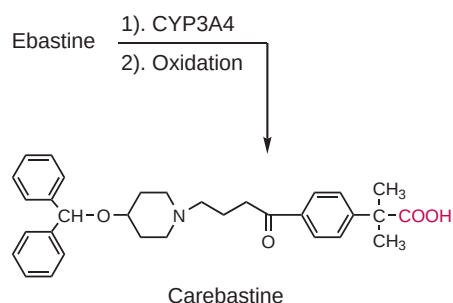


FIGURE 32.16 Metabolism of ebastine to carebastine.

chronic idiopathic urticaria. Comparison with cetirizine, fexofenadine, and desloratadine has shown similar effectiveness and safety characteristics. It has a long duration of action allowing once-daily dosing (57). Bilastine has recently received approval for use in Europe.

Topical H₁ Antihistamines

Therapeutic Applications

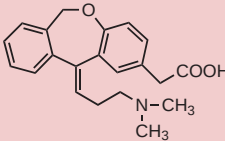
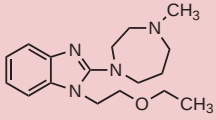
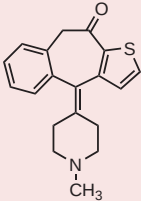
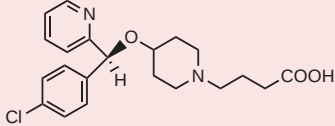
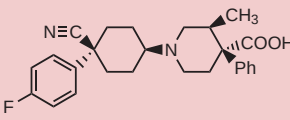
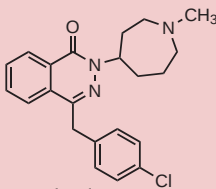
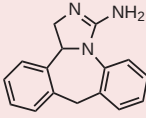
Topical application of H₁ antihistamines to the eye relieves itching, congestion of the conjunctiva, and erythema (19,58). The density of mast cells in the conjunctiva is high, and the histamine concentrations in tear film are significant in the ocular allergic response. From eye drops, only small amounts of the antihistamine (1% to 5%) penetrate the cornea. More of the compound is absorbed via the conjunctiva and nasal mucosa, and still more ends up swallowed from tear duct and nasal drainage. Until recently, topical ocular antihistamines were limited to two classical agents: antazoline (Table 32.2) from the ethylenediamine series and pheniramine (Fig. 32.9) from the alkylamine series. Both are used in combination with sympathomimetic vasoconstrictors.

A slow rate of receptor dissociation of H₁ antagonists is associated with long duration of action systemically, which occurs with the more recently available ocular antihistamines. Based on correlations of pK_a values and lipophilicity data, it appears that compounds with a logD (the sum of the partition coefficients of both the ionized and nonionized species) near 1.0 ± 0.5 at pH 7.4 are most efficacious and their water-soluble salts also show a low incidence of ocular irritation. Relationships between partitioning characteristics of these and other antihistamines indicate receptor affinity (moderate at least) and a particular range of lipophilicity is optimum for topical ocular antihistamines with minimum ocular irritation (59). Some of these compounds are currently available (Table 32.9) or are being evaluated as nasal sprays, and some are also occasionally used as systemic antihistamines.

Olopatadine (Patanol)

Olopatadine (Table 32.9) is structurally related to the tricyclic antihistamines. It is available as an eye drop and more recently also as a nasal spray. It has a long duration of action when applied topically, and it appears to also inhibit the release of inflammatory mediators (e.g., histamine, tryptase, PGD₂) from mast cells. Its selectivity for H₁ receptors in tissue assays (over H₂ and H₃ receptors) is very high, and its selectivity for H₁-receptor blockade over α -adrenergic, dopaminergic, serotonergic, and muscarinic receptors is also very high. Olopatadine is reported to have a rapid onset of action and a long duration of action, consistent with high histamine receptor affinity and a slow rate of receptor dissociation. The presence of the carboxylic acid side chain is apparently responsible for the observed lack of muscarinic receptor affinity. This feature may also be

TABLE 32.9 Topical Antihistamines

	
Olopatadine	Emedastine
	
Ketotifen	Bepotastine
	
Levocabastine	Azelastine
	
Epinastine	

responsible for limited penetration. Olopatadine is also a mast cell stabilizer. The mechanism for this activity has not been delineated, but it has been shown to stabilize model cell membranes by interaction with phospholipid monolayers.

Levocabastine (Livostin)

Levocabastine (Table 32.9) is a potent selective H₁-receptor antagonist used topically in eye drops for seasonal allergic conjunctivitis. A small amount of systemic absorption of the compound is reported. The agent also prevents release of transmitters from mast cells. A nasal spray used for allergic rhinitis is available outside the United States.

Emedastine (Emadine)

Emedastine (Table 32.9) is also a newer antihistamine used topically in the eye for conjunctivitis. It has very high H₁-receptor selectivity characteristics and is structurally related to the benzimidazoles such as astemizole. Inhibition of mast cell release of inflammatory mediators has been noted.

Azelastine (Astelin)

Azelastine, although not a close structural analog to the benzimidazoles, has some structural similarities to them (Table 32.9). It is used as a nasal spray for allergic rhinitis and as eye drops for allergic conjunctivitis. Like olopatadine, azelastine also stabilizes mast cells, preventing degranulation and subsequent release of histamine, leukotrienes, and PGD_2 . It is available in Europe for systemic use for the treatment of asthma and seasonal allergies. Besides antihistaminic effects, it may also block release of histamine and other inflammatory mediators from mast cells. When administered orally, the *N*-dealkylated metabolite appears to contribute significantly to its pharmacologic effects.

Ketotifen (Zaditor)

Ketotifen is a potent selective H_1 antihistamine that also prevents release of transmitters from mast cells (Table 32.9). It is approved in the United States for topical use to prevent itching of the eye due to allergic conjunctivitis. It is used as a systemic antiallergy agent in several countries outside the United States for the treatment of seasonal allergic rhinitis, hay fever, and asthma. Being structurally analogous to the cyproheptadine-like antihistamines, differences in activity of the two enantiomers (atropisomers) have been noted, being about six- to sevenfold in ligand displacement and rodent-based assays (60). Ketotifen has been shown to stabilize mast cells and to inhibit degranulation of eosinophils. Like olopatadine, it has been shown to interact with model membranes, stabilizing them by interaction with phospholipids monolayers.

Epinastine (Elestat)

Epinastine (Table 32.9) is a potent, long-acting H_1 antihistamine and an inhibitor of release of histamine and other transmitters from mast cells. It has some affinity for H_2 receptors, as well. It is used as an eye drop for allergic conjunctivitis. It does not penetrate into the CNS and is classified as a nonsedating antihistamine.

Bepotastine (Bepreve)

Bepotastine is a potent selective H_1 antihistamine that decreases the production of proinflammatory cytokines from mast cells and prevents eosinophil migration (Table 32.9). In the United States, it is approved for topical use to prevent itching of the eye due to allergic conjunctivitis. It is approved in Japan as a systemic antihistamine used for allergic rhinitis, urticaria, and pruritus.

Ant ulcer Agents

Background

The secretion of gastric acid occurs at the level of parietal cells of the oxyntic gland in the gastric mucosa (Fig. 32.17), producing 2 to 3 L of gastric juice daily (pH 1 in hydrochloric acid). Ultimately, this secretory process occurs via an H^+/K^+ -ATPase that exchanges hydronium ion (H_3O^+) with uptake of a potassium ion.

Several mediators regulate this secretion by way of receptor systems on the basolateral membrane. The H_2 histaminergic pathway is cAMP dependent. Gastrin and muscarinic receptors also regulate the secretion of gastric acid through calcium ion-dependent pathways. In parietal cells, E series prostaglandins work in opposition to the histaminergic pathway, inhibiting histamine-stimulated adenylyl cyclase activity. Other epithelial cells in the mucosal lining under the influence of prostaglandin-mediated pathways secrete bicarbonate and mucus, both of which are important in protecting the gastric lining from the effects of acid secretion. In many cases, hypersecretion of gastric acid appears to be associated with *Helicobacter pylori* infection, which may contribute to defects in mucosal protective defenses. There is evidence that some H_2 antihistamines, particularly cimetidine and ranitidine, have regulatory effects on T-cell lymphocyte proliferation by augmenting cytokine production and immunoglobulin production. These effects may not be associated with histamine receptors and may not be shared by nizatidine and famotidine.

Therapeutic Applications of H_2 Antihistamines

H_2 antihistamines are used in the treatment of duodenal ulcers, gastric ulcers, gastroesophageal reflux disease (GERD), pathologic hypersensitivity disorders, and upper gastrointestinal bleeding in critically ill patients and are sold OTC for acid indigestion (61). They are also included in multidrug treatment protocols for eradication of *H. pylori* in treatment of peptic ulcers and prior to surgery to prevent aspiration pneumonitis. Like H_1 antihistamines, H_2 antihistamines are inverse agonists that block the basal level of activity at this receptor. Combinations of H_1 and H_2 antihistamines are useful in idiopathic urticaria not responding to H_1 antihistamines alone and to treat itching and flushing of anaphylaxis, pruritus, and contact dermatitis.

STRUCTURAL REQUIREMENTS H_2 antihistamines specifically designed to decrease the secretion of gastric acid are based on an extensive investigative approach to drug design that began from the structures of partial agonist molecules very closely related to histamine (62). Ultimately, this work resulted in the development of cimetidine (Table 32.10), in which the imidazole ring like that of histamine is maintained. The imidazole ring is substituted with a C-4 methyl group, which in histamine agonists affords H_2 selectivity, a four-atom side chain that includes one sulfur atom (the sulfur atom increases potency compared to carbon and oxygen congeners), and a terminal polar non-basic unit, in this case an N-cyanoguanidine substituent. Guanidines substituted with electron-withdrawing groups have significantly decreased basicity compared to guanidine, and they are neutral (nonprotonated) at physiologic pH. Thus, these are logical substituents to replace the terminal thiourea feature in unsuccessful earlier homologous candidates, metiamide and burimamide. The former agent was not marketed due to untoward effects, including agranulocytosis, and the latter agent lacked significant

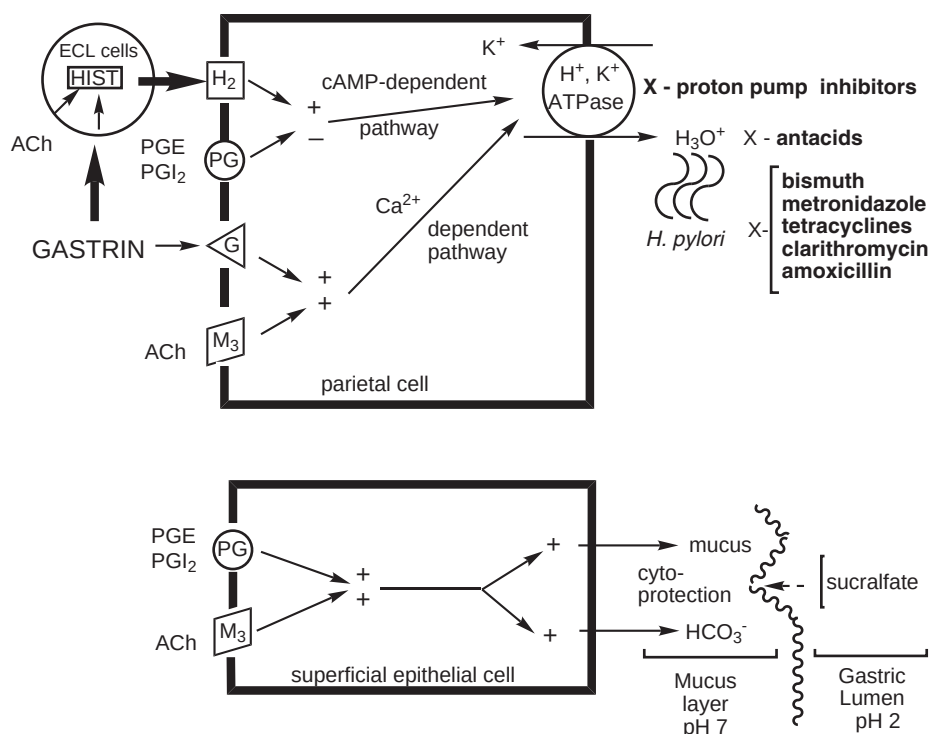


FIGURE 32.17 Secretion of gastric acid and peptic ulcer disease. Histamine is secreted from an endochromaffin-like (ECL) cell, which is innervated by muscarinic receptors (M) via the enteric nervous system and by gastrin receptors (G). Agonist occupation of histamine H_2 receptors in parietal cells leads to gastric acid secretion. Other input at parietal cells includes the prostaglandins (PGs), gastrin, and muscarinic receptors. (Adapted from Hoogerwerf WA, Pasricha PJ. Pharmacotherapy of gastric acidity, peptic ulcers, and gastroesophageal reflux disease. In: Hardman JG, Limbird LE, Mollinoff PB, et al., eds. Goodman and Gilman's the Pharmacological Basis of Therapeutics, 11th Ed. New York: McGraw-Hill, 2005:968, with permission.)

oral bioavailability. Subsequently, the nitromethylene unit was a replacement of the *N*-cyanoimino group in the substituted guanidine analogs, affording compounds of increased potency. Replacement for the imidazole ring with other heteroaromatic rings resulted in other useful analogs.

METABOLISM Cimetidine, ranitidine, and famotidine are subject to first-pass metabolism, and each has oral bioavailability of about 50%. The oral bioavailability of nizatidine is about 90%. All have half-lives of 1.5 to 4 hours, with that of nizatidine being the shortest. Significant amounts of each of these H_2 antihistamines are excreted unchanged, with small amounts of urinary products of sulfoxidation being a common metabolic feature. As expected, hydroxylation of the imidazole C-4 methyl group of cimetidine occurs. Ranitidine is excreted largely unchanged, but minor metabolic pathways include *N*-demethylation and *N*- and *S*-oxidation. The metabolites are not thought to contribute to the therapeutic properties of the parent drugs, with the exception of nizatidine from which the *N*-desmethyl metabolite retains H_2 antihistamine activity (63).

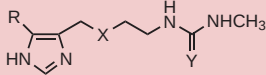
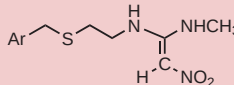
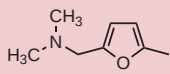
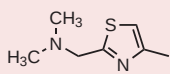
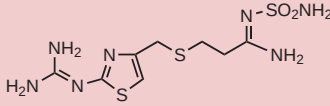
SIDE EFFECTS AND DRUG INTERACTIONS Cimetidine, the earliest of these agents, shows the greatest number of drug interactions (64). Among them are somnolence and confusion in elderly patients with decreased renal function.

Gynecomastia, presumably related to increased prolactin secretion, has been reported. Cimetidine inhibits CYP450-dependent metabolic processes, affording increased concentration of several agents, the most important being those having narrow therapeutic concentration windows (e.g., phenytoin, theophylline, some benzodiazepines, warfarin, quinidine). Inhibition of several CYP450 oxidative processes is associated with the presence of the imidazole ring of cimetidine, which apparently replaces the histidine that serves as a ligand to the porphyrin iron in CYP450 enzymes. Other agents in this group contain heterocyclic rings other than imidazole and do not show this effect. Cimetidine is an inhibitor of renal tubular secretion of some drugs including procainamide. These tubular secretion effects are less prevalent or absent with other agents in this class. The other agents in the group are more potent than cimetidine, and significant differences are noted among them. Of these, ranitidine is the most widely used. The H_2 antihistamines have OTC status and are widely available for gastric hyperacidity.

Proton Pump Inhibitors

Proton pump inhibitors (PPIs) are widely used in the treatment of duodenal and gastric ulcers, erosive esophagitis, GERD, GERD-related laryngitis, and hypersecretory conditions (Zollinger-Ellison syndrome) (65,66). Some of the generic agents are available OTC for the treatment

TABLE 32.10 H₂-Receptor Antihistamines

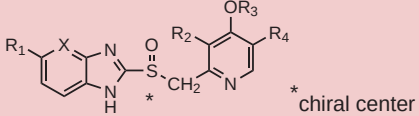
				
Drug	Trade name	R	X	Y
Burimamide		H	CH ₂	S
Metiamide		CH ₃	S	S
Cimetidine	Tagamet	CH ₃	S	N-C≡N
				
Drug	Trade name	Ar		
Ranitidine	Zantac			
Nizatidine	Axid			
 Famotidine (Pepcid)				

of acid reflux disorders. The final step in acid secretion in parietal cells of the gastric mucosa is a process mediated by H⁺/K⁺-ATPase, the gastric proton pump, an enzyme with significant homology to Na⁺/K⁺-ATPase. This H⁺/K⁺-ATPase has some similarities to the H⁺/K⁺-ATPase in osteoclasts, which is involved in bone resorption. Gastric acid secretion can be inhibited in many ways. These include by antagonists at muscarinic, gastrin, or histamine H₂ receptors; by agonists at inhibitory receptors for prostaglandins and somatostatin; by PPIs; or by carbonic anhydrase inhibitors.

MECHANISM OF ACTION Omeprazole and related analogs (Table 32.11), produce inhibition of stimulated gastric acid secretion irrespective of the receptor stimulation process. Nearly all of the compounds are close structural relatives, being weakly basic 2-pyridylmethylsulfinylbenzimidazoles. An analog, tenatoprazole, which is an imidazopyridine isostere, is in clinical trial.

Only a few successful changes of the heterocyclic rings are possible (67). These agents have irreversible effects on the secretion of gastric acid because the molecule rearranges in the strongly acidic environment of the parietal cell. Covalent binding of the rearranged inhibitor to the H⁺/K⁺-ATPase results in inactivation of the catalytic function of the proton pump. A number of cysteine sulphydryl groups are potential sites. There is evidence

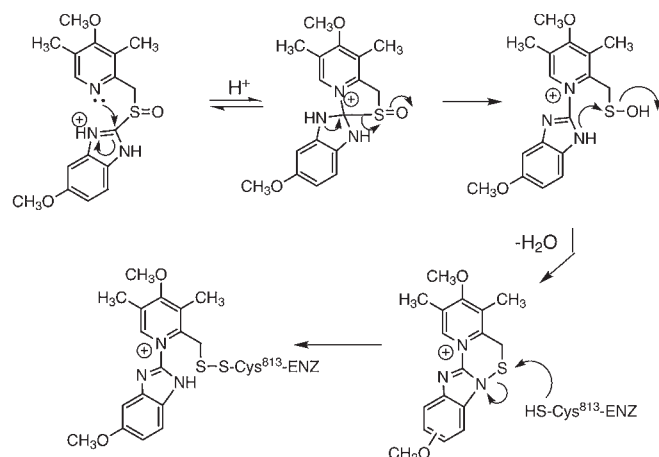
TABLE 32.11 H⁺/K⁺-ATPase Proton Pump Inhibitors

 * chiral center						
Drugs	Trade name	X	R ₁	R ₂	R ₃	R ₄
Omeprazole	Prilosec	CH	OCH ₃	CH ₃	CH ₃	CH ₃
Esomeprazole (S-enantiomer)	Nexium	CH	OCH ₃	CH ₃	CH ₃	CH ₃
Tenatoprazole		N	OCH ₃	CH ₃	CH ₃	CH ₃
Lansoprazole	Prevacid	CH	H	CH ₃	CH ₂ CF ₃	H
Dexlansoprazole (R-enantiomer)	Dexilant	CH	H	CH ₃	CH ₂ CF ₃	H
Rabeprazole	Aciphex	CH	H	CH ₃	(CH ₂) ₃ OCH ₃	H
Pantoprazole	Protonix	CH	OCHF ₂	OCH ₃	CH ₃	H

that two molecules of the intermediate from omeprazole are bound to the active site, and one of these sites has been identified as cysteine-813 (and probably cysteine-892 and/or -822) of the cysteine-rich H⁺/K⁺-ATPase (68). These cysteines are in different environments (e.g., exposed to the lumen or in the membrane), and different PPIs bind differentially to them and other sulphydryl groups. In the covalent binding, disulfide bonds with it are formed. Analogous, but slightly different results are reported for lansoprazole, pantoprazole, and rabeprazole (69). A chemical mechanism for the process is shown in Figure 32.18.

Because the initial rearrangement only occurs at a strongly acidic pH, acid-stable oral dosage forms are used that allow for dissolution, release, and absorption of drug in the duodenum (enteric-coated granules in capsules or enteric-coated tablets). More recently, granular preparations of lansoprazole (enteric-coated granules) and omeprazole (with sodium bicarbonate) have become available. Intravenous dosage forms of lansoprazole and pantoprazole are also available. The acid-catalyzed rearrangement of absorbed drug then occurs selectively in the acidic environment of the canaliculus as it is secreted into the gastric lumen from the parietal cells. Some differences may occur in the molecular sites of binding of the agents, and differences have been noted in recovery times, with rabeprazole having a shorter duration of action (70). Tenatoprazole has the longest plasma half-life and, in testing, appears to offer better control of nocturnal hyperacidity (71).

CYP450 METABOLISM Metabolism of omeprazole and other PPIs occurs primarily in the liver (Fig. 32.19). The sulfone, sulfide, and hydroxylated and O-demethylated metabolites have been reported as products.



covalently bound species from omeprazole

FIGURE 32.18 Acid-catalyzed activation of omeprazole to reactive sulfenamide. At the parietal cell, H^+/K^+ -ATPase, a cysteine residue, reacts to form disulfide-attached enzyme inhibitor.

Omeprazole is a substrate primarily for CYP2C19 and may elevate concentrations of other substrates for this enzyme (diazepam) when given concurrently. CYP3A4 contributes to a lesser extent. Further oxidation of the sulfone affords additional metabolites, which are excreted in the feces. Lansoprazole is metabolized by analogous routes (72). Fewer drug interactions with lansoprazole have been reported, although it is also a substrate for CYP2C19.

These sulfoxides have a chiral sulfur atom, and recent work has been reported on the effects of stereochemistry on pharmacologic and dispositional characteristics. The oxidative metabolism of omeprazole is catalyzed principally by CYP2C19 (primarily 5'-hydroxylation and to a lesser extent benzimidazole *O*-demethylation) (73). In human liver microsomes, the *R*-(+)-enantiomer is cleared more rapidly, and it is almost exclusively metabolized by CYP2C19. The clearance of the *S*-(-)-enantiomer is dependent both on oxidation by CYP2C19 and to a lesser extent on CYP3A4-mediated metabolism, primarily sulfone formation. The marketed single enantiomer, esomeprazole, provides greater bioavailability in CYP2C19 extensive metabolizers and less interindividual

variation between extensive and slow CYP2C19 metabolizer populations (~3% of Caucasians and up to 15% to 20% of Asians). Thus the impact of variant alleles of CYP2C19 is less on the *S*-(-)-enantiomer than on the parent racemate. Higher blood levels and greater AUCs are observed in extensive metabolizers, and increases in the duration of gastric pH >4.0 are observed, which are correlated with healing rates. Esomeprazole, the *S*-(-)-enantiomer of omeprazole, has been marketed.

Different PPIs depend differently on CYP2C19 for oxidative metabolism, and the enantiomers show variation in dependence on CYP2C19 and other pathways (principally CYP3A4). Pantoprazole and lansoprazole show greater metabolism via CYP2C19, with the enantiomers being affected differently, and rabeprazole is metabolized to a lesser extent by oxidative CYP450 enzymes. Lansoprazole, pantoprazole, and rabeprazole all lack the pyridyl 5-methyl group, a major site of CYP2C19 oxidation, but CYP2C19 oxidations occur at other sites in each of them.

Risk of adverse outcomes associated with concomitant use of clopidogrel with PPIs following acute coronary syndrome has been reported. Platelet antiaggregatory activity of clopidogrel is dependent on CYP2C19 oxidation to a reactive metabolite; thus PPIs are competitive substrates with clopidogrel via this pathway (see Chapter 26).

Single-Enantiomer Proton Pump Inhibitors

The development of single enantiomers has been of interest in this class of compounds for more than 15 years. The primary reasons for such development are to obtain more optimal treatment and specificity of effects, potentially resulting in exposure of nontarget tissues to lower drug concentrations. The development of improved stereoselective syntheses and assay techniques has facilitated these efforts. A quantitative difference in pharmacologic effects of the two enantiomers provides impetus to develop the enantiomer with higher affinity or potency, assuming it has no increased adverse pharmacologic effects or toxicity and/or no undesirable pharmacokinetic properties (74). When only one enantiomer has desirable pharmacologic properties, the term "isomeric ballast" has been used to characterize the much less active or inactive enantiomer, because it contributes little to the observed effects. However, for the benzimidazole sulfoxide PPIs, the enantiomers are approximately equiactive in vitro. The chemical process to the reactive intermediate occurs with loss of the chiral center (Fig. 32.18).

For omeprazole, exposure of the target (H^+/K^+ -ATPase at the parietal cell) as represented by the AUC was correlated to inhibition of gastric acid secretion, as expected in a normal log-linear dose-effect relationship curve (75). Differential metabolism of the enantiomers of omeprazole by the important isozymes CYP2C19 and CYP3A4 occurs. From racemic omeprazole, the AUCs of the *S*-enantiomer differed about threefold between extensive and slow metabolizer genotypes and about 7.5-fold for the *R*-enantiomer, thus arguing that esomeprazole

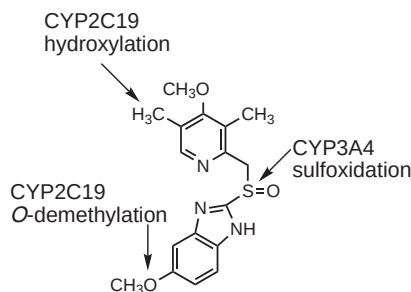


FIGURE 32.19 Oxidation sites and CYP450 isozymes in the metabolism of omeprazole.

TABLE 32.12 Examples of Prokinetic Agents

R_1	R_2	
	$-\text{CH}_2\text{CH}_2\text{NEt}_2$	CH_3 Metoclopramide
		CH_3 Cisapride
	$-\text{CH}_2\text{CH}_2\text{N(CH}_2\text{CH}_2\text{O)}\text{CH}_2\text{C}_6\text{H}_4\text{F}$	C_2H_5 Mosapride
		Prucalopride
		Levosulpiride
		Domperidone
		Itopride
		Tegaserod

antagonist. Metoclopramide and some other prokinetic drugs also have 5-HT₃ antagonism effects. Metoclopramide is used in GERD, diabetic gastroparesis, and nausea and vomiting caused by emetogenic cancer chemotherapy.

Cisapride was removed from the US prescription market because of metabolism-based interactions. In the presence of competing CYP3A4 substrates, high concentrations of the parent molecule lead to life-threatening cardiac arrhythmias via interaction with the hERG K⁺ channel, similar to H₁ antihistamines terfenadine and astemizole. It is available only through an investigational limited access program.

Structurally related agents, some of which are available outside the United States, are in clinical trial. Among them are prucalopride, a potent and selective 5-HT₄ receptor agonist used for chronic constipation (85); mosapride, which is used primarily for gastritis (86); and itopride, which is also being tried for gastritis (87). Tegaserod is also a selective 5-HT₄ partial agonist that is used in irritable bowel syndrome in women and has been used successfully in critically ill patients with gastroparesis.

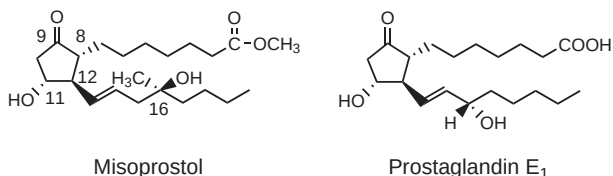
Other agents (e.g., domperidone, a specific peripheral D₂-receptor antagonist) also have effects on the upper gastrointestinal tract similar to metoclopramide and are used for antiemetic effects and the treatment of gastroparesis and pediatric gastroesophageal reflux. Domperidone is available as an investigational drug for patients with severe gastrointestinal disorders who are refractory to other therapies.

Erythromycin, a macrolide antibiotic (Chapter 33), is also used as a prokinetic agent. It is a motilin receptor agonist. Motilin is a polypeptide hormone in endocrine cells of the distal stomach and the duodenum. It increases lower esophageal sphincter pressure and is responsible for the migrating motor complex (i.e., the process of irregular and subsequently regular contractions that clear the stomach of undigested food).

Prostaglandins

Prostaglandins have antisecretory effects on gastric acid (Fig. 32.17). Besides inhibiting adenylyl cyclase activity in parietal cells that results in secretion of gastric acid, prostaglandins stimulate secretion of mucus and bicarbonate in adjacent superficial cells. Cytoprotective effects of endogenous E series prostaglandins and of other more stable synthetic congeners are observed. The only available oral prostaglandin in the United States is misoprostol. The orally administered carboxylic acid ester is hydrolyzed to the pharmacologically active carboxylic acid. It is a synthetic analog of prostaglandin E₁, in which structural changes at C-13, C-14, C-15, and C-16 are made to prevent rapid metabolic conversion to inactive products. The presence of the tertiary alcohol one carbon removed to C-16 obviates the usual conversion of the allylic secondary alcohol ($\Delta^{13,14}$ -alcohol) of prostaglandins to the corresponding saturated ketones. A mixture of diastereomers of misoprostol is used; most of the activity arises from the 11*R*,16*S*-isomer (88). Misoprostol

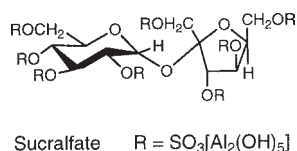
reduces basal levels of gastric acid secretion, but it has considerable smooth muscle contraction effects.



THERAPEUTIC APPLICATIONS AND SIDE EFFECTS In the United States, misoprostol is administered with some current NSAIDs to reduce the risk of complications of gastric ulceration and bleeding. A primary mechanism of action of NSAIDs is derived from their inhibition of formation of prostaglandins from arachidonic acid. A prostaglandin may be added for the duration of NSAID therapy. A combination product of diclofenac (an NSAID) and misoprostol is available. Unlabelled uses of misoprostol include treatment of duodenal ulcers, for which it appears effective, and treatment of duodenal ulcers unresponsive to H₂ antagonists. In other countries, it has been used in the treatment of duodenal ulcers.

Significant side effects are those associated with its abortifacient properties and other smooth muscle contraction effects (e.g., diarrhea and abdominal pain). Misoprostol is also an effective cervical ripening agent (by vaginal application) for the induction of labor. Other unlabeled uses include the treatment of postpartum hemorrhage and, with mifepristone, pregnancy termination.

Sucralfate and Insoluble Bismuth Preparations



Sucralfate is a complex of the sulfuric acid ester of sucrose and aluminum hydroxide. Secondary polymerization with aluminum hydroxide forms intermolecular bridges between molecules of sulfate esters with aluminum (89). Limited dissociation of the complex occurs in gastric acid, but these anionic sulfate esters form insoluble adherent complexes with the proteinaceous exudate at the abraded surface of a crater of the ulcerated area in the stomach. This physical complex protects the ulcer from the erosive action of pepsin and bile salts. Sucralfate also stimulates synthesis and release of prostaglandins, bicarbonate, and epidermal and fibroblast growth factors. Significant ulcer healing effects are noted in placebo-controlled trials. Only small amounts of sucralfate are absorbed systemically. In renal impairment, there is a risk of accumulation of absorbed aluminum from the drug. Sucralfate reduces absorption of other drugs including H₂ antihistamines, quinolone antibiotics, phenytoin, and perhaps warfarin (90).

Bismuth-containing preparations (those containing colloidal salts of bismuth subsalicylate) have effects similar to sucralfate apparently due to their similar physical properties and coating effects. The combination of an H₂ antihistamine along with a three-drug combination of metronidazole, tetracycline, and potassium bismuth subsalicylate is used as a second-line approach to eradication of *H. pylori* in the treatment and prevention of recurrence of duodenal ulcers.

H₃-RECEPTOR AGONISTS AND ANTAGONISTS

Physiologic Role of H₃ Receptors

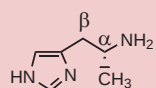
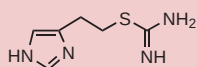
The H₃ receptor was identified as an autoreceptor that regulates the release of histamine. Histaminergic neurons are located primarily in the hippocampus projecting to all major areas of the brain. These neurons are involved with the regulation of sleep and wakefulness and feeding and memory (17). The H₃ receptor is also a heteroreceptor in the CNS, where it is involved in the regulation of synthesis and release of other neurotransmitters. In addition, H₃ heteroreceptors have been identified in peripheral tissues, including the airway and gastrointestinal tract.

H₃ Agonists and Antagonists

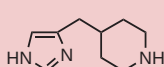
Several H₃ agonists and antagonists have been studied (Table 32.13). *R*-α-Methylhistamine is a more potent agonist than is histamine. The addition of methyl groups to the side chain or to the aliphatic amine nitrogen of histamine usually results in potent H₃ agonists (e.g., α,α-dimethylhistamine, *R*-α,β-dimethylhistamine, and *N*-methyl- and *N*-ethylhistamine), unlike the deleterious effect of these changes on H₁ and H₂ agonist activity. Other selective agonists are the isothiourea imetit, immepip and its *N*-methyl analog methimepip, both substituted piperidines, immethridine, a pyridine congener of immepip, and SCH 50971. All retain the imidazole ring of histamine.

Early H₃ antagonists, like thioperamide and clobenpropit analogs of immepip and imetit, respectively, are primarily pharmacologic tools. Like the H₃ agonists, these agents retain the imidazole group but possess widely varying *N*-substituents. Some of the imidazole-containing agonist and antagonist analogs have affinity for other receptors (e.g., some α-adrenergic receptor subtypes), and they have the potential to interact with CYP450 enzymes as an iron-porphyrin ligand. Thus, non-imidazole-containing ligands have been sought.

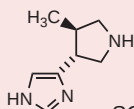
Centrally acting H₃ antagonists are under study by several drug companies. Agents are sought for a variety of disorders. These include treatment of depression, mild cognitive impairment, Alzheimer disease, schizophrenia, narcolepsy, obesity, and attention-deficient hyperactivity disorder. Examples of lead compounds, some of which have entered clinical trial, are tiprolisant, JNJ 1018146, and GSK-189254 (91,92).

TABLE 32.13 Examples of H_3 Agonists and Antagonist **H_3 Agonists***R*- α -Methylhistamine

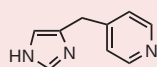
Imetit



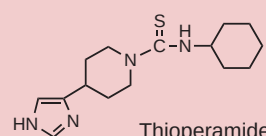
Immepip



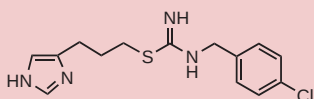
SCH 50971



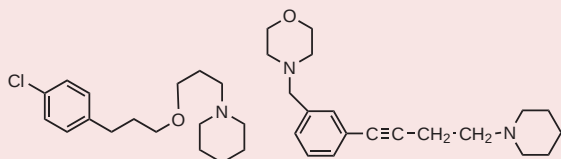
Immethridine

 H_3 Antagonists

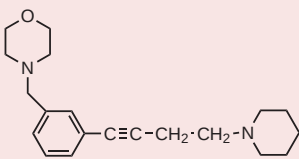
Thioperamide



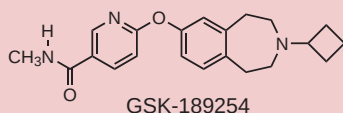
Clobenpropit



Tiprolisant



JNJ-10181457



GSK-189254

sequence homology is greatest with H_3 receptor (35% to 40%) and is very low versus H_1 and H_2 receptors (93).

The localization of H_4 receptors largely in hematopoietic cells such as dendritic cells, mast cells, monocytes, eosinophils, basophils, and T cells, with a role in chemotaxis and mediator release, led to reevaluation of their role in physiologic effects in altered responses to inflammatory stimuli. As such, histamine has taken on a larger role in T-cell responses (94,95).

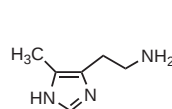
 H_4 Agonists and Antagonists

Classical leukocyte chemoattractant effects that occur are blocked by thioperamide, an H_3/H_4 antihistamine. Other H_3 receptor ligands such as *R*- α -methylhistamine and imetit are also agonists at H_4 receptors, but they have lower affinity than at H_3 receptors (Table 32.13). Clobenpropit is a partial agonist at H_4 receptors, but it is an H_3 antihistamine (inverse agonist). Thioperamide is an inverse agonist at H_4 receptors and at H_3 receptors. 4-Methylhistamine (Fig. 32.21) has greater affinity (>100-fold) for H_4 receptors than other histamine receptor subtypes (96).

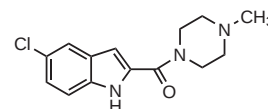
Selective H_4 antihistamines have been reported, beginning from JNJ 7777120 (Fig. 32.21). It blocks many mediated functions, including histamine-induced chemotaxis in mast cells and eosinophils (97). Conditions in which H_4 antagonists may be useful include several autoimmune inflammatory and allergic disorders, including rheumatoid arthritis, asthma, allergic rhinitis, pruritus, and inflammatory bowel disease. Nasal stuffiness and blockage in allergic rhinitis, conditions that are poorly treated with H_1 and H_2 antihistamines, can also possibly be treated using H_4 antihistamines. Analogs of JNJ 7777120 and congeners in related nitrogen-containing heterocyclic systems are being investigated (98,99).

 H_4 -RECEPTOR AGONISTS AND ANTAGONISTS**Physiologic Role of H_4 Receptors**

The H_4 receptor is expressed in mast cells, eosinophils, and other cells of hematopoietic lineage, including basophils and T cells. It is a relatively recent discovery; thus, much is not yet understood about its physiologic role. It is thought to play a role in inflammatory responses, especially in the mediation of chemotactic responses. Its



4-Methylhistamine



JNJ 7777120

FIGURE 32.21 Other ligands at the H_4 receptor.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

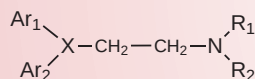
Anne Pace, PharmD

The pharmacist assesses the patient as having seasonal allergies and recommends one of the over-the-counter second-generation antihistamines, loratadine or cetirizine. The pharmacist does not recommend one of the first-generation antihistamines because of a drug–disease interaction. CP has benign prostatic hyperplasia, which causes urinary retention that will be worsened if CP uses a first-generation antihistamine.

Chemical Analysis

S. William Zito and Victoria Roche

H₁ antihistamines were thought of as reversible competitive inhibitors of the interaction of histamine with the H₁ receptors. However, it is now recognized that they are inverse agonists that bind to the inactive conformation of the H₁ receptor shifting the active-inactive equilibrium of the receptor toward the inactive state. All H₁-receptor antagonists have the following pharmacophore:



The X group defines the chemical classes:

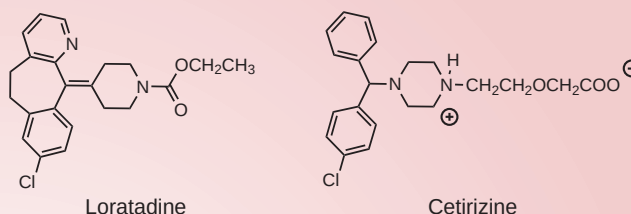
Aminoalkyl ether X = CH₂-O-

Ethylenediamines X = -N-, phenothiazines, piperazines

Alkylamines X = -C-(sp³), -C= (sp²)

Both aromatic rings are necessary for antagonistic activity at the H₁ receptor. Having only one aromatic ring makes these compounds more like histamine, and therefore, they might have agonist activity. Sometimes the two aromatic rings are bridged,

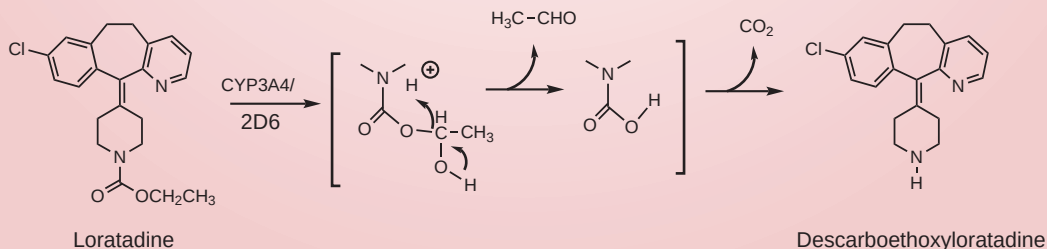
as in loratadine, or the aminoethyl function may be part of a ring structure, as in cetirizine.



The first-generation antihistamines have adverse central effects (sedation, drowsiness, decreased cognitive ability, and somnolence) with the following general order of sedative action: aminoalkyl ethers > ethylenediamines > alkylamines. In addition, they also have significant antimuscarinic action including blurred vision, dry mouth, urinary retention, and constipation.

The second-generation antihistamines are nonsedating with little or no effect on muscarinic receptors. These agents possess selective peripheral H₁ antihistaminic effects and apparently do not penetrate the blood–brain barrier. Cetirizine does not cross into the CNS because it exists primarily as a zwitterion in the body. However, it is not clear why loratadine or its active metabolite descarboethoxyloratidine do not achieve significant concentration in the CNS. Perhaps conversion of loratadine to the more active descarboethoxyloratidine, which requires CYP3A4 and CYP2D6, does not easily occur in the CNS.

In this scenario, CP has benign prostatic hyperplasia. Symptoms of benign prostatic hyperplasia are those of bladder outlet obstruction (i.e., weak stream, hesitancy, urinary frequency, urgency, nocturia, incomplete emptying, terminal dribbling, overflow or urge incontinence, and complete urinary retention). Therefore, the first-generation H₁ antihistamines, especially the aminoalkyl ethers, are contraindicated because of their antimuscarinic activity, which can slow urine flow and exacerbate urinary retention.



CASE STUDY

S. William Zito and Victoria Roche

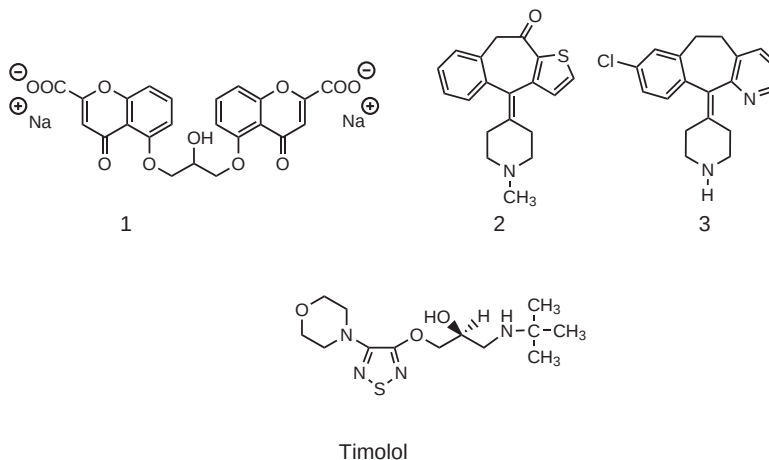
FT is a 45-year-old man who visits his ear, nose, and throat specialist complaining of severe allergic rhinitis and itching, burning, and tearing in both eyes that is affecting his vision. FT is a New York City taxi driver and is worried about his ability to

drive safely. A review of FT's medical records reveals he has a family history of allergic diseases. He has glaucoma and often suffers from bouts of eczema. The glaucoma is treated with 0.25% timolol maleate (1 gtt bid), and eczema flares respond

well to treatment with an over-the-counter topical steroid. FT states that these recent symptoms began shortly after he picked up a woman at the local airport who was relocating from the Midwest and transporting four cats to her new home in New York. His physician diagnoses FT with an exacerbation of atopic keratoconjunctivitis (AKC). He prescribes fluticasone (nasal spray) to treat the allergic rhinitis but is reluctant to prescribe steroid eye drops to treat the AKC because ocular steroids can

promote further development of FT's glaucoma. Which of the following antihistamines would be the best choice in this case?

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.



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Chapter 33

Antibiotics and Antimicrobial Agents

ELMER J. GENTRY

Drugs Covered in This Chapter*

ANTIBACTERIALS

- Fosfomycin
- Methenamine
- Metroindazole
- Nitrofurantoin
- Quinolone class
 - Besifloxacin
 - Ciprofloxacin
 - Gatifloxacin
 - Gemifloxacin
 - Levofloxacin, ofloxacin
 - Moxifloxacin
 - Norfloxacin
- Sulfonamide class
 - Silver sulfadiazine
 - Sulfacetamide
 - Sulfamethoxazole
 - Sulfisoxazole
- Trimethoprim

ANTIBIOTICS

- Penicillin class
 - Amoxicillin
 - Ampicillin
 - *Bacampicillin*
 - Benzylpenicillin
 - *Carbenicillin and indanyl carbenicillin*
 - Dicloxacillin
 - *Methicillin*
 - *Mezlocillin*

- Nafcillin
- Oxacillin
- Phenoxymethylpenicillin
- Piperacillin
- Ticarcillin
- β -Lactamase Inhibitors
 - Clavulanic acid
 - Sulbactam
 - Tazobactam
- Cephalosporin class
 - Cefaclor
 - Cefadroxil
 - Cefazolin
 - Cefdinir
 - Cefditoren pivoxil
 - Cefepime
 - Cefixime
 - Cefotaxime
 - Cefotetan
 - Cefoxitin
 - Cefpodoxime proxetil
 - Cefprozil
 - Ceftaroline
 - Ceftazidime
 - Ceftibuten
 - Ceftizoxime
 - Ceftriaxone
 - Cefuroxime
 - Cephalixin

- Carbapenems
 - Doripenem
 - Ertapenem
 - Imipenem/cilastatin
 - Meropenem
- Monobactams
 - Aztreonam
- Aminoglycosides
 - Amikacin
 - Gentamicin
 - Kanamycin
 - Neomycin
 - *Spectinomycin*
 - Streptomycin
 - Tobramycin
- Macrolides and ketolides
 - Azithromycin
 - Clarithromycin
 - *Erythromycin estolate*
 - Erythromycin ethylsuccinate
 - Erythromycin stearate
 - Telithromycin
- Lincosamides
 - Clindamycin
 - Lincomycin
- Tetracyclines and glycylcyclines
 - Demeclocycline
 - Doxycycline
 - Minocycline

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

- Tetracycline
- Tigecycline
- Special Purpose Antibiotics
 - Bacitracin
 - Chloramphenicol
- Daptomycin
- Linezolid
- Mupirocin
- Polymyxin B
- Quinupristin/dalfopristin
- Retapamulin
- Telavancin
- Vancomycin

Abbreviations

AAC , aminoglycoside acetylase	CNS , central nervous system	MRSA , methicillin-resistant <i>Staphylococcus aureus</i>
ANT , aminoglycoside nucleotide transferase	GABA , γ -aminobutyric acid	MTT , methylthiotetrazole
6-APA , 6-aminopenicillanic acid	GI , gastrointestinal	PABA , <i>p</i> -aminobenzoic acid
APH , aminoglycoside phosphorylase	HABA , L-hydroxyaminobutyryl amide	PBP , penicillin-binding protein
ATP , adenosine triphosphate	MIC , minimum inhibitory concentration	

SCENARIO

Elizabeth Coyle, PharmD

ER is a 24-year-old man who is in the intensive care unit after being in a motor vehicle accident 5 days ago. He sustained a laceration of his spleen, which was removed during surgery shortly after his arrival at the hospital. ER has been mechanically ventilated since surgery and has been stable for the last 96 hours; however, on rounds this morning the nurse states that ER spiked a fever (103°F) at 6 AM and his respiratory secretions have become more mucopurulent compared with yesterday. The team reviews the chest radiograph and appreciates new diffuse, bilateral infiltrates. Cultures are taken from his respiratory secretions,

and ER is started empirically on cefepime 2 g every 8 hours and vancomycin 1 g every 12 hours. Two days later ER is not improving and the culture results come back as *Escherichia coli* that is sensitive only to carbapenems and aminoglycosides and has intermediate susceptibility to cefepime and piperacillin/tazobactam. The results also note that the *E. coli* is an extended spectrum beta-lactamase (ESBL)-producing organism.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

Antibiotics are microbial metabolites or synthetic analogs inspired by them that, in small doses, inhibit the growth and survival of microorganisms without serious toxicity to the host. Selective toxicity is the key concept. Antibiotics are among the most frequently prescribed medications today, although microbial resistance due to evolutionary pressures and misuse threatens their continued efficacy. In many cases, the utility of natural antibiotics has been improved through medicinal chemical manipulation of the original structure, leading to broader antimicrobial spectrum, greater potency, lesser toxicity, more convenient administration, and additional pharmacokinetic advantages. Synthetic substances that are unrelated to natural products but still inhibit or kill microorganisms are referred to as antimicrobial agents.

All of the parts of our bodies that are in contact with the environment support microbial life. It is estimated that our body contains approximately 10 times more bacterial cells than human cells. These organisms are generally harmless to the host and actually provide a number of benefits. However, all of our internal fluids, organs, and body structures are sterile under normal circumstances, and the presence of bacteria, fungi, viruses, or other organisms in these places is diagnostic evidence of infection. When mild

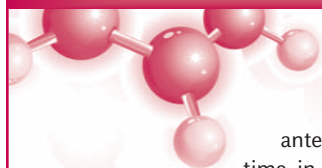
microbial disease occurs, an otherwise healthy patient will often recover without requiring treatment because their immune system is called upon to kill invasive microorganisms. When this is insufficient to protect us, appropriate therapeutic intervention is indicated.

HISTORY

Humankind has been subject to infection by microorganisms since before recorded history. One presumes that mankind has been searching for suitable therapy for nearly as long. This was a desperately difficult enterprise given the acute nature of most infections and the nearly total lack of understanding of their origins until the 19th century. Although one can find indications in ancient medical writings of folkloric use of plant and animal preparations, these factors were inefficiently applied, and they often failed. Until the discovery of bacteria by van Leeuwenhoek in 1676 and subsequent understanding of their role in infection about 150 years later, there was no hope for rational therapy.

In the 19th century, Robert Koch showed that specific microorganisms could always be isolated from the excreta and tissues of people with particular infectious diseases and that these same microorganisms were usually absent in healthy individuals. They could then be

CLINICAL SIGNIFICANCE



The treatment of bacterial infections is one of the few disease states that all clinicians are guaranteed to be challenged with at some time in their career. Because bacteria are constantly changing, the selection of appropriate antimicrobial therapy is crucial in providing efficacious treatment of infections. It is of the utmost importance that clinicians understand the medicinal chemistry of antimicrobial agents to choose the most effective therapy for their patients. Antimicrobial resistance trends are constantly changing and vary from institution to institution; therefore, a complete understanding of the structural relationship differences between antibiotics, even within the same class, is helpful when selecting the most appropriate therapy for an individual patient. In addition, side chains and subtle structural differences within antimicrobial classes can change the side effect profiles of these agents. The informed clinician can optimize treatment choices from patient to patient while avoiding severe adverse effects.

The development of newer and more effective antimicrobial agents is essential in the fight against infectious diseases. Understanding the medicinal chemistry of the older and newer antimicrobial agents helps the clinician to comprehend their differences in antimicrobial spectrum and side effect profile. Newer antibiotics are continuously being developed; however, many of them are from antimicrobial classes already established with small changes on the functional groups and/or side chains, leading to dramatic differences in their antimicrobial spectra. As stated previously, bacteria are constantly changing, as are antimicrobial susceptibility patterns. To optimize patient care, the clinician needs to stay informed of the effects of the medicinal chemistry differences as newer drugs come to market and antimicrobial resistance trends change.

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grown on culture media and be administered to healthy individuals so as to reproduce in healthy individuals all of the classic symptoms of the same disease. Finally, the identical microorganism could then be isolated from this deliberately infected person. Following these rules, at long last, a chain of evidence connecting cause and effect was forged between certain microorganisms and specific infectious diseases. This work laid the foundation for rational prevention and therapy of infectious diseases.

Louis Pasteur reported in 1877 that when what he termed “common bacteria” were introduced into a pure culture of anthrax bacilli, the bacilli died, and that an injection of deadly anthrax bacillus into a laboratory animal was harmless if “common bacteria” were injected along with it. This did not always work but led to the appreciation of antibiosis, wherein two or more microorganisms competed with one another for survival (1). However, it was more than a half century later that the underlying mechanisms began to be appreciated and applied to achieve successful therapy.

The modern anti-infective era opened with the discovery of the sulfonamides in France and Germany in 1936 as an offshoot of Paul Ehrlich’s earlier achievements in treating infections with organometallics and his theories of vital staining (2). The well-known observation of a clear zone of inhibition (lysis) in a bacterial colony surrounding a colony of contaminating airborne *Penicillium* mold by Alexander Fleming in England in 1929 and the subsequent purification of penicillin from it in the late 1930s and early 1940s by Florey, Chain, Abraham, and Heatley, provided important additional impetus (3,4). With the first successful clinical trial of crude penicillin in 1941 and the requirements of war times, an explosion of successful activity ensued that continues into the 21st

century. In rapid succession, deliberate searches of the metabolic products of a wide variety of soil microbes led to discovery of tyrothricin (1939), streptomycin (1943), chloramphenicol (1947), chlortetracycline (1948), neomycin (1949), and erythromycin (1952). These discoveries ushered in the age of the so-called miracle drugs. The discovery of antibiotics is widely considered to be one of the top five discoveries/inventions of the 20th century.

Microbes of soil origin remain to this day one of the more fruitful sources of antibiotics, although the specific means employed for their discovery are more sophisticated today than those employed 70 years ago. Initially, extracts of fermentations were screened simply for their ability to kill pathogenic microorganisms *in vitro*. Those that did were pushed along through ever more complex pharmacologic and toxicologic tests in attempts to discover clinically useful agents. Today, many thousands of such extracts of increasingly unusual microbes are tested each year, and the tests now include sophisticated assays for agents operating through particular biochemical mechanisms or possessing particular properties.

Combinatorial chemical synthesis coupled with high-throughput screening today make it possible to screen hundreds of thousands of compounds in a short time for antimicrobial activity. This is coupled with dramatic advances in all of the relevant sciences. One would logically suppose that this would lead to the emergence of a large number of new antimicrobial agents. That this is yet to happen is a measure of the complexity of the task. The impact of genomics and proteomics is predicted to have a larger effect on this effort. The genome of *Haemophilus influenzae* was determined in 1995, and currently more than 1,000 microbial genomes have been deciphered and are publicly available (5,6). Of the more than 1,700 genes

of *H. influenzae*, it is thought that 642 are essential and thus are potential targets for antimicrobial drug development (7). These exciting new possibilities have yet to yield practical results due to the inherent complexity of the task.

This picture has an increasing dark side, however, because of the increasing impact of bacterial resistance. Intrinsic resistance to antimicrobial agents (resistance present before exposure to antibiotics) was recognized from the beginning. Some bacteria are immune to treatment from the outset because they do not take up the antibiotic or lack a susceptible target. Starting in the 1940s, however, and encountered with increasing frequency to this day, bacteria that were previously expected to respond were found to be resistant, many bacteria became resistant during the course of chemotherapy, and others were simultaneously resistant to several different antibiotics. The organisms were found to be capable of passing this trait on to other bacteria, even to those belonging to different genera. The spread of this phenomenon is aided by microorganisms' short generation time (sometimes measured in fractions of an hour) and genetic versatility, as well as by poor antibiotic prescribing and utilization practices. Some authorities predict an impending return to the defenseless days of the preantibiotic era. An understanding of these phenomena and the devising of appropriate practical response measures are important contemporary priorities.

GENERAL THERAPEUTIC APPROACH

Drug Nomenclature

The names given to antimicrobials and antibiotics are as varied as their inventor's taste, and yet some helpful unifying conventions are followed. For example, the penicillins are derived from fungi and have names ending in the suffix -cillin, as in the term ampicillin. The cephalosporins are likewise fungal products, although their names mostly begin with the prefix cef- (or sometimes, following the English practice, spelled ceph-). The synthetic fluoroquinolones mostly end in the suffix -floxacin. Although helpful in many respects, this nomenclature does result in many related substances possessing quite similar names. This can make remembering them quite difficult. Most of the remaining antibiotics are produced by fermentation of soil microorganisms belonging to various *Streptomyces* species. By convention, these have names ending in the suffix -mycin, as in streptomycin. Some prominent antibiotics are produced by fermentation of various soil microbes known as *Micromonospora* sp. These antibiotics have names ending in -micin (e.g., gentamicin).

In earlier times, the terms "broad spectrum" and "narrow spectrum" had specific clinical meaning. The widespread emergence of microbes resistant to single agents and multiple agents has made these terms less meaningful. It is, nonetheless, still valuable to remember that some antimicrobial families have the potential of inhibiting a wide range of bacterial genera belonging to both gram-positive and gram-negative cultures and so

are called broad spectrum (such as the tetracyclines). Others inhibit only a few bacterial genera and are termed narrow spectrum (such as the glycopeptides, typified by vancomycin, which are used almost exclusively for a few gram-positive and anaerobic microorganisms).

The Importance of Identification of the Pathogen Empiric-Based Therapy

Fundamental to appropriate antimicrobial therapy is an appreciation that individual species of bacteria are associated with particular infective diseases and that specific antibiotics are more likely to be useful than others for killing them. Sometimes this can be used as the basis for successful empiric therapy. For example, first-course community-acquired urinary tract infections in otherwise healthy individuals are commonly caused by gram-negative *Escherichia coli*. Even just knowing this much can give the physician several convenient choices for useful therapy. Likewise, skin infections, such as boils, are commonly the result of infection with gram-positive *Staphylococcus aureus*. In most other cases, the cause of the disease is less obvious and so likewise is the agent that might be useful against it. It is important to determine the specific disease one is dealing with in these cases and what susceptibility patterns are exhibited by the causative microorganism. Knowing these factors enables the clinician

GRAM STAIN

Hans C. J. Gram, a Danish microbiologist, developed the Gram staining method for staining bacteria so that they were more readily visible under the microscope. The term has proven to be particularly useful in describing antibiotics as well because antibiotics are conveniently classified by their activity against microorganisms depending on their reaction to this method. Gram-positive microorganisms are stained purple by contact with a methyl violet-iodine process. This is largely a consequence of their lack of an outer membrane and the nature of the thick cell wall surrounding them. Gram-negative microorganisms do not retain the methyl violet-iodine stain when washed with alcohol but rather are colored pink when subsequently treated with the red dye safranin. The lipopolysaccharides on their outer membrane apparently are responsible for the staining behavior of gram-negative cells.

Since the Gram stain is dependent on the outer layers of bacterial cells and this also strongly influences the ability of antimicrobial agents to reach their cellular targets, knowing the Gram staining behavior of infectious bacteria helps one decide which antimicrobial might be effective in therapy.

Not all bacteria can be stained by the Gram procedure. These often require special staining processes for visualization. Among the more prominent of these for our purposes are the mycobacteria (the causative agents of tuberculosis, for example). These very waxy cells are called acid-fast and are stained by carbol fuchsin instead.

to narrow the range of therapeutic choices. The only certainty, however, is that inability of a given antibiotic to kill or inhibit a given pathogen *in vitro* is a virtual guarantee that the drug will fail *in vivo*. Unfortunately, activity *in vitro* often also results in failure to cure *in vivo*, but in these cases, at least, there is a significant possibility of success. Before the emergence of widespread bacterial resistance, identification of the causative microorganism often was sufficient for selecting a useful antibiotic. Now this is only a useful first step, and much more detailed laboratory studies are needed in order to make a successful choice.

Experimentally Based Therapy

The modern clinical application of Koch's discoveries to the selection of an appropriate antibiotic involves sampling infectious material from a patient before instituting anti-infective chemotherapy, culturing the microorganism on suitable growth media, and identifying its genus and species. The bacterium in question is then grown in the presence of a variety of antibiotics to see which of them will inhibit its growth or survival and what concentrations will be needed to achieve this result. This is expressed in minimum inhibitory concentration (MIC) units. The term MIC refers to the concentration that will inhibit 99% or more of the microbe in question and represents the minimum quantity that must reach the site of the infection in order to be useful. These concepts are illustrated in Figure 33.1. To "cure" the infection, it is usually

desirable to have several multiples of the MIC at the site of infection. This requires not only an understanding of the MIC, but also an understanding of pharmacokinetic and pharmacodynamic considerations as well as the results of accumulated clinical experience. The choice of anti-infective agent is made from among those that are active. One of the most convenient experimental procedures is that of Kirby and Bauer. With this technique, sterile filter paper disks impregnated with fixed doses of commercially available antibiotics are placed on the seeded Petri dish. The dish is then incubated for a period of time. If the antibiotic is active against the particular strain of bacterium isolated from the patient, a clear zone of inhibition will be seen around the disk. If a given antimicrobial agent is ineffective, the bacterium may even grow right up to the edge of the disk. The diameter of the inhibition zone is directly proportional to the degree of sensitivity of the bacterial strain and the concentration of the antibiotic in question. Currently, a given zone size in millimeters is dictated above which the bacterium is sensitive and below which it is resistant. When the zone size obtained is near this break point (the break point represents the maximum clinically achievable concentration of an anti-infective agent), the drug is regarded as intermediate in sensitivity, and clinical failure can occur. This powerful methodology gives the clinician a choice of possible antibiotics to use. This method is illustrated in Figure 33.2. The widespread occurrence of resistance of certain strains of bacteria to given antibiotics reinforces the need to perform susceptibility testing. Other laboratory methods can be employed for similar purposes. Of particular note is the Epsilon test (E test), which uses the same idea but employs a gradient of drug concentrations on a filter paper strip. This high level of scientific medicine requires significant expertise and equipment and thus is practiced mainly in medical centers. In

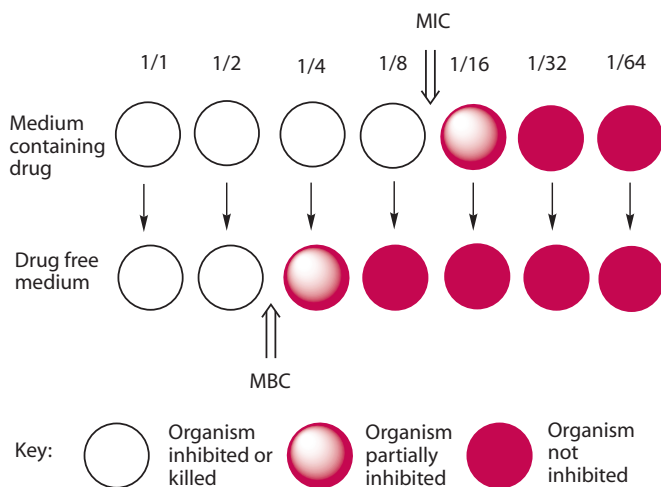


FIGURE 33.1 In the top tubes (viewed from the top), a serially decreasing amount of antimicrobial agent is added to a suitable growth medium inoculated with a microorganism. Following incubation, microbial growth is detected by turbidity. The last concentration that produces no visible growth is scored as the minimum inhibitory concentration (MIC) (1/8). Next, a loopful is taken from each tube and placed in fresh medium (bottom row). In tubes where the organisms were killed by the drug, there is no resumption of growth. Where the organisms were inhibited but not killed, removal of drug allows for resumption of growth. The last concentration that produces no visible growth under these conditions is scored as the minimum bactericidal concentration (MBC) (1/2).

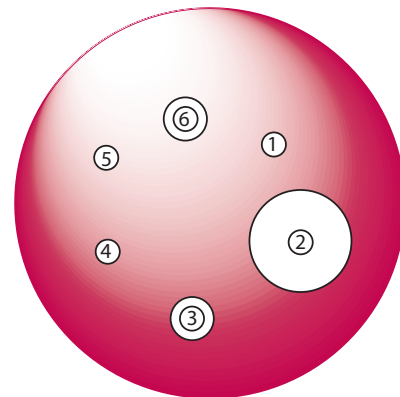


FIGURE 33.2 Looking down on a Petri dish containing solidified nutrient agar to which had been added a suspension of a bacterial species. Next, six filter paper disks containing six different antimicrobials were added, followed by overnight incubation. The antimicrobials in disks 1, 4, and 5 were inactive. Of the active agents in disks 2, 3, and 6, antibiotic 2 was much more active because the microorganism was not able to grow as near this impregnated disk.

outpatient practice, the choice of antimicrobial agents is more commonly made empirically.

Bactericidal Versus Bacteriostatic

Almost all antibiotics have the capacity to be bactericidal *in vitro*; that is, they will kill bacteria if the concentration or dose is sufficiently high. In the laboratory, it is almost always possible to use such doses. Subsequent inoculation of fresh, antibiotic-free media with a culture that has been so treated will not produce growth of the culture because the cells are dead. When such doses are achievable in live patients, such drugs are clinically bactericidal. At somewhat lower concentrations, bacterial multiplication is prevented even though the microorganism remains viable (bacteriostatic action).

The smallest concentration that will kill a bacterial colony is the minimum bactericidal concentration. The difference between a minimum bactericidal dose and a bacteriostatic dose is characteristic of given families of antibiotics. With gentamicin, for example, doubling or quadrupling the dose changes the effect on bacteria from bacteriostatic to bactericidal. Such doses are usually achievable in the clinic, so gentamicin is termed bactericidal. However, the difference between bactericidal and bacteriostatic doses with tetracycline is approximately 40-fold, and it is not possible to achieve such doses safely in patients, so tetracycline is referred to as bacteriostatic. If a bacteriostatic antibiotic is withdrawn prematurely from a patient, the microorganism can resume growth, and the infection can reestablish itself because the organism is still viable. When a patient is immunocompetent or the infection is not severe, a bacteriostatic concentration will break the fulminating stage of the infection (when bacterial cell numbers are increasing at a logarithmic rate). With *E. coli*, for example, the number of cells doubles approximately every 2 hours. A bacteriostatic agent will interrupt this rapid growth and give the immune system a chance to deal with the disease. Cure usually follows if the numbers of live bacteria are not excessive at this time. Obviously, in immunocompromised patients who are unable to contribute natural defenses to fight their own disease, having the drug kill the bacteria is more important for recovery. Thus, although it is preferred that an antibiotic be bactericidal, bacteriostatic antibiotics are widely used and are usually satisfactory.

Microbial Susceptibility

Resistance

Resistance is the failure of microorganisms to be killed or inhibited by antimicrobial treatment. Resistance can either be intrinsic (be present before exposure to drug) or acquired (develop subsequent to exposure to a drug). Resistance of bacteria to the toxic effects of antimicrobial agents and to antibiotics develops fairly easily both in the laboratory and in the clinic and is an ever-increasing public health hazard. Challenging a culture in the laboratory with sublethal quantities of an antibiotic kills the

most intrinsically sensitive percentage of the strains in the colony. Those not killed or seriously inhibited continue to grow and have access to the remainder of the nutrients. A mutation to lower sensitivity also enables individual bacteria to survive against the selecting pressure of the antimicrobial agent. If the culture is treated several times in succession with sublethal doses in this manner, the concentration of antibiotic required to prevent growth becomes ever higher. When the origin of this form of resistance is explored, it is almost always found to be due to an alteration in the biochemistry of the colony so that the molecular target of the antibiotic has become less sensitive, or it can be due to decreased uptake of antibiotic into the cells. This is genomically preserved and passes to the next generation. The altered progeny may be weaker than the wild strain so that they die out if the antibiotic is not present to give them a competitive advantage. In some cases, additional compensatory mutations can occur that restore the vigor of the resistant organisms. Resistance of this type is usually expressed toward other antibiotics with the same mode of action and thus is a familial characteristic; most tetracyclines, for example, show extensive cross-resistance with other agents in the tetracycline family. This is very enlightening with respect to discovery of the molecular mode of action but is not very relevant to the clinical situation.

In the clinic, resistance more commonly takes place by resistance (R) factor mechanisms. In this case, enzymes are elaborated that attack the antibiotic and inactivate it. Mutations leading to resistance occur by many mechanisms. They can result from point mutations, insertions, deletions, inversions, duplications, and transpositions of segments of genes or by acquisition of foreign DNA from plasmids, bacteriophages, and transposable genetic elements. The genetic material coding for this form of resistance is often carried on extra chromosomal elements consisting of small circular DNA molecules known as plasmids. A bacterial cell may have many plasmids or none. The plasmid may carry DNA for several different enzymes capable of destroying structurally dissimilar antibiotics. Such plasmid DNA may migrate within the cell from plasmid to plasmid or from plasmid to chromosome by a process known as transposition. Such plasmids may migrate from cell to cell by conjugation (passage through a sexual pilus), transduction (carriage by a virus vector), or transformation (uptake of exogenous DNA from the environment). These mechanisms can convert an antibiotic-sensitive cell to an antibiotic-resistant cell. This can take place many times in a bacterium's already short generation time. The positive selecting pressure of inadequate levels of an antibiotic favors explosive spread of R-factor resistance. This provides a rationale for conservative but aggressive application of appropriate antimicrobial chemotherapy. Bacterial resistance is generally mediated through one of three mechanisms: failure of the drug to penetrate into or stay in the cell, destruction of the drug by defensive enzymes, or alterations in the cellular target of the drug. It is rarely an all-or-nothing effect. In many cases, a resistant microorganism can still

be controlled by achievable, although higher, doses than are required to control sensitive populations.

Persistence

Sensitive bacteria may not all be killed. Survivors are thought to have been resting (not metabolizing) during the drug treatment time and are still viable. These bacteria are still sensitive to the drug even though they survived an otherwise toxic dose. Some bacteria also can aggregate in films. A poorly penetrating antibiotic may not reach the cells lying deep within such a film. Such cells, although intrinsically sensitive, may survive antibiotic treatment. Bacteria living in host cells, living in cysts, or existing as an abscess are also harder to reach by drugs and thus are more difficult to control.

Combination Therapy

It could be assumed that use of combinations of antibiotics would be superior to the use of individual antibiotics because this would broaden the antimicrobial spectrum and make less critical the accurate identification of the pathogen. It has been found, however, that many times such combinations are antagonistic. A useful generalization, but one that is not always correct, is that one can often successfully combine two bactericidal antibiotics, particularly if their molecular mode of action is different. A common example is the use of a β -lactam antibiotic and an aminoglycoside for empiric therapy of overwhelming sepsis of unknown etiology. Therapy must be instituted as soon as a specimen is obtained or the patient may die. This often does not allow the microbiologic laboratory sufficient time to identify the offending microorganism or to determine its antibiotic susceptibility. Both of the antibiotic families applied in this example are bactericidal in readily achievable parenteral doses. As will be detailed later in this chapter, the β -lactams inhibit bacterial cell wall formation, and the aminoglycosides interfere with protein biosynthesis and membrane function. Their modes of action are supplementary. Because of toxicity considerations and the potential for adverse effects, this empiric therapy is replaced by suitable specific monotherapy at the first opportunity after the sensitivity of the bacterium is experimentally established.

One may also often successfully combine two bacteriostatic antibiotics for special purposes, for example, a macrolide and a sulfonamide. This combination is occasionally used for the treatment of an upper respiratory tract infection caused by *Haemophilus influenzae* because the combination of a protein biosynthesis inhibitor and an inhibitor of DNA biosynthesis results in fewer relapses than the use of either agent alone. However, the use of a bacteriostatic agent, such as tetracycline, in combination with a bactericidal agent, such as a β -lactam, is usually discouraged. The β -lactam antibiotics are much more effective when used against growing cultures, and a bacteriostatic agent interferes with bacterial growth, often giving an indifferent or antagonistic response when such

agents are combined. Additional possible disadvantages of combination chemotherapy are higher cost, greater likelihood of adverse effects, and difficulties in demonstrating synergism. The rising rate of clinical failure due to antibiotic resistance is overcoming these reservations, and combination therapy is becoming more common.

Serum Protein Binding

The influence of serum protein binding on antibiotic effectiveness is fairly straightforward. It is considered in most instances that the percentage of antibiotic that is protein bound is not available at that moment for the treatment of infections so must be subtracted from the total blood level in order to get the effective blood level. The tightness of the binding is also a consideration. Thus a heavily and firmly serum protein-bound antibiotic would not generally be a good choice for the treatment of septicemias or infections in deep tissue, even though the microorganism involved is susceptible in *in vitro* tests. If the antibiotic is rapidly released from the protein, however, this factor decreases in importance, and the binding becomes a depot source. Distinguishing between these two types of protein binding is accomplished by comparing the percentages of binding to the excretion half-life. A highly bound but readily released antibiotic will have a comparatively short half-life and work well for systemic infections. An antibiotic that is not significantly protein bound will normally be rapidly excreted and have a short half-life. Thus, some protein binding of poorly water-soluble agents is normally regarded as helpful.

Preferred Means of Dosing

Under ideal circumstances, it is desirable for an antibiotic to be available in both parenteral and oral forms. Whereas there is no question that the convenience of oral medication makes this ideal for outpatient and community use, very ill patients often require parenteral therapy. It would be consistent with today's practice of discharging patients sooner to send them home from the hospital with an efficacious oral version of the same antibiotic that led to the possibility of discharge in the first place. In this way, the patient would not have to come back to the hospital at intervals for drug administration and would not have to risk treatment failure by starting therapy with a new drug.

COST

Antibiotics are often expensive, but so is morbidity and mortality. For many patients, however, cost is a significant consideration. The pharmacist is in an ideal position to guide the physician and the patient on the question of possible alternative equivalent treatments that might be more affordable. The most frequent comparisons are based on the cost of the usual dose of a given agent for a single course of therapy (usually the wholesale cost to the pharmacist for 10 days' worth of drug).

Agricultural Use of Antibiotics

It is estimated that more than half of the antibiotics of commerce are used for agricultural purposes. Their use for treatment of infections of plants and animals is not to be discouraged so long as drug residues from the treatment do not contaminate foods. In contamination, problems such as penicillin allergy or subsequent infection higher up the food chain by drug-resistant microbes can occur. Animals demonstrably grow more rapidly to marketable size when antibiotics are added to their feed even though the animals have no apparent infection. This is believed to be due in large part to suppression of subclinical infections that would consequently divert protein biosynthesis from muscle and tissue growth into proteins needed to combat the infection. Under appropriate conditions, antibiotic feed supplementation is partly responsible for the comparative wholesomeness and cheapness of our food supplies. This practice has the potential, however, to contaminate the food we consume or to provide reservoirs of drug-resistant enteric microorganisms. Occasionally, infections are traced to this cause, and resistance genes can originate in this manner and pass from strain to strain and even to other species.

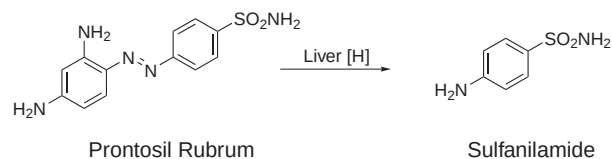
THERAPEUTIC CLASSES

Synthetic Antimicrobial Agents

Synthetic antimicrobial agents have not been modeled after any natural product, so they may not properly be called “antibiotics.” Some synthetics are extremely effective for treatment of infections and are widely used. Very few antibiotics are known to work in precisely the same way as these agents in antibacterial action.

Sulfonamides

The antibacterial properties of the sulfonamides were discovered in the mid-1930s after an incorrect hypothesis, but after observing the results carefully and drawing correct conclusions. Prontosil rubrum, a red dye, was one of a series of dyes examined by Gerhard Domagk of Bayer of Germany in the belief that it might be taken up selectively by certain pathogenic bacteria and not by human cells, in a manner analogous to the way that the Gram stain works, and thus serve as a selective poison to kill these cells (2). The dye, indeed, proved active in vivo against streptococcal infections in mice. Curiously, it was not active in vitro. Trefouel and others soon showed that the urine of prontosil rubrum-treated animals was bioactive in vitro (8). Fractionation led to identification of the active substance as *p*-aminobenzenesulfonic acid amide (sulfanilamide), a colorless cleavage product formed by reductive liver metabolism of the administered dye. Today, we would call prontosil rubrum a prodrug.



The discovery of sulfanilamide's in vivo antibacterial properties ushered in the modern anti-infective era, and Domagk was awarded a Nobel Prize for medicine in 1939.

Once mainstays of antimicrobial chemotherapy, the sulfonamides have decreased enormously in popularity and are now comparatively minor drugs. The relative inexpensiveness of the sulfonamides is one of their most attractive features and accounts for much of their persistence on the market.

MECHANISM OF ACTION The sulfonamides are bacteriostatic when administered to humans in achievable doses. They inhibit the enzyme dihydropteroate synthase, an important enzyme needed for the biosynthesis of folic acid derivatives and, ultimately, the thymidine required for DNA (9). They do this by competing at the active site with *p*-aminobenzoic acid (PABA), a normal structural component of folic acid derivatives. PABA is otherwise incorporated into the developing tetrahydrofolic acid molecule by enzyme-catalyzed condensation with 6-hydroxymethyl-7,8-dihydropterin-pyrophosphate to form 7,8-dihydropteroate and pyrophosphate. Thus, sulfonamides may also be classified as antimetabolites (Fig. 33.3). Indeed, the antimicrobial efficacy of sulfonamides can be reversed by adding significant quantities of PABA into the diet (in some multivitamin preparations and as metabolites of certain local anesthetics) or into the culture medium. Most susceptible bacteria are unable to take up preformed folic acid from their environment and convert it to a tetrahydrofolic acid but, instead, synthesize their own folates *de novo*. Folates are essential intermediates for the biosynthesis of thymidine without which bacteria cannot multiply. Thus, inhibition of the dihydropteroate synthase is bacteriostatic. Humans are unable to synthesize folates from component parts, lacking the necessary enzymes (including dihydropteroate synthase), and folic acid is supplied to humans in our diet. Sulfonamides consequently have no similarly lethal effect on human cell growth, and the basis for the selective toxicity of sulfonamides is clear.

In a few strains of bacteria, however, the picture is somewhat more complex. Here, sulfonamides are attached to the dihydropteroate diphosphate in place of the normal PABA. The resulting unnatural product, however, is not capable of undergoing the next necessary reaction, condensation with glutamic acid. This false metabolite is also an enzyme inhibitor, and the net result is inability of the bacteria to multiply when the folic acid in their cells is used up, and further nucleic acid biosynthesis becomes impossible. The net result is the same, but the molecular basis of the effect is somewhat different in these strains (Fig. 33.3).

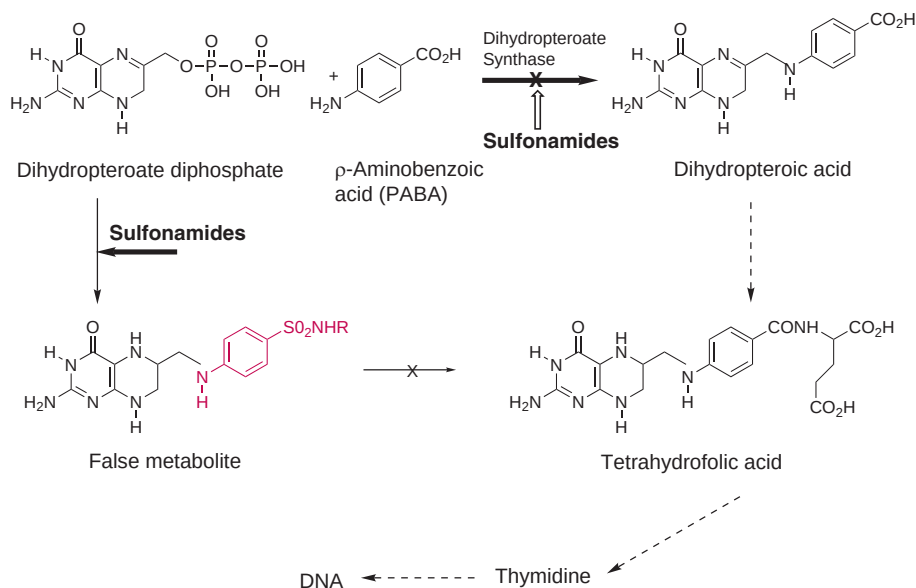
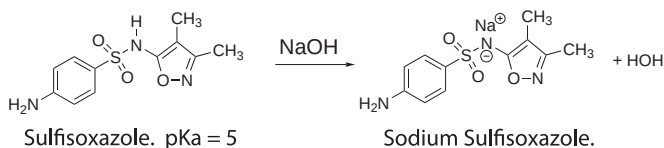


FIGURE 33.3 Microbial biosynthetic pathway leading to tetrahydrofolic acid synthesis and major site of action ($\rightleftharpoons$) of sulfonamides as well as the site of action seen in some bacteria ($\leftarrow$), resulting in incorporation of sulfonamide as a false metabolite.

Bacteria that are able to take up preformed folic acid into their cells are intrinsically resistant to sulfonamides.

STRUCTURE–ACTIVITY RELATIONSHIPS The basis of the structural resemblance of sulfonamides to PABA is clear. The functional group that differs in the two molecules is the carboxyl of PABA and the sulfonamide moiety of sulfanilamide. The strongly electron-withdrawing character of the aromatic SO_2 group makes the nitrogen atom to which it is directly attached partially electropositive. This, in turn, increases the acidity of the hydrogen atoms attached to the nitrogen so that this functional group is slightly acidic ($\text{p}K_a = 10.4$). The $\text{p}K_a$ of the carboxyl group of PABA is approximately 4.9. It was soon found that replacement of one of the NH_2 hydrogens by an electron-withdrawing heteroaromatic ring enhanced the acidity of the remaining hydrogen and dramatically enhanced potency. With suitable groups in place, the $\text{p}K_a$ is reduced to the same range as that of PABA itself. Not only did this markedly increase the antibacterial potency of the product, but it also dramatically increased the water solubility under physiologic conditions. The $\text{p}K_a$ of sulfisoxazole, one of the sulfonamides in present use, is approximately 5.0. The poor water solubility of the earliest sulfonamides led to occasional crystallization in the urine (crystalluria) and resulted in kidney damage because the molecules were un-ionized at urine pH values. It is still recommended to drink increased quantities of water to avoid crystalluria when taking certain sulfonamides, but this form of toxicity is now comparatively uncommon with the more important agents used today because they form sodium salts that are at least partly ionized and hence reasonably water soluble at urinary pH values. They are poorly tolerated on injection, however, because these salts are corrosive to tissues.

Structural variation among the clinically useful sulfonamides is restricted primarily to installation of various heterocyclic aromatic substituents on the sulfonamide nitrogen.



Pharmacokinetics

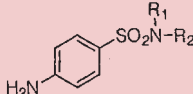
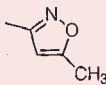
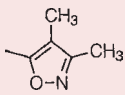
The orally administered sulfonamides are well absorbed from the gastrointestinal (GI) tract, distributed fairly widely, and excreted by the kidney. The drugs are bound to plasma protein (sulfisoxazole 30% to 70%, sulfamethoxazole 70%) and, as such, may displace other protein-bound drugs as well as bilirubin. The latter phenomenon disqualifies them for use in late-term pregnancy because they can cause neonatal jaundice. Sulfonamides are partly deactivated by acetylation at N-4 and glucuronidation of the anilino nitrogen in the liver (10).

Plasmid-mediated resistance development is common, particularly among gram-negative microorganisms and usually takes the form of decreased sensitivity of dihydropteroate synthase or increased production of PABA (11).

Therapeutic Applications

Of the thousands of sulfonamides that have been evaluated, only a few are still available and are often used in combination with other agents. The surviving sulfonamides (Table 33.1) include sulfisoxazole, which is used in combination with erythromycin. It has a comparatively

TABLE 33.1 Clinically Relevant Sulfonamides

Generic Name		pKa	
	R ₁	R ₂	
Sulfamethoxazole	H		5.6
Silver Sulfadiazine	Ag [⊕]		
Sulfisoxazole	H		5.0
Sulfacetamide	H	COCH ₃	5.4

broad antimicrobial spectrum *in vitro*, especially against gram-negative organisms, but clinical use is generally restricted due to the development of bacterial resistance. Susceptible organisms may include *Enterobacteriaceae* (*E. coli*, *Klebsiella*, and *Proteus*) and *Streptococcus pyogenes*, *Streptococcus pneumoniae*, and *Haemophilus* (12). Sulfamethoxazole in combination with trimethoprim is more commonly seen and is further described in the discussion of trimethoprim.

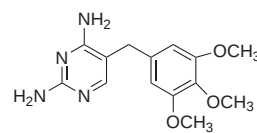
The remaining sulfonamides are not used systemically. Sulfadiazine in the form of its silver salt is used topically for treatment of burns and is effective against a range of bacteria and fungus, whereas sulfacetamide is used ophthalmically for treatment of eye infections caused by susceptible organisms. Sulfasalazine stands out from the typical sulfonamide because although it is administered orally, the drug is not absorbed in the gut, so the majority of the dose is delivered to the distal bowel. In addition, the drug is a prodrug that undergoes reductive metabolism by gut bacteria converting the drug into sulfapyridine and 5-aminosalicylic acid, the active component (Fig. 33.4). The liberation of 5-aminosalicylic acid (mesalamine), an anti-inflammatory agent, is the purpose for administering this drug. This agent is used to treat ulcerative colitis and Crohn disease (13). Direct administration of salicylates is otherwise irritating to the gastric mucosa.

Adverse Effects

Allergic reactions are the most common and take the form of rash, photosensitivity, and drug fever. Less common problems are kidney and liver damage, hemolytic anemia, and other blood problems. The most serious adverse effect is the Stevens-Johnson syndrome characterized by sometimes fatal erythema multiforme and ulceration of mucous membranes of the eye, mouth,

and urethra (10). Fortunately, these effects are comparatively rare.

Trimethoprim



Trimethoprim (Proloprim, Trimplex)

MECHANISM OF ACTION A further step in the pathway leading from the pterates to folic acid and on to DNA bases requires the enzyme dihydrofolate reductase. Exogenous folic acid must be reduced stepwise to dihydrofolic acid and then to tetrahydrofolic acid, an important cofactor essential for supplying a one-carbon unit in thymidine

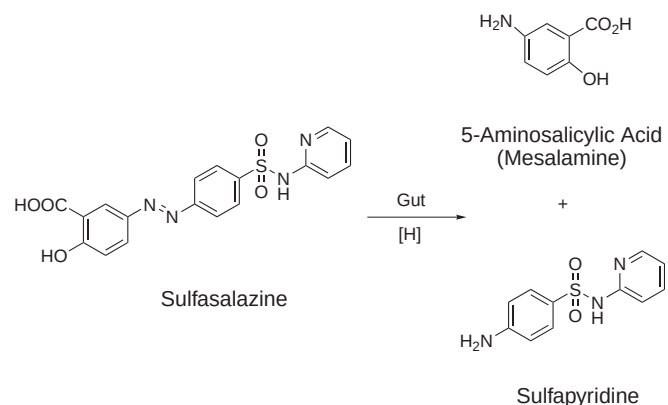


FIGURE 33.4 Activation of sulfasalazine to 5-aminosalicylic acid.

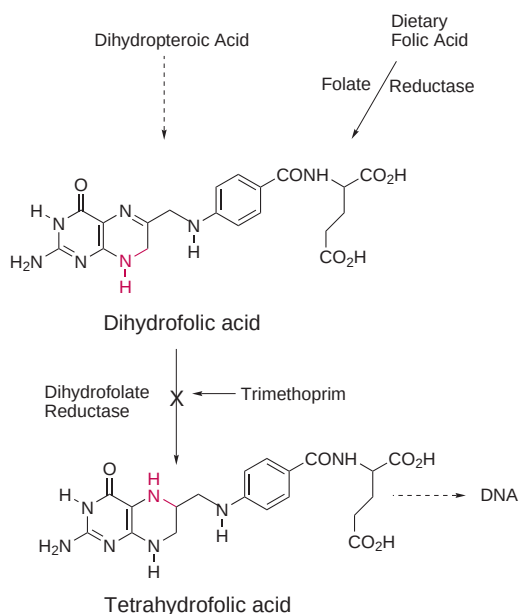


FIGURE 33-5 Site of action of trimethoprim.

biosynthesis and ultimately for DNA synthesis (Fig. 33.5). The same enzyme must also reduce endogenously produced dihydrofolate. Inhibition of this key enzyme had been widely studied in attempts to find anticancer agents by starving rapidly dividing cancer cells of needed DNA precursors. Antifolates such as methotrexate arose from such studies. Methotrexate, however, is much too toxic to be used as an antibiotic. Subsequently, however, trimethoprim was developed in 1969 by George Hitchings and Gertrude Elion (who shared a Nobel Prize for this and other contributions to chemotherapy in 1988). This inhibitor prevents tetrahydrofolic acid biosynthesis and results in bacteriostasis. Trimethoprim selectivity between bacterial and mammalian dihydrofolate reductases results from the subtle but significant architectural differences between these enzyme systems. Whereas the bacterial enzyme and the mammalian enzyme both efficiently catalyze the conversion of dihydrofolic acid to tetrahydrofolic acid, the bacterial enzyme is sensitive to inhibition by trimethoprim by up to 100,000 times lower concentrations than is vertebrate enzyme (14). This difference explains the useful selective toxicity of trimethoprim.

THERAPEUTIC APPLICATION Trimethoprim can be used as a single agent clinically for the oral treatment of uncomplicated urinary tract infections caused by susceptible bacteria (predominantly community-acquired *E. coli* and other gram-negative rods) (15). It is, however, most commonly used in a 1:5 fixed concentration ratio with the sulfonamide sulfamethoxazole (Bactrim, Septra). This combination is not only synergistic in vitro but is less likely to induce bacterial resistance than either agent alone. It is rationalized that microorganisms not completely inhibited

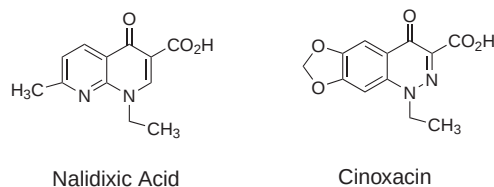
by sulfamethoxazole at the pterate condensation step will not likely be able to push the lessened amount of substrates that leak through past a subsequent blockade of dihydrofolate reductase. Thus these agents block sequentially at two different steps in the same essential pathway, and this combination is extremely difficult for a naive microorganism to survive. It is also comparatively uncommon that a microorganism will successfully mutate to resistance at both enzymes during the course of therapy. However, if the organism is already resistant to either drug at the outset of therapy, much of the advantage of the combination is lost.

Pairing these two particular antibacterial agents was based on pharmacokinetic factors. For such a combination to be useful in vivo, the two agents must arrive at the necessary tissue compartment where the infection is at the correct time and in the right ratio. In this context, the optimum ratio of these two agents in vitro is 1:20. Administration of the 1:5 combination of the two drugs orally produces the desired 1:20 ratio in the body once steady-state is reached (10). The combination is used for oral treatment of urinary tract infections, shigellosis, otitis media, traveler's diarrhea, community-acquired methicillin-resistant *Staphylococcus aureus* (MRSA), acute exacerbations of chronic bronchitis, and pneumocystis pneumonia (10). The pneumonia-causing fungus *Pneumocystis jirovecii* (previously classified as *Pneumocystis carinii*) is an opportunistic pathogen for immunocompromised individuals.

The most frequent side effects of trimethoprim-sulfamethoxazole are rash, nausea, and vomiting. Blood dyscrasias are less common, as is pseudomembranous colitis (caused by non-antibiotic-sensitive opportunistic gut anaerobes, often *Clostridium difficile*) (10). Despite a significant effort, no structurally related analog has emerged to compete with trimethoprim.

RESISTANCE Bacterial resistance to trimethoprim is increasingly common. In pneumococcal infections, it can result from a single amino acid mutation (Ile-100 to Leu) in the dihydrofolate reductase enzyme. Overexpression of dihydrofolate reductase by *Staphylococcus aureus* has also been reported in resistant strains (11).

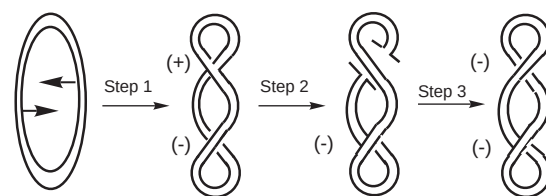
Quinolones



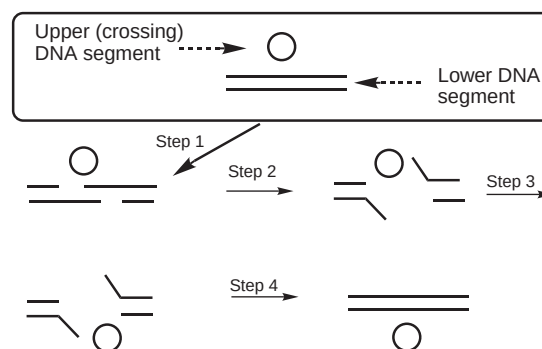
The quinolone antimicrobials comprise a group of synthetic substances possessing in common an *N*-1-alkylated 3-carboxypyrid-4-one ring fused to another aromatic ring, which itself carries other substituents. The first quinolone to be marketed was nalidixic acid, which has since been discontinued. Nalidixic acid was classified

as a first-generation quinolone based on its spectrum of activity and pharmacokinetic properties. The spectrum of activity was limited to a small number of gram-negative organisms. Thus, the quinolones were of little clinical significance until the discovery that the addition of a fluoro group to the 6 position of the basic nucleus greatly increased the biologic activity (16). Agents that contain the 6-fluoro substitution are referred to as fluoroquinolones and represent an important therapeutic class of antimicrobials. Norfloxacin was approved for use in 1986 and represents the first of the second-generation quinolones; it is considered to be broad spectrum and equivalent in potency to many of the fermentation-derived antibiotics (16). Following its introduction, intense research ensued, and over a thousand analogs have now been made. Ciprofloxacin, gemifloxacin, norfloxacin, ofloxacin, levofloxacin, and moxifloxacin are currently marketed for systemic use in the United States. In addition, ciprofloxacin, levofloxacin, moxifloxacin, and ofloxacin along with besifloxacin and gatifloxacin are available for ophthalmic use (Fig. 33.6).

MECHANISM OF ACTION The quinolones are rapidly bactericidal, largely as a consequence of inhibition of DNA gyrase and topoisomerase IV, key bacterial enzymes that dictate the conformation of DNA (17). Using the energy generated by adenosine triphosphate (ATP) hydrolysis, DNA is progressively wound about itself in a positive supercoil. In the absence of ATP, the process is reversed, relaxing the molecule. It must also be partially unwound so that the cell has access to the genetic information it contains. This requires reversible conformational changes so that it can be stored properly, unwound, replicated, repaired, and transcribed on demand. DNA gyrase alters the conformation of DNA by catalyzing transient double-strand cuts, passing the uncut portion of the molecule through the gap, and resealing the molecule back together (Fig. 33.7) (17). In this way, DNA



A. View from the top: Step 1. Stabilize positive node. Step 2. Break both strands of the back segment. Step 3. Pass unbroken segment through the break and reseal on the front side.



B. View from the side: Step 1. Staggered cuts in each strand. Step 2. Gate opens. Step 3. Transverse segment passed through the break. Step 4. Reseal cut segment.

FIGURE 33.7 Schematic depicting supercoiling of circular DNA catalyzed by DNA gyrase.

gyrase alters the degree of twisting of DNA by introducing negative DNA supercoils releasing tensional stress in the molecule. DNA topoisomerase IV, on the other hand, decatenates (unties) enchain daughter DNA molecules produced through replication of circular DNA (17). Inhibition of DNA gyrase and topoisomerase IV makes a cell's DNA inaccessible and leads to cell death, particularly if the cell must deal with other toxic effects at the same time. Different quinolones

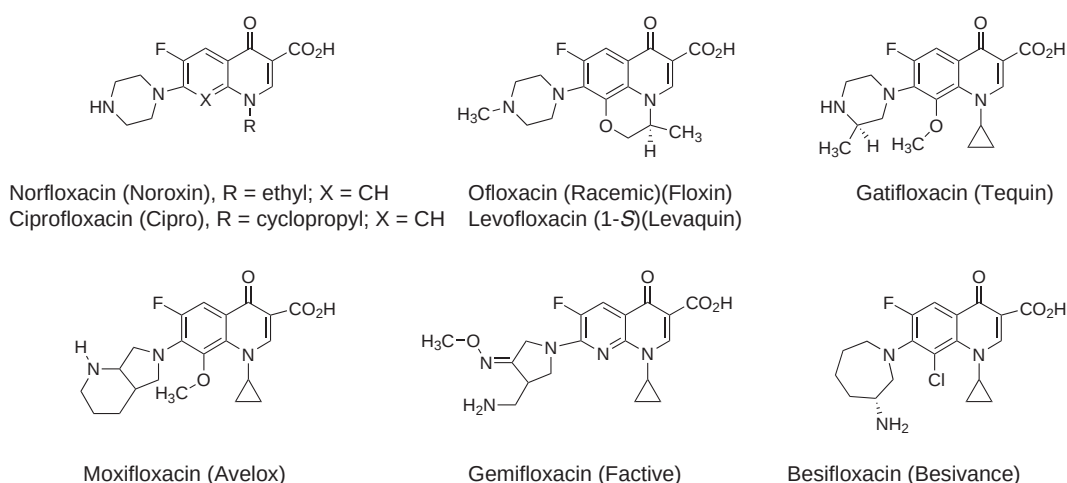


FIGURE 33.6 Second-, third-, and fourth-generation quinolones.

inhibit these essential enzymes to different extents, which explains some of the differences in the spectrum of activity of the fluoroquinolones. Topoisomerase IV seems more important to some gram-positive organisms, and DNA gyrase seems more important to some gram-negative organisms.

Humans shape their DNA with a topoisomerase II, an analogous enzyme to DNA gyrase that, however, does not bind quinolones at normally achievable doses, so the quinolones of commerce do not kill host cells.

STRUCTURE–ACTIVITY RELATIONSHIP The structural features of the quinolones strongly influence the antimicrobial and pharmacokinetic properties of this class of drugs (18). The essential pharmacophore for activity is the carboxy-4-pyridone nucleus (Fig. 33.8). Apparently, the carboxylic acid and the ketone are involved in binding to the DNA/DNA-gyrase enzyme system. Reduction of the 2,3-double bond or the 4-keto group inactivates the molecule, and substitution at C-2 interferes with enzyme–substrate complexation. Fluoro substitution at the C-6 position greatly improves antimicrobial activity by increasing the lipophilicity of the molecule, which in turn improves the drug's penetration through the bacterial cell wall. C-6 fluoro also increases the DNA gyrase/topoisomerase IV inhibitory action. An additional fluoro group at C-8 further improves drug absorption and half-life, but also increases drug-induced photosensitivity. Substitution of a methoxy group at C-8 reduces the photosensitivity (moxifloxacin and gatifloxacin). Heterocyclic substitution at C-7 improves the spectrum of activity especially against gram-negative organisms. The piperazinyl (ciprofloxacin) and pyrrolidinyl (moxifloxacin) represent the most significant antimicrobial improvement. Unfortunately, the piperazinyl group at C-7 also increases binding to central nervous system (CNS) γ -aminobutyric acid (GABA) receptors, which accounts for CNS side effects. Alkyl substitution on the piperazine nitrogen (ofloxacin and levofloxacin) is reported to decrease binding to GABA. The cyclopropyl substitution at N-1 appears to broaden activity of the quinolones to include activity against atypical bacteria including *Mycoplasma*, *Chlamydia*, and *Legionella* species. Substitution of a 2,4-difluorophenyl at N-1 also improves antimicrobial potency, but agents with this substitution (trovafloxacin and temafloxacin) have been withdrawn from the market due to serious adverse effects.

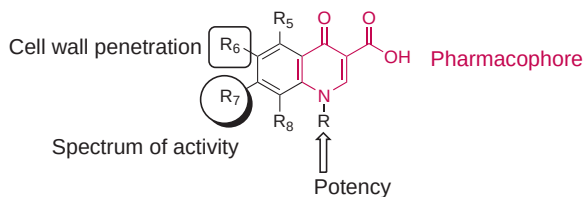
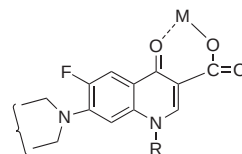


FIGURE 33.8 Major structure–activity relationship features of 4-quinolones.

The introduction of a third ring to the nucleus of the quinolones gives rise to ofloxacin. Additionally, ofloxacin has an asymmetric carbon at the C-3' position. The *S*-(–)-isomer (levofloxacin) is twice as active as ofloxacin and 8 to 128 times more potent than the *R*-(+)-isomer resulting from increased binding to the DNA gyrase.

Finally, a chemical incompatibility common to all of the quinolones involves the ability of these drugs to chelate polyvalent metal ions (Ca^{2+} , Mg^{2+} , Zn^{2+} , Fe^{2+} , Al^{3+}), resulting in decreased solubility and reduced drug absorption. Chelation occurs between the metal and the 3-carboxylic acid and 4-keto groups. Agents containing polyvalent metals should be administered separately from the quinolones.



PHARMACOKINETICS The fluoroquinolones are well absorbed following oral administration, with excellent bioavailability. The maximum plasma concentration is usually reached within a few hours, and the drugs are moderately bound to plasma protein, leading to comparatively long half-lives (Table 33.2). Earlier quinolones were rapidly excreted into the urine, which limited their therapeutic application to urinary tract infections, whereas the newer drugs are distributed to alveolar macrophages, bronchial mucosa, epithelial lining fluid, and saliva, improving the use in various systemic infections.

THERAPEUTIC APPLICATIONS The quinolones therapeutically fall into one of four classifications (Table 33.3). The specific drugs within each classification include nalidixic acid and cinoxacin as first-generation agents with utility limited to uncomplicated urinary tract infections. The second-generation quinolones include norfloxacin and

TABLE 33.2 Pharmacokinetic Properties for Select Quinolones

Drug	Bioavailability (%)	Protein Binding (%)	Half-Life (hours)
Ciprofloxacin	70	30	3.5
Enoxacin	90	40	3–6
Gatifloxacin	96	20	8.0
Gemifloxacin	71	60–70	8.0
Levofloxacin	99	31	6.9
Moxifloxacin	86	47	12.1
Norfloxacin	30–40	10–15	3–4
Ofloxacin	98	32	9

TABLE 33.3 Therapeutic Classification of Quinolones

Generation	Characteristics
First generation	Poor serum and tissue concentration Not valuable for systemic infections Lack activity against <i>Pseudomonas aeruginosa</i> , gram-positive organisms, and anaerobes
Second generation	Adequate serum and tissue concentration Good for systemic infections Active against gram-negative organisms including <i>P. aeruginosa</i> ; weak activity against <i>Streptococcus pneumoniae</i> ; and no activity against anaerobes
Third generation	Once-daily dosing Active against <i>S. pneumoniae</i> and atypical bacteria; less active against <i>P. aeruginosa</i>
Fourth generation	Active against anaerobes and aerobic gram-positive and gram-negative organisms

ciprofloxacin. Norfloxacin is mainly used for urinary tract infections (*Enterobacter*, *Enterococcus*, or *Pseudomonas aeruginosa*), prostatitis, and gonorrhea, whereas ciprofloxacin is also used for upper respiratory tract infections, bone infections, septicemia, staphylococcal and pseudomonadal endocarditis, meningitis, and anthrax infections (19,20). The third-generation quinolones include levofloxacin, gatifloxacin, and gemifloxacin. These agents find use in the treatment of bacterial exacerbation of chronic bronchitis and community-acquired pneumonia (21). Levofloxacin is also used to treat respiratory infections caused by *Legionella*, *Chlamydia*, and *Mycoplasma* as well as other nosocomial pneumonias. Additional indications for levofloxacin include skin and skin structure infections and acute sinusitis caused by *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis* (22). The fourth-generation quinolones include moxifloxacin and have a spectrum of activity that includes *Bacteroides fragilis* (23). Moxifloxacin and levofloxacin are also recommended as second-line agents for tuberculosis as an off-label use (22,23).

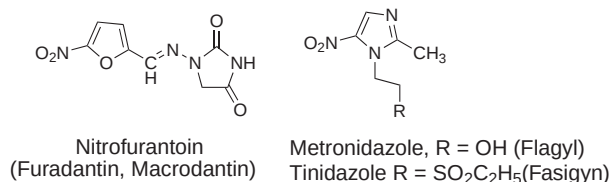
RESISTANCE Resistance to the quinolones is becoming more frequent and is associated with spontaneous mutations in genes (*gyrA* and *gyrB*) that encode for the quinolone target protein, DNA gyrase, and genes (*parC* and *parE*) that encode for topoisomerase IV (17). A single mutation can lead to low-level resistance, whereas mutations in more than one gene lead to high-level resistance. This mechanism of resistance would be expected to produce differing levels of cross-resistance within the class of quinolones. In addition, there are suggestions that resistance may be associated with an increase in drug efflux or a decrease in outer membrane permeability affecting drug influx. Such a mechanism of resistance would be expected to be more common in gram-negative organisms with a more complex cell wall than in gram-positive organisms with their cell envelope.

ADVERSE EFFECTS The quinolone class is associated with more side effects than the β -lactam and macrolide classes, but nonetheless see very widespread medicinal use. All of the fluoroquinolones have a black box warning for possible tendonitis and tendon rupture. They are also not for use in myasthenia gravis patients. Another side effect associated with quinolones is a proconvulsant action, especially when coadministered with nonsteroidal anti-inflammatory drugs. Other CNS problems include hallucinations, insomnia, and visual disturbances. Some patients also experience diarrhea, vomiting, abdominal pain, and anorexia. The quinolones are associated with erosion of the load-bearing joints of young animals. As a precaution, these drugs are not used casually in children less than 18 years of age or in sexually active females of childbearing age. They are also potentially damaging in the first trimester of pregnancy because of a risk of severe metabolic acidosis and of hemolytic anemia. Some of the quinolones may potentiate the action of theophylline and should be monitored closely. They also have been linked to QT prolongation and may increase the risk of torsades de pointes when used with other QT-prolonging agents such as some antiarrhythmic agents (20–24).

SEVERE TOXICITIES Certain members of the fluoroquinolone family were marketed for a while but were subsequently severely limited in use or withdrawn because of the unacceptable toxicities experienced by some patients. These agents, because of their breadth of spectrum and potency against resistant microorganisms, were introduced with great hope. Temafloxacin, for example, was removed from the market because of hemolysis, renal failure, and thrombocytopenia (the hemolytic uremic syndrome). These effects only became apparent when large numbers of patients received the drug. Severe liver toxicity led to the removal from the market of trovafloxacin. Grepafloxacin was introduced on the market in late 1997 as a broad-spectrum fluoroquinolone and withdrawn from the market in 1999 because of cardiovascular toxicity. Analogs with a C-8 chloro substituent, such as clinafloxacin and sitafloxacin, were also very potent but have been withdrawn due to excessive phototoxicity.

Miscellaneous Agents

NITROHETEROAROMATIC COMPOUNDS

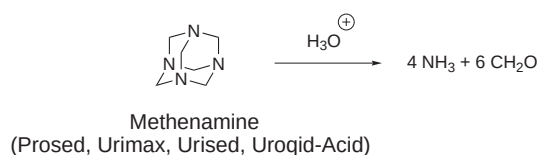


Nitrofurantoin, a widely used oral antibacterial nitrofuran, has been available since World War II. It is used for prophylaxis or treatment of acute urinary tract infections when kidney function is not impaired, and it inhibits kidney stone growth. Nausea and vomiting are common

side effects. This is avoided in part by slowing the rate of absorption of the drug through use of wax-coated large particles (Macrodantin). Nitrofurantoin inhibits DNA and RNA functions through mechanisms that are not well understood, although bioreductive activation is suspected to be an important component of this. There is little acquired resistance to this agent. Severe side effects can be experienced when using this drug (acute pulmonary reactions, peripheral neuropathy, hemolytic anemia, liver toxicity, and fertility impairment) (25).

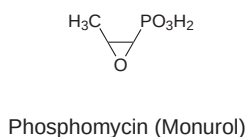
Metronidazole was initially introduced for the treatment of vaginal infections caused by amoeba. This nitroimidazole is also useful orally for the treatment of trichomoniasis, giardiasis, and *Gardnerella vaginalis* infections. It has found increasing use of late in the parenteral treatment of anaerobic infections and for treatment of pseudomembranous colitis due to *Clostridium difficile*. *C. difficile* is an opportunistic pathogen that occasionally flourishes as a consequence of broad-spectrum antibiotic therapy, and infections can be life-threatening. The drug is believed to be metabolically activated by reduction of its nitro group to produce metabolites that interfere with DNA and RNA function. Metronidazole is also a component of a multidrug cocktail used to treat *Helicobacter pylori* infections associated with gastric ulcers (26). Tinidazole, another nitroimidazole, has been introduced primarily as an antiprotozoal agent (27). Because of the similarities in structure and action, tinidazole is believed to have a similar spectrum to metronidazole. Both drugs can cause disulfiram-like adverse reactions when alcohol is consumed. Metronidazole use is associated with allergic rashes and CNS disturbances, including convulsions, in some patients. It is carcinogenic in rodents (26).

METHENAMINE



Methenamine is a drug that can be used for the disinfection of acidic urine. Structurally it is a low molecular weight polymer of ammonia and formaldehyde which reverts to its components under mildly acid conditions. Formaldehyde is the active antimicrobial component. Methenamine can be used for recurrent urinary tract infections. The drug is available in various dosage forms as well as various salts including the hippurate and mandelate.

FOSFOMYCIN



Fosfomycin (also known as phosphomycin) inhibits enolpyruvyl transferase, an enzyme catalyzing an early step in bacterial cell wall biosynthesis. Inhibition results in reduced synthesis of peptidoglycan, an important component in the bacterial cell wall. Fosfomycin is bactericidal against *Escherichia coli* and *Enterobacter faecalis* infections. It is used for treatment of uncomplicated urinary tract infections by susceptible organisms.

Antibiotics: Inhibitors of Bacterial Cell Wall Biosynthesis

THE BACTERIAL CELL WALL Bacterial cells are enclosed within a complex and largely rigid cell wall. This differs dramatically from mammalian cells that are surrounded by a flexible membrane whose chemical composition is dramatically different. This provides a number of potentially attractive targets for selective chemotherapy of bacterial infections. For one thing, enzymes that have no direct counterpart in mammalian cells construct the bacterial cell wall. Three of the main functions of the bacterial cell wall are: 1) to provide a semipermeable barrier interfacing with the environment through which only desirable substances may pass; 2) to provide a sufficiently strong barrier so that the bacterial cell is protected from changes in the osmotic pressure of its environment; and 3) to prevent digestion by host enzymes. The initial units of the cell wall are constructed within the cell, but soon the growing and increasingly complex structure must be extruded; final assembly takes place outside of the inner membrane. This circumstance makes the enzymes involved in late steps more vulnerable to inhibition because they are at or near the cell surface. Whereas individual bacterial species differ in specific details, the following generalized picture of the process is sufficiently accurate to illustrate the process.

Gram-Positive Bacteria The cell wall of gram-positive bacteria, although complex enough, is simpler than that of gram-negative organisms. A schematic representation is shown in Figure 33.9. On the very outside of the cell is a set of characteristic carbohydrates and proteins that together make up the antigenic determinants that differ from species to species and that also cause adherence to particular target cells. There may also be a lipid-rich capsule surrounding the cell (not shown in the diagram). The next barrier that the wall presents is the peptidoglycan layer. This is a spongy, gel-forming layer consisting of a series of alternating sugars (*N*-acetylglucosamine and *N*-acetylmuramic acid) linked (1,4)- β in a long chain (Fig. 33.10). To the lactic acid carboxyl moieties of the *N*-acetylmuramic acid units is attached, through an amide linkage, a series of amino acids of which L-alanyl-D-glutamyl-L-lysyl-D-alanine is typical of *Staphylococcus aureus*. One notes the D-stereochemistry of the glutamate and the terminal alanine. This feature is presumably important in protecting the peptidoglycan from hydrolysis by host peptidases, particularly in the GI tract.

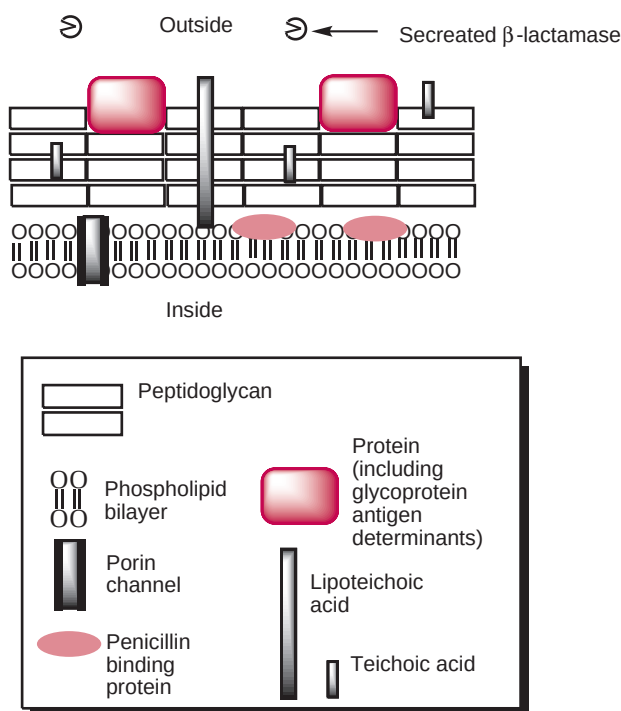


FIGURE 33.9 Schematic of some features of the gram-positive bacterial cell wall.

The early steps in the biosynthesis of the peptidoglycan unit result in formation of a complex polymeric sheet. This is then cross-linked to form a thickened wall. This bonds the terminal D-alanyl unit to the lysyl unit

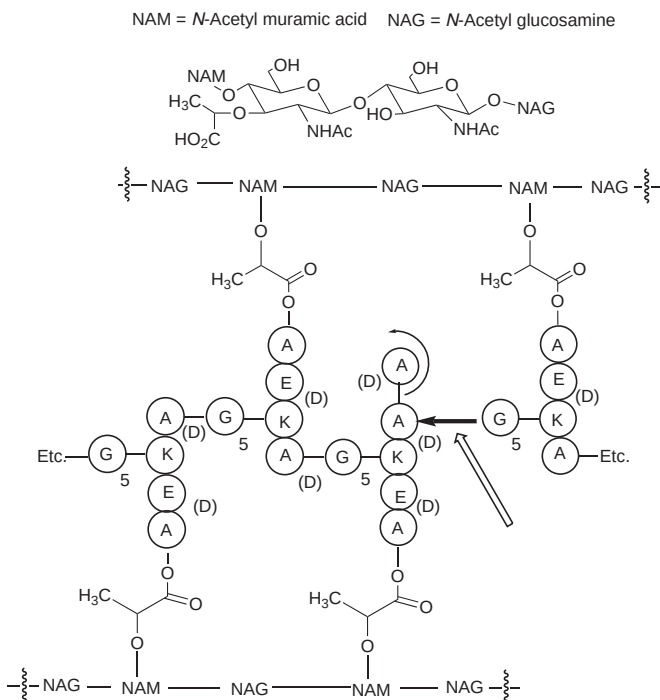


FIGURE 33.10 Schematic of cell wall cross-linking. Pentaglycyl group replaces terminal D-alanine.

of an adjacent tetrapeptide strand through a pentaglycyl unit. This last step is an enzyme-catalyzed transamidation by which the terminal amino moiety on the last glycine unit of the A strand displaces the terminal D-ala unit on the nearby B strand. The cell wall transamidase, one of the penicillin-binding proteins (PBPs), forms a transient covalent bond during the synthesis phase with a particular serine hydroxyl on the enzyme. Completion of the catalytic cycle involves displacement of the enzyme by a glycine residue, which regenerates the enzyme. This process gives the wall additional rigidity. This strong barrier protects against osmotic stress and accounts for the retention of characteristic morphologic shape of gram-positive bacteria (globes and rods, for example). This step is highly sensitive to inhibition of β -lactam antibiotics. It is also the target of the glycopeptide antibiotics (such as vancomycin), as will be discussed later.

The peptidoglycan layer is traversed by complex glycopospholipids called teichoic and teichuronic acids. These are largely responsible for the acid mantle of gram-positive bacteria. Beneath the peptidoglycan layer is the lipoidal cytoplasmic cell membrane in which a number of important protein molecules float in a lipid bilayer. Among these proteins are the β -lactam targets, the PBPs (28). These are enzymes that are important in cell wall formation and remodeling. In gram-positive bacteria, the outer layers are relatively ineffective in keeping antibiotics out. The inner membrane and its protein components provide the principal barrier to uptake of antibiotics. There are a number of different types of PBPs (PBP-1a, PBP-1b, PBP-2, PBP2a, PBP-3, etc), and their function is dependent on the species of bacteria and has been more fully reviewed elsewhere (29). The functions of all of the PBPs are not entirely understood, but they are important in construction and repair of the cell wall. β -Lactam antibiotics bind to these proteins and kill bacteria by preventing the biosynthesis of a functional cell wall. Various β -lactam antibiotics display different patterns of binding to the PBPs. These proteins must alternate in a controlled and systematic way between their active and inert states so that bacterial cells can grow and multiply in an orderly manner. Selective interference by β -lactam antibiotics with their functioning prevents normal growth and repair and creates serious problems for bacteria, particularly young cells needing to grow and mature cells needing to repair damage or to divide.

Gram-Negative Bacteria With the gram-negative bacteria, the cell wall is more complex and more lipoidal (Fig. 33.11). These cells usually contain an additional outer lipid membrane that differs considerably from the inner membrane. The outer layer contains complex lipopolysaccharides that encode antigenic responses, cause septic shock, provide the serotype, and influence morphology. This exterior layer also contains a number of enzymes and exclusionary proteins. Important among these are the porins. These are transmembranal supermolecules made up of two or three monomeric proteins.

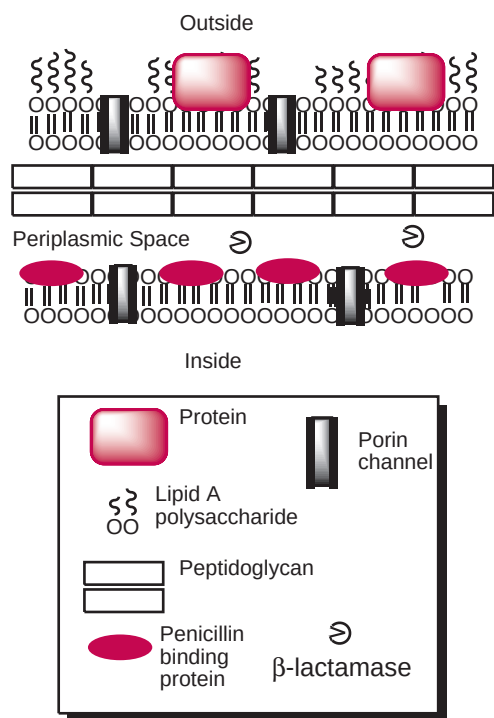
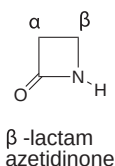


FIGURE 33.11 Schematic of some features of the gram-negative bacterial cell wall.

The center of this array is a transmembranal pore of various dimensions. Some allow many kinds of small molecules to pass, and others contain specific receptors that allow only certain molecules to come in. The size, shape, and lipophilicity of drugs are important considerations controlling porin passage. Antibiotics have greater difficulty in penetrating into gram-negative bacterial cells as a consequence. Next comes a periplasmic space containing a somewhat less impressive and thinner, as compared to gram-positive organisms, layer of peptidoglycan. Also present is a phospholipid-rich cytoplasmic membrane in which floats a series of characteristic proteins with various functions. The β -lactam targets (PBPs) are found here.

Other inner membrane proteins are involved in transport, energy, and biosynthesis. In many such cells, there are proteins that actively pump out antibiotics and other substances at the expense of energy and that may require the simultaneous entrance of oppositely charged materials to maintain an electrostatic balance.

β -LACTAM ANTIBIOTICS



The name lactam is given to cyclic amides and is analogous to the name lactone given to cyclic esters. In an

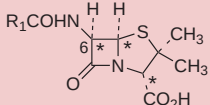
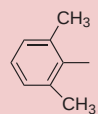
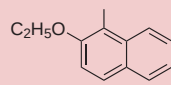
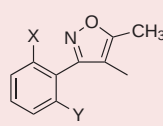
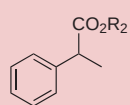
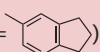
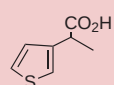
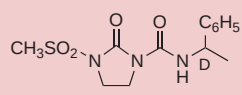
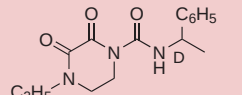
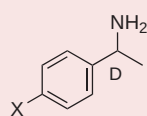
older nomenclature, the second carbon in an aliphatic carboxylic acid was designated alpha, the third beta, and so on. Thus a β -lactam is a cyclic amide with four atoms in its ring. The contemporary name for this ring system is azetidinone. This structural feature was very rare when it was found to be a feature of the structure of the penicillins, so the name β -lactam came to be a generic descriptor for the whole family. Ultimately, this ring proved to be the main component of the pharmacophore, so the term possesses medicinal as well as chemical significance. The penicillin subclass of β -lactam antibiotics is characterized by the presence of a substituted five-membered thiazolidine ring fused to the β -lactam ring. This fusion and the chirality of the β -lactam ring result in the molecule roughly possessing a "V" shape. This drastically interferes with the planarity of the lactam bond and inhibits resonance of the lactam nitrogen with its carbonyl group. Consequently the β -lactam ring is much more reactive and thus more sensitive to nucleophilic attack when compared with normal planar amides.

History The general story of the discovery of the penicillins was previously discussed. The earliest penicillins were produced by fungi from media constituents. The bicyclic heterocyclic nucleus of 6-aminopenicillanic acid is constructed by a process catalyzed by enzymes. The side chain was added essentially intact from media constituents. It was discovered that certain arylacetic acids, when added to the medium, were used to form the side chain amide moiety and that this was very important for stability and breadth of spectrum (30). It was later discovered that exclusion of such materials from the medium allowed the production of 6-aminopenicillanic acid without a side chain (31). Chemists could then add a wider variety of side chains without being limited by the specific requirements of the fungal enzymes. With this breakthrough, the penicillin field expanded to include orally active, broad-spectrum, and enzymatically stable penicillins. The cephalosporins were discovered as secondary metabolites of a different fungal species (32). Because it was stable to many activity-destroying β -lactamases, its core nucleus, 7-aminocephalosporanic acid was substituted with a wide variety of unnatural side chains, and three generations of clinically useful analogs have resulted. Later work produced the carbapenems, monobactams, and β -lactamase inhibitors (33–35). Many thousands of these compounds have been prepared by partial or total chemical synthesis, and a significant number of these remain on the market many years after their discovery.

Penicillins The medicinal classifications, chemical structures, and generic names of the penicillins currently available are provided in Table 33.4.

Preparation of Penicillins. The original fermentation-derived penicillins were produced by growth of the fungus *Penicillium chrysogenum* on complex solid media with

TABLE 33.4 Commercially Significant Penicillins and Related Molecules

		
Generic Name	Trade Name	R ₁
Fermentation-Derived Penicillins		
6-Aminopenicillanic acid		H
Benzylpenicillin (Penicillin G)	Generic	C ₆ H ₅ -CH ₂
Phenoxymethylpenicillin (Penicillin V)	Generic	C ₆ H ₅ -OCH ₂
Semi-Synthetic Penicillinase-Resistant Parenteral Penicillins		
Methicillin ^a		
Nafcillin	Nallpen, Unipen	
Semi-Synthetic Penicillinase-Resistant Oral Penicillins		
Oxacillin (X = Y = H)	Bactocill } Generic }	
Dicloxacillin (X = Y = Cl)		
Semi-Synthetic Penicillinase-Sensitive, Broad Spectrum, Parenteral Penicillins		
Carbenicillin ^a (R ₂ = H)		
Carbenicillin indanyl ^a (R ₂ = )		
Ticarcillin (in combination with clavulanic acid)	Timentin	
Mezlocillin ^a		
Piperacillin (in combination with tazobactam)	Zosyn	
Semi-Synthetic Penicillinase-Sensitive, Broad-Spectrum, Oral Penicillins		
Ampicillin (X = H)	Generic Amoxil, Trimox }	
Amoxicillin (X = OH)		

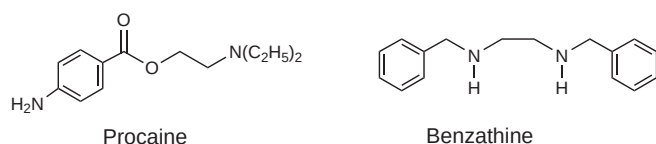
^aNot currently available in U.S.

the result that they were mixtures differing from one another in the identity of the side chain moiety. When a sufficient supply of phenylacetic acid is present in liquid media, this is preferentially incorporated into the molecule to produce mainly benzylpenicillin (penicillin G in the old nomenclature). Use of phenoxyacetic acid instead leads to phenoxymethyl penicillin (penicillin V).

More than two dozen different penicillins have been made in this way, but these two are the only ones that remain in clinical use. The complete exclusion of side chain precursor acids from the medium produces the fundamental penicillin nucleus, 6-aminopenicillanic acid (6-APA), but in poor yield. By itself, 6-APA has only very weak antibiotic activity, but when substituted on its

primary amino group with a suitable acid to give amide side chains, its potency and antibacterial spectrum are profoundly enhanced. With this key precursor isolated, limitations caused by enzyme specificities in biosynthesis could be overcome by use of partial chemical synthesis.

The sodium and potassium salts of penicillins are crystalline, hygroscopic, and water soluble. They can be employed orally or parentally. When dry, they are stable for long periods, but they hydrolyze rapidly when in solution. Their best stability is noted at pH values between 5.5 and 8, especially at pH 6.0 to 7.2. The procaine and benzathine salts of benzylpenicillin, on the other hand, are water insoluble. Because they dissolve slowly, they are used for repository purposes following injection when long-term blood levels are required.



Nomenclature. The nomenclature of the penicillins, as with most antibiotics, is complex. The Chemical Abstracts system is definitive and unambiguous but too complex for ordinary use (Fig. 33.12). For example, the chemical name for benzylpenicillin sodium is monosodium (2*S*,5*R*,6*R*)-3,3-dimethyl-7-oxo-6-(2-phenylacetamido)-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylate. A simpler system has stood the test of time and involves taking the repeating unit, carbonyl-6-APA, and adding to this the chemical trivial name for the added radical that completes the structure. The use of the names benzylpenicillin and phenoxymethylpenicillin makes practical sense. There are three asymmetric centers in the benzylpenicillin molecule as indicated by the asterisk in Table 33.4. This absolute stereochemistry must be preserved for useful antibiotic activity.

Clinically Relevant Chemical Instabilities. The most unstable bond in the penicillin molecule is the highly strained and reactive β -lactam amide bond. This bond cleaves moderately slowly in water unless heated, but breaks down much more rapidly in alkaline solutions to produce penicilloic acid, which readily decarboxylates to produce penilloic acid (Fig. 33.13). Penicilloic acid has a negligible tendency to reclose to the corresponding penicillin, so this reaction is essentially irreversible under physiologic conditions. Because the β -lactam ring is an essential portion of the pharmacophore, its hydrolysis deactivates the antibiotic. A fairly significant degree of hydrolysis also takes place in the liver. The bacterial enzyme, β -lactamase, catalyzes this reaction also and is a principal cause of bacterial resistance in the clinic. Alcohols and amines bring about the same cleavage reaction, but the products are the corresponding esters and amides. These products are inactive.

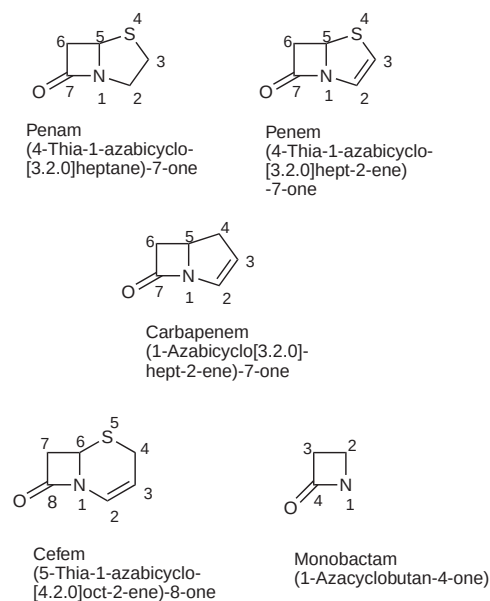


FIGURE 33.12 Ring and numbering systems of clinically available β -lactam antibiotic types.

A reaction with a specific primary amino group of aminoglycoside antibiotics is of clinical relevance as it inactivates penicillins and cephalosporins and will be discussed later in the chapter. When proteins serve as the nucleophiles in this reaction, the antigenic conjugates that cause many penicillin allergies are produced. Small molecules that are not inherently antigenic but react with proteins to produce antigens in this manner are called haptens. Commercially available penicillin salts may be contaminated with small amounts of these antigenic penicilloyl proteins derived from reaction

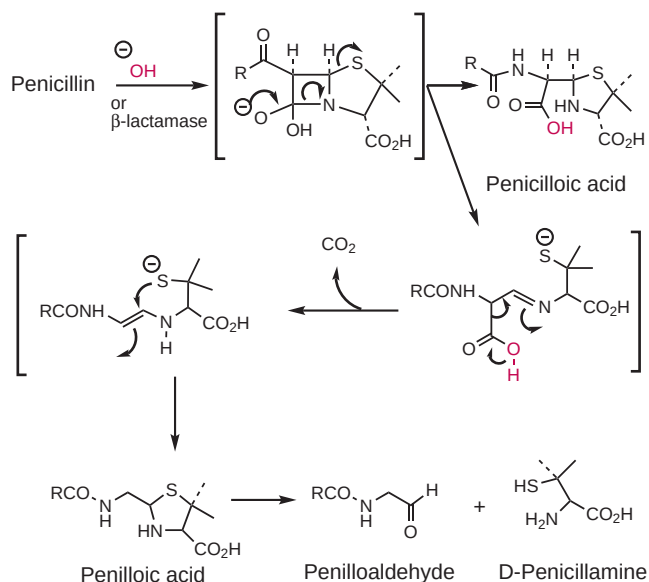


FIGURE 33.13 Instability of β -lactams to nucleophiles.

with proteins encountered in their fermentative production or by high molecular weight self-condensation-derived polymers resulting when penicillins are concentrated and react with themselves. Both of these classes of impurities are antigenic and may sensitize some patients.

In acidic solutions, the hydrolysis of penicillins is complex. Hydrolysis of the β -lactam bond can be shown through kinetic analysis to involve participation of the side chain amide oxygen because the rate of this reaction differs widely depending on the nature of the R group. The main end products of the acidic degradation are penicillamine, penilloic acid, and penilloaldehyde (Fig. 33.14). The intermediate penicillenic acid is highly unstable and undergoes subsequent hydrolysis to the corresponding penicilloic acid. An alternate pathway involves sulfur ejection to a product that in turn fragments to liberate penicilloic acid also. Penicilloic acid readily decarboxylates to penilloic acid. The latter hydrolyzes to produce penilloaldehyde and penicillamine (itself used clinically as a chelating agent). Several related fragmentations to a variety of other products take place. None of these products has antibacterial activity. At gastric pH (~ 2.0) and temperature of 37°C , benzyl penicillin has a half-life measured in minutes. The less water-soluble amine salts are more stable.

Structure–Activity Relationship. The chemical substituents attached to the penicillin nucleus can greatly influence the stability of the penicillins as well as the spectrum of activity. It is important to recognize whether the

structural changes affect drug stability on the shelf or in the GI tract (in vivo), improve stability toward bacterial metabolism, or enlarge the spectrum of activity.

The substitution of a side-chain R group on the primary amine of 6-APA with an electron-withdrawing group decreases the electron density on the side-chain carbonyl and protects these penicillins in part from acid degradation. This property has clinical implications because these compounds survive passage through the stomach better and many can be given orally for systemic purposes. The survival of passage and degree of absorption under fasting conditions is shown in Table 33.5.

In addition, in vitro degradation reactions of penicillins can be retarded by keeping the pH of solutions between 6.0 and 6.8 and by refrigerating them. Metal ions such as mercury, zinc, and copper catalyze the degradation of penicillins so they should be kept from contact with penicillin solutions. The lids of containers used today are routinely made of inert plastics in part to minimize such problems.

The more lipophilic the side chain of a penicillin, the more serum protein bound is the antibiotic (Table 33.6). Although this has some advantages in terms of protection from degradation, it does reduce measurably the effective bactericidal concentration of the drug in whole blood. The degree of serum protein binding of the penicillins has comparatively little influence on their half-lives. The penicillins are actively excreted into the urine via an active transport system for weak acids, and the rate of release from their protein-bound form is sufficiently rapid that the controlling rate is the

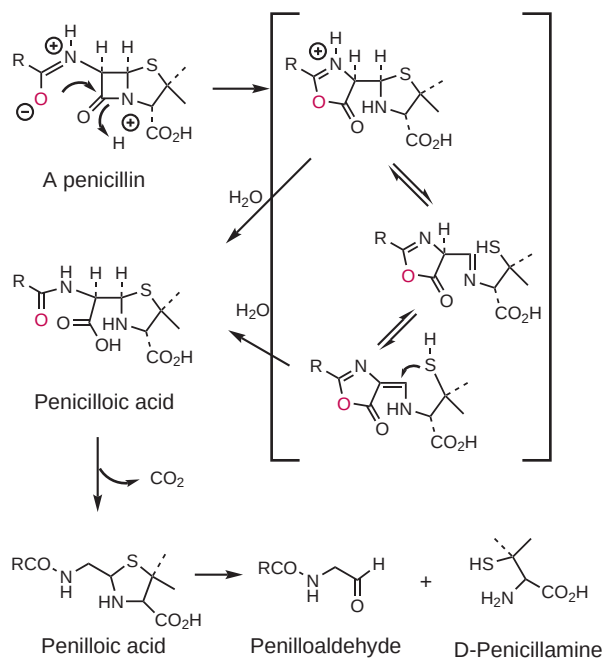


FIGURE 33.14 Instability of penicillins in acid. Hydrolysis involves the C-6 side chain.

TABLE 33.5 Improved Acid Stability and Absorption of Substituted Penicillins

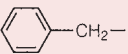
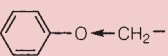
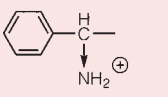
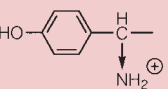
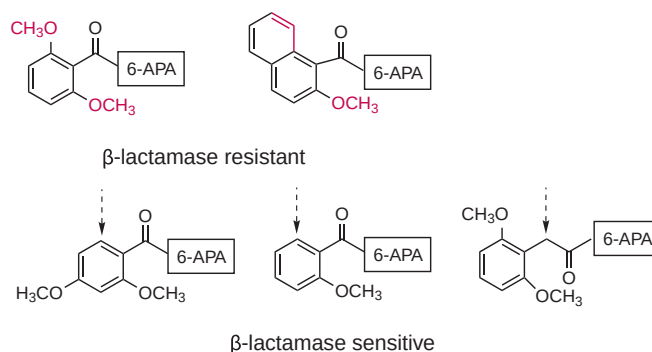
Drug		% Absorption intact drug
Benzylpenicillin		15–30
Pen V		60–73
Ampicillin		30–50
Amoxicillin		75–90

TABLE 33.6 Protein Binding of Penicillins

Penicillin	Protein Binding (%)
Benzyl penicillin	45–68
Phenoxymethyl penicillin	75–89
Methicillin	35–80
Ampicillin	25–30
Amoxicillin	25–30
Carbenicillin	~50
Oxacillin	>90
Cloxacillin	>90

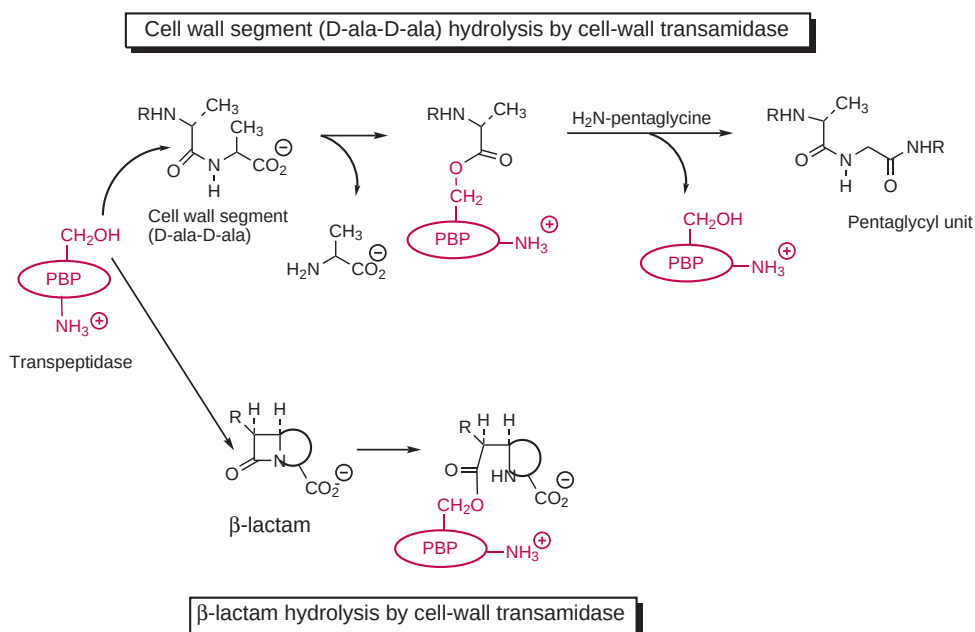
kidney secretion rate. The serum half-life of penicillin G is about 0.4 to 0.9 hours, and that of phenoxymethyl penicillin is about 0.5 hours. Both are excreted into the urine by tubular excretion. Probenecid, when present, competes effectively for excretion and thus prolongs the half-life.

Stability of the penicillins toward β -lactamase is influenced by the bulk in the acyl group attached to the primary amine. β -Lactamases are much less tolerant to the presence of steric hindrance near the side-chain amide bond than are the PBPs. The stability of methicillin to β -lactamases is an example of this. When the aromatic ring is attached directly to the side chain carbonyl and

**FIGURE 33.15** β -Lactamase resistant/sensitive structural features.

both ortho-positions are substituted by methoxy groups, β -lactamase stability results (Fig. 33.15). Movement of one of the methoxy groups to the para position or replacing one of them with a hydrogen results in an analog sensitive to β -lactamases. Putting in a methylene between the aromatic ring and 6-APA likewise produces a β -lactamase-sensitive agent (Fig. 33.15). These findings provide strong support for the hypothesis that its resistance to enzyme degradation is based on differential steric hindrance. Prime examples of this effect are seen in nafcillin, oxacillin, and dicloxacillin (Table 33.4).

Mechanism of Action. The molecular mode of action of the β -lactam antibiotics is a selective and irreversible inhibition of the enzymes processing the developing peptidoglycan layer (Fig. 33.16). Just before cross-linking

**FIGURE 33.16** Cell wall cross-linking and mechanism of action of β -lactams.

occurs, the peptide pendant from the lactate carboxyl of a muramic acid unit terminates in a D-alanyl-D-alanine unit. The terminal D-alanine unit is exchanged for a glycine unit on an adjacent strand in a reaction catalyzed by a cell wall transamidase. This enzyme is one of the PBPs (carboxypeptidases, endopeptidases, and transpeptidases) that normally reside in the bacterial inner membrane and perform construction, repair, and house-keeping functions, maintaining cell wall integrity and playing a vital role in cell growth and division. They differ significantly from bacterium to bacterium, and this is used to rationalize different potency and morphologic outcomes following β -lactam attack on the different bacteria. The cell wall transamidase uses a serine hydroxyl group to attack the penultimate D-alanyl unit forming a covalent ester bond, and the terminal D-alanine, which is released by this action, diffuses away. The enzyme-peptidoglycan ester bond is attacked by the free amino end of a pentaglycyl unit of an adjacent strand, regenerating the transpeptidase's active site for further catalytic action and producing a new amide bond, which connects two adjacent strands together.

The three-dimensional geometry of the active site of the enzyme perfectly accommodates to the shape and separation of the amino acids of its substrate. Because the substrate has unnatural stereochemistry at the critical residues, this enzyme is not expected to attack host peptides or even other bacterial peptides composed of natural amino acids.

The penicillins and the other β -lactam antibiotics have a structure that closely resembles that of acylated D-alanyl-D-alanine. The enzyme mistakenly accepts the penicillin as though it were its normal substrate. The highly strained β -lactam ring is much more reactive than a normal amide moiety, particularly when fused into the appropriate bicyclic system. The intermediate acyl-enzyme complex, however, is rather different structurally from the normal intermediate in that the hydrolysis does not break penicillin into two pieces as it does with its normal substrate. In the penicillins, a heterocyclic residue is still covalently bonded and cannot diffuse away as the natural terminal D-alanine unit does. This presents a steric barrier to approach by the nearby pentaglycyl unit and thus keeps the enzyme's active site from being regenerated and the cell wall precursors from being cross-linked. The result is a defective cell wall and an inactivated enzyme. The relief of strain that is obtained on enzymatic β -lactam bond cleavage is so pronounced that there is virtually no tendency for the reaction to reverse. Water is also an insufficiently effective nucleophile and cannot hydrolyze the complex either. Thus, the cell wall transamidase is stoichiometrically inactivated. The gaps in the cell wall produced by this covalent interruption are not filled in because the enzyme is now inactivated. The resulting cell wall is structurally weak and subject to osmotic stress. Cell lysis can result, and the cell rapidly dies assisted by another class of bacterial enzymes, the autolysins (36).

Resistance. The first literature reports of a penicillinase were published in 1940 (37). This phenomenon was rare at the time and caused no particular alarm. Resistance to β -lactam antibiotics is unfortunately increasingly common and is rather alarming. It can be intrinsic and involve decreased cellular uptake of drug, or involve lower binding affinity to the PBPs. This is particularly the case with MRSA. MRSA produces a mutated PBP-2 (PBP-2a) that does not efficiently bind methicillin any longer. More common, however, is the elaboration of a β -lactamase. β -Lactamases are enzymes (usually serine proteases) elaborated by microorganisms that catalyze hydrolysis of the β -lactam bond and inactivate β -lactam antibiotics to penicilloic acids before they can reach the PBPs (Fig. 33.17). They somewhat resemble the cell wall transamidase, which is the usual target. Hydrolytic regeneration of the active site is dramatically more facile with β -lactamases than is the case with cell wall transamidase, so that the enzyme can turn over many times and a comparatively small amount of enzyme can destroy a large amount of drug. With gram-positive bacteria, such as staphylococci, the β -lactamases are usually shed continuously into the medium and meet the drug outside the cell wall. They are biosynthesized in significant quantities. With gram-negative bacteria, a more conservative course is followed. Here the β -lactamases are secreted into the periplasmic space between the inner and outer membrane so, while still distal to the PBPs, they do not readily escape into the medium and need not be resynthesized as often. Numerous β -lactamases with various antibiotic substrate specificities are now known. Various classification systems are used for β -lactamases and have been extensively reviewed (29). Elaboration of β -lactamases is often R-factor mediated and, in some cases, is even induced by the presence of β -lactam antibiotics.

Allergenicity. It is estimated that 2% to 8% of the U.S. population is allergic to β -lactam antibiotics. The actual number is very difficult to determine due to overreporting. Most commonly, the allergy is expressed as a mild drug rash or itching and is of delayed onset. Occasionally, the reaction is immediate and profound. It may include cardiovascular collapse and shock and can result in death. Sometimes penicillin allergy can be anticipated

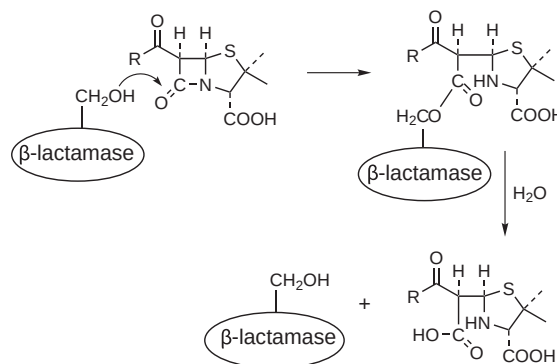


FIGURE 33.17 β -Lactamase-catalyzed hydrolysis of penicillins.

by taking a medication history, and often, patients likely to be allergic are those with a history of hypersensitivity to a wide variety of allergens (e.g., foods and pollens). A prior history of allergy to penicillins is a contraindicating factor to their use. Skin tests are available when there is doubt. When an allergic reaction develops, the drug must be discontinued and, because cross-sensitivity is relatively common, other β -lactam families should be used carefully. Considering all therapeutic categories, penicillins are probably the drugs most associated with allergy. Erythromycin and clindamycin are useful alternate choices for therapy in many cases of penicillin allergy.

In some cases, the patient may have become sensitized without knowing it due to prior passive exposure through contaminated foodstuffs or cross-contaminated medications. Penicillins are manufactured in facilities separate from those used to prepare other drugs in order to prevent cross contamination and possible sensitization. Animals treated with penicillins are required to be drug free for a significant time before products prepared from them can be consumed. Because the origin of the allergy is a haptenic reaction with host proteins and the responsible bond in the drug is the β -lactam moiety, this side effect is caused by the pharmacophore of the drug and is unlikely to be overcome by molecular manipulation.

Individual Penicillins. The penicillins are usually discussed under various groups based on spectrum of activity and sensitivity or resistance toward β -lactamase. One of the earliest and still most commonly used penicillins is benzylpenicillin.

Benzylpenicillin Group.

BENZYLPENICILLIN (PENICILLIN G, TABLE 33.4) With the exception of *Neisseria gonorrhoeae* and *Haemophilus influenzae*, and a few bacteria encountered less frequently, the useful antimicrobial spectrum of benzylpenicillin is primarily against gram-positive cocci. Because of its cheapness, efficacy, and lack of toxicity (except for acutely allergic patients), benzylpenicillin remains a remarkably useful agent for treatment of diseases caused by susceptible microorganisms. As with most antibiotics, susceptibility tests must be performed because many formerly highly sensitive microorganisms are now comparatively resistant. Infections of the upper and lower respiratory tract and the genitourinary tract are the particular province of benzylpenicillin. Infections caused by group A β -hemolytic streptococci (pharyngitis, scarlet fever, cellulitis, pelvic infections, and septicemia) are commonly responsive. Group B hemolytic streptococci infections, especially of neonates (acute respiratory distress, pneumonia, meningitis, septic shock, and septicemia), also may respond. Pneumococcal pneumonia, *H. influenzae* pneumonia of children, *Streptococcus pneumoniae*- and *Streptococcus pyogenes*-caused otitis media and sinusitis, meningococcal meningitis and brain abscess, meningococcal and pneumococcal septicemia, streptococcal endocarditis (often by *Streptococcus viridans*), pelvic inflammatory disease

(often by *Neisseria gonorrhoeae* and *S. pyogenes*), uncomplicated gonorrhea (*N. gonorrhoeae*), meningitis (*Neisseria meningitidis*), syphilis (*Treponema pallidum*), Lyme disease (*Borrelia burgdorferi*), gas gangrene (*Clostridium perfringens*), and tetanus (*Clostridium tetani*) are among the diseases that may respond to benzylpenicillin therapy, either alone or sometimes with other drugs used in combination. Nonpenicillinase-producing *Staphylococcus aureus* and *Staphylococcus epidermidis* are quite sensitive but are very rare today. Other, less common, bacterial diseases also respond, such as those caused by *Bacillus anthracis* (anthrax) and *Corynebacterium diphtheriae* (diphtheria).

Because of its low cost, mild infections with susceptible microorganisms can be treated with comparatively large oral doses, although the most effective route of administration is parenteral because five times the blood level can be regularly achieved in this manner. As previously discussed (see previous section *Clinically relevant chemical instabilities*), penicillin G is unstable under the acidic conditions of the stomach.

Very water-insoluble penicillin salts form with procaine and with *N,N*-bibenzyl ethylenediamine. These find therapeutic application for intramuscular injections. This produces lower but prolonged levels of penicillin as the drug slowly diffuses from the injection site.

The need to improve defects in benzylpenicillin stimulated an intense research effort that persists to this day. Overcoming such negative features as comparative instability (particularly to acid), comparatively poor oral absorption, allergenicity, sensitivity to β -lactamases, and relatively narrow antimicrobial spectrum has been an objective of this work.

PHENOXYMETHYL PENICILLIN (PENICILLIN V, TABLE 33.4) Penicillin V is produced by fermentation where the medium is enriched in phenoxyacetic acid. It can also be prepared by semi-synthesis. It is considerably more acid stable than benzylpenicillin as indicated by oral absorption (Table 33.5). This is rationalized as being due to the electronegative oxygen atom in the C-7 amide side chain inhibiting participation in β -lactam bond hydrolysis. In any case, penicillin V was the first of the so-called oral penicillins giving higher and more prolonged blood levels than penicillin G itself. Its antimicrobial and clinical spectrum is roughly the same as that of benzylpenicillin, although it is somewhat less potent and is not used for acutely serious infections. Penicillin V has approximately the same sensitivity to β -lactamases and allergenicity as penicillin G.

Penicillinase-Resistant Penicillins.

METHICILLIN (TABLE 33.4) Methicillin is now archaic but was the first of the penicillinase-resistant agents to reach the clinic (38). It is unstable to gastric acid, having a half-life of 5 minutes at pH 2, so it has to be administered via injection. As discussed previously, increased bulk resulting from the addition of the dimethoxybenzoyl group to 6-APA leads to methicillin being a β -lactamase-resistant drug. Methicillin has significantly narrower antimicrobial spectrum and less potency, so it was restricted to

clinical use primarily for parenteral use in infections due to β -lactamase-producing *S. aureus* and a few other infections. An increasing number of infections are caused by MRSA. In these organisms, an altered PBP is formed that has a very low affinity for β -lactams, excluding ceftazoline. Furthermore, methicillin is an efficient inducer of penicillinases. Consequently this drug fell out of favor, and methicillin has now been replaced.

NAFCILLIN (TABLE 33.4) Nafcillin has a 2-ethoxynaphthyl side chain. This bulky group serves to inhibit destruction by β -lactamases analogous to methicillin. Although slightly more acid stable than methicillin, it is clinically virtually identical to it.

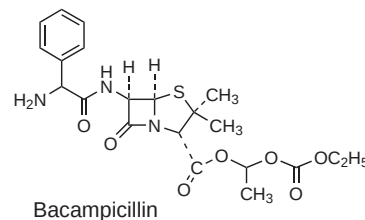
OXACILLIN AND DICLOXACILLIN Using a substituted isoxazoyl ring as a bioisosteric replacement for the benzene ring of penicillin G produces the isoxazoyl penicillins. Currently, oxacillin and dicloxacillin (Table 33.4) are available. Chemically, they differ from one another by chlorine substituents on the benzene ring. Like methicillin, these are generally less potent than benzylpenicillin against gram-positive microorganisms (generally staphylococci and streptococci) that do not produce a β -lactamase but retain their potency against those that do. An added bonus exists in that they are somewhat more acid stable; thus they may be taken orally, and they are more potent as well. Because they are highly serum protein bound (Table 33.6), they are not good choices for treatment of septicemia. Microorganisms resistant against methicillin are also resistant to the isoxazoyl group of penicillins. Like nafcillin, the isoxazoyl group of penicillins is primarily used against penicillinase-producing *Staphylococcus aureus* (39,40).

Penicillinase-Sensitive, Broad-Spectrum, Oral Penicillins.

AMPICILLIN The first member of this group, ampicillin, is a benzylpenicillin analog in which one of the hydrogen atoms of the side chain methylene has been replaced with a primary amino group to produce an *R*-phenylglycine moiety (Table 33.4). In addition to significant acid stability enhancing its successful oral use, the antimicrobial spectrum is shifted so that many common gram-negative pathogens are sensitive to ampicillin. This is believed to be due to greater penetration of ampicillin into gram-negative bacteria. The acid stability is generally believed to be caused by the electron-withdrawing character of the protonated primary amine group reducing participation in hydrolysis of the β -lactam bond as well as to the comparative difficulty of bringing another positively charged species (H_3O^+) into the vicinity of the protonated amino group. The oral activity is also enhanced, in part, to active uptake by the dipeptide transporters (41). It unfortunately lacks stability toward β -lactamases, and resistance is increasingly common. To assist with this, several β -lactamase inhibitors for coadministration (discussed below) have been developed that restore many penicillinase-producing strains to the clinical spectrum of the aminopenicillins.

Ampicillin is essentially equivalent to benzyl penicillin for pneumococcal, streptococcal, and meningococcal infections and many strains of gram-negative *Salmonella*, *Shigella*, *Proteus mirabilis*, and *Escherichia coli*, as well as many strains of *Haemophilus influenzae* and *Neisseria gonorrhoeae* respond well to oral treatment with ampicillin.

BACAMPICILLIN

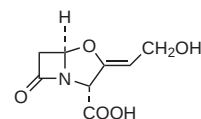


Although comparatively well absorbed, ampicillin's oral efficacy for systemic infections can be enhanced significantly through the preparation of prodrugs. In contrast to ampicillin itself, which is amphoteric, bacampicillin is a weak base and is very well absorbed in the duodenum. Enzymatic ester hydrolysis in the gut wall liberates carbon dioxide and ethanol followed by spontaneous loss of acetaldehyde and production of ampicillin. The acetaldehyde is metabolized oxidatively by alcohol dehydrogenase to produce acetic acid, which joins the normal metabolic pool. Amoxicillin, which has better oral availability than ampicillin has made the clinical use of bacampicillin unnecessary, and it is no longer available in the United States.

AMOXICILLIN (TABLE 33.4) Amoxicillin is a close analog of ampicillin in which a para-phenolic hydroxyl group has been introduced into the side-chain phenyl moiety. This adjusts the isoelectric point of the drug to a more acidic value and is believed to be partially responsible, along with the intestine dipeptide transporter, for the enhanced blood levels obtained with amoxicillin compared with ampicillin itself (Table 33.5). Better oral absorption leads to less disturbance of the normal GI flora and, therefore, less drug-induced diarrhea. The antimicrobial spectrum and clinical uses of amoxicillin are approximately the same as those of ampicillin.

The addition of clavulanic acid (below) to amoxicillin gives a combination (Augmentin) in which the clavulanic acid serves to protect amoxicillin to a considerable extent against β -lactamases. This expands the spectrum of activity to include organisms and strains that produce β -lactamases.

CLAVULANIC ACID



Clavulanic acid

Clavulanic acid is a mold product that has only weak intrinsic antibacterial activity but is an excellent irreversible

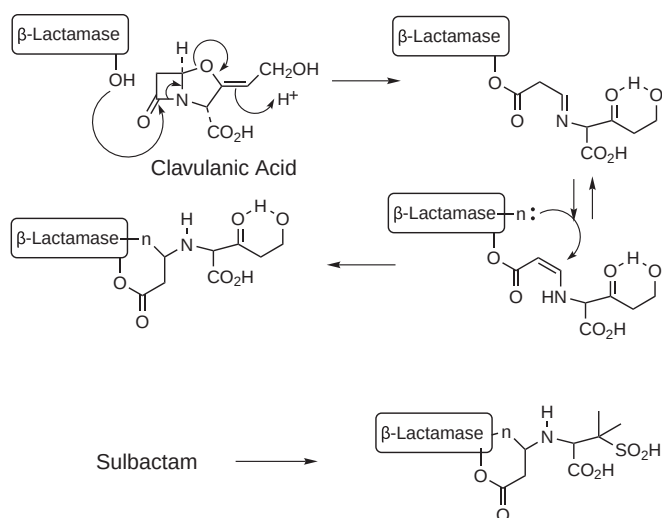
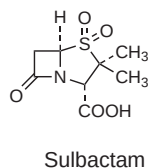


FIGURE 33.18 Speculative mechanism for irreversible inactivation of β -lactamase by clavulanic acid and sulbactam.

inhibitor of most β -lactamases. It is believed to acylate the active site serine by mimicking the normal substrate. While hydrolysis occurs with some β -lactamases, in many cases, subsequent reactions occur that inhibit the enzyme irreversibly. This leads to its classification as a mechanism-based inhibitor (or so-called suicide substrate). The precise chemistry is not well understood (Fig. 33.18), but when clavulanic acid is added to amoxicillin and ticarcillin preparations, the potency against β -lactamase-producing strains is markedly enhanced.

SULBACTAM



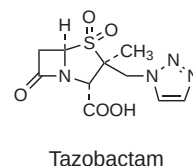
Another β -lactamase inhibitor is sulbactam. Sulbactam is prepared by partial chemical synthesis from penicillins. The oxidation of the sulfur atom to a sulfone greatly enhances the potency of sulbactam. The combination of sulbactam and ampicillin (Unasyn) is also clinically popular. Not all β -lactamases are sensitive to the presence of clavulanic acid or sulbactam.

Penicillinase-Sensitive, Broad-Spectrum, Parenteral Penicillins.

MEZLOCILLIN AND PIPERACILLIN Mezlocillin and piperacillin are ampicillin derivatives in which the D-side chain amino group has been converted by chemical processes to a variety of substituted urea analogs (Table 33.4). These are known as acylureidopenicillins and preserve the useful anti-gram-positive activity of ampicillin but have higher anti-gram-negative potency. Some strains of *Pseudomonas aeruginosa* are sensitive to these agents. It is speculated that the added side-chain moiety mimics a longer segment of the peptidoglycan chain than ampicillin does. This cell

wall fragment is usually a tetrapeptide, so there is expected to be room for an extension in this direction. This would give more possible points of attachment to the penicillin-binding proteins, and perhaps these features are responsible for their enhanced antibacterial properties. These agents are used parenterally with particular emphasis on gram-negative bacteria, especially *Klebsiella pneumoniae* and the anaerobe, *Bacteroides fragilis* (42). Resistance due to β -lactamases is a prominent feature of their use, so disk testing and incorporation of additional agents (such as an aminoglycoside) for the treatment of severe infections are advisable. Mezlocillin is not marketed in the United States.

PIPERACILLIN AND TAZOBACTAM COMBINATION (ZOSYN)



Tazobactam is often coadministered with piperacillin because of tazobactam's ability to inhibit β -lactamases. Tazobactam, like other β -lactamase inhibitors, has little or no antibacterial activity. This effect is analogous to that of clavulanic acid and sulbactam discussed earlier. This combination is the broadest spectrum penicillin currently available (43).

CARBENICILLIN AND INDANYL CARBENICILLIN Carbenicillin is a benzyl penicillin analog in which one of the methylene hydrogens of the side chain has been substituted with a carboxylic acid moiety (Table 33.4). The specific stereochemistry of this change is not very important as both diastereoisomers are configurationally unstable and mutarotate with time to produce the same mixture of epimers. The introduction of the side-chain carboxyl produces enhanced anti-gram-negative activity. In fact, carbenicillin is intrinsically one of the broadest-spectrum penicillins. Carbenicillin is an order of magnitude less potent than the acylureidopenicillins. The drug is susceptible to β -lactamases and is acid unstable and thus must be given by injection.

Because it is a malonic acid hemiamide with a carbonyl (amide) moiety beta to the carboxyl group, carbenicillin can decarboxylate readily to produce benzyl penicillin (Fig. 33.19). While still an antibiotic, this degradation product has no activity against the organisms for

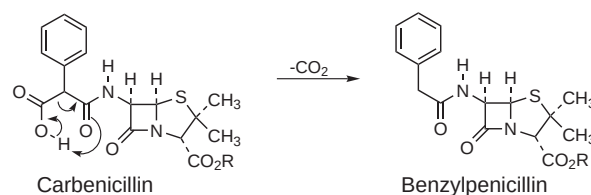


FIGURE 33.19 Decarboxylation of carbenicillin to benzylpenicillin.

which carbenicillin would be indicated. In addition, the large doses of carbenicillin sodium that have to be used (multigrams per day) result in ingestion of a significant amount of sodium ion, which could be a consideration for patients. Indanyl carbenicillin is an oral prodrug form. Due to these issues, neither form is currently available, and these drugs have been replaced by ticarcillin.

TICARCILLIN Ticarcillin is a sulfur-based bioisostere of carbenicillin that cannot decarboxylate as the carboxyl group of carbenicillin does (Table 33.4). This agent is somewhat more potent against pseudomonads than is indanyl carbenicillin.

When potassium clavulanate is added to ticarcillin (Timentin), the combination has enhanced spectrum due to its enhanced stability to lactamases (44).

Summary. The penicillins ushered in the era of powerful antibiotics, and their use transformed the practice of antimicrobial chemotherapy. A significant percentage of the population that is alive today owes their longevity and relative freedom from morbidity to the use of these agents. The pace of discovery has fallen off dramatically, and no new penicillin has been introduced into the market for many years. Instead, although retaining their important place in contemporary medicine, research has turned elsewhere for novel agents.

Cephalosporins

History. In contrast to the discovery of the penicillins, in which the first agent had such outstanding biologic antibiotic properties that it entered clinical use with comparatively little modification, the cephalosporins are remarkable for the level of persistence required before their initial discovery yielded economic returns. Abraham and Newton described the structure of the first cephalosporin, cephalosporin C (32). The compound was interesting because, although it was not very potent, it had activity against some penicillin-resistant cultures due to its stability to β -lactamases. Cephalosporin C is not potent enough to be a useful antibiotic, but removal, through chemical means, of the natural side chain produced 7-aminocephalosporanic acid, which, analogous to 6-APA, could be fitted with unnatural side chains (Fig. 33.20). Many of the compounds produced in this way are remarkably useful antibiotics. They differ from one another in antimicrobial spectrum, β -lactamase stability, absorption from the GI tract, metabolism, stability, and side effects as detailed below.

Chemical Properties. The cephalosporins have their β -lactam ring annealed to a six-membered dihydrothiazine ring, in contrast to the penicillins where the β -lactam ring is fused to a five-membered thiazolidine ring. As a consequence of the bigger ring, the cephalosporins should be less strained and less reactive/potent. However, much of the reactivity loss is made up by possession of an olefinic

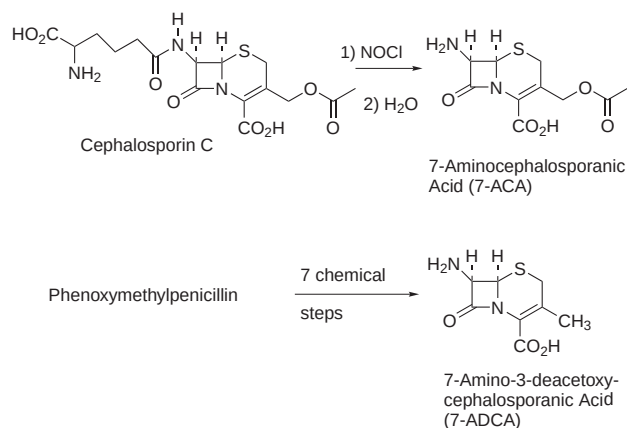
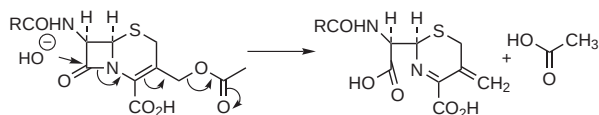


FIGURE 33.20 Chemical preparation of 7-aminocephalosporanic acid (7-ACA) and 7-amino-3-deacetoxycephalosporanic acid (7-ADCA).

linkage at C-2,3 and a methyleneacetoxy or other leaving group at C-3. When the β -lactam ring is opened by hydrolysis, the leaving group can be ejected, carrying away the developing negative charge. This greatly reduces the energy required for the process. Thus the facility with which the β -lactam bond of the cephalosporins is broken is modulated both by the nature of the C-7 substituent (analogous to the penicillins) as well as the nature of the C-3 substituent and its ability to serve as a leaving group. Considerable support for this hypothesis comes from the finding that isomerization of the olefinic linkage to C-3,4 leads to great losses in antibiotic activity. In practice, most cephalosporins are comparatively unstable in aqueous solutions, and the pharmacist is often directed to keep injectable preparations frozen before use. Being carboxylic acids, they form water-soluble sodium salts, whereas the free acids are comparatively water insoluble.

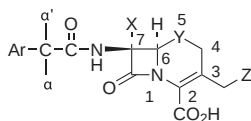
Clinically Relevant Chemical Instabilities. The principal chemical instability of the cephalosporins is associated with β -lactam bond hydrolysis. The role of the C-7 and C-3 side chains in these reactions was discussed previously. Ejection of the C3 substituent following β -lactam-bond cleavage is usually drawn for convenience as though this is an unbroken (concerted) process, although ejection of the side chain may at certain times and with specific cephalosporins involve a discrete intermediate with the β -lactam bond broken, but the C-3 substituent not yet eliminated, while other cephalosporins have nonejectable C-3 substituents. The methylthiotetrazole (MTT) group, found in a number of cephalosporins, is capable of elimination. When this happens, this moiety is believed to be responsible in part for clotting difficulties and acute alcohol intolerance in certain patients. The role of the C-7 side chain in all of these processes is clearly important, but active participation of the amide moiety in a manner analogous to the penicillins is rarely specifically invoked. The same considerations that modulate the chemical

stability of cephalosporins are also involved in dictating β -lactamase sensitivity, potency, and allergenicity.



Metabolism. Those cephalosporins that have an acetyl group in the side chain are subject to enzymatic hydrolysis in the body. The result is molecules with a hydroxymethyl moiety at C-3. A hydroxy moiety is a poor leaving group, so this change is considerably deactivating with respect to breakage of the β -lactam bond. In addition, the particular geometry of this part of the molecule leads to facile lactonization with the carboxyl group attached to C-2 (Fig. 33.21). In principle, this should result in formation of a different but reasonable leaving group. The result is, instead, inactivation of the drugs involved. The PBPs have an absolute requirement for a free carboxyl group to mimic that of the terminal carboxyl of the D-alanyl-D-alanine moiety in their normal substrate. Lactonization masks this docking functional group, and as a result, blocks affinity of the inhibitor for the enzyme.

Structure–Activity Relationship.



As with the penicillins, various molecular changes in the cephalosporin can improve in vitro stability, antibacterial activity, and stability toward β -lactamases (45). The addition of an amino and a hydrogen to the α and α' positions, respectively, results in a basic compound that is protonated under the acidic conditions of the stomach.

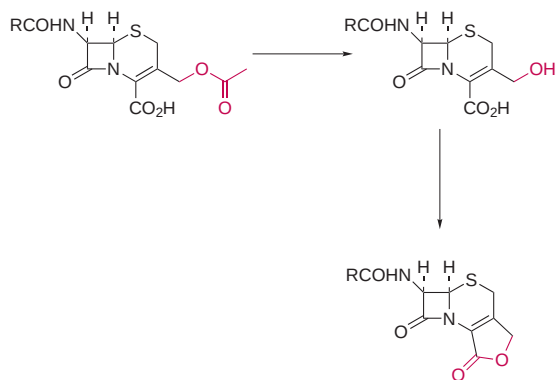


FIGURE 33.21 Metabolism of C-3-acetyl-substituted cephalosporins.

The ammonium ion improves the stability of the β -lactam of the cephalosporin, leading to orally active drugs.

The 7 β amino group is essential for antimicrobial activity ($X = H$), whereas replacement of the hydrogen at C-7 ($X = H$) with an alkoxy ($X = OR$) results in improvement of the antibacterial activity of the cephalosporin. Within specific cephalosporin derivatives, the addition of a 7 α methoxy also improves the drugs stability toward β -lactamase. The derivatives where $Y = S$ exhibit greater antibacterial activity than if $Y = O$, but the reverse is true when stability toward β -lactamase is considered. The 6 α hydrogen is essential for biologic activity. And finally, antibacterial activity is improved when Z is a five-membered heterocycle versus a six-membered heterocycle.

In a study examining stability of cephalosporins toward β -lactamase, it was noted that the following changes improved β -lactamase resistance: 1) the L-isomer of an α amino α' hydrogen derivative of a cephalosporin was 30 to 40 times more stable than the D-isomer; 2) the addition of a methoxyoxime to the α and α' positions increased stability nearly 100-fold; and 3) the Z-oxime was as much as 20,000-fold more stable than the E-oxime (Fig. 33.22) (45). These changes have been incorporated into a number of marketed and experimental cephalosporins (e.g., cefuroxime, ceftizoxime, ceftazidime, cefixime).

Mechanism of Action. The cephalosporins are believed to act in a manner analogous to that of the penicillins by binding to the penicillin-binding proteins followed by cell lysis (Fig. 33.16). Cephalosporins are bactericidal in clinical terms.

Resistance. Analogous to the penicillins, susceptible cephalosporins can be hydrolyzed by β -lactamases before they reach the penicillin-binding proteins. Certain β -lactamases are constitutive (chromosomally encoded) in certain strains of gram-negative bacteria (*Citrobacter*, *Enterobacter*, *Pseudomonas*, and *Serratia*) and are normally repressed. These are induced (or derepressed) by certain β -lactam antibiotics (e.g., imipenem, cefotetan, and cefoxitin). As with the penicillins, specific examples will be seen below wherein resistance to β -lactamase hydrolysis is conveyed by strategic steric bulk near the side-chain amide linkage. Penetration barriers to the cephalosporins are also known.

Allergenicity. Allergenicity is less commonly experienced and is less severe with cephalosporins than with penicillins. Cephalosporins are frequently administered to patients

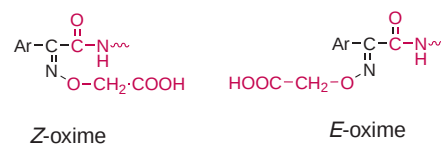


FIGURE 33.22 Z- and E-oxime configuration.

who have had a mild or delayed penicillin reaction; however, cross allergenicity is possible, and this should be done with caution for patients with a history of allergies. Patients who have had a rapid and severe reaction to penicillins should not be treated with cephalosporins.

Nomenclature and Classification. Most cephalosporins have generic names beginning with cef- or ceph-. This is convenient for classification but makes discriminating between individual members a true memory test. The cephalosporins are classified by a trivial nomenclature system loosely derived from the chronology of their introduction but more closely related to their antimicrobial spectrum. The first-generation cephalosporins are primarily active in vitro against gram-positive cocci (penicillinase-positive and -negative *Staphylococcus aureus* and *Staphylococcus epidermis*), group A β -hemolytic streptococci (*Streptococcus pyogenes*), group B streptococci (*Streptococcus agalactiae*), and *Streptococcus pneumoniae*. They are not effective against MRSA. They are not significantly active against gram-negative bacteria, although some strains of *Escherichia coli*, *Klebsiella pneumoniae*, *Proteus mirabilis*, and *Shigella* sp. may be sensitive (46). The second-generation cephalosporins generally retain the anti-gram-positive activity of the first-generation agents but include *Haemophilus influenzae* as well and add to this better anti-gram-negative activity, so that some strains of *Acinetobacter*, *Citrobacter*, *Enterobacter*, *E. coli*, *Klebsiella*, *Neisseria*, *Proteus*, *Providencia*, and *Serratia* are also sensitive (47). Cefotetan and cefoxitin also have antianaerobic activity as well (48). The third-generation cephalosporins are less active against staphylococci than the first-generation agents but are much more active against gram-negative bacteria than either the first- or the second-generation drugs (49). They are frequently particularly useful against nosocomial multidrug-resistant hospital-acquired strains. *Morganella* sp. and *Pseudomonas aeruginosa* can also be added to the list of species that are often sensitive. The fourth-generation cephalosporins have an antibacterial spectrum like the third-generation drugs but add some enterobacteria that are resistant to the third-generation cephalosporins. They are also more active against some gram-positive organisms (50).

Therapeutic Application. The incidence of cephalosporin resistance is such that it is usually preferable to do sensitivity testing before instituting therapy. Infections of the upper and lower respiratory tract, skin and related soft tissue, urinary tract, bones, and joints, as well as septicemias, endocarditis, intra-abdominal, and bile tract infections caused by susceptible gram-positive organisms are usually responsive to cephalosporins. When gram-positive bacteria are involved, a first-generation agent is preferable. When the pathogen is gram-negative and the infection is serious, parenteral use of a third-generation agent is recommended.

Adverse Effects. Aside from mild or severe allergic reaction, the most commonly experienced cephalosporin

toxicities are mild and temporary nausea, vomiting, and diarrhea associated with disturbance of the normal flora. Rarely, a life-threatening pseudomembranous colitis diarrhea associated with the opportunistic and toxin-producing anaerobic pathogen, *Clostridium difficile*, can be experienced. Rare blood dyscrasias, which can even include aplastic anemia, can also be seen. Certain structural types (details below) are associated with prolonged bleeding times and a disulfiram-like acute alcohol intolerance.

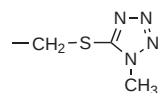
First-generation Cephalosporins (Table 33.7).

Cefazolin. Cefazolin has the natural acetyl side chain at C-3 replaced by a thio-linked thiadiazole ring. Although this group is an activating leaving group, the moiety is not subject to the inactivating host hydrolysis reaction. At C-7, it possesses a tetrazolymethylene unit. Its dosing should be reduced in the presence of renal impairment. It is comparatively unstable and should be protected from heat and light.

Cephalexin. Use of the ampicillin-type side chain conveys oral activity to cephalexin. Whereas it no longer has an activating side chain at C-3, and as a consequence is somewhat less potent, it does not undergo metabolic deactivation and thus maintains potency. It is rapidly and completely absorbed from the GI tract and has become quite popular. Somewhat puzzling is the fact that the use of the ampicillin side chain in the cephalosporins does not result in a comparable shift in antimicrobial spectrum. Cephalexin, like the other first-generation cephalosporins is active against many gram-positive aerobic cocci but is limited against gram-negative bacteria. It is a widely used drug, particularly against gram-negative bacteria causing urinary tract infections, gram-positive infections (*S. aureus*, *S. pneumoniae*, and *S. pyogenes*) of soft tissues, pharyngitis, and minor wounds (46).

Cefadroxil. Cefadroxil has an amoxicillin-like side chain at C-7 and is orally active. The prolonged biologic half-life of cefadroxil allows for once-a-day dosage.

Second-Generation Cephalosporins (Table 33.8). Some of the second-generation cephalosporins contain an *N*-methyl-5-thiotetrazole (commonly referred to by the acronym, MTT) at the C-3 position.

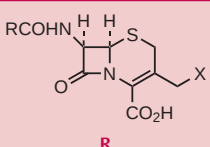
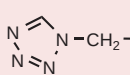
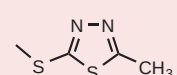
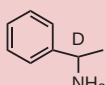
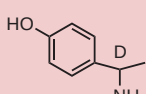


MTT

Loss of the MTT group is associated with prothrombin deficiency and bleeding problems, as well as with a disulfiram-like acute alcohol intolerance (51). However, this grouping enhances potency and prevents metabolism by deacetylation.

Cefuroxime. Cefuroxime has a *Z*-oriented methoxyimino moiety as part of its C-7 side chain (Fig. 33.22). This

TABLE 33.7 First Generation Cephalosporins

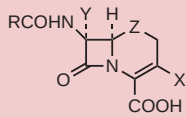
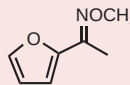
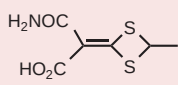
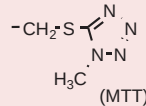
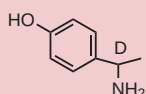
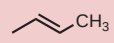
Generic Names	Trade Names	 R	X	Salt
Parenteral agents: Cefazolin	Ancef, Kefzol, Zolicef			Na
Oral agents: Cephalexin	Keflex, Biocef Keftab		H	HCl
Cefadroxil	Duricef		H	—

conveys considerable resistance to attack by many β -lactamases but not by all. This is believed to result from the steric demands of this group. Resistance by *Pseudomonas aeruginosa*, on the other hand, is attributed to lack of penetration of the drug rather than to enzymatic hydrolysis. The carbamoyl moiety at C-3 is intermediate in metabolic

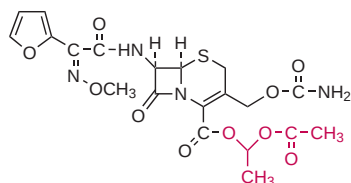
stability between the classic acetyl moieties and the thiotetrazoles.

In the form of its axetil ester (1-[acetyloxy]ethyl ester) prodrug, cefuroxime axetil, a more lipophilic drug is produced that results in satisfactory blood levels on oral administration. The ester bond is cleaved metabolically,

TABLE 33.8 Second Generation Cephalosporins

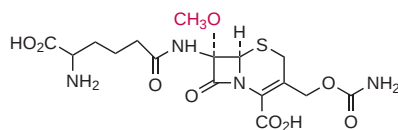
Generic Name	Trade Name	 R	X	Y	Z	Salt
Parenteral agents Cefuroxime	Ceftin Kefurox Zinacef		$-\text{CH}_2\text{OCONH}_2$	H	S	Na
Cefoxitin	Mefoxin		$-\text{CH}_2\text{OCONH}_2$	OCH_3	S	Na
Cefotetan	Cefotan		 (MTT)	OCH_3	S	diNa
Oral agents Cefaclor	Ceclor		Cl	H	S	—
Cefprozil	Cefzil			H	S	—

and the resulting intermediate form loses acetaldehyde spontaneously to produce cefuroxime itself. The axetil is officially labeled for treatment of Lyme disease (although doxycycline is often the first choice) (47).



Cefuroxime axetil

Cefoxitin. The most novel chemical feature of cefoxitin is the possession of an α -oriented methoxyl group in place of the normal H-atom at C-7. This increased steric bulk conveys very significant stability against β -lactamases. The inspiration for these functional groups was provided by the discovery of the naturally occurring antibiotic cephamycin C derived from fermentation of *Streptomyces lactamdurans* (52). Cephamycin C itself has not seen clinical use but provided the structural clue that led to useful agents such as cefoxitin. Agents that contain this 7 α methoxy group are commonly referred to as cephamycins.



Cephamycin C

Cefoxitin has useful activity against gonorrhea and against some anaerobic infections (including *Bacteroides fragilis*) as compared with its second-generation relatives. On the negative side, cefoxitin has the capacity to induce certain broad-spectrum β -lactamases.

Cefotetan. Cefotetan is also a cephamycin but has a rather unusual sulfur-containing C-7 side-chain amide. Possession of two carboxyl groups leads to its marketing as a disodium salt. The C-3 MTT side chain suggests caution in monitoring bleeding as well as care in ingesting alcohol when using this agent. Like cefoxitin, cefotetan has better activity against anaerobes than the rest of this group. Like cefoxitin, it is stable to a wide range of β -lactamases but is also an inducer in some bacteria.

Cefaclor. Cefaclor differs from cephalixin primarily in the bioisosteric replacement of methyl by chlorine at C-3 and is quite acid stable, allowing for oral administration. It is also quite stable to metabolism. It is less active against gram-negative bacteria than the other second-generation cephalosporins but is more active against gram-negative bacteria than the first-generation drugs.

Cefprozil. Cefprozil has an amoxicillin-like side chain at C-7, but at C-3, there is now a 1-propenyl group conjugated with the double bond in the six-membered ring.

The double bond is present in its two geometric isomeric forms, both of which are antibacterially active. Fortunately, the predominant *trans* form (illustrated in Table 33.8) is much more active against gram-negative organisms. Cefprozil most closely resembles cefaclor in its properties but is a little more potent.

Third-Generation Cephalosporins (Table 33.9).

Cefotaxime. Cefotaxime, like cefuroxime, has a Z-methoxyimino moiety at C-7 that conveys significant β -lactamase resistance. The oxime moiety of cefotaxime is connected to an aminothiazole ring. Like other third-generation cephalosporins, it has excellent anti-gram-negative activity. It has a metabolically vulnerable acetoxy group attached to C-3 and loses about 90% of its activity when this is hydrolyzed. This metabolic feature also complicates the pharmacokinetic data because both structures are present and have different properties. Cefotaxime should be protected from heat and light and may color slightly without significant loss of potency. Like other third-generation cephalosporins, cefotaxime has less activity against staphylococci but has greater activity against gram-negative organisms (53).

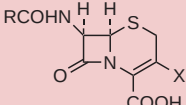
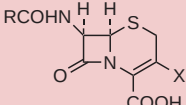
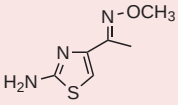
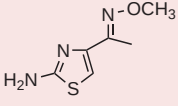
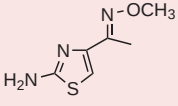
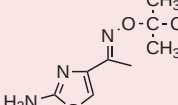
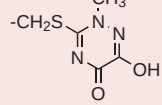
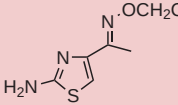
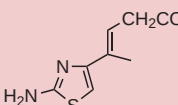
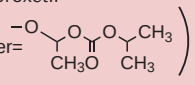
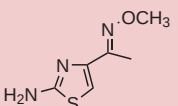
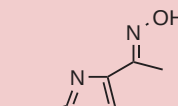
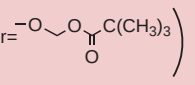
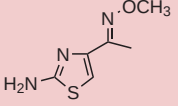
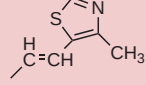
Ceftizoxime. In ceftizoxime, the whole C-3 side chain has been omitted to prevent deactivation by hydrolysis. It rather resembles cefotaxime in its properties; however, not being subject to metabolism, its pharmacokinetic properties are much less complex.

Ceftriaxone. Ceftriaxone has the same C-7 side-chain moiety as cefotaxime and ceftizoxime, but the C-3 side chain consists of a metabolically stable and activating thiotriazinedione in place of the normal acetyl group. The C-3 side chain is sufficiently acidic that at normal pH, it forms an enolic sodium salt, and thus the commercial product is a disodium salt. It is useful for many severe infections and notably in the treatment of some meningitis infections caused by gram-negative bacteria. It is quite stable to many β -lactamases but is sensitive to some inducible chromosomal β -lactamases (49).

Ceftazidime. In ceftazidime, the oxime moiety is more complex, containing two methyl groups and a carboxylic acid. This assemblage conveys even more pronounced β -lactamase stability, greater anti-*Pseudomonas aeruginosa* activity, and increased activity against gram-positive organisms. The C-3 side chain has been replaced by a charged pyridinium moiety. The latter considerably enhances water solubility and also highly activates the β -lactam bond toward cleavage. It is not stable under some conditions, such as in the presence of aminoglycosides and vancomycin. It is also attacked readily in sodium bicarbonate solutions. Resistance is mediated by chromosomally mediated β -lactamases and also by lack of penetration into target bacteria. Otherwise, it has a very broad antibacterial spectrum (54).

Cefixime. In cefixime, in addition to the β -lactamase stabilizing Z-oximino acidic ether at C-7, the C-3 side chain

TABLE 33.9 Third Generation Cephalosporins

Generic Name	Trade Name	 R	 X	Salt
Parenteral agents				
Cefotaxime	Claforan		CH ₂ OAc	Na
Ceftizoxime	Cefizox		H	Na
Ceftriaxone	Rocephin			diNa
Ceftazidime	Fortaz Tazicef			H or Na
Oral agents				
Cefixime	Suprax		—CH=CH ₂	—
Ceftibuten	Cedax		H	—
Cefpodoxime proxetil (2-carboxyester= )	Vantin		—CH ₂ OCH ₃	—
Cefdinir	Omnicef		—CH=CH ₂	—
Cefditoren pivoxil (2-carboxyester= )	Spectracef			—

is a vinyl group analogous to the propenyl group of cefprozil. This is believed to contribute strongly to the oral activity of the drug. Cefixime has anti-gram-negative activity intermediate between that of the second-generation and third-generation agents described previously. It is poorly active against staphylococci because it does not bind satisfactorily to a specific PBP (PBP-2).

Ceftibuten. Ceftibuten has a Z-ethylidene-carboxyl group at C-7 instead of the Z-oximino ether linkages seen previously. This conveys enhanced β -lactamase stability and may contribute to oral activity as well. Ceftibuten has no C-3 side chain and thus is not measurably metabolized. It is highly (75% to 90%) absorbed on oral administration, but this is decreased significantly by food. Being lipophilic and acidic, it is significantly serum protein bound (65%). Some isomerization of the geometry of the olefinic linkage appears to take place in vivo before excretion. It is mainly used for respiratory tract infections, otitis media, and pharyngitis, as well as urinary tract infections by susceptible microorganisms (55).

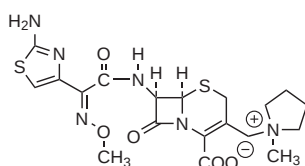
Cefpodoxime Proxetil. Cefpodoxime proxetil is a prodrug. It is cleaved enzymically to isopropanol, carbon dioxide, acetaldehyde, and cefpodoxime in the gut wall. It has better anti-*Staphylococcus aureus* activity than cefixime and is used to treat pharyngitis, urinary tract infections, upper and lower respiratory tract infections, otitis media, skin and soft tissue infections, and gonorrhea.

Cefdinir. Cefdinir has an unsubstituted Z-oxime in its C-7 side chain, the consequence of which is attributed to its somewhat enhanced anti-gram-positive activity—its main distinguishing feature. It has a vinyl moiety attached to C-3 that is associated with its oral activity. It has reasonable resistance to β -lactamases.

Cefditoren Pivoxil. Cefditoren pivoxil is an orally active prodrug. Similar to cefpodoxime proxetil, the pivoxil ester is hydrolyzed following intestinal absorption to release the active drug, cefditoren, along with formaldehyde and pivalic acid. Cefditoren pivoxil should not be administered with drugs that reduce stomach acidity because this may result in decreased absorption. The bioavailability of cefditoren pivoxil is increased if taken with food. Cefditoren pivoxil is indicated for mild to moderate infections in adults and adolescents with chronic bronchitis, pharyngitis/tonsillitis, and uncomplicated skin infections associated with gram-negative bacteria (56).

Fourth-Generation Cephalosporins.

Cefepime.

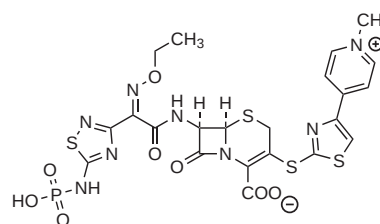


Cefepime (Maxipime)

Cefepime is a semisynthetic agent containing a Z-methoxyimine moiety and an aminothiazolyl group at C-7, broadening its spectrum, increasing its β -lactamase stability, and increasing its antistaphylococcal activity. The quaternary N-methylpyrrolidine group at C-3 seems to help penetration into gram-negative bacteria. The fourth-generation cephalosporins are characterized by enhanced antistaphylococcal activity and broader anti-gram-negative activity than the third-generation group. Cefepime is used intramuscularly and intravenously against urinary tract infections, skin and skin structure infections, pneumonia, and intra-abdominal infections (50).

Newer Cephalosporins.

Ceftaroline Fosamil.

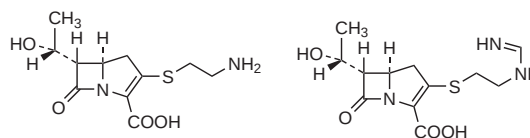


Ceftaroline fosamil (Teflaro)
(prodrug of ceftaroline)

Ceftaroline fosamil was approved in 2010. It is currently unique among the marketed β -lactams because of its affinity for “abnormal” PBPs, in addition to having affinity for normal PBPs. It is bactericidal against MRSA due to its affinity for PBP-2a and against *Streptococcus pneumoniae* due to its affinity for PBP-2x. It can be used to treat MRSA infections of skin and soft tissues. It is administered as a water-soluble prodrug that is readily hydrolyzed to the active ceftaroline (loss of the phosphate ester). The most common adverse effects are nausea, diarrhea, and rash (57).

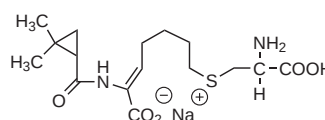
Summary. With their broader spectrum including many very dangerous bacteria, the cephalosporins have come to dominate β -lactam chemotherapy, despite often lacking oral activity. Because the cephalosporin field is still being very actively pursued, the student can expect additional introduction of new agents in the future.

Carbapenems.



Thienamycin

Imipenem



Cilastatin sodium

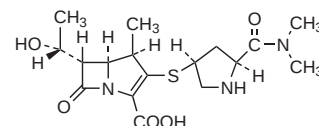
Thienamycin, the first of the carbapenems, was isolated from *Streptomyces cattleya* (58). Because of its extremely intense and broad-spectrum antimicrobial activity and its ability to inactivate β -lactamases, it combines in one molecule the functional features of the best of the β -lactam antibiotics as well as the β -lactamase inhibitors. It differs structurally in several important respects from the penicillins and cephalosporins. The sulfur atom is not part of the five-membered ring but rather has been replaced by a methylene moiety at that position. Carbon is roughly half the molecular size of sulfur. Consequently, the carbapenem ring system is highly strained and very susceptible to reactions cleaving the β -lactam bond. The sulfur atom is now attached to C-3 as part of a functionalized side chain. The endocyclic olefinic linkage also enhances the reactivity of the β -lactam ring. Both make thienamycin unstable, and this caused difficulties in the original isolation studies. The terminal amino group in the side chain attached to C-3 is nucleophilic and attacks the β -lactam bond of nearby molecules destroying activity (Fig. 33.23). Ultimately, this problem was overcome by changing the amino group to a less nucleophilic *N*-formiminoyl moiety by a semisynthetic process to produce imipenem. At C-6, there is a 2-hydroxyethyl group attached with α -stereochemistry. Thus the absolute stereochemistry of the molecule is 5*R*,6*S*,8*S*. With these striking differences from the penicillins and cephalosporins, it is not surprising that thienamycin analogs bind differently to the penicillin-binding proteins (especially strongly to PBP-2), but the result is very potent broad-spectrum activity (59). However, none of the current carbapenems have activity against MRSA.

Imipenem. Imipenem, as well as thienamycin, penetrates very well through porins and is very stable, even inhibitory, to many β -lactamases. Imipenem is not, however, orally active. Renal dehydropeptidase-1 hydrolyzes imipenem through hydrolysis of the β -lactam and deactivates it. An inhibitor for this enzyme, cilastatin, is coadministered with imipenem to protect it (59). Inhibition of human dehydropeptidase does not seem to have deleterious consequences to the patient, making this combination highly efficacious.

The combination of imipenem and cilastatin (Primaxin) is about 25% serum protein bound. On injection, it penetrates well into most tissues, but not cerebrospinal fluid, and it is subsequently excreted in the urine. This very potent combination is especially useful for treatment of serious infections by aerobic gram-negative bacilli, anaerobes, and *Staphylococcus aureus*. It is used clinically for a number of significant infections. The more

common adverse effects are irritation at the infusion site, nausea, vomiting, diarrhea, and pruritus. Of greater concern is the ability of imipenem to induce seizures. The risk factors for seizure development include impaired renal function, preexisting CNS disease or infection, and use of large doses (60).

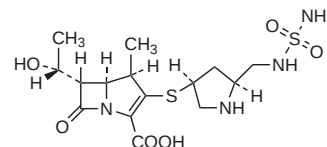
Meropenem.



Meropenem (Merrem)

Meropenem is a synthetic carbapenem possessing a complex side chain at C-3. It also has a chiral methyl group at C-4. This methyl group conveys intrinsic resistance to hydrolysis by dehydropeptidase-1 (61). As a consequence, it can be administered as a single agent for the treatment of severe bacterial infections. While the common adverse effects are similar to imipenem/cilastatin, the risk of seizures is significantly less (62).

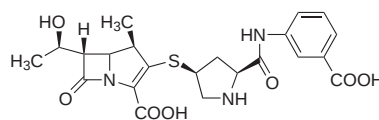
Doripenem.



Doripenem (Doribax)

Doripenem is the newest of the approved carbapenems. It also contains the 4- β -methyl group, which confers stability toward dehydropeptidase-1, so it is given as a single agent. It is similar in spectrum to imipenem and meropenem but is considered more potent against *Pseudomonas* species (63).

Ertapenem.



Ertapenem (Invanz)

Ertapenem is another synthetic carbapenem with a rather complex side chain at C-3. As with meropenem and doripenem, the 4- β -methyl group confers stability toward dehydropeptidase-1. It is not active against *Pseudomonas* or *Acinetobacter* and thus should not be substituted for other carbapenems for these organisms (63). This class of antimicrobial agents is under intensive investigation,

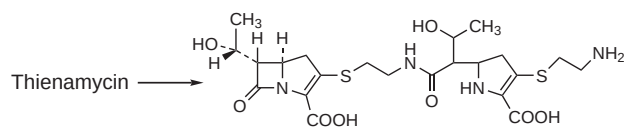
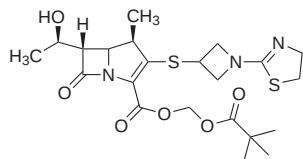


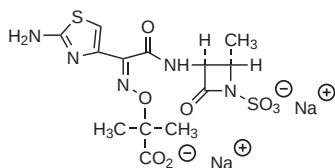
FIGURE 33.23 Intermolecular instability reaction of thienamycin.

and several analogs are currently in various phases of pre-clinical investigation, including tebipenem, an oral pro-drug carbapenem (64).



Tebipenem pivoxil

Monobactams. Aztreonam.



Aztreonam disodium (Azactam)

Fermentation of unusual microorganisms led to the discovery of a class of monocyclic β -lactam antibiotics, named monobactams (34,35). None of these natural molecules have proven to be important, but the group served as the inspiration for the synthesis of aztreonam. Aztreonam is a totally synthetic parenteral antibiotic whose antimicrobial spectrum is devoted almost exclusively to gram-negative microorganisms, and it is capable of inactivating some β -lactamases. Its molecular mode of action is closely similar to that of the penicillins, cephalosporins, and carbapenems, the action being characterized by strong affinity for PBP-3, producing filamentous cells as a consequence. Whereas the principal side chain closely resembles that of ceftazidime, the sulfamic acid moiety attached to the β -lactam ring was unprecedented. Remembering the comparatively large size of sulfur atoms, this assembly may sufficiently spatially resemble the corresponding C-2 carboxyl group of the precedent β -lactam antibiotics to confuse the PBPs. The strongly electron-withdrawing character of the sulfamic acid group probably also makes the β -lactam bond more vulnerable to hydrolysis. In any case, the monobactams demonstrate that a fused ring is not essential for antibiotic activity. The α -oriented methyl group at C-2 is associated with the stability of aztreonam toward β -lactamases.

The protein binding is moderate (~50%), and the drug is nearly unchanged by metabolism. Aztreonam is given by injection and is primarily excreted in the urine. The primary clinical use of aztreonam is against severe infections caused by gram-negative microorganisms, especially those acquired in the hospital. These are mainly urinary tract, upper respiratory tract, bone, cartilage, abdominal, obstetric, and gynecologic infections and septicemias. The drug is well tolerated, and adverse effects are infrequent (65). Interestingly, allergy would not be unexpected, but cross-allergenicity with penicillins and cephalosporins has not often been reported (65).

ANTIBIOTICS: INHIBITORS OF PROTEIN BIOSYNTHESIS

Basis for Selectivity Once the bacterial cell wall is traversed, complex cellular machinery deeper within the cell becomes available to antibiotics. Some of the most successful antibiotic families exert their lethal effects on bacteria by inhibiting ribosomally mediated protein biosynthesis. At first glimpse, this may seem problematic because eukaryotic organisms also construct their essential proteins on ribosomal organelles and the sequence of biochemical steps is closely analogous to that in prokaryotic microorganisms. At a molecular level, however, the apparent anomaly resolves itself because the detailed architecture of prokaryotic ribosomes is rather different. In *Escherichia coli*, for example, the 70S ribosomal particle is composed not only of three RNA molecules, but also of 55 different structural and functional proteins arranged in a nonsymmetrical manner. The small (30S) subunit has a 16S rRNA molecule and about 20 different proteins. The large subunit has a 23S and a 5S rRNA and over 30 proteins. The x-ray crystal structure of the components of the bacterial ribosome has been determined (66,67). This was a landmark achievement given the size and complexity of this organelle. The picture is still not complete but allows one to unravel the molecular details not only of how proteins are biosynthesized but also where important antibiotics bind when they interrupt this process. The functioning parts of the ribosome organize and catalyze some portions of the biosynthetic cycle. The tRNA molecules bind roughly at the interface where the 50S and 30S subparticles come together. The codon-anticodon interaction with mRNA takes place in the 30S subunit, and the incoming amino acid and the growing peptide chain being made lies in the 50S subunit. Upsetting the view held for decades that the antibiotics bind to ribosomal proteins, it is now known that they bind to the rRNA instead. Aside from this important fact, most of the other key beliefs are still valid. Of key importance is the repeated movement of the tRNA bases, which mostly take place near interfaces between the individual rRNA molecules where this would be easiest. In agreement with this theory, it has been found that this region is comparatively disordered, which is consistent with that movement (Fig. 33.24).

At normal doses, antibiotics do not bind to or interfere with the function of eukaryotic 80S ribosomal particles. The basis for the selective toxicity of these antibiotics is apparent. Interference with bacterial protein biosynthesis prevents repair, cellular growth, and reproduction and can be clinically bacteriostatic or bactericidal.

Aminoglycosides and Aminocyclitols

Introduction. The aminoglycoside/aminocyclitol class of antibiotics contains a pharmacophoric 1,3-diaminoinositol moiety consisting either of streptomycin, 2-deoxystreptomycin, or spectinomycin (Fig. 33.25). Several of the alcoholic functions of the 1,3-diaminoinositol are substituted through glycosidic bonds with characteristic amino sugars to form pseudo-oligosaccharides. The chemistry, spectrum, potency, toxicity, and pharmacokinetics of these agents are a function of the specific identity of the diaminoinositol

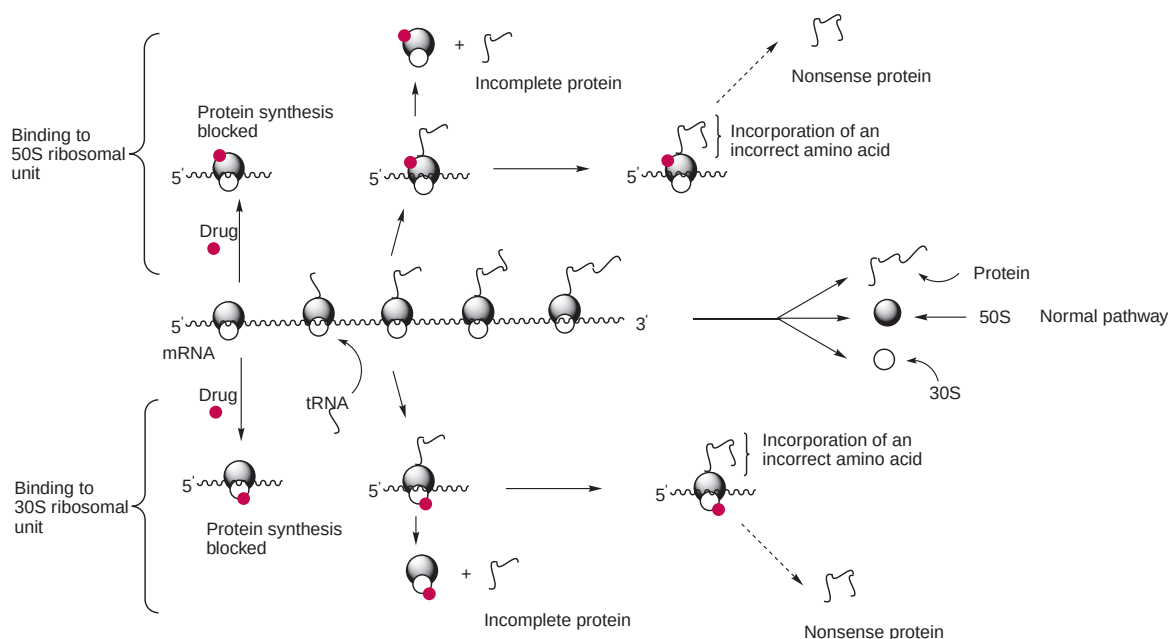


FIGURE 33.24 General mechanism of action of drugs that block protein synthesis by binding to ribosomal units.

unit and the arrangement and identity of the attachments. The various aminoglycoside antibiotics are freely water soluble at all achievable pHs, are basic and form acid salts, are not absorbed in significant amounts from the GI tract, and are excreted in active form in fairly high concentrations in the urine following injection. When the kidneys are not functioning efficiently, the dose must be reduced to prevent accumulation to toxic levels. When given orally, their action is confined to the GI tract. Aminoglycosides are used topically as solutions or ointments for ophthalmic infections. They can be given intramuscularly but are more commonly delivered by intravenous infusion. Tobramycin is available for inhalation to treat *Pseudomonas aeruginosa* infections in cystic fibrosis patients (68). This route of administration results in significantly reduced toxicity to the patient. These agents have intrinsically broad antimicrobial spectra, but their toxicity potential limits their clinical use to infections

by gram-negative bacteria. The aminoglycoside antibiotics are widely distributed (mainly in extracellular fluids) and have low levels of protein binding.

Mechanism of Action. The aminoglycosides are bactericidal due to a combination of toxic effects. At less than toxic doses, they bind to the 16S ribosomal DNA portion of the 30S ribosomal subparticle, impairing the proof-reading function of the ribosome. A conformational change occurs in the peptidyl A site of the ribosome upon aminoglycoside binding. This leads to mistranslation of RNA templates and the selection of wrong amino acids and formation of so-called nonsense proteins (Fig. 33.24). The most relevant of these unnatural proteins are involved in upsetting bacterial membrane function. Their presence destroys the semipermeability of the membrane, and this damage cannot be repaired without de novo programmed protein biosynthesis. Among the substances that are admitted by the damaged membrane are large additional quantities of aminoglycoside. At these increased concentrations, protein biosynthesis ceases altogether. These combined effects are devastating to the target bacterial cells. Given their highly polar properties, the student may wonder how these agents can enter bacterial cells at all. Aminoglycosides apparently bind initially to external lipopolysaccharides and diffuse into the cells in small amounts. The uptake process is inhibited by Ca^{2+} and Mg^{2+} ions. These ions are, then, partially incompatible therapeutically. Passage through the cytoplasmic membrane is dependent on electron transport and energy generation. At high concentrations, eukaryotic protein biosynthesis can also be inhibited by aminoglycoside/aminocyclitol antibiotics (69,70).

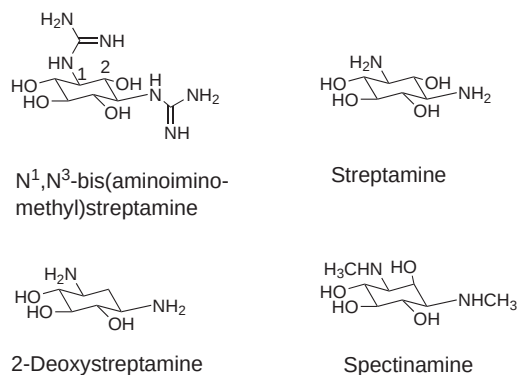


FIGURE 33.25 1,3-Diaminoinositol moieties present in aminoglycosides.

Bacterial Resistance. Bacterial resistance to aminoglycoside antibiotics in the clinic is most commonly due to bacterial elaboration of R-factor-mediated enzymes that *N*-acetylate (aminoglycoside acetylase [AAC]), *O*-phosphorylate (aminoglycoside phosphorylase [APH]), and *O*-adenylate (aminoglycoside nucleotide transferase [ANT]) specific functional groups, preventing subsequent ribosomal binding (Fig. 33.26). In some cases, chemical deletion of the functional groups transformed by these enzymes leaves a molecule that is still antibiotic but no longer a substrate; thus agents with intrinsically broader spectrum can be made semisynthetically in this way. In other cases, novel functional groups can be attached to remote functionality, which converts these antibiotics to poorer substrates for these R-factor-mediated enzymes, and this expands their spectra, as discussed later in this chapter. It is important to note that resistance mediated in this way is not necessarily class-wide, so susceptibility testing with multiple aminoglycosides should be used. Resistance can also involve point mutations of the ribosomal A site. These involve single nucleotide residues at specific positions (70). The substituent at position 6' of the aminoglycoside, the number of protonated amino groups, and the linkage between the sugar rings and the central deoxystreptamine moiety are particularly important in the interactions with the rRNA.

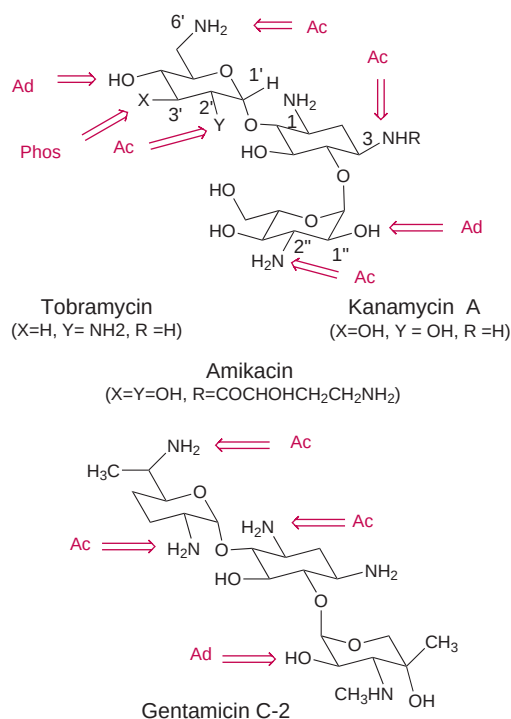


FIGURE 33.26 Commercially important 2-desylstreptamine-containing aminoglycosides. Some points of inactivating attack by specific R-factor-mediated enzymes are indicated by the following: Ac, acetylation; Ad, adenylation; Phos, phosphorylation. APH(3')-1, for example, is an acronym for an enzyme that phosphorylates aminoglycosides at the 3'-OH position.

Resistance due to decreased aminoglycoside/aminocyclitol uptake into bacterial cells is also encountered.

Therapeutic Application. Intrinsically, aminoglycosides have broad antibiotic spectra against aerobic gram-positive and gram-negative bacteria but are reserved for use in serious infections caused by gram-negative organisms because of serious toxicities that are often delayed in onset. They are active against gram-negative aerobes such as *Acinetobacter* sp., *Citrobacter* sp., *Enterobacter* sp., *Escherichia coli*, *Klebsiella* sp., *Proteus vulgaris*, *Providencia* sp., *Pseudomonas aeruginosa*, *Salmonella* sp., *Serratia marcescens*, *Shigella* sp. and gram-positive aerobes such as *Staphylococcus epidermidis* (70).

Streptomycin and spectinomycin differ from the others in their useful antimicrobial spectra. Streptomycin is most commonly used for the treatment of tuberculosis and spectinomycin for treatment of gonorrhea, although it is no longer marketed in the United States. Some aminoglycoside antibiotics in present clinical use are illustrated in Figure 33.26, along with some of their sites of enzymatic inactivation.

Adverse Effects. The toxicities associated with the aminoglycosides involve toxicity to functions mediated by the eighth cranial nerve, such as hearing loss and vertigo. Their use can also lead to kidney tubular necrosis, producing decreases in glomerular function. These toxic effects are related to blood levels and are apparently mediated by the special affinity of these aminoglycosides to kidney cells and to the sensory cells of the inner ear (71). The effects may have a delayed onset, making them all the more dangerous because the patient can be injured significantly before symptoms appear. Less common is a curare-like neuromuscular blockade believed to be caused by competitive inhibition of calcium ion-dependent acetylcholine release at the neuromuscular junction. This effect can exaggerate the muscle weakness of myasthenia gravis and parkinsonian patients (72). In current practice, all of these toxic phenomena are well known; therefore, creatinine clearance should be determined and the dose adjusted downward accordingly so that these adverse effects are less common and less severe.

Specific Agents.

Kanamycin (Kantrex). Kanamycin is a mixture of at least three components (A, B, and C, with A predominating; Fig. 33.26), isolated from *Streptomyces kanamyceticus*. In addition to typical aminoglycoside antibiotic properties, it, along with gentamicin, neomycin, and paromomycin, is among the most chemically stable of the common antibiotics. Kanamycin is, however, unstable to R-factor enzymes, being *O*-phosphorylated on the C-3' hydroxyl by enzymes APH(3')-I and APH(3')-II and is also *N*-acetylated on the C-6' amino group, among others (Fig. 33.26). These transformation products are antibiotically inactive. Kanamycin is used parenterally against some gram-negative bacteria, but *Pseudomonas aeruginosa* and anaerobes are usually resistant

(73). Although it can also be used in combination with other agents against certain mycobacteria (*Mycobacterium kansasii*, *Mycobacterium marinum*, and *Mycobacterium intracellulare*), its popularity for this use is fading.

Amikacin (Amikin). Amikacin is made semisynthetically from kanamycin A (Fig. 33.26). Interestingly, the L-hydroxyaminobutyryl amide (HABA) moiety attached to N-3 inhibits adenylation and phosphorylation in the distant amino sugar ring (at C-2' and C-3') even though the HABA substituent is not where the enzymatic reaction takes place. This effect is attributed to decreased binding to the R-factor-mediated enzymes. With this change, potency and spectrum are strongly enhanced, and amikacin is used for the treatment of sensitive strains of *Mycobacterium tuberculosis*, *Yersinia tularensis*, and severe *Pseudomonas aeruginosa* infections resistant to other agents (74).

Tobramycin (Nebcin). Tobramycin is one component of a mixture produced by fermentation of *Streptomyces tenebrarius* (Fig. 33.26). Lacking the C-3' hydroxyl group, it is not a substrate for APH(3')-I and -II, and thus has an intrinsically broader spectrum than kanamycin. It is a substrate, however, for adenylation at C-2' by ANT(2') and acetylation at C-3 by AAC(3)-I and -II and at C-2' by AAC(2') (Fig. 33.26). It is used parenterally for difficult infections, especially those by gentamicin-resistant *Pseudomonas aeruginosa* (75).

Gentamicin (Garamycin). Gentamicin is a mixture of several antibiotic components produced by fermentation of *Micromonospora purpurea* and other related soil microorganisms (hence its name is spelled with an "i" instead of a "y"). Gentamicins C-1, C-2, and C-1a are most prominent (gentamicin C-2 is shown in Fig. 33.26). Gentamicin was one of the first antibiotics to have significant activity against *Pseudomonas aeruginosa* infections. This water-loving opportunistic pathogen is frequently encountered in burns, pneumonias, and urinary tract infections. It is highly virulent. As noted earlier, some of the functional groups that serve as targets for R-factor-mediated enzymes are missing in the structure of gentamicins, so their antibacterial spectrum is enhanced. They are, however, inactivated through C-2' adenylation by enzyme ANT(2') and acetylation at C-6' by AAC(6'), at C-1 by AAC(1)-I and -II, and at C-2' by AAC(2'). Gentamicin is often combined with other anti-infective agents, and an interesting incompatibility has been uncovered. With certain β -lactam antibiotics, the two drugs react with each other so that N-acylation on C-1 of gentamicin by the β -lactam antibiotic takes place, thus inactivating both antibiotics (Fig. 33.27). The two agents should not, therefore, be mixed in the same solution and should ideally be administered into different tissue compartments to prevent this. This incompatibility is likely to be associated with other aminoglycoside antibiotics as well. Gentamicin is used for urinary tract infections, burns, some pneumonias, and bone and joint infections caused by susceptible gram-negative bacteria (72).

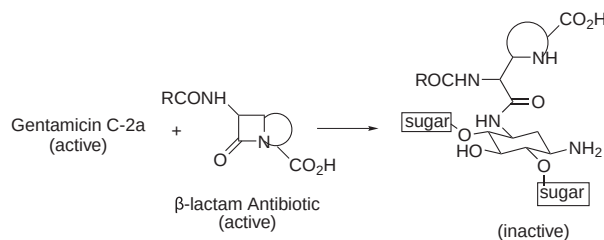
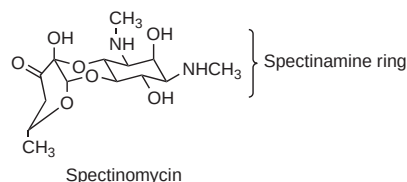


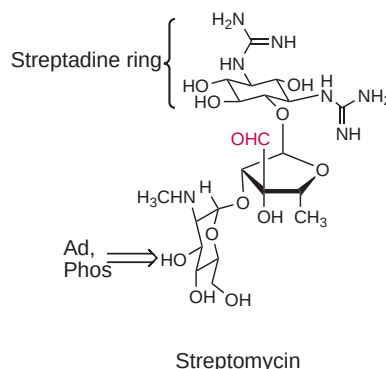
FIGURE 33.27 A chemical drug–drug incompatibility between gentamicin C-2a and β -lactams.

Spectinomycin (Trobicin).



An unusual aminoglycoside antibiotic, spectinomycin is produced by fermentation of *Streptomyces spectabilis* and differs substantially in its clinical properties from the others. The diaminoinositol unit (spectinamine) contains two mono-*N*-methyl groups, and the hydroxyl between them has a stereochemistry opposite to that in streptomycin. The glycosidically attached sugar is also unusual in that it contains three consecutive carbonyl groups, either overt or masked, and is fused by two adjacent linkages to spectinamine to produce an unusual fused three-ring structure. Spectinomycin is bacteriostatic as normally employed. It is almost exclusively used in a single bolus injection intramuscularly against *Neisseria gonorrhea*, especially penicillinase-producing strains, in cases of urogenital or oral gonorrhea and does not apparently produce any serious oto- or nephrotoxicity when used in this way. It is particularly useful for the treatment of patients allergic to penicillin and patients not likely to comply well with a medication scheme. It would likely be more widely used except that syphilis and chlamydia do not respond to it. It causes significant mistranslation following ribosomal binding but does not cause much inhibition of overall programmed protein biosynthesis. As mentioned previously, spectinomycin is no longer marketed in the United States.

Streptomycin.



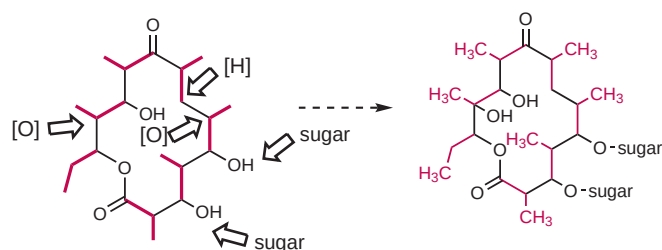


FIGURE 33.29 Biosynthetic pathway to erythromycins from propionic acid units. $\rightarrow$ refers to modifications to the basic ring skeleton.

character is weakly basic ($pK_a \sim 8$). They are not very water soluble as free bases, but salt formation with certain acids (glucoheptonic and lactobionic acids in Fig. 33.28) increases water solubility, whereas other salts decrease solubility (laurylsulfate and stearic). Macrolide antibiotics with 16-membered rings are popular outside the United States, but one example, tylosin, finds extensive agricultural use in the United States. The 14-membered ring macrolides are biosynthesized from propionic acid units so that every second carbon of erythromycin, for example, bears a methyl group and the rest of the carbons, with one exception, are oxygen bearing. Two carbons bear so-called “extra” oxygen atoms introduced later in the biosynthesis (not present in a propionic acid unit), and two hydroxyls are glycosylated (Fig. 33.29) (78).

Chemical Properties. The early macrolides of the erythromycin class are chemically unstable due to rapid acid-catalyzed internal cyclic ketal formation, leading to inactivity (Fig. 33.30). This reaction, which occurs in the GI tract, is clinically important. Most acid-susceptible macrolides are administered in coated tablets to minimize this. Analogs have been prepared semisynthetically that are structurally incapable of undergoing this reaction and have become very popular in the clinic (79).

Many macrolides have an unpleasant taste, which is partially overcome with water-insoluble dosage forms that also reduce acid instability and the gut cramps. Enteric coatings are also beneficial in reducing these adverse effects.

Mechanism of Action. The macrolides inhibit bacteria by interfering with programmed ribosomal protein biosynthesis by binding to the 23S rRNA in the polypeptide exit tunnel adjacent to the peptidyl transferase center in the 50S ribosomal subunit (Fig. 33.24). This prevents the growing peptide from becoming longer than a few residues, resulting in the dissociation of peptidyl tRNA molecules. Clindamycin, lincomycin, and chloramphenicol bind in the same vicinity, leading to extensive cross-resistance between them.

Resistance. Developed bacterial resistance is primarily caused by bacteria possessing R-factor enzymes that methylate a specific guanine residue on their own ribosomal RNA

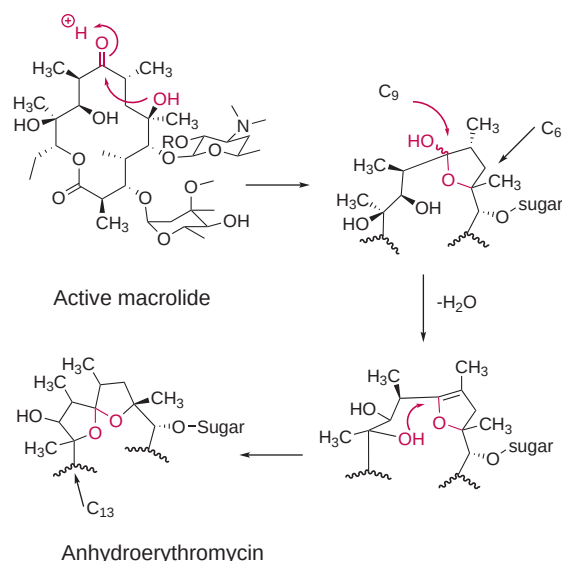


FIGURE 33.30 Acid-catalyzed intramolecular ketal formation with erythromycin.

making them somewhat less efficient at protein biosynthesis but comparatively poor binders of macrolides (80). The erythromycin-producing soil organism uses the same ribosomal methylation technique to protect itself against the toxic effects of its own metabolite (80). This leads to the speculation that the origin of some antibiotic resistance genes may lie in the producing organism itself and that this genetic material is acquired by bacteria from this source. A second mechanism of resistance is associated with the mutation of adenine to guanine that occurs in domain V at A2058. This change results in a 10,000-fold reduction of binding capacity of erythromycin and clarithromycin to the 23S rRNA (80). Some bacterial strains, however, appear to be resistant to macrolides due to the operation of an active efflux process in which the drug is expelled from the cell at the cost of energy (80). Intrinsic resistance of gram-negative bacteria is primarily caused by lack of penetration as the isolated ribosomes from these organisms are often susceptible.

Drug Interactions. Drug–drug interactions with macrolides are comparatively common and usually involve inhibition of CYP3A4 of the cytochrome P450 oxidase family (81). There is also evidence suggesting that these interactions are intensified by the inhibition of P-glycoprotein (82). These interactions can have severely negative consequences for the patient. The result of this interaction is a longer half-life and enhanced potential toxicity by increasing the effective dose over time. The main product of liver metabolism of erythromycin is the *N*-demethylated analog.

Therapeutic Application. The macrolides are among the safest of the antibiotics in common use and are often used for the treatment of upper and lower respiratory tract and soft tissue infections primarily caused by gram-positive microorganisms like *Streptococcus pyogenes* and *Streptococcus pneumoniae*, Legionnaire disease, prophylaxis of bacterial endocarditis by *Streptococcus viridans*, upper respiratory tract

and lower respiratory tract infections, otitis media caused by *Haemophilus influenzae* (with a sulfonamide added), mycoplasmal pneumonia, and *Mycobacterium avium* complex infections in AIDS patients (in combination with rifabutin). They are also used for certain sexually transmitted diseases, such as gonorrhea and pelvic inflammatory disease, caused by mixed infections involving cell wall-free organisms such as *Chlamydia trachomatis*. Clarithromycin is also used to treat gastric ulcers due to *Helicobacter pylori* infection as a component of multidrug cocktails. The macrolides have a comparatively narrow antimicrobial spectrum, reminiscent of the medium-spectrum penicillins, but the organisms involved include many of the more commonly encountered community-acquired microorganisms. The utility of the macrolides against upper respiratory tract infections is enhanced by their particular affinity for these tissues. Tissue levels in the upper respiratory tract are often several times those seen in the blood. They are considered bacteriostatic against most organisms in achievable concentrations.

Specific Agents.

Erythromycin Esters and Salts.

ESTOLATE One of the erythromycin prodrugs, erythromycin estolate, is a C-2'-propionyl ester, *N*-laurylsulfate salt (Fig. 33.28). Administration of erythromycin estolate produces higher blood levels following metabolic regeneration of erythromycin. In a small number of cases, a severe, dose-related, cholestatic jaundice occurs in which the bile becomes granular in the bile duct, impeding flow so that the bile salts back up into the circulation. This seems to be partly allergic and partly dose-related. If the drug causes hepatocyte damage, perhaps this releases antigenic proteins that promote further damage. When cholestatic jaundice occurs, the drug must be replaced by another, nonmacrolide antibiotic such as one of the penicillins, one of the cephalosporins, or clindamycin. This form is no longer available in the United States.

ETHYLSUCCINATE (ERYPED, EES) Erythromycin ethyl succinate is a mixed double-ester prodrug in which one carboxyl of succinic acid esterifies the C-2' hydroxyl of erythromycin and the other ethanol (Fig. 33.28). This prodrug is frequently used in an oral suspension for pediatric use largely to mask the bitter taste of the drug. Film-coated tablets are also used to deal with this. Some cholestatic jaundice is associated with the use of erythromycin ethyl succinate (83).

STEARATE Erythromycin stearate is a very insoluble salt form of erythromycin. The water insolubility helps mask the taste of the drug and enhances its stability in the stomach.

LACTOBIONATE Erythromycin lactobionate is a salt with enhanced water solubility that is used for injections.

Clarithromycin (Biaxin). Clarithromycin differs from erythromycin in that the C-6 hydroxy group has been converted semisynthetically to a methyl ether (Fig. 33.28).

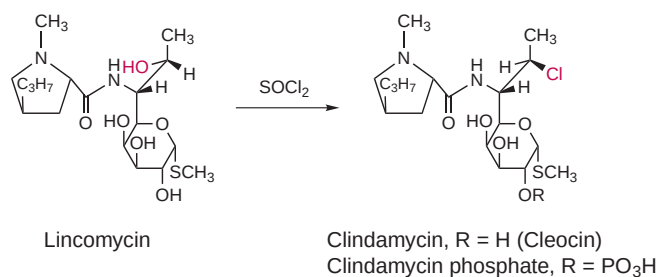
The C-6 hydroxy group is involved in the process, initiated by protons, leading to internal cyclic ketal formation in erythromycin that results in drug inactivation (Fig. 33.30). This ketal, or one of the products of its subsequent degradation, is also associated with GI cramping. Conversion of the molecule to its more lipophilic methyl ether prevents internal ketal formation, which not only gives better blood levels through chemical stabilization, but also results in less gastric upset. An extensive saturable first-pass liver metabolism of clarithromycin leads to formation of its C-14 hydroxy analog, which has even greater antimicrobial potency, especially against *Haemophilus influenzae* (84). The enhanced lipophilicity of clarithromycin also allows for lower and less frequent dosage for mild infections.

Azithromycin (Zithromax, Zmax). Azithromycin, called an "azalide," has been formed by semisynthetic conversion of erythromycin to a ring-expanded analog in which an *N*-methyl group has been inserted between carbons 9 and 10 and the carbonyl moiety is thus absent (Fig. 33.28). Azithromycin has a 15-membered lactone ring. This new functionality does not form a cyclic internal ketal. Not only is azithromycin more stable to acid degradation than erythromycin, but it also has a considerably longer half-life, attributed to greater and longer tissue penetration, allowing once-a-day dosage (85). A popular treatment schedule with azithromycin is to take two tablets on the first day and one a day for the following 5 days and then to discontinue treatment. This is convenient for patients who comply poorly. The drug should be taken on an empty stomach. Azithromycin tends to be broader spectrum than either erythromycin or clarithromycin. Azithromycin has a significant postantibiotic effect against a number of pathogens (85). Azithromycin is now commonly the first choice for treatment of infections that require a macrolide.

Ketolides. Research activity in the macrolide antibiotic class has been intense recently in attempts to reduce side effects and to broaden their antimicrobial spectra. The ketolides are a group of agents that are characterized by oxidation of the 3 position from an alcohol to a ketone. They are active against a significant number of erythromycin-resistant microorganisms.

Telithromycin (Ketek). Telithromycin (Fig. 33.28) is orally effective in the treatment of community-acquired pneumonia, acute bacterial exacerbations of chronic bronchitis, and acute sinusitis. Its principle advantage is activity against macrolide-resistant infections mediated by the methylation mechanisms. Significant controversy about the safety of this agent arose after approval by the U.S. Food and Drug Administration and ultimately led to additional warning information about liver toxicity and a black box warning regarding use in myasthenia gravis patients (86). The toxicity reported for telithromycin appears to be associated with the pyridine portion of the molecule and the antagonist action of the drug on cholinergic receptors. Ketolides under investigation that do not contain a pyridine are devoid of the reported toxicities.

Lincosamides



The lincosamides contain an unusual 8-carbon sugar, a thiomethyl amino-octoside (*O*-thio-lincosamide), linked by an amide bond to an *n*-propyl substituted *N*-methylpyrrolidylcarboxylic acid (*N*-methyl-*n*-propyl-*trans*-hygric acid). Lincosamides are weakly basic and form clinically useful hydrochloric acid salts. They are chemically distinct from the macrolide antibiotics but possess many pharmacologic similarities to them. The lincosamides bind to 50S ribosomal subparticles at a site partly overlapping with the macrolide site and are mutually cross-resistant with macrolides and work through essentially the same molecular mechanism of action (Fig 33.24). The lincosamides undergo extensive liver metabolism resulting primarily in *N*-demethylation. The *N*-desmethyl analog retains biologic activity.

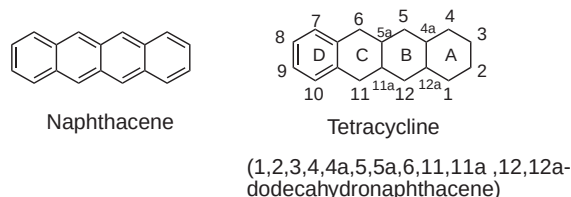
Specific Agents.

Lincomycin (Lincocin). Lincomycin is a natural product isolated from fermentations of *Streptomyces lincolnensis*. It is active against gram-positive organisms including some anaerobes. It is indicated for treatment of serious infections caused by sensitive strains of streptococci, pneumococci, and staphylococci (87). It is generally reserved for penicillin-allergic patients due to the increased risk of pseudomembranous colitis (described below). It also serves as the starting material for the synthesis of clindamycin (by an S_N-2 reaction that inverts the *R* stereochemistry of the C-7 hydroxyl to a C-7 S-chloride).

Clindamycin (Cleocin). The substitution of the chloride for the hydroxy group consequently makes clindamycin more bioactive and lipophilic than lincomycin, and thus, it is better absorbed following oral administration. It is about 90% absorbed when taken orally. Clindamycin has a clinical spectrum rather like the macrolides, although it distributes better into bones. Clindamycin works well for gram-positive coccal infections, especially in patients allergic to β -lactams, and also has generally better activity against anaerobes. However, as with lincomycin, it is associated with GI complaints (nausea, vomiting, cramps, and drug-related diarrhea) (88). The most severe of these (black box warning) is pseudomembranous colitis caused *Clostridium difficile*, an opportunistic anaerobe. Its overgrowth results from suppression of the normal flora whose presence otherwise preserves a healthier ecologic balance. The popularity of

clindamycin in the clinic has decreased even though pseudomembranous colitis is comparatively rare and is also associated with several other broad-spectrum antibiotics. A less common side effect is exudative erythema multiform (Stevens-Johnson syndrome). Clindamycin has excellent activity against *Propionibacterium acnes* when applied topically (88). A very water-insoluble palmitate hydrochloride prodrug of clindamycin is also available (lacks bitter taste).

Tetracyclines and Glycylcyclines The tetracycline family is widely but not intensively used in office practice. Of the agents in this family, minocycline and doxycycline are still frequently prescribed in the United States. This family of antibiotics is characterized by a highly functionalized, partially reduced naphthacene (four linearly fused six-membered rings) ring system from which both the family name and numbering system are derived.

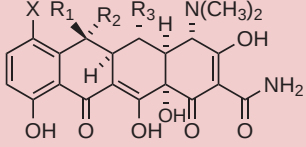


They possess a number of adverse effects, although most of them are annoying rather than dangerous. Because their antimicrobial spectrum is broad enough to include many of the pathogens encountered in a community setting, they were once very widely used. The advent of other choices and the high incidence of resistance that has developed have greatly decreased their medicinal prominence in recent years. Presently, they are recommended primarily for use against *Rickettsia*, *Chlamydia*, *Mycoplasma*, anthrax, plague, and *Helicobacter*.

Chemical Properties. The tetracyclines are amphoteric substances with three pK_a values revealed by titration (2.8 to 3.4, 7.2 to 7.8, and 9.1 to 9.7) and have an isoelectric point at about pH 5. The basic function is the C-4 α -dimethylamino moiety. Commercially available tetracyclines (Table 33.10) are generally administered as comparatively water-soluble hydrochloride salts. The conjugated phenolic enone system extending from C-10 to C-12 is associated with the pK_a at about 7.5, whereas the conjugated trione system extending from C-1 to C-3 in ring A is nearly as acidic as acetic acid (pK_a ~3) (89). These resonating systems can be drawn in a number of essentially equivalent ways with the double bonds in alternate positions. The formulae normally shown are those settled upon by popular convention.

Chelation. Chelation is an important feature of the chemical and clinical properties of the tetracyclines. The acidic functions of the tetracyclines are capable of forming salts through chelation with metal ions. The salts of polyvalent metal ions, such as Fe²⁺, Ca²⁺, Mg²⁺, and Al³⁺, are all quite insoluble at neutral pHs (Fig. 33.31). This

TABLE 33.10 Commercially Available Tetracyclines

Generic Name	Trade Name	X	R ₁	R ₂	R ₃
					
Tetracycline	Generic	H	OH	CH ₃	H
Demeclocycline	Declomycin	Cl	OH	H	H
Minocycline	Minocin	N(CH ₃) ₂	H	H	H
Doxycycline	Vibramycin	H	H	CH ₃	OH

insolubility is not only inconvenient for the preparation of solutions, but also interferes with blood levels on oral administration.

Consequently, the tetracyclines are incompatible with coadministered multivalent ion-rich antacids, and concomitant consumption of dairy products rich in calcium ion is also contraindicated (89). When concomitant oral therapy with tetracyclines and incompatible metal ions must be done, the ions should be given 1 hour before or 2 hours after the tetracyclines. Further, the bones, of which the teeth are the most visible, are calcium-rich structures at nearly neutral pHs and so accumulate tetracyclines in proportion to the amount and duration of therapy when bones and teeth are being formed. Because the tetracyclines are yellow, this leads to a progressive and essentially permanent discoloration in which, in advanced cases, the teeth are even brown. The intensification of discoloration with time is thought to be a photochemical process. This is cosmetically unattractive but does not seem to be deleterious except in extreme cases where so much antibiotic is taken up that the structure of bone is mechanically weakened. To avoid this, tetracyclines are not normally given to children (89).

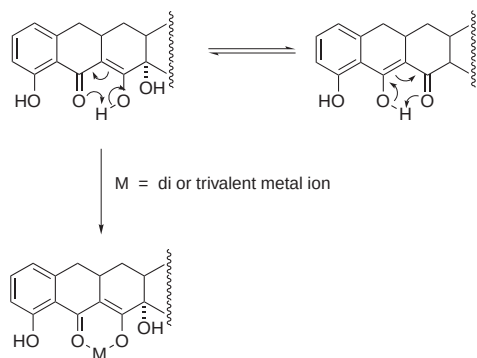


FIGURE 33.31 Metal chelation with the tetracyclines.

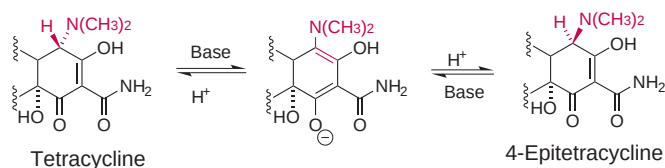


FIGURE 33.32 Epimerization of tetracyclines

Epimerization. The α -stereo orientation of the C-4 dimethylamino moiety of the tetracyclines is essential for their bioactivity. The presence of the tricarbonyl system of ring A allows enolization involving loss of the C-4 hydrogen (Fig. 33.32). Reprotonation can take place from either the top or bottom of the molecule. Reprotonation from the top of the enol regenerates tetracycline. Reprotonation from the bottom, however, produces inactive 4-epitetracycline. At equilibrium, the mixture consists of nearly equal amounts of the two diastereomers.

Dehydration. Most of the natural tetracyclines have a tertiary and benzylic hydroxyl group at C-6. This function has the ideal geometry for acid-catalyzed dehydration involving the C-5a α -oriented hydrogen (antiperiplanar *trans*). The resulting product is a naphthalene derivative, so there are energetic reasons for the reaction proceeding in that direction (Fig. 33.33). C-5a,6-Anhydrotetracycline is much deeper in color than tetracycline and is biologically inactive.

Not only can inactive 4-epitetracyclines dehydrate to produce 4-epianhydrotetracyclines, but also anhydrotetracycline can epimerize to produce the same product. This degradation product is toxic to the kidneys and produces a Fanconi-like syndrome that, in extreme cases, has been fatal (90). Tetracyclines that have no C-6-hydroxyl groups, such as minocycline and doxycycline, cannot undergo dehydration and thus are completely free of this toxicity.

Cleavage in Base. Another untoward degradation reaction involving a C-6-hydroxyl group is cleavage of the C-ring in alkaline solutions at or above pH 8.5 (Fig. 33.34).

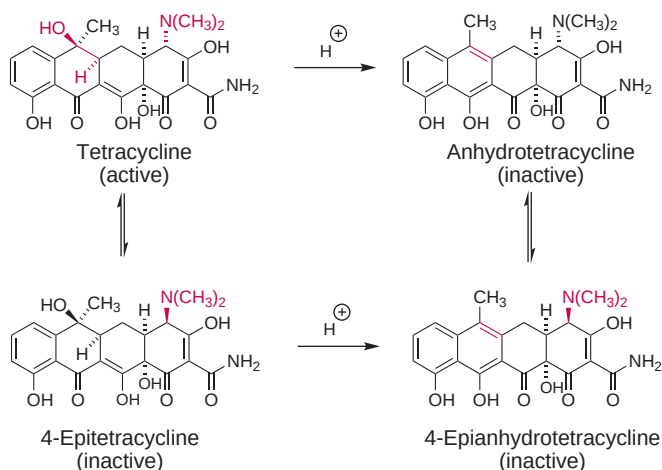


FIGURE 33.33 Acid-catalyzed instability of tetracyclines.

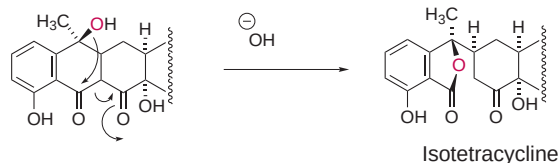


FIGURE 33-34 Base-catalyzed instability of tetracyclines.

The lactonic product, an isotetracycline, is inactive. The clinical impact of this degradation under normal conditions is uncertain.

Phototoxicity. Certain tetracyclines, most notably those with a C-7-chlorine, absorb light in the visible region, leading to free radical generation and potentially causing severe erythema to sensitive patients on exposure to strong sunlight. Patients should be advised to be cautious about such exposure for at least their first few doses to avoid potentially severe sunburn. This effect is comparatively rare with most currently popular tetracyclines (90).

Mechanism of Action. The tetracyclines of clinical importance interfere with protein biosynthesis at the ribosomal level, leading to bacteriostasis. Tetracyclines bind to rRNA in the 30S subparticle with the possible cooperation of a 50S site by a process that remains imprecisely understood despite intensive study (Fig. 33.24). There is more than one binding site, but only one is believed to be critical for its action. The points of contact with the rRNA are those associated with antibiosis, with the puzzling exception of the dimethylamino function (Fig. 33.35). This latter is known to be essential for activity but does not appear to bind in the x-ray structures that are available (91–93). Once the tetracycline binds, it inhibits subsequent binding of aminoacyltransfer-RNA to the ribosomes, resulting in termination of peptide chain growth. The more lipophilic tetracyclines, typified by minocycline, are also capable of disrupting cytoplasmic membrane function, causing leakage of nucleotides and other essential cellular components from the cell, and have bactericidal properties.

Resistance. Resistance to tetracyclines results in part from an unusual ribosomal protection process involving elaboration of bacterial proteins. These proteins associate with the ribosome, thus allowing protein biosynthesis to proceed even in the presence of bound tetracycline, although exactly how this works is not well understood. Another important resistance mechanism involves R-factor-mediated, energy-requiring, active efflux of tetracyclines from the bacterial cells (94). Some of the efflux proteins have activity that is limited to older tetracyclines, whereas others confer resistance to the entire family with the exception of glycylcyclines. Certain other microbes, such as *Mycoplasma* and *Neisseria*, seem to have modified membranes that either accumulate fewer tetracyclines or have porins through which tetracyclines have difficulty passing. Because resistance is now widespread,

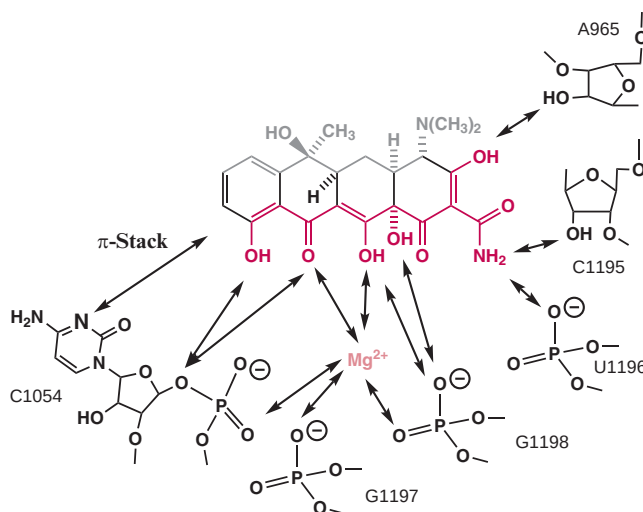


FIGURE 33-35 Schematic representation of the primary binding site for a tetracycline and the sugar phosphate groups of 16S rRNA, which also involves a magnesium ion and the critical functional groups on the “southern” and “eastern” face of the tetracycline. (Redrawn and modified from Brodersen DE, Clemons WM Jr, Carter AP, et al. The structural basis for the action of the antibiotics tetracycline, pactamycin, and hygromycin B on the 30S ribosomal subunit. *Cell* 2000;103:1143–1154).

these once extremely popular antibiotics are falling into comparative disuse.

Therapeutic Application. The tetracyclines possess very wide bacteriostatic antibacterial activity. Because of the resistance phenomenon and the comparative frequency of troublesome side effects, they are rarely the drugs of first choice. The differences between the antimicrobial spectra of various tetracyclines are not large, although greater resistance to older agents limits their use. They are used for low-dose oral and topical therapy for acne, first-course community-acquired urinary tract infections (largely due to *Escherichia coli*), brucellosis, borreliosis, sexually transmitted diseases (especially chlamydia), rickettsial infections, mycoplasmal pneumonia, prophylaxis of malaria, prevention of traveler’s diarrhea, cholera, *Enterobacter* infections, as part of *Helicobacter* cocktails, Lyme disease, Rocky Mountain spotted fever, anthrax, and many other less common problems (90). The tetracyclines are also widely used for agricultural purposes.

Adverse Effects. In addition to the adverse effects mentioned earlier (tooth staining, phototoxicity, and potential kidney damage with outdated drug), the tetracyclines are associated with nausea, vomiting, diarrhea, and some CNS effects (dizziness and vertigo). Rapid administration or prolonged intravenous use can lead to thrombophlebitis, so tetracyclines are generally administered orally because they are well absorbed. They also distinguish imperfectly between the bacterial 70S ribosomes and the mammalian 80S ribosomes, so in high doses or special

situations (i.e., intravenous use during pregnancy), these drugs also demonstrate a significant antianabolic effect. In cases of significant renal impairment, higher serum levels of tetracyclines can lead to azotemia. Additionally, inducers of CYP450 metabolism (i.e., rifampin, barbiturates, carbamazepine) increase metabolism of tetracyclines (especially doxycycline), so the dose of the tetracycline may require adjustment (90).

Specific Agents (Table 33.10).

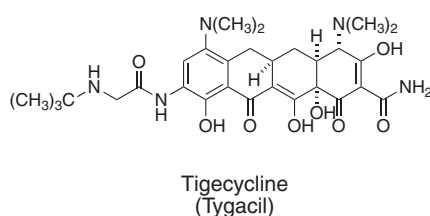
Tetracycline (Sumycin). Tetracycline is produced by fermentation of *Streptomyces aureofaciens* and related species or by catalytic reduction of chlortetracycline. The blood levels achieved on oral administration are often irregular. Food and milk lower absorption by about 50% (89).

Demeclocycline (Declomycin). Demeclocycline lacks the C-6-methyl of tetracycline and is produced by a genetically altered strain of *S. aureofaciens*. Because it is a secondary alcohol, it is more chemically stable than tetracycline against dehydration. Food and milk co-consumption decrease absorption by half, although it is 60% to 80% absorbed by fasting adults. It is the tetracycline most highly associated with phototoxicity and also has been shown to produce dose-dependent, reversible diabetes insipidus with extended use. Demeclocycline is commonly used off label for treatment of inappropriate secretion of antidiuretic hormone.

Minocycline (Minocin and Others). An important antibiotic produced by semisynthesis from demeclocycline is minocycline. It is much more lipophilic than its precursors, gives excellent blood levels following oral administration (90% to 100% available), and can be given once a day. Its absorption is lowered by about 20% when taken with food or milk. It is less dependent on active uptake mechanisms and has a somewhat broader antimicrobial spectrum. It has vestibular toxicities (e.g., vertigo, ataxia, and nausea) not generally shared by other tetracyclines (95).

Doxycycline (Vibramycin and Others). It is produced by semisynthesis from other tetracycline molecules and is the most widely used of the tetracycline family. Doxycycline is well absorbed on oral administration (90% to 100% when fasting; reduced by 20% by co-consumption with food or milk), has a half-life permitting once-a-day dosing for mild infections, and is excreted partly in the feces and partly in the urine.

Tigecycline (Tygacil).

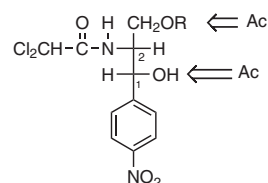


The increased incidence of resistance to the tetracyclines led to a renewed research effort to find novel

agents within this class. This effort led to the discovery of a new class of antibiotics, the glycylcyclines, that are closely structurally related to the tetracyclines but lack many of the clinical resistance issues (96,97). They are characterized by having an additional glyclamido substitution at the 9 position. Tigecycline is the first of these agents to be marketed. Although it has limited indications, treatment of complicated skin/skin structure and complicated intra-abdominal infections, this agent is a broad-spectrum antibiotic based on the in vitro data. It is administered intravenously and, like the tetracyclines, can cause injection site pain and thrombophlebitis. It has other adverse effects similar to the tetracyclines.

Special Purpose Antibiotics This group of antibiotics consists of a miscellaneous collection of structural types whose toxicities or narrow range of applicability give them a more specialized place in antimicrobial chemotherapy than those covered to this point. They are generally reserved for special purposes.

Chloramphenicol (Chloromycetin).



R = H	Chloramphenicol (Chloromycetin)
R = COCH ₂ CH ₂ CO ₂ H	Chloramphenicol hemisuccinate
R = CO(CH ₂) ₁₄ CH ₃	Chloramphenicol palmitate

Chloramphenicol was originally produced by fermentation of *Streptomyces venezuelae*, but its comparatively simple chemical structure soon resulted in several efficient total chemical syntheses. With two asymmetric centers, it is one of four diastereomers, only one of which (1*R*,2*R*) is significantly active. Because total synthesis produces a mixture of all four, the unwanted isomers must be removed before use. Chloramphenicol is a neutral substance only moderately soluble in water because both nitrogen atoms are nonbasic under physiologic conditions (one is an amide and the other a nitro moiety). It was the first broad-spectrum oral antibiotic used in the United States (1947) and was once very popular. Severe potential blood dyscrasia has greatly decreased its use in North America, although its cheapness and efficacy make it still very popular in much of the rest of the world where it can often be purchased over the counter without a prescription.

Mechanism of Action. Chloramphenicol is bacteriostatic by virtue of inhibition of protein biosynthesis in both bacterial and, to a lesser extent, in the host ribosomes. Chloramphenicol binds to the 50S subparticle in a region near where the macrolides and lincosamides bind (Fig. 33.24).

Resistance is mediated by several R-factor enzymes that catalyze acetylation of the secondary and, to some extent,

the primary hydroxyl groups in the aliphatic side chain (98). These products no longer bind to the ribosomes and thus are inactivated. *Escherichia coli* is frequently resistant due to chloramphenicol's lack of intercellular accumulation.

Metabolism. When given orally, chloramphenicol is rapidly and completely absorbed but has a fairly short half-life. It is mainly excreted in the urine in the form of its metabolites, which are C-3 and C-1 glucuronides, and, to a lesser extent, its deamidation product and the product of dehalogenation and reduction. These metabolites are all inactive. The aromatic nitro group is also reduced metabolically, and this product can undergo amide hydrolysis. The reduction of the nitro group, however, does not take place efficiently in humans but primarily occurs in the gut by the action of the normal flora (99). Chloramphenicol potentiates the activity of some other drugs by inducing liver metabolism. Such agents include anticoagulant coumarins, sulfonamides, oral hypoglycemics, and phenytoin (99).

Prodrug Forms. Two prodrug forms of chloramphenicol are available (only the injectable form is available in the United States). The drug is intensively bitter. This can be masked for use as a pediatric oral suspension by use of the C-3 palmitate, which is cleaved in the duodenum to liberate the drug.

Chloramphenicol's poor water solubility is largely overcome by conversion to the C-3 hemisuccinoyl ester, which forms a water-soluble sodium salt. This is cleaved in the body by lung, liver, kidney, and blood esterases to produce active chloramphenicol. Because cleavage in muscles is slow, this prodrug is used intravenously rather than intramuscularly.

Therapeutic Applications. Despite potentially serious limitations, chloramphenicol is an effective drug when used carefully. Its special value is in typhoid fever, *Haemophilus* infections (especially epiglottitis and meningitis, when given along with ampicillin), and rickettsial infections, and in cases where susceptible organisms have proven resistant to other therapies. Safer antibiotics should be used whenever possible. It is about 60% serum protein bound and diffuses well into tissues, especially into inflamed cerebrospinal fluid, and is, therefore, of value for meningitis. It also penetrates well into lymph and mesenteric ganglions, rationalizing its particular value in typhoid fever.

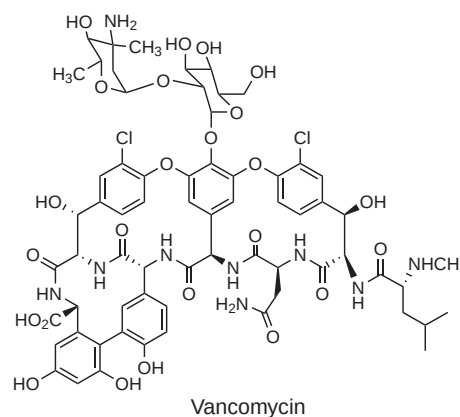
Adverse Effects. Toxicities prevent chloramphenicol from being more widely used. Blood dyscrasias are seen in patients predisposed to them (99). The more serious form is a pancytopenia of the blood that is fatal in about 70% of cases and is believed to be caused by one of the reduction products of the aromatic nitro group. This side effect is known as aplastic anemia and has even occurred following use of the drug as an ophthalmic ointment. There seems to be a genetic predisposition toward this in a very small percentage

of the population. Less severe, but much more common, is a reversible inhibition of hematopoiesis, seen in older patients or in those with renal insufficiency. If cell counts are taken, this can be controlled because it is dose-related and marrow function will recover if the drug is withdrawn.

The so-called "gray" or "gray baby" syndrome, a form of cardiovascular collapse, is encountered when chloramphenicol is given to young infants (especially premature infants) when liver glucuronidation is underdeveloped, and successive doses will lead to rapid accumulation of the drug due to impaired excretion. A dose-related profound anemia accompanied by an ashen gray pallor is seen, as are vomiting, loss of appetite, and cyanosis. Deaths have resulted, often involving cardiovascular collapse.

A variety of cyclic peptides are utilized for their antibiotic properties. The usual physiologically significant peptides are linear. Several bacterial species, however, produce antibiotic mixtures of cyclic peptides, some with uncommon amino acids and some with common amino acids but with the D absolute stereochemistry. These cyclic substances often have a pendant fatty acid chain as well. One of the consequences of this unusual architecture is that these glycopeptide agents are not readily metabolized. These drugs are usually water soluble and are highly lethal to susceptible bacteria because they attach themselves to the bacterial membranes and interfere with their semipermeability so that essential metabolites leak out and undesirable substances pass in. Unfortunately, they are also highly toxic in humans, so their use is reserved for serious situations where there are few alternatives or to topical uses. Bacteria are rarely able to develop significant resistance to this group of antibiotics. They are generally unstable, so solutions should be protected from heat, light, and extremes of pH.

Vancomycin (Vancocin, Generic).



Vancomycin is produced by fermentation of *Amycolatopsis orientalis* (formerly called *Nocardia orientalis*). It has been available for about 40 years, but its popularity has

increased significantly with the emergence of MRSA in the early 1980s. Chemically, vancomycin has a glycosylated hexapeptide chain rich in unusual amino acids, many of which contain aromatic rings cross-linked by aryl ether bonds into a rigid molecular framework.

MECHANISM OF ACTION Vancomycin is a bacterial cell wall biosynthesis inhibitor. There is evidence that the active species is a homodimer of two vancomycin units. The binding site for its target is a peptide-lined cleft having high affinity for acetyl-D-alanyl-D-alanine and related peptides through five hydrogen bonds (100). It inhibits both transglycosylases (inhibiting the linking between muramic acid and acetyl glucosamine units) and transpeptidase (inhibiting peptide cross-linking) activities in cell wall biosynthesis (Fig. 33.10). Thus, vancomycin functions like a peptide receptor and interrupts bacterial cell wall biosynthesis at the same step as does the β -lactams but by a different mechanism. By covering the substrate for cell wall transamidase, it prevents cross-linking, resulting in osmotically defective cell walls.

RESISTANCE Only recently, despite decades of intensive use, have some vancomycin-resistant bacteria emerged (vancomycin-resistant *Enterococcus* and vancomycin-resistant *Staphylococcus aureus*) (101). The mechanism of resistance appears to be alteration of the target D-alanyl-D-alanine units on the peptidoglycan cell wall precursors to D-alanyl-D-lactate. This results in lowered affinity for vancomycin. It is greatly feared that this form of resistance will become common in the bacteria for which vancomycin presently is used for successful chemotherapy.

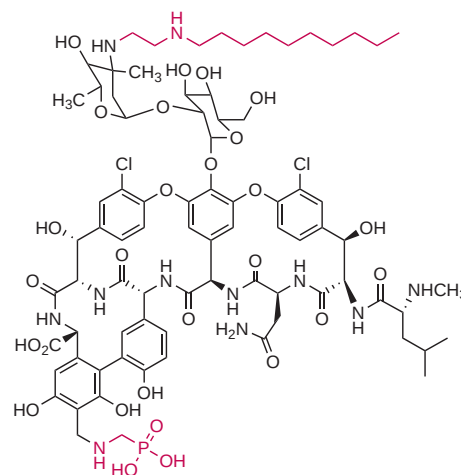
THERAPEUTIC APPLICATIONS Although a number of adverse effects can result from intravenous infusion (see below), vancomycin has negligible oral activity. It can be used orally for action in the GI tract, especially in cases of *Clostridium difficile* overgrowth. The useful spectrum is restricted to gram-positive pathogens with particular utility against multiply resistant coagulase-negative staphylococci and MRSA, which causes septicemias, endocarditis, skin and soft tissue infections, and infections associated with venous catheters (102).

ADVERSE EFFECTS Vancomycin is highly associated with adverse infusion-related events. These are especially prevalent with higher doses and a rapid infusion rate. Rapid infusion rate has been shown to cause anaphylactoid reactions, including hypotension, wheezing, dyspnea, urticaria, and pruritus. A significant drug rash (the so-called red man syndrome) can also occur. These events are much less frequent with a slower infusion rate.

In addition to the danger of infusion-related events, higher doses of vancomycin can cause nephrotoxicity

and auditory nerve damage. The risk of these effects is increased with elevated, prolonged concentrations, so vancomycin use should be monitored, especially in patients with decreased renal function. The ototoxicity may be transient or permanent and more commonly occurs in patients on high doses, those who have underlying hearing loss, and those treated concomitantly with other ototoxic agents (i.e., aminoglycosides) (102).

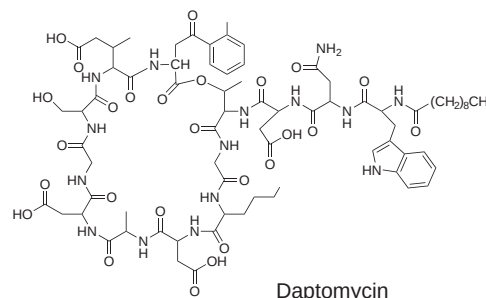
Telavancin (*Vibativ*).



Telavancin

Telavancin is a semisynthetic derivative of vancomycin that is more properly called a lipoglycopeptide. Although it does inhibit cell wall biosynthesis in the same manner as vancomycin, it appears to also disrupt cell membrane integrity as part of its mechanism. This is suspected to be why telavancin retains some activity against vancomycin-resistant strains (103). Telavancin is indicated for treatment of complicated skin and soft tissue infections by gram-positive organisms including MRSA. It should not be used in pregnancy, and like vancomycin, it requires dose adjustment with renal impairment. Telavancin also produces infusion-related toxicities, so slow infusion is recommended (104).

Daptomycin (*Cubicin*).

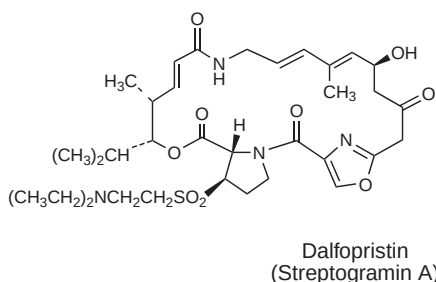
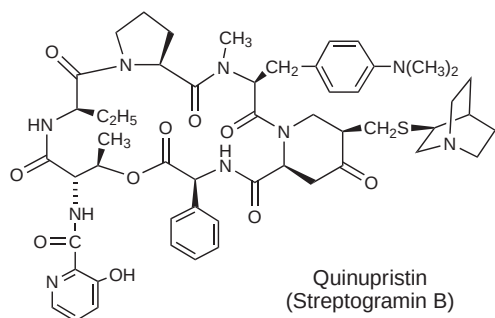


Daptomycin

Daptomycin is a fermentation product having a cyclic lipopeptide structure (105). It is primarily active against

gram-positive infections, especially skin and skin structure infections. It is given intravenously but must be administered over a period of 30 minutes or more. It binds to cell membranes and causes depolarization, which interrupts protein, DNA, and RNA synthesis. Daptomycin is bactericidal. Although resistance can be achieved *in vitro*, resistance has been slow to emerge in the clinic. Patients should be monitored for muscle pain or weakness because some incidence of elevated serum creatinine phosphokinase is associated with its use. Daptomycin is eliminated primarily by the kidney, so dose adjustment may be necessary in cases of renal insufficiency (106).

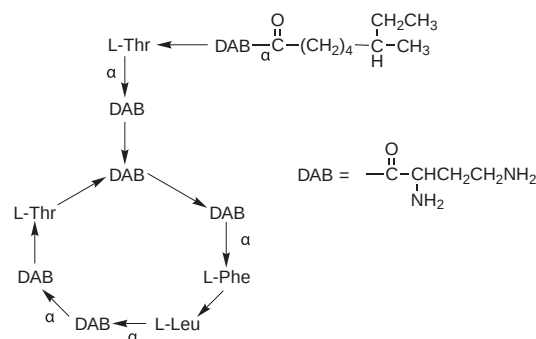
Quinupristin/Dalfopristin (Synercid).



A combination of the streptogramins quinupristin and dalfopristin is approved for intravenous use in the treatment of infections caused by vancomycin-resistant *Enterococcus faecium* bacteremia as well as skin and skin structure infections caused by methicillin-sensitive *Staphylococcus aureus* and *Streptococcus pyogenes*. Certain strains of *E. faecium* are resistant to essentially all other antibiotics, including vancomycin. Dalfopristin binds to the 70S ribosomal subparticle at the A and P sites, thus interfering with substrate attachment. Quinupristin binds to the A and P sites, also causing a misalignment of the tRNA in the P site (107). Thus, they bind to nearly the same place in the ribosome (Fig. 33.24). The two drugs are bacteriostatic when administered individually but are found to act synergistically when combined (quinupristin:dalfopristin [30:70 w/w]) to produce a bactericidal effect. It appears that dalfopristin binding creates a high-affinity binding site for quinupristin, accounting for the synergy. The combination is found to inhibit protein synthesis by binding

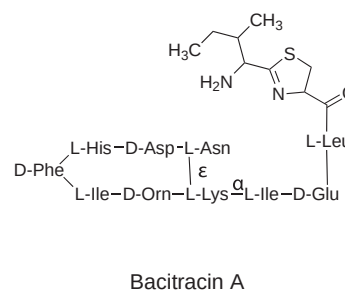
to the 70S ribosome that is thought to account for the mechanism of action. Synercid is a strong inhibitor of the CYP3A4 isozyme, and a number of drug interactions are to be expected. Although the combination does not appear to prolong the QT_c interval, it inhibits the metabolism of a number of agents that have been shown to have this effect, so concomitant administration should be avoided. Synercid also can produce a number of other adverse effects including infusion site reactions (pain, edema, and inflammation), arthralgia and myalgia, and hyperbilirubinemia (108).

Polymyxin B.



Polymyxin B is produced by fermentation of *Bacillus polymyxa*. It is separated from a mixture of related cyclic peptides and is primarily active against gram-negative microorganisms. It apparently binds to phosphate groups in bacterial cytoplasmic membranes and disrupts their integrity. It is used intramuscularly or intravenously as a sulfate salt to treat serious urinary tract infections, meningitis, and septicemia primarily caused by *Pseudomonas aeruginosa*, but some other gram-negative bacteria will also respond. It is also used orally to treat enteropathogenic *E. coli* and *Shigella* sp. diarrheas. Irrigation of the urinary bladder with solutions of polymyxin B sulfate is also employed by some to reduce the incidence of infections subsequent to installation of indwelling catheters. When given parenterally, the drug is neuro- and nephrotoxic and thus is used only after other drugs have failed.

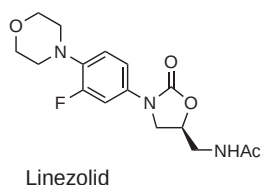
Bacitracin.



Bacitracin is a mixture of similar peptides produced by fermentation of the bacterium *Bacillus subtilis*. The A

component predominates. Its mode of action is to inhibit both peptidoglycan biosynthesis at a late stage (probably at the dephosphorylation of the phospholipid carrier step) and disruptions of plasma membrane function. It is predominantly active against gram-positive microorganisms, and parenteral use is limited to intramuscular injection for infants with pneumonia and empyema caused by staphylococci resistant to other agents. It is rather neuro- and nephrotoxic and thus is used in this manner with caution. Bacitracin is also widely used topically to prevent infection in minor cuts, scrapes, and burns.

Linezolid (Zyvox).



Linezolid is a bacteriostatic (bactericidal against some bacteria) antibacterial agent effective against primarily gram-positive bacteria. This drug is a member of the oxazolidinone class of agents and represents a new synthetic class of antibacterials. It is available in injectable, tablet, and oral suspension dosage forms; is effective against MRSA; and is used to treat nosocomial pneumonia, community-acquired pneumonia, complicated and uncomplicated skin and skin structure infections, and vancomycin-resistant *Enterococcus faecium* infections (109).

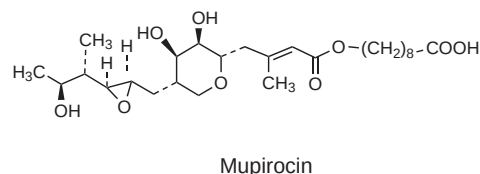
The mechanism of action of linezolid is associated with inhibition of protein synthesis, but at a stage different from that of other protein synthesis inhibitors. The oxazolidinones inhibit the initiation of protein synthesis by binding to the 50S subparticle, preventing the formation of a functional initiation complex including *N*-formylmethionyl-tRNA (tRNA^{Met}), the 30S subunit, mRNA, and initiation factors. It is believed that the drug distorts the binding site for the initiator-tRNA, which overlaps both 30S and 50S ribosomal subunits (110). The result of this is that translation of the mRNA is prevented. The other prominent ribosomal protein biosynthesis inhibitors inhibit after a functional initiation complex forms. In those cases, mRNA translation begins, but elongation of the growing peptide chain is blocked.

Resistance to oxazolidinones is encountered in the clinic due to a mutation in the 23S rRNA associated with the 50S ribosomal subunit (111). This is believed to distort the linezolid binding site. Gram-negative microorganisms are intrinsically resistant to linezolid due to the presence of endogenous efflux pumps that keep them from accumulating in the cells.

Linezolid is well absorbed orally and is generally well to moderately well tolerated; however, some severe cases of reversible blood dyscrasias have been noted, resulting

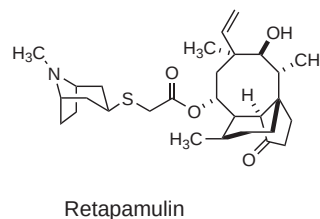
in a package insert warning that complete blood counts should be monitored weekly, especially in patients with poorly draining infections and who are receiving prolonged therapy with the drug (109). Linezolid is an inhibitor of monoamine oxidase, so patients should be cautious about eating tyramine-containing foods. Coadministration with adrenergic and serotonergic agents is also inadvisable. Significant oxidative metabolism occurs by oxidation of the morpholine ring, and the antimicrobially inactive metabolites are excreted primarily in the urine (109).

Mupirocin (Bactroban).



Mupirocin is a member of a group of lipid acids produced by fermentation of *Pseudomonas fluorescens*. It can only be used topically because of hydrolysis in vivo that inactivates the drug. It is intrinsically broad spectrum, but its primary indication is topically against staphylococcal and streptococcal skin infections. Mupirocin binds to bacterial isoleucyl tRNA synthase, preventing incorporation of isoleucine into bacterial proteins. Resistance is due to alterations of the synthase target such that the enzyme still functions but does not bind mupirocin.

Retapamulin (Altabax).



Retapamulin is approved for use in adults and children over the age of 9 months for the treatment of impetigo caused by methicillin-susceptible *Staphylococcus aureus* and *Streptococcus pyogenes*. Impetigo is a highly contagious infection found most commonly in children involving the upper layers of the skin and spread through close contact. Retapamulin is a semisynthetic diterpene derivative of the naturally occurring pleuromutilins, a group of chemicals produced by the fungus *Pleurotus mutilus* (also referred to as *Clitopilus scyphoides*) (112).

Retapamulin is a bacteriostatic topically used drug that inhibits protein synthesis by binding to the 50S ribosomal subunit. It is a semisynthetic derivative of the natural product mutilin (alcohol at C-14) and pleuromutilin (C-14 hydroxyacetate). The drug is

selective for inhibiting translational bacterial protein syntheses and is inactive in eukaryotic translation. Retapamulin is active at a site distinct from that occupied by macrolide antibiotics, both of which inhibit protein synthesis at the 50S ribosomal subunit. It inhibits ribosomal peptidyl transferase, thus preventing peptide bond formation. The C-14 sulfamylacetate moiety appears important for binding to the peptidyl transferase center in the 50S ribosomal subunit of the susceptible bacteria. By occupying this site, the pleuromutilins block peptide bond formation, thus inhibiting bacterial protein synthesis. The pleuromutilins interact with the 23S RNA domain V. There are

indications because of the unique binding of retapamulin to the 50S ribosomal subunit that bacterial resistance will be low. Additionally, it has been reported that retapamulin inhibits the biosynthesis of the 50S ribosomal subunit itself in susceptible gram-positive bacteria, leading to the suggestion of a dual antibacterial action (113–115).

Retapamulin is well tolerated, with minimal adverse events. The most common adverse effects are pruritus, diarrhea, and nasopharyngitis. The drug does not appear to be absorbed to any great degree when applied topically, although it is a substrate for CYP3A4 (112).

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Elizabeth Coyle, PharmD

Although the organism has intermediate susceptibility to cefepime, the treatment would likely fail because the ability of ESBL to hydrolyze the cephalosporin. Carbapenems are the drugs of choice for infections caused by ESBL-producing organisms because of their stability in the presences of ESBLs. The team changes ER's antibiotics to meropenem 1 g every 8 hours and discontinues the vancomycin.

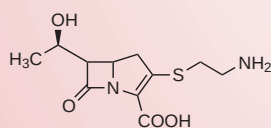
Chemical Analysis

Victoria Roche and S. William Zito

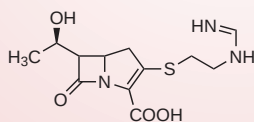
The antibacterial agent decided upon in this scenario is a second generation carbapenem. Carbapenems are novel β -lactam antibiotics derived from thienamycin which was isolated from fermentations of *Streptomyces cattleya*. Aside from meropenem, there are other carbapenems, i.e., imipenem, biapenem (approved in Japan in 2001), ertapenem, and doripenem (approved in the United States in 2007). Carbapenems are similar to penicillins and cephalosporins in that they have a fused bicyclic ring

system containing a β -lactam ring and an equivalent 3-carboxyl group. However, they differ in that the fused system contains a 5-membered ring that has a methylene group instead of the sulfur atom found in penicillins and cephalosporins. The sulfur atom is attached to C-3 as part of the side chain. In addition, the 5-membered ring contains a double bond. Both of these structural changes result in a strained carbapenem that can be hydrolyzed easily at the β -lactam bond.

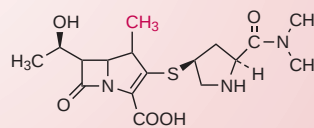
Early carbapenems suffered from both chemical and metabolic instability. The amino group in thienamycin can attack the β -lactam bond of another molecule through an intermolecular reaction deactivating activity. Imipenem and all the other carbapenems do not have this problem; however, imipenem was found to be hydrolyzed by renal dehydropeptidase-1, causing loss of antibacterial activity. Meropenem, on the other hand, is resistant to both chemical and biological deactivation. Its resistance to renal dehydropeptidase-1 is caused by the methyl group at C-4 of the bicyclic ring system. Meropenem is not active orally and must be administered parenterally. It is excreted primarily unchanged (70–80%) in the urine; the remainder is the inactive β -lactam hydrolysis product.



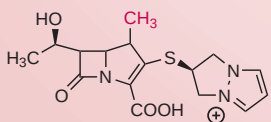
Thienamycin



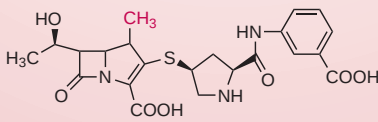
Imipenem



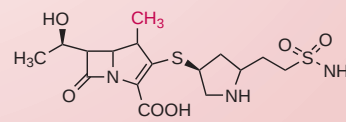
Meropenem (Merrem)



Biapenem



Ertapenem



Doripenem

CASE STUDY

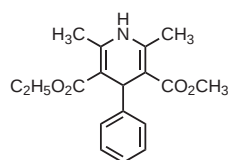
S. William Zito and Victoria Roche

YU, a 42-year-old man, presents to the clinic where you work. He states that he has a sore throat and lumps in his neck. He had some nausea and vomiting the previous evening. YU states that these symptoms have been going on for more than a week and he "just feels really bad." He denies any shortness of breath, difficulty breathing, chest pain, light headedness, dizziness, or loss of consciousness.

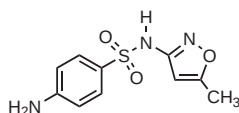
YU is 5'4" and weighs 142 lb. His blood pressure is 135/92 mm Hg, his pulse is 121 beats per minute, and his temperature is 100.1°F. YU's sinuses are not tender and his nares are normal. His throat is very red. His neck is supple, with two tender lymph nodes on the right. During questioning, YU states that he has been taking felodipine (10mg/daily) for hypertension for the past 2

years and has no side effects. YU recalls that several years ago he developed a rash when he was treated with bactrim (sulfamethoxazole/trimethoprim) for a bladder infection and had to be switched to another antibiotic. The rapid strep test is positive, and therefore YU is assessed with a Strep throat. Given the following three therapeutic choices, which would you recommend?

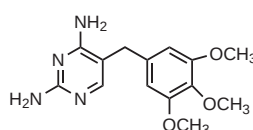
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.



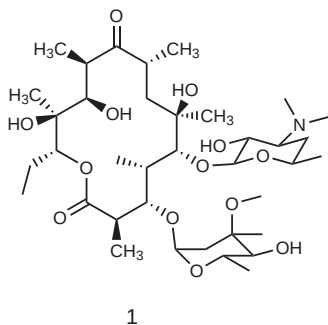
Felodipine



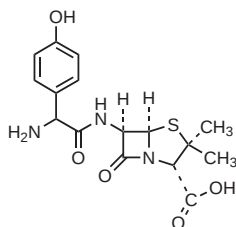
Sulfamethoxazole



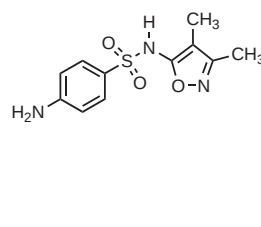
Trimethoprim



1



2



3

ACKNOWLEDGMENT

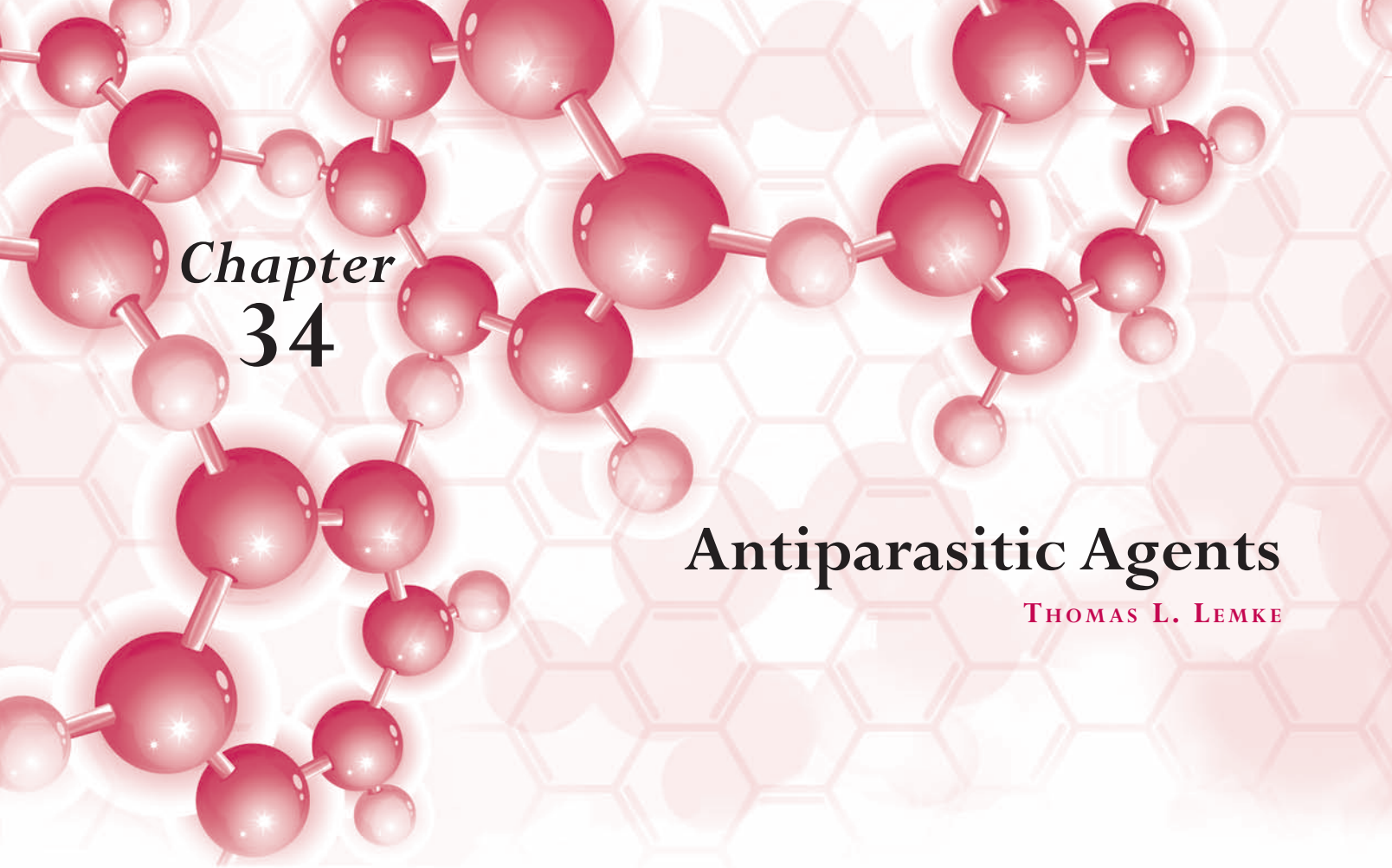
This chapter is dedicated to Dr. Lester A. Mitscher on the occasion of his retirement from the University of Kansas. Dr. Mitscher's contributions to the antimicrobial field and to previous editions of this text have served to educate thousands of pharmacy and medicinal chemistry students on the significance and intrigue of antibiotics. Thank you for your many years of commitment to the education of others.

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Chapter 34

Antiparasitic Agents

THOMAS L. LEMKE

Drugs Covered in This Chapter

DRUG TREATMENT OF AMEBIASIS, GIARDIASIS, TRICHOMONIASIS

- Diloxanide furoate
- Metronidazole
- Nitazoxanide
- Tinidazole

TREATMENT OF PNEUMOCYSTIS

- Atovaquone
- Pentamidine isethionate
- Sulfamethoxazole–trimethoprim
- Trimetrexate glucuronate

TREATMENT OF TRYPANOSOMIASIS

- *Benznidazole*
- Eflornithine
- Melarsoprol
- Nifurtimox

- Pentamidine isethionate
- Suramin sodium

TREATMENT OF LEISHMANIASIS

- *Sodium stibogluconate*
- Miltefosine

ANTIMALARIALS

- Artemisinins (arteether, artemether, artesunate, dihydroartemisinin)
- Atovaquone–proguanil
- Chloroquine
- Halofantrine
- Lumefantrine
- Mefloquine
- Pyrimethamine
- Quinine

ANTHELMINTICS

- Albendazole
- Diethylcarbamazine
- Ivermectin
- Mebendazole
- Oxamniquine
- Praziquantel
- Pyrantel pamoate
- Thiabendazole

SCABICIDES AND PEDICULOCIDES

- Crotamiton
- Lindane
- Permethrin
- Pyrethrin
- Spinosad

Abbreviations

ACT, artemisinin-based combination therapy

ADH, alcohol dehydrogenase

ALDH, aldehyde dehydrogenase

CDC, Centers for Disease Control and Prevention

CDI, *Clostridium difficile* infection

CNS, central nervous system

DDT, dichlorodiphenyltrichloroethane

DEC, diethylcarbamazine

DFMO, difluoromethyl ornithine

DHFR, dihydrofolate reductase

FDA, U.S. Food and Drug Administration

GABA, γ -aminobutyric acid

GI, gastrointestinal

HM, 2-hydroxymethylmetronidazole

IV, intravenous

IVM, ivermectin

NTZ, nitazoxanide

ODC, ornithine decarboxylase

PCP, pneumocystis pneumonia

PABA, *p*-aminobenzoic acid

PZQ, praziquantel

RBC, red blood cell

ROS, reactive oxygen species

TIZ, tizoxanide

TMQ, trimetrexate

WHO, World Health Organization

SCENARIO

Elizabeth B. Hirsch, PharmD, BCPS

JB, a 67-year-old white man, was recently hospitalized for 5 weeks following complications after coronary artery bypass graft surgery. He received multiple courses of antibiotics while in the hospital and developed *Clostridium difficile* infection (CDI). He was discharged home with a prescription to continue 7 days of treatment with oral metronidazole for CDI. After 4 days of metronidazole, he presents to his primary

care physician complaining of cold symptoms (cough and congestion) for which he had been taking over-the-counter cough medicine. Last night, he reported experiencing severe nausea and vomiting with some dizziness.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter.)

GENERAL CONSIDERATIONS

An introduction to the topic of parasitic diseases usually emphasizes two points. First, parasitic infections affect huge numbers of individuals. It is estimated that well over 1 billion people are infected with parasitic diseases worldwide. Second, the majority of these parasitic infections are found in developing nations, in which the cost of health care is the dominant factor that determines whether the patient is (or is not) treated. The incidence of some parasitic diseases can exceed 80% of the population. The high cost of drug discovery and the low incidence of many of the parasitic infections in affluent Western countries have combined to reduce the incentive for both the study of the diseases and the development of effective therapy. This may be changing, however, because of global travel, improved communications, and growth of the developing countries, leading to an increased demand for more effective treatments.

The diseases associated with parasitic infections represent a large and diverse number of conditions, some common and some relatively unheard of by the general population. Included under the title of parasitic infections are the numerous types of protozoal infections: amebiasis, giardiasis, babesiosis, Chagas disease, leishmaniasis, malaria, sleeping sickness, toxoplasmosis, trichomoniasis, and pneumocystosis (also considered to be a fungal infection). Helminth infections (worms) are also considered to be parasitic infections and can be caused by any of three classes of helminths: nematodes, cestodes, and trematodes. Insect infections, such as scabies, lice (pediculosis), chiggers, and bedbugs (Cimicidae family), are also considered to be parasitic infections.

PROTOZOAL DISEASES

Amebiasis

Amebiasis is a disease of the large intestine caused by *Entamoeba histolytica*. The disease occurs mainly in the tropics, but it is also seen in temperate climates. Amebiasis can be carried without significant symptoms or can lead to severe, life-threatening dysentery. The organism exists in one of two forms, the motile trophozoite form or the

dormant cyst form. The trophozoite form is found in the intestine or in the wall of the colon and can be expelled from the body with the feces. The cyst form is encased by a chitinous wall that protects the organism from the environment, including chlorine used in water purification; thus, the organism can be transmitted through contaminated water and foods. The cyst form is responsible for transmission of the disease. The cyst is spread by direct person-to-person contact and is commonly associated with living conditions in which poor personal hygiene, poor sanitation, poverty, and ignorance exist. The hosts can be rendered susceptible to infection by preexisting conditions, such as protein malnutrition, pregnancy, HIV infection, or high-carbohydrate intake. Under these conditions, the organism is capable of invading body tissue. The protozoal invasion is not well understood, but it does appear to involve the processes indicated in Table 34.1. Symptoms can range from intermittent diarrhea (foul-smelling loose/watery stools) to tenderness and enlargement of the liver (with the extraintestinal form) to acute amoebic dysentery. Many patients can experience no symptoms, and the organism remains in the bowels as a commensal organism.

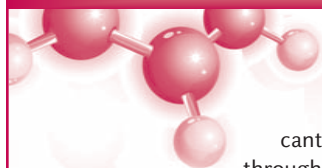
Giardiasis

Giardiasis is a disease that shows considerable similarity to amebiasis. It is caused by *Giardia lamblia*, an organism that can be found in the duodenum and jejunum. The organism exists in a motile trophozoite form and an infectious cyst form. The cyst form can be deposited in water (lives up to 2 months), and the contaminated

TABLE 34.1 *Entamoeba histolytica* Invasion of Host

Intestinal form	a. Disintegration of cyst wall in small intestine b. Movement of trophozoites into the colon c. Adhesion of trophozoite to cells of the host, which involves a change in composition and production of mucus
Extraintestinal form	d. Penetration of intestinal lining and entrance into portal circulation e. Invasion of liver tissue

CLINICAL SIGNIFICANCE



Parasitic infections affect more than half the world's population and are responsible for significant health complications, particularly throughout the developing world. Efforts to control parasitic infections often use a broad approach that includes strategies for prevention, treatment, and control. Drug therapy for parasitic infections can be quite challenging to practitioners for multiple reasons. They are often more prevalent in specific high-risk groups, so practitioners may lack experience with these agents and may be unfamiliar with the toxicities and monitoring parameters associated with these drugs. Certain agents are available in only a limited number of countries, and opinions regarding safety and efficacy vary greatly among practitioners. Understanding the medicinal

chemistry, pharmacokinetics, and pharmacodynamics of these agents is of utmost importance.

Despite efforts of vaccine development, drug therapy remains the most effective means to control parasitic infections. Limited introduction of new antiparasitic agents and drug shortages make the treatment of parasitic infestations challenging. Understanding which drugs work at different parts of the life cycle of the parasite must also be taken into consideration. Lastly, clinicians must stay up to date on the rapid evolution of drug resistance to ensure their patients receive the most efficacious treatment regimens.

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water can then be ingested by the human. The trophozoite, if expelled from the gastrointestinal (GI) tract, normally will not survive. *Giardia lamblia* is the single most common cause of waterborne diarrhea in the United States. Giardiasis is a common disease among campers who drink water from contaminated streams. It can also be spread between family members, children in day care centers, and dogs and their masters. The organism can attach to the mucosal wall via a ventral sucking disk, and similar to amebiasis, the patient can be asymptomatic or develop watery diarrhea, abdominal cramps, distention and flatulence, anorexia, nausea, and vomiting. Usually, the condition is self-limiting in 1 to 4 weeks.

Trichomoniasis

Trichomoniasis is a protozoal infection caused by *Trichomonas vaginalis*, which exists only in a trophozoite form. The organs most commonly involved in the infection include the vagina, urethra, and prostate; thus, the disease is considered to be a venereal infection. The condition is transmitted by sexual contact, and it is estimated that trichomoniasis affects 180 million individuals worldwide. Infections in the male can be asymptomatic, whereas in the female, the symptoms can consist of vaginitis, profuse and foul-smelling discharge, burning and

soreness on urination, and vulvar itching. Diagnosis is based on microscopic identification of the organism in fluids from the vagina, prostate, or urethra.

Pneumocystis

The disease pneumocystis, commonly referred to as pneumocystis pneumonia (PCP), incorrectly derived its name from what was thought to be the causative organism, *Pneumocystis carinii* and the disease state of pneumonia. The organism *P. carinii* was originally isolated and reported to grow in both humans and rats. The organism itself was difficult to characterize since it had morphologic characteristics of a protozoan (i.e., lack of ergosterol in its cell membrane), but its rRNA and mitochondrial DNA pattern resembles that of fungi. It was only later recognized that the organism infecting humans and responsible for the disease PCP was actually *Pneumocystis jirovecii*, a yeast-like fungus that can only be cultured in humans. Acute pneumocystis rarely strikes healthy individuals, although the organism is harbored in most humans without any apparent adverse effect. *P. jirovecii* becomes active only in individuals who have a serious impairment of their immune systems. Thus, the organism is considered to be an opportunistic pathogen. More recently, this disease has appeared in patients with HIV/AIDS, 80% of whom ultimately contract *P. jirovecii* pneumonia, as one of the main causes of death. The disease also occurs in those receiving immunosuppressive drugs to prevent rejection following organ transplantation or for the treatment of malignant disease. Additionally, pneumocystis is seen in malnourished infants whose immunologic systems are impaired. The disease is thought to be transmitted via an airborne route. PCP is characterized by a severe pneumonia caused by rapid multiplication of the organisms, almost exclusively in lung tissue, with

ORGANISMS THAT COMMONLY CAUSE VAGINITIS

Vaginitis may also be caused by *Haemophilus vaginalis* (bacteria) or *Candida albicans* (fungus), which are treated differently from the protozoal infection.

the organism lining the walls of the alveoli and gradually filling the alveolar spaces. Untreated, the acute form of the disease is generally fatal. Even patients who recover from pneumocystosis are at risk of recurrent episodes. Patients with AIDS experience a recurrence rate of approximately 50%.

Extrapulmonary pneumocystosis—that is, pneumocystosis outside of the lungs—is also known to exist and can be more common than presently recognized. This infection can be complicated by the presence of coinfectious organisms. Fortunately, drug therapy used for treatment of the pulmonary infection is beneficial for the extrapulmonary condition, although intravenous (IV) administration of the drugs can be necessary.

The common therapy for PCP uses antibacterial and antiprotozoal drugs.

Tritryps

Three protozoan pathogens that belong to the family Trypanosomatidae, the order Kinetoplastida, and the genus *Trypanosoma* are *Leishmania major*, which is responsible for leishmaniasis; *Trypanosoma brucei*, which is responsible for African trypanosomiasis (African sleeping sickness); and *Trypanosoma cruzi*, which is responsible for Chagas disease. Referred to as the “trityps,” these eukaryotic organisms share characteristic subcellular structures of a kinetoplast and glycosomes, are unicellular motile protozoa, are transmitted by various insect vectors, and infect mammalian hosts. The genomes of trityps have recently been reported (1–3). Together, they infect hundreds of millions of people annually.

Trypanosomiasis (4)

There are two distinct forms of trypanosomiasis: Chagas disease and African sleeping sickness.

CHAGAS DISEASE Chagas disease, also known as American trypanosomiasis, is caused by the parasitic protozoa *T. cruzi* and is found only in the Americas, primarily in Brazil but also in the southern United States. The protozoa lives in mammals and is spread by the bloodsucking insect known as the reduviid bug, assassin bug, or kissing bug. The insect becomes infected by drawing blood from an infected mammal and releasing the protozoa with discharged feces. The pathogen then enters the new host through breaks in the skin. Inflammatory lesions are seen at the site of entry. The disease can also be spread through transfusion with contaminated blood. Signs of initial infection can include malaise, fever, anorexia, and skin edema at the site where the protozoa entered the host. The disease ultimately can invade the heart, where after decades of infection with chronic Chagas disease, the patient can experience an infection-associated heart attack. It is estimated that 5% of the Salvadorian and Nicaraguan immigrants to the United States can have chronic Chagas disease.

AFRICAN TRYPANOSOMIASIS African trypanosomiasis, or sleeping sickness, is caused by several subspecies of *T. brucei* (*T. brucei rhodesiense* [east African sleeping sickness] and *T. brucei gambiense* [west African sleeping sickness]). In this case, the infected animal is bitten by the bloodsucking tsetse fly, which in turn transmits the protozoa via inoculation during a subsequent bite of a human. The protozoa, initially present in the gut of the vector, appear in the salivary gland for inoculation during the subsequent biting of a human. It is estimated that some 50 million people are at risk of African sleeping sickness, with 300,000 to 500,000 cases occurring in sub-Saharan Africa each year. The infection progresses through two stages. Stage I can present as fever and high temperatures lasting several days; hematologic and immunologic changes occur during this stage. Stage II occurs after the organism enters the central nervous system (CNS) and can involve symptoms suggesting the disease name—daytime somnolence, loss of spontaneity, halting speech, listless gaze, and extrapyramidal signs (e.g., tremors and choreiform movements). A breakdown of neurologic function leading to coma and death can occur. Death can occur within weeks if untreated (*T. brucei rhodesiense*) or only after several years (*T. brucei gambiense*).

It should be noted that the sole source of energy for the trypanosomal organism is glycolysis, which in turn can account for the hypoglycemia seen in the host. In addition, the migration of the organism into the CNS can be associated with the organism’s search for a rich source of available glucose.

Leishmaniasis

Leishmaniasis is a disease caused by a number of protozoa in the genus *Leishmania*. The protozoa can be harbored in diseased rodents, canines, and various other mammals and transmitted from the infected mammal to man by bites from female sandflies of the genus *Phlebotomus* and then appear in one of four major clinical syndromes: visceral leishmaniasis, cutaneous leishmaniasis, mucocutaneous leishmaniasis, or diffuse cutaneous leishmaniasis. The sandfly, the vector involved in spreading the disease, breeds in warm, humid climates; thus, the disease is more common in the tropics. As many as 12 million individuals worldwide are infected by this organism.

The visceral leishmaniasis, also known as kala azar (black fever), is caused by *Leishmania donovani*. This form of the disease is systemic and is characterized in patients by fever, typically nocturnal, diarrhea, cough, and enlarged liver and spleen. The skin of the patient can become darkened. Without treatment, death can occur in 20 months and is commonly associated with diarrhea, superinfections, or GI hemorrhage. Visceral leishmaniasis is most commonly found in India and Sudan.

Both cutaneous and mucocutaneous leishmaniasis are characterized by single or multiple localized lesions. These slow-healing and, possibly, painful ulcers can lead to secondary bacterial infections. The Old World

cutaneous leishmaniasis is caused by *Leishmania topica*, which is found most commonly in children and young adults in regions bordering the Mediterranean, the Middle East, Southern Russia, and India. *Leishmania major* is endemic to desert areas in Africa, the Middle East, and Russia, whereas *Leishmania aethiopica* is found in the Kenyan highlands and Ethiopia. The New World disease caused by *Leishmania peruviana*, *Leishmania braziliensis*, and *Leishmania panamensis* is found in South and Central America, whereas *Leishmania mexicana* can be endemic to south-central Texas. The incubation period for cutaneous leishmaniasis ranges from a few weeks to several months. The slow-healing lesions can be seen on the skin in various regions of the body depending on the specific strain of organism. Usually, these conditions exhibit spontaneous healing, but this can also occur over an extended period of time (1 to 2 years).

Malaria

Malaria is transmitted by the infected female *Anopheles* mosquito. The specific protozoan organisms causing malaria are from the genus *Plasmodium*. Only 4 of approximately 100 species cause malaria in humans. The remaining species affect birds, monkeys, livestock, rodents, and reptiles. The four species that affect humans are *Plasmodium falciparum*, *Plasmodium vivax*, *Plasmodium malariae*, and *Plasmodium ovale*. Concurrent infections by more than one of these species are seen in endemically affected regions of the world. Such multiple infections further complicate patient management and the choice of treatment regimens.

It is estimated that 2 billion individuals are at risk of developing malaria and that the disease affects as many as 500 million humans globally and causes more than 2 million deaths annually. It is estimated that a third of these fatalities occur in children younger than 5 years. Although this disease is found primarily in the tropics and subtropics, it has been observed far beyond these boundaries. Malaria was virtually eradicated from the United States between 1947 and 1951 through the use of the insecticide DDT (dichlorodiphenyltrichloroethane), which destroyed the insect vector, and in 1955, the World Health Organization (WHO) launched a program titled the Global Malaria Eradication Program. The program intended to eradicate the disease worldwide through the use of DDT and disease treatment with chloroquine, but due to the development of DDT-resistant *Anopheles* mosquitoes, drug-resistant plasmodium, and political resistance, WHO abandoned its program in 1972. Recently, there has been an increase in global interest in fighting the disease through the use of preventative measures and development of multidrug treatments, although at present, nearly nothing new has reached the market.

While malaria has essentially been eradicated in most temperate-zone countries, more than 1,000 cases of malaria were documented recently in U.S. citizens returning from travel abroad. Today, malaria is found

in most countries of Africa, Central and South America, and Southeast Asia. It is reported to be on the increase in Afghanistan, Bangladesh, Brazil, Burma, Cambodia, Colombia, China, Iran, India, Indonesia, Mexico, the Philippines, Thailand, and Vietnam. Infection from plasmodia can cause anemia, pulmonary edema, renal failure, jaundice, shock, cerebral malaria, and if not treated in a timely manner, even death.

Types of Malaria

Malarial infections are known according to the species of the parasite involved.

PLASMODIUM FALCIPARUM Infection with *Plasmodium falciparum* has an incubation period (time from mosquito bite to clinical symptoms) of 1 to 3 weeks (average, 12 days). The *P. falciparum* life cycle in humans begins with the bite of an infected female mosquito. The parasites in the sporozoite stage enter the circulatory system, through which they can reach the liver in approximately 1 hour. These organisms grow and multiply 30,000- to 40,000-fold by asexual division within liver cells in 5 to 7 days. Then, as merozoites, they leave the liver to reenter the blood stream and invade the erythrocytes, or red blood cells (RBCs), where they continue to grow and multiply further for 1 to 3 days. Specific receptors on the surface of the erythrocytes serve as binding sites for the merozoite. These infected RBCs rupture, releasing merozoites in intervals of approximately 48 hours. Chemicals released by the ruptured cell in turn cause activation and release of additional substances associated with the patient's symptoms. The clinical symptoms include chills, fever, sweating, headaches, fatigue, anorexia, nausea, vomiting, and diarrhea. Some of the released merozoites are sequestered in vital organs (brain and heart), where they continue to grow. Recurrence of the clinical symptoms on alternate days leads to the terminology of tertian malaria. The *P. falciparum* parasite can also cause RBCs to clump and adhere to the wall of blood vessels. Such a phenomenon has been known to cause partial obstruction and, sometimes, restriction of the blood flow to vital organs like the brain, liver, and kidneys. Reinfection of RBCs can occur, allowing further multiplication and remanifestation of the malaria symptoms. Some merozoites develop into male and female sexual forms, called gametocytes, which can then be acquired by the female mosquito after biting the infected human. Gametocytes mature in the mosquito's stomach to form zygotes. Growth of the zygotes leads to the formation of oocysts (spherical structures located on the outside wall of the stomach). Sporozoites develop from the oocysts, are released into the body cavity of the mosquito, and migrate to the salivary gland of the insect, from which they can be transmitted to another human following a mosquito bite. The life cycle of the malaria parasites is shown in the Figure 34.1. The genome of the *P. falciparum*, as well as that of the *Anopheles* mosquito, is now known and is expected to provide potential new avenues

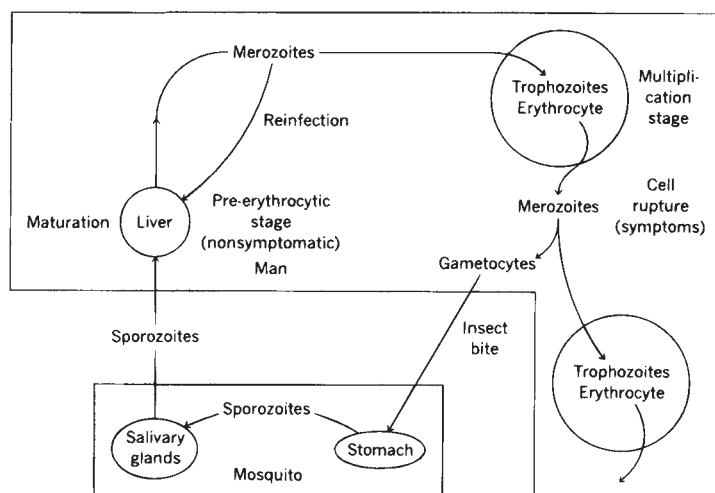


FIGURE 34.1 Life cycle of malarial protozoa.

for drug development. Genome information is also expected to give insight regarding the mechanisms of resistance and improve drug treatment.

PLASMODIUM VIVAX *Plasmodium vivax* (benign tertian) is the most prevalent form of malaria. It has an incubation period of 1 to 4 weeks (average, 2 weeks). This form of malaria can cause spleen rupture and anemia. Relapses (renewed manifestations of erythrocytic infection) can occur. This results from the periodic release of dormant parasites (hypnozoites) from the liver cells. The erythrocytic forms are usually considered to be susceptible to treatment.

PLASMODIUM MALARIAE *Plasmodium malariae* is responsible for quartan malaria. It has an incubation period of 2 to 4 weeks (average, 3 weeks). The asexual cycle occurs every 72 hours. In addition to the usual symptoms, this form also causes nephritis. This is the mildest form of malaria and does not relapse. The RBC infection associated with *P. malariae* can last for many years. The *P. malariae* is quite unlikely to become resistant.

PLASMODIUM OVALE Infection with *Plasmodium ovale* has an incubation period of 9 to 18 days (average, 14 days). Relapses have been known to occur in individuals infected with this plasmodium. The relapse can be indicative of ovale tertian malaria and is associated with the ability of the organism to lie dormant (hypnozoites) in hepatic tissue for extended periods of time.

Types of Chemotherapy

DRUG CLASSIFICATION Generally, the antimalarial drugs are classified by the life cycle stage (see Fig. 34.1) that the drugs are effective against. Listed in the following sections are the theoretical approaches of how drugs might be able to intervene in destroying the organism.

Presently, the majority of drugs act on the intracellular erythrocytic form of the disease.

Tissue Schizonticides These drugs eradicate the exoerythrocytic liver-tissue stages of the parasite, which prevents the parasite's entry into the blood. Drugs of this type are useful for prophylaxis. Some tissue schizonticides can act on the long-lived tissue form (hypnozoites of *P. vivax* and *P. ovale*) and, thus, can prevent relapses.

Blood Schizonticides These drugs destroy the erythrocytic stages of parasites and can cure cases of falciparum malaria or suppress relapses. This is the easiest phase to treat, because drug delivery into the bloodstream can be accomplished rapidly.

Gametocytocides Agents of this type kill the sexual forms of the plasmodia (gametocytes), which are transmittable to the *Anopheles* mosquito, thereby preventing transmission of the disease.

Sporontocides (Sporozooitocides) These drugs act against sporozoites and are capable of killing these organisms as soon as they enter the bloodstream following a mosquito bite.

It should be noted that antimalarials can operate against more than one form of the organism and can be effective against one species of plasmodium but lack efficacy against others. In addition, antimalarial drugs can be classified according to their structural types.

Despite the history of malaria, the epidemiology and clinical features of the disease are not that well documented. In many parts of Africa, the diagnosis of the disease is not routinely done, and therefore, the success of prevention and treatment cannot be known. For more details on these difficulties, the reader is

directed to the *Suggested Readings*, especially the article by Greenwood et al.

GENERAL APPROACHES TO PROTOZOAL THERAPY

Amebiasis and Giardiasis

The most appropriate approach for treatment of this type of protozoal infection is through prevention. Because the infection usually occurs by consumption of contaminated drinking water and food, avoidance is the key to prevention. Drinking bottled water, or boiling or disinfecting the water, will reduce the risk. Improvement in personal hygiene and general sanitation are also beneficial.

Trypanosomiasis, Leishmaniasis, and Malaria

For these diseases that are spread by insect vectors, the use of insecticides, protective clothing, and insect repellents can greatly reduce the incidence of the disease. Unfortunately, many of these protozoal infections can also infect other hosts beside humans; thus, even the most successful insect irradiation methods cannot destroy all the reservoirs of the protozoa. The use of insect repellents and protective clothing can be useful for visitors to regions with endemic infections, but these procedures can prove to be ineffective for those living in the area. For such individuals, early detection and drug therapy are the methods of treatment.

DRUG THERAPY FOR PROTOZOAL INFECTIONS

Treatment of Amebiasis, Giardiasis, and Trichomoniasis

Metronidazole (Flagyl, Metryl, Satric)



Metronidazole was initially introduced for the treatment of vaginal infections caused by *Trichomonas vaginalis* but has since been shown to be effective for treatment of amebiasis, giardiasis, and anaerobic bacterial infections, including *Clostridium difficile*.

MECHANISM OF ACTION Despite the availability of metronidazole since the late 1950s, the mechanism of action of the drug is still unknown. It is generally agreed that metronidazole is a prodrug and that anaerobic organisms reduce the nitro group in metronidazole to a hydroxylamine, as shown in Figure 34.2, during which a reactive derivative or reactive species is produced that causes destructive effects on cell components (i.e., DNA, proteins, and membranes). Specifically, DoCampo (5) has

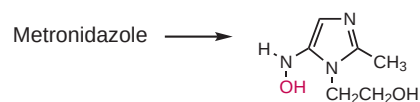


FIGURE 34.2 Metabolic activation of metronidazole.

reported that nitroaryl compounds (nitroimidazoles, metronidazole, nitrofurans, nifurtimox) are reduced to nitro radical anions, which in turn react with oxygen to regenerate the nitroaryl and the superoxide radical anion (Fig. 34.3). Further reduction of superoxide radical anion leads to hydrogen peroxide, and homolytic cleavage of the latter leads to hydroxyl radical formation. Superoxide radical anion, hydrogen peroxide, and hydroxyl radicals are referred to as reactive oxygen species (ROS) and are the reactive substances that are implicated in damage to critical cellular components of the parasite.

METABOLISM Liver metabolism of metronidazole leads to two major metabolites: hydroxylation of the 2-methyl group to 2-hydroxymethylmetronidazole (HM) and its oxidation to metronidazole acetic acid (6). Both compounds possess biologic activity. Additionally, HM is found in the urine as glucuronide and sulfate conjugates. In addition, a small amount of metronidazole is oxidized to acetamide, a known carcinogen in rats but not in humans, and to the oxalate derivative shown in Figure 34.4 (7).

PHARMACOKINETICS (6) Metronidazole is available in a variety of dosage forms, including IV, oral, rectal, and vaginal suppositories. The bioavailability of metronidazole is nearly 100% when administered orally but is significantly less when administered via the rectal route (67% to 82%) or the vaginal route (19% to 56%). The drug is not bound to plasma protein. Distribution of the drug is fairly uniform throughout the body, including mother's milk.

THERAPEUTIC APPLICATION Metronidazole is considered to be the drug of choice for treatment for the protozoal infections amebiasis (intestinal and extraintestinal), giardiasis, and trichomoniasis (8). It is the drug of choice for treatment of the gram-positive bacilli *Clostridium difficile* and, in combination, is an alternative therapy for *Helicobacter pylori* infections (9,10). The common side effects exhibited with metronidazole include abdominal distress, a metallic taste,

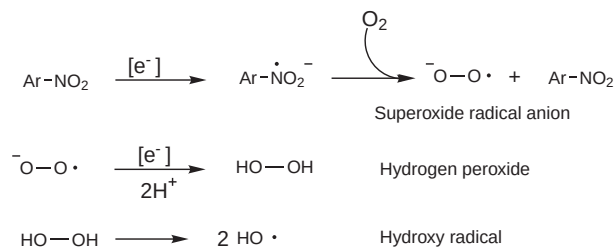


FIGURE 34.3 Formation of ROS from nitroaryl compounds.

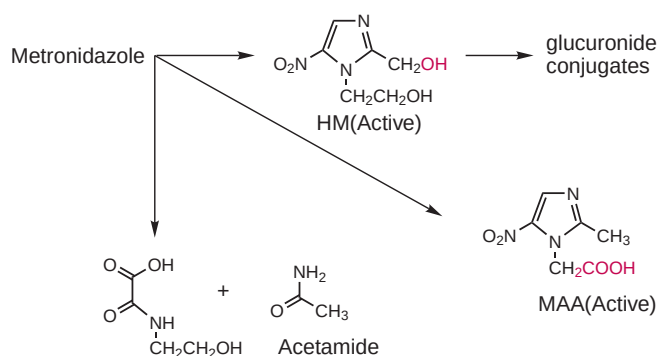


FIGURE 34.4 Metabolism of metronidazole.

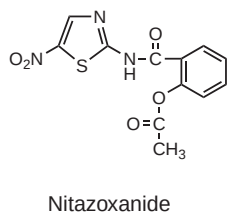
and a disulfiram-like effect if taken with alcohol. The drug is reported to be carcinogenic in mice, possibly related to the metabolite acetamide, and as a result should not be used during the first trimester of pregnancy.

Tinidazole (Tindamax)



Tinidazole has been approved by the U.S. Food and Drug Administration (FDA) for the treatment of amebiasis, giardiasis, and trichomoniasis. It appears also to be highly effective against *Helicobacter pylori* infections, although it is not approved for this use. The drug is rapidly and completely absorbed following oral administration and can be administered with food to reduce GI disturbance. Tinidazole has a mechanism of action that parallels that of metronidazole as well as a similar metabolic pathway leading to hydroxylation at the 2-methyl group catalyzed by CYP3A4. Basically, tinidazole appears to mimic the actions of metronidazole, although there are reports that it is effective against some protozoa that are resistant to metronidazole.

Nitazoxanide (Alinia)



Nitazoxanide (NTZ) has been approved as an orphan drug for the treatment of diarrhea in children (age 1 to 11 years) associated with giardiasis, but it is also approved for diarrhea caused by cryptosporidiosis in patients with AIDS. Cryptosporidiosis is a protozoal infection caused by *Cryptosporidium parvum*. The condition is uncommon in healthy individuals but can be life-threatening

in immunosuppressed patients and those with HIV infections.

MECHANISM OF ACTION (10) NTZ is a prodrug that is metabolically converted into the deacetylated drug tizoxanide (TIZ) (Fig. 34.5). TIZ then undergoes a four-electron reduction of the 5-nitro group giving various short-lived intermediates, which can include the hydroxylamine derivative. These reduced products represent the active forms of NTZ. Whereas these intermediates would suggest that NTZ has the same mechanism of action as metronidazole, this does not appear to be the case. NTZ is thought to inhibit the enzyme pyruvate:ferredoxin oxidoreductase in *Trichomonas vaginalis*, *Entamoeba histolytica*, and *Clostridium perfringens*. The result of this inhibition is disruption of the bioenergetics of these organisms. Unlike metronidazole and tinidazole, which fragment DNA and are suspected mutagenic agents, NTZ and TIZ do not cause DNA fragmentation and are not considered to be mutagenic. This might be associated with the higher redox potential found for NTZ, a nitrothiazole, in comparison with very low redox potential found for the nitroimidazoles, such as metronidazole and tinidazole. Additional metabolites of TIZ include the glucuronide, which shows some biologic activity, and small amounts of an aromatic hydroxylation product (Fig. 34.5).

PHARMACOKINETICS NTZ is available as powder that is reconstituted and dispensed as an oral suspension. The drug is well absorbed from the GI tract and rapidly metabolized, with elimination products appearing in the urine and feces. The only identified products in the plasma are TIZ and its glucuronide (11). The product can be taken with food.

THERAPEUTIC APPLICATION Although NTZ has only been approved for treatment of diarrhea in children caused by *Giardia lamblia* and diarrhea caused by *Cryptosporidium*

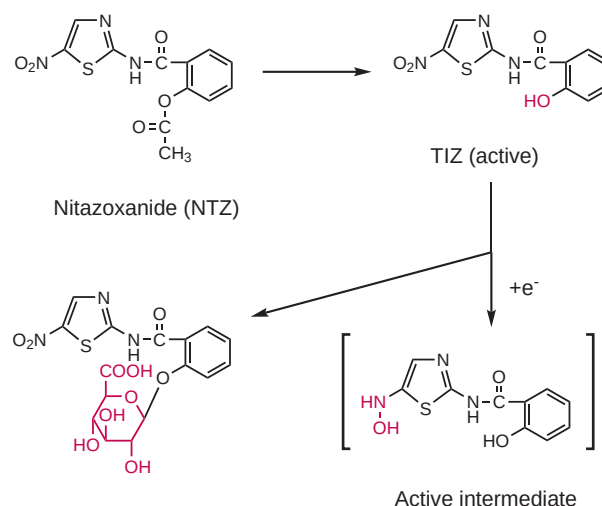
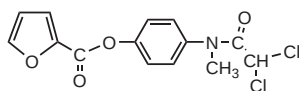


FIGURE 34.5 Metabolic activation of nitazoxanide.

parvum, the drug might soon be approved for adults suffering from diarrhea caused by *Giardia lamblia*. In addition, the drug has been shown to be effective against the protozoa *Entamoeba histolytica* and *Trichomonas vaginalis*, the bacteria *Helicobacter pylori* and *Clostridium perfringens*, and various helminths, including *Ascaris lumbricoides*, *Enterobius vermicularis*, *Ancylostoma doudenale*, and *Strongyloides stercoralis* (12).

Diloxanide Furoate



Diloxanide furoate

Diloxanide furoate (available from the Centers for Disease Control and Prevention [CDC]) is prescribed for the treatment of asymptomatic amebiasis but is ineffective as a single agent for the extraintestinal form of the disease. The drug is administered orally and is hydrolyzed in the gut to give diloxanide, which is considered to be the active drug. Diloxanide is the only form identified in the bloodstream. The drug is found in the urine as the glucuronide (Fig. 34.6).

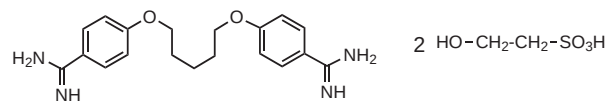
Treatment of Pneumocystis (13,14)

Sulfamethoxazole–Trimethoprim; Cotrimoxazole (Bactrim, Septra, Cotrim)

The combination of sulfamethoxazole and trimethoprim has proven to be the most successful method for treatment and prophylaxis of pneumocystis in patients with AIDS. This combination was first reported as being effective against PCP in 1975, and by 1980, it had become the preferred method of treatment, with a response rate of 65% to 94%. The combination is effective against both pneumocystic pneumonia and the extrapulmonary disease. *P. jirovecii* appears to be especially susceptible to the sequential blocking action of cotrimoxazole, which inhibits both the incorporation of *p*-aminobenzoic acid (PABA) into folic acid as well as the reduction of dihydrofolic acid to tetrahydrofolic acid by dihydrofolate reductase (DHFR). (A detailed discussion of the mechanism of action and the structure–activity relationship of these drugs can be found in Chapter 33.) Depending on the

severity of the infection, the combination is administered in doses of 20 mg/kg/d of trimethoprim and 100 mg/kg/d of sulfamethoxazole in four divided doses over a period of 14 to 21 days. The incidence of side effects of this combination is high and reflects generally the effects of the sulfa drug component. Side effects can be significant enough to terminate treatment.

Pentamidine Isethionate (Pentam 300, Nebupent)



Pentamidine isethionate

Pentamidine is available as the water-soluble isethionate salt, which is used both IV and as an aerosol. The drug can be used via the intramuscular route, but significant complications have been reported, and therefore, this route of administration is not recommended. The drug has fungicidal and antiprotozoal activity but is used primarily for treatment of PCP.

MECHANISM OF ACTION The mechanism of action of pentamidine is not known with certainty, but strong evidence supports various mechanisms of action for pentamidine. Pentamidine selectively binds to the DNA in the *Trypanosoma* parasite (see below). Pentamidine has also been shown to inhibit topoisomerase in *P. jirovecii*, which leads to double-strand cleavage of DNA in *Trypanosoma* (12–14). It has been suggested that pentamidine's mechanism of action might be different in different organisms and, therefore, that the actions reported for *Trypanosoma* might not carry over to pneumocystis.

PHARMACOKINETICS Pentamidine must be administered IV and, after multiple injections daily or on alternate days, accumulates in body tissue. Plasma concentrations were measured up to 8 months following a single, 2-hour IV infusion. The accumulation aids in treatment as well as in prophylaxis. The drug shows poor penetration of the CNS.

THERAPEUTIC APPLICATION Pentamidine is used as a second-line agent either by itself or in combination for the treatment and prophylaxis of PCP. For prophylaxis, the aerosol form of the drug is indicated and has minimum

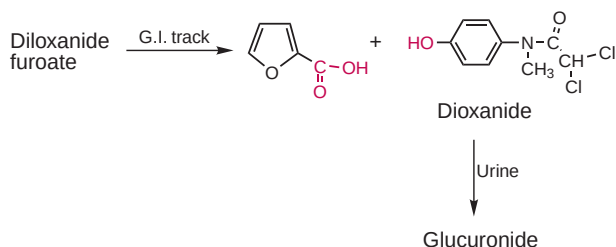


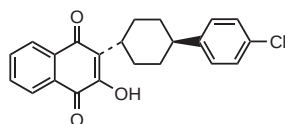
FIGURE 34.6 Metabolism of diloxanide furoate.

ORPHAN DRUG PRODUCT

Dapsone plus trimethoprim also has been used for the treatment of pneumocystis, with effectiveness nearly equal to that of cotrimoxazole.

toxicity. The limitation of pentamidine—that is, the need for IV administration—can be associated with the potential for severe toxicity, which includes breathlessness, tachycardia, dizziness, headache, and vomiting. These symptoms can occur in as many as 50% of the patients. These effects are thought to be associated with a too rapid IV administration, resulting in the release of histamine.

Atovaquone (Mepron)



Atovaquone

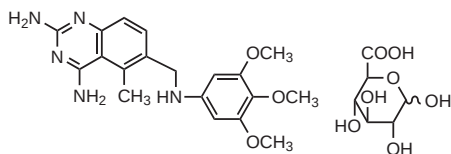
Atovaquone, a chemical with structural similarity to the ubiquinone metabolites, was initially synthesized and investigated as an antimalarial, a use for which it has recently gained acceptance when used in combination therapy with other antimalarial agents. Today, its usefulness is primarily directed toward the treatment of PCP.

MECHANISM OF ACTION Atovaquone is thought to produce its antiparasitic action by virtue of its ability to inhibit the mitochondrial respiratory chain. More specifically, atovaquone is a ubiquinone reductase inhibitor, inhibiting at the cytochrome bc_1 complex (15). This action leads to a collapse of the mitochondrial membrane potential. The compound shows stereospecific inhibition, with the *trans* isomer being more active than the *cis* isomer.

PHARMACOKINETICS Atovaquone is poorly absorbed from the GI tract because of its poor water solubility and high fat solubility, but the absorption can be significantly increased if taken with a fat-rich meal. The drug is highly bound to plasma protein (94%) and does not enter the CNS in significant quantities. It is not significantly metabolized in humans and is exclusively eliminated in feces via the bile.

THERAPEUTIC APPLICATIONS With as many as 70% of patients with AIDS developing pneumocystis and, of these, nearly 60% of the patients on cotrimoxazole developing serious side effects to this combination, atovaquone is an important alternative drug (16). Atovaquone also has been reported to be effective for the treatment of toxoplasmosis caused by *Toxoplasma gondii*, although it has not been approved for this use.

Trimetrexate Glucuronate (Neutrexin)



Trimetrexate glucuronate

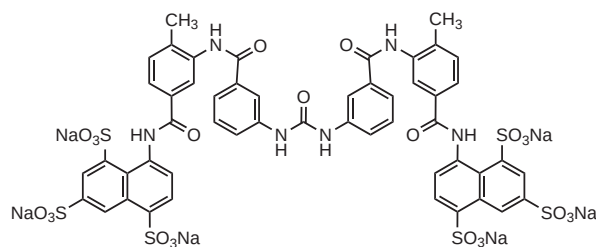
Trimetrexate (TMQ) has been approved for the treatment of *P. jirovecii* in patients with AIDS and also exhibits anti-protozoal activity against *T. cruzi*. The drug is available as a single-ingredient medication, but it can be administered along with folinic acid in much the same way that methotrexate is administered with calcium leucovorin in cancer chemotherapy. TMQ is a derivative of methotrexate.

MECHANISM OF ACTION TMQ is considered to be a non-classical folate antagonist, whereas methotrexate, the structurally similar analog of TMQ, is a classical folate antagonist. The difference between these two drugs is that methotrexate, with its polar glutamate side chain, is transported into the cell via a carrier-mediated transport system, whereas TMQ, without the glutamate moiety, is absorbed by the cell via a passive diffusion. Once in the cell, TMQ inhibits DHFR. TMQ binds to *P. jirovecii* DHFR 1,500 times more strongly than trimethoprim and somewhat more strongly than methotrexate. It has also been reported that TMQ readily enters the *P. jirovecii* cell because of the lipophilic nature of this drug (17). Methotrexate and leucovorin are not able to enter the cell, however, because the cell membrane of *P. jirovecii* does not possess the transporter protein (17).

THERAPEUTIC APPLICATION TMQ, when combined with the cytoprotective agent leucovorin (folinic acid), is more effective and better tolerated than pentamidine in the treatment of PCP (18). Because the first- and second-line agents are successful in only 50% to 75% of these cases and because adverse reactions severely limit the use of some of the older agents, TMQ can offer some advantages in treatment. TMQ is administered by IV infusion over 60 to 90 minutes and should be combined with the cytoprotective drug leucovorin. The leucovorin protects against bone marrow suppression and against renal and hepatic dysfunction. Leucovorin administration should continue for 72 hours after the last dose of TMQ. Additionally, TMQ has been reported to be effective in the treatment of Chagas disease.

Treatment of Trypanosomiasis (19)

Suramin Sodium (Available from the CDC)



Suramin Sodium

Introduced into therapy for the treatment of early trypanosomiasis in the 1920s, suramin, a bis-hexasulfonated naphthylurea, is still considered to be the drug

of choice for treatment of non-CNS-associated African trypanosomiasis.

MECHANISM OF ACTION The mechanism of action of suramin is unproven, but the drug is known to have a high affinity for binding to a number of critical enzymes in the pathogen. Among the enzymes to which suramin has been shown to bind are several dehydrogenases and kinases. As a result of binding, suramin has been shown to be an inhibitor of DHFR, a crucial enzyme in folate metabolism, and thymidine kinase. In addition, suramin is an inhibitor of glycolytic enzymes in *Trypanosoma brucei*, with binding constants much lower than those seen in mammalian cells. Inhibition of glycolysis would be expected to block energy sources of the pathogen, leading to lysis. Whether one or more of these inhibitory actions represent the toxic action of suramin on the pathogen remains unproven.

PHARMACOKINETICS Suramin sodium is a water-soluble compound that is poorly absorbed via oral administration and must be administered IV in multiple injections. Because of its highly ionic nature, suramin will not cross the blood-brain barrier and, therefore, is ineffective for the treatment of trypanosomal infections that reach the CNS. In addition, suramin is tightly bound to serum albumin. Despite this binding, the drug is preferentially absorbed by trypanosomes through a receptor-mediated endocytosis of serum protein. Because the drug remains in the bloodstream for an extended period of time, suramin has value as a prophylactic drug.

THERAPEUTIC APPLICATION Suramin sodium is effective against east African trypanosomiasis, but it has limited value against west African trypanosomiasis. As indicated, because the drug will not enter the CNS, the drug is only useful for the treatment of early stages of the disease. The drug exhibits a wide variety of side effects, which can be severe in debilitated individuals, and include nausea, vomiting, and fatigue.

Pentamidine, Isethionate (Pentam 300, Nebupent)

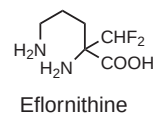
First introduced as a therapy for trypanosomiasis in 1937, pentamidine is now used in a variety of protozoal and fungal infections and finds use in the treatment of trypanosomiasis, leishmaniasis, and pneumocystis (PCP). The drug is primarily used for treatment of PCP. When used for trypanosomiasis, pentamidine is only effective against *Trypanosoma brucei rhodesiense* (east African sleeping sickness) and, even then, only during the early stage of the disease, because the drug does not readily cross the blood-brain barrier.

MECHANISM OF ACTION As indicated earlier, several biochemical actions have been reported for pentamidine. The drug has been shown to bind to DNA through hydrogen-bonding of the amidine proton with the adenine-thymine-rich regions of DNA. More specifically,

pentamidine binds to the N-3 of adenine, spans four to five base pairs, and binds to a second adenine to form interstrand cross-bonding (20). In addition to and, possibly, separate from this action, pentamidine appears to be a potent inhibitor of type II topoisomerase of mitochondria DNA (kinetoplast DNA) of the *Trypanosoma* parasite (21). The mitochondrial DNA is a cyclic DNA. This inhibition leads to double-strand breaks in the DNA and linearization of the DNA. The relationship between binding to specific regions of the DNA and inhibition of topoisomerase is unclear.

In the case of *T. brucei*, resistant strains are common. It is thought that resistance develops through an inability of the drug to reach the mitochondrial DNA (22). Transport into the mitochondria is a carrier-mediated process, with the absence of carrier in the resistant strains.

Eflornithine (Ornidyl)



Metcalf et al. (23) reported the synthesis of eflornithine (difluoromethyl ornithine [DFMO]) in 1978. Their interest arose from the desire to prepare ornithine decarboxylase (ODC) inhibitors as tools for studying the role of polyamines as regulators of growth processes. ODC catalyzes the conversion of ornithine to putrescine (1,4-diaminobutane), which in turn leads to the formation of the polyamines, spermine, and spermidine. It was not until 1980 that Bacchi et al. (24) demonstrated the potential of DFMO in the treatment of trypanosomiasis.

MECHANISM OF ACTION DFMO is a suicide inhibitor of ODC, a pyridoxal phosphate-dependent enzyme, as shown in Figure 34.7. Evidence suggests that cysteine-360 in ODC is the site of eflornithine alkylation (25). Alkylation of ODC blocks the synthesis of putrescine, the rate-determining step in the synthesis of polyamines. Mammalian ODC can also be inhibited, but because the turnover of ODC is so rapid in mammals, eflornithine does not produce serious side effects.

PHARMACOKINETICS Eflornithine can be administered either IV or orally. Administration IV requires large doses and frequent dosing, whereas poor oral absorption and rapid excretion because of the zwitterionic nature of the drug (an amino acid) has limited the oral route of administration. The drug does not bind to plasma protein and enters the CNS readily, most likely via an amino acid transport system. As a result, the drug can be used for both early and late stages of trypanosomiasis.

THERAPEUTIC APPLICATION Eflornithine is indicated for the treatment of west African trypanosomiasis caused by *T. brucei gambiense* but has proven to be ineffective

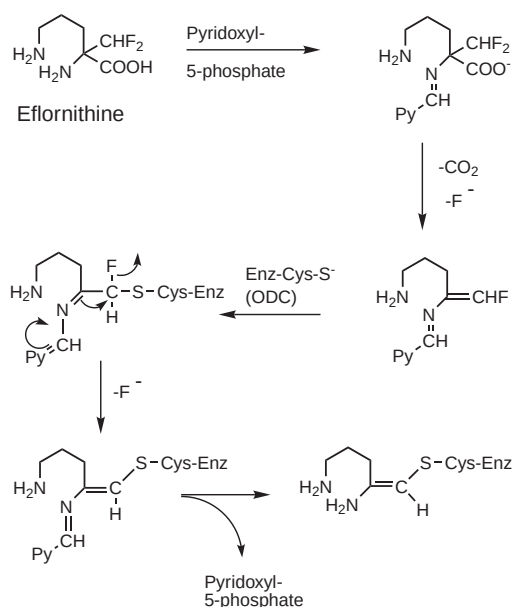
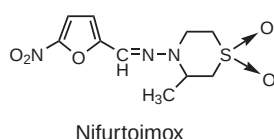


FIGURE 34.7 Inhibition of ornithine decarboxylase (Enz-Cys-SH) by eflornithine.

against east African trypanosomiasis. The cause of this ineffectiveness remains a mystery, although evidence suggests that in the resistant organism, endogenous ornithine plus increased activity of *S*-adenosylmethionine decarboxylase allows sufficient synthesis of spermidine and spermine to support cell division, thus bypassing the need for organism-synthesized ornithine (26). Side effects reported for eflornithine consist of anemia, diarrhea, and leukopenia.

Nifurtimox (Lampit)



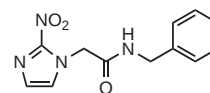
Nifurtimox

Another of the nitroaryl compounds, nifurtimox has proven to be useful as a drug for the treatment of trypanosomiasis. It is one of two drugs approved for use in treatment of Chagas disease.

MECHANISM OF ACTION As discussed for metronidazole, nifurtimox is thought to undergo reduction followed by oxidation and, in the process, generate ROS, such as the superoxide radical anion, hydrogen peroxide, and hydroxyl radical (Fig. 34.3) (5). These species are potent oxidants, producing oxidative stress that can produce damage to DNA and lipids that can affect cellular membranes. In addition, Henderson et al. (27) have reported that nifurtimox inhibits trypanothione reductase, which results in the inhibition of trypanothione formation (93% inhibition). Trypanothione is a critical protective enzyme found uniquely in trypanosomal parasites.

THERAPEUTIC APPLICATION Nifurtimox is the drug of choice for the treatment of acute Chagas disease. The drug is not effective for the chronic stages of the disease. In the acute stage, the drug has an 80% cure rate. Side effects of the drug include hypersensitivity reactions, GI complications (nausea and vomiting), myalgia, and weakness.

Benznidazole (Rochagan)



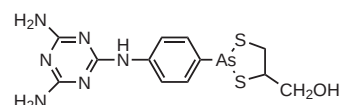
Benznidazole

Benznidazole is the second of the drugs approved for treatment of Chagas disease. Like nifurtimox, it is effective against the circulating form of *Trypanosoma cruzi* during the acute phase of the disease, but also like nifurtimox, it is ineffective during the chronic stage of the disease.

MECHANISM OF ACTION Studies suggest that benznidazole does not catalyze the formation of ROS and, therefore, has a mechanism of action different from that of nifurtimox. It has been proposed that benznidazole undergoes a one-electron transfer to the nitro group, which in turn dismutates to give back the nitroimidazole and a nitrosoimidazole (28). The latter product can then undergo an electrophilic addition to trypanothione, which leads to depletion of trypanothione, an essential enzyme system in *T. cruzi* (Fig. 34.8).

Benznidazole is not available in the United States but is available in South American countries. It is administered orally in a tablet form.

Melarsoprol (Available from the CDC)



Melarsoprol

Knowingly or unknowingly, arsenic-containing drugs have been used for treatment of parasitic conditions for thousands of years. In the late 1800s and early 1900s, Paul Ehrlich introduced the use of trivalent arsenicals. Melarsoprol, an organoarsenical, came into use in the late 1940s, and it remains the first-choice drug in the treatment of trypanosomiasis. Until 1990, it also was the only treatment for late-stage sleeping sickness.

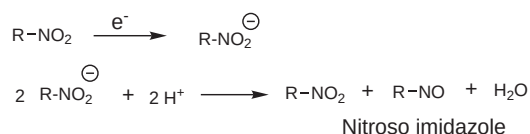


FIGURE 34.8 Proposed mechanism of action of benznidazole.

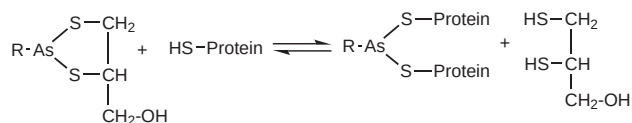


FIGURE 34.9 Mechanism of action of trivalent arsenic compounds with trypanosome organism.

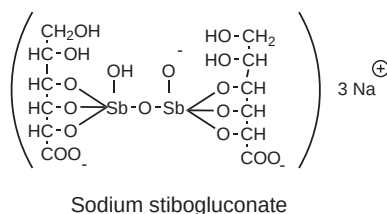
MECHANISM OF ACTION It is known that trivalent arsenic reacts rapidly and reversibly with sulfhydryl-containing proteins, as shown in Figure 34.9. It is generally accepted that the enzyme with which melarsoprol reacts is an enzyme involved in glycolysis, and as a result, inhibition of pyruvate kinase occurs. It is argued, however, that the inhibition might not occur at pyruvate kinase but, rather, at a step before the pyruvate kinase. Blockage of glycolysis would be expected to lead to loss of motility and cell lysis. More recently, Fairlamb et al. (29) have proposed a mechanism of action that results in the inhibition of trypanothione reductase through the formation of a stable complex between melarsoprol and trypanothione. Melarsoprol reacts with the cysteine sulfhydryl of trypanothione to form the stable adduct shown in Figure 34.10. Supportive of this mechanism is the synergistic action of melarsoprol with eflornithine, two drugs that produce sequential blockage of the synthesis of trypanothione.

PHARMACOKINETICS Melarsoprol is administered IV in multiple doses and multiple sessions. Its major metabolite in humans is the lipophilic melarsen oxide, which can penetrate into the CNS. This metabolite apparently is responsible for the protein-binding characteristic for melarsoprol.

THERAPEUTIC APPLICATION Melarsoprol is the drug of choice for the treatment of late-stage meningoencephalitic trypanosomiasis caused by the west and east African strains of the disease. Because the drug has the potential for serious nervous system toxicities (e.g., convulsions, acute cerebral edema, and coma), the drug is usually administered in a hospital setting with supervision. An additional problem with melarsoprol is the development of resistance by the parasite.

Treatment of Leishmaniasis

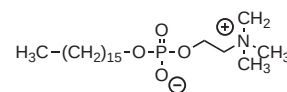
Sodium Stibogluconate (Pentostam; Available from the CDC)



Leishmaniasis was first described in the medical literature by Deishman and Donovan in 1903, and shortly

after that, antimony-based drugs were introduced as therapeutic agents to treat this disease (30). Although the structure of sodium stibogluconate is commonly drawn as shown, the actual compound probably is much more complex. The drug is a water-soluble preparation that is administered intramuscularly or IV. Pentavalent antimony compounds are thought to inhibit bioenergetic processes in the pathogen, with catabolism of glucose and inhibition of glycolytic enzymes being the primary sites of action (glucose catabolism is 86% to 94% inhibited). This in turn results in inhibition of adenosine triphosphate/guanosine triphosphate formation. Sodium stibogluconate is the drug of choice for the treatment of most forms of leishmaniasis (or meglumine antimonate, another pentavalent antimony agent). The recommended dose is 20 mg antimony/kg/d, not to exceed 850 mg antimony/d. A number of other drugs have been reported to be effective in the treatment of leishmaniasis, and these include pentamidine, amphotericin B, paromomycin, alkylphosphocholine analogs, rifampicin, and ketoconazole (31,32).

Miltefosine (Impavido)



Miltefosine

Miltefosine is an orally active drug that is effective against visceral and cutaneous leishmaniasis. Although the drug has not been approved by the FDA, a compassionate use protocol is pending. This drug is especially important since the only other drug available is sodium stibogluconate, which is administered intramuscularly or IV, thus limiting its utility, and has a high potential for toxicity associated with the antimony component of the drug. Miltefosine has been reported to exhibit activity in vitro against a broad range of leishmanial strains. Although the mechanism of action of the drug remains unknown, it has been suggested that the drug acts directly on the promastigote and amastigote stages of the parasite and not through stimulation of the immune system (33). This distinction is important because patients who are immunocompromised may also suffer from leishmaniasis.

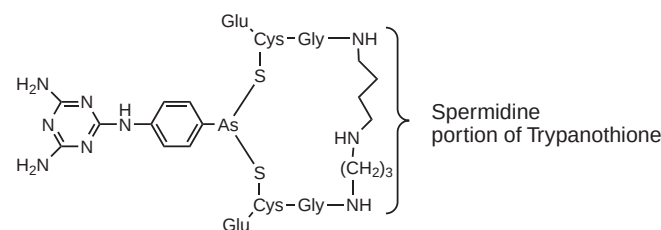
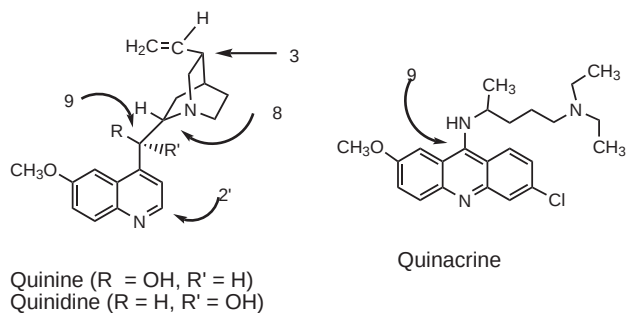


FIGURE 34.10 Structure of melarsoprol trypanothione complex.

The drug is well absorbed via the oral route and well distributed, with a half-life of 6 to 8 days.

Treatment of Malaria

Historical Background



Quinine was the first known antimalarial. It is a 4-quinolinemethanol derivative bearing a substituted quinuclidine ring. The use of quinine in Europe began in the 17th century, after the Incas of Peru informed the Spanish Jesuits about the antimalarial properties of the bark of an evergreen mountain tree they called quinquina (later called cinchona, after Dona Franciscoa Henriquez de Ribera [1576–1639], Countess of Chinchon and wife of the Peruvian Viceroy). The bark, when made into an aqueous solution, was capable of curing most forms of malaria. It was listed in the London Pharmacopeia of 1677. The alkaloid derived from the bark, quinine, was isolated in the mid-1820s. Quinine, a very bitter substance, has been used by millions of malaria sufferers. Recently, it has been employed successfully to treat chloroquine-resistant strains of *Plasmodium falciparum* and is considered to be the drug of choice for these resistant strains.

Quinidine, the C-9 stereoisomer of quinine, is also an effective antimalarial, but suffers from the fact that it is considerably more toxic than quinine. Its use is usually limited to treatment of severe malarial infections. Quinidine is also used as an antiarrhythmic agent.

A second class of chemicals that played a role in the development of synthetic antimalarials were the 9-aminoacridines. 9-Aminoacridine was known to exhibit antibacterial activity, whereas a derivative of 9-aminoacridine synthesized in 1934, quinacrine, was found to possess weak antimalarial activity.

With the beginning of World War II and concern about an interruption in the supply of cinchona bark from the East Indies, a massive effort was begun to search for synthetic alternatives to quinine and to develop more effective antimalarial agents than quinacrine. With a basic understanding of the structure–activity relationship of quinine (see *Quinine*) and the chemical similarities seen with quinacrine, it is easy to visualize the relationship between these agents and the synthetic antimalarials. The 4-aminoquinolines, chloroquine and hydroxychloroquine, are structurally similar to the right half of quinacrine (Fig. 34.11). The 8-aminoquinolines,

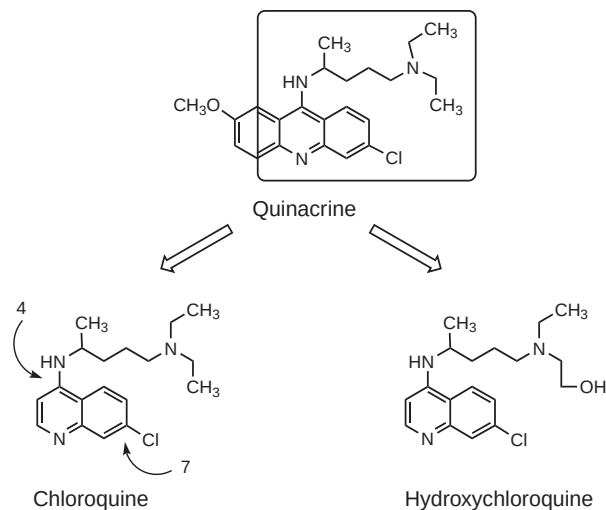


FIGURE 34.11 Structural similarity between quinacrine and the 4-aminoquinolines.

pamaquine and primaquine, retain the methoxyquinoline nucleus of quinine and quinacrine (Fig. 34.12). The quinoline-4-methanols, mefloquine and halofantrine, show similarity to the 4-quinolinemethanol portion of quinine (Fig. 34.12).

4-Substituted Quinolines

Six compounds can be considered within this class of drugs: quinine, chloroquine, and hydroxychloroquine, mefloquine, halofantrine, and lumefantrine (Figs. 34.11 and 34.12). These compounds not only share a structural similarity but are also thought to have similar mechanisms of action, are effective on the same stage of the parasite, and can share similar mechanisms of resistance.

MECHANISM OF ACTION The mechanism of action of chloroquine has been studied in depth, and the results of these studies have been assumed to be applicable to the other 4-substituted quinolines (34). Various mechanisms of actions have been offered to explain the action of this class of drugs, including the DNA intercalation mechanism, the weak base hypothesis, and the ferriprotoporphyrin hypothesis. The present understanding about the mechanism of action would appear to use various aspects of each of these previous mechanisms. It is known that hemoglobin is transported into the food vacuoles of the plasmodium (Figs. 34.13 and 34.14), where digestion of the hemoglobin supplies the organism with a source of amino acids. One of the products of this digestion is a free heme called hematin. Hematin is toxic to the plasmodium cell, but it has been demonstrated that within the plasmodium vacuole the organism is capable of converting the hematin to nontoxic hemozoin. While it was initially thought that hemozoin was a polymeric form of hematin, it is now thought that the detoxification occurs through biocrystallization (35,36). The dimeric hemozoin as a biocrystal is insoluble and chemically inert and

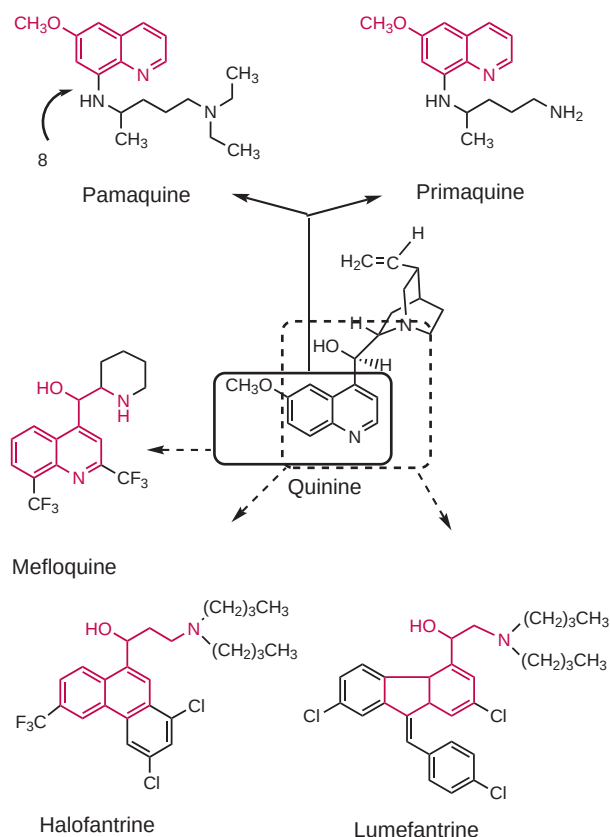


FIGURE 34.12 Structural similarity between quinine and the 8-aminoquinolines (--->) and between quinine and the quinoline-4-methanols (--->).

therefore has no biologic effect on the plasmodium. The quinolines bind to hematin through a drug-heme complex in which the aromatic quinoline ring π -bonds to the porphyrin nucleus (37). This drug-heme complex blocks further biocrystallization. The result of this complexation is that newly formed, free toxic hematin is now present, which leads to the death of the plasmodium. Additionally since heme is not a component of plasmodium, it appears unlikely that the organism will be capable of developing a resistance mechanism to this type of activity. The accumulation of the 4-substituted quinolines in the acidic food vacuoles (pH 4.8 to 5.2) is based on the fact that these drugs are weak bases, as indicated by their pK_a values. The extracellular fluid of the parasite is at pH 7.4, and as a result, the weak base will move toward the more acidic pH of the vacuoles, reaching concentrations hundreds of times those in the plasma. Additionally, the binding of the quinoline to the heme draws additional quantities into the vacuole.

MECHANISM OF RESISTANCE A limiting factor for most of the antimalarial drugs is the development of resistant strains of plasmodium. It should be noted that resistance differs from region to region, and in some cases, a resistant strain can develop to a particular drug without that

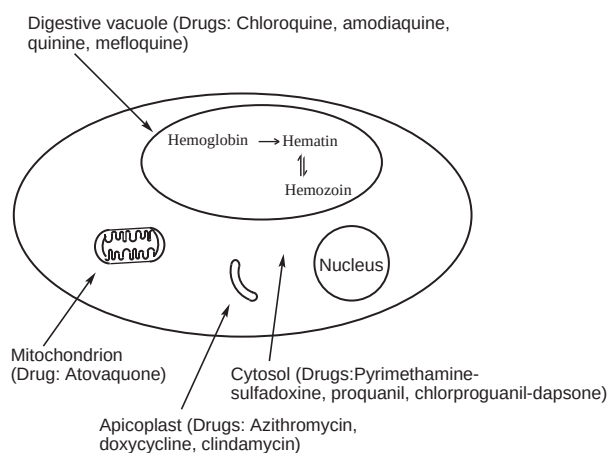


FIGURE 34.13 *Plasmodium* parasite cell as present within the erythrocyte and sites of drug action.

drug ever having been introduced to the region (possible cross-resistance). The development of resistance is thought to be a spontaneous gene mutation. Several mechanisms of resistance appear to be operating. One of these mechanisms is based on the *Plasmodium falciparum* chloroquine-resistance transporter (*pfcr*) mechanism, which is sufficient and necessary to impart resistance (38). A gene encodes for a transmembrane transporter protein found in the membrane of the food vacuole. Multiple mutations within a specific region in this gene result in reduced accumulation of chloroquine, resulting from the increased efflux of the drug. Additional transporter proteins may also be involved in resistance.

Rapid metabolism of the antimalarials by resistant strains of plasmodium also might be considered to play a significant role in the development of resistance. It has been shown that cytochrome P450 activity parallels increased resistance to specific drugs.

THERAPEUTIC APPLICATION The 4-substituted quinolines are referred to as rapidly acting blood schizonticides, with activity against plasmodium in the erythrocytic stage. Chloroquine is the drug of choice, but unfortunately, the incidence of chloroquine-resistant infections is extremely common today. The spread of chloroquine resistance has reached almost all malarious areas of the world. In addition, multidrug-resistant and cross-resistant strains of plasmodium are now common. The drug of choice for the treatment of malaria caused by *Plasmodium falciparum*, *Plasmodium ovale*, *Plasmodium vivax*, and *Plasmodium malariae* in regions infected by chloroquine-resistant *P. falciparum* is quinine, in combination with traditional antibiotics, mefloquine, or various other combinations as alternative treatment agents (Table 34.2). Of interest is the observation that after years of nonuse of chloroquine, a reemergence of chloroquine-sensitive parasites has been found.

The 4-substituted quinolines, depending on the specific drug in question, can also be used for prophylaxis

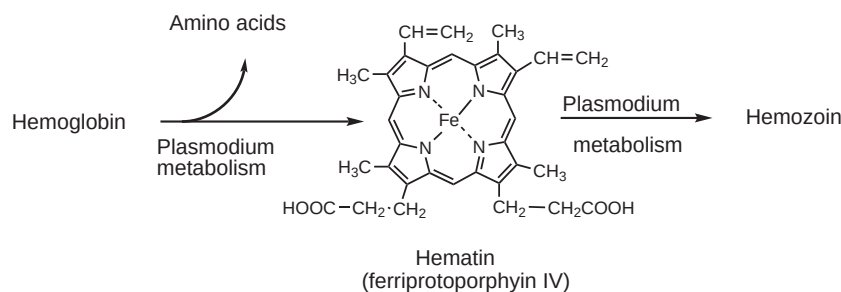


FIGURE 34.14 Proteolytic degradation of hemoglobin by the *Plasmodium* organism to the potentially toxic hematin and then to the nontoxic dimer hemozoin.

of malaria. Two types of prophylaxis are possible: causal prophylaxis and suppressive prophylaxis. The former prevents the establishment of hepatic forms of the parasite, whereas the latter eradicates the erythrocytic parasites but has no effect on the hepatic forms. Several of the 4-substituted quinolines are effective suppressive prophylactics.

Specific 4-substituted Quinolines

Quinine. Quinine is the most prevalent alkaloid present in the bark extracts (~5%) of cinchona. Four stereochemical centers exist in the molecule (at C-3, C-4, C-8, and C-9) (Fig. 34.12). Quinine (absolute configuration of 3R:4S:8S:9R), quinidine (absolute configuration of 3R:4S:8R:9S), and their optical isomers all have antimalarial activity, whereas their C-9 epimers (i.e., the epi-series having either 3R:4S:8R:9R or 3R:4S:8S:9S configurations) are inactive. Modification of the secondary alcohol at C-9, through oxidation, esterification, or

similar processes, diminishes activity. The quinuclidine portion is not necessary for activity; however, an alkyl tertiary amine at C-9 is important.

Quinine is metabolized in the liver to the 2'-hydroxy derivative, followed by additional hydroxylation on the quinuclidine ring, with the 2,2'-dihydroxy derivative as the major metabolite. This metabolite has low antimalarial activity and is rapidly excreted. The metabolizing enzyme of quinine is CYP3A4. With the increased use of quinine and its use in combination with other drugs, the potential for drug interactions based on the many known substrates for CYP3A4 (see Chapter 4) is of concern (39).

A quinine overdose causes tinnitus and visual disturbances; these side effects disappear on discontinuation of the drug. Quinine can also cause premature contractions during the late stages of pregnancy. Although quinine is suitable for parenteral administration, this route is considered to be hazardous because of its ability to cause hemolysis. Quinidine, the (+)-isomer of quinine, has been

TABLE 34.2 Guidelines for Treatment of Malaria in the United States^a

Clinical Diagnosis	Sensitivity	Drug Recommendation
Uncomplicated malaria	Chloroquine sensitive	Chloroquine phosphate
<i>P. falciparum</i>	Chloroquine resistant or unknown	A. Atovaquone–proguanil (Malarone) B. Artemether–lumefantrine (Coartem) C. Quinine sulfate + one of the following: Doxycycline Tetracycline Clindamycin D. Mefloquine
Uncomplicated malaria <i>P. malariae</i> , <i>P. knowlesi</i>	Chloroquine sensitive	A. Chloroquine B. Hydroxychloroquine
Uncomplicated malaria <i>P. vivax</i> or <i>P. ovale</i>	Chloroquine sensitive	Chloroquine phosphate + primaquine phosphate
Uncomplicated malaria <i>P. vivax</i>	Chloroquine resistant	A. Quinine sulfate + doxycycline or tetracycline B. Atovaquone–proguanil C. Mefloquine + primaquine phosphate
Severe malaria	Chloroquine sensitive/resistant	Quinidine gluconate (parenteral) + one of the following: Doxycycline Tetracycline Clindamycin

^aInformation taken from the CDC Guideline for Treatment of Malaria in the United States. For more details, including infectious region and dosing, go to http://www.cdc.gov/malaria/diagnosis_treatment/index.html.

shown to be more effective in combating the disease, but it has undesirable cardiac side effects. When used for severe malaria, the quinidine gluconate is administered IV.

Chloroquine (Aralen). Chloroquine is the most effective of the hundreds of 4-aminoquinolines synthesized and tested during World War II as potential antimalarials. Structure–activity relationships demonstrated that the chloro at the 8-position increased activity, whereas alkylation at C-3 and C-8 diminished activity. The replacement of one of its *N*-ethyl groups with a hydroxyethyl produced hydroxychloroquine, a compound with reduced toxicity that is rarely used today except in cases of rheumatoid arthritis.

Chloroquine is commonly administered as the racemic mixture, because little is gained by using the individual isomers. The drug is well absorbed from the GI tract and distributed to many tissues, where it is tightly bound and slowly eliminated. The drug is metabolized by *N*-dealkylation by CYP2D6 and CYP3A4 isoforms. It has been reported that the level of metabolism correlates closely with the degree of resistance. The suggestion has been made to coadminister chloroquine with CYP2D6 and CYP3A4 inhibitors to potentiate activity and reduce resistance. Although this can be possible, it is not commonly practiced.

Chloroquine is an excellent suppressive agent for treating acute attacks of malaria caused by *Plasmodium vivax* and *Plasmodium ovale*. The drug is also effective for cure and as a suppressive prophylactic for the treatment of *Plasmodium malariae* and susceptible *Plasmodium falciparum*.

Chloroquine is generally a safe drug, with toxicity occurring at high doses of medication if the drug is administered too rapidly via parenteral routes. With oral administration, the side effects primarily are GI effects, mild headache, visual disturbances, and urticaria.

Mefloquine (Lariam) (40). Mefloquine, which was synthesized with the intent of blocking the site of metabolism in quinine with the chemically stable CF_3 group, exists as four optical isomers of nearly equal activity. The drug is active against chloroquine-resistant strains of plasmodium, yet cross-resistance is not uncommon. Metabolism is cited as the possible mechanism of resistance. Mefloquine is slowly metabolized through CYP3A4 oxidation to its major inactive metabolite, carboxymefloquine (Fig. 34.15). Most of the parent drug is excreted unchanged into the urine. Its coadministration with CYP3A4 inhibitors (e.g., ketoconazole) has increased the area under the curve for mefloquine by inhibiting its metabolism to carboxymefloquine.

Mefloquine is only available in an oral dosage form, which is well absorbed. The presence of food in the GI tract affects the pharmacokinetic properties of the drug, usually enhancing absorption. The lipophilic nature of the drug accounts for the extensive tissue binding and low clearance of total drug, although the drug does not accumulate after prolonged administration. The drug has a high affinity for erythrocyte membranes.

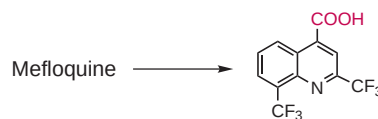


FIGURE 34.15 *Plasmodium falciparum* metabolism of mefloquine.

Mefloquine is an effective suppressive prophylactic agent against *Plasmodium falciparum* both in nonimmune populations (travelers coming into regions of malaria) and in resident populations. The drug has also high efficacy against falciparum malaria, with a low incidence of recrudescence. The drug is ineffective against sexual forms of the organism.

The incidence of side effects with mefloquine is considered to be high. The effects are classified as neuropsychiatric, GI, dermatologic, and cardiovascular. The neuropsychiatric effects can be serious (e.g., suicidal tendencies or seizures) or minor (e.g., dizziness, vertigo, ataxia, and headaches). GI side effects included nausea, vomiting, and diarrhea, whereas the dermatologic effects include rash, pruritus, and urticaria. Finally, cardiovascular side effects can include bradycardia, arrhythmias, and extrasystoles.

Halofantrine (Hafan). Halofantrine (40–42), a member of the 9-phenanthrenemethanol class (Fig. 34.12), originally came out of a synthesis program dating to World War II, but this particular agent was not fully developed until the 1960s. Halofantrine has one chiral center and has been separated into its enantiomers. There appears to be little difference between the enantiomers; thus, the drug is used as a racemic mixture.

Halofantrine is considered to be an alternative drug for treatment of both chloroquine-sensitive and chloroquine-resistant *Plasmodium falciparum* malaria, but its efficacy in mefloquine-resistant malaria can be questionable. It is generally believed that halofantrine acts through a mechanism similar to that of the 4-aminoquinolines. The drug is metabolized via *N*-dealkylation to desbutylhalofantrine by CYP3A4 (Fig. 34.16). The metabolite appears to be several-fold more active than the administered drug.

At present, halofantrine is only available in a tablet form, which has significant implications as it relates to its insolubility and drug absorption (bioavailability). Animal studies have shown that following oral administration, the drug is eliminated in feces, suggesting poor oral absorption. Its oral suspensions leads to as much as

ADDITIONAL THERAPEUTIC INDICATIONS FOR CHLOROQUINE

Chloroquine is also prescribed for treatment of rheumatoid arthritis, discoid lupus erythematosus, and photosensitivity diseases.

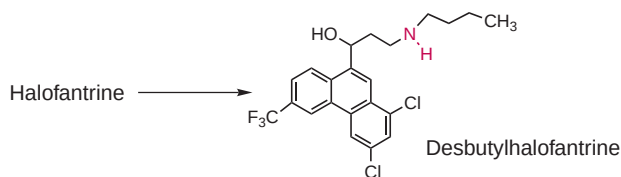


FIGURE 34.16 Metabolism of halofantrine.

30% lower plasma levels of the drug in comparison with the tablet. A micronized form of the drug has shown improved bioavailability. Its administration with or without food in the stomach also leads to considerable variation in plasma levels. A high lipid content in a meal taken 2 hours before dosing leads to substantial increases in the rate and extent of absorption. Several cases of drug treatment failure appear to be related to poor absorption. Incomplete absorption and, as a result, low plasma levels can play a role in the development of organism resistance. The elimination half-lives of halofantrine and desbutylhalofantrine tend to be prolonged, which can be another factor in the development of resistance. Low levels of the drug can increase the likelihood of augmenting the emergence of halofantrine resistance.

Absorption problems with halofantrine cannot be solved by increasing the dosage of halofantrine because of significant toxicity problems. Toxicity, although minimal with short-term low doses, can be severe with high doses of halofantrine. GI side effects include nausea, vomiting, diarrhea, and abdominal pain. Cardiovascular toxicity includes orthostatic hypotension and dose-dependent lengthening of QTc intervals.

Lumefantrine. Lumefantrine, an effective erythrocytic schizonticide, is a derivative of halofantrine that has been reported to exhibit antimalarial activity when combined with artemether in the treatment of multidrug-resistant *Plasmodium falciparum* (Fig. 34.12). The drug is only used in combination with artemether (artemisinin-based combination therapies) in which a synergistic effect has been noted (see below). Lumefantrine has a relatively long half-life of 3 to 6 days, which makes it ideal for combining with the short half-life artemisinins. The drug is quite lipophilic, and a diet rich in fat increases the bioavailability of the drug. Unfortunately, many patients suffering from malaria do not tolerate food, and therefore, the bioavailability of lumefantrine is not very high. No evidence of cardiotoxicity has been reported with this combination, which can offer promise for successful treatment of resistant organisms.

8-AMINOQUINOLINES.

Pamaquine/primaquine Pamaquine, an 8-aminoquinoline, was first introduced for treatment of malaria in 1926 and has since been replaced with primaquine (Fig. 34.12). Primaquine is active against latent tissue forms of *Plasmodium vivax* and *Plasmodium ovale*, and it is active against the hepatic stages of *Plasmodium falciparum*. The drug is not active against erythrocytic stages of the

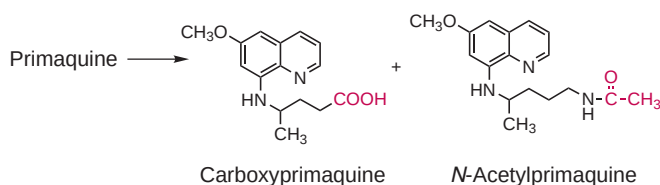


FIGURE 34.17 Metabolism of primaquine.

parasite but does possess gametocidal activity against all strains of plasmodium.

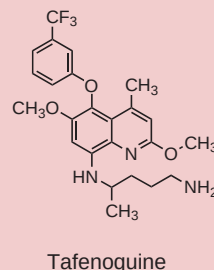
Mechanism of Action. The mechanism of action of the 8-aminoquinolines is unknown, but primaquine can generate ROS via an auto-oxidation of the 8-amino group. The formation of a radical anion at the 8-amino group has been proposed by Augusto et al. (43). As a result, cell-destructive oxidants, such as hydrogen peroxide, superoxide, and hydroxyl radical, can be formed, as shown in Figure 34.3, leading to oxidative damage to critical cellular components.

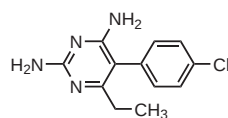
Metabolism. Primaquine is almost totally metabolized by CYP3A4 (99%), with the primary metabolite being carboxyprimaquine (Fig. 34.17) (44). Trace amounts of N-acetylprimaquine plus aromatic hydroxylation and conjugation metabolites also have been reported.

Therapeutic Application. Primaquine is classified as the drug of choice for the treatment of relapsing vivax and ovale forms of malaria and will produce a radical cure of the condition. It is recommended that the drug be combined with chloroquine to eradicate the erythrocytic stages of malaria. Primaquine is not given for long-term treatment because of potential toxicity and sensitization. The sensitivity appears most commonly in individuals who have glucose-6-phosphate dehydrogenase deficiency. In these cases, hemolytic anemia can develop.

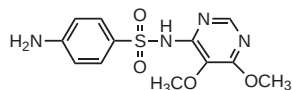
TAFENOQUIN

Tafenoquine is a new 8-aminoquinoline derived from primaquine which is reported to exhibit reduced toxicity, a prolonged half-life (2-3 weeks) and is active against the erythrocytic stages of the malaria including chloro-quine resistant strains. The drug may also have activity against liver stages of the disease and gametocytes.



Pyrimethamine (Daraprim)

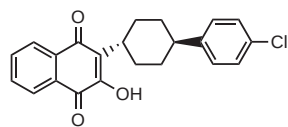
Pyrimethamine



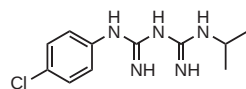
Sulfadoxine

Pyrimethamine is a potent inhibitor of DHFR (45). The drug has been shown to have a significantly higher affinity for binding to the DHFR of *Plasmodium* than to the host enzyme (>1,000 times in *Plasmodium berghei*) and, as a result, has been used to selectively treat plasmodium infections. The combination of pyrimethamine with a long-acting sulfonamide, sulfadoxine, which blocks dihydrofolate synthesis by blocking incorporation of PABA into the dihydrofolate, is called Fansidar, which produces sequential blockage of tetrahydrofolate synthesis similar to that reported for treatment of bacterial infections (see Chapter 33). It should be noted that the site of action of this combination is in the cytosol of the plasmodium (Fig. 34.13). *Plasmodium* enzymes catalyzing folic acid synthesis differ from those enzymes found in other organisms. A single bifunctional protein present in *Plasmodium* sp. catalyzes the phosphorylation of 6-hydroxymethyl-7,8-dihydropterin (a pyrophosphokinase) and the incorporation of PABA into dihydropterotic acid. A second bifunctional enzyme catalyzes the reduction of dihydropterotic acid and thymidylate synthesis. As a result, the drug combination (Fansidar) appears to have improved drug-mediated disruption of folic acid in *Plasmodium* sp. (34,46). This combination has been used with quinine for the treatment and prevention of chloroquine-resistant malaria (*Plasmodium falciparum*, *Plasmodium ovale*, *Plasmodium vivax*, and *Plasmodium malariae*). The combination therapy (Fansidar) has the added advantage of being inexpensive, which is essential for successful therapy in developing countries. When used on its own, pyrimethamine is a blood schizonticide without effects on the tissue stage of the disease.

The mechanism of resistance to the folate inhibitor combination has been shown to be associated with point mutations in both DHFR and the dihydropterotate synthase enzymes (35).

Atovaquone–proguanil

Atovaquone

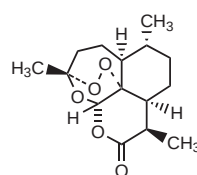


Proguanil

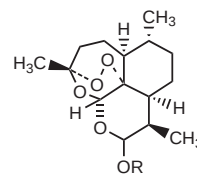
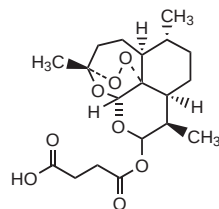
Atovaquone was originally developed as an antimalarial, but because of the high failure rate (~30%), it is not prescribed as a single chemical entity but, rather, is used to treat pneumocystis (see page XX). More recently, however, atovaquone has been combined with proguanil as an effective prophylactic and therapeutic antimalarial

(38). The two drugs together (Malarone) exhibit synergy in which proguanil reduces the effective concentration of atovaquone needed to damage the mitochondrial membrane and atovaquone increases the effectiveness of proguanil but not its active metabolite (for the mechanism of action of atovaquone, see page XX). Proguanil was developed decades earlier as a folic acid antagonist and functions as a prodrug. The active form of proguanil is cycloguanil, which acts as a DHFR inhibitor (Fig. 34.18). Later, this discovery led to the development of pyrimethamine. The sites of action of this combination involve two different locations. Atovaquone is active within the mitochondrion of the plasmodium, while proguanil as an antifolate is active in the cytosol (Fig. 34.13).

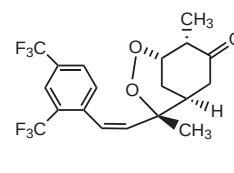
Resistance to atovaquone used as a monotherapy might have been associated with the pharmacokinetics of the drug. Atovaquone is quite lipophilic and has slow uptake, resulting in the pathogen experiencing low concentrations of the drug over an extended period of time, both of which encourage the development of resistance. A single point mutation appears to be sufficient for resistance (47). To date, resistance to the combination has not been reported.

ARTEMISININS (48–51)

Artemisinin

Artemether (R = CH₃)
Arteether/Artemotil (R = C₂H₅)
Dihydroartemisinin (R = H)

Artesunate



Arteflene

1,2,4-Trioxane (X = O) and 1,2-Dioxane (X = CH₂) rings

The most recent additions to the drug therapy for malaria are artemisinin and its derivatives. Isolated from *Artemisia annua* (qinghao, sweetworm wood), this material has been used by Chinese herbalists since 168 BC. Artemisinin and the synthetic and semisynthetic derivatives, artemether, arteether, arteflene, and artesunate, are active by virtue of the endoperoxide. The artemisinin derivatives are built on the 1,2,4-trioxanes or 1,2-dioxane ring systems.

Mechanism of Action Two mechanisms have been suggested to account for the antimalarial action of the

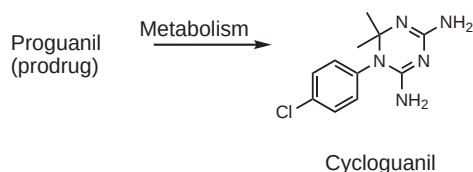


FIGURE 34.18 Activation of proguanil leading to cycloguanil.

artemisinins. In the first mechanism, the artemisinins appear to kill the parasite by a free radical mechanism—not by the generation of ROS but, rather, by virtue of a free radical associated with the endoperoxide, possibly involving a carbon radical. It is proposed that the heme in the form of hemozoin within the digestive vacuole is a source of Fe^{II} , which reacts with the peroxide to generate an oxy radical and Fe^{III} . A carbon radical is formed from the oxy radical, which is then lethal to the plasmodium within the erythrocyte. The second mechanism suggests endoperoxide activation via an iron-dependent mechanism, but the activated artemisinins target sarcoplasmic/endoplasmic reticulum Ca^{2+} -ATPase of the *Plasmodium falciparum* (PfATP6), altering calcium stores (52,53). The artemisinins actually can form covalent adducts to specific membrane-associated proteins after concentrating in infected erythrocytes.

Therapeutic Application The artemisinins are hydrophobic in nature with the exception of artesunate, which is available as a water-soluble hemisuccinate salt, and are partitioned into the membrane of the plasmodium. These compounds have gametocytocidal activity as well as activity against all asexual stages of the parasites. These agents are short acting, with relatively short half-lives. Little or no cross-resistance has been reported, with the drugs rapidly clearing the blood of parasites. The drugs have limited availability in the United States, but they are being used elsewhere as commercial or experimental agents, often in combination therapy (Table 34.3). Combination therapy has the goal of reducing resistance with the hope for synergism and, when combined with longer-acting drugs, an improved therapy.

Among the combinations reportedly used are artesunate–fosmidomycin, artemether–lumefantrine (Coartem), amodiaquine–artesunate, chloroquine–artemisinin, and artesunate–sulfadoxine–pyrimethamine (54). These combinations are referred to as artemisinin-based combination therapy (ACT). These ACTs have been reported to show cure rates of greater than 90%. The fixed-dose combination Coartem has been used in more than 10 million treatments, with significant increases being forecast.

The use of the artemisinins and artemisinin derivatives in other parasitic infections has received recent attention (53). Specifically, several derivatives of artemisinin exhibited activity as antileishmaniasis agents, in treatment of schistosomiasis, and in treatment of trypanosomiasis. Combinations of artesunate–sulfamethoxypyrazine–pyrimethamine and artemether–lumefantrine are effective against *Schistosoma mansoni*, and artemisinin was reported to show activity against *Trypanosoma brucei rhodesiense*, *Trypanosoma cruzi*, and *Leishmania donovani* in vitro.

Antibiotics

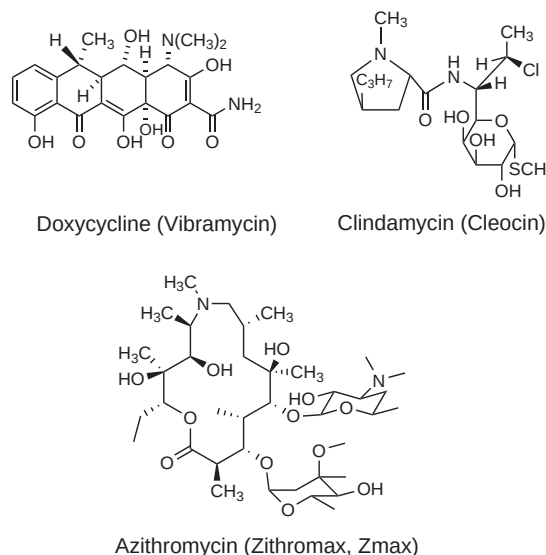


TABLE 34.3 Examples of Commercially Available Artemisinins

Artemisinin	Trade Name	Dose/Route of Administration	Treatment Condition	Patients
Arteether or artemotil	Artecef	50 mg/IM 150 mg/IM	Severe <i>P. falciparum</i>	Children and adolescents
Artemether–lumefantrine	Coartem	20 mg artemether/120 mg lumefantrine PO	Acute uncomplicated <i>P. falciparum</i>	Weight >5 kg
Artesimate	IND in United States	IV used in combination	Severe <i>P. falciparum</i>	
Artemether	Gvither, Artenam	300 mg/100 mL PO and injectable 180 mg/60 mL PO	Multidrug-resistant <i>P. falciparum</i>	
Dihydroartemisinin	Alaxin	60 mg PO tablet 160 mg/80 mL PO suspension 20/40/80 mg suppository	<i>P. falciparum</i> <i>P. falciparum</i> , <i>vivax</i> , <i>malariae</i>	Adults and children

IM, intramuscular; IND, investigational new drug; IV, intravenous; PO, oral.

Various antibiotics have proven useful in the treatment of malaria. They can be used alone as prophylaxis or in combination with quinine or artesunate as treatment measures. The most commonly used antibiotics are doxycycline, clindamycin, and azithromycin (55).

Their mechanisms of action are the same as those seen when treating a bacterial infection and involve interference in protein synthesis. What is unique about the use of the antibiotics is the site of action. The antibiotics affect protein synthesis within the apicoplast (Fig. 34.13). The apicoplast is an organelle found in most Apicomplexa (a group of eukaryotic microorganisms which include plasmodium). The apicoplast is a nonphotosynthesizing plastid with undefined function, although it does contain tRNAs, rRNAs, RNA polymerases, and ribosomal protein and is essential to the parasite's survival. Interference with protein synthesis blocks replication of the apicoplast organelle and therefore prevents a second replication of the life cycle of the plasmodium and thus the prophylactic action. Limitations on the use of doxycycline include children and pregnant women, two major groups of malarial patients; thus, clindamycin and azithromycin are used as back-up drugs. Azithromycin has proven valuable in prophylaxis against *Plasmodium vivax*.

HELMINTH INFECTIONS

Helminthiasis, or worm infestation, is one of the most prevalent diseases—and one of the most serious public health problems—in the world. Many worms are parasitic in humans and cause serious complications. Hundreds of millions (if not billions) of human infections by helminths exist worldwide, and with increased world travel and immigration from developing countries, one might expect to see this pattern of infection continue. It is estimated that one-fourth of the world population may be infected. It is interesting to note that helminths differ from many other parasites in that these organisms multiply outside of the definitive host and have the unique ability to evade host immune defenses for reasons that are not fully understood. As a result, helminth infections tend to be chronic, possibly lasting for the entire lifetime of the host (for a discussion of the uniqueness of helminth infections, see Maizels et al. in *Suggested Readings*). Helminths that infect human hosts are divided into two categories, or phyla: Platyhelminthes (flatworms), and Aschelminthes or nematodes (roundworms). The flatworms include the classes Cestode (tapeworms) and Trematode (flukes or schistosomes). The nematode class includes helminths common to the United States: roundworm, hookworm, pinworm, and whipworm. These worms are cylindrical in shape, with significant variations in size, proportion, and structure.

Nematode Infections

Ancylostomiasis or Hookworm Infection

The two most widespread types of hookworm in humans are the American hookworm (*Necator americanus*) and the “Old World” hookworm (*Ancylostoma doudeanale*). The life cycles of both are similar. The larvae are found in the soil

and are transmitted either by penetrating the skin or being ingested orally. The circulatory system transports the larvae via the respiratory tree to the digestive tract, where they mature and live for 9 to 15 years if left untreated. These worms feed on intestinal tissue and blood. Infestations cause pulmonary lesions, skin reactions, intestinal ulceration, and anemia. The worms are most prevalent in regions of the world with temperatures of 23°C to 33°C, abundant rainfall, and well-drained, sandy soil.

Enterobiasis or Pinworm Infection (*Enterobius vermicularis*)

These worms are widespread in temperate zones and are a common infestation of households and institutions. The pinworm lives in the lumen of the GI tract, attaching itself by the mouth to the mucosa of the cecum. Mature worms reach 10 mm in size. The female migrates to the rectum, usually at night, to deposit her eggs. This event is noted by the symptom of perianal pruritus. The eggs infect fingers and contaminate nightclothes and bed linen, where they remain infective for up to 3 weeks. Eggs resist drying and can be inhaled with household dust to continue the life cycle. Detection of the worm in the perianal region can be accomplished by means of a cellophane tape swabbed in the perianal region in the evening. The worms can be visible with the naked eye. The eggs can be collected in a similar manner but can only be seen under a microscope.

Ascariasis or Roundworm Infections (*Ascaris lumbricoides*)

These roundworms are common in developing countries, with the adult roundworm reaching 25 to 30 cm in length and lodging in the small intestine. Some infections are without symptoms, but abdominal discomfort and pain are common with heavy infestation. Roundworm eggs are released into the soil, where they incubate and remain viable for up to 6 years. When the egg is ingested, the larvae are released in the small intestine, penetrate the intestinal walls, and are carried via the blood to the lungs. The pulmonary phase of the disease lasts approximately 10 days, with the larvae passing through the bronchioles, bronchi, and trachea before being swallowed and returning to the small intestine. Some patients have reported adult worms exiting the esophagus through the oral cavity, and it is not unusual for live ascariis to be expelled with a bowel movement. Poor or lacking sanitary facilities expose the population to infestation through contaminated foods and beverages.

Trichuriasis or Whipworm Infections (*Trichuris trichiura*)

Infections by this parasite are caused by swallowing eggs from contaminated foods and beverages. The eggs are passed with the feces from an infected individual. These eggs can live in the soil for many years. The ingested eggs hatch in the small intestine, and the larvae embed in the intestinal wall. The worms then migrate to the large intestine, where they mature. Adult worms, which reach approximately 5 cm in length, thread their bodies into the epithelium of the colon. They feed on tissue fluids and blood. Infections from this worm cause symptoms

of irritation and inflammation of the colonic mucosa, abdominal pain, diarrhea, and distention. Infections can last 5 or more years if not treated. Whipworm infections are commonly seen in individuals returning from visits to the subtropics and are more common in rural areas of the southeastern United States.

Trichinosis or Trichina Infection (*Trichinella spiralis*)

Trichinella spiralis produces an infection that can be both intestinal and systemic. The worm is found in muscle meat, where the organism exists as an encysted larvae. Traditionally, the worm has been associated with domestic pork that feeds on untreated garbage. More recently, outbreaks have occurred in individuals eating infected game, such as wild boar, bear, or walrus. Trichinosis infections are more likely to occur after consumption of homemade pork or wild-game sausages. After ingestion, the larvae are released from the cyst form and then migrate into the intestinal mucosa. After maturation and reproduction, the newly released larvae penetrate the mucosal lining and are distributed throughout the body, where they enter skeletal muscle. During the adult intestinal stage, diarrhea, abdominal pain, and nausea are the most common symptom, whereas the muscular form of the disease has symptoms that can include muscle pain and tenderness, edema, conjunctivitis, and weakness.

Filariasis

The term “filariasis” denotes infections with any of the Filarioidea, although it is commonly used to refer to lymphatic-dwelling filariae, such as *Wuchereria bancrofti*, *Brugia malayi*, and *Brugia timori*. Other filarial infections include *Loa loa* and *Onchocerca volvulus*. The latter two are known as the eyeworm and the river blindness worm, respectively. Elephantiasis is the most common disease associated with filariasis. These parasites vary in length from 6 cm for *Brugia* to 50 cm for *Onchocerca*. The incubation periods also vary from 2 months for *Brugia* to 12 months for *bancrofti* filaria. It is estimated that 400 million persons are infected with human filarial parasites. Depending on the specific organism, various intermediate hosts are involved in spreading the infection. Mosquitoes are involved with the spread of *Wuchereria bancrofti*, *Brugia malayi*, and *Brugia timori*, whereas the female blackfly spreads river blindness. The larvae released by the female filaria are referred to as microfilariae and commonly can be found in the lymphatics.

Cestode and Trematode Infections

Cysticercosis or Tapeworm Infection

Helminths of this class that are of concern as potential parasites in humans include:

- **Beef tapeworm (*Taenia saginata*).** This worm is found worldwide and infects people who eat undercooked beef. The worm reaches a length of more than 5 m, and it contains approximately 100 segments/m. Each of these segments contains its own reproductive organs.

- **Pork tapeworm (*Taenia solium*).** Pork tapeworms sometimes are called bladder worms and occasionally are found in uncooked pork. The worm attaches itself to the intestinal wall of the human host. The adult worm reaches 5 m in length and, if untreated, survives in the host for many years.
- **Dwarf tapeworm (*Hymenolepis nana*).** This infection is transmitted directly from one human to another without an intermediate host. *Hymenolepis nana* reaches only 3 to 4 cm in length. It is found in temperate zones, and children are most frequently infected.
- **Fish tapeworm (*Diphyllobothrium latum*).** The fish tapeworm reaches a length of 10 m and contains approximately 400 segments/m. These tapeworms attach themselves to the intestinal wall and rob the host of nutrients. They especially absorb vitamin B₁₂ and folic acid. Depletion of these critical nutrients, especially vitamin B₁₂, can lead to pernicious anemia. Tapeworm eggs are passed in the patient's feces, and contamination of food and drink can result in transmission of the infection.

Schistosomiasis or Blood Flukes

Three primary trematode species cause schistosomiasis in humans: *Schistosoma hematobium*, *Schistosoma mansoni*, and *Schistosoma japonicum*. Infections result from the penetration of normal skin by living (free-swimming) cercariae (the name given to the infectious stage of the parasite) with the aid of secreted enzymes. The cercariae develop to preadult forms in the lungs and skin. Then, these parasites travel in pairs via the bloodstream and invade various tissues. The adult worm reaches approximately 2 cm in length. The female deposits her eggs near the capillary beds, where granulomas form. Some of the eggs will move into the lumen of the intestines, bladder, or ureters and are released into the environmental surrounding, where the parasite will seek out the intermediate snail vector. Asexual reproduction occurs in the snail. After a period of time, the cercariae are again released from the snail to continue the cycle. The patients might experience headache, fatigue, fever, and GI disturbances during the early stages of the disease. Hepatic fibrosis and ascites occur during later stages. Untreated patients can harbor as many as 100 pairs of worms. Untreated worms can live 5 to 10 years within the host. As many as 200 million persons worldwide are estimated to be infected with schistosomes. Depending on the species of schistosome, the disease is found in parts of South America, the Caribbean Islands, Africa, and the Middle East.

Drug Therapy for Helminth Infections (56)

Helminths represent a biologically diverse group of parasitic organisms differing in size, life cycle, site of infection (local and systemic), and susceptibility to chemotherapy. With such variation in infectious organisms, it is not surprising that the drugs used to control helminth infections also represent a varied group of chemical classes. As indicated in Table 34.4, the drugs can have fairly narrow spectra of activity (pyrantel pamoate) or a broad spectra of activity (benzimidazoles).

TABLE 34.4 Therapeutic Application of Anthelmintics for Specific Helminth Infections

	M	A	DEC	IVM	PZQ	PP
Nematode Infections:						
<i>Necator americanus</i>	✓	✓				
<i>Ancylostoma doudeale</i>	✓	✓				✓
<i>Enterobius vermicularis</i>	✓			✓		✓
<i>Ascaris lumbricoides</i>	✓	✓		✓		✓
<i>Trichuris trichiura</i>	✓	✓		✓		
<i>Trichinella spiralis</i>	✓	✓				
<i>Wuchereria bancrofti</i>			✓	✓		
<i>Brugia malayi</i>			✓	✓		
<i>Brugia timori</i>			✓			
<i>Loa Loa</i>		✓	✓	✓		
<i>Onchocerca volvulus</i>		✓	✓	✓		
Cestode Infections:						
<i>Taenia saginata</i>	✓	✓				
<i>Taenia solium</i>	✓	✓			✓✓	
<i>Hymenolepis nana</i>	✓					
<i>Diphyllobothrium latum</i>					✓✓	
Trematode Infections:						
<i>Schistosoma hematobium</i>						
<i>Schistosoma mansoni</i>					✓✓	
<i>Schistosoma japonicum</i>					✓	

Benzimidazoles: A, albendazole; DEC, diethylcarbamazine; IVM, ivermectin; M, mebendazole; PP, pyrantel pamoate; PZQ, praziquantel.

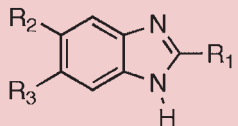
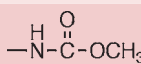
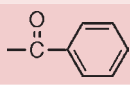
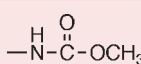
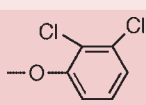
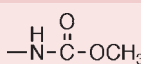
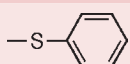
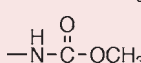
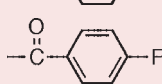
Benzimidazoles

The benzimidazoles (Table 34.5) are a broad-spectrum group of drugs discovered in the 1960s with activity against GI helminths. Several thousand benzimidazoles have been synthesized and screened for anthelmintic activity, with albendazole, mebendazole, and thiabendazole representing the benzimidazoles marketed today. The development and chemistry of this class of agents have been reviewed by Townsend and Wise (57).

MECHANISM OF ACTION Two mechanisms have been proposed to account for the action of the benzimidazoles. Fumarate reductase is an important enzyme in helminths that appears to be involved in oxidation of NADH to NAD. The benzimidazoles are capable of inhibiting fumarate reductase (58). Inhibition of fumarate reductase ultimately uncouples oxidative phosphorylation, which is important in adenosine triphosphate production.

A second mechanism and, probably, the primary action of the benzimidazoles is associated with the ability of these drugs to bind to the protein tubulin and, thus, prevent tubulin polymerization to microtubules (59,60). Tubulin is a dimeric protein that is in dynamic equilibrium with the polymeric microtubules. Binding to the tubulin prevents the self-association of subunits and creates a “capping” of the microtubule at the associating end of the microtubule. The microtubulin continues to dissociate from the opposite end, with a net loss of microtubule length. What is interesting is the unique selectivity of the benzimidazoles. Benzimidazole can also bind to mammalian tubulin, but when used as anthelmintics, these drugs are destructive to the helminth, with minimal toxicity to the host. It has been suggested that the

TABLE 34.5 BENZIMIDAZOLE ANTHELMINTICS

				
Drugs	Trade name	R1	R2	R3
Thiabendazole	Mintezol		—H	H
Mebendazole	Vermox			H
Albendazole	Zental		—SCH ₂ CH ₂ CH ₃	H
Triclabendazole	Egaten ¹ Fasinex*	—S—CH ₃		Cl
Fenbendazole	Several brand names*			H
Flubendazole				

¹Egaten has recently been shown to be useful for treatment of fascioliasis by WHO. *Used in veterinary practice for protection and treatment of parasite and worm infections.

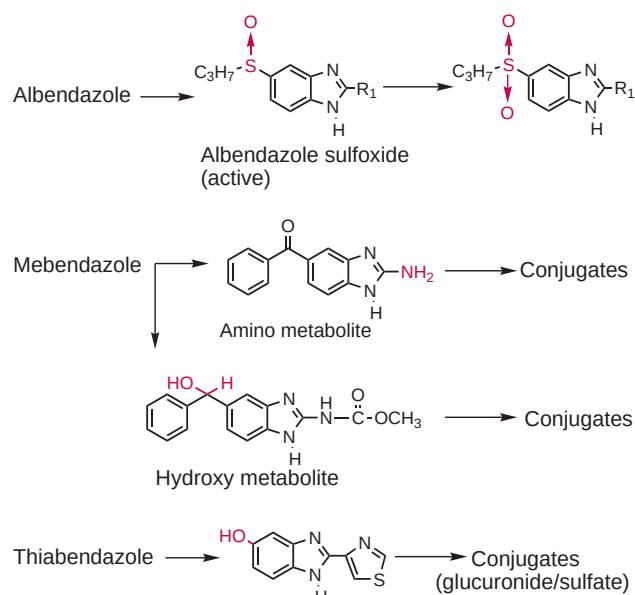


FIGURE 34.19 Metabolism of benzimidazoles.

selectivity is associated with differing pharmacokinetics between binding to the two different tubulin proteins.

METABOLISM The benzimidazoles have limited water solubility and, as a result, are poorly absorbed from the GI tract (a fatty meal will increase absorption). Poor absorption can be beneficial, because the drugs are used primarily to treat intestinal helminths. To the extent that the drugs are absorbed, they undergo rapid metabolism in the liver and are excreted in the bile (Fig. 34.19) (61,62). In most cases, the parent compound is rapidly and nearly completely metabolized with oxidative and hydrolytic processes predominating. The phase 1 oxidative reaction commonly is a cytochrome P450-catalyzed reaction, which can then be followed by a phase 2 conjugation.

Albendazole is unique in two ways. First, the presence of a thioether substituent at the five position increases the likelihood of sulfur oxidation. Second, the initial metabolite, albendazole sulfoxide, is a potent anthelmintic. This initial oxidation is catalyzed principally (70%) by CYP3A4 and CYP1A2 and (30%) by flavin-containing monooxygenase, giving rise to a compound that is bound to plasma protein. This intermediate has an expanded utility in that it has been shown to be active against the hydatid cyst found in echinococcosis, a tapeworm disease (63). Further oxidation by cytochrome P450 leads to the inactive sulfone. Additional metabolites of the sulfone have been reported that include carbamate hydrolysis to the amine and oxidation of the 5-propyl side chain. These reactions occur only to a minor extent.

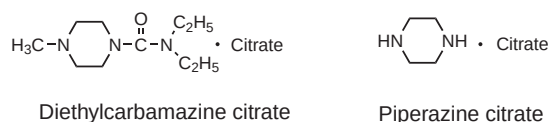
Metabolism of mebendazole occurs primarily by reduction of the 5-carbonyl to a secondary alcohol, which greatly increases the water solubility of this compound. An additional phase 1 metabolite resulting from carbamate hydrolysis has been reported as well. Both the secondary alcohol and the amine are readily conjugated

(a phase II metabolism). Evidence would suggest that the anthelmintic activity of mebendazole resides in the parent drug and none of the metabolites.

Thiabendazole is metabolized through aromatic hydroxylation at the five position catalyzed by CYP1A2. The resulting phenol is conjugated to 5-hydroxythiabendazole glucuronide and 5-hydroxythiabendazole sulfate, respectively. The initial metabolite and minor amount of N_1 -methylthiabendazole (from a methylation phase 2 reaction) have been reported to be teratogenic in mice and rats.

THERAPEUTIC APPLICATION As indicated in Table 34.4, mebendazole and albendazole have a wide spectrum of activity against intestinal nematodes. The drugs are useful and effective against mixed infections. The adverse reactions are commonly GI in nature (nausea, vomiting, and diarrhea). Both drugs have been reported to be teratogenic in rats and, therefore, should not be used during the first trimester of pregnancy. A third drug of this class is thiabendazole, which remains of some value in treatment of strongyloidiasis, as an alternate drug, and cutaneous larva migrans (creeping eruption), for which it is the drug of choice. Thiabendazole is commonly used in veterinary medicine. The drug is less commonly used because of associated toxicity. Thiabendazole has been reported to cause Stevens-Johnson syndrome and has the potential for hepatotoxicity and crystalluria.

Diethylcarbamazine (Hetrazan)



Discovered in the 1940s, diethylcarbamazine (DEC) has proven to be especially effective as a filaricidal agent. The incidence of filariasis among American troops during World War II necessitated a search for drugs with an antifilarial spectrum of activity. The once-popular piperazine also was discovered during these initial screenings. Although chemically similar, the activity against helminths is quite different. Piperazine is active against nematodes, whereas DEC is active against falaria and microfalaria (64).

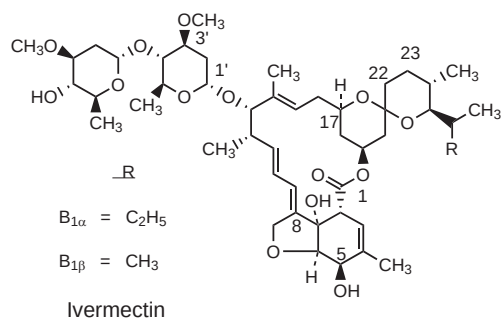
MECHANISM OF ACTION Although studied extensively, the mechanism of action of DEC remains unknown. DEC appears to be the active form of the drug, with a very rapid onset of action (within minutes), but of interest is the fact that the drug is inactive *in vitro*, suggesting that activation of a cellular component is essential to the filaricidal action. Three mechanisms have been suggested. The first is involvement of blood platelets triggered by the action of filarial excretory antigens. A complex reaction is thought to occur between the drug, the antigen, and platelets (65). Although these authors were unable to show a direct action of the drug on the microfalaria, a more recent study showed that DEC produced

morphologic damage to the microfilaria. The damage consisted of the loss of the cellular sheath, exposing antigenic determinants to immune defense mechanisms. Severe damage then occurred to microfilaria organelles, leading to death (66). The second is inhibition of microtubule polymerization and disruption of preformed microtubules (67). The third is interference with arachidonic acid metabolism (68). DEC is known to have anti-inflammatory action, which appears to involve blockage at cyclooxygenase and leukotriene A_4 synthase (leukotriene synthesis). This action appears to alter vascular and cellular adhesiveness and cell activation. This latter action would suggest a possible relationship between the first and third mechanism.

METABOLISM The metabolism of DEC leads to the compounds shown in Figure 34.20 plus trace amounts of methylpiperazine and piperazine. Nearly all of the metabolites appear in the urine. As much as 10% to 20% of the drug is excreted unchanged. As indicated by the rapid action of the drug, it would appear that none of the metabolites are involved in the therapeutic action of DEC.

THERAPEUTIC APPLICATION DEC citrate is freely soluble in water, is rapidly absorbed, and is effective against microfilariae. The drug does not appear to be effective against the adult worm. In general, the drug has mild adverse effects, but under some conditions, it can produce severe adverse reactions, including anaphylactic reactions, intense pruritus, and ocular complications (69). The severe anaphylactic reaction is known as the Mazzotti reaction, and it appears to be an immune response related to the presence of dead microfilariae. This reaction is more common in individuals who have a high-load microfilarial infection, and it can preclude the use of DEC in some patient populations (56).

Ivermectin (Stromectol)



Extracted from the soil actinomycete *Streptomyces avermitilis*, the natural avermectins are 16-membered macrocyclic lactones that are an 80:20 mixture of avermectin $B_{1\alpha}$ and $B_{1\beta}$, respectively, and are used under the generic name abamectin. As such the product is used for the control of various insects, mite pests, fire ants and as a veterinary antihelminthic. Reduction of the C_{22-23} double bond gives

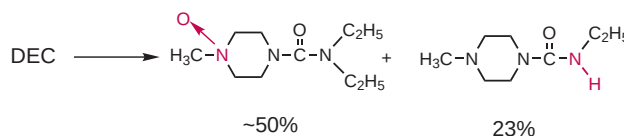


FIGURE 34.20 Metabolism of diethylcarbamazine (DEC).

rise to ivermectin (IVM), which is an 80:20 mixture of dihydroavermectin $B_{1\alpha}$ and $B_{1\beta}$, respectively. The natural avermectins have minimal biologic activity, but IVM has proven to be quite beneficial in the treatment of various nematode infections.

MECHANISM OF ACTION Two mechanisms of action are thought to be involved in the action of IVM (56,70). The first is an indirect action in which motility of microfilaria is reduced, which in turn allows cytotoxic cells of the host to adhere to the parasite, resulting in elimination from the host. This action can occur by virtue of the ability of IVM to act either as a γ -aminobutyric acid (GABA) agonist or as an inducer of chloride ion influx, leading to hyperpolarization and muscle paralysis. The chloride ion influx appears to be the more plausible mechanism (71). Recently, it has been shown that IVM binds irreversibly to the glutamate-gated chloride channel of the nematode *Haemonchus contortus*, whereas the channel is in an open conformation. The binding then remains locked in the open conformation, allowing ions to cross the membrane, leading to the paralytic action of IVM (72). The result of this action is a rapid decrease in microfilarial concentrations.

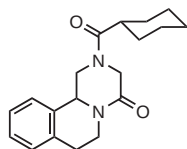
A second action of IVM leads to the degeneration of microfilariae in utero. This action would result in fewer microfilariae being released from the female worms, and it occurs over a longer period of time. The presence of degenerated microfilariae in utero prevents further fertilization and production of microfilariae.

METABOLISM IVM is rapidly absorbed, is bound to a great extent to plasma protein, and is excreted in the urine or feces either unchanged or as the 3'-O-demethyl-22,23-dihydroavermectin $B_{1\alpha}$ or as the dihydroavermectin $B_{1\alpha}$ monosaccharide. The absorption of IVM is significantly affected by the presence of alcohol. Administration of IVM as an alcoholic solution can result in as much as a 100% increase in absorption.

THERAPEUTIC APPLICATION Although IVM has activity against a variety of microfilaria, including *Wuchereria bancrofti*, *Brugia malayi*, *Loa loa*, and *Mansonella ozzardi*, as well as activity against *Strongyloides stercoralis*, the drug is used primarily in the treatment of onchocerciasis (African river blindness) caused by *Onchocerca volvulus*. It is estimated that 20 million people are affected by this condition and an additional 123 million are at risk of the infection. The drug is effective against both the eyeworm as well as skin

infections of *O. volvulus*. IVM has the distinct advantage over DEC in that IVM can be used as a single dose (150 µg/kg) once a year (although there is support for dosing every 6 months), has far less likelihood of causing the potentially fatal anaphylactic reaction (Mazzotti reaction), and can be used for mass treatment programs.

Praziquantel (Biltricide)



Praziquantel

Praziquantel (PZQ) is an isoquinoline derivative with most of the biologic activity found in the levo enantiomer. The compound has no activity against nematodes, but it is highly effective against cestodes and trematodes.

MECHANISM OF ACTION More than one mechanism of action can exist for PZQ, possibly depending on the type of parasite being treated. The mechanism of action appears to involve Ca^{2+} redistribution either directly or indirectly. In the case of helminths found in the lumen of the host (cestode infection), the drug leads to muscle contraction and paralysis, leading in turn to worm expulsion. Additionally, PZQ has been shown to inhibit phosphoinositide metabolism, which by an undetermined mechanism leads to the worm paralysis (73). With intravascular-dwelling schistosomes, PZQ leads to drug-induced damage of the tegument of the worm. As a result, antigens in the helminth are subject to attack by immune antibodies of the host (74,75). An antigen–antibody immunologic reaction leads to the death of the parasite. Finally, PZQ affects glycogen content and energy metabolism (76,77).

METABOLISM PZQ is rapidly absorbed and undergoes hepatic first-pass metabolism. The metabolites are either less active or inactive and consist of hydroxylated compounds. In the serum, the major metabolite appears to be the monohydroxylated 4-hydroxycyclohexylcarboxylate, whereas in the urine, 50% to 60% of the initial PZQ exists as dihydroxylated products (Fig. 34.21) (78). These hydroxylation reactions are catalyzed by CYP2B6 and CYP3A4. The metabolites would be expected to exist in the conjugated form in the urine.

THERAPEUTIC APPLICATION PZQ is the drug of choice for treatment of schistosomiasis and liver flukes (trematode and cestode infections). The drug is stage specific, with activity against the invasive stages, which includes the cercariae and young schistosomula and the mature worms, but not against the liver stages. Although an approved drug, PZQ is considered to be an investigational drug by the FDA in the treatment of schistosomiasis and liver

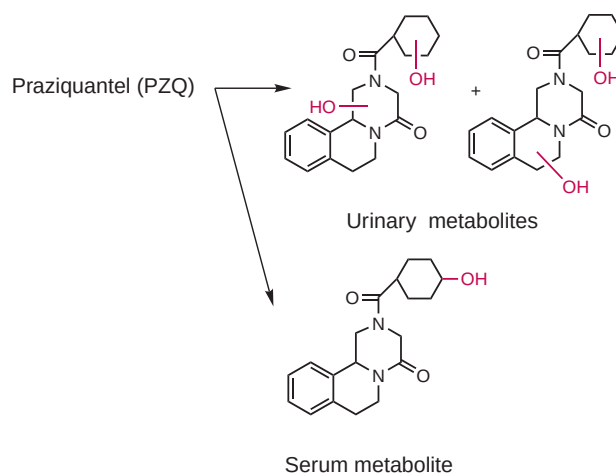
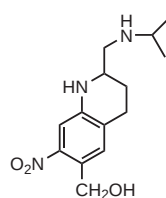


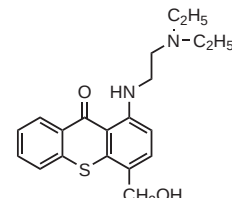
FIGURE 34.21 Metabolism of praziquantel (PZQ).

flukes. The drug has a bitter taste and, therefore, should not be chewed. The side effects are usually not severe and consist of abdominal discomfort (pain and diarrhea). Mounting evidence suggests that resistance can become a significant problem.

Oxamniquine (Mansil, Vansil)



Oxamniquine



Hycanthone

Oxamniquine was originally investigated in the 1960s and was found to have limited antiprotozoal activity, with activity against *Schistosoma mansoni* but no activity against the other two schistosomal organisms. In addition, the drug is stage specific, with activity against cercariae and very young schistosomula and adult worms. For unknown reasons, the drug is more effective against adult male worms than against female worms. The drug has structural similarity to hycanthone, which is no longer used because of severe toxicity and teratogenic effects.

MECHANISM OF ACTION Oxamniquine is activated by the organism via esterification to a biologic ester that spontaneously dissociates to an electrophile, which alkylates the helminth DNA, leading to irreversible inhibition of nucleic acid metabolism (Fig. 34.22) (77). Resistant helminths do not esterify oxamniquine; therefore, activation does not occur. Other metabolic reactions consist of oxidative reactions, leading to inactivation (Fig. 34.22). The metabolites are excreted primarily in the urine.

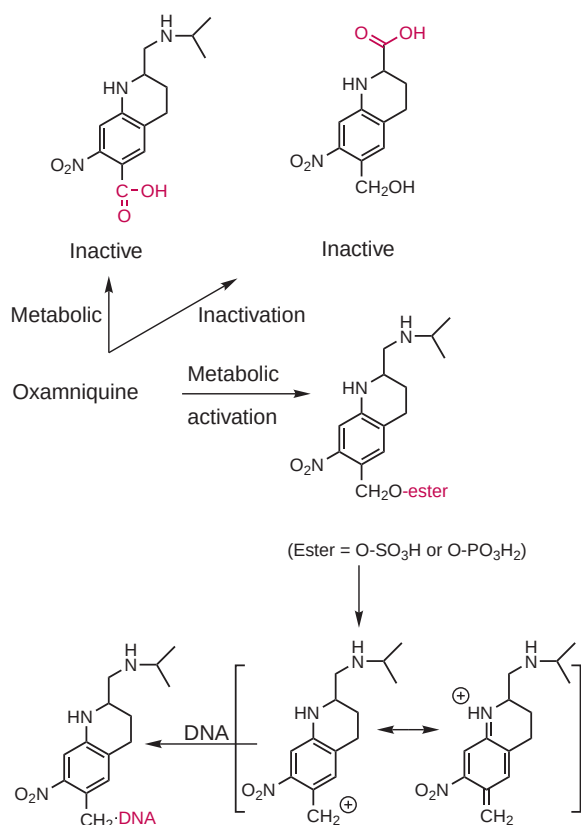
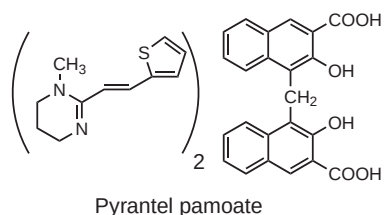


FIGURE 34.22 Metabolism of oxamniquine accounting for the mechanism of action and inactivation.

THERAPEUTIC APPLICATION Oxamniquine is readily absorbed following oral administration and has a relatively short half-life. The drug has been highly effective against *Schistosoma mansoni* native to Brazil, where it is marketed under the trade name Mansil. It is also beneficial against west African *S. mansoni* and is supplied under the trade name Vansil. Side effects are minimal, with transient dizziness being reported. The major drawback is its high cost. Encouraging outcomes have been reported with the combination of oxamniquine and PZQ.

Pyrantel Pamoate (Antiminth)



Pyrantel was first reported for its anthelmintic activity in 1966 (79). Although it has activity against most intestinal roundworm infections, it has not been approved by the FDA for treatment of these infestations. It is considered to be the drug of choice in the treatment of pinworms.

The drug is used as the pamoate salt, which is quite insoluble and, as a result, is not readily absorbed. This property improves the usefulness of the drug for treatment of intestinal helminths. In addition to its value in treating enterobiasis, the drug is effective for hookworm and roundworm (ascariasis) infections. Pyrantel acts as a depolarizing neuromuscular blocking agent that activates nicotinic receptors and inhibits cholinesterase, ultimately leading to worm paralysis.

ECTOPARASITIC INFECTIONS

Three parasitic organisms that cause common topical infections are *Sarcoptes scabiei*, which is responsible for scabies, *Pediculus humanus*, which is responsible for lice infections, and *Cimex lectularius*, which is the common bedbug, an insect living exclusively on the blood of warm-blooded animals. This latter organism has shown a recent reemergence, and although its bite normally is not responsible for a primary or secondary infection, the topical irritation and the social implications can be quite disturbing to the patient. The only treatment is the use of antipruritics and the hiring of a professional exterminator.

Scabies

Scabies, commonly referred to as the “seven-year itch,” is a condition caused by *Sarcoptes scabiei*, or the itch mite. The condition is commonly spread by direct, person-to-person contact, although the organism is capable of living for 2 to 3 days in clothing, bedding, or house dust. Sharing of clothing is a common means whereby the condition spreads. The organism burrows into the epidermis, usually in the folds of the skin of the fingers, the elbows, female breast, penis, scrotum, and buttocks. The female parasite lays eggs in the skin, which then hatch and mature to adults. The itch mite can live for 30 to 60 days. The infections are most common in children, but they may also be found in adults in institutional settings. The primary symptom of severe itching can foster secondary infections at the site of scratching. Because of the potential for spread to other members of a family, it is common to treat all members of the family. This will prevent reinfection from a second family member after successful therapy of the first family member.

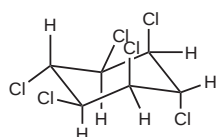
Lice

Pediculosis, or lice, is caused by any of the following parasites: *Pediculus humanus capitis*, the head louse; *Pediculus humanus corporis*, the body louse; or *Phthirus pubis*, the crab louse (found in the genital area). Lice are bloodsucking insects that live for 30 to 40 days on the body of the host. The organisms reproduce, and the female lays her eggs, the nits, which become attached to hair. The nits are white in color and hatch in 8 to 10 days. For the parasite to live, it must feed on blood, which it sucks through punctures in the skin. A hypersensitivity

reaction occurs at these puncture sites, which then leads to pruritus, host scratching, and possible secondary infection. In addition to the scalp and skin, the eyebrows, eyelids, and beard can become sites of infection. The transfer of infection can occur through person-to-person contact and from infected clothing, on which the organism can survive for up to 1 week. The sharing of clothing is a common means for the spread of body lice. Head lice are quite common among children in grade school, whereas crab lice are common among individuals who are sexually active. Treatment of family members is recommended, and clothing and bed linens should be removed and washed in very hot water.

Drug Therapy for Scabies and Pediculosis

Lindane (Kwell)



γ -Benzene hexachloride

Chlorination and reduction of benzene leads to a mixture of hexachlorocyclohexanes. The insecticidal activity resides primarily in the γ -isomer of hexachlorocyclohexane (γ -benzene hexachloride). The compound is thought to produce its insecticidal action by virtue of a CNS stimulatory action that occurs by blockage of GABA. The compound is readily absorbed through the chitinous exoskeleton of the parasite. Unfortunately, lindane is also readily absorbed through intact human skin, especially the scalp, and has the potential for systemic neurotoxicity in the host. Infants and children and, possibly,

the elderly are most prone to the neurotoxic effects of the drug. Because the lindane is quite lipophilic and is applied to the scalp as a shampoo, it can be absorbed, where upon it can readily enter the CNS of the patient producing signs of neurotoxicity (convulsions, dizziness, clumsiness, and unsteadiness).

The drug is available in a lotion and a shampoo and is recommended for the treatment of both pediculosis and scabies. When using the lotion topically, it should be applied to dry skin, covering the entire surface and being left in place for 8 hours. The lindane then should be removed by washing thoroughly. If the shampoo is used for *Pediculosis capitis*, the hair should be cleaned of oil and dried before application of the lindane shampoo. The shampoo is then worked into the hair and scalp, being applied in such a way as to prevent other parts of the body from coming into contact with the drug. After approximately 4 minutes, the drug is removed by washing with water, and the hair is dried and then combed with a fine-toothed comb to remove nits.

Pyrethrum and Pyrethroids

The naturally occurring pyrethrums have been used as insecticides since the 1800s. These compounds are extracted from the flowering portion of the chrysanthemum plant. The flowers produced in Kenya have, on average, 1.3% pyrethrins. These pyrethrum extracts are a major agricultural product for that country.

CHEMISTRY The chrysanthemum extract is a mixture of esters consisting of the acids chrysanthemic and pyrethric and the alcohols pyrethrolone and cinerolone (Fig. 34.23). The esters are prone to hydrolysis and oxidation and, as a result, should be stored in the cold and protected from light. Because of the high cost, limited availability, and rapid degradation, synthetic derivatives

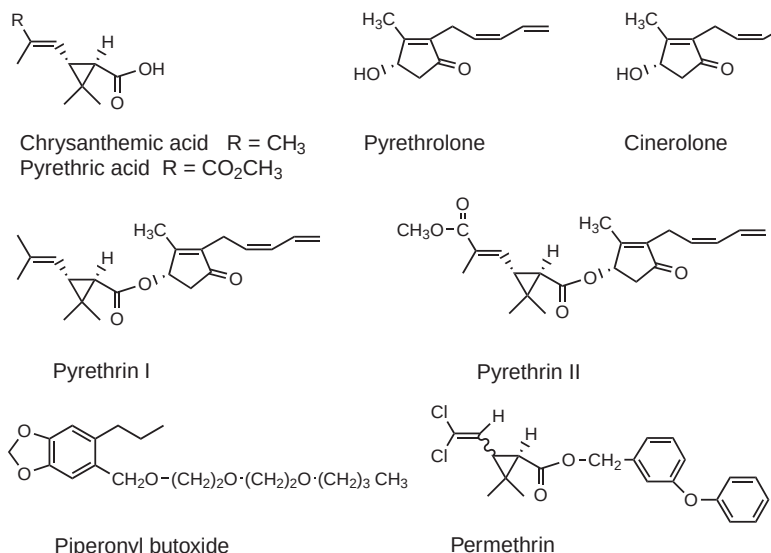


FIGURE 34.23 Structures of pyrethrum and pyrethroid.

have been investigated. The result has been the preparation of pyrethroids, the synthetic derivatives of pyrethrins. The compound used therapeutically is permethrin, which exists as a 60:40 mixture of *trans*:*cis* isomers.

MECHANISM OF ACTION (80–83) The pyrethrins and pyrethroids (permethrin) are nerve membrane sodium channel toxins that do not affect potassium channels. The compounds bind to specific sodium channel proteins and slow the rate of inactivation of the sodium current elicited by membrane depolarization and, as a result, prolong the open time of the sodium channel. At low concentrations, the pyrethroids produce repetitive action potentials and neuron firing; at high concentrations, the nerve membrane is depolarized completely and excitation blocked.

The receptor interaction of the pyrethrums with the sodium channel complex is stereospecific and dependent on the stereochemistry of the carboxylic acid. In the case of the pyrethroids, the most active isomers are the *1R,3-cis*- and *1R,3-trans*-cyclopropanecarboxylates. The *1S-cis*- and *-trans*-isomers are inactive and actually are antagonists to the action of the *1R*-isomers.

METABOLISM A property that enhances the usefulness of the pyrethrums and pyrethroids is that these compounds are highly toxic to the ectoparasites but relatively nontoxic to mammals if absorbed. The apparent lack of

toxicity is associated with the rapid metabolism of these drugs through hydrolysis and or oxidation (Fig. 34.24) (84,85). The nature of the metabolism (i.e., hydrolysis vs. oxidation) is dependent on the structure of the pyrethrins or pyrethroids. Oxidation of the *trans*-methyl of the isobutylene in the carboxyl moiety initially gives an alcohol, which then proceeds to the carboxylic acid, whereas epoxidation of the terminal alkene of the alcohol portion of pyrethrin I gives either the 1,2-diol or the 1,4-diol. No ester hydrolysis is reported. Permethrin is hydroxylated on the terminal aromatic ring at either the 4- or 2-position, is oxidized on the methyl group of the dimethylcyclopropane, and is hydrolyzed at the ester moiety. The rapid breakdown of these agents also accounts for their low persistence in the environment.

THERAPEUTIC APPLICATION

Pyrethrins (A-200, RID) Because of the high cost and rapid degradation of the pyrethrins, they are usually combined with piperonyl butoxide, a synergist (Fig. 34.23). Piperonyl butoxide has no insecticidal activity in its own right but is thought to inhibit the cytochrome P450 enzyme of the insect, thus preventing an oxidative inactivation of the pyrethrins by the parasite. The combination is used in a 10:1 ratio of piperonyl butoxide to pyrethrins. The mixture is used for treatment of *Pediculus humanus capitis*, *Pediculus humanus corporis*, and *Phthirus*

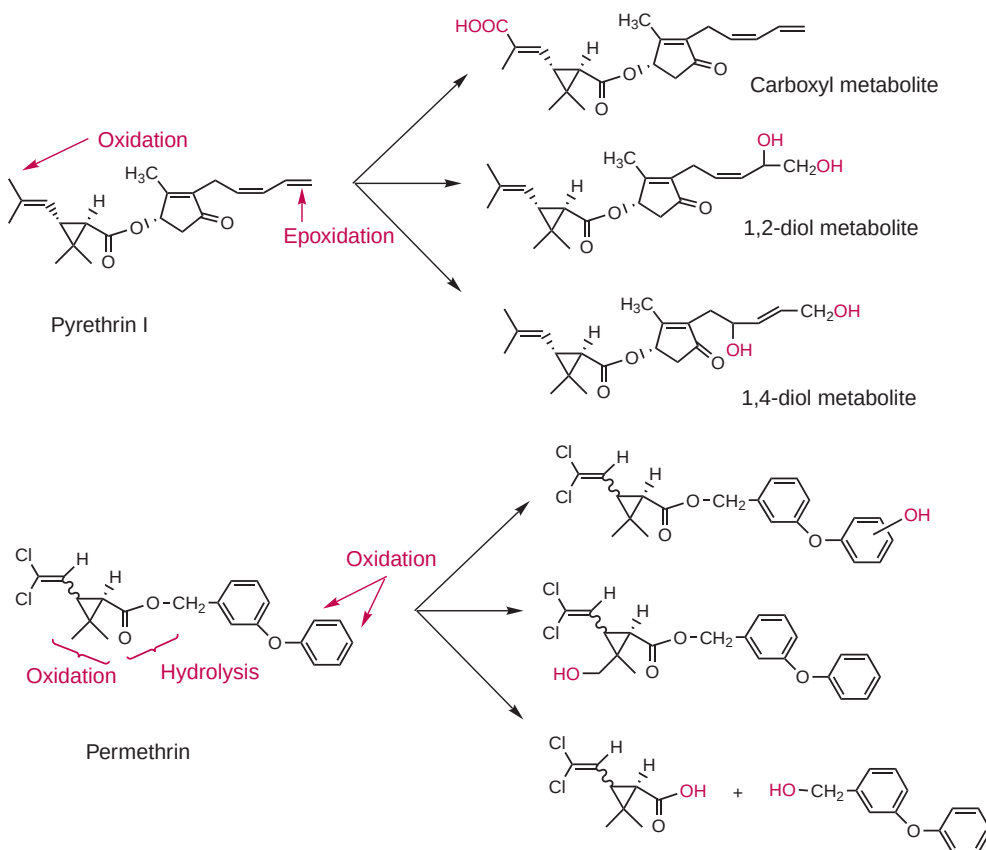


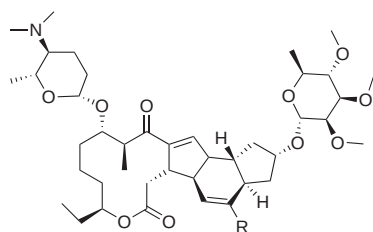
FIGURE 34.24 Metabolism of pyrethrin I and permethrin.

pubis. Various dosage forms are available, including a gel, shampoo, and topical solution.

Permethrin (Nix-1% lotion, Elimate-5% Cream) Permethrin, because of its increased stability and its availability synthetically, is not used with a synergist. The compound is used in a 1% lotion for the treatment of pediculosis capitis and in a 5% cream as a scabicide.

CROTAMITON Crotamiton is available as a 10% cream for the treatment of scabies, although it is less effective than pyrethrins or permethrin (86,87). Because crotamiton may need to be applied a second time for successful treatment of scabies but the pyrethrins or permethrin require a single application, poor patient compliance with crotamiton can reduce its effectiveness. The advantage of crotamiton over lindane comes from the fact that lindane has potential neurotoxicity if absorbed, especially in infants and children, whereas crotamiton has less systemic neurotoxicity. The most common side effect reported for crotamiton is skin irritation.

SPINOSAD (NATROBA)



Spinosad (Natroba)

(Spinosyn A R = H, Spinosyn D, R = CH₃)

Spinosad is a newly FDA-approved treatment for head lice infestation (*Pediculus humanus capitis*). The product is available as a 0.9% topical suspension, which is applied to a dry scalp and hair for a period of 10 minutes and is then rinsed with water. Because spinosad has ovicidal activity, nit combing is not required. Studies suggest that spinosad, which requires a single treatment, is more effective (~80%) than permethrin (~45%), especially after a single treatment. Fewer topical adverse events were noted with spinosad versus permethrin, and presently resistance, a growing problem with the permethrin, has not been reported for spinosad (88).

Spinosad is a naturally occurring 12-member macrolide that consists of a mixture of spinosyn A and D produced by the bacterium *Saccharopolyspora spinosa*, a member of the actinomycete family. The product was a serendipitous discovery isolated from a soil sample found in an abandoned rum distillery on a Caribbean island by an employee of Eli Lilly. The mixture has a broad spectrum of activity and was approved for agricultural use by the Environmental Protection Agency in 1995 (89).

The mechanism of action has been investigated and found to lead to death of the insect via rapid excitation of the insect's nervous system. Spinosad is an insecticide by virtue of its ability to act as an agonist at the nicotinic acetylcholine receptors and, more specifically, at the Da6 subunit (90,91). Thus, spinosad has a novel mechanism of action compared to other insecticides. The results are that spinosad creates prolonged hyperexcitation of the CNS of the insects and eventually paralysis and death of the organism.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Elizabeth B. Hirsch, PharmD, BCPS

After reviewing JB's medication list, it was discovered that the over-the-counter nighttime cough syrup he started taking yesterday contained a small amount of alcohol (ethanol). It is believed the oral metronidazole in combination with the ethanol caused a disulfiram-like reaction, resulting in nausea, vomiting, and dizziness. The pharmacist advised JB to discontinue the offending cough syrup and recommended another safer non-ethanol-containing medication.

Chemical Analysis

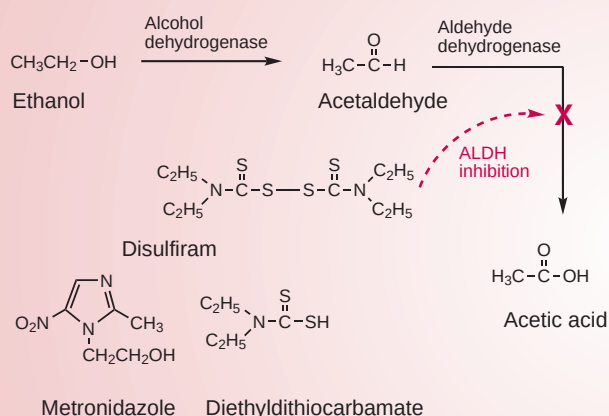
Victoria Roche and S. William Zito

Ethanol is primarily metabolized by sequential hepatic oxidation, first to acetaldehyde by alcohol dehydrogenase (ADH) and

then to acetic acid by aldehyde dehydrogenase (ALDH). Certain antibacterials, including metronidazole, have been implicated in the ability to act similarly to disulfiram, causing a disulfiram-like reaction. Disulfiram irreversibly inhibits the oxidation of acetaldehyde, causing a considerable increase in acetaldehyde concentrations following ethanol ingestion; this is known as the disulfiram-ethanol reaction. Disulfiram's ability to react with sulfhydryl groups on essential proteins (generating diethyldithiocarbamate in the process) is believed to be important to its activity. Signs and symptoms of a disulfiram-ethanol reaction are due to high acetaldehyde concentrations and include facial and body warmth and flushing, conjunctival injection, pruritus, urticaria, diaphoresis, lightheadedness, vertigo, headache, nausea, vomiting, and abdominal pain. Cardiac effects include dyspnea, palpitations, and chest pain.

(Continued)

SCENARIO: OUTCOME AND ANALYSIS (Continued)



The true incidence of disulfiram-like reactions during metronidazole treatment is difficult to estimate. Unfortunately, most reports citing this adverse effect arise from case reports, which may be confounded by patient comorbidities, polypharmacy, and/or use

of other drugs of abuse. More recent literature has called the mechanism of this reaction (with metronidazole) into question. Certainly, metronidazole's lack of a disulfide moiety capable of engaging in a redox reaction with free sulfhydryl groups on biologically relevant proteins would argue against an identical mechanism of action. A relatively recent article confirms that metronidazole "neither inhibits ALDH of the liver nor increases blood acetaldehyde" (92). However, the slight inhibition of ADH by metronidazole may explain some of the alcohol intolerance symptoms observed when alcohol is consumed during therapy (nausea, dizziness, blurred vision). It was also demonstrated that the "disulfiram-like reaction" induced by metronidazole may be due to the synergistic activity it has with ethanol in elevating central serotonin levels, which has been termed "serotonergic duplication" by these authors (92).

Despite the lack of ALDH inhibiting action by metronidazole, most clinicians continue to counsel patients on the avoidance of alcohol (or alcohol-containing medications) during metronidazole treatment, as the unpleasant symptoms observed are real despite the difference in mechanism.

CASE STUDY

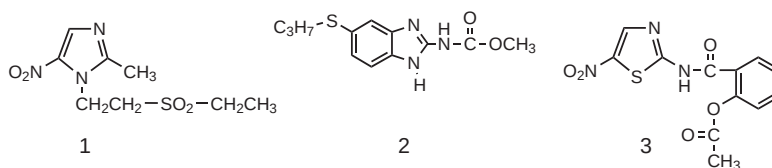
Victoria Roche and S. William Zito

DD is a 28-year-old woman who recently completed a PhD in Public Health. A person with a strong social conscience, she has been very troubled by the continuing suffering of the Haitian people more than 1 year after the devastating 2010 earthquake. She and her husband, who finished his PhD in civil engineering a month after DD defended her thesis, decided to spend the next year in Port-au-Prince using their professional expertise and commitment to global health to help in any way possible with the rebuilding.

Three months into their Haitian service immersion they found out that DD was pregnant, but she felt certain she could continue in her service mission until it was time to deliver. However, 2 weeks later, with the onset of the rainy season, she fell ill with what was ultimately diagnosed as "water-borne

diarrhea" caused from exposure to *Giardia* parasites in standing water. In light of her early pregnancy, they have flown home to ensure that she receives the proper medical treatment and totally recovers before they return to Haiti to complete their mission. She is staying well-hydrated and is now about to be placed on an antiprotozoal agent to help eradicate the offending *Giardia lamblia* organism.

Because of her pregnancy, DD is taking no medications other than the vitamins and supplements recommended by her obstetrician. As the pharmacist consulting with her medical team, consider the structure of the three potential therapeutic choices below and make the best antiprotozoal drug product selection for this patient.



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 35

Antifungal Agents

ROBERT K. GRIFFITH

Drugs Covered in This Chapter*

POLYENES

- Amphotericin B
- Natamycin
- Nystatin

AZOLES

- Butoconazole
- Clotrimazole
- Econazole
- Miconazole
- Oxiconazole

- Sertaconaole
- Sulconazole
- Terconazole
- Tioconazole

ALLYLAMINES

- Butenafine
- Naftifine
- Terbinafine

MORPHOLINES

- *Amorolfine*

ECHINOCANDINS

- Anidulafungin
- Caspofungin
- Micafungin

MISCELLANEOUS AGENTS

- Ciclopirox
- Flucytosine
- Griseofulvin
- Haloprogin
- Tolnaftate
- Undecylenic acid

Abbreviations

AUC, area under the curve
CNS, central nervous system

5-FU, 5-fluorouracil
IV, intravenous

SCENARIO

Douglas Slain, Pharm.D. BCPS

A 55 year old Caucasian female with a history of atrial fibrillation, rheumatoid arthritis, and diabetes mellitus is diagnosed with moderate pulmonary histoplasmosis. The patient is started on itraconazole in the hospital and sent home after a three day stay on itraconazole 200 mg orally twice a day for 8–10 weeks. Her other medications are metformin, prednisone, and digoxin. After about a week of her itraconazole therapy she started to feel very nauseated and tired. She called the pharmacist to see

if this could be from the itraconazole. The patient's daughter also told the pharmacist that her mother's pulse rate was very low (50 BPM). The pharmacist recognized that this was more consistent with too much digoxin and referred her to her physician.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

INTRODUCTION

Until recently, chemotherapy of fungal infections has lagged far behind chemotherapy of bacterial infections. This lack of progress has resulted, in part, because the most common fungal infections in humans have been relatively superficial infections of the skin and mucosal membranes, and potentially lethal deep-seated infections have been quite rare. Because most humans with a normally functioning immune system are able to ward off invading fungal pathogens with little difficulty, the demand for improvements in antifungal therapy has been small. Immunocompromised patients, however, are very susceptible to invasive fungal infections. The onset of the AIDS epidemic, combined with the increased use of powerful immunosuppressive drugs for organ transplants and cancer chemotherapy, has resulted in a greatly increased incidence of life-threatening fungal infections and a corresponding increase in demand for new agents to treat these infections. The number of effective antifungal agents available is quite small compared to those available to treat bacterial infections, but research in this area is quite active. Several new agents have been introduced in the last few years.

FUNGAL DISEASES

The fungal kingdom includes yeasts, molds, rusts, and mushrooms. Most fungi are saprophytic, which means that they live on dead organic matter in the soil or on decaying leaves or wood. A few of these fungi can cause opportunistic infections if they are introduced into a human through wounds or by inhalation. Some of these infections can be fatal. There are relatively few obligate animal parasites (i.e., microorganisms that can only live on mammalian hosts) among the fungi, although *Candida albicans* is commonly found as part of the normal flora of the gastrointestinal tract and vagina. The obligatory parasites are limited to dermatophytes that have evolved to live on/in the keratin-containing hair and skin of mammals, where they cause diseases such as ringworm and athlete's foot. (Ringworm is not caused by a parasitic worm but, rather, is named for the ring-like appearance of this fungal infection of the skin.) A detailed description of fungal infections is beyond the scope of this book, but comprehensive treatises are available (1).

Most fungal infections are caused primarily by various yeasts and molds. Yeasts, such as the opportunistic pathogen *Candida albicans* and the baker's yeast *Saccharomyces cerevisiae*, typically grow as single oval cells and reproduce by budding. *C. albicans* and some other pathogenic yeasts also can grow in multicellular chains called hyphae. Infection sites may contain both yeast and hyphal forms of the microorganism. Molds, such as *Trichophyton rubrum*, one of the causative agents of ringworm, grow in clusters of hyphae called a mycelium. All fungi produce

spores, which may be transported by direct contact or through the air. Although most topical fungal infections are readily treated, the incidence of life-threatening systemic fungal infections, including those caused by yeasts such as *C. albicans* and molds such as *Aspergillus fumigatus* are increasing, and mortality remains high (2).

Dermatophytes

Dermatophytes are fungi causing infections of skin, hair, and nails (3). The dermatophytes obtain nutrients by attacking the cross-linked structural protein keratin, which other fungi cannot use as a food source. Dermatophytic infections, known as tinea, are caused by various species of three genera (*Trichophyton*, *Microsporum*, and *Epidermophyton*) and are named for the site of infection rather than for the causative organism. Tinea capitis is a fungal infection of the hair and scalp. Tinea pedis refers to infections of the feet, including athlete's foot, tinea manuum to fungal infection of the hands, tinea cruris to infection of the groin (jock itch), and tinea unguium to infection of the fingernails. Athlete's foot in particular may be an infection involving several different fungi, including yeasts. Tinea unguium, also known as onychomycosis, whether of the fingernails or toenails, can be particularly difficult to treat, because the fungi invade the nail itself. Appropriate drug therapy prevents the fungus from spreading to the newly formed nail. Penetration of drugs into previously existing nail is problematic, however, and with some drug regimens, the infection is not cured until an entirely new, fungus-free nail has grown in. Because this can take months, patient compliance with a lengthy drug regimen can be a problem.

Yeasts

The most common cause of yeast infections is *Candida albicans*, which is part of the normal flora in a significant portion of the population where it resides in the oropharynx, gastrointestinal tract, vagina, and surrounding skin (4). It is the principal cause of vaginal yeast infections and oral yeast infections (thrush). These commonly occur in mucosal tissue when the normal population of flora has been disturbed by treatment of a bacterial infection with an antibiotic or when growth conditions are changed by hormonal fluctuations, such as occur in pregnancy. *C. albicans* can cause infections of the skin and nails, although the latter are not common. In persons with healthy immune systems, *Candida* infections are limited to superficial infections of the skin and mucosa. In persons with impaired immune systems, however, *C. albicans* also may cause deep-seated systemic infections, which can be fatal. Several other infections with *Candida* species occur, including *C. tropicalis*, *C. krusei*, *C. parapsilosis*, and *C. glabrata* (also known as *Torulopsis glabrata*). These organisms are becoming more common and often do not respond to antifungal therapy as readily as *C. albicans*.

CLINICAL SIGNIFICANCE



Antifungal agents include diverse compounds with varied actions. A few key examples can highlight the importance of medicinal chemistry to clinical practice. Knowledge about the molecular structure of polyene antifungal agents, such as amphotericin B, is essential for understanding how they work. These agents are macrocyclic lactones with distinct hydrophilic and lipophilic regions. One of the putative mechanisms of polyene action involves the formation of pores in the fungal cell membrane. The lipophilic regions of the polyene molecules facilitate the binding to the cell membrane sterols. The hydrophilic portions of the molecule align to create a hydrophilic pore in the sterol-containing cell membrane. As a result, there is membrane depolarization and increased membrane permeability and, eventually, fungal cell death. The lipophilic regions of amphotericin B also contribute to its poor solubility in aqueous solutions.

The traditional intravenous formulation of amphotericin B includes a dispersing agent, deoxycholate, which facilitates formation of the required micellar dispersion when administered in a 5% dextrose in water solution. 5-Flucytosine (5-FC) is an analogue of the natural pyrimidine cytosine that is converted

to 5-fluorouracil (5-FU) in susceptible fungi. Formation of 5-FU is essential to the antimycotic effect of 5-FC; 5-FU acts as a pyrimidine antimetabolite and is phosphorylated to the cytotoxic agent 5-fluorodeoxyuridine monophosphate. All of these facts are commonly emphasized in a medicinal chemistry sequence, and readers probably are aware that 5-FU is a chemotherapeutic agent that causes myelosuppression as its major toxicity. Therefore, it should not be surprising that the same side effect can be seen in patients receiving 5-FC. The newest antifungals, echinocandins, are macromolecular structures with high molecular weights (1,000 dalton), which can be visualized easily by looking at the chemical structure. Their relatively low volume of distribution in the body can be partially explained by their size. Clinically, these agents have not achieved high concentrations in the central nervous system and the vitreous chamber of the eye, two compartments that are subject to fungal invasion.

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Cryptococcus neoformans is a yeast commonly found in bird droppings, particularly pigeon droppings (5). When dust contaminated with spores is inhaled by persons with a competent immune system, the organism causes a minor, self-limiting, lung infection. Such infections frequently are mistaken for a cold, and medical treatment is not sought. In immunocompromised persons, however, the organism can be carried by the circulatory system from the lungs to many other organs of the body, including the central nervous system (CNS). Infection of the CNS is uniformly fatal unless treated.

Although most yeast infections are caused by various species of *Candida* or *Cryptococcus*, other yeasts also can cause infections in humans, including *Malassezia furfur*, *Trichosporon beigeli*, and *Blastoschizomyces capitatus* (6). These infections are relatively rare and are difficult to treat.

Thermally Dimorphic Fungi (Endemic Mycoses)

Thermally dimorphic fungi are saprophytes that grow in one form at room temperature and in a different form in a human host at 37°C (7). The most common infectious agents are *Blastomyces dermatitidis*, *Paracoccidioides brasiliensis*, *Coccidioides immitis*, and *Histoplasma capsulatum*, the causative agents of blastomycosis, paracoccidioidomycosis, coccidioidomycosis (valley fever), and histoplasmosis, respectively. All these organisms live in soil and cause disease through inhalation of contaminated dust. The resulting lung infections are often mild and self-limiting, but they may progress to a serious lung infection. The circulatory

system may transport the organisms to other tissues, where the resulting systemic infection may be fatal. *B. dermatitidis* is endemic to south central United States and *P. brasiliensis* to Central and South America, where it is the most common cause of fungal pulmonary infections. *C. immitis* is endemic to the dry areas of the southwestern United States and northern Mexico. It is particularly prevalent in the San Joaquin Valley of California, hence the name valley fever. *H. capsulatum* is endemic to the Mississippi and Ohio River valleys of the United States, where nearly 90% of the population tests positive for exposure to the organism.

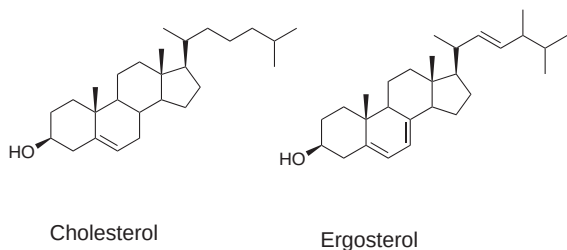
Molds

Various *Aspergillus* species are found worldwide and are virtually ubiquitous in the environment. The most common organisms causing disease are *A. fumigatus*, *A. niger*, and *A. flavus*. Several other *Aspergillus* species are known to cause infection, and some, such as *A. nidulans*, are becoming more common. *Aspergillus* spp. very rarely cause disease in persons with normal immune systems but are very dangerous to persons with suppressed immune systems. Because *Aspergillus* spores are everywhere, inhalation is the most common route of inoculation, but infection through wounds, burns, and implanted devices (e.g., catheters) also is possible. Nosocomial (hospital-derived) aspergillosis is a major source of infection in persons with leukemia and in those receiving organ or bone marrow transplants. Aspergillosis of the lungs may be contained, but systemic aspergillosis has a high mortality rate.

Zygomycosis (mucormycosis) is a term used to describe infections caused by the genera *Rhizopus*, *Mucor*, and *Absidia* of the fungal order Mucorales (8). As with several other opportunistic fungal pathogens, these soil microorganisms generally are harmless to those with a competent immune system but can cause rapidly developing, fatal infections in an immunosuppressed patient. These organisms can infect the sinus cavity, from which they spread rapidly to the CNS. Blood vessels also may be attacked and ruptured. Zygomycoses spread rapidly and are often fatal.

BIOCHEMICAL TARGETS FOR ANTIFUNGAL CHEMOTHERAPY

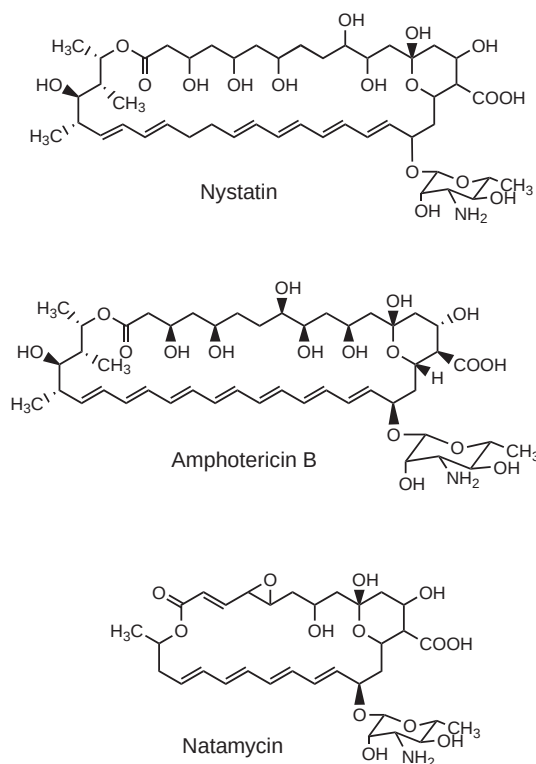
Antifungal chemotherapy depends on biochemical differences between fungi and mammals (9,10). Unlike bacteria, which are prokaryotes, both fungi and mammals are eukaryotes, and the biochemical differences between them are not as great as one might expect. At the cellular level, the greatest difference between fungal cells and mammalian cells is that fungal cells have both a cell membrane and an outer cell wall whereas mammalian cells have only a cell membrane. The fungal cell wall therefore is a logical target for a similar class of drugs, which would be expected to be potent antifungals yet have little human toxicity. Only recently, however, have a few potent inhibitors of fungal cell wall biosynthesis become available for clinical use (11). Other targets for antifungal agents include inhibitors of DNA biosynthesis, disruption of mitotic spindles, and general interference with intermediary metabolism. The difference between fungal and mammalian cells that is most widely exploited, however, is that the cell membranes of fungi and mammals contain different sterols. Sterols are important structural components of both fungal cell and mammalian cell membranes and are critical to the proper functioning of many cell membrane enzymes and ion-transport proteins. Mammalian cell membranes contain cholesterol as the sterol component, whereas fungal cell membranes contain ergosterol (12).



Although the two sterols are quite similar, the side chains are slightly different, and when three-dimensional models are constructed, the ring system of ergosterol is slightly flatter because of the additional double bonds in the B ring. Nevertheless, with only a few exceptions, this difference in sterol components provides the biochemical basis of selective toxicity for most of the currently available antifungal drugs.

Polyene Membrane Disruptors: Amphotericin B, Nystatin, and Congeners

Before the mid-1950s, effective antifungal therapy was limited to topical applications of undecylenic acid derivatives, mixtures of benzoic acid and salicylic acid, and a few other agents of modest efficacy. No reliable treatments existed for the few cases of deep-seated systemic fungal infections that did occur. The discovery of the polyene antifungal agents, however, provided a breakthrough into both a new class of antifungal agents and the first drug to be effective against deep-seated fungal infections (13). The polyenes are macrocyclic lactones with distinct hydrophilic and lipophilic regions. The hydrophilic region contains several alcohols, a carboxylic acid, and usually, a sugar. The lipophilic region contains, in part, a chromophore of four to seven conjugated double bonds. The number of conjugated double bonds correlates directly with antifungal activity in vitro and, inversely, with the degree of toxicity to mammalian cells. That is, not only are the compounds with seven conjugated double bonds, such as amphotericin B, approximately 10-fold more fungitoxic, they are the only ones that may be used systemically (14).



Mechanism of Action

The polyenes have an affinity for sterol-containing membranes, insert into the membranes, and disrupt membrane functions. The lipophilic polyene portion crosses the cell lipid bilayer forming a pore in the cell membrane. The membranes of cells treated with polyenes become leaky, and eventually, the cells die because of the loss of essential cell constituents, such as ions and small

organic molecules (15). Polyenes have a demonstrably higher affinity for membranes containing ergosterol over cholesterol-containing membranes (16). This is the basis for their greater toxicity to fungal cells. Some evidence suggests that the mechanism of insertion differs between the types of cells. Polyene molecules may insert individually into ergosterol-containing membranes but require prior formation of polyene micelles before inserting into cholesterol-containing membranes.

Specific Drugs

NYSTATIN Nystatin, the first clinically useful polyene antifungal antibiotic, is a conjugated tetraene isolated from cultures of the bacterium *Streptomyces noursei* in 1951 (13). Nystatin is an effective topical antifungal against a wide variety of organisms and is available in a variety of creams and ointments. Nystatin is too toxic to be used systemically, but because very little drug is absorbed following oral administration, it may be administered by mouth to treat fungal infections of the mouth and gastrointestinal tract (Table 35.1). Although nystatin itself was not a breakthrough in systemic antifungal therapy, the search for other polyenes led to the discovery of a polyene that can be used systemically.

AMPHOTERICIN B Amphotericin B, which as a heptaene has low enough toxicity to mammalian cells to permit intravenous (IV) administration, was discovered in 1956 (13). Amphotericin B is nevertheless a very toxic drug and must be used with caution. Adverse effects include fever, shaking chills, hypotension, and severe kidney toxicity. Despite its toxicity, amphotericin B is considered to be the drug of choice for many systemic, life-threatening fungal infections. The drug cannot cross the blood–brain barrier and must be administered intrathecally for treatment of fungal infections of the CNS. Closely related heptaenes are candicidin, hamycin, and trichomycin.

The nephrotoxicity of amphotericin B has been a serious drawback to the use of this drug since its introduction. Recently, however, the toxicity of the drug has been decreased substantially by changes in formulation (Table 35.2). The polyenes are only sparingly soluble in water, and amphotericin B has long been formulated as a complex with deoxycholic acid for IV administration. More recently developed formulations of amphotericin B, such as liposomal encapsulation and lipid complexes, have dramatically decreased the toxicity of the drug to humans, which permits higher plasma levels to be employed (17). The mechanisms by which the new

TABLE 35.1 Topical Antifungals

Chemical Class	Generic Name	Trade Name	Dosage Form
Allylamine	Butenafine	Lotrimin Ultra	1% cream
		Mentax	
	Naftifine	Naftin	1% cream, gel
	Terbinafine	Lamisil	1% cream
		Lamisil DermGel	10 mg/g gel
	Tolnaftate (thiocarbamate)	Various	1% cream, solution
		Absorbine Athlete's Foot Cream	1% cream
		Genaspor	1% cream
		Tinactin	1% cream, solution, powder, spray powder, spray liquid
		Aftate	1% gel, spray powder, spray liquid
Imidazole	Butoconazole	Femstat-3	2% cream
		Gynazole 1	2% cream
	Clotrimazole	Cruex	1% cream
		Lotrimin AF	1% cream, lotion, solution
		Desenex	1% cream
	Econazole	Various	1% cream
		Spectazole	1% cream
	Sertaconazole	Ertaczo	2% cream
	Ketoconazole	Nizoral	2% cream, shampoo
	Miconazole	Micatin	2% cream, powder, spray powder, spray liquid

(Continued)

TABLE 35.1 Topical Antifungals (*Continued*)

Chemical Class	Generic Name	Trade Name	Dosage Form
		Monistat-Derm	2% cream
		Maximum Strength Desenex	2% cream
		Fungoid Creme	2% cream
		Lotrimin AF	2% cream, spray powder
		Zeasorb AF	2% cream
		Prescription Strength Desenex	2% spray liquid
		Funcoide Tincture	2% solution
	Oxiconazole	Oxistat	1% cream, lotion
	Sulconazole	Exelderm	1% cream, solution
	Tioconazole	Vagistat-1	6.5% ointment
Triazole	Terconazole	Terazol	Topical 0.4%–0.8% cream, suppository
Polyene	Nystatin	Mycostatin	100,000 U/g cream, ointment, powder
		Nilstat	100,000 U/g cream, ointment
		Nystex	100,000 U/g cream, ointment
	Natamycin	Alcon	5% ophthalmic solution
Miscellaneous	Ciclopirox	Loprox	0.77% cream, gel, suspension
			1% shampoo
		Penlac	8% nail lacquer
	Haloprogin	Halotex	1% cream, solution
	Undecylenic acid	Protectol Medicated	15% powder
		Caldesene	10% powder
		Cruex	10% powder
		Cruex Aerosol	19% powder
		Desenex	25% powder
		Phicon F	8% cream
		Various	2%–25% creams, powders, ointments

formulations decrease the toxicity are not entirely clear, but altered distribution is clearly a factor. Because the blood vessels at the site of infection are more permeable than those of normal tissue, the large suspended particles of the lipid formulations can penetrate the site of infection more readily than they can penetrate healthy tissue. The result is selective delivery of drug to the site of infection. Some evidence also indicates that the newer formulations transfer amphotericin B to ergosterol-containing fungal cells more efficiently than to cholesterol-containing mammalian cells (17).

NATAMYCIN Natamycin, a tetraene, is available in the United States as a 5% suspension applied topically for the treatment of fungal infections of the eye (Table 35.1).

Ergosterol Biosynthesis Inhibitors

A schematic of fungal ergosterol biosynthesis starting from squalene is shown in Figure 35.1. The biosynthetic pathway has been simplified to emphasize steps important to the action of currently employed antifungal drugs (12). The last nonsteroidal precursor to both ergosterol and cholesterol is the hydrocarbon squalene. Squalene is converted to squalene epoxide by the enzyme squalene epoxidase. Squalene epoxide is then cyclized to lanosterol, the first steroid in the biosynthetic pathway. The steps involved in converting the side chain of lanosterol to the side chain of ergosterol, and the steps in removal of the geminal dimethyl groups on position 4, are not shown, because none of these reactions is targeted by clinically employed antifungal agents.

TABLE 35.2 Systemic Antifungal Agents

Chemical Class	Generic Name	Trade Name	Dosage Form
Allylamine	Terbinafine	Lamisil	250-mg tablets
Azole – imidazole	Ketoconazole	Various	200-mg tablets
		Nizoral	200-mg tablets
Azole – triazole	Fluconazole	Diflucan	50-, 100-, 150-, 200-mg tablets; 350-mg (10 mg/mL reconst.) powder for oral suspension; 100-, 200-mL (2 mg/mL) solution for injection
	Voriconazole	Vfend	50- and 200-mg tablets; 200-mg powder for injection; 45 g (40 mg/mL reconst.) powder for oral suspension
	Itraconazole	Various	100-mg capsules
		Sporanox	100-mg capsules; 10-mg/mL injection solution; oral solution 10 mg/mL
	Posaconazole	Noxafil	105-mL bottle, 40-mg/mL oral suspension
Echinocandins	Caspofungin	Cancidas	50- and 70-mg powder for injection
	Anidulafungin	Eraxis	50-mg powder for injection
	Micafungin	Mycamine	50-mg powder for injection
Polyene	Amphotericin B	Amphocin (desoxycholate)	50-mg powder for injection
		Fungizone (desoxycholate)	50-mg powder for injection
		Abelcet (lipid complex)	Suspension 100 mg/20 mL, single-use vials
		Amphotec (cholesteryl)	50 mg/20 mL and 100 mg/50 mL powder, single-use vials
		AmBisome (liposomal)	Powder, single-use vials
Miscellaneous	Flucytosine	Ancobon	250- and 500-mg capsules
	Griseofulvin	Fulvicin, Grifulvin, Grisactin	250- and 500-mg microsize tablets; 125- to 330-mg ultramicrosize tablets

A key step in conversion of lanosterol to both cholesterol and ergosterol is removal of the 14α -methyl group. This reaction is carried out by a cytochrome P450 enzyme, 14α -demethylase, also known as CYP51 (18). The mechanism of this reaction involves three successive hydroxylations of the 14α -methyl group, converting it from a hydrocarbon through the alcohol, aldehyde, and carboxylic acid oxidation states (Fig. 35.2). The methyl group is eliminated as formic acid to afford a double bond between C-14 and C-15 of the D ring. This enzyme is the primary target of the azole antifungal agents discussed below.

Eventually, either before or after modification of the side chain, the Δ^{14} double bond is reduced by a Δ^{14} -reductase to form a *trans* ring juncture between the C and D rings. Several steps later, the double bond between C-8 and C-9 is isomerized to a Δ^7 double bond by the enzyme Δ^8, Δ^7 -isomerase. Many of the steps are identical to those involved in mammalian cholesterol biosynthesis, and the basis for selective toxicity to fungal cells will be discussed under the specific agents.

Azoles—Imidazoles and Triazoles

Azole antifungal agents are the largest class of antimycotics available today, with more than 20 drugs on the market. Some are primarily used topically to treat superficial dermatophytic and yeast infections (Table 35.1), whereas others are administered orally for the treatment of systemic fungal infections (Table 35.2). The oral bioavailability of some azoles, in contrast to amphotericin B, combined with their generally broad spectrum of activity has led to their widespread use in treating a variety of serious infections. The characteristic chemical feature of azoles from which their name is derived is the presence of a five-membered aromatic ring containing either two or three nitrogen atoms. Imidazole rings have two nitrogens and triazoles three. In both cases, the azole ring is attached through N₁ to a side chain containing at least one aromatic ring. Imidazole-containing agents are shown in Figure 35.3 (for triazoles, see Fig. 35.6).

MECHANISM OF ACTION All the azoles act by inhibiting ergosterol biosynthesis through inhibition of the

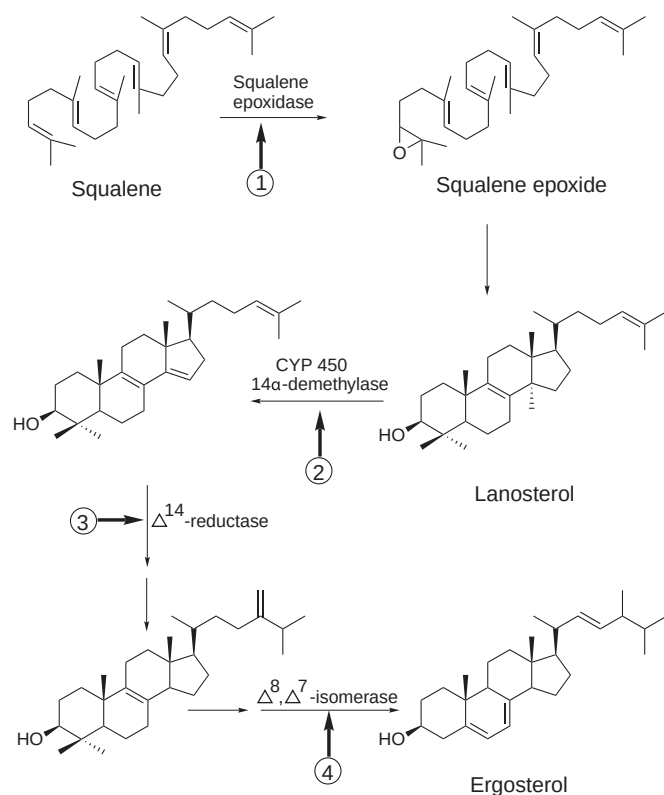


FIGURE 35.1 Ergosterol biosynthesis from squalene with key steps shown in this simplified figure. Enzymatic steps known to be the site of action of currently employed antifungal agents are indicated by a heavy black arrow and a number.

14 α -demethylase, CYP51, discussed earlier under ergosterol biosynthesis (Fig. 35.1, site 2). The basic N3 atom of the azole forms a bond with the heme iron of the CYP450 prosthetic group in the position normally occupied by the activated oxygen (Fig. 35.4). The remainder of the azole antifungal forms bonding interactions with the apoprotein in a manner that determines the relative selectivity of the drug for the fungal demethylase versus other CYP450 enzymes.

Inhibition of the 14 α -demethylase results in accumulation in the fungal cell membrane of sterols still bearing a 14 α -methyl group. These sterols do not have the exact shape and physical properties of the normal membrane sterol ergosterol. This results in permeability changes, leaky membranes, and malfunction of membrane-embedded proteins. These effects taken together lead to fungal cell death. Because biosynthesis of the mammalian membrane sterol cholesterol also employs a CYP450 14 α -demethylase, why do 14 α -methyl sterols not accumulate in human cell membranes? The reason is in the relative strength of inhibition of the same enzyme from different species. For example, the median inhibitory concentration (IC_{50}) for ketoconazole against the enzyme from *Candida albicans* is approximately 10^{-8} mol/L versus approximately 10^{-6} mol/L for mammalian enzymes (19). This two orders of magnitude difference in strength of

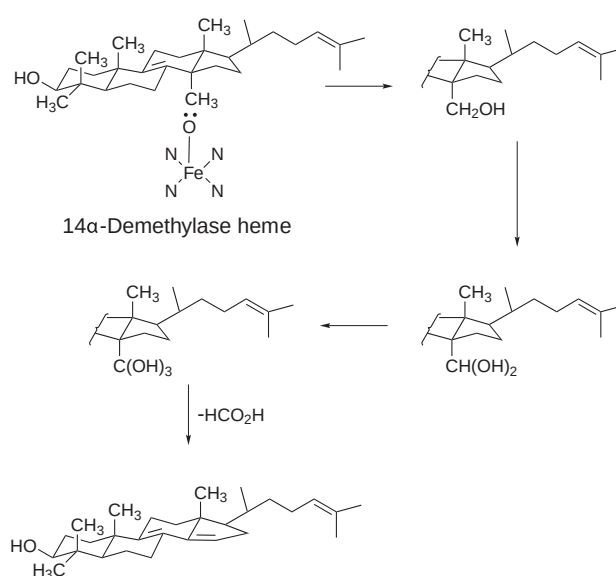


FIGURE 35.2 Demethylation of the 14 α -methyl group from lanosterol carried out by the cytochrome P450 enzyme sterol 14 α -demethylase, CYP51. The mechanism involves three successive heme-catalyzed insertions of activated oxygen into the three carbon-hydrogen bonds of the 14 α -methyl group which raises the oxidation state of the methyl group to a carboxylic acid. The group is finally eliminated as formic acid to create a double bond between carbons 14 and 15.

inhibition provides the therapeutic index with respect to this particular enzyme. As discussed later, however, many of the azoles are powerful inhibitors of other mammalian CYP450 enzymes.

The early azole antifungal drugs were all either extensively and rapidly degraded by first-pass metabolism or too toxic for systemic use. As a result, only those drugs with reduced or slow first-pass metabolism (ketoconazole, fluconazole, itraconazole, voriconazole, and posaconazole) are used systemically. The other azoles (clotrimazole, tioconazole, terconazole, butoconazole, econazole, oxiconazole, sulconazole, miconazole, and ketoconazole) are available in a variety of creams and ointments for topical treatments of dermatophytic infections and intravaginal use for vaginal yeast infections (Table 35.1).

Specific Drugs

KETOCONAZOLE Ketoconazole (Fig. 35.3), an imidazole antifungal, was the first orally active antifungal azole to be discovered and, as a consequence, has been widely studied and employed for the treatment of systemic fungal infections, primarily candidiasis. Ketoconazole has little effect on *Aspergillus* or *Cryptococcus*. Ketoconazole is highly dependent on low stomach pH for absorption, and antacids or drugs that raise stomach pH will lower the bioavailability of ketoconazole. As with other azoles, it is extensively metabolized by microsomal enzymes (Fig. 35.5). All the metabolites are inactive. Evidence that CYP3A4 plays a significant role in metabolism of

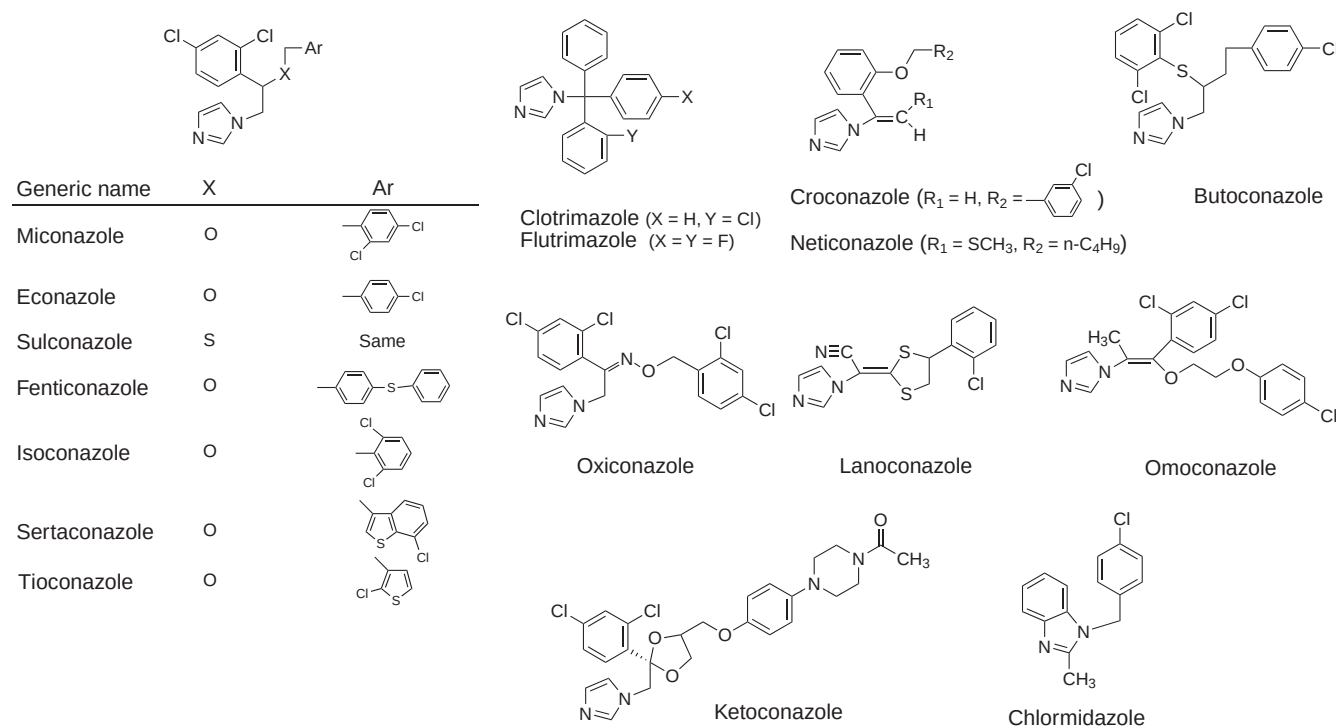


FIGURE 35.3 Imidazole antifungal agents.

ketoconazole is that coadministration of CYP3A4 inducers, such as phenytoin, carbamazepine, and rifampin, can cause as much as a 50% reduction in levels of ketoconazole (20). Ketoconazole also is a powerful inhibitor of human CYP3A4 and, as a consequence, has many serious interactions with other drugs. For example, coadministration of ketoconazole with the hypnotic triazolam results in a 22-fold increase in triazolam's area under the curve (AUC) and a sevenfold increase in half-life. CYP3A4 also is present in the gut and may contribute substantially to the metabolism of many drugs, such as the immunosuppressant agent cyclosporine, thus the potential for drug–drug interaction. Ketoconazole is a weak inhibitor of CYP2C9,

which is the enzyme responsible for the metabolism of several narrow therapeutic index drugs, such as warfarin and phenytoin. As better systemic agents have become available, ketoconazole's clinical use has become limited to topical applications in a variety of dosage forms, including creams, lotions, suppositories, and shampoos.

ITRACONAZOLE Itraconazole was, along with fluconazole, one of the first triazoles introduced into clinical use (Fig. 35.6) (21). Itraconazole's oral bioavailability is variable and is influenced by food and stomach pH, a strongly acidic pH being required for good absorption. Like ketoconazole, itraconazole is extensively metabolized by CYP3A4 following oral administration, and levels are markedly reduced by coadministration of the CYP3A4-inducers phenytoin, carbamazepine, and rifampin (20). Additionally, like ketoconazole, itraconazole has been demonstrated to be a strong inhibitor of CYP3A4 (22). This interaction has proven to be of clinical significance because of the risk of developing rhabdomyolysis following lovastatin or simvastatin therapy with coadministration of itraconazole (23–25). Therefore, itraconazole is likely to have serious interactions with any other drug metabolized by CYP3A4. Again like ketoconazole, itraconazole appears to have little or no effect on CYP2C9-mediated metabolism of warfarin and phenytoin.

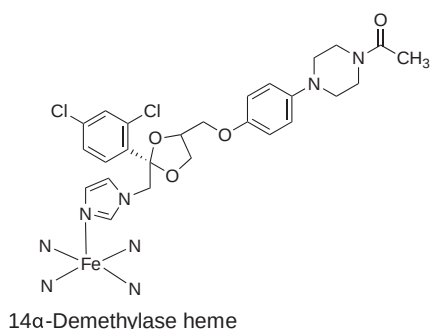


FIGURE 35.4 Mechanism of azole/CYP450 binding. The basic nitrogen of azole antifungal agents forms a bond to the heme iron of CYP450 enzymes, preventing the enzyme from oxidizing its normal substrates. Ketoconazole is representative of the azole antifungals.

TERCONAZOLE Terconazole (Fig. 35.6) is a close analog of ketoconazole and itraconazole. It is approved only for the treatment of vaginal candidiasis and is not used systemically (Table 35.1) (26,27).

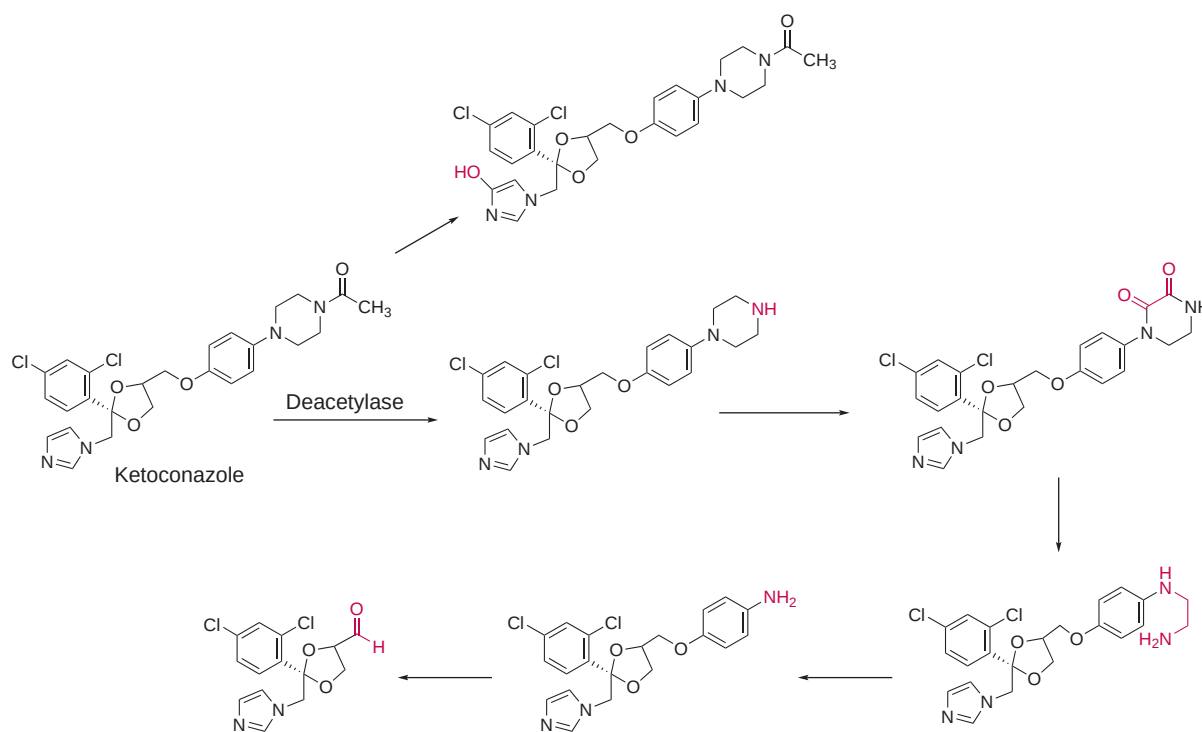


FIGURE 35.5 Extensive metabolism of ketoconazole involving hydrolysis of the *N*-acetyl by a deacetylase. The oxidation reactions are catalyzed by CYP3A4 and flavin-linked mixed-function oxidase. All metabolites are inactive.

FLUCONAZOLE Fluconazole (Fig. 35.6), which was introduced at the same time as itraconazole, differs from ketoconazole and itraconazole in that it is equally bioavailable when given orally or IV. Two major advantages of fluconazole over other antifungal agents are that it can cross the blood–brain barrier and has efficacy against *Cryptococcus neoformans* (28). Fluconazole also differs in that it is only a weak inhibitor of CYP3A4 but a strong inhibitor of CYP2C9 (29). For instance, fluconazole doubles the AUC of (*S*)-warfarin (the active enantiomer) and greatly prolongs the prothrombin time in patients receiving warfarin anticoagulant therapy (30). Because warfarin has such a narrow therapeutic index and excessive anticoagulation can be extremely harmful, this interaction is considered to be of major clinical significance. Fluconazole also decreases the metabolism of the CYP2C9 substrate phenytoin, an antiepileptic agent that also has a narrow therapeutic index (31). Depending on the dose of fluconazole, coadministration with phenytoin can result in a 75% to 150% increase in the phenytoin AUC, and numerous case reports have documented substantial adverse effects following this regimen. Fluconazole also will inhibit CYP3A4, although not to the same degree as ketoconazole and itraconazole. Fluconazole exhibits a dose-dependent inhibition of triazolam metabolism (a CYP3A4 reaction) causing as much as a fourfold increase in triazolam AUC.

Fluconazole has limited water solubility, and current IV formulations of only 2 mg/mL require high-volume infusions, which is a drawback for rapid administration in

seriously ill patients. For this reason, a highly water-soluble fluconazole prodrug, fosfluconazole (Fig. 35.6), is being developed with a water solubility greater than 100 mg/mL (32). Fosfluconazole is a phosphate ester that is rapidly converted to fluconazole in vivo by alkaline phosphatases. This new prodrug allows much lower volume bolus injections to be administered, with over 90% of the prodrug rapidly converted to active fluconazole (33,34).

VORICONAZOLE Voriconazole (Fig. 35.6) is a fluconazole analog that was developed to overcome some of the limitations of fluconazole (35) and does, indeed, have a broader spectrum of activity than fluconazole, having activity against *Aspergillus* and fluconazole-resistant strains of *Candida* and *Cryptococcus* (36). Voriconazole is orally absorbed and penetrates the blood–brain barrier. Unfortunately, voriconazole is extensively metabolized by CYP450 enzymes (Fig. 35.7) and is an inhibitor of CYP2C19, CYP2C9, and CYP3A4, leading to many drug interactions (37,38). Voriconazole exhibits non-linear, saturable kinetics, and because CYP2C19 exhibits genetic polymorphisms, plasma levels can be higher in poor metabolizers versus extensive metabolizers (39,40).

POSACONAZOLE Posaconazole (Fig. 35.6), a recently introduced triazole, has a number of advantages over previous agents (41,42). Posaconazole has a wide spectrum of activity compared to other azoles, particularly against *Aspergillus* and other increasingly common nosocomial

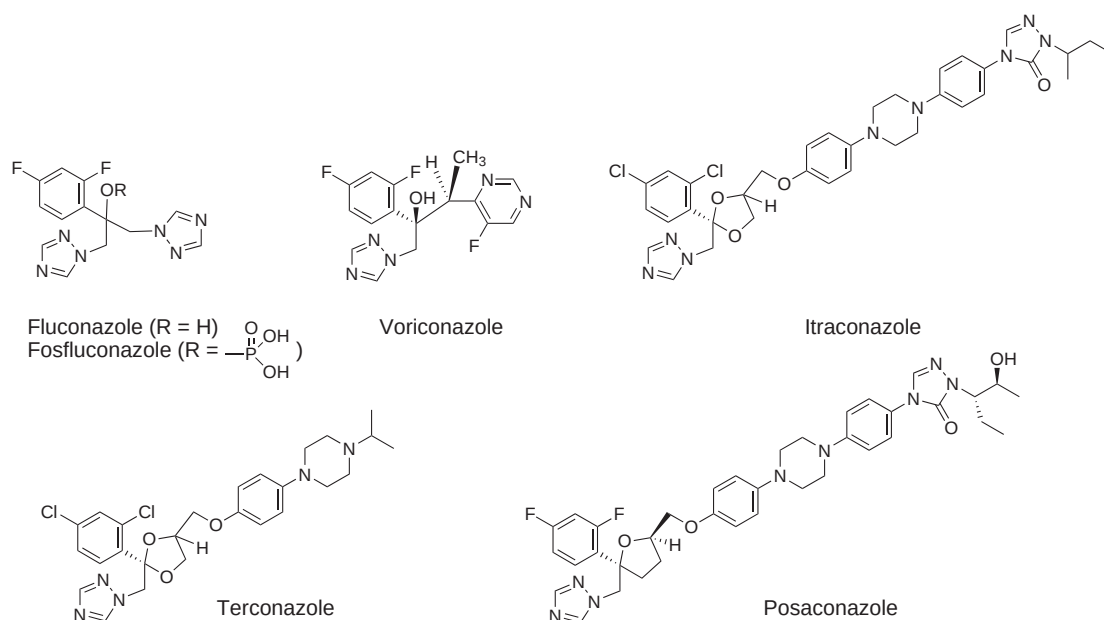


FIGURE 35.6 Triazole antifungal agents.

infections resistant to treatment by other antifungal drugs. *Aspergillus* resistance to azole antifungals has been attributed to the unique presence in *Aspergillus* of two distinct 14 α -demethylases CYP51A and CYP51B (43). *Aspergillus* CYP51B is sensitive to fluconazole and itraconazole, but CYP51A remains functional, allowing the

fungus to synthesize needed ergosterol. Posaconazole is active against both *Aspergillus* CYP51 enzymes accounting for its greater activity against the fungus compared to other azoles (44).

Posaconazole is metabolized primarily by phase 2 glucuronide conjugation and has little interaction with most oxidative CYP450 drug-metabolizing enzymes, but is an inhibitor of CYP3A4 (45,46). Posaconazole is structurally similar to itraconazole and saperconazole, but it contains a tetrahydrofuran ring in place of the dioxolan ring of those agents, which may account for some of its unique properties.

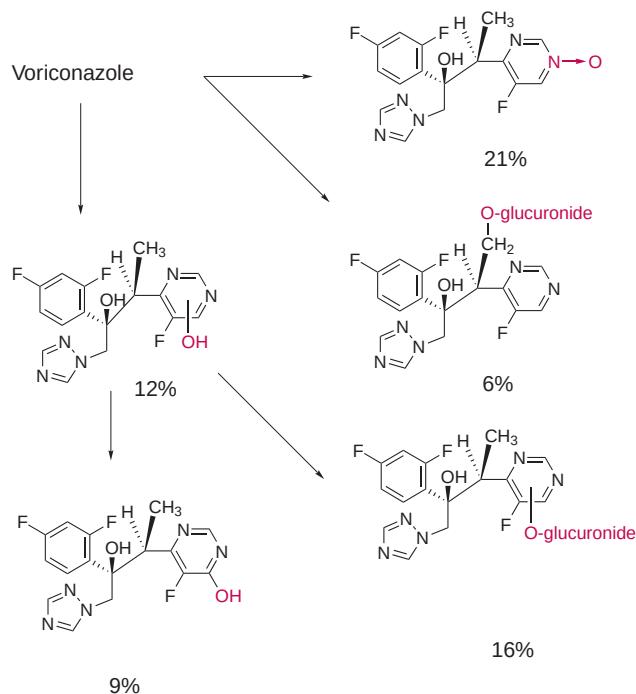


FIGURE 35.7 Major metabolic products resulting from CYP450 metabolism of voriconazole.

Azole Antifungals and P-Glycoprotein

In addition to being substrates and inhibitors of various cytochrome P450 enzymes, azole antifungals can also be inhibitors and/or substrates for the multi-drug resistance enzyme P-glycoprotein. As discussed in Chapters 4 and 5, P-glycoprotein is an adenosine triphosphate-dependent pump that expels some drugs from blood circulation into the lumen of the small intestine and is also a component part of the blood-brain barrier. Therefore, P-glycoprotein is an important contributor to drug distribution and plasma levels and can be a major factor in drug-drug interactions. Fluconazole (47) and voriconazole (48) have little or no effect on P-glycoprotein, being neither substrates nor inhibitors. Itraconazole and ketoconazole bind tightly to P-glycoprotein and appear to be substrates for the transporter as well as strong inhibitors of its actions on other drugs (47). The most recently introduced azole, posaconazole, is also both a substrate and inhibitor of P-glycoprotein (49).

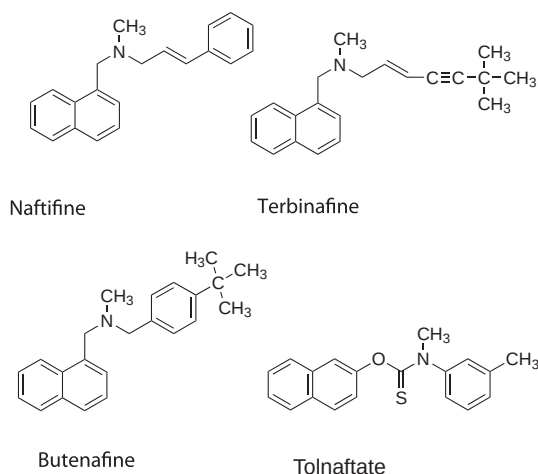


FIGURE 35.8 The squalene epoxidase inhibitors, allylamines. Naftifine was the first drug shown to act by inhibition of squalene epoxidase as was the much older thiocarbamate, tolnaftate.

Allylamines and Other Squalene Epoxidase Inhibitors

The group of agents generally known as allylamines (50) strictly includes only naftifine and terbinafine, but because butenafine and tolnaftate function by the same mechanism of action, they are included in this class and are shown in Figure 35.8. One can, of course, consider the benzyl group of butenafine to be bioisosteric with the allyl group of naftifine and terbinafine. Tolnaftate, a much older drug, is chemically a thiocarbamate but has the same mechanism of action as the allylamines. These drugs have a more limited spectrum of activity than the azoles and are effective only against dermatophytes. Therefore, they are employed in the treatment of fungal infections of the skin and nails (51).

MECHANISM OF ACTION All of the drugs in Figure 35.8 act through inhibition of the enzyme squalene epoxidase (Fig. 35.1, site 1). Inhibition of this enzyme has two effects, both of which appear to be involved in the fungitoxic mechanism of this class (52). First, inhibition of squalene epoxidase results in a decrease in total sterol content of the fungal cell membrane. This decrease alters the physicochemical properties of the membrane, resulting in malfunctions of membrane-imbedded transport proteins involved in nutrient transport and pH balance. Second, inhibition of squalene epoxidase results in a buildup within the fungal cell of the hydrocarbon squalene, which is itself toxic when present in abnormally high amounts. Mammals also employ the enzyme squalene epoxidase in the biosynthesis of cholesterol, but a desirable therapeutic index arises from the fact that the fungal squalene epoxidase enzyme is far more sensitive to the drugs than the corresponding mammalian enzyme. Terbinafine has a K_i of 0.03 $\mu\text{mol/L}$ versus squalene epoxidase from *Candida*

albicans but a K_i of only 77 $\mu\text{mol/L}$ versus the same enzyme from rat liver—a 2,500-fold difference (53).

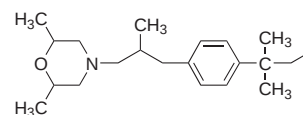
Specific Drugs

NAFTIFINE Naftifine (Fig. 35.8) was the first allylamine to be discovered and marketed (50). Because of its extensive first-pass metabolism, naftifine is not used orally and is only available in topical preparations (Table 35.1). The widest use of naftifine is against various tinea infections of the skin.

TERBINAFINE Terbinafine (Fig. 35.8) is available in both topical and oral dosage forms (Tables 35.1 and 35.2) and is effective against a variety of dermatophytic infections when employed topically or systemically (53). A unique property of terbinafine is its effectiveness in the treatment of onychomycoses (nail infections) (54). Given orally, the highly lipophilic drug redistributes from the plasma into the nail bed and into the nail itself, where the infection resides (55,56), making terbinafine superior to other agents for treating this particular type of infection. Terbinafine is extensively metabolized by several CYP450 enzymes, including CYP1A2, CYP2C19, CYP2C9, CYP2C8, CYP3A4, and CYP2B6 (57). Because there are so many pathways for terbinafine metabolism, inhibition of any one has very little effect on overall clearance of the drug, although drugs that inhibit several CYP450 enzymes, such as cimetidine, can increase terbinafine plasma levels. Although not a substrate for the enzyme, terbinafine is a strong inhibitor of CYP2D6 and can have significant interactions with drugs that are metabolized by this enzyme, such as codeine and desipramine (29,58).

BUTENAFINE AND TOLNAFTATE Butenafine and tolnaftate, like naftifine (Fig. 35.8), are only available in topical preparation for the treatment of dermatophytic infections (Table 35.1). Tolnaftate has been marketed in a variety of nonprescription drug preparations for decades. Butenafine, discovered more recently, has a somewhat wider spectrum of activity than tolnaftate. For example, butenafine is active against superficial *Candida albicans* infections, which are not affected by tolnaftate (59).

Morpholines



Amorolfine

Amorolfine is the only drug in this class that is employed clinically in the treatment of human fungal infections. Amorolfine is not currently available in the

United States, but it is marketed in Europe and Asia for the topical treatment of dermatophytic infections (60).

Morpholine antifungals inhibit ergosterol biosynthesis by acting on the enzymes Δ^{14} -reductase and Δ^8, Δ^7 -isomerase (Fig. 35.1, sites 3 and 4) (61). Inhibition of these enzymes results in incorporation into fungal cell membranes of sterols retaining a Δ^{14} double bond, a Δ^8 double bond, or both. None of these will have the same overall shape and physicochemical properties as the preferred sterol, ergosterol. As with the antifungals already discussed, this results in membranes with altered properties and malfunctioning of membrane-embedded proteins.

Inhibitors of Cell Wall Biosynthesis—Echinocandins

The most notable difference between fungal and mammalian cells is that fungi have a cell wall and mammals do not. Drugs interfering with cell wall biosynthesis would be expected to be relatively nontoxic to mammals. Such drugs have been the foundation of antibacterial therapy since the discovery of penicillin and the development of dozens of effective penicillins and cephalosporins. Only recently, however, have a few drugs affecting fungal cell wall biosynthesis become available. Echinocandins, a group of cyclic peptides with long lipophilic side chains and sometimes called lipopeptides, have been under investigation for a number of years (Fig. 35.9) (62). Echinocandins interfere with cell wall biosynthesis through inhibition of the enzyme β -1,3-glucan synthase. β -Glucan is an important polymer component of many fungal cell walls, and reduction in the glucan content severely weakens the cell wall, leading to rupture of the fungal cell.

Caspofungin, Anidulafungin, and Micafungin

Recently, three semisynthetic echinocandins have been approved for use in treating life-threatening systemic fungal infections (63). These are caspofungin, anidulafungin, and micafungin (Fig. 35.9). These

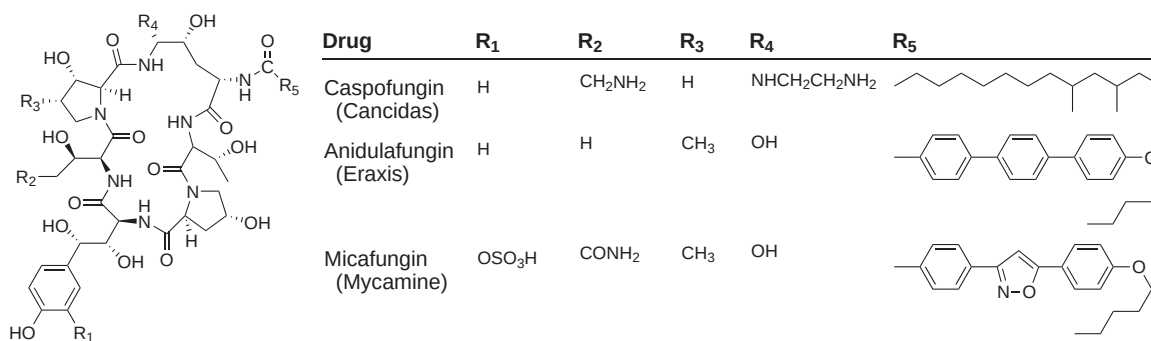


FIGURE 35.9 Echinocandins.

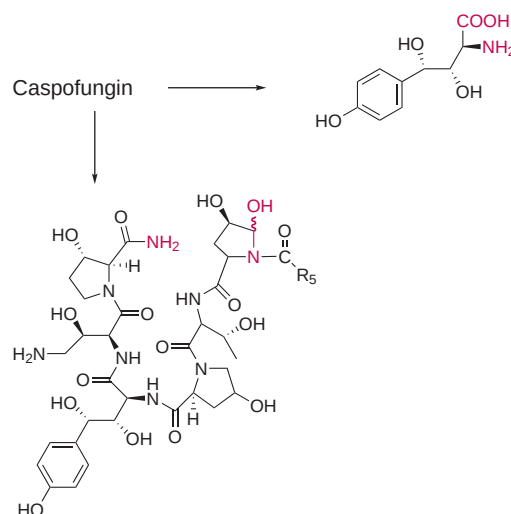


FIGURE 35.10 Metabolic products formed from caspofungin.

drugs represent the first class of antifungal agents with a novel mechanism of action to be marketed in more than 35 years and are a very valuable contribution to therapy for systemic fungal infections. They are effective against a variety of *Candida* species that have proven to be resistant to other agents as well as effective against azole-resistant *Aspergillus*. Therefore, these drugs are truly life saving for those afflicted with these previously resistant fungi. Unfortunately, these echinocandins are not effective against *Cryptococcus neoformans*. None of these drugs is orally active, and all must be administered by IV infusion. Because of limited hepatic metabolism, drug-drug interactions are not a problem. Caspofungin is metabolized by hydrolysis in two portions of the hexapeptide ring (Fig. 35.10) (64), whereas anidulafungin does not appear to be actively metabolized but rather slowly degrades (65). Micafungin is metabolized by a sulfotransferase and by catechol-*O*-methyltransferase, but no significant drug interactions are known (66).

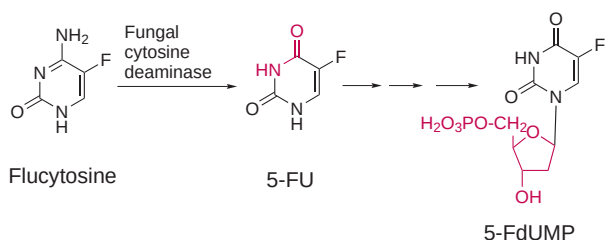


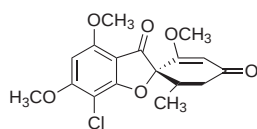
FIGURE 35.11 Flucytosine, a prodrug, is converted by fungal cytosine deaminase to 5-fluorouracil (5-FU). This reaction does not occur in mammalian cells. A further transformation of 5-FU to the actual cytotoxic agent 5-fluorodeoxyuridinemonophosphate (5-FdUMP) also occurs.

Drugs Acting Through Other Mechanisms

Flucytosine

Flucytosine is a powerful antifungal agent used in the treatment of serious systemic fungal infections, such as *Cryptococcus neoformans* and *Candida* spp. (Table 35.2). Flucytosine itself is not cytotoxic but, rather, is a prodrug that is taken up by fungi and metabolized to 5-fluorouracil (5-FU) by fungal cytosine deaminase (Fig. 35.11) (67). Then, 5-FU is converted to 5-fluorodeoxyuridine, which is a thymidylate synthase inhibitor that interferes with both protein and RNA biosynthesis. 5-Fluorouracil is cytotoxic and is employed in cancer chemotherapy (see Chapter 37). Human cells do not contain cytosine deaminase and, therefore, do not convert flucytosine to 5-FU. Some intestinal flora, however, do convert the drug to 5-FU, so human toxicity does result from this metabolism. Resistance rapidly develops to flucytosine when used alone, so it is almost always used in conjunction with amphotericin B. Use of flucytosine has declined since the discovery of fluconazole.

Griseofulvin

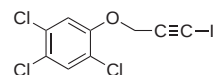


Griseofulvin

Griseofulvin is an antifungal antibiotic produced by an unusual strain of *Penicillium* (68). It is used orally to treat superficial fungal infections, primarily fingernail and toenail infections, but it does not penetrate skin or nails if used topically (Table 35.2). When given orally, however, plasma-borne griseofulvin becomes incorporated into keratin precursor cells and, ultimately, into keratin, which cannot then support fungal growth. The infection is cured when the diseased tissue is replaced

by new, uninfected tissue, which can take months. The mechanism of action of griseofulvin is through binding to the protein tubulin, which interferes with the function of the mitotic spindle and, thereby, inhibits cell division. Griseofulvin also may interfere directly with DNA replication. Griseofulvin is gradually being replaced by newer agents (69).

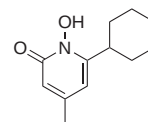
Haloprogin



Haloprogin

Haloprogin is an iodinated acetylene active against dermatophytes (70). Haloprogin is only used for topical applications (Table 35.1). The mechanism of haloprogin is not clear, but it appears to lead to nonspecific metabolic disruption. It has been demonstrated to interfere with DNA biosynthesis and cell respiration.

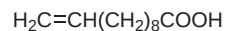
Ciclopirox



Ciclopirox

Ciclopirox is a hydroxylated pyridinone that is employed for superficial dermatophytic infections, principally onychomycosis. Ciclopirox has a unique mechanism of action through chelation of polyvalent cations, such as Fe^{3+} , which causes inhibition of a number of metal-dependent enzymes within the fungal cell. Although ciclopirox has been available for more than 30 years, a new formulation of an 8% lacquer has been recently introduced for treating nail infections (71).

Undecylenic Acid



Undecylenic acid

Undecylenic acid is widely employed, frequently as the zinc salt, in over-the-counter preparations for topical treatment of infections by dermatophytes (Table 35.1) (72). Undecylenic acid is a fungistatic that acts through a nonspecific interaction with components in the fungal cell membrane.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Douglas Slain, PharmD, BCPS

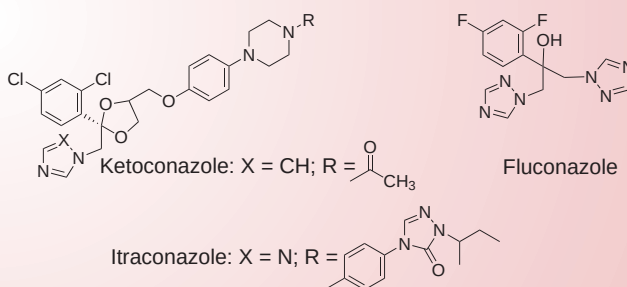
The patient was told to go to the emergency room at the hospital. They measured her plasma digoxin concentration, which was found to be very elevated (3.3 ng/mL). The pharmacist at the emergency department found reports in the medical literature of interactions between itraconazole and digoxin. The team monitored and stabilized the patient. They decided to lower her digoxin dose by 50%. A week later, digoxin concentration was again tested and was within the desired range (1.9 ng/mL), and there were no signs of toxicity. The patient was able to remain on the reduced digoxin dose while finishing the course of itraconazole. Fluconazole does not have an interaction with digoxin, but it is not as effective for treating histoplasmosis.

Chemical Analysis

Victoria Roche, PhD, and S. William Zito, PhD

The interaction is due to the fact that itraconazole is a P-glycoprotein inhibitor. Digoxin is a substrate of P-glycoprotein. From a chemical standpoint, itraconazole and ketoconazole bind tightly to P-glycoprotein and serve as significant inhibitors and substrates. Structures that bind to P-glycoprotein are often lipophilic entities with molecular weights between 300

and 1,000. Conazole antifungal agents commonly have a structural pattern of three electron donor groups separated by 4.6 Angstroms, a motif known to be recognized by this efflux protein (73). Fluconazole does not appear to bind to P-glycoprotein. It is a much smaller structure than either itraconazole or ketoconazole, and it is significantly less lipophilic. As a result, it has been claimed to distribute ineffectively across membranes and experience difficulty reaching and interacting with the P-glycoprotein binding site (47). Therefore, a P-glycoprotein-mediated interaction with digoxin would not be expected from fluconazole.



Agents that are CYP3A4 substrates and inhibitors are often P-glycoprotein substrates and inhibitors. Newer azoles like voriconazole and posaconazole appear to bind to P-glycoprotein but have not been as well studied.

CASE STUDY

Victoria Roche and S. William Zito

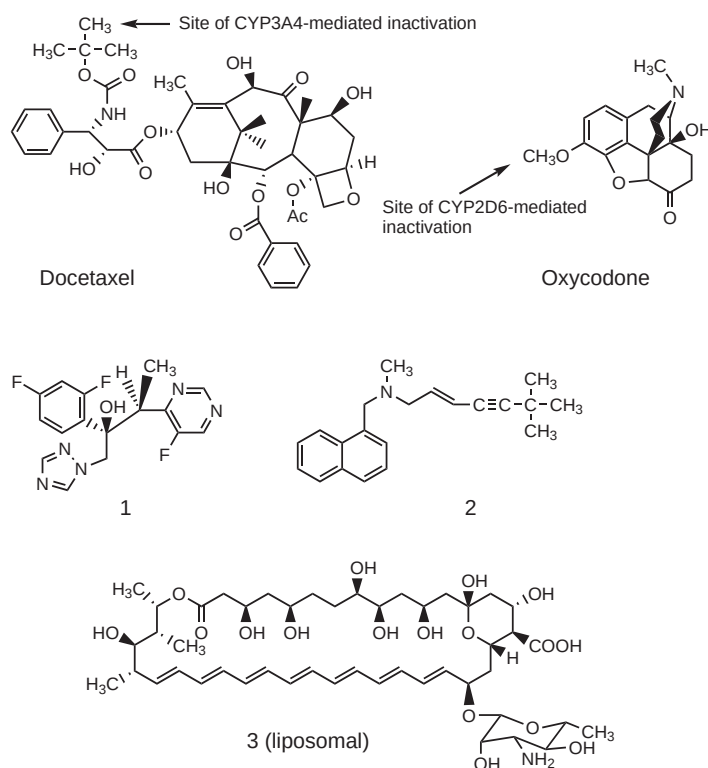
HN is a 64-year-old custodian at a brick plant located in the deep South. He is currently being treated for prostate cancer, and the antineoplastic drug he's on (docetaxel) has left him feeling weak and tired. He knows this drug is toxic to the bone marrow and leaves him vulnerable to infection. Despite feeling lethargic, he really needs the money (not to mention the company's insurance) and is currently cleaning six nights a week. When he's around people he wears a surgical mask to minimize his exposure to bacteria- and virus-laden environmental hazards. However, he works the night shift and is often alone, so he doesn't usually wear the mask while he's cleaning.

From a weather-related perspective, this was a horrible spring, with many devastating tornadoes and monsoon-like rainstorms pelting the region. A tornado-damaged roof allowed several areas of the large plant to flood, and the steamy late June heat that's now descended is close to intolerable. For the past several weeks HN had noticed a musty smell in several areas of the

plant, and told his supervisor about a greenish-brown discoloration that was forming on some ceiling tiles and drywall in the more remote maintenance-related workrooms. He has been trying to clean the drywall but has not had much permanent success. Recently he began to experience new and disturbing symptoms, including shortness of breath, headaches, and nosebleeds. When he reported this to his oncologist, he was sent to an infectious disease specialist who diagnosed an invasive aspergillosis infection.

As the pharmacist working with the infectious disease team, you know that systemic mold infections are tough to treat, and HN's immunocompromised status is not helping matters (in fact, it contributed significantly to his fungal susceptibility). Taking into account that his anticancer drug (docetaxel) is inactivated exclusively by CYP3A4-mediated aliphatic hydroxylation and his pain medication (oxycodone) is activated by CYP2D6-mediated O-dealkylation, you consider the structure of the following antifungal candidates.

(continued)



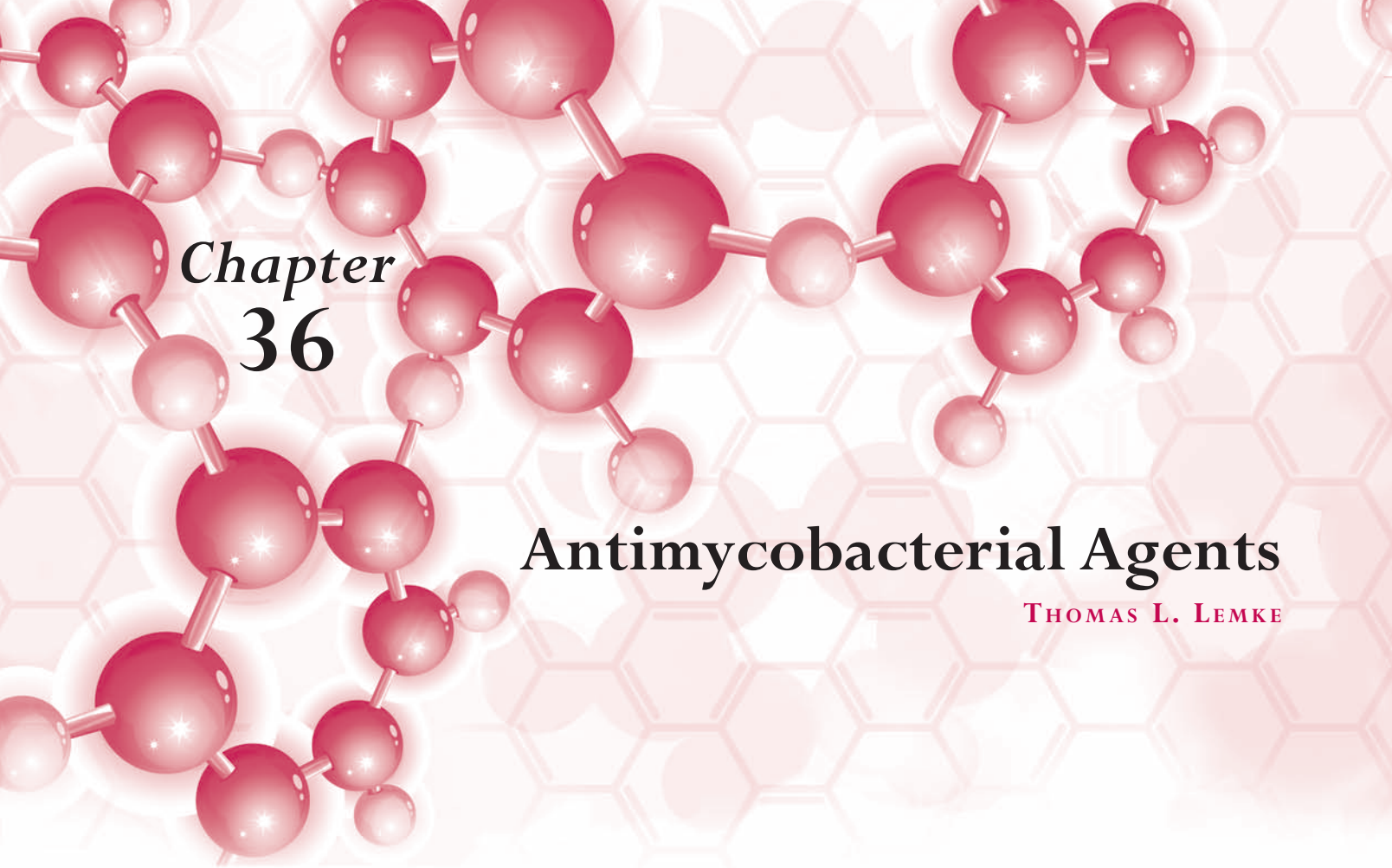
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 36

Antimycobacterial Agents

THOMAS L. LEMKE

Drugs Covered in This Chapter

ANTITUBERCULIN DRUGS

- Capreomycin
- Cycloserine
- Ethambutol
- Ethionamide
- Isoniazid
- Kanamycin
- Para-aminobenzoic acid

- Pyrazinamide
- Rifabutin
- Rifampin
- Rifapentine
- Streptomycin

MAC THERAPY

- Azithromycin
- Clarithromycin

LEPROSTATIC DRUGS

- Clofazimine
- Dapsone
- Rifampin
- Thalidomide

Abbreviations

AG, arabinogalactan

ATP, adenosine triphosphate

CDC, Centers for Disease Control and Prevention

CNS, central nervous system

DDRP, DNA-dependent RNA polymerase

DOTS, direct observed treatment, short-course

EMB, ethambutol

ENL, erythema nodosum leprosum

FDA, U.S. Food and Drug Administration

GI, gastrointestinal

G6PD, glucose-6-phosphate dehydrogenase

INH, isoniazid

LAM, lipoarabinomannan

MAC, *Mycobacterium avium*–intracellulare complex

MDR-TB, multidrug-resistant tuberculosis

PAS, *p*-aminosalicylic acid

RIF, rifampin

SAR, structure–activity relationship

STM, streptomycin

TB, tuberculosis

XTR-TB, extensively drug-resistant tuberculosis

SCENARIO

Elizabeth Coyle, PharmD

RG is a 48-year-old man who had a positive tuberculin skin test (PPD). One week ago he began taking isoniazid 300 mg daily for 9 months for the treatment of latent tuberculosis (TB). RG's medical history is unremarkable except for a right lower extremity deep vein thrombosis (DVT) 3 months ago, for which he is taking warfarin 5 mg daily. His anticoagulation has been fine with his International Normalization Ratio (INR)

therapeutic at 2.5 when it was taken 2 weeks ago. However, over the last week he has noticed increased bruising, and his INR is now 5.5. He is taking no other medications, and his diet has stayed the same.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

GENERAL CONSIDERATIONS

Mycobacteria are a genus of acid-fast bacilli belonging to the Mycobacteriaceae, which include the organisms responsible for tuberculosis and leprosy as well as a number of other, less common diseases. Characteristic of mycobacteria is that these organisms tend to be slow-growing and difficult to stain and, when they are stained with basic dye, can resist decolorization with acid alcohol. The staining characteristics relate to the abnormally high lipid content of the cell wall. In fact, the cell wall or cell envelope of the mycobacterium holds the secret to many of the characteristics of this genus of organisms. The cell envelope is unique in both structure and complexity. It has been suggested that the cell envelope is responsible for mycobacterium pathogenicity or virulence, multiple drug resistance, cell permeability, immunoreactivity, and inhibition of antigen responsiveness, as well as disease persistence and recrudescence. In addition, several of the successful chemotherapeutic agents are known to inhibit the cell envelope synthesis as their mechanism of action. It is no wonder that significant effort has been put forth to define the chemical structure of the mycobacterium cell envelope.

A series of papers were presented and reported in 1991 dealing with the topic of the structure and functions of the cell envelope of mycobacterium (1). As illustrated in Figure 36.1, the mycobacterial cell envelope contains, on the interior surface, a plasma membrane similar to that found in most bacteria. A conventional peptidoglycan layer affording the organism rigidity appears next. This layer is composed of alternating *N*-acetyl- β -glucosamines (Glu) linked to *N*-glycyl- β -muramic acids (Mur) through 1–4 linkages that, in turn, is attached to the peptido chain of β -alanine (A), β -glutamine (G), meso-diaminopimelic acid (DP), and L-alanine (A). A novel disaccharide phosphodiester linker made up of *N*-acetyl- β -glucosamine and rhamnose connects the muramic acid to a polygalactan and polyarabinose chain. The latter polysaccharides are referred to as the arabinogalactan (AG) portion of the cell envelope. The manner in which the arabinosyl and galactosyl residues are arranged is still under investigation. It is known that the arabinosyl chains terminate in mycolic acid residues. (The mycolates will be discussed in more detail later in this chapter.) Noncovalently bound

to the mycolates are a number of free nonpolar and polar lipids (the phthiocerol lipids and the glycopeptidolipids, respectively). Finally, spanning from the interior, embedded in the plasma membrane, to the exterior is the lipopolysaccharide (LPS) polymer. As indicated, this unit is composed of polyarabinose, polymannan, and various lipids attached through a phosphatidylinositol moiety (2,3).

SPECIFIC DISEASES

Leprosy (Hansen Disease)

Throughout the Bible, one finds reference to the condition of leprosy, such as that described in *Leviticus*: “[I]f there be any flesh in the skin of which there is a burn by fire and the quick flesh of the burn becomes a bright spot, reddish white or white, ... and if the hair in the bright spot is turned white and it appears deeper than the skin, it is leprosy broken out in the burn.” Associated with the disease was a belief that individuals suffering from this disease were unclean. Today, leprosy (Hansen disease) is recognized as a chronic granulomatous infection caused by *Mycobacterium leprae*. The disease can consist of lepromatous leprosy, tuberculoid leprosy, or a condition with characteristics between these two poles and referred to as borderline leprosy. The disease is more common in tropical countries but is not limited to warm climate regions. It is thought to afflict some 10 to 20 million individuals. Children appear to be the most susceptible population, but the signs and symptoms do not usually occur until much later in life. The incubation period is usually 3 to 5 years. The disease is contagious, but the infectiousness is quite low. Person-to-person contact appears to be the means by which the disease is spread, with entrance into the body occurring through the skin or the mucosa of the upper respiratory tract. Skin and peripheral nerves are the regions most susceptible to attack.

The first signs of the disease consist of hypopigmented or hyperpigmented macules. Anesthetic or paresthetic patches can be additionally experienced by the patient. Neural involvement in the extremities leads ultimately to muscle atrophy, resorption of small bones, and spontaneous amputation. When facial nerves are involved, corneal ulceration and blindness can occur. The identification of

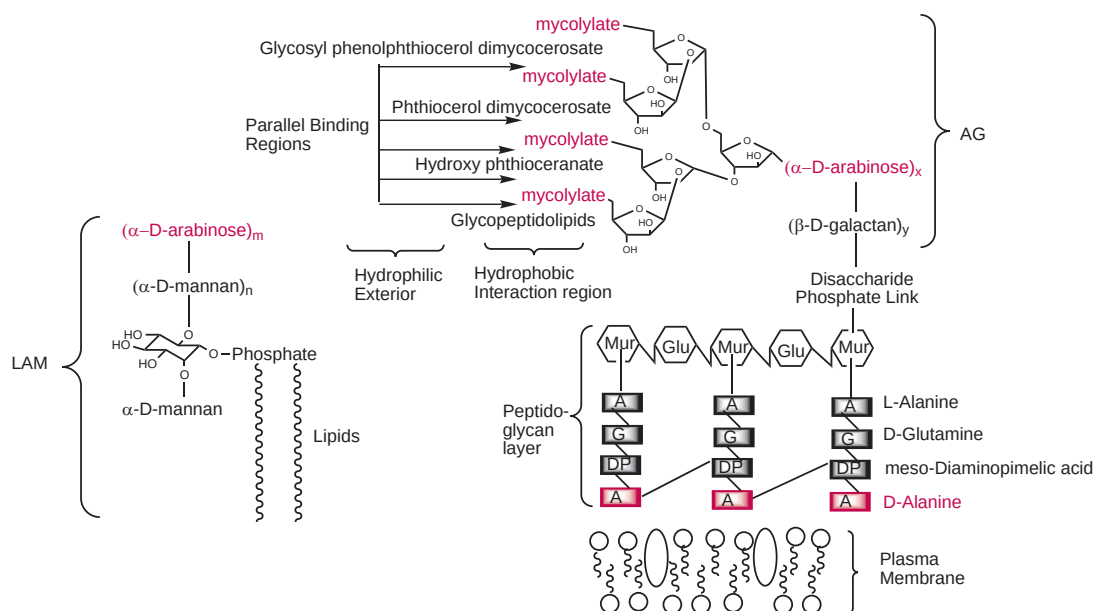


FIGURE 36.1 Diagrammatic representation of the cell wall/cell envelope of mycobacterium with drug sites of action highlighted in red.

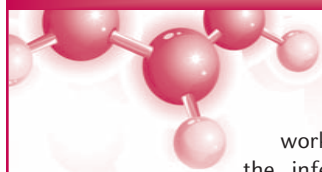
M. leprae in skin or blood samples is not always possible, but the detection of the antibody to the organism is an effective diagnostic test, especially for the lepromatous form of the disease.

Tuberculosis

Tuberculosis (TB) is a disease that has been known from the earliest recorded history. It is characterized as a chronic bacterial infection caused by *Mycobacterium tuberculosis*, an acid-fast, aerobic bacillus with the previously

discussed, unusual cell wall. The cell wall has a high lipid content, resulting in a high degree of hydrophobicity and resistance to alcohol, acids, alkali, and some disinfectants. After staining with a dye, the *M. tuberculosis* cell wall cannot subsequently be decolorized with acid wash, thus the characteristic of being an acid-fast bacillus. It is estimated that today, one-third to one-half of the world population is infected with *M. tuberculosis*, leading to approximately 6% of all deaths worldwide (~2 million deaths) (4,5). TB is the leading worldwide cause of mortality resulting from an infectious bacterial

CLINICAL SIGNIFICANCE



Diseases caused by mycobacteria such as TB and MAC are of great concern to both healthcare workers and the public. Understanding the infectivity and pathophysiology of mycobacteria and the medicinal chemistry of the antimicrobials in the complex treatment of mycobacteria infections can help facilitate a clinician's pharmacotherapy decisions. Mycobacteria are very slow-growing organisms that grow both intracellularly and extracellularly. Because of their growth, the treatment of mycobacterial infections, especially TB, requires long-term treatment with a combination of agents to effectively eradicate the disease and prevent resistance. In addition, many of these agents are associated with significant side effects and drug interactions. A good understanding of the medicinal chemistry of antimycobacterial agents helps the clinician to determine the most effective and safe combination for an individual patient.

Continuing interest in the study of the medicinal chemistry of antimycobacterial agents is prudent in the development of newer and safer agents. Research efforts are continuously being made, looking at agents which have more rapid antimycobacterial activity as well as lower resistance and fewer side effects. The discovery of such agents would have a significant effect on increasing the eradication of this worldwide epidemic. Shorter-course therapies would lead to an increase in patient compliance and possible decrease in mycobacterial resistance. As newer and more effective therapies are developed, clinicians will need to stay informed about their pharmacokinetic and pharmacodynamic profiles so patients can receive the most optimal and effective treatment.

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agent. A steady decline in the reported cases of TB had been occurring in the United States from the 1950s until 1985. From 1985 until 1988, however, this decline leveled off, but beginning in 1989, an increase was noted. In 1991, the Centers for Disease Control and Prevention (CDC) reported 25,701 new cases of TB. Today, the press and professional publications announce the “epidemic” spread of TB. The resurgence has been linked to urban crowding, homelessness, immigration, drug abuse, the disappearance of preventive-medicine health clinics, crowded prisons, and the AIDS epidemic. “Most alarming is the emergence of multidrug-resistant TB” (MDR-TB) (6). Before 1984, only 10% of the organisms isolated from patients with TB were resistant to any drug. In 1984, 52% of the organisms were resistant to at least one drug, and 32% were resistant to more than one drug. MDR-TB can have a fatality rate as high as 50%. The World Health Organization estimates that as many as 5% of all TB cases are MDR-TB. MDR-TB is associated with resistance to the two most potent first-line anti-TB drugs (i.e., isoniazid and rifampicin). More patients are being diagnosed with extensively drug-resistant TB (XTR-TB), a condition in which the organism is resistant to either isoniazid or rifampicin, any fluoroquinolone, and at least one of the three second-line injectable drugs (i.e., capreomycin, kanamycin, and amikacin). As a result of MDR-TB and XTR-TB, isolates of *M. tuberculosis* should be tested for antimicrobial susceptibility. In fact, drug resistance is encountered in patients who have never been treated with any of the TB drugs.

M. tuberculosis is transmitted primarily via the respiratory route. The organism appears in water droplets expelled during coughing, sneezing, or talking. Either in the droplet form or as the desiccated airborne bacilli, the organism enters the respiratory tract. The infectiousness of an individual will depend on the extent of the disease, the number of organisms in the sputum, and the amount of coughing. Within 2 weeks of beginning therapy, the infected individual will usually no longer be infectious. TB is a disease that mainly affects the lungs (80% to 85% of the cases), but *M. tuberculosis* can spread through the bloodstream and the lymphatic system to the brain, bones, eyes, and skin (extrapulmonary TB). In pulmonary TB, the bacilli reach the alveoli and are ingested by pulmonary macrophages. Substances secreted by the macrophages stimulate surrounding fibroblasts to enclose the infection site, leading to formation of granulomas or tubercles. The infection, thus contained locally, can lie dormant, encapsulated in a fibrotic lesion, for years and then reappear later. Extrapulmonary TB is much more common in HIV-infected patients (40% to 75%).

Because of the effect of the AIDS virus on the immune system, all HIV-infected individuals should be screened for TB, and if infected, the patient should be treated for TB before an active infection develops. Patients with HIV infection and TB are 100-fold more likely to develop an active infection than noninfected

patients. Individuals diagnosed with active TB should be counseled and tested for HIV because TB could have developed in conjunction with the weakened immune system seen in the patient with HIV infection. TB is the leading cause of death in patients who are HIV positive. Complicating treatment of HIV–TB coinfections is that drug interactions are quite common between the standard TB treatment regimens and antiretroviral drugs.

***Mycobacterium Avium–Intracellulare* Complex**

Mycobacterium avium and *Mycobacterium intracellulare* are atypical acid-fast bacilli that are ubiquitous in the environment and considered usually to be nonpathogenic in healthy individuals. Unfortunately, in immunocompromised individuals, these and possibly other, unidentified mycobacteria cause severe, life-threatening infections. Disseminated *Mycobacterium avium–intracellulare* complex (MAC) is the most common bacterial opportunistic infection seen in patients with AIDS and the third most common opportunistic infection behind candidal esophagitis and primary *Pneumocystis jirovecii* pneumonia (PCP) reported in patients with AIDS. Between 1981 and 1987 and before the availability of effective antiretroviral medication, the incidence of MAC was reported as 5%. Today, approximately half of all patients with AIDS develop an infection caused by MAC. The lungs are the organs most commonly involved in patients without AIDS, but the infection can involve bone marrow, lymph nodes, liver, and blood in patients with AIDS. The CD4 T-lymphocyte count is used as a predictor for risk of disseminated MAC; a count of less than 50 cells/mm³ in an HIV-infected person (adult or adolescent) is an indication of a potential infection and a recommendation for chemoprophylaxis. The MAC organisms grow within macrophages; therefore, the drug must be capable of penetration of the macrophage. Treatment of MAC, both prophylactically and for diagnosed infections, requires the use of multidrug therapy, and for disseminated MAC, this treatment is for the life of the patient.

GENERAL APPROACHES TO DRUG THERAPY

The mycobacteria have a number of characteristics in common, but it is important to recognize that the species vary widely in their susceptibility to the different drugs and that, in turn, this can relate to significant differences in the organisms. Some species, such as *M. tuberculosis*, are very slow-growing, with a doubling time of approximately 24 hours, whereas others, such as *Mycobacterium smegmatis*, double in 2 to 3 hours. The pathogenic mycobacterial organism can be divided into organism that are actively metabolizing and rapidly growing; semidormant bacilli, which exhibit spurts of metabolism; bacilli that exist at low pH and exhibit low metabolic activity; and dormant or persisters. The latter characteristic is the most problematic and responsible for treatment failures. Most current TB

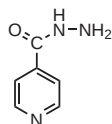
drugs are those that are effective against actively metabolizing and rapidly growing bacilli. Thus, successful chemotherapy calls for drugs with bactericidal action against all stages of the organisms—but especially against the persisters. The use of combination therapy over an extended period of time is one answer to successful treatment.

DRUG THERAPY FOR TUBERCULOSIS

Drug therapy for the treatment of TB has been greatly hampered by the development of MDR-TB (and more recently XDR-TB) and the lack of new classes of drugs. In fact, no new drugs have been developed in the last 45 years. The only change in the treatment of TB has been the strategy of using direct observed treatment, short-course (DOTS), with an emphasis on patient-centered care (7). Additionally, although the course of treatment has been reduced through the use of drug combinations to 6 months, patient compliance continues to be a serious problem, which in turn can be associated with the development of bacterial resistance.

First-Line Agents

Isoniazid (Isonicotinic Acid Hydrazide, Nydrazid, Laniazid)



Isoniazid (INH) is a synthetic antibacterial agent with bactericidal action against *M. tuberculosis*. The drug was discovered in the 1950s as a beneficial agent effective against intracellular and extracellular bacilli, and it is generally considered to be the primary drug for treatment of *M. tuberculosis*. Its action is bactericidal against replicating organisms, but it appears to be only bacteriostatic at best

against semidormant and dormant populations. After treatment with INH, *M. tuberculosis* loses its acid fastness, which can be interpreted as indicating that the drug interferes with cell wall development.

MECHANISM OF ACTION Although extensively investigated, the mechanism of action of INH has remained unknown until recently. New investigations into mechanisms of bacterial resistance have shed light on the molecular mechanism of action of INH (8). It is generally recognized that INH is a prodrug that is activated through an oxidation reaction catalyzed by an endogenous enzyme (9). This enzyme, *katG*, which exhibits catalase-peroxidase activity, converts INH to a reactive species capable of acylation of an enzyme system found exclusively in *M. tuberculosis*. Evidence in support of the activation of INH reveals that INH-resistant isolates have decreased catalase activity and that the loss of catalase activity is associated with the deletion of the catalase gene, *katG*. Furthermore, reintroduction of the gene into resistant organisms results in restored sensitivity of the organism to the drug. Reaction of INH with catalase-peroxidase results in formation of isonicotinaldehyde, isonicotinic acid, and isonicotinamide, which can be accounted for through the reactive intermediate isonicotinoyl radical or isonicotinic peroxide, as shown in Figure 36.2 (10). Evidence has been offered both for and against the reaction of catalase-peroxidase activated INH with a portion of the enzyme *inhA*, which is involved in the biosynthesis of the mycolic acids (Fig. 36.3) (11–13). The mycolic acids are important constituents of the mycobacterial cell wall in that they provide a permeability barrier to hydrophilic solutes. The enzyme *inhA*, produced under the control of the *inhA* gene, is an NADH-dependent, enoyl reductase protein thought to be involved in double-bond reduction during fatty acid elongation (Fig. 36.4). Isoniazid specifically inhibits long-chain fatty acid synthesis (>26 carbon atoms). It should be noted that the mycolic acids are α -branched lipids having a “short” arm of 20 to 24 carbons and a “long” arm of

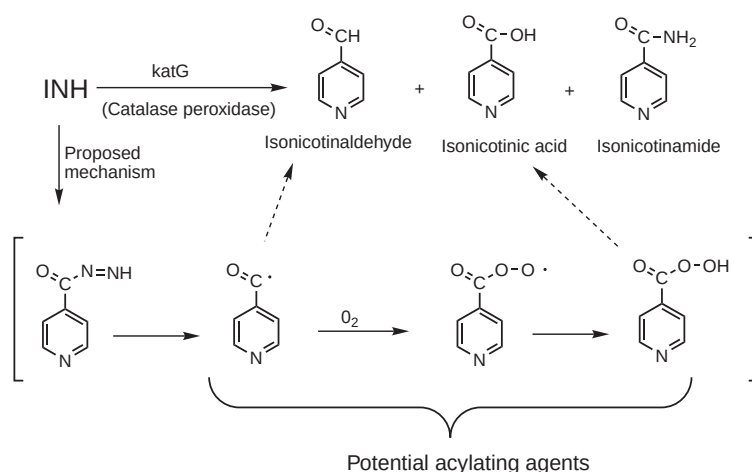


FIGURE 36.2 Reaction products formed from catalase-peroxidase reaction with isoniazid (INH).

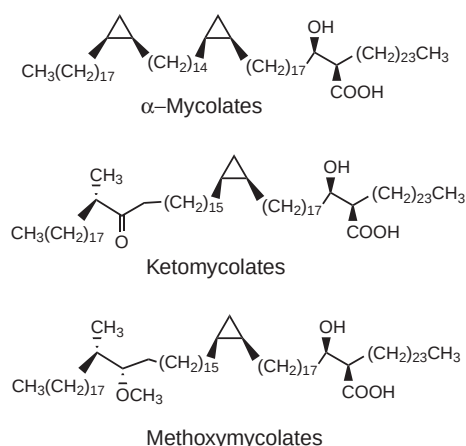
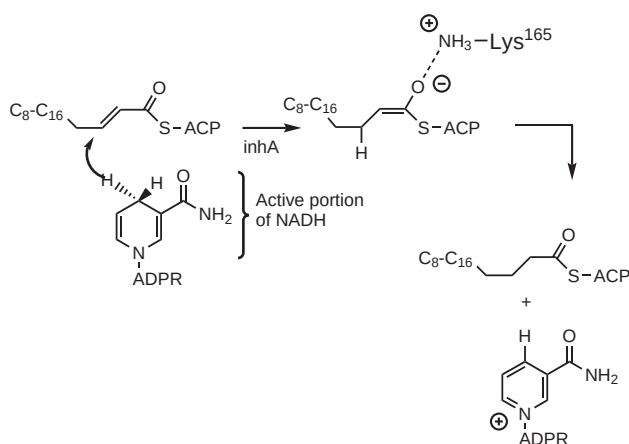
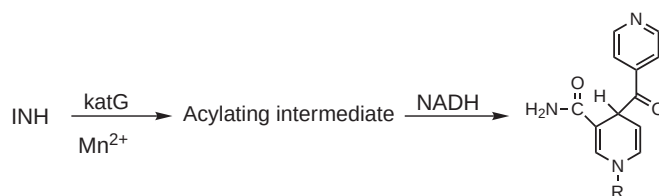


FIGURE 36.3 Mycolic acids.

50 to 60 carbons. It has been proposed that INH is activated to an electrophilic species that acylates the four position of the NADH (Fig. 36.5). The acylated NADH is no longer capable of catalyzing the reduction of unsaturated fatty acids, which are essential for the synthesis of the mycolic acids (14–16).

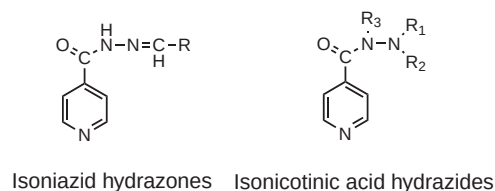
The mechanism of resistance to INH appears to be a very complicated process, possibly involving mutations in multiple genes including *katG* and *inhA*. It is possible that different mutations result in different degrees of resistance. This seems to be especially true in MDR-TB. A high level of resistance is reported to occur through *katG* mutation, which leads to a Ser-315 to Thr-315 in *katG* catalase-peroxidase and prevents INH activation (17).

STRUCTURE–ACTIVITY RELATIONSHIP An extensive series of derivatives of nicotinaldehyde, isonicotinaldehyde, and substituted isonicotinic acid hydrazides have been prepared and investigated for their tuberculostatic activity. Isoniazid hydrazones were found to possess activity, but these compounds were shown to be unstable in the

FIGURE 36.4 Enoylthioester. ACP, acyl carrier (protein) reduction catalyzed by NADH and *inhA*.FIGURE 36.5 Acylation of NADH and NADH-dependent enoylacyl protein (*inhA*).

gastrointestinal (GI) tract, releasing the active isonicotinic acid hydrazide (i.e., INH). Thus, it would appear that their activity resulted from the INH and not from the derivatives (18,19). Substitution of the hydrazine portion of INH with alkyl and aralkyl substituents resulted in a series of active and inactive derivatives (20–23).

Substitution on the *N*-2 position resulted in active compounds (R1 and/or R2 = alkyl; R3 = H), whereas any substitution of the *N*-1 hydrogen with alkyl groups destroyed the activity (R1 and R2 = H; R3 = alkyl). None of these changes produced compounds with activity superior to that of INH.



METABOLISM Isoniazid is extensively metabolized to inactive metabolites (Fig. 36.6) (24,25). The major metabolite is *N*-acetylisoniazid. The enzyme responsible for acetylation, cytosolic *N*-acetyltransferase, is produced

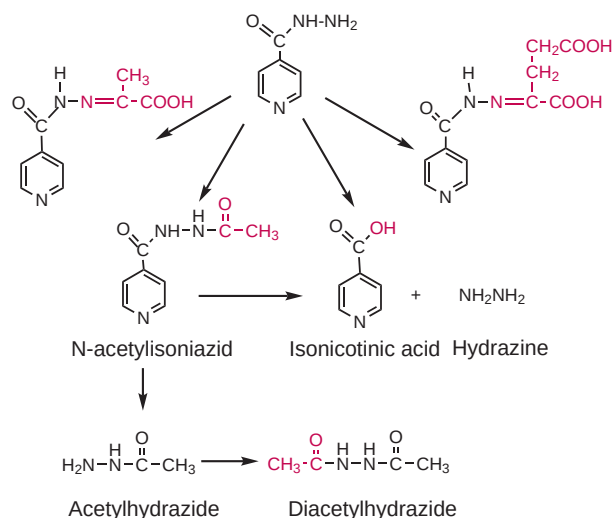


FIGURE 36.6 Metabolism of isoniazid.

under genetic control in an inherited autosomal fashion. Individuals who possess high concentrations of the enzyme are referred to as rapid acetylators, whereas those with low concentrations are slow acetylators. This can result in a need to adjust the dosage for fast acetylators. The *N*-acetyltransferase is located primarily in the liver and small intestine. Other metabolites include isonicotinic acid, which is found in the urine as a glycine conjugate, and hydrazine. Isonicotinic acid can also result from hydrolysis of *N*-acetylisoniazid, but in this case, the second product of hydrolysis is acetylhydrazine. Mono-*N*-acetylhydrazine is acetylated by *N*-acetyltransferase to the inactive *N*-diacetyl product. This reaction occurs more rapidly in rapid acetylators. The formation of *N*-acetylhydrazine is significant in that this compound has been associated with hepatotoxicity, which can occur during INH therapy. *N*-Acetylhydrazine has been postulated to serve as a substrate for microsomal P450, resulting in the formation of a reactive intermediate that is capable of acetylating liver proteins, in turn resulting in liver necrosis (26). It has been suggested that an *N*-hydroxylamine intermediate is formed that results in a reactive acetylating agent (Fig. 36.7). The acetyl radical/cation acylates liver protein (see Chapter 4).

PHARMACOKINETICS Isoniazid is readily absorbed following oral administration. Food and various antacids, especially aluminum-containing antacids, can interfere with or delay the absorption; therefore, it is recommended that the drug be taken on an empty stomach. The drug is well distributed to body tissues, including infected tissue. A long-standing concern about the use of INH during preventive therapy for latent TB has been the high incidence of hepatotoxicity. Recent studies have concluded that, excluding patients over 35 years of age, if relevant clinical monitoring is employed, the rate of hepatotoxicity is quite low (27). The risk of hepatotoxicity is associated with increasing age and, for unknown reasons, appears to be higher in women than in men.

Rifamycin Antibiotics

The rifamycins are members of the ansamycin class of natural products produced by *Streptomyces mediterranei*. This chemical class is characterized as molecules with an aliphatic chain forming a bridge between two nonadjacent positions of an aromatic moiety.

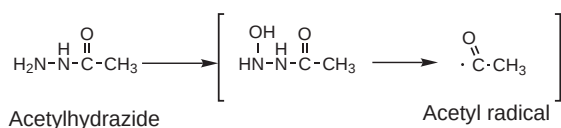
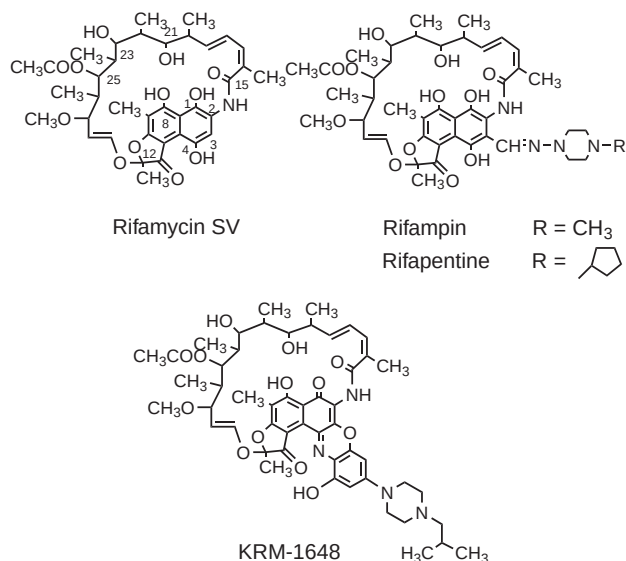


FIGURE 36.7 Acylating metabolite of isoniazid.



While investigating the biologic activity of the naturally occurring rifamycins (B, O, and S), a spontaneous reaction gave the biologically active rifamycin SV, which was later isolated from natural sources. Rifamycin SV was the original rifamycin antibiotic chosen for clinical development (28). Semisynthetic derivatives are prepared via conversion of the natural rifamycins to 3-formylrifamycin, which is derivatized with various hydrazines to give products such as rifampin (RIF) and rifapentine. RIF and rifapentine have significant benefit over previously investigated rifamycins in that they are orally active, are highly effective against a variety of both gram-positive and gram-negative organisms, and have high clinical efficacy in the treatment of TB. The rifamycin antibiotics are active against both growing and slow-metabolizing, non-growing bacilli.

MECHANISM OF ACTION The rifamycins inhibit bacterial DNA-dependent RNA polymerase (DDRP) by binding to the β -subunit of the enzyme and are highly active against rapidly dividing intracellular and extracellular bacilli. RIF is active against DDRP from both gram-positive and gram-negative bacteria, but because of poor penetration of the cell wall of gram-negative organisms by RIF, the drug has less value in infections caused by such organisms. Inhibition of DDRP leads to blocking the initiation of chain formation in RNA synthesis. It has been suggested that the naphthalene ring of the rifamycins π - π bonds to an aromatic amino acid ring in the DDRP protein (29). The DDRP is a metalloenzyme that contains two zinc atoms. It is further postulated that the oxygens at C-1 and C-8 of a rifamycin can chelate to a zinc atom, which increases the binding to DDRP, and finally, the oxygens at C-21 and C-23 form strong hydrogen bonds to the DDRP. The binding of the rifamycins to DDRP results in the inhibition of the RNA synthesis. Specifically, RIF has been shown to inhibit the elongation of full-length transcripts, but it

has no effect on transcription initiation (8). Resistance develops when a mutation occurs in the gene responsible for the β -subunit of the RNA polymerase (*rpoB* gene), resulting in an inability of the antibiotic to readily bind to the RNA polymerase (30).

STRUCTURE–ACTIVITY RELATIONSHIP A large number of derivatives of the naturally occurring rifamycins have been prepared (31). From these compounds, the following generalizations can be made concerning the structure–activity relationship (SAR): 1) Free $-\text{OH}$ groups are required at C-1, C-8, C-21, and C-23; 2) these groups appear to lie in a plane and to be important binding groups for attachment to DDRP, as previously indicated; 3) acetylation of C-21 and/or C-23 produces inactive compounds; 4) reduction of the double bonds in the macro ring results in a progressive decrease in activity; and 5) opening of the macro ring also gives inactive compounds. These latter two changes greatly affect the conformational structure of the rifamycins, which in turn decreases binding to DDRP. Substitution at C-3 or C-4 results in compounds with varying degrees of antibacterial activity. The substitution at these positions appears to affect transport across the bacterial cell wall. A compound incorporating such substitution is the benzoxazinorifamycin KRM-1648, which is proceeding through clinical investigation. In vitro studies have shown rapid tissue sterilization and encouraging results concerning combination therapy for TB and, possibly, MAC.

METABOLISM RIF and rifapentine are readily absorbed from the intestine, although food in the tract can affect absorption. RIF's absorption can be reduced by food in the intestine; therefore, the drug should be taken on an empty stomach (25). Intestinal absorption of rifapentine has been reported to be enhanced when taken after a meal (32). Neither drug appears to interfere with the absorption of other antituberculin agents, but there are conflicting reports on whether INH affects absorption of RIF. The major metabolism of RIF and rifapentine is deacetylation, which occurs at the C-25 acetate (Fig. 36.8). The resulting products, desacetylriofampin and desacetylriofapentine, are still active antibacterial agents. The majority of both desacetyl products are found in the feces, but desacetylriofampin glucuronide can be found in the urine as well. 3-Formylrifamycin SV has been reported as a second metabolite after administration of both RIF and rifapentine. This product is thought to arise in the gut from an acid-catalyzed hydrolysis reaction. Formylrifamycin is reported to possess a broad spectrum of antibacterial activity (33).

PHYSICOCHEMICAL PROPERTIES RIF and rifapentine are red-orange, crystalline compounds with zwitterionic properties. The presence of the phenolic groups results in acidic

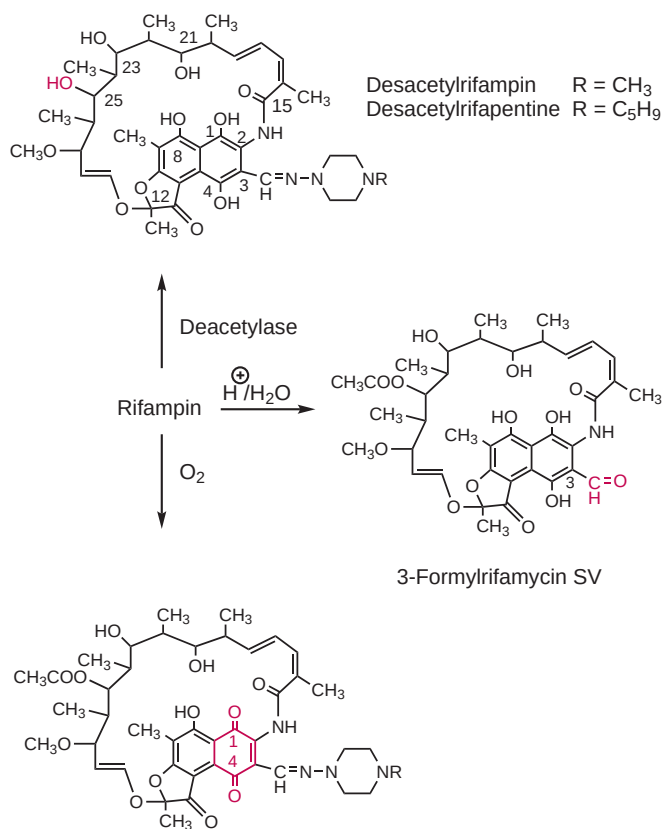


FIGURE 36.8 Metabolism and in vitro reactions of rifampin.

properties ($\text{p}K_a \sim 1.7$), whereas the piperazine moiety gives basic properties ($\text{p}K_a \sim 7.9$). These compounds are prone to acid hydrolysis, giving rise to 3-formylrifamycin SV, as indicated earlier. RIF and presumably rifapentine are prone to air oxidation of the *p*-phenolic groups in the naphthalene ring to give the *p*-quinone (C-1,4 quinone) (Fig. 36.8). RIF, rifapentine, and their metabolites are excreted in the urine, feces (biliary excretion), saliva, sweat, and tears. Because these agents have dye characteristics, one can note discoloration of the body fluids containing the drug. Notably, the tears can be discolored, and permanent staining of contact lenses can occur.

THERAPEUTIC APPLICATION

Rifampin (Rifadin, Rimactane) With the introduction of RIF in 1967, the duration of combination therapy for the treatment of TB was significantly reduced (from 18 to 9 months). RIF is nearly always used in combination with one or more other antituberculin agents. The drug is potentially hepatotoxic and can produce GI disturbances, rash, and thrombocytopenic purpura. RIF is known to be a potent inducer of the hepatic CYP3A4 and can decrease the effectiveness of oral contraceptives, corticosteroids, warfarin, quinidine,

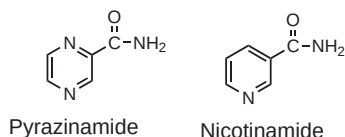
methadone, zidovudine, clarithromycin, and the azole antifungal agents (see Chapter 4) (34). In addition, the rifamycins appear to increase the expression of P-glycoproteins, which increase the efflux of various drugs.

Because of the decreased effectiveness of protease inhibitors and nonnucleoside reverse transcriptase inhibitors used in the treatment of HIV, the CDC has recommended avoidance of RIF in treatment of HIV-infected patients presently on these HIV therapies.

Rifapentine (Priftin) Rifapentine is the first new agent introduced for the treatment of pulmonary TB in the last 25 years. The drug's major advantage over RIF is the fact that when used in combination therapy, rifapentine can be administered twice weekly during the "intense" phase of therapy, followed by once-a-week administration during the "continuous" phase. In contrast, RIF normally is administered daily during the "intense" phase, followed by twice-a-week dosing during the "continuous" phase. Because relapse and the emergence of resistant strains of bacteria are associated with poor patient compliance, reduced dosing is expected to increase compliance. Initial clinical studies actually showed that the relapse rates in patients treated with rifapentine (10%) were higher than those in patients treated with RIF (5%). It was found that poor compliance with the nonrifamycin antituberculin agents was responsible for the increased relapse (32).

Rifapentine is readily absorbed following oral administration and is highly bound to plasma protein (97.7% vs. 80% for RIF). Related to the higher plasma binding, rifapentine has a longer mean elimination half-life (13.2 hours in healthy male volunteers) in comparison with the half-life reported for RIF (~2 to 5 hours). Greater than 70% of either drug is excreted in the feces. Rifapentine is generally considered to be more active than RIF and can be used in patients with varying degrees of hepatic dysfunction without the need for dose adjustment (35). This drug, similar to what is seen with RIF, induces hepatic microsomal enzymes (CYP3A4 and CYP2C8/9) but appears to have less of an inducing effect on cytochrome P450 than RIF. Rifapentine has been reported to be teratogenic in rats and rabbits (32).

Pyrazinamide

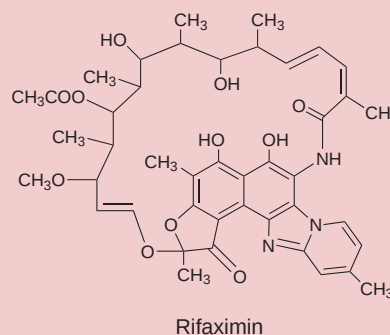


Pyrazinamide (pyrazinecarboxamide) was discovered while investigating analogs of nicotinamide. Pyrazinamide is a bioisostere of nicotinamide and possesses bactericidal

action against *M. tuberculosis*. Pyrazinamide has become one of the more popular antituberculin agents despite the fact that resistance develops quickly. Combination therapy, however, has proven to be an effective means of reducing the rate of resistant strain development. The activity of pyrazinamide is pH dependent with good in vivo activity at pH 5.5, but the compound is nearly inactive at neutral pH.

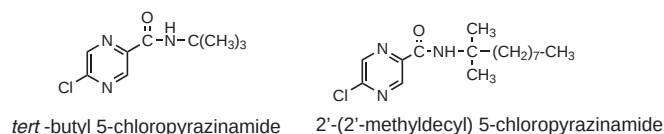
MECHANISM OF ACTION The mechanism of action of pyrazinamide is unknown, but recent findings suggest that pyrazinamide can be active either totally or in part as a prodrug. Susceptible organisms produce pyrazinamidase, which is responsible for conversion of pyrazinamide to pyrazinoic acid intracellularly (8). Mutation in the pyrazinamidase gene (*pncA*) results in resistant strains of *M. tuberculosis* (37). Pyrazinoic acid has been shown to possess biologic activity at a pH 5.4 or lower, in contrast in vitro tests that show pyrazinoic acid is 8- to 16-fold less active than pyrazinamide (38). Pyrazinoic acid can lower the pH in the immediate surroundings of the *M. tuberculosis* to such an extent that the organism is unable to grow, but this physicochemical property appears to account for only some of the activity. The protonated pyrazinoic acid can also permeate the mycobacterial membrane to lower the pH of the cytoplasm. Recent evidence suggests that pyrazinoic acid decreases membrane potential in older, nonreplicating bacilli, thus decreasing membrane transport, and interferes with the energetics of the membrane (39).

RIFAXIMIN (XIFAXAN)



The U.S. Food and Drug Administration (FDA) approved the release of rifaximin, a rifamycin antibiotic, for the treatment of travelers' diarrhea caused by enterotoxigenic *Escherichia coli*. Rifaximin is administered orally, with less than 0.4% absorption; therefore, its actions are limited to the GI tract. Its mechanism of action is essentially the same as that of other rifamycin antibiotics (36).

STRUCTURE–ACTIVITY RELATIONSHIP Previous structural modification of pyrazinamide has proven to be ineffective in developing analogs with increased biologic activity. Substitution on the pyrazine ring or use of alternate heterocyclic aromatic rings has given compounds with reduced activity (40). More recently, using quantitative SAR, a series of analogs have been prepared with improved biologic activity. The requirements for successful analogs include 1) provision for hydrophilicity to allow sufficient plasma concentrations such that the drug can be delivered to the site of infection, 2) lipophilicity to allow penetration into the mycobacterial cell, and 3) susceptibility to hydrolysis such that the prodrug is unaffected by the “extracellular” enzymes but is readily hydrolyzed at the site of action. Two compounds have been found that meet these criteria: *tert*-butyl 5-chloropyrazinamide and 2'-(2'-methyldecyl) 5-chloropyrazinamide (41).



METABOLISM Pyrazinamide is readily absorbed after oral administration, but little of the intact molecule is excreted unchanged (Fig. 36.9). The major metabolic route consists of hydrolysis by hepatic microsomal pyrazinamidase to pyrazinoic acid, which can then be oxidized by xanthine oxidase to 5-hydroxypyrazinoic acid. The latter compound can appear in the urine either free or as a conjugate with glycine (24).

THERAPEUTIC APPLICATION Pyrazinamide has gained acceptance as an essential component in combination therapy for the treatment of TB (component of Rifater with INH and RIF). The drug is especially beneficial in that it is active against semidormant intracellular tubercle bacilli that are not affected by other drugs (7,35). Evidence suggests that pyrazinamide is active against nonreplicating persister bacilli. The introduction of pyrazinamide combinations has reduced treatment regimens to 6 months from the previous 9-month therapy. The major serious side effect of pyrazinamide is the potential for hepatotoxicity. This effect is associated with dose and length of treatment. Pyrazinamide is not affected by the presence of food in the GI tract or by the use of aluminum-magnesium antacids (42).

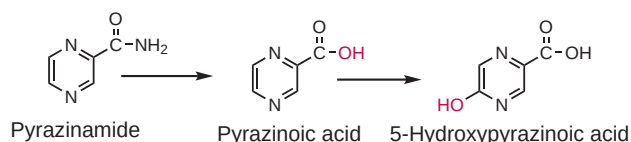
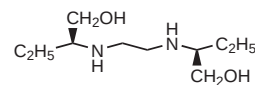


FIGURE 36.9 Metabolism of pyrazinamide.

Ethambutol (Myambutol)



Ethambutol

Ethambutol (EMB), an ethylenediiminobutanol, is administered as its (+)-enantiomer, which is 200- to 500-fold more active than its (–)-enantiomer. The difference in activity between the two isomers suggests a specific receptor for its site of action. EMB is a water-soluble, bacteriostatic agent that is readily absorbed (75% to 80%) after oral administration.

MECHANISM OF ACTION The mechanism of action of EMB remains unknown, although mounting evidence suggests a specific site of action for EMB. It has been known for some time that EMB affects mycobacterial cell wall synthesis; however, the complicated nature of the mycobacterial cell wall has made pinpointing the site of action difficult. In addition to the peptidoglycan portion of the cell wall, the mycobacterium have a unique outer envelope consisting of arabinofuranose and galactose (AG), which is covalently attached to the peptidoglycan and an intercalated framework of LAM (Fig. 36.1). The AG portion of the cell wall is highly branched and contains distinct segments of galactan and distinct segments of arabinan. At various locations within the arabinan segments (terminal and penultimate), the mycolic acids are attached to the C-5' position of arabinan (43,44). Initially, Takayama et al. (45) reported that EMB inhibited the synthesis of the AG portion of the cell wall. More recently, it has been reported that EMB inhibits the enzyme arabinosyl transferase. One action of arabinosyl transferase is to catalyze the polymerization of d-arabinofuranose, leading to AG (46,47). EMB mimics arabinan, resulting in a buildup of the arabinan precursor β -d-arabinofuranosyl-1-monophosphoryldecaprenol and, as a result, a block of the synthesis of both AG and LAM (Fig. 36.10) (48). The mechanism of resistance to EMB involves a gene overexpression of arabinosyl transferase, which is controlled by the *embAB* gene (49).

This mechanism of action also accounts for the synergism seen between EMB and intracellular drugs, such as RIF. Damage to the cell wall created by EMB improves the cell penetration of the intracellular drugs, resulting in increased biologic activity.

STRUCTURE–ACTIVITY RELATIONSHIP An extensive number of analogs of EMB have been prepared, but none has proven to be superior to EMB itself. Extension of the ethylene diamine chain, replacement of either nitrogen, increasing the size of the nitrogen substituents, and moving the location of the alcohol groups are all changes that drastically reduce or destroy biologic activity (Note: The drug candidate SQ109, which is in phase I/II of the

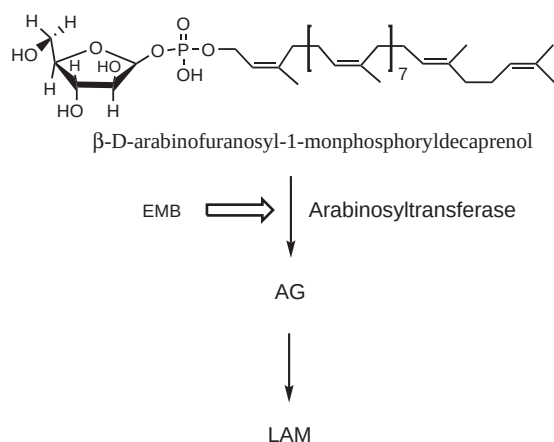
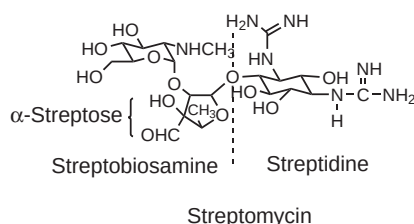


FIGURE 36.10 Site of action of ethambutol (EMB) in cell wall synthesis.

TB pipeline, is considered an analog of EMB; see the Antituberculin Drugs in the Pipeline box).

METABOLISM The majority of the administered EMB is excreted unchanged (73%), with no more than 15% appearing in the urine as either Metabolite A or Metabolite B (Fig. 36.11). Both metabolites are devoid of biologic activity.

Streptomycin



Streptomycin (STM) was first isolated by Waksman and coworkers in 1944 and represented the first biologically active aminoglycoside. The material was isolated from a manure-containing soil sample and, ultimately, was shown to be produced by *Streptomyces griseus*. The structure was proposed and later confirmed by Kuehl et al. (50) in 1948. STM is water soluble, with basic properties. The compound is usually available as the trihydrochloride or sesquisulfate salt, both of which are quite soluble in water. The hydrophilic nature of STM results in very poor absorption from the GI tract. Orally administered STM is recovered intact from the feces, indicating that the lack of biologic activity results from poor absorption and not chemical degradation.

MECHANISM OF ACTION The mechanism of action of STM and the aminoglycosides in general has not been fully elucidated. It is known that the STM inhibits protein synthesis, but additional effects on misreading of an mRNA

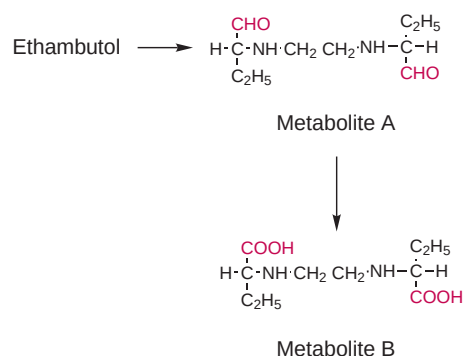


FIGURE 36.11 Metabolism of ethambutol.

template and membrane damage can contribute to the bactericidal action of STM. STM is able to diffuse across the outer membrane of *M. tuberculosis* and, ultimately, to penetrate the cytoplasmic membrane through an electron-dependent process. Through studies regarding the mechanism of drug resistance, it has been proposed that STM induces a misreading of the genetic code and, thus, inhibits translational initiation. In STM-resistant organisms, two changes have been discovered: First, S12 protein undergoes a change in which the lysine present at amino acids 43 and 88 in ribosomal protein S12 is replaced with arginine or threonine; and second, the pseudoknot conformation of 16S rRNA, which results from intramolecular base pairing between GCC bases in regions 524 to 526 of the rRNA to CGG bases in regions 505 to 507, is perturbed (51). It is thought that S12 protein stabilizes the pseudoknot, which is essential for 16S rRNA function. By some yet-to-be-defined mechanism, STM interferes with one or both of the normal actions of the 16S protein and 16S rRNA.

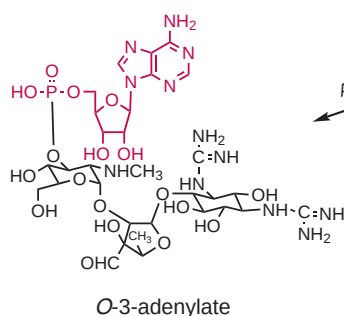
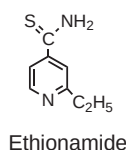
STRUCTURE-ACTIVITY RELATIONSHIP All of the aminoglycosides have very similar pharmacologic, pharmacodynamic, and toxic properties, but only STM and, to a lesser extent, kanamycin and possibly amikacin are used to treat TB. This is an indication for the narrow band of structurally allowed modifications giving rise to active analogs. Modification of the α -streptose portion of STM has been extensively studied. Reduction of the aldehyde to the alcohol results in a compound, dihydrostreptomycin, that has activity similar to STM but with a greater potential for producing delayed, severe deafness. Oxidation of the aldehyde to a carboxyl group or conversion to Schiff base derivatives (oxime, semicarbazone, or phenylhydrazone) results in inactive analogs. Oxidation of the methyl group in α -streptose to a methylene hydroxy gives an active analog that has no advantage over STM. Modification of the aminomethyl group in the glucosamine portion of the molecule by demethylation or by replacement with larger alkyl groups reduces activity; removal or modification of either guanidine in the streptidine nucleus also decreases activity.

METABOLISM No human metabolites of STM have been isolated in the urine of patients who have been administered the drug, with approximately 50% to 60% of the dose being recovered unchanged in the urine (25). Metabolism appears to be insignificant on a large scale, but it is implicated as a major mechanism of resistance. One problem with STM that was recognized early was the development of resistant strains of *M. tuberculosis*. Combination drug therapy was partially successful in reducing this problem, but over time, resistance has greatly reduced the value of STM as a chemotherapeutic agent for treatment of TB. Various mechanisms can lead to the resistance seen in *M. tuberculosis*. Permeability barriers can result in STM not being transported through the cytoplasmic membrane, but the evidence appears to suggest that enzymatic inactivation of STM represents the major problem. The enzymes responsible for inactivation are adenylyltransferase, which catalyzes adenylation of the C-3 hydroxyl group in the *N*-methylglucosamine moiety to give the *O*-3-adenylated metabolite, and phosphotransferase, which phosphorylates the same C-3 hydroxyl to give the *O*-3-phosphorylate metabolite (Fig. 36.12). This latter reaction appears to be the most significant clinically. The result of these chemical modifications is that the metabolites produced will not bind to ribosomes.

Second-Line Agents

A number of drugs, including ethionamide, *p*-aminosalicylic acid, cycloserine, capreomycin, kanamycin, and amikacin are considered to be second-line agents (it should be noted that some authorities classify STM as a second-line agent). These agents are active antibacterial agents, but they are usually less well tolerated or have a higher incidence of adverse effects. These agents are used in cases of resistance (MDR-TB), retreatment, or intolerance to the first-line drugs.

Ethionamide (Trecator-SC)

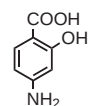


The synthesis of analogs of isonicotinamide resulted in the discovery of ethionamide and a homolog in which the ethyl group is replaced with a propyl (prothionamide). Both compounds have proven to be bactericidal against *M. tuberculosis* and *Mycobacterium leprae*.

MECHANISM OF ACTION Evidence has been presented suggesting that the mechanism of action of ethionamide is similar to that of INH (see *Mechanism of Action* under *Isoniazid*) (11,15). Similar to INH, ethionamide is considered to be a prodrug that is converted via oxidation by catalase-peroxidase to an active acylating agent, ethionamide sulfoxide, which in turn inactivates the inhA enoyl reductase enzyme (Fig. 36.13). In the case of ethionamide, it has been proposed that the ethionamide sulfoxide acylates Cys-243 in inhA protein.

METABOLISM Ethionamide is orally active but is not well tolerated in a single large dose (>500 mg). The GI irritation can be reduced by administration with meals. Additional side effects can include central nervous system (CNS) effects, hepatitis, and hypersensitivities. Less than 1% of the drug is excreted in the free form, with the remainder of the drug appearing as one of six metabolites. Among the metabolites are ethionamide sulfoxide, 2-ethylisonicotinamide, and the *N*-methylated-6-oxodihydropyridines (Compounds A, B, and C in Fig. 36.14) (52).

p-Aminosalicylic Acid



Para-aminosalicylic acid

Once a very popular component in TB therapy, *p*-aminosalicylic acid (PAS) is used as a second-line agent today. A combination of bacterial resistance and severe side effects has greatly reduced its value.

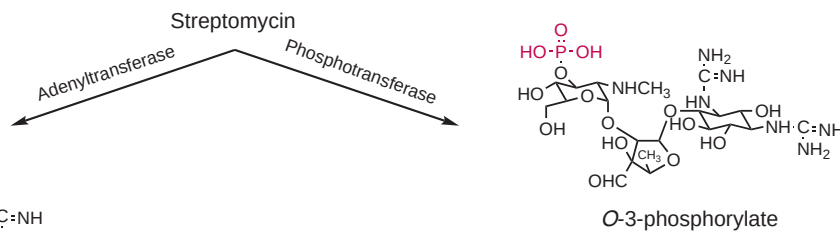


FIGURE 36.12 Metabolism of streptomycin (STM) as a mechanism of resistance.

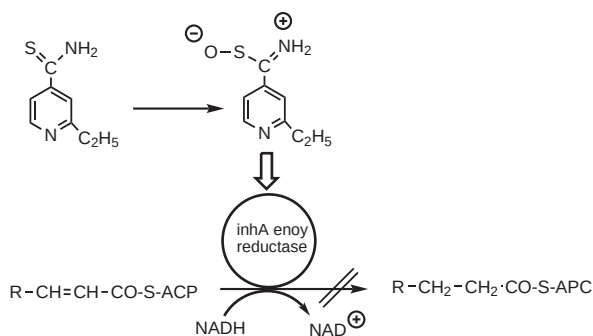


FIGURE 36.13 Mechanism of action of ethionamide.

As a bacteriostatic agent, PAS is used at a dose of up to 12 g/day, which causes considerable GI irritation. In addition, hypersensitivity reactions occur in 5% to 10% of the patients, with some of these reactions being life-threatening.

MECHANISM OF ACTION PAS is thought to act as an antimetabolite interfering with the incorporation of *p*-amino-benzoic acid into folic acid. When coadministered with INH, PAS is found to reduce the acetylation of INH, itself being the substrate for acetylation, thus increasing the plasma levels of INH. This action can be especially valuable in patients who are rapid acetylators.

METABOLISM PAS is extensively metabolized by acetylation of the amino group and by conjugation with glucuronic acid and glycine at the carboxyl group. It is used primarily in cases of resistance, retreatment, and intolerance of other agents and is available from the CDC.

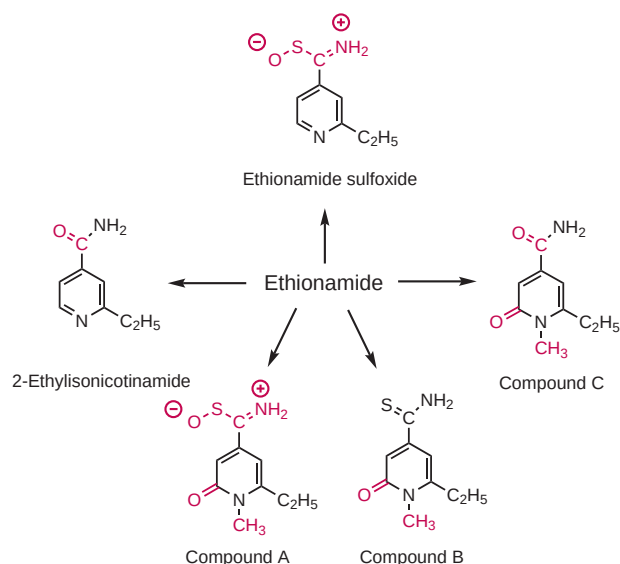
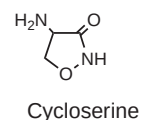


FIGURE 36.14 Metabolism of ethionamide.

Cycloserine (Seromycin)



Cycloserine is a natural product isolated from *Streptomyces orchidaceus* as the D-(+)-enantiomer.

MECHANISM OF ACTION D-Cycloserine is considered to be the active form of the drug, having its action associated with the ability to inhibit two key enzymes, D-alanine racemase and D-alanine ligase. D-Alanine is an important component of the peptidoglycan portion of the mycobacterial cell wall. Mycobacteria are capable of using naturally occurring L-alanine and converting the L-alanine to D-alanine via the enzyme D-alanine racemase. The resulting D-alanine is coupled with itself to form a D-alanine–D-alanine complex under the influence of D-alanine ligase, and this complex is incorporated into the peptidoglycan of the mycobacterial cell wall (Fig. 36.15). D-Cycloserine is a rigid analog of D-alanine; therefore, it competitively inhibits the binding of D-alanine to both of these enzymes and its incorporation into the peptidoglycan (Fig. 36.15) (53). Resistance is associated with an overexpression of D-alanine racemase.

SIDE EFFECTS Cycloserine is readily absorbed after oral administration and is widely distributed, including the CNS. Unfortunately, D-cycloserine binds to neuronal N-methylaspartate receptors and, in addition, affects synthesis and metabolism of γ -aminobutyric acid, leading to a complex series of CNS effects. As a second-line agent, cycloserine should only be used when retreatment is necessary or when the organism is resistant to other drugs. Cycloserine should not be used as a single drug; it must be used in combination.

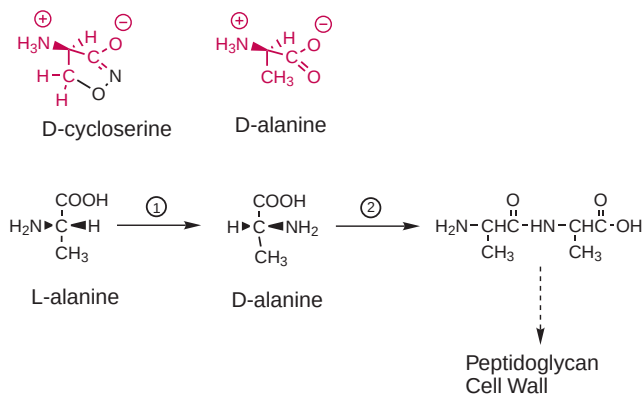
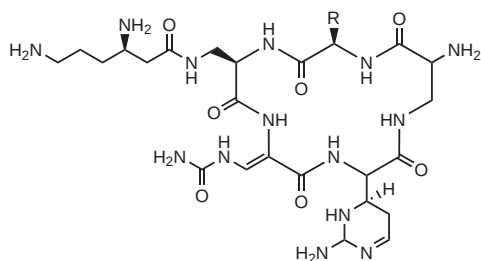
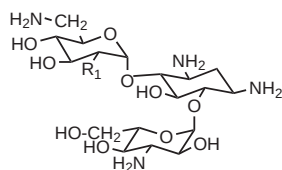


FIGURE 36.15 Sites of action of D-cycloserine: 1, D-alanine racemase; 2, D-alanine ligase.

Capreomycin (Capastat)

Capreomycin

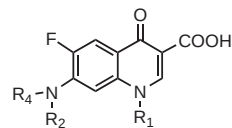
Capreomycin is a mixture of four cyclic polypeptides, of which capreomycin Ia ($R = OH$) and Ib ($R = H$) make up 90% of the mixture. Capreomycin is produced by *Streptomyces capreolus* and is quite similar to the antibiotic viomycin. Little is known about its mechanism of action, but if the chemical and pharmacologic similarity to viomycin carries over to the mechanism of action, then one might expect something similar. Viomycin is a potent inhibitor of protein synthesis, particularly that which depends on mRNA at the 70S ribosome (54). Viomycin blocks chain elongation by binding to either or both the 50S or 30S ribosomal subunits. As a polypeptide, the drug must be administered parenterally, with the preferred route of administration being intramuscular. As a second-line bacteriostatic antituberculin drug, it is reserved for “resistant” infections and cases of treatment failure. The drug should not be given as a single agent; rather, it should be used in combination with EMB or INH. Reported toxicity of capreomycin includes renal and hepatic damage, hearing loss, and allergic reactions.

Kanamycin (Kanamycin A, $R = OH$; Kanamycin B, $R = NH_2$; Kantrex)

Kanamycin

A member of the aminoglycoside class, kanamycin is a second-line agent with very limited use in the treatment of *M. tuberculosis*. The drug is only used to treat resistant organisms and then should be used only in combination with other effective agents. The parenteral form of the drug is used, because as an aminoglycoside, the drug is poorly absorbed via the oral route. The narrow range of effectiveness and the severe toxicity, especially if the drug is administered over a long period of time, have limited its usefulness. (For additional information

on kanamycin and aminoglycosides in general, see Chapter 33.)

Fluoroquinolones

6-Fluoro- 4-quinolones

The fluoroquinolones are a broad-spectrum class of antibacterials that have been demonstrated to have activity against a wide range of gram-negative and gram-positive pathogens, including *M. tuberculosis*, *Mycobacterium kansasii*, *Mycobacterium xenopi*, *Mycobacterium fortuitum*, MAC, and *M. leprae*. The quinolones are attractive in that they are active at low concentrations, concentrate within macrophages, and have a low frequency of side effects.

MECHANISM OF ACTION The mechanism of action of the fluoroquinolones is reported in detail in Chapter 33 and basically involves binding to DNA gyrase–DNA complex (GyrA and GyrB), inhibiting bacterial DNA replication and transcription. As a result, these drugs exhibit bactericidal activity.

STRUCTURE–ACTIVITY RELATIONSHIP The structural requirements for activity against mycobacterium and, specifically, for activity against the MAC have been explored (55,56). It is known that nonfluorinated quinolones are inactive against mycobacteria. In addition, it has been reported that certain fragments or substructures within the quinolones improve activity toward the MAC (biophores), whereas other fragments deactivate the quinolones (biophobes). The important structural features acting as biophores include: 1) a cyclopropyl ring at the *N*-1 position, 2) fluorine atoms at positions C-6 and C-8, and 3) a C-7 heterocyclic substituent. Excessive lipophilicity at *N*-1 can decrease activity (i.e., 2,4-difluorobenzene). The C-7 substituents with greatest activity against mycobacteria include the substituted piperazines and pyrrolidines (Fig. 36.16). Two biophobes have also been reported and are shown in Figure 36.17.

Several C-8 methoxy-substituted fluoroquinolones have been reported with superior activity over earlier quinolones (57,58). Moxifloxacin (Fig. 36.18) is reported to be active against *M. tuberculosis* when combined with INH.

THERAPEUTIC APPLICATION Fluoroquinolone therapy for TB is predominantly used in patients infected with multidrug-resistant organisms. Resistance to the quinolones has been reported and appears to be associated with mutations in the *gyrA* and *gyrB* genes, leading to single-amino-acid substitution in the DNA

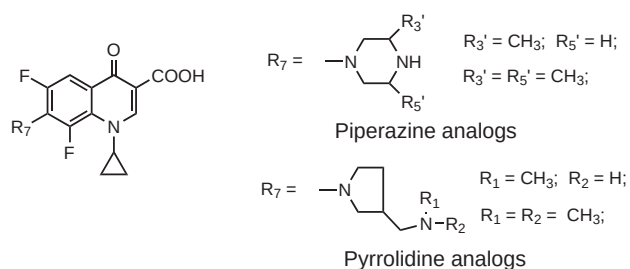


FIGURE 36.16 4-Quinolones demonstrating high activity against mycobacteria.

gyrase protein (8). As a result, use of the fluoroquinolones must be monitored in the treated patient population. The most active fluoroquinolones available for treatment of TB are ofloxacin, levofloxacin (the levo isomer of ofloxacin), and moxifloxacin (Fig. 36.18) (59).

Therapeutic Considerations for Treatment of Tuberculosis

Overview

Various stages of infectious organisms have been identified that can require special consideration for chemotherapy. The organism can be in a dormant stage, which is usually not affected by drugs. The continuously growing stage of the organism can find the bacteria either in an extracellular or an intracellular location. A stage of the organism, which is classified as the very slowly metabolizing bacteria, exists also in a relatively acidic environment. Finally, the organism can exhibit a stage in which it is dormant, followed by spurts of growth. As noted in the discussion of specific drugs, one stage or another can be more or less susceptible to a particular drug based on the above characteristics. It is also recognized that organisms from some geographic regions can show a low incidence of drug resistance, whereas those from other regions have a high incidence of drug resistance.

For patients with TB likely to be infected with organisms suspected of showing low rates of drug resistance, the American Thoracic Society currently recommends a minimum 26-week treatment period, consisting of an initial 2-month (8-week) phase, followed by a continuation phase of either 4 or 7 months. Four basic regimens are recommended for treatment of susceptible *M. tuberculosis* infections in adults. During the initial phase, three of the four regimens use a combination

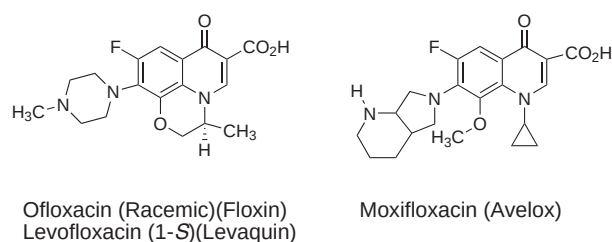


FIGURE 36.18 Fluoroquinolones active against *Mycobacterium tuberculosis*.

of INH, RIF, pyrazinamide, and EMB, given either 7, 5, or 3 days per week. The fourth regimen uses INH, RIF, and EMB given 7 or 5 days per week (when drugs are administered DOTS, the drugs can be given less often). The four-drug regimen is administered based on the assumption that a proportion of the organisms are resistant to INH. For treatment of children when visual acuity cannot be tested, EMB is usually not recommended. Several options for drug treatment exist for the continuation phase of treatment (60). In a majority of cases, the continuation phase will last 4 months. Here again, if DOTS is used, the patient might only need to be treated two or three times weekly; without DOTS, the treatment is daily. The drug combination of INH and RIF is used during the continuation phase. Typical daily doses are 300 mg of INH, 600 mg of RIF, 2 g of pyrazinamide, and 1 g of EMB. The addition of pyrazinamide to the drug regimen results in a reduction of treatment time from 9 to 6 months. Individuals on any of these regimens are considered to be noninfectious after the first 2 weeks. This same group of drugs is recommended for patients with both TB and AIDS. The "cardinal rules" for all TB regimens are: 1) get drug susceptibility information as soon as possible; 2) always begin therapy with at least three drugs; 3) at all costs, avoid a regimen using only one effective drug; and 4) always add at least two drugs to a failing regimen (6,34,61). In addition, it is recommended that consideration be given to treating all patients with DOTS.

The only proven treatment for prophylaxis of TB (patients with a positive skin test or a high-risk factor) is INH used for 6 or 12 months. High-risk persons are considered to be adults and children with HIV infection, close contacts of infectious cases, and those with fibrotic lesions on chest radiographs. Adverse effects when using INH over a long treatment period can be a serious problem. Isoniazid can cause severe liver damage, and the drug should be removed if serum aminotransferase activity increases to three- to fivefold the normal level or the patient develops symptoms of hepatitis. Peripheral neuropathy can be seen with INH therapy. This condition can be prevented by coadministration of pyridoxine. Persons who are presumed to be infected with INH-resistant organisms should be treated with RIF rather than INH. Hepatitis, thrombocytopenia, and nephrotoxicity can be

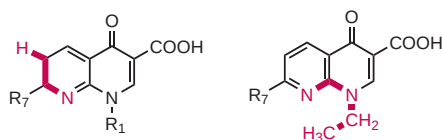


FIGURE 36.17 Biophobes (shown in *red*) that inactivate 4-quinolones toward mycobacteria.

seen with RIF therapy. RIF is thought to potentiate the hepatitis caused by INH. GI upset and staining effects caused by RIF are of minor importance.

Although drug-susceptible TB has been successfully treated, the long period of treatment and low patient adherence have in part led to the development of drug-resistant TB, both MDR-TB and XDR-TB. In such situations, a new treatment paradigm has become necessary. For treatment of MDR-TB a five- to six-drug regimen is commonly used (Table 36.1) (62,63). The particular regimen should be based on drug susceptibility testing or previous treatment history. With dual mycobacterium resistance to INH and RIF, a combination of an aminoglycoside from group III drugs, a fluoroquinolone from group II drugs, along with EMB and pyrazinamide and ethionamide or prothionamide from groups I and IV, respectively, can be used at the doses indicated. If resistance to all group I drugs occurs, then *p*-aminosalicylic acid and cycloserine are added to the previous indicated drug list of an aminoglycoside, a fluoroquinolone, and ethionamide or prothionamide. The treatment should continue

for up to 18 months after smear conversion to negativity as recommended by the World Health Organization.

DRUG THERAPY FOR MAC

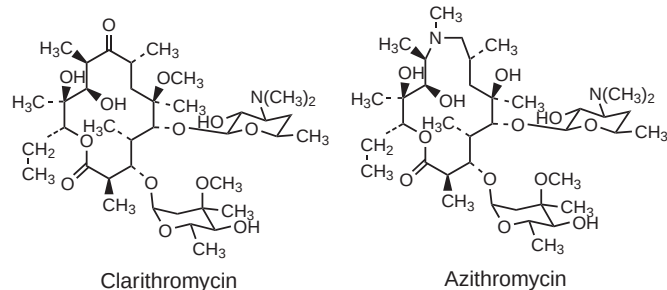
Drug therapy for the treatment of MAC is complicated. It underwent significant changes in the early 1990s, but little has changed during the last 15 years. Recommendations for treatment are presently based on small and, in some cases, incomplete studies; more changes can be expected in the future. For the most up-to-date information, the reader is referred to the CDC's homepage (www.cdc.gov). The 1997 guidelines for prophylaxis of MAC advise that all adults and adolescents with HIV infection and a CD4 lymphocyte count of less than 50 cells/mL receive clarithromycin 500 mg twice a day or azithromycin 1,200 mg once a week. This recommendation is considered to be a standard of care (64–66). For treatment of MAC, it is recommended that a combination therapy be used that includes at least two drugs (either clarithromycin or azithromycin plus EMB

TABLE 36.1 Grouping of Drugs Used to Treat MDR-TB and XDR-TB

Drug Group	Drugs	Doses
I. First-line oral TB drugs	Isoniazid	5 mg/kg
	Rifampicin	10 mg/kg
	Ethambutol	15–25 mg/kg
	Pyrazinamide	30 mg/kg
II. Fluoroquinolones	Ofloxacin	15 mg/kg
	Levofloxacin	15 mg/kg
	Moxifloxacin	7.5–10 mg/kg
III. Injectable TB drugs	Streptomycin	15 mg/kg
	Kanamycin	15 mg/kg
	Amikacin	15 mg/kg
	Capreomycin	15 mg/kg
IV. Second-line TB drugs	Ethionamide/prothionamide	15 mg/kg
	Cycloserine/terizidone	15 mg/kg
	<i>p</i> -Aminosalicylic acid	150 mg/kg
V. Less effective drugs (efficacy uncertain)	Clofazimine	100 mg
	Amoxicillin/clavulanate	875/125 mg every 12 hours
	Linezolid	600 mg
	Imipenem	500–1,000 mg every 6 hours
	Clarithromycin	500 mg every 12 hours
	Thioacetazone	150 mg

for life). Other drugs that can be added to the combination consist of rifabutin, fluoroquinolones, and amikacin. It should be noted that INH and pyrazinamide are ineffective in the treatment of disseminated MAC.

Macrolides



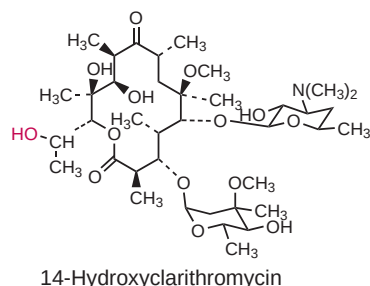
Both clarithromycin and azithromycin are considered to be first-line agents for the prevention and treatment of MAC and have replaced rifabutin. Both macrolides are concentrated in macrophages (clarithromycin concentration is 17.3-fold higher in macrophage cells than in extracellular fluid) and appear to be equally effective, although clarithromycin has a lower minimum inhibitory concentration. Azithromycin has an intra-alveolar macrophage half-life of 195 hours, compared to a 4-hour half-life for clarithromycin. For prevention, the macrolides can be used as single agents, although there is a risk of resistant organisms forming and of a cross-resistance between clarithromycin and azithromycin. In one study, the combination of azithromycin and rifabutin proved to be more effective than either drug used alone. For the treatment of MAC, combination therapy is recommended.

Mechanism of Action

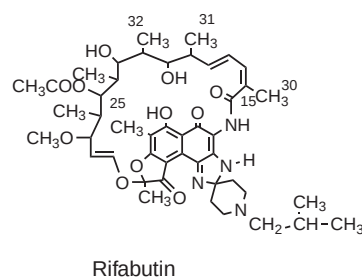
The macrolide antibiotics are bacteriostatic agents that inhibit protein synthesis by binding to the 50S ribosomal units. (For a more detailed discussion, see Chapter 33.)

Metabolism

Clarithromycin is metabolized in the liver to an active metabolite, 14-hydroxyclearithromycin, which is less active than the parent molecule. In addition, the drug is an inhibitor of CYP3A4, which could lead to increased concentrations of some drugs, such as rifabutin (see below). Azithromycin is primarily excreted unchanged in the gut, and at present, there is no evidence of CYP3A4 induction or inhibition.



Rifamycins



Various rifamycin derivatives have been investigated or are under investigation for use in the prevention and treatment of MAC. Up until 1997, rifabutin (Mycobutin) was considered to be the drug of choice for prophylaxis of MAC-infected patients. Studies since 1995, however, have suggested that the macrolides are more effective (survival rates), present fewer side effects, and cause less drug interactions than rifabutin. Early treatment of MAC bacteremia consists of multidrug regimens, usually involving four or five drugs. Drug interactions and, in some studies, exceptionally high drug doses have given confusing results. It is generally agreed that rifabutin should be used in treatment when macrolides have failed or can be combined with azithromycin for prophylaxis or treatment when clarithromycin is unsuccessful.

Drug Interactions

The most significant drug interaction identified with rifabutin is associated with the fact that the drugs in this class are inducers of CYP3A4 and the CYP2C family. As a result, certain drugs that are substrates for these isoforms will show reduced activity. Rifabutin has been shown to reduce the area under the curve and the maximum concentration of clarithromycin and most HIV protease inhibitors. This action could lead to inactivity or resistance to these agents. In addition, because the HIV protease inhibitors are inhibitors of CYP3A4, a combination of rifabutin plus an HIV protease inhibitor is expected to increase the rifabutin area under the curve and maximum concentration, thus increasing the risk of rifabutin side effects. The most serious side effect of rifabutin is uveitis (inflammation of the iris). Under these conditions, appropriate changes in dosing are required. If combination therapy is desirable for the treatment of MAC, the combination of azithromycin and rifabutin is recommended, because no significant change in mean serum drug concentration is reported to occur with either agent when used in combination.

Drug Metabolism

The hepatic metabolism of rifabutin is complex, with as many as 20 metabolites having been reported. The structure of most of the metabolites remains unknown, but several have been identified, including 25-desacetyl-rifabutin, 25-desacetyl-rifabutin-*N*-oxide, 31-hydroxy-rifabutin, 32-hydroxy-rifabutin, and

32-hydroxy-25-desacetyl rifabutin. The metabolites appear in the urine (50%) and in the feces (30%). Based on the activity of other rifamycins, it might be expected that one or more of the metabolites possess antimycobacterial activity.

Additional Drugs

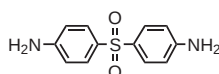
Various other agents have been combined with the macrolides or RIF for the prophylaxis and treatment of MAC. As indicated earlier, the effectiveness of each component in multidrug treatment is not easily defined. The additional drugs used include EMB, ciprofloxacin, amikacin, and clofazimine. EMB and ciprofloxacin appear to have good activity against MAC, but clofazimine has shown unfavorable results. The FDA has advised against the use of clofazimine during initial therapy for MAC (65).

LEPROSY

Sulfones

The diaryl sulfones represent the major class of agents used to treat leprosy. The initial discovery of the sulfones came about as a result of studies directed at exploring the SAR of sulfonamides (Fig. 36.19). A variety of additional chemical modifications have produced several other active agents, but none has proved to be more beneficial than the original lead, 4,4'-diaminodiphenylsulfone. Dapsone was first introduced into the treatment of leprosy in 1943.

Dapsone



4,4'-Diaminodiphenylsulfone (Dapsone)

Dapsone, a diaminodiphenylsulfone, is a nearly water-insoluble agent that is very weakly basic ($pK_a \sim 1.0$). The lack of solubility can account, in part, for the occurrence of GI irritation. Despite the lack of solubility, the drug is efficiently absorbed from the GI tract. Although dapsone is bound to plasma protein (~70%), it is distributed throughout the body.

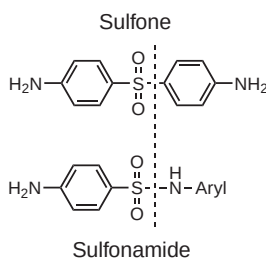
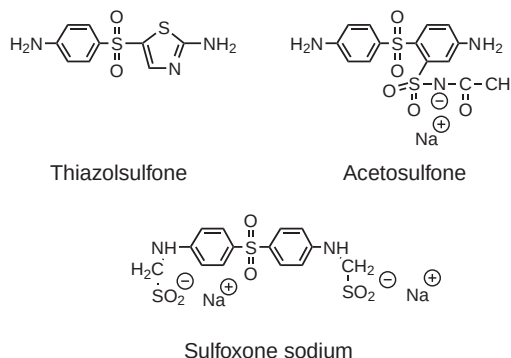


FIGURE 36.19 Structural comparison of sulfones versus sulfonamide.

MECHANISM OF ACTION Dapsone, a bacteriostatic agent, is thought to act in a manner similar to that of the sulfonamides—namely, through competitive inhibition of *p*-aminobenzoic acid incorporation into folic acid (see *Sulfonamides* in Chapter 33). Bacteria synthesize folic acid, but host cells do not. As a result, coadministration of dapsone and *p*-aminobenzoic acid will inactivate dapsone. Both dapsone and clofazimine have significant anti-inflammatory actions, which may or may not play a role in the antimicrobial action. The anti-inflammatory action can also be a beneficial side effect, offsetting the complication of erythema nodosum leprosum seen in some patients. The anti-inflammatory action can come about by inhibition of myeloperoxidase-catalyzed reactions (69).

STRUCTURE–ACTIVITY RELATIONSHIP Several derivatives of dapsone have been prepared in an attempt to increase the activity. Isosteric replacement of one benzene ring resulted in the formation of thiazolsulfone. Although still active, it is less effective than dapsone. Substitution on the aromatic ring, to produce acetosulfone, reduced activity while increasing water solubility and decreasing GI irritation. A successful substitution consists of adding methanesulfinate to dapsone to give sulfoxone sodium. This water-soluble form of dapsone is hydrolyzed in vivo to produce dapsone. Sulfoxone sodium is used in individuals who are unable to tolerate dapsone because of GI irritation, but it must be used in a dose threefold that of dapsone because of inefficient metabolism to dapsone. The chemical modification of dapsone derivatives continues to be pursued with the intent of finding newer agents useful for the treatment of resistant strains of *Mycobacterium leprae* (70).



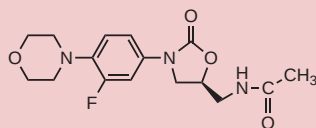
METABOLISM The major metabolic product of dapsone results from *N*-acetylation in the liver by *N*-acetyltransferase. Dapsone is also *N*-hydroxylated to the hydroxylamine derivative. These metabolic reactions are catalyzed by CYP3A4 isoforms. Neither of these compounds possesses significant leprostatic activity, although *N*-acetyldiaminodiphenylsulfone can be deacetylated back to dapsone. Products found in

ANTITUBERCULIN DRUGS IN THE PIPELINE

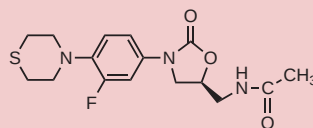
The need for new drugs to treat TB continues to mount, and as many as 10 compounds are presently moving through the clinical testing and approval process. Several of these drugs are already marketed for treatment of bacterial infections and include the fluoroquinolones (moxifloxacin and gatifloxacin), the oxazolidinone

linezolid and a close analog, PNU-100480, and an analog of EMB (SQ109). Several new chemical entities offer hope in that they appear to act by unique mechanisms and include the nitroimidazoles PA-824 and OPC-67863. A drug candidate presently in phase II clinical studies is a diarylquinoline TMC207.

Oxazolidinones:

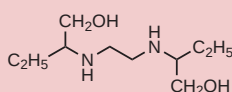


Linezolid

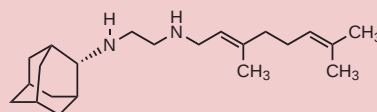


PNU-100480

Ethylene diamines:

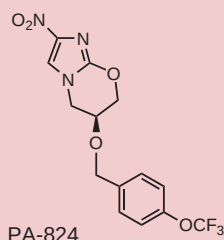


Ethambutol

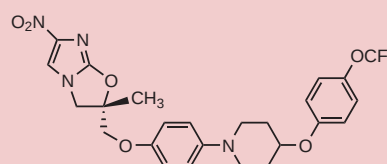


SQ109

Nitroimidazole:

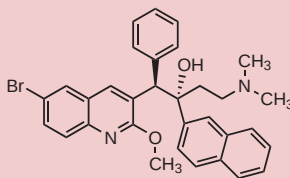


PA-824



OPC-67863

Diarylquinoline (DARQ):



TMC207

Early studies with PNU-100480 are encouraging and show that it is more active than linezolid in treatment of TB. The chemical is in phase I investigation (62). SQ109 has been reported to have a mechanism different from that of EMB, possibly acting on the cell wall of the mycobacterium. The drug is presently in early phase II studies, but no recent information is available. The nitroimidazoles, PA-824 and OPC-67863, appear to be pro-drugs and are reduced to the active drug, which in turn interferes with mycolic acid synthesis and thus cell wall synthesis. The two drugs are at various stages of early clinical investigation

(63). The uniqueness of the nitroimidazoles is that they are active against both replicating and nonreplicating bacteria (67). Probably the most interesting compound is the diarylquinoline TMC207 due to its novel mechanism of action, which involves inhibition of F₁F₀ proton adenosine triphosphate (ATP) synthase and the enzyme involved in ATP synthesis (61,67). The drug is active against nonreplicating persisters. Recently it has been shown that the inhibitory action is selective against mycobacterium with only weak inhibitory activity against mammalian cells (68).

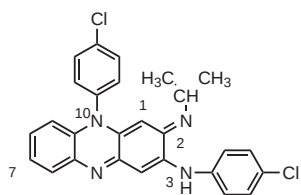
ADDITIONAL USES FOR DAPSONE

Dapsone has a variety of uses in addition to the treatment of leprosy. These include the treatment of dermatitis herpetiformis as well as an unlabeled use in the treatment of brown recluse spider bites, inflammatory bowel disorders, leishmaniasis, malaria prophylaxis, relapsing polychondritis, rheumatic and connective tissue disorders, and prophylaxis of *Pneumocystis jirovecii* pneumonia in HIV-infected patients and organ transplant patients. Recently, dapsone has been approved for the treatment of acne. Under the trade name of Aczone, the drug is available as a 5% gel. Glucose-6-phosphate dehydrogenase (G6PD) levels, blood hemoglobin, and reticulocyte counts should be followed in patients with G6PD deficiency anemia and predisposed to increased hemolytic effects.

the urine consist of small amounts of dapsone, the metabolites *N*-acetyldiaminodiphenylsulfone and *N*-hydroxydiaminodiphenylsulfone, and glucuronide and sulfates of each of these substances (Fig. 36.20).

Although the acetylated metabolites of dapsone are inactive, there is growing concern over the adverse hematologic effects of the hydroxylated metabolite. The specific adverse effect reported for the *N*-hydroxydiaminodiphenylsulfone metabolite is methemoglobinemia (71).

Clofazimine (Lamprene)



Clofazimine

Although classified as a secondary drug for the treatment of leprosy and commonly used as a component of multidrug therapy, clofazimine appears to be increasing in use. Clofazimine was first used to treat advanced leprosy unresponsive to dapsone or STM in 1966. The chemical, a phenazine derivative, is a water-insoluble dye (dark-red crystals) that leads to pigmentation of the skin. In addition, discoloration (pink, red, or brownish-black) of the feces, eyelid lining, sputum, sweat, tears, and urine is seen.

MECHANISM OF ACTION The mechanism of action remains unclear at the present time. The molecule possesses direct antimycobacterial and immunosuppressive properties. It has been shown that clofazimine increases prostaglandin synthesis and the generation of antimicrobial

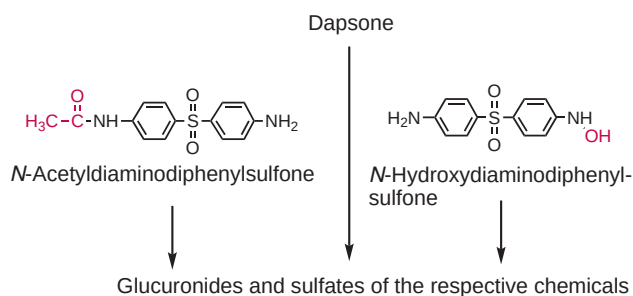


FIGURE 36.20 Metabolites of dapsone.

reactive oxidants from neutrophils, which can play a role in the antileprosy effects. The host cell defense can be stimulated by clofazimine, resulting in the generation of oxidants, such as the superoxide anion, which in turn could have a lethal effect on the organism (72,73).

STRUCTURE–ACTIVITY RELATIONSHIP Several investigators have reported studies directed toward an understanding of the SAR of clofazimine (74–76). Substituents on the imino group at position 2, *p*-chloro substitution on the phenyls at C-3 and N-10, and substituents at position 7 have been investigated. The imino group at C-2 appears to be essential, with activity increased when the imino group is substituted with alkyl and cycloalkyl groups. Halogen substitution on the para position of the two phenyls at C-3 and N-10 enhance activity but are not essential to activity. The following order of activity has been reported: Br > Cl > CH₃ > EtO > H or F. In the analogs studied, the increased activity correlates well with pro-oxidative activities of the molecule (e.g., ability to generate superoxide anion) as well as increased lipophilicity.

METABOLISM Various metabolites of clofazimine have been identified, but these account for less than 1% of the administered dose. The lack of higher concentrations of the metabolites can, in part, result from the very slow elimination of clofazimine from the body, which has an estimated half-life of 8.8 to 69 days. The lipophilic nature of clofazimine results in distribution and storage of the drug in fat tissue. There appears to be some discrepancy as to the structures of the metabolites (76,77). The most recent studies suggest the presence of two conjugates, with the possibility of intermediates (Fig. 36.21). Clofazimine is thought to undergo hydroxylic dehalogenation on the 3-chloroaniline, followed by sulfate conjugation and 4-hydroxylation, followed by glucuronic acid conjugation.

Rifampin (Rifadin, Rimactane)

RIF, an antituberculin drug, has already been discussed. Its actions against *Mycobacterium leprae* parallel those effects reported for *M. tuberculosis*. Today, RIF is considered to be an effective antileprosy agent when used in combination with the sulfones.

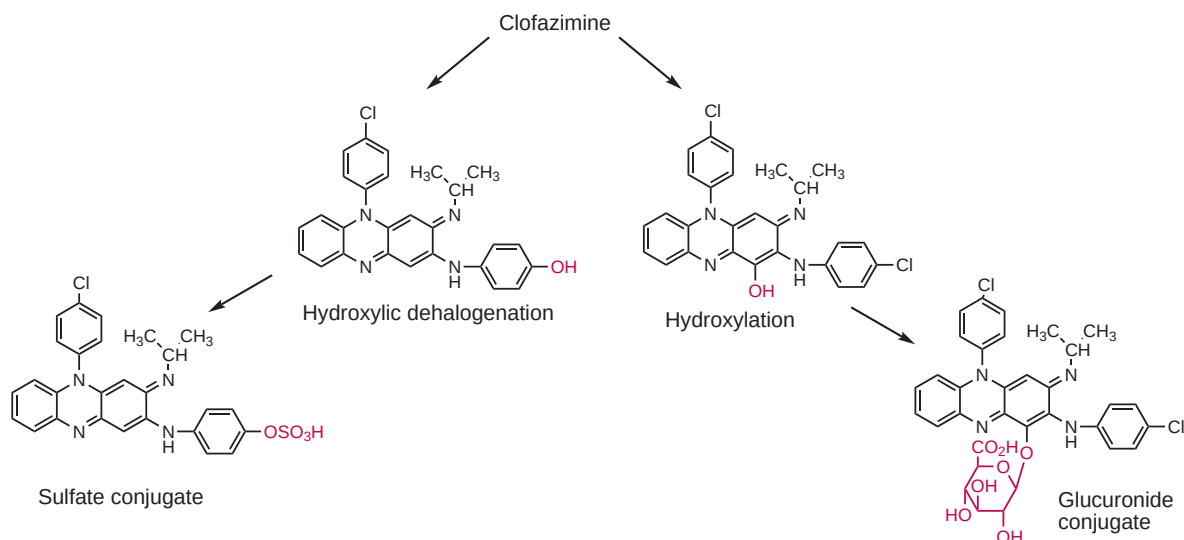
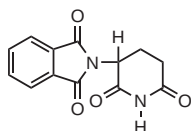


FIGURE 36.21 Human metabolic products of clofazimine.

Thalidomide (Thalomid)



Thalidomide

The development of painful, tender, inflamed, subcutaneous nodules that can last a week or two but can reappear and last for long periods is seen in a number of diseases. In the case of leprosy, the condition is referred to as erythema nodosum leprosum (ENL). The condition appears to be a hypersensitivity reaction, and although it can appear in nontreated patients, it commonly is seen as a complication of the chemotherapy of leprosy. In addition to painful nodules, the patient can experience fever, malaise, wasting, vasculitis, and peripheral neuritis. This condition has been successfully treated with thalidomide. Recently, thalidomide has been approved by the FDA for treatment of ENL and is considered to be the drug of choice (78,79). The mechanism whereby thalidomide produces relief is thought to be associated with the drug's ability to control inflammatory cytokines. Specifically, thalidomide inhibits the synthesis and release of tumor necrosis factor- α , which is synthesized and released by blood mononuclear cells and appears in the serum during ENL; concentrations drop when the patient is treated with thalidomide. In addition to the treatment of ENL, thalidomide has been reported to exhibit beneficial effects in the treatment of aphthous ulcers in HIV-positive patients, Behcet disease, chronic graft-versus-host disease, rash caused by systemic or cutaneous lupus erythematosus, pyoderma gangrenosum, and multiple myeloma.

Thalidomide is a very potent teratogenic agent, with a history of an estimated 10,000 deformed infants born to mothers who used the drug during pregnancy. It can be used safely in postmenopausal women, but strict controls are required for women of childbearing age. Although no evidence suggests that men can transmit the drug during sex, the use of condoms by male patients will be required.

Therapeutic Considerations

Since its introduction into the chemotherapy of leprosy in 1947, dapsone has proved to be the single most effective agent. This drug was used as a monotherapeutic agent despite the recognition that resistant strains were beginning to emerge. Since 1977, monotherapy with dapsone is no longer recognized as an acceptable method for the treatment of leprosy. Today, combination chemotherapy is the method of choice. The combination consists of RIF (600 mg monthly), dapsone (100 mg daily), and clofazimine (300 mg monthly, with 50 mg daily added for patients with multibacillary leprosy, which is defined as five or more skin lesions). Therapy is usually continued for at least 2 years or as long as skin smears are positive. The patient is kept under supervision for 5 years following completion of chemotherapy. A similar treatment regimen is recommended for treatment of paucibacillary leprosy (defined as five or fewer skin lesions) except that treatment is continued for 6 months and the patient is kept under observation for an additional 2 years (80,81). It should be noted that the patient is noninfectious within 72 hours of starting treatment. Other combinations that have been reported include RIF plus ofloxacin and minocycline or ofloxacin plus minocycline. The new regimens allow a shortened treatment period and a reduced rate of relapse.

An important aspect of therapy for leprosy involves the treatment of peripheral nerve damage. This nerve damage can be treated with steroids, such as prednisolone. For severe cases, however, thalidomide is used.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

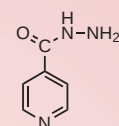
Elizabeth Coyle, PharmD

After reviewing RG's medication list, it is realized that the increase in RG's INR is most likely due to a drug interaction between isoniazid and warfarin. Both drugs are metabolized through the liver, and isoniazid is a known inhibitor of many hepatically metabolized drugs. The pharmacist decides to decrease RG's warfarin dose to 2.5 mg daily, to recheck his INR in 2 days, and to continue to decrease the warfarin until RG's INR is again therapeutic at 2.5. The pharmacist will make sure to increase the warfarin dose once RG is finished taking 9 months of isoniazid.

Chemical Analysis

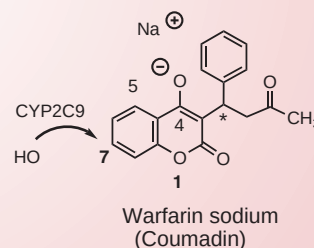
S. William Zito and Victoria Roche

Isoniazid is an aromatic hydrazide that kills replicating *M. tuberculosis* organisms and halts the replication of dormant organisms. Its mechanism of action is to acylate NADH and an inhA enzyme that generates mycolic acids required for bacterial cell wall synthesis. The specific acylating intermediates are produced by the action of a catalase-peroxidase known as katG. The most important ones are the phenylacetyl radical, phenylacetylperoxy radical, and isonicotinic peroxide. The terminal nitrogen of isoniazid can be substituted with lipophilic alkyl groups without loss of activity, but the greatest potency is realized in the unsubstituted primary hydrazide. Isoniazid is metabolized via N-acetylation by an N-acetyltransferase to a metabolite that, in turn, generates an extremely hepatotoxic acetyl radical via a hydroxylamine intermediate. Patients who are slow acetylators express lower amounts of this transferase enzyme than those who are fast acetylators, and therefore they are less likely to experience hepatotoxicity from this agent. Oral absorption of this drug is good, but it is slowed by food and metal-containing antacids.



Isoniazid

Warfarin is a 3-arylalkyl-4-hydroxycoumarin derivative. It was developed from structure activity relationship studies of the bis-hydroxycoumarin (dicoumarol) that was discovered to cause bleeding in cattle that ingested fermented sweet clover. Its anticoagulant activity derives from its ability to interfere with the cyclic inter-conversion of vitamin K and vitamin K 2,3-epoxide. Vitamin K is an essential cofactor necessary for the γ -carboxylation of the glutamic acid residues on the N-terminal of clotting factors II, VII, IX, and X. The 4-hydroxy substituent confers water solubility and the substituent at position 3 affects the pharmacokinetic parameters of onset, duration, and half-life. Warfarin is the only coumarin used in the United States. It is rapidly absorbed and highly protein bound.



Warfarin sodium
(Coumadin)

Warfarin is metabolized by CYP2C9 to both 6- and 7-hydroxy-warfarins, with the 7-hydroxy metabolite being the major oxidation product. When one thinks of warfarin drug-drug interactions, it is usually that, because warfarin is highly protein bound, other coadministered drugs will displace it and lead to higher blood levels and therefore enhanced anticoagulant activity. However, in this case it is more likely that the isoniazid is inhibiting the action of CYP2C9 on warfarin, leading to potentiation of its anticoagulant activity.

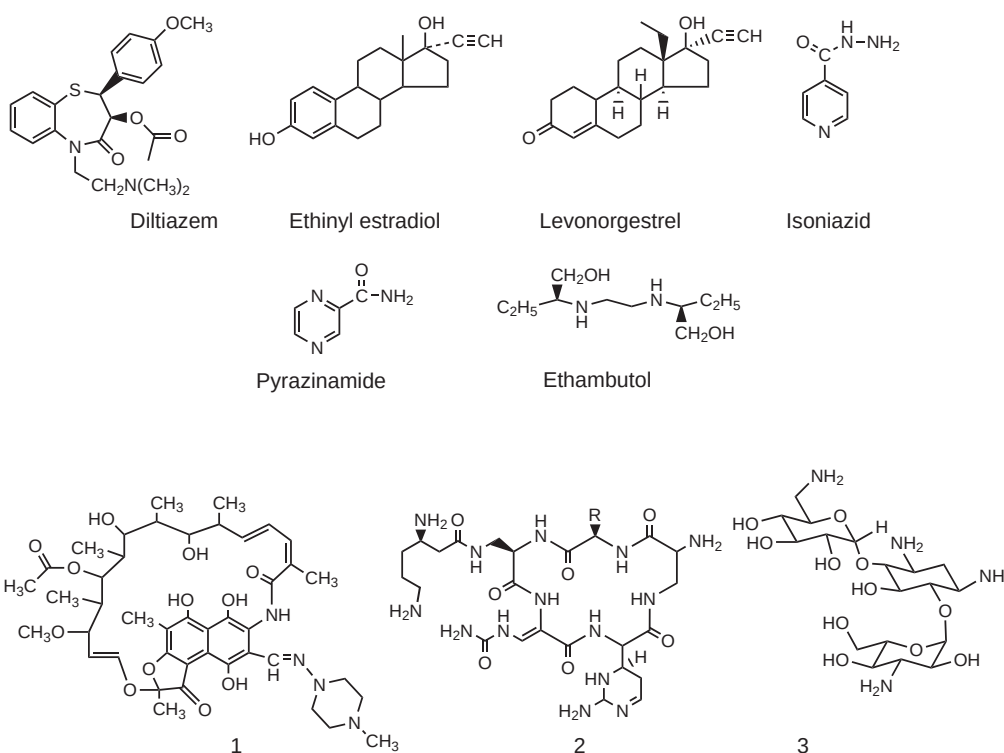
CASE STUDY

S. William Zito and Victoria Roche

LT is a 35-year-old female veterinary pathologist who was recently involved with the clinical and pathological evaluation of a dog that had presented with symptoms of a central nervous system disease. However, before a diagnosis could be made, the dog rapidly deteriorated and died. The postmortem examination revealed systemic mycobacteriosis, and *Mycobacterium tuberculosis* was identified by polymerase chain reaction amplification of DNA samples. Infection in companion animals is mainly acquired from close contact to a diseased human. LT, who performed the necropsy on the dog, tested positive for *M. tuberculosis*. Her accidental infection was most likely caused by inhalation of

M. tuberculosis-containing aerosols created by using an electric saw to open the brain cavity.

LT presents to her physician with all the signs and symptoms of early infection (persistent cough, pain in the chest, fatigue, and loss of appetite). A thorough medical workup reveals that LT has hypertension, which is treated with extended-release diltiazem (120 mg/daily); no history of liver or kidney disease; and is taking oral contraceptives (BCs) that contain ethinyl estradiol and levonorgestrel. Her physician wants to treat LT with a four-drug regimen, and decides to use isoniazid, pyrazinamide, and ethambutol; however, he is undecided about which of the following antimycobacterial drugs to use as the fourth agent. What is your opinion?



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 37

Cancer and Chemotherapy

VICTORIA F. ROCHE

Drugs Covered in This Chapter

DNA CROSS-LINKING AGENTS

Miscellaneous DNA alkylating agents

- Altretamine
- Busulfan

Nitrogen mustards

- Bendamustine
- Chlorambucil
- Cyclophosphamide
- Ifosfamide
- Mechlorethamine
- Melphalan
- Thiotepa

Nitrosoureas

- Carmustine
- Lomustine
- Streptozocin

Organoplatinum complexes

- Carboplatin
- Cisplatin
- Oxaliplatin
- Picoplatin
- Satraplatin

Procarbazine and triazenes

- Dacarbazine
- Procarbazine
- Temozolomide

ANTIMETABOLITES

Antifolates

- Methotrexate
- Pemetrexed

- Pralatrexate

DNA methyltransferase inhibitors

- Azacitidine
- Decitabine
- Nelarabine

DNA polymerase inhibitors

- Cladribine
- Clofarabine
- Cytarabine
- Fludarabine
- Gemcitabine

Miscellaneous antimetabolites

- Hydroxyurea
- Pentostatin

Pyrimidine antagonists

- Capecitabine
- Floxuridine
- Fluorouracil

Purine antagonists

- Mercaptopurine
- Thioguanine

HISTONE DEACETYLASE INHIBITORS

- Romidepsin
- Vorinostat

IMMUNOMODULATORS

- Lenalidomide
- Thalidomide

MISCELLANEOUS ANTIBIOTICS

- Bleomycin
- Dactinomycin

- Mitomycin

MISCELLANEOUS ANTICANCER AGENTS

- Arsenic trioxide
- Bortezomib

MITOSIS INHIBITORS

- Cabazitaxel
- Docetaxel
- Estramustine
- Ixabepilone
- Paclitaxel
- Vinblastine
- Vincristine
- Vinorelbine

TOPOISOMERASE POISONS

- Daunorubicin
- Doxorubicin
- Epirubicin
- Etoposide
- Idarubicin
- Irinotecan
- Mitoxantrone
- Teniposide
- Topotecan
- Valrubicin

TYROSINE KINASE AND RELATED INHIBITORS

- Dasatinib
- Erlotinib
- Everolimus

- Gefitinib
- Imatinib
- Lapatinib
- Nilotinib
- Sorafenib
- Sunitinib
- Temsirolimus

Abbreviations

ABC , ATP-binding cassette	GAR , glycine amide ribonucleotide	PDGFR , platelet-derived growth factor receptor
ABL , Abelson	GI , gastrointestinal	P-gp , P-glycoprotein
AIC , 5-aminoimidazole-4-carboxamide	GIST , gastrointestinal stromal tumors	Ph , Philadelphia chromosome
Ala , alanine	GMP , guanosine monophosphate	Phe , phenylalanine
AMP , adenosine monophosphate	GSH , glutathione	Pt , platinum
APL , acute promyelocytic leukemia	HER2 , human epidermal growth factor receptor 2	RCC , renal cell carcinoma
Asn , asparagine	HGPRT , hypoxanthine guanine phosphoribosyl transferase	ROS , reactive oxygen species
Asp , aspartate	His , histidine	SNP , single nucleotide polymorphism
ATP , adenosine triphosphate	hTS , human thymidylate synthase	SPF , sun protection factor
BCR , breakpoint cluster	Ile , isoleucine	STEPS , System for Thalidomide Education and Prescribing Safety
BCRP , breast cancer resistance protein	ITPA , inositol triphosphate pyrophosphatase	TEPA , triethylenephosphoramidate
CLL , chronic lymphocytic leukemia	Leu , leucine	THF , tetrahydrofolate
CML , chronic myelogenous leukemia	Lys , lysine	Thr , threonine
CNS , central nervous system	MAP , microtubule-associated protein	TK , tyrosine kinase
CYP , cytochrome P450	MDR , multidrug resistance	TKI , tyrosine kinase inhibitor
Cys , cysteine	Met , methionine	TNM , tumor-node-metastasis
DACH , diaminocyclohexane	MMR , mismatch repair	TopI , topoisomerase I
DHF , dihydrofolate	MoAb , monoclonal antibody	TopIIα , topoisomerase II α
DHFR , dihydrofolate reductase	MTIC , 3-methyl-(triazene-1-yl)imidazole-4-carboxamide	TPMT , thiopurine methyl transferase
DPD , dihydropyrimidine dehydrogenase	mTOR , mammalian target of rapamycin	Trp , tryptophan
dTMP , deoxythymidine monophosphate	NER , nucleotide-excision repair protein	Tyr , tyrosine
dUMP , deoxyuridine monophosphate	NHL , non-Hodgkin lymphoma	Val , valine
EGCG , epigallocatechin-3-gallate		VEGFR , vascular endothelial growth factor receptor
EGFR , epidermal growth factor receptor		
FDA , U.S. Food and Drug Administration		
FPGS , folyl polyglutamate synthase		

SCENARIO

Kelly Nystrom, PharmD, BCOP

DT, a 37-year-old white man, was diagnosed with precursor B-cell acute lymphoblastic leukemia approximately 1 year ago. He was treated with a R-Hyper-CVAD program (cyclophosphamide, mesna, vincristine, doxorubicin and dexamethasone alternating with rituximab, high-dose methotrexate and cytarabine) and achieved a complete remission. He presented to the office 1 month ago with complaint of mild headaches (treated

with ibuprofen 800 mg orally every 6 hours as needed) and left upper quadrant pain. A workup showed disease relapse and DL was admitted for cycle 2 of re-induction with a R-Hyper-CVAD program.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter.)

INTRODUCTION

Healthy cells are under strict biochemical control for growth and differentiation. Cells divide and proliferate under the influence of various growth stimulators and are subject to arrested growth (senescence) and

programmed cell death (apoptosis). In cancer, these regulatory processes have gone awry, and cells grow and divide uncontrollably, consuming energy and losing both structure and function due to an inability to adequately differentiate. To add insult to injury, rampant cell division

is accompanied by disabled cell-death processes, leading first to cellular immortality and, eventually, to genetic instability. The causes of cancer are many and varied (e.g., chemical, environmental, viral, and mutagenic), but all ultimately lead to an aberration in the expression of proto-oncogenes, the products of which control normal cell life. When these genes mutate to become oncogenes in a sequential, multistep process, cancer results. Oncogenes (e.g., *myc* and *ras*) can either overexpress or underexpress regulatory biochemicals, resulting in preferential and accelerated cellular growth. Concomitantly, tumor suppressor genes (e.g., anti-oncogenes like *p53*, *p21*, *p^{INK4A}*, and *retinoblastoma*) can be inhibited (1).

Initially, tumors grow exponentially, taking a consistent amount of time for every doubling of the tumor cell population. In fact, the majority of a cancer cell's lifetime is spent before the tumor presents clinically. Initially, growth is very rapid (doubling time measured in days), but doubling time can slow to weeks or months as the tumor ages due to increasingly poor vascularization and the resulting decrease in access to blood and essential nutrients (2).

SELECTED DEFINITIONS

Oncogenes and Tumor Suppressor Genes

Oncogenes are regulators of cellular communication with the outside environment. They are derived through the mutation of proto-oncogenes, which are normal and ubiquitous genes involved in the regulation of homeostatic cellular functions. Mutations in proto-oncogenes can occur as spontaneous point mutations, inherited germline mutations, chromosomal rearrangements or through augmentation of gene expression. Regardless of the mutational mechanism, when the mutated oncogenes are stimulated by exposure to chemical, environmental, or viral carcinogens, they produce proteins that are either wrongly expressed within their normal cell or expressed in inappropriate tissues. In either case, cellular proliferation leading to cancer results (1,3).

Tumor suppressor genes are intended to keep oncogenes in check by halting uncontrolled cellular growth. In direct opposition to oncogenes, which induce cancer when stimulated or amplified, tumor suppressor genes promote cancer when inactivated or attenuated. Two of the most prevalent tumor suppressor genes involved in the generation of cancer are *p53* and *retinoblastoma*, or *Rb*. When either of these two suppressor genes loses function, the negative control on cellular proliferation is lifted and cells gain immortality (an essential quality of a cancer cell). The loss or disruption of function of the *p53* tumor suppressor gene is found in approximately half of human cancers and is a harbinger of a poor prognosis.

Oncogenes and tumor suppressor genes that have been linked to specific types of cancer are identified in Table 37.1 (1,3). Table 37.2 relates oncogenic markers of selected cancers to disease prognosis and treatment strategy (1).

TABLE 37.1 Oncogenic Origin of Selected Cancers

Cancer Type	Common Oncogenic or Tumor Suppressor Gene Origin
Chronic myelogenous leukemia	<i>bcr-abl</i> proto-oncogene translocation
Follicular lymphoma	<i>bcl-2</i> amplification, <i>myc</i> mutation
Sporadic thyroid cancer	<i>ret</i> mutation
Colorectal and gastric cancer	<i>APC</i> gene mutation
Familial breast and ovarian cancer	<i>BRCA1</i> , <i>BRCA2</i> mutation
Invasive ductal breast cancer	<i>HER2</i> amplification
Familial melanoma	<i>p16^{INK4A}</i> mutation
Childhood neuroblastoma, small cell lung cancer	<i>N-myc</i> amplification
Leukemia, breast, colon, gastric, and lung cancer	<i>c-MYC</i> amplification
Renal cell cancer	<i>VHL</i> dysfunction

Cell Cycle

When cells reproduce, they do so via a very specific game plan known as the cell cycle. Cell division (mitosis) kicks off the cycle, and after a period of 30 to 60 minutes, the cells go into either a resting phase (called G_0) or a pre-synthetic (gap) phase (called G_1), in which enzyme production occurs in preparation for de novo nucleic acid synthesis. Production of DNA then occurs in an S phase that can last up to 20 hours. The S phase is followed by a gap phase (G_2), in which RNA, critical proteins, and the mitotic spindle apparatus are generated for the next mitotic (M) phase (3,4).

This is important to our discussion since some anti-cancer agents are specific for a certain phase of the cell cycle. For example, antimetabolite antineoplastics damage cells in the S phase, whereas mitosis inhibitors pack their greatest cell-killing punch in the M phase. The administration of cell cycle phase-specific antineoplastics is carefully planned so that the drug encounters cancer cells at their most vulnerable moments. This often involves continuous infusion therapy or treatments spread over several days. Other antineoplastic agents are toxic to cells regardless of cycle phase (e.g., DNA alkylating agents and most antineoplastic antibiotics). These cell cycle phase-nonspecific agents can often be administered as a single bolus injection and/or at any time that is feasible for the provider and convenient for the patient.

In general, cancer cells undergoing rapid division are most vulnerable to the cytotoxic action of antineoplastic agents, and antineoplastic therapy holds its greatest promise for positive outcomes if initiated when the tumor is small but growing aggressively. Conversely, slow-growing tumors with a high percentage of cells remaining in the

CLINICAL SIGNIFICANCE



Chemotherapy has significantly changed the treatment of cancer since the first agents were studied in the 1940s. Understanding the chemical mechanism of action for traditional chemotherapy agents, including whether the agent is cell-cycle specific or cell-cycle nonspecific, is important so administration can be planned accordingly and co-administration of agents with similar toxicities can be avoided.

The disadvantage of traditional chemotherapy agents is the inability of the agent to recognize the difference between normal and cancer cells. So, although the agents may shrink or eliminate the tumor, the treatment is accompanied by many unwanted side effects. There are numerous examples where the chemical understanding of the chemotherapy agent is necessary. For example, when an anthracycline, such as doxorubicin, is given, free radicals are formed through an iron-dependent process that begins with the reduction of the anthracycline quinone ring to a hydroquinone. These free radicals can cause significant cardiac damage if left unmonitored. Dexrazoxane is a product that is predictably hydrolyzed to an electron-rich metabolite that binds to iron and prevents these toxic free radicals from being formed, thus reducing damage to the heart. Another example can be found when using cyclophosphamide or ifosfamide. The dissociation of aldophosphamide produces the active compound

phosphoramidate mustard, but also produces acrolein, an electrophilic aldehyde, which can cause significant damage to the bladder by forming covalent bonds with cysteine residues of essential proteins. This bladder damage, or hemorrhagic cystitis, can be prevented with the use of mesna, a sulfhydryl-containing cysteine decoy that inactivates the effects of acrolein in the bladder. In addition, deficiencies found in individual patients can impact toxicities of certain chemotherapy agents.

Dihydropyrimidine dehydrogenase (DPD) is needed to deactivate both fluorouracil and capecitabine, which is a prodrug of fluorouracil. About 5% of the population has a deficiency of DPD, which results in accumulation of the drug and leads to severe toxicity if fluorouracil or capecitabine are administered. Recognizing the metabolic profile of these agents allows pharmacists to identify pharmacogenetically “at-risk” patients so that the safest dose of these highly toxic drugs can be administered.

Understanding the chemical basis for the toxicities seen with chemotherapy is imperative in managing or preventing them in clinical practice.

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G₀ phase (e.g., non-small cell lung cancer) are often non-responsive to cell cycle-specific chemotherapy (4). If the tumor is not detected until it is quite large, therapy can also be compromised by inefficient or substandard drug delivery due to poor tumor vascularization.

Metastasis

Metastasis refers to the process by which malignant cells leave the parent tumor, migrate to distant sites, and invade new tissue. The primary metastatic highways used by meandering cancer cells are the blood and lymph fluids. Sloughed cells must find a biologic environment with all of their essential growth factors in place before

they can put down roots and evolve into a full-fledged metastatic tumor. Since many distinct and interdependent steps must be accomplished to establish metastatic disease, the process has been termed the “metastatic cascade” (5). Fortunately, there are many opportunities within the cascade for the body to mount a successful defense and destroy the potential invaders.

Cancer Staging

Clinicians need to have a common language through which to communicate about disease severity to make the best team-based decisions about the relative risks and benefits of treatment options. In the tumor-node-metastasis

TABLE 37.2 Oncogenic Markers and Therapeutic Strategies in Selected Cancers

Cancer Type	Oncogenic Marker	Prognosis/Responsiveness to Chemotherapy	Approach
Breast cancer	<i>HER2</i> amplification	Poor	Aggressive chemotherapy, targeted therapy
Acute myelogenous leukemia	t(8;21) or inv(16) translocation	Good	Standard chemotherapy
Acute lymphocytic leukemia	<i>bcr-abl</i> rearrangement	Poor	Bone marrow transplantation

(TNM) cancer staging classification, the severity of solid tumor neoplastic growth is characterized by the size of the tumor mass (T_1 to T_4), the extent of lymph node involvement (N_0 to N_3), and whether distant metastasis has occurred (M_0 or M_1). The larger the subscripted number in each of these parameters, the more advanced and/or disseminated the disease. Taken together, the TNM characteristics of a tumor can be translated into a comprehensive staging scale ranging from I (localized) to IV (metastatic). The intermediate disease severity stages indicate local (stage II) or regional (stage III) tissue invasion (3). Staging is an essential prerequisite for the prediction of prognosis and the identification of the most appropriate treatment plan and optimal dosing regimen (2).

Response Criteria

In this era of patient-centered, team-based care, it is equally beneficial to quantify a patient's clinical response to therapy in a manner that is consistent and universally understood by all health care providers. Five discrete anticancer therapy response categories have been defined, with criteria established for each (3). Whereas *cure* is obviously the most noble goal, it is very difficult to achieve in most types of cancer. A cure for all cancers except breast and melanoma equates to no evidence of disease for a minimum of 5 years. More commonly, the response category viewed as the pinnacle is *complete response*, in which the patient has no evidence of cancer for at least 1 month following the cessation of therapy, but where relapses are still possible. A *partial response* is claimed when tumor size has been reduced by 30% or more and there is no evidence of new lesions at the primary site or elsewhere for a minimum of 1 month. If this level of clinical improvement is not reached, yet the patient has experienced significant attenuation of symptoms and/or enhancement of quality of life, the response is termed *clinical benefit*. A less optimistic response category is *stable disease*, in which tumor size has either increased by less than 20% or decreased by less than 30%. Most dire is *progression*, a category that is characterized by tumor growth at the 20% or higher level and/or the formation of new lesions during therapy.

HISTORICAL BACKGROUND (6)

"Those who have not been trained in chemistry or medicine, which after all is only applied chemistry, may not realize how difficult the problem of [cancer] treatment really is. It is almost, not quite, but almost as hard as finding some agent that will dissolve away the left ear, say, yet leave the right ear unharmed: so slight is the difference between the cancer cell and its normal ancestor."

Thus wrote noted cancer researcher and physician Dr. William H. Woglom in a monograph published by the American Association for the Advancement of Science in 1947 (7). Although somewhat predictive of what we now know to be true regarding the

relationship between resident genes and oncogenes, Dr. Woglom's rather gloomy prognosis of our ability to meet cancer on its own ground and beat it was underpinned by centuries of unsuccessful attempts to treat neoplastic disease with toxic metals, including lead, arsenic, silver, zinc, antimony, mercury, and bismuth. However, the era of more promising chemotherapy was just on the horizon even as Woglom penned his words of therapeutic woe.

Among the first nonmetallic therapeutic agents to show benefit in the treatment of cancer was cortisone and, later, prednisone. In the 1940s, these glucocorticoids were shown to induce tumor regression in a laboratory cancer model (murine lymphosarcoma) and in acute leukemia. In the same decade, the retrospective recognition that World War I soldiers exposed to sulfur mustard gas, used as an agent of war, suffered from damaged lymphoid tissue and bone marrow led to the development of the cytotoxic nitrogen mustards for the treatment of lymphoma. Chemists then used their scientific understanding of mustard reactivity to design agents that were either "superpotent" and nonselective (e.g., highly toxic) or of lower reactivity so as to provide oral activity and less systemic toxicity.

The discovery in 1940 that *p*-aminobenzenesulfonamide was effective against streptococcal infections ushered in the era of antimetabolite chemotherapy. The development of antifolate antineoplastics, which were shown to be effective in combating childhood leukemias, got its start in the late 1940s. In the mid to late 1950s, on the heels of the success of antifolates, came the development of antimetabolites based on the structures of endogenous purine and pyrimidine bases. Perhaps the most exciting discovery in this regard was the recognition that a very simple analog of the endogenous pyrimidine uracil (5-fluorouracil) was a potent inhibitor of deoxythymidine monophosphate biosynthesis and that inhibiting the production of this essential nucleotide produced positive results in patients suffering from colon, stomach, pancreatic, and breast cancers. Antimetabolites that target DNA polymerase (e.g., cytarabine) were conceptualized and synthesized in the late 1950s and subsequently shown to be effective in acute myeloblastic leukemia.

The antibiotic antineoplastics came into clinical utility when the highly toxic actinomycin (discovered in the 1940s) was found to be effective in the treatment of human testicular cancer and uterine choriocarcinoma. Other natural anticancer antibiotics, such as bleomycin, subsequently were found to be active against various hematologic cancers and solid tumors (1960s), which led in more recent times to the development of semisynthetic analogs with both high potency and wider margins of safety. The antimitotic vinca alkaloids vincristine and vinblastine were shown to have activity against Hodgkin's disease and acute lymphoblastic leukemia around the same time that the antibiotic antineoplastics were being developed.

Cancer chemotherapy appears to have come full circle since the “metal-intense Renaissance,” because some of the newer anticancer drugs to join the U.S. market are organometallic platinum (Pt) complexes. The activity of cisplatin (the first such complex to be commercially available) against lymphosarcoma and solid tumors of the head, neck, and reproductive organs was first noted in the early 1970s. The fortuitous discovery of organoplatinum complexes in the treatment of cancer is attributed to Dr. Barnett Rosenberg, who was studying the impact of electromagnetic radiation on bacterial cell growth using platinum electrodes. He followed up on the astute observation that the bacteria exposed to the electrodes experienced profound changes in cellular structure, which ultimately were attributed to the *in situ* generation of cisplatin. Both Pt(II) and Pt(IV) analogs of cisplatin, which offer high potency coupled with lower resistance potential and fewer use-limiting side effects (e.g., oto-, nephro-, and hematotoxicity), are currently on the market and in clinical trials. In addition to organometallics, the efficacy of sex hormones and hormone antagonists in fighting hormone-dependent cancers (e.g., estrogen receptor-positive breast cancer or prostate cancer) and the advent of therapeutic biologic response modifiers with direct antiproliferative effects (e.g., interferons) have added significantly to the therapeutic options available to providers and the cancer patients for whom they care.

The later 1990s saw the introduction of the tyrosine kinase inhibitors (TKIs) to the antineoplastic armamentarium. The recognition in 1960 that a mutant chromosome known as BCR-ABL (or the Philadelphia [Ph] chromosome) appeared consistently in the cells of patients with chronic myelogenous leukemia (CML) represented the first time a chromosomal aberration had been directly linked to a neoplastic disorder. The product of this abbreviated chromosome, the Bcr-Abl protein, is an unregulated tyrosine kinase that promotes cellular proliferation at the expense of apoptosis. Imatinib, the first rationally designed drug in the TKI class, was made available in 2001 and dramatically changed the treatment and clinical outcome of Ph-positive leukemias. Although imatinib selectively targets the Bcr-Abl protein, other inhibitors with selectivity for epidermal growth factor receptor (EGFR) and vascular endothelial growth factor receptor (VEGFR) kinases soon followed. These growth factor-selective TKIs show efficacy in the treatment of solid tumors of the lung, breast, pancreas, and kidney.

Despite the wide range of antineoplastic agents currently available, it has been estimated that approximately 40% of patients with cancer ultimately succumb to their disease (3). Novel therapies based on an in-depth understanding of the molecular mechanisms involved in the complex cascade of events we call cancer are urgently needed. Fortunately, molecular targets for focused chemotherapeutic interventions are being discovered with increasing regularity, opening the door for the scientifically grounded development of new drugs. The critical role of computer-based technology in facilitating the ability of

chemists to conceptualize and visualize molecular interactions between potential drugs and putative receptor targets that lead to rational drug design and development, as well as in analyzing and managing the overwhelming amounts of data that are generated from these studies, cannot be overestimated. Likewise, the availability of viable tumor cell lines has facilitated a disease-specific orientation to the hunt for more effective therapies. Currently, there are tumor cell lines for lung, colon, breast, ovarian, brain, and kidney cancers, as well as for melanoma and leukemia (8).

Several monoclonal antibodies targeted to tumor cell antigens or proteins critical to cellular proliferation (e.g., human epidermal growth factor, vascular endothelial growth factor, tyrosine kinase, and proteasomes) have found their way to the U.S. market. In addition, several new targets for anticancer drug development currently are being actively explored by biomedical scientists (1,8). For example, cancer cells overexpress the enzyme telomerase, which inhibits the natural destruction of chromosomal telomeres (DNA caps), leading to unwanted cellular immortality. Telomerase inhibitors would be expected to reestablish cellular senescence and to halt uncontrolled cell division by maintaining the integrity of the telomeres and are being pursued as a new biochemical approach to disease attenuation or control (9). Other potential anti-neoplastic drug targets being seriously investigated are aberrant genes or enzymes unique to specific tumors and P-glycoprotein (P-gp), which is overexpressed in many cancers as a result of an amplified *mdr-1* gene and responsible for the rapid ejection of antineoplastic agents from target cells. Other multidrug resistance-associated proteins (the MRP family) involved in this devastating rebound of the cancer cell are also being investigated as potential sites of therapeutic intervention. The intense focus on resistance molecules such as P-gp is warranted because patients whose tumors express this efflux-promoting protein respond poorly to chemotherapy and have a poor prognosis (2).

It is hoped that clinicians will one day be able to generate a genetic expression profile for each patient to help them assess the likelihood of response to all possible therapies and to guide pharmacotherapy selection. Pharmacogenomics-based predictors of therapeutic response to anticancer drugs currently being explored include single nucleotide polymorphisms (SNPs) and the multiplicity of genes and gene products within a single biochemical pathway (10). While there are multiple scientific, regulatory, and ethical barriers to overcome before the full power of pharmacogenomics can positively impact the care of every cancer patient, these issues are being actively addressed at the national level, giving hope that the age of individualized cancer chemotherapy may indeed be close at hand.

DISEASE STATE

Cancers can usually be classified as lymphatic, epithelial, nerve, or connective tissue related, and tumor nomenclature is based on tissue of origin as follows: carcinoma

(epithelial origin), sarcoma (muscle or connective tissue origin), leukemia and lymphoma (lymphatic or hematologic origin), and glioma (neural origin). The risk of developing epithelial-derived cancers increases with age.

Incidence

In 2007, approximately 1.4 million people received a diagnosis of cancer, and approximately 560,000 died of the disease (3). The American Cancer Society estimates that over 1.5 million new cases of cancer will be diagnosed in 2010 (11). If the ratio of new diagnoses to deaths observed in 2007 holds steady, over 555,000 lives will be lost. The most commonly acquired cancers include those of the prostate, breast, lung, colon, and rectum. Lung cancer is the most fatal and will be responsible for approximately 157,000 U.S. deaths in 2010. These prominent cancers (prostate/breast, lung, and colorectal) occur with very similar frequency in men and women, and few gender-related differences in mortality have been noted (Table 37.3) (11). Some geographical differences in incidence have been observed, with lung cancer being more prevalent in rural southern U.S. states, and breast and colon cancer more commonly diagnosed in the “northeast corridor” of the United States (12).

Signs and Symptoms

The clinical manifestations of cancer can vary widely depending on type and stage of neoplastic disease. The American Cancer Society has been promulgating its list of the major warning signs of cancer for decades (Table 37.4) (3,13). Patients are well-served by being familiar with these early warning signs because cancer is most effectively treated when diagnosed before advanced disease develops. Recognizing that the first letters of each sign spell the word “caution” can help adults remember them. One readily recognized symptom of cancer is persistent weight loss (especially in children), and severe, unrelenting pain is a hallmark symptom of cancer in the later stages. Solid tumors can become palpable or observable masses when the cancer is advanced.

Biochemical Bases and Causes of Cancer

Currently, it is understood that cancer is caused by mutations in “resident” or normal genes rather than by the introduction of foreign genes into otherwise healthy systems (1,3). The single-gene theory of cancer (where a single mutation could result in neoplastic disease) has been

TABLE 37.4 The American Cancer Society’s Major Warning Signs of Cancer

Cancer Warning Signs in Adults	Cancer Warning Signs in Children
Change in bowel or bladder habits	Continued unexplained weight loss
A sore that does not heal	Frequent headaches, with vomiting
Unusual bleeding or discharge	Persistent pain in bones or joints
Thickening or lump in breast or elsewhere	Any unusual mass or swelling
Indigestion or difficulty in swallowing	Sudden eye or vision changes
Obvious change in a wart or mole	Recurrent or unexplained fever
Nagging cough or hoarseness	Excessive bruising or bleeding
	Noticeable paleness or loss of energy

abandoned in favor of the multiple mutation prerequisite, and complex gene pathways, interactions, and communications are now the focus of study in the understanding of malignant processes and their treatments. Once determined, the “mutational profile” of malignant cells may very well predict such parameters as disease severity, most promising therapeutic interventions, and clinical outcome.

The development of cancer occurs in four discrete steps or phases. In the **initiation** phase, exposure to a precipitating carcinogen prompts irreversible mutation in a number of different genes. The **promotion** phase is a time during which mutated cells arising from altered genes grow preferentially compared to normal cells. This preferential growth may result from continued exposure to the original carcinogen or from promotion by environmental “accelerants.” This stage is reversible, so cancer sometimes can be avoided with appropriate changes in diet and lifestyle. The **transformation** phase is the 5- to 20-year progression of a mutated cell to a cancer cell. Cellular proliferation, clonal colony development, tissue invasion and destruction, and metastasis define the final **progression** phase of cancer development (3).

TABLE 37.3 Estimated 2010 U.S. Incidence and Mortality of Common Cancers

	Prostate/Breast		Lung and Bronchus		Colorectal	
	Men	Women	Men	Women	Men	Women
Incidence	217,730 (28%)	207,090 (28%)	116,750 (15%)	105,770 (14%)	72,090 (9%)	70,480 (10%)
Mortality	32,050 (11%)	39,840 (15%)	86,220 (29%)	71,080 (26%)	26,580 (9%)	24,790 (9%)

As previously mentioned, the genetic mutations leading to the diseases that we call cancer can be stimulated by a variety of chemical, environmental, and viral triggers. Both RNA retroviruses and DNA viruses have been implicated in human cancer causation (Table 37.5), although many more DNA than RNA viruses have oncogenic potential (14,15).

Individuals in certain occupations may be at enhanced risk for the development of some cancers due to unavoidable exposure to carcinogenic chemicals (12). Perhaps the best-known example of occupationally induced cancer involves exposure to asbestos, which has been conclusively linked with the development of lung, pleural, and peritoneal malignancies. Miners exposed to radon are also at a significantly enhanced risk for the development of lung cancer, as are individuals exposed through their work to soot, tars, hexavalent chromium, and nickel-containing compounds. The aromatic amines β -naphthylamine and 4-aminobiphenyl are known to induce bladder cancer, and exposure to the common organic solvent benzene has been linked to the development of leukemia.

Environmental carcinogens are all around us (Table 37.6) (12). Fortunately, individuals can do many things to protect themselves from exposure or from negative consequences of limited exposure. The chemicals deposited in the lungs from inhaling cigarette smoke are the primary cause of lung cancer in the United States, but smokers who quit decrease their risk for this often-fatal cancer by 67% or more after 10 smoke-free years (12). Nonsmokers can protect themselves from the cancer-promoting effects of secondhand smoke by removing themselves from smoke-filled environments. Smoking combined with alcohol has a synergistic effect in promoting the development of oral cancer. In addition to quitting smoking, abstaining from alcohol or drinking in

TABLE 37.5 RNA and DNA Viruses Associated with Cancer Development

Virus	Cancer
<i>RNA Virus</i>	
Human T-lymphotrophic virus	Adult T-cell leukemia
Hepatitis C	Hepatocellular carcinoma
<i>DNA Virus</i>	
Hepadnavirus	Hepatocellular carcinoma
Papillomavirus	Skin cancer, cervical cancer, anogenital cancer
Epstein-Barr virus	Burkett lymphoma, Hodgkin's disease, anaplastic nasopharyngeal carcinoma, gastric cancer
Herpes	Kaposi sarcoma

TABLE 37.6 Some Environmental Precipitants of Cancer

Environmental Cancer Precipitant	Cancer Type
Tobacco	Lung, oral, bladder, pancreatic, stomach, and renal cancer
Alcohol	Liver, rectal, and breast cancer
Tobacco plus alcohol	Oral cancers
Radon	Lung cancer
Halogenated compounds	Bladder cancer
Immunosuppressive agents	Lymphoma
Herbicides	Lymphoma
Ionizing or ultraviolet radiation	Leukemia, breast, thyroid, lung, and skin cancer

moderation is a choice that individuals can make in an effort to decrease their overall risk of developing cancer. Other measures that patients can take to minimize the risk of cancer from environmental causes include protection from damaging ultraviolet rays through the use of high-SPF (sun protection factor) sunscreens and the consumption of foods that are low in fat but rich in carotenoids, vitamins A and C, folate, selenium, and/or fiber (12).

GENERAL THERAPEUTIC APPROACHES

Cancer treatment can be comprised of surgery, radiation, antineoplastic chemotherapy, and/or therapy with biologic response modifiers, which stimulate the patient's own immunologic defense mechanisms. Surgery and radiation (ionizing, thermal, or photodynamic) are favored for isolated or localized cancers; chemotherapy and biologic response modifiers (with or without surgery and/or radiation) are reserved for disseminated or systemic cancers. Chemotherapy can also be used after surgery and/or radiation as an "insurance policy" against microscopic metastatic disease (adjuvant therapy) or before surgery to decrease the size of the mass to be removed (neoadjuvant therapy).

Unfortunately, cancer cells do not simply lie down in the face of chemotherapeutic intervention. Rather, these aggressive cells fight back in an attempt to retain their immortality. Some cancer cells acquire resistance to anti-cancer drugs by downregulating enzymes essential for drug transport or for the activation of antineoplastic prodrugs, or by upregulating enzymes involved in inactivating biotransformation. As noted previously, other mechanisms of biochemical retaliation include downregulation of target enzymes, altered drug uptake and efflux mechanisms (e.g., amplification of the gene that encodes for P-gp or the multidrug resistance-associated protein), inhibition of cellular repair proteins, and apoptosis inhibition (2–4).

Cancer Chemotherapy

The word *antineoplastic* means “against new growth.” In general, the mechanism of cytotoxic action for all antineoplastic agents is interference with cellular synthesis or the function of RNA, DNA, and the proteins that sustain life. All antineoplastic agents are poisons because they are designed to kill cells. Currently available anticancer drugs are often highly and generally toxic, especially for cells with short half-lives. For example, nonspecific destruction of the rapidly dividing cells of the gastrointestinal (GI) tract leads to the severe nausea and vomiting associated with cancer chemotherapy, whereas alopecia and fatigue (as well as susceptibility to infection) are the result of the destruction of rapidly dividing cells in hair follicles and bone marrow, respectively. Factors such as the extent and severity of the disease, individual sensitivity to the antineoplastic mechanism employed by the drugs selected for use, the kinetics controlling drug transport, and cell cycle specificity all impact the chance for chemotherapeutic success (2).

Because cancer chemotherapy is most often given in several courses or “rounds,” with an interval of several days or weeks in between to permit attenuation of side effects, three distinct aspects of drug dosing must be considered when determining the impact of antineoplastic therapy on overall patient welfare. First, the dose that ideally should be given per course has been identified for each commonly employed antineoplastic agent but can be altered significantly by individual patient health status (e.g., hepatic, renal, cardiovascular, hematopoietic, pulmonary, and/or other comorbidities), activity/performance status, genetic makeup, and the nature and anticipated severity of side effects. Each round of antineoplastic therapy kills a given percentage of cancer cells with each administration (“cell kill hypothesis”), and the percentage killed rises proportionally with the dose of drug. If chemotherapy can shrink tumors to 10^4 or fewer cells, normal host defense systems are usually capable of eradicating them (2,3). Therefore, the dose of drug that comes as close as possible to the recommended dose is the goal. Expect significant interpatient variation in response to the same chemotherapeutic regimen secondary to individual genetics, level of debilitation, extent of tissue invasion, critical organ system function (including bone marrow), and past exposure to chemotherapeutic agents.

As alluded to earlier, an ever-growing understanding of genetic polymorphism and its impact on the biosynthesis of target proteins and metabolizing enzymes is helping health care providers make wiser decisions about antineoplastic therapy and drug regimens. The length of the “drug-free” interval is the second important drug-dosing consideration because a shorter interval (or higher dose intensity—the “one-two punch”) is associated with a more aggressive inhibition of tumor growth. Most often, patients cannot tolerate the debilitating side effects (e.g., myelosuppression) without a prolonged interval between rounds. The advent of genetically engineered biologic response modifiers, such as granulocyte

colony-stimulating factor, which boosts the ability of bone marrow to produce neutrophils, has had a positive impact on optimizing dosing intensity/density.

Finally, many chemotherapeutic agents produce serious chronic or delayed toxicities that may be irreversible, particularly in heart, lung, and kidneys, which demands that the total cumulative dose be taken into account when designing the regimen. The ultimate balancing act is to give the patient as much antineoplastic drug as is normally recommended in the time frame most likely to kill the greatest percentage of cancer cells without inducing intolerable or life-threatening toxicity in healthy organs and tissues. Armed with the knowledge of the biochemical and/or molecular basis of toxicity, the pharmacist is in an excellent position to employ appropriate pharmacotherapeutic agents to attenuate unavoidable side effects.

One approach for minimizing unwanted toxicity is to employ a chemotherapeutic regimen of several drugs that act by distinct mechanisms and/or precipitate different side effects. Attacking the tumor with different therapeutic “guns” should target a larger variety of the mutant cells that comprise the tumor and permit a lower dose of each to be used compared to single-agent therapy. Minimizing side effect overlap provides a greater chance that the patient will be able to tolerate therapy and accommodate a shorter interval between courses.

It is essential that the oncology pharmacist be well versed in the pharmacotherapy-based management of severe pain, infection, and the nausea, vomiting, and fatigue associated with chemotherapy. The provision of contemporary and valid drug information to patients and families is essential, as is assistance in helping with the interpretation of information that patients and loved ones secure either through their health care providers or independently (e.g., from the Internet).

THERAPEUTIC CLASSES OF ANTICANCER DRUGS

DNA Cross-Linking Agents (Alkylators and Organometallics)

The primary target of DNA cross-linking agents is the actively dividing DNA molecule. The DNA cross-linkers are all extremely reactive electrophilic (δ^+) structures. When encountered, the nucleophilic groups on various DNA bases (particularly, but not exclusively, the N^7 of guanine) readily attack the electrophilic drug, resulting in irreversible alkylation or complexation of the DNA base.

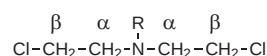
Some DNA alkylating agents, such as the nitrogen mustards and nitrosoureas, are bifunctional, meaning that one molecule of the drug can bind two distinct DNA bases. Most commonly, the alkylated bases are on different DNA molecules, and interstrand DNA cross-linking through two guanine N^7 atoms results. The DNA alkylating antineoplastics are not cell cycle specific, but they are more toxic to cells in the late G_1 or S phases of the cycle.

This is the time when DNA is unwinding and exposing its nucleotides, increasing the chance that vulnerable DNA functional groups will encounter the electrophilic antineoplastic drug and launch the nucleophilic attack that leads to its own destruction. The DNA alkylators have a great capacity for inducing both mutagenesis and carcinogenesis; in other words, they can promote cancer in addition to treating it.

Organometallic antineoplastics (platinum coordination complexes) also cross-link DNA, and many do so by binding to adjacent guanine nucleotides, called diguanosine dinucleotides, on a single strand of DNA. This leads to intrastrand DNA cross-linking. The anionic phosphate group on a second strand of DNA stabilizes the drug–DNA complex and makes the damage to DNA replication irreversible. Some organometallic agents also damage DNA through interstrand cross-linking.

Nitrogen Mustards and Aziridine-Mediated Alkylators

Nitrogen mustards are bis(β -haloalkyl)amines. The term “bis” means two, and the “halo” (short for halogen) in the nomenclature is invariably chlorine. The two chlorine atoms dramatically decrease the basic strength of the amino nitrogen through a strong negative inductive effect. As a result, the unionized conjugate of these drugs predominates at physiologic pH. This is intentional because it is the unionized amine (with its lone pair of electrons) that allows the formation of the highly electrophilic aziridinium ion, which is the reactive DNA-destroying intermediate generated by all true mustards.



Bis- β -haloalkylamine

MECHANISM OF ACTION The mechanism of action of the nitrogen mustards (16) is depicted in Figure 37.1. In step 1, the lone pair of electrons on the unionized amino group conducts an intramolecular nucleophilic attack at the β -carbon of the mustard, displacing chloride anion and forming the highly electrophilic aziridinium ion intermediate, a quaternary amine. The carbon atoms of this strained cyclic structure are highly electrophilic due to the strong negative inductive effect of the positively charged nitrogen atom.

In step 2, a DNA nucleophile conducts an intermolecular nucleophilic attack, which breaks the aziridine ring and alkylates DNA. Although guanine is the preferred nucleic acid base involved in the alkylation reaction, adenine is also known to react. Of critical importance is the fact that the lone pair of electrons on the mustard nitrogen is regenerated when the aziridine ring cleaves.

Steps 3 and 4 are simply repetitions of steps 1 and 2, respectively, involving the second arm of the mustard and a second molecule of DNA. Ultimately, two molecules of DNA will be cross-linked through the carbon atoms of what was once the nitrogen mustard. Finally, hydrolytic depurination (step 5) cleaves the bound guanine residues from the DNA strand. This is an attempt to liberate the DNA from the mustard’s covalent “stranglehold,” but the DNA released from this mustard trap is damaged and unable to replicate. Cell death is the inevitable result. If

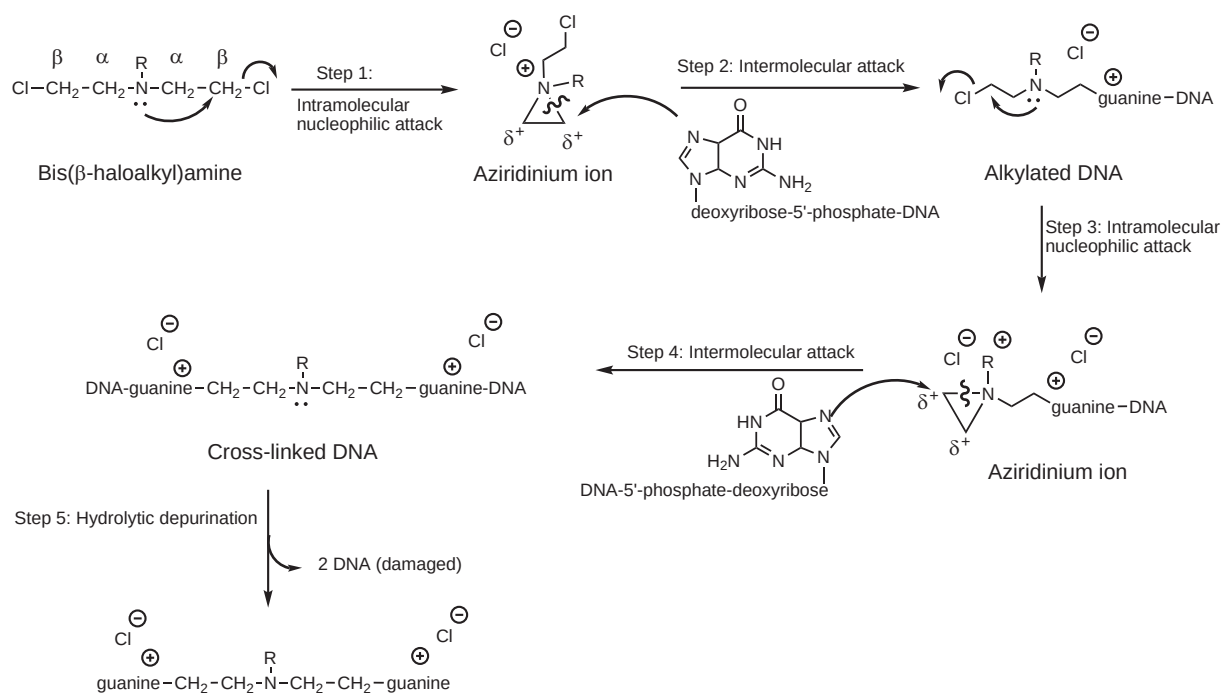


FIGURE 37.1 DNA destruction through nitrogen mustard-mediated alkylation.

this is happening in a tumor cell, the therapeutic goal has been accomplished. If it is happening in a healthy cell, particularly one with a short half-life, then the patient may experience side effects that can be use-limiting.

CHEMISTRY The structure of nitrogen mustards differs only in the nature of the third group (R) attached to the amino nitrogen. This group, which can be either aliphatic or aromatic, is the prime determinant of chemical reactivity, oral bioavailability, and the nature and extent of side effects.

An aliphatic nitrogen substituent (e.g., CH_3) will release electrons to the amine through σ bonds. This electronic enrichment enhances the nucleophilic character of the lone pair of electrons and increases the speed at which the δ^+ β -carbon of the mustard will be attacked. Whether in a tumor cell or a healthy cell, as soon as the aziridinium ion forms, it will react with unpaired DNA and/or other cell nucleophiles, such as electron-rich SH (mercapto or sulfhydryl), OH (hydroxyl), and NH (amino) groups of amino acids on enzymes or membrane-bound receptors. The body's water can also react with (and inactivate) the aziridinium ion. The intra- and intermolecular reactions designated as steps 1 through 4 in Figure 37.1 happen so rapidly that almost no chance exists for tissue or cell specificity, which means a greatly increased risk of serious side effects and use-limiting toxicity.

Conversely, an aromatic nitrogen substituent (e.g., phenyl) conjugated with the mustard nitrogen will stabilize the lone pair of electrons through resonance. Resonance delocalization significantly slows the rate of intramolecular nucleophilic attack, aziridinium ion formation, and DNA alkylation. Aromatic mustards have a reactivity sufficiently controlled to permit oral administration and attenuate the severity of side effects. The higher stability also provides the chance for enhanced tissue selectivity by giving the intact mustard time to reach malignant cells before generating the electrophilic aziridinium ion.

Nitrogen mustards can decompose in aqueous media through formation of the inactive dehalogenated diol shown in Figure 37.2. Both the mustard nitrogen (pathway a) and the oxygen of water (pathway b) can act as nucleophiles to advance this degradative process. The decomposition reactions can be inhibited if the nucleophilic character of these atoms is eliminated through protonation, so buffering solutions to a slightly acidic pH helps to enhance stability in aqueous solution.

SPECIFIC DRUGS (FIG. 37.3)

Mechlorethamine Hydrochloride Mechlorethamine is the only aliphatic nitrogen mustard currently on the U.S. market. Its use is limited by extremely high reactivity, which leads to rapid and nonspecific alkylation of cellular nucleophiles and excessive toxicity. It is a severe vesicant, and if accidental skin contact occurs, the drug must be inactivated with 2% sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) solution. This reagent reacts with the mustard to create

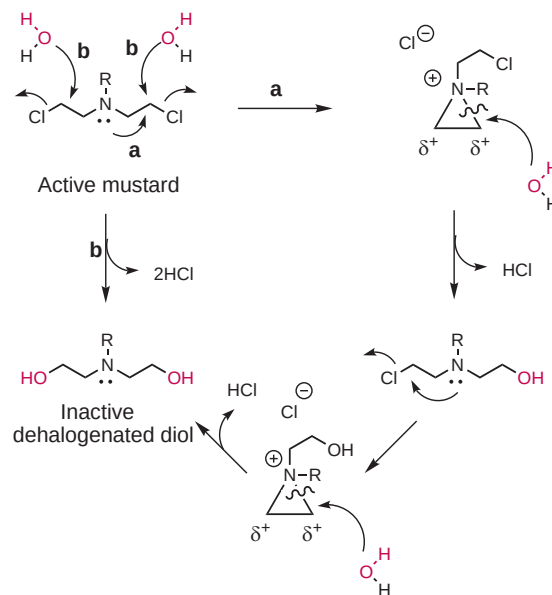
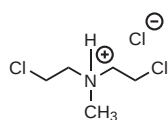


FIGURE 37.2 Aqueous decomposition of nitrogen mustards.

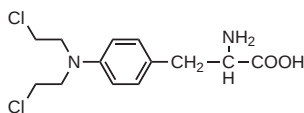
an inactive, highly ionized, and water-soluble thiosulfate ester that can be washed away (Fig. 37.4). The affected tissue should also be treated with an ice compress for 6 to 12 hours.

Mechlorethamine is marketed in hydrochloride salt form to provide water solubility for intravenous or intracavitary administration. The strong electron-withdrawing effect of the two chlorine atoms reduces the $\text{p}K_a$ of mechlorethamine to 6.1, which gives a ratio of un-ionized to ionized drug forms of approximately 20:1 at pH 7.4. This agent is too reactive for oral administration and too toxic to use alone. In addition to severe nausea and vomiting, myelosuppression (lymphocytopenia and granulocytopenia), and alopecia, it can cause myelogenous leukemia with extended use due to its mutagenic/carcinogenic effects on bone marrow stem cells. Mechlorethamine is still used in regimens for cancers of the blood (e.g., Hodgkin's disease, chronic myelocytic leukemia, chronic lymphocytic leukemia); fortunately, safer and still highly potent antineoplastic agents are now available.

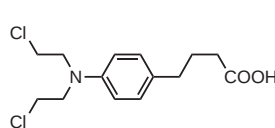
Melphalan This aromatic mustard, used primarily in the treatment of multiple myeloma, is able to stabilize the lone pair of electrons on the mustard nitrogen through resonance with the conjugated phenyl ring, slowing the formation of the reactive aziridinium ion. The L-isomer of the amino acid phenylalanine (L-Phe) was purposefully incorporated into this antineoplastic agent because naturally occurring L-amino acids are preferentially transported into cells by the action of specific amino acid carrier proteins. It was assumed that the L-Phe would act as a homing device and actively transport the toxic mustard inside the tumor cells, but some studies indicate that melphalan enters cells through facilitated diffusion rather than by active transport (17).

Nitrogen mustards and aziridine-mediated alkylators:

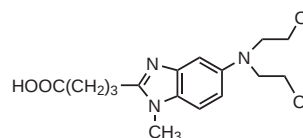
Mechlorethamine
hydrochloride
(Mustargen)



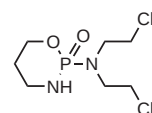
Melphalan
(Alkeran)



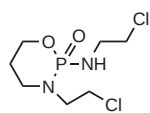
Chlorambucil
(Leukeran)



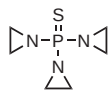
Bendamustine
(Treanda)



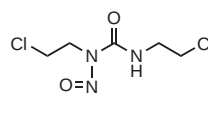
Cyclophosphamide
(Cytoxan)



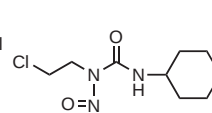
Ifosfamide
(Ifex)



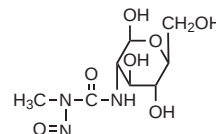
Thiotepa
(Thioplex)

Nitrosoureas:

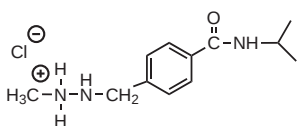
Carmustine
(BiCNU)



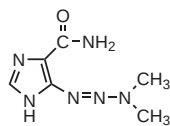
Lomustine
(CeeNU)



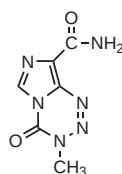
Streptozocin
(Zanosar)

DNA methylators:

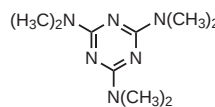
Procarbazine hydrochloride
(Matulane)



Dacarbazine
(DTIC-Dome)



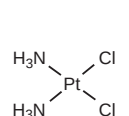
Temozolomide
(Temodar)

Miscellaneous DNA alkylators:

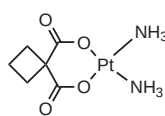
Altretamine
(Hexalen)



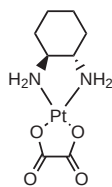
Busulfan
(Myleran)

Organoplatinum complexes:

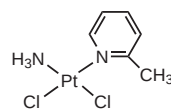
Cisplatin
(Platinol-AQ)



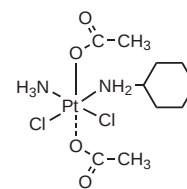
Carboplatin
(Paraplatin)



Oxaliplatin
(Eloxatin)



Picoplatin



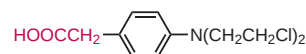
Satraplatin
(investigational)

FIGURE 37.3 DNA cross-linking agents.

Because the lone pair of electrons of melphalan (and other aromatic mustards) is less reactive than the lone electron pair on aliphatic mustards, there is a greater opportunity for distribution to cancer cells and a decreased incidence of severe side effects. There is a lower incidence of nausea and vomiting compared to mechlorethamine, but patients still experience myelosuppression, which can be severe. This drug is also mutagenic and can induce leukemia.

Melphalan is orally active, but absorption can be erratic. Absorption is decreased with food, but dosing regimens do not demand an empty stomach. The hydrochloride salt is available for intravenous administration, but the risk of serious side effects is higher. Melphalan distributes into body water, so toxicity can be pronounced in dehydrated patients or in those with renal dysfunction. Dehydration can be corrected, but dosage adjustments should be considered in patients with renal disease.

Chlorambucil Like melphalan, chlorambucil has good oral bioavailability (which is decreased in the presence of food) and the potential to induce nonlymphocytic leukemia. This drug is active intact and also undergoes β -oxidation to provide an active phenylacetic acid mustard metabolite, which is responsible for some of the observed antineoplastic activity. It is used in the palliative treatment of chronic lymphocytic leukemia, malignant lymphoma, and Hodgkin disease.



Phenylacetic acid mustard (an active chlorambucil metabolite)

Bendamustine Hydrochloride Bendamustine is an excellent chemical example of the adage “everything old is new again.” First synthesized in 1963, serious interest in it as a

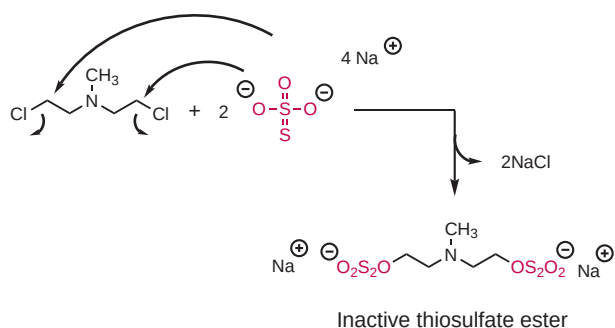
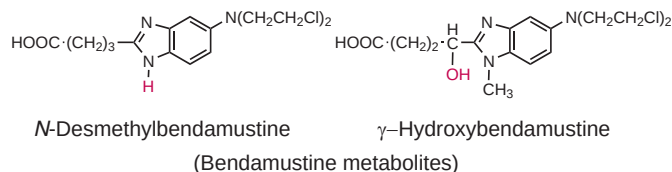


FIGURE 37.4 Mechlorethamine inactivation by sodium thiosulfate.

viable therapeutic agent surfaced only recently after several well-designed and appropriately executed clinical trials documented its value in the treatment of hematologic cancers, specifically chronic lymphocytic leukemia (CLL) and non-Hodgkin's lymphoma (NHL). Bendamustine is the *N*-methylbenzimidazole analog of chlorambucil, and the substitution of this purine-like aromatic ring was purposefully done to promote an antimetabolite mechanism along with DNA alkylation. DNA damage is more extensive and less repairable than that induced by other alkylating agents, and the drug is unique in its ability to stimulate *p21* and *p53*-induced apoptosis, S-phase cell cycle arrest, and "mitotic catastrophe" (18). The risk of acquired resistance and cross-resistance appears lower than with other DNA alkylators (19).

Unlike chlorambucil, bendamustine is given only intravenously on days 1 and 2 of a 21-day (NHL) or 28-day (CLL) cycle. It can be given alone or, in the case of slow-growing, refractory, and/or relapsed (indolent) lymphomas, in combination with other antineoplastic

agents (e.g., the anticancer monoclonal antibody rituximab and/or the topoisomerase poison mitoxantrone). It undergoes minor CYP1A2-catalyzed *N*-demethylation and γ -hydroxylation. While active, these metabolites are clinically insignificant. There is currently no evidence of serious metabolism-based interactions or toxicities associated with bendamustine; however myelosuppression, hypersensitivity/anaphylaxis, and skin reactions have been noted with its use. Pretreatment with antihistamines and corticosteroids can help to minimize infusion reactions, a major cause of drug discontinuation.



Cyclophosphamide Cyclophosphamide is a chiral pro-drug antineoplastic agent requiring activation by metabolic and nonmetabolic processes (20) (Fig. 37.5). The initial metabolic step is mediated primarily by CYP2B6 (and, to a lesser extent, by CYP3A4 and CYP2C isoforms) and involves regioselective hydroxylation at C₄ of the oxazaphosphorine ring to generate a carbinolamine (20,21). This hydroxylation reaction must occur before the molecule will be transported into cells, and approximately 90% of an administered dose will be appropriately converted (20). CYP3A4 and CYP2B6 stereospecifically catalyze an inactivating *N*-dechloroethylation reaction on the *R* and *S* isomers, respectively, which yields highly nephrotoxic and neurotoxic chloroacetaldehyde (20,21). Chloroacetaldehyde toxicity is accompanied by glutathione depletion, indicating that, as expected, this

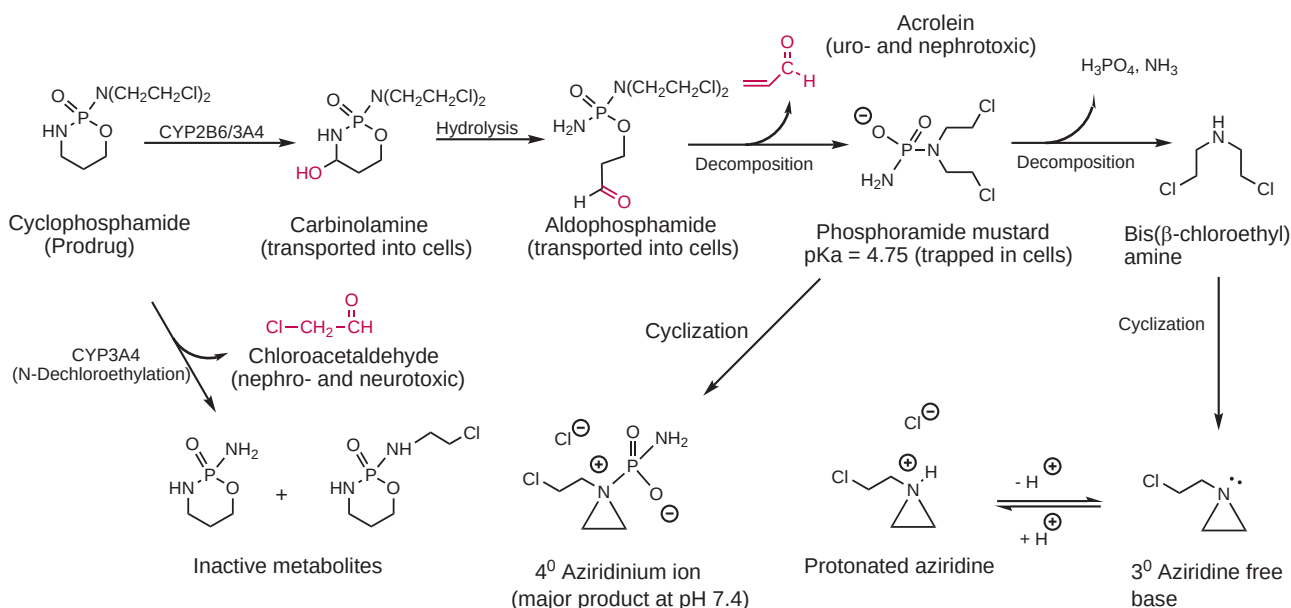


FIGURE 37.5 Cyclophosphamide metabolism.

electrophilic by-product alkylates cysteine (Cys) residues of critical cell proteins (22). Alkylation of lysine, adenosine, and cytidine residues is also possible.

The CYP-generated carbinolamine undergoes non-enzymatic hydrolysis to provide the aldophosphamide either in the bloodstream or inside cells. If this hydrolysis occurs extracellularly, the aldophosphamide is still able to penetrate cell membranes to reach the intracellular space. Once inside the cell, acrolein (a highly reactive α,β -unsaturated aldehyde) is cleaved via spontaneous β -elimination, generating phosphoramidate mustard. With a pK_a of 4.75, the mustard will be persistently anionic at intracellular pH and trapped inside the cell.

The fate of phosphoramidate mustard is varied. Most of it cyclizes to form the quaternary aziridinium ion, which alkylates DNA in the manner of all mustards. Some of it will decompose, losing phosphoric acid (H_3PO_4) and ammonia (NH_3) and leaving the naked bis(β -chloroethyl) amine mustard. This secondary amine cyclizes in a manner similar to the tertiary phosphoramidate mustard, forming a tertiary aziridine (rather than a quaternary aziridinium) species. The free tertiary aziridine can protonate at intracellular pH to provide the cationic aziridine species, which is in equilibrium with the free base form. Some electrophilic character is lost, but the carbon atoms in both forms are still δ^+ enough to attract DNA nucleophiles, albeit less vigorously. The net result is DNA alkylation and cell death. Oxidation of oxazaphosphorine intermediates along the metabolic pathway by cytosolic alcohol or aldehyde dehydrogenase is inactivating (20).

The need for metabolic activation in the liver means lowered GI toxicity and less nonspecific toxicity for cyclophosphamide compared with other DNA alkylating agents, but cyclophosphamide is not without its toxic effects. Acrolein, generated during the formation of phosphoramidate mustard, is a very electrophilic and highly reactive species, and it causes extensive damage to cells of the kidney and bladder. While acrolein can be produced in kidney via CYP3A4-mediated metabolism, it is predominantly generated in liver, where it readily conjugates with reduced glutathione (GSH) as expected (23, 24). However, when the acrolein-GSH or mercapturic acid conjugate is delivered to the bladder for excretion, the conjugate can cause direct toxicity or cleave to release electrophilic, reactive acrolein to the cells (23). Without additional GSH to re-scavenge liberated toxin, the acrolein will be attacked at its δ^+ terminal carbon by the nucleophilic SH of bladder cell Cys residues (Fig. 37.6). More complex biochemical events involving increased levels of preoxynitrite and subsequent lipid peroxidation also contribute to the observed urotoxicity (20). Physiologic results can include severe hemorrhage, sclerosis, and, on occasion (e.g., 5%), induction of bladder cancer. Acrolein also damages the nephron, particularly when used in high doses, in children, in patients with only one kidney, or when coadministered with other nephrotoxic agents (e.g., cisplatin). To minimize

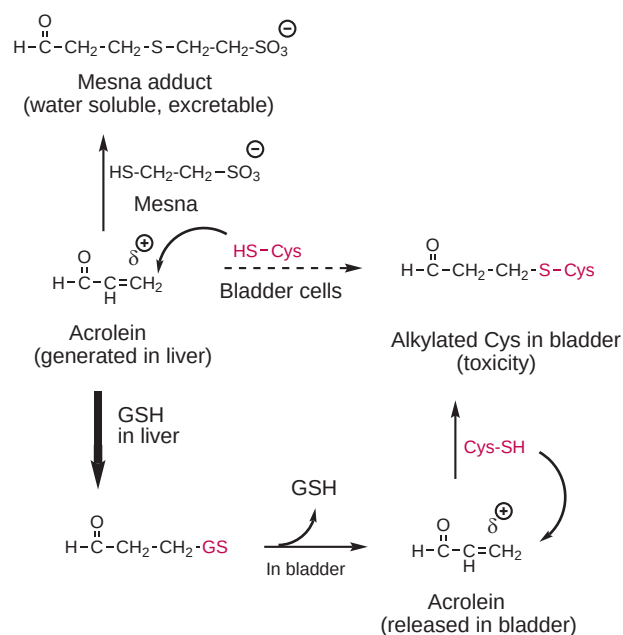


FIGURE 37.6 Sulfhydryl alkylation by acrolein.

the risk of bladder toxicity from acrolein, fluids should be forced and the bladder irrigated. Mesna (Mesnex) is available as adjuvant therapy in case of overt toxicity or as a prophylactic protectant. A sulfhydryl reagent, mesna is transported in the bloodstream as the inactive oxidized disulfide (dimesna) and selectively reduced to the reactive sulfhydryl in the proximal tubules (20). Once excreted into the bladder, the reactive SH group competes with Cys residues for the alkylating acrolein, as shown in Figure 37.6. Mesna concentrates in the bladder and will prevent damage to those cells. It does not concentrate to any appreciable extent in the nephron and, therefore, is not good protection against cyclophosphamide-induced nephrotoxicity.

As effective as mesna is for preventing acrolein-induced urotoxicity, it does little to spare the kidney and nerve cells from chloroacetaldehyde, the other toxic by-product of cyclophosphamide metabolism (25). Luckily, only approximately 10% of a standard dose of cyclophosphamide undergoes the dechloroethylation reaction, and most of the chloroacetaldehyde generated can be scavenged by GSH. However, since the competing hydroxylation reaction is saturable, this percentage can rise if higher doses are used (21).

Cyclophosphamide is most commonly used in combination with other antineoplastic agents to treat a wide range of neoplasms, including leukemias and malignant lymphomas, multiple myeloma, ovarian adenocarcinoma, and breast cancer. The drug is metabolized in the liver and eliminated via the kidney, with approximately 15% of a given dose being excreted unchanged. Doses should be reduced in patients with creatinine clearance levels less than 30 mL/min. Interestingly, hepatic dysfunction does not seem to alter the metabolism of this drug, but caution

should be exercised in patients with inhibited CYP450 enzymes or with a combination of factors that could negatively impact drug activation/inactivation pathways. Significant stereochemistry-based variation in the metabolic, enzyme inhibition, excretion, and toxicity profiles of cyclophosphamide and related oxazaphosphorine antineoplastic agents have been described in the literature (20).

Ifosfamide This cyclophosphamide analog has the two arms of the mustard on different nitrogen atoms. Ifosfamide also requires metabolic activation (Fig. 37.7), but this time, it is the CYP3A4 isoform that converts the majority of the dose to the carbinolamine, with CYP2B6 taking on a minor supporting role (26). Because ifosfamide has a lower affinity for the hydroxylating CYP3A4 and CYP2B6 enzymes, presumably as a result of steric hindrance, bioactivation proceeds at a slower rate (27). Doses three- to fourfold higher than cyclophosphamide are required to achieve the same antineoplastic result.

Unlike cyclophosphamide, dechloroethylation is a significant metabolic pathway for ifosfamide, and up to 60% of a standard dose will undergo this toxicity-inducing biotransformation (20,28). CYP3A4/3A5 catalyzes approximately 70% of ifosfamide dechloroethylation, with CYP2B6 taking care of the remainder (21). So much chloroacetaldehyde is generated that the endogenous GSH

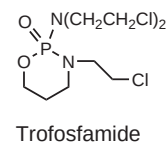
decoy (available in limited quantities) just cannot keep up. The fact that this reaction can occur in the renal tubule, generating chloroacetaldehyde right in the nephron, contributes to a significantly higher nephrotoxicity that can result in glomerular and renal tubular failure (27). Neurotoxicity is most commonly central in origin (e.g., mental status dysfunction, seizures). Ultimately, both chloroalkyl groups are lost before the compound is excreted.

It bears repeating that there is a significantly higher risk of bladder toxicity and nephrotoxicity with ifosfamide than with cyclophosphamide. The higher risks of these toxicities result because significantly more chloroacetaldehyde is generated through CYP3A4- and CYP2B6-mediated dechloroethylation. Although this toxic by-product has been claimed to have some antitumor activity (20), the biotransformation can take place in the nephron, which generates chloroacetaldehyde right where it will do the most unwanted cellular damage. Ifosfamide is also more water soluble than cyclophosphamide and will concentrate in the renal system. In addition, higher doses must be administered to achieve the same degree of antineoplastic action, so more molecules of acrolein and chloroacetaldehyde will be produced.

Because acrolein is generated during the bioactivation of ifosfamide, the same precautions against hemorrhagic cystitis that were previously outlined for cyclophosphamide must be taken: hydrate well, irrigate thoroughly, and administer with mesna. As previously stated, mesna will not prevent chloroacetaldehyde-induced toxicity. GSH-based rescue agents such as *N*-acetylcysteine may have some benefit in attenuating ifosfamide-induced nephrotoxicity, but because they do not penetrate the blood–brain barrier, they would be of little value in neurotoxicity prophylaxis (20).

Ifosfamide is currently used as third-line therapy in testicular cancer, although it has also shown activity in a number of other solid tumors and hematologic cancers. Patients on ifosfamide (but not cyclophosphamide) commonly exhibit cerebral neuropathy attributed to the significantly higher levels of chloroacetaldehyde generated by this drug (20). In severe forms, CNS depression can progress to coma and death.

A new chiral oxazaphosphorine, trofosfamide, is currently undergoing clinical trials for the palliative care of solid tumor and NHL patients. Trofosfamide is rapidly dechloroethylated at the mustard and oxazaphosphorine nitrogen atoms to form ifosfamide (predominant) and cyclophosphamide, respectively (20).



Trofosfamide

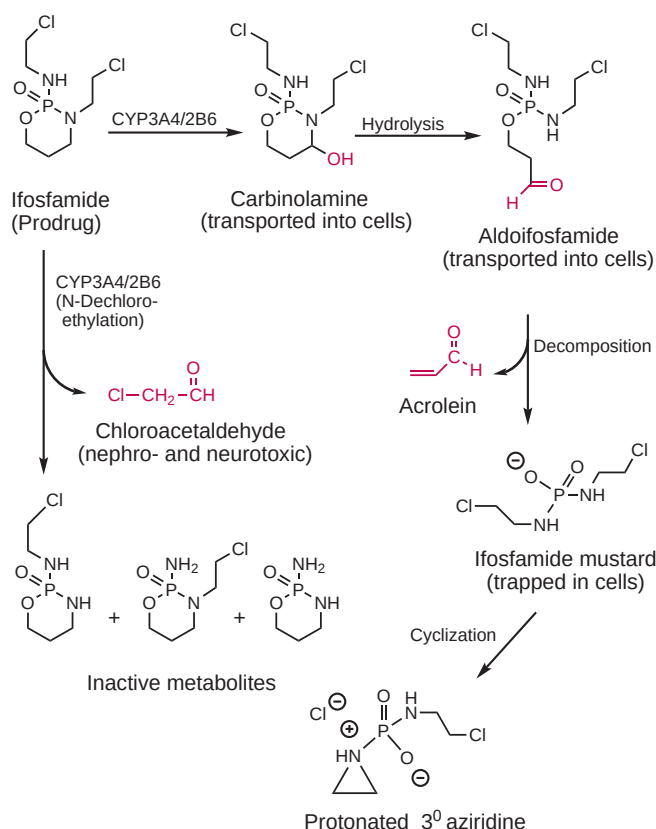
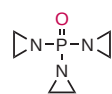


FIGURE 37.7 Ifosfamide metabolism.

Thiotepa Thiotepa, a tertiary aziridine, is less reactive than quaternary aziridinium compounds and is classified as a weak alkylator. It is possible for the

nitrogen atoms to protonate before reacting with DNA. However, the electron-withdrawing effect of the sulfur atom decreases the pK_a to approximately 6, which keeps the percentage ionized at pH 7.4 relatively low. Thiotepa undergoes oxidative desulfuration, forming an active cytotoxic metabolite known as TEPA (triethylenephosphoramidate). Both thiotepa and TEPA have been proposed to alkylate DNA through the hydrolytic release of aziridine, a more easily protonated structure, which is, therefore, a more attractive target for DNA nucleophiles. In this mechanistic model, the parent drug and the TEPA metabolite serve as carriers to transport aziridine-releasing drug across tumor cell membranes. (Fig. 37.8). Thiotepa, but not TEPA, is capable of direct, regioselective DNA alkylation due to its slower rate of intracellular hydrolysis (29).



Triethylenephosphoramidate
(an active thiotepa metabolite)

Thiotepa is most commonly employed in the treatment of ovarian and breast cancers, as well as papillary carcinoma of the bladder. Thiotepa and its TEPA metabolite readily enter the central nervous system (CNS) after systemic administration, leading to dizziness, blurred vision, and headaches. More critically, these agents are also severe myelosuppressants and can induce leukopenia, thrombocytopenia, and anemia. Patients treated with thiotepa are at high risk for infection and hemorrhage. Oral absorption is unreliable, so thiotepa is either given intravenously, or instilled intravesically in the treatment of bladder cancer. Even when administered locally in bladder cancer, high levels of this lipophilic drug reach the systemic circulation, resulting in bone marrow depression. Patients have died from myelosuppression after intravesically administered thiotepa. The drug also causes damage to the hepatic and renal systems. Dose

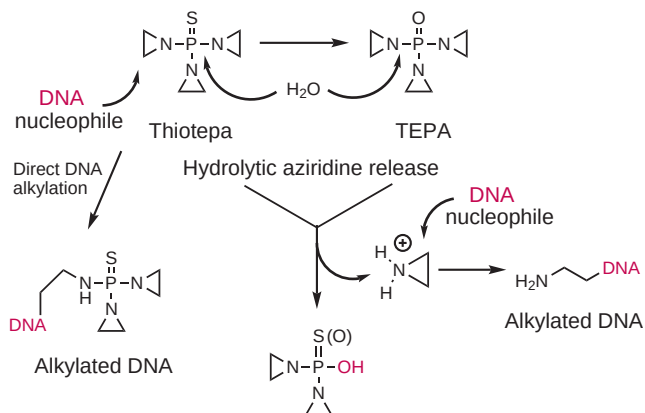


FIGURE 37.8 Mechanism of thiotepa DNA alkylation.

and/or administration frequency should be increased slowly, even if the initial response to the drug is sluggish, or unacceptable toxicity may result.

Nitrosoureas

MECHANISM OF ACTION The nitrosoureas are unstable structures that decompose readily in the aqueous environment of the cell. Nonenzymatic fragmentation is stimulated by the loss of proton from the urea moiety. Cyclization of the resultant anion to an unstable oxazolidine (pathway A) is followed by decomposition to vinyl diazotic acid and a substituted isocyanate, both of which release a gaseous fragment (nitrogen and carbon dioxide, respectively) to generate cytotoxic electrophiles (Fig. 37.9). Vinyl carbocation, acetaldehyde, and 2-chloroethylamine generated from the 2-chloroethylisocyanate moiety of carmustine are all capable of alkylating DNA in the standard manner (30). A second decomposition mechanism (pathway B)

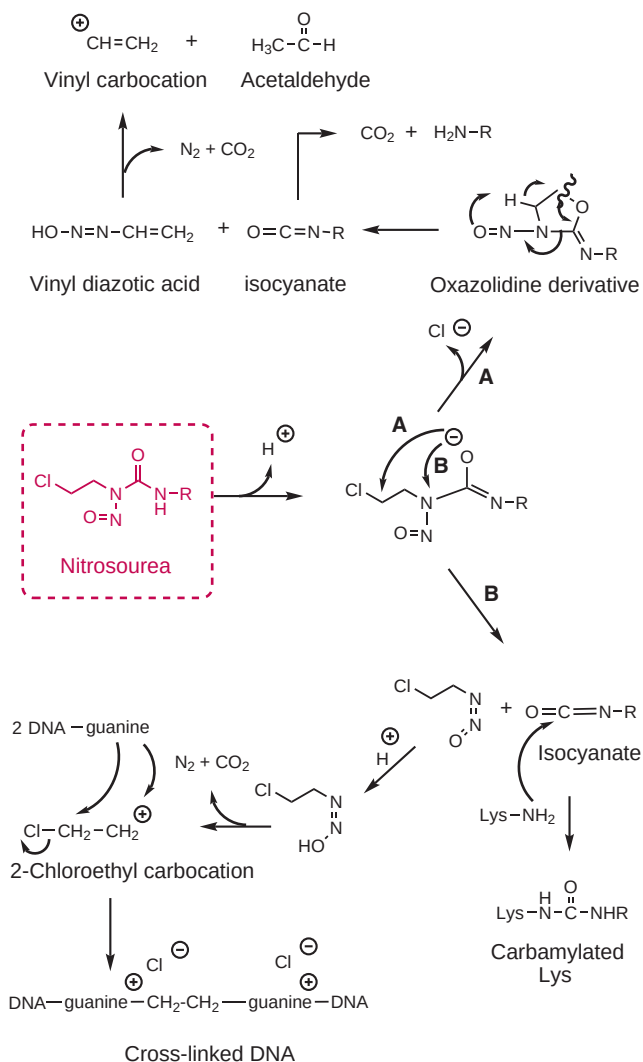


FIGURE 37.9 Nitrosourea decomposition to cytotoxic electrophiles.

ultimately produces an electrophilic 2-chloroethylcarbo-cation capable of DNA alkylation at guanine- N^7 and O^6 , as well as an isocyanate that can carbamylate amino acid residues (e.g., lysine [Lys]).

SPECIFIC DRUGS (FIG. 37.3)

Carmustine and Lomustine Carmustine and lomustine are both highly lipophilic chloroethylnitrosourea analogs marketed for use in brain tumors and Hodgkin's disease. Carmustine has also shown value in the treatment of NHL and multiple myeloma, and it is given intravenously or incorporated into biodegradable wafers that are implanted directly into the CNS after tumor resection. The high lipophilicity of carmustine precludes a totally aqueous intravenous formulation, and the drug is administered in 10% ethanol. Although carmustine degrades within 15 minutes of intravenous administration, lomustine is stable enough for oral use and is marketed in capsule form. Carmustine can also decompose in vitro if exposed to temperatures around 90°F. Pure carmustine is a low-melting solid, but the decomposed product is an oil and, therefore, readily detected. Vials of carmustine that appear oily should be discarded.

Both carmustine and lomustine can induce thrombocytopenia and leukopenia, leading to hemorrhage and massive infection. Acute (as well as potentially fatal delayed) pulmonary toxicity is also a risk. Pulmonary toxicity is dose-related, and individuals who received the drug in childhood or early adolescence are at higher risk for the delayed reaction. The grand mal seizures that are possible from the wafer formulation of carmustine appear to result from the wafer rather than from the nitrosourea. Resistance to carmustine involves upregulation of O^6 -methylguanine-DNA methyltransferase with the subsequent repair of drug-induced DNA damage, and possibly sequestration by neuroprotective metallothionein proteins (31).

Streptozocin The glucopyranose moiety of streptozocin confers both islet cell specificity and high water solubility to this nitrosourea-based antineoplastic. As a result, it is used exclusively in metastatic islet cell carcinoma of the pancreas and is administered intravenously in D5W or normal saline. Lacking the 2-chloroethyl substituent of carmustine and lomustine, it is much less reactive as a DNA alkylating agent. Myelotoxicity, while not unknown, is relatively rare. However, cumulative, dose-related renal toxicity may be severe or fatal, and 67% of patients receiving this drug will exhibit some kidney-related pathology. Good hydration is essential to successful therapy, and kidney function should be monitored weekly.

Procarbazine and Triazines

MECHANISM OF ACTION Procarbazine and the triazines dacarbazine and temozolomide act by different mechanisms, but they all exert an antineoplastic effect through the O^6 -methylation of guanine nucleotides. O^6 -Methylguanine pairs preferentially with thymine, and

these “mispairs” prompt point mutations during subsequent DNA replication cycles and trigger cell destruction through the activation of the normal postreplication mismatch repair system. Patients who are able to repair this damage through the action of O^6 -alkylguanine-DNA-alkyltransferase, which transfers the offending CH_3 group to a Cys residue on the alkyltransferase protein, will exhibit resistance to these agents, whereas those who underexpress this protein should respond well (32). Since the alkyltransferase is irreversibly inactivated in the DNA rescue process, enzyme depletion with subsequent loss of DNA repair capability is a significant risk.

Procarbazine metabolism involves CYP1A and CYP2B enzymes (33), and DNA alkylation operates through a free radical mechanism (Fig. 37.10). The major degradation pathway involves benzylic oxidation of azoprocarbazine, producing methylhydrazine that generates a methyl radical through an unstable diazene intermediate (34,35). In addition to O^6 , the reactive methyl radical formed can

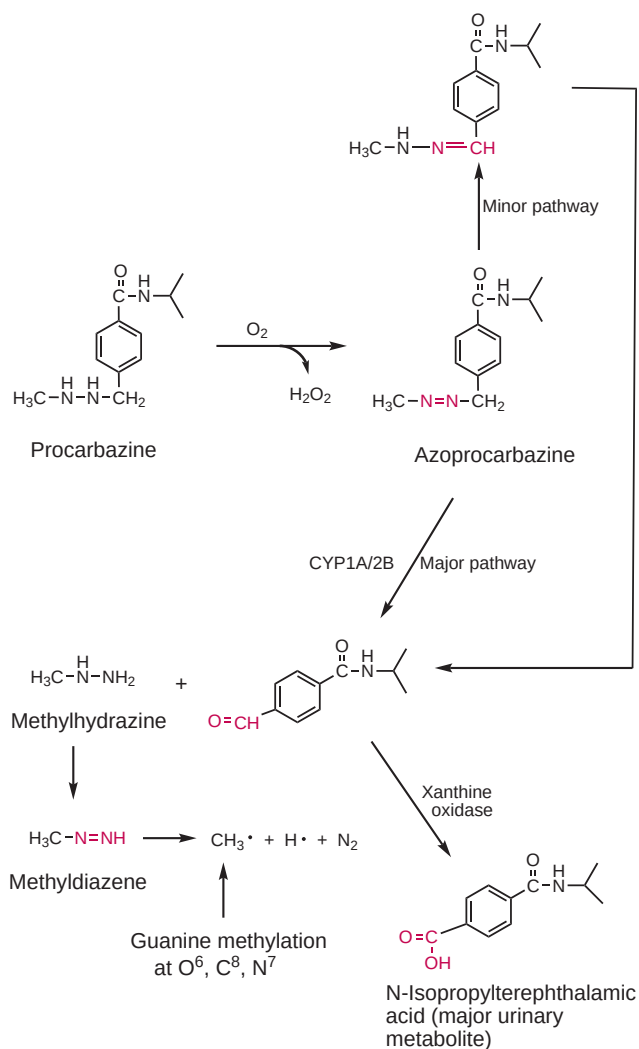


FIGURE 37.10 Procarbazine metabolism and mechanism of action.

alkylate the C^8 and N^7 positions of guanine. In contrast, the triazenes methylate DNA guanine via diazomethane and/or methyl carbocation generated in situ. Although temozolomide is converted to the diazomethane precursor 3-methyl-(triazene-1-yl)imidazole-4-carboxamide (MTIC) through nonenzymatic mechanisms, the conversion of dacarbazine to MTIC depends on the action of CYP1A1 and CYP1A2 enzymes, with a smaller contribution by CYP2E1 (Fig. 37.11) (33, 36). The O^6 and N^7 positions of guanine are the most vulnerable to triazene methylation.

SPECIFIC DRUGS (FIG. 37.3)

Procarbazine Hydrochloride This methyl radical generator is used predominantly in the treatment of Hodgkin's disease. It is administered as part of a multidrug regimen that also includes a nitrogen mustard (mechlorethamine), a mitosis inhibitor (vincristine), and prednisone. It is administered as capsules and is well absorbed after oral administration. Procarbazine is extensively metabolized in the liver, and 70% of an administered dose is excreted in the urine as *N*-isopropylterephthalamic acid (Fig. 37.10). In addition to methylating DNA guanine residues, it is proposed to inhibit the *de novo* synthesis of proteins and nucleic acids. Procarbazine inhibits monoamine oxidase, leading to several significant and potentially fatal drug–drug and drug–food interactions. Facial flushing and other disulfiram-like symptoms are

noted when alcohol is concomitantly consumed because the drug also inhibits enzymes involved in ethanol metabolism.

Dacarbazine This DNA methylating agent is administered intravenously as a single agent in the treatment of malignant melanoma and in combination with other agents in the treatment of metastatic melanoma. Approximately 40% of the drug is excreted unchanged, but both the 5-aminoimidazole-4-carboxamide (AIC, formed through the action of CYP1A enzymes) and the carboxylic acid (AIC hydrolysis product) are major urinary metabolites (Fig. 37.11). Leukopenia and thrombocytopenia are the most common side effects and can be fatal. Patients are also at risk for hepatotoxicity, including hepatocellular necrosis.

Temozolomide This imidazolotetrazine derivative is administered orally in capsule form for the treatment of glioblastoma multiforme or in patients with anaplastic astrocytoma who have not responded to procarbazine or the nitrosoureas. Oral absorption is rapid and complete. While CYP450 enzymes are not extensively involved in temozolomide metabolism, less than 6% of the drug is excreted unchanged in the urine. Women clear the drug less effectively than men and have a higher incidence of severe neutropenia and thrombocytopenia in the initial therapy cycle. Food decreases temozolomide absorption, and myelosuppression is the most significant adverse effect. Resistance involves drug-induced damage reversal by O^6 -methylguanine-DNA methyltransferase, and some authors are advocating the design of inhibitors of this potential target for use in combination with temozolomide and other guanine O^6 -methylating antineoplastics (37).

Miscellaneous DNA Alkylating Agents (Fig. 37.3)

ALTRETAMINE This unique hexamethylmelamine structure is believed to damage tumor cells through the production of the weakly alkylating species formaldehyde, a product of CYP450-mediated *N*-demethylation. Administered orally, altretamine is extensively metabolized on first pass, producing primarily mono- and didemethylated metabolites. Additional demethylation reactions occur in tumor cells, releasing formaldehyde in situ before the drug is excreted in the urine. The carbinolamine (methylol) intermediates of CYP450-mediated metabolism can also generate electrophilic iminium species that are capable of reacting covalently with DNA guanine and cytosine residues, as well as protein (Fig. 37.12). Iminium-mediated DNA cross-linking and DNA-protein interstrand cross-linking, mediated through both the iminium intermediate and formaldehyde, have been demonstrated (38,39), although the significance of DNA cross-linking on altretamine antitumor activity is uncertain. Resistance to altretamine has been shown to parallel resistance to formaldehyde-induced cytotoxicity (38). Its use currently is restricted to patients with ovarian cancer who have not responded to organoplatinum therapy. The toxicities

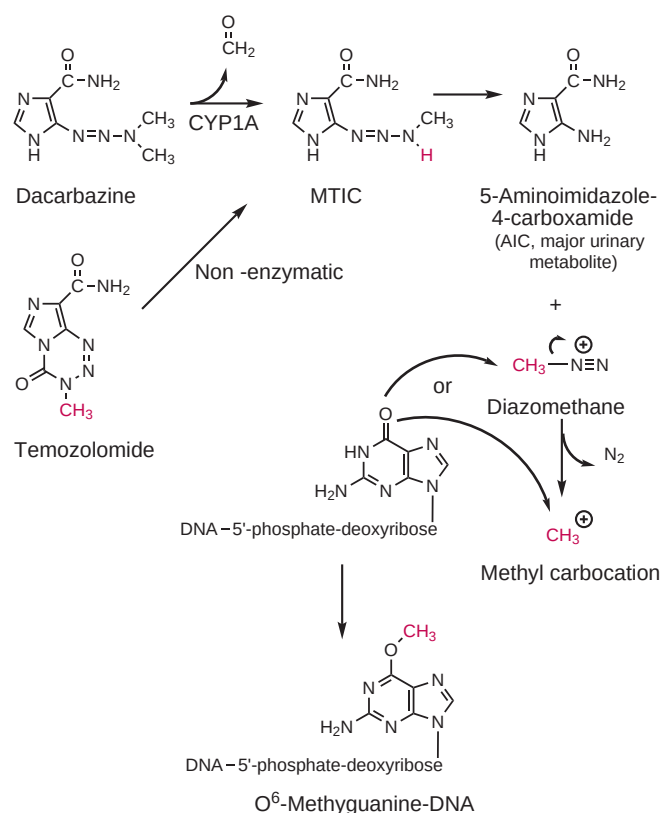


FIGURE 37.11 Metabolic activation of triazenes.

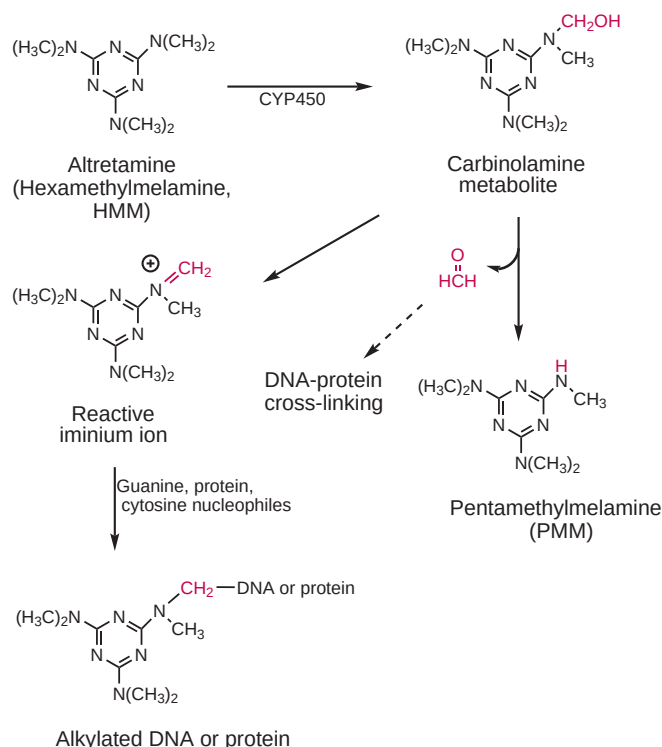


FIGURE 37.12 Altretemine metabolism and mechanism of action.

induced by altretamine are GI, neurologic, and hematologic in nature. An orally active liposomal formulation containing sodium deoxycholate has been investigated but is not currently commercially available (40).

BUSULFAN Chemically, busulfan is classified as an alkyl sulfonate. One or both of the methylsulfonate ester moieties can be displaced by the nucleophilic N^7 of guanine, leading to monoalkylated and cross-linked DNA as shown in Figure 37.13. The extent of alkyl sulfonate-mediated

DNA interstrand cross-linking has been shown to vary with the length of the alkyl chain between sulfonate esters, with the tetramethylene-containing busulfan showing less interstrand cross-linking capability than hexamethylene, methylene, or octamethylene analogs (41). Intrastrand cross-linking also occurs, preferentially at 5'-GA-3' but also at 5'-GG-3' sequences (42). Alkylation of Cys sulfhydryl groups is yet another mechanism of cytotoxicity. Busulfan is used in the treatment of CML and can be administered either orally or by intravenous infusion. Serious bone marrow hypoplasia and myelosuppression are possible with this agent, and recovery from busulfan-induced pancytopenia can take up to 2 years.

Organoplatinum Complexes

MECHANISM OF ACTION Organoplatinum antineoplastic agents contain an electron-deficient metal atom that acts as a magnet for electron-rich DNA nucleophiles. Like nitrogen mustards, organoplatinum complexes are bifunctional and can accept electrons from two DNA nucleophiles. Intrastrand cross-links most frequently occur between adjacent guanine residues referred to as diguanosine dinucleotides (60% to 65%) or adjacent guanine and adenine residues (25% to 30%) (43). Interstrand cross-linking, which occurs much less frequently (1% to 3%), usually involves guanine and adenine bases (44).

CHEMISTRY All the currently marketed organoplatinum anticancer agents are Pt(II) complexes with square planar geometry, although an octahedral Pt(IV) complex currently is undergoing clinical trials. Platinum is inherently electron deficient, but the net charge on the organometallic complex is zero due to the contribution of electrons by two of the four ligands bound to the parent structure. Most commonly, the electron-donating ligand is chloride. Before reacting with DNA, the electron-donating ligands are displaced through nucleophilic attack by cellular

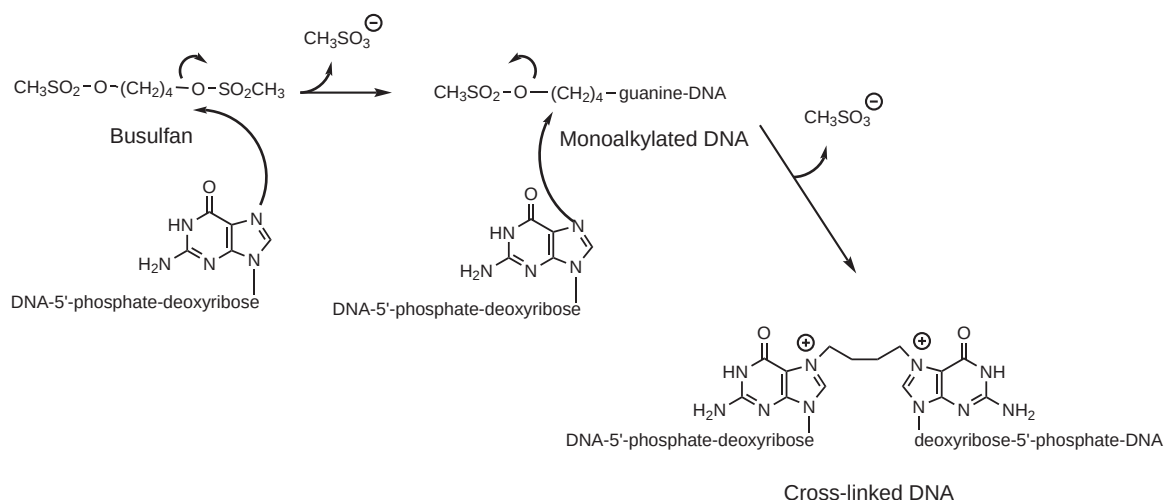
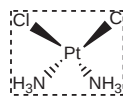


FIGURE 37.13 Busulfan-mediated DNA alkylation.

water. When the displaced ligands are chloride anions (e.g., cisplatin), the chloride-poor environment of the tumor cell facilitates the process, driving the generation of the active, cytotoxic hydrated forms (Fig. 37.14). Since the original ligands leave the metal with their electrons, the hydrated organoplatinum molecule has a net positive charge (45).



Cisplatin (square planar geometry)

The hydrated platinum analogs are readily attacked by DNA nucleophiles (e.g., the N^7 of adjacent guanine residues) due to the net positive charge that has been regained on the Pt atom (Fig. 37.14). The DNA bases become coordinated with the platinum, and in the cis configuration, DNA repair mechanisms are unable to permanently correct the damage. The net result is a major change in DNA conformation such that base pairs that normally engage in hydrogen bond formation are not permitted to interact. The two ammine ligands of the complex are bound irreversibly to the Pt atom through very strong coordinate covalent bonds. They cannot be displaced by DNA nucleophiles, but they do stabilize the cross-linked DNA–platinum complex by forming strong ion-dipole bonds with the anionic phosphate residues on DNA. The DNA distortion prompts a futile cycle of damage recognition and repair before giving up the ghost and succumbing to cell cycle arrest and apoptosis.

SPECIFIC DRUGS (FIG. 37.3)

Cisplatin The simplest of the organometallic antineoplastic agents, cisplatin is used intravenously in the treatment of metastatic testicular and ovarian cancer and advanced bladder cancer. It is rapidly hydrated, resulting

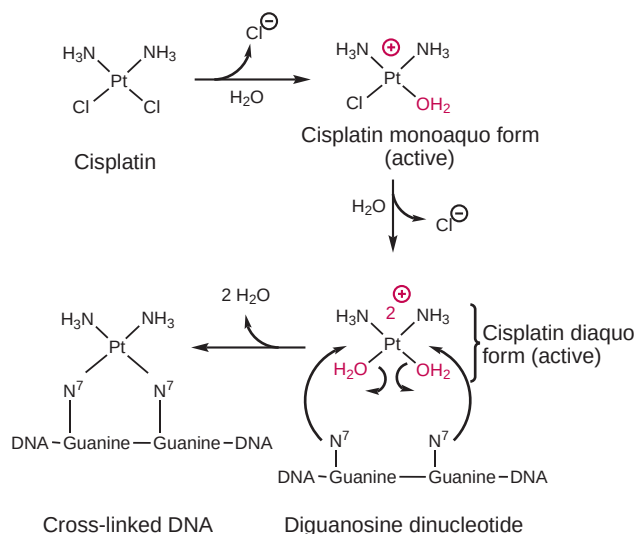
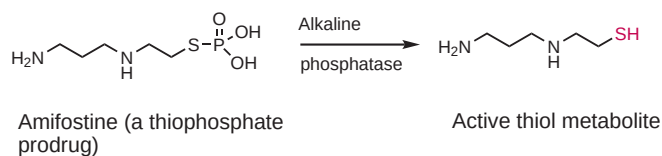


FIGURE 37.14 Cisplatin activation and DNA cross-linking.

in a short plasma half-life of less than 30 minutes. It is eliminated predominantly via the kidney, but approximately 10% of a given dose undergoes biliary excretion. It is highly nephrotoxic and can cause significant damage to the renal tubules, especially in patients with preexisting kidney disease, with one kidney, or who are concurrently receiving other nephrotoxic drugs (e.g., cyclophosphamide or ifosfamide). Dosages should be reduced in any of these situations. Clearance decreases with chronic therapy, and toxicities can manifest at a later date.

To proactively protect against kidney damage, patients should be aggressively hydrated with chloride-containing solutions. Saline or mannitol diuretics can be administered to promote continuous excretion of the drug and its hydrated analogs. Sodium thiosulfate, which accumulates in the renal tubules, has also been used to neutralize active drug in the kidneys in an effort to avoid nephrotoxicity (Fig. 37.15). The reaction of sodium thiosulfate with cisplatin in the serum is much less significant because the drug does not concentrate there, and what is there is very strongly bound to serum proteins. The very strong protein binding explains why dialysis, even when prolonged, cannot rescue patients from cisplatin toxicity.

Cisplatin is a very severe emetogen, and vomiting almost always occurs unless antiemetic therapy is coadministered. Myelosuppression and ototoxicity that can lead to irreversible hearing loss can also occur with cisplatin use. A recent study has shown that, in medulloblastoma patients 3 to 21 years of age, bolus administration of the thiol-generating prodrug amifostine, both immediately before and again during cisplatin infusion, significantly decreased ototoxicity and the need for a hearing aid after therapy (46). Alkaline phosphatase, amifostine's activating enzyme, shows a higher activity in normal tissue compared to tumor tissue. This allows for higher levels of drug destruction in nontarget healthy cells and provides protection against unwanted toxicity without compromising cisplatin's antineoplastic action.



Resistance to cisplatin therapy can be intrinsic in colorectal, lung, and breast cancer, or acquired after multiple courses of therapy (e.g., in ovarian cancer).

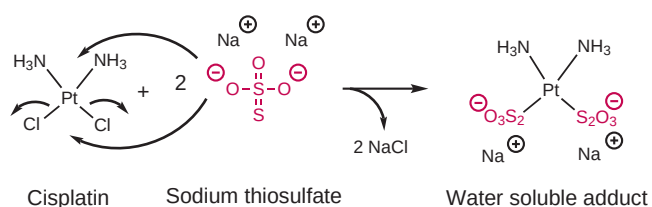


FIGURE 37.15 Cisplatin inactivation by sodium thiosulfate.

Resistance is mediated through several distinct mechanisms, including: 1) compromised carrier-mediated cellular transport via the copper transporting protein CTR1; 2) enhanced intracellular inactivation through drug trapping in vesicles; 3) drug inactivation through conjugation to Cys and/or methionine (Met)-containing GSH and metallothionein proteins; and 4) increased DNA repair and/or tolerance to cisplatin-induced DNA damage (47). Regarding the latter mechanism, cisplatin damage can be successfully repaired by nucleotide-excision repair proteins (NERs), which remove platinum-damaged segments from the DNA, and these proteins are often upregulated in cisplatin-resistant tumors. Cisplatin (and carboplatin)-DNA adducts are also recognized by mismatch repair proteins (MMRs). The downregulation of MMRs in cisplatin (and carboplatin)-treated cancer cells induces resistance through the loss of an apoptotic response that normally follows several ill-fated attempts to repair organoplatinum-induced damage. Testicular tumors are particularly responsive to cisplatin due to their inherent deficiency in DNA repair processes. Cellular efflux mediated by P-gp (a product of the *ABCB1* gene), a resistance mechanism common to natural product anticancer agents, is not believed to be a component of resistance to organometallics (47).

Cisplatin and the other organoplatinum anticancer agents react with aluminum and cannot be administered through aluminum-containing needles. The drug is photosensitive, is packaged in amber bottles, and must be protected from light.

Carboplatin Carboplatin, another square planar Pt(II) complex, forms the same cytotoxic hydrated intermediate as cisplatin but does so at a 10-fold slower rate, making it a 20- to 40-fold less potent chemotherapeutic agent (47). The ultimate damage done to cells as a result of carboplatin use approaches that of cisplatin, but the side effect profile is significantly milder. Suppression of platelets and white blood cells is the most significant toxic reaction, and nonhematologic toxicities (e.g., emesis, nephrotoxicity, and ototoxicity) are rare. The plasma half-life of carboplatin is 3 hours, and the drug is less extensively bound to serum proteins than cisplatin. Excretion is still predominantly renal, however, and doses must be reduced in patients with kidney disease.

Carboplatin is only approved for use in the treatment of advanced ovarian cancer, although clinical trials have shown that it may have a future in the treatment of hormone-refractory prostate cancer. Carboplatin can be used in combination with docetaxel (often considered the standard of care for prostate cancer) and estramustine (a less commonly used mitosis inhibitor). This drug has provided therapeutic benefit as a single agent in patients whose cancer has progressed after docetaxel therapy. Unlabeled uses for carboplatin include combination therapy in lung and head and neck cancers.

Oxaliplatin This Pt(II) complex loses oxalate dianion (OOC-COO^-) in vivo to form the mono- and dihydrated diaminocyclohexane (DACH) platinum analogs shown in Figure 37.16. The trans-(*R,R*)-DACH structure serves as the carrier for the cytotoxic hydrated platinum and extends into the major groove of DNA when the DNA-Pt complex forms (48). Hydrophobic DNA intrusion is believed to contribute to the cytotoxicity of this organometallic. Oxaliplatin engages primarily in intrastrand cross-linking with diguanosine dinucleotides, adjacent A-G nucleotides, and guanines that are separated by one nucleotide (G-X-G). Interstrand cross-linking, although less common, also occurs.

The adduct formed between oxaliplatin and DNA diguanosine dinucleotides is conformationally distinct from the adduct formed with cisplatin or carboplatin. Specifically, whereas the cisplatin diguanosine dinucleotide adduct bends the DNA by 60 to 80 degrees and presents a relatively wide minor groove, the oxaliplatin adduct produces a 31-degree bend with a comparatively narrow minor groove (49). This distinct oxaliplatin conformation is believed to result from the steric impact of the (*R,R*)-DACH carrier, which permits the *cis*- NH_3 moieties to hydrogen bond with a guanine- O^6 , a bond that the inactive (*S,S*)-isomer cannot make (50). The conformation of the oxaliplatin-DNA adduct is much less likely to be recognized by MMR proteins, and the effectiveness of oxaliplatin in MMR-deficient cells is, at least in part, responsible for the lack of resistance that has plagued cisplatin and carboplatin (51,52). Oxaliplatin is also less dependent on CRT1 active transporting proteins for intracellular access and often retains activity in patients who are no longer responding to the first-generation organometallics. It is significantly less mutagenic, nephrotoxic, hematotoxic, and ototoxic than cisplatin. Excretion is via the kidney. Oxaliplatin decomposes in alkaline media and should not be coadministered with drugs that will increase the pH of the intravenous solution.

Oxaliplatin is used in the treatment of metastatic colon or rectal cancer, either alone or in combination with fluorouracil. Pulmonary fibrosis and peripheral sensory

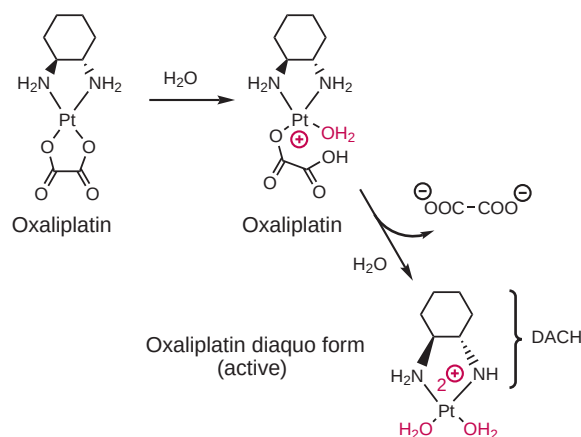
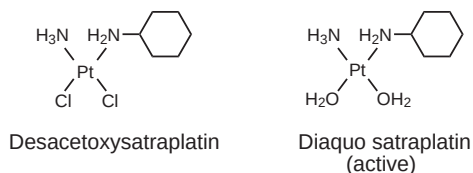


FIGURE 37.16 Activation of oxaliplatin.

neuropathies that can be life-threatening are known to occur. It has been proposed that the latter adverse effect is caused by oxalate-based chelation of intracellular Ca^{2+} , which inhibits voltage-gated sodium channels in sensory nerve cells (53). This hypothesis is supported by the observation that infusions of calcium or magnesium salts can significantly attenuate oxaliplatin-induced neuropathy without compromising therapeutic efficacy (54). In the future, exploitation of genetic differences in the expression of various repair proteins, growth factors, and metabolizing enzymes may allow the tailoring of oxaliplatin therapy based on an individual's pharmacogenetic profile (52).

Picoplatin Picoplatin is a Pt(II) organometallic currently designated as an orphan drug for the treatment of small cell lung cancer. It is also being investigated for use in platinum-sensitive ovarian cancer (in combination with paclitaxel), prostate cancer (in combination with docetaxel), and colorectal cancer (in combination with fluorouracil/leucovorin). The name of this antineoplastic is derived from the 2-methylpyridine (picoline) ring associated with the platinum atom. The intentional incorporation of bulk around the platinum decreases intracellular inactivation through attack by the sulfhydryl-containing glutathione peptide and metallothionein proteins, allowing continued efficacy in cisplatin-resistant tumors. This platin is active by the intravenous and oral routes. Like carboplatin, the drug is not nephro- or neurotoxic, but does induce myelosuppression that reverses upon drug discontinuation (47,55).

Satraplatin This asymmetrical organometallic agent is a Pt(IV) complex. As with the Pt(II) complexes, the platinum has no net charge in the parent drug since its original +4 charge has been "neutralized" by the donation of electrons from the chloride and acetate ligands. Unlike most marketed square-planar Pt(II) complexes, it is active by the oral route. Oral efficacy is enhanced by administering the drug on an empty stomach, using a 5 consecutive day regimen within a 28- to 35-day cycle. It is metabolized quickly in whole blood, producing up to six metabolites. The major metabolite is the desacetoxo analog. As with other organoplatinum complexes, the diaquo form is active.



Satraplatin is currently in clinical trials as a second-line agent for the treatment of hormone-refractory prostate cancer. Trials pairing satraplatin with prednisolone have been particularly promising, with patients on the combination therapy showing a 40% reduction in progression compared to patients on the corticosteroid alone. Other

trials in patients with small cell or non-small cell lung cancer are investigating the benefits of satraplatin as a single agent and satraplatin-paclitaxel combination therapy, respectively. Data analyzed to date suggest that satraplatin (administered as a single agent) is as effective as cisplatin or carboplatin in the treatment of small cell lung cancer and ovarian cancer. Satraplatin is also being paired with the pyrimidine antagonist capecitabine, the DNA polymerase inhibitor gemcitabine, or the mitosis inhibitor docetaxel in clinical trials in patients with advanced solid tumors.

Satraplatin retains activity in cisplatin-resistant cancers. Satraplatin-induced DNA damage is not recognized by MMR proteins, and its higher lipophilicity liberates it from dependence on CTR1 for cellular uptake. Resistance to satraplatin therapy can occur, but it is mediated through the intracellular sequestering of the drug in cytoprotective vesicles, a resistance pathway it shares with other organometallics. Its toxicity profile is mild, with dose-related, but noncumulative, myelosuppression (particularly neutropenia and thrombocytopenia), anemia, and diarrhea being the major use-limiting side effects (47,51,55,56).

Topoisomerase Poisons

Topoisomerases are enzymes that control the degree of DNA supercoiling and, in so doing, maintain proper DNA structure during replication and transcription to RNA (57). Topoisomerase II α (TopII α) cleaves double-stranded DNA during the replication phase via a transesterification reaction involving a topoisomerase tyrosine residue and a terminal 5' phosphate but, through a reverse transesterification, repairs its own damage after replication is complete (58). Topoisomerase I (TopI) functions in essentially the same way, but cuts and religates a single DNA strand. Antineoplastic agents that function as topoisomerase poisons stimulate the DNA cleavage reaction but inhibit the DNA resealing activity of the enzymes, leaving the DNA irreversibly damaged and unable to replicate.

Three chemically distinct classes of anticancer agents can be classified as topoisomerase poisons: camptothecins, epipodophyllotoxins, and anthracyclines (discussed under antineoplastic antibiotics).

Camptothecins

MECHANISM OF ACTION Camptothecins are chiral, extensively conjugated, amine-containing pentacyclic lactones (Fig. 37.17). The biologic target of camptothecins is TopI, rather than the TopII α enzyme that serves as the receptor for the epipodophyllotoxins and anthracyclines. However, as noted earlier, the mechanism of antineoplastic action of both enzyme inhibitors is stabilization of a cleavable ternary DNA-enzyme complex that does not permit the resealing of nicked DNA. Although the fragmented DNA is capable of resealing in the absence of drug, when DNA replication forks encounter the fragmented DNA, a double-stranded DNA break occurs, killing the cell.

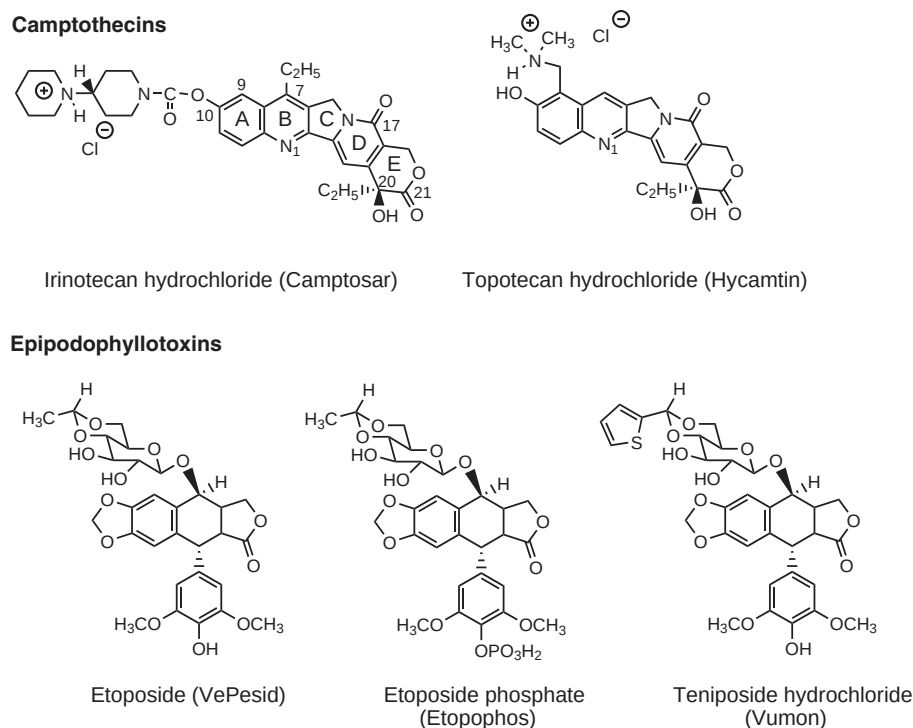


FIGURE 37.17 Camptothecins and epipodophyllotoxins: topoisomerase poisons.

The binding of camptothecins occurs in such a way as to stabilize a covalent DNA–topoisomerase bond at the point of single-strand breakage (tyrosine [Tyr] at 723 on the human enzyme) but sterically keep a TopI Lys532 residue from catalyzing the DNA religation reaction (59,60). The binding pocket, located within the DNA strand, is revealed only after the normal DNA nicking has occurred, explaining why these poisons preferentially bind to the enzyme–DNA complex rather than to unoccupied DNA or enzyme. The flat camptothecin ring system intercalates DNA at the site of cleavage, mimicking a DNA base pair (59). The crystal structures of human ternary complexes involving the parent alkaloid, camptothecin, and the semisynthetic analog, topotecan (Fig. 37.17), have been solved, and important drug–protein interacting entities are noted in Table 37.7 (59–61). The bulky substituents at C₇, C₉, and C₁₀ of the marketed compounds, which project into the major groove of DNA, do not hinder binding.

Camptothecins are most toxic to cells undergoing active DNA replication and cell division (e.g., they are S-phase specific). Mechanisms of resistance are similar to those operational in many other classes of antineoplastic drugs and include downregulation or mutation of the target enzyme, downregulation of enzymes needed for drug activation, and cellular efflux. Breast cancer resistance protein and multidrug resistance (MDR)–associated proteins, such as MAP-2 and MAP-3, rather than P-gp, appear to mediate resistance to these agents (62,63).

CHEMISTRY The parent camptothecin alkaloid, isolated from the bark of *Camptotheca acuminata* (the Chinese xi shu or “happy tree”) (64), has antitumor activity, but its limited water solubility necessitates delivery as the sodium salt of the significantly less active hydrolyzed lactone. Lactonization of the hydroxy acid in acidic urine is significant, and elevated levels of active intact alkaloid in the kidney account for the hemorrhagic cystitis induced by this compound.

Camptothecin’s quinoline ring system is unsubstituted, but currently marketed analogs have a basic side chain incorporated at either C₉ (topotecan) or C₁₀ (irinotecan), allowing the formation of water-soluble salts of the intact semisynthetic alkaloid. At pH 7.4, the active lactone exists in equilibrium with the hydroxy acid hydrolysis product, with the direction dictated by the extent of binding to serum albumin. The preferential protein

TABLE 37.7 Topotecan–Topoisomerase I Interactions

Topotecan Functional Group	Topoisomerase I Residue
Pyridine N ₁	Arg364
C ₁₀ -OH	Enzyme-associated water (H-bond)
C ₁₇ -pyridone carbonyl	Asn722
C ₂₀ -OH	Asp533 (H-bond)
C ₂₁ -lactone carbonyl	Tyr723-phosphate, Lys532

binding of the lactone, which occurs with irinotecan, shifts the equilibrium to favor the production of the more active lactone, thus enhancing potency.

Some authors have recently pondered whether camptothecins are making a clinical comeback (64). A quantitative study on camptothecin structure–property relationships has recently been published (65), as has a review summarizing the current literature describing outcomes in preliminary, preclinical, and clinical in vivo studies for camptothecins and various small and macromolecular analogs of the natural products (66).

SPECIFIC DRUGS (FIG. 37.17)

Irinotecan Hydrochloride In combination with fluorouracil, this prodrug camptothecin analog is considered first-line therapy in the treatment of metastatic colorectal cancer. It has also shown efficacy in small cell and non-small cell lung cancers when used in combination with cisplatin. Given intravenously, the drug is slowly bioactivated in the liver through hydrolysis of the C₁₀-carbamate ester. The catalyzing enzyme is a saturable carboxylesterase known as irinotecan-converting enzyme. Levels of active metabolite, known as SN-38 (Fig. 37.18), are 50- to 100-fold lower than the parent drug, but preferential protein binding of the lactone (95%) permits significant plasma levels of the optimally active SN-38 compared to the hydroxy acid metabolite. SN-38 has a terminal

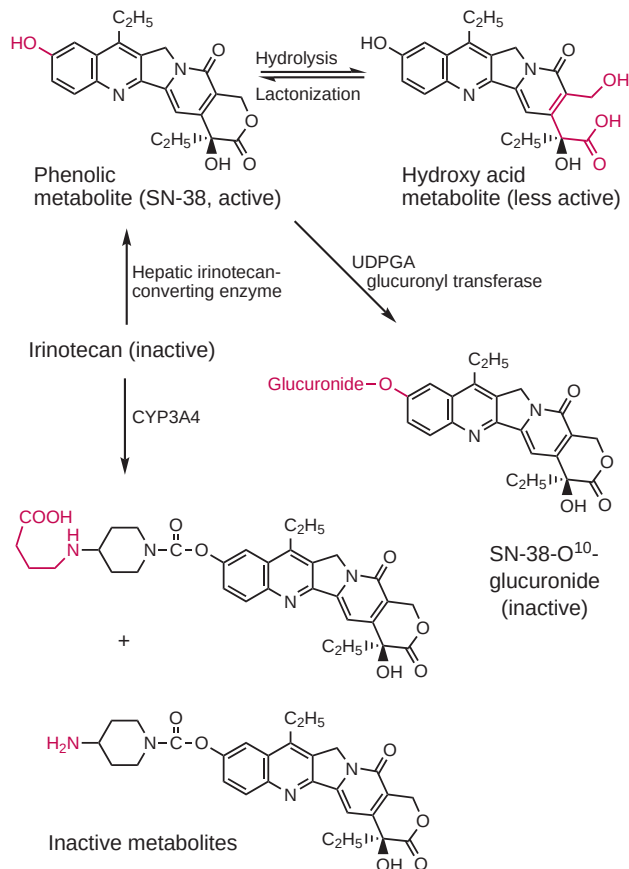


FIGURE 37.18 Irinotecan metabolism.

half-life of 11.5 hours (compared with 5.0 to 9.6 hours for the prodrug parent) and is glucuronidated or sulfated at the C₁₀ phenol before elimination. CYP3A4 also cleaves the terminal piperidine ring through oxidation at the α -carbons. This is followed by hydrolysis of the resultant amides, which produces inactive metabolites. Excretion of the parent drug and metabolites is renal (14% to 37%) and, to a lesser extent, biliary.

Delayed diarrhea induced by irinotecan is dose-limiting and potentially fatal, and vigorous loperamide therapy should be instituted at the first sign of symptoms. Acute diarrhea is attributed to the drug's ability to inhibit acetylcholinesterase and can be addressed through anticholinergic pretreatment. Pretreatment also helps patients to avoid "cholinergic syndrome," a collection of annoying side effects that include flushing, sweating, blurred vision, lacrimation, and less commonly, bradycardia. Camptothecins are also myelosuppressive, and leukopenia can be severe, particularly in patients with elevated bilirubin levels. Extensive biotransformation also demands cautious use of irinotecan in patients with hepatic dysfunction. Prophylactic antiemetic therapy should be given at least 30 minutes before the administration of irinotecan to minimize the nausea and vomiting associated with this anticancer agent.

It is now recognized that patients, particularly those of Asian heritage, may be pharmacogenetically predisposed to life-threatening toxicity from camptothecin therapy. Variations in expression of gene(s) involved in the inactivating glucuronidation of irinotecan (specifically overexpression of the low activity UGT1A1*6 allele) are deemed responsible, and genotyping efforts are being made to more safely and effectively individualize therapy in "at-risk" individuals (67,68).

Topotecan Hydrochloride This active camptothecin analog is used by the intravenous route in the treatment of ovarian and small cell lung cancer that has not responded to first-line therapy. Myelosuppression, particularly neutropenia, is use-limiting and has precluded combination therapy with other bone marrow-suppressing drugs. Thrombocytopenia and anemia occur in approximately one-third of treated patients. Schedules that call for daily (for 5 days) administration can also result in serious mucositis and diarrhea.

Topotecan elimination is biphasic with a terminal half-life of 2.0 to 3.5 hours. Lactone hydrolysis is rapid, and binding to serum proteins is limited to between 25% and 40%. CYP3A4-mediated *N*-dealkylation to mono- and di-dealkylated metabolites occurs to a limited extent, and the *O*-glucuronides that form at multiple points along the metabolic path are excreted via the kidney (Fig. 37.19). Extensive renal clearance demands dosage adjustment in patients with kidney disease. A nanoliposomal formulation of topotecan with an enhanced cytotoxic and pharmacokinetic profile is currently in development (69).

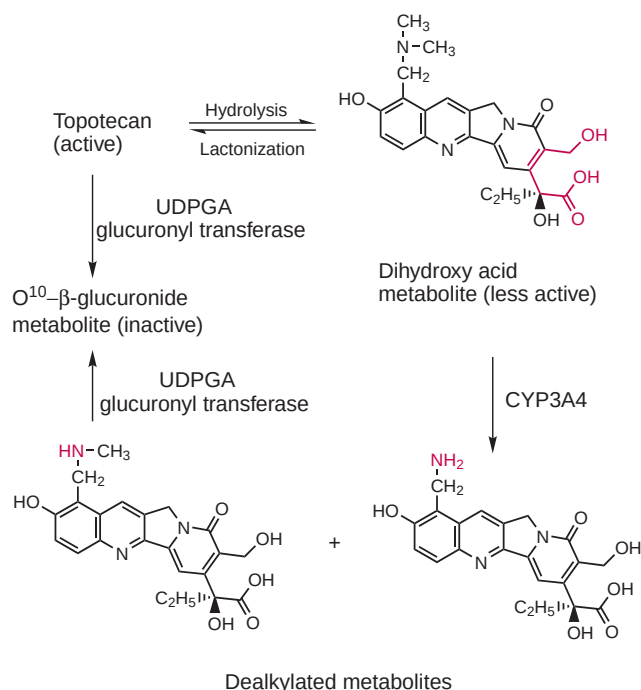


FIGURE 37.19 Topotecan metabolism.

Because both topotecan and irinotecan are metabolized by CYP3A4, the potential for drug–drug interactions must be evaluated. Reduced clearance was noted when azole antifungal agents and cyclosporine were coadministered with irinotecan, and accelerated clearance was observed when topotecan was coadministered with CYP3A4 inducers such as phenobarbital and phenytoin. Unlike irinotecan, at the time of this writing, there have been no published reports correlating the risk of topotecan toxicity to UGT1A1 polymorphism.

Epipodophyllotoxins

The epipodophyllotoxins (Fig. 37.17) are semisynthetic glycosidic derivatives of podophyllotoxin, the major component of the resinous podophyllin isolated from the dried roots of the American mandrake or mayapple plant (*Podophyllum peltatum*). Although these compounds are capable of binding to tubulin and inhibiting mitosis, their primary mechanism of antineoplastic action is TopIIα poisoning, a mechanism that they share with anthracyclines (see antineoplastic antibiotics). TopIIα has two distinct DNA-independent binding sites for the epipodophyllotoxins, one within the catalytic domain and a second within the N-terminal adenosine triphosphate (ATP)-binding domain (70). Once bound, the toxins stabilize the cleavable ternary drug–enzyme–DNA complex, stimulating DNA ligation but inhibiting resealing. The DNA-topoisomerase fragments accumulate in the cell, ultimately resulting in apoptosis. The RNA transcription processes are also disrupted by the interaction of epipodophyllotoxins with TopIIα (63). The epipodophyllotoxin binding site has recently been probed with

a carbene-generating diazirine photoaffinity label (71), and a virtual library of 143 epipodophyllotoxin derivatives has been docked to a three-dimensional human TopIIα receptor model in order to identify key drug–enzyme interactions (Fig. 37.20) (72). This valuable information should serve as the starting point for the rational development of more potent and target-specific epipodophyllotoxins.

Epipodophyllotoxins are cell cycle specific and have their most devastating impact on cells in the S or early G₂ phase. For this reason, doses are divided and administered over several days. Resistance is multifaceted and involves downregulation of TopIIα, attenuation of enzymatic activity levels, development of novel DNA repair mechanisms, and P-gp-mediated cellular efflux.

CHEMISTRY Structurally, the two marketed epipodophyllotoxins, etoposide and teniposide, differ only in the nature of one β-D-glucopyranosyl substituent (methyl or thienyl, respectively). Both are highly water insoluble, but teniposide's higher lipophilicity facilitates cellular uptake and results in a 10-fold enhancement of potency (63). The need for solubility enhancers, such as polysorbate 80 (Tween, etoposide) or polyoxyethylated castor oil (Cremophor EL, teniposide), in intravenous formulations puts patients at risk for hypersensitivity reactions that can manifest as hypotension and thrombophlebitis. Epinephrine, antihistamines, and corticosteroids are often coadministered to minimize risk. A water-soluble phosphate ester analog of etoposide can be administered in standard aqueous vehicles, permitting higher doses than the oil-modified formulations would allow. The phosphate ester is rapidly cleaved to the free alcohol in the blood.

METABOLISM Epipodophyllotoxins are subject to metabolic transformation before renal and nonrenal elimination (Fig. 37.21). Etoposide is stable enough for oral administration, although a dose approximately twice that of the intravenous formulation must be administered. Teniposide is more extensively metabolized, presumably due to its enhanced ability to penetrate into hepatocytes, and no oral dosage form is marketed. Both

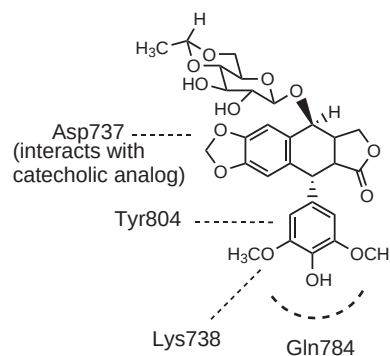


FIGURE 37.20 Proposed etoposide–TopIIα binding interactions.

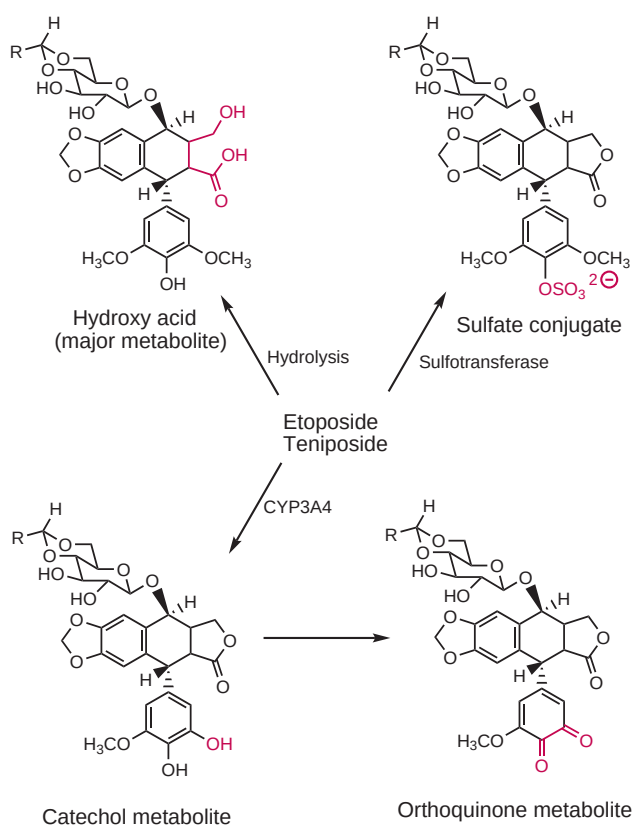


FIGURE 37.21 Etoposide metabolism.

drugs undergo lactone hydrolysis to generate the inactive hydroxy acid as the major metabolite, but the parent drugs can also be transformed by CYP3A4-catalyzed *O*-demethylation and phase 2 glucuronide or sulfate conjugation. Phase 2 metabolism accounts for between 5% and 22% of the dose. Clinically significant interactions between epipodophyllotoxins and CYP3A4 inducers, such as phenytoin, phenobarbital, and St. John's wort, have been documented, and coadministration can enhance antineoplastic drug clearance by as much as 77%. Conversely, CYP3A4 inhibitors, such as cyclosporine or macrolide antibiotics, can decrease clearance, leading to unwanted toxicity.

The catechol metabolite can oxidize to a reactive orthoquinone, and both have been proposed to promote topoisomerase-mediated DNA cleavage, potentially enhancing the risk of the translocations that result in therapy-induced acute myeloid leukemia in children treated with these drugs. Epipodophyllotoxin-induced leukemia occurs in 2% to 12% of patients and is believed to result from translocation of the *MLL* gene at chromosome band 11q23. The mean latency period of 2 years is shorter than the 5- to 7-year latency for leukemia induced by DNA alkylators, and the drug-induced cancer is often resistant to standard treatment (including bone marrow transplantation) (73). Other serious adverse effects include dose-limiting mucositis and myelosuppression, particularly leukopenia. Alopecia is common, and nausea

and vomiting, most noticeable with the oral dosage form, are generally mild.

SPECIFIC DRUGS (FIG. 37.17)

Etoposide Etoposide is used in the treatment of small cell lung cancer and in combination with other agents in refractory testicular cancer. Both intravenous and oral formulations are available. Oral bioavailability is concentration-dependent and runs approximately 50% for the 50-mg capsule. Little first-pass metabolism is noted with the gelatin capsule dosage form. The drug is more than 96% protein bound, undergoes biphasic elimination, and has a terminal half-life of 4 to 11 hours. Approximately 35% to 45% of a dose is eliminated via the kidneys, with less than 6% excreted in feces. The drug should be used with caution in patients with renal or liver disease. Specifically, doses should be decreased in patients with creatinine clearance of less than 50 mL/min or bilirubin levels of greater than 1.5 mg/dL, and the drug should not be used in patients with bilirubin levels of greater than 3.1 mg/dL. Organoplatinum anticancer agents (e.g., cisplatin) decrease etoposide clearance, especially in children. If used in combination, administration must be separated by at least 2 days.

Teniposide Teniposide is used in combination with other agents for the treatment of refractory childhood acute lymphoblastic leukemia. Compared to etoposide, it is more tightly protein bound (>99%), more extensively metabolized, more slowly cleared (terminal half-life, 5 to 40 hours), and less dependent on renal elimination (10% to 21%). Exposure to heparin can cause teniposide to precipitate, so lines must be thoroughly flushed before and after teniposide administration. The drug can also spontaneously precipitate, particularly if solutions are overagitated, and patients receiving 24-hour infusions should be monitored for blockage of access catheters. Teniposide and etoposide are Category D teratogens and, if at all possible, should not be used in women of childbearing age.

Antibiotics

The antibiotic antineoplastics (Fig. 37.22) are a broad category of natural or semisynthetic compounds that block DNA transcription by inhibiting enzymes critical to the DNA replication process and/or by nicking and/or inducing point mutations in the DNA strand. Antibiotic antineoplastics that interact directly with DNA first intercalate the double-stranded helix by inserting between the base pairs and forming strong noncovalent interactions with DNA bases. The highly stabilized complex deforms and uncoils the DNA, prohibiting proper replication. To bulldoze its way between the bonded DNA strands, a segment of the antibiotic must have the trigonal coplanar geometry guaranteed by aromaticity.

The antineoplastic anthracycline antibiotics function as TopII α poisons. Another mechanism of cytotoxic

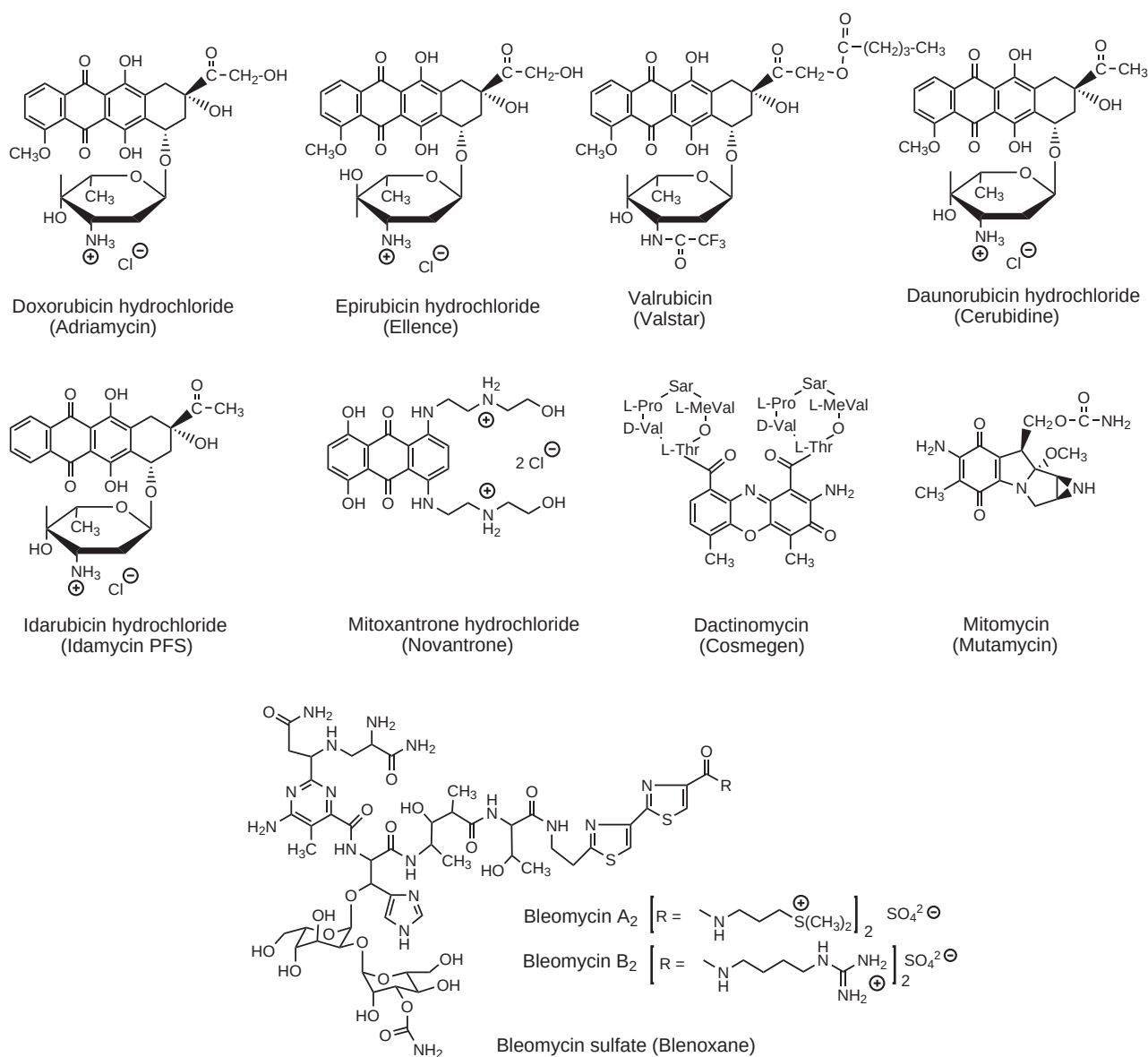
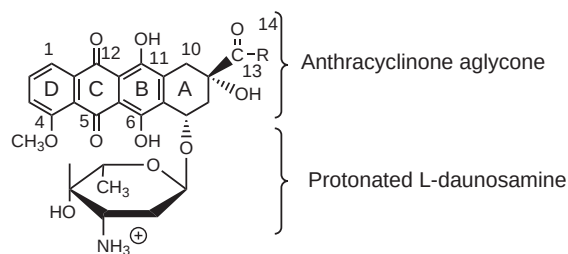


FIGURE 37.22 Anticancer antibiotics.

action, particularly for one antibiotic (bleomycin), is the generation of cytotoxic free radicals that cause single-strand breaks in DNA. Another antibiotic (mitomycin) is capable of alkylating DNA, a mechanism more commonly associated with the nitrogen mustard antineoplastics but that is predictable from the nucleophilic aziridine ring found within the structure of this anticancer agent.

Anthracyclines and Anthracenediones

Anthracycline antineoplastic antibiotics are very closely related to the tetracycline antibacterials. Structurally, they are glycosides and contain a sugar portion (L-daunosamine) and a nonsugar organic portion. The nonsugar portion of glycosides is generically referred to as an aglycone. In anthracyclines, the aglycone moiety is specifically called anthracyclinone or anthraquinone.



MECHANISM OF ACTION Anthracycline-based antineoplastic agents act by poisoning TopII α through the stabilization of the ternary drug–enzyme–DNA cleavable complex. Like the topoisomerase poisons discussed earlier, they allow DNA to be cut and covalently linked to the conserved topoisomerase Tyr residue but inhibit the resealing reaction. The flat, aromatic portion of

the anthracyclinone ring system and the daunosamine sugar bind to DNA, whereas the anthracyclinone A ring is believed to bridge the gap between DNA and enzyme (57,74). Since a small number of anthracycline-induced DNA breaks can result in a high level of cell death, it has been hypothesized that the site of DNA cleavage, which contains an essential thymine-adenine (T-A) dinucleotide, is particularly lethal to the cell (75).

CHEMISTRY DNA intercalation initiates the antineoplastic action of the anthracyclines (58). Rings B, C, and D slide between the two DNA strands, orienting the anthracyclinone moiety in a perpendicular fashion relative to the long axis of DNA and stabilizing the complex through π stacking and other affinity-enhancing interactions. A recent study in which doxorubicin was docked in a modeled DNA “post-cleavage” intercalation site proposed highly efficacious H-bonds between top Ser740 and the C₅ quinone oxygen (ring C), top Thr744 and the C₄-OCH₃ (ring D), and a DNA thymine base and the C₉-OH (ring A) (58). If present, a C₁₄-OH should H-bond with the carbonyl oxygen of a DNA thymine base (Fig. 37.23). Interestingly, although the C₄-OCH₃ helps hold drug to TopII α enzyme, its removal increases antitumor activity by enhancing anthracyclinone planarity (thereby facilitating intercalation) and by directing the binding of the daunosamine sugar to stabilize the inhibitory ternary cleavable complex (76).

The daunosamine sugar is known to bind in the minor groove of DNA at the DNA-topoisomerase interface and subsequently orchestrate the DNA sequence specificity of the intercalation and overall DNA poisoning process (58,74). Binding roles for the protonated 3'-amino group have run the gamut from ion-ion paring with a DNA phosphate to covalently linking to the C₂-NH₂ of guanine via a formaldehyde methylene bridging unit (77). The aforementioned molecular modeling study suggested that the cationic daunosamine amino group binds with high affinity to the carbonyl oxygen of a DNA thymine base when in the naturally occurring α configuration. In the epimerized β configuration, there is an increased distance between these two moieties and an unfavorable steric interaction with other DNA residues. While the loss or epimerization of the daunosamine 3'-amino

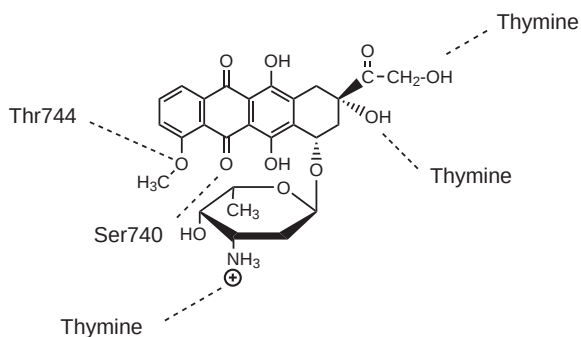
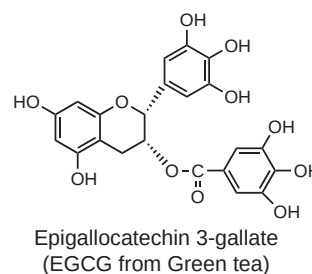


FIGURE 37.23 Proposed interaction between doxorubicin, DNA, and topoisomerase II α .

moiety decreases DNA binding, it does not destroy it. In fact, it has been stated that the antitumor activity of anthracyclines is related more to the proper positioning and stabilization of the drug within the “cleavable ternary complex” of drug-topoisomerase-DNA than to the actual affinity of the drug for the DNA (74,76).

Resistance to anthracycline chemotherapy can be intrinsic or acquired. The major mechanisms through which cancer cells fight back include compromised drug transport across cell membranes, active efflux via P-gp and MDR transporters, changes in tumor cell responsiveness to apoptotic triggers, alterations in TopII α expression and activity, and augmented biochemical defenses against anthracycline-induced oxidative stress (78).

Interestingly, some authors have shown that the polyphenol epigallocatechin-3-gallate (EGCG, found in green tea) can inhibit cellular efflux of the anthracycline doxorubicin (79,80) and sensitize doxorubicin-treated/resistant human colon carcinoma cells (81). In addition to the scientific evidence supporting the positive doxorubicin-EGCG interaction, Sadzuka and colleagues (80) holistically state, “We think that the intake of a favorite beverage favors a positive mental attitude of the patient and increases the efficacy of the chemotherapeutic index, and that this efficacy is useful for improving the quality of life on cancer chemotherapy.”



CHEMICAL MECHANISM OF CARDIOTOXICITY An important mechanism of use-limiting anthracycline cardiotoxicity involves the formation of cytotoxic free radicals. A free radical is a highly reactive species with an unpaired electron. Of particular importance is the formation of superoxide radical anion (O_2^-) and hydroxyl radical ($\cdot\text{OH}$), both of which are formed via a one-electron reduction of the anthracyclinone quinone (ring C) to hydroquinone by NADPH/CYP450 reductase. The mechanism by which these reactive oxygen species (ROS) are generated is shown in Figure 37.24.

When NADPH/CYP450 reductase reduces the quinone ring to a hydroquinone, superoxide radical anions (O_2^-) are generated. Superoxide radical anions react to generate hydrogen peroxide (H_2O_2), a reaction that requires protons and is catalyzed by the enzyme superoxide dismutase in a Cu²⁺-mediated process. The fate of this hydrogen peroxide dictates the degree of cytotoxicity observed from the anthracycline.

In the presence of the enzyme catalase, hydrogen peroxide is rapidly converted to water and oxygen, which

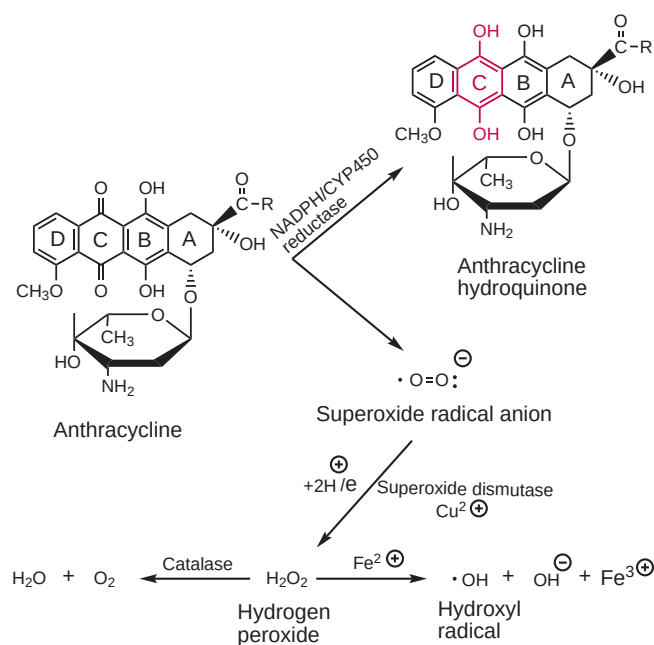
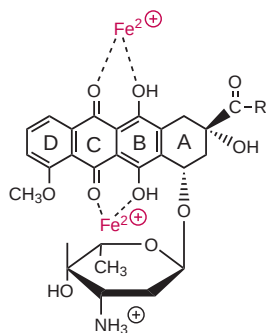


FIGURE 37.24 Anthracycline-mediated free radical formation.

obviously are harmless chemicals as far as the body is concerned. However, in the presence of ferrous ion (Fe^{2+}), hydrogen peroxide is converted into the highly toxic hydroxyl radical through a process called the Fenton reaction. Hydroxyl radicals promote single-strand breaks in DNA, which is therapeutically desirable to treat the uncontrolled growth of cancer cells. Anthracycline anticancer agents are also known to interfere with normal ferritin-iron mobilization, resulting in iron accumulation (82). The anthracyclines chelate strongly with di- and trivalent cations, including intracellular Fe^{2+} , so the generation of cytotoxic hydroxyl radicals after the initial NADPH/CYP450 reductase reduction is essentially guaranteed. Hydroxide anion and Fe^{3+} are also formed during the production of hydroxyl radicals. Anthracycline-iron complexes (particularly those involving Fe^{3+}) can also spontaneously generate ROS without the aid of enzymatic catalysis (83).



The generation of hydroxyl radicals inside the tumor cell could augment the antineoplastic effect of the anthracyclines, but such generation is uncommon at standard antineoplastic doses (75). These cytotoxic

radicals are generated within the heart, however, and can lead to acute, albeit reversible, cardiotoxicity. Cardiac tissue is particularly vulnerable to free radical damage by the anthracyclines because it does not contain significant amounts of catalase and other relevant cytoprotective enzymes (84). When hydrogen peroxide forms in the myocardium, it has no choice but to go down the Fenton pathway. Cardiac toxicity is the major use-limiting side effect of anthracycline use, but coadministration of dexrazoxane (an antioxidant and iron chelator) has been shown to lower its incidence (85).

A role for nitric oxide metabolism, particularly nitric oxide synthase, in anthracycline-mediated cardiotoxicity has also been proposed (83,86). The reactive nitrogen species of greatest concern is peroxynitrite (ONOO^-), a strong oxidant that shows no selectivity in its destruction of life-sustaining macromolecules.

Although the rate of quinone metabolism influences the risk of acute anthracycline-induced cardiotoxicity, metabolism at C_{13} is believed to be responsible for the more life-threatening chronic cardiotoxicity that some patients experience. The C_{13} -carbonyl is reduced via a two-electron mechanism to a commonly less active (87,88) or inactive (83) secondary alcohol via cytosolic aldoketoreductase enzymes (Fig. 37.25), and the larger the R group, the slower this reaction and the longer the

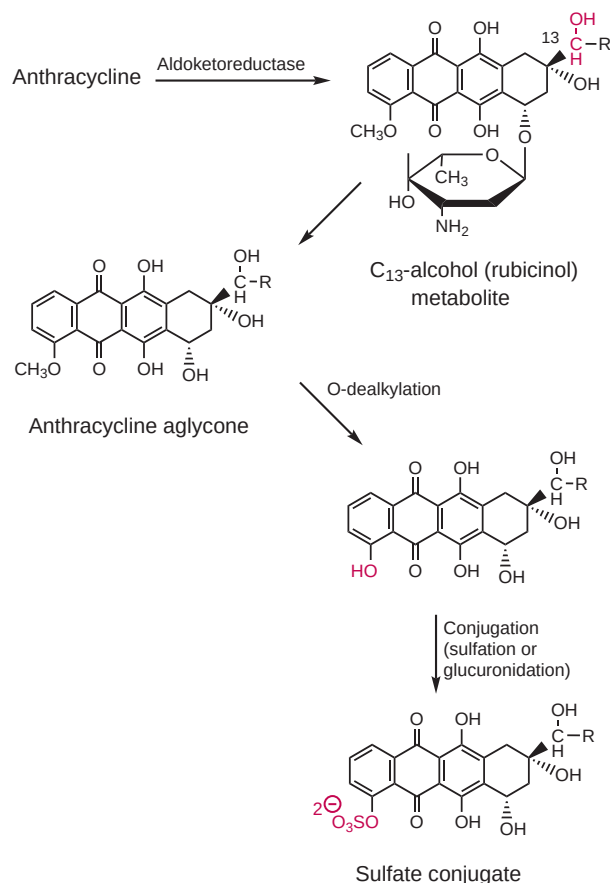
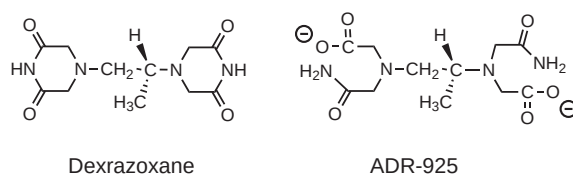


FIGURE 37.25 Anthracycline metabolism.

duration of antineoplastic action. The C_{13} -substituents found on most marketed anthracyclines include CH_3 (daunorubicin and idarubicin) and CH_2OH (doxorubicin and epirubicin). Before excretion, anthracyclines are further metabolized via hydrolytic or reductive deglycosidation to their 7-hydroxy or 7-deoxy aglycones, respectively, followed by *O*-dealkylation of the C_4 methoxy ether (if present) and conjugation with either glucuronic acid or sulfate. The aglycones may also have cardiotoxic properties (83).

The secondary alcohol (“rubicinol”) metabolites formed by aldoketoreductase concentrate in cardiomyocytes and induce a prolonged inhibition of calcium loading, open a selective ion channel, inhibit Ca^{2+} , Mg^{2+} -ATPase leading to increased cytosolic levels of Ca^{2+} in the sarcoplasmic reticulum, and inhibit Na^+ , K^+ -ATPase action in the sarcolemma. Collectively, these cellular events can induce a chronic cardiomyopathy that presents as severe congestive heart failure involving systolic and diastolic dysfunction (82). As the rubicinol metabolites form a “long-lived reservoir” of cardiotoxic drug within the myocardium, chronic anthracycline-induced congestive heart failure can manifest without warning years after therapy, and it is often unresponsive to therapeutic intervention (83). It has been estimated that more than half of the patients diagnosed with chronic anthracycline-induced congestive heart failure will die within 2 years (83). Elevated risk has been related to age (both very young and very old), genetic polymorphisms impacting ROS production and/or anthracycline transport, underlying cardiovascular disease, high cumulative doses, and cyclophosphamide co-therapy. Females and blacks appear to be at risk for increased incidence or severity of drug-induced cardiomyopathy. Because toxicity is dose-dependent, patients with liver dysfunction who cannot adequately metabolize and clear anthracyclines are also at risk. Dosage adjustments in patients with liver disease must be made to avoid life-threatening toxicity.

Although the acute and chronic phases of anthracycline-induced cardiomyopathy appear metabolically distinct, a “unifying hypothesis” has recently been put forward to suggest that induction of ROS-mediated oxidative stress that characterizes acute cardiac toxicity may upregulate aldoketoreductase, thereby facilitating the development of rubicinol-induced chronic toxicity (83). As noted earlier, the pharmacotherapeutic approach currently used to attenuate anthracycline-induced cardiotoxicity is coadministration of dexrazoxane, an antioxidant and prodrug iron chelating agent. Dexrazoxane readily enters cells and is hydrolyzed to the active Fe^{2+} and Fe^{3+} chelating form (ADR-925). Although the affinity of ADR-925 for iron surpasses that of doxorubicin, iron-independent mechanisms for this cardioprotectant have also been proposed (83). This prodrug chelator can also be used to prevent serious tissue injury following accidental anthracycline extravasation (89).



OTHER TOXICITIES In addition to cardiac toxicity, all anthracycline antineoplastics can cause severe myelosuppression (especially leukocytopenia) as well as moderate to severe nausea and vomiting, mucositis leading to hemorrhage and potentially fatal infection, and alopecia. Side effects are dose-dependent. Most of the anthracyclines are orally inactive and must be given by intravenous injection. They are highly necrotic to skin and, if extravasation occurs, can cause such severe blistering and ulceration that skin excision, followed by plastic surgery, may be indicated. The anthracyclines contain photosensitive phenolic groups that must be protected from light and air. The highly conjugated structure imparts a reddish-orange color to these compounds (implied in the name “rubicin”), which is maintained when these compounds are excreted in the urine. Patients should be warned that the reddish urine they will experience is not hemorrhagic but, rather, simply the result of the conjugated chemistry of this class of drugs.

The risk of serious or life-threatening adverse reactions can potentially outweigh the therapeutic benefits of anthracyclines in some patients, which has fueled the search for biomarkers that will predict tumor responsiveness to these powerful antineoplastic agents (90).

SPECIFIC DRUGS (FIG. 37.22)

Doxorubicin Hydrochloride The C_{13} substituent of doxorubicin is hydroxymethyl, which retards the action of cytosolic aldoketoreductase and slows the conversion to the less active and chronically cardiotoxic doxorubicinol. This contributes to the longer duration of action compared to analogs that have CH_3 at this position (e.g., daunorubicin). Doxorubicin is highly lipophilic and concentrates in the liver, lymph nodes, muscle, bone marrow, fat, and skin. Elimination is triphasic, and the drug has a terminal half-life of 30 to 40 hours. The majority of an administered dose is excreted in the feces, approximately half of it unchanged (83). Doxorubicin is used either alone or in combination therapy to treat a wide range of neoplastic disorders, including hematologic cancers and solid tumors in breast, ovary, stomach, bladder, and thyroid gland.

A liposomal formulation of doxorubicin, marketed as Doxil, is used in the treatment of AIDS-related Kaposi sarcoma and organoplatinum-resistant ovarian cancer. Liposomes are taken up selectively into tumor cells, presumably due to their persistence in the bloodstream and enhanced permeability of tumor vascular membranes. In liposomal form, the drug is protected against enzymes that generate cardiotoxic free radicals and is less likely to concentrate in the heart (91). However, because this

form of the drug can still induce potentially fatal congestive heart failure, all precautions outlined for the use of doxorubicin are employed when the liposomal formulation is used. The half-life of Doxil is extended to approximately 55 hours, and it is administered in doses ranging from 20 to 50 mg/m² every 3 to 4 weeks. The area under the curve of the liposomal formulation is approximately triple that of the free drug formulation. It is cleared more slowly than conventional doxorubicin and generates very little of the doxorubicinol metabolite. Significant side effects have occurred when the liposomal formulation is erroneously dispensed, so pharmacists must be vigilant when interpreting therapeutic orders.

Epirubicin Hydrochloride This stereoisomer of doxorubicin has the 4'-hydroxy group of the daunosamine sugar oriented in the unnatural β -position. However, this relatively modest structural change has a large impact on pharmacokinetic properties. Epirubicin is reduced to the C₁₃ alcohol (epirubicinol) to a much lower (60%) extent than doxorubicin, and it is not highly susceptible to ROS-generating one-electron oxidation. The overall cardiotoxicity has been estimated at 30% lower than doxorubicin, but the margin of safety is mitigated somewhat by epirubicin's greater propensity to accumulate in myocardiocytes (83). The epirubicinol metabolite has an antitumor potency one-tenth that of the parent drug and does not contribute significantly to the therapeutic action. Both parent drug and metabolite readily undergo O-dealkylation/glucuronidation, resulting in a shortened terminal half-life compared to doxorubicin. Cleavage to the aglycone will occur prior to elimination (57,83). Although excretion is primarily biliary, dosage reduction in severe renal impairment, as well as in hepatic dysfunction, is warranted.

Epirubicin is indicated for use in breast cancer, and the starting dose is 100 to 120 mg/m² (compared to a starting dose of 60 to 75 mg/m² for doxorubicin). The side effects and precautions are as outlined previously for doxorubicin, although, as noted, there is a lower risk of serious myocardial toxicity or myelotoxicity.

Valrubicin Chemically, valrubicin differs from its doxorubicin parent by the addition of a C₁₄-valerate ester and a 3'-trifluoroacetamide moiety. The carbon-rich valerate is obviously lipophilic, and acylation of the daunosamine amino group makes the 3'-substituent nonionizable. Both of these structural changes promote a more rapid and extensive penetration into tumor cells.

Valrubicin currently has orphan drug status in the treatment of bacille Calmette-Guérin-refractory bladder cancer (the total patient population is ~1,000 individuals) and is used in patients for whom surgical intervention would result in high morbidity or death. It is administered directly into the bladder through a catheter (intravesically). The lipophilic drug is water insoluble, but it dissolves in an aqueous vehicle that includes polyethoxylated castor oil (Cremophor EL) and ethanol.

The patient retains the drug in the bladder for 2 hours and then voids the solution in the normal fashion.

Valrubicin is an active antineoplastic as administered, although it does not function as a TopII α poison. Rather, the intact drug inhibits the incorporation of nucleosides into DNA and RNA, which induces chromosomal damage and cell cycle arrest (92). Despite the fact that hydrolysis of the ester and trifluoroacetamide can be readily envisioned, it is excreted essentially unchanged. Less than 1% of an administered dose is absorbed systemically, so there is essentially no exposure to metabolizing enzymes. The reduced C₁₃-alcoholic metabolite does not form to any appreciable extent during the 2-hour treatment period. Therapy is considered to be almost exclusively local, and there is little risk for cardiac toxicity, bone marrow suppression, drug-drug interactions, or other side effects. Systemic exposure to the drug and its hydrolyzed (and cardiotoxic) metabolites N-trifluoroacetyldoxorubicin and N-trifluoroacetyldoxorubicinol would, of course, be greater in patients whose bladder wall integrity has been compromised by disease. These patients should not receive valrubicin. The most commonly reported adverse reactions to valrubicin are abdominal pain, urinary tract infection, hematuria, and dysuria. Some patients experience severe allergic reactions, most probably due to the Cremophor EL solubilizer that is notorious for inducing hypersensitivity reactions. Unlike other anthracyclines, valrubicin is not necrotic to skin.

Daunorubicin Hydrochloride The absence of the OH group at C₁₄ in daunorubicin results in a faster conversion to the less active and chronically cardiotoxic C₁₃-ol metabolite (daunorubicinol) compared to hydroxymethyl-substituted anthracyclines like doxorubicin. The 18.5-hour terminal half-life of daunorubicin is approximately half that of doxorubicin, and the terminal half-life of the daunorubicinol metabolite is 26.7 hours. Excretion is approximately 40% biliary and 25% urinary. Daunorubicin is administered intravenously at a dose of 45 mg/m² for the treatment of lymphocytic and nonlymphocytic leukemia. The toxicity and side effect profile of this anthracycline is similar to that of doxorubicin, and all previously identified precautions apply.

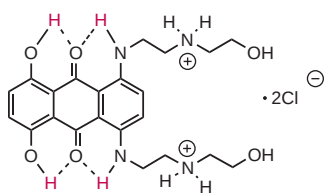
The citrate salt of daunorubicin is marketed as a liposomal formulation, which promotes the use of this agent in solid tumors. Like Doxil (the liposomal formulation of doxorubicin), DaunoXome is indicated for use in AIDS-related Kaposi sarcoma and is administered intravenously at a dose of 40 mg/m² every 2 weeks. The pharmacokinetic profiles of Doxil and DaunoXome are similar.

Idarubicin Hydrochloride Idarubicin is the 4-desmethoxy analog of daunorubicin. The loss of the C₄-ether flattens the D ring, facilitating intercalation between DNA base pairs. In turn, this orients the daunosamine sugar in the minor groove in a way that better stabilizes the ternary complex between drug, DNA, and topoisomerase (76).

The loss of the 4-methoxy moiety also makes this compound more lipophilic than either doxorubicin or daunorubicin. This results in a better penetration into tumor cells and an enhanced antineoplastic potency. Increased rates of remission have been noted with the use of idarubicin compared to other anthracycline antineoplastic agents. Unlike its congeners, idarubicin shows significant oral bioavailability and is lipophilic enough to penetrate the blood–brain barrier. Currently, it is given only by the intravenous route and is not used in the treatment of brain cancer. Its primary indication is in acute myeloid leukemia, and it is administered in combination with other antileukemic drugs.

Idarubicin is reduced by aldoketoreductases to idarubicinol which, unlike other rubicins, is as active an anti-tumor agent as the parent drug (88). Because there is no aromatic methoxy group, there is no *O*-dealkylation to the C₄-phenol. The major metabolite is free, unconjugated idarubicinol. The half-lives of idarubicin and idarubicinol are 22 and 45 hours, respectively. Idarubicin is administered intravenously at a dose of 10 to 12 mg/m²/day for 3 to 4 days, and the idarubicinol metabolite can still be found in therapeutic concentrations in the blood 8 days after administration. Like other anthracyclines, excretion primarily is fecal, with a lesser dependence on renal elimination. Some authors have shown that idarubicin is transported into cardiac tissue via a saturable transporter and that the coadministration of methylxanthines (e.g., caffeine) can increase both myocardial drug concentrations and the risk of idarubicin-induced cardiotoxicity (93).

Mitoxantrone Hydrochloride Chemically, mitoxantrone is classified as an anthracenedione. The sugar moiety is missing, but the cationic side-chain amino nitrogens could bind to the anionic phosphate residue of the DNA backbone in the same fashion that the cationic L-daunosamine amino group of the true anthracyclines has been presumed to do. This molecule has the structural features needed to intercalate DNA and inhibit TopII α , but the enhanced stability of the quinone ring (possibly through an increased potential for intramolecular hydrogen bonding) makes the ring highly resistant to NADPH/CYP450 reductase. This limits the formation of the highly toxic ROS. In addition, there is no active cardiotoxic metabolite to induce chronic toxicity by disrupting the movement of myocardial cations. The chance of cardiovascular toxicity from mitoxantrone is significantly decreased, although patients who have been previously treated with anthracycline antineoplastics may still be at risk. It is thought that any observed myocardial toxicity may be operating through mechanisms other than the generation of cytotoxic radicals.



In addition, the risk of ulceration and necrosis on extravasation, as well as of non-marrow-related toxicities such as nausea, vomiting, mucositis, and alopecia, is significantly less than observed with true anthracyclines. There is a significant risk of bone marrow suppression, however. The risk of myelosuppression increases with dose, but it can be observed even when low doses are used.

Mitoxantrone excretion primarily is biliary. Both the unchanged drug and inactive metabolites resulting from *N*-dealkylation, deamination, and oxidation of the resultant aldehyde to the carboxylic acid are observed. Both arms of the structure can be metabolized, leading to mono- or dicarboxylic acid metabolites (Fig. 37.26), which are excreted as the glucuronide conjugate. The conjugated metabolites are an intense, dark blue in color and will result in blue-green excrement. The whites of the eyes and, in some cases, the skin may also take on a bluish cast.

Mitoxantrone is used in combination with other agents during the initial treatment of acute nonlymphocytic leukemia and hormone-refractory prostate cancer. Mitoxantrone also decreases the rate of relapse and disease progression in patients with multiple sclerosis (94). Although too toxic for use in patients with primary progressive disease, it is available for the treatment of chronic progressive, progressive relapsing, or deteriorating relapsing-remitting multiple sclerosis.

Miscellaneous Antibiotics (Fig. 37.22)

DACTINOMYCIN Dactinomycin has two pentapeptide lactones attached to an aromatic (and, therefore, flat) actinocin (or phenoxazinone) structure. It is capable of intercalating DNA and binds preferably between guanine and cytosine residues on a single DNA strand. This interaction results in DNA elongation and distortion, commonly referred to as a point mutation. When sliding between adjacent DNA base pairs, the actinocin orients itself perpendicular to the main DNA axis, allowing the pentapeptide lactone units to bind to residues in the minor groove of DNA through hydrophobic and

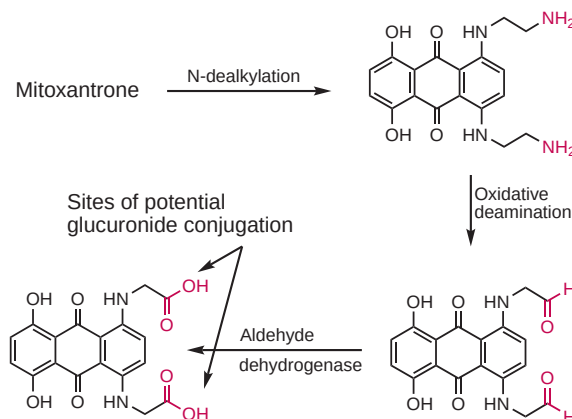
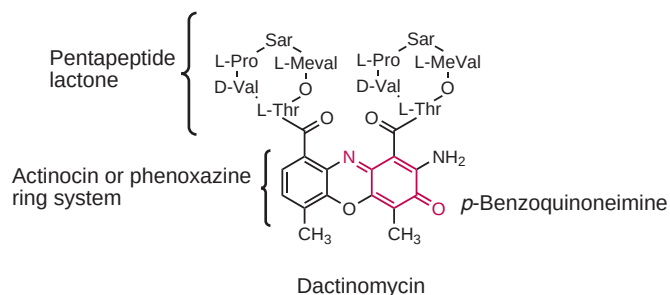


FIGURE 37.26 Mitoxantrone metabolism.

hydrogen bonds. An affinity-enhancing bond between the threonine carbonyl oxygen and a protonated C₂-amino group of guanine also forms. Other hydrogen, hydrophobic, and π -stacking interactions form between the lactone and DNA residues, particularly guanine and cytosine.

The binding of the actinocin and polypeptide lactone portions of dactinomycin to DNA is cooperative, meaning that the binding of one unit facilitates the binding of the other, most likely by promoting an optimal orientation. This significantly enhances drug-DNA affinity. The binding of dactinomycin to DNA, although noncovalent, is much stronger than that observed with the anthracyclines. Drug dissociation from DNA is slow, leading to a pseudoirreversible effect. Dactinomycin blocks gene transcription and translation processes, and RNA polymerase is inhibited, resulting in a decrease in de novo RNA (especially mRNA) and protein synthesis. A recent study has also documented the drug's ability to inhibit the polymerization of oligonucleotides containing the oncogenic c-Myc promoter G-quadruplex sequence through direct binding to this DNA segment (95).

The *p*-benzoquinoneimine segment of dactinomycin renders the molecule vulnerable to NADPH/CYP450 reductase. Free radicals can be generated, and additional single-strand DNA breaks can result. The loss of either aromatic methyl group results in a loss of activity. The reason for this profound impact on pharmacologic action and therapeutic utility is unknown.



Dactinomycin is used for the treatment of various solid tumors and muscle-related cancers. It induces severe side effects, and nausea and vomiting can be use-limiting. Myelosuppression is also common and, most often, is the dose-limiting toxic effect. The drug is usually given by the intravenous route, but toxicity can be limited if the tumor can be perfused with drug (assuming minimal distribution into the general circulation). Dactinomycin is a severe blistering agent, and extravasation can cause irreversible and profound tissue damage. The side effects of radiation therapy are significantly exaggerated by the concurrent use of dactinomycin. The drug's 36-hour half-life is the result of a very high affinity for DNA, a large volume of distribution, and minimal metabolic breakdown. Dactinomycin is photosensitive and must be protected from light.

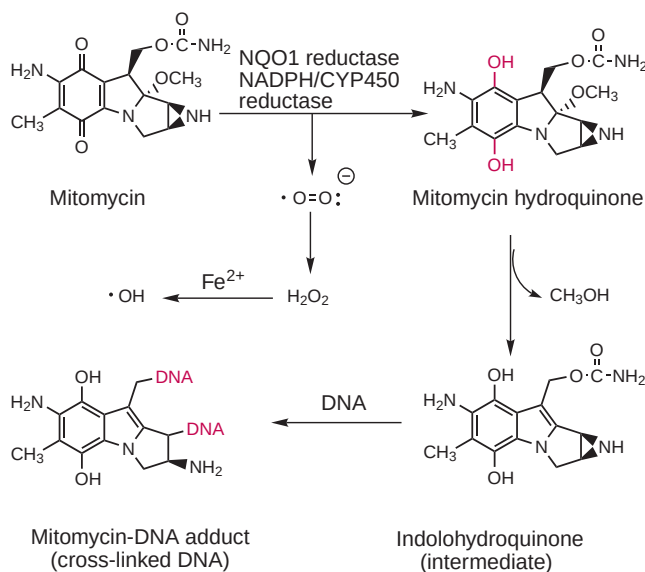


FIGURE 37.27 Mitomycin metabolism.

MITOMYCIN As shown in Figure 37.27, mitomycin is activated through a two-electron bioreductive process using NADPH/CYP450 reductase and/or NAD(P)H quinone oxidoreductase 1 (NQO1 reductase), an enzyme extensively expressed in many neoplastic cells (96–99). Through these enzymes, the quinone ring of mitomycin is readily reduced to the hydroquinone, generating superoxide radicals in the process that ultimately will be converted to cytotoxic hydroxyl radicals through the Fenton reaction. As previously discussed, hydroxyl radicals induce single-strand breaks in DNA. Formation of the hydroquinone is followed by aromatization to the indole ring through the loss of methanol. Both the electrophilic aziridine ring and the δ^+ methylene group adjacent to the carbamate ester are vulnerable to attack by DNA nucleophiles, such as the 2-NH₂ group of guanine or the 4-NH₂ group of cytosine, and can conceivably result in cross-linked DNA and cell death. The fact that mitomycin activating enzymes are predominantly cytosolic has brought into question the nuclear mechanisms long thought operational with this antibiotic. A recent report has provided evidence for 18S rRNA as the true therapeutically relevant target of mitomycin. Through the inhibition of cytosolic rRNA, mitomycin would induce cell death through “genome-wide translational silencing” (98).

Mitomycin is administered intravenously in the treatment of disseminated adenocarcinoma of the stomach or pancreas, and it has been used intravesically in superficial bladder cancer. Biotransformation pathways are saturable, and approximately 10% of an administered dose is eliminated unchanged via the kidneys. Myelosuppression is the major use-limiting side effect of this drug, which is slow to manifest but quite prolonged in duration. Severe skin necrosis can occur on extravasation, and potentially fatal pulmonary toxicities have been noted as well. Mitomycin can induce hemolytic uremia accompanied

by irreversible renal dysfunction and thrombocytopenia, and the drug should not be administered to patients with serum creatinine levels greater than 1.7 mg/dL. Severe bronchospasm has also been noted in patients treated with vinca alkaloids who are also receiving (or who have previously received) mitomycin.

BLEOMYCIN The commercially available bleomycin drug product is a mixture of naturally occurring glycopeptides, predominantly bleomycin A₂. Through DNA intercalation, the aromatic bithiazole ring system positions bleomycin for DNA destruction via cytotoxic free radicals. The disaccharide, polyamine, imidazole, and pyrimidine structures are very electron rich and readily chelate intracellular Fe²⁺. Once chelated, Fe²⁺ is oxidized to Fe³⁺ with a concomitant reduction of bound oxygen. The ferric hydroperoxide bleomycin complex is considered the cytotoxic form (100). Through the direct abstraction of a hydrogen atom from 4' of deoxyribose, a free radical is generated that subsequently decomposes to a highly electrophilic base propenal that inactivates essential cellular proteins via Cys alkylation (Fig. 37.28). Reduced GSH is proposed to serve a protective role by acting as propenal scavenger and, until depleted, saves cellular proteins from alkylation (101).

Bleomycin is a natural product isolated from *Streptomyces verticillus*. It is normally chelated with Cu²⁺, which must be removed via catalytic reduction before marketing. This increases the cost of the drug, but it frees up the critical bleomycin functional groups for chelation with intracellular Fe²⁺.

The action of bleomycin is terminated through the action of bleomycin hydrazase, a cytosolic aminopeptidase that cleaves the terminal amide moiety to form the inactive carboxylate metabolite (Fig. 37.29). The metabolic replacement of the electron-withdrawing amide with an electron-donating carboxylate increases the pK_a of

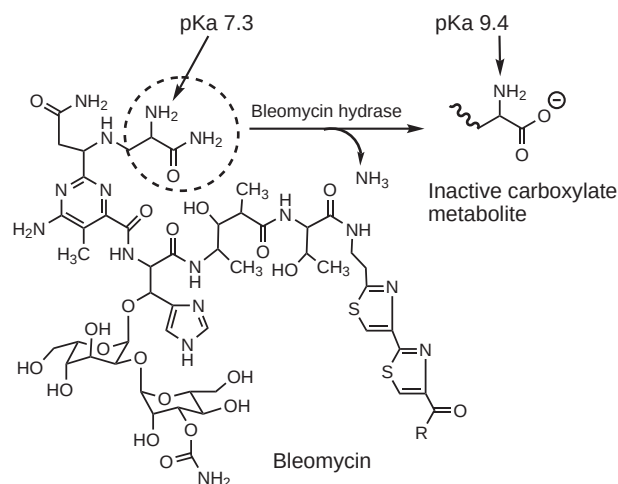


FIGURE 37.29 Bleomycin hydrazase-mediated inactivation of bleomycin.

the α -amino group, which normally interacts with DNA in the un-ionized conjugate form. After hydrolysis, the ratio of ionized to un-ionized forms of this critical amine increases approximately 126-fold, destroying DNA affinity and leading to the loss of therapeutic action. Drug destruction via the bleomycin hydrazase pathway is rapid, and tumors will be resistant to bleomycin if they contain high concentrations of the enzyme. Conversely, tumors that are poor in bleomycin hydrazase (e.g., squamous cell carcinoma) respond well to this agent.

Bleomycin hydrazase is found in all tissues except skin and lung. Approximately 10% of patients who are administered bleomycin will experience potentially fatal pulmonary fibrosis, which can occur during therapy or several months following termination of therapy, often without warning. The copper-complexing agent tetrathiomolybdate may reduce the risk of bleomycin-induced fibrosis by inhibiting the action of copper-dependent inflammatory cytokines (102). A recent report also supports the protective effect of inhibitors of the N-terminal catalytic site of angiotensin-converting enzyme (e.g., N-acetyl-Ser-Asp-Lys-Pro or AcSDKP) (103). Erythema and hypertrophic modifications in skin are also common side effects that manifest after 2 to 3 weeks of bleomycin therapy.

Bleomycin is used intravenously in the palliative treatment of squamous cell head and neck cancers, testicular and other genital carcinomas, Hodgkin's lymphoma, and NHL. It is excreted via the kidneys, and serum concentrations of active drug are increased in patients with renal disease. The elimination half-life can rise from 2 to 4 hours to more than 20 hours in renal failure, resulting in significant toxicity, especially pulmonary toxicity. Dosage adjustments are warranted. Unlike many anti-neoplastic agents, bleomycin does not suppress the bone marrow, and it is often given in combination with compounds that do so that the dose of all drugs can be optimized. Nausea and vomiting are also relatively mild, but approximately 1% of lymphoma patients who are treated

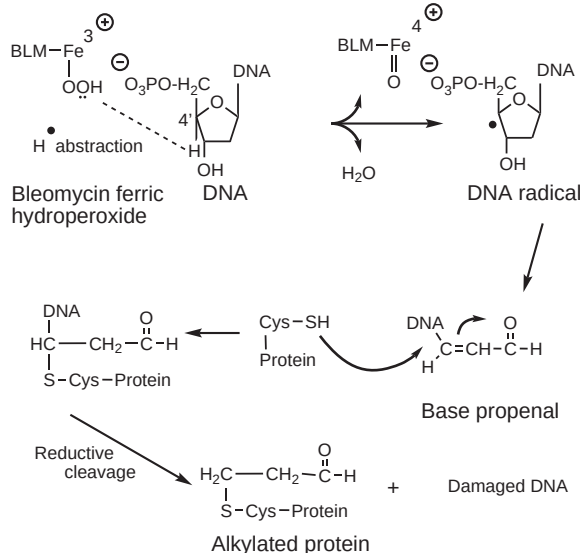


FIGURE 37.28 Mechanism of bleomycin-induced DNA cleavage.

with bleomycin will experience an immediate or delayed severe idiosyncratic reaction that mimics anaphylaxis.

Antimetabolites

MECHANISM OF ACTION The antineoplastic agents discussed thus far have all damaged existing DNA (or, less commonly, RNA) and inhibited its ability to replicate. The antimetabolites, on the other hand, most commonly stop the *de novo* synthesis of DNA by inhibiting the formation of the nucleotides that make up these life-sustaining polymers. We will see that the rate-limiting enzymes of nucleotide biosynthesis are often the primary targets for the antimetabolites since inhibition of these key enzymes is the most efficient way to shut down any biochemical reaction sequence. Antimetabolites are also capable of inhibiting other enzymes required in the biosynthesis of DNA, and many can arrest chain elongation by promoting the incorporation of false nucleotides into the growing DNA strand.

The antimetabolites serve as false substrates for critical nucleotide biosynthesis enzymes. These enzyme inhibitors are structurally “dolled up” to look like a super attractive version of the normal (endogenous) substrate. Speaking anthropomorphically, through a form of chemical entrapment, they entice the enzymes to choose them over the endogenous substrate and, once they do, the antimetabolites bind them irreversibly or pseudoirreversibly. If the building block nucleotides cannot be synthesized, then DNA synthesis (and tumor growth) is stopped dead in its tracks. If tumor growth is arrested, then metastasis slows, and the patient has a fighting chance for remission and/or cure.

Many antimetabolite antineoplastics are categorized by the class of nucleotide they inhibit. Purine antagonists inhibit the synthesis of the purine-based nucleotides adenosine monophosphate (AMP) and guanosine monophosphate (GMP), and the pyrimidine antagonists stop the production of the pyrimidine-based nucleotides, primarily deoxythymidine monophosphate (dTMP).

Pyrimidine Antagonists: dTMP Synthesis Inhibitors

dTMP BIOSYNTHESIS Looked at simply, dTMP is produced via C₅-methylation of deoxyuridine monophosphate (dUMP). The rate-limiting enzyme of the dTMP synthetic pathway is the sulfhydryl-containing thymidylate synthase, with 5,10-methylenetetrahydrofolate (5,10-methylene-THF) serving as the methyl-donating cofactor. All dTMP synthesis inhibitors will inhibit thymidylate synthase either directly or indirectly, and this will result in a “thymineless death” in actively dividing cells. Without dTMP and its deoxythymidine triphosphate metabolite, DNA will fragment, and the cell will die.

To understand how an antimetabolite inhibits a biochemical pathway, we must first understand completely how the pathway normally functions. A quick look at the dTMP synthesis pathway (Fig. 37.30) will confirm that our “simple methylation reaction” is comprised of several important steps, each of which is analyzed in turn below.

The synthase enzyme is very large and contains a deep pocket for the binding of both substrate and cofactor. It may be illuminating to think of this binding pocket like a big cooking pot. Once the “ingredients” are added (substrate and cofactor), the process of making the product (dTMP) can begin. The active site binding motifs for both substrate and cofactor are highly conserved among all thymidylate synthase enzymes, regardless of source (104). Whereas early studies on binding were conducted with bacteria-derived synthases, the human enzyme (human thymidylate synthase [hTS]) has now been crystallized and some binding residues identified (105,106).

As shown in Figure 37.31 (hTS sequence numbers are given where known), Asp226 forms essential hydrogen bonds with the dUMP pyrimidine 4-oxo and N₃-H moieties, whereas His196 forms a hydrogen bond with the pyrimidine 4-oxo moiety. The main-chain amide of Asp218 forms a hydrogen bond with the pyrimidine 2-oxo moiety. Four arginine residues (50, 175', 176', and 215) form electrostatic bonds with the anionic deoxyribose-5'-phosphate, and Tyr (H-donor) and histidine (His) (H-acceptor) residues form hydrogen bonds with the deoxyribose 3'-OH. As noted in the dTMP synthesis pathway (Fig. 37.30), Cys195 forms a transient covalent bond with pyrimidine C₆.

The glutamate tail of 5,10-methylene-THF cofactor binds to Phe80, Lys, and His residues of the synthase, whereas Leu221, Ile108, Phe80, and Phe225 interact with the *p*-aminobenzoic acid component of the folate. Asp218, Leu221, and Phe225 are known to interact with the pteridine portion of the cofactor (Fig. 37.31) (105–107).

The binding of both substrate and cofactor promotes a conformational change in the synthase protein and causes the N-terminal portion of the synthase to change its location, which covers the opening of the binding “pot” like a big lid. The conformational change positions the pteridine ring of the folate cofactor “face to face” with the dUMP substrate, permitting the ring stacking that properly orients all key functional groups for the reaction to come (106).

CHEMISTRY The C₆ position of the dUMP substrate is surrounded by electron-withdrawing nitrogen and oxygen atoms, leaving it highly electrophilic (δ^+) and ready to be attacked by the nucleophilic Cys195 of the synthase. The Cys sulfhydryl (SH) group launches an intermolecular nucleophilic attack, forming a new covalent bond between the sulfur and C₆ of the substrate (step 1). The bond that breaks in response to this attack is the 5,6-double bond of dUMP, which attacks the methylene group of the cofactor (step 2). With the release of the N₁₀ nitrogen, the cofactor imidazolidine ring breaks (step 3). Taken together, steps 1 to 3 generate a ternary complex of enzyme, substrate, and cofactor (Fig. 37.30).

A series of reactions involving bond breaking and bond making are shown in Figure 37.30, leading to formation of dTMP, 7,8-dihydrofolate (7,8-DHF), and

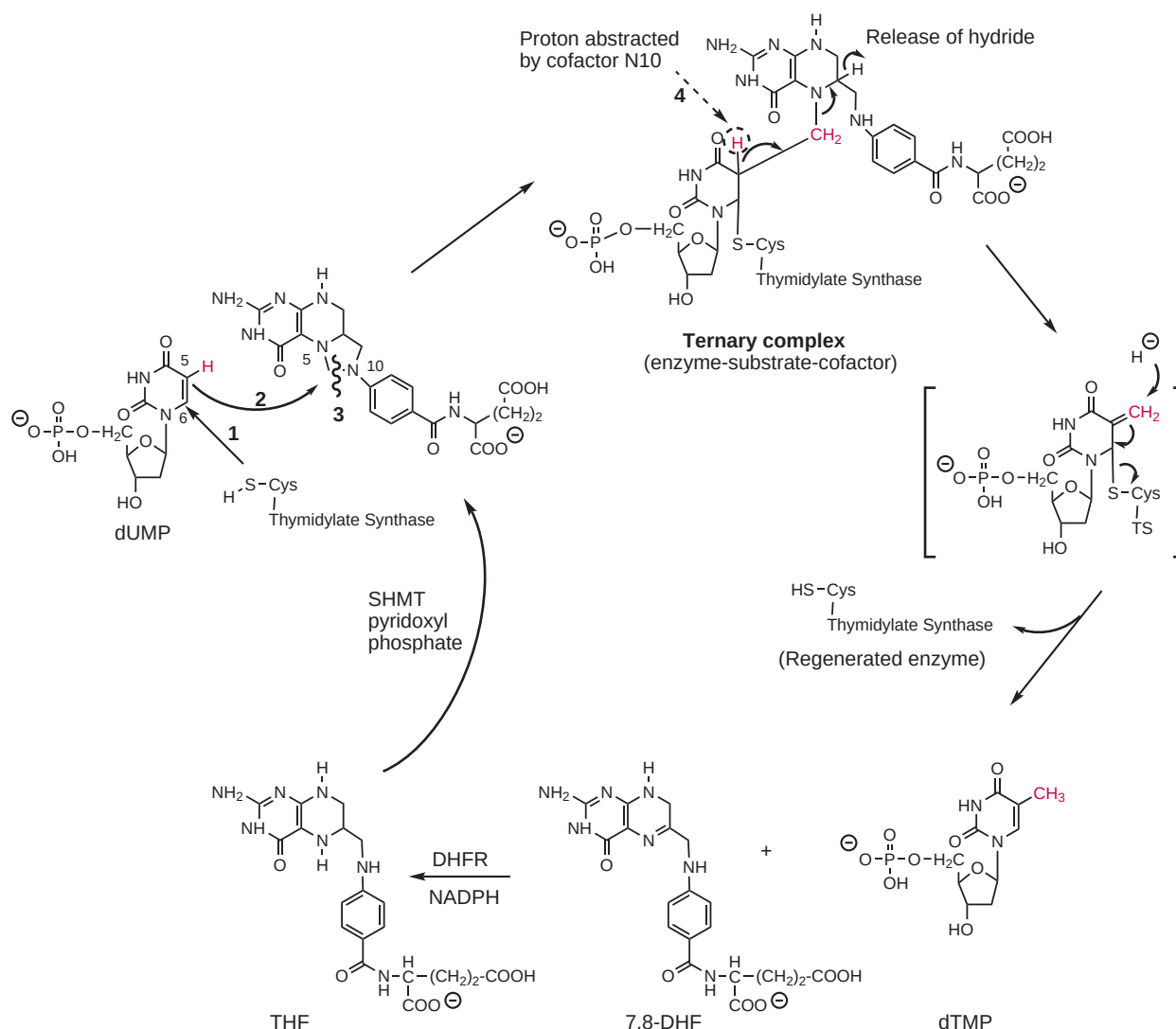


FIGURE 37.30 Synthesis of deoxythymidine monophosphate (dTMP).

regenerated thymidylate synthase. The C₅-H abstraction by N₁₀ of the cofactor (step 4) is essential for synthesis of dTMP. To complete the biochemical cycle, 7,8-DHF must be reduced to tetrahydrofolate (THF) via dihydrofolate reductase (DHFR) using NADPH. Finally, THF is converted to 5,10-methylene-THF through the action of serine hydroxymethyltransferase and vitamin B₆.

With the enzyme and cofactor both regenerated and with plenty of dUMP stored in cellular pools, the cell is ready to synthesize another molecule of dTMP. This happens at a regular pace in healthy cells and at an uncontrolled rate in tumor cells.

SPECIFIC DRUGS: PYRIMIDINE ANALOGS (FIG. 37.32)

Fluorouracil To bind to thymidylate synthase, this fluorinated pyrimidine prodrug must be converted to its deoxyribonucleotide form (Fig. 37.33). The active form of fluorouracil differs from the endogenous substrate only by the presence of the 5-fluoro group, which holds the

key to the cell-killing action of this drug. The C₆ position of the false substrate is significantly more electrophilic than normal due to the strong electron-withdrawing effect of the C₅ fluorine. This greatly increases the rate of attack by Cys195, resulting in a very fast formation of a fluorinated ternary complex (Fig. 37.34). The small size of the fluorine atom assures no steric hindrance to the formation of this false complex.

The next step in the pathway required the abstraction of the C₅-H (as proton) by N₁₀ of the cofactor, but this is no longer possible. Not only is the C₅-fluorine bond stable to cleavage, the fluorine atom and N₁₀ would repel one another because they are both electron rich. The false ternary complex cannot break down, no product is formed, no cofactor is released, and most importantly, the rate-limiting enzyme (thymidylate synthase) is not regenerated. With thymidylate synthase directly and irreversibly inhibited, dTMP can no longer be synthesized and the cell will die.

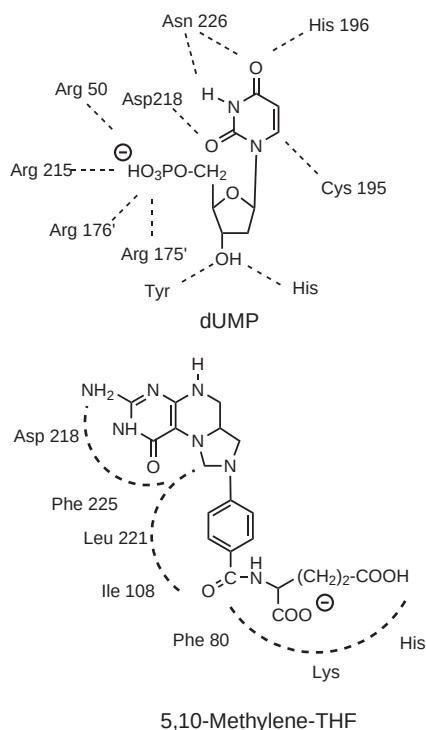


FIGURE 37.31 dUMP and 5,10-methylene-THF binding to thymidylate synthase.

Fluorouracil is administered intravenously in the palliative treatment of colorectal, breast, stomach, and pancreatic cancers. Patients are treated for 4 consecutive days, followed by treatment on odd-numbered days up to a maximum of 12 days. Fluorouracil is rapidly cleared from the bloodstream, and although up to 20% of a dose is excreted unchanged in the urine, most undergoes hepatic catabolism via a series of enzymes that includes the polymorphic dihydropyrimidine dehydrogenase (DPD) (Fig. 37.35). Patients who are genetically deficient in this enzyme (~5% of the population) will experience a more pronounced effect from this drug and are at significant risk for use-limiting or life-threatening toxicity unless doses are appropriately adjusted (108). It has been estimated that between 40% and 60% of patients who experience severe toxicity from fluorouracil are deficient in DPD (109). The incidence of DPD deficiency in African Americans is threefold higher than in Caucasians, and black women have a threefold higher incidence of DPD deficiency than black men (109). In patients with normal DPD activity, dosage adjustments are usually not required in hepatic or renal dysfunction. Major toxicities are related to bone marrow depression, stomatitis/esophagopharyngitis, and potential GI ulceration. Nausea and vomiting are common. Solutions of fluorouracil are light sensitive, but discolored products that have been properly stored and protected from light are still safe to use.

Floxuridine This deoxyribonucleoside prodrug is bioconverted via 2'-deoxyuridine kinase-mediated

phosphorylation to the same active 5-fluoro-dUMP structure generated in the multistep biotransformation of fluorouracil (Fig. 37.33). It is given by intra-arterial infusion for the palliative treatment of GI adenocarcinoma that has metastasized to the liver and that cannot be managed surgically. Because floxuridine does not generate fluorouracil, its kinetic profile is not impacted by DPD pharmacogenetic status.

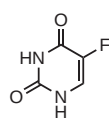
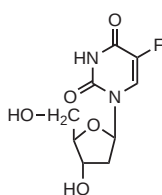
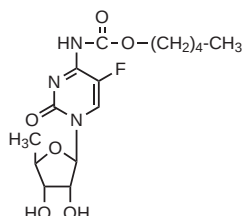
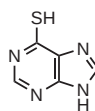
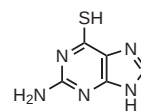
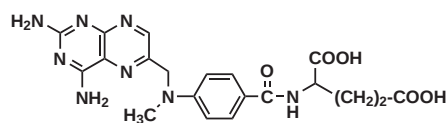
Capecitabine Although capecitabine is a carbamylated analog of cytidine, the drug actually is another 5-fluoro-dUMP prodrug (Fig. 37.36). Given orally, it is extensively metabolized to fluorouracil, which is then converted to the active fluorinated deoxyribonucleotide as previously described. Thymidine phosphorylase, an enzyme involved in this biotransformation, is much more active in tumors than in normal tissue, which improves the tumor-selective generation of fluorouracil. Levels of active drug in the tumor can be up to 3.5-fold higher than in surrounding tissue, leading to a lower incidence of side effects compared to fluorouracil therapy (110). Because capecitabine is biotransformed to fluorouracil, it follows the same catabolic and elimination pathways reported for fluorouracil (Figs. 37.33 and 37.35). Doses should be attenuated in moderate to severe renal impairment, and the caution relative to the augmented risk of toxicity in patients with dihydropyrimidine dehydrogenase deficiency applies.

Capecitabine is indicated for use as first-line therapy in patients with colorectal cancer. It is also used alone or in combination with docetaxel in patients with metastatic breast cancer who have experienced disease progression or recurrence after anthracycline therapy. Given twice daily in tablet form, the total daily dose is calculated based on patient body surface area and is taken 30 minutes after eating to avoid food-induced decreases in absorption. In addition to bone marrow suppression, nausea, and vomiting, the drug can induce severe diarrhea and a potentially disabling disorder termed “hand-and-foot syndrome” (palmar-plantar erythrodysesthesia).

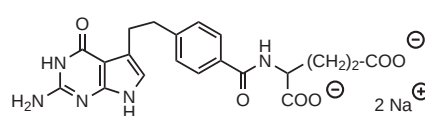
Capecitabine inhibits CYP2C9 and, along with competition for serum protein binding sites, results in clinically significant drug-drug interactions with both warfarin and phenytoin. The interaction with warfarin can result in potentially fatal bleeding episodes, which can appear within days of combination therapy or be delayed up to 1 month after discontinuation of capecitabine therapy.

SPECIFIC DRUGS: ANTIFOLATES (FIG. 37.32)

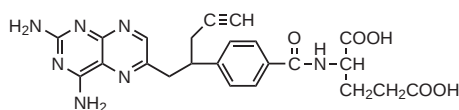
Methotrexate Methotrexate is a folic acid antagonist structurally designed to compete successfully with 7,8-DHF for the DHFR enzyme. The direct inhibition of DHFR causes cellular levels of 7,8-DHF to build up, which in turn results in feedback (indirect) inhibition of thymidylate synthase. Methotrexate is also effective in inhibiting glycine amide ribonucleotide (GAR) transformylase

Pyrimidine antagonists:Fluorouracil
(Adrucil)Floxuridine
(FUDR)Capecitabine
(Xeloda)**Purine antagonists:**Mercaptopurine
(Purinethol)Thioguanine
(Tabloid)**Folate antagonists:**

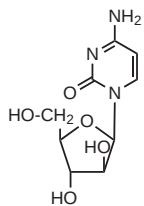
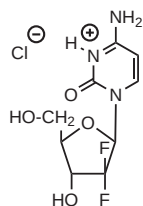
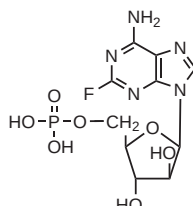
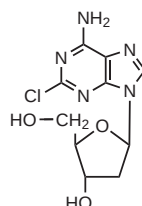
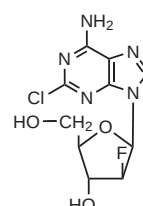
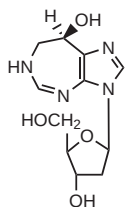
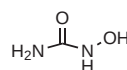
Methotrexate (Trexall)



Pemetrexed disodium (Alimta)



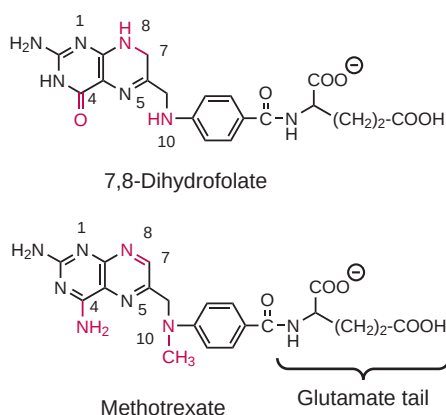
Pralatrexate (Folotylin)

DNA polymerase and chain elongation inhibitors:Cytarabine
(Tarabine PFS,
DepoCyt)Gemcitabine
hydrochloride
(Gemzar)Fludarabine phosphate
(Fludara)Cladribine
(Leustatin)Clofarabine
(Clolar)**Miscellaneous antimetabolites:**Pentostatin
(Nipent)Hydroxyurea
(Hydrea)**FIGURE 37.32** Antimetabolites.

(see Fig. 37.38), a key enzyme in the synthesis of purine nucleotides.

Methotrexate's C_4 - NH_2 substituent, along with its lack of a 7,8-double bond, hold the key to its DHFR-inhibiting action. It has been proposed that the N_5 position of DHF is protonated by Glu30 of DHFR (111) and, in cationic form, binds to DHFR Asp27 through an electrostatic bond

(Fig. 37.37). N_5 is the strongest base in the DHF structure, in part due to attenuating the impact of the C_4 carbonyl on electron density around N_1 . Additional affinity-enhancing interactions between enzyme and substrate have also been identified (112,113), and once bound, the substrate's 5,6-double bond is positioned close to the NADPH cofactor so that the transfer of hydride can proceed.



In contrast, the C₄-NH₂ substituent of methotrexate enriches electron density at N₁ through π -electron donation, increasing its basic character between 10- and 1,000-fold and promoting protonation by Glu30 at the expense of N₅. Because N₁ and N₅ are across the pteridine ring from one another, the interaction of N₁ with the DHFR Asp27 will effectively stand the false substrate

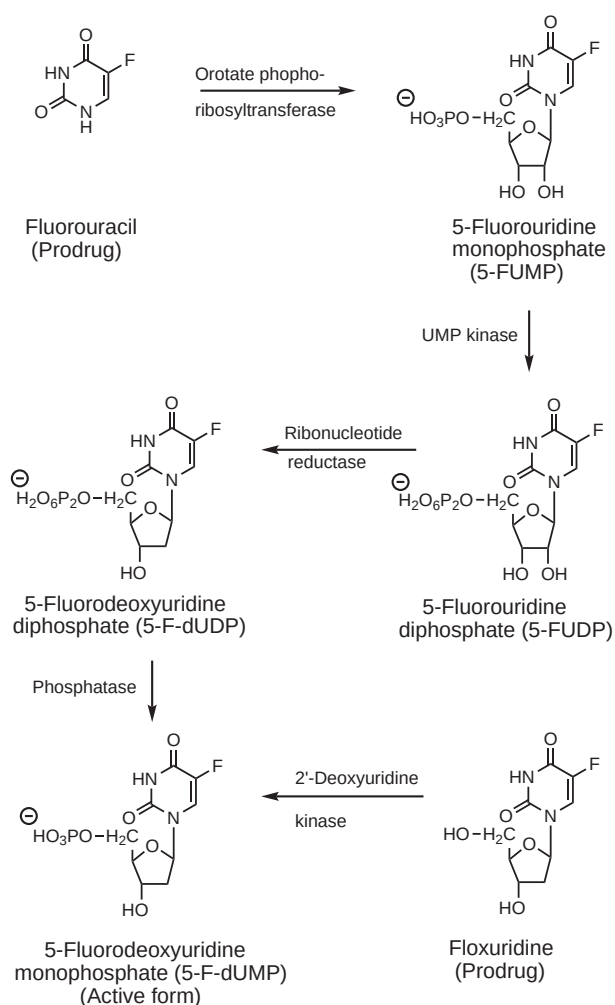


FIGURE 37-33 Activation of fluorouracil and floxuridine.

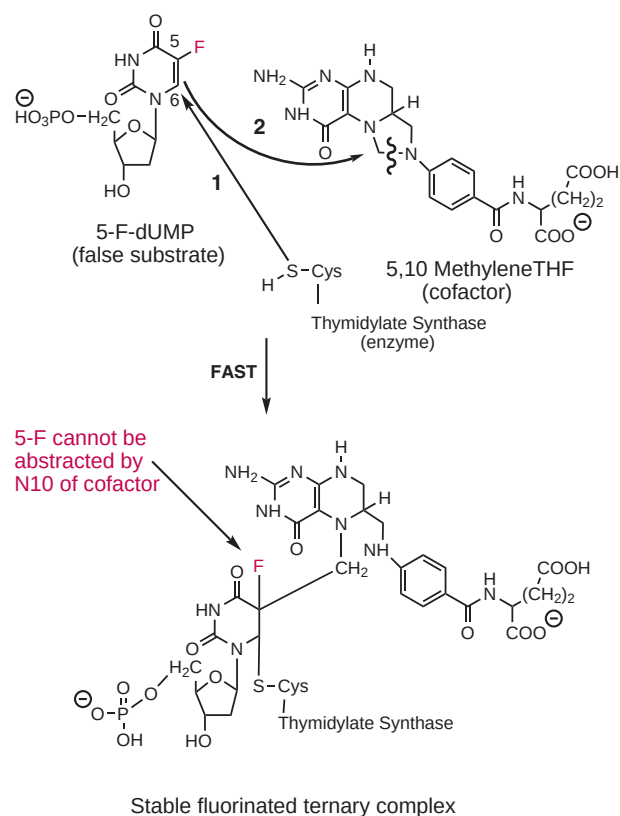


FIGURE 37-34 Mechanism of action of fluorouracil.

“on its head” relative to the orientation of 7,8-DHF (Fig. 37.37) (113,114). With the 5,6-double bond of methotrexate 180 degrees away from the bound NADPH cofactor and stabilized by the fully aromatic pteridine ring, the possibility for reduction is eliminated (113). The DHFR enzyme will be pseudoirreversibly bound to a molecule it cannot reduce, which ties up the DHFR enzyme and prevents the conversion of DHF to THF. In turn, this halts the synthesis of the 5,10-methylene-THF cofactor required for dTMP biosynthesis and causes feedback inhibition of the thymidylate synthase enzyme. The cell will die a “thymineless death.”

Methotrexate is given orally in the treatment of breast, head and neck, and various lung cancers as well as in NHL. The sodium salt form is also marketed for intravenous, intramuscular, intra-arterial, or intrathecal injection. Oral absorption is dose-dependent and peaks at 80 mg/m² due to site saturation. The monoglutamate tail of methotrexate permits active transport into cells via a reduced folate carrier (RFC1), which predominates at serum concentration levels lower than 100 μ mol/L. Methotrexate undergoes intracellular folyl polyglutamate synthase (FPGS)-catalyzed polyglutamation, which adds several anionic carboxylate groups to the molecule and traps the drug at the site of action. Polyglutamation is more efficient in tumor cells than in healthy cells, which promotes the selective toxicity of this drug. The polyglutamated drug will be hydrolyzed back

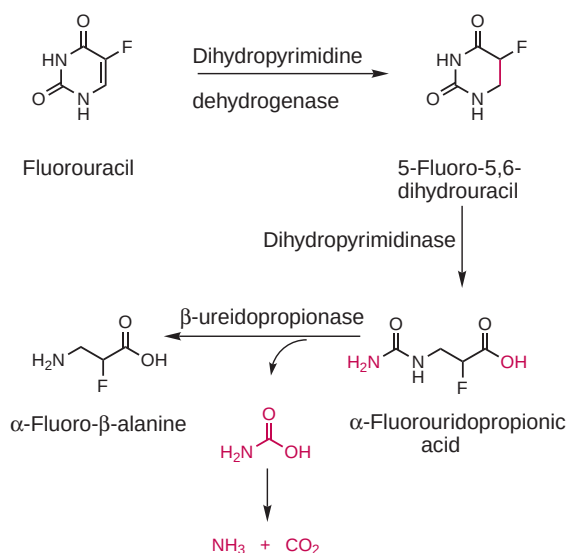


FIGURE 37.35 Fluorouracil metabolism.

to the parent structure before renal elimination. Up to 90% of an administered dose of methotrexate is excreted unchanged in the urine within 24 hours.

Methotrexate toxicity occurs with high doses if “third space” fluids allow drug to accumulate in ascites

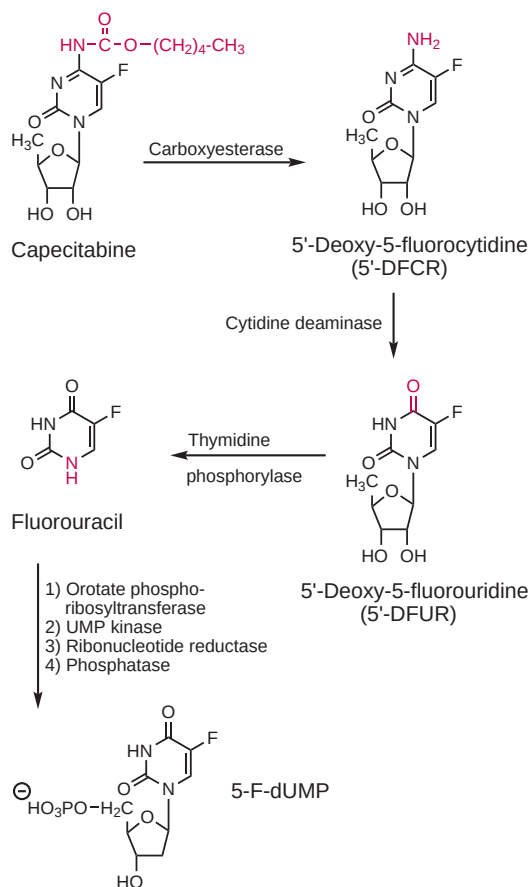


FIGURE 37.36 Capecitabine activation.

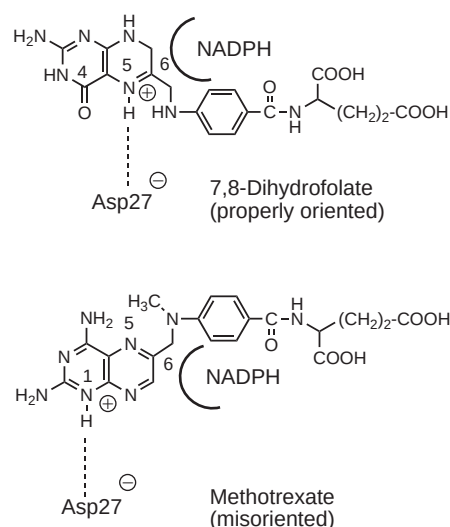
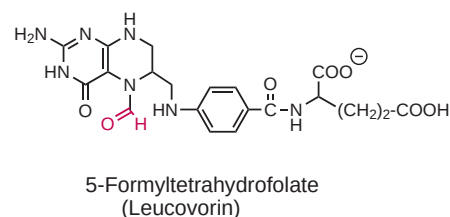


FIGURE 37.37 Misorientation of methotrexate at DHFR.

and pleural effusions and/or when renal excretion is impaired by kidney disease. When used in high doses, methotrexate and its 7-hydroxymetabolite (which has a three- to fivefold lower water solubility) can precipitate in the renal tubule, causing damaging crystalluria. Methotrexate-induced lung disease is a particularly critical problem because it arises at any time and at any dose and can be fatal. Methotrexate use also precipitates severe GI side effects, including ulcerative stomatitis and hemorrhagic enteritis, leading to intestinal perforation. Potentially fatal skin reactions are a risk as well. As a Category X teratogen, this drug should not be given to women who are pregnant or planning to become pregnant.

If severe methotrexate toxicity occurs, reduced folate replacement therapy with 5-formyltetrahydrofolate (leucovorin) must be initiated as soon as possible. Leucovorin generates the folate cofactors needed by DHFR and GAR transformylase to ensure the continued synthesis of pyrimidine and purine nucleotides in healthy cells. “Leucovorin rescue” therapy is often given as prophylaxis after high-dose methotrexate therapy.



Cancer cells can become resistant to methotrexate over time. Acquired resistance mechanisms include increased DHFR expression, impaired transport, active cellular efflux, and/or attenuated intracellular polyglutamation.

Pemetrexed Disodium Pemetrexed is a novel multitarget antifolate administered by the intravenous route for the treatment of advanced or metastatic nonsquamous non-small cell lung cancer (115) and in combination with cisplatin in malignant pleural mesothelioma. Like methotrexate, it is actively transported into tumor cells via RFC1. Pemetrexed has a higher affinity for FPGS than methotrexate, so more polyglutamated drug is generated (and trapped) inside the cancer cell. Tetra- and pentaglutamates predominate. Both mono- and polyglutamated forms of pemetrexed are capable of inhibiting DHFR, and they do so with comparable activity (116,117). Unlike methotrexate, pemetrexed does not distribute centrally to any significant extent (117).

In addition to DHFR, polyglutamated pemetrexed (but not the monoglutamated parent) binds tightly to thymidylate synthase and GAR transformylase (116,118,119). Since intracellular polyglutamation of pemetrexed is so efficient, this drug realizes a significant portion of its potent anticancer activity through the inhibition of these two enzymes. In fact, thymidylate synthase is considered its primary target. Fortunately, the affinities of the polyglutamated forms of pemetrexed for this enzyme are higher than that of the monoglutamate parent form, so efficacy is not sacrificed for enhanced localization at the site of action. Pemetrexed is often used in combination with the organometallic cisplatin (116). A synergistic effect with gemcitabine (a DNA polymerase inhibitor) in the treatment of lung cancer patients has also been noted as long as gemcitabine is administered immediately prior to the antifolate (117).

Patients on pemetrexed must take folic acid (commonly 400 μg daily) and vitamin B₁₂ (1 g on an established schedule) to reduce the risk of bone marrow suppression (neutropenia, thrombocytopenia, and anemia) and use-limiting GI side effects like mucositis and stomatitis. Pretreatment with corticosteroids can reduce the risk of drug-induced skin rash. Antiemetic therapy to proactively combat drug-induced nausea and vomiting is also warranted (117). Pemetrexed has a half-life of 3.5 hours and is excreted primarily unchanged via the kidneys.

Resistance to pemetrexed is mediated through a decrease in FPGS activity, enhanced hydrolysis of pemetrexed polyglutamates via γ -glutamyl hydrolase enzymes, and upregulation of the thymidylate synthase target. Significant cross-resistance has been noted between pemetrexed and other pyrimidine and folate antagonists (116,120). A recent study has shown that pharmacogenetic differences in ATP-binding cassette (ABC) C11 transporters may be an important predictor of sensitivity to pemetrexed chemotherapy (121). In the ABCC11 SNP 538G>A, the A/A variant (unlike the G/G or G/A) does not promote pemetrexed efflux from the cancer cell. ABCC11 genotype also impacts the nature of earwax, and it has been noted that East Asians (80% to 95% of whom express the A/A variant and have dry earwax) are particularly responsive to pemetrexed–cisplatin chemotherapy. Patients expressing the other ABCC11 variants would be at risk for pemetrexed resistance due to excessive drug efflux.

Pralatrexate The only structural difference between pralatrexate and methotrexate is in the area of N₁₀; pralatrexate is a 10-deaza analog where the carbon atom replacing N₁₀ has been substituted with a triple bond-containing propargyl group. This structural alteration results in a significantly enhanced tumor cell uptake via RFC1 and FPGS-mediated polyglutamation compared to methotrexate and pemetrexed without compromising affinity for target enzymes (122,123). The rate of active transport of pralatrexate into tumor cells by RFC1 has been measured at approximately 12 to 14 times that of methotrexate, and a 10-fold higher FPGS polyglutamation efficacy has been estimated (124,125). This translates to more active drug inside the tumor cell for a longer period of time. Pralatrexate is excreted intact in the urine.

Unlike pemetrexed, which is also more extensively polyglutamated than methotrexate, pralatrexate's primary antineoplastic enzymatic target is DHFR. Although it interacts with and inhibits the isolated DHFR enzyme 2- to 3-fold less vigorously than methotrexate (K_i apparent = 26 and 45 nmol/L, respectively), the cellular influx and polyglutamation first-order rate constants (V_{max}/K_m) are 12- and 10-fold greater, respectively (126). Higher concentrations of active polyglutamated drug trapped within the cell provide a more potent DHFR inhibitory effect compared to methotrexate and a greater antitumor response than either methotrexate or pemetrexed (123,126).

Pralatrexate dosing is based on body surface area in order to standardize exposure to active drug. Some T-cell lymphoma patients treated with pralatrexate have achieved complete remission (124). In addition to T-cell lymphoma (127), pralatrexate might eventually find use in the treatment of non-small cell lung cancer and, in combination with the DNA polymerase inhibitor gemcitabine, in NHL. When used in combination with gemcitabine, the synergistic apoptotic effect is maximized when the antifolate is administered before the polymerase inhibitor (126). Pralatrexate is not effective against B-cell lymphoma (125). As with other antifolates, patients on pralatrexate should receive daily folic acid supplementation and vitamin B₁₂ to reduce the risk of mucositis/stomatitis and bone marrow suppression. Whereas the latter adverse effect is viewed as a minor risk (126), mucositis has been termed the major dose-limiting adverse reaction, particularly in patients with high levels of methylmalonic acid (128). Unlike methotrexate, acquired resistance to pralatrexate is believed to be more highly dependent on a reduced expression of RFC1 than on an increased expression of DHFR (129).

Purine Antagonists: Amidophosphoribosyl Transferase Inhibitors

AMP AND GMP BIOSYNTHESIS Purine antagonists inhibit the de novo biosynthesis of AMP and GMP. The rate-limiting enzyme in the synthesis of these purine nucleotides is amidophosphoribosyl transferase (also known as phosphoribosylpyrophosphate amido transferase), which is a major target for one of the two thiol-containing purine anticancer antimetabolites on the U.S. market (mercaptopurine).

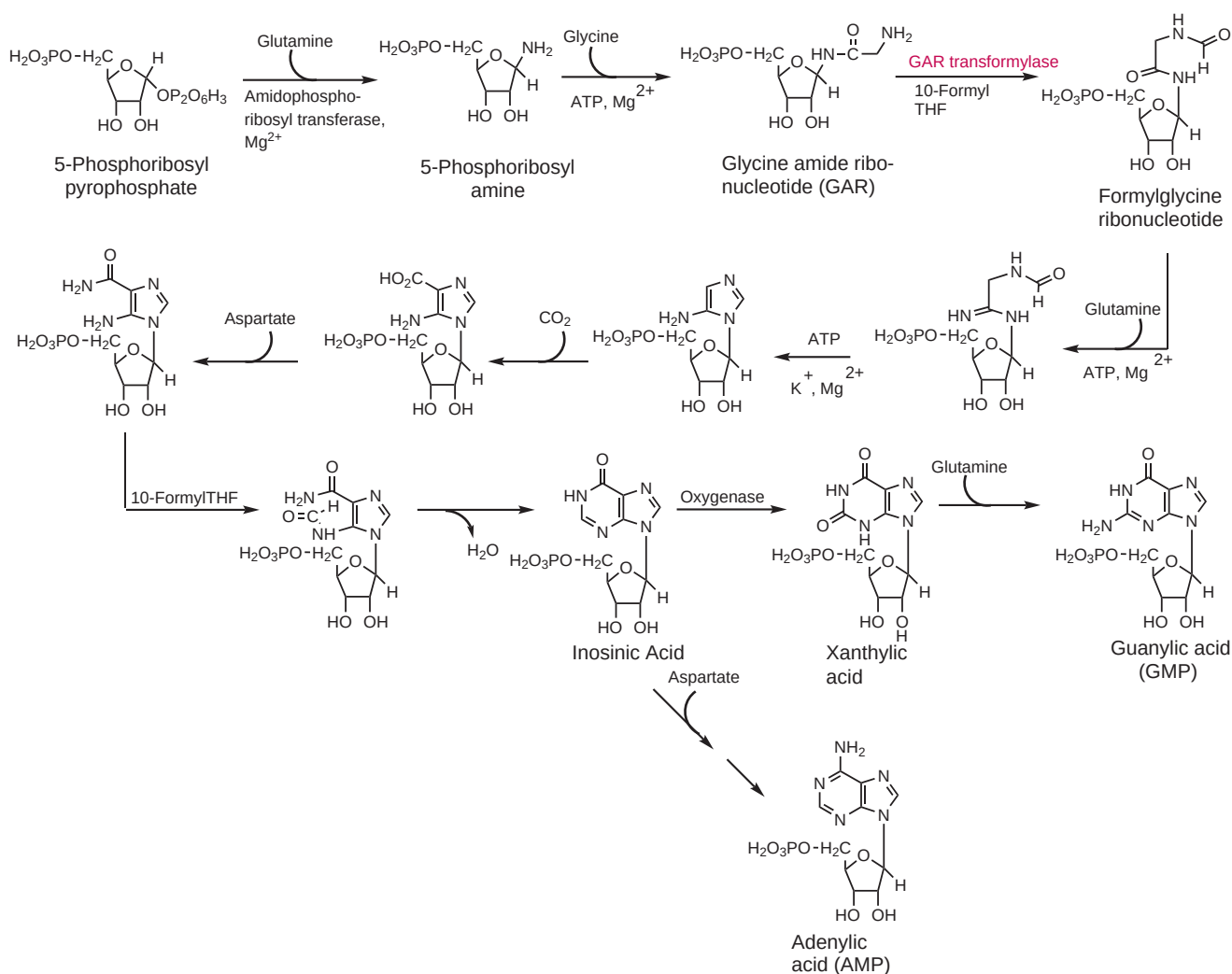


FIGURE 37.38 Biosynthesis of purine nucleotides.

The pathway outlining the synthesis of AMP and GMP is provided in Figure 37.38. It is important to recognize that the rate-limiting step is the first of the pathway; if that step is inhibited, no other step can proceed. Since the rate-limiting transferase enzyme works on a phosphorylated ribose substrate, no enzyme in the sequence will function without its presence. The formylation reaction catalyzed by GAR transformylase requires the methyl-donating 10-formyltetrahydrofolate and can be inhibited by the antifolates methotrexate and pemetrexed.

CHEMISTRY The two currently marketed purine anticancer agents are both 6-thio analogs of the endogenous purine bases guanine and purine, also known as inosine (Fig. 37.32). They are prodrugs and must be converted to ribonucleotides by hypoxanthine guanine phosphoribosyl transferase (HGPRT) before they can exert their cytotoxic actions (Fig. 37.39). Mercaptopurine, acting through a methylated ribonucleotide metabolite, inhibits the target amidophosphoribosyl transferase enzyme, leading to the true antimetabolic effect of lowered AMP

and GMP biosynthesis. A second mechanism of antineoplastic activity for mercaptopurine (and the predominant mechanism for thioguanine) involves the incorporation of di- and triphosphate deoxy- and ribonucleotides generated within the tumor cell into DNA and RNA, respectively (130). This illicit substitution inhibits further elongation of the strands and promotes apoptosis.

Thiopurines are metabolized by *S*-methylation via the polymorphic enzyme thiopurine methyl transferase (TPMT) with *S*-adenosylmethionine serving as cofactor. The methylated thiopurine bases cannot react with HGPRT and, therefore, cannot form the active false ribonucleotides. Drug manufacturers take this into account when establishing dosing regimens. The active false ribonucleotide 6-thioinosinic acid is also subject to extensive TPMT-catalyzed methylation. The *S*-methyl-6-thioinosinic acid metabolite is a potent inhibitor of the amidophosphoribosyl transferase enzyme and contributes to the cytotoxic action of the parent drug (Fig. 37.39). In contrast, little or no 6-methylthioguanic acid is produced inside the cell (130,131).

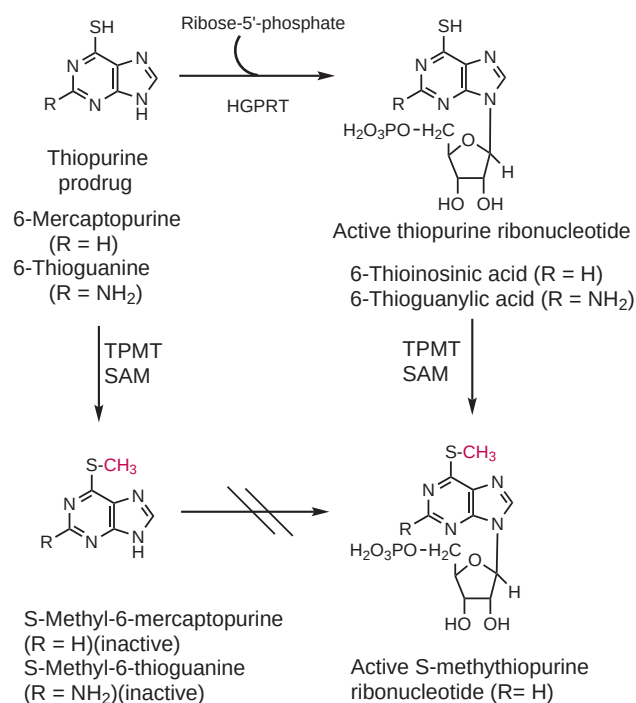


FIGURE 37.39 Thiopurine metabolism leading to activation and inactivation.

TPMT is polymorphic in humans, and some individuals do not express this protein to any significant extent (132). Patients who are poor TPMT metabolizers (e.g., 10% of Caucasians, but also evident in other races) will not experience the activity-attenuating metabolic effect and will generate more active ribonucleotide per dose than patients with normal or excessive levels of the enzyme. The TPMT genotype of patients should be assessed before initiating thiopurine therapy because poor metabolizers are at a high risk of life-threatening myelosuppression from elevated levels of false ribonucleotides, even when standard doses are administered (131). In addition, the accumulation of mutagenic thiopurine-based ribonucleotides puts these patients at higher risk for secondary malignancies (130). Thiopurines can still be used in poor TPMT metabolizers, but the dose should be decreased significantly (e.g., 10- to 15-fold) and white blood cell counts monitored vigilantly. Mercaptopurine appears to be more significantly impacted by TPMT genotype than thioguanine (133).

Genes that encode for inositol triphosphate pyrophosphatase (ITPA) are also known to impact the metabolic and toxicity profiles of mercaptopurines. Carriers of the rs41329251 ITPA allele appear to be at significantly higher risk for the development of mercaptopurine-induced febrile neutropenia even after TPMT genotype-based dosage adjustment (132,134).

Extensive TPMT metabolizers, who represent up to 90% of patients on thiopurine therapy, will form lower amounts of apoptotic 6-thiolated ribonucleotides. In the case of mercaptopurine, the molecules of ribonucleotide generated will be methylated very rapidly to the

antimetabolic 6-methylthioinosinic acid, thus enhancing sensitivity to the drug (131). In contrast, extensive TPMT metabolizers show a decreased sensitivity to thioguanine because there is no compensatory increase in the formation of methylated ribonucleotide to offset the decreased production of 6-thioguanilic acid (130).

Xanthine oxidase competes with TPMT for mercaptopurine (but not for thioguanine) and converts it to inactive 6-thiouric acid, which is excreted in the urine (Fig. 37.40) (135). 6-Thioinosinic acid is also subject to metabolism via the xanthine oxidase pathway, ultimately forming the same inactive metabolite. Allopurinol, which inhibits xanthine oxidase and increases levels of active 6-thioinosinic acid, can be coadministered with mercaptopurine to increase its duration of action and antineoplastic potency. The dose of mercaptopurine can be cut approximately in half when coadministered with allopurinol. Coadministration of allopurinol with thioguanine is not warranted, since the impact of xanthine oxidase on its metabolic degradation is minor.

SPECIFIC DRUGS (Fig. 37.32)

Mercaptopurine Mercaptopurine is used in the treatment of acute lymphatic and myelogenous leukemia. It is available in an oral dosage form, but absorption can be erratic and is reduced by the presence of food. The drug is extensively metabolized on first pass and excreted by the kidneys. Bone marrow suppression is the major use-limiting toxicity, although the drug can be hepatotoxic in high doses. Dosage adjustments should be considered in the face of renal or hepatic impairment. Since the major mechanism of action of mercaptopurine is inhibition of de novo purine nucleotide biosynthesis rather than apoptosis secondary to the incorporation of false nucleotides into DNA, there is a lower risk for mutagenesis and secondary malignancy compared to thioguanine (130).

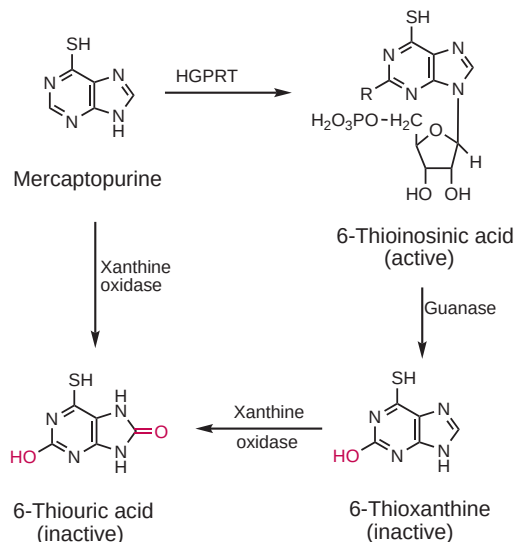


FIGURE 37.40 Xanthine oxidase inactivation of mercaptopurine and 6-thioinosinic acid.

Resistance to mercaptopurine (and thioguanine) therapy may have as much to do with attenuated active uptake via nucleoside transporters as with the more commonly cited deficiency in the activating HGPRT enzyme (136).

Thioguanine Thioguanine is administered orally in the treatment of nonlymphocytic leukemias. Like mercaptopurine, absorption is incomplete and variable, and the toxicity profiles are similar except where previously noted.

DNA Polymerase/DNA Chain Elongation Inhibitors

MECHANISM OF ACTION Five halogenated and/or ribose-modified DNA nucleoside analogs are marketed for the treatment of a wide variety of hematologic cancers and solid tumors (Fig. 37.32). These agents have complex and multifaceted mechanisms. All include inhibition of DNA polymerase and/or DNA chain elongation among their actions and all nucleosides must be converted to triphosphate nucleotides before activity is realized.

CHEMISTRY As nucleosides, the DNA polymerase inhibitors are actively taken up into cells via a selective nucleoside transporter protein, so tumors deficient in this transporter system will be resistant to these anticancer agents. Once inside the cell, specific kinases conduct the essential phosphorylation reactions. In active triphosphate form, they can be mistakenly incorporated into the growing DNA chain, thus arresting further elongation, and/or inhibit enzymes essential for DNA synthesis. All drugs in this group are administered intravenously, are excreted via the kidneys, and induce myelosuppression as their major use-limiting side effect. Resistance can involve aberrations in the expression of metabolizing enzymes as well as of transporting and efflux proteins. Some in vitro evidence points to the loss of functional nucleoside transporter proteins (specifically hENT1/SLC29A1) and deoxycytidine kinase enzymes as the primary causes of acquired resistance to DNA polymerase inhibitors (137).

As alluded to earlier, there is significant commonality in the enzymes involved in the activation and inactivation of these five antineoplastics, and the search is on for biomarkers that can reliably predict the degree of sensitivity to their therapeutic and potentially fatal toxic effects. Several polymorphic genes are under active investigation as candidates to assist practitioners in successfully individualizing antimetabolite chemotherapy (138).

SPECIFIC DRUGS (Fig. 37.32)

Cytarabine and Gemcitabine Both of these cytidine-based anticancer agents undergo initial phosphorylation by deoxycytidine kinase to the monophosphate with subsequent phosphorylations catalyzed by pyrimidine monophosphate and diphosphate kinases. Cytarabine, an arabinoside, is catabolized by cytidine and deoxycytidylate (deoxycytidine monophosphate) deaminases to inactive uracil analogs (Fig. 37.41). The significantly longer half-life of gemcitabine (19 hours) compared to conventional cytarabine (3.6 hours) is due to the inhibitory action of the difluorodeoxycytidine triphosphate

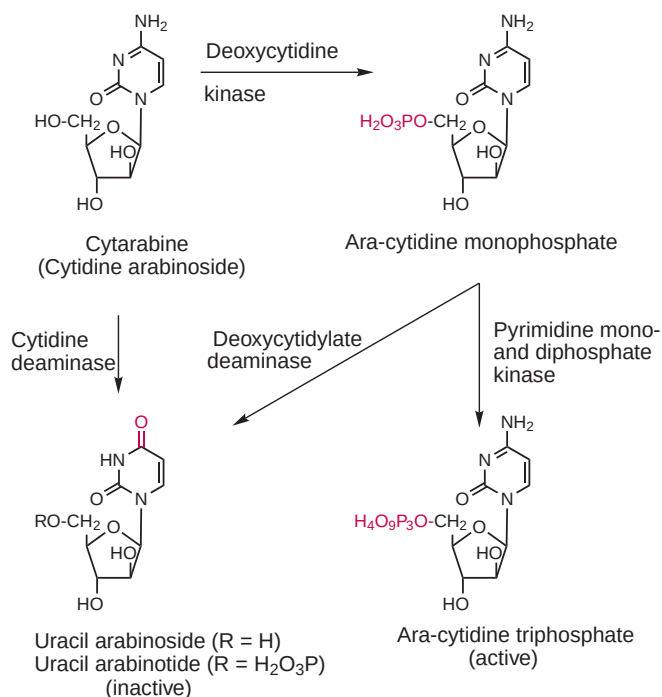


FIGURE 37.41 Cytarabine metabolism.

metabolite on the potentially degradative deoxycytidine monophosphate deaminase enzyme (110). Gemcitabine elimination is gender-dependent, with women having the greater risk for toxicity due to lower renal clearance. Gemcitabine is indicated in the treatment of breast, pancreatic, and non-small cell lung cancers. Cytarabine, which can be administered subcutaneously and intrathecally in addition to intravenously, is used in the treatment of various leukemias. A liposomal formulation of cytarabine is available for the treatment of lymphomatous meningitis.

Fludarabine, Cladribine, and Clofarabine Like their pyrimidine counterparts, these 3-halogenated adenosine-based nucleosides undergo conversion to the active triphosphate nucleotides after active transport into tumor cells. All are initially phosphorylated by deoxycytidine kinase, and cells with high levels of this enzyme should respond well to these agents. The C₂-halogen renders the molecules relatively resistant to the degradative action of adenosine deaminase, and a significant fraction of the dose is eliminated unchanged via the kidneys. Fludarabine, an arabinoside, is marketed as the monophosphate nucleotide to enhance water solubility for intravenous administration, but this group is cleaved rapidly in the bloodstream, allowing the free nucleoside to take advantage of the nucleoside-specific transporting proteins.

Cladribine is indicated in the treatment of hairy cell leukemia, whereas fludarabine phosphate is used in chronic lymphocytic leukemia. In addition to myelosuppression, fludarabine phosphate can induce hemolytic anemia, and severe CNS toxicity has been noted with high doses.

Clofarabine is used in acute lymphoblastic leukemia patients who are 21 years or less and who have failed with

at least two previous regimens. In addition to inhibiting DNA chain elongation, this drug also inhibits ribonucleotide reductase and facilitates the release of proapoptotic proteins from mitochondria. The rapid attenuation of leukemia cells after administration of this agent can result in a condition known as tumor lysis syndrome, and respiratory and cardiac toxicities can occur secondary to cytokine release. Toxicity can progress to potentially fatal capillary leak syndrome and organ failure, and patients should receive intravenous fluids for the entire 5-day course of therapy to minimize risk of serious adverse events.

DNA Methyltransferase Inhibitors

MECHANISM OF ACTION In contrast to the DNA alkylating agents discussed earlier in this chapter, three nucleic acid–based chemotherapeutic agents, azacitidine, decitabine, and nelarabine (Fig. 37.42), block abnormal cellular proliferation by inhibiting DNA alkylation (specifically methylation) on genes responsible for differentiation and growth. The hypomethylation effect, mediated through the inhibition of DNA methyltransferase, can sometimes restore normal gene function while selectively killing cells that have stopped responding to the body's cellular proliferation control processes.

All of the marketed DNA methyltransferase inhibitors are nucleoside analogs that, once converted to triphosphate nucleotides, are mistakenly incorporated into DNA in lieu of their cytidine or guanine nucleotide counterparts. Interaction of the false nucleotide with methyltransferase results in an irreversible inhibition of the enzyme.

CHEMISTRY All of these nucleoside analogs hydrolyze in aqueous solution and must be administered soon after the dose is constituted. Their vulnerability to deaminase enzymes explains their short elimination half-lives of less than or equal to 4 hours. Nelarabine will also be *O*-demethylated prior to DNA incorporation and can be further metabolized to guanine, xanthine, and uric acid. Anemia, neutropenia, and thrombocytopenia are among the most common side effects of this class of drugs.

SPECIFIC DRUGS (FIG. 37.42)

Nelarabine Nelarabine is considered third-line treatment for T-cell acute lymphoblastic leukemia or lymphoma. The drug can induce severe and potentially irreversible neurologic symptoms including convulsions,

severe central depression, and peripheral neuropathy that can mimic Guillain-Barré syndrome. These adverse effects are considered dose-limiting. As noted earlier, nelarabine can be metabolized to uric acid. Along with hydration and urine basification, the xanthine oxidase inhibitor allopurinol can be given prophylactically to minimize risk of nelarabine-induced hyperuricemia.

Azacitidine and Decitabine Azacitidine and decitabine are given intravenously (and, in the case of azacitidine, subcutaneously) for the treatment of myelodysplastic syndrome. Patients should be monitored for hematologic and renal toxicities while undergoing therapy with either agent, although renal toxicity is a more serious concern with azacitidine use. Both drugs are known to cause fetal harm, and patients should be actively counseled to take appropriate reproductive precautions.

MISCELLANEOUS ANTIMETABOLITES (FIG. 37.32)

Pentostatin Pentostatin is a ring-expanded purine ribonucleoside that inhibits adenosine deaminase and is primarily used in the treatment of hairy cell leukemia. The elevated levels of deoxyadenosine triphosphate that result from inhibition of this degradative enzyme inhibit the action of ribonucleotide reductase (the enzyme that converts ribose diphosphate to deoxyribose diphosphate), thus halting DNA synthesis within the tumor cell. When used in chronic lymphocytic leukemia, some authors claim pentostatin offers a therapeutic efficacy comparable to fludarabine, but with a lower risk of toxicity (139).

Hydroxyurea Hydroxyurea, a drug with a 100-plus year history, blocks the synthesis of DNA by trapping a tyrosyl free radical species at the catalytic site of ribonucleotide reductase, thereby inhibiting the enzyme that converts ribonucleotide diphosphates into their corresponding deoxyribonucleotides. It is used orally for the treatment of melanoma, metastatic or inoperable ovarian cancer, resistant chronic myelocytic leukemia, and as an adjunct to radiation in the treatment of squamous cell carcinoma and cancer of the head and neck. Hydroxyurea increases the effectiveness of radiation therapy through its selective toxicity to cells in the radiation-resistant S phase and by stalling the cell cycle in the G₁ stage, in which radiation therapy does the greatest damage. In addition, hydroxyurea thwarts the normal damage-repair mechanisms of surviving cells. A review of its chemical, pharmacologic, metabolic, and therapeutic properties has recently been published (140).

Hydroxyurea has excellent oral bioavailability (80% to 100%), and serum levels peak within 2 hours of consuming the capsules. If a positive response is noted within 6 weeks, toxicities are mostly mild enough to permit long-term or indefinite therapy on either a daily or every-3-day basis. Leukopenia and, less commonly, thrombocytopenia and/or anemia are the most serious adverse effects. Excretion of the unchanged drug and the urea metabolite is via the kidneys. The carbon dioxide produced as a by-product of hydroxyurea metabolism is excreted in the expired air.

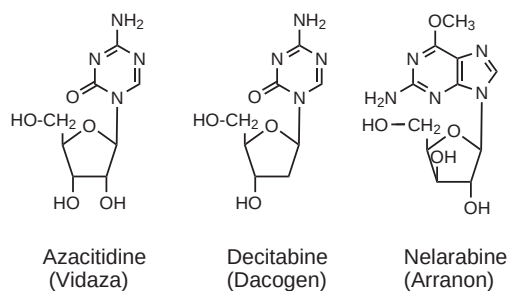


FIGURE 37.42 DNA methyltransferase inhibitors.

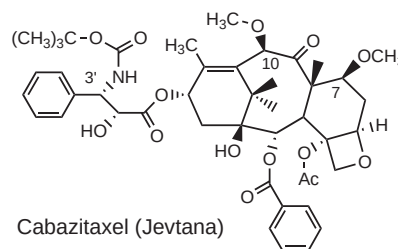
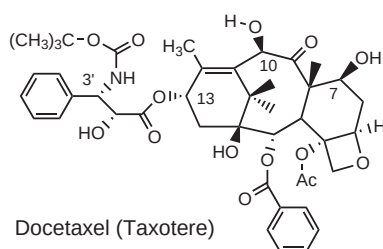
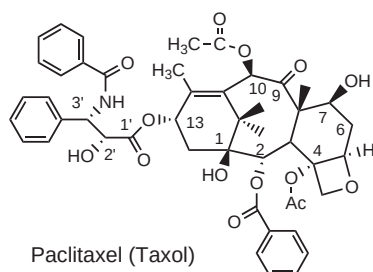
Mitosis Inhibitors

The mitotic process depends on the structural and functional viability of microtubules (polymeric heterodimers consisting of isotypes of α - and β -tubulin proteins). These distinct but nearly identical 50-kD proteins lie adjacent to one another within the tubule and roll up to form an open, pipe-like cylinder akin to a hollow peppermint candy stick. A γ -tubulin protein is found at the organizational center of the microtubule. The tubulin isotypes found in the microtubule are conserved throughout specific tissues within a given species and will impact the cell's sensitivity to mitosis inhibitors.

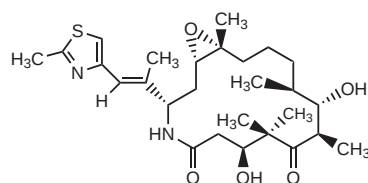
During cell division, tubulin undergoes intense, sporadic, and alternating periods of structural growth and erosion known as "dynamic instability." The proteins alternatively polymerize and depolymerize through guanosine triphosphate- and Ca^{2+} -dependent processes.

Polymerization involves the addition of tubulin dimers to either end of the tubule, although the faster-growing (+) end is more commonly involved. Polymerization results in tubular elongation, whereas depolymerization results in the shortening of the structure. The frenetic alteration in structure, facilitated by microtubule-associated proteins (MAPs), ultimately allows the formation of the mitotic spindle and the attachment to chromosomes that are prerequisites to cell division. Inhibiting the essential hyperdynamic changes in microtubular structure results in mitotic arrest and apoptosis. Two general chemical classes of mitosis inhibitors have historically been marketed for the treatment of cancer: taxanes, and vinca alkaloids (141). They have been joined by the single agents estramustine, an estrogen-based nitrogen mustard-like carbamate originally thought to act via DNA alkylation, and the epothilone ixabepilone (Fig. 37.43).

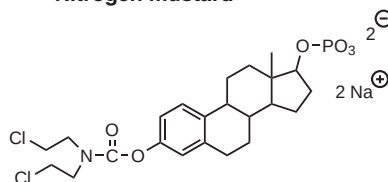
Taxanes



Epothilone



Nitrogen mustard



Vinca alkaloids:

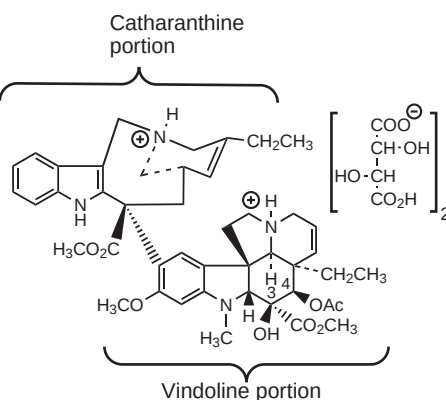
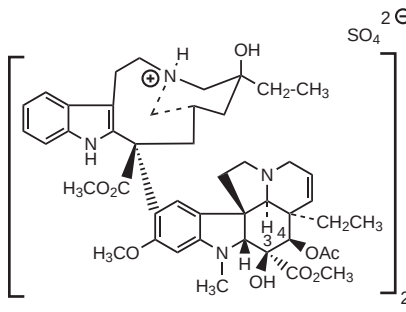
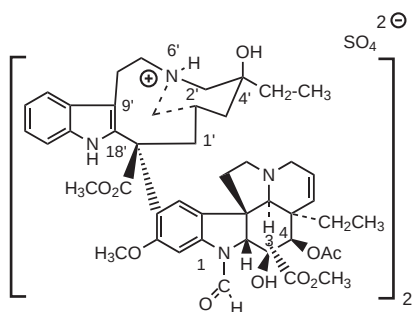


FIGURE 37.43 Mitosis inhibitors.

Taxanes

MECHANISM OF ACTION Anticancer taxanes initially were isolated from the bark of the Pacific yew (*Taxus brevifolia*) but are now produced semisynthetically from an inactive natural precursor (10-deacetylbaicatin III) found in the leaves of the European yew (*Taxus baccata*), a renewable resource. Taxanes bind to polymerized (elongated) β -tubulin at a specific hydrophobic receptor site comprised of the 31 N-terminal residues located deep within the tubular lumen (142). At standard therapeutic doses (which should lead to intracellular concentrations of 1 to 20 μ M), taxane-tubulin binding promotes a stable tubulin conformation similar to that of the guanosine triphosphate-bound protein, which renders the microtubules resistant to depolymerization and prone to polymerization (143). This promotes the elongation phase of microtubule dynamic instability at the expense of the shortening phase and inhibits the disassembly of the tubule into the mitotic spindle. In turn, this interrupts the normal process of cell division. At these concentrations, extensive polymerization causes the formation of large and dense aberrant structures known as asters, which contain stabilized microtubule bundles.

Taxanes are substrates for P-gp, and cellular efflux via this carrier protein is a major mechanism of taxane resistance. It has also been demonstrated that patients who express a mutant variety of the oncogene *p53* that causes the overexpression of MAP4, and subsequently promotes microtubular polymerization/elongation, show enhanced sensitivity to taxane chemotherapy and a resistance to vinca alkaloids (144).

CHEMISTRY Chemically, diterpenoid taxanes consist of a 15-membered tricyclic taxane ring system (tricyclo[9.3.1.0]pentadecane) fused to an oxetane (D) ring and contain an esterified β -phenylisoserine side chain at C₁₃. As shown in Figure 37.43, the three marketed

taxane antineoplastics differ in substitution pattern at C₁₃ (benzamido or *t*-butoxycarboxamido), C₁₀ (secondary alcohol, acetate ester, or methoxy ether), and/or C₇ (secondary alcohol or methoxy ether). The taxane ring system is often conceptualized as having “northern” and “southern” halves. The “southern” segment is critical to receptor binding, whereas the “northern” section ensures the proper conformation of essential functional groups, including the C₁₃-isoserine side chain [with its C₁-carbonyl, free C₂-(*R*)-OH and C₃-(*S*)-benzamido or *t*-butoxycarboxamido groups], the benzoyl and acetyl esters at C₂ and C₄, respectively, and the intact oxetane ring (145–147).

The key taxane–tubulin binding interactions are identified in Table 37.8 using paclitaxel as ligand (133,148,149). Paclitaxel interacts at the β -tubulin binding site in a folded (“T” or “butterfly”) conformation that places C₂-benzoyl and the C₃-benzamido groups in close proximity (145,149). Their independent intermolecular engagement with a critical β -tubulin His residue perfectly positioned between them keeps them from interacting with one another. The oxetane ring of taxanes, although capable of enhancing receptor affinity through hydrogen bonding (146,148), is believed to serve a more critical role in properly orienting the C₄-acetyl moiety for interaction within its hydrophobic binding pocket (146). The C₁-OH also promotes conformational stability through intramolecular interaction with the carbonyl oxygen of the C₂ benzoyl moiety (145). The areas of the paclitaxel structure where steric influences are most critical to receptor binding have been identified (147).

METABOLISM The taxanes are metabolized to significantly less cytotoxic metabolites by CYP450 enzymes (Fig. 37.44). In humans, CYP2C8 bioconverts paclitaxel to 6 α -hydroxypaclitaxel, the major metabolite, which is 30-fold less active than the parent structure

TABLE 37.8 Paclitaxel– β -Tubulin Binding Interactions (142,145,146)

Paclitaxel Functional Group	β -Tubulin Binding Residues	Interaction
C ₂ -benzoyl phenyl	Leu217, Leu219, His229, Leu230	Hydrophobic
C ₂ -benzoyl carbonyl	Arg278	Hydrogen bond
C ₃ -benzamido NH	Asp26	Hydrogen bond
C ₃ -benzamido carbonyl	His229	Hydrogen bond
C ₃ -phenyl	Ala233, Ser236, Phe272	Hydrophobic
C ₄ -acetyl	Leu217, Leu230, Phe272, Leu275	Hydrophobic
C ₇ -OH	Thr276 Ser277, Arg278	Hydrogen bond
C ₁₂ -CH ₃	Leu217, Leu230, Phe272, Leu275	Hydrophobic
C ₃ -OH	Arg369, Gly370 (NH)	Hydrogen bond
C ₃ -carbonyl	Gly370 (NH)	Hydrogen bond
Oxetane oxygen	Thr276 (NH)	Hydrogen bond

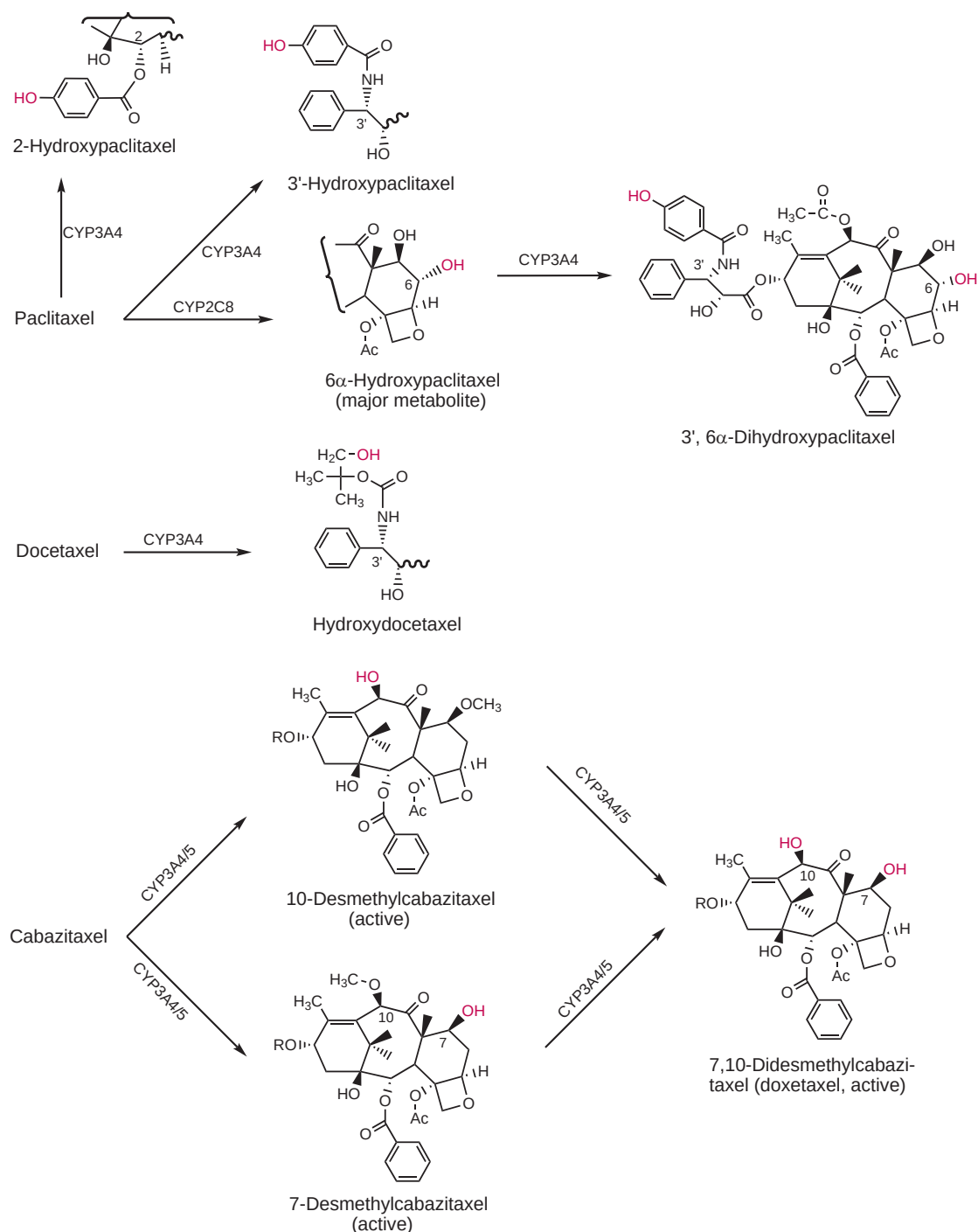


FIGURE 37.44 Metabolism of the taxane analogs.

(150–153). CYP3A4 mediates the formation of additional minor *p*-hydroxylated metabolites of the benzamido and benzoyl moieties at C₃' and C₂' respectively, and the 10-desacetyl metabolite has been documented in urine and plasma. Docetaxel is oxidized exclusively by CYP3A4/5, with CYP3A4 having a 10-fold higher affinity for the drug than CYP3A5. The major metabolite, known as hydroxydocetaxel, is the hydroxymethyl derivative of the 3'-*t*-butoxycarboxamide side chain (153).

Hydroxydocetaxel is further oxidized and cyclized to isomeric oxazolidinediones before excretion. Cabazitaxel is metabolized predominantly (80% to 90%) by CYP3A4/5, with CYP2C8 taking on a minor biotransformation role. Three active metabolites (including docetaxel) result from *O*-demethylation at C₇ and/or C₁₀. The elimination of taxanes is predominantly biliary.

The metabolic patterns of these closely related structures are distinct, and it has been hypothesized that

the C₁₃ side chain plays a major role in positioning the compounds in the catalytic site of CYP450 enzymes. Specifically, the 3'-phenyl ring of paclitaxel has been proposed to properly orient C₆ for hydroxylation through π -stacking interactions with CYP2C8 active site residues while decreasing affinity for CYP3A4 binding groups. The hydrophobic character of the 10-acetoxy group, found in paclitaxel, enhances CYP450-mediated hydroxylation two- to fivefold by facilitating substrate binding or augmenting catalytic capability. Both isoforms are impacted by the presence of this ester, often to the same extent (153).

Epothilones

CHEMISTRY Low water solubility is a significant drawback to the therapeutic utility of the taxanes. This is particularly true of paclitaxel, which has a more lipophilic acetate moiety at C₁₀ compared to docetaxel's more polar hydroxyl group. Paclitaxel must be administered in a vehicle of 50% alcohol/50% polyoxyethylated castor oil (Cremophor EL), which can lead to an enhanced risk of hypersensitivity reactions (dyspnea, hypotension, angioedema, and urticaria) in patients not pretreated with H₁ and H₂ antagonists and dexamethasone (154). As noted previously, high P-gp-mediated cellular efflux of paclitaxel and docetaxel can result in drug resistance. To overcome these problems, epothilones, 16-membered macrolides structurally unrelated to the taxanes but with functional groups properly positioned to mimic critical tubulin-binding groups, are being actively investigated for use in a variety of solid tumor and hematologic cancers (Fig. 37.45). Epothilone B, one of the original structures investigated, binds with very high affinity to the taxane binding site on polymerized β -tubulin, and it acts through the same cytotoxic mechanism. In addition to enhanced water solubility and a lack of P-gp affinity, epothilone is more efficiently produced through fermentation with the myxobacterium *Sorangium cellulosum* and

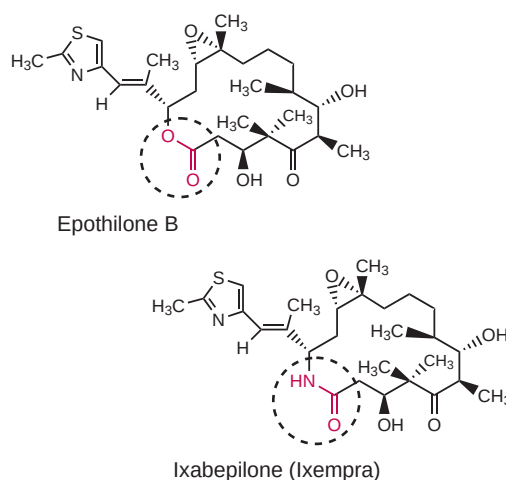


FIGURE 37.46 Epothilones.

has a higher antineoplastic potency (149,155,156). The lactam analog ixabepilone has a comparable anticancer activity with an even higher water solubility and better in vivo and in vitro stability (Fig. 37.46). The story of the discovery and subsequent development of this lactam as the lead compound in the search for a paclitaxel alternative makes for interesting reading (157).

Despite some conformational differences in ring system substituents between epothilones and taxanes, they are presumed to share a common or overlapping tubulin binding site, an assumption supported by the discovery of tubulin mutants that are resistant to both classes of antimetotics (158). Comparing crystal structures of tubulin-bound epothilone A (the 12-desmethyl derivative of epothilone B) and paclitaxel documents that the smaller macrolide fills only about half the binding site volume of the larger taxane ligand. The binding site is plastic, and residues adjust their side-chain conformations to accommodate either drug. Despite being surrounded by identical tubulin residues, the only common binding interaction involves a hydrogen bond between Arg282 and the 7-OH group of taxanes or epothilones.

SPECIFIC DRUGS (FIG. 37.43)

Paclitaxel Paclitaxel, which is claimed to be “the best-selling anticancer drug in history” (145), is indicated for intravenous use in combination with cisplatin as first-line therapy for advanced ovarian and non-small cell lung cancer. It is also used alone or in combination with the fluorouracil prodrug capecitabine in anthracycline-resistant metastatic breast cancer. Paclitaxel's ability to upregulate thymidine phosphorylase, one of capecitabine's activating enzymes, is the rationale behind the combination therapy (159). Solution (Taxol, Onxol) and albumin-bound (Abraxane) formulations are available and cannot be used interchangeably. Abraxane, which does not require the hypersensitivity-inducing Cremophor EL in its formulation, has also been used in various solid tumors of the GI and genitourinary tracts. Solution-based infusions usually are

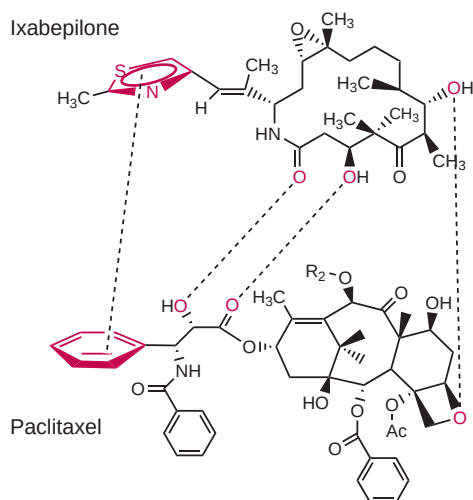


FIGURE 37.45 Complementary ixabepilone and paclitaxel functional groups.

administered over 3 to 24 hours and can be passed through an in-line, 0.22- μ m filter to reduce vehicle-related cloudiness. The albumin-bound form is given over 30 minutes and should be well mixed (but not filtered) to ensure complete suspension of the protein-drug particles. A paclitaxel receptor targeting peptide conjugate designed for selective drug delivery to the CNS is in clinical trials (142).

Besides hypersensitivity reactions common with the non-albumin-bound formulations, the major use-limiting adverse effect of paclitaxel is dose-dependent myelosuppression, particularly neutropenia, and first doses might need to be decreased in patients with hepatic dysfunction. Subsequent dose reductions, if any, should be tailored to individual response. The drug should not be given to patients who have baseline neutrophil counts below 1,500 cells/mm³. The albumin-bound formulation is also associated with sensory neuropathy. As noted earlier, all patients receiving solution-based paclitaxel should be pretreated with antihistamines and a corticosteroid to minimize the risk of potentially fatal hypersensitivity reactions. Paclitaxel is a Category D teratogen and carries a high risk of fetal intrauterine mortality. Both male and female patients are advised not to attempt conception while on this drug. Due caution should be observed when coadministering paclitaxel with drugs that inhibit or compete for metabolizing enzymes, particularly CYP2C8 (e.g., 17 α -ethinylestradiol and diazepam).

Docetaxel The indications for docetaxel as a rule mirror those of paclitaxel, although docetaxel is not used in ovarian cancer. It has greater water solubility than paclitaxel due to presence of the free C₁₀-OH group, and it is formulated with polysorbate 80 rather than with polyoxyethylated castor oil. Hypersensitivity reactions, while less likely, are still possible, and all patients should receive corticosteroid premedication. In addition to neutropenia and teratogenicity, this taxane can induce significant fluid retention, and 2-kg weight gains are not uncommon. Although rare, onycholysis has also been reported. Drug-drug interactions have been noted when docetaxel is coadministered with drugs that inhibit or compete for CYP3A4 enzymes (e.g., "azole" antifungals, erythromycin, and cyclosporine) (160).

Cabazitaxel Cabazitaxel is the 7,10-dimethoxy analog of docetaxel. Conversion of the two secondary alcohol groups to methoxy ethers dramatically lowers affinity for P-gp, resulting in sustained retention in tumor cells along with a higher blood-brain barrier penetration (161). Cabazitaxel has shown efficacy in docetaxel-resistant cell lines where resistance was due to P-gp overexpression. Although currently approved for use only in docetaxel-resistant metastatic prostate cancer, cabazitaxel has shown activity against a wide variety of human tumors including colon, lung, pancreas, squamous cell, head and neck, and metastatic breast cancers (161–163). Neutropenia is the most problematic dose-limiting reaction, although its incidence and severity are no worse than observed with other taxanes. Cabazitaxel does not promote the fluid

retention commonly observed with docetaxel, but peripheral motor and/or sensory neuropathy can be persistent. Its relatively high incidence of diarrhea (~50%) may be explained by the accumulation of the drug in enterocytes, cells that constitutively express P-gp and, therefore, actively evict other taxanes. Despite the conversion of the two secondary alcohols to more lipophilic ethers, cabazitaxel's aqueous solubility is on a par with docetaxel's. Like docetaxel, it is formulated with polysorbate 80 rather than the more hypersensitivity-inducing Cremophor EL, although antihistamine/corticosteroid pretreatment is still recommended (160). It is administered by intravenous infusion in doses of 25 mg/m² every 3 weeks.

Ixabepilone The epothilone ixabepilone is used in combination with the thymidylate synthesis inhibitor capecitabine in anthracycline- and/or taxane-resistant advanced or metastatic breast cancer, or when these alternative drugs are contraindicated. Some phase II clinical trials have shown overall response rates to this agent as high as 57% in previously untreated breast cancer patients and up to 30% in patients who had been heavily pretreated (164). As noted previously, the lactam moiety provides stability to in vivo hydrolysis by carboxylesterases, but the drug is extensively metabolized by CYP3A4 to over 30 inactive metabolites prior to predominately fecal excretion. Drug-drug interactions with CYP3A4 substrates, inducers, or inhibitors have been reported, and dosage adjustments may be warranted if coadministration cannot be avoided. Like cabazitaxel, ixabepilone's serious use-limiting adverse reactions include peripheral neuropathy (67%) and neutropenia. Like paclitaxel, it requires Cremophor EL for solubilization, so hypersensitivity reactions are likely, and prophylactic premedication is required. The most common intravenous dosage regimen is 40 mg/m² administered over 3 hours every third week.

Vinca Alkaloids

MECHANISM OF ACTION Several alkaloids found naturally in *Catharanthus roseus* (periwinkle) have potent antimitotic activity. In opposition to the taxoids, vinca alkaloids halt cell division by inhibiting polymerization. They bind at the interface of two heterodimers within the inner tubular lumen at a single high-affinity site on β -tubulin in the vicinity of the guanosine triphosphate binding site on the (+) end of the tubules. Once bound, these alkaloids attenuate the uptake of the guanosine triphosphate essential to tubule elongation (165). Simultaneous binding to α - and β -tubulin results in protein cross-linking, which promotes a stabilized protofilament structure (166). Inhibition of microtubule elongation occurs at substoichiometric concentrations, at which alkaloid occupation of only 1% to 2% of the total number of high-affinity sites can result in up to a 50% inhibition of microtubule assembly (167,168). At high concentrations, when alkaloid binding to high-affinity sites becomes stoichiometric and lower-affinity binding sites on the tubule

wall are also occupied, microtubular depolymerization is stimulated, leading to the exposure of additional alkaloidal binding sites and resulting in dramatic changes in microtubular conformation. Spiral aggregates, protofilaments, and highly structured crystals form, and the mitotic spindle ultimately disintegrates (143,159). The loss of the directing mitotic spindle promotes chromosome “clumping” in unnatural shapes (balls and stars), leading to cell death (154). Other nonmitotic toxicities related to the microtubule-disrupting action of the vinca alkaloids include inhibition of axonal transport and secretory processes and disturbances in platelet structure and function (167).

As noted earlier, the mutant *p53* oncogene is associated with resistance to vinca alkaloid-induced cytotoxicity due to its augmentation of MAP4-mediated microtubulin polymerization, which counteracts the depolymerizing mechanism of the alkaloids. In addition, the mutant oncogene gives a degree of immortality to stathmin, a cytosolic protein that must be inactivated for mitosis to begin. Finally, *p53* upregulates the MRP-1 efflux protein that ejects vinca alkaloids from cells. It has been suggested that *p53* phenotype could be harnessed to better predict a patient's anticipated susceptibility to various mitosis inhibitor antineoplastic therapy options (144).

CHEMISTRY The specific chemical nature of the vinca binding site remains elusive due to difficulties encountered in binding assay development and implementation, as well as in data analysis. It is known that the active site is close to residue 339 and residue 390 on α - and β -tubulin, respectively (168). Of the three marketed vinca alkaloids (vincristine, vinblastine, and vinorelbine), vincristine binds most tightly, whereas vinblastine has the lowest affinity (167). Because vinca alkaloids enter cells by simple passive diffusion, unbound vinorelbine and vinblastine (being more lipophilic than vincristine) may be more extensively taken up into tissues. Vincristine, however, is cleared more slowly from the system and has the longest terminal half-life of the three agents, resulting in a more prolonged tumor cell exposure (143,167). Like the taxanes, tumor resistance to vinca alkaloids is mediated, in part, through P-gp.

The vinca alkaloids are complex structures composed of two polycyclic segments known as catharanthine (or velbanamine) and vindoline (Fig. 37.43), both of which are essential for high-affinity tubulin binding. The three commercially available anticancer alkaloids differ in the length of the alkyl chain bridging positions 6' and 9' of the catharanthine moiety (methylene or ethylene), in the substituents at position 4' (olefin or tertiary alcohol), and in the N_1 vindoline indole nitrogen (methyl or formyl). Although subtle, these structural changes lead to significant differences in clinical spectrum, potency, and toxicity. For example, vincristine's relative lack of bone marrow toxicity at standard therapeutic doses makes it popular in combination therapy with more myelosuppressive anticancer agents, whereas vinblastine's relative

lack of neurotoxicity permits its coadministration with cisplatin. It is known that acetylation of either hydroxyl group destroys antineoplastic activity and reduction of the vindoline olefinic linkage greatly attenuates antineoplastic action (154). The C_{18} -methoxycarbonyl and the stereochemistry at positions 18' and 2' are also believed to be critical to activity (169).

Vinca alkaloids undergo O_4 -deacetylation to yield metabolites equal to or more active than the parent drug. They are also subject to extensive CYP3A4/5-mediated metabolism before biliary excretion, although the structures of these metabolites are currently unknown (170–173).

SPECIFIC DRUGS (FIG. 37.43)

Vincristine Sulfate It has been estimated that over half of U.S. children with cancer who receive chemotherapy will be given vincristine, as will more than 300,000 adult cancer patients per year (170). The drug is given by the intravenous bolus or continuous infusion routes in the treatment of acute leukemia and various Hodgkin's lymphomas and NHLs. Toxicity is often more pronounced by the latter route.

Elimination of vincristine is triphasic, with the first phase (5 minutes) representing rapid uptake into tissues and the last phase (85 hours) representing release back to the plasma from tubulin-containing cells. Since the drug is extensively metabolized by O_4 -deacetylation and CYP3A5-catalyzed oxidation in the liver, patients with hepatic dysfunction are at an increased risk for toxicity, and dosage reductions should be considered. The most significant dose-limiting adverse effect is peripheral neuropathy, which is initially manifested as numbness and painful paresthesias in the extremities and progresses to muscular pain, severe weakness, and loss of coordination. Patients can also experience constipation secondary to intestinal neurotoxicity, which may require treatment with cathartics. African American patients, who commonly carry the CYP3A5*1 allele that expresses a catalytically active CYP3A5 enzyme, are at lower risk for severe and/or prolonged neurotoxicity than Caucasians, who carry less active/inactive alleles (170). Myelosuppression is not particularly problematic because it occurs at doses higher than those that can be tolerated. As previously noted, coadministration with mitomycin can induce acute or delayed pulmonary toxicity characterized by severe bronchospasm.

All vinca alkaloids are severe vesicants that can induce necrosis, cellulitis, and/or thrombophlebitis. Proper needle placement before administration should be assured to eliminate the risk of extravasation. Unlike the tissue damage caused by the vesicant action of nitrogen mustards and antibiotic antineoplastics, cold exacerbates tissue destruction. If extravasation occurs, apply heat for 1 hour 4 times a day for 3 to 5 days, coupled with local hyaluronidase injections. Vinca alkaloids are all Category D teratogens and are fatal if administered by the intrathecal route.

Vinblastine Sulfate In addition to the hematologic indications that it shares with vincristine, vinblastine has found utility in the treatment of advanced testicular carcinoma (often in combination with bleomycin), advanced mycosis fungoides, Kaposi sarcoma, and histiocytosis X. Leukopenia is the dose-limiting side effect, and dose reductions are warranted in patients with serum bilirubin levels greater than 3 mg/dL. The drug-related impact on erythrocyte and thrombocyte levels is usually insignificant. Like vincristine, it is administered as an intravenous bolus or infusion. The initial elimination half-life of 3.7 minutes is similar to vincristine, but the 24.8-hour terminal half-life is significantly shorter.

Vinorelbine Tartrate Vinorelbine is used alone or in combination with cisplatin for first-line treatment of non-small cell lung cancer. This semisynthetic alkaloid is unique in having oral bioavailability (167), but it is currently available only for intravenous injection. The initial phase elimination half-life is on a par with that observed for vincristine and vinblastine, and the terminal phase half-life is between 28 and 44 hours. Although dose-limiting granulocytopenia is the major adverse effect, potentially fatal interstitial pulmonary changes have been noted. Patients with symptoms of respiratory distress should be promptly evaluated. Biotransformations include deacetylation and *N*-oxidation. As with all vinca alkaloids, elimination is primarily hepatobiliary, and dosage reduction should be considered in patients with liver dysfunction.

Estramustine Phosphate Sodium Because this anticancer agent contains a carbamylated nitrogen mustard moiety, it was originally thought to function as a DNA alkylator; however, it is now known that its primary mechanism of antineoplastic action is inhibition of mitosis. Estramustine binds to MAP-4, prompting dissociation of this protein from the microtubule and promoting depolymerization and disassembly. It can also bind directly to α - and β -tubulin at a site distinct from the vinca alkaloid and taxane binding sites, although paclitaxel exerts a noncompetitive inhibition of estramustine binding to tubulin. A specific estramustine binding protein in prostate tissue is believed to facilitate its action in the treatment of metastatic carcinoma of the prostate. Estramustine has a low affinity for the β_m tubulin isotype, which is often overexpressed in estramustine-resistant prostatic neoplasms as one defense against this therapeutic intervention (159,174).

Estramustine's resonance-stabilized, mustard-like β -haloalkylamine carbamate structure uses an estradiol carrier to selectively deliver drug to steroid-dependent prostate tissue, and its use is limited to the palliative treatment of progressive prostate cancer. The ionized sodium phosphate ester makes the compound water soluble and able to distribute in the blood. The phosphate ester is readily cleaved during absorption to provide the

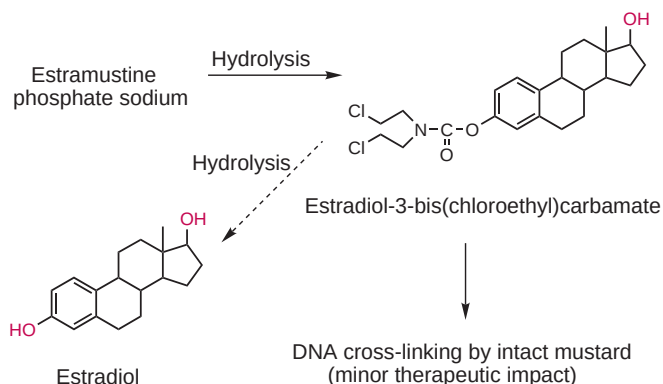


FIGURE 37.47 Estramustine metabolism.

active 17β -OH. The 3-carbamate group is also cleaved in vivo to generate estradiol (Fig. 37.47), which explains why this drug is not used to treat estrogen-dependent tumors (e.g., estrogen-dependent breast cancer). The liberated estradiol may also increase blood pressure and induce blood clots, leading to myocardial infarction. Fortunately, the myocardial infarctions are usually non-fatal, but the drug should be used with extreme caution in men who are predisposed to clotting disorders or who have a history of cerebral vascular disease or coronary artery disease. Hepatotoxicity is also associated with estramustine use.

Tyrosine Kinase and Related Inhibitors

Aberrations in the activity of protein tyrosine kinases (TKs) are associated with several neoplastic disorders, and it has been estimated that more than 80% of human oncogenes and proto-oncogenes direct the expression of these essential phosphorylating enzymes. When functioning normally, TKs regulate cell proliferation, differentiation, and survival. When functioning in a deregulated manner, they accelerate cell signaling cascades and cellular growth, induce tumors, augment antiapoptotic processes, and, in so doing, confer resistance to many chemotherapeutic drugs.

MECHANISM OF ACTION TKs are of two general types, receptor-associated and cellular (nonreceptor), both of which are ATP-dependent. The highly conserved ATP binding domain of the TKs serves as the receptor for antineoplastic TK inhibitors (TKIs, or tyrphostins). This hydrophobic domain, a depression or groove rich in isoleucine (Ile), leucine (Leu), alanine (Ala), and valine (Val) residues, is found in the "hinge region" that connects the amino terminal (N) and carboxy terminal (C) lobes of the catalytic unit. A minimum of five potential binding pockets surround this site, which may help explain the otherwise surprising degree of selectivity exhibited by many anticancer TKIs (175). Van der Waals, hydrophobic, hydrogen-bonding, and electrostatic interactions are all of prime importance in holding ATP and the TKIs to this enzymatic domain (175,176).

TKs that serve as targets for anticancer TKIs include EGFR, VEGFR, human epidermal growth factor receptor 2 (HER2), platelet-derived growth factor receptor (PDGFR), Bcr-Abl (a product of the translocated break-point cluster [BCR]-Abelson [ABL] gene known as the Philadelphia, or Ph, chromosome), and Src. Bcr-Abl and Src are nonreceptor kinases, whereas the remaining are receptor-associated enzymes. Type 1 inhibitors bind to the active conformation of the kinase, whereas type 2 TKIs inhibit the enzyme in its inactive conformation. Since the inactive conformation of TKs differs in structure to a greater degree than the active conformer, it has been proposed that type 2 inhibitors may have a better opportunity for selectivity (175,177). Promiscuous

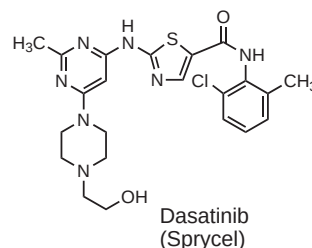
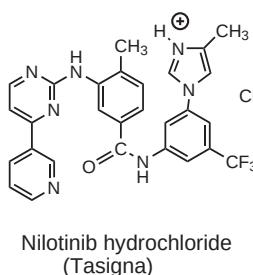
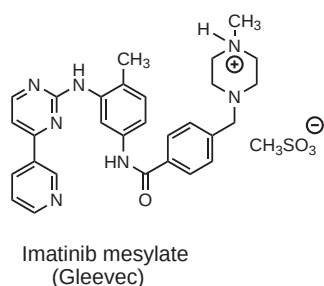
TKIs inhibit several kinase enzymes and are referred to as “multikinase inhibitors.”

CYP3A4 is a common TKI-metabolizing isoform, and many TKIs inhibit a variety of cytochrome P450 (CYP) enzymes. Several TKIs are also substrates for, and inhibitors of, cellular efflux pumps like P-gp and breast cancer resistance protein (BCRP). All are excreted predominantly in the feces. The structures of the currently marketed TKIs are shown in Figure 37.48.

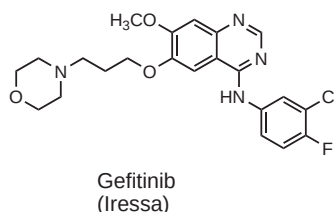
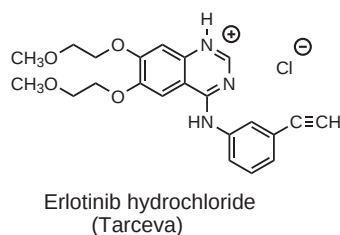
Bcr-Abl Inhibitors: Imatinib, Nilotinib, and Dasatinib

CHEMISTRY Bcr-Abl inhibitors were the first TKIs to be introduced, and they literally changed the face of CML therapy. The aberrant Ph chromosome is viewed as the

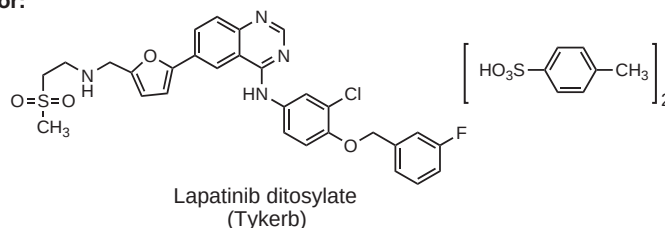
Bcr-Abl kinase inhibitors:



EGFR kinase inhibitors:



EGFR and HER2 kinase inhibitor:



VEGFR kinase inhibitors:

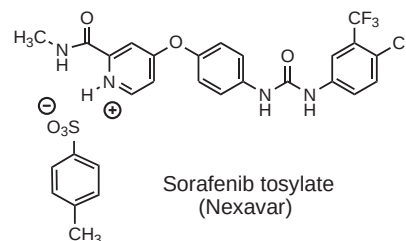
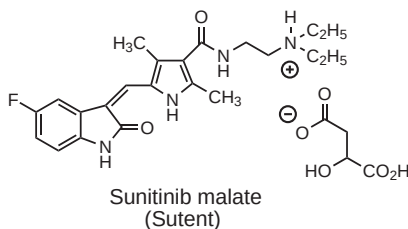
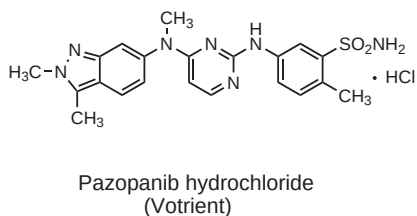


FIGURE 37.48 Tyrosine kinase inhibitors.

single cause of more than 90% of adult CML. The clinical availability of imatinib, discovered serendipitously in a search for new anti-inflammatory agents, allowed patients to realistically anticipate a 5-year survival, as opposed to the 2- to 3-year prognosis for patients with untreated disease. The drug has its greatest effect in the initial (chronic) phase of CML and is significantly less effective in the accelerated or highly fatal blastic phases.

As is the case with so many chemotherapeutic agents, acquired resistance has undermined the positive clinical outcomes this drug first promised. Resistance mechanisms are often associated with point mutations in the Abl (kinase) domain, in particular a threonine (Thr) to Ile mutation at position 315 (T315I). Residue 315 is known as the “gatekeeper” to the hydrophobic binding pocket, and the more significant bulk of the Ile residue blocks imatinib’s access to this receptor through steric interference with the benzyl moiety (175,178). Alternate mechanisms of resistance are related to *BRC-ABL* gene amplification, P-gp overexpression, underexpression of an organic cation transporter-1 protein that helps imatinib (but not nilotinib) gain access to cells, and activation of Src kinases. Although nilotinib can show activity in some imatinib-resistant mutants, it cannot overcome the resistance induced by T315I. Twelve percent of all Bcr-Abl mutations are T315I, and CML patients carrying this mutation have a poor prognosis.

The Bcr-Abl inhibitors imatinib and nilotinib contain a 2-phenylaminopyrimidine pharmacophore. They are relatively large, extended structures that bind to both the ATP hinge region and to adjacent hydrophobic subdomains of the Bcr-Abl protein. As type 2 inhibitors, they bind to the inactive conformation of the kinase known as the “DFG-out state” (177). In this conformation, an enzymatic aspartate (Asp¹⁷⁵) (D)-Mg⁺² interaction is broken, and a Phe residue is pushed out toward the aqueous environment, occluding the approach of ATP and generating a new binding area that attracts type 2 Bcr-Abl inhibitors. This conformation is also stabilized by an unphosphorylated Tyr³⁹³, which orients toward the center of the enzyme and H-bonds with Asp³⁶³ (179).

The *o*-methyl substituent found on imatinib and nilotinib confers selectivity for the cytosolic Bcr-Abl protein in its inactive conformation, although the benzamide moiety permits some inhibition of PDGFR kinase. It is the PDGFR-related component of the activity profile that is believed responsible for the fluid retention induced by these two TKIs. These agents also contain a pyridyl substituent at C₄ of the pyrimidine ring, which enhances affinity through H-bonding with the amide NH of Met³¹⁸ within the hinge region of the kinase (Fig. 37.49). The amide moiety, inverted in nilotinib compared to imatinib, forms H-bonds with Glu²⁸⁶ and Asp³⁸¹ (180,181). Yet another H-bond is formed between gatekeeper Thr³¹⁵ and the NH group connecting the pyrimidine and phenyl rings. Water solubility and the favorable oral bioavailability profiles of imatinib and nilotinib are conferred by the substituted piperazine and imidazole

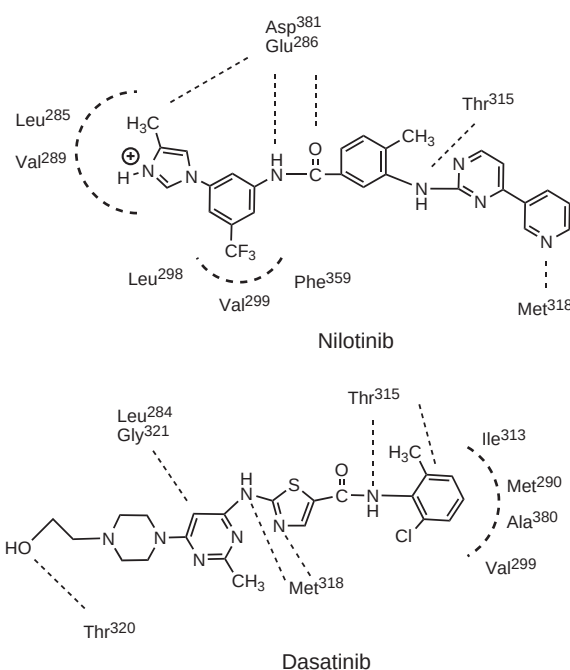


FIGURE 37.49 Binding interactions between tyrosine kinase and selected Bcr-Abl inhibitors.

rings, respectively. The lower pK_a of imidazole compared to piperazine explains why nilotinib is not a substrate for the organic cation transporter-1 protein that ferries imatinib into cells and contributes to imatinib resistance (182).

The trifluoromethyl moiety of nilotinib engages in unique affinity-enhancing hydrophobic interactions with Leu²⁹⁸, Val²⁹⁹, and Phe³⁵⁹, which are, in part, responsible for nilotinib’s 30- to 50-fold increase in TK-inhibiting potency compared to imatinib. The methylimidazole moiety also augments affinity through hydrophobic interactions with Leu²⁸⁵, Glu²⁸⁶, and Val²⁸⁹, while leaving the basic nitrogen exposed to the aqueous environment (175,177–179).

In contrast to imatinib and nilotinib, dasatinib is a mixed type 1 and type 2 Bcr-Abl kinase inhibitor that also has significant affinity for cellular Src kinases. Although it contains a pyrimidine ring, this moiety does not bind in the hinge region as the pyrimidine rings of imatinib and nilotinib do. Rather, this binding honor goes to aminothiazole, and it is facilitated by a critical hydrogen bond between the thiazole nitrogen and the amide NH of Met³¹⁸. Met³¹⁸’s carbonyl oxygen also binds to the hydrogen of the amino group connecting the thiazole and pyrimidine rings. A third hydrogen bond between the drug’s amide nitrogen and the hydroxyl of gatekeeper residue Thr³¹⁵ ensures high affinity for the active (DFG-in) state of the kinase (178).

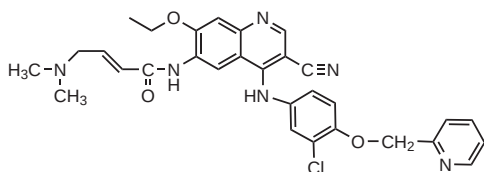
The 2-chloro-6-methylbenzamide segment of dasatinib binds deep within the hydrophobic pocket of the kinase in an area distinct from the ATP-binding site. Interactions between this aromatic moiety and Thr³¹⁵, Met²⁹⁰, Val²⁹⁹,

Ile³¹³, and Ala³⁸⁰ are known (183). Hydrophobic interactions with Leu²⁴⁸ and Gly³²¹ also hold the pyrimidine ring to the kinase. As in imatinib, dasatinib's water solubility is assured through the hydroxyethyl-substituted piperazine ring, which is oriented toward the hinge region and is solvent-exposed. An H-bond between this moiety and the backbone carbonyl oxygen of Thr³²⁰ is known to occur.

Dasatinib's ability to bind to the active kinase conformation has been attributed to the fact that it does not insert into the hydrophobic pocket containing Phe(F)³⁸², as the substituted benzyl moiety of imatinib does (175). Because it can bind to both the active and inactive conformations of Bcr-Abl kinase, dasatinib has a potency approximately 325 times that of imatinib, and it can be used in patients resistant to imatinib or nilotinib. Most mutations that confer resistance to imatinib occur in the P-loop of the kinase, an area not important to dasatinib binding (183). Dasatinib is effective against all imatinib-resistant mutants except T315I, retains activity in cells made resistant to imatinib via activation of Src kinases, and does not bind to P-gp.

EGFR and EGFR/HER2 Inhibitors: Erlotinib, Gefitinib, and Lapatinib

CHEMISTRY EGFR and HER2 are closely related membrane-bound TKs. EGFR expression in solid tumors of the breast, lung, bladder, esophagus, and oral cavity has been clearly correlated with decreased life expectancy (184), and it is a consistent presence in almost all epithelial-derived cancers. Likewise, HER2 overexpression is a classic feature of treatment-resistant breast, ovarian, lung, and gastric cancers, and it endows tumors with what's been called an "antiapoptotic shield" (185). A unique Cys residue is located within the ATP-binding domain, and TKIs capable of irreversibly inactivating the protein through covalent bond formation with Cys⁷⁹⁷ (e.g., neratinib) are now being investigated (175,176).



Neratinib (Irreversible tyrosine kinase inhibitor)

The currently marketed reversible EGFR TKIs all contain a 4-anilinoquinazoline pharmacophore with an oxygen-containing substituent at C₆. Ether-containing moieties of varying size are also permitted at C₇. An electron-withdrawing substituent at position 3' of the aniline phenyl ring provides high selectivity for EGFR kinase as long as the 4' position remains unsubstituted (erlotinib) or is modified with a very small substituent such as fluorine (gefitinib). Erlotinib and gefitinib are quintessential type 1 TKIs, with the *m*-substituted phenyl ring of the aniline moiety enhancing affinity through binding

with back pocket residues of the active (open) enzyme. The H-bonds formed between the quinazoline nitrogen atoms and hinge residues Met⁷⁹³ (N₁) and Thr⁷⁹⁰ (N₃) are crucial to activity. The halogenated and acetylene-substituted phenyl rings of gefitinib and erlotinib, respectively, are maintained at a 42-degree angle relative to the quinazoline ring and, in this orientation, bind well in the hydrophobic pocket near Thr⁷⁹⁰ as long as this gatekeeper residue remains small (175). A T790M mutation is believed responsible for acquired resistance to these highly selective TKIs (186,187). Unlike the acquired Bcr-Abl kinase resistance, which was steric in nature, an unfavorable TKI/ATP binding affinity ratio in the T790M mutant has been proposed to explain acquired resistance to EGFR TKIs (175).

Increasing the size of the 4' substituent with groups like *m*-fluorobenzyloxy (lapatinib) restricts access to only the inactive conformation of EGFR (175) and broadens TK specificity to include HER2. The entire substituted aniline structural component of lapatinib binds within a lipophilic pocket of the HER2 kinase in its inactive (closed) conformation. The quinazoline ring of EGFR/HER2 inhibitors again affiliates with residues in the hinge region of the ATP-binding domain. Selectivity for the inactive conformation of its target kinases (which requires a conformational change to dislodge the inhibitor) may contribute to the very slow (300 minutes) enzyme dissociation half-life compared to erlotinib and gefitinib (30 minutes) (175).

Lapatinib's 2-furanyl substituent can be further substituted at position 5 with long, unbranched chains that extend out into the aqueous environment. The addition of a methylsulfone moiety to the chain terminus enhances water solubility. The ionizable morpholine ring of gefitinib serves a similar purpose. The water solubility of erlotinib is predictably low given the relative lack of polar functional groups in this area of the molecule.

VEGFR Inhibitors: Sunitinib, Sorafenib, and Pazopanib

CHEMISTRY VEGF2 is a key enzymatic player in the generation of new blood vessels. Inhibition of this important TK starves tumors by inhibiting angiogenesis and keeping oxygen and essential nutrients from supporting their continued uncontrolled growth. This deprivation, coupled with the buildup of cellular waste materials, kills cells treated with TKIs that target this kinase. Augmenting the cytotoxic effect is the fact that VEGF inhibition decreases the permeability of tumor cell vasculature and eases intracellular delivery of chemotherapeutic agents (188).

The binding of sorafenib to VEGFR in its inactive conformation is believed to be similar to its interaction with its originally recognized kinase target B-RAF, a component of the RAS signal transduction network (175). Assuming so, the pyridine nitrogen in its un-ionized form and the nearby methylamide NH moiety H-bond with the amide group of Cys919 in the hinge region (NH and carbonyl oxygen, respectively). The trifluoromethylphenyl moiety binds in a hydrophobic pocket, possibly

occupying the site normally reserved for Phe1047 in the “DFG-in” (active) conformation, and the urea moiety forms H-bonds with several residues, including the Asp1046 of the DFG trio.

Unlike sorafenib, a type 2 TKI, sunitinib can inhibit VEGFR in both its active and inactive conformations. The indolinone (or oxindole) moiety of sunitinib has been shown to confer high affinity for a hydrophobic pocket within the ATP-binding domain of kinases (189). However, the selectivity of VEGFR TKIs is low, since this kinase domain is replicated in many closely related enzymes, such as PDGFR α and c-kit, that are associated with GI stromal tumors (GISTs) (190). Broad activity against several kinases is expected, particularly from inhibitors that are retained exclusively within this conserved nucleotide-binding region and that avoid neighboring residues that permit differentiation of kinase affinity profiles (e.g., sunitinib). Sorafenib's pyridine ring interacts with amino acids in the kinase ATP-binding domain, whereas the urea component of the bisarylurea moiety augments affinity through H-bonding.

Geometric isomerism is important to the TKI activity of sunitinib, with the lower energy *Z* isomer being over 100 times as potent as the higher energy *E* isomer. Exposure to light will prompt a *Z* to *E* isomeric inversion, with a return to the lower energy *Z* state over time in the dark (191).

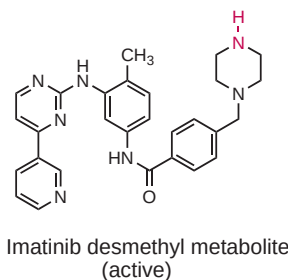
Sunitinib and sorafenib enter cells via passive diffusion as opposed to carrier-mediated transport (192). All currently marketed VEGFR TKIs are used in the treatment of renal cell carcinoma (RCC), a highly vascularized tumor (177).

SPECIFIC DRUGS (FIG. 37.48)

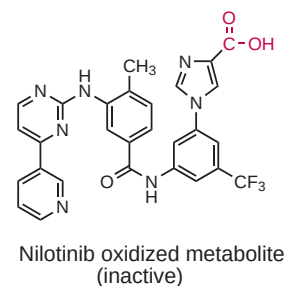
Imatinib Mesylate Imatinib is indicated in Ph⁺ CML, acute lymphoblastic leukemia, GIST (a tumor expressing a c-kit kinase mutation), and myeloproliferative diseases. Available in 100- and 400-mg tablets, recommended doses in patients with adequate renal function run mostly between 400 and 600 mg daily. The drug should be taken with a large glass of water and with food to minimize GI distress.

Imatinib has a 98% mean oral bioavailability, and maximum serum concentrations are achieved within 4 hours. It is metabolized predominantly by CYP3A4-mediated *N*-dealkylation to an equally active desmethyl metabolite (191). Serum levels of imatinib will increase if coadministered with CYP3A4 inhibitors, and the drug should not be given with grapefruit juice. Conversely, the dose of imatinib should be increased by 50% if coadministration with potent CYP3A4 inducers (e.g., cyclosporine) is warranted. Imatinib is a competitive inhibitor of CYP3A4, CYP2C9, and CYP2D6, and care should be taken when substrates of these isozymes are coadministered. The parent drug and active desmethyl metabolite have elimination half-lives of 18 and 40 hours, respectively. Approximately 25% of a dose is excreted as the unchanged drug. Although well tolerated, common adverse effects include edema,

diarrhea, nausea, and rash. Myocardial toxicity and hepatotoxicity, while potentially severe, are rare.



Nilotinib Hydrochloride Nilotinib is indicated in newly diagnosed or imatinib-resistant Ph⁺ CML. It is supplied as 150- and 200-mg capsules, and twice-daily doses of 300 mg (new diagnosis) and 400 mg (resistant disease) are standard. The oral bioavailability of nilotinib is much lower than imatinib (30% vs. 98%, respectively). Taking the drug within 30 minutes of a high-fat meal increases oral bioavailability to 50% and the area under the curve by 82% (193), and this can result in an increased risk of serious toxicity. No food should be eaten 2 hours before or 1 hour after administration. Biotransformation is limited, with the major metabolite being a carboxylic acid arising from CYP3A4-mediated hydroxylation of an aromatic methyl group. No metabolites are active. Sixty-nine percent of a dose is excreted unchanged (191).



Nilotinib is associated with life-threatening toxicities, including QT interval prolongation that can progress to torsades de pointes, sudden death, and myelosuppression. The risk of potentially fatal myocardial toxicity is elevated if the drug is taken with food or coadministered with potent CYP3A4 inhibitors. Nilotinib doses should be cut in half if CYP3A4 inhibitors must be coadministered and reduced in patients with hepatic impairment. Patients should also be counseled to avoid grapefruit juice and St. John's wort.

Dasatinib Dasatinib is available as 20-, 50-, 70-, and 100-mg tablets for use in Ph⁺ CML and acute lymphoblastic leukemia patients who are resistant or intolerant to other therapies, including imatinib. The starting dose is 140 mg once daily, and patients should be titrated up or down to the maximum dose tolerated. Doses higher than 180 mg daily are not recommended. Oral bioavailability is low due

to poor absorption and rapid first-pass CYP3A4-mediated metabolism. Biotransformations include aromatic hydroxylation, benzylic hydroxylation, *N*-dealkylation, *N*-oxidation (at piperazine N_4), and oxidation of the hydroxyethyl moiety (Fig. 37.50).

FMO-3 and uridine diphosphate glucuronyl transferase also catalyze minor metabolic reactions (193). All five CYP3A4-generated metabolites retain activity, but represent only 5% of the parent area under the curve, so their clinical relevance is questionable. The *p*-hydroxylated metabolite can be oxidized to a potentially hepatotoxic quinoneimine (191). The mean elimination half-life of dasatinib is 3 to 5 hours, and 19% of a dose is excreted unchanged (191). Doses should be decreased (20 to 40 mg daily) or increased (as tolerated) if CYP3A4 inhibitors or inducers, respectively, must be coadministered. Like other drugs in this class, dasatinib also inhibits CYP3A4, and it should not be taken with grapefruit juice.

Dasatinib can be taken with or without food, and myelosuppression, peripheral edema, and GI distress are the most commonly encountered adverse effects. Unlike nilotinib, there is no black box warning related to prolongation of the QT interval, but the potential of dasatinib to exacerbate the toxicity of agents that do should not be ruled out.

Erlotinib Hydrochloride Erlotinib is used in the treatment of non-small cell lung cancer in patients whose disease has either stabilized after four rounds of organoplatinum therapy or progressed after completion of a non-TKI-based chemotherapeutic regimen. It is also used in combination with gemcitabine as first-line therapy in advanced or metastatic pancreatic cancer. Doses of

100 to 150 mg daily are common, and the product is marketed as 25-, 100-, and 150-mg tablets. Because administration with food significantly increases oral absorption from 60% to 100%, patients are instructed to take the drug 1 hour before or 2 hours after eating to minimize toxicity risks.

Extensive CYP3A4-mediated *O*-dealkylation of the terminal methoxy group of the C_6 side chain occurs, and some molecules of the primary alcohol are further oxidized by cytosolic enzymes to the carboxylic acid. CYP1A1 and CYP1A2 can also catalyze this reaction, and erlotinib's half-life and area under the curve are dramatically decreased in cigarette smokers (191). Coadministration of strong CYP3A4 inducers has the same effect. Conversely, erlotinib doses must be decreased in 50-mg increments if serious side effects occur with coadministration of CYP3A4 inhibitors. The potentially fatal hepatotoxicity that can be induced by this drug might be due, at least in part, to the formation of an electrophilic quinoneimine from a CYP3A4- and CYP1A1-generated phenolic metabolite (Fig. 37.51) (191).

In addition to hepatotoxicity, erlotinib can cause diarrhea and a maculopapular skin rash that has been positively correlated with therapeutic efficacy. The rash worsens when exposed to sunlight, so patients should be counseled to take appropriate precautions. Erlotinib is both a substrate and an inhibitor of P-gp and, like other TKIs, it inhibits CYP isoforms, specifically CYP3A4 and CYP3A5.

Gefitinib Gefitinib is used exclusively as a single agent in the treatment of organoplatinum- and docetaxel-refractory non-small cell lung cancer. One 250-mg tablet

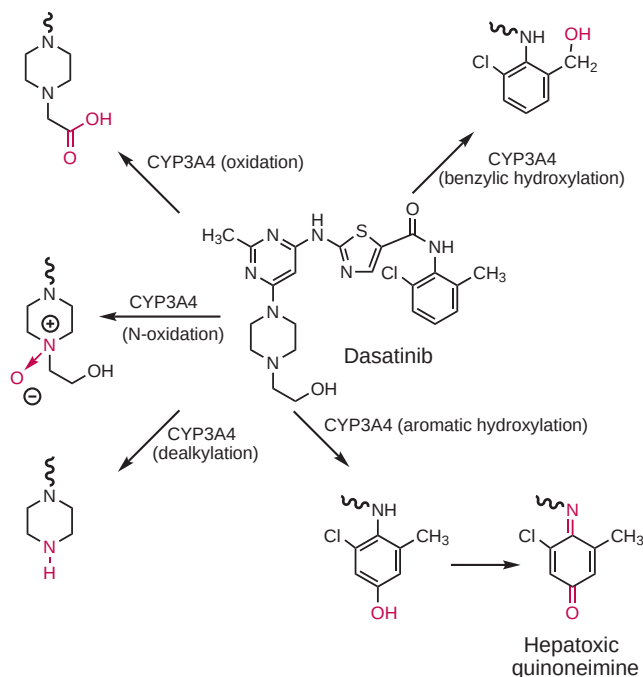


FIGURE 37.50 Dasatinib metabolism.

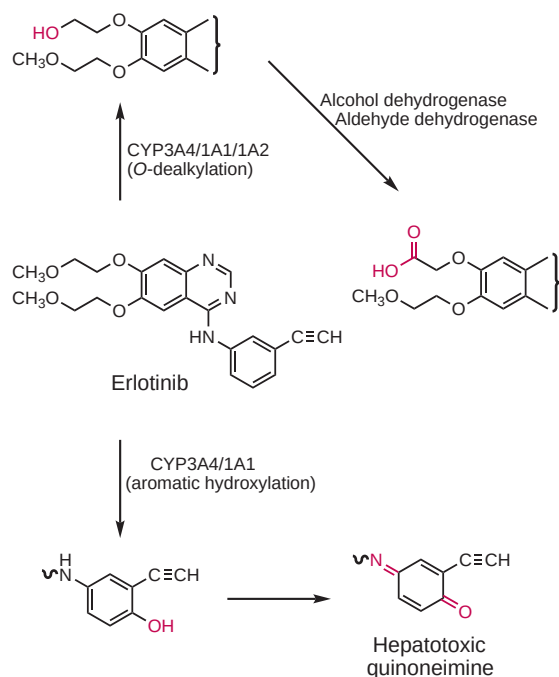


FIGURE 37.51 Erlotinib metabolism.

is administered daily without regard to meals. Higher doses induce more toxicity without additional therapeutic benefit.

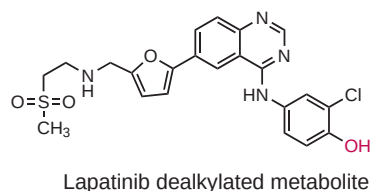
CYP2D6, along with CYP3A4/3A5, catalyzes the dealkylation of the quinazoline 7-methoxy ether to a phenolic metabolite with equal *in vitro* EGFR-inhibiting action but limited therapeutic relevance due to polarity-induced difficulties in tumor cell penetration (193). Enzymatic cleavage and subsequent loss of the morpholine ring occur to a lesser extent, as do defluorination and subsequent *p*-hydroxylation of the aniline phenyl moiety. As noted with other TKIs, oxidation of the *p*-phenol to a reactive quinoneimine may occur in both liver (CYP3A4) and lung (CYP1A1) (Fig. 37.52). Smokers can generate up to 12 times more of this reactive metabolite than non-smokers (191).

Like erlotinib, gefitinib induces rash (a consequence of its EGFR kinase specificity) and diarrhea. Between 40% and 50% of patients on the 250-mg daily dose will experience these two side effects. Infrequent but serious toxicities include potentially fatal interstitial lung disease (possibly related to quinoneimine formation), interstitial pneumonia, worsening pulmonary fibrosis, and corneal ulceration. The drug induces aberrant eyelash growth, which can induce eye pain.

Lapatinib Ditosylate This dual kinase inhibitor is used in combination with capecitabine as second-line therapy in the treatment of HER2-positive advanced or metastatic breast cancer. Patients receiving this drug should have previously received anthracycline, taxane, and trastuzumab therapy. Five 250-mg tablets are administered daily on an empty stomach. The drug is very expensive

(\$2,500 to \$3,000 per month), and the fact that the area under the curve can be increased threefold by taking the drug with a high-fat meal has caused some to question whether this pharmacokinetic profile could be safely harnessed to lower costs without compromising clinical outcomes (191).

CYP3A4 is the major lapatinib-metabolizing enzyme and catalyzes the oxidative removal of the *m*-fluorobenzyl moiety. Doses are decreased to 500 mg daily when potent CYP3A4 inhibitors are coadministered, and titrated up to a maximum of 4,500 to 5,500 mg daily (as tolerated) when exposure to potent CYP3A4 inducers is required.



Severe diarrhea is the most common dose-limiting adverse reaction, but potentially fatal hepatotoxicity was the prompt for a black box toxicity warning for this agent. Given that the *o*-chlorinated phenolic metabolite generated by CYP3A4 could generate a quinoneimine as electrophilic and hepatotoxic as the one formed from gefitinib (Fig. 37.52), along with the high (1,250 mg) daily dose, the warning is not surprising. Other serious side effects are cardiovascular in nature and include decreased left ventricular ejection fraction and arrhythmia. The characteristic EGFR kinase-related rash is experienced by approximately 28% of patients taking this drug. Lapatinib's 24-hour half-life is along the lines of other TKIs. Like other drugs in this class, lapatinib inhibits CYP isoforms (CYP3A4 and CYP2C8), P-gp, and BCRP efflux proteins.

Sunitinib Malate Sunitinib is indicated in the treatment of advanced RCC and imatinib-resistant kit-positive GIST. Its introduction in 2006 more than doubled the rate of positive therapeutic outcomes previously achieved with interferon- α or interleukin-2 in RCC patients (40% vs. 2% to 20% response rates) (191). The drug is supplied as 12.5-, 25-, and 50-mg capsules and is administered daily on a 4 weeks on/2 weeks off regimen. Patients are usually started on 50 mg and then titrated up or down in 12.5-mg increments until an acceptable balance of benefit-to-toxicity is achieved. Absorption is independent of food, but could possibly be impacted by body mass index, which could partially explain the wide individual variability in bioavailability (193).

CYP3A4 deethylates sunitinib to an equally active secondary amine metabolite whose serum levels are approximately one-third of the parent drug (191). All precautions outlined for other CYP3A4-vulnerable kinase inhibitors should be taken.

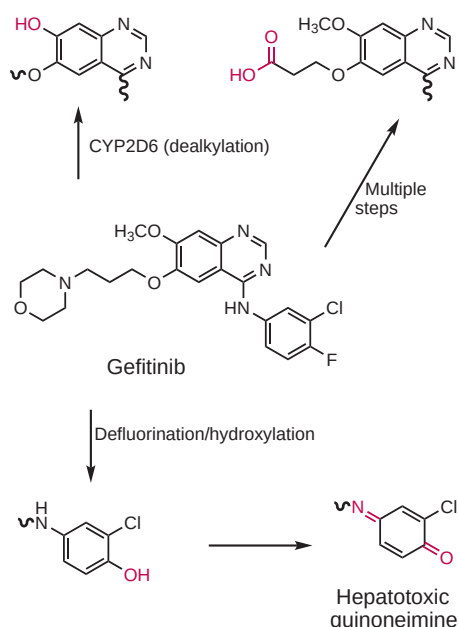
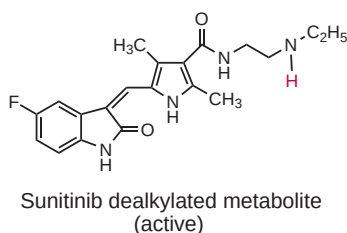


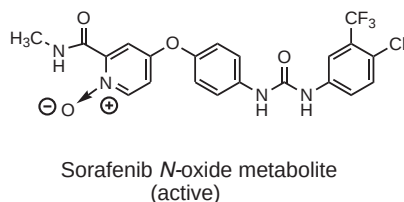
FIGURE 37.52 Gefitinib metabolism.



Like many TKIs, sunitinib is a substrate for P-gp. The terminal half-life of sunitinib and its *N*-deethylated metabolite can extend up to 60 and 100 hours, respectively. The highly conjugated structure imparts a yellow color to the drug and its metabolites, which can be transferred to skin and body fluids. Patients often experience diarrhea, hand-foot syndrome (a consequence of drug leakage from palmar and plantar capillaries), and fatigue. More serious treatment-associated adverse events include potentially fatal hepatotoxicity, left ventricular dysfunction and arrhythmias, and elevated blood pressure. It has been estimated that almost one-fifth of patients on this drug discontinue it, with women and the elderly being at highest risk for adverse events (191).

Sorafenib Tosylate This TKI is indicated for the treatment of advanced RCC and unresectable hepatocellular carcinoma. For either indication, the dosing regimen is one 200-mg tablet twice daily on an empty stomach. Nonlinear kinetics manifest when doses of more than 400 mg/day are administered. Important to its use in hepatocellular carcinoma, doses are cut in half in patients with moderate hepatic impairment, and patients with severe hepatic impairment cannot usually take this drug. Interestingly, when hepatic dysfunction progresses to the very severe stage, 200-mg maximum daily doses are again tolerated (191). Slow tablet disintegration in the GI tract and enterohepatic circulation may contribute to interpatient variability in serum concentrations, and about half of an administered dose of drug will be lost to direct fecal elimination.

Between 9% and 16% of a dose of sorafenib is bio-transformed by CYP3A4-mediated *N*-oxidation to an equally active metabolite, which subsequently undergoes glucuronic acid conjugation. Side effects are similar to those induced by sunitinib. Sorafenib is known to inhibit a number of CYP isoforms including CYP3A4, CYP2D6, CYP2B6, CYP2C19, and CYP2C8, but there appear to be no clinically significant interactions with substrates or modifiers of these enzymes (193,194).

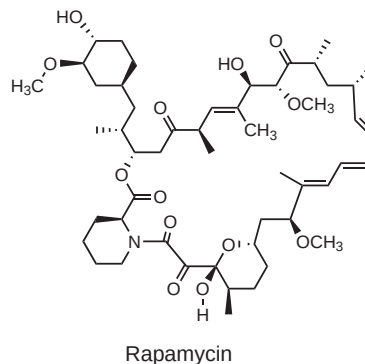


Pazopanib Hydrochloride Like others in its class, pazopanib is indicated in RCC and administered at 800 mg

once daily on an empty stomach. Nonlinear kinetics are observed when doses exceed this maximum. The drug is a substrate for CYP3A4 and P-gp, and it has a half-life of approximately 31 hours. Drug-induced hepatotoxicity can be severe, and patients with moderate hepatic dysfunction should not take more than 200 mg daily. Serious cardiovascular and GI toxicities have also been noted.

mTOR Inhibitors

MECHANISM OF ACTION Mammalian target of rapamycin (mTOR) is a serine/threonine kinase that is regulated through the action of phosphatidylinositol (175). When activated, it phosphorylates kinases that ultimately result in the *de novo* synthesis of proteins (including VEGF) that promote growth. The kinase domain of this 289-kD protein is in the C-terminal area, with the 100-residue macrolide-binding domain (known as FKBP12-FRB) located just to its N-terminal side. The two marketed mTOR inhibitors are O₁₃ analogs of rapamycin, a bacterium isolated from soil (Fig. 37.53).



CHEMISTRY Rapamycin binds tightly to FKBP12 and FRB and, by burrowing into the domain “gap” between these proteins, promotes an interaction that is not observed in the absence of the macrolide. This inhibits the ability of the kinase to function, albeit in a manner not yet understood. Van der Waals interactions with a number of Tyr, Phe, and tryptophan (Trp) residues in the FKBP12

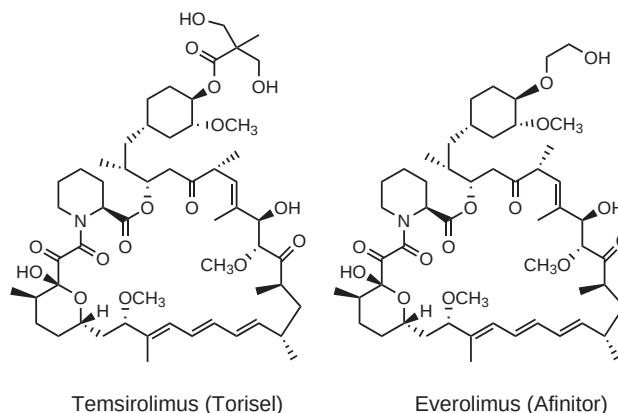


FIGURE 37.53 mTOR inhibitors.

domain along with H-bonds with Asp and Tyr side chains promotes high-affinity binding between inhibitor and enzyme. Hydrophobic interactions between the carbon-rich macrolide and the kinase are known to occur and are the only interactions believed to form with the FBR domain (175). Rapamycin is 92% buried when bound to these two mTOR domains, and only O₁₃ (the site of modification in the commercially available rapamycin-based inhibitors), C₄₀, and C₄₁ (of the cyclohexane ring) contact the surrounding environment. The polar nature of the O₁₃ ester and alcoholic substituents of temsirolimus and everolimus would promote solubility in these aqueous fluids.

SPECIFIC DRUGS (FIG. 37.53)

Temsirolimus Temsirolimus is given as a 25-mg intravenous infusion once weekly to patients with advanced RCC. Patients receiving this agent should be pretreated with 25 to 50 mg of diphenhydramine (or a related antihistamine) 30 minutes prior to administration of the macrolide to minimize the risk of hypersensitivity reactions. In patients still responding positively after 6 months of therapy, the drug should be continued for an additional 6 months or for 2 months after complete remission, whichever occurs first. CYP3A4-mediated metabolism to rapamycin (an active metabolite) demands avoidance of inducers or inhibitors of this isoform if at all possible. If coadministration of these agents is essential, appropriate dosage adjustments should be made.

As an immunosuppressive agent, temsirolimus increases the risk of infection and delays wound healing. It also elevates blood glucose, cholesterol, and triglyceride levels. Potentially fatal adverse events associated with temsirolimus include interstitial lung disease, bowel perforation, renal failure, and cerebral hemorrhage.

Everolimus Everolimus is an orally active rapamycin analog that is supplied as 0.25-, 0.5-, 0.75-, 5-, and 10-mg tablets. Ten-milligram doses are administered once daily in the treatment of advanced RCC resistant to sunitinib or sorafenib, whereas the lower doses (<1 mg) are used to stop rejection of transplanted kidneys. In patients with moderate hepatic dysfunction, the antineoplastic dose is cut to 5 mg once daily.

Unlike temsirolimus, therapy can continue for as long as positive therapeutic outcomes are being achieved. However, like temsirolimus, CYP3A4-mediated metabolism requires appropriate dosage adjustments if strong inducers or inhibitors of this isoform must be coadministered. Major biotransformations involve monohydroxylation, O-dealkylation, and cleavage of the large lactone ring. None of these metabolites are active. Everolimus shares many of the same adverse effects as temsirolimus (immunosuppression, blood glucose elevation, and dyslipidemias) and can also induce angioedema and ulceration of the oral cavity.

Histone Deacetylase Inhibitors

MECHANISM OF ACTION Several structural prototypes can inhibit the deacetylation of Lys residues of the highly basic

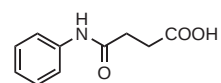
histone proteins involved in regulating gene transcription. Exposure to these inhibitors is associated with an increase in the antioncogene *p21*, induction of cell cycle arrest, and subsequent apoptosis (195). The fact that a wide variety of proteins can undergo acetylation helps explain the significant impact of this therapy on cellular homeostasis (196). These agents are currently used intravenously in the treatment of cutaneous T-cell lymphoma.

SPECIFIC DRUGS (FIG. 37.54)

Romidepsin Romidepsin is a depsipeptide histone deacetylase inhibitor that induces a caspase-dependent apoptosis and retains efficacy in cells overexpressing the prosurvival protein Bcl-2 (197). It is a substrate for CYP3A4 and P-gp. Side effects are, as a rule, manageable, with fatigue, fever, and sepsis being the most commonly observed adverse reactions.

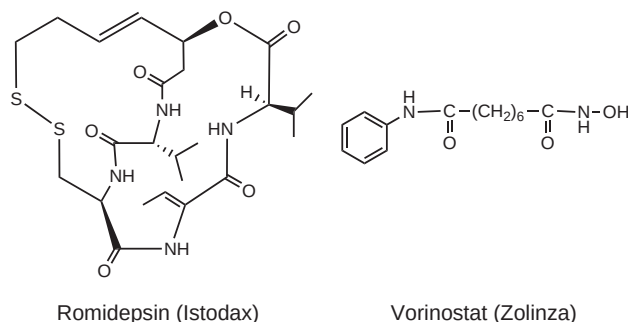
Myelosuppression and GI distress are rare. The belief that this agent induces QT interval prolongation has recently been challenged, although serum electrolytes, particularly potassium and magnesium, should be brought within normal range before instituting therapy with this agent (198).

Vorinostat This hydroxamic acid histone deacetylase inhibitor has an overlapping mechanism with romidepsin, but is not vulnerable to P-gp-mediated efflux and loses its caspase-dependent apoptotic action in cells that overexpress Bcl-2 (197). It is inactivated by direct glucuronic acid conjugation and/or hydrolysis at the hydroxamic acid end of the molecule and subsequent β -oxidation to 4-anilino-4-oxobutanoic acid (199). Side effects are classified as GI, constitutional (e.g., chills, fever), hematologic (thrombocytopenia, anemia), and taste-related. Pulmonary embolism is also a risk of vorinostat therapy.



4-Anilino-4-oxobutanoic acid

Vorinostat is available only to patients who are enrolled in a manufacturer-sponsored program entitled Accessing Coverage Today. Patients must specify their pharmacy so



Romidepsin (Istodax)

Vorinostat (Zolinza)

FIGURE 37.54 Histone deacetylase inhibitors.

that the medication can be provided for distribution by those specific pharmacists. Pharmacists dispensing this drug to approved patients do not require special certification or program enrollment (200).

Immunomodulators

SPECIFIC DRUGS (FIG. 37-55)

Thalidomide Anyone of the “baby boomer” or World War II generation will recognize the name of the drug thalidomide because it induced perhaps the most well-known incident of drug-associated birth defects in recent history (201). Used in the late 1950s and early 1960s as an antiemetic and sedative in pregnant women, more than 10,000 European children were born with flipper-like arms and legs, a condition called phocomelia or “seal limb.” Approximately half of those afflicted survived. Although it is known that the teratogenic and sedative actions of chiral thalidomide are mediated by different stereoisomers (202), the *in vivo* conversion of the therapeutic *R*-isomer to its cytotoxic *S*-enantiomer prohibited the use of this drug in women who were or could become pregnant. Although not licensed in the United States during these early years, the drug was banned by the U.S. Food and Drug Administration (FDA) until 1997 (203).

Despite this troubled beginning, the therapeutic potential of thalidomide, when used wisely and rationally, is significant (204). Its action as a tumor necrosis factor α inhibitor prompted the lifting of the FDA ban to allow its use in the treatment of leprosy. Its antiproliferative and proapoptotic actions have now allowed it to be approved for use in combination with dexamethasone in multiple myeloma (205,206). In addition to arresting uncontrolled cell growth, thalidomide denies multiple myeloma cells access to bone marrow stromal cells and various growth factors needed for tumor cell survival and augments circulating levels of natural killer cells, interleukin-2, and interferon- γ (202). The benefit of adding thalidomide to other chemotherapeutic regimens for multiple myeloma, and in treating other neoplastic disorders, is under active investigation (207).

Thalidomide is marketed as 50-, 100-, and 200-mg capsules. If tolerated, a 200-mg dose is administered at least 1 hour after the evening meal, preferably at bedtime. It undergoes spontaneous hydrolysis of the phthalimide and glutarimide moieties to generate 12 distinct hydrolysis products prior to renal elimination (208). Thalidomide has a minor vulnerability to stereoselective CYP2C19-catalyzed metabolism, and three metabolites

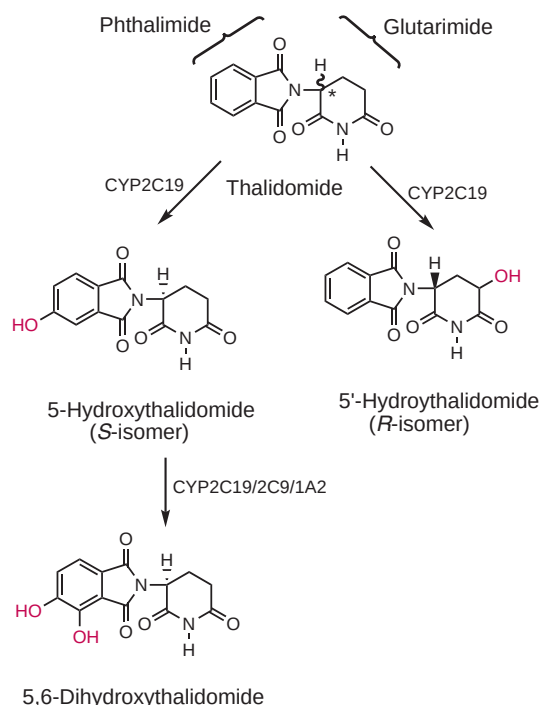


FIGURE 37-56 CYP2C19-mediated thalidomide metabolism.

have been identified (Fig. 37.56). However, the extent of formation shows significant individual variability, and the clinical significance of this potential metabolic pathway *in vivo* is highly questionable.

Thalidomide can only be dispensed by pharmacists or other qualified providers who are registered with the System for Thalidomide Education and Prescribing Safety (STEPS) distribution program (209), and women of childbearing age may be given this drug only if no viable alternative exists, and then only with ongoing weekly (first therapeutic month) and monthly (thereafter) proof of nonpregnant status. Because the drug also finds its way into semen, male thalidomide patients capable of having sexual relations must agree to safety-promoting restrictions on reproductive activities. Use-limiting adverse effects include peripheral neuropathy that can be irreversible (most commonly with chronic use) and thromboembolic events (e.g., deep vein thrombosis, pulmonary emboli). The latter side effect is exacerbated by the coadministered dexamethasone. Other potentially problematic, but less serious, side effects include sedation, fatigue, and constipation (202).

Lenalidomide Lenalidomide has a similar activity profile to its predecessor immunomodulator, thalidomide, but is effective at a little more than one-tenth of the dose (25 mg daily as opposed to 200 mg). The side effect profile is milder, although because the drug is also coadministered with dexamethasone, significant thromboembolic events can still occur. Neutropenia and thrombocytopenia can also be dose-limiting. The potential for serious

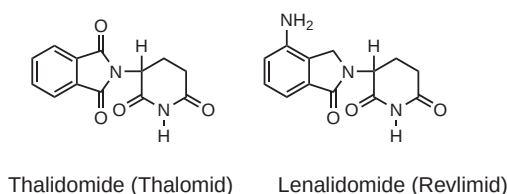


FIGURE 37-55 Thalidomide-based immunomodulators.

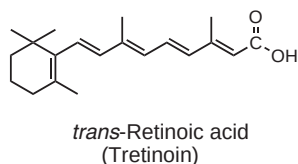
dose-independent teratogenicity is the basis for restricted use in people of reproductive potential. Practitioners dispensing lenalidomide must be registered with RevAssist, a distribution program similar to STEPS (210).

Despite having retained the glutarimide moiety that was a prime target for nonenzymatic hydrolysis in thalidomide, approximately two-thirds of a dose of lenalidomide is excreted unchanged in the urine. The drug is not a substrate for or an inhibitor or inducer of CYP enzymes (211).

Miscellaneous Anticancer Agents

SPECIFIC DRUGS

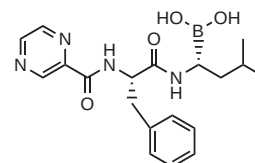
Arsenic Trioxide (Proapoptotic) As noted at the start of this chapter, the toxic effects of arsenic have been recognized for millennia (212). Arsenic trioxide (As_2O_3 , known commercially as Trisenox) was originally introduced into Western medicine in the late 19th century for the treatment of leukemia and then fell out of favor as newer chemotherapeutic and radiation-based approaches to care became available. The drug experienced a pharmacotherapeutic renaissance in the late 20th century, and it is currently used intravenously to induce remission in patients with acute promyelocytic leukemia (APL). APL is characterized by the reciprocal translocation of chromosomes 15 and 17, resulting in the abnormal joining of the promyelocytic gene and retinoic acid receptor α (213). This, in turn, results in the generation of immature leukemia cells, differentiation arrest, and the induction of serious/fatal hemorrhagic and other coagulation-related disorders. Arsenic trioxide induces remission in APL patients through the destruction of the offending fusion protein, promotion of promyelocyte differentiation, and stimulation of apoptosis in malignant cells. One documented chemical apoptotic mechanism is inhibition of intracellular catalase and glutathione peroxidase, with the resultant accumulation of the free radical precursor H_2O_2 . Another is the downregulation of the antiapoptotic protein Bcl-2 (212). Arsenic trioxide's indication specifies use after therapy with an anthracycline and a retinoid (e.g., tretinoin, *trans*-retinoic acid) has failed, but there is evidence to show its efficacy as first-line therapy in newly diagnosed or previously untreated patients (213). Although patient numbers were relatively small, complete remission rates ranging between 73% and 92% have been reported.



The pentavalent As(V) in the parent drug is reduced to the active trivalent As(III) by arsenate reductase. Prior to excretion, As(III) is methylated to mono- and dimethylarsonic acid. Although arsenic trioxide is usually well tolerated, the possibility of QT interval prolongation,

torsades de pointes, and complete atrioventricular block demands an assessment of serum electrolytes (particularly magnesium and potassium) prior to the initiation of therapy. APL differentiation syndrome (also known as retinoic acid syndrome and characterized by pleural/pericardial/pulmonary infiltrates or effusions, dyspnea, weight gain, and fatigue) responds to corticosteroid intervention. Reversible hepatotoxicity has also been noted, and the strong emetogenic potential of As_2O_3 warrants coadministration of drugs to control nausea and vomiting.

Bortezomib (Proteasome Inhibitor) Facilitated by ubiquitin tagging, proteasomes selectively clear cells of cytoplasmic regulatory proteins by cleaving them into short peptides. Inhibition of this homeostatic process induces apoptosis, at least in part through the dysregulation of unfolded protein response gene expression (214) and the inhibition of the activation of antiapoptotic factor nuclear factor- κB by proteasome substrate $\text{I}\kappa\text{B}\alpha$ (215). Bortezomib binds to the 20S proteasome (or “core particle”) component of the 26S proteasome, a highly conserved protein that can cleave proteins essentially anywhere along the chain (216). The binding selectivity is quite high, as the estimated affinity ratio for this proteasome and the next most attractive target protein is over 1,500 (215). The crystal structure of the bortezomib-20S proteasome complex from yeast has been reported, and specific drug-proteasome interactions have been identified (217). The multiplicity of potential binding sites encompassing multiple 20S subunits confirms the complex nature of its activity and may pave the way for the development of more selective inhibitors with higher affinities and more desirable pharmacokinetic properties.



Bortezomib (Velcade)

Bortezomib is used intravenously in the treatment of multiple myeloma and mantle cell lymphoma. The boronic acid moiety is essential, as CYP-mediated deboronation yields a metabolite that does not inhibit the 26S proteasome. The drug is hydroxylated at various positions prior to excretion, and the CYP isoforms involved in the biotransformations include CYP3A4, CYP2C19, and CYP1A2. Like many other anticancer agents, the drug suppresses bone marrow, leading to neutropenia and thrombocytopenia, and can induce serious sensory (and less commonly motor) peripheral neuropathy. Neuropathy can often be managed with dose reduction, although drug discontinuation is required with grade 4 dysfunction (215). Cardiac and pulmonary complications have also been reported.

OTHER CHEMOTHERAPEUTIC APPROACHES

Hormone Therapy

Glucocorticoids (e.g., prednisone and methylprednisolone) are commonly administered with anticancer agents to suppress lymphocytic activity and to enhance the chance of success in the treatment of leukemias and lymphomas. In addition, some tumors, such as estrogen-dependent breast cancer, endometrial cancer, and metastatic cancer of the prostate, depend on the presence of sex hormones for viability. In these neoplastic diseases, the use of steroid receptor antagonists, synthesis inhibitors, or gonadotropin secretion inhibitors, either alone or in combination with other antineoplastic drugs, is a common approach to chemotherapeutic care. Antiestrogens (e.g., tamoxifen), antiandrogens (e.g., flutamide), progestins (e.g., megestrol acetate), aromatase inhibitors (e.g., exemestane), and luteinizing hormone-releasing hormone agonists (e.g., leuprolide acetate) are all available for use in managing hormone-dependent tumors and are often employed after surgery, radiation therapy, and/or other chemotherapy.

Enzyme Therapy

Exogenous asparagine (Asn) is essential to the survival of malignant lymphocytic leukemia cells, because these cells lack asparagine synthetase enzymes. L-Asparaginase (also known as L-asparagine amidohydrolase) or its derivatives can be added to the chemotherapeutic regimen of patients with leukemia to deplete serum Asn by hydrolysis to aspartate and ammonia. Being deprived of this avenue for Asn acquisition, tumor cells die from an inability to synthesize essential proteins. Normal cells, which contain asparagine synthetase, are able to synthesize this essential nutrient and can withstand therapy.

Monoclonal Antibodies

Seven human, humanized, or chimeric monoclonal antibodies (MoAbs), as well as three MoAbs conjugated

with either a cytotoxic calicheamicin-based antibiotic or γ - and/or β -emitting radioisotopes, are also commercially available for the treatment of various neoplasms (Table 37.9) (218). The two murine MoAbs on the market (ibritumomab tiuxetan and tositumomab) have a momab ending in their names (mouse), as opposed to the humanized antibodies, which most commonly end in umab (human). Whether future marketed anticancer MoAbs will support this helpful hint to antibody origin remains to be seen.

MoAbs target tumor cell antigens, and in unconjugated form, the complex recruits endogenous immunomodulator and cell-destroying cytokines (e.g., natural killer cells, macrophages) to stop the advancement of the cancer. The protein components of conjugated MoAbs are homing devices designed to selectively deliver cell-killing chemicals attached to the antibody to tumor cells. It should be noted that some endogenous immunomodulator/biologic response modifying proteins (e.g., interleukin-2) are also available as drug products to treat selected solid and hematologic cancers.

The selected targeting of tumor cells over healthy cells (which do not express the target antigen to the extent malignant cells do) should promote a higher margin of safety for MoAbs when compared to administering toxins that can distribute as widely as their chemical properties allow. The engineering of humanized or chimeric MoAbs has significantly attenuated the risk of therapy-sabotaging immunologic responses to the murine-derived proteins found in the early agents. However, black box warnings for all MoAb anticancer agents except one (ofatumumab) caution against such potentially fatal events as infusion/hypersensitivity reactions, cytopenias, hemorrhage, hepatotoxicity, infection, tumor lysis syndrome, cardiomyopathy, and cardiopulmonary arrest.

The interested reader is directed to several reviews on the clinical use of anticancer MoAbs that have recently appeared in the literature (219–222).

TABLE 37.9 Anticancer MoAbs (217)

MoAb	Drug/Nuclide Conjugate?	Cellular Target	Cancer
Alemtuzumab	No	CD52	B-cell CLL
Bevacizumab	No	VEGF	Colorectal, lung
Cetuximab	No	EGFR	Colorectal, head and neck
Gemtuzumab ozogamicin	Calicheamicin	CD33	Acute myeloid leukemia
Ibritumomab tiuxetan (murine)	^{90}Y (β -emitting isotope)	CD20	NHL
Ofatumumab	No	CD20	CLL
Panitumumab	No	EGFR	Colorectal
Rituximab	No	CD20	NHL
Tositumomab (murine)	^{131}I (γ - and β -emitting isotope)	CD20	NHL
Trastuzumab	No	HER2	Breast

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Kelly Nystrom, PharmD, BCOP

Despite maintaining urinary pH at or above 7.0, DT's 24-hour level of methotrexate (MTX) was reported at 3.57 μM , and leucovorin (prophylactic cells rescue from excessive MTX toxicity) was increased from 15 mg to 50 mg IV every 6 hours. The pharmacist reviewed DT's medications and, noting that he was taking ibuprofen around the clock for headache, discontinued that medication. Subsequently, DL's 48-hour level of MTX dropped to an acceptable 0.03 μM , and he was discharged.

Chemical Analysis

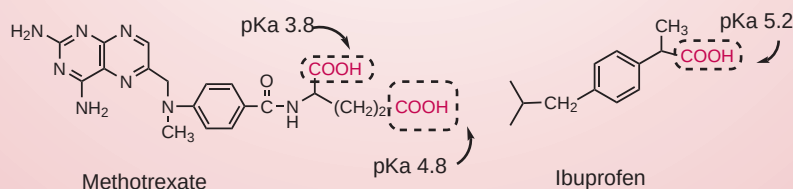
Victoria Roche and S. William Zito

The stronger of methotrexate's two glutamate COOH groups (α -COOH) would exist predominately in water-soluble anionic conjugate base form at DL's urinary pH ($i/u = 1587/1$ at pH 7.0). However, DL's chronic use of ibuprofen is compromising his renal function and allowing MTX to rise to toxic levels in the blood, necessitating additional rescue therapy with leucovorin.

Ibuprofen, an arylpropionic acid nonsteroidal anti-inflammatory drug (NSAID), inhibits COX-1 and COX-2

isoforms of cyclooxygenase that are responsible for the production of prostaglandin. COX-1 serves homeostatic functions in many tissues, including the kidney, and inhibition can result in a serious (e.g., up to 40%) decrease in glomerular filtration rate and potentially fatal renal insufficiency. Ibuprofen also inhibits proximal tubule transporter proteins (MRP2 and MRP4) used by MTX, resulting in a lower renal tubular efflux of MTX. COX-2 selective inhibitors (sulfonamide or sulfone-containing diaryl-heteroaromatic structures) do not alter MTX renal clearance kinetics, but COX-2 selective inhibitors are not used to treat headache.

MTX is approximately 50% bound to serum proteins; its 7-hydroxy metabolite is ~93% protein bound with a small volume of distribution. Ibuprofen's serum protein binding approaches 99%. All drugs are anionic in the bloodstream, contributing to their ability to compete for limited protein binding sites. Displacement of MTX and its cytotoxic 7-hydroxy metabolite from serum proteins secondary to ibuprofen co-therapy can result in abnormally high blood levels of free drug. Because ibuprofen also inhibits the urinary excretion of MTX, life-threatening toxicity results. The toxicity is addressed by increasing the dose of leucovorin ("rescue") and discontinuing use of the offending NSAID.



CASE STUDY

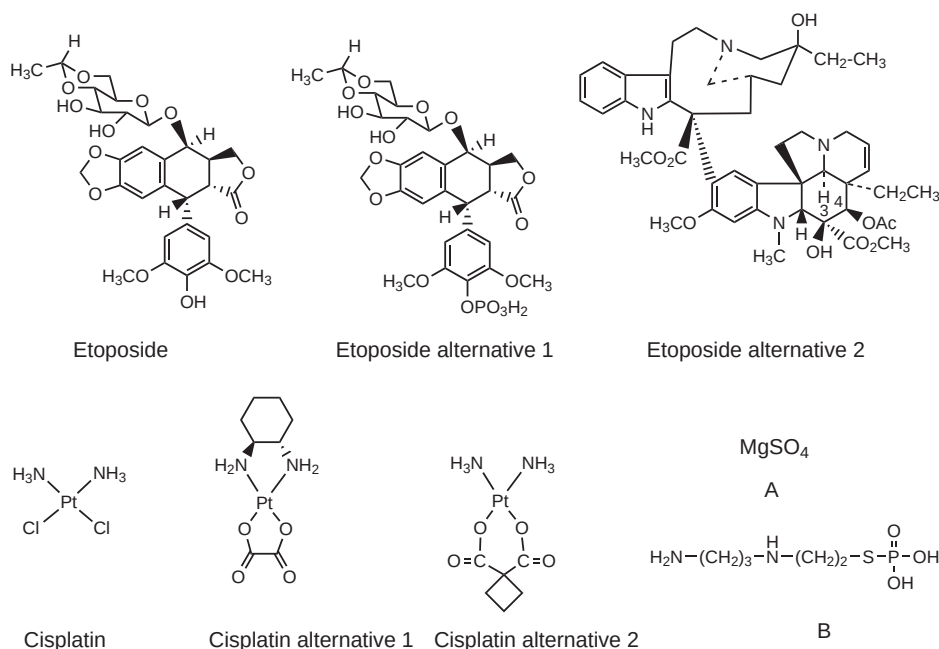
Victoria Roche and S. William Zito

KV is a 48-year-old firefighter who has been called into action over the past several years fighting wildfires in Arizona, Montana, Wyoming, and California. An increased incidence of testicular cancer has been noted among men in KV's line of work, and he received this difficult diagnosis a little more than a month ago. Fortunately his cancer is still in the early stages (stage IB).

KV's oncologist has elected the "BEP regimen," which combines bleomycin, etoposide, and cisplatin. He has been through one round of chemotherapy but is not responding as well as anticipated. KV's oncologist suspects intrinsic resistance to cisplatin. In addition, KV read on the internet about the possibility of hearing loss with this platinum coordination complex, and he has verbalized his fear that this permanent toxic effect could keep him from returning to work once he is (hopefully)

in full remission. He also experienced hypersensitivity to the etoposide infusion, which, while initially managed with an infusion rate adjustment, corticosteroids, and antihistamines, progressed to the point where epinephrine was required to reverse bronchospasm.

The oncologist is consulting with you about appropriate next steps that could be taken to develop a chemotherapy regimen for KV that would give him the best chance for recovery with a high quality of life. A decision is made to change both the etoposide and the platinum complex in the regimen. Which of the structures in each pair of molecules below might be viable alternatives to accomplish the desired outcome? If you made any of these alterations in therapy, would you also recommend the coadministration of either compounds A or B?



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 38

Antiviral Agents and Protease Inhibitors

PATRICK M. WOSTER

*Drugs Covered in This Chapter**

ACYCLIC NUCLEOSIDE ANALOGUES

- Acyclovir
- Adefovir dipivoxil
- Cidofovir
- 6-Deoxyacyclovir
- Fanciclovir
- Ganciclovir
- Valacyclovir

AGENTS AFFECTING THE RIBOSOME

- *Methisazone*

ANTIRETROVIRAL AGENTS: NUCLEOSIDE REVERSE TRANSCRIPTASE INHIBITORS

- Abacavir
- Didanosine
- Emtricitabine
- Lamivudine
- Stavudine
- Tenofovir disoproxil
- Zalcitabine
- Zidovudin

ANTIRETROVIRAL AGENTS: NONNUCLEOSIDE REVERSE TRANSCRIPTASE INHIBITORS

- Delavirdine
- Efavirenz
- Nevirapine

CONVENTIONAL NUCLEOSIDE ANALOGUES

- Cytarabine
- Idoxuridine
- Ribavirin
- Trifluorothymidine
- Vidarabine

FUSION INHIBITORS

- Enfuvirtide
- Maraviroc

HIV INTEGRASE INHIBITORS

- Raltegravir

HIV PROTEASE INHIBITORS

- Amprenavir
- Atazanavir
- Darunavir

- Fosamprenavir
- Indinavir
- Lopinavir
- Nelfinavir
- Ritonavir
- Saquinavir
- Tipranavir

INHIBITORS OF VIRAL ATTACHMENT, PENETRATION, OR EARLY REPLICATION

- Amantadine
- Interferon
- Rimantadine

NEURAMINIDASE INHIBITORS

- Oseltamivir
- Peramivir
- Zanamivir

NON NUCLEOSIDE ANALOGUES

- Boceprevir
- *Fomivirsen*[†]
- Foscarnet
- Telaprevir

Abbreviations

AIDS, acquired immunodeficiency syndrome

ara-HX, arabinosyl hypoxanthine

ARC, AIDS-related complex

AUC, area under plasma concentration–time curve

CCD, catalytic core domain

cdNA, complementary DNA

CMV, cytomegalovirus

CNS, central nervous system

CSF, cerebrospinal fluid

D4T, stavudine

* Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

[†] Fomivirsen has been withdrawn from the U.S. market.

DANA, 2-deoxy-2,3-dehydro-*N*-acetylneuraminic acid

ddI, didanosine

ddC, zalcitabine

DMSO, dimethylsulfoxide

EBV, Epstein-Barr virus

FDA, U.S. Food and Drug Administration

GI, gastrointestinal

HA, hemagglutinin

HAART, highly active antiretroviral therapy

HAV, hepatitis A virus

HBV, hepatitis B virus

HCV, hepatitis C virus

HEV, hepatitis E virus

HHV, human herpesvirus

HIV, human immunodeficiency virus

HPV, human papilloma virus

HSV, herpes simplex virus

HTLV-1, human T-cell leukemia virus 1

IC₅₀, half maximal inhibitory concentration

IV, intravenous

MHC, major histocompatibility complex

NA, neuraminidase

RNAi, RNA interference

RSV, respiratory syncytial virus

RT, reverse transcriptase

siRNA, short interfering RNAs

3TC, lamivudine

VZV, varicella-zoster virus

ZDV, zidovudine

SCENARIO

Douglas Slain, PharmD, BCPS

WM is a 37-year-old white woman who is employed as a nurses' aid in an elder care nursing home. She presents to the hospital with extreme flu-like symptoms and is having difficulty breathing. She is placed in the intensive care unit where she ultimately requires intubation and mechanical ventilation to maintain her respiratory function. WM is also given empiric broad-spectrum antibiotics in case of bacterial pneumonia. A rapid influenza test is positive. She needs anti-influenza therapy, but oral oseltamivir and inhaled zanamivir cannot be reliably delivered given her sedation, intubation, and mechanical ventilation. Many of

the residents of her nursing home had an H1N1 strain of influenza that has the neuraminidase-resistant H274Y mutation. This mutation will drastically reduce the effectiveness of neuraminidase inhibitors that are dependent on C6 hydrophobic binding. Most circulating strains have also been resistant to rimantadine. Treatment options to consider are investigational intravenous zanamivir and investigational intravenous peramivir.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter.)

INTRODUCTION

Viruses are the smallest of the human infectious agents and range in size from about 20 nm to about 300 nm in diameter (1,2). They contain one kind of nucleic acid, either RNA or DNA, as their entire genome, which codes for a variety of enzymes and other proteins used in replication and transmission of the organism. It can be argued that a virus does not qualify as a true life form, since it is nothing more than a nucleic acid strand with associated proteins, and it cannot move on its own power. However, when it attaches itself to a host cell, it internalizes itself and forces the host to make additional copies of the virus, demonstrating a clear reproductive plan. During replication, it uses host cellular biochemicals and processes and, thus in a sense, takes in "nutrients" in order to survive and multiply. In some cases, viruses respond to external conditions and escape the immune response by integrating into the host DNA, demonstrating the ability to respond to external stimuli. Although viruses are simple organisms, they are a significant causative agent for numerous human diseases and, as such, represent one of the major challenges in the area of drug discovery. Agents that are used clinically for a variety of viral diseases act by targeting processes that are specific to the virus, such as a unique viral enzyme or a necessary process such as transcription. However, to date, no drug has been discovered that is truly curative for viral infection. In addition, because viruses have the ability to

undergo mutations, resistance to existing therapies can develop. The discovery of new antiviral agents is thus an important ongoing effort in medicinal chemistry.

VIRUS STRUCTURE AND CLASSIFICATION (1,2)

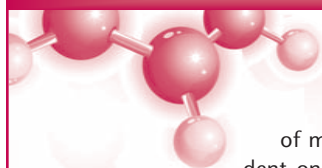
Numerous species of virus that infect bacteria, plants, and animals have been identified, and they exhibit a remarkable range of diversity. All viruses exist as obligate cellular parasites, and as such, they do not possess the complex biochemical machinery that is characteristic of higher organisms. However, they do have a defined macromolecular structure that is designed to protect them from the environment and facilitate their entry into cells. The basic subunit of a virus is its genome, which can be made up of either DNA or RNA. The nucleic acid portion of a virus can be single or double stranded, and may be present in linear or circular form. Viral genomic DNA or RNA is often associated with basic nucleoproteins and may be surrounded by a symmetrical protein known as a capsid. The capsid is made up of repeating structural units known as protomers, which themselves are made up of nonidentical protein subunits. The combination of the nucleic acid core and the capsid is termed the nucleocapsid, and in some cases, this comprises the entire virus. In other cases, the nucleocapsid structure is surrounded by a lipid-containing membrane that is

derived during viral maturation, when the virus undergoes budding through the host cell membrane. The complete virus particle, with or without an envelope, is termed the virion. Viral architecture can be grouped into three types based on the arrangement of morphologic subunits, and each virus exhibits cubic (icosahedral) symmetry, helical symmetry, or has a complex structure. Icosahedral virions are symmetrical structures that contain 20 surfaces, each of which is an equilateral triangle. A sufficient number of capsid structural units must be employed in the icosahedron to make a capsid large enough to encapsulate the viral genome. Morphologic units called capsomeres are seen on the surface of icosahedral virus particles. These structures are clusters of polypeptides, but they do not necessarily correspond to the chemically defined structural units. Some viruses arrange their structural subunits into a standard helical formation. In viruses with helical symmetry, protein subunits systematically bind to the viral nucleic acid, ultimately forming a nucleocapsid helix. The filamentous nucleocapsid is then coiled inside a lipid-containing envelope. Unlike icosahedral virions, regular, periodic interaction between capsid protein

and viral nucleic acid prevents the formation of “empty” helical particles. Finally, some virus particles, such as the large and complex poxviruses, do not exhibit cubic or helical symmetry, but instead form more complicated structures that can be spherical, brick-shaped, or ovoid. A subset of complex viruses is termed pleomorphic, in that they assume multiple complex morphologies. Advances in the visualization of viral capsid structure suggest that the assembly and processing of viral proteins could serve as a new target for antiviral drug development (3).

Viral taxonomy is complex, and viruses are classified according to a number of factors, including morphology, properties of the genome (i.e., DNA vs. RNA, single strand vs. double strand, linear vs. circular, sense vs. antisense), physicochemical properties, structure of associated proteins, replication strategy, and so on. Viruses are separated into major groups called families, with names that end in the suffix *-viridae*, and then into genera that end in *-virus*. Thus, pox viruses are in the family *Poxviridae*, and in the genus *Poxvirus*. A comparison of the genetic and structural features of viral families with members that can infect humans appears in Table 38.1.

CLINICAL SIGNIFICANCE



Antiviral agents include diverse compounds with varied actions.

A thorough clinical understanding of most of these compounds is dependent on a basic appreciation of biochemistry and medicinal chemistry.

Viruses are mainly composed of either RNA or DNA nucleic acid strands. As such, one of the most ideal targets for treating viral pathogens has been the inhibition of viral replication. Successful viral replication is dependent on enzyme-mediated transcription of viral RNA or DNA. A significant number of antiviral agents, including HIV nucleoside reverse transcriptase inhibitors, and a majority of antiherpes agents are designed to be “false” substrates of viral transcription enzymes. It is through competitive inhibition that these agents are able to fulfill their intended effect. Their chemical structures are similar to naturally occurring substrate nucleoside purines (adenosine and guanosine) and pyrimidines (thymidine and cytidine). However, as they are purposely modified in key positions with functional groups designed to foil polymerase enzymes, their incorporation will lead to termination of replication. While these nucleoside antiviral agents are often actively carried into cells via specific nucleoside transporting proteins, they typically require triphosphorylation to become active intracellularly. It is important to recognize that resistance to these antiviral medications can develop when genetic mutations alter the ability for viral enzymes to phosphorylate the drugs. Drugs like cidofovir and tenofovir are nucleotide analogs, which means they are already monophosphorylated. As a result,

these agents can maintain activity against viruses that have developed resistance to other nucleoside agents through certain resistant mutations.

In clinical practice, HIV antiretroviral regimens usually require a pair of nucleoside reverse transcriptase inhibitors in the regimen. It is of paramount importance that combinations of “like” nucleosides analogs (e.g., two thymidine, two cytosine) are not used together as they have displayed antagonism and reduced viral load suppression. Nonnucleoside reverse transcriptase inhibitors are not structurally related to nucleic purines and pyrimidines; therefore, they do not act as substrates of the target enzyme.

Protease inhibitors, which are the most potent antiretroviral agents available, were the end result of coordinated chemical design and structure-based computational analysis of the protease enzyme. With identification of the protease crystalline structure and identification of active binding sites, scientists were able to create compounds that would fit the protease enzyme with strong affinity, and cause an inhibition of protease function. The clinical impact of these man-made drugs is regarded by many as the greatest single advance in the treatment of HIV infection.

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TABLE 38.1 Characteristics of Virus Families Containing Members That Infect Humans

Family	Examples	Genome	Capsid Symmetry	Size (nm ⁻¹)	Envelope	Diseases
Parvoviridae	Parvovirus B19	ssDNA sense or antisense	Icosahedral	18–26	No	Erythema infectiosum (fifth disease); polyarthralgia arthritis; aplastic crisis, anemia
Papillomaviridae	Human papilloma (wart) virus; polyoma virus; SV 40	ds, circular DNA	Icosahedral	55	No	Warts; salivary gland infection; multifocal leukoencephalopathy; tumors (i.e., cervical)
Adenoviridae	Multiple types (40 adenoviruses and mastadenovirus)	dsDNA	Icosahedral	70–90	No	Infections of the eye and respiratory tract; tumors
Hepadnaviridae	Hepadnavirus, hepatitis B virus	dsDNA, circular, one ss region	Icosahedral	40–48	Yes	Hepatitis B; tumors
Herpesviridae	Herpes simplex I and II; varicella zoster; herpes zoster; cytomegalovirus; Epstein-Barr virus	dsDNA	Icosahedral	150–200	Yes	Eye, skin and genital infection; chickenpox; shingles; mononucleosis; tumors
Poxviridae	Variola; vaccinia	dsDNA	Complex	230–400	Complex	Smallpox; cowpox; chickenpox; tumors
Picornaviridae	Hepatitis A virus; poliovirus; enterovirus; rhinovirus, coxsackie virus A and B	ssRNA, sense	Icosahedral	28–30	No	Respiratory diseases; gastrointestinal diseases; polio; aseptic meningitis
Astroviridae	Astrovirus	ssRNA, sense	Icosahedral	28–30	No	Diarrhea in infants and immunocompromised patients
Caliciviridae	Norwalk virus	ssRNA, sense	Icosahedral	27–40	No	Epidemic gastroenteritis
Togaviridae	Rubella virus; alphavirus, arbovirus	ssRNA, sense	Icosahedral	50–70	Yes	Measles (rubella)
Flaviviridae	Hepatitis C virus; arbovirus; yellow fever virus; dengue virus; West Nile virus	ssRNA, sense	Complex	40–60	Yes	Hepatitis C; yellow fever; dengue fever; encephalitis; tumors
Coronaviridae	Coronavirus	ssRNA, sense	Complex	120–160	Yes	Colds; gastroenteritis in infants; SARS
Retroviridae	HIV I and II; lentivirus; human T-cell lymphotropic viruses	ssRNA as dimer	Complex	80–100	Yes	AIDS; AIDS-related complex; breast cancer; human T-cell leukemia; nasopharyngeal carcinoma
Arenaviridae	Arenavirus	ssRNA, antisense	Complex	50–300	Yes	Lassa fever; hemorrhagic fever; choriomeningitis
Orthomyxoviridae	Influenza virus A, B, C	ssRNA, antisense	Helical	80–120	Yes	Influenza
Bunyaviridae	Hantavirus	ssRNA, antisense	Helical	80–120	Yes	Hemorrhagic fever
Rhabdoviridae	Rhabdovirus; rabies virus; encephalitis virus	ssRNA, antisense	Helical	75–180	Yes	Rabies; encephalitis
Paramyxoviridae	Syncytial virus; parainfluenza virus	ssRNA, antisense	Helical	150–300	Yes	Mumps; measles (rubeola)
Filoviridae	Marburg virus; Ebola virus	ssRNA, antisense	Helical	80–800	Yes	Marburg viral fever; ebola hemorrhagic fever
Reoviridae	Reovirus; rotavirus; orbivirus	dsRNA in 10–12 pieces	Icosahedral	60–80	No	Mild respiratory and gastrointestinal infection; Colorado tick fever
Prion	Prion proteinaceous material	None	NA	NA	No	Bovine spongiform encephalopathy; Creutzfeldt-Jakob disease

ds, double stranded; NA, not applicable; SARS, severe acute respiratory syndrome; ss, single stranded.

VIRAL REPLICATION, CELLULAR EFFECTS, AND PATHOGENESIS (2,4)

As mentioned earlier, all viruses exist as obligate intracellular parasites, and as such, they rely on the cellular machinery of the host for their growth, development, and replication. In order to synthesize the proteins needed for viral replication, the organism must be capable of producing usable mRNA in sufficient quantities to compete with host mRNA for protein synthesis. During viral replication, all of the macromolecules required by the virus are synthesized in a highly organized sequence. The replication cycle (Fig. 38.1) begins when the intact virion binds to a host cell through electrostatic adsorption to a specific “receptor” site. This process is known as the attachment phase. Attachment is most likely a fortuitous event resulting from structural complementarity between the exterior structure of the virion and a normal cell surface structure on the host cell. For example, HIV binds to the CD4 receptor on cells of the immune system, rhinoviruses bind ICAM-1, and Epstein-Barr virus recognizes the CD21 receptor on B cells. Recent studies suggest that some mammalian cells can develop proteins that restrict the binding of viral capsid structures, thus preventing entry into noninfected cells (5). When attachment has been achieved, the virion enters the penetration phase, the process by which it gains entry into the host cell. Penetration may occur by receptor-mediated

endocytosis, fusion of the viral envelope with the cell membrane, or in some cases direct penetration of the membrane. Following penetration of the cell, viruses must be uncoated, resulting either in the naked nucleic acid or in the nucleocapsid form, which usually contains polymerase enzymes. After they have been uncoated, viruses are no longer infectious.

Once the virus has penetrated the cell and uncoated, it enters a segment of its life cycle known as the eclipse period, the length of which varies with the type of virus. It is during this time that the virus utilizes host resources to replicate and produce necessary viral proteins. Cells that can support viral reproduction are termed permissive, and as a result, the infection is known as productive, since it results in new viral particles. When new infectious viral particles are produced, host cellular metabolism may be completely directed to the production of viral products, resulting in destruction of the cell. In other cases, host cell metabolism is not dramatically altered, and the infected cell can survive. During viral reproduction, up to 100,000 new virions can be produced, and the replication cycle can vary from a few hours to more than 3 days. Some cells types, called nonpermissive, are unable to support the reproduction of an infective virion, resulting in an abortive infection. Abortive infections also occur when the virus itself is defective. Either situation can lead to a latent infection, where the viral genome may persist in a surviving host cell. As will be described below, such

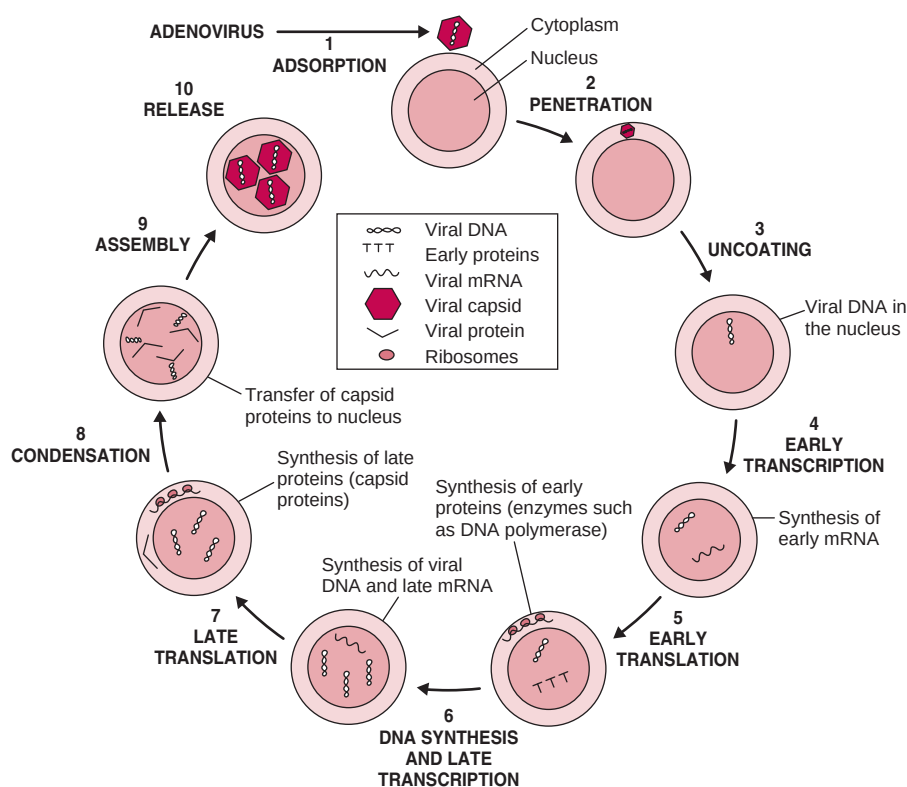


FIGURE 38.1 Steps involved in the viral life cycle. (Used with permission from Brooks GF, Butel JS, Morse SA, et al. Jawetz, Melnick and Adelberg's Medical Microbiology, 23rd Ed. New York: McGraw-Hill Companies, 2004.)

an infection can lead to the transformation of a cell from normal to malignant.

DNA Virus

The strategies used by various viruses to replicate vary widely, but all are characterized by the need to transcribe mRNA that is suitable for translation of viral proteins. There are several pathways leading to the required mRNA, after which the host enzymes and raw materials are used to make viral proteins. Early viral proteins used in replication are synthesized immediately after infection, while late proteins used to produce the complete virion structure are synthesized after viral nucleic acid synthesis. Most DNA viruses contain double-stranded DNA as their genome and thus can replicate using host cell machinery to produce mRNA directly. Papillomavirus, adenovirus, and herpesvirus are replicated in the host nucleus, and thus use transcriptional enzymes of the host (i.e., DNA-dependent RNA polymerase) to synthesize mRNA. This mRNA is then translated to form proteins needed by the virus, including enzymes (e.g., DNA-dependent DNA polymerase) used to produce progeny DNA copies. These progeny DNA strands are infectious. By contrast, poxviruses replicate in the cytoplasm using a mechanism that is not well understood, wherein the genome is initially transcribed by a viral enzyme in the virion core. Parvoviruses contain a single-stranded DNA genome, and must synthesize double-stranded DNA in the host nucleus prior to synthesis of mRNA and translation of proteins. This process may or may not require a helper virus such as herpes simplex. The hepatitis B viral genome, comprised of double-stranded DNA, contains numerous gaps that must be repaired using a DNA polymerase packaged in the virion before transcription to form mRNA.

RNA Virus

Compared to DNA viruses, those viruses with RNA-based genomes have evolved a wide variety of reproductive strategies. The single-stranded RNA viruses may be divided into three groups that differ in the method by which the RNA genome is utilized. In all three groups, the RNA genome must serve two functions: to be translated to form protein and to be replicated to form progeny RNA strands. The first group is comprised of viruses such as picornaviruses, flaviviruses, and togaviruses that have an RNA genome that can be used directly as mRNA. Viral RNA that can be used as mRNA is by convention termed (+) or sense-strand RNA. In most cases (e.g., picornaviruses), this sense-strand RNA binds to the host ribosome shortly after entering the cell, where it is read and used to produce a single polypeptide called the polyprotein. The polyprotein is then processed by autocatalysis and various proteolytic enzymes to produce the required viral proteins. In some cases (e.g., togaviruses), only a portion of the RNA genome is available to

be translated by the host ribosome. Following the initial translation of the sense strand, it serves a second function, namely to serve as a template for the synthesis of a (–) or antisense strand via an RNA-dependent RNA polymerase. This antisense strand then can be used to produce additional sense-strand RNAs that are infectious and can also serve as mRNA. These progeny sense RNA strands are then packaged into an intact virion prior to transmission to another host cell.

The second group of single-stranded RNA viruses, including orthomyxoviruses, bunyaviruses, arenaviruses, paramyxoviruses, filoviruses, and rhabdoviruses, all contain an antisense RNA genome that can only be used for transcription of new RNA. All antisense RNA viruses contain an RNA transcriptase as part of their virion, because the host cell does not have this type of RNA-dependent RNA polymerase. In the first round of genome expression, a series of short sense-strand RNAs are made and then translated to form the required viral enzymes for replication. Ultimately, these enzymes are used to produce a full-length sense RNA strand, which is then used to make multiple copies of the antisense viral genome. The progeny antisense DNA strands by themselves are not infectious, because they have not yet been packaged with the required RNA transcriptase. When the progeny antisense RNA has been synthesized, it is packaged into an intact virion, in which form it becomes infectious prior to transmission to another cell.

The third group of RNA viruses is the retroviruses, in which single-stranded RNA exists as a dimer of a sense and antisense strand. The genomic RNA strands can be base-paired, although the structure of this complex is not well understood, or the strands can be hydrogen bonded to other macromolecules in the virion. Retroviral genomic RNA serves a single function, namely, to act as a template for the formation of double-stranded viral DNA. Host cells do not contain an enzyme that can form DNA from viral RNA, and thus the virion of a retrovirus must contain a reverse transcriptase (RT) enzyme, as well as various host tRNA molecules. Transcription of the genome begins when a complex of RT and tRNA binds to the viral genome. A complementary DNA strand is then synthesized using one of the host tRNAs as a primer, and the original RNA strand is digested by RNase H, and viral ribonuclease packaged in the virion. A complementary DNA strand is then synthesized, and the resulting double-stranded DNA is translocated to the nucleus, where viral enzymes incorporate genome-length viral DNA into the host genome. In some cases, the viral portion of the genome can remain dormant for long periods, or it may be immediately used to make progeny viral RNA, a process that is catalyzed by host RNA polymerase II. Transcription produces both shortened segments that are used to make polyproteins and full length progeny RNA. The polyproteins are processed to form various viral proteins, while the full-length RNA is packaged into an infectious virion.

Virus Protein

In addition to replication of the viral genome, a number of other structures associated with the complete virion can also be made. A number of viral proteins may be synthesized that have important functions in the structure, transmission, and survival of the virus. These proteins can protect the genetic material in the virus from destruction by nucleases, participate in the attachment process, and provide structure and symmetry to the virion. In addition, in certain cases where the virus requires an enzymatic process for which there is no host enzyme, a virion may include enzymes such as RNA polymerases or an RT. Some viruses require a lipid envelope that contains transmembrane proteins specifically coded for by the virus and that envelops the genetic material during viral budding. Viral envelopes contain glycoproteins that are involved in cell recognition during attachment to the host cell. These glycoproteins often reflect the composition of glycoproteins in the host cell. They are a determinant of the antigenic nature of viruses, and thus facilitate recognition by the immune system of the host. However, depending on their composition, they can also help the virus elude neutralization by the immune system.

Cellular Egress

Viruses use one of two strategies for exiting infected cells. Nonenveloped viruses (picornaviruses, rheoviruses, etc.) complete their maturation by assembling into their corresponding virion within the cell nucleus or the cytoplasm. For example, picornaviruses assemble by clustering 60 copies of each of three viral proteins, called VP0, VP1, and VP3, into a structure called a procapsid. Viral RNA is then packaged into the procapsid, and proteolytic cleavage of VP0 produces two new viral proteins called VP2 and VP4. The resulting conformational change produces a stable and symmetrical structure that shields the genome from degradation by host nucleases. In most cases, destruction of the host cell is required when the virion exits. Enveloped viruses (all antisense RNA viruses, togaviruses, flaviviruses, coronaviruses, hepadnaviruses, herpesviruses, and retroviruses) contain proteins that carry signal sequences and markers that cause them to be inserted into the inner and outer surface of the host cell cytoplasmic membrane. Viral proteins on the outer surface are glycosylated using host enzymes and then displace host cell surface proteins and collect into patches. Viral nucleocapsids recognize proteins on the inner surface of the membrane, where they bind and are engulfed by the patch area of the membrane. The completed virion exits the cell by budding and release into the extracellular space. Viral egress can have a variety of effects on the host cell, ranging from destruction of the cell to minimal noncytolytic effects. Herpesviruses differ from other enveloped viruses in the manner in which they form their envelope. The nucleocapsid is formed in the nucleus, and final maturation of the virion occurs only on the inner surface of the host cell membrane, forming

vesicles that are stored in between the inner and outer aspect of the cell membrane. Egress of the herpesvirus vesicle always occurs through destruction of the host cell.

Virus Pathogenesis

A complete discussion of viral pathogenesis is beyond the scope of this chapter. However, in general it may be considered that the symptoms of a viral infection arise from the response to viral replication and cell injury in the host. These responses range from asymptomatic or subclinical to severe clinical manifestations, and may be either local or systemic. Understanding the biochemical events that produce viral diseases can aid in the design of effective and specific therapies. Not surprisingly, viral pathogenesis occurs in distinct steps: 1) viral entry into the host and primary viral replication; 2) viral spread; 3) cellular injury and host immune response; 4) viral clearance or establishment of persistent infection; and 5) viral shedding. Most viruses enter the host through the respiratory or gastrointestinal tract, but may also penetrate the skin, urogenital tract, or conjunctiva. In a few cases, virus particles can enter through direct injection (e.g., HIV and hepatitis) or through insect bites (arboviruses). When a local infection occurs, the virus replicates near the site of entry, and the underlying tissue is not affected. However, some viruses are able to migrate to other sites, usually through the bloodstream or lymphatic system, to produce systemic infections. When infection occurs at a remote site, most viruses demonstrate tissue or organ preference (e.g., herpesvirus localizes in nerve ganglia, and the rabies virus migrates to the CNS). Localization of a virus in a particular tissue can be the result of cell receptor specificity or can arise because a virus may be activated by proteolytic enzymes in a specific cell type.

Clinical disease develops through a complex series of events when virus-infected cells are destroyed or their function is impaired, and some symptoms such as malaise and anorexia can result from host responses such as cytokine release. Disease-mediated damage may become chronic when cell types that do not regenerate (e.g., brain tissue) are involved. Ultimately, the host either succumbs to the infection; develops a chronic, latent, or subclinical infection; or completely recovers. In chronic infections, the virus can be continuously detected, at low levels, and either mild or no clinical symptoms may present. By contrast, latent infections are those in which the virus persists for extended periods of time in an inactive form or a location not exposed to the immune response. Intermittent flare-ups of clinical disease can occur, during which time infectious virus can be detected. Subclinical infections are those that give no overt sign of their presence. Humoral and cell-mediated immunity, interferon and other cytokines, and other host defense factors, depending on the type of virus, are common mediators of recovery, and begin to develop very soon after infection. Infiltration with mononuclear cells and lymphocytes is responsible for the inflammatory reaction in uncomplicated viral lesions. Virus-infected cells can be lysed by

T lymphocytes through recognition of viral polypeptides on the cell surface, and humoral immunity protects the host against reinfection by the same virus. Neutralizing antibodies that are directed against capsid proteins can prevent viral infection by disrupting viral attachment or uncoating. Interestingly, viruses have evolved a variety of survival tactics that serve to suppress or evade the host immune response. Because viruses are obligate intracellular parasites, a method of transmission from one host to another is required for survival of the species. Thus, during an active infection, shedding of the infectious virion into the environment is a required step in the life cycle of the virus and ensures transmission of the virus to new hosts. Shedding usually occurs at the same site where the infection was initiated and can occur at various stages of the disease course.

VIRAL DISEASES (6,7)

HIV

The human immunodeficiency virus (HIV-1) was first identified in 1979 and was found to be the cause of acquired immunodeficiency syndrome (AIDS) in 1981 (8). Since that time, AIDS had become a serious worldwide epidemic that continues to expand. The Joint United Nations Program on HIV/AIDS estimated that by the end of 2005, a total of 40.3 million people worldwide were living with HIV/AIDS, the majority having been infected by heterosexual contact. After four decades, the number of infected individuals has begun to drop slightly, and that number stands at 33.3 million in 2010. Only 5 million of these patients are receiving treatment for the disease. In 2005, more than 3.1 million people died of AIDS, and 4.9 million new cases of HIV were diagnosed, including more than 700,000 children (9). The number of new infections dropped to 2.9 million in 2009, and annual deaths decreased to 1.8 million (10). The incidence of the disease varies by location, with sub-Saharan Africa having the highest incidence of the disease. In the Third World, HIV infection is often comorbid with neglected tropical diseases (trypanosomiasis, leishmaniasis), malaria, and tuberculosis, causing a higher level of mortality (11). Because it is sexually transmitted, a high percentage of infected individuals are young adult workers, and as such, the disease has a significant economic impact in some regions. In addition, infected mother-to-fetus transmission occurs between 13% and 40% of the time. Although a variety of drugs have been developed for treating AIDS patients, none have proven successful in curing the disease. The basic difficulty experienced with this viral infection is the ability of virus to mutate, leading to rapid drug resistance.

The HIV-1 genome consists of two identical 9.2-kb single-stranded RNA molecules within the virion, each of which contains information for only nine genes. Following infection of the host cell, the persistent form of the HIV-1 genome is proviral double-stranded DNA.

Mature HIV virions are spherical and consist of a lipid bilayer membrane surrounding a nucleocapsid that contains genomic RNA, a viral protease, RT, an integrase, and some other cellular factors. The HIV life cycle is depicted in Figure 38.2, and begins when the viral extracellular protein gp120 attaches to the CD4 receptor on T lymphocytes. Following attachment, the viral envelope and host cell membrane are fused, and the nucleocapsid is released into the cytoplasm. The nucleocapsid is uncoated, and the resulting RNA serves as a substrate for RT, producing a proviral double-stranded DNA that migrates to the nucleus and is incorporated into host DNA by integrase. This DNA is not expressed in resting

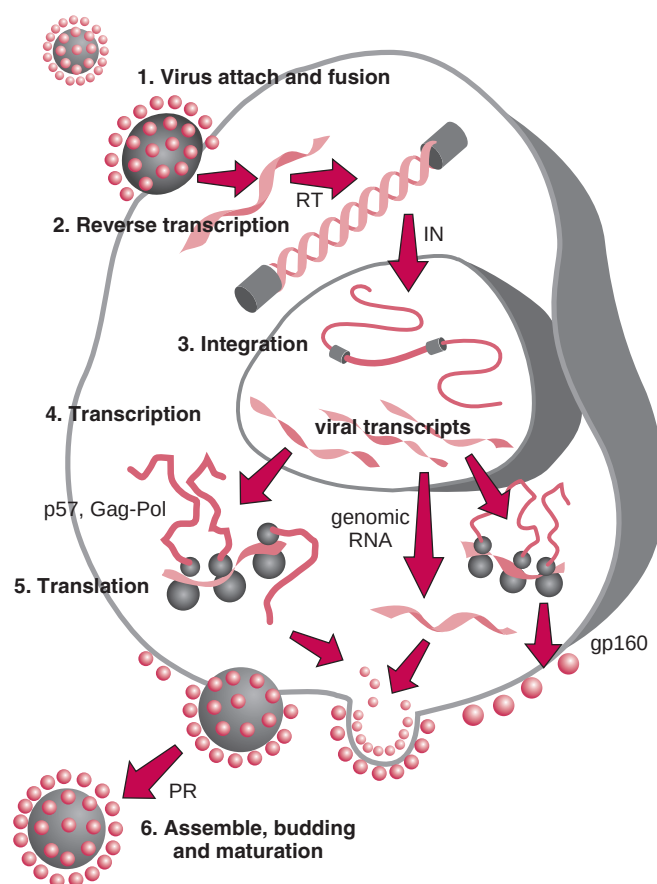


FIGURE 38.2 Replicative cycle of HIV. (1) The virus gp120 protein binds to CD4, resulting in fusion of the viral envelope and the cellular membrane and the release of viral nucleocapsid into the cytoplasm. (2) The nucleocapsid is uncoated, and viral RNA is reverse transcribed by reverse transcriptase (RT). (3) The resulting double-stranded proviral DNA migrates into the cell nucleus and is integrated into the cellular DNA by integrase (IN). (4) Proviral DNA is transcribed by the cellular RNA polymerase II. (5) The mRNAs are translated by the cellular polysomes. (6) Viral proteins and genomic RNA are transported to the cellular membrane and assemble. Immature virions are released. Polypeptide precursors are processed by the viral protease (PR) to produce mature viral particles. (Used with permission from Tyler KL, Fields BN. *Fields Virology*, 2nd Ed. New York: Raven Press, 1990:191–239.)

T lymphocytes, but when the cell is activated, proviral DNA is transcribed by host RNA polymerase II. The viral RNA and proteins are transported to the cell membrane, where they assemble, form a viral bud, and are released from the lymphocyte membrane. This produces an immature virion, and processing of the viral surface proteins by HIV protease then affords the mature, infectious virus.

About 50% of HIV infections are asymptomatic, whereas the other 50% produce flu-like symptoms within 4 weeks. During this initial phase, viral titers are very high and decline as specific antibodies are formed. The latent period is mediated by factors that are not well understood and can last several years. During this time, viral replication continues, and the level of CD4-positive T lymphocytes steadily declines. Eventually, the patient immune system becomes compromised, resulting in a variety of opportunistic infections. RT sometimes makes mistakes reading the viral RNA sequence, leading to mutations in the virus and changes in the structure of the surface proteins. Thus vaccines, which induce the production of antibodies that recognize and bind to very specific viral surface molecules, are unlikely to be effective in HIV therapy. As will be discussed below, other viral processes, such as reverse transcription and proteolytic processing, are viable targets for small-molecule therapy.

Kaposi sarcoma is a common complication of AIDS and has been shown to arise from a complex interaction between HIV and the human herpesvirus 8 (HHV-8). This disease presents as a reddish or purple lesion on various areas of the skin, and in advanced state may involve the lungs or gastrointestinal tract. HHV-8 does not work alone, but rather in combination with a patient's altered response to cytokines and the HIV-1 transactivating protein Tat, which promotes the growth of endothelial cells. HHV-8 can then encode interleukin-6 viral proteins, specific cytokines that stimulate cell growth in the skin. HHV-8 destroys the immune system further by directing a cell to remove the major histocompatibility complex (MHC-1) proteins that protect it from invasion. These proteins are then transferred to the interior of the cell and are destroyed, leaving the cell unguarded and vulnerable to pathogens that would normally be cleared by the immune system. It has recently been discovered that xCT, the 12-transmembrane light chain of the human cystine/glutamate exchange transporter system, serves as a receptor for internalization of HHV-8 (12).

Herpesvirus (2)

The herpesvirus family contains several of the most important human pathogens, which are responsible for causing a spectrum of common diseases. A common and significant feature of the herpesviruses is their ability to establish lifelong persistent infections in their hosts and to undergo periodic reactivation. Herpesviruses possess a large number of genes, and some of the resulting proteins are targets for antiviral chemotherapy. The herpesviruses that commonly infect humans include herpes

simplex virus (HSV) types 1 and 2, varicella-zoster virus, cytomegalovirus, Epstein-Barr virus, human herpesviruses 6 (HHV-6) and 7 (HHV-7), and Kaposi sarcoma-associated herpesvirus HHV-8 (also known as KSHV). Herpesviruses are large viruses that have identical morphology and consist of a core of linear, double-stranded DNA surrounded by a protein coat that exhibits icosahedral symmetry and has 162 capsomeres. The genome is large and encodes at least 100 different proteins, including polypeptides involved in viral structure, the viral envelope, and enzymes involved in nucleic acid metabolism, DNA synthesis, and protein regulation. Oral cold sores and genital herpes infections are caused by HSV1 and HSV2, respectively. These viruses can establish a latent infection in the ganglia of nerves that supply the site of the primary infection, and the latent disease is reactivated by a number of stress factors. It is estimated that virtually 100% of adult humans are infected with HSV1, although many infections are subclinical and asymptomatic. The varicella virus is the cause of chickenpox, and the herpes zoster virus is responsible for shingles. Human cytomegalovirus rarely causes disease in healthy people, but when infection occurs in adulthood, it may cause an infectious mononucleosis-like illness. Primary infection with the Epstein-Barr virus is the cause of infectious mononucleosis, and this virus is thought to be a factor in the development of Burkitt lymphoma and other malignancies. HHV-6 is thought to cause roseola and mononucleosis, whereas HHV-7 is probably not involved in any human diseases. The role of HHV-8 in the pathogenesis of Kaposi sarcoma has been discussed earlier.

Hepatitis (2)

Viral hepatitis is a systemic disease, but primarily involves the liver. Hepatitis A virus (HAV) is responsible for infectious hepatitis, hepatitis B virus (HBV) is associated with serum hepatitis, and hepatitis C virus (HCV) is a common cause of posttransfusion hepatitis. Another viral agent, hepatitis E virus (HEV), causes an enterically transmitted form of hepatitis. On occasion, disease can arise from hepatic infection with yellow fever virus, cytomegalovirus, Epstein-Barr virus, herpes simplex virus, rubella virus, and the enteroviruses. Viral hepatitis usually involves acute inflammation of the liver, fever, nausea, vomiting, and jaundice, and all forms of hepatitis produce identical histopathologic lesions in the liver during acute disease. HAV is a member of the picornavirus family and carries a single-stranded RNA genome; only one strain of the virus exists. The onset of HAV hepatitis occurs within 24 hours, in contrast to the slower onset of clinical symptoms with HBV and HCV infection. Complete recovery occurs in most HAV cases, and chronic infection never occurs. HBV is classified as a hepadnavirus with a double-stranded circular DNA genome. The outcome of HBV infection ranges from complete recovery to progression to chronic hepatitis and, rarely, death. HBV establishes chronic infections, especially in infants; 80% to 95% of infants and young children infected with HBV become

chronic carriers and are at high risk of developing hepatocellular carcinoma. In adults, 65% to 80% of infections are asymptomatic, and 90% to 95% of all patients recover completely. HCV is a positive-stranded RNA flavivirus and exists in at least six major genotypes. Most cases of posttransfusion hepatitis are caused by HCV, and these infections are usually subclinical with minor elevation of liver enzymes and a low incidence of jaundice. However, 70% to 90% of HCV patients develop chronic hepatitis, and many are at risk of progressing to chronic active hepatitis and cirrhosis decades later. HEV is transmitted enterically and occurs in epidemic form in developing countries, where water supplies are sometimes contaminated with feces. The disease is more severe in adults than in children, who are usually asymptomatic.

Influenza (2)

Respiratory illnesses commonly known as colds and flu account for more than half of all acute illnesses in the United States each year. Influenza viruses belong to the Orthomyxoviridae family and are a major source of morbidity and mortality caused by respiratory disease. Outbreaks of infection can occur in worldwide epidemics that have resulted in millions of deaths worldwide. Genetic mutations often cause antigenic changes in the structure of viral surface glycoproteins, making influenza viruses extremely difficult to control. Three immunologic types of influenza viruses are known and are termed influenza A, B, and C. Antigenic changes are very common in the type A group of influenza viruses, which are responsible for the majority of influenza epidemics. Influenza type B undergoes more infrequent antigenic changes and is less often the cause of an influenza epidemic, whereas influenza type C is antigenically stable and causes only mild illness in immunocompetent individuals. The viruses carry a single-stranded, negative-sense RNA genome that has eight segments in influenza A and B. Influenza C viruses contain only seven segments of RNA and lack a neuraminidase gene (see below). The complete virion in each type contains nine different structural proteins. A nucleoprotein associates with viral RNA to form a ribonucleoprotein structure that makes up the viral nucleocapsid. Three other large proteins are bound to the viral ribonuclear protein and are responsible for RNA transcription and replication. A matrix protein is also included in the virion that forms a shell underneath the viral lipid envelope and comprises about 40% of all viral protein.

The influenza virion structure also includes a lipid envelope derived from the host cell. This envelope contains two viral surface glycoproteins called hemagglutinin (HA) and neuraminidase (NA). Mutations cause antigenic changes in the structure of these two surface glycoproteins, and thus, they are the main determinants of antigenicity and host immunity. The ability of the virus to change the structure of HA on the virus surface is primarily responsible for the continual evolution of new strains, sometimes leading to subsequent influenza epidemics.

NA is an enzyme that removes sialic acid from surface glycoproteins during viral maturation and is required to produce infectious particles and lower the viscosity of the mucin layer of the respiratory tract. Influenza virus spreads through airborne droplets or contact with contaminated hands or surfaces and has an incubation period that varies from 1 to 4 days. However, transmission of the virus begins to occur 24 hours prior to the onset of symptoms. Interferon is detectable in respiratory secretions at the onset of symptoms, and the host response to interferon contributes to recovery. Antibodies and other cell-mediated responses are seen 1 to 2 weeks after infection. It is well established that secondary infections from other viruses or bacteria can occur, and Reye syndrome, an acute encephalopathy occurring in children and adolescents, is a rare complication of influenza B, influenza A, and herpesvirus varicella-zoster infections. The chances of contracting Reye syndrome are increased when salicylates are used in children suffering from influenza and related respiratory diseases.

Tumor Viruses (2,13–17)

Viruses are etiologic factors in the development of several types of human tumors, most notably cervical cancer and liver cancer. At least 15% of all human tumors worldwide have a viral cause. Tumor viruses can be found in both the RNA and DNA virus kingdoms (18). The list of human viruses presently known to be involved in tumor development includes four DNA viruses (Epstein-Barr virus [EBV], certain papilloma viruses, HBV, and the Kaposi sarcoma herpesvirus HHV-8) and two RNA viruses, (human T-cell leukemia virus 1 [HTLV-1] and HCV). Nearly every case of cervical cancer is known to be caused by human papillomavirus (HPV), a fact that spurred the development of the HPV vaccines Gardasil and Cervarix (19). Tumor viruses alter cellular behavior through the use of a small amount of genetic information, using two general strategies. The tumor virus either introduces a new “transforming gene” into the cell (direct-acting), or the virus alters the expression of a pre-existing cellular gene or genes (indirect-acting). In both cases, normal regulation of cellular growth processes is lost. Viruses alone cannot act as carcinogens, and other events are necessary to disable regulatory pathways and checkpoints in order to produce transformed, malignant cells. The processes used in the transformation of host cells by human tumor viruses are very diverse.

Cellular transformation may be defined as a stable, heritable change in the growth control of cells that results in tumor formation. Transformation from a normal to a neoplastic cell generally requires the retention of viral genes in the host cell. In the majority of cases, this is accomplished by the integration of certain viral genes into the host cell genome. Retroviruses incorporate their proviral DNA, formed through the action of reverse transcriptase, into host cell DNA. By contrast, DNA tumor viruses integrate a portion of the DNA of the viral genome into the host cell chromosome.

All RNA tumor viruses are members of the retrovirus family and belong to one of two classes (14,15,17). Class I RNA viruses are direct-transforming and carry an oncogene obtained through accidental incorporation from the host cell. No class I RNA viruses are known to produce tumors in humans. Class II or chronic RNA tumor viruses are weakly transforming and do not carry host cell–derived oncogenes. The two known cancer-causing retroviruses in humans act indirectly. They often act by inserting their proviral DNA into the immediate neighborhood of a host cellular oncogene. The human adult T-cell leukemia virus HTLV-1 acts in this manner, thus increasing the number of preneoplastic cells and facilitating secondary cellular changes leading to transformation.

DNA tumor virus strains exist among the papilloma-, polyoma-, adeno-, herpes-, hepadna-, and poxvirus groups (16). DNA tumor viruses encode viral oncoproteins that are important for viral replication, but also affect cellular growth control pathways. For example, inactivation of Rb and the p53 pathway by viral transforming proteins is a common strategy used by papillomaviruses and adenoviruses. As mentioned earlier, all DNA tumor viruses kill their host cell when the infectious virion is released to infect other cells. Thus, transformation and tumorigenicity are entirely dependent on a host cell interaction with the virus that does not involve viral spread to other cells, and cells transformed by DNA tumor viruses depend on the continued expression of the virally encoded oncogene.

Recent studies have revealed that the human tumor viruses EBV, HHV-8, HPV, HBV, HCV, and HTLV-1 express proteins that are targeted to the mitochondria (13). Because the mitochondria play a critical role in energy production, cell death, calcium homeostasis, and redox balance, these proteins have profound effects on host cell physiology. Further study of these proteins and their interactions with mitochondria will aid in the understanding of the mechanisms of viral replication and tumorigenesis and could reveal important new targets for antitumor therapy.

Prion Diseases (20–22)

Although they are not viruses, the infective proteins known as prions have sufficient similarities to viruses to warrant their discussion in this chapter. Prions are small proteins that have been shown to cause a variety of transmissible spongiform encephalopathies, which are rare neurodegenerative disorders typified by symptoms in the central nervous system (CNS) such as spongiform changes, neuronal loss, glial activation, and the accumulation of amyloid aggregates of an abnormally folded host protein. Human prion diseases include kuru, Creutzfeldt-Jakob disease (variant, sporadic, familial, and iatrogenic), Gerstmann-Sträussler-Scheinker syndrome, and fatal familial insomnia. The disease in cattle known as bovine spongiform encephalopathy and the related disease scrapie exhibit similar pathologic

features. Following exposure, prions accumulate in lymphoid tissue such as the spleen, lymph nodes, and tonsils and in Peyer's patches (specialized lymphoid follicles located in the submucosa of the small intestine). This accumulation of the infectious agent is necessary for invasion of the CNS. In humans, the incubation period of the disease can vary between 18 months and 40 years. Prions appear to be variant, improperly folded versions of a normal cellular protein called PrP^c, a 30- to 35-kd protein with two sites for N-glycosylation that is anchored in the neuronal cell membrane. PrP^c protein contains three α -helices, a short β -pleated sheet region, and a long, unstructured portion that comprises half of the molecule. The variant, infectious form of the protein is known as PrP^{sc} and is produced autocatalytically from PrP^c. Prion diseases are always fatal, with no known cases of remission or recovery. The host shows no inflammatory response or immune response and no production of interferon, and there is no effect on host B-cell or T-cell function. At present, there are no effective agents to treat prion diseases.

VIRAL CHEMOTHERAPY

General Approaches (23,24)

The principles involved in the design of antiviral agents are similar to those used in the design of all chemotherapeutic agents. Drugs in this category are targeted to some process in the virus that is not present in the host cell. The earliest examples of antiviral agents did not achieve this goal, and these drugs were toxic at therapeutic levels or had a limited spectrum of activity. A variety of factors make the design of effective antiviral agents difficult, including their ability to undergo antigenic changes, the latent period during which there are no symptoms, and their reliance on host enzymes and other processes. This problem is compounded by the fact that host immunity is not well understood and that symptoms of viral infection may not appear until replication is complete and the viral genome has been incorporated into infected cells. Nonetheless, the continuing identification of new targets for antiviral agents provides new avenues for the discovery of small-molecule therapies. The following section includes information on currently marketed antiviral compounds that have been designed in eight general areas:

1. Agents that disrupt virus attachment to host cell receptors, penetration, or uncoating.
2. Agents that inhibit virus-associated enzymes such as DNA polymerases and others.
3. Agents that inhibit viral transcription.
4. Agents that inhibit viral translation.
5. Agents that interfere with viral regulatory proteins.
6. Agents that interfere with glycosylation, phosphorylation, sulfation, and so on.
7. Agents that interfere with the assembly of viral proteins.

8. Agents that prohibit the release of viruses from cell surface membranes.

The remainder of this chapter deals primarily with small-molecule antiviral agents that have been approved by the U.S. Food and Drug Administration (FDA) and are clinically effective in the treatment of viral infection. Immunizing biologic agents such as vaccines, as well as antineoplastic agents with antiviral activity, are not covered. Some compounds used primarily in the treatment of bacterial infections, such as rifampicin, bleomycin, doxorubicin, and actinomycin, also inhibit viral replication. However, these antibiotics do not affect the transcription or translation of viral mRNA and are only effective in high concentrations. Therefore, such antibiotics are not commonly used for viral infections. There is a continuing need for new antiviral agents, primarily because viral infections are not curable after the virus invades the host cell and begins to replicate. Vaccines are effective, but they are only able to prevent an infection, and only in cases where specific virus strains are involved. For example, immunization against influenza, which is a yearly routine in many parts of the world, can only provide immunity against specific strains that are represented in that preparation. New virulent strains may arise from nonhuman sources, such as the so-called swine flu or avian flu viruses, and currently available vaccines would have no effect against these new strains. With regard to small-molecule antiviral agents, the ideal drug would have

broad-spectrum antiviral activity, completely inhibit viral replication, maintain efficacy against mutant viral strains, and reach the target organ without interfering with normal cellular processes or the immune system of the host.

ANTIVIRAL AGENTS (23,24)

Agents Inhibiting Virus Attachment, Penetration, and Early Viral Replication (Table 38.2)

Amantadine

MECHANISM OF ACTION Amantadine hydrochloride (1-adamantanamine hydrochloride) is a symmetrical tricyclic primary amine that inhibits penetration of RNA virus particles into the host cell (Fig. 38.3) (25). It also inhibits the early stages of viral replication by blocking the uncoating of the viral genome. Recent studies suggest that amantadine inhibits viral replication by blocking the influenza A virus M2 proton-selective ion channel.

CLINICAL APPLICATION Amantadine is clinically effective in preventing and treating all A strains of influenza, particularly A2 strains of Asian influenza virus and, to a lesser extent, German measles (rubella) or togavirus. It also shows in vitro activity against influenza B, parainfluenza (paramyxovirus), respiratory syncytial virus (RSV), and some RNA viruses (murine, Rous, and Esh sarcoma viruses). Many prototype influenza A viruses

TABLE 38.2 Antiviral Agents Interfering with Cellular Penetration and Early Replication

Generic Name	Trade Name	Spectrum of Activity	Dosage Form
Amantadine	Symmetrel	Influenza A	Cap (100 mg), syrup (50 mg/5 mL)
Rimantadine	Flumadine	Influenza A	Cap (100 mg)
Interferon α -2a	Roferon A Alferon Intron Wellferon	Chronic hepatitis, CMV, HSV, papillomavirus, rhinovirus, and others	Injectable (3, 5, 10, 18, 25, and 50 million units/mL)
Interferon α -2b	Interon A	Chronic hepatitis B and C, many other viruses	Injectable (3, 5, 10, 18, 25, and 50 million units/mL)
γ -Interferon	Actimmune	Chronic hepatitis B and C, many other viruses	Injectable (100 mcg/0.5 mL)
Tilorone	Amixin IC	Induces the production of interferons	Variable single doses given by injection
Zanamivir	Relenza	Influenza A and B	Inhaled powder (5 mg)
Oseltamivir	Tamiflu	Influenza A and B	Cap (75 mg)
Peramivir	Experimental	H1N1	600 mg daily by IV administration
Enfuvirtide	Fuzeon	HIV	Adult: 90 mg SQ BID Age 6–16: 2 mg/kg SQ BID
Maraviroc	Selzentry	HIV	300 mg orally twice daily

BID, twice a day; Cap, capsule; SQ, subcutaneous.

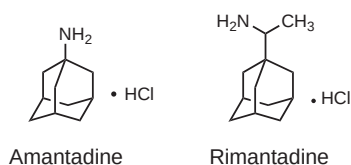


FIGURE 38.3 Agents that inhibit virus attachment, penetration, and early viral replication.

of different human subtypes (H1N1, Fort Dix, H2N2, Asian type, and H3N2, Hong Kong type) are also inhibited by amantadine hydrochloride *in vitro* and in animal model systems. If given within the first 48 hours of onset of symptoms, amantadine hydrochloride is effective in respiratory tract illness resulting from influenza A but not influenza B virus infection, adenoviruses, and RSV.

PHARMACOKINETICS Amantadine is well absorbed orally, and the usual dosage for oral administration is 100 mg twice daily. A 100-mg oral dose produces blood serum levels of 0.3 mg/mL within 1 to 8 hours. Maximum tissue concentration is reached in 48 hours when a 100-mg dose is given every 12 hours. In healthy adults receiving 25-, 100-, and 150-mg doses of the drug twice daily, steady-state trough plasma concentrations were 110, 302, and 588 mg/mL, respectively. Usually, no neurotoxicity is observed if the plasma level of amantadine is no more than 1.00 mg/mL. Amantadine crosses the blood-brain barrier and is distributed in saliva, nasal secretions, and breast milk (26). Approximately 90% of the drug is excreted unchanged by the kidney, primarily through glomerular filtration and tubular secretion, and there are no reports of metabolic products. Acidification of urine increases the rate of amantadine excretion. The half-life of the drug is 15 to 20 hours in patients with normal renal function.

SIDE EFFECTS Generally, the drug has low toxicity at therapeutic levels but may cause severe CNS symptoms such as nervousness, confusion, headache, drowsiness, insomnia, depression, and hallucinations. Gastrointestinal (GI) side effects include nausea, diarrhea, constipation, and anorexia. Convulsions and coma occur with high doses and in patients with cerebral arteriosclerosis and convulsive disorders. Chronic toxicity with amantadine is unexpected since few side effects have been experienced when the drug has been used for long-term therapy for Parkinson disease. Some serious reactions, however, include depression, orthostatic hypotension, psychosis, urinary retention, and congestive heart failure. Amantadine hydrochloride should be used with caution in patients who have a history of epilepsy, severe arteriosclerosis, liver diseases, and eczematoid dermatitis. Because amantadine does not appear to interfere with the immunogenicity of inactivated influenza A virus vaccine, patients may continue the use of amantadine

for 1 week after influenza A vaccination. A virus resistant to amantadine has been obtained in cell culture and from animals, but these reports are not confirmed in humans.

Rimantadine

MECHANISM OF ACTION Rimantadine hydrochloride (α -methyl-1-adamantanemethylamine hydrochloride) is a synthetic adamantane derivative that is structurally and pharmacologically related to amantadine (Fig. 38.3) (27,28). It appears to be more effective than amantadine hydrochloride against influenza A virus with fewer CNS side effects. Rimantadine hydrochloride is thought to interfere with virus uncoating by inhibiting the release of specific proteins. It may act by inhibiting RT or the synthesis of virus-specific RNA but does not inhibit virus adsorption or penetration. It appears to produce a virustatic effect early in the virus replication. It is used widely in Russia and Europe.

CLINICAL APPLICATION Rimantadine hydrochloride has activity against most strains of influenza A including H1N1, H2N2, and H3N2 but has no activity against influenza B virus. It is used for prevention of infection caused by various human, animal, or avian strains of influenza A virus in adults and children. The side effects are nightmares, hallucinations, and vomiting. The most common side effects of rimantadine are associated with the CNS and GI tract.

PHARMACOKINETICS Rimantadine is metabolized in the liver, and about 90% is excreted in the urine within 24 hours. The major urinary metabolites are ring-hydroxylated derivatives, a small percentage of which are glucuronidated. The half-life of rimantadine in adults ranges from 24 to 36 hours. Over 90% of rimantadine doses are absorbed in 3 to 6 hours. Steady-state plasma concentrations range from 0.10 to 2.60 mg/mL at doses of 3 mg/kg/d in infants to 100 mg twice daily in the elderly. Nasal fluid concentrations of rimantadine at steady-state were 1.5 times higher than plasma concentration.

Interferon (29,30)

Isaacs and Lindenmann discovered interferon in 1957. When they infected cells with viruses, interference with the cellular effects of viral infection was observed. Interferon was subsequently isolated and found to protect the cells from further infection. When interferon was administered to other cells or animals, it displayed biologic properties such as inhibition of viral growth, cell multiplication, and immunomodulatory activities. The results led to the speculation that interferon may be a natural antiviral factor, possibly formed before antibody production, and may be involved in the normal mechanism of resistance displayed against viral infection. Some investigators relate interferon to the polypeptide hormones and suggest that interferon functions

in cell-to-cell communication by transmitting specific messages. Recently, antitumor and anticancer properties of interferon have evoked worldwide interest in the possible use of this agent in therapy for viral diseases, cancer, and immunodeficiency disorders. Host cells in response to various inducers synthesize interferons.

INTERFERON STRUCTURE Interferon consists of a mixture of small proteins with molecular weights ranging from 20,000 to 160,000 daltons. They are glycoproteins that exhibit species-specific antiviral activity. Human interferons are classified into three types: alpha (α), beta (β), and gamma (γ) (31). The α -type is secreted by human leukocytes (white blood cells, non-T lymphocytes), and the β -type is secreted by human fibroblasts. Lymphoid cells (T lymphocytes), which either have been exposed to a presensitized antigen or have been stimulated to divide by mitogens, secrete α -interferon. γ -Interferon is also called "immune" interferon. Interferons are active in extremely low concentrations.

MECHANISM OF ACTION Although, interferons are mediators of immune response, different mechanisms for the antiviral action of interferon have been proposed. α -Interferon possesses broad-spectrum antiviral activity, and acts on virus-infected cells by binding to specific cell surface receptors. It inhibits the transcription and translation of mRNA into viral nucleic acid and protein. Studies in cell free systems have shown that the addition of adenosine triphosphate and double-stranded RNA to extracts of interferon-treated cells activates cellular RNA proteins and a cellular endonuclease. This activation causes the formation of translation inhibitory protein, which terminates production of viral enzyme, nucleic acid, and structural proteins (32). Interferon may also act by blocking synthesis of a cleaving enzyme required for viral release.

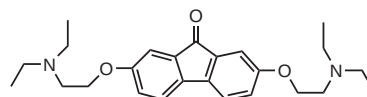
CLINICAL APPLICATION Interferon has been tested for use in chronic HBV infection, herpetic keratitis, herpes genitalis, herpes zoster, varicella-zoster, chronic hepatitis, influenza, and common cold infections. Other uses of interferon are in the treatment of cancers, such as breast cancer, lung carcinoma, and multiple myeloma. Interferon has had some success when used as a prophylactic agent for cytomegalovirus (CMV) infection in renal transplant recipients. The scarcity of interferon and the difficulty in purifying it have limited clinical trials. Supplies have been augmented by recombinant DNA technology, which allows cloning of the interferon gene (33), although the high cost still hinders clinical application. The FDA has approved recombinant interferon α -2a and α -2b and γ -interferon for the treatment of hairy cell leukemia (a rare form of cancer), AIDS-related Kaposi sarcoma, and genital warts (condyloma acuminatum). Subcutaneous injection of recombinant interferon α -2b has been approved for the treatment of chronic HCV. Some foreign countries have approved α -interferon for the treatment of cancers such as multiple myeloma (cancer of plasma cells), malignant melanoma

(skin cancer), and Kaposi sarcoma (cancer associated with AIDS). Both β - and γ -interferons and interleukin-2 may be commercial drugs of the future for the treatment of cancers and viral infections, including genital warts and the common cold.

PHARMACOKINETICS The pharmacokinetics of interferon are not well understood. Maximum levels in blood after intramuscular injection were obtained in 5 to 8 hours. Interferon does not penetrate well into cerebrospinal fluid (CSF). Oral administration of interferon does not indicate a detectable serum level, and as such, oral administration is clinically ineffective. After intramuscular or subcutaneous injection, drug concentration in the plasma is dose related. Clinical use of interferon is limited to topical administration (nasal sprays) for prophylaxis and treatment of rhinovirus infections. Adverse reactions and toxicity include influenza-like syndrome of fever, chills, headache, myalgias, nausea, vomiting, diarrhea, bone marrow suppression, mental confusion, and behavioral changes. Intranasal administration produces mucosal friability, ulceration, and dryness. Two long-acting forms of interferon have recently become available. The addition of polyethylene glycol to the formulation of interferon α -2a and interferon α -2b affords an injectable form of each drug that has an extended half-life.

Because viruses were found to induce the release of interferon, efforts were made to induce the production or release of interferon in humans by the administration of chemical "inducers" (34). Various small molecules (substituted propanediamine) and large polymers (double-stranded polynucleotides) were used to induce interferons. Statolon, a natural double-stranded RNA produced in *Penicillium stoloniferum* culture, and a double-stranded complex of polyriboinosinic acid and polyribocytidylic acid (poly I:C) have been used as non-viral inducers for releasing preformed interferons. A modification of poly I:C stabilized with poly-L-lysine and carboxymethylcellulose (poly ICLC) has been used experimentally in humans. Clinically, it prevented coryza when used locally in the nose and conjunctival sacs. This substance was found to be a better interferon inducer than poly I:C. Another interferon inducer is Ampligen, a polynucleotide derivative of poly I:C with spaced uridines. It has anti-HIV activity in vitro and is an immunomodulator.

Other chemical inducers, such as pyran copolymers, tilorone, diethylaminoethyl dextran, and heparin, have also been used. Tilorone is an effective inducer of interferon in mice but is relatively ineffective in humans. Initial use of interferon and its inducers instilled intranasally after rhinovirus exposure was successful in the prevention of respiratory diseases.



Tilorone

The clinical success of interferon and its inducers has not yet been established, although they may play a significant role in cell-mediated immunity to viral infections and cancer. Disadvantages of interferon use include unacceptable side effects, such as fever, headache, myalgias, leukopenia, nausea, vomiting, diarrhea, hypotension, alopecia, anorexia, and weight loss.

Neuraminidase Inhibitors (35)

Virus Activation

The role played by the surface glycoproteins HA, an enzyme important for viral binding to host cell receptors via a terminal sialic acid residue, and NA, an enzyme involved in various aspects of activation of influenza viruses, was discussed earlier. Freshly shed virus particles are coated with sialic acid residues. NA is found in both influenza A and B viruses and is thought to be involved in catalytically cleaving glycosidic bonds between terminal sialic acid residues and adjacent sugars on HA. The cleavage of sialic acid bonds facilitates the spread of viruses by enhancing adsorption to cell surface receptors, and thus increases the infective level of the virus. In the absence of sialic acid cleavage from HA, viral aggregation or inappropriate binding to HA will occur, interfering with the spread of the infection. NA also appears to play a role in preventing viral inactivation by respiratory mucus (Fig. 38.4).

Mechanism of Action

Since NA plays such an important role in the activation of newly formed viruses, it is not surprising that the development of NA inhibitors has become an important

potential means of inhibiting the spread of viral infections. X-ray crystallography of NA has shown that while the amino acid sequence of NA from various viruses is considerably different, the sialic acid binding site is quite similar for type A and B influenza viruses. In addition, it is believed that the hydrolysis of sialic acid proceeds through an oxonium cation–stabilized carbonium ion as shown in Figure 38.4. Mimicking the transition state with novel carbocyclic derivatives of sialic acid has led to the development of transition state–based inhibitors (36). The first of these compounds, 2-deoxy-2,3-dehydro-N-acetylneuraminic acid (DANA; Fig. 38.5), was found to be an active neuraminidase inhibitor but lacked specificity for viral neuraminidase. Upon determination of the crystal structure of neuraminidase, more sophisticated measurements of the binding site for sialic acid led to the development of zanamivir and, later, oseltamivir.

Specific Drugs

ZANAMIVIR Crystallographic studies of DANA bound to NA defined the receptor site to which the sialic acid portion of the virus binds. These studies suggested that substitution of the 4-hydroxy with an amino group or the larger guanidino group should increase binding of the inhibitor to NA. The 4-amino derivative was found to bind to a glutamic acid (Glu 119) in the receptor through a salt bridge, whereas the 4-guanidino derivative was able to form both a salt bridge to Glu 119 and a charge–charge interaction with a glutamic acid at position 227. The result of these substitutions was a dramatic increase in binding capacity of the 4-amino and 4-guanidino derivatives to NA, leading to effective competitive inhibition of the enzyme. The result has been the development of the 4-guanidino analog as zanamivir (Fig. 38.5), which has become an effective agent against influenza A and B virus.

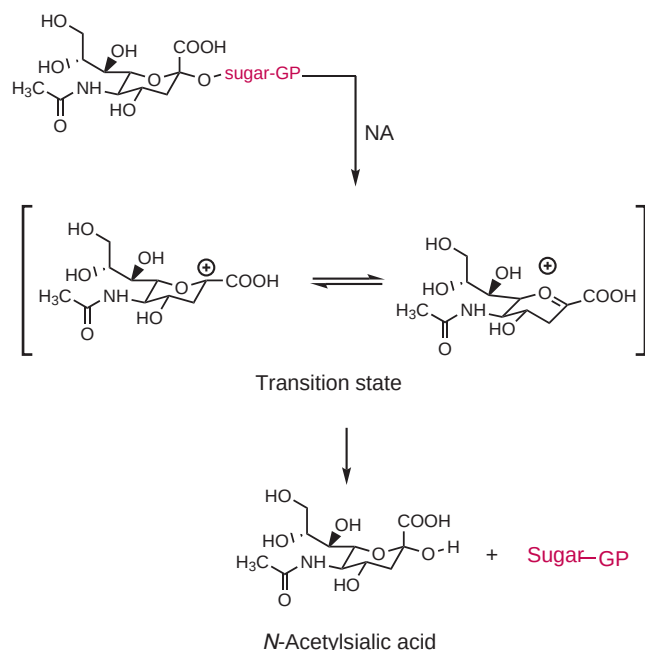


FIGURE 38.4 Neuraminidase-catalyzed removal of a sialic acid residue from a glycoprotein chain. GP, glycoprotein; NA, neuraminidase.

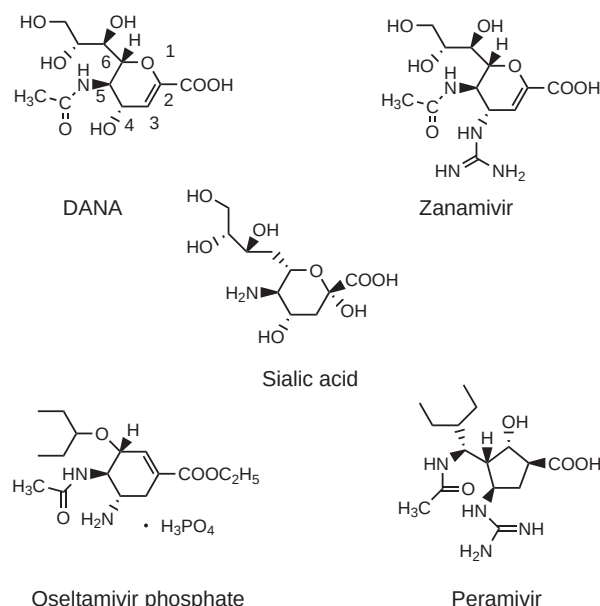


FIGURE 38.5 Neuraminidase inhibitors based on sialic acid.

Zanamivir is effective when administered via the nasal, intraperitoneal, and intravenous routes but is inactive when given orally (Table 38.2). Animal studies have shown 68% recovery of the drug in the urine following intraperitoneal administration, 43% urinary recovery following nasal administration, and only 3% urinary recovery following oral administration. Human data gave similar results to those obtained in animal models. Human efficacy studies with nasal drops or sprays demonstrated that the drug was effective when administered before exposure and after exposure to influenza A or B virus. When given before viral inoculation, the drug reduced viral shedding, infection, and symptoms. When administered beginning at either 26 or 32 hours after inoculation, there was a reduction in shedding, viral titer, and fever. Presently the drug is available as a dry powder for oral inhalation by adults and adolescents who have been symptomatic for no more than 2 days. Zanamivir is able to more rapidly resolve influenza symptoms and improve recovery (from 7 days with placebo to 4 days with treatment). Additional studies have suggested the prophylactic benefit of zanamivir when administered to family members after one member of the family developed flu-like symptoms. As a result, the manufacturer has submitted an application for the use of the drug for the prevention of influenza A and B.

OSELTAMIVIR X-ray crystallographic studies further demonstrated that additional binding sites exist between NA and substrate involving the C-5 acetamido carbonyl of DANA and an arginine (Arg 152), the C-2 carboxyl and arginines as 118, 292, and 371, and the potential for hydrophobic binding to substituents at C-6 (with glutamic acid, alanine, arginine, and isoleucine). Structure–activity relationship studies showed that maximum binding occurred to NA when C-6 (based on DANA numbering) was substituted with the 3-pentyloxy side chain such as the one found in oseltamivir (Fig. 38.5). In addition, esterification with ethanol gave rise to a compound that is orally effective. Oseltamivir phosphate was approved as the first orally administered NA inhibitor used against influenza A and B (Table 38.2). The drug is indicated for the treatment of uncomplicated acute illness due to influenza infection. Recently, the manufacturer has submitted a request for approval of the drug for prevention of influenza A and B for use in adults, adolescents, and children 1 year of age and older. The drug is effective in treating influenza if administered within 2 days after onset of symptoms. The recommended dose is 75 mg twice daily for 5 days. The prophylactic dose is 75 mg taken once daily for 7 days. Oseltamivir is readily absorbed from the GI tract following oral administration. It is a prodrug that is extensively metabolized in the liver undergoing ester hydrolysis to the active carboxylic acid (Fig. 38.6). Two oxidative metabolites have also been isolated, with the major oxidation product being the ω -carboxylic acid (37). Side effects with oseltamivir are minor, consisting of nausea and vomiting that occur primarily in the first 2 days of therapy.

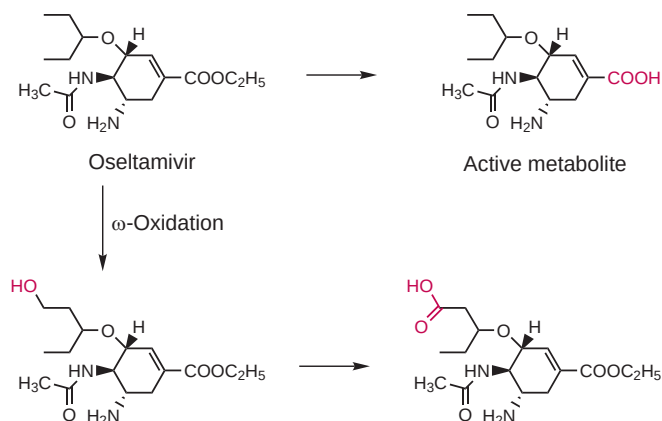


FIGURE 38.6 Metabolism of oseltamivir by hydrolysis and ω -oxidation.

PERAMIVIR In 2009, the experimental NA inhibitor peramivir (Fig. 38.5, Table 38.2) received emergency FDA approval for use in treatment of certain hospitalized patients with known or suspected H1N1 influenza infections. This represents the first time in which an experimental drug in phase III clinical trials has received an emergency use approval. The drug is only available by electronic request from the U.S. Strategic National Stockpile. The use of peramivir is restricted to severe H1N1 influenza cases where other antiviral agents (including NA inhibitors) have failed (38). For example, a patient infected with an H1N1 strain of influenza that is resistant to oseltamivir and who is unable to inhale zanamivir would be a candidate for peramivir treatment. Peramivir is structurally distinct from oseltamivir and zanamivir in that it contains a cyclopentane ring structure that replaces the sialic acid core and because it does not contain a glycerol side chain, which is thought to reduce the oral bioavailability of oseltamivir and zanamivir. The structure of peramivir was proposed using the technique of structure-based design. In this approach, in silico docking of proposed NA inhibitors suggested that peramivir bound most efficiently to the four known binding pockets in the viral NA catalytic site. Peramivir is administered intravenously as a single dose (0.5 to 8 mg/kg) or as two daily doses (4 mg/kg). It has a serum half-life of 7.7 to 20 hours, and 90% of the drug is excreted unchanged in the urine. Because peramivir is still in phase III clinical trials, efficacy data have not yet been released.

Entry (Fusion) Inhibitors

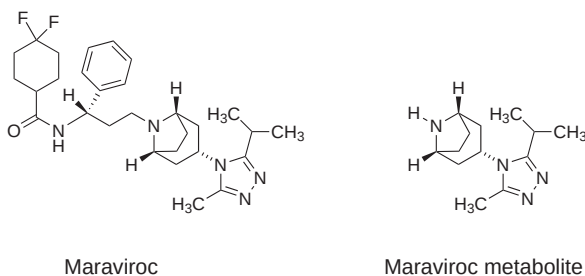
Enfuvirtide (39,40)

N-acetyl-Tyr-Thr-Ser-Leu-Ile-His-Ser-Leu-Ile-Glu-Glu-Ser-Gln-Asp-Gln-Gln-Glu-Lys-Asp-Glu-Gln-Glu-Leu-Leu-Glu-Leu-Asp-Lys-Try-Ala-Ser-Leu-Try-Asp-Try-Phe-NH₂

Enfuvirtide

Entry inhibitors, also known as fusion inhibitors, are a new class of drugs for the treatment of HIV infection, and enfuvirtide is the first compound of this

family to be approved for clinical use. Enfuvirtide is an oligopeptide consisting of 36 amino acids. It is a synthetic peptide that mimics an HR2 fragment of gp41, blocking the formation of a six-helix bundle structure that is critical in the fusion of the HIV-1 virion to a CD4-positive T lymphocyte. Specifically, it binds to the tryptophan-rich region of the gp41 protein. Enfuvirtide is used in combination with other antiretrovirals and works against a variety of HIV-1 variants, but it is not active against HIV-2. Resistance to enfuvirtide can develop when the virus produces changes in a 10-amino acid domain between residues 36 to 45 in the gp41 HIV surface glycoprotein. The drug is administered twice daily as a subcutaneous injection and has a complex absorption pattern. Enfuvirtide is highly bound to plasma protein (~92%) and is prone to proteolytic metabolism. Adverse reactions include pain, erythema, and pruritus at the site of injection and insomnia.



Maraviroc

CCR5 is a protein produced by humans that is a member of the chemokine coreceptor family and is located in the cell membrane of specific human cells (i.e., T cells, macrophages, dendritic cells, microglia). Although there are a number of natural substrates for this receptor protein, HIV can use this G protein-coupled receptor in part as a coreceptor for entry into its targeted cells. In addition, individuals who have a mutation in the *CCR5* gene (*CCR5-Δ32*) have a deletion of a portion of the *CCR5* gene, which in turn leads to reduced or absent expression of *CCR5* in heterozygous or homozygous genotypes, respectively, with no apparent deleterious consequences. Individuals homozygous with respect to *CCR5-Δ32* have a high degree of resistance to HIV infection, whereas heterozygotes have a reduced rate of disease progression. As a result, CCR5 antagonists represent a promising new class of cell entry inhibitor drugs under development for the treatment of HIV-1 infection (fusion inhibitor). The CCR5 antagonist maraviroc has potent anti-HIV activity in vitro and is safe and well-tolerated at multiple doses up to 300 mg twice daily in healthy volunteers. Maraviroc was recently approved by the FDA (Selzentry; Pfizer). Maraviroc is an orally administered drug available as 150- and 300-mg film-coated tablets. The current FDA-approved dosage of maraviroc is 300 mg twice daily, and it must be used in combination with other antiretroviral medications. Maraviroc is actively absorbed following oral administration but has a low bioavailability

(~23%) and is a substrate for P-glycoprotein, possibly accounting for the poor bioavailability. The drug is highly bound to plasma proteins (~76%) and is metabolized to inactive metabolites via CYP3A4. The major metabolite results from *N*-dealkylation.

Agents Interfering with Viral Nucleic Acid Replication (Table 38.3)

Acyclic Nucleoside Analogs (Fig. 38.7)

ACYCLOVIR

Mechanism of Action Acyclovir is a synthetic analog of deoxyguanosine in which the carbohydrate moiety is acyclic (Fig. 38.7) (41). Because of this difference in structure as compared to other antiviral compounds (idoxuridine, vidarabine, and trifluridine, see below), acyclovir possesses a unique mechanism of antiviral activity. The mode of action of acyclovir consists of three consecutive mechanisms (42). The first of these mechanisms involves conversion of the drug to active acyclovir monophosphate within cells by viral thymidine kinase (Fig. 38.8). This phosphorylation reaction occurs faster within cells infected by herpesvirus than in normal cells because acyclovir is a poor substrate for the normal cell thymidine kinase. Acyclovir is further converted to di- and triphosphates by a normal cellular enzyme called guanosine monophosphate kinase. In the second mechanism, viral DNA polymerase is competitively inhibited by acyclovir triphosphate with a lower half maximal inhibitory concentration (IC_{50}) concentration than that for cellular DNA polymerase. Acyclovir triphosphate is incorporated into the viral DNA chain during DNA synthesis. Because acyclovir triphosphate lacks the 3'-hydroxyl group of a cyclic sugar, it terminates further elongation of the DNA chain. The third mechanism depends on preferential uptake of acyclovir by herpes-infected cells as compared to uninfected cells, resulting in a higher concentration of acyclovir triphosphate and leading to a high therapeutic index between herpes-infected cells to normal cells. Acyclovir is active against certain herpes virus infections. These viruses induce virus-specific thymidine kinase and DNA polymerase, which are inhibited by acyclovir. Thus, acyclovir significantly reduces DNA synthesis in virus-infected cells without significantly disturbing the active replication of uninfected cells.

Clinical Application Acyclovir has potent activity against several DNA viruses including HSV-1, the common cause of labial herpes (cold sore), and HSV-2, the common cause of genital herpes (Table 38.3) (43). Varicella-zoster virus (VZV) and some isolates of EBV are affected to a lesser extent by acyclovir. However, CMV is less sensitive to acyclovir, which has no activity against vaccinia virus, adenovirus, and parainfluenza infections. An ointment containing 5% acyclovir has been used in a regimen of five times a day for up to 14 days for the treatment of herpetic keratitis and primary and recurrent infections of herpes genitalis. Mild pain, transient burning, stinging, pruritus, rash, and vulvitis have been noted. The

TABLE 38.3 Antiviral Agents Interfering with Viral Nucleic Acid Replication

Generic Name	Common Name	Trade Name	Spectrum of Activity	Dosage Form (mg/unit)
Acyclic nucleoside analogs (Fig. 38.7)				
Acyclovir	Acyclo-G	Zovirax	HSV-1; HSV-2; VZV; EBV	5% Oint, Inj (5mg/ml), Cap (200), Tab (400, 800), Susp (200/5 mL)
Valacyclovir		Valtrex	HSV-1; VZV; CMV	Tab (500)
Cidofovir	HPMPC	Vistide	CMV; HSV-1; HSV-2; VZV; CMV; EBV	Inj (75/mL)
Famciclovir	FCV	Famvir	HSV; VZV; EBV; chronic HBV	Tab (125, 250, 500)
Ganciclovir	DHPG	Cytovene Vitrasert	CMV retinitis	Inj (50/mL), Cap (250, 500), Insert (4.5/insert)
Adefovir dipivoxil		Hepsera	Hepatitis B	Tab (10)
Conventional nucleoside analogs (Fig. 38.11)				
Idoxuridine	5-IUDR	Herplex Stoxil	HSV keratitis	0.1% Sol, 0.5% Oint
Ribavirin		Virazole	RSV; influenza A and B; HIV-1; parainfluenza	Aerosol (20/mL)
Trifluorothymidine	TFT, FT ₃	Viroptic	HSV-1	1% Sol
Vidarabine	Ara-A	Vira A	HSV-1; HSV-2	3% Oint (monohydrate)
Cytarabine	Ara-C	Cytosar	Herpes zoster	Inj (10, 20, 50, 100/mL)
Non-nucleoside analogs (Fig. 38.13)				
Boceprevir		Victrelis	HCV	Tab (200)
Fomivirsen			CMV retinitis	
Foscarnet	PFA	Foscavir	CMV retinitis; HSV	Inj (24/mL)
Telaprevir		Incivek	HCV	Tab (375)

CMV, cytomegalovirus; EBV, Epstein-Barr virus; HBV, hepatitis B virus; HSV, herpes simplex virus; VZV, varicella-zoster virus.

FDA has approved topical and intravenous (IV) acyclovir preparations for initial herpes genitalis and HSV-1 and HSV-2 infections in immunocompromised patients (44). In these individuals, early use of acyclovir shortens the duration of viral shedding and lesion pain. Oral doses of 200 mg of acyclovir, taken five times a day for 5 to 10 days, have not proven successful because of the low bioavailability of current preparations. Oral doses of 800 mg of the drug given five times daily for 7 to 10 days have been approved by the FDA for treatment of herpes zoster infection. This treatment shortens the duration of viral shedding in chickenpox and shingles. The IV injection of the drug (10 mg/kg three times daily for 10 to 12 days) has been approved for the treatment of herpes simplex encephalitis (45). Excessive and high doses of acyclovir have, however, caused viruses to develop resistance to the drug. This resistance results from reduction of virus-encoded thymidine kinase, which does not effectively activate the drug.

Pharmacokinetics Pharmacokinetic studies show that IV dose administration of acyclovir 2.5 mg/kg results in peak plasma concentrations of 3.4 to 6.8 mg/mL (46,47). The bioavailability of acyclovir is 15% to 30%, and it is metabolized to 9-carboxymethoxymethylguanine, which is inactive. Plasma protein binding averages 15%, and approximately 70% of acyclovir is excreted unchanged in the urine by both glomerular filtration and tubular secretion. The half-life of the drug is approximately 3 hours in patients with normal renal function. In an individual with renal diseases, the half-life of the drug is prolonged. Therefore, acyclovir dosage adjustment is necessary for patients with renal impairment. Acyclovir easily penetrates the lung, brain, muscle, spleen, uterus, vaginal mucosa, intestine, liver, and kidney.

VALACYCLOVIR Valacyclovir hydrochloride is an amino acid ester prodrug of acyclovir, which exhibits antiviral activity only after metabolism in the intestine or liver to

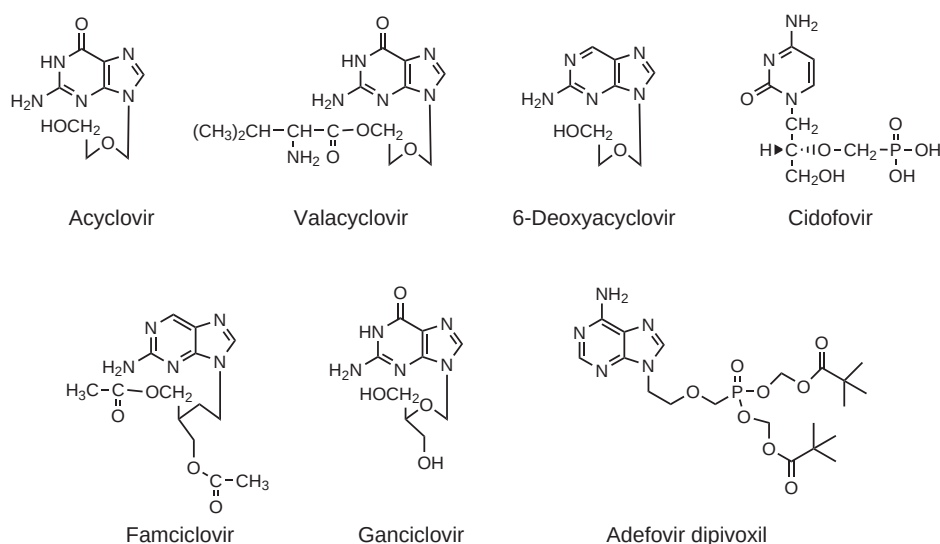


FIGURE 38.7 Agents that interfere with viral nucleic acid replication: acyclic nucleosides.

acyclovir followed by conversion to the triphosphate as shown in Figure 38.8 (48). Structurally, it differs from acyclovir by the presence of the amino acid valine attached to the 5'-hydroxyl group of the nucleoside. Valacyclovir's benefit comes from an increased GI absorption resulting in higher plasma concentrations of acyclovir, which is normally poorly absorbed from the GI tract. As with acyclovir, valacyclovir is active against HSV-1, VZV, and CMV because of its affinity for the viral form of the enzyme thymidine kinase. Oral valacyclovir is used for the treatment of acute, localized herpes zoster (shingles) in immunocompetent patients and may be given without meals. It is also used for the initial and recurrent episodes of genital herpes infections (Table 38.3). The adverse effects are similar to acyclovir and include nausea, headache, vomiting, constipation, and anorexia. The binding of valacyclovir to human plasma proteins ranges from 13.5% to 17.9%. The plasma elimination half-life of acyclovir is 2.5 to 3.3 hours. The bioavailability of valacyclovir hydrochloride is 54% compared to approximately 20% for oral acyclovir. It is as effective as acyclovir in decreasing the duration of pain associated with posttherapeutic neuralgia and episodes of genital lesion healing. The related analog 6-deoxyacyclovir (Fig. 38.8) is a prodrug form of acyclovir that is activated through metabolism by xanthine oxidase. This drug, which has improved solubility characteristics when compared to acyclovir, is used for the treatment of VZV infection.

CIDOFOVIR

Mechanism of Action Cidofovir (1-[(S)-3-hydroxy-2-(phosphonomethoxy)propyl]cytosine) is a synthetic acyclic purine nucleotide analog of cytosine (Fig. 38.7) (49). It is a phosphorylated nucleotide that is additionally phosphorylated by host cell enzymes to its active intracellular metabolite, cidofovir diphosphate. This reaction occurs without initial virus-dependent phosphorylation

by viral nucleoside kinases. It produces antiviral effects through interfering with DNA synthesis and inhibiting viral replication.

Clinical Application Cidofovir is active against herpes viruses including HSV-1 and HSV-2, VZV, CMV, and EBV. It is effective against acyclovir-resistant strains of HSV and ganciclovir-resistant strains of CMV. Cidofovir is a long-acting drug for the treatment of CMV retinitis in AIDS patients given as IV infusion or intravitreal injection. It is not a curative drug, and its benefit over foscarnet or ganciclovir is yet to be determined. The major adverse effect is nephrotoxicity, which appears to result in renal tubular damage. Concomitant administration of cidofovir with probenecid is contraindicated because of increased risk of nephrotoxicity. Topical cidofovir (0.2%) is as effective as trifluridine (1%) in reducing HSV-1 shedding and healing time in rabbits with dendritic keratitis. Cidofovir is administered IV, topically, and by ocular implant (Table 38.3). Peak plasma concentration of 3.1 to 23.6 mg/mL is achieved with doses of 1.0 to 10.0 mg/kg, respectively. The terminal plasma half-life is 2.6 hours, and 90% of the drug is excreted in the urine. It has a variable bioavailability (2% to 26%).

FAMCICLOVIR

Mechanism of Action Famciclovir is a synthetic purine nucleoside analog related to guanine (50,51). It is the diacetyl 6-deoxy ester of penciclovir, which is structurally related to ganciclovir. Its pharmacologic and microbiologic activities are similar to acyclovir. Famciclovir is a prodrug of penciclovir (Fig. 38.9), which is formed in vivo by hydrolysis of the acetyl groups and oxidation at the 6-position by mixed function oxidases. Penciclovir and its metabolite penciclovir triphosphate possess antiviral activity resulting from inhibition of viral DNA polymerase.

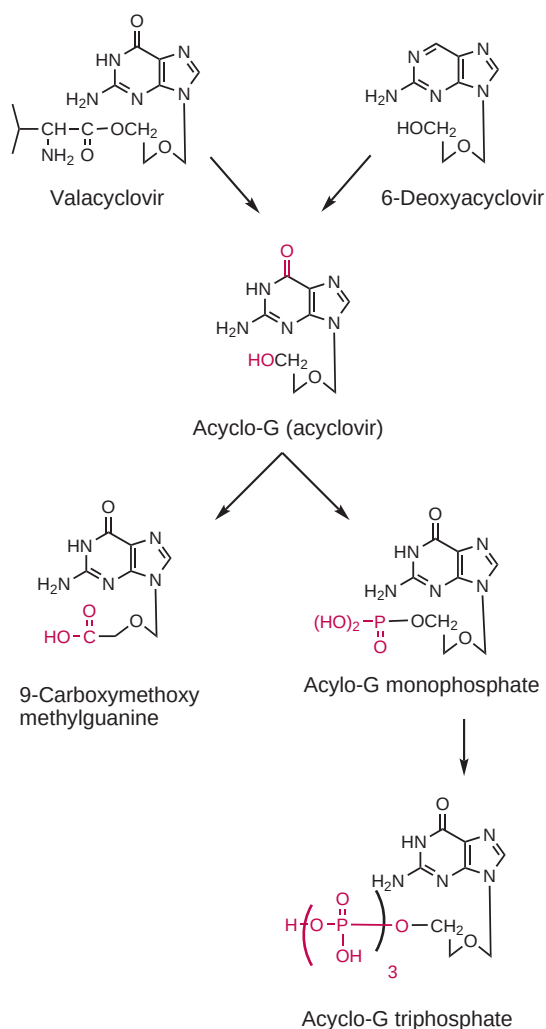


FIGURE 38.8 Metabolic reactions of acyclovir and valacyclovir.

Clinical Application Famciclovir is active against recurrent HSV (genital herpes and cold sores), VZV, and EBV but less active against CMV (Table 38.3). It is used in the treatment of recurrent localized herpes zoster and genital herpes in immunocompetent adults and is also promising for the treatment of chronic HBV reinfection after liver transplantation.

Pharmacokinetics Famciclovir can be given with or without food. The most common adverse effects are headache and GI disturbances. Concomitant use of famciclovir with probenecid results in increased plasma concentrations of penciclovir. The recommended dose of famciclovir is 500 mg every 8 hours for 7 days. The absolute bioavailability of famciclovir is 77%, and area under plasma concentration–time curve (AUC) is 86 mcg/mL. Famciclovir with digoxin increased plasma concentration of digoxin to 19% as compared to digoxin given alone.

GANCICLOVIR

Mechanism of Action Ganciclovir sodium is an acyclic deoxyguanosine analog of acyclovir (52). Its active form

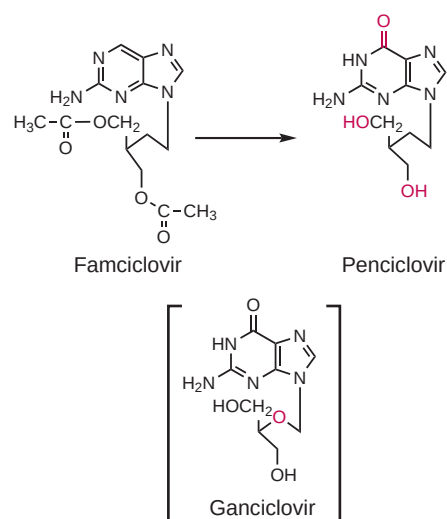


FIGURE 38.9 Metabolic activation of famciclovir.

is ganciclovir triphosphate, which is an inhibitor of viral rather than cellular DNA polymerase. The phosphorylation of ganciclovir does not require a virus-specific thymidine kinase for its activity against CMV. The mechanism of action is similar to that of acyclovir; however, ganciclovir is more toxic to human cells than is acyclovir. Ganciclovir has greater activity than acyclovir against CMV and EBV infection in immunocompromised patients. It is also active against HSV infection and in some mutants resistant to acyclovir. In AIDS patients, ganciclovir stopped progressive hemorrhagic retinitis and symptomatic pneumonitis related to CMV infection.

Clinical Application Ganciclovir is absorbed and phosphorylated by infection-induced kinases of HSV and VZV infections. Common side effects are leukopenia, neutropenia, and thrombocytopenia. Ganciclovir with zidovudine causes severe hematologic toxicity. Ganciclovir is available only as an IV infusion because oral bioavailability is poor (Table 38.3). It is given in doses of 5 mg/kg twice daily for 14 to 21 days. When ganciclovir is given by IV administration, concentrations of the drug in the CSF and the brain vary from 25% to 70% of the plasma concentration. After minimal metabolism, ganciclovir is excreted in the urine. In adults with normal renal function, the serum half-life of the drug is approximately 3 hours. Ganciclovir has been approved by the FDA for the treatment of CMV retinitis in immunocompromised and AIDS patients.

ADEFOVIR DIPVOXIL (53)

Mechanism of Action Adefovir dipivoxil is an orally active prodrug indicated for the treatment of chronic HBV (Fig. 38.7). The drug is hydrolyzed by extracellular esterases to produce adefovir, which in turn is phosphorylated by adenylate kinase to adefovir phosphate, which inhibits HBV DNA polymerase. Incorporation of adefovir into viral DNA also leads to DNA chain termination.

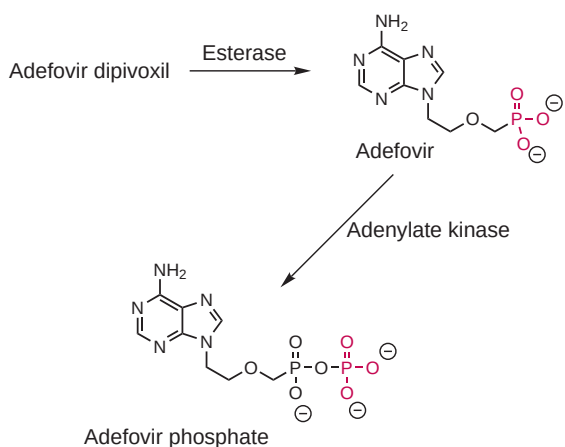


FIGURE 38.10 Activation of the prodrug adefovir dipivoxil by esterase and adenylate kinase.

As shown in Figure 38.10, adefovir dipivoxil is activated in two steps involving an esterase that exposes a free phosphate group (adefovir), followed by addition of a second phosphate by adenylate kinase to form adefovir phosphate.

Pharmacokinetics Adefovir is poorly absorbed orally, but the bioavailability of adefovir dipivoxil reaches approximately 59%. The drug is absorbed to an equal extent with or without the presence of food, and it is excreted unchanged by the kidney. Adefovir dipivoxil joins interferon and lamivudine in the treatment of chronic HBV. It can be used singly or in combination with lamivudine. Early clinical studies indicate benefit of the use of adefovir dipivoxil to treat lamivudine-resistant HBV with a low level of resistant virus developing to monotherapy with adefovir dipivoxil.

Conventional Nucleoside Analogs (Fig. 38.11)

IDOXURIDINE

Mechanism of Action Idoxuridine is a nucleoside analog of thymidine containing a halogenated pyrimidine (54). It acts as an antiviral agent against DNA viruses by interfering with their replication based through competitive inhibition. Idoxuridine is first phosphorylated by the host cell virus-encoded enzyme thymidine kinase to an active triphosphate form. The phosphorylated drug inhibits cellular DNA polymerase to a lesser extent than HSV DNA polymerase, which is necessary for the synthesis of viral DNA. The triphosphate form of the drug is then incorporated during viral nucleic acid synthesis by a false pairing system that replaces thymidine. When transcription occurs, faulty viral proteins are formed, resulting in defective viral particles (55).

Clinical Application Idoxuridine is available as ophthalmic drops (0.1%) and ointment (0.5%) for the treatment of HSV keratoconjunctivitis (Table 38.3). Because of its poor solubility, the drug is ineffective in labial or

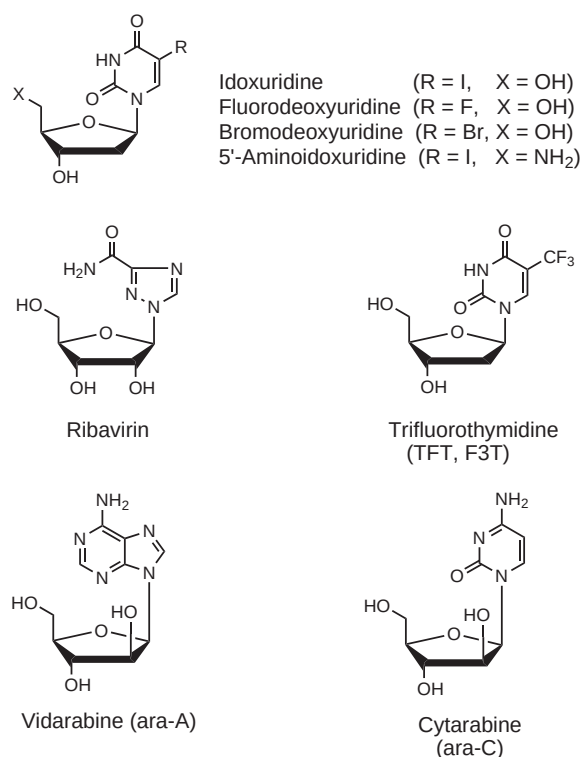


FIGURE 38.11 Agents that interfere with viral nucleic acid replication: conventional nucleosides.

genital HSV or for cutaneous herpes zoster infection. Idoxuridine in dimethylsulfoxide (DMSO), however, has been used in mucocutaneous HSV infection of the mouth and nose. Because DMSO facilitates drug absorption and also has some therapeutic effect, a 40% solution of idoxuridine in DMSO is more effective than idoxuridine used without this vehicle. Therefore, the FDA approved idoxuridine only for topical treatment of herpes simplex keratitis, and it is more effective in epithelial than in stromal infections. It is less effective for recurrent herpes keratitis, probably because of the development of drug-resistant virus strains.

Side Effects Adverse reactions of idoxuridine include such local reactions as pain, pruritus, edema, burning, and hypersensitivity. Systemic administration of idoxuridine by IV injection may be given in an emergency but leads to bone marrow toxicities such as leukopenia, thrombocytopenia, and anemia. It may also induce stomatitis, nausea, vomiting, abnormalities of liver functions, and alopecia. Idoxuridine has a plasma half-life of 30 minutes and is rapidly metabolized in the blood to idoxuracil and uracil.

Other halogenated uridine derivatives can also be used as antiviral agents (Fig. 38.11). Fluorodeoxyuridine has in vitro antiviral activity but is not used in clinical practice. Bromodeoxyuridine has been used in subacute sclerosing panencephalitis, a deadly virus-induced CNS disease. This agent appears to interfere with DNA synthesis in the

same way as idoxuridine. The 5'-amino analog of idoxuridine (5-iodo-5'-amino-2',5'-dideoxyuridine) is a better antiviral agent than idoxuridine and is less toxic. It is metabolized in herpesvirus-infected cells only by thymidine kinase to di- and triphosphoramidates. These metabolites inhibit HSV-specific late RNA transcription, causing reduction of less infective abnormal viral proteins. Although all of these analogs of idoxuridine have been studied for various clinical applications, none are presently marketed in the United States.

REBIVIRIN

Mechanism of Action Ribavirin, a guanosine analog, has broad-spectrum antiviral activity against both DNA and RNA viruses (Fig. 38.11) (56,57). It is phosphorylated by adenosine kinase to the triphosphate, resulting in inhibition of viral-specific RNA polymerase, disrupting messenger RNA and nucleic acid synthesis.

Clinical Application Ribavirin is highly active against influenza A and B and the parainfluenza group of viruses, genital herpes, herpes zoster, measles, and acute hepatitis types A, B, and C. Aerosolized ribavirin has been approved by the FDA for the treatment of lower respiratory tract infections (bronchiolitis and pneumonia) and serious RSV infections, but it can cause cardiopulmonary and immunologic disorders in children. Ribavirin inhibits in vitro replication of HIV-1; clinically, ribavirin was shown to delay the onset of full-blown AIDS in patients with early symptoms of HIV infection. Some viruses are less susceptible, for example, poliovirus, herpes viruses excluding varicella, vaccinia, mumps, reovirus, and rotavirus. A randomized double-blind study of aerosolized ribavirin treatment of infants with RSV infections indicated significant improvement in the severity of infection with a decrease in viral shedding (58).

Pharmacokinetics Oral or IV forms of ribavirin are useful in the prevention and treatment of Lassa fever. The oral bioavailability is approximately 45%, and serum half-life is 9 hours. Peak plasma level after 1 hour is 1 to 3 mg/mL. IV administration of the drug has higher peak plasma levels. Aerosol preparation delivery of drug (0.8 mg/kg/hour) produced drug levels in respiratory secretions of 50 to 200 mg/mL (Table 38.3). The clinical benefits of this agent have yet to be confirmed. Its few side effects are generally limited to GI disturbances, such as nausea, vomiting, and diarrhea. The drug is contraindicated in asthma patients because of deterioration of pulmonary function. Viral strains susceptible to ribavirin have not been found to develop drug resistance, as is the case with other antiviral agents, such as acyclovir, idoxuridine, and bromovinyldeoxyuridine.

TRIFLUOROTHYIMIDINE

Mechanism of Action Trifluorothymidine is a fluorinated pyridine nucleoside structurally related to idoxuridine (Fig. 38.11) (59). It has been approved by the FDA

and is a potent, specific inhibitor of replication of HSV-1 in vitro. Its mechanism of action is similar to that of idoxuridine. Like other antiherpes drugs, it is first phosphorylated by thymidine kinase to mono-, di-, and triphosphate forms, which are then incorporated into viral DNA in place of thymidine to stop the formation of late virus mRNA and subsequent synthesis of the virion proteins.

Clinical Application Because of its greater solubility in water, it is active against HSV-1 and HSV-2. It is also useful in treating infections caused by human CMV and VZV infections. The advantage of use of this agent over idoxuridine is its high topical efficacy in the cure of primary keratoconjunctivitis and recurrent epithelial keratitis. It is also useful for difficult cases of herpetic iritis and established stromal keratitis.

Pharmacokinetics Trifluorothymidine is available as a 1% ophthalmic solution, which is effective in dendritic ulcers (Table 38.3). Generally, a 1% eye solution of trifluorothymidine is well tolerated. Cross-hypersensitivity and cross-toxicity between trifluorothymidine, idoxuridine, and vidarabine are rare. The most frequent side effects are temporary burning, stinging, localized edema, and bone marrow toxicity. It is less toxic but more expensive than idoxuridine. Trifluorothymidine, given intravenously, shows a plasma half-life of 18 minutes and is excreted in urine unchanged or as the inactive metabolite 5-carboxyuracil.

VIDARABINE

Mechanism of Action Vidarabine is an adenosine nucleoside obtained from cultures of *Streptomyces antibioticus* (Fig. 38.11) (60). Cellular enzymes convert vidarabine to mono-, di-, and triphosphate derivatives that interfere with viral nucleic acid replication, specifically inhibiting the early steps in DNA synthesis. This agent was used originally as an antineoplastic drug. Its antiviral effect is, in some cases, superior to that of idoxuridine or cytarabine.

Clinical Application Vidarabine is used mainly in human HSV-1 and HSV-2 encephalitis, decreasing the mortality rate from 70% to 30%. Whitley et al. (61) reported that early vidarabine therapy is helpful in controlling complications of localized or disseminated herpes zoster in immunocompromised patients. Vidarabine is also useful in neonatal herpes labialis or genitalis, vaccinia virus, adenovirus, RNA viruses, papovavirus, CMV, and smallpox virus infections. Given the efficacy of vidarabine in certain viral infections, the FDA approved a 3% ointment for the treatment of herpes simplex keratoconjunctivitis and recurrent epithelial keratitis and a 2% IV injection for the treatment of herpes simplex encephalitis and herpes zoster infections (Table 38.3). A topical ophthalmic preparation of vidarabine is useful in herpes simplex keratitis but shows little promise in herpes simplex labialis or genitalis. The monophosphate esters of vidarabine are more water-soluble and can be used in smaller volumes

and even intramuscularly. These esters are under clinical investigation for the treatment of HBV, systemic and cutaneous herpes simplex, and herpes zoster virus infections in immunocompromised patients.

Pharmacokinetics Vidarabine is deaminated rapidly by adenosine deaminase, which is present in serum and red blood cells. The enzyme converts vidarabine to its principal metabolite, arabinosyl hypoxanthine (ara-HX), which has weak antiviral activity (Fig. 38.12) (62). The half-life of vidarabine is approximately 1 hour, whereas ara-HX has a half-life of 3.5 hours. The drug is detected mostly in the kidney, liver, and spleen because 50% of it is recovered in the urine as ara-HX. Levels of vidarabine in CSF are 50% of those in the plasma. The most common side effects of vidarabine are GI disturbances such as anorexia, nausea, vomiting, and diarrhea. CNS side effects include tremors, dizziness, pain syndromes, and seizures. Bone marrow suppression is reported at higher doses. Because vidarabine is reported to be mutagenic, carcinogenic, and teratogenic in animal studies, its use in pregnant women is to be avoided. Allopurinol and theophylline may interfere with the metabolism of vidarabine at higher doses because of the xanthine oxidase metabolism of vidarabine. Therefore, this agent should be avoided or given with caution to patients receiving these medications concurrently. Also, adjustment of the dose is necessary in patients with renal insufficiency.

CYTARABINE

Mechanism of Action Cytarabine (1- β -arabinofuranosylcytosine) is a pyrimidine nucleoside related to idoxuridine (Fig. 38.11) (63). It is used primarily as an anticancer rather than an antiviral agent (see Chapter 37). Cytarabine acts by blocking the utilization of deoxycytidine, thereby inhibiting the replication of viral DNA. The drug is first converted to mono-, di-, and triphosphates, which interfere with DNA synthesis by inhibiting both DNA polymerase and the reductase that promotes the conversion of cytidine diphosphate into its deoxy derivatives.

Clinical Application Cytarabine is used to treat Burkitt lymphoma and both myeloid and lymphatic leukemias. Its antiviral use is in the treatment of herpes zoster

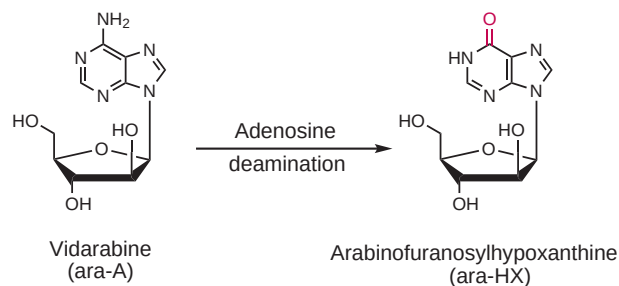


FIGURE 38.12 Metabolism of vidarabine by adenosine deaminase.

(shingles) infection. It is also used to treat herpetic keratitis and viral infections resistant to idoxuridine. The drug is usually used topically, but it has been given by IV injection to individuals with serious herpes infection (64). Cytarabine is deaminated rapidly in the body to an inactive compound, arabinosyluracil, which is excreted in the urine. The half-life of the drug in plasma is 3 to 5 hours. The toxic effects of cytarabine are chiefly on bone marrow, the GI tract, and the kidney. The drug is not given in the early months of pregnancy because of its teratogenic and carcinogenic effects in animals.

Nonnucleoside Analogs (Fig. 38.13)

FOMIVIRSEN SODIUM Fomivirsen was the first antisense oligonucleotide agent approved as an alternative medicine for patients with CMV retinitis for whom other agents did not work. Fomivirsen is a 21-mer phosphorothioate oligodeoxynucleotide (Fig. 38.13). Fomivirsen was used to treat CMV, which causes opportunistic retinitis in patients with AIDS. Such patients respond to fomivirsen but not to other treatment for CMV retinitis, which leads to blindness (65). It works by inhibiting synthesis of proteins responsible for the regulation of viral gene expression that is involved in infection of CMV retinitis. Fomivirsen has recently been withdrawn from the market by the drug manufacturer Novartis and is thus no longer available.

FOSCARNET SODIUM Foscarnet is a trisodium phosphonate hexahydrate that inhibits DNA polymerase of herpes viruses including CMV and retroviral RT (Fig. 38.13) (66). It is not phosphorylated into an active form by viral host cell enzymes. Therefore, it has the advantage of not requiring an activation step before attacking the target viral enzyme. Foscarnet sodium was approved by the FDA

5'-GCG TTT GCT CTT CTT CTT GCG-3'

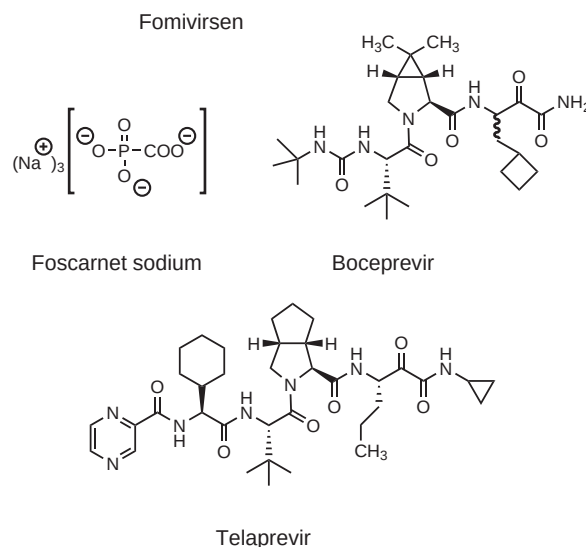
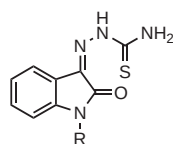


FIGURE 38.13 Nonnucleoside inhibitors of viral replication.

for the treatment of CMV retinitis in AIDS patients. In combination with ganciclovir, the results have been promising, even in progressive disease with ganciclovir-resistant strains. Foscarnet sodium is also effective in the treatment of mucocutaneous diseases caused by acyclovir-resistant strains of HSV and VZV in AIDS patients. The drug is administered intravenously (60 mg/kg) three times a day for initial therapy and at a dose of 90 to 120 mg/kg daily for maintenance therapy (Table 38.3). The plasma-half life is 3 to 6 hours. Foscarnet sodium penetrates into the CSF and the eye. The drug is neurotoxic, and common adverse effects include phlebitis, anemia, nausea, vomiting, and seizures. Foscarnet sodium carries risk of severe hypocalcemia, especially with concurrent use of IV pentamidine. Foscarnet sodium used with zidovudine has an additive effect against CMV and acts synergistically against HIV.

BOCEPREVIR AND TELAPREVIR Boceprevir (SCH 503034, Victrelis) (67) and telaprevir (VX-950, Incivek) (68) represent significant advances in the treatment of HCV infection (Fig. 38.13) (Table 38.3). HCV is an enveloped, single-stranded sense RNA virus that encodes a polyprotein precursor of about 3,000 amino acids. The essential HCV enzyme NS3-4A protease is responsible for cleaving this HCV polyprotein at four sites to generate four peptides called NS4A, NS4B, NS5A, and NS5B. Boceprevir and telaprevir act as potent inhibitors of the HCV NS3-4A protease, and this inhibition prevents the processing of the HCV polyprotein and prevents the virus from replicating. Both compounds have recently been approved by the FDA and can be used in combination with pegylated interferon and ribavirin (69). For untreated HCV infection, telaprevir is administered for the first 12 weeks of therapy at 750 mg three times per day in combination with pegylated interferon α -2a 180 μ g/week, and ribavirin 1.0 to 1.2 g/day according to body weight. After 12 weeks, telaprevir is discontinued, and pegylated interferon α -2a and ribavirin are continued until HCV RNA is undetectable in serum. Boceprevir must be administered using a different regimen. For untreated HCV infection, pegylated interferon α -2b and ribavirin are given alone for 4 weeks, and then boceprevir (800 mg three times a day) is given in combination with pegylated interferon α -2b (1.5 μ g/kg/week) and ribavirin (0.8 to 1.4 g/day according to body weight) for 24 weeks. Total treatment duration is 28 weeks if patients test negative for HCV RNA but may need to be continued for an additional 20 weeks to clear the infection if HCV RNA is still detected after 28 weeks.

Agents Affecting Translation by the Ribosome



Methisazone (R = CH₃)

Methisazone (70)

Methisazone interferes with the translation of mRNA message at the ribosome, preventing the synthesis of protein synthesis. Ultimately, it produces a defect in protein incorporation into the virus. Although viral DNA increases and host cells are damaged, an infectious virus is not produced. Methisazone is active against poxviruses, including variola and vaccinia (71). Some RNA viruses, such as rhinoviruses, echoviruses, reoviruses, influenza, parainfluenza, and polioviruses are also inhibited. Therapeutically, methisazone is given in 1.5- to 3.0-g doses, twice daily by mouth. It has also been used as a prophylactic agent against smallpox. Historically, methisazone was one of the first antiviral compounds used in clinical practice. It is orally absorbed, with nausea and vomiting as the principal side effects. The drug is also used in vaccinia gangrenosa and disseminated vaccinia infections. This drug is not available in the United States, but it has been used in Europe for some time. Several analogs of methisazone (R = H, R = ethyl) possess activity against variola, neurovaccinia, smallpox in mice, Rous sarcoma virus, and vaccinia generalisata.

Antiretroviral (Anti-HIV) Agents Including Protease Inhibitors (72,73)

While there can be no permanent cure of AIDS without prevention or elimination of HIV infection, AIDS patients can prolong their life if the disease is diagnosed early and treatment is promptly initiated. Initial HIV treatment requires specific drugs that inhibit RT and HIV protease. In advanced HIV infection, AIDS is complicated by other organisms that proliferate in immunocompromised hosts, known as opportunistic infections. Such patients are treated symptomatically with a variety of drugs depending on the opportunistic infections (74–76). Anti-HIV agents have side effects, but patients can be managed by a careful monitoring of the drugs. Opportunistic diseases include infections by parasites, bacteria, fungi, and viruses. Neoplasms including Kaposi sarcoma and Burkitt lymphoma also commonly occur.

Anti-HIV agents are classified according to the mode of action. The drugs inhibiting RT interfere with replication of HIV and stop synthesis of infective viral particles. They are further classified into nucleoside and nonnucleoside RT inhibitors. The drugs inhibiting HIV protease inactivate RT activity and block release of viral particles from the infected cells. The chemistry, pharmacokinetics, side effects, toxicity, and drug interactions of RT inhibitors and protease inhibitors are discussed in the following sections.

Nucleoside Reverse Transcriptase Inhibitors (77,78)

The synthesis of viral DNA under the direction of RT requires availability of purines and pyrimidine nucleosides and nucleotides. Therefore, it is not surprising that a variety of chemical modifications of natural nucleosides have been investigated. Two such modifications

have resulted in active drugs. Removal of the ribosyl 3'-hydroxyl group of the deoxynucleosides has given rise to dideoxyadenosine (didanosine is the prodrug for this derivative) (79,80), dideoxycytidine (81–83), and dideohydideoxythymidine (84). Replacement of the 3'-deoxy with an azido group has given 3'-azidothymidine (85–88) and 3'-azidouridine (no longer used as a drug) (Fig. 38.14 and Table 38.4) (89). All of these drugs have similar mechanisms of action in that their incorporation into the viral DNA will ultimately lead to chain-terminating blockade due to the lack of a 3'-hydroxyl needed for the DNA propagation.

ZIDOVUDINE

Mechanism of Action Zidovudine (ZDV) is an analog of thymidine in which the azido group is substituted at the 3-carbon atom of the dideoxyribose moiety (Fig. 38.14). It is active against RNA tumor viruses (retroviruses) that are the causative agents of AIDS and T-cell leukemia. Retroviruses, by virtue of RT, direct the synthesis of a provirus (DNA copy of a viral RNA genome). Proviral DNA integrates into the normal cell DNA, leading to the HIV infection. ZDV is converted to 5'-mono-, di-, and tri-phosphates by the cellular thymidine kinase. These phosphates are then incorporated into proviral DNA, because RT uses ZDV-triphosphate as a substrate. This process prevents normal 5',3'-phosphodiester bonding, resulting in termination of DNA chain elongation due to the presence of an azido group in ZDV. The multiplication of HIV is halted by selective inhibition of RT and thus viral DNA polymerase by ZDV-triphosphate at the required dose concentration. ZDV is a potent inhibitor of HIV-1 but also inhibits HIV-2 and EBV.

Clinical Application ZDV is used in AIDS and AIDS-related complex (ARC) to control opportunistic infections by raising absolute CD4⁺ lymphocyte counts. ZDV

was first synthesized by Horwitz (1964) (90), biologic activity was reported by Ostertag et al. (1974) (91), and in 1986, Yarchoan et al. (92) demonstrated application of ZDV in clinical trials of AIDS and related diseases. ZDV is recommended in the control of the disease in asymptomatic patients in whom absolute CD4⁺ lymphocyte counts are less than 200/mm³. It prolongs the life of patients affected with *Pneumocystis jirovecii* pneumonia and improves the condition of patients with advanced ARC by reducing the severity and frequency of opportunistic infections. Substantial benefits are obtained when the drug is given after the CD4⁺ cell counts fall below 500/mm³. Therefore, ZDV is used in early and advanced symptomatic treatment of AIDS or ARC patients. ZDV with other RT inhibitors or in combination with protease inhibitors is more beneficial when resistance to ZDV occurs.

HIV attacks susceptible cells and interacts mainly with CD4⁺ cell surface proteins of helper T cells. As discussed earlier, the viral glycoprotein gp120 forms a complex with CD4 receptor on host cells and enters the cells by endocytosis. The sequence of events is shown in Figure 38.2. Ultimately, the immune system of the host is altered and AIDS symptoms appear. AIDS patients have symptoms such as high fever, weight loss, lymphadenopathy, chronic diarrhea, myalgias, fatigue, and night sweats. ZDV is given in such conditions. However, the drug is toxic to the bone marrow and causes macrocytic anemia, neutropenia, and granulocytopenia. Other adverse reactions include headache, insomnia, nausea, vomiting, seizures, myalgias, and confusion.

Pharmacokinetics ZDV is available in 100-mg capsules for oral administration. For asymptomatic adults, the initial recommended dosage is 1,200 mg daily (200 mg every 4 hours) reducing to 600 mg daily (100 mg every 4 hours) for patients with advanced disease (Table 38.4).

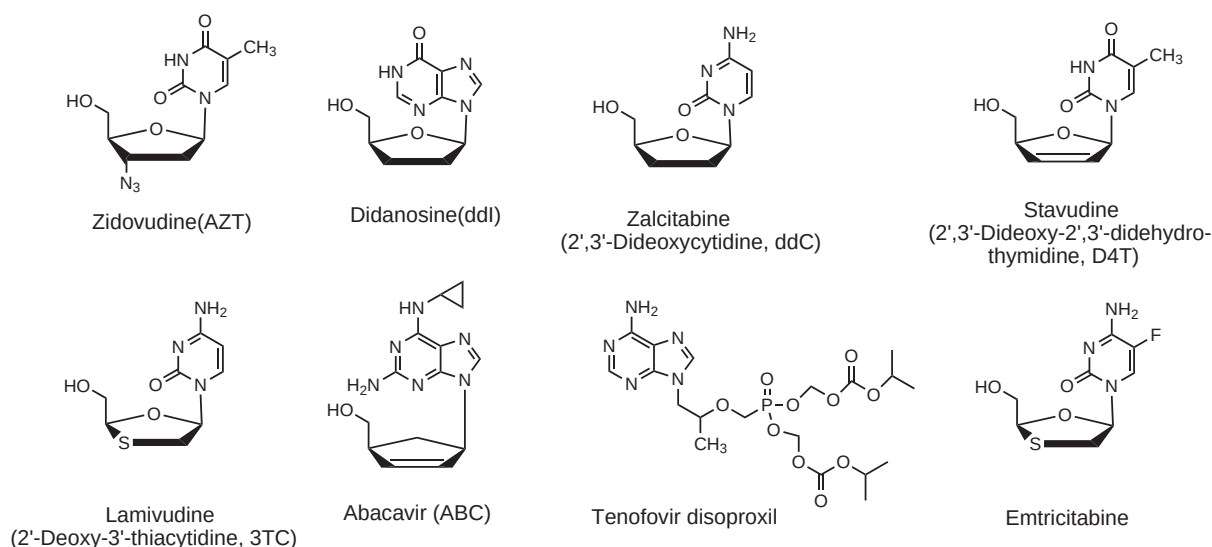


FIGURE 38.14 Nucleoside reverse transcriptase inhibitors (NRTIs).

TABLE 38.4 HIV Reverse Transcriptase Inhibitors

Generic Name	Common Name	Trade Name	Dosage Form (mg/unit)
Nucleoside reverse transcriptase inhibitors			
Zidovudine	AZT ZDV	Retrovir	Tab (300), Cap (100), Syrup (50/5 mL), Inj (10/mL)
Didanosine	ddI	Videx	Tab (25, 50, 100, 150, 200)
Dideoxyadenosine	ddA		Powder for oral Sol (100, 167, 250)
Zalcitabine	ddC	Hivid	Tab (0.375)
Stavudine	D4T	Zerit	Cap (15, 20, 30, 40) Powder for oral Sol (1/mL)
Lamivudine	3TC	Epivir Epivir HBV	Tab (150), Sol (10/mL) Tab (100), Sol (5/mL)
Abacavir	ABC	Ziagen	Tab (300), Sol (20/mL)
Tenofovir disoproxil		Viread	Tab (300)
Emtricitabine		Emtriva	Cap (200)
Nonnucleoside reverse transcriptase inhibitors			
Nevirapine		Viramune	Tab (200)
Delavirdine		Rescriptor	Tab (100)
Efavirenz		Sustiva	Cap (50, 100, 200)

Cap, capsule; Inj, injection; Sol, solution; Tab, tablet.

The maintenance dose is 600 mg daily in symptomatic patients. ZDV is sensitive to heat and light because of its azide group and should be stored in colored bottles at 15° to 25°C. ZDV is well absorbed through the GI tract. It concentrates in the body tissues and fluids, including CSF. The bioavailability of the drug was found to be approximately 65%. Its half-life is approximately 1 hour. IV doses of 2.5 mg/kg or oral doses of 5 mg/kg yielded peak plasma concentrations of 5 mmol/L. Plasma protein binding was approximately 30%. Most of the drug is converted to its inactive glucuronide metabolite and is excreted unchanged through urine. ZDV also crosses the blood–brain barrier. Pentamidine, dapsone, amphotericin B, flucytosine, and doxorubicin may increase the toxic effects of ZDV. Probenecid prolongs the plasma half-life of the drug.

DIDANOSINE

Mechanism of Action Didanosine (ddI) is a purine dideoxynucleoside analog of inosine. Chemically, it is 2',3'-dideoxyinosine and differs from inosine by having hydrogen atoms in place of 2'- and 3'-hydroxyl groups on the ribose ring (Fig. 38.14). ddI is a prodrug that is bioactivated by metabolism to dideoxyadenosine triphosphate. Dideoxyadenosine triphosphate is a competitive inhibitor of viral RT and is incorporated into the developing viral DNA in place of deoxyladenosine triphosphate.

As such this agent causes chain termination due to the absence of a 3'-hydroxyl group. ddI inhibits HIV RT and exerts a virustatic effect on the retroviruses. Combined with ZDV, antiretroviral activity of ddI is increased.

Pharmacokinetics ddI has a plasma half-life of 1.5 hours and is given in 200-mg dose twice daily. Oral bioavailability of the drug is approximately 25% at doses of 7 mg/kg or less. It significantly decreases p24 antigen levels and increases CD4⁺ cell counts. In some cases, viral resistance to ddI has been known to occur after treatment for 1 year. It is given in advanced HIV infection, ZDV intolerance, or significant clinical/immunologic deterioration. The major side effects of ddI are painful peripheral neuropathy and pancreatitis. Some of the minor side effects include abdominal pain, nausea, and vomiting. The use of products, such as pentamidine, sulfonamides, and cimetidine should be avoided with ddI.

ZALCITABINE Zalcitabine (ddC) is a useful alternate to ZDV and is a synthetic pyrimidine nucleoside analog. It differs from 2'-deoxycytidine in that the 3'-hydroxyl group of the 2'-deoxyribose moiety is replaced with a hydrogen atom (Fig. 38.14). It is given in combination with ZDV when CD4⁺ cell counts fall below 300 cells/mm³. Monotherapy with ddC was more active than ZDV. Its oral bioavailability is 87%, and plasma half-life is approximately 1 hour.

It has side effects such as stomatitis, rash, fever, malaise, arthritis, and arthralgia. In low doses (0.005 mg/kg every 4 hours), ddC produced sustained decrease in p24 antigen level and increase in CD4⁺ cell counts. The CSF fluid/plasma ratio of ddC is 0.2.

After oral administration, bioavailability of ddC was less than 80%, which was further reduced with food. The mean maximum plasma concentration of the drug was also reduced from 25.2 to 15.5 ng/mL when the drug was taken with food. Dideoxyuridine is the major metabolite in urine and feces. The drug demonstrated penetration through blood–brain barrier. The major toxicity of ddC is peripheral neuropathy, in which case it should be discontinued. Pancreatitis occurs in some cases when given alone or in combination with ZDV.

STAVUDINE Stavudine (D4T) is a pyrimidine nucleoside analog that has significant activity against HIV-1 after intracellular conversion of the drug to a D4T-triphosphate. It differs in structure from thymidine by the replacement of the 3'-hydroxyl group with a hydrogen atom and a double bond in the 2' and 3' positions on the deoxyribose ring (Fig. 38.14). It decreased p24 antigen and raised CD4⁺ cell counts. D4T is beneficial for patients in whom CD4⁺ cell counts do not decrease below 300 cells/mm³ with ZDV and ddI. It was shown to be more effective than ZDV or ddC in delaying the progression of HIV infection. It is recommended for patients with advanced HIV infection.

Pharmacokinetics D4T is rapidly absorbed, and absolute bioavailability in adults is 85% at an oral dose of 4 mg/kg. A peak plasma concentration in a dose-dependent manner occurs within an hour. It can be taken with food. The apparent volume of distribution after oral dose is 66 L. The plasma half-life of D4T is approximately 1.5 hours, and the intracellular half-life of D4T-triphosphate is 3.5 hours. It is less toxic to bone marrow but causes peripheral neuropathic toxicity. The side effects include pain, tingling, and numbness in the hands and feet.

LAMIVUDINE Lamivudine (3TC) is an analog of ddC in which 3'-methylene group is replaced with a sulfur (S) atom in the ribose ring (Fig. 38.14) (93). It exerts virustatic effect against retroviruses by competitively inhibiting HIV RT after intracellular conversion of the drug to its active 5'-3TC-triphosphate form. It is usually given with other antiretroviral agents, such as ZDV or D4T. 3TC in a 600 mg/day dose reduced HIV cells by 75%, and in combination with ZDV, the reduction in viral load was 94%. 3TC is rapidly absorbed through the GI tract. Its bioavailability is approximately 86% after oral administration of 2 mg/kg twice daily; peak serum 3TC concentration was approximately 2 mg/mL. 3TC binding to human plasma was approximately 36%. In vivo, it is converted to *trans* sulfoxide metabolite, and a majority of the drug is eliminated unchanged in urine. The FDA approved 3TC in combination with ZDV for the treatment of disease

progression caused by HIV infection. The combinations of 3TC with ddI, ddC, or D4T also are used for advanced HIV infection. Such combinations have the ability to delay resistance to ZDV and restore ZDV sensitivity in AIDS patients. Recently, oral therapy in lower doses of 3TC (Table 38.4) has been approved by the FDA for treatment of chronic HBV. Peripheral neuropathy and GI disturbances are the major side effects of 3TC. The minor side effects are nausea, vomiting, and diarrhea.

ABACAVIR SULFATE Abacavir was approved in 1998 as a nucleoside reverse transcriptase inhibitor to be used in combination with other drugs for the treatment of HIV and AIDS. The drug is extensively metabolized via stepwise phosphorylation to 5'-mono-, di-, and triphosphate. Abacavir is well absorb (>75%) and penetrates the CNS and can be taken without regard to meals. The drug does not show any clinically significant drug–drug interactions but has been reported to produce life-threatening hypersensitivity reactions. The major use of abacavir appears to be in combination with other nucleoside reverse transcriptase inhibitors. A fixed combination product has recently been approved by the FDA under the trade name of Trizivir and consists of 300 mg of abacavir, 150 mg of 3TC, and 300 mg of ZDV. The combination has been shown to be superior to other combinations in reducing viral load as well as showing improvement in CD4⁺ cell count. The most common adverse effects reported with abacavir include headache, nausea, vomiting, malaise, and diarrhea.

TENOFOVIR DISOPROXIL (94)

Mechanism of Action Tenofovir is a prodrug with a bioavailability of 25%, which is improved in the presence of food (35%). The drug is approved for the treatment of HIV infections in adult patients. Tenofovir diphosphate is an HIV reverse transcriptase inhibitor. The drug is hydrolyzed via plasma esterase to tenofovir, which is then phosphorylated to the active tenofovir diphosphate, as shown in Figure 38.15. The diphosphate competes with

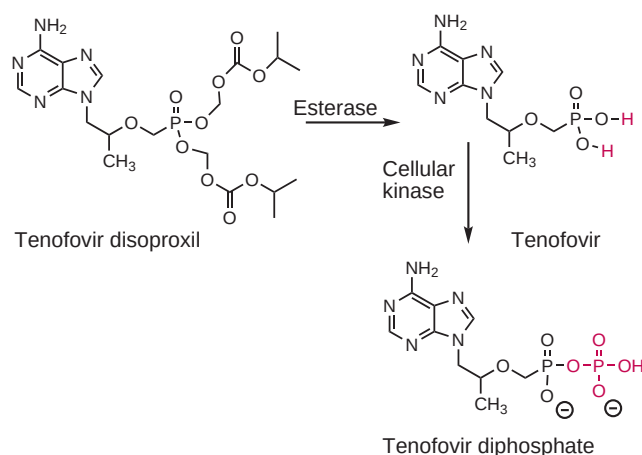


FIGURE 38.15 Metabolic activation of tenofovir disoproxil.

deoxyadenosine triphosphate for incorporation into viral DNA, and when incorporated, tenofovir diphosphate results in premature termination of DNA growth and inhibition of DNA polymerase. Tenofovir disoproxil is indicated for treatment-experienced patients with HIV-1. The drug appears to also be effective in treatment-naïve patients, but initial approval is for treatment-experienced patients. The drug is administered as one tablet once daily (300 mg). It is recommended that the drug be combined with other reverse transcriptase inhibitors or HIV protease inhibitors, which results in additive or synergistic activity. At present, two fixed combinations exist that contain tenofovir: tenofovir disoproxil fumarate plus emtricitabine (Truvada) and a fixed triple combination of tenofovir, emtricitabine, and efavirenz (Atripla). Other fixed-dose products may be approved in the future for treatment of HIV.

EMTRICITABINE (95) Emtricitabine is an orally active nucleoside RT inhibitor (Fig. 38.14). It is metabolized in vivo to the 5'-triphosphate, which in turn competes with the normal substrate (deoxycytidine-5'-triphosphate) for incorporation into DNA. In addition, incorporation of emtricitabine into viral DNA inhibits further chain elongation, resulting in chain termination. The drug is administered orally once daily. The (–)-enantiomer is the most active form of the drug, although the (+)-isomer is also active. Emtricitabine is not bound to plasma protein, and approximately 86% is excreted unchanged in the urine. The only metabolites identified consist of the 3'-sulfoxide and the 2'-O-glucuronide. Emtricitabine is reported to be more active than 3TC with a low level of resistance developing when used in combination therapy with efavirenz and ddI. As indicated earlier, the drug is also available in fix-dosed combinations with tenofovir.

Nonnucleoside Reverse Transcriptase Inhibitors

The FDA has recently approved several nonnucleosides that inhibit RT activity (Fig. 38.16 and Table 38.4). They are used with nucleoside drugs to obtain synergistic

activity in decreasing the viral load and increasing CD4⁺ cell count. These drugs are primarily designed and synthesized by protein structure-based drug design methodologies. Their use as monotherapy may be limited because of rapid onset of resistance and hypersensitivity reactions. However, interaction of nonnucleoside drugs with other protease inhibitors, such as saquinavir, indinavir, and ritonavir, is being investigated. In addition, interaction of these drugs with clarithromycin, ketoconazole, rifabutin, and rifampin is under study.

NEVIRAPINE

Mechanism of Action Nevirapine and its analogs exhibit antiretroviral effects against AZT-resistant HIV strains (96). Nevirapine in combination with ZDV and ddI produced approximately 18% higher CD4⁺ cell counts and a decrease in viral load compared with ZDV and ddI. Nevirapine is recommended with nucleosides for HIV-1-infected patients who have experienced clinical or immunologic deterioration. The significant side effects of nevirapine are liver dysfunction and skin rashes. Nevirapine is a dipyrindiazepinone derivative, which binds directly to RT. Thus, it blocks RNA- and DNA-dependent polymerase activities by causing a disruption of the RT catalytic site. HIV-2 RT and human DNA polymerases are not inhibited by nevirapine. The 50% inhibitory concentration ranged from 10 to 100 nmol/L against HIV-1.

Pharmacokinetics Nevirapine is rapidly absorbed after oral administration, and its bioavailability is approximately 95%. Peak plasma nevirapine concentrations of 2 ± 0.4 mg/mL (7.5 mmol/L) are obtained in 4 hours after a single 200-mg dose (Table 38.4). Following multiple doses, nevirapine concentrations appear to increase linearly in the dose range of 200 to 400 mg/day. Nevirapine is about 60% bound to plasma proteins in the plasma concentration range of 1 to 10 mg/mL. It readily crossed the placenta and was found in breast milk. Nevirapine is metabolized as glucuronide conjugates of hydroxylated metabolites, which are excreted in urine. In vivo, ketoconazole did not produce any significant inhibitory effect on nevirapine metabolism. The plasma concentrations of nevirapine were elevated or reduced in patients receiving cimetidine or rifabutin, respectively.

DELAVIRDINE Delavirdine, a bisheteroaryl piperazine derivative, is a potent nonnucleoside RT inhibitor with activity specific for HIV-1 (97). The FDA has approved this drug in combination with other anti-HIV agents (Table 38.4). In phase I/II trials, it demonstrated sustained improvements in CD4⁺ cell counts, p24 antigen levels, and RNA viral load. Promising results were obtained when the drug was used in two- or three-drug combination with nucleoside drugs. Combination of delavirdine with ddI, ddC, or ZDV demonstrated additive or synergistic effect. However, delavirdine with ZDV was more beneficial in

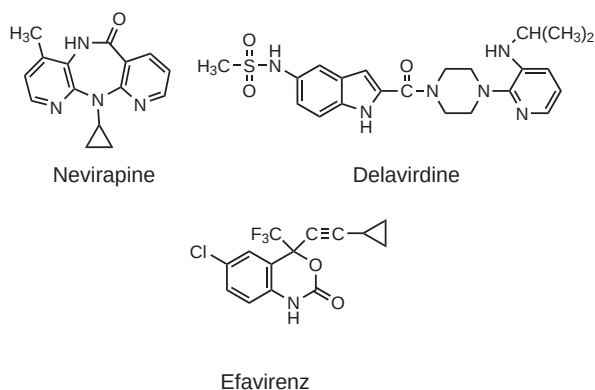


FIGURE 38.16 Nonnucleoside reverse transcriptase inhibitors (NNRTI).

early HIV infection. Combinations of nevirapine and delavirdine have an antagonistic effect on HIV-1 RT inhibition. Delavirdine directly inhibits RT and DNA-directed DNA polymerase activities of HIV-1 after the formation of the enzyme–substrate complexes, thereby causing chain termination effects.

Pharmacokinetics Delavirdine is rapidly absorbed by oral administration, and peak plasma concentration is obtained in 1 hour. The single-dose bioavailability of delavirdine tablets relative to oral solution was approximate 85%. The 50% inhibitory concentration for delavirdine against RT activity was 6.0 nmol/L. Delavirdine is extensively bound to plasma protein (~98%). Delavirdine is metabolized to its inactive *N*-desisopropyl metabolite in liver, and the pharmacokinetics is non-linear. This reaction is catalyzed by CYP3A4, and delavirdine appears to be an inhibitor of this enzyme, increasing the risk for drug–drug interactions with other drugs that are substrates for this cytochrome isoform. Skin rashes are the major side effect of delavirdine therapy. Cross-resistance between delavirdine and protease inhibitors, such as indinavir, nelfinavir, ritonavir, and saquinavir, is unlikely because they act at different target enzymes.

EFAVIRENZ Efavirenz is a new nonnucleoside RT inhibitor that was approved by the FDA (Table 38.4) as a potent inhibitor of wild-type and resistant mutant HIV-1, which is inhibited up to 95% with efavirenz concentrations of 1.5 mmol/L (98). In combination with indinavir, a mean reduction in HIV-RNA of 1.68 log and an increase in CD4⁺ cell counts of 96 cells/mm³ were reported. Coadministration of efavirenz with indinavir reduced indinavir concentration (AUC) by approximately 35%.

Pharmacokinetics Efavirenz is administered once a day and can be used as a substitute for indinavir in combination therapy with standard drugs, such as ZDV and 3TC (also as a fixed combination of tenofovir, emtricitabine, and efavirenz [Atripla]). Because it is given once a day, this cuts down the number of pills that an AIDS patient has to swallow. In the current cocktail therapy of AIDS patients, efavirenz is a good option for reducing the many side effects of cocktail therapy. It is administered to both adults and children and may be less expensive than indinavir. The side effects of efavirenz include dizziness, insomnia, impaired concentration, abnormal dreams, and drowsiness. The most common adverse effect is a skin rash. Other side effects are diarrhea, headache, and dizziness. Efavirenz is recommended to be taken at bedtime with or without food. Avoiding driving or operating machinery and intake of high-fat meals are recommended. It should always be taken in combination with at least one other anti-HIV agent. Efavirenz is contraindicated with midazolam, triazolam, or ergot derivatives.

TABLE 38.5 HIV Protease Inhibitors

Generic Name	Trade Name	Dosage Form (mg/unit)
Saquinavir	Invirase	Cap (200)
	Fortovase	Cap (200)
Ritonavir	Norvir	Cap (200)
Indinavir	Crixivan	Cap (200, 400)
Nelfinavir	Viracept	Tab (250), Powder (50/g)
Amprenavir	Agenerase	Cap (50, 150) Sol (15/mL)
Lopinavir/ritonavir	Kaletra	Cap (133.3/33.3) Sol (80/20 per mL)
Atazanavir	Reyataz	Cap (150, 200)
Fosamprenavir	Lexiva	Tab (700)
Tipranavir	Aptivus	Cap (250)
Darunavir	Prezista	Tab (75, 150, 300, 400 or 600)

Cap, capsule; Sol, solution; Tab, tablet.

HIV Protease Inhibitors (Table 38.5) (99)

Mechanism of Action

The HIV genome contains various structural genes, such as the *gag* and *gag-pol* genes, that are translated into precursor polypeptides that form immature viral particles. These precursor protein molecules are processed (cleaved) by an essential viral pol-encoded aspartic proteinase, HIV protease, to form the desired structural proteins of the mature viral particle that are necessary for virus replication and survival. For example, the structural proteins p7, p9, p17, and p24, which play important roles in infectivity of HIV, are products of the *pol* gene. HIV protease also activates RT and plays an important role in the release of infectious viral particles. Thus, an area of considerable interest has been the development of drugs that act as inhibitors of HIV protease. Such inhibitors act on HIV protease and prevent posttranslational processing and budding of immature viral particles from the infected cells. This group of drugs represents a major breakthrough in treatment of HIV when used in combination with RT inhibitors, and their development is one of the most significant advances in medicinal chemistry.

Functional HIV protease exists as a dimer in which each monomer contains one of two conserved aspartate residues at the active site. There are several HIV protease cleavage sites on the protein precursors, but the enzyme prefers to perform the hydrolysis of the peptide bond on the amino terminal side of a proline (100). As shown in Figure 38.17, the amino acid R groups flanking the scissile bond are designated P1, P2, and so on, on the amino terminus, and P1', P2', and so on, on the carboxyl terminus of the cleavage site (Fig. 38.17A). The corresponding pockets on the enzyme that are responsible for binding

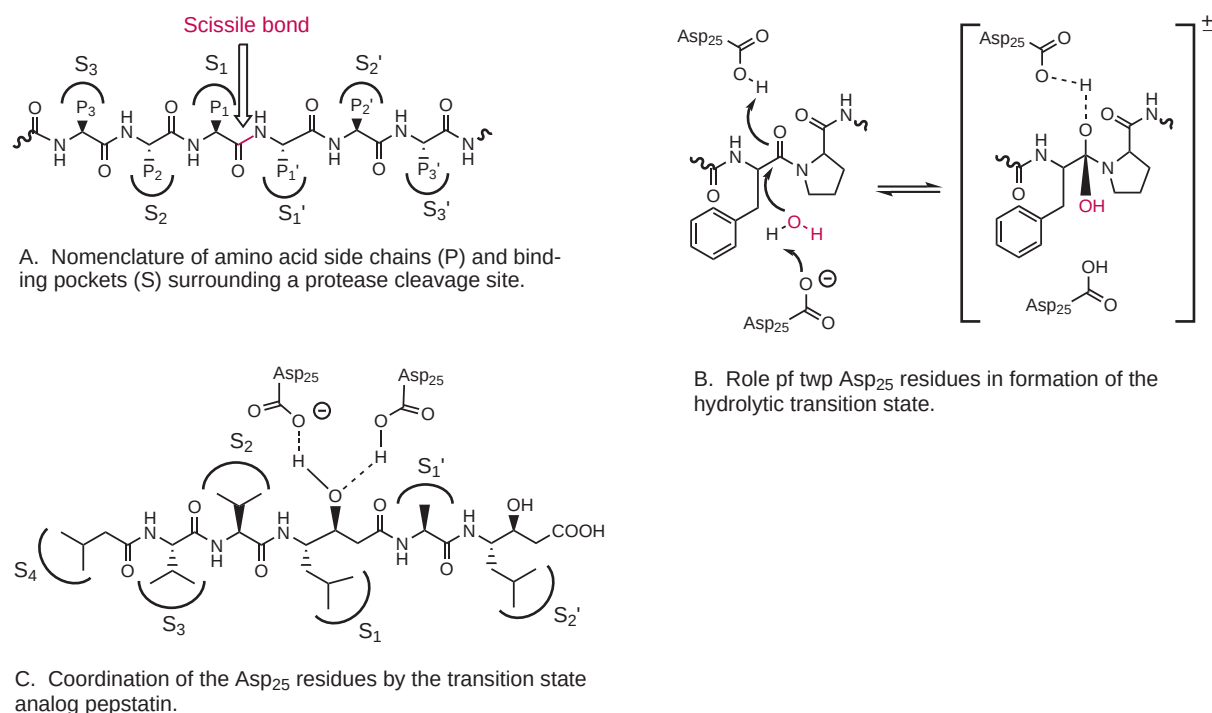


FIGURE 38.17 Principles of the design of transition-state analog inhibitors of HIV protease.

the P groups are termed S1, S2, S1', S2', and so on. In the active site, a pair of aspartates (the Asp₂₅ on each of the two subunits of the protein) work together to complete the reaction (Fig. 38.17B). One coordinates the water that performs the hydrolysis, while the other coordinates the carboxyl group on the P1' amino acid. Rationally designed drugs that inhibit HIV protease are designed as transition-state mimics that align at the active site of HIV-1 protease, as defined by three-dimensional crystallographic analysis of the protein structure. Figure 38.17C shows the aspartic protease inhibitor pepstatin bound in the HIV protease active site. Pepstatin is an inhibitor of all aspartic proteases, but can be used to illustrate the mechanism by which transition-state analogs inhibit HIV protease (101). Pepstatin contains an unnatural amino acid core known as statine (Fig. 38.18), and the hydroxyl-bearing sp³ carbon of this core mimics the tetrahedral transition state that occurs during hydrolysis of the peptide bond. Other transition-state cores have been used in the design of highly potent aspartic protease inhibitors, including the hydroxyethylene and hydroxyethylamine cores (Fig. 38.18). The crystal structures of several HIV protease inhibitors bound to wild-type and mutant HIV proteases have been solved by x-ray crystallography. The structure of the HIV protease inhibitor lopinavir bound to wild-type HIV protease is depicted in Figure 38.19. Note that the hydroxyl group of the transition-state core is hydrogen bonded to the two Asp₂₅ carboxylate moieties at a distance of about 3 Å. A number of oligopeptide-like analogs have been synthesized that differentially inhibit viral and mammalian aspartic proteases, and the

most useful of these are selective for the viral enzyme HIV-1 protease. Structurally, these agents are either peptidomimetic or nonpeptide compounds. Their effectiveness is related to their ability to inhibit the processing of the gag-pol precursor polyprotein to p24, p55, and p160. Consequently, the infectivity of HIV-1 is diminished. Although, some compounds exhibit *in vitro* and *in vivo* antiviral activities, optimization of their pharmacokinetic and pharmacodynamic properties has presented major problems. In view of the great demand for successful anti-AIDS drugs, the FDA has approved the protease inhibitors described in the following sections using the accelerated approval process. The structures of currently marketed HIV protease inhibitors are shown in Figure 38.20.

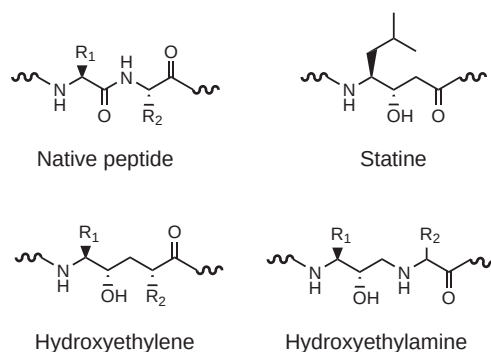


FIGURE 38.18 Transition-state mimics used in the design of HIV protease inhibitors.

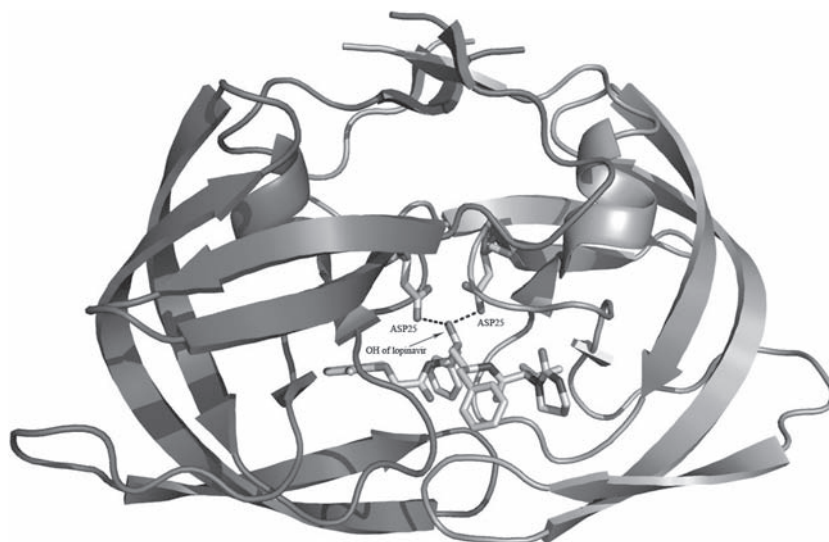


FIGURE 38.19 Model of the HIV protease inhibitor lopinavir bound to wild-type HIV protease.

Potential Drug Interactions

Protease inhibitors have a high potential for drug interactions, stemming from the fact that they are substrates for and inhibitors of the CYP3A4 enzyme system. As a result, concurrent use of protease inhibitors with other drugs metabolized by CYP3A4 may be contraindicated, and in some cases, the resulting drug interactions can be life-threatening. The most potent CYP450 inhibitor in this class is ritonavir (used to advantage in combination with lopinavir and other HIV protease inhibitors), followed by indinavir, nelfinavir, and amprenavir (moderate inhibitors) and saquinavir (the least potent inhibitor). Drug interactions have been reported with bepridil, dihydroergotamine, and a number of benzodiazepines. Marked increases in the activity of amiodarone, lidocaine (systemic), quinidine, the tricyclic antidepressants, and warfarin might be expected (102). Other interactions have been reported with rifampin, rifabutin, phenobarbital, phenytoin, dexamethasone, or carbamazepine. Because the protease inhibitors are themselves metabolized by CYP450, their action may be altered by other agents that induce or inhibit this system. In the case of rifabutin, which inhibits CYP3A4 in the gut, relative bioavailability of the protease inhibitors is increased, and the dose of the protease inhibitor may need to be decreased.

HIV Protease Inhibitors that are Used Clinically (Fig. 38.20)

SAQUINAVIR MESYLATE

Clinical Application Saquinavir was the first protease inhibitor approved by the FDA in December 1995 (103,104). It is a carboxamide derivative used in the treatment of advanced HIV infection in selected patients. Saquinavir is used concomitantly in either ZDV-untreated patients or ddC-treated patients previously treated with prolonged ZDV therapy. Although combined therapy did

not slow progression of disease, CD4⁺ cell counts were increased in patients infected with HIV in the United States and European countries. Triple therapy with saquinavir, ZDV, and ddC has been more effective than double therapy with saquinavir plus ZDV or ddC. Thus, combination therapy slowed disease progression and mortality.

The IC₅₀ concentration of saquinavir in both acutely and chronically infected cells was 1 to 30 nmol/L. In combination with ZDV, ddC, or ddI, the activity of saquinavir was increased without increased cytotoxicity. The resistance of HIV isolates to saquinavir was observed due to substitution mutations in the HIV protease at amino acid positions 48 (glycine to valine) and 90 (leucine to methionine).

Pharmacokinetics The bioavailability of saquinavir in a single 600-mg dose following a high-fat meal was shown to be about 4%. Approximately 30% of a 600-mg dose of saquinavir reached the liver, where it showed first-pass metabolism. The metabolites, various mono- and dihydroxylated compounds of which the major metabolites are the monohydroxylated products shown in Figure 38.21, are not active. Approximately, 88% and 19% of a 600-mg oral dose was found in the feces and urine, respectively. The volume of distribution after IV administration of a 12-mg dose of saquinavir was 700 L. The drug is 98% bound to plasma proteins, and a very low concentration of saquinavir was found in CSF. The steady-state AUC was 2.5 times higher than that observed after a single dose of 600 mg in HIV-infected patients after a meal as compared to multiple dosing. Saquinavir has a plasma half-life of approximately 1.8 hours. Although saquinavir hard-gel capsule in combination with other antiretroviral drugs reduced the risk of disease progression or death, it has limited bioavailability. To overcome this limitation, the FDA has approved saquinavir soft-gel capsules. Saquinavir is well tolerated in combination with ZDV and/or ddC

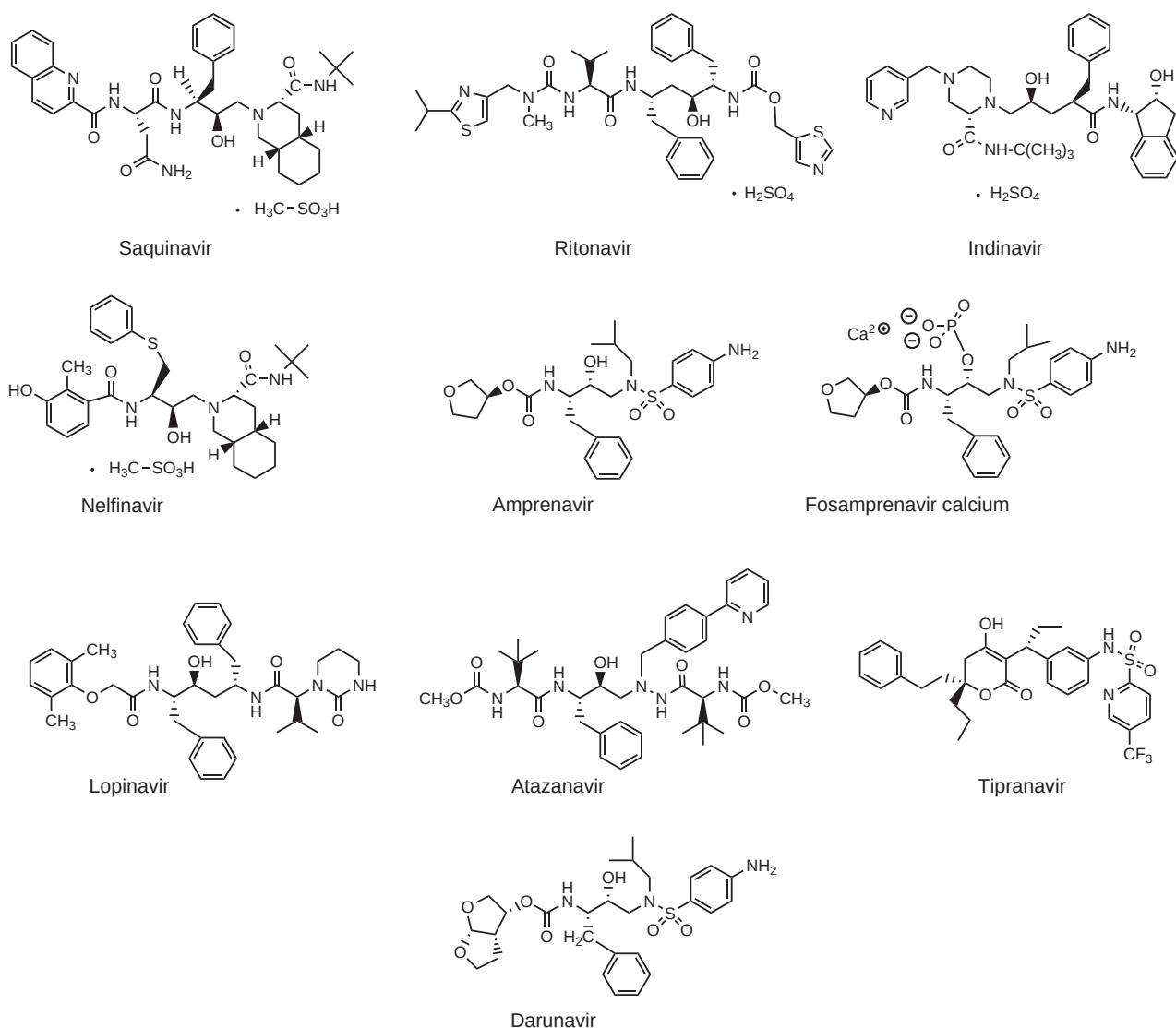


FIGURE 38.20 Structures of HIV protease inhibitors that are used clinically.

and has few side effects, but GI disturbances were common adverse effects. Saquinavir has a few mild side effects, such as headache, rhinitis, nausea, and diarrhea.

RITONAVIR Ritonavir is another HIV protease inhibitor that was approved by the FDA in March 1996 (Fig. 38.20) (105). Ritonavir is a peptidomimetic inhibitor of both the HIV-1 and HIV-2 proteases. A 50% reduction in viral replication was obtained at a 3.8 to 153 nmol/L concentration of ritonavir.

Pharmacokinetics After a 600-mg dose of an oral solution, peak concentrations of ritonavir were obtained in approximately 2 or 4 hours under fasting or nonfasting conditions, respectively. Under nonfasting conditions, peak ritonavir concentrations decreased 23%, and the extent of absorption decreased 7% relative to fasting conditions. In two separate studies, the capsule and oral

solution indicated AUC values of 129.5 ± 47.1 and 129.0 ± 39.3 mg/hour/mL, respectively, when a 600-mg dose was given under nonfasting conditions. Five ritonavir metabolites have been isolated from human urine and feces. The isopropylthiazole oxidation product was the major active metabolite (M2) (Fig. 38.22).

As with saquinavir, ritonavir is metabolized by CYP3A4 and is an inhibitor of the CYP450 system. Ritonavir is contraindicated with several compounds, such as clarithromycin, desipramine, ethinyl estradiol, rifabutin, sulfamethoxazole, and trimethoprim because of increased concentrations of these drugs in the plasma due to inhibited oxidative metabolism. Ritonavir alone or in combination with 3TC, ZDV, saquinavir, or ddC increased CD4⁺ cell counts and decreased HIV RNA particle levels. Cross-resistance between ritonavir and RT inhibitors is unlikely because of the different mode of action and enzyme involved. Common adverse reactions, such as nausea,

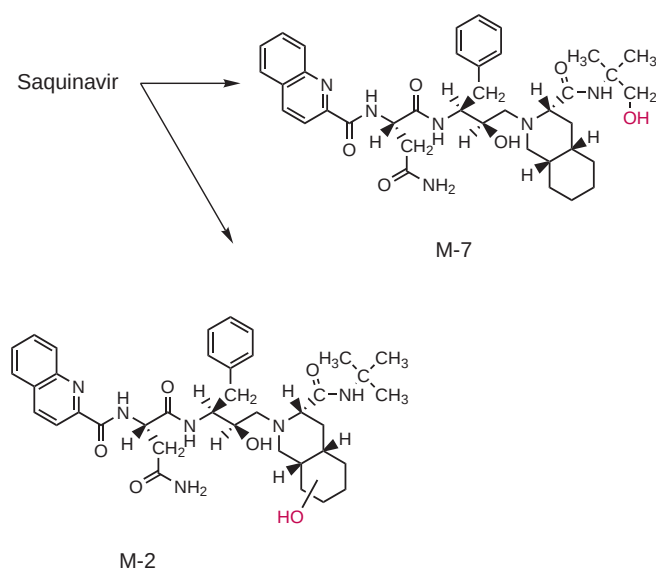


FIGURE 38.21 Major metabolic products from saquinavir.

diarrhea, vomiting, anorexia, abdominal pain, and neurologic disturbances, were reported with the use of ritonavir alone or in combination with other nucleoside analogs. Ritonavir is used for the treatment of advanced HIV infection including opportunistic infections. In combination with nucleoside drugs, ritonavir reduced the risk of mortality and clinical progression.

INDINAVIR SULFATE Indinavir, a pentanoic acid amide derivative, was approved by the FDA in March 1996 (Fig. 38.20) (106). The 95% inhibitory concentration against laboratory-adapted HIV variants, primary clinical

isolates and clinical resistant virus to indinavir analogs, is 25 to 100 nmol/L in drug combination studies with ZDV and ddI. However, HIV has shown ritonavir resistance in some patients. This resistance is due to mutation of the virus that is correlated with the expression of amino acid substitutions in the viral protease. Cross-resistance to indinavir is observed with other protease inhibitors but not with the RT inhibitor. For this reason, indinavir is beneficial with ZDV and other nucleoside drugs.

Pharmacokinetics Indinavir is rapidly absorbed in fasting patients, and plasma peak concentration is observed in about 1 hour. At a dose of 800 mg every 8 hours, peak plasma concentration is approximately 300 nmol/L. The drug is approximately 60% bound to human plasma proteins. Indinavir is metabolized via oxidation and glucuronide conjugation. These metabolites were recovered in feces and urine, with about 20% of the drug excreted in the urine. The half-life of indinavir is approximately 1.8 hours. Because of indinavir's metabolism, a number of drug interactions are possible. Indinavir interacts with rifabutin or ketoconazole, leading to increased or decreased indinavir concentration, respectively, in the blood plasma. Administration of drug combinations of indinavir with antiviral nucleoside analogs, cimetidine, quinidine, trimethoprim-sulfamethoxazole, fluconazole, or isoniazid resulted in an increased activity of indinavir. Indinavir is contraindicated in patients taking triazolam or midazolam because inhibition of metabolism of these drugs may result in prolonged sedation, nephrolithiasis, asymptomatic hyperbilirubinemia, and GI problems (anorexia, constipation, dyspepsia, and gastritis). The usual oral dose for indinavir alone or in combination with other antiviral agents is one 800-mg capsule every 8 hours. The drug is well absorbed if given on an empty stomach or 1 hour before or 2 hours after a light meal with water. The dose is reduced to 600 mg every 8 hours if given concurrently with ketoconazole. Indinavir activity is increased when combined with RT inhibitors.

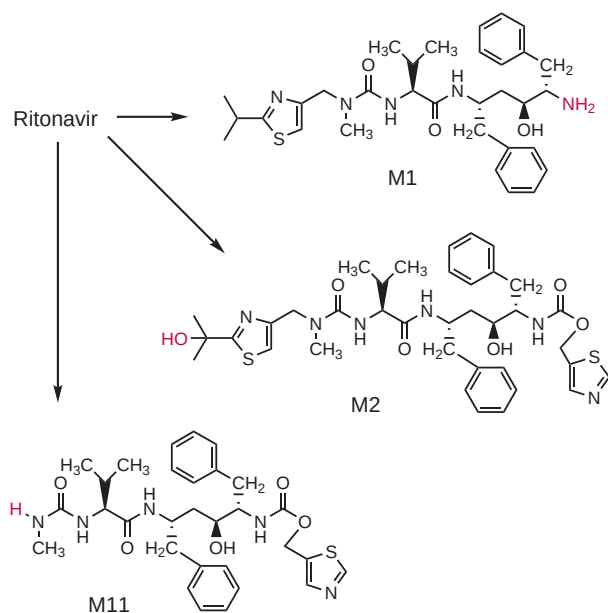


FIGURE 38.22 Major metabolic products from CYP3A4 metabolism of ritonavir.

NELFINAVIR MESYLATE Nelfinavir is a peptidomimetic drug that is effective in HIV-1 and HIV-2 wild-type and ZDV-resistant strains (Fig. 38.20). ED_{50} concentrations range from 9 to 60 nmol/L (95% effective dose was 0.04 mg/mL) (107). After IV administration, the elimination half-life of nelfinavir was about 1 hour. In combination with D4T, nelfinavir reduced HIV viral load by about 98% after 4 weeks. It is well tolerated when used with azole antifungals (ketoconazole, fluconazole, or itraconazole) or macrolide antibiotics (erythromycin, clarithromycin, or azithromycin). However, it causes diarrhea and other side effects common to nonnucleoside drugs. Following oral administration, nelfinavir peak levels in plasma ranged from 0.34 mg/mL (10 mg/kg in the dog) to 1.7 mg/mL (50 mg/kg in the rat). Nelfinavir was slowly absorbed, and bioavailability was 47% in the dog. The drug appeared to be metabolized in the liver, and the major excretory route was in feces.

AMPRENAVIR Amprenavir is the fifth in a series of protease inhibitors to be approved for marketing in the United States. While structurally unique from the previous agents, its pharmacologic profile does not appear to differ significantly from the previously marketed agents. Early studies suggest that a different resistance profile may exist and that the drug may be effective against some resistant strains of HIV. Side effects appear to be more common than with other protease inhibitors and include nausea, vomiting, paresthesia, depression, and rash. Because amprenavir is a sulfonamide, there is some concern for cross-sensitivity with antibacterial sulfonamides. Although this has not been reported, care should be taken if sensitivity to trimethoprim-sulfamethoxazole, used in *Pneumocystis jirovecii* pneumonia, is reported.

Pharmacokinetics Amprenavir is rapidly absorbed following oral administration and may be taken with or without food. High-fat meals decrease the absorption of the drug and therefore should be avoided. The product is available in capsule and liquid form. The recommended adult and adolescent dose of 1,200 mg twice daily requires the patient to take eight capsules (150 mg) twice daily. The liquid preparation is recommended for children between 4 and 12 years of age or for patients 13 to 16 years of age who weigh less than 50 kg. The dose is 22.5 mg/kg twice daily or 17 mg/kg three times a day. Since this preparation contains the excipient propylene glycol, it is not recommended for children less than 4 years of age and certain other individuals who are unable to metabolize this alcohol.

FOSAMPRENAVIR CALCIUM (108) Fosamprenavir has been approved for the treatment of HIV in adults when used in combination with other anti-HIV drugs. It is a prodrug that, upon hydrolysis by serum phosphatases, gives rise to amprenavir, which as indicated earlier is a peptidomimetic transition-state inhibitor that targets HIV-1 protease and reduces viral replication and, thus, infectiousness of HIV-1. It is commonly administered in combination with RT inhibitors to produce excellent efficacy in AIDS patients. The drug is administered as two 700-mg tablets twice daily or, in combination with ritonavir, can be given as two 700-mg tablets once daily or one 700-mg tablet twice daily. Fosamprenavir is a slow-release version of amprenavir and, as a result, decreases the pills required for the patient and lowers the “pill burden” in AIDS patients.

LOPINAVIR/RITONAVIR Recently, the FDA has approved the release of lopinavir/ritonavir combination in patients who have not responded to other regimens for treatment of HIV. The product is available in a soft gelatin capsule containing 133.3 mg of lopinavir and 33.3 mg of ritonavir, as well as oral solutions containing 80 mg/mL of lopinavir and 20 mg/mL of ritonavir. The small amount of ritonavir is not expected to have antiretroviral activity, but rather the ritonavir is meant to increase the plasma concentrations of lopinavir by inhibiting lopinavir's metabolism by CYP3A4. These drugs, in combination with other antiretroviral

agents, have been approved for use in adults and patients between the ages of 6 months and 12 years. This is the first protease inhibitor to be indicated for the very young.

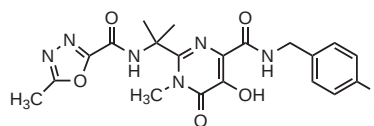
ATAZANAVIR CALCIUM (109) Atazanavir is an antiretroviral agent approved for use in combination with other antiretroviral agents for the treatment of HIV infections (Fig. 38.20). It targets HIV-1 protease and reduces viral replication and thus virulence of HIV-1. Similar to saquinavir, ritonavir, indinavir, nelfinavir, amprenavir, and lopinavir, the drug is used in combination with RT inhibitors to produce excellent efficacy in AIDS patients. Atazanavir is dosed orally once daily, thus reducing “pill burden,” and seems to have a minimum impact on lipid parameters, but does increase total bilirubin. The drug is well absorbed when administered orally with food (bioavailability ~68%). The drug is highly bound to plasma protein (86%) and is metabolized by CYP3A isoenzyme. Atazanavir is a moderate inhibitor of CYP3A, and potential drug-drug interactions are possible with CYP3A inhibitors and inducers.

TIPRANAVIR Tipranavir is a recently approved, nonpeptidic inhibitor of HIV protease (Fig. 38.20) (110). It is indicated for patients who have developed resistance to other antiretroviral drugs, including other HIV protease inhibitors. Although it is very potent, it also exhibits more severe side effects than other HIV protease inhibitors, including intracranial hemorrhage, hepatitis, and diabetes mellitus. It is dosed at 500 mg twice daily in combination with 200 mg of ritonavir to suppress metabolism by cytochrome P450 (111). There is evidence that tipranavir induces its own metabolism, but using twice-daily coadministration of tipranavir and ritonavir at varying doses, tipranavir levels were 24- to 70-fold higher than levels achieved with administration of tipranavir alone.

DARUNAVIR Darunavir is a second-generation HIV protease inhibitor that can be used in treatment-naïve and treatment-experienced adult and pediatric patients (Fig. 38.20) (112). It was designed to bind tightly and specifically to HIV protease and thereby overcome the problems associated with first-generation drugs, including toxicity, high dose, expensive manufacture, and susceptibility to resistance. Like other HIV protease inhibitors, darunavir is metabolized by cytochrome P450, and thus, the drug is dosed at 800 mg once daily in combination with 100 mg of ritonavir (113).

HIV Integrase Inhibitors

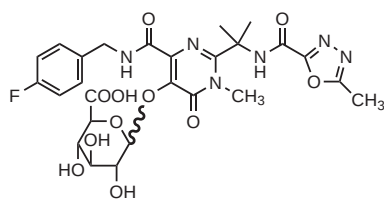
RALTEGRAVIR



Mechanism of Action Raltegravir (Isentress) is the first (and thus far only) FDA-approved treatment for HIV that

acts by inhibition of HIV integrase (114). A second HIV integrase inhibitor, elvitegravir, is currently in phase III clinical trials. Like HIV protease, HIV integrase is essential for replication of the HIV virus and, as such, has become a validated target for drug discovery. Each HIV particle contains 40 to 100 integrase molecules the role of which is to facilitate the insertion of viral complementary DNA (cDNA) into the host cell genome. Following host cell infection, the viral RNA is converted to cDNA via the action of RT. The preparation of cDNA for insertion requires integrase-mediated trimming of the 3'-ends of the viral DNA (3'-processing). The 3'-processing occurs within the cytoplasm, and the processed cDNA plus integrase complex, which also contains viral and cellular protein, is referred to as the preintegration complex. This complex is transported into the nucleus of the host cell where integration into the host DNA occurs through the catalytic action of integrase. The 3'-OH of the cDNA is joined to the 5'-phosphate of the host chromosome with both ends of the cDNA being joined to the host DNA. This very complex integration of cDNA into the host genome involves numerous viral and host proteins. Integrase itself is a 32-kd protein with three structural domains: 1) amino-terminal domain; 2) catalytic core domain (CCD); and 3) carboxy-terminal domain. All three domains are essential for integration, whereas only the CCD is required for the reverse reaction referred to as disintegration. The amino-terminal domain binds zinc, while the CCD binds manganese (Mn^{2+}) and magnesium (Mg^{2+}). It has been proposed that raltegravir acts by inhibiting cDNA integration via chelation to the divalent cations in the CCD at the interface the integrase-donor-acceptor complex (Fig. 38.23).

Pharmacokinetics Raltegravir is readily absorbed following oral administration. The drug is highly bound to plasma protein (~83%). Food does not appear to affect the rate of absorption. The major route of metabolism is glucuronidation mediated by uridine diphosphate-glucuronosyl-transferase.



Approximately 51% of the drug is recovered in the feces unchanged, and 32% is recovered in the urine where 23% is the glucuronide. Glucuronidation inhibitors have the potential for drug-drug interactions, although there was no clinically significant effect when administered with atazanavir. Raltegravir is recommended for use in combination therapy and in patients who have demonstrated resistance to previous regimens.

Raltegravir was approved in 2008 for use in individuals whose infection has proven resistant to highly active

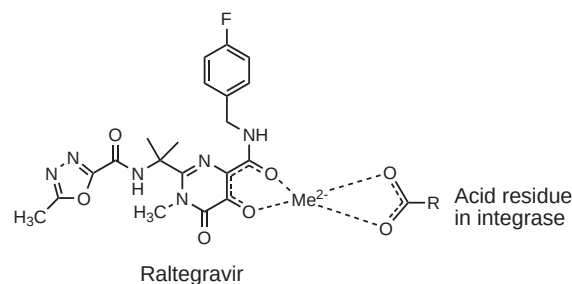


FIGURE 38.23 Chelation complex between raltegravir and integrase.

antiretroviral therapy drugs (see next section), and in 2009, the FDA expanded approval for use in all patients. It is taken twice daily, and doses of 200, 400, and 600 mg have been used.

Combination Drug Therapy (115,116)

When several antiretroviral drugs, typically three or four, are taken in combination for the treatment of HIV, the approach is known as highly active antiretroviral therapy (HAART). Multiple HAART therapies have been undertaken, and the National Institutes of Health recommends that all infected patients should be treated in this manner (114). The synergistic antiviral effects of rimantadine with ribavirin and tiazofurin against influenza B virus and ganciclovir with foscarnet against HSV-1 and HSV-2 are noteworthy. The synergistic action of either trifluorothymidine or acyclovir with leukocyte interferon has been used in the topical treatment of human herpetic keratitis. During the past decade, research into combination antiretroviral therapy for AIDS patients has made remarkable progress. ZDV, the first approved drug for HIV-infected patients, produced bone marrow toxicity. To overcome toxic effects, combinations of ZDV with foscarnet, ddC, or ddI have been used. Such combination therapy indicated improved efficacy and decreased side effects as compared to either drug used alone. The combination of ZDV with α -interferon has been used to treat patients with AIDS-related Kaposi sarcoma. This combination drug therapy delayed emergence of ZDV-resistant HIV strains.

A combination of granulocyte-macrophage colony-stimulating factor with ZDV and α -interferon has been successful in managing treatment-related cytopenia in HIV-infected patients. The advantages of combination therapy include therapeutic antiviral effect, decreased toxicity, and low incidence of drug-resistant infection. In recent years, emergence of drug resistance has been demonstrated in patients receiving single antiviral agent therapy. Resistance to amantadine, acyclovir, ribavirin, ganciclovir, ZDV, and other antiviral agents is noteworthy.

Combined antiretroviral drug therapy serves different purposes. It prolongs the life of AIDS patients, removes drug resistance, and/or reduces toxicity of drugs. With these objectives, successful combinations of ZDV have been reported with ddC, ddI, 3TC, or D4T. Recently,

combinations of nucleoside drugs (ZDV, ddC, ddI, 3TC) are used with protease inhibitors (saquinavir, indinavir, ritonavir) for delaying HIV infection. Combined nucleoside drugs are known to delay progression of HIV infection.

Antiretroviral therapy includes nucleosides or non-nucleoside RT inhibitors and protease inhibitors. These drugs inhibit HIV replication at different stages of viral infection. Nucleoside and nonnucleoside drugs inhibit RT by preventing RNA formation or viral protein synthesis. Nonnucleoside drugs inhibit RT by inactivating the catalytic site of the enzyme. Protease inhibitors act after HIV provirus has integrated into the human genome. These drugs inhibit protease, which is an enzyme responsible for cleaving viral precursor polypeptides into effective virions. Thus, protease inhibitors combined with RT inhibitors act by a synergistic mechanism to interrupt HIV replication.

Two-drug combinations, such as ZDV plus ddI or ddC, 3TC plus ZDV, and D4T plus ddI, have been successful in raising CD4⁺ cell counts and decreasing HIV RNA viral load. Triple-drug therapy consisting of ZDV, 3TC, and a protease inhibitor (indinavir, ritonavir, nelfinavir) has been more effective than two-drug therapy consisting of two nonnucleoside analog combinations. In addition, fewer opportunistic infections were noted when patients took the three-drug combination. ZDV can also be combined with immunomodulators to increase immunologic response in AIDS patients. ZDV has been combined with α -interferon to obtain synergistic activity of the drug. An ideal approach of combined antiretroviral drug therapy would be drugs acting at different stages of HIV cell replication.

Investigational Antiviral Agents: Short Interfering RNA (117)

The field of directed RNA interference (RNAi) has rapidly developed into a highly promising approach for specifically interrupting gene function in order to alleviate disease pathology. This technique was first described in 1998, and the inventors of this technology, Andrew Fire and Craig C. Mello, were awarded a Nobel Prize in 2006 for their work with RNAi. RNAi is a natural mechanism for silencing gene expression that has been conserved through evolution in eukaryotes ranging from plants to humans. In this process,

double-stranded duplexes 21 to 25 nucleotides in length are created from a parent double-stranded RNA molecule by an enzyme known as dicer. These short interfering RNAs (siRNA) bind to specific mRNAs that feature a complementary sequence to form an RNA-induced silencing complex, and thereby block the expression of predetermined specific proteins at the posttranscriptional level. This binding also appears to direct the cell to cleave target mRNA/siRNA complexes. Many research groups are attempting to develop RNAi therapies that induce the degradation of target mRNA involved in inherited or acquired disorders. This technology is especially well suited to treating viral infections, and numerous examples now illustrate that a wide range of viruses can be inhibited with RNAi, both in vitro and in vivo. Antiviral RNAi therapies can be tailored to the biochemical characteristics of each pathogen and can be made more specific through choice of delivery vehicle, route of administration, selection of gene targets, and regulation of RNAi induction. As mentioned earlier, successful antiviral therapeutics possess the ability to discriminate virus from host. However, because viruses rely extensively on host cell machinery for many functions and activities involved in viral replication, they offer a very limited number of therapeutic targets. Because RNAi specifically targets a short stretch of viral nucleic acids rather than a viral protein, even a small viral genome can provide a large number of potential targets.

Researchers are continuing to look for suitable siRNA treatments for viral infections such as HBV, HCV, and HIV. Despite extensive research, the development of an siRNA-based therapy for viral infection faces significant barriers including poor siRNA stability, inefficient cellular uptake, widespread biodistribution, activation of immune response, and occurrence of nonspecific (i.e., off-target) effects (118). The most notable of these barriers is the delivery of intact siRNA molecules to the interior of virus-infected cells. A number of approaches have been tried, including bioconjugation of siRNA with lipids, incorporation into nanoparticles, and in situ biosynthesis of siRNAs. Although this strategy for antiviral development shows great promise, a reliable siRNA-based therapy for viral infection will likely not be realized in the near future.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Douglas Slain, PharmD, BCPS

WM was started on the investigational zanamivir intravenous formulation. After about 5 days of therapy, the patient became clinically stable and was able to come off of mechanical ventilation. She ultimately recovered. Resistance testing did show that she had an H1N1 strain that had the H274Y mutation.

Chemical Analysis

Victoria Roche and S. William Zito

These three antiviral molecules all take advantage of the research conducted to map the binding site of the influenza virus activating enzyme neuraminidase. Neuraminidase facilitates the spread of shed viruses by cleaving sialic acid (*N*-acetylneuraminic acid) from the saccharide-coated viral particles, which promotes their receptor-mediated absorption into unaffected cells.

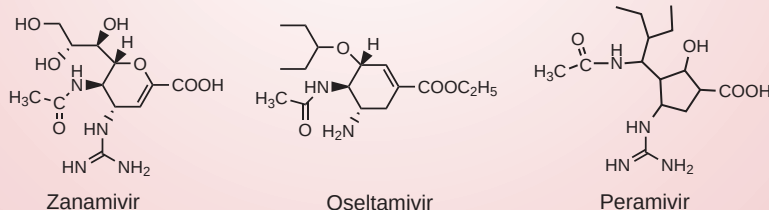
SCENARIO: OUTCOME AND ANALYSIS (*Continued*)

The potentially cationic amino (oseltamivir) and guanine (zanamivir and peramivir) moieties bind to one or more anionic glutamate residues on the neuraminidase enzyme. Neuraminidase is rich in Arg residues, and the acetamido carbonyl oxygen atom found on all three antiviral agents forms an ion-dipole bond with cationic Arg152. Several Arg residues also interact with the anionic carboxylate moieties of zanamivir and peramivir. Presumably, the nonionizable ester of oseltamivir is able to form an ion-dipole bond with this cationic collection of residues.

The major affinity difference among these neuraminidase inhibitors relates to their ability to interact with a hydrophobic area of the neuraminidase that is known to bind with the 3-pentyloxy side chain found at position 6 of oseltamivir and related inhibitors. Peramivir has a lipophilic 3-pentyl

moiety that extends from the carbon atom connecting the acetamido moiety to the cyclopentane ring. This lipophilic group presumably interacts hydrophobically with the neuraminidase enzyme in a manner similar to 3-pentyloxy group of oseltamivir.

Since WM was infected with a strain of H1N1 that is resistant to drugs that interact hydrophobically with neuraminidase, the selection of an inhibitor without this capability was justified. Zanamivir has an oxygen-rich trihydroxypropyl (glycerol) moiety at position 6, which significantly detracts from its lipophilic character and, therefore, its ability to bind within the hydrophobic cavity of neuraminidase. As a neuraminidase inhibitor that is not dependent on hydrophobic bonding, it would stand the best chance of helping this patient successfully overcome her H274Y-mutated influenza infection.



CASE STUDY

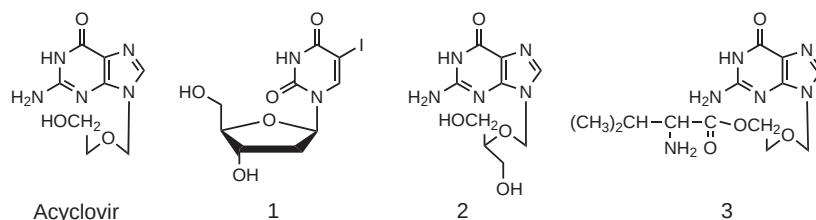
Victoria Roche and S. William Zito

LP is a 29-year-old man from a disadvantaged background who recently found new hope for his future through his affiliation with Homeboy Industries, an LA-based organization founded by Fr. Greg Boyle that assists former gang members in redirecting their lives through job training, education, and holistic support (www.homeboy-industries.org). Though he is regularly urged to return to the streets by his former “friends,” LP is now seeing the potential for a better life and is trying to stick it out despite his apprehension. Through frequent unprotected sexual activity, LP contracted genital herpes several years ago and routinely had been given oral acyclovir (when available) and some acyclovir ointment by physicians working pro bono at a downtown free clinic. In the past, LP was minimally compliant with his regimen, and he has admitted to both overusing the topical preparation and stopping oral therapy prematurely. Now this drug is no longer effective in managing his outbreaks, and partial resistance to this therapy is presumed. He has presented to the clinic with

painful genital lesions, most likely exacerbated by the stress he is currently undergoing as he tries to turn his life around.

As a new pharmacist donating your professional services to the patients of this clinic, you are speaking with LP about his disease state. This young man is now involved in a serious relationship, and he says he is committed to taking his medication as directed and practicing safe sex. Though your choice of therapies is limited to what’s been donated to the clinic, three options (drawn below) present themselves. Although none may be truly optimal, which would be your first choice to help this patient through his latest HSV-2 outbreak?

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient’s specific needs to make a therapeutic recommendation.



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Part
IV

**DISEASE STATE
MANAGEMENT**

CHAPTER 39 Asthma and Chronic Obstructive Pulmonary Disease 1309

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CHAPTER 41 Women's Health 1386

CHAPTER 42 Nutrition and Obesity 1434

Chapter 39

Asthma and Chronic Obstructive Pulmonary Disease

S. WILLIAM ZITO

Drugs Covered in This Chapter*

ADRENOCORTICOIDS

- Beclomethasone dipropionate
- Budesonide
- *Ciclesonide*
- Flunisolide
- Fluticasone propionate
- Hydrocortisone
- Methylprednisolone
- Mometasone furoate
- Prednisolone
- Triamcinolone acetonide

β_2 -ADRENERGIC AGONISTS

- Albuterol
- Bitolterol mesylate

- Epinephrine (Adrenalin)
- *Fenoterol hydrobromide*
- Formoterol fumarate (arformoterol)
- Isoetharine hydrochloride
- Metaproterenol sulfate
- Salmeterol xinafoate
- Terbutaline sulfate

ANTIMUSCARINICS

- *Acidinium bromide*
- Ipratropium hydrobromide
- Tiotropium bromide

LEUKOTRIENE MODIFIERS

- Montelukast

- Zafirlukast
- Zileuton

MAST CELL DEGRANULATION INHIBITORS

- Cromolyn sodium (*sodium cromoglycate*)
- Nedocromil sodium

METHYLYXANTHINES

- Dyphylline
- Theophylline

MONOCLONAL ANTI-IGE ANTIBODY

- Omalizumab

PHOSPHODIESTERASE INHIBITORS

- Roflumilast

Abbreviations

AMP, adenosine monophosphate
ATT, α_1 -antitrypsin
cAMP, cyclic adenosine monophosphate
cGMP, cyclic guanosine monophosphate
COMT, catechol O-methyl transferase
COPD, chronic obstructive pulmonary disease
CRF, corticotrophin-releasing factor
cysLT, cysteinyl leukotriene
EPI, epinephrine
Fc ϵ RI, high-affinity Fc immunoglobulin E receptor
FDA, U.S. Food and Drug Administration

FEV₁, forced expiratory volume in 1 second
FLAP, 5-lipoxygenase-activating protein
FVC, forced vital capacity
GI, gastrointestinal
GOLD, Global Initiative for Chronic Lung Disease
HDAC2, histone deacetylase 2
HPA, hypothalamus-pituitary-adrenal
HRE, hormone response element
Ig, immunoglobulin
IL, interleukin
IV, intravenous

L-DOPA, L-dihydroxyphenylalanine
MAO, monoamine oxidase
MIH, macrophage migration inhibitory factor
NE, norepinephrine
NIH, National Institutes of Health
PDE, phosphodiesterase
PEF, peak expiratory flow
SRS-A, slow-reacting substance of anaphylaxis
TSLP, thymic stromal lymphopoietin

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

SCENARIO

Joseph V. Etzel, PharmD

JP is a 42-year old man who presents to his community pharmacy for a Medication Therapy Management consultation for his moderate-persistent asthma. Besides asthma, JP's medical history is significant for hypertension (controlled with ramipril 5 mg PO daily) and perennial allergic rhinitis (well controlled with fexofenadine 60 mg PO twice daily). For the management of his asthma, JP is currently being treated with fluticasone propionate 110 mcg inhalation powder at a dose of two inhalations twice daily and albuterol sulfate HFA inhaler at a dose of two inhalations every 4 to 6 hours as needed for bronchospasm. JP reports inadequate control of his asthma despite

being adherent to the above medication regimens for the past several months. He states that he is required to use his albuterol inhalation three to four times per week for the relief of acute asthma exacerbations. In addition, he admits to nighttime awakenings due to asthmatic symptoms at least once a week. Peak expiratory flow rate (peak flow or FEV₁) evaluations demonstrate values approximately 70% of his personal best.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

ASTHMA

Epidemiology

Asthma has been known since antiquity. The earliest reference to asthma can be found in Homer's *Iliad* where it is a noun used to denote breathlessness or panting. The earliest use of the term as a medical condition dates back to Ancient Greece, in Hippocrates' *Corpus Hippocraticum*, and the best clinical description of asthma from antiquity was offered by the physician Aretaeus of Cappadocia who practiced in the first century AD (1).

Over the period of 1980 to 1999, asthma prevalence increased in the United States. However, since then, asthma prevalence has stopped increasing, and in addition, asthma-related mortality and hospitalizations have decreased, which perhaps is an indication of improved disease state management (2,3). According to U.S. population data gathered by the National Health Interview Survey, the lifetime estimate of the number of people who were ever diagnosed with asthma is 34 million, or 11.5% of the population. The current percentages for asthma prevalence as of 2007 are 9.1% for children less than 18 years old and 7.3% for adults over age 18 years. Boys have a higher incidence of asthma than girls (9.7% vs. 8.5%); however, this reverses in adulthood, when females show a higher prevalence compared to males (9% vs. 5.4%) (4). In a recent study, over a 1-year period (2004–2005), 15% of all children in the study population (>4 million) age less than 17 years were dispensed an asthma-related medication (5). Race/ethnicity plays a significant role in the prevalence of asthma. Blacks have the highest prevalence, followed by whites and Hispanics (10.2% vs. 7.6% vs. 6.8%, respectively) (4). If asthma is not controlled, it can result in death. Over a 3-year period (2001–2003), 4,210 asthma-related deaths occurred, of which over 50% occurred in persons older than age 65. These prevalence, morbidity, and mortality data demonstrate the significant public health burden of asthma in the United States and why the Centers for Disease Control and Prevention (CDC) have established an asthma public

health approach, which includes the collection of standardized comprehensive surveillance data (3).

Etiology, Signs, and Symptoms

Asthma is a common chronic, complex, airway disorder that is characterized by airflow obstruction, bronchial hyperresponsiveness, and an underlying inflammation that leads to variable degrees of symptoms, such as difficulty breathing (paroxysmal dyspnea), wheezing, and cough (6). The National Institutes of Health (NIH) Expert Panel Report 3, "Guidelines for the Diagnosis and Management of Asthma," simply defines asthma as a chronic inflammatory response of the airways (6,7).

The most common form of asthma is allergic asthma (atopic or extrinsic asthma), and it is associated with environmental allergens, such as plant pollens, house dust mites (*Dermatophagoides farinae*), domestic pet dander, molds, and foods. The less common form, intrinsic asthma, has no known allergic cause and usually occurs in adults older than 35 years. Intrinsic asthma may result from an autonomic dysfunction characterized by excess cholinergic and/or tachykinin activity, but this hypothesis has never been proven (8). Aside from environmental allergens, an asthmatic attack may be precipitated by respiratory infection, strenuous physical exercise, polyps, drugs (e.g., aspirin and β -adrenergic antagonists), and environmental pollutants, primarily cigarette smoke. Strong emotional stress and breathing cold dry air can also precipitate an asthmatic onset (9).

Why has there been a marked increase in asthma in affluent industrialized countries? To answer this question, recent thought has focused on the "hygiene hypothesis," which implicates an imbalance of TH₁ and TH₂ lymphocytes as a major cause of the increased prevalence of asthma (10). The TH₁ lymphocytes are the type of CD4⁺ T lymphocyte associated with defense against bacterial infection, whereas the TH₂ type predominates in allergic inflammation. The hypothesis claims that because bacterial infections have significantly decreased in industrialized nations, there is an imbalance, in

CLINICAL SIGNIFICANCE



The therapeutic approach to the management of asthma and chronic obstructive pulmonary disease (COPD) has changed dramatically over the past several decades. Treatment guidelines have evolved based on a better understanding of these disease states as well as on the development of newer and more efficacious treatment modalities through the application of structure-activity relationships (SARs) of chemical lead compounds. Although both disease states are characterized by pulmonary obstruction and chronic inflammation, the nature of such pulmonary abnormalities differ between the two conditions. Traditional therapy had focused on the symptomatic relief of airflow obstruction through the use of bronchodilators, such as adrenergic agonists and anticholinergics. Today, however, evidence-based consensus guidelines for the treatment of asthma emphasize interventions to manage the underlying inflammation of this condition and minimize disease progression. Likewise, the goal of managing COPD is to use agents to achieve and maintain clinical control by preserving airway patency.

The current therapeutic approach to managing these diseases includes the use of rapidly acting drugs to relieve acute symptoms as well as the use of maintenance medications either to minimize inflammation or control long-term symptoms, thereby limiting the need for symptomatic relief. Short-acting adrenergic agonists are those most commonly used to manage acute exacerbations of these disease states. Though effective, older agents, such as epinephrine and isoproterenol, were

limited in their pharmacokinetic profile as well as in their lack of pulmonary selectivity. Modifying the chemical structure of these compounds, however, has resulted in the development of agents with significant clinical advantages in terms of duration of action and adverse effects profiles. Similarly, the application of SAR principles to the development of anti-inflammatory agents, such as corticosteroids, leukotriene modifiers, and mast-cell stabilizers, has led to the availability of superior long-term controlling agents in terms of potency, pharmacokinetic profile, and safety.

When treating patients with asthma and COPD, clinicians should be mindful of the SARs of those agents being employed. The application of these principles should be used to determine the most appropriate drug therapy in light of patient-specific needs and desired outcomes. In addition, clinicians can look forward to the availability of newer and more clinically appropriate agents for the management of asthma and COPD as our understanding of these conditions evolve and as advances in medicinal chemistry and biotechnology result in the development of increasingly selective drug entities and improved receptor targeting modalities.

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susceptible children, in favor of the TH₂-type lymphocytes, and this imbalance therefore favors allergic asthma (11). However, it is an oversimplification to claim that asthma inflammation is simply an overproduction of TH₂-type lymphocytes. Recent studies have discovered that a number of other T-helper cell subtypes are also involved in allergic inflammation, including TH₉, TH₁₇, TH₂₅, and TH₃ cells, and that regulatory T cells also contribute to asthma pathogenesis (12).

Asthma frequently occurs in families, and studies have shown that this results, at least in part, from mutually shared genes (13,14). Asthma is a complex disorder and lacks a mendelian genetic pattern, so it is difficult to study. To date, however, genetic research indicates that what is inherited is the susceptibility to develop asthma. It is clear that genes alone are not responsible for the development of asthma, because environmental factors also play a major role. Numerous genes on various chromosomes have been linked to asthma. Table 39.1 shows several gene products that may influence the development of asthma (13–19). In reality, however, there is little correlation between gene expression and clinical symptoms. The one exception to this is the low-expression allele of macrophage migration inhibitory

factor (MIF), which has been shown to have a strong association with patients who have mild asthma (20). Drug development of MIF inhibitors might lead to future treatments for asthma as well as other inflammatory diseases (21).

Pathogenesis of Asthma

For a long time, asthmatic symptoms were thought to be the result of airway smooth muscle abnormalities, resulting in episodic bronchoconstriction. Today, however, it is clear that the constriction of bronchial smooth muscle is only one of many effects of chronic airway inflammation. Evidence of inflammation appears very soon after the onset of symptoms; therefore, treatment algorithms for asthma now emphasize quick relief of the bronchoconstriction and the amelioration of the underlying inflammation (22).

Inflammation in asthma is characterized by mucous plugging, epithelial shedding, basement membrane thickening, inflammatory cell infiltration, and smooth muscle hypertrophy and hyperplasia. An acute extrinsic asthmatic attack begins when allergens interact with lung epithelia, which causes the release of cytokines (thymic stromal lymphopoietin [TSLP], interleukin

TABLE 39.1 Putative Asthma-Associated Gene Products

Gene Products
β_2 -Adrenergic receptor
Interleukin-4, -5, -9, and -13
Interleukin-4 receptor α (IL-4R α)
β Chain of the high-affinity immunoglobulin E receptor (Fc ϵ RI β)
Tumor necrosis factor α (TNF α)
Major histochemical complex (MHC)
T-cell receptor α/δ complex
ADAM33 (a disintegrin and metalloproteinase)
Dipeptidyl peptidase 10
PHD finer protein 11
Prostanoid DP1 receptor
Macrophage migration inhibitory factor (MIF)

[IL]-33, and IL-25). TSLP mediates migration and maturation of dendritic cells and production of IL-4 from competent T-helper cells (T_{FH}), leading to the activation of B cells to produce immunoglobulin (Ig) E antibody, which in turn binds to high-affinity Fc IgE receptors (Fc ϵ RI) on the mast cells. Activation of mast cells occurs when an antigen cross-links with IgEs on their surface. This IgE complex triggers mast cell degranulation, leading to the rapid release of inflammatory mediators, such as histamine, prostaglandins, leukotrienes, and cytokines (including tumor necrosis factor- α and ILs). The initial attack generally resolves within an hour; however, a second phase begins 4 to 6 hours after exposure and can last up to 24 hours. This late phase is the result of recruitment of additional inflammatory cells, primarily eosinophils, by the release of cytokines (IL-4, IL-13, IL-5, IL-9, and IL-3) from macrophages and TH_2 -type lymphocytes in the lower lung (Fig. 39.1) (23,24).

Clinical Evaluation

Most patients with asthma are asymptomatic between acute exacerbations. The onset of symptoms can be sudden or gradual and frequently occurs during the night or early in the morning. Acute symptoms include shortness of breath, wheezing or whistling at the end of exhalation, cough, and chest tightness. Patients with chronic and poorly controlled asthma develop barrel chest and diminished diaphragm movement, both of which are evidence of chronic pulmonary hyperinflation.

Acute asthmatic attacks may be mild, moderate, or severe depending on the degree of airway obstruction. The determination of the degree of airflow obstruction is

accomplished by pulmonary function testing. The most common pulmonary function test uses spirometry, which measures the rate at which the lung changes volume during forced breathing maneuvers. The most important spirometric measure is the forced vital capacity (FVC). This requires the patient to exhale as rapidly and as completely as possible after a maximal inhalation. Normal lungs generally can empty their volume in 6 seconds or less. When airflow is obstructed, however, the expiratory time may increase by as much as fivefold. In practice, the forced expiratory volume in 1 second (FEV_1) is measured and compared to the FVC and then recorded as the ratio of FEV_1 to FVC (FEV_1/FVC) (Fig. 39.2) (measurement is done by means of a spirometer). Healthy persons normally can expel at least 75% of their FVC in the first second. Decreases in the FEV_1/FVC ratio indicate obstruction, and a decrease below 40% indicates severe asthma (22,23).

A more convenient way to measure airway obstruction is to determine the peak expiratory flow (PEF) rate. The PEF rate correlates well with the FEV_1 ; can be determined using inexpensive, handheld, peak flow meters; and is easily and simply measured at home. The PEF rate is the maximal rate of flow that is produced through forced expiration. Peak flow meters come with a chart that lists predicted PEF rates based on the patient's age, gender, and height. The patient or clinician can then compare the determined PEF rate with the predicted PEF rate and make an evaluation regarding the severity of an asthmatic attack. The PEF rate or the FEV_1/FVC ratio, along with the frequency of daytime and nighttime symptoms, forms the basis for the classification of the severity of an asthmatic attack. Table 39.2 shows the severity classification of asthma established by the NIH Expert Panel Report 3, "Guidelines for the Diagnosis and Management of Asthma" (6).

GENERAL THERAPEUTIC APPROACHES TO THE TREATMENT AND MANAGEMENT OF ASTHMA

Asthma symptoms are caused by bronchoconstriction and inflammation, and approaches to treatment are directed at both of these physiologic problems. Therefore, drugs that affect adrenergic/cholinergic bronchial smooth muscle tone and drugs that inhibit the inflammatory process are used to treat and control asthma symptoms. In the normal lung, bronchiole smooth muscle tone results from the balance between the bronchoconstrictive effects of the cholinergic system and the bronchodilating effects of the adrenergic system on the smooth muscles of the bronchioles. Pharmacologic treatment of asthmatic bronchoconstriction consists of either increasing adrenergic tone with an adrenergic agonist or inhibiting cholinergic tone with an anticholinergic agent.

The inflammatory effects seen in asthma are the result of the release of physiologically active chemicals from a variety of inflammatory cells. Pharmacologic treatment, therefore, uses anti-inflammatory drugs

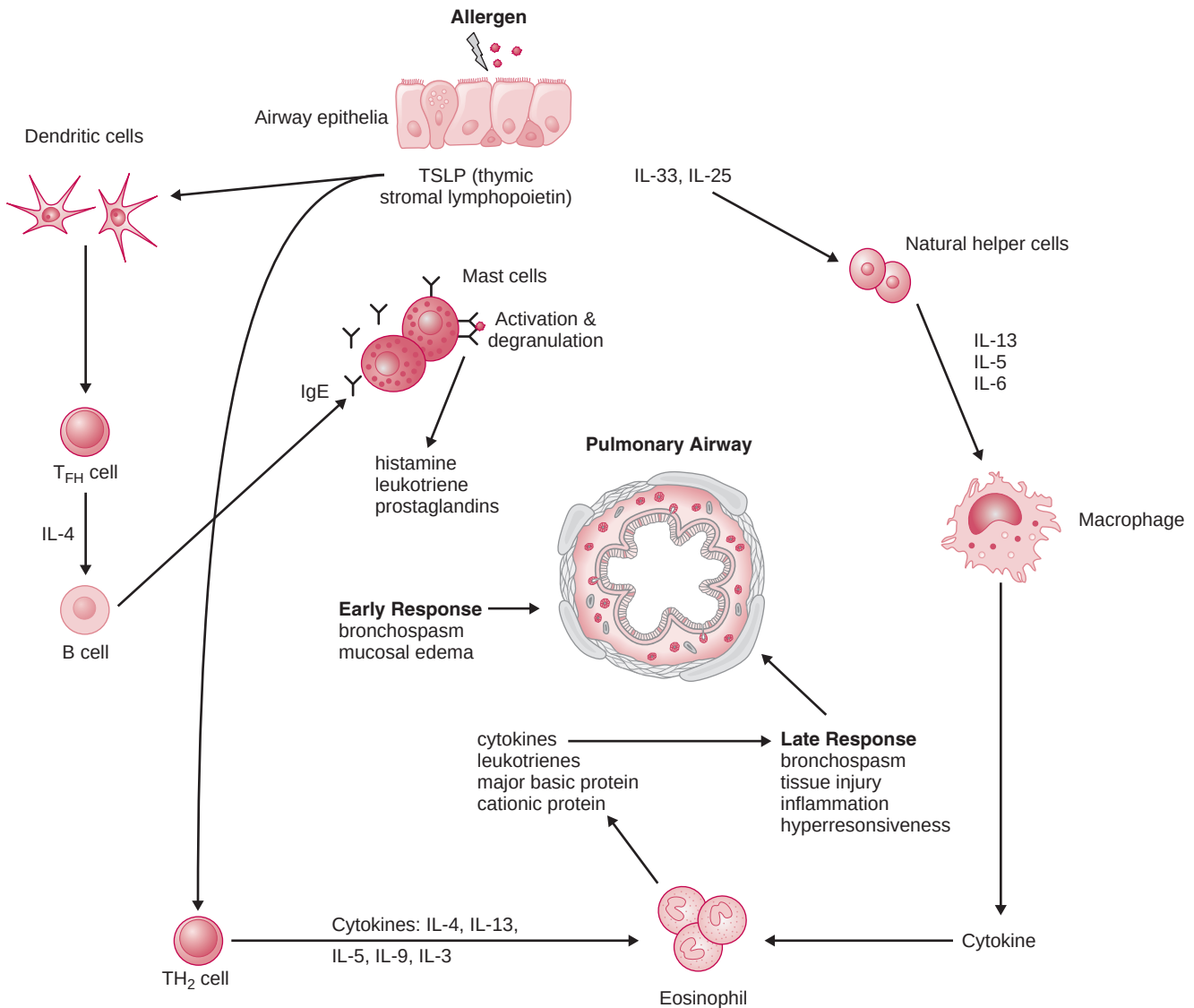


FIGURE 39.1 Pictorial summary of asthma pathogenesis.

(corticosteroids), mast cell stabilizers, leukotriene modifiers, and IgE monoclonal antibodies. Figure 39.3 depicts the overall approach to the pharmacologic treatment of asthma.

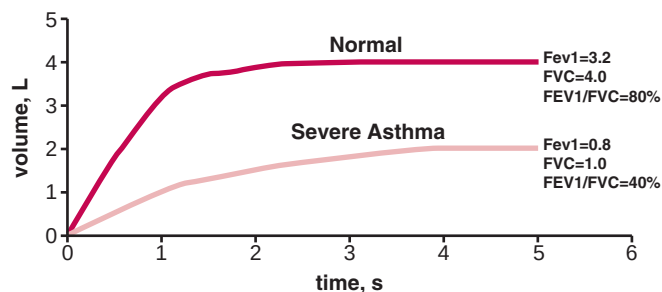


FIGURE 39.2 Spirometric comparison of normal and severe asthma. FEV₁, forced expiratory volume in 1 second; FVC, forced vital capacity.

Therapeutic management of asthma requires the use of quick-acting drugs to relieve an acute attack as well as drugs that control symptoms over the long-term. The current approach to asthma management uses a stepwise approach (6). The quick-reliever medication is almost always an inhaled short-acting β_2 -adrenergic agonist, whereas controller drugs are inhaled corticosteroids, long-acting β_2 -agonists, leukotriene modifiers, cromolyn sodium, and/or methylxanthines. The dose, route of administration, and number of controller drugs depend on the severity of the patient's disease. Recently, the U.S. Food and Drug Administration (FDA) has recommended that long-acting β_2 -agonists should never be used alone in the treatment of asthma in children or adults based on analyses of studies showing an increased risk of severe worsening of asthma symptoms, leading to hospitalizations and even death in some patients (25). Table 39.3 shows the stepwise approach to asthma management based on disease severity.

TABLE 39.2 Classification of Asthma Severity (6)

Components of Severity		Intermittent Step 1		Persistent	
			Mild Step 2	Moderate Step 3	Severe Step 4 or 5
	Symptoms	≤ 2 days/week	>2 days/week but not daily	Daily	Throughout the day
Impairment	Nighttime awakenings	≤ 2x/month	3-4x/month	>1x/week but not nightly	Often 7x/week
Normal FEV ₁ /FVC:	Rescue inhaler use	≤ 2 days/week	> 2 days/week but not daily and not > 1x on any day	Daily	Several times per day
8-19 yr	85%				
20-39 yr	80%				
40-59 yr	75%	Interference with normal activity	None	Minor limitation	Some limitation
60-80 yr	70%				Extremely limited
	Lung function	Normal FEV ₁ between exacerbation			
		FEV ₁ >80% predicted	FEV ₁ >80% predicted	FEV ₁ >60% but <80% predicted	FEV ₁ <60% predicted
		FEV ₁ /FVC normal	FEV ₁ /FVC normal	FEV ₁ /FVC reduced 5%	FEV ₁ /FVC reduced >5%

FEV₁, forced expiratory volume in 1 second; FVC, forced vital capacity

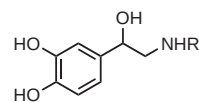
THERAPEUTIC CLASSES OF DRUGS USED TO TREAT ASTHMA AND CHRONIC OBSTRUCTIVE PULMONARY DISEASE

β₂-Adrenergic Agonists

What structural features make a drug an adrenergic agonist? What features make it a selective β₂-agonist? The answers to these questions lie in the structural relationship of a drug to that of norepinephrine (NE), the physiologic neurotransmitter of the sympathetic branch of the autonomic nervous system. Drugs that act on postsynaptic sympathetic receptors in the same way as NE are called sympathomimetics or, more commonly, adrenergic agonists. The related natural agonist epinephrine (EPI; adrenalin) is the predominant adrenergic hormone produced in the chromaffin cells of the adrenal medulla. EPI interacts just like NE at all adrenergic receptors.

Chemistry and Biochemistry of Norepinephrine and Epinephrine

Chemically, NE is classified as a catecholamine. A catechol is a 1,2-dihydroxybenzene, and NE is a β-hydroxyethylaminodihydroxybenzene. EPI is the *N*-methyl derivative of NE, and they both have acidic and basic functional groups. Physiologically, however, they behave as a base, being more than 90% protonated at pH 7.4 ($pK_a = 9.6$) and functioning as an ionized acid.



Norepinephrine R = H
Epinephrine R = CH₃

NE is biosynthesized in the neurons of both the central nervous system and the autonomic nervous system, whereas EPI is formed in the chromaffin cells of the adrenal medulla. Both NE and EPI are derived from L-tyrosine by a series of enzyme-catalyzed reactions (Fig. 39.4 depicts the overall pathway). L-Tyrosine hydroxylase hydroxylates the meta position of L-tyrosine, producing L-dihydroxyphenylalanine (L-DOPA), and is the rate-limiting step. The L-DOPA is then decarboxylated by L-aromatic amino acid decarboxylase to form dopamine, which is converted to NE by the action of dopamine β-hydroxylase. Dopamine β-hydroxylase occurs in storage vesicles of the nerve ending, and the NE formed is stored there until it is released into the synaptic cleft. In the chromaffin cells, the formed NE is converted to EPI by *N*-methylation catalyzed by phenylethanolamine *N*-methyltransferase.

Termination of Neurotransmission

Stimulated adrenergic neurons release NE into the synaptic cleft, which then binds reversibly with receptors to produce a characteristic adrenergic response. Termination

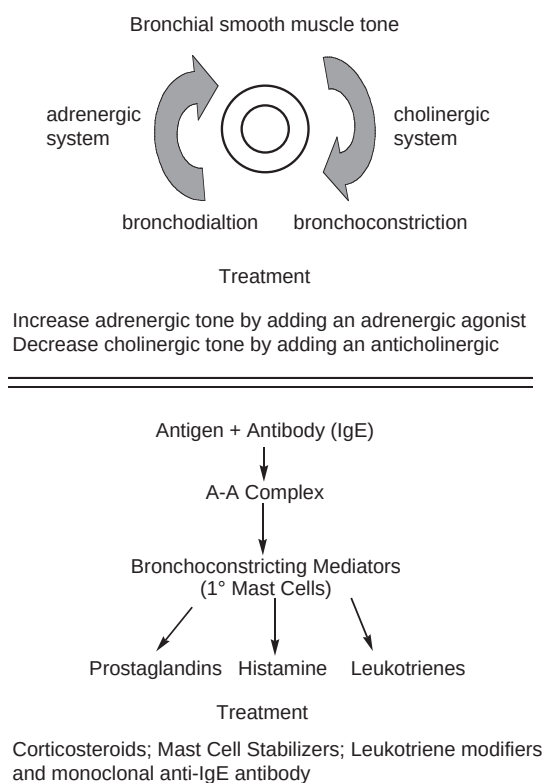


FIGURE 39.3 Overview of pharmacologic treatment for asthma.

of the adrenergic response occurs primarily by reuptake (uptake-1) into the presynaptic neuron; however, diffusion away from the receptors and extracellular metabolism also occur to a limited extent. The NE that is taken back up into the presynaptic neuron is either used again as a neurotransmitter or is metabolized by mitochondrial monoamine oxidase (MAO). The extraneuronal NE that diffuses away from the neurons is either metabolized by catechol *O*-methyl transferase (COMT) in situ or reaches the circulatory system and is metabolized by COMT and MAO in various tissues, most importantly the liver, gastrointestinal (GI) tract, and the lungs. Figure 39.5 depicts the possible metabolic pathways for both NE and EPI. It is important to note that agonists that are resistant to MAO and/or COMT have greater oral availability and longer duration of action.

Adrenergic Receptors

The adrenergic receptors have long been pharmacologically classified as α or β based on their interaction with NE, EPI, and the adrenergic prototype, isoproterenol (26).

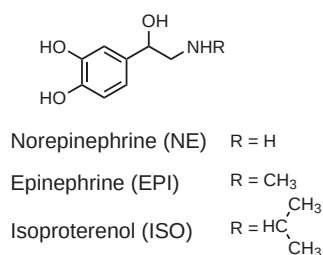


TABLE 39.3 Stepwise Medication Management of Asthma

All Asthma Patients: Short-acting inhaled β_2 -agonist as needed for acute exacerbations (SABA)

Severity Classification	Long-term Control
Step 1: Intermittent Asthma Persistent Asthma	<i>Preferred:</i> SABA PRN
Step 2: Mild	<i>Preferred:</i> Low-dose Inhaled corticosteroid (ICS) <i>Alternative:</i> Cromolyn, Nedocromil, Leukotriene receptor antagonist (LTRA) or theophylline
Step 3: Moderate	<i>Preferred:</i> Low-dose ICS + long-acting β_2 -agonist (LABA) or medium-dose ICS <i>Alternative:</i> Low-dose ICS + either LTRA, Theophylline or Zileutin
Step 4: Severe	<i>Preferred:</i> Medium-dose ICS + LABA <i>Alternative:</i> Medium-dose ICS + either LTRA, Theophylline or Zileutin
Step 5: Severe	<i>Preferred:</i> High-dose ICS + LABA and <i>Consider:</i> Omalizumab for patients who have allergies
Step 6: Severe	<i>Preferred:</i> High-dose ICS + LABA + oral corticosteroid and <i>Consider:</i> Omalizumab for patients who have allergies

NE and EPI are nonselective and interact with all adrenergic receptors, but isoproterenol selectively interacts only with the β -receptors. The adrenergic receptors have been further divided into three groups, α_1 , α_2 , and β , each of

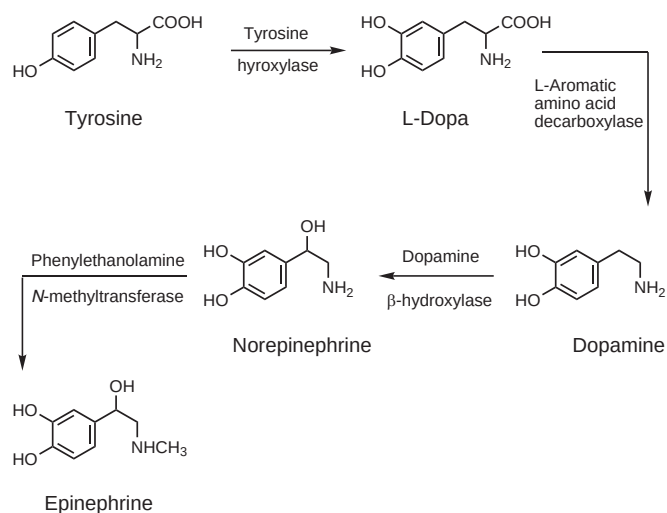


FIGURE 39.4 The biosynthesis of norepinephrine and epinephrine from tyrosine.

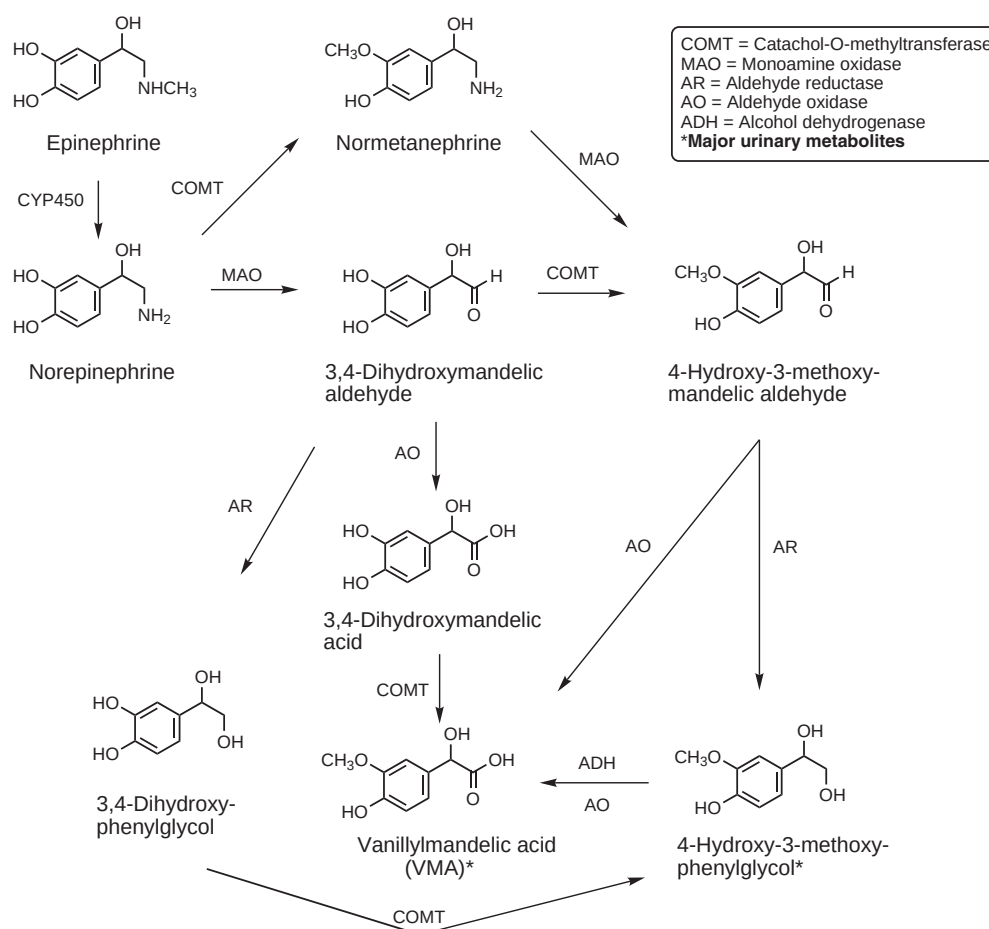


FIGURE 39.5 Metabolic pathways for norepinephrine and epinephrine.

which has been further divided into three receptor subtypes based on their organ distribution and physiologic activities (27). Therefore, there are now a total of nine adrenergic receptor subtypes, but the most important in relation to the treatment of asthma and chronic obstructive pulmonary disease (COPD) are the β_1 and β_2 subtypes that are found primarily in the heart and the lung, respectively. As

may be deduced from Table 39.4, adrenergic agonists that are selective for the β_2 subtype will cause bronchial dilation and might be expected to relieve the bronchospasm of an asthmatic attack. Nonselective β -agonists, however, will have stimulatory cardiac effects and, therefore, would have limited use in cardiac patients with asthma.

Adrenergic Receptor Structure and Agonist Interactions

The adrenergic receptors are members of the guanine nucleotide binding regulatory protein-coupled receptor family more commonly referred to as G protein-coupled receptors. They affect biologic activity by releasing secondary messenger molecules inside the cell after they bind an extracellular agonist. This process usually is referred to as signal transduction and is common to hormone and neurotransmitter receptors found in the muscarinic, serotonergic, dopaminergic, and adrenergic systems. All the G protein-coupled receptors are structurally similar, being comprised of seven transmembrane α -helix bundles. The helices are connected by short stretches of hydrophilic residues, which form multiple loops in the intracellular and extracellular domains. The G proteins generally are bound to the third intracellular loop and imbedded in the inner membrane (Fig. 39.6).

TABLE 39.4 Physiologic Response in Relationship to β -Receptor Subtype and Organ Site

Receptor Subtype	Organ Location	Response
β_1	Heart	Increased rate and force
		Increase conduction velocity
β_2	Bronchiole smooth muscle	Dilation
	Intestine	Decreased motility
	Liver	Increased gluconeogenesis
		Increased glycogenolysis
	Uterus	Contraction
	Lungs	Bronchial dilation

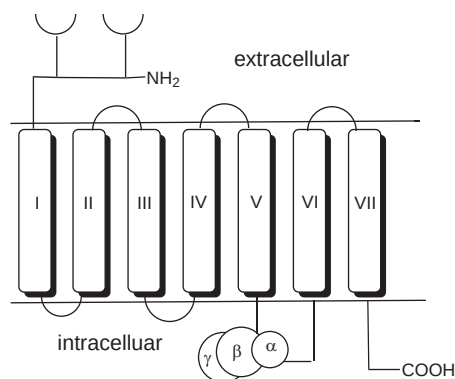


FIGURE 39.6 Representations of the G protein-coupled receptor.

All the β -adrenergic receptors are coupled to adenylate cyclase via specific G stimulatory proteins (Gs). When agonist binds to the β -adrenergic receptors, the α -subunit migrates through the membrane and stimulates adenylate cyclase to form cyclic adenosine monophosphate (cAMP) from adenosine triphosphate. Once formed in the cell, cAMP activates protein kinase A, which catalyzes the phosphorylation of numerous proteins, thereby regulating their activity and leading to characteristic cellular responses. The intracellular enzyme phosphodiesterase (PDE) hydrolyzes cAMP to form adenosine monophosphate (AMP) and terminates its action (Fig. 39.7).

A great deal of research has been done to identify the binding residues of the adrenergic receptors. Molecular modeling methods have been used to construct three-dimensional models for agonist complexes with the β -adrenergic receptors. The picture that has emerged is that NE binds ionically via its protonated amine to Asp-113 in helix 3 and hydrogen bonds to both hydroxyls of the catechol ring with Ser-204 and Ser-207 in helix 5. That binding limits configurational and rotational freedoms, which allows reinforcing van der Waals interactions between the aromatic ring with residues Phe-290 in helix 6 and Val-114 in helix 3. The *N*-alkyl substituents are believed to fit into a pocket formed between aliphatic residues in helix 6 and helix 7. Stereochemistry also plays an important role in receptor binding. The β -carbon in NE/EPI is chiral and can

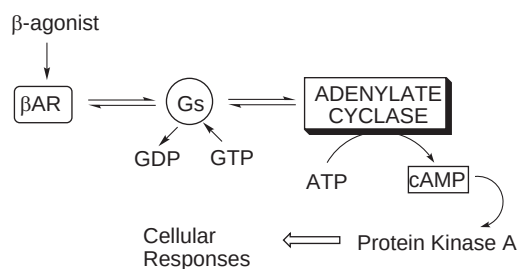


FIGURE 39.7 β -Adrenergic receptor G protein coupling to adenylate cyclase.

be either *R* or *S* in configuration. Endogenous NE/EPI exists in the *R* configuration so that the β -hydroxyl is oriented toward the receptor Asn-293 (28,29) (Fig. 39.8).

Adrenergic Agonist Structure-Activity Relationships

The fundamental pharmacophore for all adrenergic agonists is a substituted β -phenylethylamine (Table 39.5). The nature and number of substituents on the pharmacophore influences whether an analog will be direct-acting or indirect-acting or have a mixture of direct and indirect action. In addition, the nature and number of substituents also influences the specificity for the β -receptor subtypes. Direct-acting adrenergic agonists bind the β -adrenergic receptors just like NE/EPI, producing a sympathetic response. Indirect-acting agonists cause their effect by a number of mechanisms. They can stimulate the release of NE from the presynaptic terminal, inhibit the reuptake of released NE, or inhibit the metabolic degradation of NE by neuronal MAO (i.e., MAO inhibitors). Mixed-acting agonists work as their name implies (i.e., they have both direct and indirect abilities). Table 39.5 shows the relationship between substituents and the mechanisms of action by adrenergic agonists.

Relationship of Structure to α - or β -Receptor Selectivity

The substituents on the amino group (R_1) determine α - or β -receptor selectivity. As was noted earlier, when the *N*-substituent was changed from hydrogen (NE) to methyl (EPI) to isopropyl, the receptor affinity went from nonselective for NE/EPI to β -selective for isopropyl. Therefore, one can say that the larger the bulk of the *N*-substituent, the greater the selectivity for the β -receptor. As a matter of fact, if R_1 is *t*-butyl or aralkyl, there is complete loss of α -receptor affinity, and the β -receptor affinity shows preference for the β_2 -receptor. It must be said that receptor selectivity is dose related, and when the dose is high enough, all selectivity can be lost.

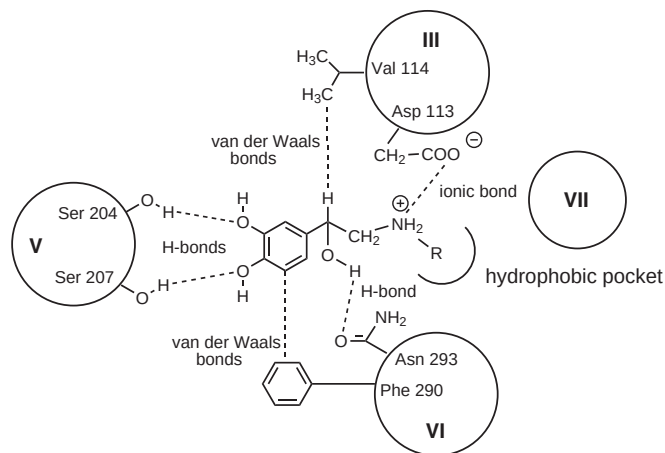
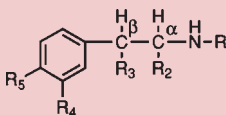


FIGURE 39.8 Ligand binding to key residues in the adrenergic receptor.

TABLE 39.5 Relationship of Substituents to Adrenergic Agonist Mechanism of Action


Action	R ₃	R ₄	R ₅
Direct ^a	OH	OH	OH
or	OH	OH	H
Indirect ^b	H	H	H
or	H	H	OH
Mixed ^c	H	OH	OH
or	OH	H	OH

^aβ-OH (R₃) necessary for direct action, optimal if in the R-configuration; *m*-OH (R₄) also necessary for direct action; *p*-OH (R₅) can be an H or OH, however catechol optimal.

^bWithout the catechol OHs (R₃ and R₄) or β-OH (R₃) there is little affinity for the β-receptor; *p*-OH does not contribute to β-receptor binding affinity.

^cβ-OH (R₃) or *m*-OH (R₄) contribute to direct action.

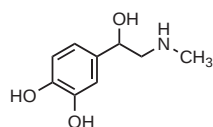
Substituents on the α-carbon (R₂) other than hydrogen will show an increased duration of action, because they make the compound resistant to metabolism by MAO. In addition, if the substituent is ethyl, there is selectivity for the β₂-receptor, which is enhanced by a bulky *N*-substituent. Interestingly, α-methyl substitution shows a slight β-receptor enhancement and only for the *S*-configuration.

For an adrenergic agonist to demonstrate significant β₂-receptor selectivity, there needs to be, in addition to the bulky *N*-substituent, an appropriately substituted phenyl ring. The currently marketed adrenergic agonists contain a resorcinol ring, a salicyl alcohol moiety, or an *m*-formamide group (Fig. 39.9). In addition, these ring configurations are resistant to COMT metabolism and will increase the duration of action.

Specific Adrenergic Drugs Used to Treat Asthma

For a more detailed discussion of the chemistry of adrenergic agents, see Chapter 10.

Epinephrine (Adrenalin)



Epinephrine (Epi Pen)

The combination of the catechol nucleus, the β-hydroxyl group, and the *N*-methyl give EPI a direct action and a strong affinity for all adrenergic receptors. Epinephrine and all other catechols are chemically susceptible to oxygen and other oxidizing agents, especially in the presence of bases and light, quickly decomposing to inactive

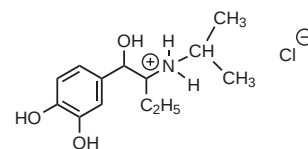
**FIGURE 39.9** Ring configurations that contribute to β₂-receptor selectivity.

quinones. Therefore, all catechol drugs are stabilized with antioxidants and dispensed in air-tight amber containers.

Epinephrine is ultimately metabolized by COMT and MAO to 3-methoxy-4-hydroxymandelic acid (vanillylmandelic acid), which is excreted as the sulfate or glucuronide in the urine (Fig. 39.5). Only a very small amount is excreted unchanged.

Epinephrine usually is administered slowly by intravenous (IV) injection to relieve acute asthmatic attacks not controlled by other treatments. IV injection produces an immediate response. Use of EPI with drugs that enhance cardiac arrhythmias (digitalis or quinidine) is not recommended. Tricyclic antidepressants and MAO inhibitors will potentiate the effects of EPI on the heart. Epinephrine should be used with caution in individuals suffering from hyperthyroidism, cardiovascular disease, hypertension, or diabetes. Adverse effects include palpitations, tachycardia, sweating, nausea and vomiting, respiratory difficulty, dizziness, tremor, apprehension, and anxiety.

Isoetharine Hydrochloride

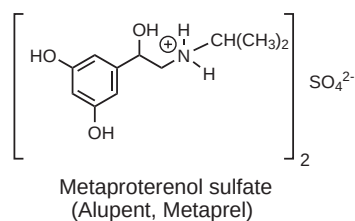


Isoetharine hydrochloride

The α-ethyl group confers β₂-selectivity, and the β-hydroxyl group and catechol nucleus makes it a direct-acting drug. It is susceptible to COMT metabolism; however, the α-ethyl group inhibits MAO. Therefore, one would expect some oral activity.

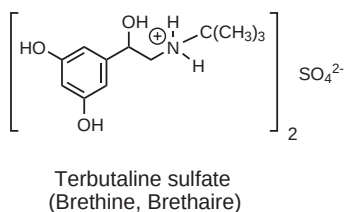
Isoetharine is dispensed as a solution only for inhalation administration to treat reversible bronchospasm of asthma. It has fallen into relative disuse, because with high doses, there is a significant incidence of cardiovascular (β₁-receptor) adverse effects, and it has a low β₂-receptor potency compared to newer β₂-selective agonists. It has a 2- to 4-minute onset of action when inhaled and a duration of action of 3 hours. Isoetharine has adverse effects similar to those of EPI, including palpitations, tachycardia, nausea and vomiting, dizziness, tremor, and headache. Isoetharine may cause decreased levels of theophylline when coadministered. Cardiovascular effects are a concern when isoetharine is taken with other asthma drugs.

Metaproterenol Sulfate



Metaproterenol is a direct-acting resorcinol analog of isoproterenol. The *N*-isopropyl is β -directing, and the combination with the resorcinol ring system enhances the selectivity for the β_2 -receptors. It is the least potent of the β_2 -selective agonists, however, most likely because of the poor β_2 -selectivity of the isopropyl group. It has good oral bioavailability being resistant to COMT and only slowly metabolized by MAO. When administered orally, it has an onset of approximately 30 minutes with a 4-hour duration. Inhaled metaproterenol can have an onset as rapidly as 5 minutes; however, it can be as long as 30 minutes in susceptible individuals. Metaproterenol is available in tablet, syrup, and inhalation dosage forms and is recommended for bronchial asthma attacks and treatment of acute asthmatic attacks in children 6 years of age and older (5% solution for inhalation only). Metaproterenol has the same adverse effect profile as other adrenergic agonists, but with a decreased incidence of arrhythmias.

Terbutaline Sulfate



Terbutaline is the *N*-*t*-butyl analog of metaproterenol and, as such, would be expected to have a more potent β_2 -selectivity. When compared to metaproterenol, terbutaline has a threefold greater potency at the β_2 -receptor. Like metaproterenol, it is resistant to COMT and slowly metabolized by MAO, and thus has good oral bioavailability, with similar onset and duration. Terbutaline is available as tablets and solutions for injection and inhalation. Adverse effects are similar to other direct-acting β_2 -selective agonists, but with a greater incidence of palpitations.

Bitolterol Mesylate

Bitolterol is a prodrug that releases colterol on activation by esterases in the lung (Fig. 39.10). Colterol is a direct-acting agonist, and the *N*-*t*-butyl group makes it β_2 -selective with a binding potency equivalent to that of isoetharine and terbutaline. The ester form is lipophilic, which helps to keep it local in the lung and resistant to COMT, which tends to increase its duration of action.

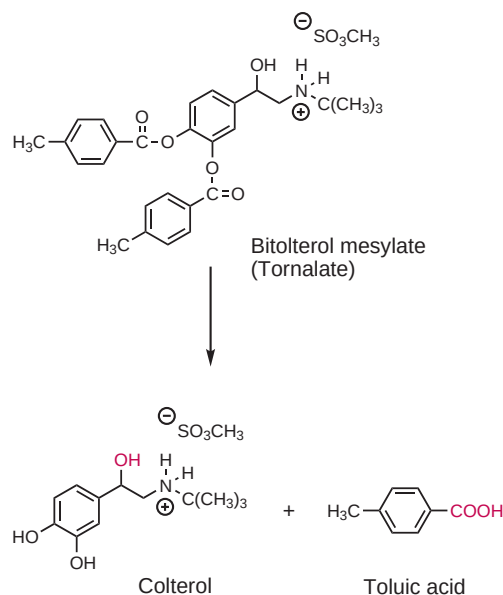
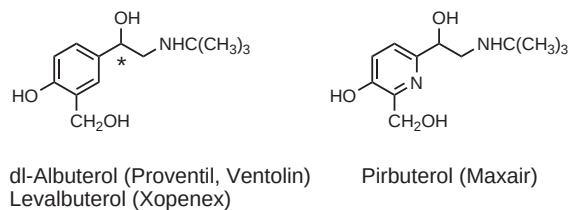


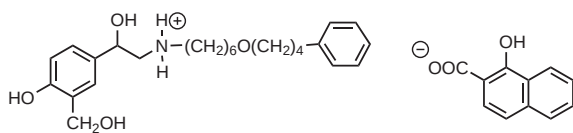
FIGURE 39.10 Esterase hydrolysis of bitolterol.

Onset begins 2 to 4 minutes after administration, and the effect can last as long as 8 hours. It has an adverse effect profile similar to that of other β_2 -selective agonists, with less drowsiness and restlessness compared to other direct-acting agonists.

Albuterol

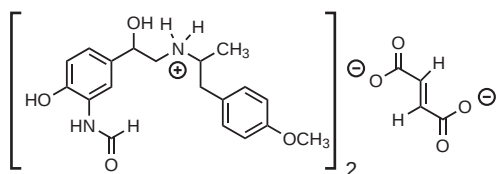


Albuterol has the *N*-*t*-butyl and a salicyl alcohol phenyl ring, which gives it optimal β_2 -selectivity. It is resistant to COMT and slowly metabolized by MAO, thus having good oral bioavailability. Its onset by inhalation is within 5 minutes, with a duration of action between 4 and 8 hours. It currently is the drug of choice for relief of the acute bronchospasm of an asthmatic attack. Levalbuterol is the *R*-(-)-isomer of albuterol and is available only in solution to be administered via nebulizer. Because it is the active isomer, the dose is fourfold less than that of albuterol. Pirbuterol is the pyridine isostere of albuterol. It has pharmacokinetics similar to albuterol but is half as potent at the β_2 -receptor. Pirbuterol is only available as an inhaler, whereas albuterol comes in tablet, syrup, solution, and aerosol formulations. Adverse effects of pirbuterol are nervousness, tremor, and headache, which are less than the profile for albuterol, which adds nausea, vomiting, dizziness, hypertension, insomnia, tachycardia, and palpitations.

Salmeterol Xinafoate

Salmeterol xinafoate
(Serevent diskus)

Salmeterol has an *N*-phenylbutoxyhexyl substituent in combination with a β -hydroxyl group and a salicyl phenyl ring for optimal direct-acting β_2 -receptor selectivity and potency. Salmeterol has the greatest receptor affinity of all the adrenergic agonists. It is resistant to both MAO and COMT and that, together with its increased lipophilicity, gives salmeterol a long duration of action. It is postulated that the phenylbutoxyhexyl *N*-substituent binds outside of the receptor site and keeps the active pharmacophore moiety in position for prolonged stimulations. It is available only as a powder for inhalation, with a 20-minute onset of action, which lasts for 12 hours. It is used as a controller for the long-term treatment of asthma and is not recommended for quick relief of an acute attack. It also is available in combination with the steroid fluticasone propionate (Advair Diskus). There was a small but significant increase in asthma-related deaths among patients receiving salmeterol during a large clinical trial, and as a result, the FDA has recommended that it not be used as monotherapy for asthma in either children or adults. Subgroup analyses suggested that the risk may be greater in black patients compared with Caucasian patients (25,30).

Formoterol Fumarate

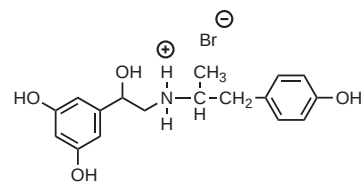
Formoterol fumarate (Foradil aerosolizer)
Arformoterol (*R,R*-Formoterol [Brovana])

Formoterol has a β -directing *N*-isopropyl-*p*-methoxyphenyl group and a unique *m*-formamide and *p*-hydroxyphenyl ring, which provides selectivity for β_2 -receptors. It is resistant to MAO and COMT, making it a long-acting agonist. The most prominent metabolic pathway involves direct glucuronidation at the phenolic hydroxyl group (31). *O*-Demethylation followed by conjugation at the phenolic 2'-hydroxyl group also occurs and appears to involve four cytochrome P450 isozymes (CYP2D6, CYP2C19, CYP2C9, and CYP2A6).

Formoterol has a more rapid onset as compared to salmeterol while maintaining the same long duration of action. This is believed to result from formoterol's

greater water solubility, allowing it to get to receptor sites faster, whereas its moderate lipophilicity keeps it in the lungs longer. It is indicated for the long-term maintenance treatment of asthma and for patients with symptoms of nocturnal asthma who require regular treatment with inhaled, short-acting, β_2 -agonists. It is not indicated for patients whose asthma can be managed by occasional use of inhaled, short-acting, β_2 -agonists. Formoterol is available only as a powder in a capsule for administration via the aerosolizer. Patients should be cautioned not to take the capsules orally and to keep them in a safe place to avoid accidental oral administration.

Formoterol has two asymmetric centers and, therefore, can exist in four possible enantiomers. The *R,R*-enantiomer is active and reported to be 1,000 times more active than the *S,S*-enantiomer and twice as potent as racemic formoterol. It is available under the name arformoterol (Brovana), undergoes the same metabolism, and is only indicated for the treatment of COPD.

Fenoterol Hydrobromide

Fenoterol hydrobromide
(Berotec)

Fenoterol is an investigational drug in the United States that has been in use in Europe since 1970. It is the *p*-hydroxyphenyl derivative of metaproterenol, and the combination of the resorcinol ring and the bulky *p*-hydroxyphenyl isopropyl group on the nitrogen gives fenoterol significant β_2 -receptor selectivity. It has approximately half the affinity for the β_2 -receptor as compared to albuterol. The resorcinol ring is resistant to COMT metabolism, and the bulky nitrogen substituent greatly retards MOA metabolism as well giving fenoterol a reasonable oral bioavailability with pharmacokinetics similar to albuterol (i.e., rapid onset and a 4- to 6-hour duration of action after oral inhalation).

Antimuscarinics

Acetylcholine is the endogenous neurotransmitter of the parasympathetic nervous system. The parasympathetic nerve fibers are found in both the autonomic and central nervous systems. These fibers are classified into those that are stimulated by muscarine and those that are stimulated by nicotine. Nicotine, an alkaloid from *Nicotiana tabacum*, stimulates preganglionic fibers in both the parasympathetic and sympathetic systems as well as the somatic motor fibers of the skeletal system. Muscarine, an alkaloid from the poisonous mushroom *Amanita muscaria*, stimulates postganglionic parasympathetic fibers

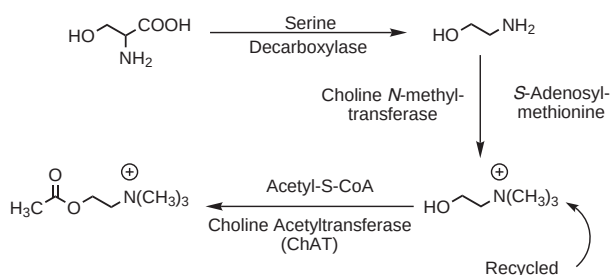


FIGURE 39.11 Biosynthesis of acetylcholine.

with receptors found on autonomic effector cells. The central nervous system has fibers that contain both nicotinic and muscarinic receptors. In this chapter, we are most interested in the drugs that block the muscarinic fibers (antimuscarinics), because blocking them results in cardiovascular, mydriatic, antispasmodic, antisecretory, and bronchodilatory effects.

Biochemistry and Metabolism of Acetylcholine

Acetylcholine is the ester formed between acetylcoenzyme A and choline by the action of choline acetyltransferase in the presynaptic cholinergic neurons. Most of the choline used to biosynthesize acetylcholine comes via uptake from the synaptic space, where it is produced from the hydrolysis of acetylcholine by acetylcholinesterase, a serine hydroxylase. Additionally, some choline is biosynthesized in the presynaptic neurons from serine (Fig. 39.11). Once formed, acetylcholine is stored in vesicles from which it is released on stimulation.

The duration of action of acetylcholine is very short, because it is rapidly hydrolyzed by the acetylcholinesterase present in the synaptic space. This hydrolysis is a straightforward splitting of the acetylcholine into acetic acid and choline; however, the way this happens is very interesting and begins by the proper binding of acetylcholine in the catalytic pocket. Binding of the cationic *N*-end to tyrosine, glutamate, and tryptophan via a combination of π -cation and electrostatic forces places the acyl head of acetylcholine in the correct position for attack by the serine hydroxyl group (Fig. 39.12).

Once properly bound, the hydrolysis actually involves two hydrolytic steps. The first step is the hydrolysis of

acetylcholine by nucleophilic attack at the carbonyl carbon by the serine hydroxyl group, which liberates choline and leaves the enzyme acetylated. A triad formed between glutamine, histidine, and the serine at the catalytic site activates the serine for the nucleophilic attack. The second step is the hydrolysis of the acetylated enzyme by water to regenerate the free enzyme. The water is activated by hydrogen-bonding to the histidine residue, which increases the nucleophilic character of the oxygen of water. The activated water attacks the electrophilic carbonyl carbon of the acetyl group to generate acetate and regenerate the free hydroxyl group of serine (Fig. 39.13) (32).

Muscarinic Receptor Structure and Agonist/Antagonist Interactions

The muscarinic receptors are considered to be part of the superfamily of G protein-coupled receptors. They consist of seven transmembrane helices and are linked to their G protein through interaction with the second and third intracellular loops (33). There are five subtypes of receptor, designated M_{1-5} , and the odd-numbered receptors (M_1 , M_3 , and M_5) are coupled to the G_q/G_{11} class. This class of receptors activates intracellular phospholipase C to hydrolyze phosphatidylinositol 4,5-diphosphate to diacylglycerol and inositol triphosphate as intracellular messengers. The even-numbered receptors (M_2 and M_4) are coupled to the G_i/G_o class, which mediates the inhibition of adenylate cyclase (Fig. 39.14).

Table 39.6 lists the physiologic action of the M_3 receptors. Because the M_3 receptors cause bronchiole constriction, they counterbalance the bronchiole dilation of the β_2 -adrenergic receptors in the lung, resulting in maintenance of bronchiole tone. This forms the basis for the therapeutic use of inhaled antimuscarinics, because they block cholinergic bronchiole constriction and allow adrenergic bronchiole dilation to help overcome the pulmonary constriction associated with an asthmatic attack.

Affinity labeling and mutagenic studies have established that acetylcholine binds to its receptor in a narrow region of the circular arrangement of the seven transmembrane helices approximately 10 to 15 angstroms away from the membrane surface. The cationic nitrogen of acetylcholine binds to the anionic carboxylate of an Asp located in helix 3 (34). As depicted in Figure 39.15, the ionic interaction is

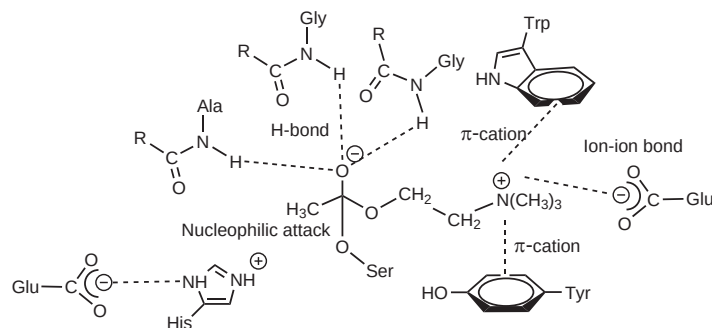


FIGURE 39.12 Binding of acetylcholine in the catalytic site of acetylcholinesterase.

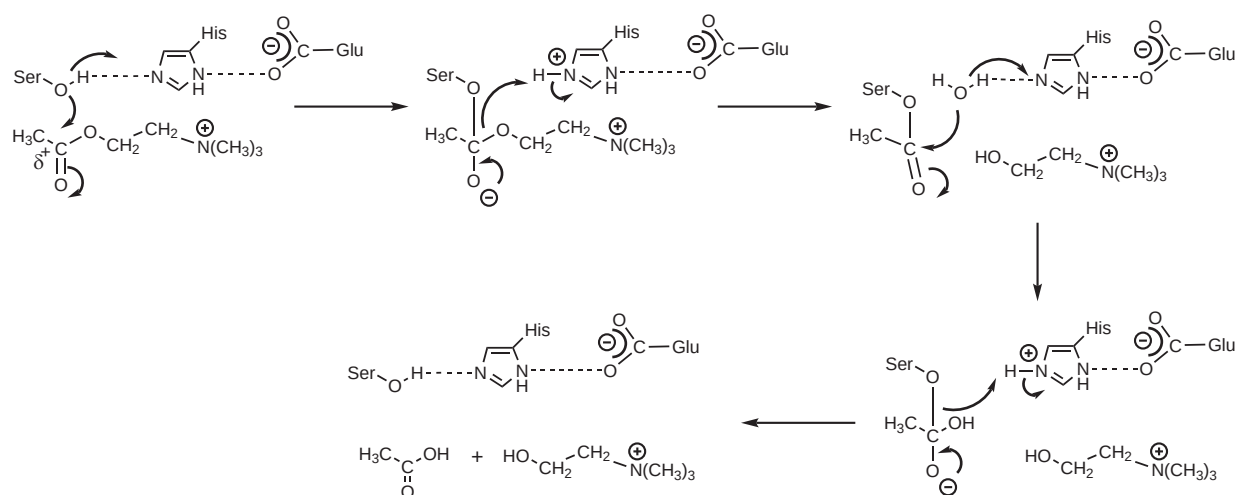


FIGURE 39.13 The role of the triad formed between glutamine, histidine, and serine in the hydrolysis of acetylcholine by acetylcholinesterase.

stabilized by hydrogen-bonding with a Tyr in helix 5 and a Thr in helix 5. It is postulated that muscarinic antagonists (see *Structure–Activity Relationships of Antimuscarinic Agents* below) bind to the Asp and contain hydrophobic substituents that bind to a hydrophobic pocket in the receptor, which does not allow the change in conformation needed to transfer the agonist signal to the coupled G protein (35).

Structure–Activity Relationships of Antimuscarinic Agents

The structural pharmacophore for all antimuscarinic drugs is an acetylcholine analog in which the acetyl methyl group is substituted with at least one phenyl ring. This pharmacophore generally is classified as an amino alcohol ester. The ester function can be replaced with different moieties to produce different classes of antimuscarinic drugs. When the ester function is replaced

by an ether function, the amino alcohol ether class is produced. When the ester function is replaced by a saturated carbon, the amino alcohols are obtained when R_1 is a hydroxyl group, and the amino amides are obtained when R_1 is an amido group (Fig. 39.16).

The classic chemical prototype for the antimuscarinics is atropine, an alkaloid from *Atropa belladonna*. Buried within its structure is the amino alcohol ester pharmacophore, where R_1 is a hydroxymethyl group, R_2 is hydrogen, and the nitrogen is part of a bicyclic ring system called tropine (Fig. 39.17).

Although atropine does not have a quaternary nitrogen, the nitrogen is protonated at physiologic pH;

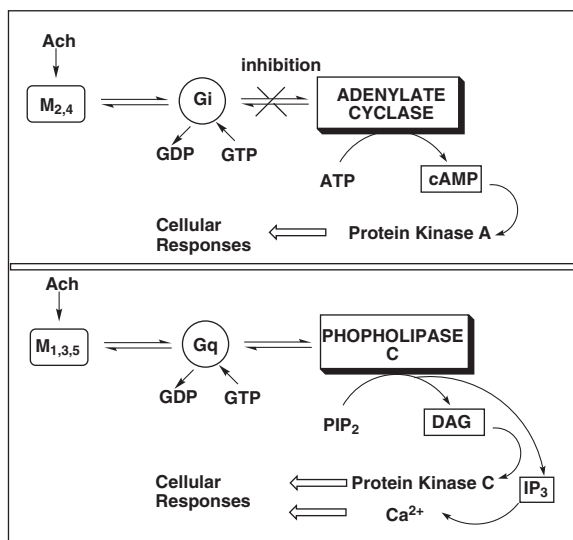


FIGURE 39.14 Comparison of the role of G protein in odd- and even-numbered muscarinic receptors.

TABLE 39.6 Physiologic Action Associated with the M_3 Muscarinic Receptors

Organ	M_3 -Receptor Effect
Eye	
Iris circular muscle	Contracts
Ciliary muscle	Contracts
Heart	
Sinoatrial node	Decelerates
Atrial contractility	Decelerates
Bronchiole smooth muscle	Contracts
Gastrointestinal tract	
Smooth muscle	Contracts
Secretions	Increases
Sphincters	Relaxes

From Katzung B. Introduction to autonomic pharmacology. In: Katzung B, ed. Basic and Clinical Pharmacology. 8th Ed. New York: Lange Medical Books/McGraw-Hill, 2001:75–191; with permission.

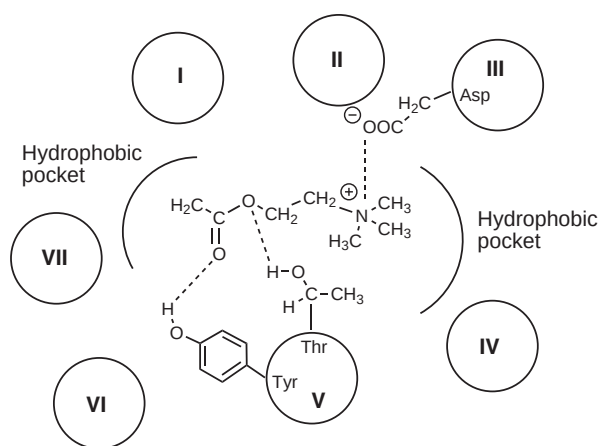


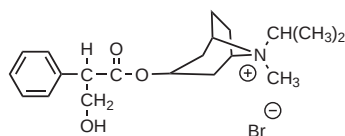
FIGURE 39.15 Acetylcholine binding to residues in the muscarinic receptor.

therefore, it can bind to the anionic Asp residue in the muscarinic receptor. The nitrogen is not absolutely necessary for activity and can be replaced with a carbon atom. This leads to a substantial loss of binding affinity, however, and there are no marketed drugs with this configuration. The nitrogen can be substituted with alkyl groups, and methyl is the optimal size. When the nitrogen is made quaternary, the molecule loses its oral availability but leads to compounds that can be administered effectively by inhalation. The asterisks in atropine (Fig. 39.17) refer to chiral carbons. When stereochemistry is present in the amino alcohol moiety (tropine), there is little difference between the activities of the *R*- and *S*-configurations. When stereochemistry is found in the acid moiety, however, the *R* configuration is approximately 100-fold more active than the *S* isomer. This indicates the importance of the binding role for the phenyl ring in causing the uncoupling of the G protein, which leads to receptor inhibition.

Specific Antimuscarinic Drugs Used to Treat Asthma

For a more detailed discussion of the chemistry of cholinergic agents see Chapter 9.

Ipratropium Hydrobromide



Ipratropium hydrobromide
(Atrovent)

Ipratropium is the *N*-isopropyl analog of atropine. Its quaternary cationic nature makes it highly hydrophilic and poorly absorbed from the lungs after inhalation via solution or aerosol. Much of an inhaled dose is swallowed. There is no significant absorption, however, and the bronchodilation effect can be considered to be a local, site-specific effect. Ipratropium is indicated primarily for the relief of

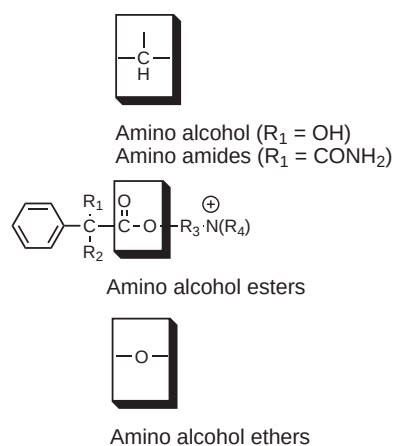
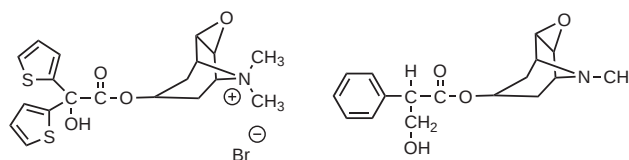


FIGURE 39.16 The pharmacophore for all classes of antimuscarinic agents.

bronchospasms associated with COPD and has seen little application for the treatment of asthma. It also is administered by nasal spray for the relief of rhinorrhea associated with the common cold and perennial rhinitis. Inhaled ipratropium has a 15-minute onset of action and a rather short duration of action (<4 hours); therefore, it is dosed four times a day. Other drugs, including adrenergic agonists, methylxanthines, steroids, and cromolyn sodium, can be coadministered with ipratropium without adverse drug reactions. The little ipratropium that reaches the circulation is minimally protein bound and is partially metabolized to by esterase to inactive products. Most adverse effects from ipratropium are common to antimuscarinics and include blurred vision, dry mouth, tachycardia, urinary difficulty, and headache. Patients should be careful not to spray ipratropium into their eyes, because its dilation effects can precipitate or exacerbate narrow-angle glaucoma.

Tiotropium Bromide



Tiotropium bromide
(Spiriva)

Scopolamine

Tiotropium is the dithienyl derivative of *N*-methyl scopolamine, a quaternary analog of naturally occurring scopolamine in *Atropa belladonna*. It is indicated primarily for

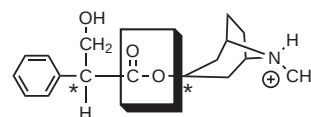


FIGURE 39.17 Atropine is an example of the amino alcohol ester class of antimuscarinic agents.

the relief of bronchospasms associated with COPD and can be considered to be a site-specific, local medication to the lung. Tiotropium is administered as a dry powder via inhalation using a HandiHaler, in which is placed the drug, contained in a green capsule. Patients should be cautioned not to be confused and take the medication orally. Systemic distribution following oral inhalation is minimal, essentially because of its hydrophilic character. If swallowed, only approximately 14% of the dose is eliminated in the urine, with the remainder being found in the feces. Inhaled tiotropium has a 30-minute onset of action but a much longer duration of action than ipratropium (24 hours vs. <4 hours, respectively). Tiotropium is metabolized by both CYP3A4 and CYP2D6, followed by glutathione conjugation to a variety of metabolites. Only a very small amount is nonenzymatically hydrolyzed to inactive products. Tiotropium has an adverse reaction profile similar to that of ipratropium, with dry mouth being the most common adverse effect; however, blurred vision, tachycardia, urinary difficulty, headache precipitation, and exacerbation of narrow-angle glaucoma have been reported.

Methylxanthines

The methylxanthines naturally occur in coffee (*Coffea arabica*), cacao (*Theobroma cacao*), and tea (*Camellia sinensis*) (39). The major methylxanthines are caffeine, theophylline, and theobromine, and they differ by the position and number of methyl groups on their xanthine ring system (Fig. 39.18).

The most common source of these xanthines is in the beverages coffee, tea, and cocoa, which are universally consumed mainly for their stimulant properties. A cup of coffee or tea contains between 60 and 85 mg of caffeine, and a cup of cocoa can have as much as 250 mg of theobromine. Caffeine frequently is added to cola drinks and to over-the-counter analgesics and stimulants. Theophylline is used for its bronchodilating effects in the treatment of asthma. Its importance has declined greatly since the development of the inhaled β_2 -adrenergic agonists and inhaled steroids and because it has a narrow therapeutic window, which requires close patient monitoring and periodic blood level determination to avoid serious side effects.

Theophylline Mechanism of Action and Metabolism

Despite a great deal of investigation, just how theophylline causes bronchodilation is not clearly understood. Inhibition of the enzyme PDE, which is responsible for the hydrolysis of cAMP and cyclic guanosine monophosphate (cGMP), generally is put forth as the mechanism

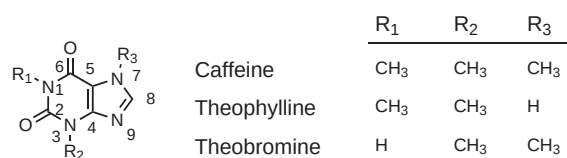
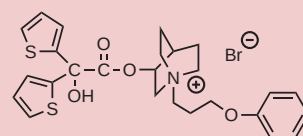


FIGURE 39.18 Structural differences between the methylxanthines.

ACLIDINIUM BROMIDE: ANTIMUSCARINIC DRUG IN THE PIPELINE

Acclidinium bromide is new antimuscarinic drug currently in phase III clinical trials. It is recommended for the inhalation treatment of COPD. Structurally, acclidinium bromide is related to tiotropium bromide in that it is a dithienyl derivative where the *N*-methyl scopolamine moiety is replaced by an *N*-phenoxypropyl-1-aza bicyclo [2,2,2]octane. This change results in a molecule with kinetic selectivity for the M₃ versus the M₂ muscarinic receptors. Acclidinium bromide is reported to be a long-acting bronchodilator with a fast onset of action. Acclidinium bromide undergoes rapid hydrolysis of its ester bond in human plasma to its constituent acid and alcohol derivatives with a half-life of 2.4 minutes in humans. In comparison, more than 70% of tiotropium bromide remains intact in plasma after 60 minutes. The acid and alcohol metabolites do not contribute to the bronchodilator activity of acclidinium bromide, and the short half-life combined with the lack of antimuscarinic activity of its hydrolysis products suggest that acclidinium may have reduced antimuscarinic side effects. Although the major metabolites of acclidinium bromide are the ester hydrolysis products, it also undergoes, to a minor extent, oxidative metabolism, yielding mono-oxygenated phenyl and the loss of one of the thienyl rings mediated by CYP3A4 and CYP2D6 isoforms. In a small clinical trial in patients with moderate to severe COPD, acclidinium bromide demonstrated significant elevation in FEV₁ from baseline, producing sustained bronchodilation over 24 hours, and no side effects were related to the treatment protocol (36–38).



Acclidinium bromide (Almirall)

of action; however, theophylline also is an adenosine antagonist and has been implicated in stimulation of the release of catecholamines. It has been clearly shown that theophylline does inhibit PDEs in vitro, and x-ray crystallographic studies have identified the binding residues that interact with the methylxanthines (Fig. 39.19).

Theophylline binds to a subpocket of the active site and appears to be sandwiched between a phenylalanine and a valine via hydrophobic bonds. Its binding affinity is reinforced by hydrogen-bonding between a tyrosine and N-7 and a glutamine and O-6 of the xanthine ring system. There are more than 11 families of PDEs, and studies have shown that theophylline binds in a similar manner to both the PDE4 and PDE5 family isoforms (40).

Chemically, theophylline is 1,3-dimethylxanthine and contains both an acidic and a basic nitrogen (N-7 and N-9, respectively). Physiologically, it behaves as an acid ($pK_a = 8.6$), and its poor aqueous solubility

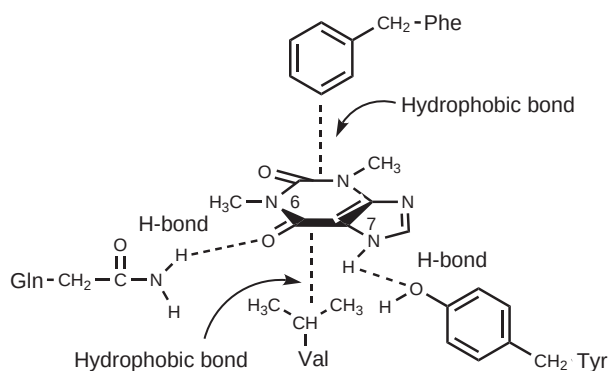
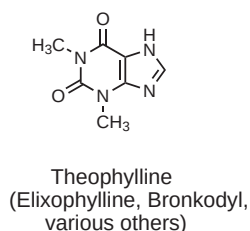


FIGURE 39.19 Methylxanthine binding interactions in the catalytic pocket of phosphodiesterase.

can be enhanced by salt formation with organic bases. Theophylline is metabolized by a combination of C-8 oxidation and *N*-demethylation to yield methyluric acid metabolites (Fig. 39.20). The major urinary metabolite is 1,3-dimethyl uric acid, which is the product of the action of xanthine oxidase. Because none of the metabolites is uric acid itself, theophylline can be safely given to patients who suffer from gout.

Methylxanthine Drugs Used to Treat Asthma

Theophylline



The primary indication for theophylline is as a controller medication for the treatment of bronchospasm of asthma and COPD. In addition to bronchodilation

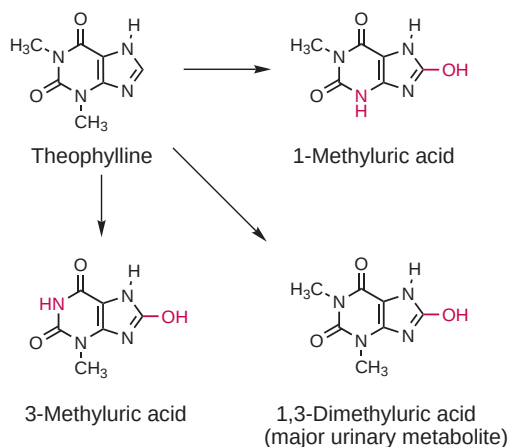
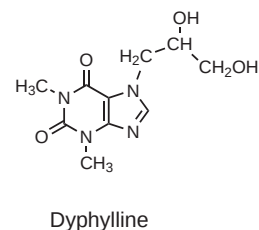


FIGURE 39.20 Metabolism of theophylline.

effects, theophylline dilates pulmonary blood vessels, acts centrally to stimulate respiration, acts as a diuretic, increases gastric acid secretion, and inhibits uterine contractions. Dosing requires the determination of plasma levels, with 10 to 20 $\mu\text{g/mL}$ associated with the least incidence of side effects. Overdose of theophylline can result in a quick onset of ventricular arrhythmias, convulsions, or even death without any previous warning. Many drugs increase the plasma concentration of theophylline, including quinolone and macrolide antibiotics, nonselective β -blockers, ephedrine, calcium channel blockers, cimetidine, and oral contraceptives. Theophylline is available in tablet, capsule, liquid, and parenteral dosage preparations. There also are combination products with guaifenesin and ephedrine available as tablet and liquid dosage forms. There are two products that are theophylline salts. Aminophylline is theophylline ethylenediamine, which contains 70% theophylline and is available in tablets, liquid, parenteral, and suppository dosage forms. Oxytriphyllyne is the choline salt of theophylline, and it contains 64% theophylline in tablets and liquid dosage forms. Care must be taken to correctly calculate the equivalent dose when switching a patient from theophylline to one of its salts.

Dyphylline (Dihydroxypropyl Theophylline)



Dyphylline is the N-7 dihydroxypropyl derivative of theophylline and is not a theophylline salt. Dyphylline does not get metabolized to theophylline in vivo, and even though it contains 70% theophylline by molecular weight ratio, the equivalent amount to theophylline is not known. Dosing must be accomplished independently by monitoring dyphylline blood levels. Dyphylline has a diminished bronchodilator effect compared to theophylline, but it may have lower and less serious side effects. Dosage forms available are an elixir and tablets.

Adrenocorticoids Used to Treat Asthma and COPD

Steroids are a large class of tetracyclic terpene compounds that are widely distributed in plants and animals. Many synthetic analogs have been made to take advantage of their various pharmacologic activities. There are four major classes of steroids: the adrenocorticoids, the sex hormones, the bile acids, and the vitamins. The adrenocorticoids are formed in the adrenal cortex and are subdivided into the glucocorticoids and mineralocorticoids. The glucocorticoids are so named because they affect glucose homeostasis, but they also have significant

anti-inflammatory activity. The mineralocorticoids affect sodium and water retention. The reproductive organs, in both the male and female, produce steroid hormones, which are responsible for the differentiation of the sex characteristics and the development of muscle and hair. The bile acids are derived from cholesterol and are an essential aid to the action of lipase in the digestion of fats, and vitamin D and its derivatives are associated with calcium homeostasis. We will limit our discussion to the glucocorticoids, because their anti-inflammatory activity makes them useful for the treatment of asthma and COPD. Of note, because the major role of the glucocorticoids is to provide the body with levels of glucose that are compatible with life, they should be used with caution in patients with diabetes. Similarly, because the major role of the mineralocorticoids is to maintain blood volume and regulate electrolyte balance, their use in patients with hypertension should be monitored carefully. (For more detail regarding the adrenal corticosteroids, see Chapter 28.)

Steroid Nomenclature

The basic structure of a steroid is a tetracyclic cyclopentanoperhydrophenanthrene, referred to as the steroid nucleus, as depicted in Figure 39.21. The addition of methyl groups at C-10 and C-13 and of an ethyl group at C-17 to the steroid nucleus gives a 21-carbon base structure called pregnane. All glucocorticoids are substituted pregnanes. Rings A, B, and C are in the chair conformation, and all ring junctions are *trans*, giving the glucocorticoids a rigid backbone with α and β faces. The glucocorticoids bind to their receptor via the β face. All glucocorticoids have at least one double bond in ring A and hydroxyl groups at C-11, C-17, and C-21 as well as 3- and 20-keto groups. For example, hydrocortisone is Δ^4 -11 β ,17 α ,21-trihydroxy-3,20-pregnenedione. A more complete discussion of steroid nomenclature and stereochemistry has been published elsewhere (41).

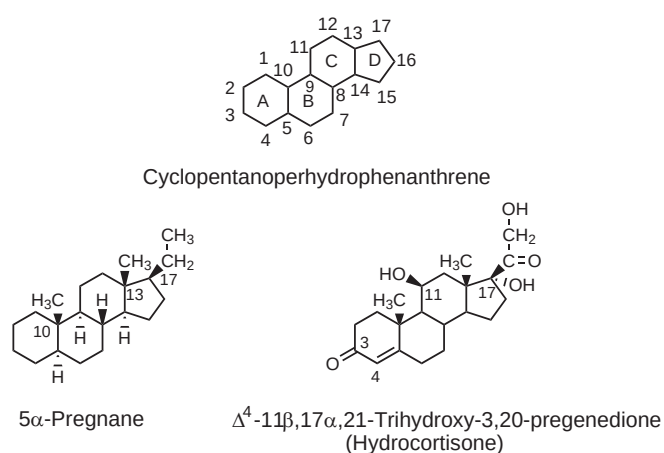


FIGURE 39.21 Basic skeletons for steroid structures, numbering, and nomenclature.

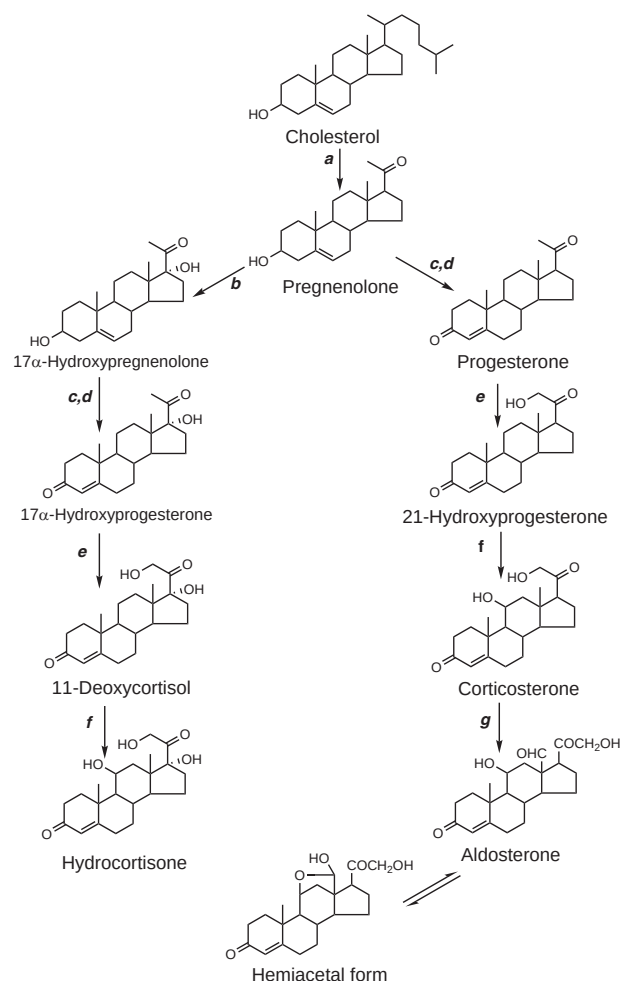


FIGURE 39.22 Biosynthesis of the adrenocorticoids from cholesterol. The enzymes involved are (a) side chain cleavage, (b) 17 α -hydroxylase, (c) 5-ene-3 β -hydroxy-steroid dehydrogenase, (d) 3-oxosteroid-4,5-isomerase, (e) 21-hydroxylase, (f) 11 β -hydroxylase, and (g) 18-hydroxylase.

Steroid Biosynthesis and Secretion

Steroids are biosynthesized from cholesterol. Cholesterol is obtained through our diet (~0.3 g/day), but the majority is biosynthesized from acetate in the liver (~1 g/day). The complete pathway can be seen in Figure 39.22. Of note is the central branching role played by pregnenolone, leading to both classes of the adrenocorticoids.

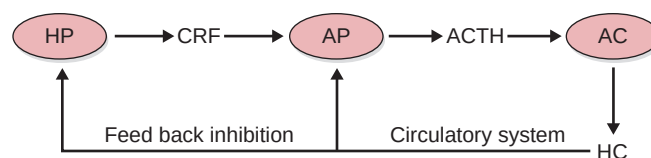


FIGURE 39.23 Mechanism of adrenocorticoid secretion and control. AC, adrenal cortex; ACTH, adrenocorticotropic hormone; AP, anterior pituitary; CRF, corticotrophin-releasing factor; HC, hydrocortisone; HP, hypothalamus.

Once formed, the adrenocorticoids are secreted into the circulatory system, and circulating levels are maintained via a feedback mechanism (Fig. 39.23). When circulating levels are low, the hypothalamus secretes corticotrophin-releasing factor (CRF). In turn, CRF stimulates the anterior pituitary to secrete adrenocorticotrophic hormone, which stimulates the adrenal cortex to synthesize and secrete the adrenocorticoids (mainly hydrocortisone and aldosterone). When circulating levels of the adrenocorticoids are sufficiently high, they feedback-inhibit CRF secretion from the hypothalamus and adrenocorticotrophic hormone release from the anterior pituitary gland. It should be noted that chronic use of steroids inhibits the adrenal cortex from producing glucocorticoids. This is known as hypothalamus-pituitary-adrenal (HPA) suppression. If steroid therapy is stopped abruptly, the lack of endogenous glucocorticoids can be life-threatening. A slow and gradual withdrawal from exogenous steroid therapy is always warranted to allow the HPA axis to recover.

Steroid Pharmacologic Action

The major pharmacologic activities of the glucocorticoids are anti-inflammation, inhibition of cytokines, and inhibition of mast cell release of autocoids. The anti-inflammatory activity of the glucocorticoids is derived from their ability to affect protein synthesis. Specifically, they stimulate the synthesis of lipocortin, a protein that inhibits phospholipase A_2 , which is an enzyme that catalyzes the breakdown of membranes to release arachidonic acid, the first step in the arachidonic acid cascade that results in

the production of inflammatory prostaglandins and leukotrienes (Figs. 39.24 and 39.25). Therefore, inhibition of phospholipase A_2 ultimately results in the reduction of the inflammatory prostaglandins and leukotrienes.

A second anti-inflammatory mechanism of glucocorticoids involves inhibition of IL-1. IL-1 stimulates the proliferation of T and B lymphocytes that are responsible for the production of the cytokines and antibodies, which in turn are important in the inflammatory and immune responses to antigens. The glucocorticoids, by their ability to inhibit IL-1, cause a decrease in T and B lymphocytes, leading to immunosuppression, and therefore must be used with caution in patients with infection. A third action of glucocorticoids is to inhibit the synthesis and release of histamine and other autocoids from mast cells. These pharmacologic activities make the glucocorticoids especially useful for treating the inflammatory processes associated with asthma and COPD.

Steroid Mechanism of Action

How do glucocorticoids and other adrenocorticoids affect the levels of proteins and other biologically important compounds? The short answer is that these steroids bind to their receptor in the cytoplasm and that the glucocorticoid–receptor complex travels into the nucleus, where it binds to DNA and affects gene transcription, which increases and, sometimes, decreases the production of important biologically active compounds. The detail of just how the glucocorticoids accomplish this is quite interesting and is depicted in Fig. 39.26.

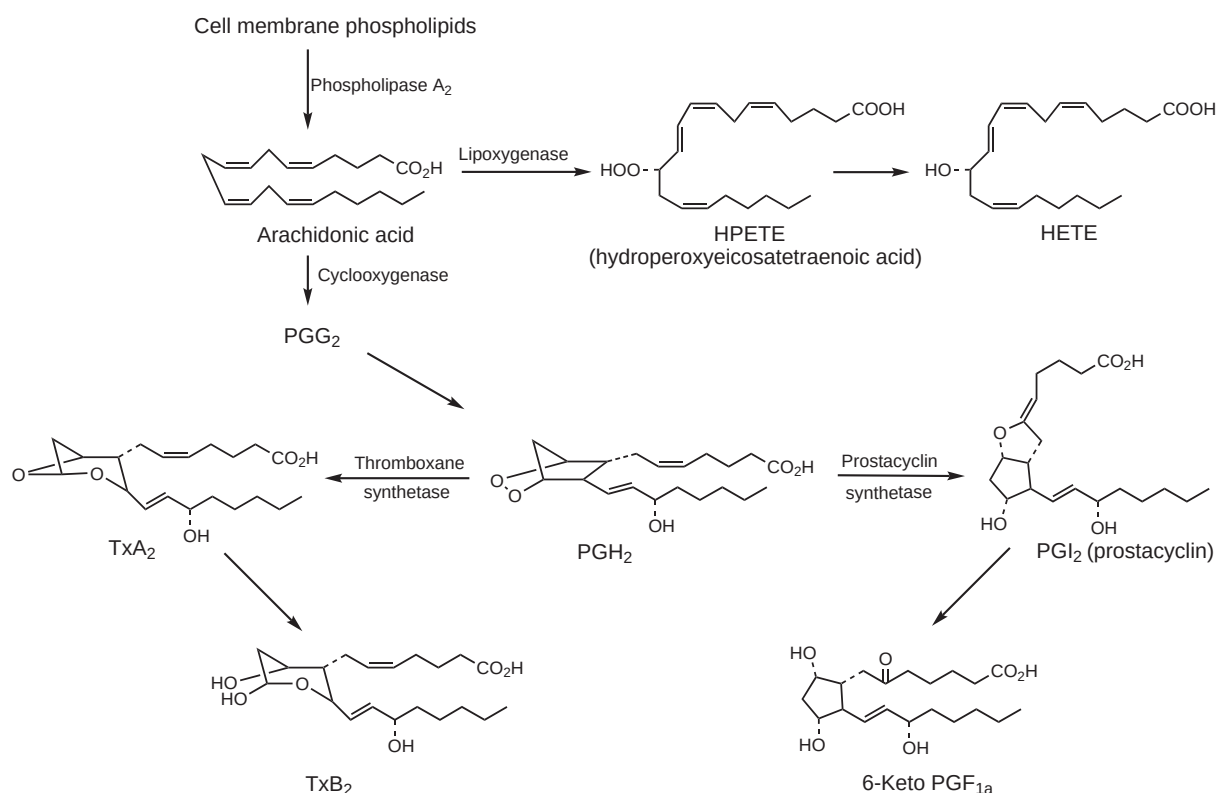


FIGURE 39.24 Biosynthesis of thromboxanes, prostacyclin, and prostaglandins.

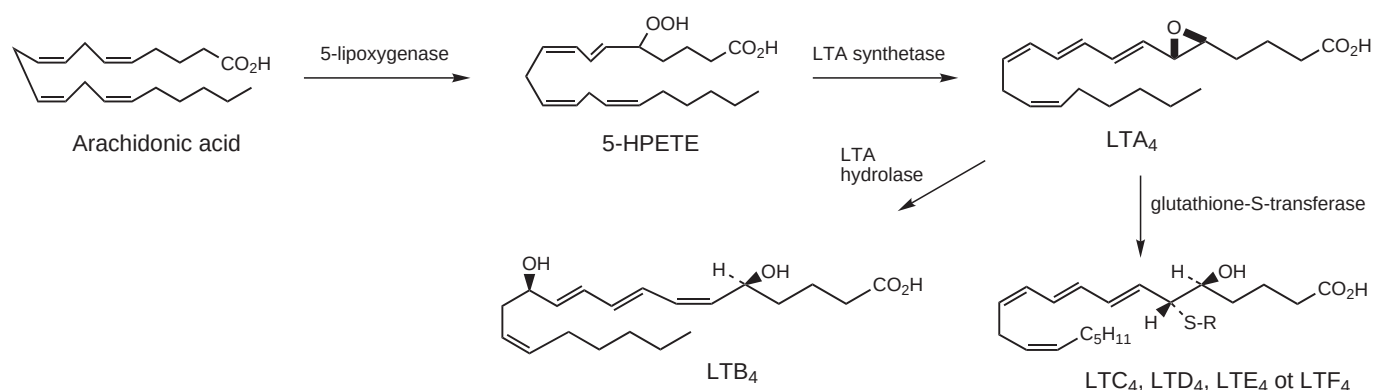


FIGURE 39.25 Biosynthesis of leukotrienes.

Circulating glucocorticoids enter the target cell by simple diffusion, because they are relatively lipophilic. The glucocorticoid receptor occurs inside the cell in combination with a heat shock protein. The glucocorticoid receptor has three distinct regions: the amino terminal for maximum activity, a middle region for binding to DNA, and the carboxy terminal, where the glucocorticoids bind. When the glucocorticoid binds to its receptor, the heat shock protein is released, with a conformational change in the steroid–receptor complex. The glucocorticoid–receptor complex then translocates into the nucleus of the cell, where it binds to specific sequences of DNA called hormone response elements (HREs), which are located in the promoter areas of hormone-responsive genes. The DNA binding region of the glucocorticoid receptor contains eight cysteine amino acid residues that coordinate two zinc ions that form peptide loops called

zinc fingers. When the zinc finger domains bind to the HRE of DNA, it dimerizes the glucocorticoid–receptor complex, placing each subunit in adjacent binding grooves. In most cases, binding causes gene transcription and protein synthesis; however, in some instances, binding will block transcription (42). A more precise understanding of just how the glucocorticoids inhibit the expression of inflammatory proteins has only recently become understood. In the inflammatory cells, such as macrophages, inflammatory protein genes are activated by the acetylation of core histones, which are entwined in DNA. This acetylation opens the chromatin structure allowing gene transcription and synthesis of inflammatory proteins. Glucocorticoids recruit histone deacetylase 2 (HDAC2) to actively transcribing genes, which reverses this process and switches off inflammatory gene transcription (43).

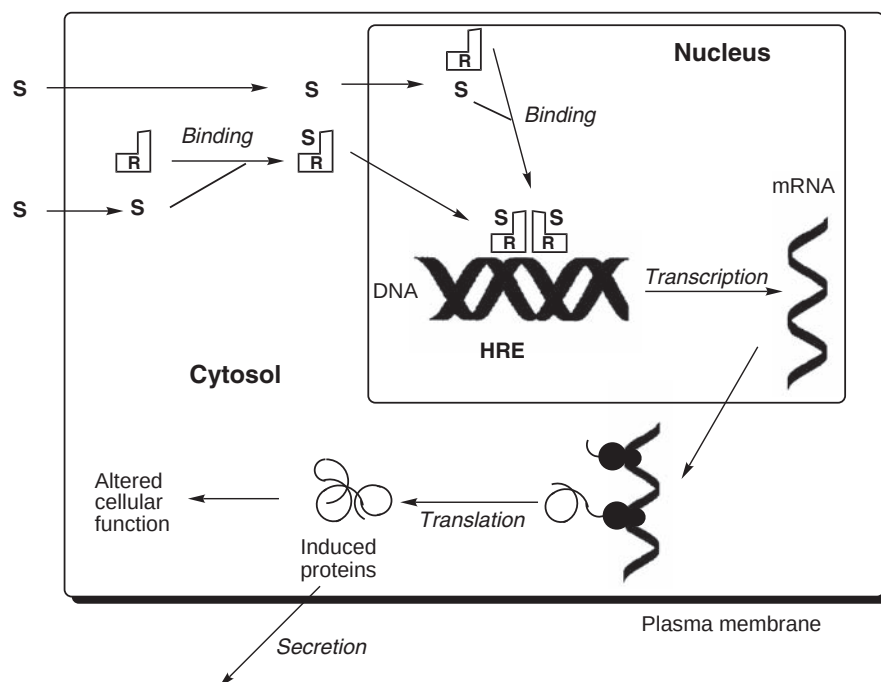


FIGURE 39.26 Mechanism of steroid hormone action. S, steroid; R, receptor.

Receptor Structure and Glucocorticoid Binding

The glucocorticoid receptor is a member of the nuclear receptor superfamily, which includes the receptors for the steroid hormones, retinoids, peroxisomal activators, vitamin D, and thyroid hormones. The steroid receptor is a soluble protein found in the cytosol of the cell. The binding domain of all the steroid receptors consists of 11 α -helices and 4 β -strands. X-ray crystallographic structural analysis, however, has revealed a unique binding pocket for the glucocorticoid receptor, which makes it distinct from the estrogen receptor, androgen receptor, and mineralocorticoid receptor. When a glucocorticoid binds to the receptor, it is completely enclosed within a pocket formed by helices 3, 4, 5, 6, 7, and 10 and β -strands 1 and 2. The strong binding affinity for the glucocorticoids is a result of the hydrophobic and hydrophilic interactions with amino acid residues in the ligand binding domain. Nearly every atom of the glucocorticoid contacts one or more amino acid residues. The most significant contacts are the hydrogen-bonding between every hydrophilic group on the glucocorticoid structure (Fig. 39.27). The carbonyl group on ring A forms hydrogen bonds with both Arg-611 and the amide of Gln-570. The hydroxyls at C-11 and C-21 hydrogen bond with the side chain of Asn-564. The C-17 hydroxyl group hydrogen bonds with Gln-642, and the C-20 carbonyl group bonds with Thr-739. Glucocorticoid binding to the ligand binding domain releases the heat shock protein and stabilizes the receptor in an active form capable of dimerization subsequent to HRE site binding (44,45).

Glucocorticoid Structure–Activity Relationships

There are essential structural features that are necessary for glucocorticoid activity. The natural glucocorticoids also interact with the mineralocorticoid receptor and, therefore, will have salt-retaining properties. A large number of synthetic analogs have been prepared to decrease the mineralocorticoid effects in favor of increasing the glucocorticoid (anti-inflammatory) effect of the steroids. In addition, many derivatives are prepared to enhance

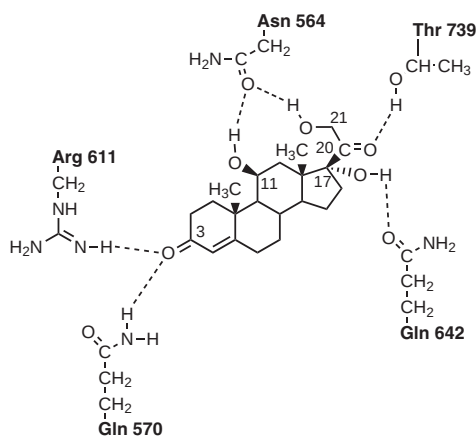


FIGURE 39.27 Important glucocorticoid receptor–ligand binding residues.

pharmacokinetic parameters, most notably the synthesis of lipophilic and hydrophilic esters.

Functional groups that are essential for both mineralocorticoid and glucocorticoid activity include the pregnane skeleton with an all-*trans* backbone, the ring A-en-one system (Δ^4 -3-one ring A), and the 17 β -ketol side chain (C-20-keto-C-21-hydroxy). The C-21 hydroxyl group must be free for both mineralocorticoid and glucocorticoid activity. An exception to this rule are the C-21 chloro and C-20 fluoromethyl thio ester ($-\text{SCH}_2\text{F}$) derivatives that retain anti-inflammatory activity when applied topically or by inhalation. Glucocorticoids that are used for their anti-inflammatory effects and that have both glucocorticoid and mineralocorticoid activity are hydrocortisone, cortisone, prednisone, and prednisolone.

Many derivatives of the fundamental glucocorticoid skeleton have been synthesized with the intention of enhancing anti-inflammatory activity and decreasing the salt-retaining properties of the glucocorticoids when administered in therapeutic doses. Figure 39.28 depicts those positions where substituents will affect anti-inflammatory and/or mineralocorticoid activity. The triangles indicate those positions where substitutions will increase the anti-inflammatory activity, the open circles indicate those positions where substituents will increase both anti-inflammatory and mineralocorticoid properties, the closed circle is the position where substituents will decrease the mineralocorticoid activity, and the open box is where specific substituents can either increase anti-inflammatory activity or decrease mineralocorticoid properties depending on the specific substituent. Therefore, the anti-inflammatory activity will be greatly increased by flattening ring A by adding a double bond at C-1 (Δ^1) and also by adding a hydroxyl group at C-11. A 17 α -hydroxy increases both activities, as does 9 α -fluoro or 9 α -chloro. Mineralocorticoid activity is substantially eliminated by 16 α - or 16 β -methyl groups, a 16 α -hydroxyl group, or a 16 α ,17 α -acetone. A C-6 α -methyl group only slightly enhances the anti-inflammatory activity, whereas a C-6 α -fluoro has twice as much anti-inflammatory effect. Both substituents, in combination with a C-16 substituent, will greatly decrease the mineralocorticoid effects.

Therefore, synthetic compounds can be prepared with marked glucocorticoid activity and little if any significant mineralocorticoid activity. These include triamcinolone, methylprednisolone, fluticasone, beclomethasone, budesonide, mometasone, and ciclesonide.

Glucocorticoid Metabolism

If one or more of the essential functional groups on the glucocorticoid skeleton are modified by metabolism, glucocorticoid activity will be lost. There are three major metabolic reactions that will eliminate glucocorticoid activity. They are ring A reductions, C-17 oxidation, and C-11 keto-enol isomerization. Reduction of the ring A 3-keto to a 3 α -hydroxy by 3 α -hydroxysteroid dehydrogenase and the reduction of the 4,5-double bond by

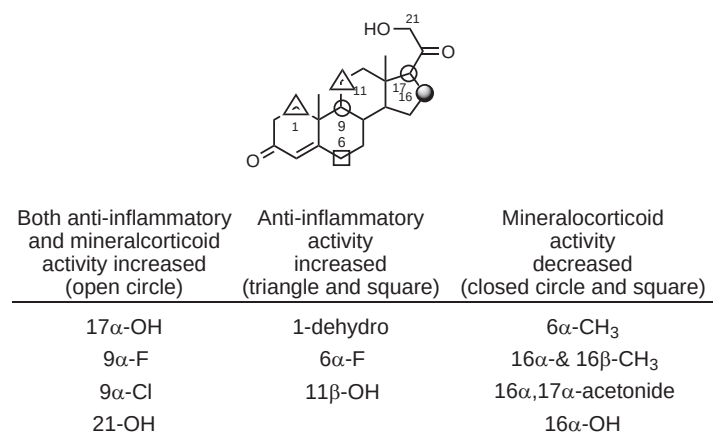


FIGURE 39.28 Glucocorticoid structure–activity relationship.

5 β -reductase to produce the A/B *cis*-fused ring give rise to inactive metabolites (Fig. 39.29). Oxidation of the C-17 side chain to produce a 17-keto steroid requires both the C-17 hydroxyl group and a free C-21 hydroxyl group. Esterification of one or both will inhibit this metabolism and prolong the duration of action. The rapid *in vivo* equilibrium that exists between cortisone (11-keto) and hydrocortisone (11 β -hydroxy) is catalyzed by 11 β -hydroxysteroid dehydrogenase and produces the keto form that is inactive but also the hydroxy form that is highly glucocorticoid enhancing and found on all glucocorticoids. Figure 39.29 summarizes the metabolism of hydrocortisone.

Synthetic Steroid Esters

The glucocorticoids are lipophilic despite having at least three hydroxyl groups. The hydroxyl groups can

be esterified with an appropriate acid that will either enhance or decrease that lipophilicity. The C-21 hydroxyl group is the most accessible and, therefore, is the easiest to esterify. The C-17 hydroxyl group also is easily esterified, but because it is slightly hindered by the C-17 side chain, it will react more slowly. The C-11 hydroxyl group is highly hindered by the C-10 and C-13 methyl groups and will not react with acids to form esters. In addition, esterification has an effect on both receptor affinity and glucocorticoid metabolism. Because the C-21 must be free to hydrogen bond to the Arg-564 in the glucocorticoid receptor, C-21 esters are prodrugs requiring hydrolysis to become active. Also, the C-17 hydroxyl group needs to be free to be metabolically oxidized; therefore, esterification of the C-17 hydroxyl group inhibits oxidation to the C-17 keto group, thus prolonging duration of action. Duration

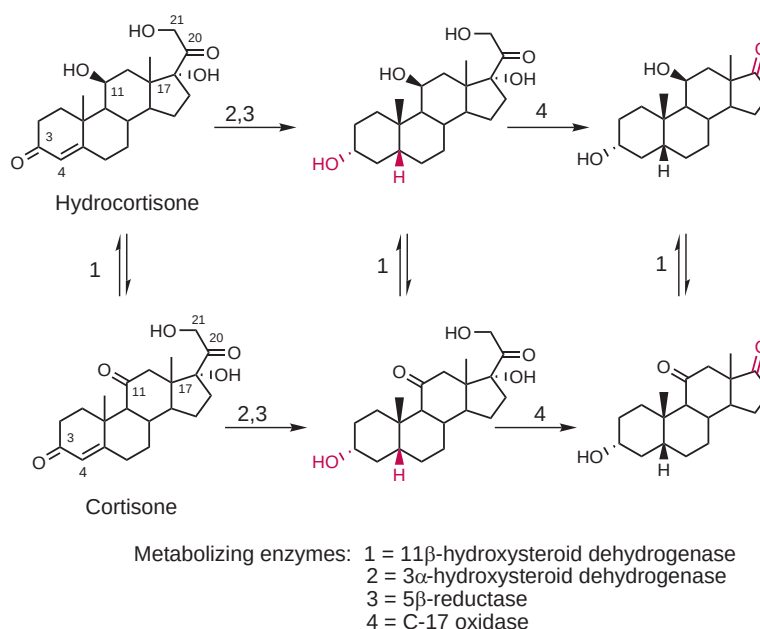


FIGURE 39.29 Metabolic pathways for hydrocortisone.

also is prolonged because of the fact that the lipophilic glucocorticoid esters do not concentrate in the urine (i.e., they undergo tubular reabsorption after glomerular filtration).

The lipophilicity of a glucocorticoid can be enhanced by esterification with lipophilic acids. Doing so results in a number of effects, including increasing the logP, which provides local activity with less systemic absorption and decreased side effects. The fact that C-21 lipophilic esters are prodrugs means that they have a longer duration of action. They also have a slower onset, because the lipophilic esters generally are bulky and retard hydrolytic enzymes.

These lipophilic prodrugs can be administered orally to treat a variety of conditions, by inhalation to treat asthma and COPD, and topically to treat various types of dermatitis. When administered orally, their longer duration of action means that they can be given less frequently, which often results in better compliance. When administered topically, C-21 lipophilic ester prodrugs are activated by hydrolysis by skin esterases (46,47), which slows their systemic absorption. Systemically absorbed glucocorticoids are further metabolized in the liver as depicted in Figure 39.29. Figure 39.30 shows the structures of the most common C-21 lipophilic esters found on commercially available glucocorticoids.

Diesters and ketals provide a further enhancement of logP (i.e., increased lipophilicity) and an increased duration of action because they are slow to metabolize. It should be noted that diesters and hexacetonides (Fig. 39.31) are prodrugs, whereas plain ketals are not (i.e., they already have a free C-21 hydroxyl group). Diesters are formed at C-21 and at either the C-17 or, if present, the C-16 hydroxyl groups. The diesters are prodrugs; however, the esters at C-17 or C-16 are active intact. The most common diesters are either the diacetate or dipropionate.

A ketal is a cyclic derivative formed between hydroxyl groups on adjacent carbons and a carbonyl carbon of a ketone. The most common ketal found in glucocorticoids is called an acetonide and is the condensation product formed between the C-16 and C-17 hydroxyl groups and the carbonyl carbon of acetone (CH_3COCH_3)

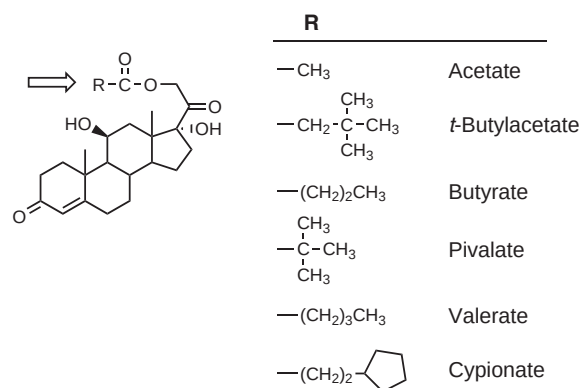
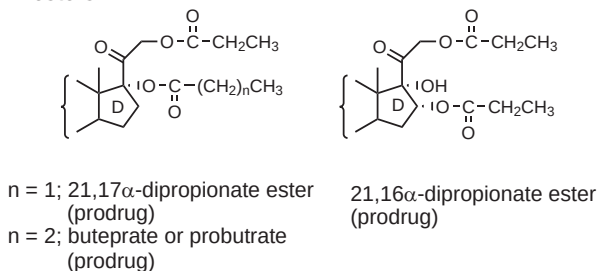
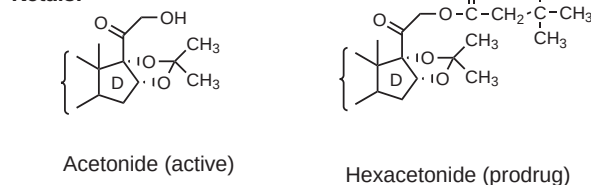


FIGURE 39.30 Common lipophilic esters of glucocorticoids.

Diesters:



Ketals:



Hydrophilic esters:

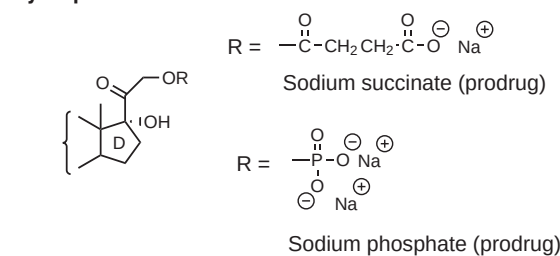


FIGURE 39.31 Examples of glucocorticoid lipophilic diesters and ketals and hydrophilic esters.

(Fig. 39.31). If the acetonide is found along with a C-21-*t*-butylacetate ester, it is named a hexacetonide; the “hex” refers to the total number of carbons found in the butylacetate group. The hexacetonides are prodrugs and must hydrolyze to become active. The bupropionate (i.e., probupionate) is a diester consisting of a propionate at C-21 and a butyrate at C-17 (Fig. 39.31). It also is a prodrug, and the C-21 ester must be hydrolyzed to yield the active form.

Esterification also can be used to increase the hydrophilicity of the glucocorticoids, making them water-soluble prodrugs. The synthetic approach here is to use either a dicarboxylic acid or phosphoric acid to condense with the C-21 hydroxyl group (Fig. 39.31). This places an ionizable group (a carboxylate or phosphate) on the prodrug, yielding water-soluble esters. These derivatives are used to prepare aqueous injectable products that can be administered intramuscularly or IV.

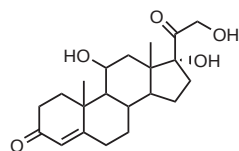
Hydrophilic glucocorticoid prodrugs have a rapid onset, because they are readily hydrolyzed by plasma esterases. In contrast to the lipophilic prodrugs, their water solubility allows them to be easily excreted through the kidney, resulting in a shorter duration of action. Hydrophilic prodrugs have more systemic side effects because of their wide distribution that comes from their high solubility in the blood. Along with their ability to be injected IV, this property makes them useful in asthmatic

emergencies (i.e., status asthmaticus) during which the patient is unable to take oral medication. Table 39.7 summarizes glucocorticoid prodrug derivatives.

Glucocorticoid Drugs Used to Treat Asthma

Systemic Glucocorticoids

HYDROCORTISONE



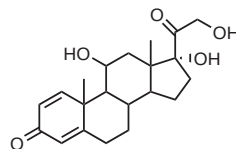
Hydrocortisone
(Cortisol)

Δ^4 -11 β ,17 α ,21-trihydroxy-3,20-pregnenedione

Hydrocortisone is endogenous, and it has both glucocorticoid and mineralocorticoid activity. It is the fundamental structure by which the glucocorticoid and mineralocorticoid activities of all other corticosteroids are judged. Functional groups that are essential for both

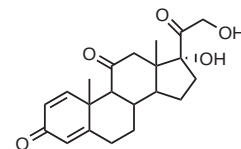
mineralocorticoid and glucocorticoid activity include the pregnane skeleton with an all-*trans* backbone, the ring A-en-one system (Δ^4 -3-one ring A), and the 17 β -ketol side chain (C-20-keto-C-21-hydroxy). The glucocorticoid activity is enhanced by the C-11 and C-17 hydroxyl groups. Hydrocortisone can be used to treat severe asthmatic attacks that do not respond to conventional treatment. It is available as various ester forms (Table 39.7).

Prednisolone



Prednisolone
(Delta-Cortef)

$\Delta^{1,4}$ -11 β ,17 α ,21-trihydroxy-
pregnadiene-3,20-dione



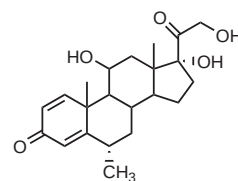
Prednisone
(Deltasone, Meticortin)

$\Delta^{1,4}$ -17 α ,21-dihydroxy-
pregnadiene-3,11,20-trione

Prednisolone is hydrocortisone to which has been added a Δ^1 double bond. This places two double bonds in ring A, which flattens it and increases glucocorticoid action at the expense of mineralocorticoid activity. Prednisolone has fourfold the glucocorticoid activity of hydrocortisone while having approximately half its mineralocorticoid activity. In addition, prednisolone has an increased duration of action compared to hydrocortisone, because the extra double bond in ring A retards its metabolic reduction.

Prednisolone can be used to treat severe asthmatic attacks that do not respond to conventional treatment, and it is available as the free alcohol for oral administration. The C-21 sodium phosphate (Hydeltrasol) ester is available for parenteral use. Various ester prodrugs are available, as listed in Table 39.7. A prodrug of prednisolone is prednisone. It is the 11-keto analog of prednisolone and must be converted in vivo to the active 11 β -hydroxy compound, which is necessary to hydrogen bond to Asn-564 in the glucocorticoid receptor. Prednisone should not be used in patients with hepatic dysfunction, because their ability to reduce the 11-keto group with 11 β -hydroxysteroid dehydrogenase to the active metabolite may be impaired.

Methylprednisolone



6-Methylprednisolone
(Medrol)

$\Delta^{1,4}$ -6 α -methyl-11 α ,17 α ,21-trihydroxy-
pregnadiene-3,20-dione

TABLE 39.7 Glucocorticoid Prodrug Derivatives

Drug	Trade Name	Site of Ester	Route of Administration
Hydrocortisone derivatives			
Hydrocortisone phosphate		C-21	IM, IV, SC
Sodium succinate	A-Hydrocort Sulucortef	C-21	IM, IV
Cypionate	Cortef	C-21	PO
Acetate	Cortaid	C-21	Topical
Butyrate	Locoid	C-21	Topical
Buteprate	Pandel	C-21	Topical
Valerate	Westcort	C-21	Topical
Prednisolone derivatives			
Phosphate	Hydeltrasol	C-21	IV, IM
Acetate	Key-Pred	C-21	Topical
Di- <i>t</i> -butylacetate (tebutate)	Prednisol TBA	C-17, 21	Topical
Methylprednisolone derivatives			
Sodium succinate	Solu-medrol	C-21	IV, IM
	A-Methapred		
Acetate	Depo-Medrol		
	Depoject	C-21	IM

IM, intramuscular; IV, intravenous; PO, oral; SC, subcutaneous.

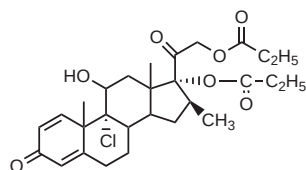
Adding a 6 α -methyl group to prednisolone increases the glucocorticoid activity and effectively abolishes mineralocorticoid action. It has fivefold the glucocorticoid activity of hydrocortisone (prednisolone has fourfold the glucocorticoid activity) and none of its mineralocorticoid properties. It is used almost exclusively as a systemic product and is available as the free alcohol for oral administration and as various esters (Table 39.7).

Side Effects

Systemic administration of glucocorticoids can result in a large number of adverse effects on prolonged use. As already stated, there is the severe risk of HPA axis suppression, requiring the slow tapering off of the dose. In addition, glucocorticoids have cardiovascular effects (thromboembolism, hypertension, and arrhythmias), central nervous system effects (convulsions and steroid psychosis), dermatologic effects (impaired wound healing and hair growth), endocrine effects (amenorrhea, postmenstrual bleeding, and suppression of growth in children), electrolyte disturbances (sodium and fluid retention, hypertension, and metabolic alkalosis), GI effects (pancreatitis, increased appetite, and peptic ulcer), musculoskeletal effects (muscle weakness, steroid myopathy, and osteoporosis), and ophthalmic effects (cataracts, increased intraocular pressure, and glaucoma).

Glucocorticoids for Inhalation

Beclomethasone Dipropionate

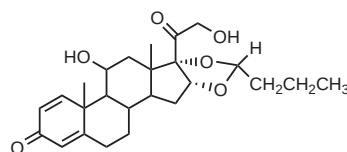


Beclomethasone dipropionate
(QVAR, Vancril)

9 α -Chloro-16 β -methyl-11 β ,17 α ,21-trihydroxypregna-1,4-diene-3,20-dione 17,21-dipropionate

Beclomethasone dipropionate is a lipophilic prodrug that, when inhaled, shows a systemic bioavailability of approximately 20% of the administered dose. The 16 β -methyl group decreases mineralocorticoid activity, and the 9 α -chloro group increases both the glucocorticoid and mineralocorticoid activity, resulting in potent anti-inflammatory activity with little or no salt-retaining effects. The main adverse effects are headache, sinusitis, and pain. Beclomethasone dipropionate is metabolized to the more active 17 α -monopropionate derivative during absorption from the lungs and then further metabolized to the free alcohol in the liver. The dipropionate also is metabolized to the inactive 21-monopropionate in the liver. Beclomethasone dipropionate and its metabolites are mainly excreted in the feces, with less than 10% excreted in the urine.

Budesonide

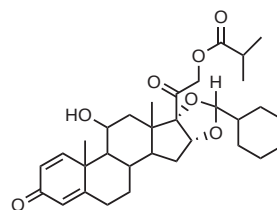


Budesonide
(Pulmacort tubuhaler, Respules)

16 α ,17-[(*R*,*S*-butyridenebis(oxy))-11 β ,21-dihydroxypregna-1,4-diene-3,20-dione

Budesonide is an acetal formed between the 16 α ,17 α -dihydroxyl groups and butanal. It is a nonhalogenated glucocorticoid with a 16,17-acetal that decreases the mineralocorticoid activity. In receptor affinity studies, the *R*-epimer was twofold more active than the *S*-epimer. Because the C-21 hydroxy is free, budesonide is not a prodrug and is active as administered. Only 34% of the metered dose of inhaled budesonide reaches the lung. Systemically absorbed budesonide is highly protein bound and metabolized to 16 α -hydroxyprednisolone and 6 β -hydroxybudesonide in the liver. Both metabolites have less than 1% of the glucocorticoid activity of the parent compound. Inhaled budesonide is excreted mainly as metabolites via both the feces (15%) and the urine (32%). Budesonide is available as a powder delivered through an aerosol or as a suspension for inhalation.

Ciclesonide



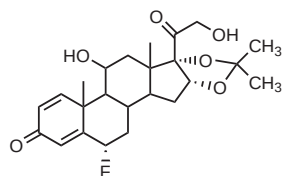
Ciclesonide (Alvesco, Omnaris)

16 α ,17-[(*R*-cyclohexylmethoxy)bis(oxy))-11 β -hydroxy-21-(2-methyl-1-oxopropoxy)-pregna-1,4-diene-3,20-dione

Ciclesonide is a glucocorticoid with an acetal formed at positions 16 and 17 with cyclohexylcarboxaldehyde. Acetals at this position decrease mineralocorticoid activity. In addition, ciclesonide has a Δ^1 double bond that retards ring A reduction, promoting a longer duration of action, and also flattens ring A, which increases glucocorticoid action at the expense of mineralocorticoid activity. Ciclesonide is a prodrug that is converted to its active form desisobutyrylciclesonide via esterases in the lung. It is reported to have a relative potency equivalent to fluticasone, mometasone, budesonide, and beclomethasone dipropionate. Lung deposition on inhalation is 52%, and the oral bioavailability is less than 1%, presumably due to low GI absorption and high first-pass metabolism. Des-ciclesonide is

metabolized in the liver to additional metabolites mainly by CYP3A4 and to a lesser extent CYP2D6. Ciclesonide is available in a hydrofluoroalkane propellant metered-dose formulation for inhalation. Ciclesonide is also used for the treatment of hay fever in the form of a nasal spray under the trade name of Omnaris.

Flunisolide

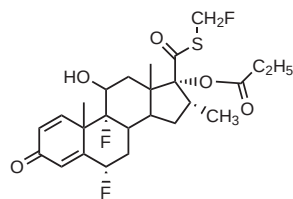


Flunisolide
(Aerobid)

6 α -Fluoro-11 β ,21-dihydroxy-16 α ,17-[(1-methylethylidene)bis(oxy)]pregna-1,4-diene-3,20-dione

Flunisolide is an acetone ketal (acetonide) with a 6 α -fluoro group and a free C-21 hydroxyl group. The acetonide decreases mineralocorticoid activity, and the 6 α -fluoro group increases glucocorticoid activity. It is not a prodrug, because it has the free hydroxyl group at C-21. Flunisolide has approximately 20% of the receptor affinity as budesonide, and approximately 40% of the inhaled dose is systemically bioavailable. Flunisolide is quickly metabolized by CYP3A4 to the 6 β -hydroxy metabolite, which has less than 1% of the activity of the parent compound. This, along with the rapid elimination of both flunisolide and its metabolite as glucuronides, greatly limits any systemic adverse effects. Flunisolide is available as a microcrystalline powder aerosol.

Fluticasone Propionate



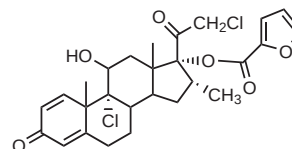
Fluticasone propionate (Flovent, Flonase)

6 α ,9 α -Difluoro-5-fluoromethyl-11 β -hydroxy-16 α -methyl-3-oxo-17 α -propionyloxyandrost-1,4-diene-17 α -carbothioate

Fluticasone propionate has a unique C-20 thiofluoromethyl group, and that, in combination with the 17 α -propionate ester, gives it 36-fold the glucocorticoid receptor affinity as compared to beclomethasone dipropionate and twofold the affinity as compared to budesonide. The 9 α -fluoro group increases both glucocorticoid and mineralocorticoid activities, and the 6 α -fluoro group enhances only the glucocorticoid action. Studies have determined that the effectiveness of inhaled fluticasone propionate results from a local rather than a systemic

effect. Interestingly, less than 1% of the swallowed dose is bioavailable, in contrast to the fact that the majority of the inhaled dose is systemically available and highly protein bound. Fluticasone propionate is not a prodrug and is extensively metabolized by the liver. The only detectable metabolite is the 17 β -carboxylic acid derived from CYP3A4 oxidation. This metabolite has a 2,000-fold less affinity for the glucocorticoid receptor than the parent drug. Elimination is through both the feces and the urine, with the relative amounts determined by the route of administration. Fluticasone propionate is available in aerosol and powder inhalation formulations.

Mometasone Furoate

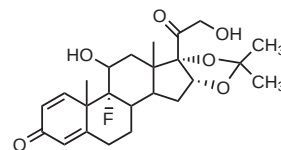


Mometasone furoate (Asmanex Twisthaler)

9 α ,21-dichloro-17 α -[(2-furanylester)oxy]-11 β -hydroxy-16 α -methylpregna-1,4-diene-3,20-dione

Mometasone furoate has a number of unique functional groups that confer enhanced glucocorticoid activity as well as pharmacokinetic advantages. The combination of the C-21 chloro and the furoic acid ester at C-17 results in the highest glucocorticoid receptor affinity of any topical corticosteroid. The 16 α -methyl decreases the mineralocorticoid effects, and the 9 α -chloro group increases both glucocorticoid and mineralocorticoid activities. Inhaled mometasone furoate acts locally as an anti-inflammatory treatment for asthma and has the least systemic bioavailability of all the inhaled glucocorticoids (<1%). It is extensively metabolized, with 6 β -hydroxymometasone and 21-hydroxymometasone being found among the metabolites. Inhaled mometasone furoate is mainly excreted in the feces (~74%) as metabolites and only minimally excreted in the urine (~8%). It has a relatively long half-life in the lung and is administered once daily, usually in the evening.

Triamcinolone Acetonide



Triamcinolone acetonide (Azcort)

9 α -Fluoro-11 β ,17 α ,21-trihydroxypregna-1,4-diene-3,20-dione-16 α ,17-acetonide

Triamcinolone acetonide has the acetonide function, which greatly decreases its mineralocorticoid activity.

The 9 α -fluoro group increases both the glucocorticoid and the mineralocorticoid activities. Triamcinolone acetonide has a receptor affinity that is 10-fold that of beclomethasone dipropionate but 6-fold less than that of mometasone furoate. After oral inhalation, as much as 25% of triamcinolone acetonide can be detected systemically; a good proportion of that is assumed to be from swallowing and absorbed from the GI tract unchanged. It is metabolized in the liver to a number of metabolites, including 6 β -hydroxytriamcinolone acetonide and the C-21 carboxy-6 β -hydroxytriamcinolone acetonide, both of which are readily excreted via the kidneys. The acetonide remains intact during metabolic reactions and, therefore, is highly resistant to hydrolytic cleavage. Triamcinolone acetonide is excreted mainly as metabolites in the urine (40%) and in the feces (60%), with less than 1% being excreted unchanged.

Mast Cell Degranulation Inhibitors

The discovery that a Middle Eastern herb, *Ammi visnaga* (Khella, Bishop's weed), had mild bronchodilation effects led to the isolation of a benzopyrone (a chromone), khellin. Khellin had only weak bronchodilator effects, so synthetic analogs were prepared in an attempt to enhance the bronchodilation. All analogs were less active; however, it was noted by Dr. Roger Altounyans, who was experimenting on himself with these analogs, that one of them, if inhaled before an asthmatic attack, gave excellent protection against the attack's severity (48). The active compound was identified as a bischromone and named cromolyn sodium in the United States and sodium cromoglycate in Europe (Fig. 39.32).

This class of mast cell degranulation inhibitors prevents the release of histamine, leukotrienes, prostaglandins,

and other inflammatory autocooids by interaction with the sensitized mast cell before antigenic challenge and does not inhibit the binding of IgE to the mast cell or the antigen to IgE. The exact mechanism of action is still not completely understood; however, it is clear that inhibition of the role of calcium in the degranulation process is involved. A number of membrane and cellular proteins that bind cromolyn sodium are known to regulate intracellular calcium levels and include basophilic membrane protein, nucleoside diphosphate kinase, calgranulins B and C, and annexins I through V (49,50). Mast cell degranulation inhibitors also are used topically in the eye to treat allergic reactions; in addition to cromolyn sodium, they include nedocromil sodium, lodoxamide tromethamine, and pemirolast potassium. (For additional details about this class of compounds, see Chapter 32.)

Mast Cell Stabilizers Used to Treat Asthma

Cromolyn Sodium

Cromolyn sodium is a bischromone that contains the fundamental benzopyrone moiety of khellin (Fig. 39.32). The two chromone rings are necessary for activity and must be coplanar, with a linking chain of no longer than six carbons. If one changes the linking chain to positions 8 and 8', coplanarity cannot be maintained, and the compound loses all activity. Cromolyn sodium is poorly absorbed from the lungs (~8%), insignificantly from the eye (~0.07%), and by approximately 1% from the GI tract. What little that finds its way into systemic circulation is eliminated intact in the urine and the bile. For the treatment of asthma, cromolyn sodium is available as a solution for both intranasal and inhalation administration. There also is an oral concentrate (100 mg/5 mL), which is administered as a 200-mg dose given four times a day.

Nedocromil Sodium

Nedocromil sodium was developed by changing the furan ring of khellin to a piperidinone ring (Fig. 39.32). In vitro, nedocromil sodium inhibits the release of inflammatory response mediators from a variety of cells, including neutrophils, mast cells, macrophages, and platelets. Inhaled nedocromil sodium is poorly absorbed into the systemic circulation, with approximately 3% of an inhaled dose excreted in the urine during the first 6 hours after administration. Only 2% of orally dosed nedocromil sodium is bioavailable, 89% of which is protein bound. When administered IV, nedocromil sodium is not metabolized and is excreted unchanged in the bile and the urine. Nedocromil sodium is available in aerosol canisters for oral inhalation via a mouthpiece.

Leukotriene Modifiers

It has been known for more than 40 years that a substance called SRS-A (slow-reacting substance of anaphylaxis) produced a slowly developing, long-lasting contraction of isolated guinea pig ileum and that this same substance was associated with the pathophysiology of asthma.

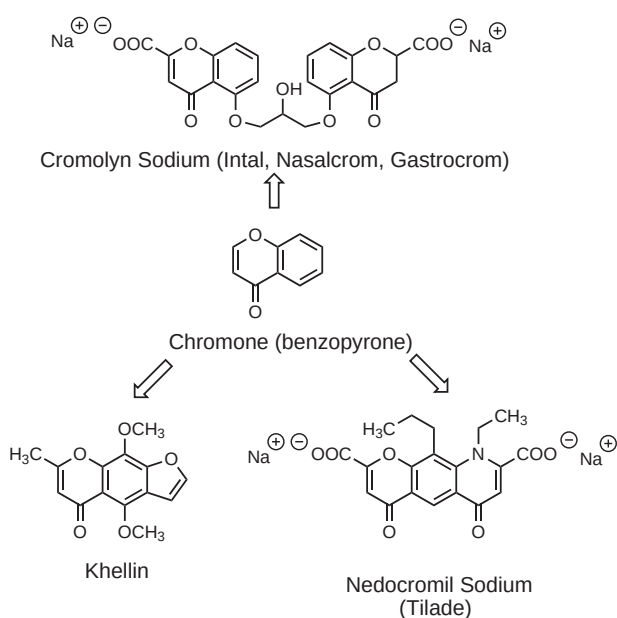


FIGURE 39.32 Structures of several chromosome mast cell stabilizers.

Subsequently, it was determined that SRS-A actually was a mixture of triene-containing lipids, designated as cysteine-leukotrienes: LTC_4 , LTD_4 , and LTE_4 . Determination of the chemical structure of these biologically active compounds led to the development of biosynthesis inhibitors as well as receptor antagonists that are useful in the treatment of asthma (51).

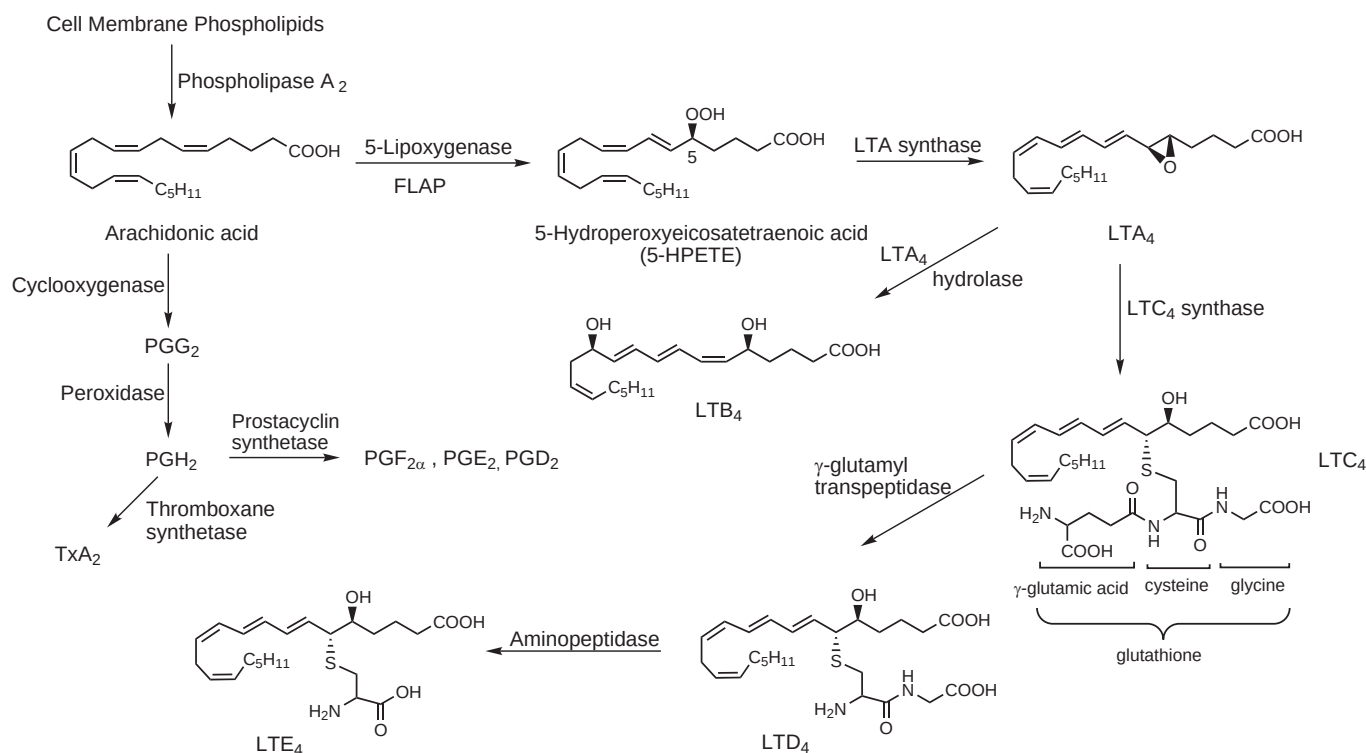
Biosynthesis of Leukotrienes

The leukotrienes occur in a variety of inflammatory cells that are abundant in asthma, including eosinophils, mast cells, and macrophages. They are derived from arachidonic acid via a branch of the common pathway to the prostacyclins and thromboxanes. Arachidonic acid itself is produced by the action of phospholipase A_2 on cell walls. Unlike the prostacyclins and thromboxanes, excess arachidonic acid does not activate the pathway. Instead, the first step in the conversion of arachidonic acid to produce the leukotrienes is controlled by an activating protein, 5-lipoxygenase-activating protein (FLAP), which regulates the interaction of 5-lipoxygenase with its substrate. Figure 39.33 depicts the biosynthetic pathway and shows that 5-lipoxygenase oxidizes arachidonic acid to the unstable peroxide intermediate, 5-hydroperoxyeicosatetraenoic acid, which is quickly dehydrated to LTA_4 , which in turn is further metabolized by LTA_4 hydrolase to LTB_4 and by LTA_4 synthase to the glutathione adduct, LTC_4 . Cleavage of γ -glutamic acid by γ -glutamyl transpeptidase converts LTC_4 to LTD_4 , which in turn is converted to LTE_4 by

the removal of glycine under the action of dipeptidase. The three leukotrienes produced from LTA_4 have strong bronchoconstrictive activity: LTD_4 is the most potent bronchoconstrictor, with the fastest onset of action; however, LTB_4 has no bronchoconstrictive activity but is a potent neutrophil chemotactic agent.

Leukotriene Receptors

The cysteinyl leukotriene (cysLT) receptors are of the rhodopsin family of the G protein-coupled receptor family. As such, they consist of seven transmembrane-spanning helices that activate intracellular signaling pathways in response to their endogenous ligands (LTC_4 , LTD_4 , and LTE_4). Studies have identified two distinct cysLT receptors, and molecular biologists have cloned the human genes for both the cysLT_1 receptor and the cysLT_2 receptor (52). Pharmacology studies distinguished between the two cysLT receptors by evaluating their interaction with known antagonists, showing that the cysLT_1 receptor is competitively inhibited from binding LTD_4 , whereas the cysLT_2 receptor was not. In addition, there is a slight difference in ligand-binding affinities between the two receptors. The cysLT_1 receptors have an affinity profile of $\text{LTD}_4 > \text{LTC}_4 = \text{LTE}_4$, whereas the cysLT_2 receptors show an affinity profile of $\text{LTC}_4 = \text{LTD}_4 > \text{LTE}_4$. Both receptor types occur in the lungs and spleen. The cysLT_1 receptor is found only in the placenta and small intestines, whereas the cysLT_2 receptor occurs only in the heart, lymph nodes, and brain. The incomplete overlap of tissue distribution along with their distinct ligand-binding



FLAP = 5-lipoxygenase activating protein; LT = Leukotriene; PG = Prostaglandin

FIGURE 39.33 Biochemical pathways leading to the leukotrienes.

properties suggests the cysLT_1 receptors and the cysLT_2 receptors might serve different functions in vivo (53), and because the cysLT_1 receptors are inhibited by selective antagonists, they have importance in the treatment of leukotriene-related bronchoconstriction in asthma.

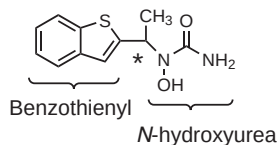
Leukotriene Modifier Drugs

Two approaches to the development of leukotriene modifiers have been taken. The first approach was to block their biosynthesis by looking for compounds that inhibit one or more of the enzymes involved in the biochemical pathway. The second approach was to identify antagonists with selective affinity for the cysLT_1 receptors.

Leukotriene Biosynthesis Inhibitors

The search for orally active 5-lipoxygenase inhibitors has resulted in only a few classes of compounds that are effective in animals and humans with the *N*-hydroxyureas, yielding a useful product (54). The 5-lipoxygenase inhibitors block the production of LTB_4 as well as LTC_4 , LTD_4 , and LTE_4 , thereby decreasing both the bronchoconstrictive and chemotactic effects of the leukotrienes.

ZILEUTON

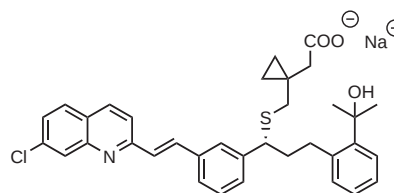


Zileuton is the first *N*-hydroxyurea 5-lipoxygenase inhibitor to be marketed. It is the ethylbenzothienyl derivative of *N*-hydroxyurea and occurs as the racemic mixture of *R*-(+)- and *S*-(-)-enantiomers, both of which are pharmacologically active. The *N*-hydroxyl group is essential for inhibitory activity, with the benzothienyl group contributing to its overall lipophilicity. Zileuton is rapidly absorbed orally and is 93% protein bound in the plasma. Metabolism occurs in the liver, with the inactive *O*-glucuronide being the major metabolite, along with less than 0.5% inactive *N*-dehydroxylated and unchanged zileuton. The glucuronidation is stereoselective, with the *S*-isomer being metabolized and eliminated more quickly (55). Greater than 90% of an oral dose is bioavailable, and 95% is excreted as metabolites in the urine, with a half-life of 2.5 hours, thus requiring four-times-a-day dosing. Zileuton increases the plasma levels of propranolol, theophylline, and warfarin, and dosing of these drugs should be reduced and the serum levels monitored carefully in patients taking both drugs. The most serious side effect of zileuton is elevation of liver enzymes; if symptoms of liver dysfunction (e.g., nausea, fatigue, pruritus, jaundice, or flu-like symptoms) occur, the drug should be discontinued.

Leukotriene Receptor Antagonists

The search for leukotriene receptor antagonists began without the aid of ligand–receptor binding data and took the form of three approaches. These included the design of leukotriene structural analogs and quinoline analogs and the random screening of compounds. The combination of these efforts led to a simple structure–activity relationship: The lipophilic tetraene tail of LTD_4 can be mimicked by a variety of more stable aromatic rings, the thioether of the glycylcysteinyl dipeptide can be replaced by an alkyl carboxylic acid, and the C_1 carboxylate of LTD_4 needs to be retained. Additional research focusing on the three-dimensional requirements for antagonist binding to the cysLT receptors further clarified that the pharmacophore needs to consist of an acidic or negative ionizable functional group, a hydrogen-bond acceptor function, and three hydrophobic regions (56). Based on this background, synthetic efforts resulted in the development of montelukast and zafirlukast as cysLT_1 receptor antagonists. Figure 39.34 demonstrates how both these antagonists fit the pharmacophore model.

MONTelukAST



Montelukast was developed from other weakly antagonistic quinoline derivatives. A number of changes can be made to the structure without the loss of activity. These include changing the double bond between the two aromatic rings to an ether linkage, reducing the quinoline ring, changing the chlorine to a fluorine, and/or exchanging the sulfur for an amide group. Montelukast is a high-affinity, selective antagonist of the cysLT_1 receptor. It is rapidly absorbed orally, with a bioavailability of 64%. Montelukast is 99% bound to plasma proteins and is metabolized extensively in the liver by CYP3A4 and CYP2C9 to oxidized products. CYP3A4 oxidizes the sulfur and the C-21 benzylic carbon, whereas CYP2C9 is selectively responsible for the methyl hydroxylation. Figure 39.35 shows the primary metabolic pathway for montelukast in humans (57). More than 86% of an oral dose is eliminated as metabolites through the bile. Montelukast did not demonstrate any significant adverse effects greater than placebo in clinical trials; however, because it is metabolized by the cytochrome P450 enzymes, its plasma levels should be monitored when coadministered with cytochrome P450-inducing drugs, such as phenobarbital, rifampin, and phenytoin. Montelukast is available in tablet, chewable tablet, and granules for administration mixed with food.

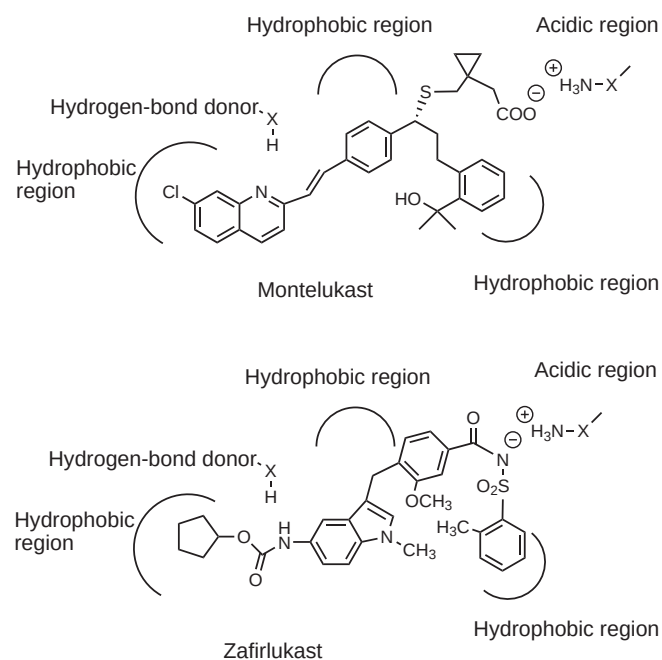
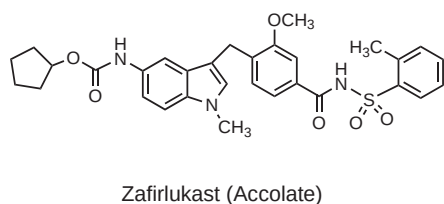


FIGURE 39.34 Interaction of montelukast and zafirlukast with cysteinyl leukotriene (cysLT) receptor model.

Zafirlukast



Zafirlukast is an indole derivative with a sulfonamide group that fulfills the need for an ionizable moiety on the pharmacophore. A large number of analogs have been prepared; however, they all resulted in a decrease in antagonist activity. Zafirlukast, like montelukast, is a selective antagonist for the cysLT_1 receptor and antagonizes the bronchoconstrictive effects of all leukotrienes (LTC_4 , LTD_4 , and LTE_4). Zafirlukast is well absorbed orally; however, food will decrease its absorption by as much as 40%. Zafirlukast is primarily metabolized in the liver by CYP2C9 and CYP3A4 to hydroxylated metabolites (58). Zafirlukast also has been shown to undergo carbamate hydrolysis, followed by *N*-acetylation. Additionally, zafirlukast is known to produce an idiosyncratic hepatotoxicity in susceptible patients. This appears to result from the formation of an electrophilic α,β -unsaturated iminium intermediate evidenced by the formation of a glutathione adduct on the methylene carbon bridging the indole ring to the methoxybenzene moiety of the molecule (59).

Figure 39.36 summarizes the metabolism of zafirlukast. More than 90% of its metabolites are excreted in the feces, with the remaining found in the urine. Zafirlukast inhibits CYP3A4 and CYP2C9 in concentrations equivalent to clinical plasma levels and, therefore, should be used with caution in patients taking drugs metabolized by these enzymes. Specifically, coadministration with warfarin results in a significant increase in prothrombin time. Other drugs metabolized by CYP2C9 are phenytoin and carbamazepine. In addition, CYP3A4-metabolized drugs are cyclosporine, cisapride, and the dihydropyridine class of calcium channel blockers. Of particular interest is the fact that aspirin increases the plasma levels of zafirlukast, and theophylline decreases the plasma levels of zafirlukast.

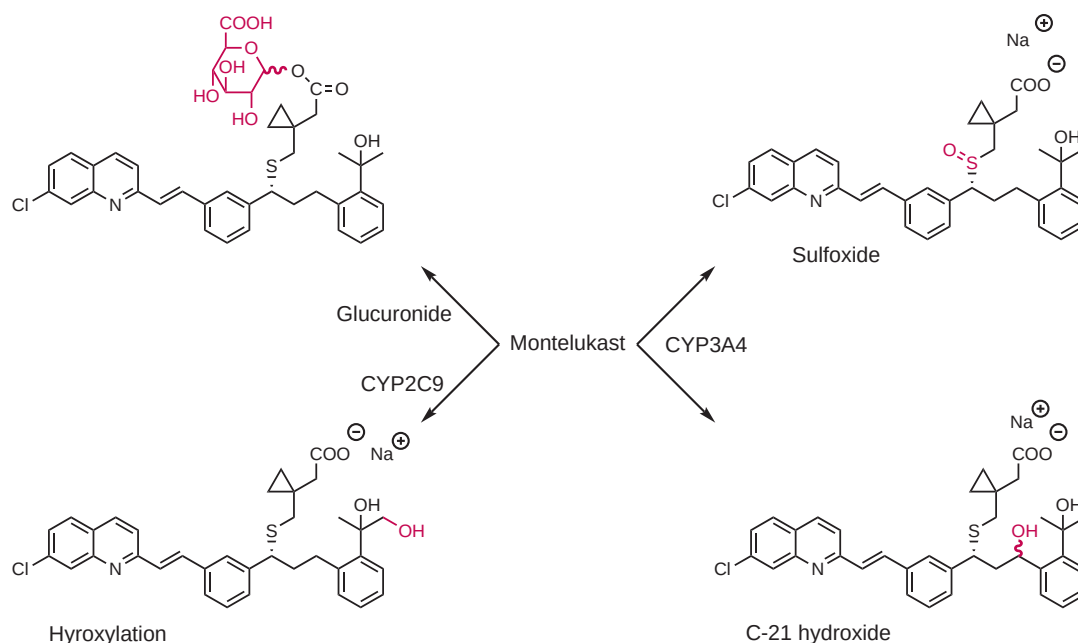


FIGURE 39.35 Metabolism of montelukast.

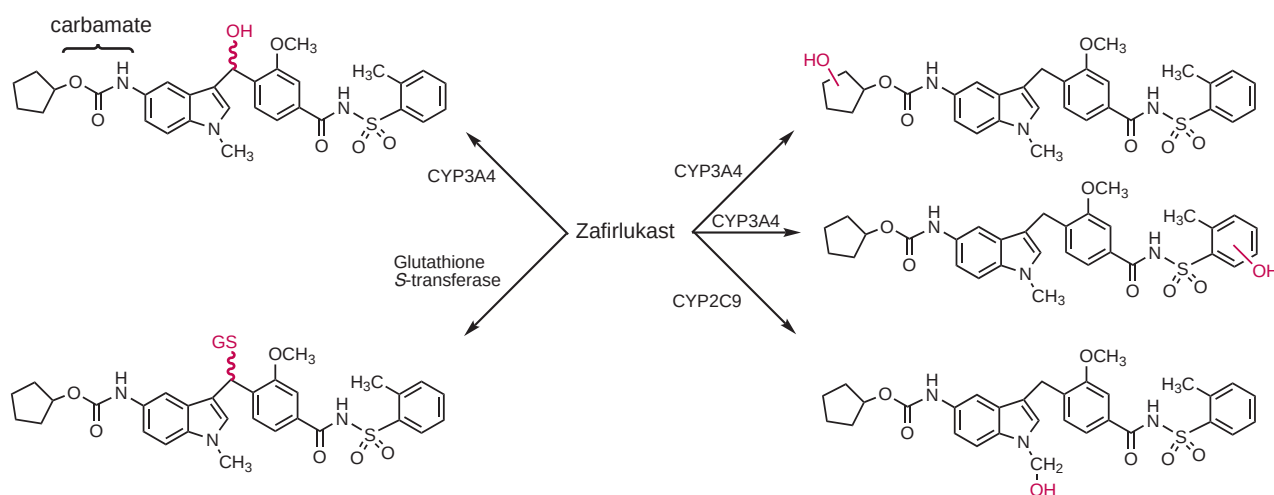


FIGURE 39.36 Metabolism of zafirlukast to oxidative products and a glutathione adduct.

Care should be taken when coadministering with erythromycin, because this decreases the bioavailability of zafirlukast. Zafirlukast is only available in tablet formulations.

Monoclonal Anti-IgE Antibody

The pathophysiology of allergic asthma, as already discussed, ultimately results in the production of allergen-specific IgE by activated B lymphocytes. The IgE binds to high-affinity (FcεRI) receptors on mast cells. The site where IgE binds to the receptor is located on the Fc fragment area of the C-ε-3 region, hence the acronym FcεRI (Fig. 39.37A). Subsequent allergen exposure causes cross-linking of bound IgE molecules, which triggers degranulation of these cells, resulting in the release of asthma mediators. Monoclonal anti-IgE antibody development is designed to moderate the role of IgE in activating mast cells, thereby decreasing the severity of allergic asthmatic attacks, and may also have beneficial effects in treating seasonal allergic rhinitis.

Omalizumab Development and Pharmacology

Omalizumab is a monoclonal antibody developed through somatic cell hybridization techniques and was identified as a murine anti-human IgE antibody, originally called MAE11 (60). It is designed to interact with the site that binds to FcεRI on mast cells. Additional amino acid sequences have been incorporated into the antibody so that a humanized product resulted that only differs by 5% nonhuman amino acid residues.

In vitro, omalizumab has been shown to complex with free IgE, forming trimers consisting of a 2:1 complex of IgE to omalizumab or a 1:2 complex of IgE to omalizumab. In addition, larger complexes also are formed, consisting of a 3:3 ratio of each (Fig. 39.37B). Omalizumab does not bind to IgE already bound to mast cells and, therefore, does not cause the degranulation that might be expected from such interaction. Thus, omalizumab effectively neutralizes free IgE and, aside from the obvious decrease of

available IgE, also causes the down-regulation of FcεRI receptors on the mast cell surface, resulting in a decrease of IgE bound to the mast cell.

Omalizumab (Xolair)

The clinical role for omalizumab is in the treatment of allergic asthma. It is approved for the treatment of adults and adolescents 12 years of age and older whose symptoms

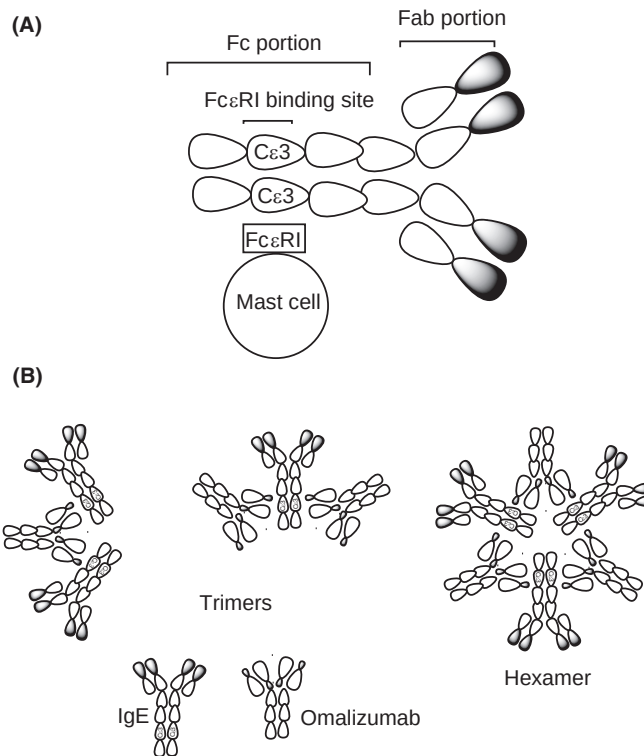


FIGURE 39.37 (A) Graphic representation of IgE binding to the FcεRI on a mast cell. (B) Graphic representation of immune complexes formed between omalizumab. Hexamers predominate when components are in a 1:1 ratio, and the trimers predominate when one of the components is in excess.

are not controlled with inhaled glucocorticoids and who have a positive skin test for airborne allergens. The bioavailability after subcutaneous administration is 62%, with slow absorption resulting in peak serum levels in 7 to 8 days from a single dose. Steady-state plasma concentration is reached in 14 to 29 days with multiple dosing regimens. The elimination of omalizumab is not clearly understood; however, studies have determined that intact IgE is excreted via the bile and that omalizumab:IgE complexes are cleared faster than uncomplexed omalizumab and slower than free IgE. This means that over time, total IgE concentrations (free and complexed IgE) increase, because the complex is cleared more slowly. The metabolism of omalizumab is not known, and the clearance of the complex is similar to the liver elimination of another immunoglobulin, IgG. The reticuloendothelial system degrades IgG, and it is believed that the same process occurs for the omalizumab:IgE complex. Omalizumab is available as a lyophilized powder for injection in single-use, 5-mL vials.

CHRONIC OBSTRUCTIVE PULMONARY DISEASE

Definition and Epidemiology

COPD is characterized by persistent breathing difficulty that is not completely reversible and is progressive over time. It usually is the result of an abnormal inflammatory response to airborne toxic chemicals. Thus, COPD is a general term and most commonly refers to chronic bronchitis and emphysema. Asthma is considered to be a disease entity unto itself and is not included in the definition of COPD. Both COPD and asthma are considered to be inflammatory diseases; however, the nature of the inflammation is different. Asthma is associated with the release of inflammatory mediators from mast cells and eosinophils, whereas chronic bronchitis is primarily associated with neutrophils and emphysema with alveoli damage. In addition, asthma is more often than not allergenic, whereas chronic bronchitis and emphysema have no allergic component. Finally, it is uncommon for asthma to be associated with smoking, whereas there is a very high incidence of both chronic bronchitis and emphysema in smokers.

Patients with COPD display a variety of symptoms, ranging from chronic productive cough to severe dyspnea requiring hospitalization. Other chronic illnesses, including cardiac, endocrine, and renal disease, often occur along with COPD in many patients. In the United States, COPD is the fourth leading cause of death, being responsible for more than 100,000 deaths per year (61). During 2000–2005, COPD was the underlying cause of death for 718,077 persons aged 25 years or older in the United States, with the mortality rate for women being almost twice that of men (11% vs. 5%) (62). It has been estimated that between 16 and 24 million people in the United States have COPD and that those with chronic bronchitis outnumber those with emphysema. Rates of COPD-related death among women

have tripled over the last 30 years, and this is being attributed to their increase in smoking. COPD is recognized as a global health problem, and in 2001, the NIH and the World Health Organization developed the Global Initiative for Chronic Lung Disease (GOLD) guidelines, which present only evidence-based recommendations for the treatment of COPD (63,64).

Pathogenesis

The single most important risk factor for the development of COPD is smoking. It is estimated that 85% of COPD cases are attributable to cigarette smoking. Not all people who smoke, however, develop the disease, which means other factors are involved. It seems that genetics, environmental pollutants, and infection along with bronchial hyperreactivity all play an important role.

Cigarette smoke attracts inflammatory cells into the lungs and stimulates the release of the proteolytic enzyme elastase. Elastase breaks down elastin, which is a needed structural component of lung tissue. Normally, the lung is protected from elastase by an inhibitor, α_1 -antitrypsin (ATT). Cigarette smoke, however, causes an abnormal amount of elastase to be produced that ATT cannot counter, leading to lung damage. Smokers with an inherited deficiency of ATT have a greatly increased risk of developing emphysema, especially at an early age. Patients with COPD show an accelerated decline in lung function (50 to 90 mL FEV₁/year) as compared to nonsmokers (20 to 30 mL/year) (see earlier discussion of FEV₁ under *General Therapeutic Approaches to the Treatment and Management of Asthma*). Patients with COPD who stop smoking slow down the progression of the disease. Unfortunately, however, they do not improve, because the symptoms are an indication of irreparable lung tissue damage.

The role of other risk factors for developing COPD is not clear. Air pollution and occupational exposure to gases and particulates from the incomplete combustion of coal, diesel, and gasoline are related to the development of cough and sputum, but development of COPD seems to occur only in susceptible individuals. Passive cigarette smoke comes from the burning end of the cigarette and actually is higher in toxic substances compared with exhaled smoke. It has been established that respiratory infections as well as cough and sputum production are more common in children who live in households where one or both parents smoke. Chinese and Afro-Caribbean races have a reduced incidence of COPD, whereas COPD in American blacks is on the rise (61,62).

Permanent destructive enlargement of the air spaces distal to the terminal bronchioles without obvious fibrosis is the pathologic characteristic of emphysema. On the other hand, chronic bronchitis is characterized by hypersecretion of mucus, most of which is produced by the trachea and first branches of the bronchi. The smaller bronchi and terminal bronchioles, however, are the site of the increased airway resistance in chronic bronchitis. In the early stages of chronic bronchitis, the mucus and inflammation contribute to “smoker’s cough,” with little

effect on airway obstruction. As inflammation continues, cell wall edema and the production of large amounts of mucus contribute to airway narrowing and the difficulty in breathing associated with COPD. Even though bronchospasm may seem to be involved in COPD, there is little related pathogenesis. Chronic obstructive pulmonary disease is a disease of the small airways and their adjacent alveoli. In chronic bronchitis, there are structural changes in the small airways as a result of persistent inflammatory irritation, which leads to airway narrowing. Emphysema on the other hand is the result of loss of lung elastic recoil because of inflammation and alveolar wall destruction. Both diseases often occur together, with one predominate over the other. An important clinical manifestation between chronic bronchitis and emphysema is that there is significant hypoxia and carbon dioxide retention with chronic bronchitis, which does not happen with emphysema. Because of this, emphysema patients are referred to as “pink puffers,” whereas patients with chronic bronchitis are called “blue bloaters” (63).

Pharmacotherapy

All the medications used to treat COPD have already been covered in the previous section on asthma. The GOLD guidelines base their treatment protocols on a classification of disease severity divided into five stages (63–65). Stages 0 and I are defined as “at risk” (0) and mild COPD (I). In stage 0, the patient has normal lung function, with chronic cough and sputum production. Treatment at this stage is to counsel the patient to reduce risks and, especially, to stop smoking. In stage I, there is minor airway limitation, characterized as $FEV_1/FVC < 70\%$ but $FEV_1 \geq 80\%$ of the predicted value. Stage I treatment is to use a short-acting bronchodilator, usually as needed, but regular use is effective in patients with concurrent asthma. The most frequently used short-acting bronchodilator is the β_2 -agonist albuterol, although pirbuterol and the anticholinergic ipratropium can be just as effective, but with a slightly longer onset of action. The patient also should take precautions to avoid bacterial or viral infections by receiving vaccinations against influenza and pneumococcal pneumonia.

Stage II COPD is a moderate disease condition in which the patient demonstrates shortness of breath on exertion and spirometry reveals $FEV_1/FVC < 70\%$ and FEV_1 between 50% and 80% of the predicted value. Stage II drug treatment requires the addition of a long-acting bronchodilator along with the short-acting bronchodilator. Salmeterol and formoterol are long-acting β_2 -agonists, and tiotropium is the long-acting anticholinergic most often used. Alternatively, the addition of ipratropium along with the short-acting β_2 -agonist also can be effective at this stage. Extended-release theophylline is an option for patients who do not receive adequate relief of symptoms or who cannot tolerate other bronchodilators.

Stage III is a severe form of COPD, with $FEV_1/FVC < 70\%$ and FEV_1 between 30% and 50% of the predicted value. The patient experiences increasing dyspnea that affects his or her ability to perform routine tasks (i.e., climbing stairs). A course of oral glucocorticoids (i.e., prednisone) may be used to control a severe attack. The role of inhaled glucocorticoids is not clearly established, and no evidence suggests that an inhaled glucocorticoid has any advantage in patients who can maintain their FEV_1 with bronchodilators. However, the combination of an inhaled glucocorticoid with a long-acting β_2 -agonist has been shown to be more effective than either drug class alone in reducing exacerbations and improving lung function. The combination formulation of fluticasone propionate and salmeterol dosage (250 μ g/50 μ g) is approved for twice-daily administration.

Stage IV is the most severe form of COPD, with airflow restriction of $FEV_1/FVC < 70\%$ and $FEV_1 < 30\%$ of the predicted value along with chronic respiratory failure. At this stage, the patient is experiencing debilitating exacerbations that are not controlled by medication and requires daily oxygen for respiratory failure. Surgery also is an option, but it is not without serious risks. Surgical options include bullectomy (removal of large blebs in the lungs), lung transplant (uncommon), and lung volume reduction surgery (removal of lung sections affected by emphysema).

Patients with emphysema that is associated with ATT deficiencies can receive weekly IV infusions of ATT to maintain acceptable antiprotease activity that can minimize their disease progression. The three approved ATT products are Aralast, Prolastin, and Zemaira. Because these protein products are derived from human plasma, there is the risk of transmission of viral infection and Creutzfeldt-Jakob disease.

New Drug Classes for Treatment of Asthma and COPD

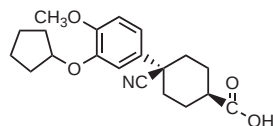
Phosphodiesterase Inhibitors

A number of important therapeutic agents owe their pharmacologic action to their ability to inhibit the enzyme PDE. In the treatment of asthma, theophylline, at least in part, relaxes bronchospasm by relaxing bronchiole smooth muscles; amrinone and milrinone are ionotropic agents that relax vasculature, causing vasodilation; sildenafil and vardenafil relax smooth muscle of the vasculature in the penis; dipyridamole inhibits platelet aggregation; and the alkaloid papaverine relieves smooth and cardiac muscle spasms through its ability to inhibit PDE. These pharmacologic effects are the result of inhibiting the ability of PDE to break down cAMP and cGMP and prolong their action as secondary messengers within a variety of cell types throughout the body. The PDE inhibitors also are implicated in an anti-inflammatory role by increasing cAMP and cGMP levels in cell types associated with the release of inflammatory chemicals from T and B cells, monocytes, neutrophils, and eosinophils. This last

discovery is the impetus to develop new PDE inhibitors to treat asthma (eosinophils) and COPD (neutrophils).

NEW PHOSPHODIESTERASE INHIBITORS Progress in the development of PDE inhibitors to treat COPD and asthma awaited the basic pharmacologic research that identified the specific PDE isoforms associated with inflammatory cells so that selective inhibitors could be designed and synthesized (66). There are 11 families of PDEs, with PDE4 being associated with inflammatory processes. Specifically, three PDE4 subtypes (PDE4A, PDE4B, and PDE4D) are found in inflammatory cells (with PDE4B being predominant). Therefore, recent efforts in this area have been directed toward the development and synthesis of selective PDE4 inhibitors. Two PDE4 inhibitors are being investigated for their possible use in treating asthma and COPD. In the United States, cilomilast is being investigated for the treatment of COPD, and in Europe, roflumilast is being investigated for the treatment of both asthma and COPD. Progress of these and other PDE4 inhibitors toward marketing has been slow because of lack of clinical efficacy and dose-limiting side effects (nausea, diarrhea, and headache). The primary problem is the low therapeutic ratio of these compounds because the tolerated dose is either subtherapeutic or at the very bottom of the effective dose–response curve (67,68).

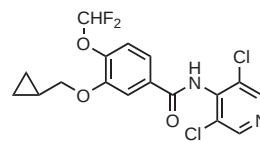
Cilomilast



Cilomilast (Ariflo)

Cilomilast contains the dialkoxyphenyl ring characteristic of selective PDE4 inhibitors. The ether oxygens hydrogen bond to a glutamine in the binding pocket, and the cyclopentyl ring adds additional hydrophobic interactions. The oxygen atoms of the carboxyl group form hydrogen bonds with water that is coordinated with Mg^{2+} located in the distal end of the binding pocket. Orally administered cilomilast is 96% bioavailable. Food does not interfere with the overall absorption; however, food does slow down the rate. Cilomilast is 99% bound to albumin in the plasma and is metabolized in the liver by CYP2C8. The metabolism is extensive and results in oxidation, carboxyl group glucuronidation, and dealkylation of the cyclopentyl group, followed by glucuronidation or sulfation. A major difficulty with cilomilast is that in therapeutic doses, patients during clinical trials have experienced significant diarrhea and nausea. These effects appear to be significant enough to keep cilomilast off the market. These side effects, theoretically, result from its inhibition of the PDE4D receptor subtype.

Roflumilast (Daliresp)



Roflumilast (Daxas)

The PDE4 receptor binding of roflumilast is similar to that of cilomilast. The dialkylphenyl oxygens hydrogen bond to a glutamine deep inside the binding pocket. The difluoromethoxy group and the cyclopropyl group contribute hydrophobic bonds; however, the cyclopentyl group on cilomilast makes far more interactions compared with the cyclopropyl group. The water molecule coordinated to the Mg^{2+} hydrogen bonds to the dichloropyridyl group just like the carboxylic acid oxygens in cilomilast. These structural differences make roflumilast a more potent inhibitor than cilomilast toward PDE4B (69). Roflumilast is well absorbed on oral administration and has a half-life of 10 hours. Roflumilast is metabolized in the liver to its *N*-oxide derivative, which also is a PDE4 inhibitor, and it has a plasma half-life of 20 hours. Roflumilast's preclinical pharmacology continues to be understood, which enables it to be an effective oral treatment for COPD with an acceptable tolerability profile (70). Roflumilast was approved for treatment of COPD associated with chronic bronchitis in the European Union in 2010 and in the U.S. in 2011.

Anticytokine Therapies

Because glucocorticoids are not effective in suppressing or controlling inflammation symptoms in asthmatics and those suffering from COPD, there continues to be a need to develop new approaches to novel therapeutic treatments. Most of these approaches are focused on the inhibition of the chemokines and cytokines associated with the inflammatory process.

The following is a brief overview, because a detailed discussion of each of these novel approaches is beyond the scope of this chapter. Anti-eosinophil attack includes targeting cytokines (anti-IL-4, -13, and -5), their receptors (βc), and the chemokine receptor (CRR3). Anti-IL-4 and anti-IL-13 therapies have been shown to be effective in allergen challenge tests. TPA ASM-8 (a mixture of two antisense oligonucleotides), which is a CCR3/ βc antagonist, also has been shown to be effective. Evaluation of both of these approaches is in very early stages, and much longer studies are needed. Anti-IL-5 had no effect on allergen challenge, which indicates that targeting just one cytokine may not be the best approach to the development of new therapies. Anti-tumor necrosis factor- α therapies have shown some clinical efficacy, but there is a lingering concern about serious side effects, such as infection and malignancy. Therefore, it is likely that inhaled glucocorticoid and β_2 -agonist treatments for both asthma and COPD are to be the standard of treatment for now and in the near future (71).

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Joseph V. Etzel, PharmD

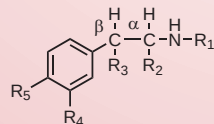
The goal of asthma treatment is to achieve and maintain clinical control of symptoms and minimize the occurrence of symptoms. After reviewing JP's case, the pharmacist concludes that JP's disease state is not well controlled, as suggested by his frequent exacerbations requiring short-acting β_2 -agonist use, occasional nighttime awakenings, and diminished FEV₁ values. After confirming the patient's medication adherence, appropriate inhalation techniques, and environmental control, the pharmacist recommends a step up in therapy to achieve better management of JP's asthma. Because JP's asthma is inadequately controlled with a medium-dose inhaled corticosteroid, the pharmacist recommends the addition of a long-acting β_2 -agonist (LABA), such as salmeterol 50 mcg inhalation twice daily. This recommendation is based on comparative studies that demonstrate significantly greater improvements in asthma control when an LABA is added to medium-dose inhaled corticosteroid therapy compared with other adjunctive therapies. Although salmeterol is available as a single entity product, the pharmacist recommends a twice-daily combination dry-powder inhalation product containing both fluticasone (250 mcg) and salmeterol (50 mcg) to improve patient adherence. Once therapy is adjusted, the patient should be re-evaluated in 2 to 6 weeks to ensure adequate response. Although comorbid disease states, such as allergic rhinitis, may exacerbate asthma, this patient's allergies appear to be well controlled and no intervention seems to be necessary.

Chemical Analysis

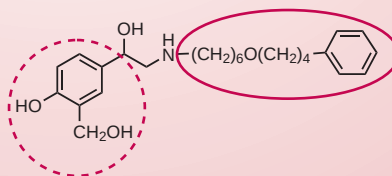
S. William Zito and Victoria Roche

It is recommended in this case to add an LABA to JP's asthma treatment. The fundamental pharmacophore for all adrenergic agonists is a substituted β -phenylethylamine.

The number and nature of substituents on the pharmacophore influences the specificity for the β -receptor subtypes. The substituents on the amino group (R_1) determines α - or β -receptor selectivity: the larger the bulk of the N-substituent, the greater the selectivity for the β -receptor. If R_1 is *t*-butyl or aralkyl, there is complete loss of α -receptor affinity, and the β -receptor affinity shows preference for the β_2 -receptor. It must be said that receptor selectivity is dose related, and when the dose is high enough, all selectivity can be lost. Substituents other than hydrogen on the α -carbon (R_2) will show an increased duration of action because they make the compound resistant to metabolism by MAO. For an adrenergic agonist to demonstrate significant β_2 -receptor selectivity there needs to be an appropriately substituted phenyl ring. The currently marketed adrenergic agonists contain a resorcinol ring, a salicyl alcohol moiety, or an *m*-formamide group. In addition, these ring configurations confer resistant to COMT metabolism and will increase their duration of action. So, as can be seen in the structure of salmeterol, there are two moieties that contribute to its β_2 -agonist activity as well as its duration of action: (1) the phenylbutoxyhexyl N-substituent, which confers β_2 -selectivity, and (2) the salicyl alcohol ring, which both reinforces β_2 -selectivity and increases duration of action due to inhibition of COMT metabolism.



Adrenergic agonist pharmacophore



Salmeterol

CASE STUDY

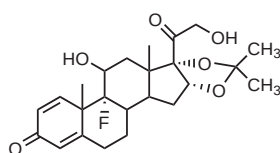
S. William Zito and Victoria Roche

TJ is a 65-year-old woman who presents to the emergency department with difficulty breathing because of uncontrollable wheezing and tightness of the chest, which was brought on as a consequence of a weekend camping and fishing trip. TJ has a long history of severe persistent asthma and does not regularly monitor peak flow levels at home. TJ's asthma is treated with fluticasone propionate (2 puffs BID), zafirlukast (20 mg BID), and albuterol (2 puffs BID; four times a day PRN).

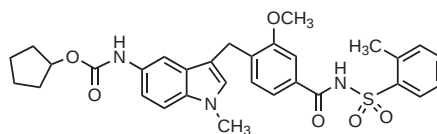
TJ says that the albuterol usually controls her acute attacks but it did not work this morning, leaving her feeling as if she could not breathe at all. On the way to the emergency department she was treated with two albuterol nebs (2.5 mg each) and 0.5 mg SQ epinephrine. In the emergency department,

TJ demonstrated a peak flow of 175 L/min (baseline of last office visit was 480 L/min). The attending physician administered supplemental oxygen by nasal cannula to achieve an O_2 saturation of $\geq 95\%$ and continued the albuterol at 2.5 mg every 1 hour as needed. In addition, the attending physician wants to add an oral corticosteroid to achieve a response to therapy above 60% of baseline peak expiratory flow sustained for at least 60 minutes. Evaluate the three structures for use in this case.

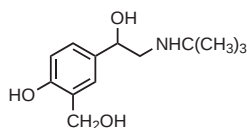
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.



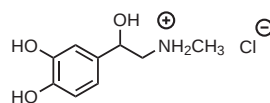
Triamcinolone acetonide



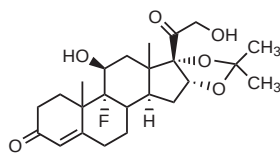
Zafirlukast



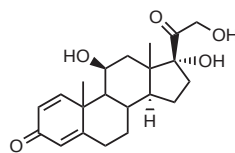
Albuterol



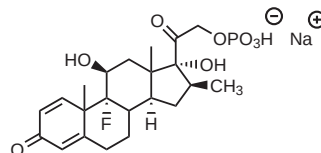
Epinephrine



1



2



3

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Chapter 40

Men's Health

DUANE D. MILLER, ROBERT W. BRUEGGEMEIER, AND
JAMES T. DALTON

*Drugs Covered in This Chapter**

ANABOLIC AGENTS

- Nandrolone
- Oxandrolone
- Oxymetholone
- Stanozolol
- Testolactone

CHEMOTHERAPY AND IMMUNOTHERAPY

- Docetaxel
- Sipuleucel-T

DRUGS FOR THE TREATMENT OF BENIGN PROSTATIC HYPERPLASIA

α_1 -Adrenergic antagonists

- Alfuzosin
- Doxazosin
- Silodosin
- Tamsulosin
- Terazosin

5 α -Reductase inhibitors

- Dutasteride
- Finasteride

DRUGS FOR TREATMENT OF ERECTILE DYSFUNCTION

Phosphodiesterase-5 inhibitors

- Sildenafil
- Tadalafil
- Vardenafil

Miscellaneous drugs

- Prostaglandin E1
- Papaverine
- Phentolamine

TESTOSTERONE PRODUCTS

- Fluoxymesterone
- Methyltestosterone

- Testosterone
- Testosterone esters

THERAPIES FOR PROSTATE CANCER

Luteinizing hormone–releasing hormone agonists

- Goserelin
- Leuprolide

Luteinizing hormone–releasing hormone antagonist

- Degarelix

Antiandrogens

- Bicalutamide
- Flutamide
- Nilutamide

Abbreviations

AR, androgen receptor

ARE, androgen response element

AUA, American Urological Association

BPH, benign prostatic hyperplasia

cAMP, cyclic adenosine monophosphate

cGMP, cyclic guanosine monophosphate

DBD, DNA binding domain

DHEA, dehydroepiandrosterone

DHT, dihydrotestosterone

ED, erectile dysfunction

FSH, follicle-stimulating hormone

IC₅₀, median inhibitory concentrations

IM, intramuscular

LBD, ligand binding domain

LH, luteinizing hormone

LHRH, luteinizing hormone–releasing
hormone

NCI, National Cancer Institute

NO, nitric oxide

PAP, prostatic acid phosphatase

PDE, phosphodiesterase

PGE₁, prostaglandin E₁

PSA, prostate-specific antigen

SARM, selective androgen receptor
modulator

SHBG, sex hormone binding globulin

TRT, testosterone replacement therapy

*Drugs listed include those available inside and outside the United States; drugs available outside of the United States are shown in italics.

SCENARIO

Autumn Stewart, PharmD

NP is 68-year-old African American man presenting to his community pharmacy with a new prescription for itraconazole 200 mg QD for 12 weeks for onychomycosis. The patient had the following prescriptions refilled 1 week earlier: dutasteride 0.5 mg QD, doxazosin 4 mg HS, and amlodipine 10 mg QD. While conducting a prospective drug review of the patient's medications, the pharmacist identifies a potential

drug interaction between itraconazole and dutasteride. After consultation with the physician, the patient is counseled to monitor for adverse effects and report any new symptoms immediately.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

This chapter focuses on the physiology, pharmacology, metabolism, and structure–activity relationships for therapeutic and emerging classes of drugs that are used almost exclusively in men. Major differences in endocrine hormones and the anatomy and physiology of the reproductive system and genitourinary tract between males and females make men uniquely susceptible to a variety of disorders, including aging-related androgen insufficiency (hypogonadism and andropause), prostate and testicular cancer, benign prostatic hyperplasia (BPH), and erectile dysfunction (ED). The majority of these disorders and their treatments are associated with the male sex hormones (i.e., androgens), their pharmacologic target (i.e., the androgen receptor [AR]), and the tissues that rely on the androgens.

Prostate problems are common in older men, particularly those age 50 years and older. A man may have prostate problems for a number of reasons, including an infection of the prostate (prostatitis), a noncancerous enlargement of the prostate (BPH), or prostate cancer, the second most common cancer in men. BPH is the most common disease of the prostate, occurring in 50% to 60% of men in their 60s and up to 80% to 90% of men over the age of 70 years (1). Risk of prostate cancer also increases with age; approximately 70% of all cases of the disease are diagnosed in men age 65 years and older. Prostate problems often are discovered by men themselves. The signs of prostate problems include frequent urge to urinate, blood in the urine, painful or burning urination, difficulty urinating, or inability to urinate.

Aging-related androgen insufficiency (male hypogonadism) is a physiologic condition characterized by the inability of the testes to produce sufficient testosterone to maintain sexual function, muscle strength, bone mineral density, and fertility (spermatogenesis). One in five men older than 50 years will exhibit symptoms of this condition. Symptoms of aging-related androgen insufficiency may include lethargy or decreased energy, decreased libido or interest in sex, ED (with loss of erections), muscle weakness and aches, inability to sleep, hot flashes, night sweats, depression, infertility, thinning of bones or bone loss, and cardiovascular disease. By the time that men are between the ages of 40

and 55 years, some may experience a phenomenon similar to the female menopause called andropause. Whether andropause is more common than hypogonadism in the aging male is a matter for debate. The decline in testosterone occurs very gradually in men over several decades and may be accompanied by loss of bone mass (osteoporosis), loss of muscle mass and strength (sarcopenia), and changes in fat distribution, cholesterol levels, spermatogenesis, sexual performance, quality of life, and impotence (i.e., ED). Studies show that a decline in testosterone actually can put men at risk for other health problems, such as heart disease, metabolic syndrome, and weak bones. Psychological stress, alcohol abuse, injuries or surgery, medications, obesity and infections, tobacco, and drugs, such as decongestants, anti-hypertensives, tranquilizers, statins, or antiseizure agents, can contribute to the onset of these conditions. There is no way of predicting who will experience the symptoms of androgen insufficiency that are of sufficient severity to seek medical help; neither is it predictable at what age the symptoms of aging-related androgen insufficiency will occur in a particular individual. Each man's symptoms also may be different. Because all this happens at a time of life when many men begin to question their values, accomplishments, and direction in life, it often is difficult to realize that the changes occurring are related to more than just external conditions. Now that men are living longer, there is heightened interest in aging-related androgen insufficiency, its risks for other health problems, and its treatment.

Testicular cancer is the most common form of cancer among males age 15 to 44 years and is approximately four times more common in white men than in African American men. After motor vehicle accidents and suicide, cancer is the leading cause of death in this age group, followed by homicide, heart disease, and HIV. Testicular cancer is known as “young man's cancer.” Early detection is the key to survival. Testicular cancer is androgen-dependent, with a very fast onset, because the tumors can be very aggressive. When the cancer is confined to the testicles, there often is no pain, but by the time pain develops, it often is a sign that the cancer has already spread. Survival rates increase significantly if treatment has begun before the cancer has a chance to metastasize. Testicular cancer is most often discovered

CLINICAL SIGNIFICANCE



The therapeutics of men's health is currently experiencing significant changes that will impact the pharmacist's role in providing care to this population. The use of treatments for male andropause is becoming more popular as men seek medical advice for the treatment of erectile dysfunction (ED) and other associated symptoms. Though androgen replacement therapy is becoming more common, it has not replaced the phosphodiesterase (PDE5) inhibitors' role as the drug of choice for the treatment of ED. Both Viagra (sildenafil) and Cialis (tadalafil) were among the Top 200 Drugs dispensed in 2009. There are advantages and disadvantages to each of these PDE5 inhibitors. Some can be taken with food whereas others cannot. Some have optimal durations of action, and others durations of action are much shorter. However, all have been known to have adverse effects and drug interactions in common. Some of these adverse effects can be rather severe, and the drug interactions with nitrates can be fatal.

The development of newer, more effective, and less deleterious agents is essential for the treatment of ED. Agents that

are more selective for PDE5 and avoid other PDEs are critical to this endeavor. By studying the structure–activity relationships between the PDEs and the inhibitors, this can be achieved. Minor changes to the side chains or functional groups of the currently available PDE inhibitors can potentially eliminate the adverse effects and drug interactions that limit the use of these agents. As new agents become available, clinicians continually compare and contrast them to the ones that are currently available in an effort to improve patient care. This is an ongoing, joint effort by chemists and clinicians alike to optimize patient care, either indirectly or directly, by continually making adjustments to either chemical structures or therapeutic treatment plans for the patient.

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by self-examination. Therefore, all men should conduct testicular self-examinations at least monthly and, preferably, every time they shower for any changes or lumps and see a doctor immediately if any changes are noted. The diagnosis is noninvasive and involves using ultrasound to look at the density, size, and shape of the testicles and other masses in the scrotum.

Men who experience any of these symptoms associated with the prostate, aging-related androgen insufficiency, and testicular cancer should see a doctor to find out the cause of the problem and to talk about possible treatment.

Testosterone and its metabolite, dihydrotestosterone (DHT), are the primary endogenous androgens and play crucial physiologic roles in establishing and maintaining the male phenotype (2). Their actions are essential for the differentiation and growth of male reproductive organs, initiation and regulation of spermatogenesis, and control of male sexual behavior. In addition, androgens are important for the development of male characteristics in certain extragenital structures, such as muscle, bone, hair, larynx, skin, lipid tissue, and kidney (3). In females, the precise physiologic roles of androgens are not completely understood, but the aging-related decline in circulating androgen levels has been linked to symptoms such as decreased libido and sexuality, lack of vigor, diminished well-being, and loss of bone mineral density in postmenopausal women (4–6).

THE SEX HORMONES

The sex steroid hormones are steroid molecules that are necessary for reproduction in females and males and that affect the development of secondary sex characteristics in

both sexes. The sex steroids are comprised of three classes: estrogens, progestins, and androgens (Fig. 40.1). The two principal classes of female sex steroid hormones are estrogens and progestins (for further discussion, see Chapter 41). Chemically, the naturally occurring estrogens are C-18 steroids and have in common a planar unsaturated A ring with a 3-phenolic group that distinguish them from the other sex steroid hormones. The most potent endogenous estrogen is estradiol (Fig. 40.1). The naturally occurring progestins are C-21 steroids and have in common a 3-keto-4-ene structure in the A ring and a ketone at the C-21 position. The most potent endogenous progestin is progesterone (Fig. 40.1). The principal class of the male sex steroid hormone is the androgens. The naturally occurring androgens are C-19 steroids and have in common oxygen atoms (as either hydroxyl or ketone groups) at both the

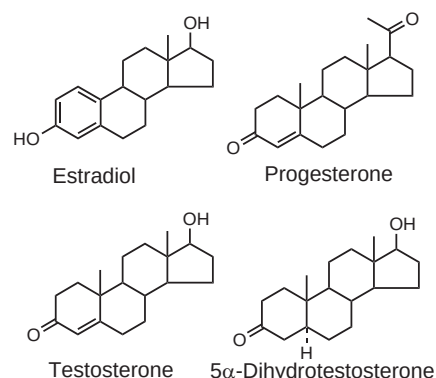


FIGURE 40.1 The sex hormones.

C-3 and C-17 positions. Testosterone is a potent androgen that is found in the blood at higher concentrations than other androgens (Fig. 40.1), whereas DHT (Fig. 40.1) is a more potent metabolite of testosterone that is formed in certain androgen target tissues including the prostate and scalp. All three classes of endogenous steroids are present in both males and females. The production and circulating plasma levels of estrogens and progestins are higher in females, however, and the production and circulating plasma levels of androgens are higher in males.

DISCOVERY OF ANDROGENS

One of the earliest and most unusual experiments with testicular extracts was carried out in 1889 by the French physiologist Charles Brown-Séquard. He administered such an extract to himself and reported that he felt an increased vigor and capacity for work (7). In 1911, Pézard showed that extracts of testicular tissue increase comb growth in capons (8). Early attempts to isolate pure male hormones from the testes failed, because only small amounts are present in this tissue.

The earliest report of an isolated androgen was presented by Butenandt (9) in 1931. He isolated 15 mg of crystalline androsterone from 15,000 L of human male urine. A second crystalline compound, dehydroepiandrosterone, which has weak androgenic activity, was isolated by Butenandt and Dannenberg (10) in 1934. During the following year, testosterone was isolated from bull testes by David et al. (11). This hormone was shown to be 6 to 10 times more active than androsterone.

Shortly after testosterone was isolated, Butenandt and Hanisch (12) reported its synthesis. In that same year, extracts of urine from males were shown to cause nitrogen retention (a measure of protein anabolism) as well as the expected androgenic effects (13). Many steroids with androgenic activity have subsequently been synthesized. Steroid hormones may have many potent effects on various tissues, and slight chemical alterations of androgenic steroids may increase some of these effects without altering others.

Testosterone was the first androgen to be used clinically for its anabolic activity. New sources of the hormone were needed, however, because only 270 mg could be isolated from a ton of bull testes (14). Commercially, testosterone is prepared from various steroids, including sarsapogenin, diosgenin, and certain androgens found in stallion urine. Because of its androgenic action, testosterone is limited in its use in humans as an anabolic steroid. Many steroids were synthesized in an attempt to separate the androgenic and the anabolic actions. Because testosterone had to be given parenterally, it also was desirable to find orally active agents.

In the United States, most of the androgens and anabolic steroid products are subject to control by the U.S. Federal Control Substances Act as amended by the Anabolic Steroid Control Act of 1990 as Schedule III drugs.

ANDROGEN PHYSIOLOGY

The overall physiologic effects of endogenous androgens are contributed by testosterone and its active metabolites, DHT and estradiol. Testosterone and DHT execute their actions predominantly through the AR, which belongs to the nuclear receptor superfamily and functions as a ligand-dependent transcription factor. Circulating testosterone is essential for the differentiation and growth of male accessory reproductive organs (e.g., prostate and seminal vesicles), control of male sexual behavior, and the development and maintenance of male secondary characteristics that involve muscle, bone, larynx, and hair. Healthy young adult men produce approximately 3 to 10 mg of testosterone per day, with circulating plasma levels ranging from approximately 500 to 1,000 ng/dL in eugonadal (normal) men. Circulating testosterone and estradiol participate in the feedback regulation of androgen production by the hypothalamus-pituitary-testis axis, as shown in Figure 40.2. Testosterone, luteinizing hormone (LH), and LH-releasing hormone (LHRH; also known as gonadotropin-releasing hormone) constitute the elements of a negative feedback control mechanism, whereby testosterone controls its own release. Low circulating testosterone levels increase the hypothalamic secretion of LHRH, which leads to increased production of LH and, consequently, increased testosterone production by the Leydig cells. More than 95% of circulating testosterone is synthesized and secreted by the Leydig cells in the testes. High testosterone levels, on the other hand, inhibit LHRH release, thus suppressing both LH secretion and testosterone secretion. The LHRH is released from the hypothalamus in short, intermittent pulses every 2 hours and at greater magnitude in the morning, which in turn stimulates the pulsatile secretion of LH and follicle-stimulating hormone (FSH) from the pituitary. FSH stimulates the Sertoli cells controlling spermatogenesis and development of the testis. Thus, testosterone

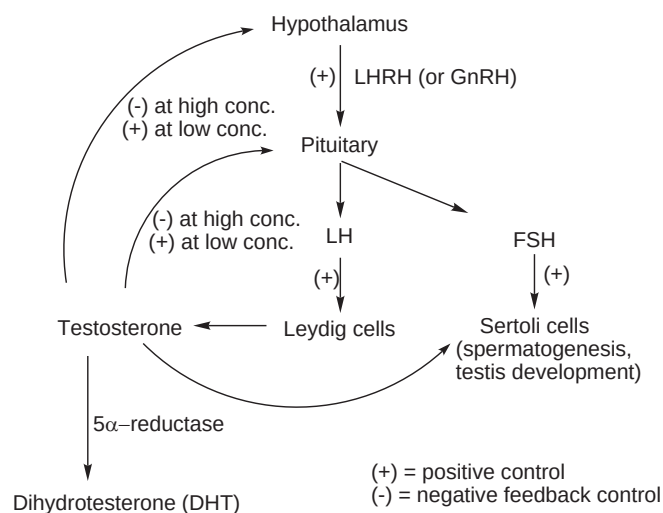


FIGURE 40.2 Hypothalamus-pituitary-testicular axis. (+), positive control; (-), negative feedback control.

secretion likewise is pulsatile and diurnal, with the highest concentration occurring at approximately 8:00 AM and the lowest at approximately 8:00 PM. In older men, especially those older than 60 years, the body does not produce enough testosterone that is needed to do all the intended work, and the testosterone levels remain relatively constant, without the morning pulses observed in young adult men. Average plasma testosterone concentrations in older men peak in the morning, with daily ranges of approximately 300 to 550 ng/dL by 70 years of age. The diurnal cycling is blunted as men age (15). In women, testosterone levels range from 15 to 100 ng/dL. Starting at approximately 40 years of age, testosterone levels drop by approximately 10% every decade in men. In the normal functioning of the male hormonal system, 97% to 98% of plasma testosterone is bound to sex hormone binding globulin (SHBG), essentially making it unavailable to the body's tissues due to the high binding affinity of testosterone for SHBG. The remaining 2% to 3% is known as "bioavailable" or free testosterone. Serum concentrations of SHBG increase with age in men, resulting in a corresponding decrease in free testosterone as men age. Free testosterone levels in serum are positively correlated with muscle strength and bone mineral density in elderly men, but negatively correlated with fat mass (16).

In the testis, FSH directly interacts with FSH receptors expressed in Sertoli cells and stimulates spermatogenesis, whereas LH indirectly stimulates spermatogenesis through testosterone synthesized by Leydig cells. Testosterone and its aromatized metabolite (estradiol) negatively regulate circulating levels of testosterone in the hypothalamus and pituitary. Activin and inhibin produced by Sertoli cells stimulate or inhibit, respectively, the secretion of FSH from the pituitary (17). High concentrations of intratesticular testosterone are essential for the initiation and maintenance of spermatogenesis, as evidenced by the infertility of hypogonadal men. Results from both animal models and humans, however, support the idea that both FSH and testosterone are required to achieve quantitative and qualitative spermatogenesis.

Androgens also are needed for the development of secondary sex characteristics. The male voice deepens because of thickening of the laryngeal mucosa and lengthening of the vocal cords. In both men and women, they play a role first in stimulating the growth of hair on the face, arms, legs, and pubic areas and later in the recession of the male hairline. The fructose content of human semen and both the size and the secretory capacity of the sebaceous glands also depend on the levels of testosterone.

Testosterone causes nitrogen retention by increasing the rate of protein synthesis and muscle mass while decreasing the rate of protein catabolism. The positive nitrogen balance therefore results from both decreased catabolism and increased anabolism of proteins that are used in male sex accessory apparatus and muscle. The actions of androgen in the reproductive tissues,

including prostate, seminal vesicle, testis, and accessory structures, are known as the androgenic effects, whereas the nitrogen-retaining effects of androgen in muscle and bone are known as the anabolic effects. Although the precise mechanism of androgen action on muscle remains unknown, the common hypothesis is that androgens promote muscle protein synthesis. Evidence supports the idea that testosterone supplementation increases muscle protein synthesis in elderly men (18) and young hypogonadal men (19). Also, androgen-induced increases in muscle mass appear to arise from muscle fiber hypertrophy rather than hyperplasia (i.e., cellular enlargement rather than cellular proliferation) (20).

The thickness and linear growth of bones are stimulated and, later, limited by testosterone because of closure of the epiphyses. Androgens affect bone mineral density by changing overall osteoblast (bone-forming cell) activity and osteoclast (bone-resorbing cell) activity, resulting from changes in the total number of each cell type and individual cell functional capacity (21,22). Androgens seem to have the ability to decelerate the bone remodeling cycle and tilt the focal balance of that cycle toward bone formation. The loss of androgens is thought to increase the rate of bone remodeling by removing the restraining effects on osteoblastogenesis and osteoclastogenesis.

ANDROGEN BIOSYNTHESIS

The major pathways for the biosynthesis of the sex steroid hormones are summarized in Figure 40.3. Cholesterol is stored in endocrine tissues and is converted to androgen, estrogen, or progesterone when the tissue is stimulated by a gonadotropic hormone. Androgens (male sex hormones) primarily are synthesized from cholesterol in the testes, whereas estrogens are biosynthesized chiefly in the ovary in mature, premenopausal women and adipose tissue in men. This is not surprising, because androgens are intermediates in the biosynthesis of estrogens. In the liver, androgens are formed from C-21 steroids. During pregnancy, the placenta is the main source of estrogen biosynthesis, and pathways for production change (23,24). Small amounts of these hormones also are synthesized by the adrenal cortex, the hypothalamus, and the anterior pituitary in both sexes. The major source of estrogens in both postmenopausal women and men is adipose tissue (25).

LH binds to its receptor on the surface of the Leydig cells to initiate testosterone biosynthesis. As in other endocrine cells, the binding of gonadotropin activates the G_s signal transduction pathway, increasing intracellular cyclic adenosine monophosphate (cAMP) levels via activation of adenylate cyclase. One of the processes influenced by elevated cAMP levels is the activation of cholesterol esterase, which cleaves cholesterol esters and liberates free cholesterol. The free cholesterol is then converted in mitochondria to pregnenolone via the side-chain cleavage reaction (Fig. 40.3).

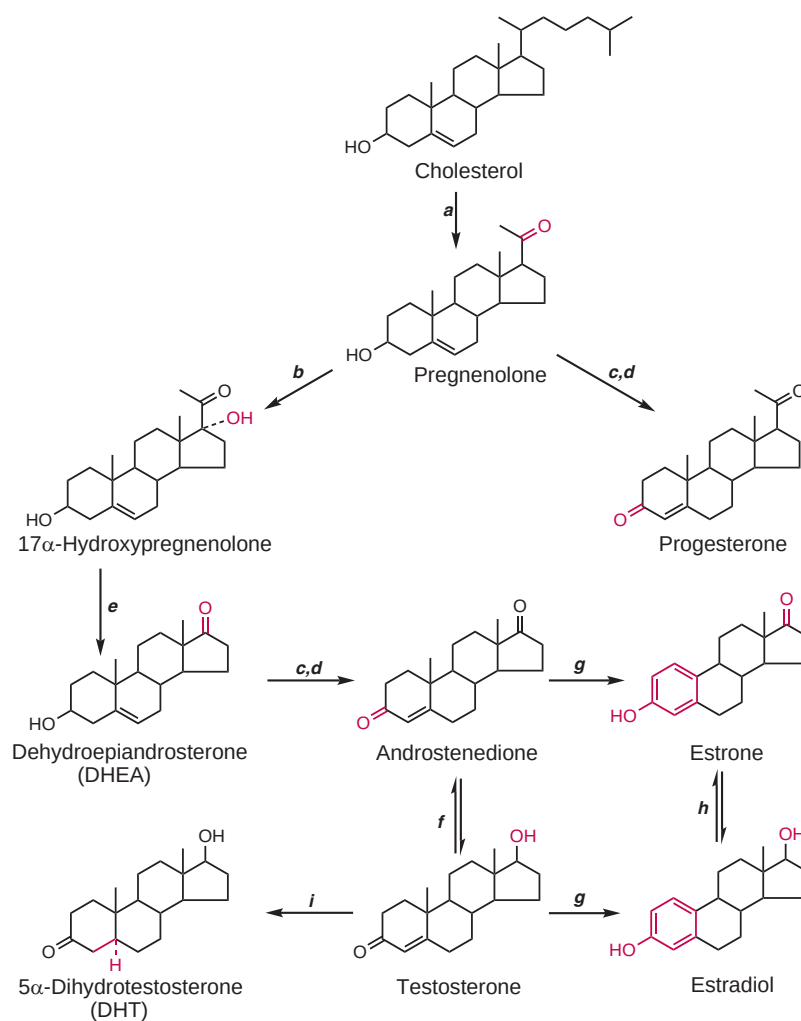


FIGURE 40.3 Biosynthesis of androgens and other sex steroid hormones. The enzymes involved in this biosynthesis are (a) side-chain cleavage, (b) 17 α -hydroxylase, (c) 5-ene-3 β -hydroxysteroid dehydrogenase, (d) 3-oxosteroid-4,5-isomerase, (e) 17,20-lyase, (f) 17 β -hydroxysteroid dehydrogenase, (g) aromatase, (h) estradiol dehydrogenase, and (i) 5 α -reductase.

Pregnenolone is converted by 17 α -hydroxylase to 17 α -hydroxypregnenolone and then to dehydroepiandrosterone (DHEA; a C-19 steroid) via 17,20-lyase, which involves cleavage of the C-17 to C-20 carbon-carbon bond and loss of the 17 β -acetyl side chain (26). The 17 α -hydroxylase is found in the endoplasmic reticulum membrane and is comprised of the CYP17A protein and the ubiquitous NADPH-P450 reductase. CYP17A is expressed by the *CYP17* gene found on chromosome 10 in humans. Alternatively, progesterone can be formed from pregnenolone via the action of 5-ene-3 β -hydroxysteroid dehydrogenase and 3-oxosteroid-4,5-isomerase. These same enzymes are responsible for the conversion of DHEA to the 17-ketosteroidal androgen, androstenedione (Fig. 40.3).

Testosterone is formed by reduction of the 17-ketone of androstenedione by 17 β -hydroxysteroid dehydrogenase type 5, also known as aldo-ketoreductase 1C3, or AKR1C3 (27,28). Testosterone and androstenedione are metabolically interconvertible (Fig. 40.3). Loss of

the C-19 angular methyl group and aromatization of the A ring of testosterone or androstenedione is catalyzed by the microsomal cytochrome P450 enzyme complex, called aromatase, and results in the C-18 steroids—namely, 17 β -estradiol or estrone, respectively—as shown in Figure 40.4. 17 β -Estradiol and estrone are also metabolically interconvertible, catalyzed by estradiol dehydrogenase. Research interests in the aromatization reaction continue to expand from basic endocrinology and reproductive biology studies to aromatase inhibition for the treatment of estrogen-dependent cancers, as illustrated in several conferences and reviews (29–32). Androstenedione is the preferred substrate for aromatization, and three molecules of NADPH and three molecules of oxygen are necessary for conversion of one molecule of androgen to estrogen (33).

The most potent endogenous androgen is the 5 α -reduced steroid, DHT, which is biosynthesized by two 5 α -reductase isoforms, type 1 and type 2. Type 1 5 α -reductase is expressed predominantly in sebaceous

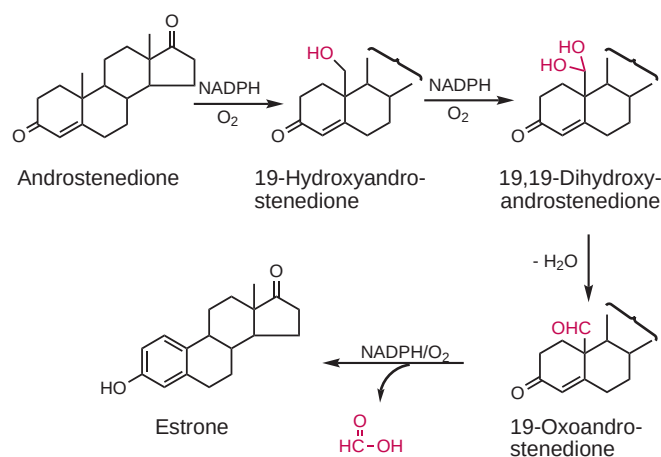


FIGURE 40.4 Aromatase mechanism.

glands of the skin, scalp, and liver and is responsible for approximately one-third of the circulating DHT. Type 2 5 α -reductase is found primarily in prostate, seminal vesicles, epididymides, genital skin (scrotum), hair follicles, and liver, and it is responsible for two-thirds of the circulating DHT. Approximately 6% to 8% of testosterone is converted to DHT. 5 α -Reductase has been

found in both the microsomal fraction and the nuclear membrane of homogenized target tissues, and it catalyzes an irreversible reduction reaction, which requires NADPH as a cofactor, that provides the α -hydrogen at C-5 (34,35). Conversion to DHT amplifies the action of testosterone by three to five times because of the greater binding affinity of DHT as compared to testosterone for the ARs (36).

ANDROGEN METABOLISM

Testosterone can be metabolized in either its target tissues or the liver (37–39), as shown in Figure 40.5. In androgen target tissues, testosterone can be converted to physiologically active metabolites. In the prostate gland, skin, and liver (40), testosterone is reduced to DHT by 5 α -reductase (types 1 and 2) (41). On the other hand, a small amount of testosterone (0.3%) also can be converted to estradiol by aromatase through cleavage of the C-19 methyl group and aromatization of ring A, which mainly occurs in adipose tissue. This process also occurs in the ovaries of women. In men, approximately 80% of the circulating estrogen arises from aromatization of testosterone in the adipose tissue (42), with the other 20% being secreted by the Leydig cells in the testes (43).

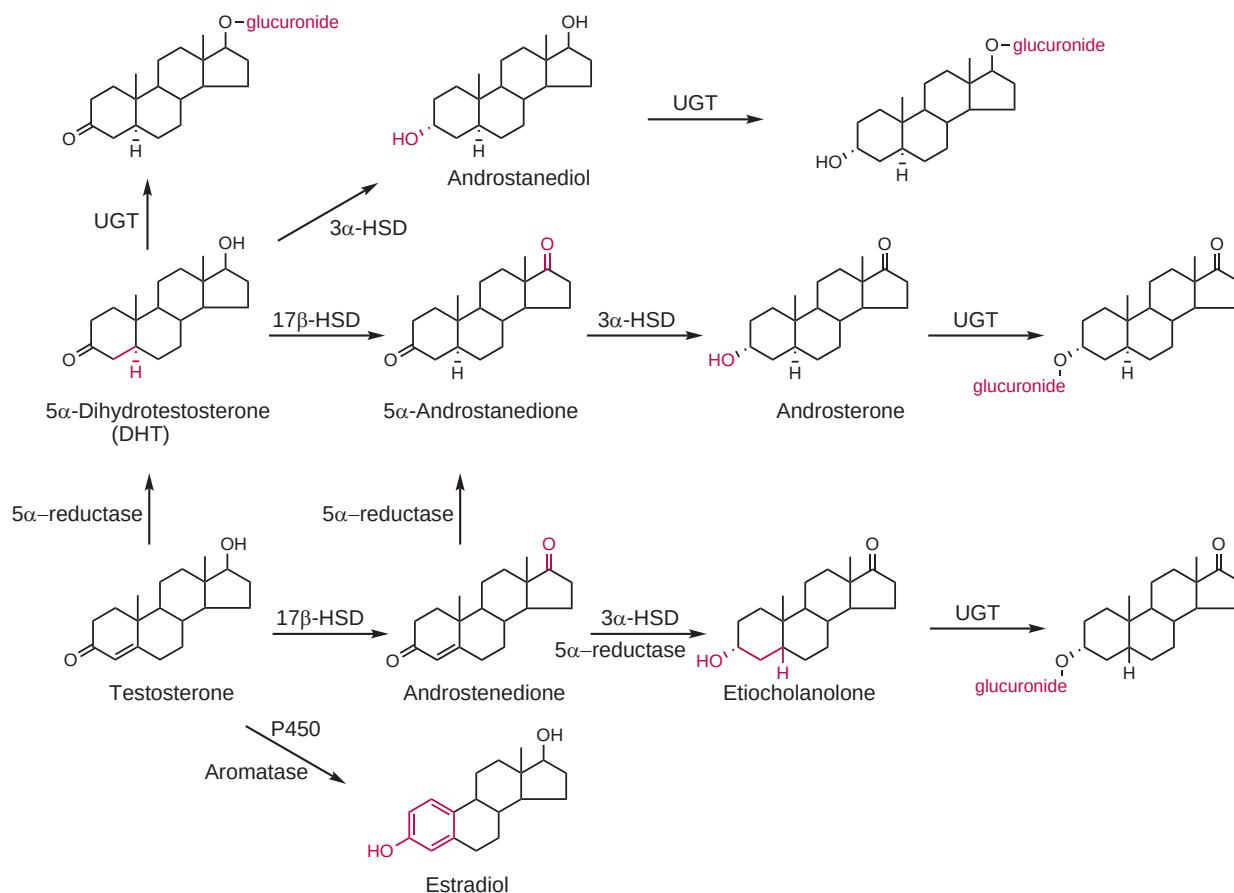


FIGURE 40.5 Testosterone metabolism. G, glucuronide; HSD, hydroxy steroid dehydrogenase; UGT, uridine diphosphoglucuronosyltransferase.

Both 5α -reduction and aromatization are irreversible processes. In addition to these pathways, testosterone also can be further inactivated in the liver through reduction and oxidation, followed by glucuronidation and renal excretion. It can be metabolized to androstenedione through oxidation of the 17β -OH group and to androstenedione with 5α -reduction of ring A. Androstenedione can be further converted to androstosterone after 3-keto group reduction. Alternatively, androstenedione also can be converted to etiocholanolone through 5β - and 3-keto reduction. Similarly, DHT can be converted to androstenedione, androstosterone, and androstenediol (44).

After the administration of radiolabeled testosterone, approximately 90% of the radioactivity is found in the urine, and 6% is recovered in the feces through enterohepatic circulation (45). Major urinary metabolites include androstosterone and its 5α -diastereoisomer etiocholanolone, both of which are inactive metabolites. They are excreted mainly as glucuronide conjugates or, to a lesser extent, as sulfate conjugates (46). The reduction of testosterone to its *cis* A/B ring juncture (5β) conformation, etiocholanolone, explains its complete loss of activity, because the *cis* A/B ring no longer has affinity for the AR, as shown in Figure 40.6. Most of the other metabolites mentioned earlier undergo extensive glucuronidation of the 3α - or 17β -OH groups as well, either in the target tissues or in the liver (46) and are further excreted in the urine. Therefore, following oral administration, the plasma testosterone half-life is less than 30 minutes because of extensive hepatic metabolism. Approximately 90% of an oral dose of testosterone undergoes first-pass metabolism before it reaches the systemic circulation. A number of minor metabolites of testosterone also have been isolated from urine and identified as 5α -androstanes and 5β -androstanes with a 3α -hydroxyl function. Most 17-ketosteroids isolated from the urine result from catabolism of the adrenocorticoids rather than from metabolism of androgens.

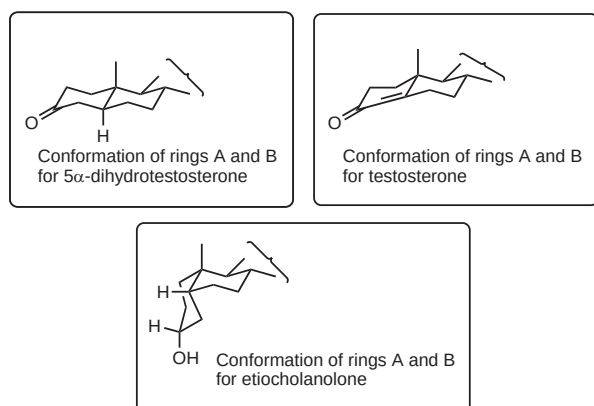


FIGURE 40.6 Conformations of dihydrotestosterone (DHT), testosterone (T), and etiocholanolone A and B rings.

MECHANISMS OF ANDROGEN ACTION

Testosterone, DHT, and other androgens execute their actions predominantly through the AR. The AR is mainly expressed in androgen target tissues, such as the prostate, skeletal muscle, liver, and central nervous system, with the highest expression level being observed in the prostate, adrenal gland, and epididymis (47). Testosterone is thought to be largely responsible for initiation of AR action in muscle, bone, brain, and bone marrow, whereas DHT plays a major role in genitalia, prostate, skin, and hair follicles due to their higher expression of 5α -reductase enzymes. The AR is a member of the steroid and nuclear receptor superfamily, which is composed of more than 100 members and continues to grow. Among this large family of proteins, only five vertebrate steroid receptors (estrogen, progesterone, androgen, glucocorticoid, and mineralocorticoid receptors) are known. Like other steroid receptors, AR is a soluble protein that functions as an intracellular transcriptional factor. The AR function is regulated by the binding of androgens, which initiates sequential conformational changes of the receptor that affect receptor-protein interaction and receptor-DNA interactions (48).

The AR gene is more than 90 kb long and codes for a protein of 919 amino acids that has three major functional domains, as illustrated in Figure 40.7. The N-terminal domain, which serves a modulatory function, is encoded by exon 1 (1,586 bp). The DNA binding domain (DBD) is encoded by exons 2 and 3 (152 and 117 bp, respectively). The ligand binding domain (LBD) is encoded by five exons, which vary from 131 to 288 bp in size. There also is a small hinge region between the DBD and LBD. Two transactivation functions have been identified. The N-terminal activation function (AF1) is not conserved in sequence and is ligand-independent (constitutively active), whereas the C-terminal activation function (AF2) is conserved in sequence and functions in a ligand-dependent manner (49). A nuclear localization signal spans the region between the DBD and the hinge region.

Similar to the other steroid receptors, unbound AR is mainly located in the cytoplasm and associated with a complex of heat shock proteins through interactions with LBD (50). On agonist binding (51), AR goes through a series of conformational changes: The heat shock proteins dissociate from AR, and the transformed AR undergoes dimerization, phosphorylation, and translocation to the nucleus, which is mediated by the nuclear localization signal, as shown in Figure 40.8. Translocated receptor then binds to androgen-response elements (AREs) in DNA, which are characterized by a six-nucleotide half-site consensus sequence 5'-TGTTCT-3' spaced by three random nucleotides and are located in the promoter or enhancer region of AR gene targets. Recruitment of other transcription coregulators (including coactivators and corepressors) and transcriptional machinery further ensures the transactivation of AR-regulated gene expression. All these complicated processes are initiated by the ligand-induced conformational changes in the LBD.

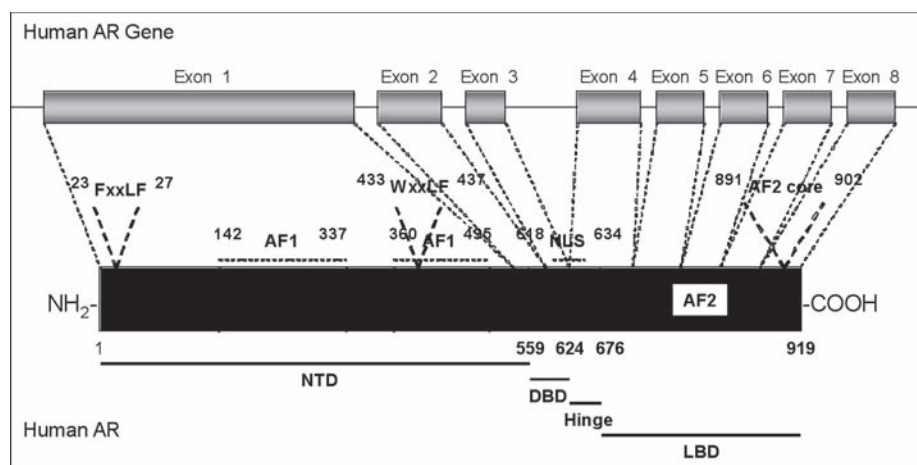


FIGURE 40.7 Structural domains of the androgen receptor gene and protein.

Ligand-induced AR conformational changes provide the structural basis for the recruitment of cofactor proteins and transcriptional machinery, which also is required for the assembly of AR-mediated transcription complexes (52), as shown in Figure 40.8. The formation of an activation complex is known to involve AR, coactivators, and RNA polymerase II recruitment to both the enhancer and promoter, whereas the formation of a repression complex involves factors bound only at the promoter and not at the enhancer. Because the formation of a functional AF2 region provides a structural basis for ligand-induced protein-protein interaction, ligand-specific recruitment of coregulators might be crucial for the agonist or antagonist activity of AR ligands.

Binding of DNA also is required for AR-regulated gene expression, which is known as the classic genomic

function of AR. The ARE half-site sequence can be arranged either as inverted repeats or as direct repeats (52,53), and AR recognizes and binds to the ARE site through two zinc fingers located in the DBD. Like other steroid receptors, ligand-bound AR forms homodimers and appears to form “head-to-head” dimers (54) even when it is bound to the direct repeats of ARE. Selective recognition of specific ARE sequences could be regulated by ligand binding (55) and/or the presence of other transcriptional factors, which bind to their own DNA binding sites as well (combinatorial regulation) (56).

Besides the genomic pathway, the nongenomic pathway of AR also has been reported in oocytes (57), skeletal muscle cells (58), osteoblasts (59,60), and prostate cancer cells (61,62). As compared to the genomic pathway, the nongenomic actions of steroid receptors are

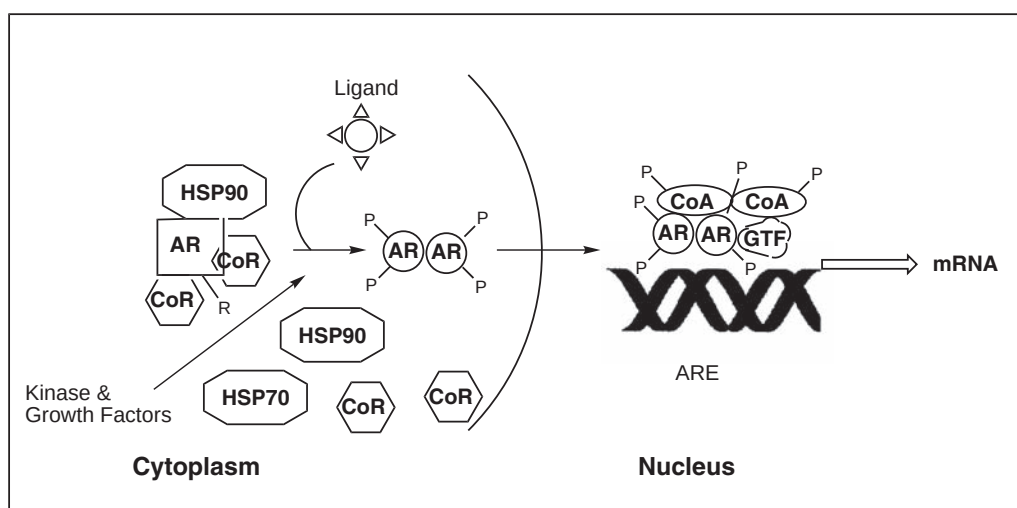


FIGURE 40.8 Mechanism of androgen action. The androgen receptor (AR) is maintained in an inactive complex by heat shock protein (HSP) 70, HSP90, and corepressors (CoR). On ligand binding, it homodimerizes and enters the nucleus. The receptor is basally phosphorylated (P) in the absence of hormone, and hormone binding increases the phosphorylation status of the receptor. The AR binds to the androgen response element (ARE) on the promoter of androgen responsive genes, leading to the recruitment of coactivators (CoA) and general transcription factors (GTF), leading to gene transcription.

characterized by the rapidity of the action, which varies from seconds to an hour or so, and by interaction with plasma membrane-associated signaling pathways (63). Nevertheless, the structural basis for nongenomic action is direct interactions between AR and cytosolic proteins from different signaling pathways, which could be closely related to the ligand-induced conformational change of the LBD or, indirectly, the N-terminal domain. Functionally, the nongenomic action of androgen involves either rapid activation of kinase-signaling cascades or modulation of intracellular calcium levels, which could be related to stimulation of gap junction communication, neuronal plasticity, and aortic relaxation (64). Separation of the genomic and nongenomic functions of steroid receptors using specific ligands also was proposed as a new strategy to achieve tissue selectivity (63,65).

DRUGS USED IN THE TREATMENT OF AGING-RELATED ANDROGEN INSUFFICIENCIES

Male Hypogonadism

Male hypogonadism (testosterone deficiency) is the inability of the testes to produce sufficient testosterone to maintain sexual function, muscle strength, bone mineral density, and fertility (spermatogenesis). In men, there is a gradual decline of approximately 1% per year in the production of testosterone beginning around 40 years of age. For most men, testosterone levels naturally decline with advancing age but still remain within the physiologic range throughout their lifetimes, causing no significant problems. Approximately 20% of men older than age 60 years and 30% to 40% of men older than 80 years have plasma testosterone levels indicative of hypogonadism (<280 to 300 ng/dL). Male hypogonadism is most commonly primary hypogonadism (testicular failure to produce testosterone for various reasons), secondary hypogonadism (hypothalamic-pituitary failure to stimulate testes to produce testosterone), or a combination of both.

Testosterone and structurally related steroidal androgens have been used for decades to treat male hypogonadism, Klinefelter syndrome (a chromosomal abnormality resulting in testicular dysfunction), anemia secondary to chronic renal failure, aplastic anemia, protein wasting diseases associated with cancer, burns, traumas, AIDS, short stature, breast cancer (as an anti-estrogen), and hereditary angioedema. However, these agents have been demoted to the therapy of final resort for anemia and cancer because of serious hepatotoxicity and the recent development of more effective therapies (e.g., erythropoietin, aromatase inhibitors, and taxanes). Although severe hypogonadism is uncommon, aging-related androgen insufficiency is much more frequent. Low endogenous testosterone concentrations are associated with sarcopenia and frailty arising from decreased

fat-free mass, lessened muscle strength, and reduced bone mineral density (osteoporosis). Low testosterone concentrations also are associated with decreased sexual libido and ED. More than 30 million men older than 40 years in the United States are estimated to suffer from ED. Although androgens are not essential for erection (66), transdermal and intramuscular (IM) testosterone replacement therapy often is employed in hypogonadal men with ED (67). Furthermore, selective phosphodiesterase (PDE5) inhibitors that increase penile blood flow are considered to be the treatment of choice for men with ED. Hormone replacement therapy with testosterone in aging men also improves body composition, bone and cartilage metabolism, and memory and cognition, and it even decreases cardiovascular risk (68).

Low testosterone concentrations frequently are seen in patients with ED, aging, type 2 diabetes, HIV/AIDS, osteoporosis, depression, obesity, alcohol abuse, anabolic steroid abuse, chronic inflammatory disease, cancer, and glucocorticoid use.

Andropause

Andropause was first described in the medical literature in the 1940s, but the ability to diagnose it is relatively new. The idea that men, as well as women, might be subject to sex hormone fluctuations in later life has been a topic of debate among endocrinologists and men's health professionals (69). Andropause affects men between the ages of 40 and 55 years, but unlike women, men do not have a clear-cut signpost, such as the cessation of menstruation, to mark this transition. Men's "transition" may be much more gradual and expand over many decades, and men will very likely experience andropausal symptoms including ED, loss of muscle mass, irritability, generalized fatigue, and even problems with memory and cognition. A decline in testosterone levels will occur in virtually all men, and there is no way of predicting who will experience andropausal symptoms of sufficient severity to seek medical help. It is estimated that 30% of men in their 50s, and up to 50% of men older than 65 years, will have testosterone levels low enough to cause noticeable symptoms. Once andropause is discovered, the process of replacing the missing testosterone is either by injection, locally applied hormone gel, transdermal patch, or implanted pellets.

Testosterone Replacement Therapy

The acceptance of testosterone replacement therapy (TRT) has been hampered by the lack of orally active preparations with good efficacy and, particularly, a safe profile (70). Progress has been limited over the last three decades in developing synthetic molecules that could separate the desirable physiologic functions normally regulated by endogenous androgens from the undesirable or dose-limiting side effects. The abuse of synthetic anabolic steroids by athletes and body builders has

contributed to the general perception of certain undesirable side effects, such as aggressive behavior, liver toxicity, acne, or impotency.

Current formulations for TRT largely are restricted to injectable formulations of testosterone esters, transdermal delivery formulations (scrotal or nonscrotal patches or gel), or buccal testosterone. Marketed injectable forms of testosterone esters (e.g., testosterone enanthate, propionate, or cypionate) produce undesirable fluctuations in testosterone blood levels, with supraphysiologic concentrations early and subphysiologic levels toward the end of the period before the next injection. These fluctuations provide an unsatisfactory benefits profile and, in some cases, undesired side effects. Skin patches provide a better blood level profile of testosterone, but skin irritation and daily application limit the usefulness and acceptability of this form of therapy. Topical gels (AndroGel and Testim) are widely used for TRT, but must be cautiously used in homes with children due to the risk of virilization. Oral preparations such as fluoxymesterone and 17 α -methyltestosterone are only sparingly used because of concerns about liver toxicity linked to the 17 α -alkyl group and because of somewhat lower efficacy (70). Thus, these oral androgens are generally considered to be obsolete and do not represent a viable form of therapy.

Benefits and Risks of Testosterone Replacement Therapy

Multiple large-scale and long-term clinical trials of TRT have been conducted in aging men to evaluate the risk-benefit ratio of TRT, but no agreement exists regarding

the benefits and risks of TRT (Table 40.1) [for review, see Hijazi and Cunningham (69) and Rhoben and Morgentaler (70)]. The potential benefits of TRT include increase in bone mineral density and improvement in muscle mass and strength, cognitive function, mood, and sexual function. The potential risks of TRT, however, including those in the cardiovascular system, blood (e.g., hematocrit and hemoglobin levels), and prostate, are routinely experienced. An emerging class of drugs known as selective androgen receptor modulators (SARMs) may soon transform the therapeutic landscape of androgen use by selectively stimulating anabolic targets and avoiding steroid-related side effects. Because of the long-term effects of TRT in otherwise healthy men remain unclear, guidelines published by The Endocrine Society recommend a general policy that TRT not be offered to older men with low serum testosterone levels and that TRT only be used in men with clinically significant symptoms of androgen deficiency and serum total testosterone concentrations approaching 200 ng/dL (71).

Clearly, TRT is beneficial for hypogonadal men with androgen insufficiency to restore sexual function and muscle strength, to prevent bone loss, and to protect against heart disease (atherosclerosis) (70). Increasing testosterone levels with TRT, however, may pose problems by stimulating the growth of the prostate. Long-term TRT could cause prostate gland enlargement, which might exacerbate BPH or fuel the growth of prostate cancer that is already present and could cause breast enlargement in men (gynecomastia). This is especially

TABLE 40.1 Testosterone Replacement Therapy

Benefits	Risks
Improved sexual performance and desire	Stimulated growth of preexisting prostate cancer
More energy and improved quality of life	Greater chance for benign prostatic hyperplasia
More energy and sense of well-being	Increased hemoglobin levels to above the physiologic range
Increased bone mineral density	Problems with voiding; symptoms includes poor urine flow and hesitancy before urinating
Improved muscle mass strength	
Improved (lower) low-density lipoprotein profile	Increased potential for liver damage from oral preparations Sleep apnea (stopping of breathing during sleep)
Decreased irritability and depression	Breast tenderness and swelling (gynecomastia)
Improved cognitive function	Testicular shrinkage (testicular atrophy)
Increased hemoglobin levels to the physiologic range	Infertility (decreased spermatogenesis) Skin reaction from patches or gel
Thickened body hair and skin	Pain, soreness, or bruising from injection Increased fluid retention Increased skin problems (acne, oily skin) Increased body hair

worrisome because of the high prevalence of BPH in older men and the possibility that many men may have prostate cancer that is undiagnosed. In fact, men with a palpable prostate nodule, a prostate-specific antigen level greater than 4 ng/mL, or severe lower urinary tract symptoms associated with BPH are usually advised to avoid TRT.

In elderly men, testosterone effects on muscle mass and strength have not been consistent or impressive, possibly because of the low dosages used in clinical trials. The high correlation between the dose (and plasma concentration) and the anabolic actions of androgen in muscle suggests that androgen administration of higher doses in elderly men may significantly increase muscle mass and strength, but high doses might increase the adverse effects and the aromatization of testosterone to estrogen (72).

Types of Testosterone Replacement Therapy

Several types of TRT exist. Choosing a specific therapy depends on the patient's preference of a particular delivery system, the side effects, and the cost. Types include injection, transdermal, buccal mucosal, and oral (Table 40.2).

INJECTION Orally administered testosterone is ineffective in the treatment of male androgen insufficiency syndromes due to extensive presystemic first-pass metabolism. IM injections bypass the problems of first-pass metabolism. IM injections are depot formulations of testosterone esters that undergo differing rates of *in vivo* ester hydrolysis to release free testosterone over an extended period of time. Typically, the depot esters are administered IM into a large muscle once every 2 to 4 weeks depending on the depot ester used (Table 40.2). They are safe, effective, and the least expensive androgen preparations available. The major disadvantage with the IM route for the depot esters is that testosterone plasma concentrations exhibit a saw-toothed pattern, with supraphysiologic levels within 2 to 4 days following the IM injection and subphysiologic levels before the next injection. A more satisfactory physiologic replacement therapy without the fluctuations in free testosterone plasma levels would be to administer a lower IM dose (i.e., 100 mg) on a weekly or biweekly schedule. Because IM injection of testosterone or its esters causes local irritation, the rate of absorption may be erratic.

Esters of testosterone available for IM administration include the 17 β -propionate, 17 β -enanthate, and the cypionate (17 β -cyclopentylpropionate) (Fig. 40.9). Testosterone enanthate and cypionate are commonly used and demonstrate comparable pharmacokinetics. Testosterone enanthate is formed by esterification of the 17 β -hydroxy group of testosterone with heptanoic acid, while testosterone cypionate is formed with cyclopentanepropionic acid. Sterile solutions of these esters are available in a suitable vegetable oil, such as cottonseed

oil. Unlike oral testosterone, with a half-life of 10 to 100 minutes, IM testosterone administration avoids first-pass metabolism and exhibits a longer duration due to the slow hydrolysis and release of testosterone from the depot injection site. Generally, the concentration of SHBG in plasma determines the distribution of testosterone between free and bound forms. The bulky cypionate and enanthate esters of testosterone have a duration of action of up to 2 to 4 weeks, whereas the shorter propionate ester has a shorter duration of action of 1 to 2 weeks. Doses may be adjusted by aiming for midphysiologic (400 to 600 ng/dL) testosterone values after 1 week or at the low end (300 to 400 ng/dL) just before the next injection is due.

IMPLANT Subcutaneous implantation of testosterone pellets (Testopel) is much less flexible for dose adjustment as compared to transdermal, oral, or IM injection. Fat-soluble pellets containing 75 mg of testosterone can be implanted subcutaneously on the anterior abdominal wall or buttocks to deliver testosterone over a 3- to 4-month period. The longer duration of action of the pellets is accompanied by a larger dose requirement as compared to weekly or biweekly IM injections. In general, two 75-mg pellets provide an equivalent dose of testosterone over 3 months to weekly 25-mg IM injections of testosterone propionate. Approximately one-third of the material from the pellet is absorbed in the first month, one fourth in the second month, and one sixth in the third month. Adequate effects from the pellets can occasionally continue for as long as 6 months.

TRANSDERMAL Transdermal TRT systems are, perhaps, the most commonly used systems for delivering testosterone to bypass the rapid first-pass metabolism associated with oral testosterone. Clinical studies have shown that these formulations are effective forms of testosterone replacement, with peak response within 3 to 6 months. Use of the transdermal formulation should be discontinued if the desired response is not reached within this time period. Skin irritation is more common with the transdermal formulations, with more than 50% experiencing some form of skin irritation at some point during the treatment. Pretreatment with corticosteroid creams (not with the ointment) has been shown to reduce the severity and incidence of skin irritation without significantly affecting testosterone absorption from the formulation. With the transdermal formulations, testosterone levels were maintained within physiologic values, and a beneficial effect was observed on general mood and sexual functioning. A plasma concentration in the midphysiologic range (400 to 600 ng/dL) is the goal.

Matrix-Type Transdermal Systems This type of patch (scrotal patch; Testoderm) must be applied to dry, clean (shaven) scrotal skin, which is 5 to 30 times more permeable to testosterone than other skin sites, every 24 hours to produce an adequate testosterone plasma concentration.

TABLE 40.2 Testosterone Products and Properties^a

Product	Trade Name	Onset of Peak Response	Duration of Action	Time to Peak Conc.	Time to Steady-State Conc.	Dose (mg)	Oral Frequency of Dosing	Bioavailability (%)	Elimination Half-life
Methyltestosterone	Android	—	24 hours	2 hours	—	10–50	Daily	70	3 hours
	Testred								
	Virilon								
Fluoxymesterone	Oreton Methyl								
	Halotestin	—	24 hours	—	—	5–40	Daily	80	9 hours
Testosterone undecanoate	Android-F								
	Andriol	6 hours	10 hours	~6 hours	—	40–120	Daily	<10	3 hours
Testosterone propionate	Testex	6–24 hours	1–2 weeks	3–36 hours	—	10–50	Two or three times per week	—	—
	Andronate Depotestosterone Depotest	6–24 hours	2–4 weeks	24 hours	—	50–400	2–4 weeks	—	8 days
Testosterone enanthate	Andro-LA	6–24 hours	2–4 weeks	24 hours	—	50–400	2–4 weeks	—	8 days
Testosterone	Andryl								
	Delatestryl								
	Delatest								
	Everone								
Testosterone	Testamone								
	Testrin-PA								
Testosterone pellets	Testopel	1–2 months	3–6 months	1 month	1–2 weeks	150–450	3–6 months	—	—
Transdermal patches	Androderm	3–6 months	24 hours	2–4 hours	2–3 days	2–5	24 hours	—	10–100 minutes
	Testoderm TTS								
Transdermal gels	Testoderm								
	AndroGel	3–6 months	5 days	4 (2–6) hours	2–3 days	50–100	24 hours	—	10–100 minutes
Buccal mucosal	Testim								
	Striant	—	24 hours	5 (0.5–12) hours	2–3 days	30	Every 12 hours	—	6 hours

^aThompson Healthcare, Inc. Micromedex Healthcare Series. Available at: <http://www.Thompsonhc.com>.

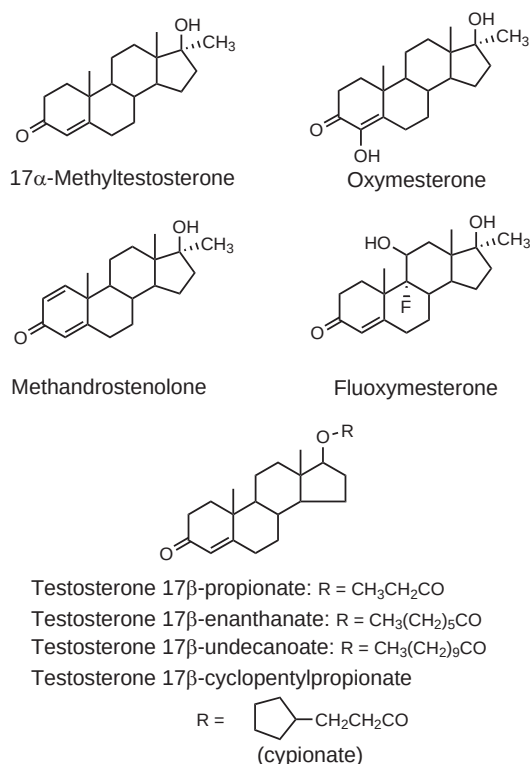


FIGURE 40.9 Testosterone esters and synthetic testosterone derivatives.

The matrix system is described as a “drug-in-adhesive film,” in which the drug is located on the adhesive layer of the film; thus, it is thinner and less bulky than the reservoir system. The advantage of the matrix system is that it produces supraphysiologic levels of DHT because of the high 5 α -reductase enzyme activity of the scrotal tissue. The patches have an occlusive backing that prevents sex partners from coming in contact with the active drug. A matrix transdermal system will not produce adequate plasma testosterone concentrations if applied to nonscrotal skin. Plasma testosterone concentrations are reached in approximately 2 to 4 hours. Although testosterone is absorbed throughout a 24-hour period, concentrations do not simulate the circadian rhythm of endogenous testosterone in normal (eugonadal) males. Within 24 hours after application of the matrix system, plasma testosterone concentration gradually falls to 60% to 80% of the peak plasma concentration, and when the system is removed, testosterone plasma concentration declines to baseline within 2 hours. Inadequate scrotal size and adherence problems are limitations. Skin irritation does occur in those with sensitive scrotal skins.

Reservoir-Type Transdermal Systems The reservoir-type patch (nonscrotal patch; Androderm, Testoderm TTS) is not applied to scrotal skin but, rather, to the abdomen, back, thighs, or upper arms every 24 hours (see Table 40.2 for dosage strength). This type of patch is membrane-controlled for the drug to diffuse continuously over 24 hours from the

reservoir into the skin. Thus, this type of patch is thicker than the matrix (scrotal) patch. The patches have an occlusive backing that prevents sex partners from coming in contact with the active drug. The site of the application is rotated at 7-day intervals between applications to lessen skin reactions at the same application site. The advantage of the reservoir transdermal system is that it achieves normal testosterone circadian rhythm as seen in younger men, peaking in the morning and decreasing throughout the rest of the day. The reservoir-type patch, when applied to nonscrotal skin, produced physiologic DHT and estradiol plasma concentrations. Steady-state plasma concentrations of testosterone, which are approximately 10 times baseline values, are reached in about 6 hours (range, 4 to 10 hours depending on application patch location), which then fall to 60% to 80% of the peak plasma concentration within 24 hours after application of the transdermal system. Thus, physiologic plasma testosterone concentrations are maintained over 24 hours with this type of patch. Drug accumulation does not occur with repeated applications. When the system is removed, testosterone plasma concentrations decline to baseline within 2 hours. A usual dose for the reservoir-type transdermal results in the systemic absorption of 2 to 10 mg daily in hypogonadal men.

Gel Testosterone gel (AndroGel, Testim) is a 1% testosterone hydroalcoholic gel that provides continuous transdermal delivery of testosterone for 24 hours once the gel is rubbed into the skin on the lower abdomen, upper arm, or shoulder. It should not be applied to scrotal tissue (Table 40.2). Because there is a continuous release of testosterone over 24 hours, the normal circadian rhythm is not observed. As the gel dries, approximately 10% of the testosterone is absorbed through the skin. Gel application of TRT appears to cause fewer skin reactions than occur with the patches. Men should avoid showering or bathing for several hours after an application to ensure adequate absorption. A potential side effect of the gel is the possibility of transferring the medication to a partner; skin-to-skin contact should be avoided either until the gel is completely dry or by covering the area after an application. Skin-to-skin contact with children is of particular concern and is included as a “black box” warning for these products. Following the application of 5 g of gel, which will deliver 50 mg of testosterone, the mean peak testosterone concentrations are reached in approximately 2 hours, which are about two to three times baseline values. For optimum results, the gel is best applied in the evening to allow maximum concentration to occur early in the morning hours. Doses of the gel may be adjusted by aiming for midphysiologic (400 to 600 ng/dL) testosterone values after 1 week. When the gel treatment is discontinued, plasma testosterone levels remain in the physiologic range for 24 to 48 hours, then return to their pretreatment levels within 5 days following the last application. An increase in plasma testosterone can be observed within 30 minutes of application. Plasma concentrations approximate the steady-state level

by the end of the first 24 hours and are at steady-state by the second or third day of dosing.

Buccal Mucosal Striant is a gel-like substance that adheres to the gumline, which softens to deliver physiologic amounts of testosterone to the systemic circulation, thereby producing circulating testosterone concentrations in hypogonadal males that approximate physiologic levels seen in healthy young men (400 to 700 ng/dL). One buccal system (30 mg) is applied to the gum region twice daily, morning and evening, approximately 12 hours apart. Because there is a continuous release of testosterone over 24 hours, the normal circadian rhythm is not observed. Peak plasma testosterone concentrations are reached within 10 to 12 hours and are stable within a few days of the buccal preparation. The buccal preparation is difficult for patients to get used to, because the side effects may include gum irritation or pain, bitter taste, and headache. A study found that this form of TRT delivers a steadier dose of testosterone throughout the day without significant adverse effects, comparable to the gel.

ORAL Orally administered testosterone is ineffective in the treatment of male androgen deficiency syndromes because of extensive presystemic first-pass metabolism, primarily to inactive 17-ketosteroid, etiocholanolone and androsterone, and androstenediol metabolites in the gastrointestinal mucosa during absorption and in the liver (Fig. 40.5). Oral administration results in supraphysiologic elevations of testosterone and undesirable variability of plasma concentrations. The plasma half-life of testosterone is less than 30 minutes. Generally, the amount of SHBG in plasma determines the distribution of testosterone between free and bound forms. Approximately 90% of a dose of testosterone is metabolized, and its metabolites are excreted in the urine primarily as glucuronide conjugates, with approximately 6% of a dose being excreted in the feces as unmetabolized testosterone. Comparative dosage ranges for testosterone and its synthetic preparations are shown in Table 40.2. Taking testosterone orally (Android, Testred), in the form of methyltestosterone, is not recommended for long-term replacement. Oral testosterone may cause an unfavorable cholesterol profile and increase the risk of blood clots and heart and liver problems.

The androgenic activity of lipophilic long-chain ester testosterone 17 β -undecanoate (Android Testocaps are not approved for use in the United States) (Fig. 40.9) has been attributed to formation of testosterone via systemic ester hydrolysis of lymphatically transported testosterone undecanoate (73). Its oral bioavailability was approximately 3%. Lymphatically transported testosterone undecanoate accounted for between 90% and 100% of the systemically available ester, and 83% to 85% of the systemically available testosterone resulted from systemic hydrolysis of lymphatically transported testosterone undecanoate. These data demonstrate that intestinal lymphatic transport of testosterone undecanoate produces increased systemic exposure of testosterone by avoiding

the extensive first-pass hepatic metabolism responsible for the inactivation of testosterone after oral administration.

Synthetic Derivatives of Testosterone

Some of the early studies with androgens included structural modifications of the naturally occurring hormones to avoid first-pass metabolism. Blocking the metabolism of the 17 β -hydroxy group with substituents in the 17 α -position resulted in androgens with an increased bioavailability and duration of action when given orally. The synthetic androgens include methyltestosterone and fluoxymesterone (Fig. 40.9). The long-term use of oral androgens has been associated with liver cancer.

17 α -METHYLTESTOSTERONE The synthesis of 17 α -methyltestosterone made available a compound that was orally active (74) in daily doses between 10 and 50 mg (Android, Testred), which is equivalent to a 400-mg oral dose of testosterone. The presence of a 17 α alkyl group reduces susceptibility to hepatic oxidative metabolism, thereby increasing oral bioavailability (~70%) by slowing metabolism. Following oral administration, methyltestosterone is well absorbed from the gastrointestinal tract, with a half-life of approximately 3 hours. This drug has the androgenic and anabolic activities of testosterone. Although orally active, it is more effective when administered sublingually. The alkylated oral androgens are seldom used due to their potential hepatotoxicity.

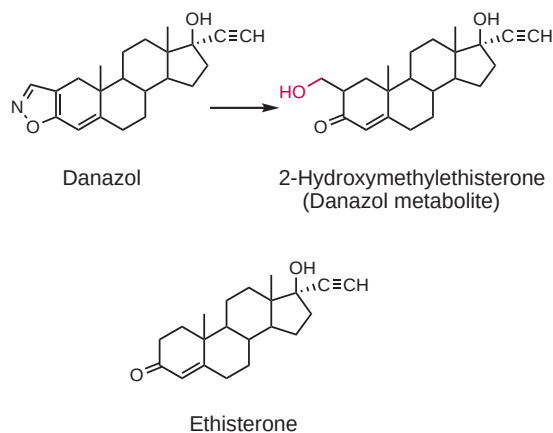
FLUOXYMESTERONE By substituting a 9 α -fluoro group onto an analog of 17 α -methyltestosterone, fluoxymesterone has 20 times the anabolic and 10 times the androgenic activity of 17 α -methyltestosterone (Fig. 40.9) (74). It has a bioavailability of approximately 80% and a mean half-life of 9 hours, and less than 5% of the drug is excreted unchanged. Daily doses of 5 to 20 mg (Androxy) are generally used for TRT, but, like methyltestosterone, concerns related to the potential hepatotoxicity of alkylated androgens have limited the clinical use of this drug. Fluoxymesterone is also associated with sodium and water retention, which could lead to edema.

Structure–Activity Relationships of Steroidal Androgens

Until the discovery of nonsteroidal androgens (discussed later), it was believed that a substance must contain a steroid skeleton to have androgenic activity (75). Oxygen functional groups normally occurring at positions 3 and 17 of the steroid ring system are not essential, because the basic nucleus, 5 α -androstane, has androgenic activity (Fig. 40.1). This appears to be the minimal structural requirement for hormonal activity of steroids. For derivatives of etiocholane, in which the hydrogen is in the 5 β -position, thereby affording a *cis* A/B ring juncture, no active androgens and anabolic agents are known (75). Generally, ring expansion (to form homo derivatives by inserting a methylene group into one of the rings in the steroid nucleus) or ring contraction (by removing a methylene group) significantly reduces or destroys the androgenic and anabolic activities.

Introduction of a 3-ketone function or a 3 α -OH group enhances androgenic activity. A hydroxyl group in the 17 α -position of androstane contributes no androgenic or anabolic activity; no known substituent can approach the effectiveness of a 17 β -OH group. Evidence indicates that the longer-acting esters of the 17 β -OH compounds are hydrolyzed in vivo to the free alcohol, which is the active species. The 17 β -oxygen atom is important for attachment to the receptor site, while 17 α -alkyl groups are important for preventing metabolic changes at this position (74). Such 17 α -substituents render the compounds orally active.

Increasing the length of the alkyl side chain at the 17 α -position, however, resulted in decreased activity, and the incorporation of other substituents, such as the 17 α -ethynyl group, produced compounds with useful progestational activity (progestins) (see Chapter 41), such as ethisterone. Attaching an isoxazole ring to ethisterone produced danazol (Danatrol, Danocrine), which exhibited potent antigonadotropic properties, weak androgen and anabolic properties, and no estrogen or progestin activity.



As a gonadotropin inhibitor, danazol suppresses the surge of LH and FSH from the pituitary, thus suppressing

ovarian steroidogenesis. For this reason, it is used in the treatment of endometriosis. Previous treatment of endometriosis had been surgical or medical, with progestins or a combination of estrogen and progestin. Danazol is metabolized by CYP3A4 to its inactive metabolite, 2-hydroxymethylethisterone.

Several modifications of 17 α -methyltestosterone lead to potent, orally active anabolic agents. Two hydroxylated analogs include oxymesterone (Fig. 40.9) and oxymetholone (Fig. 40.10). These drugs have at least three times the anabolic and half the androgenic activity of testosterone (74).

Halogen substitution produces compounds with decreased activity except when inserted into positions 4 or 9 (e.g., fluoxymesterone). Replacement of a carbon atom in position 2 by oxygen has produced the only clinically successful heterocyclic steroid (oxandrolone) among a number of azasteroids and oxasteroids. Some of the 2-oxasteroids are potent anabolic agents.

Introduction of a sp² hybridized carbon atom into the A ring (methenolone, testolactone, Fig. 40.10) renders the ring more planar, and in turn, this may be responsible for greater anabolic activity. The 19-norsteroids (nandrolone) are of interest, because these agents seem to produce a more favorable ratio of anabolic to androgenic activity. Vida (74) has extensively reviewed the replacement of various hydrogens on the androgen steroid skeleton by other functional groups. It appears that certain substitutions at positions 1, 2, 7, 17, and 18 may result in compounds with favorable activities that will be of clinical importance.

ADVERSE EFFECTS Testosterone replacement therapy can have undesirable side effects depending on the type of delivery system used. The adverse effects from oral testosterone include stomach upset, headache, acne, increased hair growth on the face or body, jaundice (liver toxicity), anxiety, change in sex drive, sleeplessness,

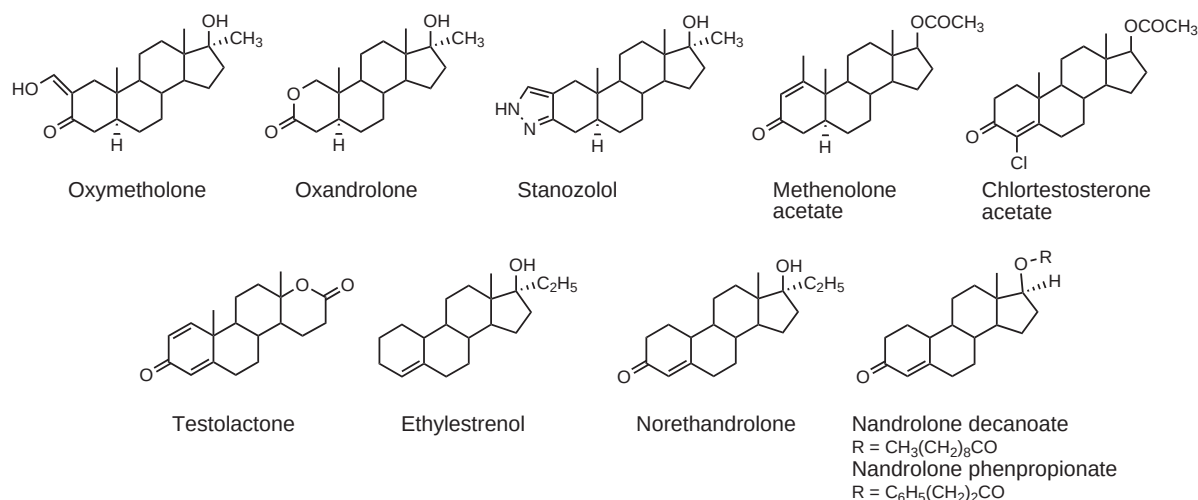


FIGURE 40.10 Anabolic steroids.

increased urination, depression, enlargement of breasts, and increased frequency and duration of erections (70–72). Breast enlargement can develop because testosterone can be converted to estradiol via aromatase. Other adverse effects include water retention, liver toxicity, cardiovascular disease, sleep apnea, and prostate enlargement. These risks are relatively uncommon when the dosage is closely monitored to maintain physiologic plasma testosterone concentrations. Testosterone replacement therapy is contraindicated in men with carcinoma of the breast or with known or suspected BPH or carcinoma of the prostate. Therefore, pretreatment screening for any prostate dysfunction is mandatory before starting TRT.

Anabolic Agents

Because complete dissociation of anabolic and androgenic effects is not possible, many of the actions of anabolic steroids are similar to those of androgens. Comparative dosage ranges for the anabolic steroids are shown in Table 40.2.

Selenium dioxide dehydrogenation of 17 α -methyltestosterone yields the 1,4-diene analog, methandrostenolone, which has several-fold the anabolic activity of the starting material. It has low androgenic activity but, apparently, can produce mammogenic effects in men. These effects are thought to result from estrogenic metabolites.

The 17 α -alkylated anabolic steroids in clinical use are oxandrolone (Anavar, Oxandrin), oxymetholone (Anadrol-50), stanozolol (Winstrol), nandrolone decanoate (Deca-Durabolin, Hybolin), and nandrolone phenpropionate, as shown in Figure 40.10.

A 2-oxasteroid analog of 17 α -methyltestosterone is oxandrolone, which contains a lactone in the A ring (oxygen bio-isostere of ring A) and, therefore, is susceptible to in vivo hydrolysis. It has three times the anabolic activity of 17 α -methyltestosterone but exhibits slight androgenic activity (76). A pyrazole heterocyclic compound used for its anabolic effects is stanozolol (76).

The anabolic steroid oxymetholone (Anadrol) was used primarily to stimulate production of erythropoietin in the treatment of anemias resulting from bone marrow failure before the advent of erythropoietin, but is seldom used today due to its association with hepatitis and liver tumors.

Testolactone (Teslac), a 18-oxasteroid, is a D-homo-oxoandrostandienedione analog, with ring D being a six-membered lactone ring. Although testolactone possesses some anabolic activity with weak androgenic effects, it was used primarily in the treatment of breast cancer as a noncompetitive irreversible inhibitor of aromatase to suppress the formation of estrogens that would stimulate the growth of breast tissue (77). It is primarily excreted in the urine unchanged, but it is metabolized in the liver by partial reduction of the 4-ene double bond in ring A to the 5 β -metabolite (*cis* A/B ring juncture). Testolactone was discontinued in 2008 and is no longer available for clinical use.

Alkylation in the 1, 2, 7, and 18 positions of the androstane molecule generally increases anabolic activity (74). One of these derivatives, methenolone acetate, is an example of a potent anabolic agent that does not have an alkyl substituent at the 17 α -position. A halogenated anabolic agent used in about the same dosage is chlortestosterone acetate (Fig. 40.10).

Androgens, having no methyl group in position 10 of the steroid nucleus, are an important class of anabolic agents often referred to as the 19-norandrogens, as shown in Figure 40.10. The removal of the 19-CH₃ group of the androgen results in reduction of its androgenic properties but retention of its anabolic, tissue-building properties. These steroids can be synthesized by the Birch reduction of the aromatic A ring of a 3-methoxy estrogen to a 2,5(10)-estradiene. Cleavage of the enol ether with HCl results in the 19-nortestosterone derivative. In animal assays, 19-nortestosterone has about the same anabolic activity as the propionate ester of testosterone, but its androgenic activity is much lower. Because 19-nortestosterone showed some separation of anabolic and androgenic activities, related analogs were synthesized and biologically investigated. Two of the more potent members of the series are norethandrolone and ethylestrenol and are shown in Figure 40.10. Norethandrolone has a better ratio of anabolic to androgenic activity than either 19-nortestosterone or 17 α -methyl-19-nortestosterone (78). Both androgenic and progestational side effects have been observed with this agent. Ethylestrenol is more potent than norethandrolone as an anabolic agent and is used in a dosage of 4 mg per day orally.

Nandrolone phenpropionate and nandrolone decanoate are esters of 19-nortestosterone, as shown in Figure 41.10. When administered IM, slow in vivo hydrolysis of the ester occurs, releasing free 19-nortestosterone over a prolonged period. Nandrolone decanoate is the longer-acting ester intended for deep IM injection, preferably into the gluteal muscle, in the treatment of anemia associated with renal insufficiency. Nandrolone phenpropionate has a shorter duration of action than the decanoate and is used in the treatment of metastatic breast cancer in women.

Abuse of Steroidal Anabolic Agents to Enhance Athletic Performance

Performance-enhancing substances are now a point of major interest for athletes, government, and news media. These substances are having a major impact on sports and the public in general. It appears that we are headed for much greater antidoping efforts in sports. A great deal of interest has recently been shown in “designer” anabolic steroids for their high-muscle-building effects, as shown in Figure 40.11. Tetrahydrogestrinone and desoxymethyltestosterone (Fig. 40.11) brought a great deal of interest to the performance-enhancing area, because their use was very difficult to detect (79,80). Tetrahydrogestrinone is thought to have been derived from gestrinone, a substance that has been used for the treatment of a variety of gynecologic disorders. Tetrahydrogestrinone also

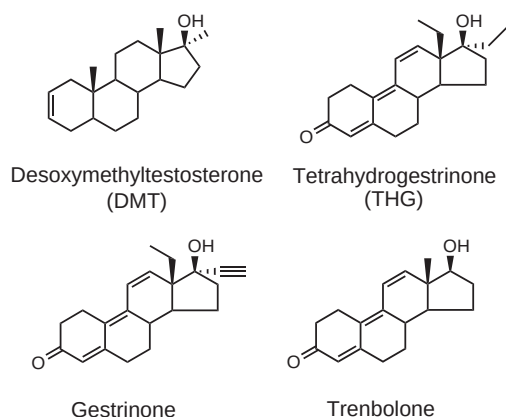


FIGURE 40.11 Illegal anabolic agents.

is related to trenbolone, which has been used by body builders and by ranchers to build up cattle before marketing. Before tetrahydrogestrinone, both gestrinone and trenbolone had been on the banned anabolic steroid list of the International Olympic Committee. Tetrahydrogestrinone was very difficult to trace, however, because it was unstable under the normal conditions of testing for anabolic steroids. Once a suitable assay was developed, it was possible to go back and test samples of athletes around the world, and several were found to have taken tetrahydrogestrinone.

Nonsteroidal Androgens

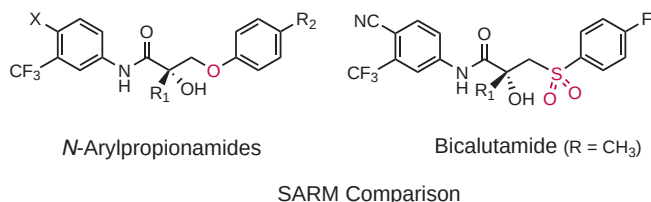
Selective Androgen Receptor Modulators (SARMs)

The successful marketing and clinical application of selective estrogen receptor modulators (see Chapter 41) raised the possibility of developing selective ligands for other members of the nuclear receptor superfamily. The concept of SARMs (81–83) recently emerged—namely, a compound that is an antagonist or weak agonist in the prostate but agonist in the bone and muscle and is orally available with low hepatotoxicity. For an ideal SARM, the antagonist or weak agonist activity in the prostate will reduce concern for the potential to stimulate nascent or undetected prostate cancer, whereas the strong agonist activity in the muscle and bone can be used to treat muscle-wasting conditions, hypogonadism, and/or aging-related frailty. Currently, research on SARMs is in its early stages—namely, preclinical discovery and the early phase of clinical development. Late-stage clinical trials planned in the next 3 to 5 years, however, should reveal the true promise of this exciting new therapeutic class of drugs.

The SARM pharmacophores can be classified into four categories: *N*-arylpropionamide (83), bicyclic hydantoin (84), tricyclic quinolines (85), and tetrahydroquinoline (86) analogs, as shown in Figure 40.12. These nonsteroidal AR ligands are not substrates for aromatase or 5 α -reductase but exhibit affinity as full AR agonists in anabolic organs (e.g., muscle and bone) or as partial AR agonists in androgenic tissues (e.g., prostate and seminal vesicles).

STRUCTURE–ACTIVITY RELATIONSHIPS FOR THE *N*-ARYLPROPIONAMIDES

The majority of published preclinical research has focused on a series of *N*-arylpropionamide analogs as AR agonists or partial agonists, using the key structural elements of bicalutamide, an androgen antagonist (83,87).



A series of chiral bicalutamide analogs, which bear electron-withdrawing groups (either a cyano or a nitro group at the 4-position and a trifluoromethyl group at the 3-position) in ring A, with a fluoro or acetamino substituents at the para position in ring B (R₂) demonstrated high in vitro AR binding affinity and in vivo androgenicity and anabolic activity in rats (88,89). The *R*-isomer analogs, which have a trifluoromethyl group instead of a methyl group at the R₂ position, exhibited higher AR binding affinity and more potent activity than their corresponding *S*-isomers (90). The sulfide analogs (replacement of ether oxygen with sulfur) exhibited greater AR binding affinity, except that hepatic oxidation of the sulfur linkage led to rapid in vivo inactivation and reduced efficacy. Partial androgen agonist activity was observed when position 3 of ring A was substituted with a trifluoromethyl group (91). Replacing the aromatic ring A with a heterocyclic ring derivative failed to retain AR binding affinity, which probably arose from steric hindrance on binding with the AR; however, a heterocyclic B ring retained AR binding affinity. Small size electron-withdrawing moieties at R₃, such as fluoro, chloro, nitro, or cyano groups, are optimum. Because aromatic nitro groups are associated with hepatotoxicity, the nitro group was replaced with a nonreducible electron-withdrawing group (i.e., a cyano group), which gave the most potent and efficacious *N*-arylpropionamide SARM with favorable pharmacokinetic properties. As evidenced from

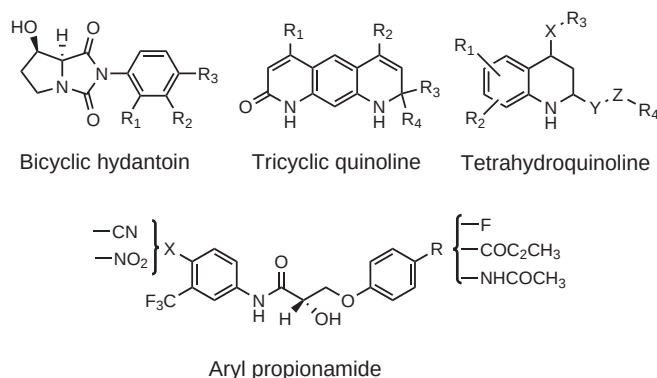


FIGURE 40.12 Selective androgen receptor modulators.

structure–activity relationship studies, minor differences in ligand structure can lead to either agonist or antagonist activity. Full or partial agonist binding to AR is influenced by stereoisomeric conformation as well as by steric and electronic effects of the substituents. Molecular modeling of *N*-arylpropionamide AR ligands was used in conjunction with pharmacology, pharmacodynamics, pharmacokinetics, and metabolism to examine and optimize structural properties.

Results from *in vitro* and *in vivo* animal studies suggest that the therapeutic promise of SARMs as treatment for muscle wasting, osteoporosis, hormonal male contraception, BPH, or other conditions associated with aging or androgen deficiency—without unwanted side effects associated with testosterone—may be soon realized (92). The AR specificity and lack of steroidal-related side effects clearly distinguish these drugs from their steroidal predecessors and open the door for expanded clinical use of androgens. As the molecular mechanisms of action of SARMs on target tissues become more fully understood, the discovery of novel SARMs and expansion into broader therapeutic applications will be more feasible. Currently, research concerning SARMs is in preclinical discovery and the early phase of clinical development with the expectations that SARMs, with the beneficial pharmacologic activity of androgens without the unwanted side effects, will provide individual patients who have various androgen-dependent disorders with a significantly improved quality of life.

TREATMENT OF PROSTATIC DISEASES

Diseases of the prostate represent some of the greatest threats to men's health. Incidence rates for BPH escalate rapidly with age, from approximately 50% of men at age 50 years to approximately 90% at age 90 years in the United States. Drugs that inhibit the metabolism of testosterone to DHT (i.e., 5 α -reductase inhibitors) (93) or block urethral constriction (α_1 -adrenergic receptor antagonists) (94) are used as front-line treatment for urinary obstruction associated with BPH. Surgery is also commonly performed as treatment for early-stage prostate cancer (i.e., prostatectomy) and transurethral resection of the prostate, making these some of the most common surgeries performed on men.

Prostate cancer is the most common noncutaneous cancer and remains the second leading cause of death from cancer in American men (95). Androgen receptor antagonists (i.e., antiandrogens) and gonadotropin-releasing hormone (LHRH) analogs are routinely used for medical management of patients with early-stage prostate cancer, whereas patients with advanced prostate cancer are treated with anticancer chemotherapy.

Benign Prostatic Hyperplasia

BPH is the noncancerous proliferation of the prostate gland. The major problem associated with BPH is lower urinary tract symptoms. Approximately 80% of men will

develop BPH within their lifetime. Although the cause of BPH is not well understood, it occurs mainly in older men, and it does not develop in men whose testes were removed before puberty. As men age, the concentration of free testosterone in the blood decreases due to increases in SHBG, while the concentrations of estradiol increase due to aromatization in adipose tissue. Animal studies have suggested that BPH may result from the increased concentration of estradiol or DHT within the gland, which promotes cell growth (96). Men who do not produce DHT do not develop BPH (97).

The symptoms of BPH stem from obstruction of the urethra by an enlarged prostate and the gradual loss of bladder function, which results in incomplete emptying of the bladder. The symptoms of BPH vary, but the most common symptoms involve changes or problems with urination. The typical symptoms of BPH are obstructive (e.g., poor urine stream, dribbling, and large residual urine volume) and irritative (e.g., hesitancy, increased frequency of urination, and nocturia), which can significantly compromise the quality of life for men. The enlarging prostate increases the adrenergic tone of the prostate in patients with BPH, which results in further tightening of the urethra. When partial obstruction is present, urinary retention also can be brought on by alcohol, cold temperatures, a long period of immobility, or the ingestion of over-the-counter cold or allergy medicines that contain a sympathomimetic decongestant drug or anticholinergics.

Severe BPH can cause serious problems over time, including urinary retention and strain on the bladder, which can lead to urinary tract infections, bladder or kidney damage, bladder stones, and incontinence.

Before and during adulthood, DHT plays a critical role in determining prostate size, and multiple lines of evidence suggest the importance of DHT in the development of BPH (98,99). For instance, BPH does not develop in males with certain type 2 5 α -reductase mutations or in males with very low levels of androgen because of prepubertal castration or hypopituitarism-related hypogonadism. Moreover, clinical treatment of BPH either by chemical or surgical castration or by type 2 5 α -reductase inhibitor (e.g., finasteride or dutasteride) induces apoptosis of epithelial cells, which in turn significantly decreases the volume of the prostate (99). Recently, the role of age-dependent changes in the intraprostatic hormonal environment in the development of BPH was evaluated (93). Despite the aging-related decrease in testosterone and intraprostatic DHT production, an increased estradiol/DHT ratio was observed in the aging human prostate, which can be relevant to the development of BPH. Furthermore, estradiol is capable of inducing precancerous lesions and prostate cancer in aging dogs (100). Therefore, TRT in older men raises concern regarding acceleration of BPH and/or prostate cancer.

Surgical procedures often are used to reduce a large prostate mass, but there are early pharmacologic treatments for BPH including α_1 -adrenergic blockers, 5 α -reductase inhibitors, and phytotherapy.

The α_1 -adrenergic antagonists treat the increased adrenergic tone of the sympathetic nervous system by relaxing the muscles at the neck of the bladder and in the prostate, thereby reducing the pressure on the urethra and increasing the flow of urine (94). They do not cure BPH but, rather, help to alleviate some of the symptoms. Approximately 60% of men find that symptoms improve significantly within the first 2 to 3 weeks of treatment with an α_1 -antagonist. Alfuzosin, tamsulosin, and silodosin are used exclusively as first-line α_1 -adrenergic antagonists for the treatment of BPH, while doxazosin and terazosin have also been used to treat high blood pressure. Prazosin, another α_1 -antagonist, is not indicated for the treatment of BPH. Tamsulosin, alfuzosin, and silodosin are uroselective α_1 -adrenergic antagonists developed specifically to treat BPH.

The 5α -reductase inhibitors work by suppressing the production of intraprostatic DHT, thereby reducing the size of the prostate (93). Finasteride and dutasteride are the most commonly used drugs for this purpose. Unlike α_1 -antagonists, 5α -reductase inhibitors are able to reverse BPH to some extent and so may delay the need for surgery. Several months of treatment may be needed before the benefit is noticed.

α_1 -Adrenergic Antagonists

MECHANISM OF ACTION α_1 -Adrenoceptors are widely distributed in the human body and play important physiologic roles (see Chapter 10). Of the three α_1 -adrenoceptor subtypes (α_{1A} , α_{1B} , and α_{1D}), α_{1A} -adrenoceptors, expressed in prostate and urethral tissue, and α_{1D} -adrenoceptors, expressed mainly in the detrusor muscle of the bladder and the sacral region of the spinal cord, have been shown mediate smooth muscle contraction and antagonists provide relief from the symptoms of BPH. The vast majority of α -adrenoceptors expressed in the prostate are of the α_{1A} (70%) and α_{1D} subtypes (27%). α_{1B} -Adrenoceptor is known to be important in the regulation of blood pressure (101). The predominant expression of the α_{1A} -adrenoceptor subtype in the prostatic and urethral smooth muscle cells led to the design of drugs with

uroselectivity for this receptor subtype. Thus, alfuzosin (an aminoquinazoline), tamsulosin (an *N*-substituted, catecholamine-related sulfonamide), and silodosin (a substituted indole carboxamide) were designed for the treatment of BPH (Fig. 40.13) (99). Doxazosin and terazosin, along with prazosin, originally were used as anti-hypertensives but also were found to be effective for the treatment of BPH based on their common mechanism of action. Comparisons of the affinities (K_i , nmol/L) of the α_{1A} -adrenoceptor antagonists (Table 40.3) did not show substantial differences for the quinazoline α_1 -antagonists (alfuzosin, doxazosin, and terazosin) but some uroselectivity for tamsulosin and silodosin (94,102,103). In vivo studies showed that those α_{1A} -adrenoceptor antagonists without in vitro adrenoceptor subtype selectivity, such as alfuzosin and doxazosin, showed uroselectivity (terazosin was not uroselective), whereas tamsulosin, which exhibited in vitro selectivity for the α_{1A} -adrenoceptor, did not show the expected in vivo uroselectivity (104). Silodosin, the most recent addition to this class of drugs, showed both in vitro and in vivo uroselectivity. However, these differences between in vitro and in vivo studies suggest that these drugs modify urethral pressures in a manner that is not directly correlated with their selectivity for the cloned α_{1A} -adrenoceptor subtypes. It is apparent that the existing α_{1A} -adrenoceptor antagonists have different in vivo pharmacologic profiles that are not yet predictable from their receptor based on the current state of knowledge regarding the α_{1A} -adrenoceptor classification (104).

Thus, tamsulosin, alfuzosin, and silodosin are first-line drugs for the treatment of BPH and have no utility in treating hypertension, because they have fewer cardiovascular effects than terazosin and doxazosin. Some of the basic physicochemical and pharmacokinetic properties of the α_1 -antagonists are summarized in Table 40.3. Improvements in urine flow occur 4 to 8 hours after the first dose and in BPH symptoms within 1 week.

ADVERSE REACTIONS In patients with BPH, the most common adverse effects for α_1 -adrenergic antagonists are

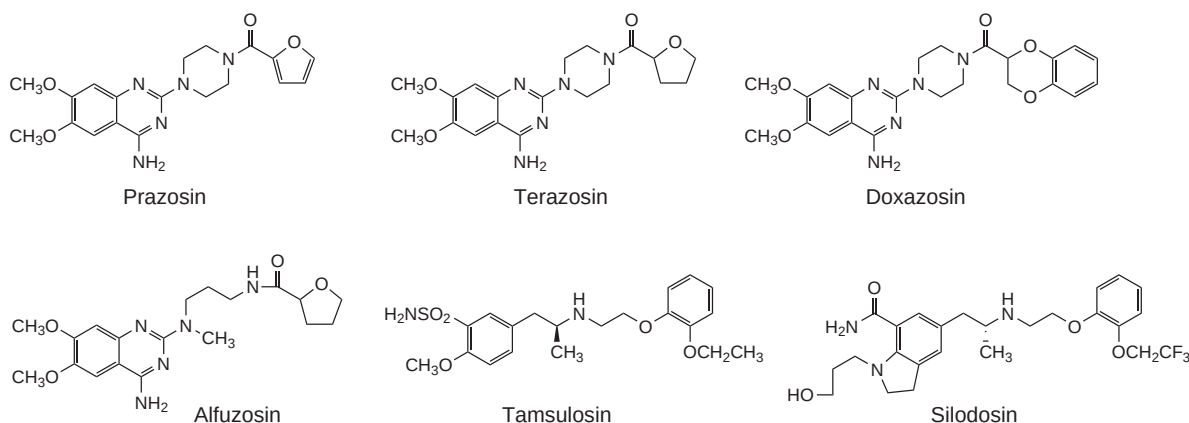


FIGURE 40.13 α_1 -Adrenergic antagonists for treatment of benign prostatic hyperplasia.

TABLE 40.3 Some Properties and Pharmacokinetics of the α_1 -Adrenergic Antagonists

Drug	Alfuzosin	Doxazosin	Silodosin	Tamsulosin	Terazosin
Trade name	Uroxatral	Cardura	Rapaflo	Flomax	Hytrin
cLogP ^a	-1.0 $\pm$ 0.4	0.7 $\pm$ 0.4	2.97	2.2 $\pm$ 0.4	-1.0 $\pm$ 0.4
logD ^a (pH 7)	-1.3	-0.5	0.5	-0.5	-1.0
Oral bioavailability (%)	65	65 (62–69)	32	<50 with food >90 fasted	90
Onset of action (weeks) ^b	<2	1–2	<1	1	2
Duration of action (hours)	>48	18–36	24	>24	>18
Protein binding (%)	82–90	98	97	94	95
Time to peak concentration (hours)	1–2	2–5	2–3	4–5	1
Volume of distribution (L/kg)	2.5–3.2	1.0–3.4	50	18	25–30
Elimination half-life (hours)	3–10	18–22	13	9–13 14–15 elderly	9–12
Cytochrome isoforms	3A4	3A4	3A4	3A4, 2D6	—
Excretion (%)	69 feces 24–30 urine (metabolites)	63–65 feces ~20 urine ~10 unchanged	33 urine 55 feces glucuronides	21 feces 76 urine 10 unchanged	55–65 feces 30 urine 10–20 unchanged
K_i (nmol/L) for α_{1A}	2.4	2.7	1.0	0.5	2.5

^aChemical Abstracts, American Chemical Society, calculated using Advanced Chemistry Development (ACD/Labs) Software V8.14 for Solaris (1994–2006 ACD/Labs).

^bTime for improvement in urine flow observed.

related to abnormal ejaculation and orthostatic hypotension, with vasodilation, dizziness, headache, and tachycardia also occurring in some patients during the first 2 weeks of treatment (105). Therefore, a dose titration usually is required, especially in patients taking doxazosin and terazosin or older than 60 years. These cardiovascular side effects are attributed to a nonselective blockade of α_1 -adrenoceptors present in vascular smooth muscle in addition to the required blockade of α_1 -adrenoceptors in prostate. No first-dose effect and fewer vasodilatory adverse events have been reported with silodosin and the sustained-release formulations of other drugs, which occur more frequently with the immediate-release formulation. At higher doses, orthostatic hypotension occurs more frequently. The first-dose phenomenon of orthostatic hypotension and syncope has been reported occasionally in elderly patients and in those concurrently receiving calcium antagonists, diuretics, and β -blockers. Concomitant use of the α_1 -adrenoceptor antagonists with PDE5 inhibitors (used commonly for ED in older men) is associated with a higher risk for dizziness and symptomatic hypotension.

CLASSES OF α_1 -ADRENERGIC ANTAGONISTS.

Quinazolines Prazosin was the first selective quinazoline α_1 -blocker to be discovered in the late 1960s as an

antihypertensive. Alfuzosin, doxazosin, and terazosin are structurally similar (a 4-amino-6,7-dimethoxyquinazoline ring system) but differ in the attached side chain as a piperazine ring or open-chain analog as shown in Figure 40.13. The other structural differences are the acyl groups attached to the second nitrogen of the piperazine or the aminopropyl chain. The differences in these groups afford dramatic differences in some of the pharmacokinetic properties for these agents (see Table 40.3). Perhaps most significant are the long half-lives and durations of action for these drugs that permit once-a-day dosing and, generally, lead to increased patient compliance.

Alfuzosin is a first-line uroselective drug for the treatment of BPH, but with no utility in treating hypertension, because it has fewer cardiovascular effects than terazosin and doxazosin. Alfuzosin is hepatically metabolized by 7-*O*-demethylation and *N*-dealkylation, primarily by CYP3A4, to inactive metabolites. In patients with moderate or severe hepatic insufficiency, a reduction in clearance resulted in a three to four times increase in its plasma concentrations, which may require a reduction in dose.

Doxazosin is primarily metabolized by 7-*O*-demethylation, hydroxylation of the benzodioxan ring, and oxidation of the piperazine ring to inactive

metabolites. In patients with renal insufficiency, the elimination half-life was not significantly different from healthy volunteers.

Terazosin is similarly metabolized via 7-*O*-demethylation and *N*-dealkylation to four metabolites: 6- and 7-*O*-demethyl terazosin, the piperazine derivative of terazosin, and the diamine metabolite of the piperazine compound.

Catecholamine-Sulfonamide Tamsulosin exhibits uroselectivity and is a first-line drug for the treatment of BPH, with no utility for treating hypertension, because of its fewer cardiovascular effects. Tamsulosin is *O*-deethylated by CYP3A4 to phenolic metabolites that are conjugated to glucuronide or sulfate before renal excretion and by *O*-demethylation and 3'-hydroxylation to catechol metabolites that also are conjugated with glucuronide and sulfate.

Indole Carboxamide Silodosin is the most uroselective of the α_1 -adrenoceptor antagonists, demonstrating 160- and 50-fold higher affinity for binding to α_{1A} - as compared to α_{1B} - and α_{1D} -adrenoceptors, respectively (103). Silodosin undergoes extensive metabolism through glucuronidation, alcohol and aldehyde dehydrogenase, and CYP3A4. The glucuronide conjugate (formed via direct glucuronidation of the alcohol) of silodosin is active, has a longer half-life (24 hours) than the parent drug, and achieves a plasma exposure (area under the curve) four times higher than that of silodosin.

5 α -Reductase Inhibitors

The development of BPH requires a combination of intraprostatic DHT and the aging process. Although not elevated in BPH, levels of DHT in the prostate remain at a physiologic levels with aging despite a decrease in plasma testosterone. Adult males with genetically inherited, type 2 5 α -reductase deficiency also have decreased DHT levels. These 5 α -reductase-deficient males have a small prostate gland throughout life and do not develop BPH. Except for the associated urogenital defects that are present at birth, no other clinical abnormalities related to 5 α -reductase deficiency have been observed in these individuals.

MECHANISM OF ACTION Inhibitors of DHT biosynthesis can result in a decrease in both circulating and target-tissue (prostate and skin) DHT concentrations, thus blocking its androgenic action in these tissues. The critical enzyme targeted for DHT inhibition is 5 α -reductase, which converts testosterone to DHT. The first agent to demonstrate 5 α -reductase inhibition was a progestin analog, medrogestone (106) (Fig. 40.14). Two azasteroid-17-amide derivatives of medrogestone have been developed as potent irreversible inhibitors of 5 α -reductase and approved for the treatment of BPH: finasteride, a selective inhibitor of type 2 5 α -reductase (107,108), and dutasteride, a nonselective inhibitor of type 1 and type

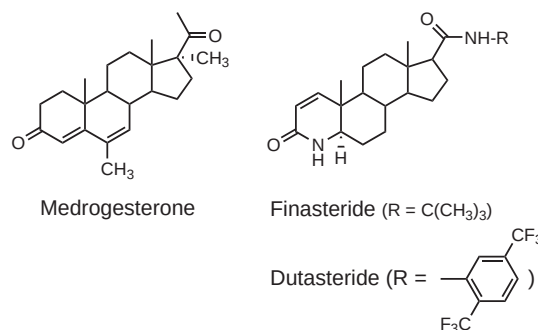


FIGURE 40.14 5 α -Reductase inhibitors for the treatment of benign prostatic hyperplasia.

2 5 α -reductase (109) (Fig. 40.14). Thus, the inhibition of type 2 5 α -reductase suppresses the metabolism of testosterone to DHT, resulting in significant decreases in plasma and intraprostatic DHT concentrations (108–110).

Finasteride and dutasteride are both mechanism-based inhibitors of type 1 and type 2 5 α -reductase isoenzymes that inactivate 5 α -reductase by an apparent irreversible modification of 5 α -reductase (111,112). The inhibition constants (median inhibitory concentrations [IC₅₀]) in Table 40.4 suggest that finasteride is 30 times more selective for type 2 5 α -reductase, whereas dutasteride appears to be approximately 2 times more potent as an inhibitor of type 2 5 α -reductase than as an inhibitor of type 1 5 α -reductase. The reduction of finasteride to dihydrofinasteride proceeds through an enzyme-bound, NADP-dihydrofinasteride adduct (see Chapter 8) (111). The mechanism-based inhibition explains the exceptional potency and specificity of finasteride and dutasteride in the treatment of BPH. This concept of mechanism-based inhibition may have application to the development of other inhibitors of pyridine nucleotide-linked enzymes.

DRUG INTERACTIONS Because finasteride and dutasteride are metabolized primarily by CYP3A4, the CYP3A4 inhibitors, such as ritonavir, ketoconazole, verapamil, diltiazem, cimetidine, and ciprofloxacin, may increase these drugs' blood levels and, possibly, cause drug–drug interactions. Clinical drug interaction studies have shown no pharmacokinetic or pharmacodynamic interactions between dutasteride and tamsulosin or terazosin, warfarin, digoxin, and cholestyramine.

FINASTERIDE The selective inhibition of the type 2 5 α -reductase isozyme by finasteride produces a rapid reduction in plasma DHT concentration, reaching 65% suppression within 24 hours of administering a 1-mg oral tablet (113). At steady-state, finasteride suppresses DHT levels by approximately 70% in plasma and by as much as 85% to 90% in the prostate. The remaining DHT in the prostate likely is the result of type 1 5 α -reductase. The mean circulating levels of testosterone and estradiol

TABLE 40.4 Some Properties and Pharmacokinetics of the 5 α -Reductases

Drugs	Finasteride	Dutasteride
Trade name	Proscar	Avodart
cLogP ^a	3.2 $\pm$ 0.4	5.6 $\pm$ 0.6
logD ^a (pH 7)	3.2	5.6
Oral bioavailability (%)	65 (26–170)	60 (40–94)
Onset of action (hours)	<24	—
Duration of action (hours)	—	>5 weeks
Protein binding (%)	90	99
Time to peak concentration (hours)	—	2–3
Volume of distribution (L/kg)	76 (44–96)	300–500
Elimination half-life (hours)	5–6 (18–60 years)	5 weeks
		>8 (60+ years)
Cytochrome isoforms	3A4	3A4
Active metabolites	None	6- <i>p</i> -OH
Excretion (%)	57 feces and 40 urine as metabolites	40 feces metabolites
		5 unmetabolized urine
IC ₅₀ (nmol/L)	313 Type 1	3.9 Type 1
	11 Type 2	1.8 Type 2

^aChemical Abstracts, American Chemical Society, calculated using Advanced Chemistry Development (ACD/Labs) Software V8.14 for Solaris (1994–2006ACD/Labs).

remained within their physiologic concentration range. Long-term therapy with finasteride can reduce clinically significant end points of BPH, such as acute urinary retention or surgery. Finasteride is most effective in men with large prostates. Finasteride has no affinity for the AR and no androgenic, antiandrogenic, estrogenic, anti-estrogenic, or progestational effects.

Pharmacokinetics The mean oral bioavailability of finasteride is 65%, as shown in Table 40.4, and is not affected by food (113). Approximately 90% of circulating finasteride is bound to plasma proteins. Finasteride has been found to cross the blood–brain barrier, but levels in semen were undetectable (<0.2 ng/mL). Finasteride is extensively metabolized in the liver, primarily via CYP3A4 to two major metabolites: monohydroxylation of the *t*-butyl side chain, which is further metabolized via an aldehyde intermediate to the second metabolite, a monocarboxylic acid (Fig. 40.15). The metabolites show approximately 20% the inhibition of finasteride for 5 α -reductase. The mean terminal half-life

is approximately 5 to 6 hours in men between 18 and 60 years of age and 8 hours in men older than 70 years of age. Following an oral dose of finasteride, approximately 40% of the dose was excreted in the urine as metabolites and approximately 57% in the feces. Even though the elimination rate of finasteride is decreased in the elderly, no dosage adjustment is necessary. No dosage adjustment is necessary in patients with renal insufficiency. A decrease in the urinary excretion of metabolites was observed in patients with renal impairment, but this was compensated for by an increase in fecal excretion of metabolites. Caution should be used during administration to patients with liver function abnormalities, because finasteride is metabolized extensively in the liver. Finasteride can be absorbed through the skin. As such, tablets, especially broken ones, should not be handled by women who are pregnant or could become pregnant due to a risk of fetal exposure.

DUTASTERIDE Similar to finasteride, dutasteride is a competitive and mechanism-based inhibitor not only of type 2 but also of type 1 5 α -reductase isoenzymes, with which stable enzyme-NADP adduct complexes are formed, inhibiting the conversion of testosterone to DHT (112). The suppression of both type 1 and type 2 isoforms results in greater and more consistent reduction of plasma DHT than that observed for finasteride (114–116). The more effective dual inhibition of type 1 and type 2 5 α -reductase isoforms lowers circulating DHT to a greater extent than with finasteride and shows advantages in treating BPH and other disease states (e.g., prostate cancer) that are DHT-dependent.

The maximum effect of 0.5-mg daily doses of dutasteride on the suppression of DHT is dose-dependent and is observed within 1 to 2 weeks. After 2 weeks of 0.5-mg daily dosing, median plasma DHT concentrations were reduced by 90%, and after 1 year, the median decrease in plasma DHT was 94% (115,116). The median increase in plasma testosterone was 19% but remained within the physiologic range. The drug also reduced serum prostatic-specific antigen (PSA) by approximately 50% at 6 months and total prostate volume by 25% at 2 years. Dutasteride produced improvements in quality of life and peak urinary flow rate and reduction of acute urinary retention without the need for surgery. The main side effects are ED, decreased libido, gynecomastia, and ejaculation disorders. Long-term use (>4 years), however, did not reveal increased onset of sexual side effects. A combination product (Jalyn) containing dutasteride (0.5 mg) and tamsulosin (0.4 mg) was approved by the U.S. Food and Drug Administration in 2010 based on clinical studies showing that the combination was more effective than either monotherapy as assessed by prostate symptom score and maximal urine flow rates. The decreased adrenergic tone associated with tamsulosin and long-term reduction in prostate size (obstruction) associated with dutasteride provide rapid and sustained symptomatic relief for men with BPH.

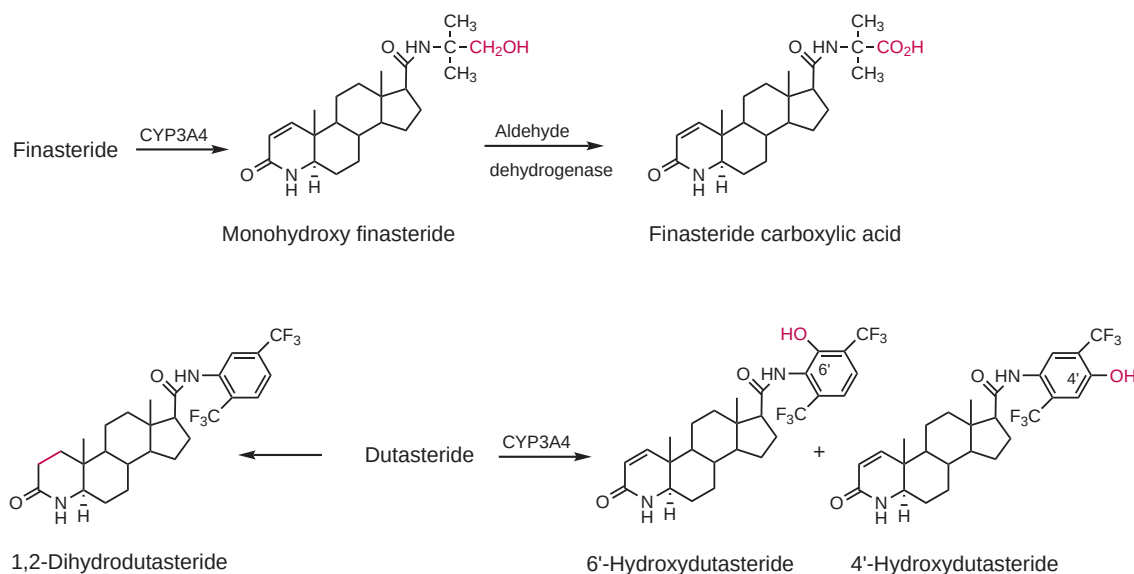


FIGURE 40.15 Metabolites of finasteride and dutasteride.

Pharmacokinetics Following oral administration, peak plasma concentrations of dutasteride occur in approximately 2 to 3 hours, with a bioavailability of approximately 60% (Table 40.4), and no meaningful reduction in absorption occurs with food (113,114). Dutasteride is highly bound to plasma proteins 99%. The concentrations of dutasteride in semen averaged approximately 3 ng/mL, with no significant effects on DHT plasma levels of sex partners. Dutasteride is extensively metabolized in humans by CYP3A4 to three major metabolites: 4'-hydroxydutasteride and 1,2-dihydrodutasteride, which are less potent than parent drug, and 6'-hydroxydutasteride, which is comparable to the parent drug as an inhibitor of both type 1 and type 2 5α-reductases (Fig. 40.15). Dutasteride and its metabolites are excreted mainly (40%) in feces as dutasteride-related metabolites. The terminal elimination half-life of dutasteride is approximately 5 weeks. Because of its long half-life, plasma concentrations remain detectable for up to 4 to 6 months after discontinuation of treatment. No dose adjustment is necessary in elderly patients, even though its half-life increased with age from approximately 170 hours in men between age 20 and 49 years to 300 hours in men older than age 70 years (113,116). No adjustment in dosage is necessary for patients with renal impairment. Like finasteride, dutasteride can be absorbed through the skin. As such, dutasteride capsules should not be handled by women who are pregnant or could become pregnant.

Phytotherapy

A number of plant extracts are popularly used to alleviate BPH, although formal evidence that they are effective is often scanty (117). Extracts of the saw palmetto berry (*Serenoa repens*) are widely used for the treatment of BPH, often as an alternative to pharmaceutical agents. In a national survey conducted in 2002, 1.1% of the adult

male population in the United States, or approximately 2.5 million males, reported using saw palmetto. The herb is widely used in Europe, where half of the German urologists prefer prescribing plant-based extracts rather than synthetic drugs. The most common nonstandardized preparation used is either the hexane-extract (Permixon) or the ethanol or carbon dioxide extraction of the dried ripe fruit from the American dwarf saw palmetto plant (*S. repens*), which is rich in fatty acids and plant sterols. The plant sterols appear to be the primary active constituents. The U.S. Pharmacopeia states that the liposterolic extract product should contain 70% to 95% fatty acids and 0.2% to 0.5% sterols. Other substances in the extracts include polyphenolic compounds and flavonoids. The usual therapeutic dose of the extracts is 320 mg daily.

Serenoa repens had been popular in the United States during the 19th century as a treatment for a variety of urogenital disorders and had been mentioned as a treatment for prostate problems as early as 1899. Research into the effects of *S. repens* in many European countries appeared to confirm a positive action on BPH. The mechanism of action for the saw palmetto is not clearly established, but the sterols may have, as one mechanism of action, the inhibition of 5α-reductase and a decrease in DHT production. Therapeutic results should be expected in 6 to 8 weeks, but clinical efficacy is observed with BPH for 6 months or longer. The liposterolic extract is largely devoid of the side effects noted for prescription BPH drugs.

Although most previous randomized trials of saw palmetto have reported small improvements in the symptoms of BPH or in urinary flow rates, these studies were limited by the small numbers of subjects enrolled, their short duration, failure to use standardized products, their failure to use standard outcome measures, and the lack of information from participants concerning how

effectively the placebo was blinded. Using widely accepted outcome measures from the American Urological Association (AUA) and a matched placebo capsule, a randomized, 1-year, double-blind saw palmetto trial (funded by the National Institutes of Health National Institute on Complementary and Alternative Medicine) was performed to determine the efficacy of saw palmetto for the treatment of BPH (118). A total of 225 men age 50 years and older with documented disease received 160 mg of a standardized saw palmetto extract twice a day or a matching-placebo capsule. Over the course of a year, the men made eight office visits and were evaluated for assessment of AUA standardized changes, which included maximal urine flow, postvoid residual urine volume, prostate size, and other health-related outcomes. In contrast to most previous studies, this study reported no significant benefit of saw palmetto on urinary symptoms in terms of objective measures of BPH over a 1-year period.

Despite the differences between these studies, the weight of evidence suggests that saw palmetto may induce mild to moderate improvements in urinary symptoms and flow measures.

Other Western herbs that have been investigated for the treatment of BPH include pumpkin seeds (*Cucurbita pepo*), nettle root (*Urtica dioica* or *Urtica urens*), bee pollen (particularly that from the rye plant), African potato (tubers of *Hypoxis rooperi*), and the African tree *Pygeum africanum*, also known as *Prunus africanum*. In most cases, but particularly with pumpkin seeds and African potato, the main active components are sterols, such as β -sitosterol, which also has been used for BPH. Triterpenoids in *Pygeum* sp. also have been proposed to be active components, potentially having the action of reducing prostate swelling. Among the Chinese herbs recommended for BPH, the iridoid glycosides may be the active components from plantago seed, catalpol from rehmannia, and morroniside from cornus (an ingredient in the rehmannia formulas). Iridoids have not been found in the Western herbal therapies for BPH and represent a potential new area for future investigation. Iridoids are the recognized active constituents of the Western herb chaste tree berry (*Vitex agnus costus*), which has been shown to reduce prolactin levels in women; elevated prolactin may be a risk factor for prostate enlargement in men. Triterpenoids found in vaccaria and alisma (an ingredient in rehmannia formulas) could contribute to their therapeutic effects in a manner similar to that suggested for pygeum.

Prostatic Cancer

Prostate cancer is the second leading cause of cancer death and the most commonly diagnosed cancer in American men (95). Prostate cancer is more common in African American males, in whom it tends to be more aggressive and progressive, leading to advanced disease. The incidence of prostate cancer increases with age. Traditional treatments for localized prostate cancer include watchful waiting, surgery (radical prostatectomy), and external-beam radiation, whereas surgical and pharmacologic

approaches to induce androgen deprivation are used for advanced prostate cancer.

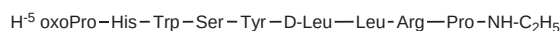
Huggins and Hodges won the 1966 Nobel Prize for describing the relationship between testosterone and prostate cancer, having shown that marked reductions in serum testosterone by castration or estrogen treatment caused metastatic prostate cancer to regress in 1941 (119). Androgen deprivation therapy has remained an important component in the treatment of prostate cancer since then. Testicular surgery (bilateral orchiectomy) to prevent testosterone production was once the most common treatment for advanced prostate cancer and is still used today. Although this surgery is not a cure, it delays the advance of the disease. Numerous refinements in therapy have occurred since this time, including androgen deprivation therapy with diethylstilbestrol (seldom used now), gonadotropin-releasing hormone agonists and antagonists, antiandrogens, and chemotherapy for advanced cases. Chemotherapy is used in the most advanced stages of the disease when the cancer is no longer responding to hormonal treatment.

Because prostate cancer typically grows slowly and causes no symptoms, digital rectal examination and the serum PSA test are most often used to screen for prostate cancer. The PSA test is also used to evaluate and manage other prostate problems, most notably BPH. However, the PSA screening test cannot tell the difference between prostate cancer and other prostate problems. PSA is a protein made by prostate tissue and prostate cancer cells. Men with prostate cancer often have elevated PSA levels, because the cancer cells make excessive amounts of this protein under the control of testosterone.

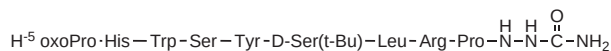
Radical prostatectomy (surgical removal of the prostate gland) is an effective treatment for cancer that remains localized to the prostate gland. However, prostate cancer may have already invaded the surrounding tissues (e.g., pelvic lymph nodes) at the time of surgery. Months, even years, may pass before this latent prostate cancer produces sufficient PSA to be detected by modern assays. When detected, increasing levels of PSA are referred to as biochemical recurrence and signal the re-emergence of the disease and the progression to advanced prostate cancer, with a median survival of approximately 10 to 15 years (120). Advanced prostate cancer in its earliest stages is thought to be dependent on androgens (i.e., testosterone and DHT) for growth. As such, androgen deprivation therapy, either by surgical castration or pharmacologic suppression of LH, is used as first-line treatment for advanced prostate cancer.

Luteinizing Hormone-Releasing Hormone Therapy

The goal of LHRH therapy for advanced prostate cancer is to suppress the production of testosterone in the testis by shutting down the production of LH by the pituitary. Both LHRH agonists and LHRH antagonists are used. The LHRH agonists (Fig. 40.16), such as leuprolide acetate (Lupron Depot and Eligard) and goserelin acetate (Zoladex), when administered in a continuous,



Leuprolide



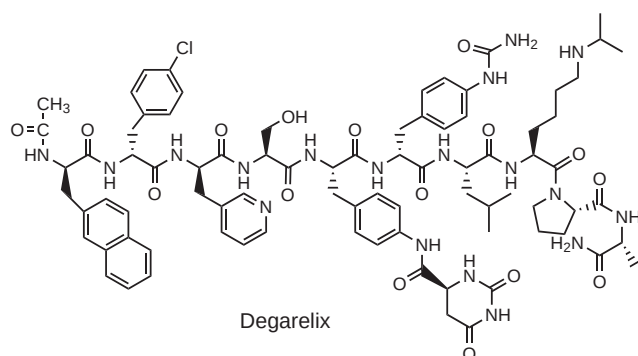
Goserelin

[Note: $\text{H}^5 \text{ oxoPro} = \text{pGlu (PyroGlu)}$]

FIGURE 40.16 Nonsteroidal antiandrogens.

nonpulsatile manner, suppress serum testosterone levels to a similar extent as surgical castration (i.e., $<50 \text{ ng/dL}$). The LHRH agonists bind with higher affinity to the LHRH receptor than does the endogenous hormone, causing desensitization of the LHRH receptor complex on the gonadotroph cells in the anterior pituitary gland (121). After a brief surge, pituitary biosynthesis and secretion of LH stops, serum LH levels fall, and testicular production of testosterone ceases (see the hypothalamus-pituitary-testis axis in Fig. 40.2). A surge in serum testosterone levels accompanies the surge in pituitary LH production within the first 3 to 7 days of initiation of therapy and can exacerbate symptoms (serum PSA levels or bone pain) in some men. Castration (i.e., serum testosterone levels $<50 \text{ ng/dL}$) is generally achieved with 2 weeks after administration of the first dose. A variety of depot formulations for intramuscular administration of leuprolide and goserelin are available, with durations of action ranging from 1 to 4 months.

Degarelix (Firmagon) is the only LHRH antagonist approved for use in the treatment of advanced prostate cancer. It competes reversibly with the endogenous LHRH for receptor binding on the pituitary gonadotroph cells and directly inhibits LH production.



Because degarelix acts as an antagonist, there is no surge in pituitary LH or testicular androgen production as is observed with the LHRH agonists. As such, approximately 96% of men achieve castration levels of serum testosterone within 3 days of receiving their first dose (240 mg as two subcutaneous injections of 120 mg) of degarelix. Lower doses of degarelix (80 mg) are administered every 28 days thereafter to maintain castration.

Androgen deprivation therapy with LHRH significantly reduces serum testosterone levels, thereby

depriving prostate cancer cells of their primary signal for growth (i.e., testosterone) and alleviating or easing the symptoms associated with advanced prostate cancer. However, LHRH agonists and antagonists are not cures for prostate cancer. They are considered as palliative care. The most common side effects associated with LHRH therapy are hot flashes, but osteoporosis (due to androgen and estrogen deprivation), impotence, loss of libido, and changes in serum lipids may also occur. Long-term use of LHRH therapy is associated with osteoporosis, decreased cognitive abilities, fatigue, and vascular stiffness. Symptoms may worsen over the first few weeks of treatment. Periodic monitoring of PSA and plasma testosterone levels is recommended. Tumor flare reactions may occur transiently but can be prevented by antiandrogens or by short-term estrogens at low dose for several weeks.

Antiandrogens (Androgen Antagonists)

Treatment for advanced prostate cancer also involves the use of hormone-blocking drugs called antiandrogens. The goal of antiandrogen therapy is to block the actions of testosterone and DHT on ARs. Antiandrogens, however, are not a cure for prostate cancer. Nonsteroidal antiandrogens (Fig. 40.17), such as flutamide, nilutamide, and bicalutamide, are referred

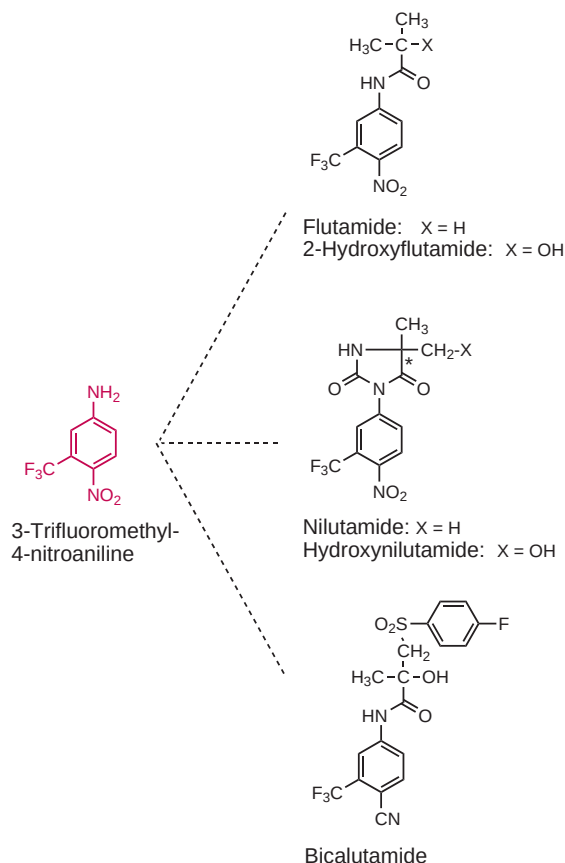


FIGURE 40.17 Steroidal antiandrogens.

to as pure antiandrogens, because they bind exclusively to AR and, thus, are devoid of antigonadotropic, antiestrogenic, and progestational effects (122). These agents have advantages over steroidal antiandrogens, such as megestrol acetate or cyproterone acetate (Fig. 40.18), in terms of specificity, selectivity, and pharmacokinetic properties.

Antiandrogens block the binding of DHT at the AR and block or diminish the effectiveness of androgens in androgen-sensitive tissues. Such compounds have shown potential therapeutic use in the treatment of acne, virilization in women, and hyperplasia and neoplasia of the prostate (123). Several steroidal and nonsteroidal agents have demonstrated antiandrogenic activity. Cyproterone acetate suppresses gonadotropin release and binds with high affinity to the AR (124,125). Oxandrolone also acts by competing for the receptor binding sites (126). A novel AR antagonist, WIN 49,596, has been described (127) that contains a fused pyrazole ring at carbons 2 and 3 of the steroid nucleus. A potent nonsteroidal antiandrogen, flutamide, has been shown to compete with DHT for the AR (128). Its hydroxylated metabolite is a more powerful antiandrogen *in vivo*, and it has a higher affinity for the receptor than the parent compound (129).

Antiandrogens are useful for the treatment of advanced prostate cancer during its early stages. Although approved as monotherapy for the treatment of advanced prostate cancer in Europe, antiandrogens are only indicated for combined use with an LHRH analog in the United States. Prostate cancer often advances to a “castration-resistant” state, in which the disease progresses in the presence of continued LHRH analog therapy or even after surgical castration, suggesting the development of prostate cancer cells that no longer require androgen for their growth or the ability of adrenal or intratumoral androgens to support tumor growth. Antiandrogens are most commonly used in the latter stages of the disease in order to achieve total androgen blockade. Instances of antiandrogen withdrawal syndrome also have been reported after prolonged treatment with antiandrogens. Antiandrogen withdrawal

syndrome is commonly observed clinically and is defined in terms of the tumor regression or symptomatic relief observed on cessation of antiandrogen therapy. The AR mutations that result in receptor promiscuity and the ability of these antiandrogens to exhibit agonist activity may account, at least in part, for this phenomenon. For example, hydroxyflutamide and bicalutamide act as AR agonists in AR mutants with an alanine residue at position 877 or leucine residue at position 741 (as opposed to threonine or tryptophan residues, respectively, which are present at these positions in the wild-type AR) (130,131).

The search for nonsteroidal antiandrogens led to the development of the substituted toluidides, flutamide and bicalutamide, and nilutamide, a hydantoin that is structurally related to the toluidides (Fig. 40.17). These compounds are pure antiandrogens and compete with DHT for the human prostate AR. They are used in combination with other drugs in the treatment of advanced prostate cancer. Although these compounds possess no intrinsic hormonal activity, their antiandrogenic mechanism of action is via competitive blockade of ARs for DHT in the hormone-sensitive tumor cells of the prostate (132). As a result of this antagonism, androgen-dependent DNA and protein synthesis is inhibited, causing arrest or regression of the prostatic tumor. Because these nonsteroidal antiandrogens are metabolized extensively in the liver, they should be used with caution in patients who have liver function abnormalities.

SPECIFIC DRUGS

Bicalutamide Bicalutamide is a nonsteroidal pure antiandrogen given at a dosage of 150 mg once daily as monotherapy for the treatment of early (localized or locally advanced) nonmetastatic prostate cancer in Europe (133). It also can be used at a lower dosage (50 mg) in combination with a LHRH analog or surgical castration for the treatment of advanced prostate cancer in the United States. Bicalutamide is a racemate, and its antiandrogenic activity resides almost exclusively in the (*R*)-enantiomer, which has an approximately four-fold higher affinity for the prostate AR than hydroxyflutamide. The (*S*)-enantiomer has no antiandrogenic activity. (*R*)-Bicalutamide is slowly absorbed, but absorption is unaffected by food (134). It has a long plasma elimination half-life of 1 week and accumulates approximately 10 times in plasma during daily administration (Table 40.5). (*S*)-Bicalutamide is much more rapidly absorbed and cleared from plasma. At steady-state, the plasma levels of (*R*)-bicalutamide are 100 times higher than those of (*S*)-bicalutamide. Although mild to moderate hepatic impairment does not affect its pharmacokinetics, evidence suggests slower elimination of (*R*)-bicalutamide in subjects with severe hepatic impairment (134). Bicalutamide metabolites are excreted almost equally in urine and feces, with little or no unchanged drug excreted in urine. Unmetabolized drug predominates in the plasma. Following oral administration, the racemate displays stereoselective oxidative metabolism

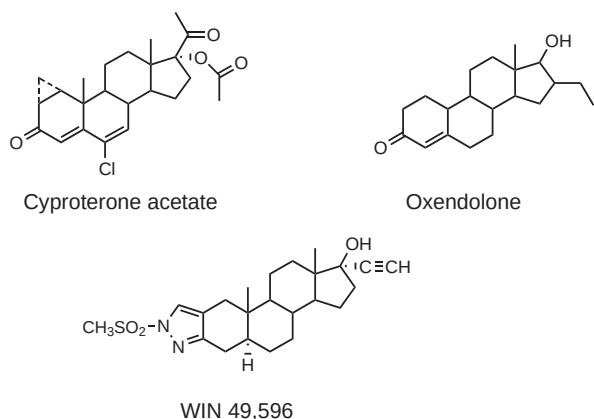


FIGURE 40.18 17-Hydroxylase/lyase inhibitors for prostate cancer.

TABLE 40.5 Some Properties and Pharmacokinetics of the Antiandrogens

Drugs	Bicalutamide	Flutamide	Nilutamide
Trade name	Casodex	Eulexin	Nilandron
cLogP ^a	4.9 ± 0.7	3.7 ± 0.4	3.3 ± 0.6
logD (pH 7)	4.9	3.7	3.3
Oral bioavailability (%)	80–90	—	—
Onset of action (weeks) ^b	8–12	2–4	1–2
Duration of action	8 days	3 months to 2.5 years	1–3 months
Protein binding (%)	96	94–96	80–84
Time to peak concentration (hours)	31	2–3	1–4
Elimination half-life (hours)	~6	8 (10 active metabolite)	40–60
	10 elderly	60–120 met	
Cytochrome isoforms	3A4	1A2	Flavin monooxygenase, CYP2C
Active metabolites	None	2-hydroxy	Yes
Excretion (%)	43 feces	<10 feces	<10 feces
	34 urine/glucunomide	~28 urine	62 urine
	metabolites	<10 unchanged	<2 unchanged

^aChemical Abstracts, American Chemical Society, calculated using Advanced Chemistry Development (ACD/Labs) Software V8.14 for Solaris (1994–2006 ACD/Labs).

^bTime for significant improvement in prostate-specific antigen or other biomarkers.

of its (*R*)-enantiomer, with an elimination half-life of approximately 6 days. (*R*)-Bicalutamide is cleared almost exclusively by CYP3A4-mediated metabolism, but glucuronidation is the predominant metabolic route for (*S*)-bicalutamide. No evidence indicates CYP3A4 induction in humans.

Flutamide After oral administration (usual dose is 250 mg every 8 hours), flutamide is completely absorbed from the gastrointestinal tract and undergoes extensive first-pass metabolism by CYP1A2 to its major metabolite, 2-hydroxyflutamide, and its hydrolysis product, 3-trifluoromethyl-4-nitroaniline (Fig. 40.17) (129). 2-Hydroxyflutamide is a more powerful antiandrogen *in vivo*, with higher affinity for the receptor than that of flutamide (135). 2-Hydroxyflutamide has an elimination half-life of approximately 8 hours. These studies show the principal role of CYP1A2 in the metabolism of flutamide to 2-hydroxyflutamide, with minor contribution from CYP3A4. 2-Hydroxyflutamide inhibits the metabolism of flutamide and both 2- and 4-hydroxylation of estradiol. Flutamide is a pure antagonist, whereas 2-hydroxyflutamide is a more potent AR antagonist but also can activate the androgenic receptor at higher concentrations (135). These findings raise the possibility that increased conversion of flutamide to 2-hydroxyflutamide or accumulation of 2-hydroxyflutamide in cells may contribute to the anomalous responses to flutamide that are observed in some advanced prostate cancers.

Rare reports of liver failure have been associated with flutamide, mandating regular monitoring of liver function tests in men taking flutamide therapy.

Nilutamide Nilutamide is a nitroaromatic antiandrogen used for the treatment of metastatic prostate carcinoma in men (124). It is a competitive antagonist of the AR. Nilutamide is a nitroaromatic hydantoin analog of flutamide, as shown in Figure 40.17, that is completely absorbed after oral administration, with a mean elimination half-life of approximately 50 hours (Table 40.5) (136). Daily oral doses of nilutamide range from 150 to 300 mg per day. One of the methyl groups attached to the hydantoin ring is stereoselectively hydroxylated to a chiral metabolite, which subsequently is oxidized to its carboxylic acid metabolite. Less than 2% of nilutamide is excreted unchanged in the urine. *In vitro*, the nitro group of nilutamide was reduced to the amine and hydroxylamine moieties by nitric oxide (NO) synthases, a flavin monooxygenase system (137). The therapeutic effects of nilutamide are overshadowed, however, by the occurrence of several adverse reactions mediated by toxic mechanisms, which are poorly investigated. The reduction of nilutamide is catalyzed by NO synthases via formation of a nitro anion free radical or via its reduction to its hydroxylamino derivative, which could explain some of the toxic effects of this drug (138). NO synthases also are involved in the formation of reactive NO and oxygen species and in the interactions with some xenobiotic compounds.

Chemotherapy and Immunotherapy

Because of the long time course for progression of prostate cancer, many men with advanced prostate cancer will die from other diseases (e.g., cardiovascular) and not from their cancer. However, advanced prostate cancer invariably progresses to a castration-resistant state, necessitating the need for increasingly harsh treatment to reduce androgen production or stop the growth of the cancer. Therapy with an LHRH agonist or antagonist alone is followed by combination therapy with an antiandrogen, which is ultimately followed by cancer chemotherapy or immunotherapy for those who are no longer responsive to any of the available hormonal therapies.

Docetaxel (Taxotere) is the only chemotherapeutic agent specifically approved for use in men with advanced prostate cancer (see Chapter 37). Infusions of 75 mg/m² of docetaxel over 1 hour once every 3 weeks in combination with 5 mg of prednisone daily was shown to increase survival to 18.9 months, or approximately 2 months longer as compared to treatment with mitoxantrone and prednisone (139).

Sipuleucel-T (Provenge) (a therapeutic cancer vaccine) was approved for the treatment of asymptomatic or minimally symptomatic metastatic castrate-resistant (hormone-refractory) prostate cancer in 2010. Sipuleucel-T is an autologous cellular immunotherapy. Although the precise mechanism of action is unknown, sipuleucel-T is designed to induce an immune response targeted against prostatic acid phosphatase (PAP), an antigen expressed in most prostate cancers. Three doses are usually administered at approximately 2-week intervals, with each dose containing a minimum of 50 million autologous CD54⁺ cells activated with PAP-granulocyte-macrophage colony-stimulating factor. Patients must be given oral acetaminophen and an antihistamine, such as diphenhydramine, approximately 30 minutes prior to administration due to the high (~70%) prevalence of acute infusion reactions. Clinical trials with sipuleucel-T demonstrated a 4-month improvement in survival as compared to placebo (140).

Inhibitors of Androgen Biosynthesis

Extragonadal (e.g., adrenal or intratumoral) androgen sources may sustain prostate tumor growth despite a castrate environment in men with castration-resistant prostate cancer. A variety of experimental approaches to better reduce androgen biosynthesis have been evaluated and are in clinical development.

17 α -HYDROXYLASE/17,20-LYASE INHIBITORS 17 α -Hydroxylase/17,20-lyase is the key enzyme that converts pregnenolone to DHEA and, subsequently, to testosterone (Fig. 40.3). Because testosterone has androgenic activity, inhibition of its biosynthesis would be useful in treating androgen-dependent diseases, such as prostate cancer (141). Inhibitors of 17 α -hydroxylase/17,20-lyase inhibitors could prevent androgen biosynthesis in the testes, adrenals, or tumor and may provide effective treatment of patients with advanced prostate cancer. Because this

is an early step in steroid biosynthesis, inhibition of this enzyme efficiently may reduce androgen biosynthesis, but lead to an accumulation of serum mineralocorticoids.

The antifungal agent ketoconazole (Fig. 40.19) is an effective inhibitor of 17 α -hydroxylase (IC₅₀ = 76 nmol/L), which has demonstrated the promise of 17 α -hydroxylase inhibitors for the treatment of metastatic prostate cancer patients (142). The nonsteroidal imidazole agent R 75251 (Liarozole) (Fig. 40.19) (143) is under development as a potentially more selective inhibitor of 17 α -hydroxylase/17,20-lyase. The steroidal compounds MDL 27,302 (144), abiraterone (CB7598), and Sa40 (145) (Fig. 40.19) were designed as mechanism-based inhibitors of 17 α -hydroxylase/17,20-lyase. For MDL 27,302, inhibition is specific to the cyclopropylamino compound, because the isopropylamino- or the cyclobutylamino-analogs were not inhibitory. Enzymatic specificity of MDL 27,302 of 17 α -hydroxylase/17,20-lyase was demonstrated by its failure to inhibit steroid 21-hydroxylase and the cholesterol side-chain cleavage enzyme (CYP450scc). Both the 17 α -hydroxylase and 17/20-lyase activities of human testicular microsomes were inhibited by MDL 27,302. Abiraterone and Sa40 are heterocyclic analogs of MDL 27,302 and potent irreversible inhibitors of 17 α -hydroxylase (IC₅₀ = 4 and 24 nmol/L, respectively) (145). Inhibition studies show that the 16,17-double bond is necessary for irreversible binding of these 17-pyridyl and 17-pyrimidyl steroids to 17 α -hydroxylase. Oxidation to an epoxide probably is not involved, however, because the epoxide was a weak inhibitor (IC₅₀ = 260 nmol/L). Recently completed clinical studies showed that abiraterone reduced serum PSA levels in men with advanced prostate cancer who had failed ketoconazole therapy. Substantial declines in serum androgen levels

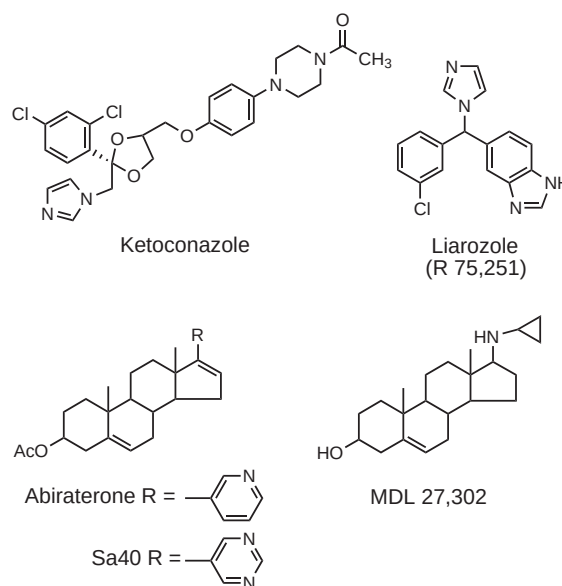


FIGURE 40.19 Luteinizing hormone-releasing hormone (LHRH) agonists used for the treatment of prostate cancer.

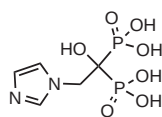
and increases in serum mineralocorticoids were seen at all doses (146).

5 α -REDUCTASE INHIBITORS The association between lifetime exposure to testosterone and DHT and the risk of developing prostate cancer suggests that chemoprevention with specific inhibitors of key enzymes associated with androgen biosynthesis may be possible (138). Two key enzymes considered for inhibition are 5 α -reductase, which inhibits the formation of DHT, and 17 α -hydroxylase/17,20-lyase, which inhibits testosterone biosynthesis (Fig. 40.3).

The 5 α -reductase inhibitors have been discussed previously (see *Benign Prostatic Hyperplasia*). The results of the Prostate Cancer Prevention Trial (147) for finasteride showed a 25% relative risk reduction in prostate cancer in men aged 55 years or older, albeit at an increased risk of invasive tumors (148). The risk of invasive tumors may outweigh the benefit of these agents. The 5 α -reductase inhibitors have not been proven to be effective as chemoprevention against clinically significant prostate cancer. Similar results were recently reported with dutasteride (149).

Bone-Protecting Treatments

The most common site for the spread of prostate cancer is bone. Most symptoms of advanced prostate cancer are caused by the presence of disease in the bone. These symptoms can be mitigated with a drug called zoledronic acid (Zometa), a bisphosphonic acid administered by intravenous infusion that inhibits osteoclastic bone resorption, which can slow the spread of disease, reducing the development of bone pain and inhibiting bone fractures.



Zoledronic acid

Zoledronic acid is most commonly given to patients whose cancer is no longer responding to hormones, but it also may be given to prevent the bone thinning and weakening that result from hormonal treatments.

Treatment of Prostatitis

Prostatitis is a broad term used to identify inflammation of the prostate gland associated with lower urinary tract symptoms in men (150). Prostatitis rarely occurs in males younger than 30 years; however, it is a common problem in older males, being described as acute bacterial prostatitis, chronic bacterial prostatitis, or nonbacterial prostatitis. Because antimicrobial drug penetration generally is poor into the prostate gland, with poor efficacy of the antimicrobial agents and long duration of treatment; a 30% to 40% failure rate occurs with common

treatment modalities. Three major factors determine the diffusion and concentration of antimicrobial agents in prostatic fluid and tissue: the lipid solubility of the antimicrobial agent, its dissociation constant (pK_a), and the percentage of plasma protein binding. The physiologic pH of human prostatic fluid is 6.5 to 6.7, but it increases in chronic prostatitis, ranging from 7.0 to 8.3 (151). A greater concentration of antimicrobial agents in the prostatic fluid occurs in the presence of a pH gradient across the membrane separating plasma from prostatic fluid. Of the available antimicrobial agents, β -lactam drugs have a low pK_a and poor lipid solubility and, thus, penetrate poorly into prostatic fluid, except for some cephalosporins. Good to excellent penetration into prostatic fluid and tissue has been demonstrated with many antimicrobial agents, including tobramycin, tetracyclines, macrolides, fluoroquinolones, sulfonamides, and nitrofurantoin. The diagnosis and therapy of prostatitis remains a challenge. Because prostatitis usually requires prolonged therapy, patients must understand the importance of compliance, and physicians should screen for drug interactions that may decrease compliance and efficacy.

Acute bacterial prostatitis is the least common of the prostate infections and, usually, is accompanied by a urinary tract infection with positive cultures. The symptoms include sudden onset of fever, chills, and low back pain, as well as complaints of urinary obstruction (e.g., dysuria, nocturia, urgency, frequency, and burning) and urinary irritation (e.g., hesitancy, straining, dribbling, weak stream, and incomplete emptying). The most commonly prescribed antimicrobials for acute bacterial prostatitis are trimethoprim-sulfamethoxazole, doxycycline, and the fluoroquinolones, ciprofloxacin, ofloxacin, and norfloxacin (see Chapter 33). The concentrations of these antimicrobial agents in the prostatic fluid are two to three times that in plasma, thus achieving adequate concentrations in prostatic tissues to eradicate the most common causative pathogens. The recommended duration of treatment for acute bacterial prostatitis is 4 to 6 weeks. A short-course therapy is not recommended because of the risk of relapse or progression to chronic bacterial prostatitis.

Chronic bacterial prostatitis occurs when acute bacterial prostatitis has been inadequately treated because of pathogen resistance, relapse, or short-course therapy or because of blocked drainage of secretions from the prostate. Most men with chronic prostatitis will have had a previous bout of acute prostatitis. The most common clinical feature of chronic prostatitis is recurrent urinary tract infections and the symptoms and complaints of acute bacterial prostatitis. Fluoroquinolones, trimethoprim-sulfamethoxazole, doxycycline, and nitrofurantoin are used in the management of chronic prostatitis. Chronic prostatitis warrants at least 10 to 12 weeks of therapy. Poor clinical outcomes, however, have been observed because of poor diffusion of antimicrobials into the prostate.

Nonbacterial prostatitis is the most common type of prostatitis. It occurs more frequently than bacterial prostatitis, with the same signs and symptoms as bacterial prostatitis except that prostatic fluid cultures are negative for presence of bacteria. Inflammation is evident on prostate gland examination. Treatment includes minocycline, doxycycline, or erythromycin. Treatment duration is approximately 2 to 4 weeks.

DRUGS USED FOR ERECTILE DYSFUNCTION

Phosphodiesterase Inhibitors

The currently available first- and second-generation oral PDE5 inhibitors, sildenafil, tadalafil, and vardenafil, have emerged as the first-line treatment for ED because of patient convenience, safety, and clinical efficacy and have markedly improved the quality of life in men with ED of various etiologies. As defined by the National Institutes of Health Consensus Development Panel on Impotence in 1993, ED is the inability to achieve or maintain an erection sufficient for satisfactory sexual performance. It is estimated that 10 to 30 million men in the United States, and more than 100 million men worldwide, experience some form of ED. This condition is strongly associated with age, and according to the community-based Massachusetts Male Aging Study, the prevalence of ED in men between 40 and 70 years of age is 52%. The general classification of ED includes psychogenic ED (e.g., depression, psychological stress, relationship problems, and performance anxiety), organic ED (e.g., diabetes, hypertension, spinal cord injuries, and some medications), and mixed psychogenic and organic ED.

Treatment options for men with ED have changed significantly over the past three decades and have progressed from psychosexual therapy and penile prostheses (1970s) through revascularization, vacuum constriction devices, and intracavernosal injection therapy (1980s) to transurethral and oral drug delivery (1990s). The introduction of the first PDE5 inhibitor, sildenafil citrate (Viagra), in 1998 revolutionized the treatment of men with ED of a broad-spectrum of etiologies and acknowledged the need for pharmacologic agents in the treatment of ED (152). With the recognition of the prevalence of ED by the public and the effectiveness of agents like sildenafil, there was an increased effort in the search for new agents with fewer side effects that led to the development of the second-generation PDE5 inhibitors, vardenafil and tadalafil, which have since been introduced into the world market at differing potencies and pharmacokinetics.

Mechanism of Action

The physiologic mechanism to achieve penile erection is mediated via an NO/cyclic guanosine monophosphate (cGMP) pathway. During sexual stimulation, parasympathetic neurons and vascular endothelial cells release NO, which activates soluble guanylate cyclase, thereby increasing the level of cGMP in the corpus cavernosum and relaxation (vasodilation) of vascular smooth muscle. PDE5 is a

cGMP-specific hydrolyzing enzyme and is present at high concentrations in the smooth muscle of the penile corpus cavernosum (153,154). One of the more effective methods for elevating cGMP levels in this tissue is to use PDE5 inhibitors. Other PDE isoenzymes also found in the human cavernous smooth muscle include PDE3 (cGMP-inhibited PDE) and PDE4 (cAMP-specific PDE). Thus, inhibitors of PDE, especially PDE5, have been shown to be an effective means for treating ED by enhancing and maintaining erections during sexual stimulation through sustaining sufficient cellular levels of cGMP in both the corpus cavernosum and the blood vessels supplying it. The increased vasodilation of the corporeal sinusoids allows more blood flow into the penis, thereby enhancing an erection.

Structure–Activity Relationships

The chemical similarities and distinct differences between sildenafil, vardenafil, and tadalafil regarding their selectivity for the inhibition of PDE5 and other PDE isoforms and their pharmacokinetic disparities largely affect the efficacy profile of these compounds. The obvious difference for the three drugs is the heterocyclic ring systems used to mimic the purine ring of cGMP. Although the heterocyclic ring systems and the *N*-substituent (ethyl vs. methyl) attached to the piperazine side chain (Fig. 40.20) are the only two structural

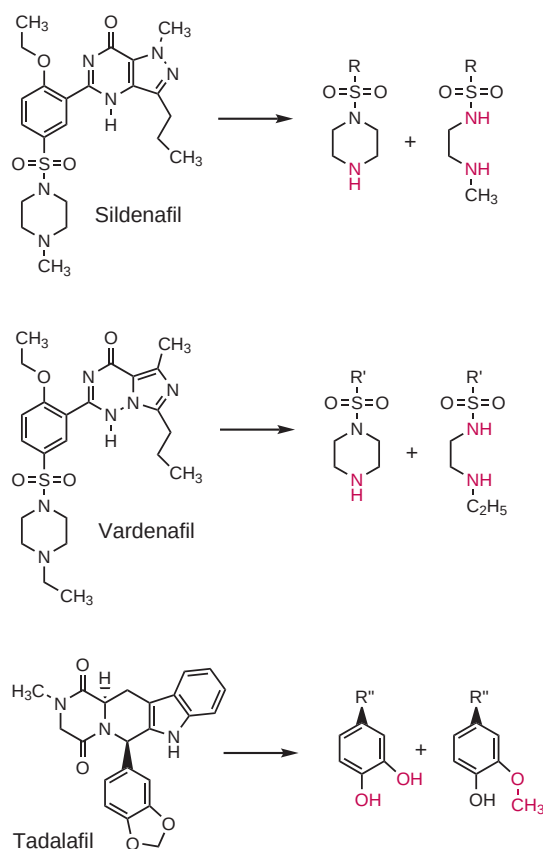


FIGURE 40.20 Phosphodiesterase 5 inhibitors for erectile dysfunction and their primary metabolites.

differences between sildenafil and vardenafil, these differences do not explain why vardenafil has a more than 20 times greater potency than sildenafil for inhibiting PDE5. A structure–activity relationship analysis for the difference in potency between sildenafil and vardenafil revealed that the methyl/ethyl group on the piperazine moiety plays very little role in the potency difference for inhibiting PDE5, whereas the differences in the heterocyclic ring systems play a critical role in higher potency for vardenafil (155).

A comparison of the in vitro inhibition values (IC_{50}) reveals that vardenafil is approximately 10 times more selective as an inhibitor of PDE5 than sildenafil and tadalafil (Table 40.6) (155). Furthermore, vardenafil also was approximately four times more selective than sildenafil for the inhibition of PDE1 and PDE6, as shown by its IC_{50} selectivity ratios: a PDE5/PDE1 ratio of 257 for vardenafil and 60 for sildenafil, and a PDE5/PDE6 ratio of 16 for vardenafil and 7.4 for sildenafil (Table 40.6). The low PDE5/PDE6 selectivity ratio suggests that at therapeutic doses, sildenafil is only approximately 10 times as potent for PDE5 as for PDE6, which

is a retinal enzyme found in the photoreceptors of the human retina. This lower PDE5/PDE6 selectivity ratio toward PDE6 for sildenafil indicates it is more likely to inhibit PDE6, which is presumed to be the cause for color vision abnormalities observed with high doses or plasma levels of sildenafil.

Pharmacokinetics

Sildenafil, vardenafil, and tadalafil have only limited oral bioavailability because of extensive presystemic metabolism in the intestine and hepatic first-pass metabolism via the CYP3A isoform family (Table 40.6) (156). The three drugs are rapidly absorbed after oral administration and reach peak plasma concentrations within 30 to 60 minutes. Rapid absorption and lipophilicity are considered to be a prerequisite for their rapid onset of efficacy and sexual satisfaction. Sildenafil and vardenafil are both rapidly absorbed, but with a significant difference in their mean bioavailability of approximately 15% for vardenafil and 40% for sildenafil. The administration of a high-fat meal had no significant effect on the rate and extent of absorption of tadalafil but did decrease the

TABLE 40.6 Some Properties and Pharmacokinetics of the Phosphodiesterase 5 Inhibitors

Drugs	Sildenafil	Vardenafil	Tadalafil
Trade Name	Viagra	Levitra	Cialis
cLogP ^a	2.3 ± 0.7	3.0 ± 0.7	1.4 ± 0.8
logD (pH 7)	2.2	3.0	1.4
Oral bioavailability (%)	38–40	15 (8–25)	~36
Onset of action (hours)	<0.5	<1	0.5–1
Duration of action (hours)	<4	<1	<36
Protein binding (%)	96	94	94
Time to peak concentration (hours)	0.5–2	0.5–3	0.5–6
Volume of distribution (L/kg)	105	208	63
Peak plasma concentration (nmol/L)	1–2	0.03	0.84
Elimination half-life (hours)	3–5	4–5	18
Cytochrome isoforms	3A4 (major) 2C9 (minor)	3A4 (major) 2C9 (minor)	3A4 (primary)
Active metabolites	<i>N</i> -desmethyl	<i>N</i> -desethyl	None
Excretion (%)	~80 feces/met ~13 urine/met	>90 feces/met <10 urine/met	~60 feces/met ~35 urine/met
PDE5 IC_{50} (nmol/L)	3.9 (3–7)	0.16 (0.09–0.7)	1.8 (0.9–5)
PDE5/PDE6 IC_{50}	7.4	16	85
PDE5/PDE1 IC_{50}	60	257	—

^aChemical Abstracts, American Chemical Society, calculated using Advanced Chemistry Development (ACD/Labs) Software V8.14 for Solaris (1994–2006 ACD/Labs). IC_{50} , median inhibitory concentration; met, metabolite; PDE, phosphodiesterase.

rate of absorption for sildenafil and vardenafil, which is consistent with their calculated lipophilicity (Table 40.6). It remains unclear whether food has any effect on their absorption and therapeutic efficacy. They are all highly protein bound, with free plasma concentration fractions of only 4% to 6%. Although the bioavailability for tadalafil has not been reported, pharmacokinetic studies and its long duration of action suggest that it is predominately metabolized by hepatic CYP3A4 to catechol metabolites, with minimal presystemic metabolism.

The major route of elimination for the three PDE5 inhibitors is hepatic metabolism, with renal excretion of unmetabolized drug accounting for 1% or less of the elimination pathways (Fig. 40.20) (156). CYP3A is the major drug-metabolizing enzymes for the three PDE5 inhibitors. CYP2C9, CYP2C19, and CYP2D6, however, also contribute to the metabolism of sildenafil, and CYP2C9 contributes to the metabolism of vardenafil. Both sildenafil and vardenafil have active metabolites that reach plasma concentrations high enough to contribute to the overall efficacy and safety profile of their parent-drug molecules. Grapefruit juice increases the bioavailability of sildenafil and vardenafil as well as delays their absorption. The pharmacokinetics of sildenafil and vardenafil may become less predictable in patients who also drink grapefruit juice. Although patients usually will not be endangered by concomitant use of grapefruit juice, it seems advisable to avoid this combination, which can cause systemic vasodilatation.

The larger differences in their volumes of distribution, together with the substantial differences in their systemic clearance, result in distinct differences in their elimination half-lives: 3 to 5 hours for sildenafil and vardenafil, compared with approximately 18 hours for tadalafil.

Hepatic CYP3A and CYP2C activity has been described as being age-dependent, with reduced activity being exhibited in elderly compared to young individuals. This decrease in metabolic activity is reflected by a corresponding increase in plasma concentrations of all three PDE5 inhibitors, warranting dose reductions for sildenafil and vardenafil in elderly patients. Similarly, ethnicity-dependent differences in the pharmacokinetics of all three PDE5 inhibitors may be expected based on known ethnic differences in CYP3A4/5 activity. Gender differences in pharmacokinetics have not been described for any of the three PDE5 inhibitors, which seems to be in agreement with the literature. Severe renal impairment resulted in an increase in plasma concentrations for all the drugs, and this warrants dose reductions for sildenafil and tadalafil in the affected patient population.

Drug Interactions

None of the three PDE5 inhibitors have been identified as CYP inhibitors, including CYP3A or CYP2C substrates. Because metabolism via CYP3A is the major elimination pathway for all three drugs (sildenafil, vardenafil, and tadalafil), all inducers and inhibitors of CYP3A activity have the potential to interfere with the elimination of

these drugs. This interaction potential has been verified clinically for inducers of CYP3A activity only for rifampicin and tadalafil (156). The strong inhibitors of CYP3A4 (ritonavir, indinavir, saquinavir, erythromycin, and ketoconazole) increased the plasma levels for sildenafil, vardenafil, and tadalafil. Grapefruit juice, a selective inhibitor of CYP3A intestinal metabolism, also increased the plasma concentrations of sildenafil and vardenafil but not of tadalafil. Ritonavir, as an inhibitor of CYP3A4 and CYP2C9, increased the plasma levels for vardenafil by 50 to 300 times, most likely as a consequence of the simultaneous inhibition of both CYP3A4 and CYP2C9, the major metabolism pathways for vardenafil. The effect of ritonavir on sildenafil was much less pronounced (11 times), because other compensatory CYP-mediated metabolism pathways were still available. Ritonavir increased the plasma levels for tadalafil (CYP3A4) by approximately three times.

Adverse Effects

A number of side effects have been reported with these drugs, including nausea, indigestion, cutaneous flushing, headache, and retinal effects, including a bluish haze and increased light sensitivity. It has been suggested that some of these side effects result from the inhibition of other PDEs, including the isoform PDE6 (156,157). The search is on for even more selective ED agents to see if additional side effects can be eliminated. If one is taking α -adrenergic blockers for high blood pressure, one should consult a physician before using the agents together (156). Other vasodilators associated with regulating the intracellular levels of cGMP, such as nitroglycerin, should not be used in combination with the PDE5 inhibitors.

Therapeutic Effects

The first line of treatment of ED is PDE5 inhibition, because the three drugs can be given orally. If the PDE5 inhibitors are not effective, then the cause may be low libido, and men should have their testosterone blood levels checked (in some instances, TRT may help to resolve ED). Other alternative drugs currently available for the treatment of ED include prostaglandin E_1 , which is given by injection at the base of the penis or by suppository into the tip of the penis, as well as the α_1 -adrenergic blocker and the nonselective PDE inhibitor papaverine. Apomorphine is a dopamine agonist that can be used for treating ED but, in humans, has the undesirable emetic side effect. Some selective dopamine D_4 agents are now being investigated for treatment of ED. Vacuum devices and penile implants are also available.

Specific Drugs for Erectile Dysfunction

SILDENAFIL In 1998, sildenafil was the first selective PDE inhibitor to be approved and found to be effective in treating ED (152). Sildenafil has approximately one-tenth the selectivity for PDE5 as for PDE6, which

is found in the photoreceptors of the human retina. In vitro metabolism studies for sildenafil have shown that the primary metabolite, *N*-desmethylsildenafil, and the minor metabolite, oxidative opening of the piperazine ring, are mediated by CYP3A4, CYP2C9, CYP2C19, and CYP2D6 (Fig. 40.20). The estimated relative contributions to clearance were 79% for CYP3A4, 20% for CYP2C9, and less than 2% for CYP2C19 and CYP2D6. These results demonstrate that CYP3A4 is the primary cytochrome mediating *N*-demethylation and that drugs inhibiting CYP3A4 likely impair sildenafil biotransformation and clearance. The pharmacokinetics of radiolabeled sildenafil were consistent with rapid absorption, first-pass metabolism, and primarily fecal elimination of *N*-demethylated metabolites. The absorption of sildenafil following oral administration was rapid (~92%), whereas the oral bioavailability was approximately 38% as a result of first-pass metabolism.

VARDENAFIL Vardenafil was the second agent to be marketed and had the advantage that its onset time was not reduced by taking the medication on a full stomach (Table 40.6) (156,158). It is 30 times more potent as an inhibitor of PDE5 (mean IC_{50} = 3.9 nmol/L) than sildenafil and 10 times more potent than tadalafil, with a greater selectivity (>1,000 times) for human PDE5 than for human PDE2, PDE3, and PDE4 and moderate selectivity (>80 times) for PDE1 (155,159) (Table 40.6). Vardenafil specifically inhibited the hydrolysis of cGMP by PDE5, with an IC_{50} of 0.7 nmol/L (sildenafil, 6.6 nmol/L). The IC_{50} of vardenafil for PDE1 was 180 nmol/L, for PDE6 11 nmol/L, and for PDE2, PDE3, and PDE4 more than 1,000 nmol/L.

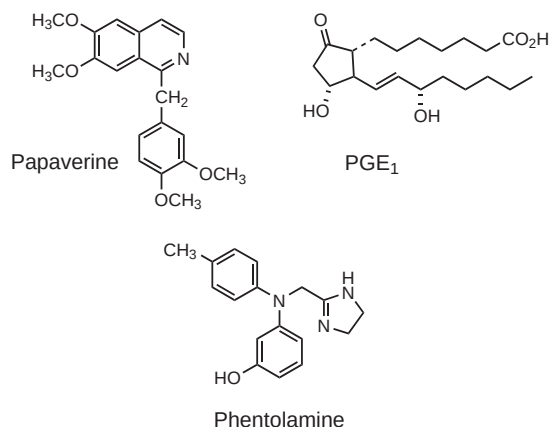
TADALAFIL Tadalafil was the latest agent to be released and can be taken on a full stomach without slowing the onset (Table 40.6) (156). It has a much longer duration of action, lasting up to 48 hours, compared with sildenafil and vardenafil, which last for approximately 4 hours. The longer half-life of tadalafil results in a lengthened period of responsiveness as compared to sildenafil and vardenafil. This longer therapeutic window requires fewer time constraints for the effectiveness of tadalafil and has been interpreted as being advantageous through providing the option for more spontaneous sexual activity. Because of its long half-life, however, tadalafil has been detected in plasma even 5 days after oral administration. This suggests the possibility of accumulation if taken regularly and in short intervals, which may result in an increased risk of side effects with the excessive use of this PDE5 inhibitor. The 3,4-methylenedioxy substitution on the phenyl ring was significant for increasing its potency as a PDE5 inhibitor. Optimization of the chain on the piperazinedione ring resulted in no significant change in IC_{50} . Tadalafil is a highly potent PDE5 inhibitor (IC_{50} = 5 nmol/L), with high selectivity for PDE5 versus PDE1 through PDE4 (160). The PDE5/PDE6 selectivity ratio is 85.

PROSTAGLANDIN E_1 Prostaglandin E_1 (PGE_1 ; Alprostadil) is approved for the intracavernosal (Caverject, Edex) or intraurethral suppository (Muse) treatment of ED. A three-drug combination of PGE_1 , papaverine, and phentolamine sometimes is used as an intracavernosal injection to achieve a synergistic action. ED that is medication-induced or caused by endocrine problems, such as hypogonadism or hyper- or hypothyroidism, should be evaluated and appropriately treated before PGE_1 treatment is considered. PGE_1 is produced endogenously to relax vascular smooth muscle and cause vasodilation by activating the adenylate cyclase/cAMP pathway. Recent studies show that cAMP is important in the PGE_1 relaxation of penile erectile tissue and vasodilation of penile resistance arteries (161). Moreover, agents that stimulate the release of cAMP also cross-activate the NO/cGMP cascade.

When administered by intracavernosal injection or as an intraurethral suppository, PGE_1 acts locally to relax the smooth muscle of the corpora cavernosa and the cavernosal arteries. Swelling, elongation, and rigidity of the penis result when arterial blood rapidly flows into the corpus cavernosum to expand the lacunar spaces. The entrapped blood reduces the venous blood outflow as sinusoids compress against the tunica albuginea. Adding papaverine and phentolamine to the PGE_1 regimen synergistically increases arterial blood flow via separate mechanisms. Papaverine relaxes the sinusoid and the smooth muscle of the helicine arteries, whereas phentolamine relaxes arterial smooth muscle and blocks both of the α -adrenergic receptors that inhibit an erection. PGE_1 is rapidly metabolized within the urethra, prostate, and corpus cavernosum to 7 α -hydroxy-5,11-diketotetranorprosta-1,16-dioic acid and 5 α ,7 α -dihydroxy-11-ketotetranorprostane-1,16-dioic acid (162). The major route of excretion of PGE_1 metabolites is via the kidney. Its elimination half-life is 5 to 10 minutes. If any alprostadil is systemically absorbed, it is metabolized by a single pass through the lungs. The onset of action is within 10 minutes, and the time to peak effect is less than 20 minutes. The duration of action is 1 to 3 hours for the intracavernosal injection and 30 to 60 minutes for the intraurethral suppository.

PAPAVERINE (INTRACAVERNOSAL) Papaverine is used, sometimes in combination with phentolamine, by intracavernosal injection to facilitate erections in men with ED. In general, papaverine is most useful in patients with organic ED (neurogenic and, to a lesser extent, vascular). It is less useful in patients with ED resulting from endocrine problems (hypogonadism or hyper- or hypothyroidism) or medications. When administered by intracavernosal injection, papaverine, a weak and nonspecific PDE inhibitor, is thought to cause relaxation of the cavernous smooth muscles and vasodilation of the penile arteries by inhibition of PDE (163). These effects result in increased arterial blood flow into the corpus cavernosa and in swelling and elongation of the penis. Venous

outflow also is reduced, possibly as a result of increased venous resistance. Adding phentolamine and PGE₁ to the papaverine regimen synergistically increases arterial blood flow via their separate mechanisms. Papaverine is slowly released into the venous circulation with minimal systemic effects. The time to peak effect usually is within 10 minutes, and the duration of action is 1 to 6 hours with concurrent administration with phentolamine.



PHENTOLAMINE (INTRACAVERNOSAL; ROGITINE)

Phentolamine is used in combination with papaverine, by intracavernosal injection, to facilitate erections in men with impotence (see *Papaverine* above). Its mechanism of action is as an α -adrenergic antagonist (see Chapter 10) of both α_1 - and α_2 -receptors, causing vasodilation and reduction in peripheral resistance. When administered by intracavernosal injection, it is thought to cause relaxation of the cavernous smooth muscles and vasodilation of the penile arteries. This results in increased arterial blood flow into the corpus cavernosa as well as swelling and elongation of the penis. Venous outflow also is reduced, possibly as a result of increased venous resistance. Phentolamine is slowly released into venous circulation with minimal, if any, systemic effects. Time to peak effect is within 10 minutes, and duration of action when used with papaverine is 1 to 6 hours.

TESTICULAR CANCER

Testicular cancer develops in the testicles and, according to the National Cancer Institute (NCI), accounts only for approximately 1% of all cancers in men. Compared with prostate cancer, testicular cancer is relatively rare. It is most common among males between 15 and 40 years of age and is approximately fourfold more common in white men than in black men.

Nearly all testicular tumors originate from germ cells, the specialized sperm-forming cells within the testicles. These tumors fall into one of two types, seminomas or nonseminomas (164). Seminomas account for approximately 40% of all testicular cancer and are made up of immature germ cells. Seminomas are slow-growing and tend to stay localized in the testicle for long periods. Nonseminomas

arise from more mature, specialized germ cells and tend to be more aggressive than seminomas. According to the American Cancer Society, 60% to 70% of patients with nonseminomas have cancer that has spread to the lymph nodes. α -Fetoprotein is a tumor-associated marker in blood for testicular cancer. Its measurement can help to show how well the chemotherapeutic drugs are working. Because seminomas are slow-growing, they tend to stay localized and usually are diagnosed at stage I (confined to testicle) or stage II (spread to lymph nodes). Treatment might be a combination of testicle removal, radiation, or chemotherapy. Most nonseminomas are not diagnosed at stage I. Advanced testicular cancer (stage III, metastasized to other tissues) seminomas, as well as stage II and stage III nonseminomas, usually are treated with multidrug chemotherapy. The majority of cases are stage I when first identified; stage III cases are relatively rare.

Chemotherapy is the standard treatment, with or without radiation, when the cancer has spread to other parts of the body. The drugs approved to treat testicular cancer include ifosfamide, etoposide, vinblastine, bleomycin, and cisplatin (Chapter 37). Cisplatin usually is given in combination with bleomycin and etoposide or other chemotherapy drugs following surgery or radiation therapy. Testicular cancer has one of the highest cure rates of all cancers, essentially 100% at stage I. Approximately 90% of men with advanced testicular cancer can be cured, according to the NCI. Because testicular cancer is curable when detected early, the NCI recommends regular monthly testicular self-examination after a hot shower, when the scrotum is looser, feeling for lumps or enlargement.

MALE OSTEOPOROSIS

Osteoporosis is a common condition in men that usually develops after the age of 70 years and affects approximately 2 million men in the United States (165). Osteoporosis occurs less frequently in men because of greater skeletal bone mass during growth (i.e., greater bone size). Approximately 20% to 25% of all hip fractures occur in men, however, and the age-adjusted prevalence of vertebral deformities appears to be similar in men and in women. Currently, bisphosphonates (alendronate and risedronate) (see Chapter 30) are the therapy of choice for increasing bone mineral density to decrease the risk of fracture in the treatment of male osteoporosis, and a short course of parathyroid hormone (1-34; teriparatide) may be indicated for men with very low bone mineral density or for those in whom bisphosphonate therapy is unsuccessful. Antiandrogen therapy for the treatment of prostate cancer, hypogonadism, and diabetes are some of the risk factors for osteoporosis and fracture. Vitamin D and calcium have been shown to help improve bone mineral density. Testosterone replacement therapy is controversial except in men who clearly have hypogonadism and low levels of testosterone; in those men, treatment with testosterone appears to increase bone density.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Autumn Stewart, PharmD

NP presents 2 weeks later complaining of sexual dysfunction. Symptoms related to benign prostatic hypertrophy remain controlled and the nail infection appears to be improving. The patient's physician is consulted and a change is made from dutasteride to finasteride 5.0 mg QD. The patient follows up 2 weeks later and reports that adverse effects related to sexual function have resolved and no new problems have arisen.

Chemical Analysis

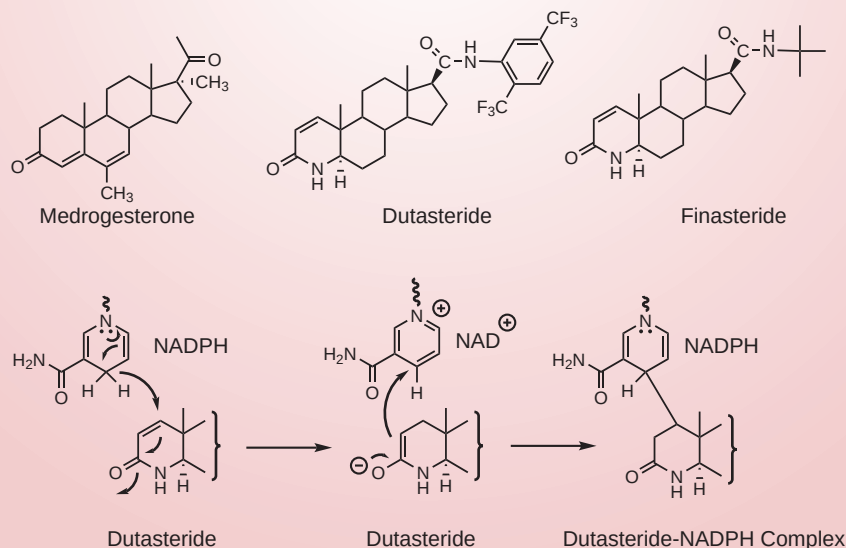
S. William Zito and Victoria Roche

NP has experienced sexual dysfunction after being treated with itraconazole for onychomycosis due to a drug-drug interaction with dutasteride, used here to treat benign prostatic hypertrophy. Itraconazole is an inhibitor of CYP3A4, and because dutasteride is metabolized by CYP3A4, inhibiting its metabolism leads to increased plasma levels and sexual dysfunction (SD). The sexual dysfunction side effect of dutasteride is most likely associated with the progestin activity of this analog. Dutasteride is a selective inhibitor of both type 1 and type 2 isoforms of 5α -reductase, which blocks conversion of testosterone to dihydrotestosterone

(DHT). DHT is an androgen primarily responsible for the growth and enlargement of the prostate gland. Switching NP to finasteride results in moderating the sexual dysfunction side effects, presumably because of a much lower incidence of SD (4.7% vs. 1.8%).

Dutasteride is a 4-azasteroid progestin analog related to medrogestosterone (6-methyl-6-dehydro-17-methylprogesterone). Specifically it is a 17β -bis(trifluoromethyl) azaandrostene-carboxamide. Its mechanism of action involves binding to the NADPH cofactor of 5α -reductase and forming a complex that is slowly released from the enzyme resulting in essentially irreversible inhibition. The key reaction is between the Δ^1 double bond of dutasteride and NADPH.

Finasteride also is a 4-azasteroid progestin analog related to medrogestosterone. It differs from dutasteride only in being a 17β -1,1-dimethylethyl carboxamide. Though dutasteride inhibits both isoforms of 5α -reductase, finasteride inhibits only the type II isozyme, which is primarily found in the prostate, seminal vesicles, epididymides, and hair follicles, as well as the liver. The type I isozyme is responsible for one third of the circulating DHT, whereas type II isozyme is responsible for two thirds. Finasteride has the same mechanism of action as dutasteride but a lower incidence of sexual dysfunction side effects.



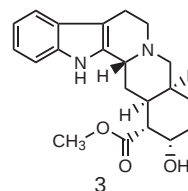
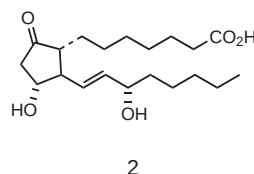
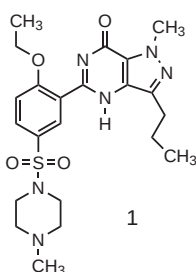
CASE STUDY

S. William Zito and Victoria Roche

BJ is a 62-year-old man. He has had chronic stable angina (CSA) for the past 3 years, which is under control with isosorbide mononitrate CR (60 mg daily). BJ reports that although he carries around sublingual nitroglycerine for acute attacks, he hasn't ever had to use it. His medical history includes type 2 diabetes mellitus since he was 45 years old, which he controls with diet and metformin (850 mg/day). He appears today at your ambulatory care clinic to speak to his primary care physician. He explains that he feels great and has become active again. He now plays tennis three times a week and swims most mornings at the local pool with his wife. He is really enjoying life but has not had an erection for the last 12 months and wants to continue to have a healthy sexual relationship with his wife. Physical examination shows a blood pressure of 125/80 mm Hg and his electrocardiogram is normal. Laboratory tests

show the following values: fasting blood glucose, 90 mg/dL; glycosylated hemoglobin, 6.0%; creatinine, 1.1 mg/dL; total cholesterol, 160; LDL, 120; HDL, 45; and triglycerides, 119 mg/dL. BJ was examined by his physician, who measured the blood flow through his penis using an echo Doppler. It was found to be below the normal range. In addition, a biothesiometry study was conducted to see if there was any nerve deterioration due to his diabetes, and the results were negative. Evaluate the following three drug choices for use in this case.

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.



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Chapter 41

Women's Health

ROBIN M. ZAVOD

Drugs Covered in This Chapter*

AROMATASE INHIBITORS

- Anastrozole
- Exemestane
- Letrozole

FULVESTRANT

INFERTILITY DRUGS

- Clomiphene citrate
- Follicle-stimulating hormone
- Gonadotropin-releasing hormone
- Lutropin alfa

PROGESTERONE RECEPTOR ANTAGONISTS

- Mifepristone
- Ulipristal acetate

PROGESTINS

- Desogestrel
- Dienogest
- Drospirenone
- *Elcometrine*
- Etonogestrel
- *Gestodene*
- Medroxyprogesterone acetate
- Megestrol acetate
- Norgestrel acetate

- Norethindrone
- Norgestimate
- Norgestrel/Levonorgestrel
- *Trimegestone*

SELECTIVE ESTROGEN RECEPTOR MODULATORS

- Tamoxifen
- Toremifene

STEROIDAL ESTROGENS

- Conjugated estrogens
- Estradiol
- Ethinyl estradiol
- Mestranol

Abbreviations

cAMP, cyclic adenosine monophosphate

CEE, conjugated equine estrogen

CoA, coactivator

CYP, cytochrome P450

DES, diethylstilbestrol

DHEA, dehydroepiandrosterone

EE, ethinyl estradiol

ER, estrogen receptor

ERE, estrogen response element

ERT, estrogen replacement therapy

FDA, U.S. Food and Drug Administration

FSH, follicle-stimulating hormone

GI, gastrointestinal

GnRH, gonadotropin-releasing hormone

hCG, human chorionic gonadotropin

hCG-H, hyperglycosylated human chorionic gonadotropin

HDL, high-density lipoprotein

HPO, hypothalamic-pituitary-ovary

HRT, hormone replacement therapy

HSDD, hypoactive sexual desire disorder

5-HT, serotonin

IUD, intrauterine device

IUI, intrauterine insemination

IVF, in vitro fertilization

LDL, low-density lipoprotein

LH, luteinizing hormone

OC, oral contraceptive

OHSS, ovarian hyperstimulation syndrome

SERM, selective estrogen receptor modulator

SHBG, sex hormone binding globulin

SSRI, selective serotonin reuptake inhibitor

VTE, venous thromboembolism

WHI, Women's Health Initiative

*Drugs listed include those available inside and outside of the United States; drugs available outside of the United States are shown in italics.

SCENARIO

Nancy Ordonez, PharmD

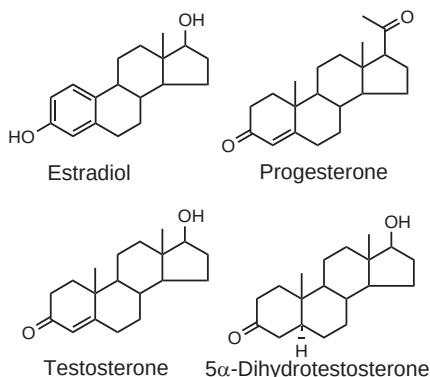
CC is a 20-year-old college student who comes to the family medicine clinic for her annual physical and gynecologic exam. She wants to discuss options for contraception. She is currently taking a combined oral contraceptive (COC) containing ethinyl estradiol 0.03 mg/levonorgestrel 0.15 mg, which she has been taking for 6 months. She is a very busy computer science major and handles a heavy academic load in addition to being a member of the winning college softball team. She is a runner and works part-time at the school's recreation center as a swimming

instructor. She admits that she has not been very diligent in taking her COC on a regular basis due to her busy schedule, and she would like to have a more long-term contraceptive agent. She is taking no other medications except a multivitamin once daily. She has no significant medical history and is negative for tobacco use. Her pregnancy test in the clinic is negative.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

The sex hormones are specific steroids necessary for reproduction and for the development of secondary sex characteristics in both sexes. The sex steroids are comprised of three classes: estrogens, progestins, and androgens. In this chapter, the two principal classes of female sex hormones, estrogens and progestins, will be discussed. The naturally occurring estrogens are C_{18} steroids and contain an aromatic A ring with a hydroxyl group at the 3 position. The most potent endogenous estrogen is estradiol. The naturally occurring progestins are C_{21} steroids and contain a 3-keto-4-ene structure in the A ring and a ketone at the C_{21} position. The most potent endogenous progestin is progesterone. The class of steroids that contains the male sex hormones is the androgen class. The naturally occurring androgens are C_{19} steroids and contain either a hydroxyl or ketone functional group at both the C_3 and C_{17} positions. The primary androgen found in the blood is testosterone, with the active metabolite 5α -dihydrotestosterone formed in certain target tissues. All three classes of endogenous steroids are present in both sexes, but the production and circulating plasma levels of estrogens and progestins are substantially higher in females.



SEX HORMONES: REPRODUCTIVE CYCLE

The female reproductive cycle is controlled by an integrated system involving the hypothalamus, anterior pituitary gland, ovary, and reproductive tract

(Fig. 41.1). The hypothalamus exerts its action on the pituitary gland via the hormone signal gonadotropin-releasing hormone (GnRH). This hormone is released by the hypothalamus in a pulsatile fashion and stimulates the release of the gonadotropins, follicle-stimulating hormone (FSH), and luteinizing hormone (LH) from the anterior pituitary.

The two main gonadotropins, FSH and LH, regulate production of the sex hormones estrogen and progesterone. As the name implies, FSH promotes the initial development and growth of immature ovarian follicles, and as these follicles mature, they secrete estradiol. During the follicular phase, when estrogen levels are elevated, the endometrium undergoes proliferation as a direct result of estrogenic stimulation. As estradiol levels increase, the production of FSH decreases via feedback inhibition, and LH release is stimulated. The level of LH rises to a sharp peak (LH surge) at the midpoint of the menstrual cycle. This serves to cause the dominant follicle to rupture and release its egg (ovulation). The luteal phase follows ovulation and ends at menses. During this phase, the endometrium shows secretory activity, and cell proliferation declines. Once ovulation has occurred, LH induces luteinization of the tissue that remains from the ruptured follicle, which leads to formation of the corpus luteum. The corpus luteum is responsible for the biosynthesis and

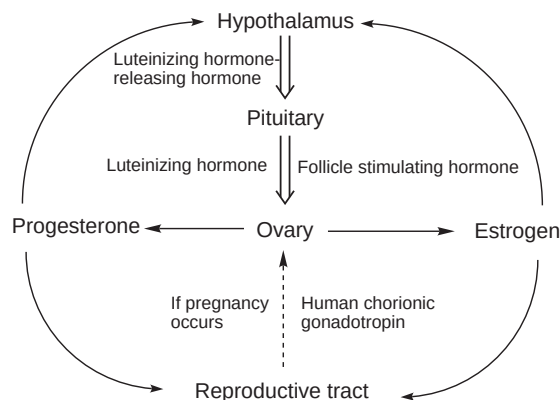


FIGURE 41.1 Hypothalamic, ovarian, and reproductive tract interrelationships.

secretion of progesterone during the luteal phase of the menstrual cycle. This phase occurs when endometrial preparation for pregnancy occurs, as well as during the early weeks of a pregnancy.

If the egg remains unfertilized, the corpus luteum degrades, and the levels of progesterone and estrogen decline. This leads to sloughing off of the endometrial lining, known as menstruation. At this point in the menstrual cycle, estrogen and progesterone levels are low, and hormonal feedback to the hypothalamus ceases. Without the influence of these hormonal signals, the hypothalamus releases additional GnRH, and the cycle begins again. A single menstrual cycle starts with the onset of menses (day 1) and ends with the start of the next menstrual period. This interval varies from 20 to 35 days, with an average length of 28 days. The major events of the ovarian cycle are summarized in Figure 41.2.

If the egg is fertilized successfully, then the menstrual cycle is interrupted because of the release of a gonadotropin synthesized in the placenta, human chorionic gonadotropin (hCG). The role of this glycoprotein is to maintain the corpus luteum. Fourteen days after egg fertilization, the level of urinary hCG reaches a concentration that can be detected by a home pregnancy test. Levels of this gonadotropin peak around the seventh week of pregnancy and then decrease to a level that is maintained throughout pregnancy.

For the first 9 weeks of pregnancy, the corpus luteum produces an adequate level of estrogen and progesterone to maintain the pregnancy. After this period, the placenta assumes these responsibilities. Both estrogen and progesterone levels increase during pregnancy, finally

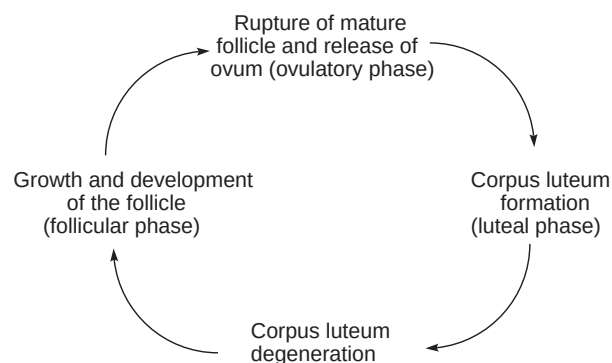
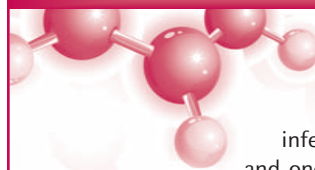


FIGURE 41.2 Ovarian cycle.

reaching their maximal concentrations a few days before labor and delivery.

Over the course of a woman's lifetime, her hormonal status and ability to reproduce vary considerably. Sexual maturation and the onset of the reproductive period are correlated with the start of cyclic menstrual bleeding. This typically occurs between the ages of 8 and 17 years (average age, 11 years). Later in life, reproductive capacity begins to wane, and women gradually lose ovarian function, which causes intermittent maturation of follicles and, therefore, irregular production of the corpus luteum. This is referred to as perimenopause. Eventually, cessation of menses occurs, resulting in menopause and the loss of reproductive capacity. This typically occurs between the ages of 45 and 55 years. Throughout a woman's reproductive life cycle, various types of hormone therapy play key roles in regulating this hormone cycle.

CLINICAL SIGNIFICANCE



Women's health issues encompass a variety of disease states and range from contraception to infertility, menopause, osteoporosis, and oncology. Pharmacologic agents treating these disease states are numerous and their uses are wide-ranging. Recent developments of drugs dealing with women's health issues not only have focused on producing better oral contraceptives but have expanded to designing drugs for the treatment of osteoporosis, infertility, and breast cancer. With so many medications from which to choose, the clinician must be knowledgeable about the drug's pharmacodynamic, pharmacokinetic, and medicinal chemistry properties to optimize therapy. For example, with so many oral contraceptives (OCs) on the market, the clinician must apply his or her knowledge about the drug to tailor the therapy. For patients with acne, OCs with low androgenic characteristics would be appropriate. Postpartum mothers who are breastfeeding and want to start taking OCs will need to use a product containing progestin only. In addition to OCs, drugs used to treat infertility must be customized to fit

the patient's clinical picture. A variety of medications are used to treat infertility, ranging from clomiphene citrate, an antiestrogen, to products containing follicle-stimulating hormone.

Advancements in the treatment of bone disorders and cancer in women have resulted from the increased knowledge of estrogen receptors and the development of selective estrogen receptor modulators (SERMs). Modifications in the structure-activity relationships of SERMs produce drugs with favorable side effect profiles that are effective for osteoporosis and breast cancer. Many ongoing research investigations continue to develop drugs for treating the assortment of health care issues faced by women. An essential component in both the development and the clinical applications of these drugs is the utilization of knowledge regarding the drugs' medicinal chemistry properties.

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HOME PREGNANCY TESTS

With the increased availability of a variety of home diagnostic tests, the sale of home pregnancy, ovulation, and fertility tests alone was a \$227 million dollar business in 2009. Home pregnancy tests have been used since 1976, with approximately 26 devices currently available in the United States. These tests claim that they can be used on the first day of a missed menstrual period (or, for certain devices, even earlier) with more than 99% diagnostic accuracy for pregnancy. Despite these claims, accurate results are obtained only a fraction of the time (Table 41.1) (1).

Home pregnancy tests are qualitative tests and simply detect the presence of the α and β subunits of hCG in a small sample of urine collected early in the day (Table 41.2) (2,3). Their detection limits range from 6.3 to 50 IU/L, and all are calibrated using regular hCG. A positive response indicates that the anti-hCG monoclonal or polyclonal antibodies found within the tests have successfully bound to a specific site along hCG. A second antibody linked to an enzyme that reacts with this anti-hCG complex causes a color change (3). What is not clear is if it is the regular hCG or the hyperglycosylated hCG (hCG-H) that is the source for the immunoreactivity observed during the weeks immediately following implantation. Compelling evidence suggests that hCG-H may be the principle source of hCG immunoreactivity, yet the detection limits for this glycoprotein are poorer than those reported for regular hCG. Only 6 of 15 home pregnancy tests evaluated were equally capable of detecting hCG-H and regular hCG (2).

False-positive results can stem from a recent birth, miscarriage, or the use of menopausal hormone therapy or hCG injections in women experiencing infertility, in whom hCG levels may still be elevated (4). Although unlikely because of the use of monoclonal antibody technology, there also may be false-positive results from pharmacologic sources, including methadone, promethazine, or chlorthalidone (3). False-negative results generally are a consequence of errors in using the device or in handling the urine sample.

This chapter will focus primarily on hormonal therapies for the treatment and/or management of contraception, endometriosis, infertility, menopause, and breast cancer.

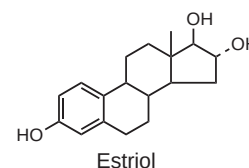
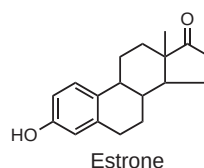
STEROID HORMONES

Estrogens

Three endogenous estrogens are present in women. Estradiol, the most potent of the three, represents 10% to 20% of the circulating estrogen. Estrone is 10-fold less potent than estradiol and accounts for 60% to 80% of the circulating estrogen. The remaining 10% to 20% is in the form of estriol, a very weak estrogen. Estrone was the first estrogen to be isolated in crystalline form from the urine of pregnant women. The two other C_{18} estrogen steroids, 17 β -estradiol and estriol, were isolated and characterized later.

TABLE 41.1 Clinical Sensitivity Evaluation of Home Pregnancy Tests (1)

Name of Device	Confidence in Positive Results (%)	Clinical Sensitivity (%)
EPT	45	16
Clear Blue Easy Earliest Results	100	80
First Response Early Result	100	>95
Fact Plus Select	24	<16
Accu-Clear	22	<16



Biosynthesis

Estrogens are biosynthesized in the maturing dominant follicle and in the corpus luteum in premenopausal women. During pregnancy, the placenta becomes the primary location of estrogen biosynthesis (5,6). Approximately 50% of estrone production occurs in the ovaries. The remaining estrone is biosynthesized from estradiol, as well as from the conversion of estrone sulfate to estrone in the adrenal gland and the aromatization of androstenedione. In contrast to premenopausal women, in whom the natural estrone-to-estradiol ratio is 1:2, postmenopausal women have an estrone-to-estradiol ratio of 2:1, which reflects the loss of ovarian function.

In endocrine tissues, cholesterol is the steroid that is stored and converted to estrogen, progesterone, or androgen when the tissue is stimulated by a gonadotropic hormone. The major pathways for the biosynthesis of sex steroid hormones are summarized in Figure 41.3. In the ovary, FSH acts on the preovulatory follicle to stimulate the biosynthesis of estrogens. The thecal cells of the preovulatory follicle convert cholesterol into androgens, whereas the granulosa cells convert androgens to estrogens.

Cleavage of the side chain of cholesterol produces pregnenolone (step a in Fig. 41.3), which can then be transformed into progesterone or, via several biosynthetic steps, to the aromatic A ring containing structures such as those found in estrogens (7). Pregnenolone is converted by 17 α -hydroxylase to 17 α -hydroxypregnenolone (step b), which then proceeds on to the intermediate dehydroepiandrosterone (DHEA) via the 17,20-lyase reaction (step c) (8). DHEA is converted by 5-ene-3 β -hydroxysteroid dehydrogenase and 3-oxosteroid-4,5-isomerase to the 17-ketosteroid, androstenedione (steps d and e), which is interconvertible with testosterone via

TABLE 41.2 Evaluation of Home Pregnancy Tests with Regular Human Chorionic Gonadotropin (hCG) and Hyperglycosylated hCG (H-hCG) Standards (2)

Name of Device	Regular hCG (IU/L)				H-hCG (IU/L)				hCG Limit (IU/L)
	6.3	13	25	50	6.3	13	25	50	
EPT	–	–	+	++	–	+	+	++	40
Clear Blue Easy	+	+	++	++	–	–	+	+	50
First Response Early Result	–	+	+	++	–	–	+	++	~50
Fact Plus Select	–	–	+	+	–	–	+	+	100
Answer	–	–	+	++	–	–	±	+	100

–, negative result observed; ±, exceptionally faint positive result observed; +, clear positive result observed; ++, strongly positive result observed.

17 β -hydroxysteroid dehydrogenase (step f). The final step in the biosynthesis is the conversion of the C₁₉ androgens to the C₁₈ estrogens via the loss of the C₁₉ angular methyl group and aromatization of ring A to form 17 β -estradiol or estrone (step g) catalyzed by aromatase. The interconversion of 17 β -estradiol and estrone is catalyzed by estradiol dehydrogenase (step h), a member of the 17 β -hydroxysteroid dehydrogenase family.

Metabolism

Metabolism of the estrogens to their water-soluble glucuronide and sulfate conjugates occurs mainly in the liver. Creation of an estrogen reservoir occurs as a result of enterohepatic recycling, a process by which estrogens undergo desulfation followed by resulfation.

Approximately 50% of an exogenous dose of 17 β -estradiol undergoes hepatic first-pass metabolism to

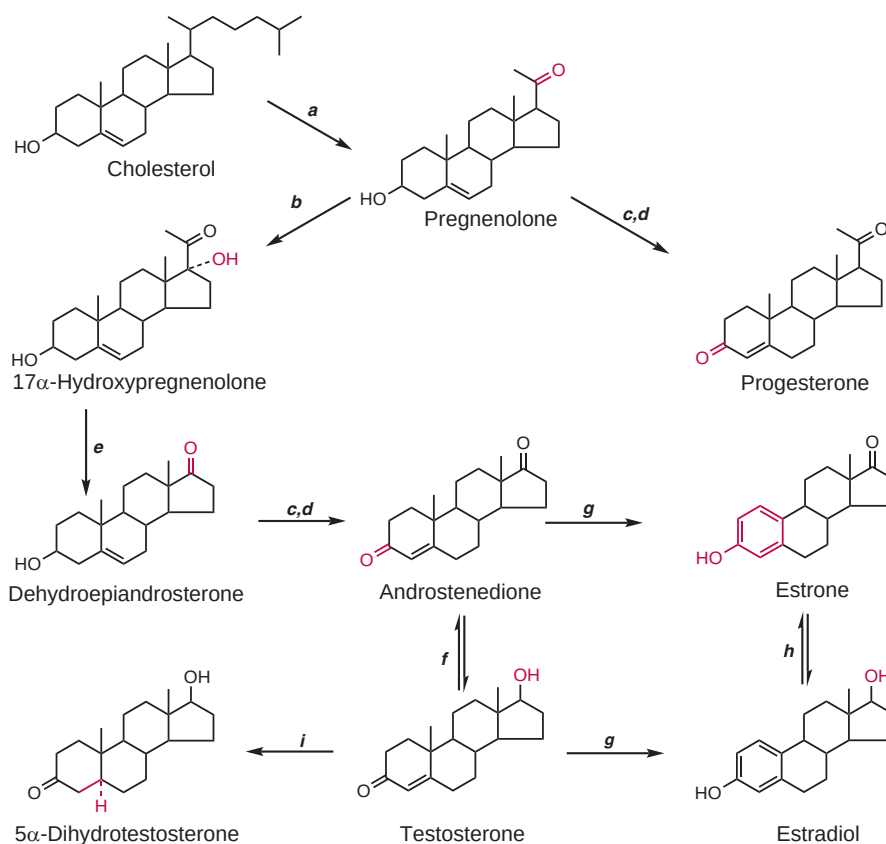


FIGURE 41.3 Biosynthesis of sex steroid hormones. The enzymes involved in this biosynthesis are (a) side-chain cleavage, (b) 17 α -hydroxylase, (c) 5-ene-3 β -hydroxysteroid dehydrogenase, (d) 3-oxosteroid-4,5-isomerase, (e) 17,20-lyase, (f) 17 β -hydroxysteroid dehydrogenase, (g) aromatase, (h) estradiol dehydrogenase, and (i) 5 α -reductase.

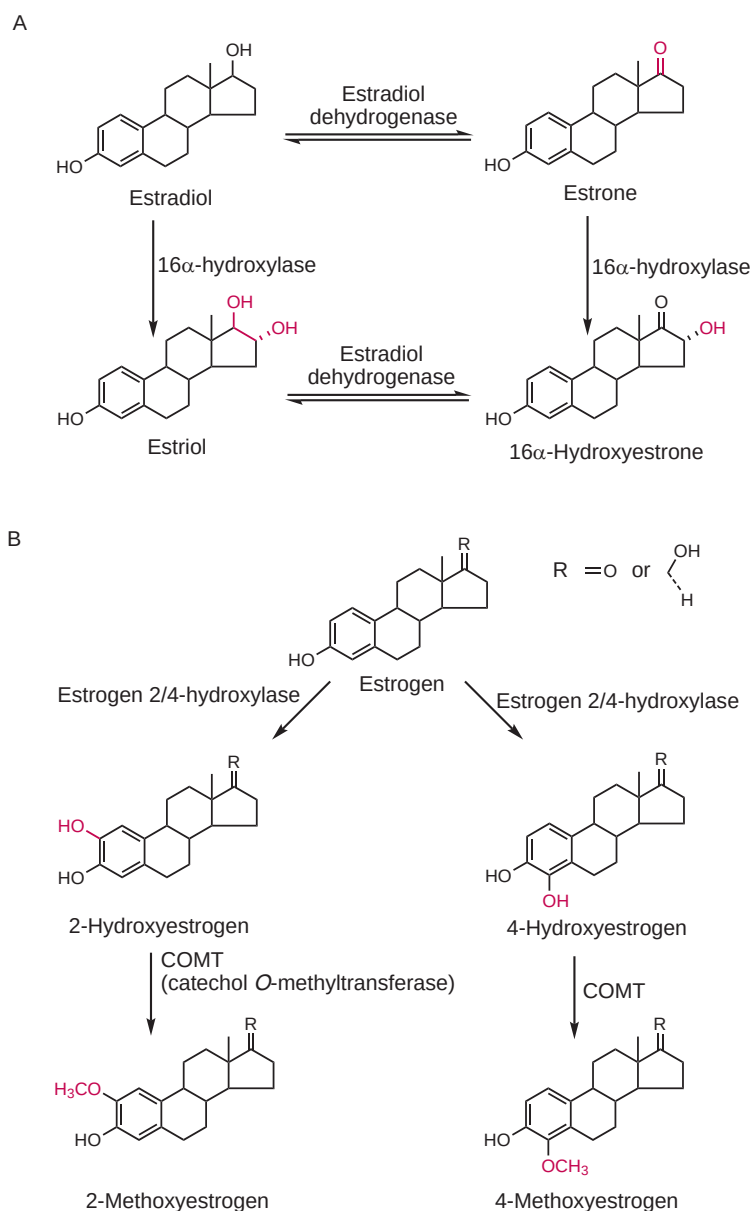


FIGURE 41.4 Estrogen metabolism.

metabolites that are readily excreted via the kidney, and the remaining 50% is found in bile fluid (9). Much of the material found in bile subsequently enters the intestine, where it is reabsorbed and returned to the liver. Only 10% of an administered dose is found in the feces (5). Both 17β -estradiol and estrone are converted by 16α -hydroxylase to yield estriol (Fig. 41.4A), which is found in the urine as the glucuronide conjugate.

Estrogens are metabolized by estrogen 2/4-hydroxylase (CYP3A4) at positions ortho to the 3-phenolic group to form the 2-hydroxyestrogens and the 4-hydroxyestrogens (Fig. 41.4B). The resulting catechol estrogens bind to estrogen receptors (ERs) and produce weak to moderate estrogenic effects. These metabolites are unstable *in vivo*, however,

and are rapidly converted to their 2-methoxy and 4-methoxyestrogen metabolites as well as to their glucuronide, sulfate, and glutathione conjugates. These metabolites are found in comparatively large amounts in the urine (9,10).

Estradiol is 37% bound to sex hormone binding globulin (SHBG) and 61% bound to albumin, leaving only 2% free in circulation (Table 41.3) (11,12). The affinity of estrone and of the sulfated esters of estrone and estradiol for SHBG is less than that of estradiol; however, they are more tightly bound to albumin than estradiol is. These proteins are vital to the transport, distribution, and metabolic clearance rate of estrogens (13). In addition, estradiol is capable of inducing the synthesis of SHBG.

TABLE 41.3 Estrogenic Affinity for Binding Proteins and Metabolic Clearance Rate (11,12)

Estrogen	RBA to SHBG	% Bound to SHBG	% Bound to Albumin	Metabolic Clearance Rate (L/day/m ²)
17 β Estradiol	50	37	61	580
Estrone	12	16	80	1050
Estriol	0.3	1	91	1100
Estrone sulfate	0	0	99	80
Equilin sulfate	0			175

RBA, relative binding affinity (testosterone, 100%); SHBG, sex hormone-binding globulin.

Mechanism of Estrogenic Action

MOLECULAR INTERACTIONS Estradiol plays a key role in several physiologic processes, including development of secondary sex characteristics during puberty and stimulation of the mammary glands during pregnancy, and has thermoregulatory capacity. Estradiol facilitates these processes via its biologic target, the ER, of which there are two subtypes (ER α and ER β). The two receptors differ in size: ER α is composed of 595 amino acids, and ER β is composed of 485 amino acids. The expression and distribution of these subtypes is inconsistent between the various tissues and organs, which may explain the wide response that is observed (11). The predominant ER in the female reproductive tract and mammary glands is ER α , whereas ER β is found primarily in vascular endothelial cells, bone, and male prostate tissues. Expression of both ER α and ER β can be regulated hormonally via estradiol. Estradiol has similar affinities for both ER α and ER β (Table 41.4), which is not the case for certain nonsteroidal estrogenic compounds and antiestrogens (14).

Within each target cell are two receptor locations, one within the nucleus, where a genomic mechanism predominates, and a second within the cell membrane, where a nongenomic mechanism prevails. Unlike the nuclear receptors, the cell membrane receptors are coupled to G proteins that are linked to a cascade of intracellular signals. Nuclear ER α and ER β are virtually identical in their DNA binding sites but are only approximately 60% homologous in the C-terminal ligand binding domain (11,13).

When estradiol binds either ER α or ER β , the receptor protein is phosphorylated and undergoes a conformational change to produce either homo- or heterodimers (ER α /ER α , ER β /ER β , or ER α /ER β) (15). The dimeric ER complex then migrates from the cytosol to the cell nucleus, where it teams up with specific estrogen response elements (EREs) found within an adaptor protein, typically a promoter, which aids in binding of the complex to estrogen activated genes (11,13). This complex also

TABLE 41.4 Relative Binding Affinities of Endogenous and Exogenous Estrogens (14)

Estrogen	ER α Binding Affinity	ER β Binding Affinity
17 β Estradiol	100	100
Estrone	60	37
Estrone sulfate	<1	<1
Estriol	14	21
Coumestrol	94	185
Genistein	5	36
Tamoxifen	7	6
Clomiphene	25	12
Diethylstilbestrol	468	295

ER, estrogen receptor.

enlists a coactivator (CoA) complex to this promoter, which regulates DNA transcription (Fig. 41.5). Because the ER can interact with other nuclear receptors as well as with transcription factors, it is not easy to tease out all the factors that control estrogen-mediated biologic responses.

PHYSIOLOGIC EFFECTS One of the principal actions of the estrogens is to promote the development of female secondary sex characteristics. These feminizing attributes include hair growth, skin softening, breast growth, and accumulation of fat in the thighs, hips, and buttocks. Estrogen also stimulates the growth and development of the female reproductive tract, including the uterine oviduct, cervix, and vagina.

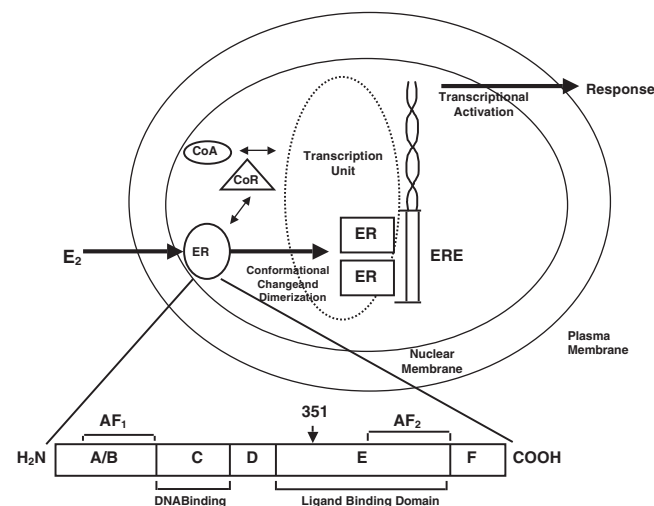


FIGURE 41.5 A subcellular model of estradiol (E₂) action in a target tissue. The abbreviations used are as described in the text.

Estrogens play a significant role in breast tissue as well. Considerable research has focused on understanding breast cancer and the factors that influence its growth. Estrogens serve as “fuel” for hormone-dependent mammary carcinoma and cause proliferation of breast cells. They also stimulate gene expression and, therefore, the production of several proteins, including intracellular proteins important for breast cell function and growth, as well as proteins that influence tumor growth and metastasis. Some of these intracellular proteins include the enzymes needed for DNA synthesis, such as DNA polymerase, thymidine kinase, thymidylate synthetase, and dihydrofolate reductase (16,17).

Steroidal Estrogens

ESTRADIOL Estradiol is the most potent endogenous estrogen, exhibiting high affinity for the ER and high potency when administered parenterally. When administered orally, estradiol is promptly conjugated in the intestine and oxidatively metabolized by the liver, resulting in its low oral bioavailability and therapeutic effectiveness.

ETHINYL ESTRADIOL AND MESTRANOL One method to increase the oral bioavailability of estradiol is to prevent metabolic oxidation of the estradiol C₁₇ hydroxyl group to the corresponding ketone (estrone). This is readily accomplished via alkylation of the C₁₇ position with a chemically inert alkyne group (e.g., ethinyl estradiol [EE]) (Fig. 41.6). This synthetic analog is several hundred-fold more potent than estradiol, with doses in the microgram rather than the milligram range (Table 41.5) (18). Following oral administration, EE is rapidly and almost completely absorbed, with an oral bioavailability of approximately 40% and an elimination half-life of 26 hours (Table 41.6). EE undergoes extensive first-pass metabolism to its 3-*O*-glucuronide and 3-*O*-sulfate metabolites and, via

TABLE 41.5 Potencies of Orally Administered Estrogens (12,13)

Type of Estrogen	FSH Suppression	Serum SHBG Suppression
Piperazine estrone sulfate	1.1	1
Micronized estradiol	1.3	1
Conjugated estrogens	1.4	3.2
Ethinyl estradiol	80–200	614

FSH, follicle-stimulating hormone; SHBG, sex hormone-binding globulin

aromatic hydroxylation, to 2-hydroxyethinylestradiol and its *O*-methyl metabolites. EE undergoes extensive enterohepatic recycling. The bacteria in the gastrointestinal (GI) tract hydrolyze the glucuronide and sulfate conjugates, thereby permitting reabsorption of EE. It is for this reason that a number of antibacterial agents have an adverse effect on oral contraceptive (OC) efficacy.

Another semisynthetic estrogen, mestranol, is the 3-*O*-methyl ether of EE (Fig. 41.6). Mestranol is a prodrug and, following oral administration, is rapidly metabolized to EE via hepatic oxidative *O*-demethylation. Mestranol and EE are used primarily in OC combination formulations (19).

ESTERS OF ESTRADIOL To deliver estrogens with a longer duration of action, 17 β -estradiol must be derivatized into an ester prodrug. In contrast to the ethinyl derivatives delivered orally, these estrogen analogs usually are administered intramuscularly. Slow hydrolysis of the ester releases free estradiol over a prolonged period of time. The therapeutically useful esters of estradiol include 17 β -valerate and 17 β -cyclopentylpropionate (cypionate) (Fig. 41.6). Estradiol cypionate (Depo-Estradiol) is available as a sterile solution of the drug in oil (e.g., cottonseed oil), with a duration of action of 14 to 28 days. Estradiol valerate (Delestrogen) is available as a sterile solution in a vegetable oil (e.g., sesame oil or castor oil), with a duration of action of 14 to 21 days. Estradiol valerate is also the estrogenic component of a newly approved combination OC (Natazia). After oral administration, estradiol valerate undergoes metabolic activation during absorption through GI mucosa and first pass through the liver via CYP3A pathways to 17 β -estradiol (20).

CONJUGATED ESTROGENS Pregnant mares produce two unique estrogenic compounds, equilenin and equilin, that are excreted in the urine as sodium sulfate conjugates. These conjugated metabolites also are used as estrogen preparations (Fig. 41.7). Conjugated Estrogens USP is a mixture of the sodium salts of the sulfate esters of estradiol derived from equine urine or prepared synthetically from estrone and equilin. These preparations (Premarin) are composed of a mixture of sodium

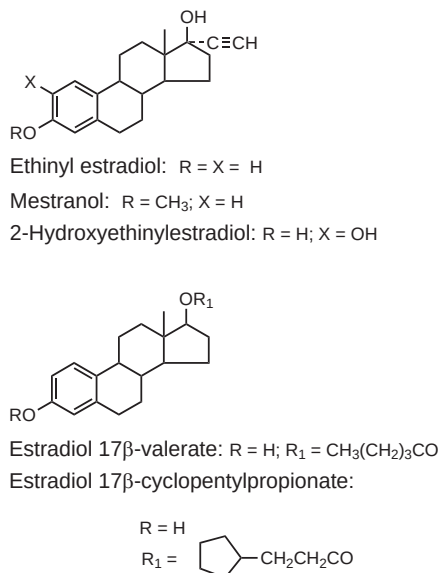


FIGURE 41.6 17 α -Ethinyl estrogens and estradiol esters.

TABLE 41.6 Pharmacokinetic Properties for Some Estrogenic and Progestinal Agents^a

Drug	Protein ^b Binding	Oral Bioavailability	Biotransformation	Elimination Half-life (hrs)	Time to Peak Conc. (hrs)	Peak Serum Conc.(ng/mL)	Elimination (%)
Estradiol	50–80%	Poor	Hepatic first pass	20 minutes		0.1–0.2	Renal: 90 ^d
Ethinyl estradiol	98%	40%	metabolism	26 (6–20)	1–2	33	
Progesterone							
Oral 200 mg	>90%	<10%	Hepatic first pass	<5 minutes	2–4	24.3	Renal: 50–60 ^d
micronized			metabolism				
IM 45 mg			Hepatic	10 weeks	28	39.1	Fecal: 10
IM 90 mg				19.6	9.2	53.8	
Vaginal gel 45 mg				34.8	6.8	14.9	
Medroxyprogesterone acetate: Oral 10 mg	>90%	High	Hepatic	30	2–4	19–35	Renal: 15–22 ^d
IM 150 mg/mL			No first-pass	50 days	3 weeks	1–7	Fecal: 45–80
every 3 months			hepatic metabolism				
Megestrol acetate							
Oral 160 mg	>90	ND	Hepatic	38 (13–104)	2–3	200	Renal: 66 ^d
Oral 600 mg					2–3	753	Fecal: 20
Norgestrel	>90%	60%	Hepatic	20	24	ND	Renal: 45 ^d Fecal: 32
Levonorgestrel 3/12/60 months implants	>90	60	No first-pass	16 (8–30)	24	1,6 first week, then 0.26–0.4	Renal: 45 ^d Fecal: 32
216 mg loading dose ^d			Hepatic metabolism				
Desogestrel (Desogen) (as 3-keto-desogestrel)	>90	76	First pass to 3-ketodesogestrel active metabolite	12–58	1–2	2–6	Renal: 43 ^d Fecal: 50
Norethindrone	>80	65	Hepatic first pass metabolism	8 (5–14)	0.5–4.0	5–10	Renal: 50 ^d Fecal: 20–40
Norethindrone acetate	>80	65	Hepatic first pass metabolism	8 (5–14)	0.5–4.0	5–10	Renal: 50 ^d Fecal: 20–40
Norgestimate as (desacetylnorgestimate)	>50–60 >90	60	First pass to desacetylnor- gestimate	37	1–2	0.5–0.7	Renal: 47 ^d Fecal: 37

^aUSP Drug Information 2000.^bSex hormone binding globulin (SHBG) synthesis is stimulated by estrogens and inhibited by androgens: levels are twice as high in women as in men.^cProgesterone binds strongly to cortisol binding globulin (CBG; 17.7%) and SHBG (0.6%) and weakly to albumin (79.3%). Absorption is the rate-limiting step for the elimination half-life of Levonorgestrel: free, 1.1–1.7%; SHBG, 92–62%; and albumin, 37.56%; but suppresses SHBG by 33%. Norethindrone: free, 3.5%; SHBG, 35.5%; and albumin, 61%. Medroxyprogesterone does not bind SHBG. 3-Keto-desogestrel, 64%; albumin, 32%. Norgestimate >90% protein bound.^dA mean dose of 35 µg levonorgestrel is released daily.

ND, no data available.

estrone sulfate (52.5% to 61.5%) and sodium equilin sulfate (22.5% to 30.5%) as well as several lesser metabolites (e.g., 17 α - and 17 β -dihydroequilenin or 17 α - and 17 β -estradiol) as sodium sulfate conjugates. Esterified Estrogens USP (Menest) is a blend of estrone sodium sulfate (75% to 85%) and sodium equilin sulfate (6% to 15%) (Fig. 41.7).

Synthetic conjugated estrogens A (Cenestin) contains a mixture of 9 of the 10 known synthetic conjugated estrogenic substances present in Conjugated Estrogens USP. This includes a mixture of the sodium salts of estrogen sulfates, including estrone sulfate and sodium equilin sulfate. Synthetic conjugated estrogens B (Enjuvia)

contains the same set of synthetic estrogens as Cenestin, as well as sodium $\Delta^{8,9}$ -dehydroestrone sulfate.

Another sulfate conjugate that is orally effective is estropipate (piperazine estrone sulfate; Ortho-Est) (Fig. 41.7). This derivative has the same actions and utility as the conjugated, naturally occurring estrogens.

Nonsteroidal Estrogens

STILBENE DERIVATIVES The steroid nucleus is not required for estrogenic action. Several derivatives of stilbene (diphenylethylene) that were used therapeutically demonstrate potent estrogenic activity. Diethylstilbestrol (DES), prepared in 1939, was one of these stilbenes (Fig. 41.8).

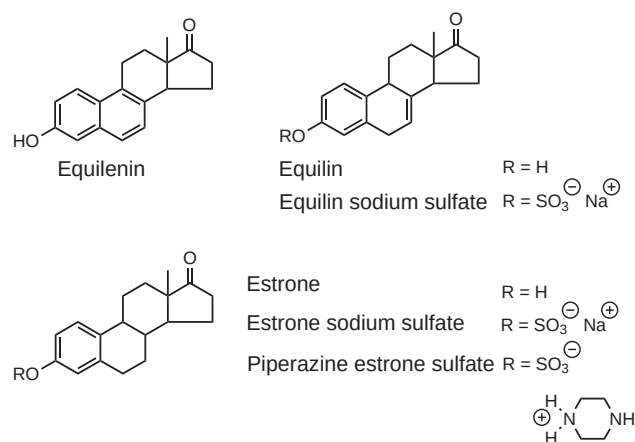


FIGURE 41.7 Conjugated and esterified estrogens.

A *trans* stilbene, DES has 10-fold the estrogenic potency of its *cis* isomer, largely because the *trans* isomer more closely resembles estradiol (21). Unfortunately, when DES was administered to relieve pregnancy-related symptoms, it was correlated with abnormal growth in the offspring. Studies are now suggesting that women exposed to DES prenatally have a higher incidence of breast cancer after age 40 (22). It has been hypothesized that the catechol metabolites of estradiol and DES are oxidized to the corresponding *o*-quinones. A quinone-DNA reaction leads to depurination that generates mutations that may represent a critical step in the initiation of cancer (Fig. 41.9) (23).

Structure–Activity Relationships

As a result of numerous studies, there is an extensive body of knowledge regarding structure–activity relationships for estrogens (24–27). The aromatic A ring and the C₃ hydroxyl group are structural features essential for estrogenic activity. The 17β-hydroxyl, the distance between the C₃ and C₁₇ hydroxyl groups, and the presence of planar hydrophobic scaffolding also are important structural contributors and help to optimize estrogenic activity. Ideally, the distance between the oxygen atoms of the C₃ and C₁₇ hydroxyl groups should range from 10.3 to 12.1 Å.

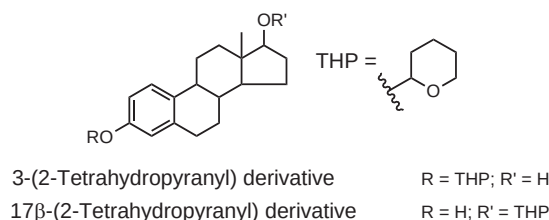
Substitution of the estrogen steroid nucleus can significantly modify estrogenic activity. Functionality at the C₁ position greatly reduces activity, and only small groups can be accommodated at the 2 and 4 positions. Addition of hydroxyl groups at positions 6, 7, and 11 reduces activity. Removal of the oxygen function from position 3 or 17 or epimerization of the 17β-hydroxyl group of estradiol to the α-configuration results in estrogenic analogs

that are less active (28). The equine estrogens contain one or two additional double bonds in the steroidal B ring (equilin and equilenin, respectively) (Fig. 41.7). The presence of this unsaturation substantially boosts the estrogenic potency of these estrogens. Substituents at the 11β position are tolerated; for example, 11β-methoxy or 11β-ethyl has significantly greater affinity for the ER as compared to estradiol.

Certain modifications at the 17α and 16 positions can lead to enhanced activity. For example, the 17α-ethynyl or 17α-vinyl groups provide the greatest activity, whereas highly polar substituents at this position are poorly tolerated. At the 16 position, moderate size and polarity are tolerated. Enlargement of the D ring (i.e., *D*-homoestradiol) greatly reduces estrogenic activity.

Pharmacokinetics

The bioavailability observed varies with the route of estrogen administration. When administered subcutaneously, the order of bioavailability of the three naturally occurring steroids is estradiol > estrone > estriol. When administered orally, this order changes to estriol > estradiol > estrone (29). Although all three of these estrogens demonstrate at least minimal bioavailability when administered orally, chemical modifications have led to estrogen analogs with improved oral activity. For example, EE is a more effective oral estrogen because of its resistance to metabolism in the GI tract and in the liver. Other types of highly active, orally bioavailable estrogen analogs include those with a labile ether [e.g., 3-(2-tetrahydropyranyl) and 17β-(2-tetrahydropyranyl) estradiol] (30). These drugs proved to be 12- and 15-fold as active, respectively, as estradiol.



Estrogen Antagonists

Agents that antagonize the actions of estrogen are of particular interest for their contraceptive utility and for the treatment of estrogen-dependent breast cancer. There are three classes of agents that either interfere with estrogen activation of its receptor (e.g., impeded estrogens or triphenylethylene antiestrogens) or limit estrogen biosynthesis (e.g., aromatase inhibitors).

IMPEDED ESTROGENS The impeded estrogens (e.g., estriol) interact with the ER in target tissues but dissociate from the receptor too rapidly to produce a significant estrogenic effect. If, however, the impeded estrogens are present in a high enough local concentration, they will compete with or impede estradiol access to its receptor site, thereby reducing the effect of estradiol on the cell (5,31).

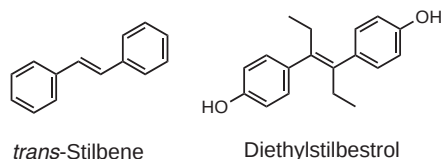


FIGURE 41.8 Nonsteroidal estrogens.

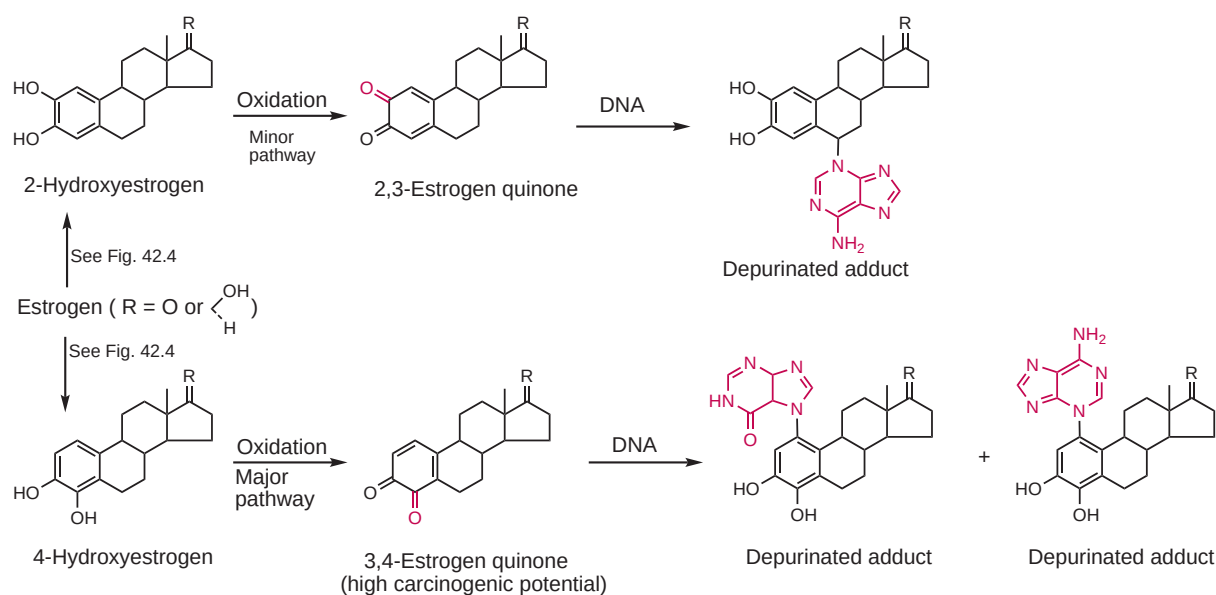
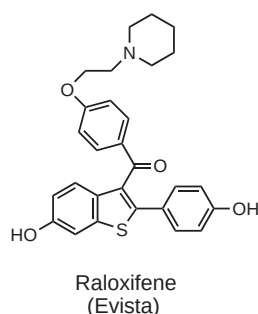


FIGURE 41.9 Metabolic pathways involving estrogens leading to DNA damage through depurination.

ANTIESTROGENS—TRIPHENYLETHYLENE ANALOGS The triphenylethylene antiestrogens (Fig. 41.10) are structurally related to the stilbene family of estrogens and exhibit high affinity for the ER. They prevent translocation of the estrogen–receptor complex into the nucleus of target cells and interfere with the binding of the receptor–hormone complex to the acceptor site of the chromatin (32,33).

Advances in the molecular pharmacology of estrogen and ERs have led to the development of selective estrogen receptor modulators (SERMs). These agents exhibit tissue-specific estrogen agonist or antagonist activity (34). The 3,4-dihydronaphthalene- and benzothiophene-containing SERMs are rigid analogs of the triphenylethylenes. The first SERM to be marketed for the treatment of osteoporosis was the benzothiophene raloxifene, an estrogen agonist at the bone (see Chapter 30). Discussion about the SERMs used in the treatment of breast cancer can be found later in this chapter.



AROMATASE INHIBITORS

Androstenedione Derivatives Inhibitors of aromatase block the conversion of androgens to estrogens and, therefore, have the therapeutic potential to control reproductive functions and aid in the treatment

of estrogen-dependent cancers, such as breast cancer. These steroidal agents compete with androstenedione for the active site of the aromatase enzyme. The structure–activity relationships for steroidal aromatase inhibitors indicate that the best agents are substrate analogs, with only small structural changes to the A ring and at C₁₉ permitted (Fig. 41.11). Analogs that contain aryl functionalities at the 7 α position have enhanced affinity for the enzyme. In addition, 4-hydroxy-androstenedione, several androst-1,4-diene-3,17-diones, and 10 β -propynyl-estr-4-ene-3,17-dione act as enzyme-activated irreversible

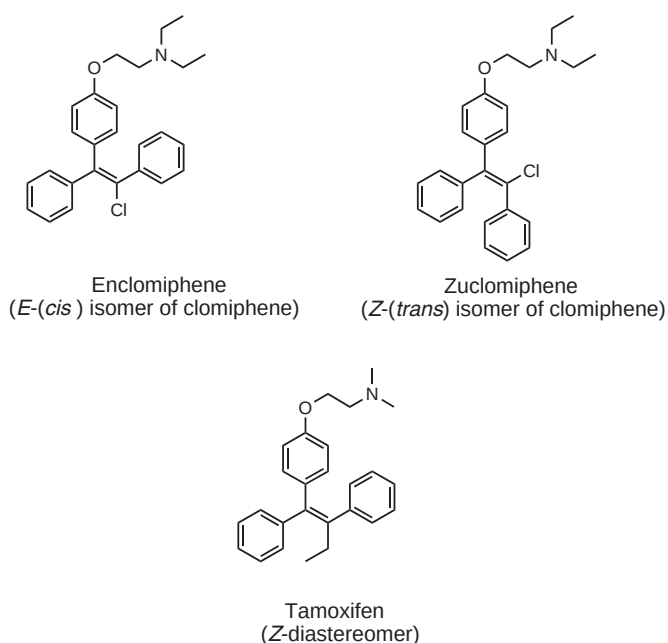


FIGURE 41.10 Antiestrogens—triphenylethylene analogs.

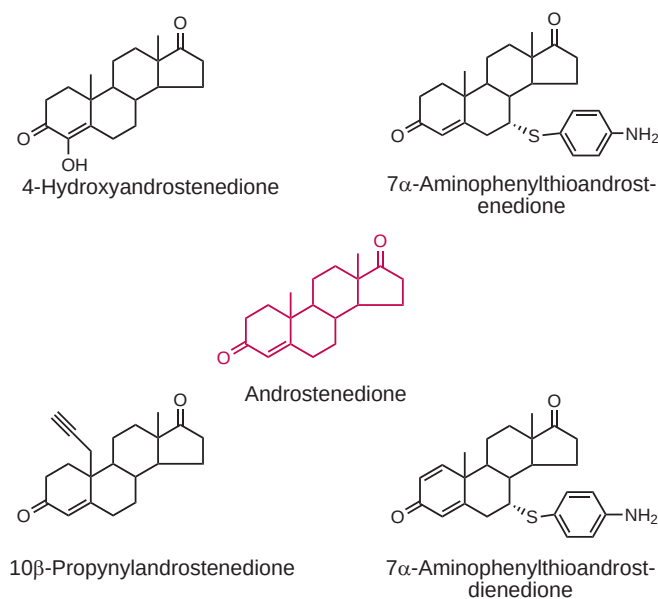


FIGURE 41.11 Androstenedione aromatase inhibitors.

inhibitors (suicide substrates) in vitro. Additional information about the use of aromatase inhibitors in the treatment of breast cancer can be found later in this chapter.

Triazole Derivatives Triazole-based aromatase inhibitors were developed based on the aromatase inhibitor aminoglutethimide (35,36) and include anastrozole (37) and letrozole (38) (Fig. 41.12). The triazoles inhibit aromatase as a result of the N-4 nitrogen of the triazole ring interaction with the heme iron atom of this CYP19 enzyme complex. Anastrozole and letrozole are competitive inhibitors of aromatase and selectively inhibit the conversion of testosterone to estrogens in all tissues and, thereby, reduce serum concentrations of circulating estrone, estradiol, and estrone sulfate. Because estrogen acts as a growth factor for hormone-dependent breast cancer cells, reduction of serum and tumor concentrations of estrogen inhibits tumor growth and delays disease progression. Ovarian production of estrogen declines in postmenopausal women, so the conversion of androstenedione and testosterone to estrone and estradiol in peripheral tissues becomes the primary source of estrogen. The structure–activity relationships of these inhibitors have

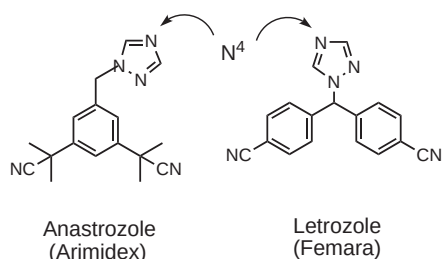


FIGURE 41.12 Triazole aromatase inhibitors.

been reviewed elsewhere (39–41). Additional information about the triazole derivatives as it relates to the treatment of breast cancer can be found later in this chapter.

Clinical Applications

Estrogens are used in the treatment of a wide variety of menstrual-related disturbances, including the management of menopausal symptoms (e.g., vulvar and vaginal atrophy); female hypoestrogenism resulting from hypogonadism, castration, or primary ovarian failure; and amenorrhea, dysmenorrhea, and oligomenorrhea. They also are effective in the management of ovarian development failure, acne, and senile vaginitis. After childbirth, estrogens have been used to suppress lactation in women who elect not to participate in breastfeeding. More detailed discussions about the clinical application of estrogens in the treatment of several disease states can be found later in this chapter.

Administration of estrogens may promote sodium retention and, as a result, cause retention of water and, possibly, edema. This effect, however, is less pronounced than that observed with the glucocorticoids. More detailed information regarding the pharmacology and toxicology of estrogens can be found in published reviews (5,24,42).

Progestins

Fraenkel (43) first observed in 1903 that removal of the corpus luteum shortly after conception resulted in pregnancy termination. In 1929, Corner and Allen (44) developed an assay method for progestogenic activity, and by 1934, the hormone progesterone had been isolated by several research groups (29).

Biosynthesis

Progesterone biosynthesis is initiated when LH, secreted from the anterior pituitary, binds to the target cell surface LH receptor. Activation of the LH receptor results in an increase of intracellular cyclic adenosine monophosphate (cAMP) levels via activation of a G protein and adenylate cyclase. In the presence of elevated concentrations of cAMP, cholesterol esterase activation occurs. This enzyme catalyzes the cleavage of cholesterol esters to free cholesterol, which is then converted in mitochondria to pregnenolone as described previously. The formation of progesterone from pregnenolone is catalyzed by 5-ene-3 β -hydroxysteroid dehydrogenase and 3-oxosteroid-4,5-isomerase (steps c and d in Fig. 41.3).

Progesterone is secreted primarily from the ovaries (30 mg/day), particularly by the corpus luteum in reproductive-age women and by the placenta during pregnancy. Compared with the ovaries, the adrenal gland produces only a small fraction of progesterone (1 mg/day).

Metabolism

Progesterone is rapidly metabolized by the liver, regardless of the route of administration, and has a half-life of 5 to 10 minutes (Fig. 41.13). Progesterone is mainly excreted renally as the glucuronide and sulfate conjugates

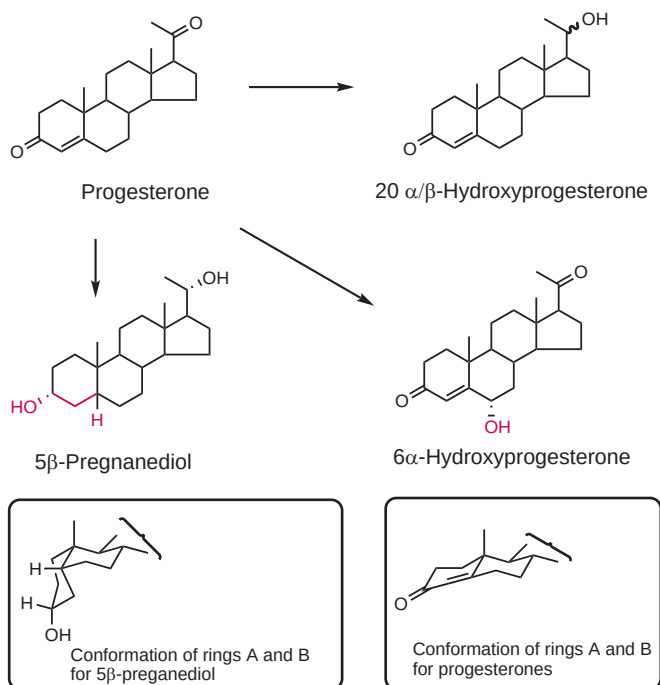


FIGURE 41.13 Metabolism of progesterone.

of 5 β -pregnanediol. The glucuronide metabolite serves as an index of the corpus luteum activity, and a premature drop in its level may be a warning of possible miscarriage.

Other routes of metabolism include 6 α -hydroxylation and reduction of the 20-ketone. These metabolites have no significant progestogenic activity. The formation of 5 β -pregnanediol from progesterone is characterized by reduction of the 4,5-double bond to create a *cis* A/B ring juncture and reduction of the 3-ketone to a 3 α -hydroxy group.

In the serum, progesterone is bound to either cortisol binding globulin or albumin, with only a small fraction being freely available. The metabolic clearance rate for progesterone is 2,100 to 2,500 L/day, for which protein binding has no role.

Mechanism of Progestin Action—Molecular Interactions

The progesterone receptor is a ligand-activated transcription factor and a member of the nuclear receptor superfamily. All nuclear receptors have three conserved functional domains, including the DNA binding domain, the N-terminal domain, and the C-terminal ligand binding domain (42). Very little is known about the structure of the N-terminal domain. The DNA binding domain contains two asymmetric zinc fingers with four cysteine residues coordinated to each zinc atom. Between the zinc fingers is an α helix, which has specific interactions with the DNA. The ligand binding domain is composed of 12 α helices and four β sheets that ultimately form a hydrophobic cavity in which the receptor agonist binds (45). When progesterone binds to its receptor, a conformational change in the receptor occurs, transforming it into an active

transcriptional factor. Phosphorylation of the receptor occurs at this point, but it is unclear what biochemical role that plays. The receptor then undergoes dimerization to form PR-A and PR-B, which are not structurally identical. These isoforms have similar activities related to steroid hormone and DNA binding, but their functional activities are unique. The dimer interacts with specific nuclear DNA sequences (hormone response elements) within progesterone-responsive genes. The receptor agonist–receptor dimer complex combines with coactivators that link the complex to gene transcription (46). The rate of gene transcription is then either increased or decreased. Once gene transcription is initiated, proteins are biosynthesized, and physiologic effects are observed in target cells and organs.

Physiologic Effects

The primary physiologic site of action of progesterone is the uterus. It acts on both the endometrium (inner mucous lining) and the myometrium (muscle mass) of the uterus. The effect of progesterone on the endometrium, already primed by estrogens, is to induce the secretory phase of the menstrual cycle. During this phase, the endometrial glands grow and secrete large amounts of carbohydrates that can be used by the fertilized ovum as an energy source. The primary function of progesterone with respect to the myometrium is to stop spontaneous rhythmic contractions of the uterus.

Progesterone often is referred to as the “hormone of pregnancy.” For the first trimester, the corpus luteum serves as the primary source of progesterone, at which point the developing placenta takes over as the major source of progesterone and estrogen. The high level of progesterone that is produced during pregnancy sends a signal to the hypothalamus via the negative feedback system to prevent release of the FSH and LH necessary for the development of new ova.

In general, the nonreproductive effects of progesterone are fairly insignificant. Progesterone is able to antagonize the actions of aldosterone, and as a result, there is an increase in sodium excretion. When sodium levels drop, aldosterone secretion is stimulated, and sodium is retained (47). It has been suggested that the temperature-raising effect of progesterone may be the result of a reduction in perspiration. This effect is not unique to progesterone, because other steroids in the pregnane and androstane series have a similar effect (48).

Ovarian biosynthesis and secretion of progesterone is controlled by the release of LH from the anterior pituitary during ovulation. The LH induces progesterone secretion from the corpus luteum during the second half of the menstrual cycle. If conception does not occur, the corpus luteum degenerates, and progesterone production decreases. As progesterone levels drop, endometrial sloughing occurs—otherwise known as menstruation.

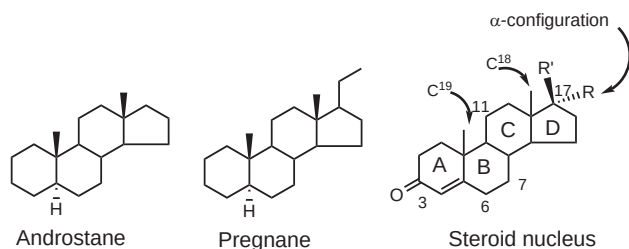
Side Effects

Characteristic progestin-related side effects include nausea and vomiting, drowsiness, and breakthrough and

irregular bleeding. With prolonged therapy, a greater incidence of these effects may occur, along with edema and weight gain, breast pain, decreased libido, and masculinization of the female fetus.

Structure–Activity Relationships

Progestin activity is restricted to those molecules with a steroid nucleus. The synthetic progestins generally can be divided into two steroidal classes: the androstanes (including the 19-norandrostanes), and the pregnanes (including the 19-norpregnanes) (46–51). In the androstane series, a 17α -substituent, such as ethynyl, methyl, ethyl, and variations of these, provides oral bioavailability.



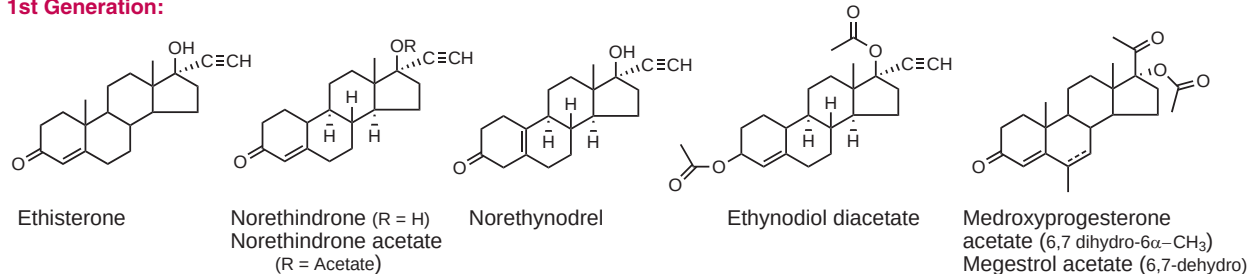
Ethisterone (17 β -hydroxy-17 α -ethynyl progesterone) (Fig. 41.14), the first androstane found to be effective, has only about one-third the activity of progesterone

when delivered subcutaneously, but is 15-fold more active than progesterone when administered orally. Closely related to testosterone, this progestin has significant androgenic activity. Removal of the CH_3 group at position 19 leads to norethindrone (norethisterone; 19-norandrostane) (Fig. 41.14), which has 5- to 10-fold more progestin activity. The activity of norethindrone is increased further by the addition of a chlorine substituent at position 21 (blocks metabolic hydroxylation) or by the addition of a methyl group at carbon 18 (norgestrel) (Fig. 41.14). Ethynodiol diacetate, another 19-norandrostane, is an extremely potent oral progestin. It is more active when administered orally than parenterally and, when combined with an estrogen, is effective as an OC.

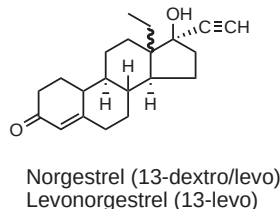
Further unsaturation of the B or C ring of androstane derivatives usually enhances progestin activity. Introduction of a halogen or methyl substituent in the 6α or 7α positions generally increases hormonal activity. Acetylation of the 17β -OH of norethindrone increases the duration of action of the drug. Removal of the 3-keto function of norethindrone allows retention of potent progestin activity without androgenic effects.

Activity of the pregnanes and 19-norpregnanes is enhanced by unsaturation at C_6 and C_7 and by substitution of a methyl group or a halogen at C_6 . This activity

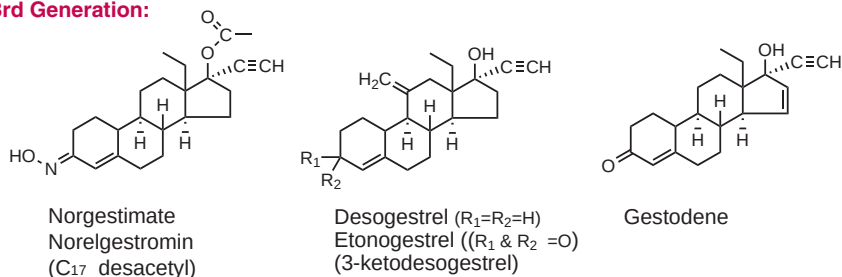
1st Generation:



2nd Generation:



3rd Generation:



4th Generation:

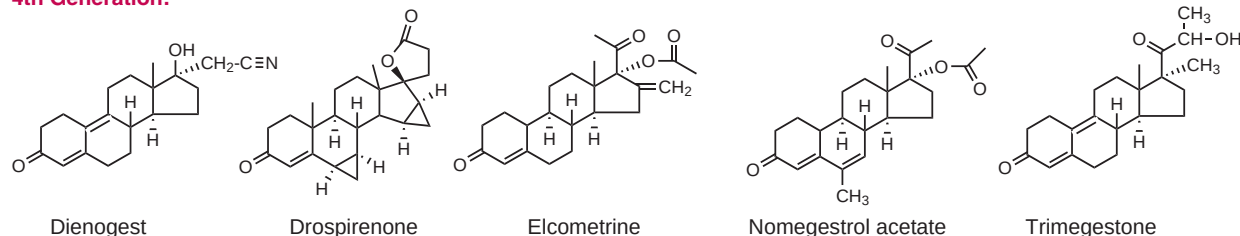


FIGURE 41.14 Progestins.

may be further increased by introducing a CH_3 group at C_{11} . These substitutions prevent metabolic reduction of the two carbonyl groups and metabolic oxidation at C_6 . Substitution of a fluoro group at C_{21} prevents metabolic hydroxylation and enhances oral effectiveness.

Because glucocorticoid receptor activity is not desirable yet is innate to those derivatives that bind with high affinity to the progesterone receptor, structural alterations have been investigated to pare out this biologic activity. Small B-ring substituents serve to decrease glucocorticoid action, as do substituents at the C_{17} position.

Synthetic Progestins

Early generations of the progestins were used primarily for contraceptive purposes, so antigonadotropic activity was considered to be desirable. Unfortunately, many of these agents were plagued by androgenic activity and the corresponding adverse effects (Table 41.7) (11). Development of newer progestins is now focused on analogs with improved progesterone receptor selectivity (Table 41.8) and little or no effect on the androgen, estrogen, or glucocorticoid receptors (43). From a structural perspective, these synthetic progestins contain either a pregnane or androstane steroid nucleus.

Not only can the progestins be classified by type of steroid ring system, they also can be organized by generation. The first-generation agents include the pregnanes (medroxyprogesterone acetate, megestrol acetate, 17α -acetoxy progesterone) (Fig. 41.14), the androstanes (ethisterone, dimethisterone), and the 19-norandrostanes (norethynodrel, norethindrone, norethindrone acetate, ethynodiol acetate) (Fig. 41.14). Norgestrel and levonorgestrel (19-norandrostanes) are two of the predominant agents in the second generation (Fig. 41.14). The levonorgestrel derivatives (19-norandrostanes) desogestrel,

gestodene, and norgestimate compose the third generation (Fig. 41.14), and the newer, fourth generation includes nomegestrol acetate (19-norpregnane) and trimegestone (pregnane) (Fig. 41.14). In addition, there is one nonethynylated 19-norandrostane, dienogest, and one spironolactone derivative, drospirenone, to round out the fourth generation (43). The antiandrogenic activity associated with several fourth-generation agents (e.g., trimegestone, drospirenone, and dienogest) makes this group of progestins unique (Table 41.7) (11). Trimegestone and elcometrine are the most potent progestins, with the third-generation levonorgestrel and etonogestrel (3-ketodesogestrel) being slightly less potent (Table 41.8).

Progesterone and Its Derivatives

Progesterone has a significant role in priming the uterine endometrium for implantation of a potential blastocyst. It also is involved in formation of the placenta postimplantation, the development of mammary glands, and by preventing contraction of the uterine musculature, pregnancy maintenance. Progesterone also has inhibitory roles, including ovulation prevention via an antigonadotropic effect and inhibition of the conversion of testosterone to dihydrotestosterone, an active metabolite, by virtue of its ability to be a substrate for 5α -reductase (46). Interestingly, progesterone reduces nuclear ER levels and induces 17-hydroxysteroid dehydrogenase, the enzyme that catalyzes the conversion of estradiol to the less potent estrone (52,53).

There are some limitations as to how progesterone can be administered, because it has relatively low bioavailability when administered orally. The pharmacokinetics for progesterone and its derivatives are listed in Table 41.6. To achieve consistent therapeutic benefit, progesterone

TABLE 41.7 Hormonal Activities of Representative Progestins (11)

Progestogen	Antiestrogenic	Androgenic	Antiandrogenic	Glucocorticoid	Antimineralocorticoid
Progesterone	++	–	+	++	++
Medroxyprogesterone acetate	++	+	–	++	–
Norethisterone (1)	++	++	–	–	–
Levonorgestrel (2)	++	++	–	–	–
Etonogestrel (3)	++	++	–	+	–
Norgestimate (3)	++	++	–	?	?
Drospirenone (4)	++	–	++	–	++
Elcometrine (4)	++	–	–	–	?
Nomegestrol acetate (4)	++	–	++	–	–
Trimegestone (4)	++	–	+	–	+
Dienogest (4)	++	–	++	–	–

The numbers in parentheses indicate the generation of progestogen. ++, effective; +, weekly effective; –, not effective; ?, unknown.

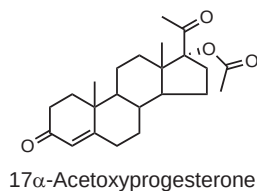
TABLE 41.8 Steroid Receptor Binding Affinities (11,43)

Progestogen	Progesterone	Androgen	Glucocorticoid	Mineralocorticoid
Progesterone	50	0	10	100
Medroxyprogesterone acetate (1)	115	5	29	160
Norethisterone (1)	75	15	0	0
Levonorgestrel (2)	150	45	1	75
Etonogestrel (3)	150	20	14	0
Norgestimate (3)	15	0	1	0
Gestodene (3)	90	85	27	290
Drospirenone (4)	35	65	6	230
Elcometrine (4)	136	0	38	?
Nomegestrol acetate (4)	125	42	6	0
Trimegestone (4)	330	1	9	120
Dienogest (4)	5	10	1	0

The numbers in parentheses indicate the generation of progestogen.

must be administered either by injection or intravaginally. Progesterone may need to be administered over a relatively long period of time to maintain a pregnancy, so it is important to note that neither of the two routes just mentioned is without significant adverse effects. As a result, development of orally active derivatives has been a significant priority.

MEDROXYPROGESTERONE ACETATE The initial structural modifications made to progesterone led to only weakly active or inactive analogs. For example, 17 α -acetoxyprogesterone had limited activity when administered orally (54). Further structural modification of 17 α -acetoxyprogesterone was aimed at limiting metabolic hydroxylation at C₆. This was accomplished by the addition of a C₆ substituent, and the resulting analog displayed improved biologic activity (51).



Among the first of these substituted 17 α -acetoxyprogesterone analogs to be used therapeutically was medroxyprogesterone acetate, a 6 α -methyl progesterone analog (Fig. 41.14) (55). This analog is 25-fold more active than ethisterone. Following oral administration, medroxyprogesterone acetate is completely and rapidly deacetylated by first-pass metabolism to medroxyprogesterone. Medroxyprogesterone is extensively metabolized

via pathways similar to those for progesterone, except for 6 α -hydroxylation. Most medroxyprogesterone acetate metabolites are excreted in the urine, primarily as glucuronide conjugates. Plasma protein binding for medroxyprogesterone is approximately 86%, primarily to serum albumin, with no binding to SHBG.

MEGESTROL ACETATE (MEGACE) Progestin activity is further enhanced when a double bond is introduced between positions 6 and 7, as is found in megestrol acetate (Fig. 41.14). Megestrol is used primarily in the treatment of breast and endometrial carcinomas and in postmenopausal women with advanced hormone-dependent carcinoma. Less than 10% of an oral dose undergoes metabolism. Several major metabolites appear in the urine (e.g., 2-hydroxy and 6-hydroxymethyl megestrol and their glucuronide conjugates).

Synthetic Progestins

ETHISTERONE AND ITS ANALOGS Ethisterone (Fig. 41.14), a 17 α -ethynyl derivative of testosterone, is one of the first synthetic progestins to be used therapeutically. In 1937, this agent was synthesized from male sex hormones (androstanes) in an attempt to find an orally active androgen (56). Ethisterone later proved to be an effective oral progestin and became useful in the treatment of menstrual dysfunctions (52). Several molecular modifications of ethisterone have improved progestogenic action, including introduction of methyl groups in the C_{6 α} and C₂₁ positions (42).

A second breakthrough was made in 1944, when Ehrenstein (57) discovered that the C₁₉ methyl group is not necessary for progestogenic activity. In fact, this

work showed that loss of the C₁₉ methyl results in analogs with activity equal to or greater than that of parenterally administered progesterone. In 1953, Djerassi et al. (58) synthesized 19-norprogesterone. This drug differed from the natural hormone only in replacement of the C₁₉ angular methyl group with a hydrogen atom. When administered parenterally, this analog was eightfold more active than progesterone and, at the time, was the most potent progestin known. While these drugs were important compounds in the development of the progestins, none are used singly or in combination today.

NORETHINDRONE/NORETHISTERONE AND NORETHYNODREL Research on 19-norsteroids as potential progestins culminated in the synthesis of two potent, orally active progestins, norethindrone and norethynodrel (Fig. 41.14) (49). These two substances were among the first 19-norsteroids to be used clinically for progesterone-related disorders. When combined with estrogens, such as EE or mestranol, these agents were effective contraceptives. Because the progestogenic action of norethynodrel is approximately one-tenth that of norethindrone, it is no longer used in OC formulations. Although norethindrone is a weak androgen, it does not exhibit any glucocorticoid or antimineralocorticoid activity (Table 41.7).

After oral administration, norethindrone acetate is completely and rapidly deacetylated by hepatic and intestinal first-pass metabolism to norethindrone, with an oral bioavailability of approximately 64%. Subsequent metabolism of norethindrone includes reduction of the Δ^4 double bond to both the 5 α - and 5 β -dihydronorethindrone products as well as reduction of the ketone. The pharmacokinetics for norethindrone acetate are indistinguishable from those of orally administered norethindrone. Roughly 36% of norethindrone is bound to SHBG, and 61% is bound to albumin (11).

Norethindrone acetate also can be administered transdermally along with estradiol when formulated as a patch. This combination of hormones can be used as part of either a continuous or a cyclic hormone replacement regimen.

NORGESTREL AND LEVONORGESTREL Norgestrel is formulated as a racemic mixture despite the fact that only its levo isomer, levonorgestrel, is pharmacologically active (Fig. 41.14). Levonorgestrel exhibits some androgenic activity but no glucocorticoid or antimineralocorticoid action. Levonorgestrel can be administered orally, transdermally (combined with estradiol and formulated as a 7-day patch), and for prolonged, continuous use, via an intrauterine device (IUD). The oral bioavailability of levonorgestrel is approximately 95% (11). From a protein binding perspective, 48% of an oral dose is bound to SHBG, and 50% is bound to albumin. Levonorgestrel undergoes metabolic reduction of its ketone and is hydroxylated.

NORGESTIMATE AND DESOGESTREL Norgestimate (Fig. 41.14) is considered to be a pro-progestin (prodrug), because it

rapidly undergoes a two-step metabolic transformation to form two active products, norelgestromine (levonorgestrel 3-oxime) and levonorgestrel (Fig. 41.14). Deacetylation occurs in the intestine and liver, whereas conversion of the 3-oxime to the corresponding ketone occurs primarily in the liver. Unlike the other progestins mentioned, norgestimate and its metabolites are not bound to SHBG.

Norgestimate exhibits high selectivity for the progesterone receptor and low androgenic activity as a result of a large drug-induced increase in the production of SHBG. As a component of both mono- and triphasic OC formulations, this progestin exhibits fewer androgenic side effects, including less of a detrimental effect on the patient's lipid profile and lack of significant weight gain (58). Rates of breakthrough bleeding were lower with a triphasic regimen including norgestimate compared to a monophasic regimen containing norethindrone acetate. Norgestimate is metabolized to norelgestromine, the primary active metabolite and the active progestin found in the contraceptive patch (Ortho Evra).

Desogestrel also is a prodrug and is rapidly metabolized in the intestinal mucosa and via first pass through the liver to its active metabolite, etonogestrel (3-ketodesogestrel) (Fig. 41.14). Following oral administration, the relative bioavailability for desogestrel is approximately 84%. Desogestrel also exhibits high selectivity for the progesterone receptor and low androgenic activity, and it does not diminish the beneficial effects of estrogen on the lipid profile. The rates of breakthrough bleeding are low with the monophasic desogestrel/EE formulation, which has improved therapy discontinuation percentages. Unfortunately, there is evidence that suggests that there is a twofold increase in the risk of venous thromboembolism with desogestrel-containing preparations (59).

GESTODENE Unlike the other members of the third generation of progestins, gestodene is not a prodrug (Fig. 41.14). It exhibits nearly 100% oral bioavailability and excellent receptor binding affinity for the progesterone receptor. Because the SHBG level is considered to be a surrogate marker for venous thromboembolism (VTE) risk, it is important to note that an increase in SHBG levels is evident both in monophasic combination OC preparations (200% to 300% increase) and in triphasic preparations (150% increase) that contain gestodene (60). Both the monophasic and triphasic preparations also cause an increase in triglyceride and total cholesterol levels, although the increase is smaller in magnitude than those associated with desogestrel and norgestimate. An increase in high-density lipoprotein (HDL) levels was not observed (61). The effectiveness of a gestodene/EE sustained-release, matrix-type transdermal patch (weekly administration) is currently under investigation. Products containing gestodene are available in Europe, but not the United States.

DENOGEST, DROSPIRENONE, ELCOMETRINE, NOMEGESTROL ACETATE, AND TRIMEGESTONE Classified as a testosterone-like, norethindrone-related progestin, dienogest is structurally

unique in that it contains an estrane skeleton, a C_{17} cyanomethyl group that replaces the C_{17} ethynyl group, and Δ^9 unsaturation (Fig. 41.14). An effective OC preparation results when dienogest is combined with EE or estradiol valerate (Fig. 41.6). The OC preparation of dienogest/estradiol valerate is the first to be used in a four-phase dosage regimen (Natazia). The pharmacologic profile for this progestin includes antiandrogenic action similar to drospirenone (see below) without the risk of hyperkalemia. Correction of menopausal symptoms results when dienogest is combined with estradiol valerate. Limited evidence suggests that this combination has a positive impact on cognition, which is in direct opposition to the Women's Health Initiative (WHI) study data regarding the combination of medroxyprogesterone acetate and conjugated estrogens. Dienogest bioavailability is approximately 90%, and it is metabolized extensively by the CYP3A4 pathway. Coadministration of inhibitors or inducers of CYP3A4 will alter plasma concentrations of dienogest and, as a result, cause deviation from therapeutic efficacy.

Derived from spironolactone, drospirenone (Fig. 41.14) is the only progestin with antimineralocorticoid activity. Its affinity for the mineralocorticoid receptor is fivefold greater than that of aldosterone. Drospirenone has progestogenic action, but only 10% that of levonorgestrel. Drospirenone also has antiandrogenic action because it blocks testosterone from binding to androgenic receptors; however, it does not exhibit estrogenic or glucocorticoid receptor activity.

This androstane-based progestin has several distinctive functional groups: two cyclopropyl groups, one that includes C_6 and C_7 and the other C_{15} and C_{16} , and a C_{17} lactone. When combined with EE, drospirenone is effective in a monophasic OC (Table 41.9). In addition, drospirenone plus estradiol (Angeliq) is used in the treatment of various menopausal symptoms. The unique antimineralocorticoid activity of drospirenone effectively negates the side effects related to angiotensinogen production (causing an increase in sodium and water retention) associated with EE. Remarkably, within the first several cycles, women actually lost weight when using this combination of hormones (46). Regrettably, this benefit was lost, and statistically significant weight gain occurred as the number of cycles approached 13 (59). Drospirenone can cause potassium retention, and therefore, hyperkalemia is a possibility in patients concurrently taking other potassium-sparing drugs including nonsteroidal anti-inflammatory drugs, diuretics, angiotensin-converting enzyme inhibitors, and/or potassium chloride.

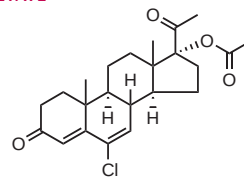
Elcometrine (Fig. 41.14) is a member of the 19-norprogesterone class of progestins. When administered subcutaneously (implant), intravaginally (vaginal ring), or transdermally, elcometrine is an exceptionally potent progestin. It is 100-fold more potent when administered subcutaneously than when delivered orally. Its oral bioavailability is only 10%, with a half-life of just 1 to 2 hours

because of rapid metabolism (11). The C_{16} methylene functionality substantially increases its affinity for the progesterone receptor. Elcometrine does not bind to the androgenic receptor and, therefore, does not possess either androgenic or antiandrogenic activity (11). Unfortunately, like medroxyprogesterone acetate, elcometrine has affinity for the glucocorticoid receptor, although adverse effects were noted only at exceptionally high doses. Because of its antiestrogenic action, elcometrine also can be administered transdermally along with estradiol as a treatment for menopausal symptoms (46). From a protein binding perspective, elcometrine does not bind to SHBG but, rather, to albumin (11). This progestin is not available in the United States but is marketed in several Spanish-speaking countries.

Nomegestrol acetate (Fig. 41.14) is structurally similar to megestrol acetate but lacks the angular C_{19} methyl group. It has substantially better selectivity for the progesterone receptor and higher potency than medroxyprogesterone acetate. Nomegestrol acetate has potent antigonadotropic action and, therefore, is an appropriate progestin to use in an OC preparation. It exhibits no glucocorticoid, antimineralocorticoid, or androgenic activity; however, it does demonstrate significant antiandrogenic action (11). This progestin is bound primarily to albumin. Like desogestrel, it does not diminish the beneficial effects of estrogen on the lipid profile (11).

Trimegestone, the most potent of the 19-norprogesterones, contains an unusual C_{21} hydroxyl group (Fig. 41.14). It has very high affinity for the progesterone receptor but only weak affinity for the mineralocorticoid receptor. Trimegestone displays no glucocorticoid, androgenic, or antiandrogenic action. This progestin undergoes metabolic hydroxylation to produce metabolites with substantial progestogenic action. Either alone or in combination with estradiol, trimegestone has shown favorable results regarding bone loss and bone turnover (46). Given with estradiol, this hormone replacement therapy (HRT) combination is effective in the treatment of postmenopausal symptoms regardless of whether it is administered continuously or as part of cyclic therapy. Like desogestrel and nomegestrol acetate, this progestin does not diminish the beneficial effects of estrogen on the lipid profile (11). Although available outside of the United States, this progestin is not presently marketed in the United States.

CHLORMADINONE ACETATE



Chlormadinone acetate was first synthesized in 1961 and is a progestin with a pharmacologic profile that

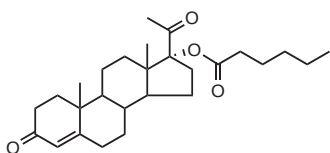
TABLE 41.9 Components of a Representative Set of Oral Contraceptives (74)

Type of Contraceptive	Trade Name	Estrogen	Progestin
Monophasic combination oral contraceptives	Lybrel (continuous)	EE (20 mcg)	Levonorgestrel (0.09 mg)
	Aviane	EE (20 mcg)	Levonorgestrel (0.1 mg)
	Loestrin-21	EE (20 mcg)	Norethindrone acetate (1 mg)
	YAZ	EE (20 mcg)	Drospirenone (3 mg)
	Yasmin	EE (30 mcg)	Drospirenone (3 mg)
	Loestrin-21	EE (30 mcg)	Norethindrone acetate (1.5 mg)
	Lo/Ovral	EE (30 mcg)	Norgestrel (0.3 mg)
	Desogen	EE (30 mcg)	Desogestrel (0.15 mg)
	Ortho-Cept	EE (30 mcg)	Desogestrel (0.15 mg)
	Levora	EE (30 mcg)	Levonorgestrel (0.15 mg)
	Nordette-28	EE (30 mcg)	Levonorgestrel (0.15 mg)
	Jolessa (3 months)	EE (30 mcg)	Levonorgestrel (0.15 mg)
	Seasonale (3 months)	EE (30 mcg)	Levonorgestrel (0.15 mg)
	Necon 1/35	EE (35 mcg)	Norethindrone (1 mg)
	Modicon	EE (35 mcg)	Norethindrone (0.5 mg)
	Femcon Fe (chewable)	EE (35 mcg)	Norethindrone (0.4 mg) + 7 ferrous fumarate tabs
	Balziva	EE (35 mcg)	Norethindrone (0.4 mg)
	Ortho-Cyclen	EE (35 mcg)	Norgestimate (0.25 mg)
	Zovia 1/35E	EE (35 mcg)	Ethinodiol diacetate (1 mg)
	Ovcon-50	EE (50 mcg)	Norethindrone (1 mg)
	Zovia 1/50E	EE (50 mcg)	Ethinodiol diacetate (1 mg)
	Ovral	EE (50 mcg)	Norgestrel (0.5 mg)
	Ortho-Novum 1/50	Mestranol (50 mcg)	Norethindrone (1 mg)
	Beyaz	EE 20 mcg	Drospirenone (3 mg) + 0.0451 mg levomefolate calcium
	Safyral	EE 30 mcg	Drospirenone (3 mg) + 451 mcg levomefolate calcium
Biphasic combination oral contraceptives	Mircette	EE (20 mcg) days 1–21	Desogestrel (0.15 mg)
	Ortho-Novum 10/11	EE (10 mcg) days 24–28	Desogestrel (0.0 mg)
	LoSeasonique (3 months)	EE (35 mcg)	Norethindrone (0.5 mg)
	Seasonique (3 months)	EE (35 mcg)	Norethindrone (1 mg)
	LO Loestrin Fe	EE (20 mcg)	Levonorgestrel (0.1 mg)
		EE (10 mcg)	Levonorgestrel (0.0 mg)
		EE (30 mcg)	Levonorgestrel (0.15 mg)
		EE (10 mcg)	Levonorgestrel (0.0 mg)
		EE (10 mcg)	Norethindrone (1 mg)
Triphasic combination oral contraceptives	Tri-Ligest	EE (20/30/35 mcg)	Norethindrone acetate (1 mg)
	Cyclessa	EE (25 mcg)	Desogestrel (0.1/0.125/0.15 mg)
	Ortho TriCyclen Lo	EE (25 mcg)	Norgestimate (0.18/0.215/0.25 mg)
	Trivora	EE (30/40/30 mcg)	Levonorgestrel (0.05/0.075/0.125 mg)
	Ortho TriCyclen	EE (35 mcg)	Norgestimate (0.18/0.215/0.25 mg)
	Ortho Novum 7/7/7	EE (35 mcg)	Norethindrone (0.5/0.75/1 mg)
	Ananelle	EE (35 mcg)	Norethindrone (0.5/1/0.5 mg)
4-Phasic combination oral contraceptive	Natazia	Estradiol valerate (30/20/20/10 mcg)	Dienogest (0/2/3/0 mg)
Progestin-only agents	Ortho Micronor		Norethindrone (0.35 mg)

EE, ethinyl estradiol.

includes a high affinity for the progesterone receptor, antiestrogenic action, and moderate antiandrogenic activity (62). Unlike progesterone, chlormadinone acetate acts as a weak glucocorticoid agonist, is a competitive inhibitor of 5 α -reductase, and has no anti-mineralocorticoid activity. Chlormadinone acetate effectively suppresses gonadotropin secretion, thereby preventing follicular growth and maturation. The mechanisms associated with its contraceptive efficacy

are similar to other progestins. It is five times more potent than norethindrone. Combined with EE (Belara; not U.S. Food and Drug Administration [FDA] approved), it is an effective OC with a good side effect profile, including excellent resolution of mild to moderate acne. The thromboembolic risk associated with this progestin is not significant. Like elcometrine and nomegestrol, chlormadinone acetate is bound to albumin.

17 α -HYDROXYPROGESTERONE CAPROATE (GESTIVA)17 α -Hydroxyprogesterone caproate

Originally marketed (Delalutin) nearly 50 years ago to prevent impending miscarriages, 17 α -hydroxyprogesterone caproate was removed from the market in 2000 when its efficacy was called into question. 17 α -Hydroxyprogesterone caproate, a synthetic injectable progesterone, was approved by the FDA in 2011 to reduce the risk of preterm delivery in women who have already experienced a preterm birth. It has not been shown to be effective in women carrying multiple fetuses. Injections begin between 16 and 21 weeks and are associated with pain, swelling, or itching at the injection site, hives, nausea, and diarrhea. Follow-up studies of the children born to women who used this drug indicate that developmental milestones were achieved at 2.5 and 5 years of age (63). A number of other pharmacologic agents are currently available for the prevention of preterm delivery including agents that can alter intracellular messengers (e.g., β -adrenergic receptor agonists, nitric oxide donors, magnesium sulfate and calcium channel blockers) and agents that modulate myometrial stimulants (e.g., inhibitors of prostaglandin synthesis and oxytocin antagonists) (64).

17 α -Hydroxyprogesterone caproate binds extensively to albumin and SHBG. From a metabolic perspective, it undergoes reduction, hydroxylation, and conjugation reactions, including becoming glucuronidated, sulfated, and acetylated. It induces several cytochrome P450 (CYP) isozymes including CYP1A2, CYP2A6, and CYP2B6 (65).

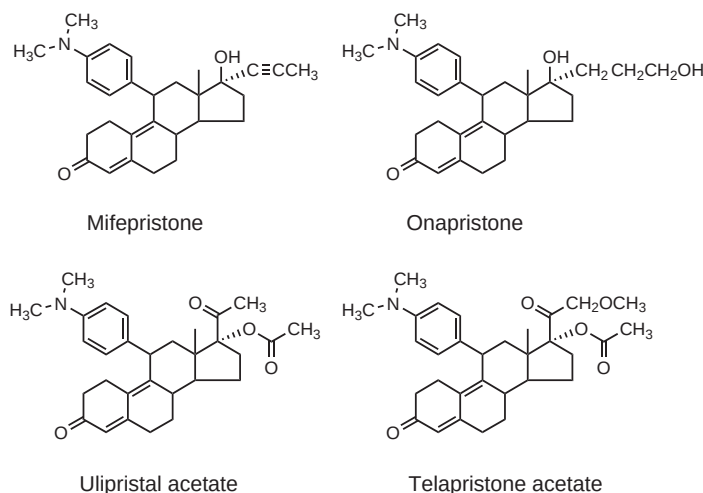
Progestin Antagonists

An antiprogesterin is a substance that competes with progesterone for its receptor and, ultimately, prevents progesterone from binding to and activating its receptor. Because progesterone is integral to the continuation of an early pregnancy, it is expected that antiprogesterins will interfere with pregnancy maintenance. In 1982, the first antiprogesterin, mifepristone (RU 486), was reported (66).

MIFEPRISTONE (MIFEPREX) Mifepristone was shown to interrupt early stages of implantation and pregnancy in humans (Fig. 41.15) (67). After oral administration, mifepristone is rapidly absorbed, with a peak plasma concentration in approximately 90 minutes, an oral bioavailability of approximately 70%, and a terminal elimination half-life of 18 hours. It is 98% protein bound, primarily to albumin and α_1 -acid glycoprotein. Mifepristone is metabolized primarily via CYP3A4 pathways involving mono- and di-oxidative *N*-demethylation and terminal hydroxylation of the 17-propynyl chain. The fact that approximately 83% of the drug is recovered in the feces and 9% in the urine suggests a biliary route of elimination. Mifepristone also demonstrates antiglucocorticoid activity (68).

Additional antiprogesterin analogs, such as onapristone (ZK 98,299), exhibit less antiglucocorticoid activity (69). These antiprogesterins also have demonstrated therapeutic potential for the treatment of hormone-dependent breast cancer (70). Telapristone acetate (Proellex) is in early clinical trial for the treatment of symptomatic uterine fibroids and amenorrhea.

ULIPRISTAL ACETATE (ELLAONE) Ulipristal acetate (FDA approved in 2010) is a synthetic progestin used similarly to levonorgestrel in the prevention of pregnancy following unprotected intercourse or suspected contraceptive failure (Fig. 41.15). It is not intended for routine use as a contraceptive.

**FIGURE 41.15** Progesterone antagonists.

The mechanism by which ulipristal acetate exerts its effect is dependent on when it is administered during the menstrual cycle. As a selective progesterone receptor modulator, it can delay or inhibit ovulation (progesterone agonist action) or alter the endometrium to prevent uterine implantation (progesterone antagonist action). If used within 120 hours of unprotected intercourse, ulipristal acetate reduces the pregnancy rate when compared to the rate in the absence of emergency contraception (71). Ulipristal acetate has very high affinity for the progesterone receptors, and it may reduce the effectiveness of hormonal contraceptive methods. Once administered as emergency contraception, it is recommended that a barrier method of contraception should be used for the remainder of that menstrual cycle.

Adverse events reported during clinical trials do not differ significantly from those found with other progestins and include headache (18%), abdominal pain (12%), and nausea (12%). Subsequent menses may occur later or earlier by a few days following use of ulipristal acetate (71).

Ulipristal acetate is metabolized to mono- and di-*N*-demethylated metabolites via CYP3A4, with the mono-*N*-demethylated metabolite retaining pharmacologic activity. Coadministration of inducers of CYP3A4 (e.g., carbamazepine, St. John's wort) will likely reduce the effectiveness of ulipristal acetate, and coadministration of inhibitors of CYP3A4 will cause an increase in therapeutic action (71).

Selective progesterone receptor modulators that are continuously administered have been evaluated as effective estrogen-free contraception and as treatment for the heavy and irregular menstrual bleeding associated with fibroids. Weekly oral administration and intrauterine and vaginal routes are under investigation (72,73).

THERAPEUTIC APPLICATIONS

A woman's reproductive life cycle spans nearly four decades. During this time, her hormonal status and ability to reproduce vary considerably. The need for hormonal intervention during this time period varies from person to person and, typically, requires highly individualized medication regimens. The remainder of this chapter will focus on hormonal and selected nonhormonal therapies for the management of contraception, endometriosis, infertility, menopause, and breast cancer.

Contraceptive Methods

Despite the availability of a wide variety of contraceptive methods and technologies, nearly half of all pregnancies in the United States are unplanned, and one-fourth are terminated. You might expect that cost, adverse effects, and inconvenience would be some of the hurdles to achieving contraceptive efficacy, but one of the greatest barriers is ineffective patient education (74). Despite ease of use, nearly 50% of those patients who have chosen

an OC regimen discontinue their therapy within a year (75). If information regarding potential side effects, dosing requirements, and how the therapy is able to prevent pregnancy is provided in an accessible fashion, however, there may be many fewer patients who discontinue their contraceptive therapy (74,76,77).

Behavioral and Barrier Methods

Behavioral methods, such as natural family planning, fertility awareness, coitus interruptus, and lactation infertility, typically result in the highest pregnancy rates of all of the contraceptive methods. To be effective, these methods require disciplined effort on the part of the patient and her partner and somewhat detailed patient training. Barrier methods, such as male and female condoms (FC2 approved by FDA in 2009; FC1 approved by FDA in 1993), diaphragms, Today Sponge (including nonoxynol-9 spermicide; FDA approved in 2005), and cervical caps, still require a very conscientious and often preplanned effort on the part of the patient as well as fairly detailed patient training. Unplanned pregnancy rates with these barrier methods are approximately 12% to 15%.

Pharmacologic contraception, including OCs, patches, vaginal rings, IUDs, and injectable medications, provides substantially lower pregnancy rates with diligent use, relatively limited patient effort, and patient education that is focused largely on adverse effects.

Combination Oral Contraceptives

Available since the early 1960s, most OC pills contain both an estrogen and a progestin component. The estrogen component, either EE or mestranol, suppresses FSH release and, therefore, prevents the formation of a dominant follicle during the follicular phase of the menstrual cycle. The dominant follicle is responsible for producing estradiol, which sends a negative feedback signal to the hypothalamus to prevent the secretion of additional gonadotropins and, therefore, prevent maturation of additional follicles. The estrogen component also is responsible for maintaining the stability of the endometrial lining (78). The progestin component, of which there are many available, suppresses the LH surge and, therefore, blocks ovulation.

In the more than 45-year evolution of OCs, the estrogen dose has dropped to one-sixth and the progestin dose to one-tenth of the original levels. This makes the newer OC agents substantially safer but also prone to causing adverse effects, such as irregular and breakthrough bleeding. In addition to these dosage changes, the progestin components have become progressively more selective in their binding to the progesterone receptor, thereby limiting the androgenic side effects that were characteristic of the older progestins (59).

There are three generations of OCs that are classified based on the progestin component and the pharmacologic profile. The first-generation OCs contain either norethisterone or norethindrone acetate along with 35 to 50 mcg of EE. Their pharmacologic profile includes

high affinity for the progesterone receptor, strong gonadotropic action, and intermediate affinity for androgen receptors (79). Second-generation OCs contain levonorgestrel (along with 20 to 50 mcg EE) and have an even higher affinity for both the progesterone and androgen receptors than the first-generation agents (79). The third-generation OCs contain desogestrel, gestodene (available in Europe), or norgestimate (along with 15 to 30 mcg of EE) and have some affinity for the androgen receptor, which results in fewer androgenic side effects. This generation was supposed to have less metabolic side effects. The fourth generation of OCs contain chlormadinone acetate (not available in the United States), drospirenone, or cyproterone acetate (along with 20 to 30 mcg of EE) (79).

OC agents are either monophasic or multiphasic in their hormone composition (Table 41.9). The multiphasic formulations more closely mimic the natural changes in both hormone concentrations that occur throughout the menstrual cycle. Most of the OC formulations are administered for 21 days, followed by 7 drug-free days to allow monthly withdrawal bleeding. Extended-cycle products (Seasonale and Seasonique) have efficacies similar to that of the standard 28-day regimen (Table 41.9) and allow the patient to experience only 7 drug-free days every 3 months. Another nonstandard combination OC consists of EE plus a progestin for 21 days, followed by 2 hormone-free days and then 5 days of EE (Mircette and Kariva) (Table 41.9) (76). A continuous regimen of low-dose EE and levonorgestrel (Lybrel) that completely eliminates the menstrual cycle is also available (80).

Another monophasic low-dose formulation (YAZ) (Table 41.9), comprised of estradiol and drospirenone, has the added benefit of diuretic action. The dosing regimen for this OC is comprised of 24 days of drug therapy, followed by drug-free days to allow for withdrawal bleeding. The 3-mg drospirenone component of this preparation is equivalent to 25 mg of spironolactone in terms of diuretic potential. Because of the diuretic action of drospirenone, the drug has the potential to cause hyperkalemia. Diminished fluid-related weight gain is an obvious advantage to this OC formulation, but caution is advised in patients with renal or adrenal insufficiency, with hepatic dysfunction, or who are receiving chronic pharmacotherapy that may increase serum potassium levels (e.g., angiotensin-converting enzyme inhibitors and potassium-sparing diuretics). This product joins a previous OC formulation that contains drospirenone (Yasmin) (Table 41.9). It should be noted that intracyclic bleeding (spotting) is more prevalent with EE doses of less than 30 mcg, as is the case with these OCs. Studies have shown that the incidence of this adverse effect was greatest during the first cycle of OC use and diminished to very low incidence within three cycles (81).

In an attempt to reduce the risk of neural tube defects in pregnancies that occur during use or shortly after discontinuation of OCs, there are two formulations available that contain levomefolate calcium.

Although the adverse effects of OC therapy include an increased risk of stroke, acute myocardial infarction, and VTE disease, the incidence of cardiovascular disease in this patient population (age, <35 years) is already low (76). In women older than 35 years, the natural incidence of cardiovascular disease increases, so these adverse effects become more important to consider. The risk of VTE in patients who use a third-generation OC (ethinyl estradiol + desogestrel or norgestimate) is about twice that of patients who use a second-generation OC (ethinyl estradiol + levonorgestrel) and five times greater than in nonusers (82). The risk of VTE is greatest in the first 3 months of OC use. From a metabolic perspective, the primary adverse effect of the estrogen component is an increase in hepatic production of proteins, including those that enhance venous and arterial thromboembolism (83). In addition, the progestin component has an adverse effect on the lipid profile, including an elevation of serum triglyceride and a decrease in HDL levels. Generally speaking, the estrogen component of a combination OC balances out this negative impact on a patient's lipid profile. The degree to which the lipid profile is altered is dependent not only on the amount of hormone delivered, but also on the route of administration (84). It has been suggested that cholesterol levels change over the course of a given menstrual cycle (~20% change), with cholesterol levels being highest during the first half of the cycle. As the level of estrogen increases over time, the level of HDL cholesterol increases to the highest point at ovulation, whereas the levels of low-density lipoprotein (LDL) cholesterol and triglycerides drop over the same interval (85). Neoplastic effects are associated with OCs as well, such as an increased risk of cervical cancer and a decreased risk of colorectal, endometrial, and ovarian cancers that persists long after the discontinuation of therapy. Women taking OCs who require thyroid hormone replacement may require a dosage adjustment of thyroid hormone because use of OCs may cause an increase in serum concentrations of thyroid-binding globulin (20). See Table 41.10 for preferred contraceptive methods with coexisting disease states (86).

Contraceptive effectiveness certainly relies on patient adherence and consistent timing of the daily dose; however, there are other factors that can contribute to poor efficacy. Low serum EE levels have been associated with elevated body weight (87). Studies that evaluated the therapeutic efficacy of OCs in obese women yielded conflicting results. Vomiting and diarrhea for longer than 24 hours, as well as any chronic malabsorption condition, can cause a decrease in the absorption of both OC components and, therefore, a significant enough decrease in effectiveness to warrant the use of an additional form of contraception. Drug efficacy may also be reduced in the presence of one or more drugs that induce liver enzymes (e.g., carbamazepine, phenytoin, topiramate, barbiturates) or that can alter the enterohepatic recycling of the hormones (e.g., certain antibiotics).

TABLE 41.10 Contraception Considerations with Coexisting Medical Conditions (86)

Medical Condition	Concern	Preferred Contraceptive Method
Bariatric surgery	Reduced absorption	Nonoral contraceptive (malabsorptive surgery); all okay (sleeve or banding surgery)
Breast cancer	Increased risk, worsening, or recurrence	All okay (+ family history); copper IUD (current or past history)
Breast feeding	Estrogen may reduce milk production	Progestin only
Diabetes	Impaired glucose control and carbohydrate metabolism	Copper IUD (no vascular disease); progestin only or copper IUD (with vascular complications)
Epilepsy	Induction of metabolic enzymes	Avoid estrogen-containing preparations (lamotrigine); depot medroxyprogesterone and IUDs (carbamazepine, phenytoin, barbiturates, topiramate)
Hypercoagulable conditions	Increased coagulation and risk of thromboembolism	All okay (+ family history); copper IUD or progestin only (personal history of DVT or PE + no anticoagulant therapy); avoid estrogen-containing preparations (personal history of DVT or PE + anticoagulant therapy)
Headache/migraine	Increased risk of stroke	All okay (nonmigraine headaches); copper IUD or progestin only (migraine without aura); copper IUD (migraine with aura); continuous or extended combined OCs (menstrual migraine)
Hyperlipidemia	Increase in triglyceride levels (estrogen) and LDL levels (progestin)	Copper IUD; other types of contraceptive forms may be acceptable depending on type of dyslipidemia
Hypertension	Increase in blood pressure and increase in cardiovascular events	No estrogen-containing types (controlled blood pressure); copper IUD or progestin only (uncontrolled/elevated blood pressure)
Obesity	Increased risk of VTE	All okay except contraceptive patch (BMI <40)
Systemic lupus erythematosus	Increased risk of ischemic heart disease, stroke, and VTE.	Copper IUD only (+ or unknown antiphospholipid antibody test)

BMI, body mass index; DVT, deep vein thrombosis; IUD, intrauterine device; LDL, low-density lipoprotein; OC, oral contraceptive; PE, pulmonary embolism; VTE, venous thromboembolism.

Progestin-Only Oral Contraceptives

A few contraceptive formulations only contain progestins (Table 41.9). Their contraceptive effects are accomplished via several mechanisms. From a hormonal perspective, progestins are able to suppress ovulation by preventing mid-menstrual cycle LH and FSH peaks via negative feedback inhibition. They also thicken cervical mucus to prevent entry of sperm into the endometrial cavity as well as transform the endometrium into an unsuitable environment for embryo implantation. Although the progestins suppress ovulation, they only do so 60% of the time, which generally is not sufficient for young women. The progestin-only OCs are an appropriate choice, however, for older women in whom fertility is diminished, for patients with contraindications to exogenous estrogen (e.g., smoking or high blood pressure), or for a lactating patient who is already experiencing prolactin-induced ovulation suppression (74). Unlike the combination oral agents, timing of administration is critical, because the effect of the progestin on the cervical mucus lasts for only 27 hours. If there is more than a 3-hour delay in the time of administration from the previous day, that dose is considered to be “missed,” and additional contraceptive methods are recommended.

The first progestin-only “mini-pills” contained either norethindrone or norgestrel. Their adverse effect profile is limited, because there is no estrogen component to increase the risk of VTE or other cardiovascular related

risks. Agents that induce metabolic enzymes (e.g., primidone, carbamazepine, topiramate, rifampin, ritonavir, St. John’s wort) may decrease the efficacy of the mini-pills by reducing the circulating levels of hormone (88). Antibiotics do not alter the enterohepatic recycling of progestins as they do with estrogens.

Transdermal Contraceptive Patch

Despite the ease of administration of the OCs, poor patient adherence rates have fueled the development of several long-acting contraceptive formulations (Table 41.11). The contraceptive patch, containing a combination of norelgestromin and EE, is applied to the upper arm, lower abdomen, upper torso, or buttocks once per week on the same day of the week for 3 weeks, followed by a patch-free week to allow for withdrawal bleeding (89). After several weeks of therapy (several patches applied), the mean and steady-state serum concentrations were in the range needed for contraceptive efficacy (90). Despite the absence of first-pass metabolism with this route of administration, the EE component undergoes hydroxylation and is conjugated to the corresponding glucuronide and sulfate derivatives. Contraceptive efficacy is compromised with coadministration of various metabolic enzyme inducers. If more than 48 hours elapse between patch removal and administration, it is recommended to start a new 4-week cycle with the newly administered patch, and an additional

TABLE 41.11 Contraceptive Methods Using Nonoral Routes of Administration

Trade Name	Estrogen	Progestin	Route of Administration
Ortho Evra	EE (0.75 mg)	Norelgestromin (6 mg)	Transdermal patch
NuvaRing	EE (0.015 mg daily)	Etonogestrel (0.12 mg daily)	Vaginal ring
Lunelle ^a	Estradiol cypionate (5 mg)	Medroxyprogesterone acetate (25 mg)	IM
Depo-Provera		Medroxyprogesterone acetate (150 mg/mL)	IM (400–1,000 mg weekly)
Norplant II		Levonorgestrel (36 mg/rod)	2 Subdermal rods (3 years)
Jadelle ^a		Levonorgestrel (75 mg/rod)	2 Subdermal rods (5 years)
Implanon		Etonogestrel (68 mg/rod)	1 Subdermal rod (3 years)
Mirena		Levonorgestrel (52 mg)	IUD (5 years)

^aNot available in the United States.

EE, ethinyl estradiol; IM, intramuscular; IUD, intrauterine device.

method of contraception should be used for 1 week (91). The patch provides equivalent efficacy to the OCs; however, it may not be as effective for women who weigh more than 90 kg (92). The adverse effects associated with the patch are headache, nausea, breast and abdominal pain, and application site irritation.

The revised Ortho Evra labels include statements that the patch can cause blood clots that could lead to a pulmonary embolism were based on the results of studies that show that this formulation exposes users to 60% more estrogen than OC users (containing 35 mcg EE) and that patients 15 to 44 years of age are at an increased risk for pulmonary embolism. Patients using this contraception formulation were twice as likely to develop blood clots and therefore at much greater risk for developing a VTE (93).

A transparent transdermal patch containing EE (0.9 mg) and gestodene (1.9 mg) is currently under investigation, as is a patch containing EE and levonorgestrel.

Contraceptive Vaginal Ring

The vaginal ring (NuvaRing), another long-acting contraceptive, continually delivers a low dose of etonogestrel and EE directly into the vagina (Table 41.11). The ring is inserted into the vagina by the patient and remains in place for 3 weeks. Again, 1 drug-free week allows appropriate withdrawal bleeding. Unscheduled removal of the ring is allowed, but the ring must be reinserted within 3 hours to ensure continued therapeutic efficacy (77). This formulation has comparable efficacy with the OCs yet provides continuous hormone delivery, thereby minimizing the side effects caused by the pulsatile delivery associated with oral dosing regimens (77,94). This formulation is not without adverse effects, including decreased libido, nausea, vaginitis, and breast tenderness.

A vaginal ring with a longer duration of action (1 year) is in phase II clinical trials. The active agents in this delivery system are nestorone and EE.

Contraceptive Injectables

An intramuscular formulation that contains an aqueous suspension of medroxyprogesterone acetate and estradiol cypionate is available but is not currently marketed in the United States (Table 41.11). The 28- to 33-day dosing interval provides excellent contraceptive effectiveness. There are several advantages to this contraceptive method, including scheduled withdrawal bleeding and a quick return to a fertile state on discontinuation of therapy. The adverse effects are similar to those seen with other combination OC formulations, including headache, weight gain, decreased libido, and breast tenderness (77).

Progestin-Only Injectables and Implants

The issue regarding timing of administration all but disappears when a progestin injection is administered every 3 months. In this formulation, medroxyprogesterone acetate is an aqueous microcrystalline suspension (Table 41.11). The mechanisms by which this progestin accomplishes contraceptive efficacy are the same as those discussed previously for other progestins. Despite the ease of use, the continuation rate after a year of injections is less than 30%, largely because of the adverse effects, including irregular and heavy bleeding, amenorrhea, and infertility that lasts for 9 to 21 months after the last injection (74). Improved patient education concerning the potential side effects may be a simple solution to discontinuation of therapy.

Several subdermal implants are available that provide contraceptive efficacy for 1 to 5 years depending on the formulation (Table 41.11). These implants consist of progestins contained within individual rods that are surgically implanted into the inner side of the upper nondominant arm. As indicated in Table 41.11, varying durations of action can be expected depending on the number of rods implanted. Implanon (etonogestrel) is available in the United States. This formulation is subject

to the same adverse effects as other progestin-only formulations. Age-appropriate fertility returns within 1 week of removal.

Intrauterine Devices

There are two forms of IUDs: one that releases the progestin levonorgestrel (Table 41.10), and one that is copper-coated. Both devices are T-shaped and must be inserted into the uterus by a physician. These devices mechanically irritate the endometrium and create an inflammatory environment that is toxic to sperm and ova and is not conducive to embryo implantation (74). The copper coating further enhances this inflammatory response. The copper-based IUD can be used continuously for 10 years, whereas the progestin-containing device is only approved for 5 years of use (95). Although generally well tolerated, ectopic pregnancy and infection (related to device insertion or to the device itself) are two serious potential complications. Discontinuation of use often is a result of breakthrough and/or heavy withdrawal bleeding.

Emergency Contraception

Emergency contraception is defined as any method that prevents pregnancy after intercourse. This type of therapy is available in either combination (Yuzpe method) or single-agent formulations. The progestin-only formulations contain either levonorgestrel (two 0.75-mg tablets = Plan B; one 1.5-mg tablet = Plan B One Step) or ulipristal acetate (Ella) (Fig. 41.15). The combination regimens (multiple tablets of 21/7 or 24/4 regimens) include EE and levonorgestrel (Table 41.12) (96). Typically, these agents are administered as one dose within 72 hours of intercourse, with another dose taken 12 hours later. Nausea and vomiting are common side effects. If the patient is hormone adverse, then insertion of a copper IUD within 5 days of ovulation (the earliest estimated date) will prevent implantation (97).

Mifepristone (Mifeprex) is a 19-nortestosterone-based antiprogesterone that competes with the endogenous agonist for the progesterone receptor (see previous discussion). If administered within 5 days of unprotected intercourse, it also will prevent pregnancy. Despite the

presence of the bulky 11 β -diethylamino phenyl group, mifepristone has higher receptor affinity for the progesterone receptor than the natural agonist. It is not completely understood how mifepristone inactivates the progesterone receptor. Of note is that mifepristone and progesterone do not interact similarly with the ligand binding domain, such that in the presence of mifepristone, the conformational change that occurs in this domain is unique from that produced by the natural agonist (45).

Endometriosis

Endometriosis, a condition present in 1 of 10 menstruating women, is characterized by implantation of endometrial tissue outside the uterus, typically within the pelvic region (98,99). During a normal menstrual cycle, hormonal signals cause the endometrial tissues to thicken in anticipation of embryo implantation. If a pregnancy does not occur, then these tissues are shed, resulting in monthly menses. When endometrial tissues implant outside the uterus, they respond identically to those within the uterus, such that when hormones fall in response to a nonpregnant state, these tissues are shed—yet have no natural exit. This tissue may form cysts, which in turn may form scar tissue and/or adhesions. Typically, the presence of one or more of these elicits significant pain and discomfort, especially during menstruation. Because the resulting scars and/or adhesions can cause infertility, treatment should be sought if the patient wishes to have children (98).

The cause of endometriosis is not completely understood. Sampson's hypothesis in 1927 suggests that menstrual fluids flow back (retrograde) through the fallopian tubes, where the endometrial cells can then implant and grow. Another theory proposes that the bloodstream is responsible for transporting the endometrial cells to regions outside the uterus. A third hypothesis is that endometriosis is a genetically acquired disease and that environmental factors trigger disease initiation (98,99). One of the hypotheses currently gaining some strength involves the immune system. Abnormal B- and T-cell function is evident in endometrial lesions, as are abnormal levels of cytokines, growth factors, and interleukins (100). Regardless of the biochemical cause, the growth of these endometrial lesions is controlled by both estrogen and progesterone (101), which means that hormonal intervention is likely to be effective in the treatment of this condition.

Pharmacologic Management

Treatment of endometriosis largely depends on the severity of the symptoms experienced by the patient. Characteristic symptoms experienced by patients include dysmenorrhea, pelvic and lower abdominal pain, and dyspareunia (101). The American Association of Reproductive Medicine has a four-tiered system to describe the severity of the endometriotic disease, ranging from stage I (mild) to stage IV (severe) (100).

TABLE 41.12 Representative Choices for Emergency Contraception (96)

Trade Name	Ethinyl Estradiol Content	Levonorgestrel	Pills per Dose; No. of Doses
Plan B	None	0.75 mg	1 tablet; 2
Ogestrel	0.50 mcg	0.25 mg	2 tablets; 2
Cryselle	0.15 mcg	0.30 mg	4 tablets; 2
Seasonale	0.15 mcg	0.30 mg	4 tablets; 2
Aviane	0.20 mcg	0.10 mg	5 tablets; 2

Pharmacologic management looks to achieve two primary goals: pain management and prevention/delay of disease progression. Nonsteroidal anti-inflammatory drugs (e.g., ibuprofen) that inhibit prostaglandin biosynthesis may be sufficient to ease menstrual pain, but this pain may not be adequately managed even at the maximum dose. Endometriotic pain is directly related to the fluctuating levels of endogenous hormones. Clear evidence indicates that endometriosis-related symptoms improve substantially during pregnancy and after menopause. Because OCs chemically produce a “pregnant” state, one might expect this type of hormone therapy to be a viable treatment option.

COMBINATION ORAL CONTRACEPTIVES There are a number of hormone-based pharmacologic options for the treatment of endometriosis. If a woman wishes to achieve complete cessation of endometriosis-related pain, a 3- to 4-month regimen using a combination OC is recommended. This forces the body into a chemically “pregnant” state, such that hormone fluctuations and ovulation cease. As a result, menstrual flow decreases, and the size of the endometrial implants diminish. No evidence suggests that one OC combination is more efficacious than another (100). After 3 to 4 months of continuous OC therapy, 1 drug-free week is required to allow withdrawal bleeding to occur. If daily administration is difficult for the patient to manage, then selection of one of several delivery systems that allow less frequent dosing (e.g., transdermal patch or vaginal ring) is appropriate. Cyclic dosing is more appropriate if the patient does not experience symptoms continually throughout the menstrual cycle. Adverse effects include nausea, headache, and breakthrough bleeding.

GONADOTROPIN-RELEASING HORMONE AGONISTS The GnRH agonists leuprolide (Lupron), goserelin (Zoladex), triptorelin (Trelstar Depot), and nafarelin (Synarel) downregulate the hypothalamic-pituitary GnRH receptors, thereby limiting the secretion of FSH and LH and causing suppression of ovarian function (Fig. 41.16) (98,100). This creates an estrogen-deficient environment that prevents menstruation and reduces the size of wayward endometrial implants. These agents have enjoyed success in the treatment of endometriosis (85% to 100% success with 6 months of therapy), but they come with a price. The GnRH agonists generate a state of artificial menopause, which can cause classical vasomotor symptoms, urogenital atrophy, and insomnia. Bone mineral density suffers substantially after prolonged use, so the duration of this type of therapy is limited to 6 months (102). One way to extend the duration of therapy is to coadminister agents that address these side effects. Suggested “add-back” therapies include a progestin alone (e.g., norethindrone), a progestin and a bisphosphonate (e.g., etidronate), or a progestin and an estrogen (e.g., conjugated equine

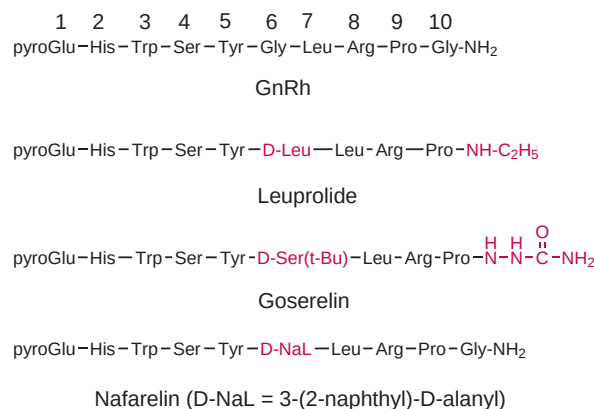
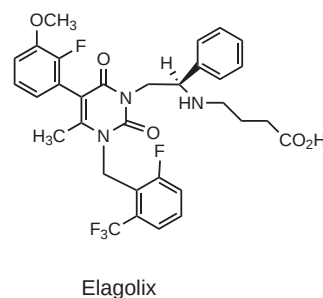


FIGURE 41.16 Structure of GnRH and its analogs.

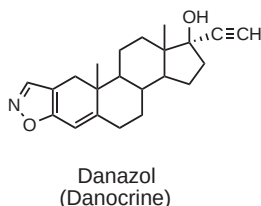
estrogen [CEE]). The amount of estrogen that is added back is enough to counteract the side effects of the GnRH agonists but insufficient to cause endometrial tissue growth (100). A serious limitation to multidrug therapy is the considerable cost associated with these agents as compared to the OCs or progestin injectables.

It should be noted that the same mechanism by which these agents are effective in the treatment of endometriosis is also applicable to the treatment of fibroids. Leuprolide acetate effectively shrinks the fibroid size by decreasing the supply of hormones that “feed” fibroids.

GONADOTROPIN-RELEASING HORMONE ANTAGONISTS Non-peptide-based GnRH antagonists offer several significant advantages over the injectable GnRH agonists in the treatment of endometriosis. Mechanistically, these agents provide rapid hormone suppression without the characteristic hormone flare. Because they are non-peptide based, they are easily formulated for oral delivery. Elagolix, an orally active nonpeptide GnRH antagonist, is in phase II clinical trial for the treatment of endometriosis. In clinical trials, it produced statistically significant reductions in dysmenorrhea, nonmenstrual pelvic pain, and dyspareunia—all of which are symptoms of endometriosis. An important advantage of an orally available agent is the ability to modulate estrogen levels to maximize pain relief while minimizing bone loss.



ANDROGENIC AGENTS Danazol, a synthetic analog of 17 α -ethinyl testosterone, induces amenorrhea, anovulation, and endometrial atrophy via suppression of the hypothalamic-pituitary-ovary (HPO) axis (98,100).



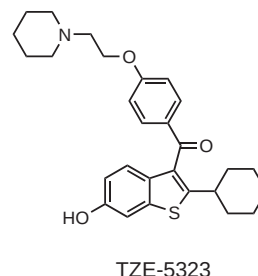
This causes an estrogen-deficient state, but it also causes an increase in androgen production. Danazol generally is not well tolerated because of its androgenic and anabolic side effects, including acne, decreased breast size, facial hair, weight gain, and oily skin (98,99). This type of therapy is not a viable option for women with liver disease or hyperlipidemia. Because danazol is teratogenic, it is recommended that effective contraception be used during treatment (98,100). Gestrinone, a 19-nortestosterone, has androgenic activity, antiestrogen activity, and antiprogesterone action (Fig. 41.14). Administered twice weekly, gestrinone decreases the secretion of LH and FSH via interaction with receptors within the HPO axis. The side effect profile includes weight gain, acne, seborrhea, and hirsutism.

PROGESTINS Medroxyprogesterone acetate (Fig. 41.14), administered by injection (Depo-Provera), orally (Provera), or via IUD, effectively suppresses the HPO axis, induces anovulation, and reduces serum estrogen levels (103). This prevents menstruation and endometrial implant growth. As a result, endometriosis-related pain is minimized in approximately 90% of the patients. Levonorgestrel intrauterine system (Mirena) has also been found to be effective in managing endometriosis. Drug therapy selection should reflect the fact that pharmacologic therapy is likely to be required on a chronic basis. The progestins are fairly well tolerated and relatively inexpensive, but they are not without adverse effects, including nausea, breakthrough bleeding, weight gain, mood swings, and depression.

AROMATASE INHIBITORS Normal endometrial tissues are devoid of aromatase activity. In patients with endometriosis, evidence suggests that high levels of aromatase are present in endometrial implants. Aromatase is the enzyme that catalyzes the conversion of androstenedione and testosterone to estrone and estradiol, respectively (Fig. 41.3). If an aromatase inhibitor, such as anastrozole (Fig. 41.12), is administered, then estrogen production will be decreased in the endometrial implants. A local state of estrogen deficiency will prevent growth of these implants (104). Unfortunately, one outcome of prolonged aromatase therapy is the development of osteoporosis. As a result, an OC typically is coadministered to

provide baseline estrogen concentrations to limit the risk of osteoporosis.

SELECTIVE ESTROGEN RECEPTOR MODULATORS To be effective in the treatment of endometriosis without therapy-limiting side effects, a SERM must behave as an estrogen antagonist at the endometrium and as an agonist at the bone. An investigational agent, TZE-5323, has antiestrogenic action by virtue of its ability to inhibit binding of estrogen to ER α and ER β and then limit subsequent transcriptional activation (105). This agent may prove to be beneficial in the treatment of endometriosis.



PROGESTERONE RECEPTOR MODULATORS

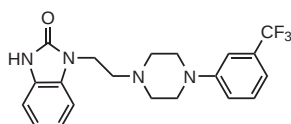
Like the SERMs, the selective progesterone receptor modulators bind to the progesterone receptors and act as either an agonist or antagonist depending on the absence or presence of progesterone and the tissue that is being targeted. In the absence of progesterone, these agents demonstrate progestin activity. In the presence of progesterone, they exhibit antiprogestin activity in some target tissues, particularly the endometrium (101). Several investigational agents, including asoprisnil, have suppressed the growth of endometrial implants while estrogen concentrations remained at similar levels to those found during the follicular phase of the menstrual cycle (106). Recently, phase III clinical trials with asoprisnil were halted due to endometrial changes found in patients.

SURGICAL INTERVENTION Laparoscopic surgical intervention to remove the endometrial implants is recommended for those patients who do not wish to use pharmacologic methods because of their immediate interest in having children. It should be noted that as many as 20% of patients do not respond to surgical intervention. This results, in part, from a failure to identify and excise all of the errant endometrial tissues (100). Adjunct therapy is not routinely used, because study data fail to consistently show efficacy (107).

Hypoactive Sexual Desire Disorder

Low libido for long periods of time is the hallmark of hypoactive sexual desire disorder (HSDD). This affects 9% to 26% of women depending on age and menopausal status. Both hormonal and nonhormonal solutions are under investigation to treat this female sexual problem.

Though originally designed and evaluated as an antidepressant, flibanserin's side effect profile included enhanced libido. Phase III clinical trial data suggested that patients derived meaningful benefit from treatment with flibanserin. Side effects included dizziness, nausea, and fatigue. In June 2010, the FDA failed to approve flibanserin for the treatment of HSDD based on lack of efficacy. A transdermal testosterone gel (LibiGel) is also in phase III clinical trials for the treatment of HSDD in menopausal women. Once-daily topical application of a pea-sized dose on the upper arm followed by rapid absorption into the skin delivers testosterone evenly to the bloodstream over time.

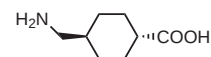


Flibanserin

Menorrhagia

Heavy menstrual bleeding (menorrhagia) is a condition reported by 3 million U.S. women annually. It is characterized by normal intervals of menstruation with an excessive volume (>80 mL) of blood loss per cycle. Tranexamic acid (Lysteda) was approved by the FDA in 2009 for the treatment of menorrhagia. As an orally available lysine receptor antagonist, tranexamic acid acts as

an antifibrinolytic by preventing the activation of plasminogen to plasmin. Plasmin is the serine protease that is responsible for the degradation of fibrin.



Tranexamic acid

Infertility

Infertility is not a disease state of the “individual” but, rather, often is considered to be a disease state of the “couple.” Defined as the inability to conceive a pregnancy after 12 months of unprotected intercourse, infertility affects 10% to 13% of couples. Identification of the cause of infertility requires thorough medical histories as well as several diagnostic tests (Table 41.13) (108,109). The causes of infertility include ovulatory disorders as well as situations in which tubal factors, uterine factors, cervical factors, and/or immunologic and thrombophilic factors play a substantial role (Table 41.14) (110).

In 2010, one in seven babies were born to women at least 35 years of age, an interesting statistic that highlights the recent trend to delay having children. With this delay comes an age-related reduction in fertility and the need to employ one or more pharmacologic therapies to achieve a live birth. As of 2006, there had been more than 3 million babies born as a result of in vitro fertilization (IVF), with nearly one in three embryo-transfer cycles

TABLE 41.13 Infertility Medical Evaluation (88,89)

Male	Female
Fertility in other relationships	Previous pregnancies, fertility in other relationships; onset of menstrual cycle
Medication use	Medication use, including contraceptive use
Alcohol or recreational drug use, cigarette smoking	Alcohol or recreational drug use, cigarette smoking
Environmental exposure (e.g., sauna, hot tub), chemotherapy or radiation exposure	Environmental exposure (e.g., DES), chemotherapy or radiation exposure
Sexual dysfunction/frequency of intercourse	Frequency of intercourse
Semen analysis	Ovulation documentation
	Screening tests: pap smear, cervical cultures, mammogram, infectious disease screening
	Day 3 FSH levels (>10 mIU/mL = low rate of pregnancy)
	Thyroid panel, including TSH levels
	Fasting Prolactin levels
	Clomiphene citrate challenge test + FSH and estradiol levels (assess ovarian reserve)
	Hysterosalpingogram (assess tubal patency, shape of intrauterine cavity)
	Laparoscopy (assess endometriosis or if adhesions are present)
	Hysteroscopy (assess interuterine lesions or abnormalities)

DES, diethylstilbestrol; FSH, follicle-stimulating hormone; TSH, thyroid-stimulating hormone.

TABLE 41.14 Infertility Classification and Causes (90)

Ovulatory Disorders	Characteristics	Therapeutic Regimen
<i>WHO Class I</i> Hypogonadotropic, hypogonadal anovulation	Low or low normal FSH, low estradiol due to decreased Gn-RH or pituitary not responding to Gn-RH	Weight gain, reduce exercise, pulsatile Gn-RH
<i>WHO Class II</i> Normogonadotropic, normoestrogenic anovulation	Normal gonadotropins and estrogens, FSH secretion during follicular stage is subnormal, includes women with PCOS	Weight loss, clomiphene with or without metformin, aromatase inhibitors
<i>WHO Class III</i> Hypergonadotropic, hypoestrogenic anovulation	Premature ovarian failure, ovarian resistance	
Hyperprolactinemic anovulation	Elevated levels of prolactin inhibits gonadotropin and therefore estrogen secretion	Dopamine agonists (bromocriptine)
Tubal Factor Infertility		Hysterosalpinogram, IVF, tubal reconstruction, laparoscopic surgery (endometriosis)
Uterine Factor Infertility		Leiomyoma, polyp removal
Cervical Factor Infertility		IUI, IVF
Immunologic and Thrombophilic Factor Infertility		Low dose aspirin and heparin, anticoagulation therapy
Unexplained Infertility		Clomiphene with or without IUI, gonadotropins with IUI, ART

ART, assisted reproduction technology; FSH, follicle-stimulating hormone; nGnRH, gonadotropin-releasing hormone; IUI, intrauterine insemination; IVF, invitro fertilization; PCOS, polycystic ovarian syndrome, WHO, World Health Organization.

resulting in a live birth (111). Many infertility treatment options and protocols are available, most of which are highly individualized (Table 41.15) (109).

Intrauterine Insemination and In Vitro Fertilization

Intrauterine insemination (IUI) involves induction of ovulation, followed by transfer of motile sperm directly into the uterine cavity. This type of procedure is indicated if erectile dysfunction or male infertility (low motility, low count, and/or poor morphology) has been identified, if the cervical environment (e.g., presence of antisperm antibodies) is not conducive for effective sperm motility and/or sustenance, and if there are any physiologic dysfunctions (112).

IVF involves hyperstimulation of the ovaries to produce multiple mature follicles. Once the follicles reach 18 to 21 mm in diameter, hCG is administered, and within 34 to 36 hours, all the mature follicles are aspirated and their eggs removed. Depending on the couple, fertilization of the oocyte by the sperm may or may not require assistance. The resulting fertilized oocytes or embryos are then sustained and monitored for 3 to 5 days. Embryo characterization is performed daily to identify the most advanced embryos of good quality for subsequent transfer into the uterine cavity. Preimplantation genetic diagnosis can be performed with a single cell biopsied from each embryo that reaches the six- to eight-cell stage by day 3. The genetic tests performed (on all chromosomes) will

help the physician and patient to identify those embryos with normal genetic makeup (including sex determination). Despite these advanced technologies and regardless of the age group, the pregnancy rate does not exceed 50%, and the risk for multiple births is substantially higher than that observed in the general population.

Clomiphene Citrate (Clomid)

Clomiphene citrate is by far the most frequently prescribed agent to stimulate ovulation. It is administered orally as a mixture of two geometric isomers. The *Z* (*cis*, zuclophene)-diastereomer displays estrogenic activity, and the *E* (*trans*, enclomiphene)-diastereomer exhibits antiestrogenic activity (Fig. 41.10). Its ability to stimulate ovulation stems from its action at the hypothalamic ERs, where it serves to block these receptors and interfere with natural feedback inhibition. When an estrogen deficiency is perceived by the hypothalamus, it responds by secreting GnRH. This peptide hormone signals the pituitary to release the gonadotropins FSH and LH. Elevated FSH levels promote follicular development, which in turn causes the maturing follicles to secrete estradiol. Clomiphene is administered orally (50 or 100 mg) for 5 consecutive days, typically on days 5 to 9 of the menstrual cycle. Approximately 7 days after the last clomiphene tablet is taken, the hypothalamus finally is able to detect that estradiol levels are elevated, and it then signals the pituitary to secrete LH. This surge in LH causes the dominant

TABLE 41.15 Representative Infertility Treatment Options (89)

Treatment Option	Patient Characteristics	Pharmacological Therapy
Intrauterine insemination	Mild endometriosis Unexplained infertility Mild male factors	Clomiphene citrate (days 5–9), monitor LH levels
Ovulation induction	Ovulatory disorder PCOS Hyperprolactinemia Hypothalamic amenorrhea Premature ovarian failure	Progestin challenge test; gonadotropin therapy (hMG and/or FSH)
In vitro fertilization (with or without donor oocytes)	Tubal factor Severe endometriosis Unexplained infertility	Ovarian hyperstimulation, hCG
Therapeutic donor insemination	Women without partners Severe male factor	Sperm donor

FSH, follicle-stimulating hormone; hCG, human chorionic gonadotropin; hMG, human menopausal gonadotropin; LH, luteinizing hormone; PCOS, polycystic ovarian syndrome.

follicle to rupture and release an egg, a process known as ovulation.

The clomiphene citrate challenge test is a diagnostic tool used to evaluate ovarian reserve in women older than 35 years of age. The protocol involves measurement of FSH and estradiol levels on day 3 of the cycle, followed by administration of 100 mg of clomiphene citrate on days 5 to 9 and measurement of FSH and estradiol on day 10 of the cycle. Elevated estrogen levels on day 3 are a good predictor of a poor pregnancy prognosis. Any FSH measurement that is high (>10 mIU/mL), even if other levels measured within the same cycle are within the normal range, is interpreted to mean a significantly limited reproductive potential because of poor ovarian reserve (108,113). The adverse effects associated with clomiphene include thickening of the cervical mucus, which creates a substantial barrier for motile sperm. A postcoital test can be performed to determine if this, in fact, represents an insurmountable hurdle. If so, an IUI is warranted to bypass this cervical challenge. Another adverse effect is endometrial thinning, which is not conducive to embryo implantation; if this is the case, additional pharmacologic therapies are recommended to thicken the endometrial lining. Other adverse effects include hot flashes and mood swings (114).

Women with low estrogen levels or hypothalamic disorders that result in irregular or absent ovulation are not candidates for clomiphene therapy. There is a 10% chance of having twins and a slightly higher risk of miscarriage in women stimulated with clomiphene (114). Clomiphene-based ovulatory stimulation should only be considered for a maximum of six cycles.

Clomiphene is readily absorbed from the GI tract following oral administration, with a half-life of approximately 5 days. It is metabolized in the liver, and its metabolites are excreted principally in the feces via enterohepatic recirculation.

Human Chorionic Gonadotropin (Pregnyl)

Human chorionic gonadotropin mimics LH both in structure and in function. It is produced by the placenta during pregnancy and is isolated from the urine of pregnant women. In the protocols for IUI or IVF, exogenous hCG must be administered to stimulate a scheduled ovulation. Between 36 and 72 hours after intramuscular administration, the dominant follicle ruptures and releases its egg. At that point, the scheduled IUI or IVF occurs. The hCG can be given in combination with clomiphene citrate to enhance ovulation (114). (For an in-depth discussion of hCG, see Chapter 6.)

Choriogonadotropin Alfa (Ovidrel [r-hCG])

The biologic activities of r-hCG are identical to those of placental and human pregnancy (urine) u-hCG. From a structural perspective, this glycoprotein has an α chain that is identical to hCG, LH, and FSH with some variation in the branching and extent of sialylation of the oligosaccharides (115). The structure and glycosylation pattern of the β chain is similar to u-hCG.

Human Menopausal Gonadotropin (Repronex, Menopur)

Human menopausal gonadotropin (menotropins) is composed of equal parts FSH and LH and is derived from the urine of menopausal women. This type of therapy is used after clomiphene-based therapy has failed or if a patient does not produce sufficient levels of FSH or LH and experiences amenorrhea. These hormones directly stimulate the ovaries to develop multiple follicles per reproductive cycle (114).

Human menopausal gonadotropin has a substantial role in several IVF protocols. It typically is administered via injection (75 or 150 U) starting on day 2 or 3 of the menstrual cycle and is administered continuously (7 to 12 days) until shortly before the patient is ready to undergo follicle retrieval. The size and number of maturing

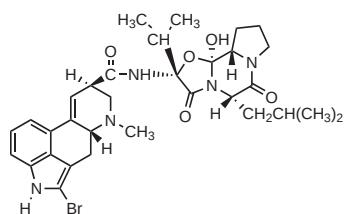
follicles, as well as the patient's estrogen level, are closely monitored to maximize the number of follicles retrieved and to minimize the risk of ovarian hyperstimulation syndrome (OHSS) (114). Adverse effects include injection site reactions and inflammation as well as the possibility of developing OHSS. Menopur, another formulation of menotropins that also contains equal parts FHS and LH, has been shown to cause fewer injection site reactions.

Follicle-Stimulating Hormone (Bravelle, Follistim, Gonal-F)

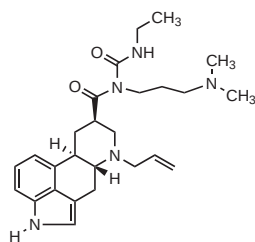
Two types of products are available that contain FSH. Bravelle contains FSH isolated from the urine of menopausal women and does not contain significant amounts of LH. Follistim and Gonal-F are products that contain only purified synthetic (recombinant) FSH. The recombinant products are more effective at stimulating the development of multiple follicles (111,112). All three agents are administered as subcutaneous injections. Administration of exogenous FSH circumvents the influence of both the hypothalamus and the pituitary on the ovaries and causes direct stimulation of ovarian follicle growth. Women who fail to achieve pregnancy with clomiphene assistance or who are diagnosed with polycystic ovarian syndrome are candidates for this type of therapy (114). (For additional discussion of FSH, see Chapters 6.)

Dopamine Agonists—Bromocriptine and Cabergoline

The hormone prolactin is responsible for stimulating the production of breast milk in mothers of newborns as well as for inducing lactation infertility. In this situation, FSH and LH secretion is inhibited; therefore, ovulation is prevented. In other populations of women, elevated levels of prolactin can result from the presence of an adenoma or a benign pituitary tumor or because the pituitary cells that produce prolactin are hyperactive. This hyperactivity may have a pharmacologic genesis (e.g., antipsychotics, tranquilizers, painkillers, or alcohol) or a pathophysiologic origin (e.g., kidney or thyroid disease). Both bromocriptine and cabergoline suppress prolactin production. Bromocriptine is available as a tablet or capsule and is administered one to four times daily. It also is available as an intravaginal formulation. Cabergoline tablets are administered one to two times weekly. Once the prolactin levels drop to normal levels, the patient should experience improved fertility. If pregnancy is not achieved despite normal prolactin levels, then addition of clomiphene or gonadotropins is warranted (114).



Bromocriptine
(Pergonal, Parlodel)



Cabergoline
(Dostinex)

Gonadotropin-Releasing Hormone

GnRH is secreted from the hypothalamus in a pulsatile fashion once roughly every 90 minutes. It is rapidly degraded by pituitary endopeptidases and has a circulatory half-life of 2 to 4 minutes. Degradation occurs primarily at the Gly⁶ residue (Fig. 41.16). Synthetic efforts to improve peptide stability have resulted in several successful GnRH agonists that contain hydrophobic D-amino acids replacing Gly⁶ (see below) (116).

When GnRH binds to and activates its pituitary cell surface receptor, it sends a signal for the pituitary to biosynthesize and secrete FSH and LH. This receptor is part of the family of G protein-coupled receptors and is composed of seven-transmembrane domains. Eventually, the receptor hormone complex is internalized, the ligand degraded, and the receptor partially recycled (116).

If a patient is unable to produce sufficient quantities of GnRH, then replacement therapy is available. To mimic normal secretion patterns, this hormone is delivered by infusion every 90 minutes via an external pump worn by the patient (114).

Gonadotropin-Releasing Hormone Agonists and Antagonists

Typically, GnRH agonists are administered continuously and not in normal physiologic pulses. As a result of constant stimulation of the GnRH receptors, an initial hyperstimulation of the pituitary occurs, causing a surge in FSH and LH secretion, which is quickly followed by complete cessation of their release. Ultimately, this causes the production of estrogen and progesterone to stop, which prevents ovulation from occurring. In patients who do not have a regular ovulatory cycle, these medications are used to squelch all internal influence over ovulation. Once internal hormonal signaling has been silenced, human menopausal gonadotropin can be administered, permitting the physician exquisite control over the levels of pituitary hormones. With exogenous FSH and LH now present, follicle development is stimulated, and oocyte retrieval for IVF is possible. When used continuously, GnRH agonists stifle the natural LH surge associated with ovulation, so hCG must be administered 34 to 36 hours before oocyte retrieval to launch ovulation.

Several GnRH agonists are available, including leuprolide acetate (Lupron) and goserelin acetate (Zoladex), which are injectable agents, and nafarelin acetate (Synarel), which is administered via nasal spray (Fig. 41.16). GnRH is rapidly degraded by peptidases (half-life = 2 to 4 minutes) between amino acids 5 and 6, 6 and 7, and 9 and 10. These analogs of GnRH have altered amino acids at these positions in order to improve analog stability. (Discussion about structure-activity relationships can be found in Chapter 6.)

Two GnRH antagonists, ganirelix acetate (Antagon) and cetrorelix acetate (Cetrotide), are administered by intramuscular injection (114). The primary advantage of the GnRH antagonists over the agonists is that they interact with and effectively block the GnRH receptor without

causing an initial surge of FSH and LH. The primary role of both the antagonists and agonists is to prevent a premature LH surge.

In an IVF cycle, the GnRH antagonist is not administered until the dominant follicle reaches 14 mm in diameter, which is nearly a week after gonadotropin therapy is initiated. The GnRH antagonists suffer from low solubility and the tendency to form gels in aqueous solutions (116). Ganirelix acetate has a short half-life and must be administered daily until hCG is administered, whereas cetrorelix has a longer half-life and can be administered once every 3 to 4 days (112). (Discussion about structure-activity relationships can be found in Chapter 6.)

Women whose ovulatory cycle has been managed by gonadotropin therapy have a 25% risk of multiple pregnancies. Some evidence suggests a lower pregnancy rate in patients using protocols that include a GnRH antagonist (116,117). Adverse effects include injection site pain and inflammation, abdominal bloating, mood swings, and menopausal symptoms, including hot flashes and vaginal dryness. The likelihood of developing OHSS is fairly small in protocols that include GnRH agonists or antagonists. In these regimens, the physician has control over the hCG injection necessary to trigger ovulation and can delay—or even withhold—this injection if estrogen levels are too high, a factor that predisposes the patient to develop OHSS (114). It should be noted that GnRH downregulation caused by these agents may have an adverse effect on oocyte quality, which has a negative impact on an IVF cycle (111).

As part of an assisted reproductive technology protocol, the use of both a GnRH agonist or antagonist and exogenous gonadotropins provides significant benefits, including enhanced stimulation of follicular development, improved oocyte quality, and LH surge suppression, all of which lead to fewer cycle cancellations and improved reproductive capability (111).

Lutropin Alfa (Luveris)

Lutropin alfa is the first recombinant human form of LH approved by the FDA. This agent serves as LH replacement therapy for women who are hypogonadotropic and hypogonadal with a substantial deficiency in LH (<1.2 IU/L). Lutropin alfa typically is used in combination with follitropin alfa to induce follicular growth and development. Both of these agents are administered subcutaneously.

The half-life of lutropin alfa is approximately 10 hours. Treatment should not exceed 14 days unless follicular development is clearly evident. Adverse effects include headache, breast and abdominal pain, and nausea. Lutropin alfa can contribute to the development of OHSS and should not be used in women with primary ovarian failure or uncontrolled thyroid or adrenal dysfunction or who are pregnant.

Progesterone (Endometrin)

Since natural progesterone levels can be compromised by IVF, progesterone supplementation is recommended

to provide luteal phase support. Formulated as a vaginal insert, micronized progesterone is administered to support embryo implantation and early pregnancy as a component of assisted reproductive technology treatment. Drug administration begins at the time of oocyte retrieval and continues two to three times per day for the following 10 weeks or until menses occur, indicating cycle failure. Higher progesterone serum concentrations were achieved more quickly with this formulation as compared to progesterone 8% gel (Crinone).

Menopause

Physiologically, menopause is described as the loss of ovarian follicle function, followed by the cessation of menses. Before menopause, the body typically is engaged in a 2- to 4-year preamble, complete with the signs and symptoms of menopause (e.g., hot flashes and night sweats). This stage is termed perimenopause and is characterized by a gradual decline in the secretion of estrogen and progesterone. When the ovary no longer prepares follicles for ovulation and the resulting corpus luteum is not formed, a dramatic decrease in the production of estrogen and progesterone occurs, along with the resulting physiologic changes, including bone loss, urogenital atrophy, and perhaps, incontinence, vasomotor symptoms, enhanced risk of cardiovascular disease, sexual dysfunction including decreased libido, and reduced skin elasticity (118). Many, if not most, of these changes have a negative effect on the quality of life for women during what is considered to be an exceptionally productive time in their lives. Identification of pharmacologic and nonpharmacologic methods to alleviate these symptoms is essential to improving quality of life during this turbulent hormonal phase at the end of a woman's reproductive lifetime.

History

As early as 1899, the Merck Manual listed several treatments for menopausal symptoms, including the drug Ovarin, which was composed of desiccated and pulverized cow ovaries flavored with vanilla. In 1932, this formulation was largely replaced by products that contained components isolated from the urine of pregnant women. By the late 1930s, these human-derived products were supplanted by Premarin, which contained components isolated from the urine of pregnant mares (119). Other formulations that take advantage of other routes of administration have been available for quite some time as well; for example, in 1928, the first estrogen patch was developed by Searle.

By the 1960s and 1970s, the first studies regarding serious adverse effects associated with estrogen therapy were reported, including severe hypertensive episodes and increased thromboembolic events. In 1978, the FDA mandated that a comprehensive warning be distributed with all estrogen products, including OCs. This document listed the approved indications and fully described the potential risks of using an estrogen-containing formulation. Despite these warnings, the use of estrogen-containing products actually increased as a result of the

FDA notice that stated that these agents were effective for the prevention of osteoporosis (119).

Hormone Replacement Therapy

Until mid-2002, HRT continued to be considered as first-line therapy for the pharmacologic management of menopausal symptoms and for health preventive purposes in postmenopausal women. At that point, the results from the WHI study and, eventually, the WHI Memory Study (120) were released and clearly identified that although HRT (combination of 0.625 mg/day of CEE and 2.5 mg/day of medroxyprogesterone acetate) reduced the incidence of colorectal and endometrial cancers as well as hip fracture, it also increased the incidence of heart attacks, stroke, and breast cancer and increased global cognitive impairment (121). Ultimately, the FDA mandated that safety warnings appear on the package labels for both combination and estrogen-only products to warn women about these risks.

Medical management of menopause largely revolves around treatment of the symptoms caused by estrogen deficiency. Without regular maturation of ovarian eggs, follicles are not consistently formed and, therefore, do not consistently secrete the same levels of estrogen as found in premenopausal women. In postmenopausal women, the primary source of endogenous estrogen (albeit a fraction of that found in premenopausal women) comes from the transformation of androstenedione into estrone. Because the release of FSH and LH by the pituitary gland is governed by estrogen concentrations via negative feedback inhibition, it is no surprise that menopausal women have FSH levels 10- to 15-fold higher and LH levels 4- to 5-fold higher than those found in premenopausal women (53). In addition, low estrogen levels prevent launch of the normal LH surge that is key to follicular rupture, followed by formation of the corpus luteum. Because the corpus luteum is responsible for the secretion of progesterone, it is, again, no surprise that the levels of this hormone also are out of balance during menopause (122). The vasomotor symptoms associated with menopause represent some of the most disturbing and, potentially, embarrassing menopause-related effects experienced by women. Since the early 1940s, estrogen replacement therapy (ERT) with CEE has been indicated for the treatment of these symptoms as well as for the treatment of urogenital atrophy and vaginal dryness. For patients with an intact uterus, progesterone was administered along with the estrogen to prevent endometrial hyperplasia and endometrial cancer. This type of combination therapy is referred to as HRT. Nearly 90% of menopausal patients experience symptom relief after initiation of either ERT or HRT (Table 41.16) (123,124). Each patient requires individual dose titration to maximize menopausal symptom relief and to minimize side effects (e.g., breast tenderness, weight gain, and vaginal bleeding). Because of the WHI findings, HRT is considered as safe to administer if used short term and if dose reduction occurs after 2 to 3 years of treatment.

Despite the WHI findings, estrogen remains the gold standard in relief of menopausal symptoms. Recently, pharmaceutical manufacturers have concentrated on the development of alternate dosage forms that deliver the minimum dose of estrogen to minimize adverse effects, to maximize therapeutic benefit, and to maintain a more normal ratio of estrone to estradiol. These formulations include several different types of transdermal products, vaginal rings, and vaginal creams. The estrogens found in these products include estradiol, EE (synthetic), piperazine estrone sulfate, and conjugated estrogens. Unfortunately, these estrogens do not have identical potencies. Conjugated equine estrogens, esterified estrogens, and estrone sodium sulfate at 0.625 mg are equivalent to 1 mg of micronized estrogen, 0.75 mcg of piperazine estrone sulfate, 5 to 10 mcg of EE, or 0.50 mcg of transdermal estrogen.

Although estradiol is the most potent endogenous estrogen, it has poor oral bioavailability. Oral administration of estrogen in a conjugated (e.g., estradiol valerate) or micronized (Estrace) form enhances absorption by the GI mucosa; however, when administered via this route, estradiol undergoes extensive hepatic metabolism (CYP3A4) to the corresponding ketone (estrone) (Fig. 41.3), which is much less potent. Only 10% of an oral dose of estradiol reaches the circulation (53). Estrone then undergoes sulfate conjugation to an inactive product. In effect, estrone sulfate represents a stable reservoir of circulating estradiol (12,13). Piperazine estrone sulfate (estropipate) also is quickly converted to estrone sulfate when administered orally. In premenopausal women, the physiologic ratio of estradiol to estrone is 2:1, but after oral administration of estradiol, this ratio shifts substantially in favor of higher circulating estrone levels.

One mechanism to increase the oral bioavailability of estradiol is to prevent metabolic oxidation of the estradiol C₁₇ hydroxyl group to the corresponding ketone. This is readily accomplished via alkylation of the C₁₇ position with a chemically inert alkyne group (e.g., EE, Fig. 41.6). This synthetic analog is several hundred-fold more potent than estradiol, with doses in the microgram rather than the milligram range.

Conjugated equine estrogen, isolated from the urine of pregnant mares, currently is the most popular form of ERT. In addition to the ring B saturated hormones already discussed (45% estrone sulfate), CEE also contains equilin estrogens that have an unsaturated B ring, including equilin sulfate (25%) and 17 α -dihydroequilin (15%). Comparison between human and equine-derived estrogens yields nearly equivalent binding properties between equilin and estrone as well as similar metabolic fates (12). Another method to keep the ratio of estrone to estradiol more equivalent is to use formulations that deliver estrogen via routes that are not subject to first-pass metabolism (e.g., transdermal, intravaginal, and intranasal). In addition, these routes permit ratios of estradiol to estrone similar to those found in premenopausal women (53). Transdermal formulations (Table 41.16) permit utilization of a lower drug dose and provide continuous systemic availability.

TABLE 41.16 Hormone Replacement Therapy for Treatment of Menopausal Vasomotor Symptoms (117,123,124)

Dosage Form	Trade Name	Estrogen	Progesterone
<i>Tablets: estrogen only</i>			
	Femtrace	Estradiol acetate	
	Premarin	Conjugated estrogens	
	Cenestin	Conjugated estrogens (plant derived)	
	Menest, Estratab	Esterified estrogen	
	Estrace	Micronized 17 β -estradiol	
	Ortho-Est, Ogen	Estropipate	
	Enjuva ^a	Conjugated estrogens $\Delta^{8,9}$ -dehydroestrone sulfate	
<i>Tablets: estrogen/progestin combination</i>			
	Angeliq	Estradiol	Drospirenone
	Prempro	Conjugated estrogens	Medroxyprogesterone acetate (daily)
	Premphase	Conjugated estrogens	Medroxyprogesterone acetate (cyclic)
	FemHRT	Ethinyl estradiol	Norethindrone acetate
	Activella	Estradiol	Norethindrone acetate
	Prefest	Estradiol	Norgestimate
<i>Transdermal or percutaneous</i>			
• Transdermal spray (solution)	Evamist	Estradiol	
• Transdermal emulsion	Estrasorb	Estradiol	
• Transdermal gel	Estrogel	Estradiol	
	Divigel	Estradiol	
	Elestrin	Estradiol	
• Patch (estrogen)	Alora (biweekly)	Estradiol	
	Climera (weekly)	Estradiol	
	Estraderm (biweekly)	Estradiol	
	Menostar (weekly)	Estradiol	
	Vivelle dot (biweekly)	Estradiol	
• Patch (combo)	Climera Pro	Estradiol	Levonorgestrel
	Combipatch	Estradiol	Norethindrone acetate
<i>Capsules</i>	Prometrium		Micronized progesterone
<i>Intravaginal delivery</i>			
• Tablets	Vagifem	Estradiol	
• Cream	Estrace Vaginal	Estradiol	
	Premarin Vaginal	Conjugated estrogens	
	SCE-A Vaginal Cream	Synthetic conjugated estrogens A	
• Ring	Ogen Vaginal	Estradiol	
	Estring	Estradiol acetate	
	Femring	Estradiol acetate	
<i>Intramuscular injection</i>	Delestrogen	Estradiol valerate	
	Depo-Estradiol	Estradiol cypionate	

Intravaginal administration of micronized estradiol tablets (Table 41.16) represents another avenue for effective estradiol absorption without concern about first-pass metabolism (12). Originally approved as a 25-mcg tablet, Vagifem (10 mcg estradiol) was approved in 2009 by the FDA as a low-dose formulation for the treatment of vaginal atrophy. Unfortunately, vaginal creams containing estradiol only achieve 25% of the absorption expected for a similar oral dose. As a result, this formulation is best used for the treatment of urogenital atrophy. A sustained-release depot formulation of estradiol is available

in the form of a vaginal ring (Femring, Estring) (Table 41.16). This type of formulation elutes estradiol or estradiol acetate over a 90-day period, thereby limiting the effort required by the patient. Estring delivers 7.5 mcg of estradiol per day and is indicated for the treatment of urogenital atrophy. Available in two strengths, Femring releases the equivalent of 50 or 100 mcg of estradiol per day via estradiol acetate, which subsequently is hydrolyzed to estradiol after release. This ring is indicated for the treatment of several urogenital symptoms associated with postmenopausal atrophy as well as for the treatment

of vasomotor symptoms as a result of delivery of sufficient systemic estrogen. Two vaginal creams have been approved by the FDA for the treatment of vulvar and vaginal atrophy due to menopause. Synthetic Conjugated Estrogens-A (plant derived) is to be administered as 1 g once daily (0.625 mg/g cream) for 1 week followed by 1 g twice weekly. Conjugated estrogens (Premarin) are administered as a cyclic regimen with delivery of 0.5 to 2 g of cream daily for 21 days and then off for 7 days. Premarin vaginal cream (0.5g), administered twice weekly or using the cyclic regimen, has also been approved for the treatment of moderate to severe dyspareunia (painful sexual intercourse).

Although easy to use, percutaneous preparations (topical emulsion) (Table 41.16) suffer from inconsistent absorption. The FDA has approved several transdermal estradiol gels (Divigel, Elestrin, EstroGel) for the treatment of moderate to severe hot flashes. Divigel, sold in individual-use packets (0.50 mg estradiol), is a quick-drying gel that is applied daily to a 5 × 7 inch area on the thigh. Elestrin is supplied as a metered-dose applicator that delivers 0.87 g of gel per actuation (0.52 mg estradiol). The gel is applied directly to the upper arm and dries within 1 to 2 minutes (125). EstroGel is also delivered in a metered-dose applicator with 1.25 g of gel supplied per actuation (0.75 mg estradiol). It is recommended that EstroGel be applied in a thin layer from shoulder to wrist (~750 cm² area). Divigel and Elestrin are the lowest dose transdermal gels approved for the relief of hot flashes.

Evamist (estradiol), approved by the FDA in 2007, is a metered-dose transdermal spray that is to be applied to the inside of the forearm. Each spray (90 µL) delivers 1.53 mg of estradiol. Dosage adjustments can be easily made by increasing or decreasing the number of sprays applied daily. The FDA has warned that children and pets should not be exposed to those areas of the skin that have been

treated, as premature puberty in girls and gynecomastia in boys developed in children of adult patients several weeks to months after initiation of Evamist therapy.

In addition to those identified by the WHI study, the adverse effects of ERT include nausea, headache, breast tenderness, and heavy withdrawal bleeding. Transdermal delivery minimizes headache and nausea, and use of a low-dose oral regimen lessens breast tenderness (5).

Micronized progesterone (Table 41.16) in combination with oral CEE provides valuable therapeutic benefit in the treatment of menopausal symptoms without the risk of endometrial hyperplasia. Unless micronized, progesterone is poorly absorbed when administered orally (12). Intravaginal administration of a controlled-release progesterone gel (Table 41.16) results in excellent absorption and more consistent serum levels than orally administered micronized progesterone. Systemic absorption is limited when progesterone is delivered by this route. Intramuscular administration of a progesterone in oil formulation produces the highest serum concentrations but is least likely to be tolerated by patients. The adverse effects associated with the use of oral progestins include irritability, depression, weight gain, and headache. These can be minimized if other routes of administration are used (e.g., IUD).

To minimize patient effort and to maximize patient adherence, a number of estrogen/progesterone combination products are commercially available (Table 41.17). Four types of combination regimens are available (oral and transdermal), including both continuous and cyclic therapeutic options. Cyclic regimens require a minimum of 12 to 14 days of progestin administration to prevent endometrial hyperplasia.

Used in perimenopause, administration of a synthetic progestin supplements the waning progesterone produced in the luteal phase of the menstrual cycle. This helps to prevent endometrial hyperplasia and to ensure

TABLE 41.17 Combination Therapies for the Treatment of Menopausal Symptoms (50)

Type of Therapy	Effect of Therapy	Regimen
Oral continuous-cyclic (sequential)	Scheduled withdrawal bleeding	CEE (1-14) + [CEE + MPA] (15-28)
Transdermal continuous-cyclic (sequential)	Scheduled withdrawal bleeding	17β-Estradiol + norethindrone acetate
Oral continuous-combined	Prevents monthly bleeding; unpredictable breakthrough bleeding likely to occur for first 6–12 months	CEE + MPA, 17β-estradiol + norethindrone acetate, ethinyl estradiol + norethindrone acetate
Transdermal continuous-combined	Prevents monthly bleeding	17β-estradiol + norethindrone acetate
Continuous long cycle (cyclic withdrawal)	Reduced monthly bleeding	Estrogen administered daily, progestin given six times/year for 12–14 days
Intermittent-combined (Continuous pulse)	Prevents monthly bleeding	3 days of estrogen alone, followed by 3 days estrogen and progestin combo, repeat continuously

CEE, conjugated equine estrogen; MPA, medroxyprogesterone acetate.

regular menses despite the irregularity of the menstrual cycle.

A unique investigational SERM RAD1901 is able to cross the blood–brain barrier and activate centrally located estrogen receptors (126). It retains ER agonist action at the bone and antagonist action in uterine and breast tissues. It is in phase II clinical trials for the treatment of menopausal hot flashes.

With such concern and confusion surrounding the use of either ERT or HRT for the treatment of menopausal symptoms and postmenopausal health benefits, women have turned to other natural or non-hormonal pharmacologic sources of relief (Table 41.18) (127,128). There are three groups of phytoestrogens that are found in over-the-counter supplements: the coumestans, lignans, and isoflavones (129). “Hormonal herbs,” such as soy, black cohosh, and red clover, are purported to contain some or all of the phytoestrogens genistein, daidzein, and coumestrol (Fig. 41.17), which possess at least weak estrogenic activity (Table 41.19) (127–130) but structurally are nonsteroidal. The isoflavones genistein and daidzein are metabolized to other compounds including equol, which exerts estrogenic action. Other sources of nonsteroidal phytoestrogens, such as resveratrol from the skin of wine grapes (e.g., *Vitis vinifera*), currently are under investigation for their purported estrogenic effects (131). Small studies have indicated that the phytoestrogens may have additional beneficial effects on lipid profile and bone density (53).

MF-101 (Menerba) is an ER β -selective agonist in phase II clinical trials for the treatment of the vasomotor symptoms associated with menopause. It does not stimulate the endometrium or breast tissue and therefore is unlikely to increase the risk of either uterine or breast cancer and thus may be a safer alternative to HRT. MF-101 is derived from 22 herbs used in traditional Chinese medicine for

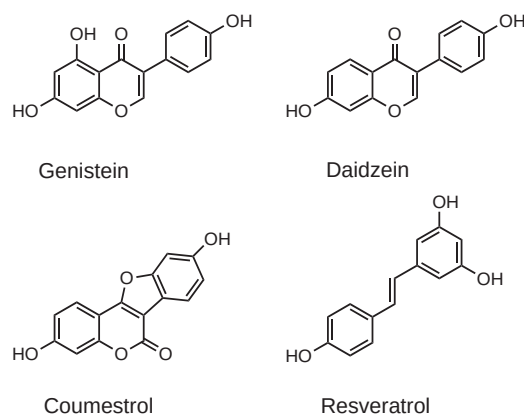
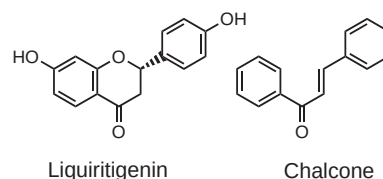


FIGURE 41.17 Naturally occurring phytoestrogens

the treatment of menopause. The active isolates include liquiritigenin and chalcone (126,132).



Several of the newer antidepressants, specifically those that modulate serotonergic neurotransmission, have become more prominent in the treatment of menopausal symptoms (Table 41.18). Some even consider this as first-line treatment when hormone therapy is contraindicated. Venlafaxine, a dual serotonin and norepinephrine reuptake inhibitor, has been found to decrease the frequency and substantially reduce the intensity of vasomotor symptoms and to improve problems of fatigue and sleep difficulty in more than 50% of the participants in clinical trials (127,133,134). With a 75-mg/day dose of venlafaxine, patients obtained relief from hot flashes within 1 to 2 weeks of treatment initiation. In comparison, fluoxetine and sertraline, which are selective serotonin reuptake inhibitors (SSRIs), only modestly improve the frequency and intensity of vasomotor symptoms. The effectiveness and onset of action of paroxetine, another SSRI, mirror those of venlafaxine. A low-dose paroxetine mesylate formulation (Mesaferm) is in phase II clinical trials. A deuterated analog of paroxetine (CTP-347) demonstrates less CYP2D6 inhibition, thereby alleviating the drug interactions characteristic of paroxetine (126). Interestingly, the SSRI citalopram not only is similarly effective as paroxetine but also is effective in patients in whom vasomotor symptoms did not resolve with venlafaxine treatment. Desvenlafaxine has been shown to decrease the frequency of moderate to severe hot flashes by 65% and also significantly reduced the severity of hot flashes (135). Mirtazapine is an α_2 -receptor antagonist and a serotonergic (5-HT₂) antagonist. The role of serotonin in modulation of the thermoregulatory processes is only starting to become clear. There is evidence that indicates that activation of central 5-HT₂

TABLE 41.18 Nonhormone Replacement Remedies for Vasomotor Symptoms of Menopause (103,104)

Serotonin Based	Adrenergic Based	GABA Based	Herbals
Venlafaxine (SNRI)	Clonidine	Gabapentin	Soy
Paroxetine (SSRI)	Mirtazepine		Black cohosh
Fluoxetine (SSRI)			Garden sage
Citalopram (SSRI)			Dong quai
Sertraline (SSRI)			Kava
Mirtazepine			Valerian
			Red clover
			Flaxseed

GABA, γ -aminobutyric acid; SNRI, serotonin and norepinephrine reuptake inhibitor; SSRI, selective serotonin uptake inhibitor

TABLE 41.19 Herbal Products with Purported Utility in Postmenopausal Women (103–105)

Herb	Purported Use	Potential Problems
Soy (beans) <i>Glycine max</i>	Contains phytoestrogens—varied evidence for success in treatment of hot flashes	No concurrent treatment with tamoxifen
Black cohosh (root) <i>Actaea racemosa</i>	Contains phytoestrogens—purported use for hot flashes, additional studies needed	Possible liver issues
Sage (leaves, flowers) <i>Salvia officinalis</i>	Reduces night sweats and hot flashes	Kidney damage (large amounts)
Kava <i>Piper methysticum</i>	Reduces anxiety—no evidence for use in hot flashes	Liver damage
Red clover (leaf) <i>Trifolium pretense</i>	Contains phytoestrogens—not clinically significant reduction in hot flashes	Interaction with warfarin that can cause bleeding issues
Dong quai (root) <i>Angelica sinensis</i>	Not effective in treating hot flashes	Interaction with warfarin that can cause bleeding issues
Valerian (root) <i>Valeriana officinalis</i>	Treats hot flashes and insomnia	Morning drowsiness
Ginseng (root) <i>Panax ginseng</i>	May help with quality of life issues (mood, sleep) but does not help with hot flashes	
Flaxseed	No significant evidence that it helps hot flashes	Alters activity of antiplatelet and anticoagulant drugs

receptors results in hyperthermia; therefore, administration of a 5-HT₂ receptor antagonist should alleviate the thermoregulatory dysfunction associated with estrogen deficiency (127). Interestingly, an endogenous estrogen deficiency causes a reduction in the expression of central 5-HT₂ receptors, which is reversible on administration of an exogenous source of estrogen.

There is limited evidence that gabapentin (including an unapproved extended-release formulation Serada) and clonidine are effective in reducing the frequency of hot flashes (45% to 49% decrease) (133). There is also tentative evidence that gabapentin improves sleep quality in menopausal women. Although not indicated for the treatment of vasomotor symptoms associated with menopause, pregabalin (Lyrica) has also demonstrated the ability to decrease the frequency and severity of hot flashes.

There are nonpharmacologic avenues to consider when trying to limit the rise in body temperature, including avoidance of caffeine, alcohol, spicy foods, anxiety, and physical contact. Maintaining a good diet and exercise program is also thought to be beneficial because obesity and a sedentary lifestyle have been related to hot flashes (129).

Hormone-Dependent Breast Cancer

Background and Introduction

In the United States, the lifetime risk of developing breast cancer is one in eight, with the greatest incidence in women older than 60 years of age (136). It is the second leading cause of death from cancer in women (lung cancer is number one). An evaluation of 51 studies that represented 52,000 women with breast cancer showed a 15% increase in the risk of breast cancer in patients who

had taken a combination of estrogen and progestin for less than 5 years. As treatment duration increased, this risk was amplified (137). In the HRT arm of the WHI study, the risk of breast cancer increased, yet in the ERT arm, there was no change in this risk (138). It should be noted that this increase in risk occurred 3 years after initiation of the study. To make matters worse, the breast cancers found in women of the HRT arm were likely to be more advanced than those found in women of the placebo arm of the study. Both arms of the WHI study were terminated prematurely, because the health risks incurred outweighed the benefits. No increased incidence of breast cancer was found in the ERT arm. The incidence of stroke did increase, however, and there was no benefit as it related to coronary heart disease (139).

According to the WHI study, there was no increased risk of endometrial cancer in patients taking HRT. If unopposed estrogen is administered to a patient with an intact uterus, then a significant increase in the risk of uterine cancer occurs within 2 years of treatment initiation (149). As the dose and duration of treatment increase, this risk grows.

Estrogen deficiency has been correlated with the development of colorectal cancer, the third most commonly diagnosed cancer. When estrogen levels are low, expression of the ER diminishes. Colorectal cells without ER expression are ripe to become cancerous. If estrogen levels are boosted by HRT, then ER expression should improve correspondingly, thereby reducing the number of colorectal cells primed to become malignant. The WHI study clearly shows a substantial decrease (37%) in the occurrence of colorectal cancer in patients treated with HRT.

MENOPAUSE AND LIBIDO

Despite what you might gather from promotional ads on the television and radio, sexual dysfunction in women occurs much more frequently than in men. Regardless of age (18 to 59 years), the prevalence of a lack of interest in sex among U.S. women is approximately 30% (140). This loss of sexual desire, a general feeling of lethargy or loss of energy, and reduced muscle strength can be attributed largely to an androgen deficiency. Urogenital atrophy and a decrease in vaginal lubrication in postmenopausal women also are potential reasons for a woman to experience an arousal disorder. These symptoms are directly linked to an estrogen deficiency. Administration of an exogenous androgen, typically testosterone, eases some of these symptoms, but it is unclear if this is a result of androgen receptor activation or of conversion into estradiol. Testosterone can be administered via several routes, including intramuscularly, subcutaneously (implants), or transdermally (patches). Estrogen replacement therapy effectively manages vasomotor symptoms as well as urogenital atrophy and dryness and diminishes some of the psychological symptoms, but it has virtually no effect on the waning libido experienced by many menopausal women. A method to address both sets of symptoms is to administer an oral estrogen/androgen combination (Estratest [methyltestosterone + esterified estrogen]) daily (53). With this type of regimen, it is imperative that the testosterone dose only generates physiologic serum testosterone levels so as to maximize therapeutic benefit and minimize undesirable side effects, including virilization and fluid retention (141).

Nonhormonal strategies also are available to treat female sexual arousal disorder. A transdermal formulation of alprostadil, used to treat male erectile dysfunction, has been shown to improve the arousal success rate. Applied 15 minutes before intercourse, this cream can cause local irritation, which is reversible on cleansing (142).

Molecular Mechanism of Action of the SERMs

The precise mechanism of SERM action is not well understood, but clearly, the ER is a primary target, as a signal transduction pathway, to modulate drug action in different tissues. Different concentrations of ER are present in breast cancers, which can be explained by heterogeneity in the tumor cell population. The more cells in the tumor that contain ERs, the higher the overall ER content, and the more likely the tumor will respond to antiestrogen therapy. Approximately 60% of ER-positive (receptor-rich) tumors are responsive to any form of additive or ablative endocrine therapy, whereas only 10% of ER-negative (receptor-poor) tumors respond to endocrine therapy (150). The current standard of care is to determine the ER status of the tumor in all patients with breast cancer.

The ER is divided into six regions (A–F, as shown in Fig. 41.5) (151). The DNA-binding domain (region C) is essential for the interaction of the ER with an ERE

(152). The ligand-binding domain (region E) is the site of estradiol binding and the site of competitive binding by antiestrogens. The activating function (AF-1 and AF-2) regions are the areas of the ER that interact with coactivator molecules to form an effective transcription unit at an estrogen-responsive gene (153).

Crystallization of the ligand-binding domain (region E) of the ER with estradiol, DES, and raloxifene (154) as well as 4-hydroxytamoxifen (128) has provided an important insight regarding the conformational changes that occur with agonist or antagonist ER complexes. Figure 41.18 illustrates the crystal structure of the estradiol and raloxifene receptor complexes. The most important difference is the position of helix 12, which is essential for the correct binding of coactivators to form a transcription complex at an estrogen-responsive gene. Estradiol causes helix 12 to seal the ligand inside the hydrophobic pocket of the ligand-binding domain. It is hypothesized that this conformation allows coactivator binding. By contrast, raloxifene prevents helix 12 from sealing the hydrophobic pocket, and gene transcription is prevented (because coactivators cannot bind) (154). A similar

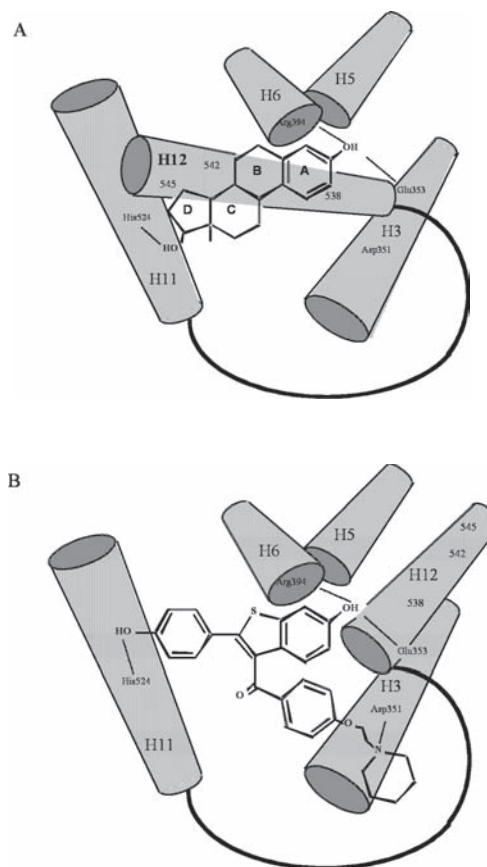


FIGURE 41.18 Comparison of the binding of estradiol (A) and raloxifene (B) in the ligand binding domain of the human estrogen receptor (ER). The key event in estrogen action is the repositioning of helix 12 (H12) to seal the steroid into the hydrophobic pocket so that coactivators can bind at the key amino acids indicated on H12. This cannot happen in the raloxifene–ER complex.

MENOPAUSE AND OSTEOPOROSIS

Osteoporosis causes 1.5 million fractures annually, including hip and spine fractures that increase both morbidity and mortality. In women older than 50 years, one in six will experience a hip fracture, and only 40% of patients older than 55 years with a hip fracture actually recover their mobility (143). (A comprehensive review of pharmacologic therapies available for the treatment and prevention of osteoporosis can be found in Chapter 30).

The estrogen deficiency associated with menopause has a detrimental effect on bone composition and quality. Before 65 years of age, postmenopausal women lose 3% to 5% of their bone mass on an annual basis. After this point, bone loss occurs at a rate of 0.5% to 1.0% annually. This results, in part, from an increase in circulating cytokines (e.g., tumor necrosis factor α , interleukin-1, and interleukin-6) when estrogen levels are low. These cytokines stimulate differentiation and proliferation of osteoclasts, the cells that normally degrade bone. The WHI study shows a positive correlation between HRT and a substantial decrease in fractures (24%). Currently, HRT (0.625 mg of CEE with 2.5 mg of medroxyprogesterone acetate) is recommended for the prevention, but not the treatment, of osteoporosis. This type of therapy should be initiated shortly after menopause and continued for at least 7 years (122). A lower dose of estrogen (0.3 mg) along with a progestin also may afford ample prevention. Despite the well-documented benefits of HRT in the prevention of osteoporosis, this should not be the primary therapeutic indication. In addition, no estrogen- or progestin-containing products have an FDA indication for the treatment of osteoporosis (119).

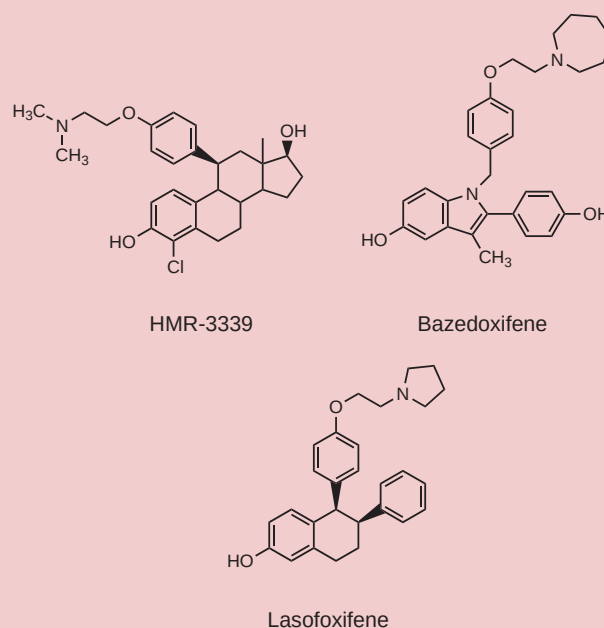
Studies that evaluate the benefit of combination OCs on bone mineral density have produced mixed results. Nearly one-third of these studies indicate no benefit, but the rest show a beneficial effect, including a 25% reduction of hip fractures in postmenopausal women (144). In patients using a combination OC containing at least 50 mcg of the estrogen component, this reduction in the risk of hip fracture improved to 44% (66). In general, the maximal benefits on bone mineral density are observed when the patients are older, have taken OCs for 5 or more years, and have used an OC preparation containing at least 50 mcg of estradiol.

Climara Pro (transdermal patch with estradiol [0.045 mg/day] and levonorgestrel [0.015 mg/day]) is indicated for the treatment of the vasomotor symptoms related to menopause. This formulation has also been approved for the prevention of postmenopausal osteoporosis.

Tamoxifen and raloxifene are SERMs and represent effective nonsteroidal therapy for the treatment of breast cancer and bone loss, respectively. The SERMs are a unique drug class in that a single agent may act as a receptor agonist in select tissues but a receptor antagonist in other tissues. In theory, it might be possible to protect a woman from osteoporosis and to reduce her risk for endometrial, breast, and colorectal cancers without worsening menopausal vasomotor symptoms with one drug. To date, no SERM can meet all these goals simultaneously. Raloxifene, a second-generation SERM, is an estrogenic

agonist at the bone and is very effective for preventing osteoporosis without negative endometrial consequences or increasing the risk of breast cancer (see Chapter 30) (53). It is classified as an antiresorptive agent and has been approved by the FDA for both the treatment and prevention of osteoporosis. It also has a beneficial role in reducing the risk of cardiovascular events in postmenopausal women because of its antiatherogenic effects (reduces LDL levels) as well as in decreasing the risk of breast cancer (145). Unlike the bisphosphonates, which decrease both hip and vertebral fractures by 30% to 50%, raloxifene reduces only vertebral fractures by the same percentage. Raloxifene only has an effect on skeletal sites composed of cancellous bone. Unfortunately, raloxifene causes or worsens vasomotor symptoms in approximately 25% of patients, which represents a limitation to its use. In addition, raloxifene causes a threefold increase in the risk for VTE, which makes it contraindicated in women with a history of thrombosis. Following oral administration, raloxifene undergoes rapid first-pass metabolism to form its glucuronide conjugates and, thus, has oral bioavailability (~2%).

Because raloxifene only reduces the risk of vertebral fracture, there is a need to identify agents that are effective at multiple skeletal sites with minimal side effects. The novel SERM HMR-3339 improves bone mineral density in adult ovariectomized rats at a variety of skeletal sites (e.g., lumbar spine, tibia, and femur), including those sites composed of cortical bone (145,146). An additional benefit afforded by HMR-3339 is a reduction in total cholesterol (10% to 15%) and LDL (10% to 24%). This antiatherogenic activity, coupled with promising reductions in fracture risk across multiple skeletal sites, makes HMR-3339 a very exciting drug candidate for the treatment of multiple postmenopausal conditions.



(Continued)

MENOPAUSE AND OSTEOPOROSIS (Continued)

Bazedoxifene

Bazedoxifene is an indole-based third-generation SERM currently in phase III clinical trials for the prevention of postmenopausal osteoporosis. Similar to raloxifene, it is an ER antagonist in both breast and uterine tissues and an ER agonist on bone and the

cardiovascular system. Bazedoxifene has been shown to reduce the risk of new vertebral fractures and reduce the risk of nonvertebral fracture in those at higher risk for fracture by 50% (147). Coadministration with conjugated estrogens has been shown effective in the management of menopausal symptoms.

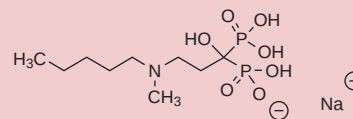
Lasofoxifene

Lasofoxifene is a naphthalene-based third-generation SERM currently in phase III clinical trials for the prevention of postmenopausal osteoporosis. Similar to raloxifene, it is an ER antagonist in both breast and uterine tissues and an ER agonist on bone and the cardiovascular system. Lasofoxifene has been shown to reduce the risk of both vertebral and nonvertebral fractures. Increases in BMD in the femoral neck (2.9%), lumbar spine (3.1%), and hip (2.7%) have been reported (147).

Additional SERMs are presently undergoing clinical evaluation for treatment of osteoporosis and menopause-related vaginal atrophy (see Chapter 30).

Ibandronate sodium (Boniva) is one example of the bisphosphonate class of antiresorptive agents and is effective in the

treatment and prevention of postmenopausal osteoporosis (see Chapter 30) (148).



Ibandronate sodium

It was first approved as a tablet (2.5 mg) to be administered daily and later approved as the first once-monthly formulation (150 mg) for the treatment and prevention of osteoporosis (148). In January 2006, the FDA approved an intravenous formulation of ibandronate sodium (3 mg) to be administered every 3 months for the treatment of osteoporosis. This type of formulation benefits patients who have difficulty swallowing oral dosage forms or are unable to remain upright for 30 to 60 minutes. The injectable formulation generated a larger increase in lumbar spine bone mineral density compared to the oral formulations. Bone mineral density in the hip, femoral neck, and trochanter was improved as well. Adverse events associated with the injectable formulation included arthralgia, back and abdominal pain, and hypertension. There is a risk of renal toxicity that is inversely related to the rate of administration of this formulation. Caution is advised in patients who have renal impairment or who are receiving other pharmacologic therapy that may negatively affect renal function (145).

model is proposed for the 4-hydroxytamoxifen ER complex, but it is clear from studies in vitro that the raloxifene- and 4-hydroxytamoxifen-receptor complexes have different efficacies (155,156).

The antiestrogenic and some of the estrogen-like actions of SERMs can be explained by the different conformations of the ER α complexes attracting novel coactivators or corepressors (Fig. 41.5) to modulate estrogen action at different sites. In this model, the receptor would be the same at each site, yet the coactivators and corepressors would be distributed differently at different targets (157).

Alternatively, ER α could be modulated by different concentrations of ER β at different sites. It has been suggested that ER β could enhance estrogen-like gene activation through a protein-protein interaction at AP-1 (*fos* and *jun*) sites (158). At present, it is not entirely clear how SERM action is modulated at each target. Indeed, more than one mechanism may occur.

Selective Estrogen Receptor Modulators

Lerner et al. (159) described the first nonsteroidal antiestrogen, ethamoxypiphetol (MER25) (Fig. 41.19). This compound is an antiestrogen, with no other hormonal or antihormonal action. Although MER25 behaved as a

“morning after” pill in laboratory animals (160), clinical trials associated with other reproductive and gynecologic applications showed that the drug had low potency and

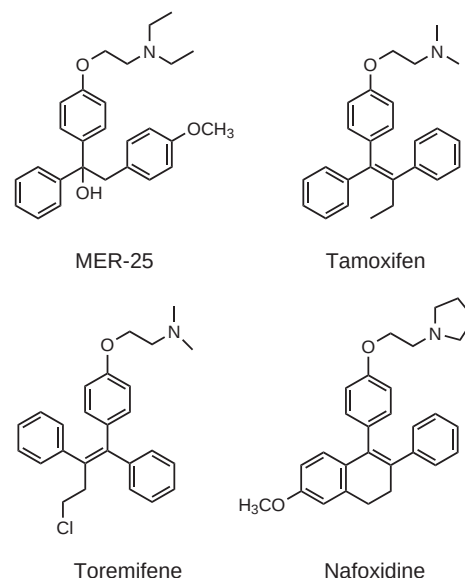


FIGURE 41.19 Nonsteroidal antiestrogens.

that the dose required caused central nervous system toxicity (161). In the search for more potent antiestrogens, clomiphene also was identified as an effective post-coital contraceptive in laboratory animals (Fig. 41.9) (162). Clinical trials in humans, however, demonstrated that clomiphene induces ovulation (163) and does not prevent implantation. Tamoxifen, a related triphenylethylene (Fig. 41.10), also was discovered as part of a fertility-control program (164). This drug is administered as the *Z*-diastereomer and, in some countries, is used for ovulation induction (165). Interestingly tamoxifen, clomiphene, and a rigid analog of tamoxifen, nafoxidine (Fig. 41.19), all inhibit the binding of [^3H]estradiol to ER; therefore, it was logical to test their efficacy as breast cancer treatments. Clomiphene and nafoxidine were not pursued after initial testing as a result of concerns about toxic side effects (166). Although a very low incidence of side effects is associated with tamoxifen, vasomotor symptoms often are evident and negatively affect patient quality of life (167–169). During the past 30 years, tamoxifen has become the endocrine treatment of choice for all stages of breast cancer and the first agent to reduce the incidence of breast cancer in high-risk pre- and postmenopausal women (170–172). Tamoxifen has now replaced endocrine ablative surgery in both pre- and postmenopausal patients with advanced breast cancer.

TAMOXIFEN (NOLVADEX) Tamoxifen (Fig. 41.19) is a SERM that is used as an antiestrogen in the treatment of estrogen-dependent breast cancer following primary treatment (chemotherapy and/or surgery). Therapy is limited to 5 years, because no additional benefit has been identified for treatment regimens of longer duration. Tamoxifen demonstrates only weak estrogenic effects at several sites, including the endometrium and bone, and on the lipid profile (169). Tamoxifen undergoes rapid oxidative *N*-demethylation to its major metabolite, *N*-demethyltamoxifen, via CYP3A4 and to its minor metabolite, 4-hydroxytamoxifen, via CYP2D6 (Fig. 41.20). Evidence suggests that 4-hydroxytamoxifen is the active metabolite of tamoxifen (173,174), with a higher binding affinity than the parent drug for the ER (173). Circulating levels of the demethylated metabolite at steady-state are up to twice the level of the parent drug, because the elimination half-life of *N*-demethyl tamoxifen is 14 days, compared with 7 days for tamoxifen (175). The aminoethyl ether side chain of the triphenylethylene antiestrogens is critical for the antiestrogenic activity of these agents.

Tamoxifen is rapidly absorbed from the GI tract with 100% bioavailability. It undergoes minimal first-pass metabolism and is highly protein bound, primarily to albumin. Tamoxifen's terminal elimination half-life is 5 to 7 days, which is indicative of enterohepatic recycling, elevated protein binding, and metabolic autoinhibition (169). For patients receiving concurrent anticoagulation therapy, a dosage reduction of the anticoagulant may be warranted.

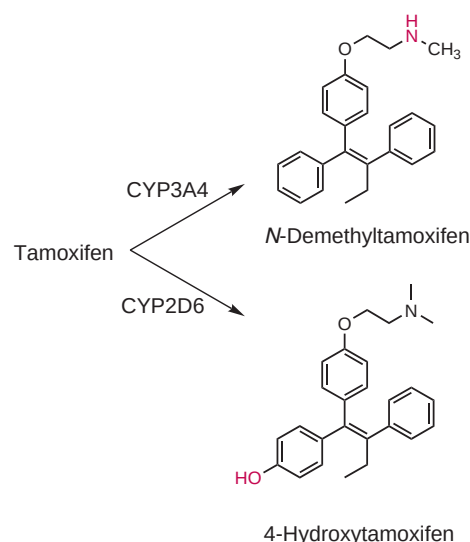


FIGURE 41.20 Metabolism of tamoxifen.

Agents that strongly inhibit CYP2D6 (e.g., paroxetine, fluoxetine, bupropion, duloxetine, thioridazine, quinidine, ticlopidine, terfenadine, cinacalcet) should be avoided in patients taking tamoxifen because these agents will prevent the metabolism of tamoxifen to its active metabolite (endoxifen). CYP2D6 genotyping will help to identify if a patient will respond to tamoxifen treatment.

After 5 years of tamoxifen therapy, a significant number of patients experience a relapse, primarily because of tamoxifen resistance. What is observed clinically is that the weak estrogenic effects of tamoxifen begin to predominate and effectively “feed” the hormone-dependent tumor.

TOREMIFENE (FARESTON) Toremifene is structurally and pharmacologically related to tamoxifen. It differs structurally from tamoxifen because of halogenation (chlorination) of the ethyl side chain, which reduces its antiestrogenic potency (Fig. 41.19). Toremifene demonstrates beneficial effects on the bone and cardiovascular system and increases HDL levels (169).

Toremifene is rapidly absorbed from the GI tract, with 100% bioavailability. It undergoes minimal first-pass metabolism and is highly protein bound, primarily to albumin. Toremifene, like tamoxifen, undergoes rapid oxidative *N*-demethylation, catalyzed by CYP3A4, to its active metabolite, *N*-demethyltoremifene. It also undergoes deamination-hydroxylation to ospemifene (Fig. 41.21) (169). Because toremifene is extensively metabolized in the liver, it should be used cautiously in patients with hepatic impairment. The terminal elimination half-life for toremifene is 5 to 6 days, which again is indicative of enterohepatic recycling and high protein binding (167).

Ospemifene (Ophena) is another SERM that is in phase III clinical trials for the treatment of postmenopausal

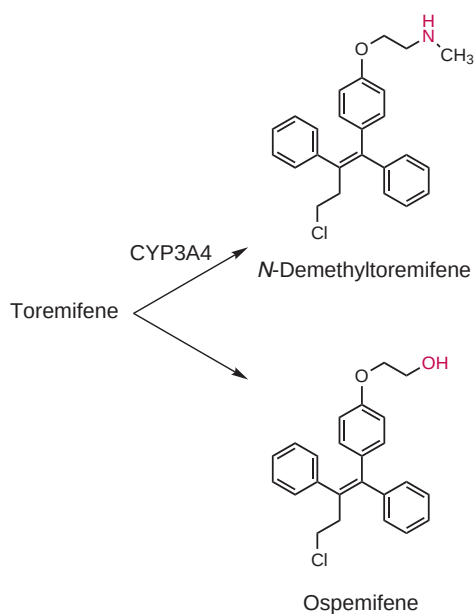
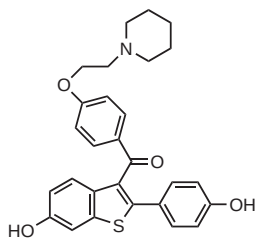


FIGURE 41.21 Metabolism of toremifene.

osteoporosis and postmenopausal vaginal syndrome (which includes vaginal atrophy, dryness, irritation, and dyspareunia) (176). As a metabolite of toremifene, it also is under evaluation as an effective treatment for breast cancer. In addition to its beneficial effects on the bone, ospemifene lowers LDL levels, improves the symptoms of vaginal dryness, and does not induce hot flashes.

RALOXIFENE HYDROCHLORIDE (EVISTA)



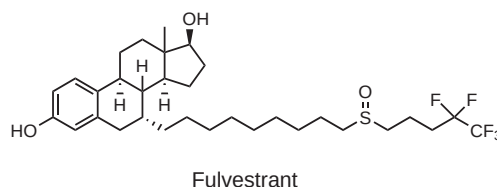
Originally indicated for the prevention and treatment of osteoporosis in postmenopausal women, raloxifene hydrochloride was approved by the FDA in 2007 to reduce the risk of invasive breast cancer in postmenopausal women with osteoporosis and postmenopausal women at high risk for invasive breast cancer (177). Raloxifene hydrochloride is a benzothiophene derivative and acts as ER antagonist in the breast and uterus while maintaining agonist action at the bone. See Chapter 30 for additional information.

Pure Antiestrogens

Despite the fact that 30% to 40% of patients with advanced breast cancer respond to tamoxifen therapy, this response only lasts for 12 to 18 months. Tamoxifen resistance is caused, in part, by its intrinsic agonist activity at the ER. In addition, tamoxifen only inhibits the

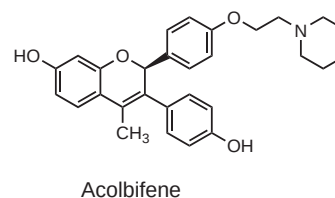
AF-2 activation pathway but does not play a role in AF-1 pathway inhibition. Pure antiestrogens do not possess any estrogenic activity in any species or target tissue and, therefore, offer an additional avenue of treatment when resistance to tamoxifen occurs. In addition, because they are devoid of estrogenic action, these agents cannot be classified as SERMs.

FULVESTRANT (FASLODEX)



Fulvestrant is effective in preventing the growth of tamoxifen-resistant breast cancers (178) both in the laboratory (179,180) and in clinical trials. It is indicated following antiestrogen therapy for the treatment of ER-positive metastatic breast cancer that has continued to progress (181). Fulvestrant is an estradiol analog with a hydrophobic side chain in the 7 α position. Oral bioavailability is poor despite metabolic protection on the end of the hydrophobic side chain. The rationale for this stems from the fact that fulvestrant is virtually insoluble. As a result, it is administered by intramuscular injection once a month. Metabolism of fulvestrant to both active and less active metabolites is similar to that observed with the endogenous steroids (181). Side effects appear to be minimal and include several GI symptoms, headache, and hot flashes. There is no clinical evidence of uterine stimulation or laboratory evidence of stimulation of endometrial carcinoma models (182). Fulvestrant should not be administered to women who are pregnant, who are taking anticoagulants, or who have thrombocytopenia.

ACOLBIFENE



Acolbifene is a nonsteroidal antiestrogen that is presently in phase II clinical studies (183). This fourth-generation antiestrogen is 200-fold more potent than tamoxifen in breast and uterine cancer cells and is effective in patients for whom tamoxifen therapy has failed. The mechanistic rationale for the effectiveness of acolbifene in tamoxifen-resistant tumors is based on the fact that it inhibits both the AF-1 and AF-2 activation pathways and does not possess any intrinsic estrogenic action. Acolbifene enjoys a 60% breast cancer cure rate (184). Structurally, acolbifene resembles raloxifene, with a benzopyran ring

substituted for the benzothiophene ring. An additional benefit observed with acolbifene is its protective effect on bone loss. Adverse effects include nausea and vomiting.

Aromatase Inhibitors

Aromatase inhibitors are considered to be first-line therapy in postmenopausal women with metastatic, hormone receptor-positive breast cancer who do not tolerate tamoxifen. Aromatase is a viable target for drug action in the treatment of breast cancer, because it is the enzyme that catalyzes the conversion of androstenedione and testosterone to estrone and estradiol, respectively (Fig. 41.3, g and h). Inhibition of this enzyme limits estrogen production and effectively starves the hormone-dependent tumor. Studies do not clearly indicate that aromatase inhibitors are better first-line agents than tamoxifen and have not clearly identified a portion of this patient population that is more likely to benefit from this class of agents. It is recommended that postmenopausal women who have completed a 5-year regimen of tamoxifen for the treatment of early-stage, hormone receptor-positive breast cancer consider extending their treatment with an aromatase inhibitor. It has not yet been definitively established what the optimum duration of treatment should be or whether an aromatase inhibitor should supplant or be sequenced after tamoxifen therapy. The aromatase inhibitors are associated with a lower risk of endometrial cancer and thromboembolic events but with higher rates of fractures and myalgia as compared to tamoxifen.

LETROZOLE (FEMARA) Letrozole (Fig. 41.12) is a nonsteroidal aromatase inhibitor that was approved for the treatment of postmenopausal women with hormone receptor-positive or hormone receptor-unknown, locally advanced or metastatic breast cancer. By binding to the heme group of aromatase, letrozole inhibits the aromatase and causes a reduction in plasma estrogen levels. Inhibition of aromatase by letrozole is competitive and highly specific, with no effect on enzymes that are responsible for the production of glucocorticosteroids and mineralocorticosteroids (185).

This agent is significantly more effective than tamoxifen in treating hormone-dependent cancer. Letrozole reduces the rate of tumor progression for 9.4 months compared with 6 months for tamoxifen. Because tamoxifen therapy is limited to only 5 years, patients with breast cancer are left vulnerable to the return of their cancer. There is a 2% to 4% chance each year that the cancer might return. Taking letrozole after this 5-year period decreases the risk of the cancer returning by half. Letrozole also is used in patients who were treated for early forms of breast cancer to prevent disease from recurring. In clinical trials, it was shown that this type of preventive treatment cut the recurrence risk in half.

Letrozole is administered orally (2.5 mg daily) and is rapidly absorbed from the GI tract. Absorption is not affected by food. It is slowly metabolized (CYP3A4 and CYP2A6) to an inactive agent that is subsequently glucuronidated and eliminated renally. Letrozole strongly inhibits CYP2A6 and moderately inhibits CYP2C19. Its

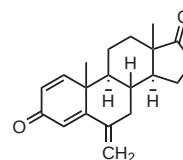
metabolism is induced by tamoxifen (169). Letrozole may increase the risk of osteoporosis and can cause hot flashes and night sweats. An increased incidence of hypercholesterolemia also has been documented.

For those women who are looking to cryopreserve oocytes or embryos prior to initiating chemotherapy for the treatment of breast cancer, letrozole can serve as an ovarian stimulant without increasing their estrogen and gonadotropin exposure. However, there are concerns that letrozole can cause birth defects and miscarriage if used off-label as a fertility drug.

ANASTROZOLE (ARIMIDEX) Anastrozole (Fig. 41.12) is a potent and highly selective, nonsteroidal aromatase inhibitor used in the treatment of advanced breast cancer that is hormone responsive. It is considered to be second-line therapy (after tamoxifen) in the treatment of postmenopausal breast cancer. Anastrozole, a benzyltriazole derivative, competes with the natural substrate for binding to the active site of the aromatase. The mechanism of enzyme inhibition resides in the coordination of the triazole ring with the heme iron atom of the aromatase enzyme complex (186,187). This coordination ultimately prevents aromatization of androgens into estrogens and, therefore, deprives the tumor of estrogen. This effect is reversible. In the presence of anastrozole, estradiol levels are reduced to undetectable levels, with no adverse effects on levels of any other hormone, including cortisol and aldosterone.

Anastrozole is well absorbed orally, with biliary elimination as its primary route (85%) and an elimination half-life of approximately 50 hours (186). Approximately 60% of an oral dose is metabolized in the liver by oxidative *N*-dealkylation, hydroxylation, and glucuronidation to inactive triazole metabolites.

EXEMESTANE (AROMASIN)



Exemestane is a steroid-based, irreversible aromatase inhibitor that is approved for the treatment of estrogen-dependent tumors and postmenopausal breast cancer. It also has been approved overseas as adjuvant therapy after 2 to 3 years of adjuvant tamoxifen treatment of ER-positive, invasive early breast cancer. The structure of exemestane is related to androstenedione. It is important to consider the woman's menstrual status when using this agent, because the synthesis of ovarian aromatase is part of a negative feedback loop such that decreased levels of circulating estrogen will promote increased biosynthesis of aromatase. As a result, exemestane should not be administered to premenopausal women or those who are pregnant. Exemestane binds irreversibly to aromatase and, therefore, is classified as a suicide inhibitor. Because

this agent has high binding affinity and specificity for aromatase, it is able to suppress the activity of this enzyme by 97.7% (anastrozole, 92% to 96%; letrozole, 98%) (188).

Exemestane is administered orally (25 mg daily); however, only 50% of a given dose is absorbed. Absorption is improved if the medication is taken following a high-fat meal. Approximately 90% of a dose is bound to plasma proteins. Exemestane undergoes extensive metabolism by CYP3A4, and dosage adjustments may be necessary if given concomitantly with a CYP3A4 inducer (188). Adverse reactions reported in patients with early breast cancer include hot flashes, arthralgia, and fatigue, whereas hot flashes and nausea were the primary adverse reactions noted in patients with advanced breast cancer.

SUMMARY

Throughout a woman's reproductive life cycle, there are a number of disease states or life circumstances that can be successfully managed with one or more hormone-related therapeutic strategies. Not only is it important to select an appropriate therapeutic regimen for each individual, but it is also essential to determine which route of administration and, therefore, which

formulation is the most therapeutically advantageous. A number of formulations are available for several single and multicomponent products, and maximizing patient compliance should be considered when recommending a particular product. There are no products devoid of adverse effects, so it is important that these be explained to the patient to minimize discontinuation of the recommended therapy.

The alarming results generated by the WHI studies have fueled the development of new estrogen-containing products designed to deliver lower doses of estrogen to minimize systemic exposure and decrease the risk of cardiovascular disease and development of cancer. In addition, a number of alternative therapies are under investigation for the treatment of menopausal symptoms. Agents that modulate serotonergic neurotransmission seem to be especially promising, because many of these therapies are already known commercial entities.

Pharmacologic management of breast cancer continues to make progress with advanced characterization techniques and the use of aromatase inhibitors and SERMs. Future generations of SERMs will need to be completely devoid of estrogenic activity to prevent the development of resistance, as is occurring with tamoxifen.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Nancy Ordonez, PharmD

CC was offered several options for contraception:

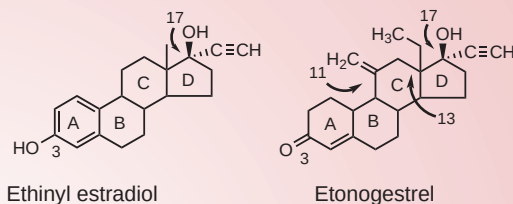
1. Depo-medroxyprogesterone is an injectable progestin-only contraceptive administered every 12 weeks.
2. Transdermal patch containing ethinyl estradiol 0.75 mg and norelgestromin 6 mg. The patch is applied to the abdomen, buttock, upper arm or upper torso and is changed on a weekly basis.
3. Vaginal contraceptive ring contains etonogestrel 0.12 mg/day and ethinyl estradiol 0.015 mg/day and is inserted vaginally for duration of 3 weeks.

She chose to use the vaginal contraceptive ring due to the longer duration of contraception and, with her job as a swimming instructor, she was afraid that the patch may become detached.

Chemical Analysis

Victoria F. Roche and S. William Zito

The vaginal contraceptive ring selected by CC and her provider is a combination product containing the estrogen ethinyl estradiol and the progestin etonogestrel. Both drugs have the 17 α -ethinyl moiety that promotes female hormonal action, protects the essential 17 β -OH group from inactivating oxidation to a 17-keto steroid, and promotes oral bioavailability.



The aromatic ring is unique to estrogenic steroids and is a defining and essential structural feature of the pharmacological class. The phenolic OH group is also essential and forms a critical hydrogen bond with estrogen receptor Arg and Glu residues. The 17 β -OH forms its own essential hydrogen bond with an estrogen receptor His residue. The remainder of the steroid hydrocarbon skeleton has been proposed to serve as a molecular scaffold to keep these two essential H-bonding hydroxyl groups separated by the appropriate distance for optimal estrogen receptor binding. The estrogenic component of a combination oral contraceptive acts by halting the secretion of follicle-stimulating hormone and other pregnancy-supporting gonadotropins.

The antiovaratory action of the C₁₈ progestin etonogestrel depends on the 17 α -ethinyl group; without it, a significant amount of androgenic activity might be observed. In addition, the C₁₃-ethyl group enhances progestational activity over the methyl substituent found on endogenous and most synthetic female sex

SCENARIO: OUTCOME AND ANALYSIS (Continued)

steroids. The unique exocyclic 11-methylene moiety adds unsaturation to ring C, another structural modification that promotes antioviulatory action in C₁₈ steroids. The Δ^4 -3-one A ring is found in all C₂₁ and most C₁₈ progestational steroids, and it has a higher receptor affinity than any other A-ring structural pattern. The 3-keto group is known to serve a H-bonding role and interacts with progesterone receptor Gln and Arg residues. Once bound to the progesterone receptor, etonogestrel prevents pregnancy by inhibiting ovulation and promoting changes in cervical and endometrial environments that are hostile to fertilization and implantation.

The use of the NuvaRing contraceptive vaginal ring would allow CC to avoid both injections and worry about the loss of a patch formulation because of sweat or chlorinated water. Because she has been noncompliant with oral medications, she could “set it and forget it” for 3 weeks, which should decrease the risk of accidental conception. The continuous release of an appropriate dose of contraceptive steroids should also result in fewer adverse effects than are commonly observed with orally administered contraceptive agents. This was a wise therapeutic choice for this patient.

CASE STUDY

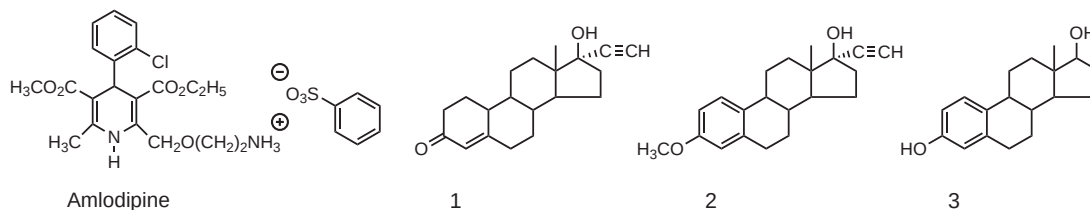
Victoria Roche and S. William Zito

AG is 40-year-old New York advertising executive who has spent her adult life advancing her career in the high-pressure world of designer fashion. She lives life in the fast lane, and she purposefully chose a single life to give herself the freedom to do what she needed to do when she needed to do it, including traveling to exotic spots around the globe to promote her clients' work. Despite her best intentions, AG fell in love with a noted photographer who contracts with her agency, and they will be married next month. Though committed to making their marriage work, neither plans to give up their career. They have agreed that children are not in their future, and AG is now seeking your advice on a suitable oral contraceptive. Having not been in any serious or regular relationships until she met her fiancé, she used a contraceptive cervical cap when needed and expected her partner to use condoms. However, she and her fiancé now want to abandon barrier-associated methods of birth control for more sustained therapeutic options.

AG is a smoker whose moderately elevated blood pressure is kept under control with amlodipine besylate (Norvasc). Both of her parents are alive and doing well, but there is cardiovascular disease in her family history, and she lost her maternal grandfather to an acute myocardial infarction when he was in his early 60s. AG is known to be a rapid 2C9 metabolizer.

After reviewing AG's medical history, you consider the structures of the following steroid-based options before deciding which should be used, either alone or in combination.

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.



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Chapter 42

Nutrition and Obesity

THOMAS L. LEMKE AND DAVID A. WILLIAMS

Nutrients/Drugs Covered in This Chapter*

DRUGS

- Amphetamine
- Benzphetamine
- Diethylpropion
- Ephedrine
- Fenfluramine
- Mazindol
- Olestra
- Orlistat
- Phendimetrazine
- Phenmetrazine
- Phentermine
- Phenylpropanolamine
- Pseudoephedrine
- Sibutramine

MACRONUTRIENTS

- Acesulfame-K
- Aspartame
- *Cyclamate*
- Fatty acids
 - Saturated
 - Unsaturated (polyunsaturated)
- Fiber
- Glycogen
- Neotame
- Rebaudioside A
- Saccharin
- Starch
- Sucralose
- Xylitol

MICRONUTRIENTS

- Sodium chloride
- Tocopherols
- Vitamin A
 - Carotenes
 - Lutein
 - Retinol
- Vitamin C (ascorbic acid)
- Vitamin D
- Vitamin E
- Vitamin K

Abbreviations

ATP, adenosine triphosphate
BMI, body mass index
BMR, basal metabolic rate
CHD, coronary heart disease
CNS, central nervous system
CoA, coenzyme A
DIT, diet-induced thermogenesis
DRI, Dietary Reference Intake
DSHEA, Dietary Supplement Health and Education Act
DV, daily value

EEA, energy of expenditure of activity
FDA, U.S. Food and Drug Administration
GI, gastrointestinal
GRAS, generally recognized as safe
HDL, high-density lipoprotein
LDL, low-density lipoprotein
NLEA, Nutrition Labeling and Educational Act
PKU, phenylketonuria
PUFA, polyunsaturated fatty acids

RAE, retinol activity equivalent
RDA, Recommended Dietary Allowance
RDI, Recommended Daily Intakes
ROS, reactive oxygen species
RS, resistant starch
SSRI, selective serotonin reuptake inhibitor
TEE, total energy expenditure
TFA, *trans* fatty acid

*Nutrients/Drugs listed include those available inside and outside of the United States; the nutrient shown in *italics* is no longer available in the United States.

SCENARIO

Beverly Lukawski, PharmD, and Jennifer Campbell, PharmD

JL, a 25-year-old Hispanic woman, presented to the emergency room after experiencing syncope during her morning workout and subsequent tightness in her chest. JL is 5'6" and 115 kg. The patient has a past medical history of migraines and currently treats them using over-the-counter aspirin with caffeine. The patient states that she also takes an oral contraceptive and recently initiated a weight-loss dietary supplement called Kwik

Loss (ingredients: bitter orange, caffeine, and vitamins B₆, B₁₂, and C) to help her lose weight for an upcoming wedding. The patient presents with a blood pressure of 141/82 mm Hg and a heart rate of 75 beats/min. All other vital signs are within normal limits.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter.)

INTRODUCTION

Every living cell in the human body and the cells of other living organisms require essential ingredients to survive. These essential ingredients are chemical substances, both organic and inorganic, that are not generally produced in the body, or if they are, they are not produced in sufficient quantities to meet the cellular or body needs. These chemical substances are commonly referred to as nutrients, and the utilization of these substances for normal body development and growth is called nutrition. Poor nutrition can result from both insufficient quantities of nutrients and overconsumption of nutrients. Examples of diseases associated with low levels of nutrients are blindness (vitamin A deficiency), osteoporosis (vitamin D, calcium, and phosphate deficiencies), scurvy (vitamin C deficiency), and kwashiorkor (protein deficiency). Excessive intake of nutrients can lead to obesity, which in turn has been associated with cardiovascular diseases, osteoarthritis, sleep apnea, respiratory problems, and type 2 diabetes, to mention only a few. Nutritional balance will always be critical to overall body health and will be emphasized throughout this chapter. Complicating the science of nutrition is the fact that nutrition is affected by age, sex, experience, religion, lifestyle, culture, economics, and a wide variety of other factors. From our standpoint, nutrition is essential for body growth and maintenance and will be presented in this context.

NUTRIENT CLASSIFICATION

The nutrients essential to man and animals can be classified as macronutrients or micronutrients. Those chemicals that must be taken in large quantities are

referred to as macronutrients, whereas those that are required in small quantities are referred to as micronutrients. The macronutrients consist of carbohydrates, fats, proteins, and water, whereas the micronutrients consist of vitamins and minerals. The quantities of macro- and micronutrients change with age. From conception until the mid-teens, the body undergoes considerable growth, and the quantities of nutrients required for good health will be high. From the teens throughout much of the remainder of life, nutrients are expected to maintain the structure of the body. In the elderly, the demand for some nutrients may actually decrease, although this may be more associated with reduced physical activity. During much of our adult life (20 years of age and older), nutrition serves the role of maintaining the body by replacing essential chemicals lost due to normal turnover and supplying energy for running the body. The energy supplied to the body is primarily met through the macronutrients and, more specifically, by carbohydrates and fats. Body composition and weight will generally be relatively stable in an individual with good health, but depending on the eating habits and physical activity, serious medical issues may arise. This chapter is intended to address the issues of energy needs, nutritional sources of energy and balance, the disease state of obesity, and the types of chemicals that can be substituted for high-caloric nutrients. The simplistic view of excessive body weight or the medical condition of obesity occurs when the intake of macronutrients exceeds the energy needs of the body and nutrients are put into storage. Carbohydrates are stored in the form of glycogen; amino acids, the absorbable form of protein, are stored in the form of protein; and fats are stored as fats or lipids (Fig. 42.1).

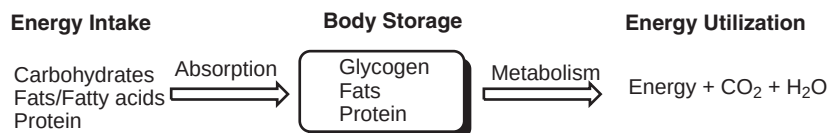


FIGURE 42.1 Diagrammatic representation of macronutrient absorption, storage, and utilization needed to run the body.

CLINICAL SIGNIFICANCE



Obesity, defined by a body mass index (BMI) of $>30 \text{ kg/m}^2$, is a current public health crisis in the United States. In 2009, Colorado was the only state to have less than a 20% prevalence of obesity, whereas in nine states, 30% or more of residents had a BMI of $>30 \text{ kg/m}^2$. Although diet and exercise are promoted as the first-line approach to weight loss for the majority of people, many are not successful in these attempts. As a result, they turn to prescription or over-the-counter medications or dietary supplements to aid in their weight loss. In 2005, more than \$1.6 billion was spent on dietary supplements for weight loss.

While the chemical relationship to therapeutic action of previously available prescription drugs is well known, the same cannot be said for the ingredients in dietary supplements. Products touted as “metabolism boosters” once contained the ingredient ephedra, a source of ephedrine alkaloids, with these nonphenolic phenethylamines showing an affinity for presynaptic norepinephrine reuptake sites. In 2004, ephedra was removed from the market due to safety concerns; a postmarketing meta-analysis detailed more than 16,000 adverse drug events attributed to ephedra-containing ingredients, ranging from liver disease and seizure to myocardial infarction and stroke (1). After the removal of ephedra, “ephedra-free” products were soon on the shelves of pharmacies and supplement stores. Many of these products had

replaced ephedra with bitter orange (*Citrus aurantium*). Bitter orange contains both octopamine and its metabolite synephrine, compounds structurally related to ephedrine. While the full chemical composition of bitter orange products is under debate, synephrine’s stimulant properties are supported by numerous case studies detailing cardiovascular incidents, including angina, syncope, and myocardial infarction, that occurred in people taking ephedra-free dietary supplements (2). Applying well-known adrenergic agonist structure–activity relationships to the synephrine molecule would have allowed health providers to accurately predict many of these effects.

As the market for these products grows, so do the number of products aimed at overweight and obese patients. It is vital that pharmacists understand the chemistry of the ingredients in dietary supplements used for weight loss, the mechanisms of action this chemistry predicts, and the physiologic impact these products may have. This knowledge will enable pharmacists to identify potential drug–herb interactions and potential adverse drug reactions and will help pharmacists adequately counsel patients who rely on their expertise to help understand when to use or avoid these products.

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BODY ENERGY NEEDS

The energy needed to meet body demands is derived from ingested organic macronutrients. Specifically, carbohydrates, fats, and proteins serve as potential sources of energy, and the oxidative breaking of carbon–carbon and carbon–hydrogen bonds converts this potential energy into the body’s energy carrier adenosine triphosphate (ATP) (Fig. 42.2). ATP in turn is used to carry out biochemical processes in the body such as muscle contraction for external work, vascular and heart contraction for blood flow, active transport of molecules and ions, biosynthesis of macromolecules, and a whole host of other cellular functions. The conversion of foods to energy (food energy) is not a very efficient process, and as a result, nearly half of the potential energy of the macronutrients is lost in the form of heat. The unit of measure of potential energy is that of the calorie (Cal or kcal) and is defined as the amount of energy

required to heat 1 kg of water 1°C from 15°C to 16°C . Note, that the terminology *Cal* (spelled with an uppercase “C”) replaces the older unit terminology *cal* (spelled with a lowercase “c”), which is 1/1,000 of a kcal. Nutritional labels report caloric content of foods in the form of calories (Cal). One may also see energy reported in joules or kilojoules (kJ) where 1 kcal equals 4.184 kJ, a standard set by the International System of Units. The hydrolysis of ATP, a phosphoric acid anhydride, releases approximately 12 kcal/mol and results from the exothermic hydrolysis of an anhydride bond and the resulting stabilization of the phosphate anion through resonance stabilization (Fig. 42.3).

The measure of potential energy present in any particular foodstuff can be determined experimentally by complete combustion of a particular foodstuff within an instrument called a calorimeter. The calorimeter consists of a combustion chamber submerged in a water bath. The foodstuff is combusted within the chamber, and the

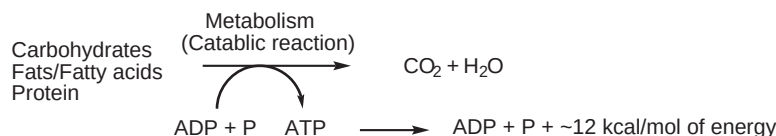


FIGURE 42.2 Conversion of foodstuff to ATP, the body energy currency.

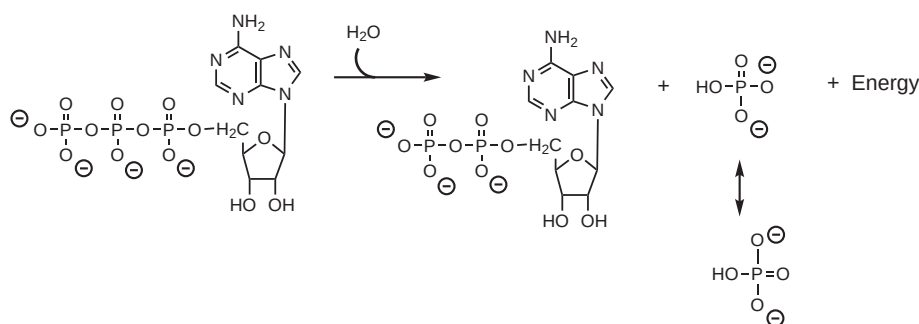


FIGURE 42.3 Energy release reaction involving ATP hydrolysis to adenosine diphosphate (ADP) plus phosphate.

rise in water temperature is measured, which in turn indicates the energy released during combustion. On average, carbohydrates release 4.1 kcal/g, proteins release 4.2 kcal/g, and fats release 9.3 kcal/g of energy (Fig. 42.4). Several attributes can be noted by viewing the chemical structures of the macronutrients. Complete combustion of the carbon atoms in these molecules gives rise to carbon dioxide and water. Nitrogen within the structure of proteins is not oxidized and is not a source of energy. The carbons in carbohydrates are already in a partially oxidized form (C-OH), as are several of the carbons in a protein (C-NH, C=O), whereas a fatty acid has a large number of carbon-carbon and carbon-hydrogen bonds that are available for metabolism. Thus by simple observation of the chemical structures of the macronutrients, one can see that fatty acids would be expected to have a significantly larger potential for generating energy.

Ideally, the amount of energy being generated from the macronutrients should equal the amount of energy needed to run and maintain the body. Actually there are two sources of macronutrient calories; those arising from the external foods that are eaten are called exogenous calories, and those calories that are found stored in the body are called endogenous calories (Fig. 42.5). In an attempt to quantify the number of calories expended on a daily basis, various calculations and formulas have been developed. In most cases, these calculations have either been too complicated to be of practical value or

are only a rough approximation of actual caloric consumption. One simplistic approximate measure of calories needed to maintain the body at a resting state (i.e., energy needed for cellular metabolism plus energy for blood circulation and respiration) is the basal metabolic rate (BMR). Combining the BMR with the energy expenditure of activity (EEA), also known as the total energy expenditure (TEE), and the diet-induced thermogenesis (DIT) can give an estimate of daily energy needs, which if combined with an estimate of calories consumed will help the individual determine their balance between calorie intake and calorie expenditure. BMR calculations can be done with the formula shown in Equation 42.1. As indicated, the BMR is dependent on the gender of the individual and the individual's weight, height, and age.

42.1

$$\begin{aligned} \text{BMR (in Cal/day for women)} &= 655 + (9.6 \times W) + (1.8 \times H) - (4.7 \times A) \\ \text{BMR (in Cal/day for men)} &= 66 + (13.7 \times W) + (5 \times H) - (6.8 \times A) \end{aligned}$$

where W = weight in kilograms of the individual (1 kg = 2.2 lb), H = height in centimeters of the individual (1 in = 2.54 cm), and A = age of the individual.

A generalization can be made for lifestyle activities; for a sedentary lifestyle, the EEA is between 400 and 800 Cal/day; for a moderately active lifestyle, the EEA would be 1,200 to 1,800 Cal/day; and for an individual involved in heavy labor, the EEA would be 1,800 to 4,500 Cal/day. An energy expenditure calculator can be found at www.healthcalc.com/diet/energy-expenditure-advanced. Table 42.1 lists a number of physical activities along with an approximation of the calorie expenditure. A third component of the calculation of daily caloric needs is DIT. The DIT is defined as the energy expenditure above basal fasting levels divided by the energy content of the food ingested over a 24-hour period and is expressed as a percentage (3). The DIT is different for each nutrient, but normally, the DIT for a mixed diet is in the range of 10% of the total amount of energy ingested. The DIT is associated with the energy required for nutrient absorption, steps involved in metabolism, and storage of the absorbed nutrients. Ideally, the sum of the BMR, EEA, and DIT should equal the dietary daily intake so as to protect against weight gain or weight loss.

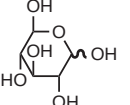
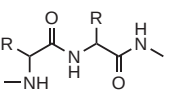
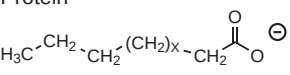
	Approximate energy released
 Carbohydrate	4.1 kcal/g
 Protein	4.2 kcal/g
 Fatty acid	9.3 kcal/g

FIGURE 42.4 Potential energy derived from various foodstuffs.

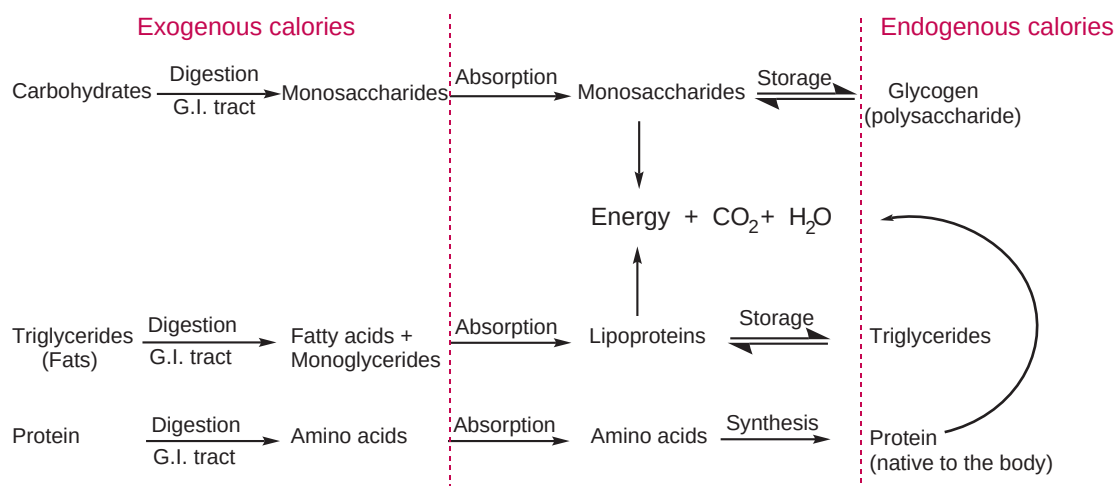


FIGURE 42.5 Digestion, absorption, and storage of macromolecules and conversion to energy.

OVERWEIGHT AND OBESITY

Worldwide, overweight and obesity have become major health issues, and nowhere can this be seen more dramatically than in the United States. In a report issued by Ogden et al. (4) in 2006, it was estimated that obesity in U.S. adults (age 20 and older) has doubled between the years of 1980 and 2002 to 32.2%, whereas overweight in children age 6 to 19 years has tripled to 17.2% (5). Today, obesity is considered to be a chronic disease, which is directly or indirectly related to the other diseases, including coronary heart disease, hypertension, stroke, and type 2 diabetes, and various other conditions, including an increase in death rates. It has been reported that even among overweight individuals, the risk of death for middle-age men and women increases by 20% to 40% compared with people who maintain a desirable weight (6). Some have

suggested that obesity should be considered an epidemic. The significance of overweight and obesity in children can only be expected to create serious problems as they grow into maturity and for public health issues in the future. In 1995, the National Health and Nutrition Examination Survey estimated that 97 million Americans, or 37% of the population, were overweight or obese, whereas the Cleveland Clinic report in 2010 estimated that 66% of the U.S. population was considered overweight or obese (7). By definition, obesity is defined as excessive body fat, whereas overweight is defined as excessive body weight composed of bone, muscle, fat, and water. It is generally agreed that men with more than 25% body fat and women with more than 30% body fat are obese. There are various methods for measuring body fat both directly and indirectly. Underwater weighing, also known as hydrodensitometry, and dual-energy x-ray absorptiometry can give direct measures of body fat content, but these methods tend to be expensive. More commonly used are the BMI calculation and waist circumference measurement, which give a good, but indirect estimate of total body fat. The BMI is a simple calculation useful for adults and is shown in Equation 42.2.

42.2

$$\text{BMI} = \text{Weight}(\text{kg}) / \text{Height}(\text{m})^2$$

$$\text{BMI} = \text{Weight}(\text{lb}) / \text{Height}(\text{in})^2 \times 703$$

Individuals with a BMI of <25 are considered normal, individuals with a BMI of 25 to 29.9 are classified as overweight, and individuals with a BMI ≥ 30 are said to be obese. With a BMI ≥ 40 , the person is considered morbidly obese. It should be noted that there are individual who have a normal amount of body fat, but because of a high muscular content may have an “obese” BMI; this may be seen in athletes, especially football players. The elderly may actually have an underestimate of body fat based on BMI calculations because of the loss of muscle mass. However, in general, the BMI gives a good estimate of total body fat. The relationship of BMI to potential

TABLE 42.1 Energy Expended During Typical Exercises^a

Activity	Calories Used per Hour
Bicycling (6 mph)	240
Bicycling (12 mph)	410
Jogging (7 mph)	920
Running in place	650
Jumping rope	720
Swimming (50 yd/min)	500
Walking (2 mph)	240
Walking (3 mph)	320
Walking (4.5 mph)	440

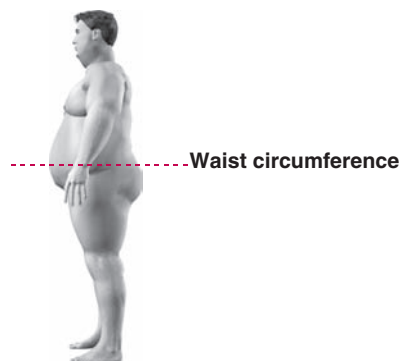
^aFor a more detailed list of activities, see Appendix B in McArdle WD, Katch FI, Katch VL. Sports and Exercise Nutrition, 3rd Ed. Baltimore: Lippincott Williams & Wilkins, 2009.

TABLE 42.2 Relationship of BMI to Health Risk

BMI Category	Health Risk	Risk Adjusted to Comorbidity
<25	Minimal	Low
25 to <27	Low	Moderate
27 to <30	Moderate	High
30 to <35	High	Very high
35 to <40	Very high	Extremely high

health risks is shown in Table 42.2. The presence of excessive abdominal fat can also be used as an indicator of health risk, and this is done through the measuring of waist circumference. The waist circumference gives a good measure of abdominal fat mass. The circumference is measured as a horizontal plane at the iliac crest (Fig. 42.6). A high-risk waist circumference is >40 inches in a man and >35 inches in a woman.

A commonly asked question is: To what extent are overweight and obesity related to genetics and environment? Current research appears to point to environment as a major contributor to overweight and obesity. A combination of increased calorie consumption and reduced physical activity over a long enough period of time leads to excessive body weight. As reported by Putnam et al. (8), between 1985 and 2000, the average consumption of calories increased by 12%, or approximately 300 calories/day. It was estimated that of this 300-calorie increase, fats and sugars accounted for 24% and 23%, respectively. An increase in 300 calories/day without a compensatory increase in physical activity could lead to a weight gain of 22 lbs. It was further reported that in 2000, an increase in fat intake occurred, while protein and carbohydrate intake remained unchanged. In another study, it was reported that between 1977 and 1996, food portions increased both inside and outside the home, resulting in increases in calories of salty snacks (93 kcal), soft drinks (49 kcal), hamburgers (97 kcal), French fries (68 kcal), and Mexican dishes (133 kcal) (9). An increase of 10 kcal/day could lead to an increase in body weight

**FIGURE 42.6** Measurement of waist circumference at iliac crest.

by 1 lb per year. All of this suggests that an imbalance between calorie consumption and physical activity may contribute to the increase in body weight that now threatens the U.S. population.

Food Labels

Food labels are meant to help individuals manage their caloric and nutrient intake, and thus, it is important to understand the terminology used on food labels. Figure 42.7 illustrates an example of a typical food label. The label will indicate a “normal” serving size and the number of calories obtained from this size serving. Following

Nutrition Facts			
Serving size		3/4 c (28 g)	
Servings per container		14	
Amount per serving			
Calories		110	
Calories from fat		9	
% Daily Value*			
Total Fat 1 g		2%	
Saturated fat 0 g		0%	
Trans fat 0 g		0%	
Cholesterol 0 mg		0%	
Sodium 250 mg		10%	
Total Carbohydrate 23 g		8%	
Dietary fiber 1.5 g		6%	
Sugars 10 g			
Protein 3 g			
Vitamin A		25%	
Vitamin C		25%	
Calcium		2%	
Iron		25%	
*Percent Daily Values are based on a 2000 calorie diet. Your daily values may be higher or lower depending on your calorie needs.			
	Calories	2000	2500
Total fat	Less than	65 g	80 g
Sat Fat	Less than	20 g	25 g
Cholesterol	Less than	300 mg	300 mg
Sodium	Less than	2400 mg	2400 mg
Total Carbohydrate		300 g	375 g
Fiber		25 g	30 g
Calories per gram:			
	Fat		9
	Carbohydrates		4
	Protein		4
INGREDIENTS: Corn, whole wheat, sugar, rolled oats, brown sugar, rice, partially hydrogenated vegetable oil (sunflower and/or canola oil), wheat flour, salt, malted barley flour, corn syrup, whey (from milk), malted corn and barley syrup, honey, artificial flavor, annatto extract (color), BHT added to packaging material to preserve product freshness.			
VITAMINS AND MINERALS: Reduced iron, niacinamide, vitamin B6, Vitamin A palmitate zinc oxide (source of zinc), riboflavin (vitamin B2), thiamin mononitrate (vitamin B1), folic acid, vitamin B12, vitamin D.			
EXCHANGE: 1-1/2 starch, exchange calculations based on <i>Exchange Lists for Meal Planning</i> ©1995, American Diabetes Association, Inc. and The American Dietetic			

FIGURE 42.7 Typical food label.

this, the actual content of total fats (including cholesterol), carbohydrates, and protein, as well as the micronutrients, which include minerals (sodium, calcium, and iron) and vitamins (vitamins A and C), is shown. The recommended calorie intake for individuals over the age of 4 years is presently set at 2,000 calories, which serves as the basis for listing the percentage of daily value (DV) for nutrients, which is also shown on the typical label. It should be noted that the U.S. Food and Drug Administration (FDA) has the authority to manage food labels and derives this authority from the Federal Food Drug & Cosmetic Act of 1938, the updated Nutrition Labeling and Educational Act of 1990 (NLEA), and the Dietary Supplement Health and Education Act of 1994 (DSHEA). The DVs are actually based on the highest Recommended Dietary Allowance (RDA) for the age or gender groups listed for RDAs and therefore would be a high-end recommendation. The RDAs are given in gram or milligram quantities, representing the average daily nutrient intake level required to meet the daily needs of an individual for good health, whereas the DV values are a percentage of the total daily requirements. These numbers are reviewed and updated by the Food and Nutrition Board of the Institute of Medicine, in partnership with Health Canada. RDAs, also referred to as the Recommended Daily Intakes (RDIs), have become part of a broader dietary guidelines known as the Dietary Reference Intake (DRI), which is revised and updated continually and serves as the basis for determination of the % DV. Based on the most up-to-date scientific findings, the RDI may be greater than, lesser than, or equal to the older RDA numbers. Presently, the RDI is 300 g for total carbohydrates, 65 g for fats, and 50 g for protein (Table 42.3). The intent of these numbers

and percentages is to help the consumer judge the nutritional value of a particular food by seeing the actual nutrient content in a serving of the food and the percentage of that nutrient toward reaching a 2,000-calorie daily intake for a cross-section of American/Canadian consumers. A further interpretation of the RDA information is that a DV of 20% or greater is considered a “high” quantity of the nutrient, whereas a DV of 5% or less is considered a “low” nutrient quantity. The expectation is that a consumer seeing a “high” DV (e.g., sodium in Fig. 42.7) might wish to choose a product with a lower value and vice versa for foods with a “low” DV (e.g., fiber in Fig. 42.7). It should be noted that carbohydrate and protein do not have a DV percent list, which is because there is no consensus on the quantitative intake of these two nutrients. However because both carbohydrates and proteins are important and essential nutrients in the diet, their RDAs/RDIs are listed to help consumers manage their intake. Nutrients present in less than 500 mg/serving may be listed as 0.0 g, and therefore may be present, but need not be listed. Finally, it was not until 2003 that *trans* fatty acids were required to be reported.

MACRONUTRIENTS

To address the issue of overweight and obesity from the standpoint of overconsumption of calories, one must first identify the sources of calories within our food. The macronutrients (carbohydrates, fats, and proteins) are the source of calories in our diet (Figs. 42.1 through 42.5). In fact, carbohydrates and fats are the usual organic nutrients from which the majority of our nutrients arise.

TABLE 42.3 Dietary Recommendations for Various Nutrients

		Micronutrients			
Macronutrients	RDI	Vitamins	RDI	Minerals	RDI
Carbohydrates	300 g	A	5,000 IU	Sodium	2.3 g
Fiber	25 g	D	400 IU	Potassium	4.7 g
Total fat	65 g	E	30 IU	Calcium	1.0 g
Saturated fat	20 g	K	80 mcg	Iron	18 mg
Protein	50 g	C	60 mg	Phosphorus	1.0 g
		Thiamin	1.2 mg	Iodine	150 mcg
		Riboflavin	1.7 mg	Magnesium	400 mg
		Niacin	20 mg	Zinc	15 mg
		B ₆	2 mg	Selenium	70 mcg
		Folate	20 mg	Copper	2 mg
		B ₁₂	400 mcg	Manganese	2 mg
		Biotin	300 mcg	Chromium	120 mcg
		Pantothenic acid	10 mg	Chloride	3.4 g

Carbohydrates

Carbohydrates are compounds composed of carbon, hydrogen, and oxygen and consist of simple monosaccharides such as glucose, galactose, and fructose; disaccharides such as sucrose (table sugar), maltose, and lactose (milk sugar); and polysaccharides such as starch, glycogen, and cellulose (Fig. 42.8). The absorbable form of a carbohydrate is the monosaccharide; disaccharides and polysaccharides must first be hydrolyzed in the gastrointestinal (GI) tract before absorption.

Monosaccharide Absorption

Absorption of the very hydrophilic monosaccharides requires facilitated absorption/transport, both from the intestine and from body fluids into the cytoplasm of cells. Two families of membrane proteins involved in monosaccharide transport are the glucose transporter proteins (GLUT1 to GLUT5) and the Na⁺-glucose cotransporters (SGLT1 to SGLT3). The GLUTs are 12 membrane-spanning proteins of approximately 500 amino acids found in red blood cells, the blood-brain barrier, liver cells, fat cells, and brush borders of the intestine. The role of these GLUT transporters is to facilitate the absorption of glucose and fructose. The SGLTs are 12 membrane-spanning proteins that are unrelated to the GLUTs, which exhibit high affinity for glucose following binding to sodium ion (Chapter 5). The SGLT1 is primarily expressed in the small intestine, whereas SGLT2 is expressed in the kidney, where they increase the absorption of glucose from the gut and

reabsorb glucose from the kidney tubules, respectively. SGLTs also show affinity for galactose transport. Because monosaccharides account for the major source of energy to human cells, the role of GLUTs and SGLTs is quite important.

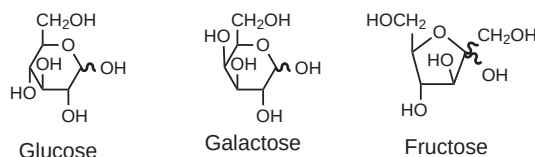
Disaccharide Metabolism

Disaccharides require hydrolysis to monosaccharides; for example, sucrose is hydrolyzed to glucose and fructose prior to absorption. Various enzymes present in the brush border of the intestine are involved in the hydrolytic process (i.e., sucrase-isomaltase, lactase-phlorizin hydrolase, maltase-glucoamylase, and trehalase).

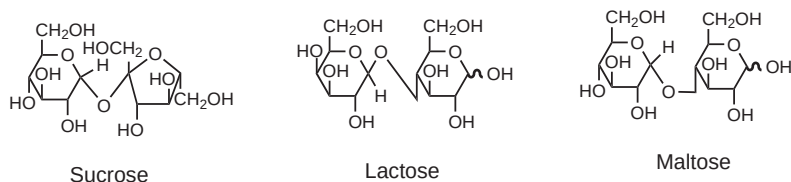
Starch Metabolism

Starch is a polysaccharide of glucose that exists as a straight chain polymer called amylose with three or more glucose monomers linked via α -1,4-linkages and amylopectin, a highly branched polymer consisting of α -1,4-linkages together with α -1,6-linkages similar to what is seen in glycogen (Fig. 42.8). Amylose exists in a helix configuration. Starch is a major constituent of plants with various ratios of amylose to amylopectin (for example, potato is 20% amylose and 80% amylopectin; rice is 18.5% amylose and 81.5% amylopectin). The hydrolytic metabolism of starch begins in the oral cavity where α -amylase, which attacks the α -1,4-linkages, is secreted by the salivary glands and later by α -amylase secreted by the pancreas in the duodenum. Glucoamylase in the small intestine is also involved

Monosaccharides:



Disaccharides:



Polysaccharides:

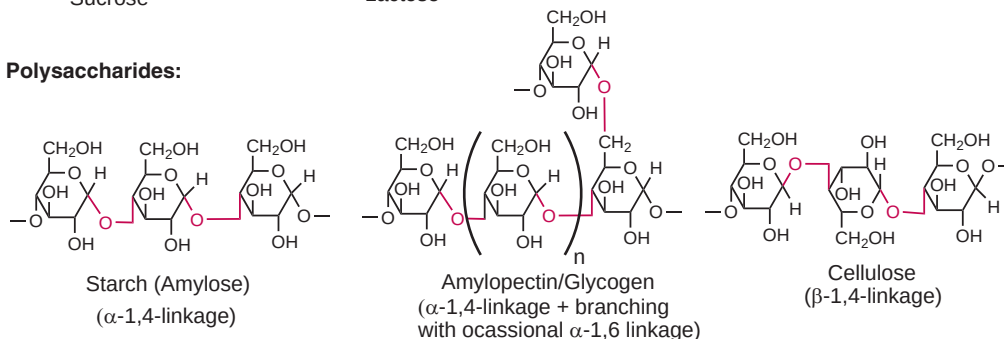


FIGURE 42.8 Structures of common carbohydrates.

in starch hydrolysis. The result of the amylase metabolism is that maltose and maltotriose (a three α -1,4 glucose unit molecule) are formed, at which point final hydrolysis occurs via the previously mentioned brush border disaccharidases, leading to formation of absorbable glucose.

Glycogen Synthesis and Metabolism

Glycogen is the human storage form for glucose, with storage occurring in the liver and muscle tissue. Once glucose is absorbed and phosphorylated to glucose-1-phosphate, it is then incorporated into glycogen through the action of glycogen synthase and glycogen branching enzyme. The structure of glycogen is similar to starch, with both α -1,4-links and α -1,6-links (Fig. 42.8). The process of glycogen formation differs from glycogen degradation. Glycogen hydrolysis occurs when the body requires energy, which involves a process of phosphorylation, transferase, and a debranching enzyme, leading to the release of glucose into the bloodstream.

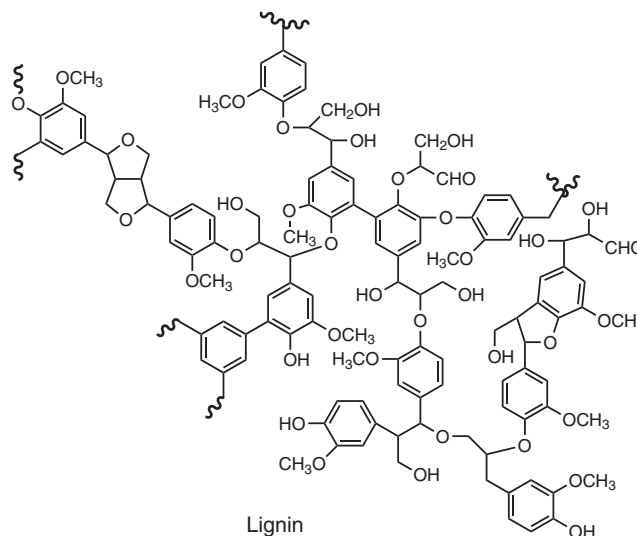
Cellulose

Cellulose is a major polysaccharide found in plants (Fig. 42.8). It is a linear polymer of glucose but differs from starch by the fact that the linkage between the glucose monomeric units has the β -1,4-linkage, which increases the stability of the polymer through internal hydrogen bonding to produce a strong linear configuration. This configuration prevents hydrolytic metabolism by human enzymes that are specific for α -1,4-linkages. The β -1,4-linkage can be hydrolyzed by cellulase found in some bacteria, protozoa, and fungi and in bovine animals.

Fiber

Fiber or, more specifically, dietary fiber is made up of nondigestible oligosaccharides usually of plant origin, such as the polysaccharides cellulose and pectin (polysaccharide containing 1,4- α -D-galactosyluronic acid monomeric units), or complex material found in wood and cell walls of plants such as lignin (Table 42.4). There is no

FDA definition of what a dietary fiber actually is. Fibers may be further classified as soluble fibers (water soluble) such as pectin or insoluble fibers such as cellulose, hemicellulose, and lignin.



As the name would imply, soluble fibers absorb and dissolve in water forming gels that are prone to fermentation by bacteria that are located in the colon, generating short-chain fatty acids, such as butyric acid, through the benefit of epithelial cells in the colon. Insoluble fibers create bulk, which is suggested to be beneficial via increased stomach transit time and a reduction of intestinal transit time. This is thought to decrease the urge to eat and cause a reduction in the ability for chemical absorption to occur from the intestine. Although evidence supporting the health benefits of dietary fibers is suggestive at best, there is general agreement that fiber in the diet can lower the incidence of obesity (creates a feeling of satiety) (10), reduce insulin resistance and the risk of type 2 diabetes, reduce the risk of heart disease (decrease cholesterol absorption from the intestine), and reduce the risk of colon cancer (decrease absorption of potential carcinogens from the gut). Insoluble fibers also act as bulk laxatives. The average American diet contains 12 to 15 g of fiber compared to 40 to 150 g of fiber in the diets of Africans and Indians. Recommendations for fiber content in the diet are presently 38 g/day for men under the age of 50, 30 g/day for men over the age of 50, 25 g/day for women under the age of 50, and 20 g/day for women over the age of 50. Recently, food manufacturers have been studying starches referred to as resistant starch (RS). Resistant starch is expected to resist digestion by amylase and amylopectinase and, therefore, would be expected to have the same health benefits claimed for the dietary fibers. Various classes of RS, some of which are natural while others are manufactured, are being investigated (informally classified as RS1 through RS5). Changes in crystalline structure of the RS, increases in branching and cross-linking within the starch, and

TABLE 42.4 Classes of Dietary Fibers

Oligosaccharides	Source
Cellulose	Bran, legumes, peas, vegetables, cabbage, seeds, and apple skin
Hemicellulose	Bran and whole grains
Polyfructose	Oligofructans and inulin
Gums	Oatmeal, barley, and legumes
Pectin	Apples, strawberries, and citrus fruits
Resistant starches	Ripe bananas and potatoes
Nonsaccharide lignin	Root vegetables, wheat, fruits, strawberries

starch–lipid complexes cause the starch to become more resistant to hydrolysis (11).

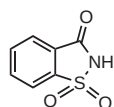
Although nondigestible fibers have little or no caloric value, the soluble fiber that is fermented in the colon can be a source of energy and is given a caloric value of 4 kcal/g.

Properties of Sugars

Sugar is defined as a sweet-tasting crystalline carbohydrate, and in most cases, when one refers to sugar, the person is referring to sucrose (table sugar), although glucose, galactose, lactose, and fructose are sugars having similar properties. Normally, carbohydrates make up 40% to 70% of the human daily caloric intake, and the majority of the carbohydrates come in the form of starch, but sizable quantities of sucrose and lactose are also commonly found in the diet. It is estimated that the average American adult consumes 64 lb of sucrose/year, while children may consume as much as 140 lb/year. Approximately one-third of sugar is ingested from the table sugar bowl, and two-thirds is found in commercially processed food, mostly as high-fructose corn syrup. The consumption of sugar has both a physiologic and a psychological basis. The perception of sweetness is detected by G protein–coupled receptors located in the taste buds on the tongue. Sweetness is one of the five basic tastes that the human body has evolved to detect, and humans crave sweet materials as a way of supplying the body with sufficient energy to meet the daily needs. However, children have been taught to crave sweet foods since sweets are commonly used to bribe children. As a result, we crave sweet foods, namely carbohydrates, in excess to what is actually needed for good health. Excessive intake of sucrose or high-fructose corn syrup is directly related to the development of tooth caries and indirectly related to the development of overweight and obesity, which in turn is related to the onset of type 2 diabetes. Although it might be assumed that a good approach to the health issues associated with excessive consumption of carbohydrates, sugars in particular, would be to reduce the intake of these macronutrients, there seems to be an ingrained pattern that replacement is more acceptable than restraint. Thus, because of the mounting concern regarding overweight and obesity and the need to reduce calorie intake and yet the desire to meet the craving for sweet substances, nonnutritive sweeteners have gained universal acceptance.

Nonnutritive Sweeteners

Saccharin (Sweet’N Low)



Saccharin

The history of saccharin began with its synthesis in 1878, and over the past 130 plus years, this food additive has

been part of the human food scene. Saccharin’s history has included a period when it was ban from the market as an adulterant only to be returned as a sugar substitute. It has had limited use directed toward diabetics, carried a warning as a possible carcinogen, and is now openly available for consumer use and a common ingredient in a host of food products. Saccharin’s colorful history has recently been presented in a book titled *Empty Pleasures: The Story of Artificial Sweeteners from Saccharin to Splenda* (12).

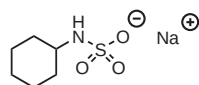
The properties of saccharin include a sweetness 300-fold that of sucrose with zero calories (Table 42.5). Saccharin does have a bitter aftertaste, a fact that makes saccharin less than the ideal nonnutritive sweetener, and instability to prolonged heating, thus limiting its use to nonbaked products. The bitter aftertaste has been masked by the addition of other sweeteners. The biggest health-related problem associated with saccharin was a scare resulting from the development of bladder cancer in male rats reported in 1977 and titled the “Canadian Rat Study.” This study resulted in the banning of saccharin by the FDA, but the passage of the Saccharin Study and Labeling Act in 1977 by the U.S. Congress allowed saccharin to remain on the market with the required label that saccharin-containing products were potentially carcinogenic. (*This product contains saccharin, which has been determined to cause cancer in laboratory animals.*) Saccharin-containing products retained this label until December

TABLE 42.5 Comparison of the Sweetness of Nutritive and Nonnutritive Sweeteners

	Sweetness	Calories per 100 g
Nutritive sweetener		
Sucrose	100%	All mono- and disaccharides ~370
Glucose	60%–70%	
Fructose	110%–180%	
Maltose	33%–50%	
Galactose	15%–40%	
Xylitol	100%	~192
Nonnutritive sweetener		
Saccharin	300× sucrose	0
Sodium cyclamate	30× sucrose	0
Aspartame	200× sucrose	~0.05
Neotame	7,000× sucrose	0
Acesulfame-K	200× sucrose	0
Sucralose	600× sucrose	0
Rebaudioside A	200× sucrose	0

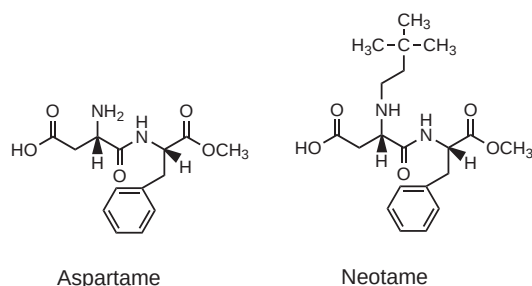
2000 when the warning label was removed following presentation of evidence that male rats are unique in that a combination of high bladder pH, high calcium phosphate concentrations, and a specific protein that favors crystal formation occurs in the presence of saccharin. The formation of microcrystals in the bladder of male rats causes damage to the bladder wall, which over an extended period of time led to tumor formation. These conditions do not exist in humans, and therefore, a threat of saccharin-induced tumors is not possible (13).

Cyclamate (Sucaryl)



Cyclamates were first synthesized in 1937 and reached the U.S. market in the 1950s. This artificial sweetener has several advantages over saccharin in that it does not possess a bitter aftertaste and is stable to heat. The compound is 30 to 50 times sweeter than sucrose with zero caloric value (Table 42.5). During the 1960s, the cyclamates became very popular as nonnutritive sweeteners when directly added to foods, used in baked products, and added to soft drinks and canned fruits. In 1958, cyclamate was added to the generally recognized as safe (GRAS) list, but in 1969, the chemical was banned by the FDA for use in the United States due to what was reported to be the development of bladder cancer in rats possibly due to metabolism of cyclamate to cyclohexylamine. While studies to repeat these results were claimed by the manufacturer to be irreproducible, the chemical has never returned to the U.S. market, although the chemical is still sold in many countries throughout the world.

Aspartame (Equal, NutraSweet) and Neotame



In 1965, a chemist working at G.D. Searle & Company discovered that a compound composed of the methyl ester of the dipeptide of two natural amino acids, L-phenylalanine and L-aspartic acid, had an outstanding sweet taste. This dipeptide was later named aspartame. Coming at a time when both saccharin and cyclamate

were facing intense scrutiny as synthetic, nonnutritive sweeteners, aspartame was seen as a potentially significant breakthrough in the nonnutritive sweetener market. The properties of aspartame included a sweetness 200-fold sucrose with no bitter aftertaste. A single serving of aspartame, equivalent to an equally sweet 180-calorie serving of sucrose, contains only 0.1 calories (Table 42.5). While still not stable to prolonged heating, aspartame's other major attraction was the fact that it was composed of L-phenylalanine and L-aspartic acid, two naturally occurring amino acids. At a time when the consumer was (and still is) concerned about synthetic, unnatural chemicals, aspartame offered a marketing windfall. Present in the form of a dipeptide methyl ester, aspartame was first approved for marketing in 1974, but concerns about tumors, brain damage, and testing procedures delayed its introduction to the market until July 1981. By 1985, aspartame was consumed to the extent of 800 million pounds/year, and 2 years later, it was estimated that aspartame appeared in more than 1,200 products.

Aspartame, being a dipeptide, is prone to hydrolytic metabolism, either in the intestine or mucus membrane of the intestinal lining, to produce phenylalanine, aspartic acid, and methanol (Fig. 42.9). This metabolism in turn raised the question of the safety of the products formed. Methanol, or wood alcohol, is a potential toxin when absorbed into the bloodstream, where it can be converted into formaldehyde and then to formic acid, which could potentially lead to systemic metabolic acidosis and blindness (14). It was found that the quantity

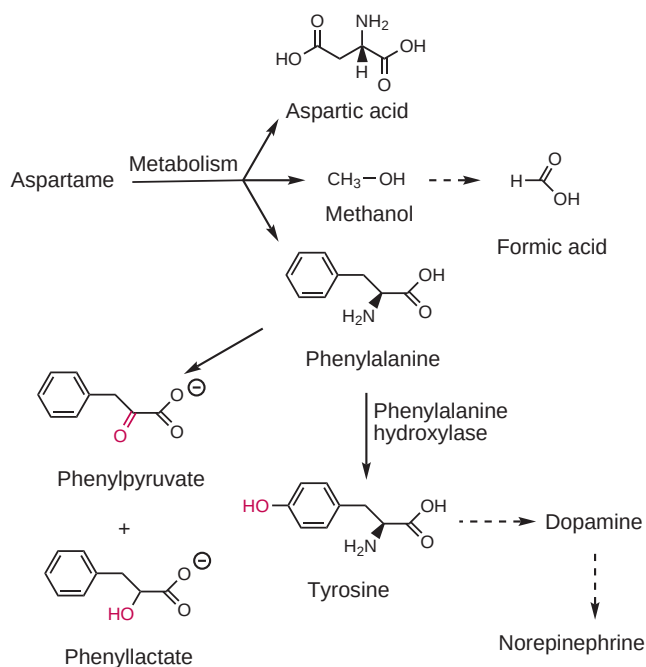


FIGURE 42.9 Metabolism of aspartame and its metabolites.

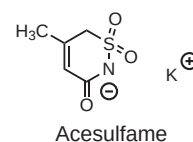
of formic acid detected in the urine (studies could not detect formic acid in the blood after aspartame administration) following large doses of aspartame generated less formic acid from the aspartame than that found after consuming some fruit juices.

The second concern for aspartame is the effect of phenylalanine on individuals who suffer from phenylketonuria (PKU). PKU is a genetic condition in which children are born with a deficiency of phenylalanine hydroxylase, an enzyme that converts phenylalanine into tyrosine. A lack of this enzyme leads to higher than normal levels of phenylalanine in the blood, which leads to formation of catabolic products such as phenylpyruvate and phenyllactate. These products can result in a reduced level of tyrosine and possibly the neurotransmitters formed from tyrosine (norepinephrine, and dopamine). If not diagnosed within a few days of birth, PKU can lead to central nervous system (CNS) damage and mental retardation. The exact cause of the mental retardation is not known, but it is assumed that one or all of the abnormal conditions cited above may have a role in PKU. Although phenylalanine is still essential in PKU patients, a diet with low phenylalanine levels is important in the prevention of the serious outcomes of excessive phenylalanine in PKU patients. As a result, the consumption of aspartame by women during pregnancy who might give birth to a PKU infant and administering aspartame-containing foods to a child diagnosed with PKU have raised concerns. Studies demonstrated that abusive doses of aspartame (200 mg/kg in an adult or 100 mg/kg in a child) or successive doses of aspartame (three 10 mg/kg doses in a child) did not lead to phenylalanine levels equal to or above those approved for adults or children with PKU (14). These studies suggested that aspartame-containing foods and beverages are unlikely to have adverse effects on patients with PKU, although there is no reason for infants with or without PKU to be using nonnutritive sweeteners. Aspartame-containing foods contain a warning indicating its presence and its potential for harm in patients with PKU.

In 2002, an analog of aspartame, neotame was approved for marketing in the United States. This compound differs from aspartame by the addition of a dimethylbutyl substituent on the nitrogen of aspartic acid. The result of this substitution is that sweetness is 30-fold greater than aspartame and from 7,000- to 13,000-fold greater than sucrose. Metabolism of neotame results in formation of methanol, but significantly less phenylalanine and *N*-dimethylbutylaspartic acid. The nitrogen substitution also decreases the hydrolysis of the dipeptide, and only 20% to 30% of the neotame is absorbed, presumably as the individual amino acids (15). Since the sweetness of neotame is so much higher than aspartame, a much lower dose would be required to meet the desired sweetness of a neotame-containing food. As a result, it is estimated that the amount of

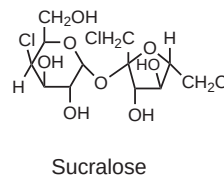
methanol absorbed after an appropriate dose of neotame is approximately 1.3 mg/L, whereas the methanol content from some juices is calculated to be approximately 140 mg/L. The amount of phenylalanine produced from neotame is estimated to be 2.6 mg/day in an adult and 1.5 mg/day in a child, whereas the restricted diet content of phenylalanine for a child with PKU is set at 0.4 to 0.6 g/day, and the average adult consumes 2.5 to 10 g/day of phenylalanine. The amount of phenylalanine obtained from a dose of neotame would have a negligible effect on a patient with PKU. Presently, there do not appear to be any food products that contain neotame.

Acesulfame-K (Sunette)

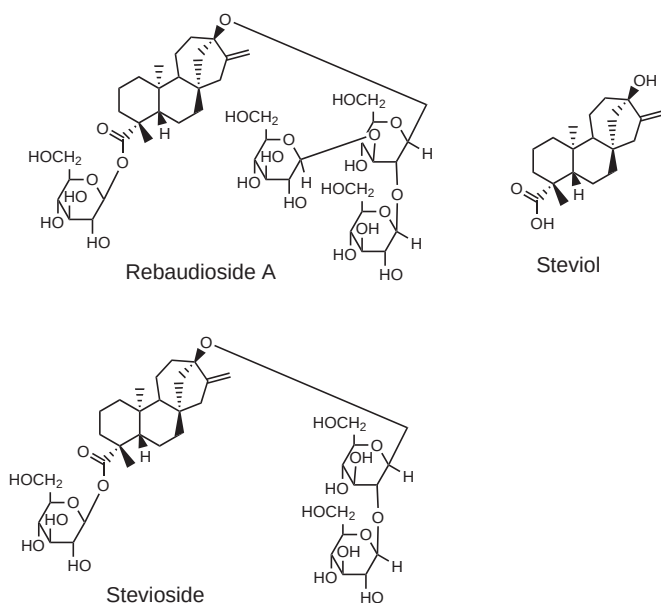


Discovered in 1967, acesulfame-K is a nonnutritive sweetener that is approximately 200 times sweeter than sucrose and heat stable and therefore can be used in cooking and baking (Table 42.5). The compound is not metabolized and shows structural similarity to saccharin. Acesulfame-K is commonly used in combination with other sweeteners due to its bitter aftertaste, a property it has in common with saccharin (both chemicals affect the same bitter receptor) (16). The chemical was approved for use in the United States in 1988 in dry food, and by 2003, it was approved as a general purpose sweetener, allowing for its added use in carbonated and noncarbonated beverages.

Sucralose (Splenda)



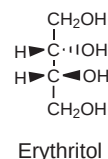
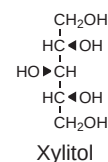
Sucralose is a synthetic derivative of sucrose in which three hydroxyl groups have been replaced with chlorine atoms. As a result, the compound is not metabolized by the body but retains the sweet taste of sucrose, and in fact, the sweetness is increased by approximate 600-fold. The chloride substitution, in addition to preventing the metabolism of Sucralose, increases stability of the chemical to heat. Sucralose can be used in baking and cooking (15). In 1999, sucralose was approved as a general purpose sweetener in the United States. Its general purpose use is indicative of its safety as judged by the FDA.

Rebaudioside A (Truvia, Rebiana, or Reb-A)

For many centuries, the leaves of the stevia plant (*Stevia rebaudiana* Bertoni) have been used for their sweet taste (15,17), but not until the last decade has commercial interest in this plant resulted in a marketed product derived from the stevia plant. Two major glycosides possess much of the sweet taste; these are stevioside and rebaudioside A, which has been developed and introduced onto the market as a nonnutritive sweetener. Rebaudioside A, known by the trade name Rebiana or Reb-A, is a diterpene glycoside that possesses a sweetness approximately 200-fold greater than sucrose (Table 42.5) and has zero calories. As a glycoside, it is prone to acid-catalyzed hydrolysis. At low pHs, stevioside can undergo hydrolysis by intestinal bacterial flora, leading to the formation of steviol, the aglycone component of the rebiosides. Human metabolic studies on stevioside have shown that only trace amounts of stevioside can be detected in blood and no steviol is detected. Because steviol has been shown to exhibit positive genetic toxicology as well as the ability to induce chromosome breakage in bacteria, stevioside, stevia extracts, and, to some extent, rebaudioside A have been studied in depth for their safety (17,18). These studies have concluded that stevioside and rebaudioside A are not genotoxic or carcinogenic in vitro or in vivo and that the only cell-damaging effects of steviol occurred in in vitro studies at excessive concentrations.

In 2008, the FDA approved inclusion of rebaudioside A in the GRAS list as well as its use in foods and beverages as a nonnutritive sweetener. Prior to these approvals, rebaudioside A was available in the United States as a dietary supplement. In the single-component product Truvia, erythritol (a four-carbon polyhydroxy sugar alcohol) is added as a sweetener to mask the aftertaste of

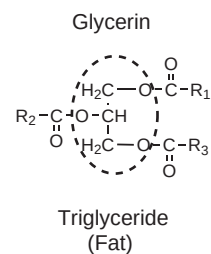
licorice, which appears to be associated with rebaudioside A. Erythritol itself has a caloric content of 0.2 kcal/g.

**Xylitol**

Xylitol is a polyol (a five-carbon polyhydroxy sugar alcohol) with structural similarity to glucose and erythritol. It has been used since the 1960s as a sugar substitute with a sweetness comparable to sucrose but with 40% fewer calories (Table 42.5). Xylitol is commonly found in chewing gums because it does not promote the development of dental caries or plaque formation and yet satisfies the need for sweetness. This property is due to the fact that the polyols are not metabolized to organic acids in the oral cavity. Xylitol can be added to the gum, and the gum can still be labeled as sugar “free” (15). Unfortunately, if consumed in large quantities, xylitol can have a laxative effect due to its poor absorption and water osmotic effect. Xylitol can be consumed by diabetics and will not affect blood sugar levels. Xylitol is found commonly in fruits such as raspberries and plums and can be prepared from the hydrogenation of xylose, which is found in corn cobs and wood pulp.

Fats and Fatty Acids**Chemistry of Fats and Fatty Acids**

By definition, a fat is a neutral molecule composed of three fatty acids attached to the alcohol glycerol, thus referred to as triglycerides, and represents a storage and transport form for fatty acids.



Fatty acids themselves play multiple biologic roles, including the following: They are components of the cell wall, usually in the form of phospholipids, glycolipids, and fatty acid acetylated proteins; a high-energy source of calories; and hormone and intracellular messengers.

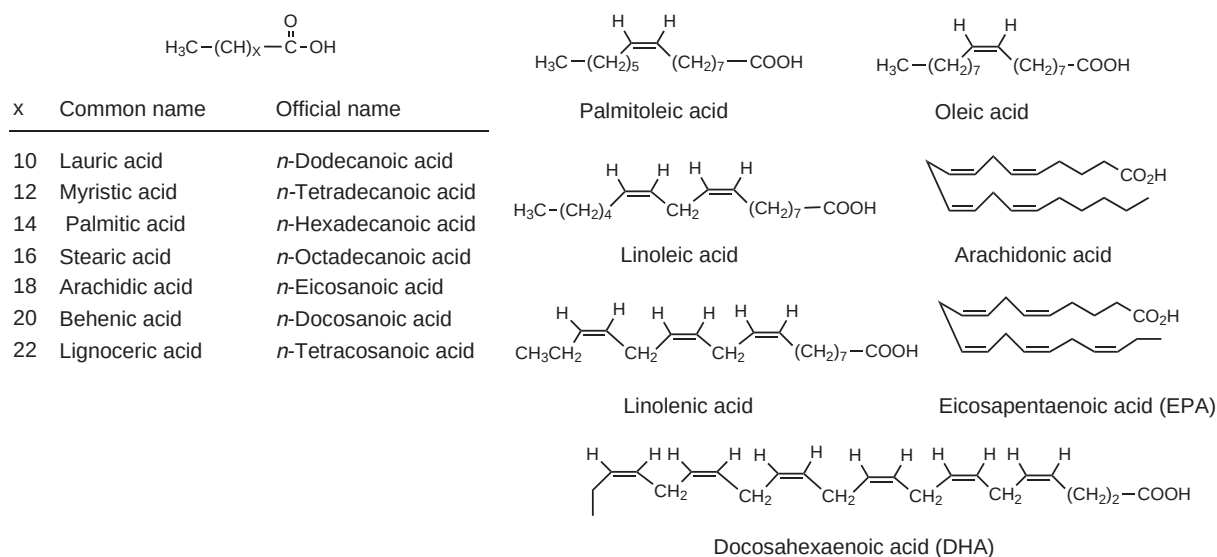


FIGURE 42.10 Structures of unsaturated, monounsaturated, and polyunsaturated fatty acids.

The fatty acids are classified based on the presence or absence of double bonds. The natural fatty acids with no double bonds are referred to as saturated fatty acids; they are totally saturated with hydrogen. The naturally occurring saturated fatty acids contain 12 to 24 carbon atoms (Fig. 42.10) and have an even number of carbon atoms, which results from the fact that they are biosynthesized from the two-carbon acetate unit. A second group of fatty acids is the monounsaturated fatty acids, consisting of palmitoleic acid (9Z-hexadecenoic acid) and oleic acid (9Z-octadecenoic acid). These acids have a single double bond, and it should be noted that the stereochemistry of these and other naturally occurring unsaturated fatty acids contain the *cis* or *Z* configuration. A third class of fatty acids are the polyunsaturated fatty acids (PUFA) and consist of linoleic and linolenic acid (9Z,12Z-octadecadienoic acid and 9Z,12Z,15Z-octadecatrienoic acid, respectively). Additional fatty acids of nutritional significance are arachidonic acid (C20:4 *cis* double bonds), eicosapentaenoic acid (EPA) (C20:5 *cis* double bonds), and docosahexaenoic acid (C22 with 6 *cis* double bonds), which are shown in Figure 42.10. A commonly used nomenclature refers to fatty acids with a double bond three carbons from the end of the fatty acid chain as the ω -3 fatty acids. Linolenic acid, eicosapentaenoic acid, and docosahexaenoic acid constitute the ω -3 fatty acids.

Physical-Chemical Properties

A distinguishing physical-chemical property of the various classes of fatty acids is that as the degree of unsaturation increases, the melting point decreases. The saturated fatty acids tend to be low-melting solids, whereas the unsaturated fatty acids and PUFAs are liquids (Table 42.6). The saturated fatty acids tend to be found in higher concentrations in animal fats, whereas the unsaturated and

polyunsaturated fatty acids are found in high concentrations in vegetable oils (Table 42.7). It is interesting that as the consumer demand moves away from saturated fats and toward unsaturated fats that seed manufacturers are turning to genetically modified plant seeds to produce plants that produce increased levels of monounsaturated fatty acids and reduced levels of saturated and polyunsaturated fatty acids (19).

TABLE 42.6 Melting Points of Saturated and Unsaturated Fatty Acids

$\text{H}_3\text{C}-(\text{CH})_x-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ I $\text{H}_3\text{C}-(\text{CH}_2)_Y-\left(\text{CH}=\text{CH}-\text{CH}_2\right)_x(\text{CH}_2)_W-\text{COOH}$ II				
Common name	x	mp (°C)		
Saturated (I)				
Lauric acid	10	44		
Myristic acid	12	58		
Palmitic acid	14	63		
Stearic acid	16	70		
Unsaturated (II)	w	x	y	
Palmitoleic acid	6	1	5	-1
Oleic acid	6	1	7	4
Linoleic acid	6	2	4	-5
Linolenic acid	6	3	1	-12
Arachidonic acid	2	5	1	-49.5

TABLE 42.7 Approximate Percentage of Fatty Acids in Various Food

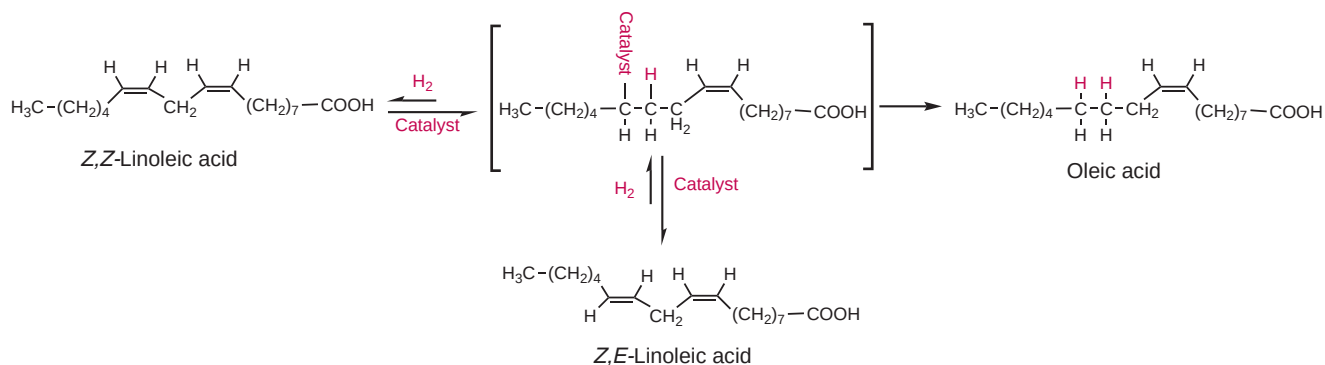
Food	Saturated Fatty Acids (%)	Monounsaturated Fatty Acids (%)	Polyunsaturated Fatty Acids (%)
Butter fat	60	36	4
Pork fat	59	39	2
Beef	53	44	2
Chicken fat	39	44	21
Margarines ^a	20	40	40
Salmon oil	24	39	36
Corn oil	15	31	53
Soybean oil	14	24	53
Olive oil	14	75	11
Sunflower	10	22	66
Canola	7	61	22

^aFatty acid content shows considerable variations between percentage of saturated fatty acids (palmitic acid and stearic acid), monounsaturated fatty acid (oleic acid), and polyunsaturated fatty acid (linoleic acid) as well as *cis/trans* ratio in various brand name margarines.

Nazir DJ, Moorecroft BJ, Mishkel MA. Fatty acid composition of margarines. *Am J Clin Nutr* 1976;29:331–339.

In part, this relates to a second physical-chemical property of the fatty acids. The fats containing saturated fatty acids generally have a more stable shelf-life and are stable at elevated temperatures to air oxidation. The PUFAs within the fat are susceptible to oxidation reactions, especially at elevated temperature (baking and frying temperatures). The oxidative reactions lead to rancidity, darkening in color, increased viscosity, and polymer formation. A commonly used method of reducing the concentration of the PUFA is through partial hydrogenation (leading to the hydrogenated oils). Catalytic hydrogenation involves a reversible

addition of hydrogen and catalyst to the double bond followed by the irreversible addition of a second hydrogen to the initial double bond (Fig. 42.11). The reversibility of the initial step in the hydrogenation leads to loss of the hydrogen from the saturated carbon and formation of both the initial alkene and a new alkene with the more stable *trans* or *E* stereochemistry and, thus, formation of a *trans* fatty acid (TFA) impurity. The quantity of TFAs formed by catalytic hydrogenation can be reduced by changes in the conditions used for the hydrogenation process (20). The 2006 report of a Canadian study, “Transforming the Food Supply: Report of the Trans Fat Task Force,” submitted to the Minister of Health reports that of the fats present in food, the TFA levels may be as high as 24% to 33% for bakery products, 20% to 40% for margarines, and as high as 40% for snack foods, nachos, and nachos cheese sauces (21). The concern is that TFA consumption increases the levels of low-density lipoprotein (LDL) cholesterol, reduces the level of high-density lipoprotein (HDL) cholesterol, and increases the ratio of total cholesterol to HDL cholesterol, which are measures that are associated with increased risk of coronary heart disease (CHD) (22). Other endogenous changes have been associated with TFA consumption including changes in triglyceride levels, promotion of inflammatory reactions, and endothelial dysfunction. Generally, *trans* fats composed of one or more TFAs attached to the glycerin backbone are not normally present in natural foods with the exception of small amounts in meats coming from cows, sheep, and other ruminants in which bacteria within the stomach of ruminants produce the TFAs. It is estimated that this accounts for approximately 0.5% of the total caloric intake. But commercially deep-fried foods, bakery products, packaged snack foods, margarines, and crackers prepared with *trans* fat-containing oils may increase the amount of *trans* fats in our diet, leading to 2% to 3% of the total caloric intake being TFAs. Thus, considerable concern has been generated in the press, health professionals, and the public about the long-term effects of *trans* fats. This has led to changes in the manufacture of cooking

**FIGURE 42.11** Catalytic hydrogenation of a polyunsaturated fatty acid to a monounsaturated fatty acid.

oils and the use of oils containing *trans* fats, including in some cases, legislation outlawing the use of *trans* fat-containing oils. It has been suggested that a 2% increase in energy intake from TFAs could result in a 23% increase in the incidence of CHD (20). Over the past several years, the presence of TFAs in food products has declined as indicated by the labeling of foods as having zero *trans* fats, although such a label does not guarantee zero grams per serving. The FDA allows a report of zero grams per serving if the food contains <500 mg/serving of *trans* fats.

Fat and Fatty Acid Absorption

The major nutritional value of fats comes from the fatty acids, which upon metabolism are a high-density source of energy. In order for this to occur, the fat must be absorbed, transported via the bloodstream either to the cells in tissue where the energy is needed or to storage sites in adipose cells for future energy needs, and finally moved into a cell's mitochondria for metabolism, with the release of energy in the form of ATP units. This process is complicated by the fact that fats themselves cannot be transported across cell membranes due to their low hydrophilicity. Rather, a sequence of hydrolysis and ester formation occurs at various stages in the absorption-transport-storage-utilization process.

Consumed fats travel to the small intestine where they are emulsified with the aid of bile acids and are then hydrolyzed through the action of pancreatic lipase. The resulting fatty acids and glycerol can now be absorbed into the intestinal mucosa. At this point, the fat is reformed and, in the presence of apoproteins and cholesterol, is converted to chylomicrons (large lipoprotein particles consisting of 85% to 92% triglycerides), which are capable of entering the bloodstream for transport to storage sites in adipocytes or muscle cells for utilization as an energy source. Once again, for transport into adipocytes or muscle cells, the fat must be hydrolyzed by lipases, absorbed across the cell membrane, and re-esterified to the triglycerol fat for storage in adipocytes or metabolized in the mitochondria of the muscle cell (myocytes) (23).

Mobilization of fats from adipocytes is a G protein-innervated process involving β -adrenergic receptors and various hormones including glucagon, epinephrine, and β -corticotropin. Release of fatty acids from the adipocytes into the bloodstream allows for the transport to myocytes with the aid of blood albumin and ultimately passive absorption into the myocytes. It should be noted that insulin can inhibit the release of fatty acids from adipocytes by inhibition of adenylcyclase. Once again, transport into the mitochondria from the cytoplasm requires esterification to a fatty acyl-coenzyme A (CoA), transport across the outer mitochondria membrane, transesterification to fatty acylcarnitine, transport across the inner mitochondria, and transesterification back to fatty acyl-CoA, which can now be metabolized

in the Krebs cycle to carbon dioxide, water, and release of ATP.

Nutritional Properties

It is estimated that 35% to 40% of the caloric intake in the United States comes in the form of fat, with greater than 90% of the fat being in the form of triglycerides (free fatty acids, cholesterol, and phospholipids represent the remaining amount of fat in the diet). This amounts to approximately 100 to 150 g/day of triglycerides. The RDI for fat is 65 g/day consisting of 20 g/day of saturated fats and 300 mg of cholesterol. This is based on a 2,000-calorie diet for an adult. Obviously the typical diet of 100 to 150 g/day of fat will lead to an excessive caloric intake of this macronutrient and an associated potential for overweight and obesity.

Fats have an essential role in the health of the body. In addition to being a major source of body energy (80% to 90%), fatty acids are an endogenous source of cholesterol via acetyl-CoA units. Cholesterol is the precursor to hormones of the body (bile salts, vitamin D, sex hormones, and adrenocortical hormones). Phospholipids and unsaturated fatty acids are components of the cell membrane, and subcutaneous fats give the body thermal insulation. However, excessive quantities of fats are associated with significant human disease.

Health Issues

As already indicated, fats and fatty acids as a source of calories are associated with overweight and obesity, but this is true of any calorie source because a calorie is a calorie irrespective of whether it comes from fats, carbohydrates, or proteins. There are particular disease states in which evidence points to dietary fats as a major contributor to the condition. CHD, which includes ischemic heart disease, myocardial infarction, and angina pectoris, is a condition in which fats and cholesterol have a significant role (Chapter 25). Countries that have diets low in saturated fats and cholesterol have a low incidence of ischemic heart disease, whereas Western industrialized countries with high dietary fat intake show significant increases in heart disease. A strong link exists between high blood lipid levels (e.g., LDL) and atherosclerosis, which in turn is tied to CHD. A reversible relationship also exists between CHD and the lowering of blood lipid levels. Thus, a goal is to reduce total fat calories to less than 35% of caloric intake.

Epidemiologic studies suggest that colorectal cancer is correlated with high calorie intake, diets rich in meat protein, and elevated dietary fats and oils. Metabolic processes in the GI tract and low fiber in the diet have been suggested as possible mechanisms.

It is generally agreed that a relationship exists between fats and insulin resistance, leading to type 2 diabetes, and diets consisting of low fat content and high carbohydrate content are commonly recommended. Obesity is certainly considered a predisposing factor to type 2 diabetes,

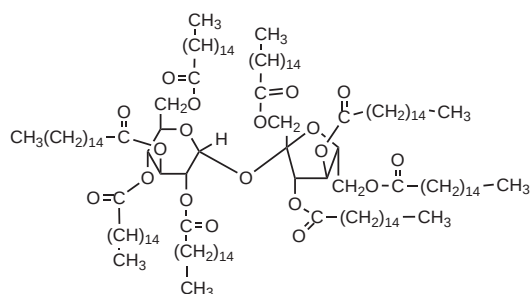
but other than fats being a significant source of calories, the relationship between diabetes and fats is tenuous. Animal studies support the relationship of saturated fats and decreased insulin sensitivity, but whether these studies can be carried over to humans is debatable. Because of the complexity of diets, human variability, and the lack of long-term studies, it is difficult to associate fats with type 2 diabetes, although evidence points to beneficial effects of low saturated fats, low *trans* fats, and high PUFA combined with high-fiber diets in reducing the risk of type 2 diabetes (24).

Fats and Fatty Acid Substitution

As with carbohydrates, replacement of fats and fatty acids with nonfats or reduced-fat calories is desirable. Other than reducing the proportion of fats in the diet, the options for fat substitutes are far less than the choices available with sugar substitutes.

SKIMMED MILK Skimmed milk is a dairy product in which nearly all of the butterfat is removed, whereas whole milk contains approximately 3.5 g of a mixture of fats in 100 g of fluid. The fatty acid content of whole milk consists of saturated fatty acids (palmitic acid [31%], myristic acid [12%], and stearic acid [11%]) and unsaturated fatty acids (oleic acid [24%], palmitoleic acid [4%], and linoleic acid [3%]). Various other milk products are available such as reduced-fat (2% fat) and low-fat milk (~1% fat). Because of the low-fat content of skimmed milk, the caloric content of 100 g of skimmed milk is approximately 34 calories compared with 64 calories in 100 g of whole milk. Ice cream prepared from cow's milk contains from 10% to 19% fat, a varying content of cholesterol (~97 mg/100 g of ice cream), and a caloric content of approximately 274 calories. In contrast, frozen yogurt made from skimmed milk has very little fat, but may have significant calorie content due to added sucrose.

OLESTRA (OLEAN)



Olestra

Olestra is composed of a sucrose nucleus with all of the alcohol groups substituted with stearic acid. As a result of the bulky substituents, the compound is not readily hydrolyzed to the fatty acids and sucrose; thus, olestra is

advertized as a “fat-free” material. The properties of olestra include a high boiling point with stability to baking and cooking and a taste and feel similar to fat. As a result, olestra is used as a frying oil to prepare potato chips. Americans purchased more than 1.6 billion pounds of potato chips, pretzels, and microwave popcorn in 1999 (25). Olestra-prepared chips have 0 g of fat and 70 calories, whereas an equivalent quantity of normally prepared chips has 10 g of fat and 160 calories. The lower metabolism is associated with poor absorption of the olestra, and as a highly fatty material, it has the potential for dissolving the fat-soluble vitamins (vitamins A, D, E, and K), increasing the potential for hypovitaminosis of these vitamins. Most significantly, olestra causes cramping and loose stools, which have seriously affected the sales of olestra-containing products. To offset the potential for fat-soluble vitamin deficiencies, olestra-containing foods commonly are fortified with small amounts of the fat-soluble vitamins.

Various nondigestible fibers from oats have been developed that can be used in foods as fat replacements products, but these appear to have low commercial value at the present time.

Protein

The third major macronutrient is protein. As indicated earlier, proteins are essential to the body as a source of amino acids. Proteins, as a polymer of amino acids, are not absorbed as such but must be metabolized by peptidases in the GI tract to the monomeric amino acids and di- and tripeptides, which can then be absorbed from the GI tract via active transport. Protein makes up 12% to 14% of our normal diet. The body's energy needs are met primarily by fat (~15 kg in the body), whereas carbohydrates in the form of glycogen give rise to immediate energy needs, but its presence in the body is quite small (~0.2 kg). Protein, in the form of muscle mass, accounts for approximately 6 kg of body weight and is a source of potential energy via metabolism to amino acids, which are converted into glucose through the biochemical process called gluconeogenesis. In general, proteins are used for body structure (muscle), hormones, and enzymes and as a source of nitrogen for the synthesis of key chemical components such as purine for RNA and DNA synthesis; only as a last resort will the body use protein as a source of energy.

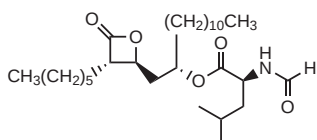
Twenty amino acids are essential to the human body and are normally obtained from the proteins in the diet. Once absorbed, these essential amino acids can be used for protein synthesis through the action of DNA and RNA (see Chapter 6). The actual RDAs for individual amino acids are not known, and it is not possible to accurately determine the RDAs for protein that have varying quantities of individual amino acids. It is generally accepted that males have an RDA of 45 to 63 g of protein/day, whereas females have an RDA of 44 to 50 g/day. These numbers are greatly affected by pregnancy and lactation in women (RDA may increase by as much as 50%), but can also be

affected in both men and women by chronic stress, where the level of protein requirement will increase.

PHARMACOTHERAPY OF OVERWEIGHT AND OBESITY

Overweight and obesity are complex chronic conditions for which successful treatment is not easily attained. Although the cause of obesity and overweight can be easily defined as resulting from a chronic intake of calories in excess to what the body needs, the involvement of social, behavioral, cultural, physiologic, metabolic, and genetic factors makes treatment extremely difficult. Dieting and physical activity can normally produce a loss of weight, but the long-term maintenance of the desired body weight is extremely difficult. Health hazards are associated with both excessive body weight and weight cycling, which commonly occurs through the loss and regaining of weight. A comprehensive study of the issues and treatment of overweight and obesity was reported in 1997 by the National Heart, Lung, and Blood Institute Obesity Education Initiative titled “Clinical Guidelines on the Identification, Evaluation, and Treatment of Overweight and Obesity in Adults” in which drug therapy was addressed as being one of several methods that can be helpful to long-term treatment (26). Presently, only one agent is available on the U.S. market, whereas other drugs are at various stages of development.

Orlistat (Alli, Xenical)



Orlistat, tetrahydrolipstatin, is a semisynthetic derivative of the natural product lipstatin produced by *Streptomyces toxytricini*. The compound acts as an irreversible inhibitor of pancreatic lipase and several other lipases and, as such, prevents the normal metabolic breakdown of fats, leading to reduced absorption of fatty acids from the GI tract (27,28). Fecal fat loss increases by as much as 30% after treatment with orlistat, thus indicating that reduced activity of intestinal lipases increases the loss of undigested fats. Studies with porcine pancreatic lipase show that acylation of serine 152 is responsible for this inhibition of lipase activity (28) (Fig. 42.12), suggesting a similar mechanism for orlistat in human intestine. Orlistat itself is not absorbed to any appreciable extent and is lost from the body via the GI tract. In addition, drug interaction studies indicate that orlistat may interfere with the absorption of a wide spectrum of drugs, such as cyclosporin, some diabetes medications, levothyroxine and warfarin, and with the fat-soluble vitamins with the exception of vitamin E (29).

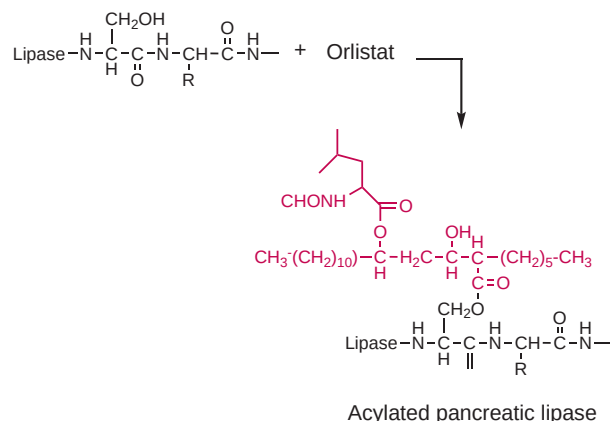


FIGURE 42.12 Irreversible acylation of a serine in pancreatic lipase by orlistat.

As might be expected, the most common adverse events were associated with the lack of absorption of fats and consisted of oily stools, spotting bowel movements, stomach pain, and flatulence. These events occurred in approximately 5% of the patients receiving orlistat at 30 to 360 mg/day. Additionally, it was noted that while on orlistat, obese patients showed a decrease in LDL and total cholesterol levels and that HbA_{1c} levels decreased in patients with type 2 diabetes.

Orlistat is available in a nonprescription product (Alli, 60 mg taken with each fat-containing meal up to 3 doses/day) and a prescription product (Xenical, 120 mg taken with each fat-containing meal up to 3 doses/day). Orlistat is used as an adjunct to diet and exercise and can be used for long-term therapy.

ANOREXIANTS AS PHARMACOLOGIC AGENTS IN THE MANAGEMENT OF OBESITY

The anorexiants constitute the starting place of pharmacologic therapy for obesity. Other terms for these agents include *anorectics* or *anorexigenics*, *antiobesity agents*, and *appetite suppressants*. These drugs are used as adjuncts in weight reduction programs under physician supervision for morbidly obese patients (BMI >30 kg/m²) who are on a caloric reduction diet, exercise program, and behavioral therapy program, who may need to reduce weight over months or years. Appetite is clearly a very important instinct to promote survival (30). Arguably any drug that would abolish appetite may carry a high mortality risk and may be unsuitable for clinical use. Because the human body uses various neurochemicals and hormones to protect its stores of fat (a reaction probably useful to our ancestors when food was scarce), a “silver bullet,” or a way to completely circumvent this natural habit of protecting excess food stores, has not yet been found. As a result, anorexiants/antiobesity drugs are not a practical long-term solution for people who are overweight. Appetite suppressants should not be used in place of proper diet.

For best results, these drugs must be used along with a medically approved diet and exercise program.

The anorectic drugs currently available can be divided into sympathomimetic amines (adrenergic agents) and serotonergic agents. The anorexiant shown in Figure 42.13 are structurally and pharmacologically related to the amphetamines (phenylisopropylamines) (see also Chapter 10). Although, anorexiant are primarily intended to suppress the appetite, most of the drugs in this class appear to be CNS stimulants, with a potential for abuse or addiction (31). Therefore, the antiobesity drugs are classified as schedule I, II, III, IV, and V as determined by the Controlled Substances Act of 1971 (Fig. 42.13). For example, schedule I drugs have a very high potential for abuse and addiction but no legitimate medical use, whereas schedule V drugs have the lowest potential for abuse and addiction and a legitimate medical use.

The anorexiant are associated with abuse and addiction because they are chemically and pharmacologically related to amphetamine. The dependence associated with these agents may be psychological and/or physiologic and is usually associated with long-term therapy or abuse. If the patient continues to lose weight and does not develop side effects or dependence, therapy may be continued past 12 weeks. If the patient does not continue to lose weight, then the dosage should not be increased, and the agent should be discontinued. Total length of therapy with the anorectic agents should not exceed 6 months.

Sympathomimetic Amines

Sympathomimetic amines are drugs that act by releasing norepinephrine from storage vesicles in the adrenergic neuron and by blocking norepinephrine reuptake from the synapse (see Chapter 10). These drugs also act as reuptake inhibitors of dopamine, which is believed to be linked with their abuse and addiction potential. Although the mechanism of action of the sympathomimetic amines in the treatment of obesity is not fully known, this class of compounds has no primary effect on the appetite brain center; rather, its appetite-suppressant action appears to be secondary to CNS stimulation. These drugs are FDA approved for short-term use and include phentermine (Ionamin), diethylpropion (Tenuate), mazindol (Sanorex), benzphetamine (Didrex), and phendimetrazine (Bontril) (Fig. 42.13). Sympathomimetic appetite suppressants are used in the short-term treatment of obesity, usually for no more than 12 weeks. Their appetite-reducing effect tends to decrease after a few weeks of treatment. As a result, the sympathomimetic amines are useful only during the first few weeks of a weight-loss program. These drugs should not be taken with other appetite suppressants due to the possibility of serious side effects. The sympathomimetic amines may cause withdrawal reactions, especially if they have been used regularly for a long time or in high doses. In such cases, withdrawal symptoms (such as depression and severe fatigue) may occur if these medications are suddenly stopped.

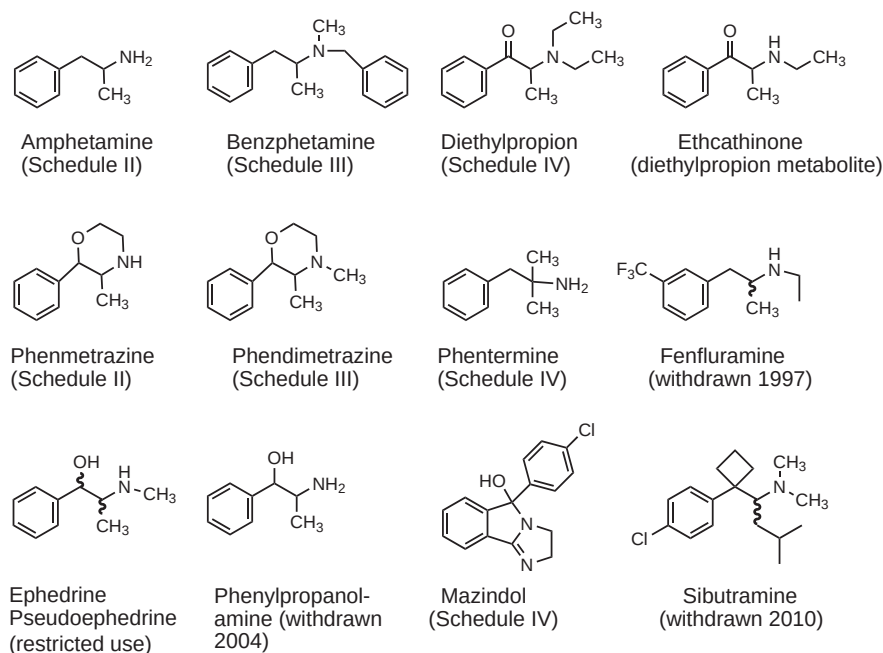


FIGURE 42.13 Anorexiant.

Amphetamine

Amphetamine is the prototypical anorexiant and a psychostimulant drug of the phenylisopropylamine family that produces increased wakefulness and decreased fatigue and appetite (Fig. 42.13). The drug has also been used recreationally and as a performance enhancer. Amphetamine is ranked as the most addictive and most harmful of the appetite suppressants (32). Amphetamine is a chiral compound, and its racemate can be divided into levo- and dextroamphetamine. Dextroamphetamine was used for treatment of obesity. Other derivatives of amphetamine used as recreational drugs are described in Chapter 19.

Amphetamine was first synthesized in 1887 as a synthetic substitute for ephedrine. No pharmacologic use was found for amphetamine until 1933 when the pharmaceutical company Smith, Kline and French began selling the volatile base form of the drug as a nasal inhaler under the trade name Benzedrine as a nasal decongestant. Shortly after marketing Benzedrine inhaler, the psychostimulant properties of amphetamine were discovered; these included a sense of well-being and a feeling of exhilaration, lessened fatigue in work, and loss of appetite. During World War II, amphetamine was extensively used to combat fatigue and increase alertness in soldiers, especially fighter and bomber crews. After decades of reported abuse, the FDA banned Benzedrine inhalers and limited amphetamine to prescription use in 1965, but nonmedical use remained common. Amphetamine became a schedule II drug under the Controlled Substances Act in 1971. An analog of amphetamine is methamphetamine; the *N*-methyl analog was also synthesized as an ephedrine derivative in 1920, but by 1941, it became a controlled substance because of abuse and was eventually banned.

Amphetamine became popular for weight loss during the late 1930s. The appetite-suppressant action for amphetamine appears to be secondary to CNS stimulation, along with other beneficial effects such as increased alertness. Use of amphetamine increased over the subsequent decades, culminating in the “rainbow pill” regimen. This was a combination of multiple pills, all thought to help with weight loss, taken throughout the day. Typical regimens included stimulants such as amphetamine, thyroxine, diuretics, digitalis, laxatives, and often a barbiturate to suppress the side effects of the stimulants. As a result of deaths attributed to diet pills and the gradual implementation of greater restrictions on diet pills, in 1979, the FDA banned the use of amphetamine in diet pills.

Benzphetamine

Benzphetamine (Didrex) is the *N*-benzyl analog of methamphetamine FDA approved in 1960 (Fig. 42.13). Because it is closely related to amphetamine, its anorexiant and pharmacologic properties appear to be similar to amphetamine. It is used as a short-term adjunct in management of morbid obesity. Benzphetamine is primarily *N*-demethylated to *N*-benzylamphetamine and to the minor metabolites methamphetamine, amphetamine, and their hydroxylated metabolites (33). Because

benzphetamine is poorly metabolized to methamphetamine and amphetamine in the liver regardless of route of administration, benzphetamine is classified as schedule III appetite suppressant because it still has the potential for abuse and addiction, but much less so than amphetamine. Benzphetamine should not be given to patients who have a history of drug abuse.

Diethylpropion

Diethylpropion (Tenuate) was developed in 1958 as an anorexiant alternative to amphetamine and phenmetrazine without the psychostimulant side effects and abuse potential and does not affect blood pressure (Fig. 42.13). The appetite-suppressant effect of diethylpropion is due to its *N*-monodeethylation to ethocathinone (34). It is used along with a medically approved, reduced-calorie diet, exercise, and behavior change program to help one lose weight. Like the other sympathomimetic appetite suppressants, taking this medication late in the day may cause trouble sleeping (insomnia). This drug is classified as schedule IV, and therefore, abuse and addiction are low but possible with this medication. Increasing the dose or frequency of use is contraindicated, as is using diethylpropion beyond the normal prescribing time.

Phenmetrazine and Phendimetrazine

PHENMETRAZINE Phenmetrazine (Preludin) is a stimulant drug of the morpholine chemical class that was previously used as an appetite suppressant but has since been withdrawn from the U.S. and world market due to its great potential for abuse and addiction. It has been replaced by its *N*-methyl analog phendimetrazine, which functions as a prodrug to phenmetrazine (Fig. 42.13). Phenmetrazine was first patented in Germany in 1952 by Boehringer-Ingelheim as an anorectic drug without the side effects of amphetamine. The drug is a cyclized analog of ephedrine. It was introduced into clinical use in 1954 and withdrawn from the market in 1965. Phenmetrazine has a high potential for abuse and addiction and therefore is classified as schedule II. In a study of the effectiveness on weight loss between phenmetrazine and dextroamphetamine, phenmetrazine was found to be slightly more effective (35). Phenmetrazine causes the release of norepinephrine and dopamine but not of serotonin and inhibits the reuptake of norepinephrine and dopamine. Phenmetrazine was used recreationally in the United States during the 1960s and the early 1970s.

PHENDIMETRAZINE Phendimetrazine (Bontril) is also a stimulant drug of the morpholine chemical class approved as an appetite suppressant. Phendimetrazine is the *N*-methyl analog of phenmetrazine with anorexigenic action resulting from a combination of its own lesser activity plus that resulting from *N*-demethylation to phenmetrazine (Fig. 42.13). Thus, phendimetrazine functions as a prodrug to phenmetrazine; approximately 30% of an oral dose is converted into phenmetrazine. Because of its slow conversion to phenmetrazine, phendimetrazine

is a less abusable version of phenmetrazine and is classified as schedule III. In addition, approximately 30% of the administered dose of phendimetrazine is excreted unchanged, and 25% is excreted as inactive *N*-oxide.

Phentermine

Phentermine (Ionamin) is structurally similar to amphetamine and is currently used as a single weight-loss agent to decrease appetite (Fig. 42.13). Phentermine received approval from the FDA in 1959 as an appetite-suppressing drug and became available in the early 1970s. Phentermine is a contraction of phenyl-tertiary-butylamine and is related to amphetamine. Phentermine is approved as an appetite suppressant to help reduce weight in obese patients when used short term (<12 weeks) and combined with exercise, diet, and behavioral modification. Phentermine has fewer psychostimulant effects and less abuse and addiction potential than amphetamine and therefore is classified as schedule IV appetite suppressant (36). Its side effects are consistent with its norepinephrine-releasing properties, but the incidence and magnitude of these effects are less than with amphetamine. After short-term use, tolerance begins and can be followed by rebound weight gain. However, phentermine was not a popular appetite suppressant drug until 1992 when Weintraub reported that combining phentermine with fenfluramine (Pondimin) (serotonergic appetite suppressant) caused a 10% weight loss, which was maintained for over 2 years (37). Thus, Fen-Phen rapidly became the most commonly prescribed diet medication. Dexfenfluramine (Redux) was developed in the mid-1990s as an alternative to fenfluramine with less side effects and received regulatory approval in 1996. However, after 24 cases of heart valve disease in Fen-Phen users were reported, fenfluramine and dexfenfluramine were voluntarily taken off the market at the request of the FDA (38). Studies later proved that the abnormal valve findings were due to the serotonergic effect of fenfluramine or dexfenfluramine on the heart valve. The FDA did not ask manufacturers to remove phentermine from the market.

Ephedrine and Pseudoephedrine

A number of over-the-counter dietary supplements and Internet products (Herbal Fen-Phen, PhenTrim, Phen-Cal, Xenadrine) contain ephedrine, which is derived from the ephedra herb, *ma huang*. Ephedra is one of a group of plants that are a source of ephedrine alkaloids, including ephedrine and pseudoephedrine (Fig. 42.13) (see Chapter 10). Ephedrine can cause a number of side effects, including rapid heartbeat, high blood pressure, psychosis, heart attacks, and seizures. Pseudoephedrine, a related stereoisomer of ephedrine, is commonly found in many antihistamines and cold medicines, has similar effects, and is sometimes used by dieters. Pseudoephedrine has also been used for its CNS stimulant properties. The structural similarity of pseudoephedrine to the amphetamines has made the drug a sought-after chemical precursor in the illicit manufacture of methamphetamine. As a result, the U.S. Congress recognized that pseudoephedrine is

used in the illegal manufacture of methamphetamine, and passed the Combat Methamphetamine Epidemic Act of 2005, restricting the sale of pseudoephedrine-containing products (i.e., Sudafed), as well as to all over the counter products containing ephedrine, pseudoephedrine, and phenylpropanolamine. Other countries also restrict the sale of pseudoephedrine. In 2004, the FDA prohibited the sale of any dietary supplement containing ephedra/ephedrine in the United States due to an unreasonable risk to those who use it. Ephedra has become an increasingly popular ingredient in dietary supplements promoted for weight loss and boosting athletic performance or energy. The FDA reports little evidence supporting ephedra's effectiveness in promoting anything but modest, short-term weight loss without any clear health benefit. In addition, ephedra raises blood pressure and puts additional stress on the circulatory system, and these actions have been linked to heart attack and stroke. In 2006, the FDA declared all dietary supplements containing ephedrine alkaloids to be adulterated, and therefore illegal for marketing in the U. S. However, ephedrine is still legal for other medical uses, and purchase is restricted from state to state. However, the safety of dietary supplements containing ephedrine alkaloids and pseudoephedrine is closely regulated by the FDA, for these hazards.

Phenylpropanolamine

Phenylpropanolamine (Dexatrim), a sympathomimetic drug and a synthetic derivative of ephedrine, was available as an over-the-counter appetite suppressant and decongestant in cold medicines (Nyquil). Individuals who took 25 to 75 mg of phenylpropanolamine experienced some weight loss (Fig. 42.13). Common adverse effects included nervousness, insomnia, dizziness, palpitations, and headaches, but not a clinically significant increase in blood pressure. In 2005, the FDA removed phenylpropanolamine from over-the-counter sale due to a proposed increased risk of stroke in younger women and because of its potential use in amphetamine manufacture; it is controlled by the Combat Methamphetamine Epidemic Act of 2005. Canada withdrew phenylpropanolamine in 2001 for the same reasons. The drug is still available in Europe by prescription and over the counter.

Mazindol

Mazindol (Sanorex) differs from the prototypical amphetamine analogs by being a fused tricyclic imidazo-isindole compound (Fig. 42.13). However, mazindol exhibits pharmacologic effects similar to those of amphetamine. Like other sympathomimetic appetite suppressants, mazindol is thought to act as a reuptake inhibitor of norepinephrine and dopamine, which stimulates the CNS and increases heart rate and blood pressure. Similar to amphetamine, patients report a decrease in appetite and an increase in energy, but mazindol has a low potential for abuse and addiction. Thus, mazindol is classified as a schedule IV appetite suppressant. As with other schedule

IV appetite suppressants, this drug is used in the short-term treatment (<12 weeks) of obesity, in combination with a weight-loss program based on caloric restriction, exercise, and behavior modification in patients with a BMI of 30 kg/m² or in patients with a BMI of 27 kg/m² in the presence of risk factors such as hypertension, diabetes, or hyperlipidemia. Its appetite-suppressing effect tends to decrease after a few weeks of treatment. Because of this effect, mazindol is useful only during the first few weeks of a weight-loss program.

Serotonergic Agents

To circumvent the feedback mechanisms that prevent most monotherapies from producing sustained large amounts of weight loss, it has been hypothesized that combinations of drugs may be more effective by targeting multiple pathways and possibly inhibiting feedback pathways that work to cause a plateau in weight loss. This was evidenced by the early success of combining phentermine and fenfluramine or dexfenfluramine, popularly referred to Fen-Phen or Redux, in producing significant weight loss. However, as previously described, Fen-Phen was removed from the market due to safety fears regarding a potential link to heart valve damage and primary pulmonary hypertension. The damage was found to be a result of activity of fenfluramine and dexfenfluramine at the 5-HT_{2B} serotonin receptor in heart valves.

Fenfluramine

Fenfluramine (Pondimin) was part of the Fen-Phen anti-obesity medication controversy as previously described. Fenfluramine was introduced on the U.S. market in 1973 as an appetite suppressant but was not very popular until combined with phentermine. It exists as the racemic mixture of two enantiomers: dextrofenfluramine and levofenfluramine (Fig. 42.13). The drug and its metabolite, norfenfluramine, cause the release of serotonin from vesicular storage sites and inhibit serotonin transporter reuptake of serotonin from the synaptic cleft, thus acting on the hypothalamus to decrease satiety. The end result is a feeling of fullness and loss of appetite. The neurotransmitter serotonin regulates mood, appetite, and other functions. Selective serotonin reuptake inhibitors (SSRIs) also suppress appetite (39) (see Chapter 18).

One mechanism to explain valvular abnormality involves heart valve serotonin receptors, which are thought to help regulate growth. Because fenfluramine and its active metabolite norfenfluramine stimulate 5-HT_{2B} serotonin receptors, this may have caused the damage by stimulation of these receptors, leading to inappropriate valve cell division (38). Fenfluramine is no longer marketed worldwide.

Sibutramine

Sibutramine (Meridia) is an adrenergic/serotonergic agent approved by the FDA in 1997 for use in the management of obesity, including weight loss and maintenance of weight loss, and should be used in conjunction with a

reduced-calorie diet (Fig. 42.13). The drug was recommended for obese patients with an initial BMI of greater than 30 kg/m². Sibutramine and its metabolite inhibit norepinephrine and serotonin reuptake, suppressing appetite similar to SSRIs. Sibutramine may also stimulate thermogenesis by activating the β_3 system in brown adipose tissue. Initially tested for its antidepressant activity, sibutramine was found to cause modest weight loss. However, weight was regained even in subjects taking high-dose sibutramine.

Since its approval in 1997, sibutramine has exhibited an increased risk of nonfatal but serious cardiovascular events in patients with a known or high risk for cardiovascular disease (40). As a result of safety concerns, minimal efficacy as an appetite suppressant, and increased risk of adverse cardiovascular events, in 2010, Abbott Laboratories withdrew sibutramine from the U.S. and European markets under pressure from the FDA and European Medicines Agency (41).

Counterfeit weight-loss products have been on the increase, prompting the FDA in 2008 and 2009 to issue safety alerts to consumers naming more than 60 different products marketed as “dietary supplements” for weight loss that illegally contain undisclosed amounts of sibutramine, sometimes containing as much as twice the dosage of the prescription drug. The safety alerts underscore the risks associated with unregulated “herbal supplements” in unsuspecting persons. This concern is especially relevant to those with underlying medical conditions incompatible with undeclared pharmaceutical adulterants (42).

The drug rimonabant was approved as an antiobesity drug in Europe that acted centrally through antagonism of cannabinoid receptors resulting in a reduced appetite. Concerns about psychiatric effects lead Sanofi-Aventis to remove the drug from FDA consideration as an antiobesity agent in 2008 and 2009.

Over-the-Counter Drugs and Herbal Remedies

People must be cautious when using any weight-loss medications, including over-the counter diet pills and herbal or so-called natural remedies. The following are examples of some products being sold for weight loss.

The Internet is a source of large numbers of products that are advertised as aids to promote weight loss. Many of these products consist of mixtures of natural products that also contain unlisted anorexiant such as ephedrine, propanolamine, or sibutramine, which the manufacturers claim will burn fats and create significant weight loss. None of these products have been approved by the FDA and are sold based on testimonials with some literature references suggesting effects on fat metabolism but, in most cases, without any documented clinical support. An example of such a product is LipoFuze, which the manufacturer states is a combination of “six patented and four clinically tested fat burning ingredients.” The list of ingredients includes evodiamine, raspberry ketone, fucoxanthin, *Coleus forskohlii* extract, guggulsterones,

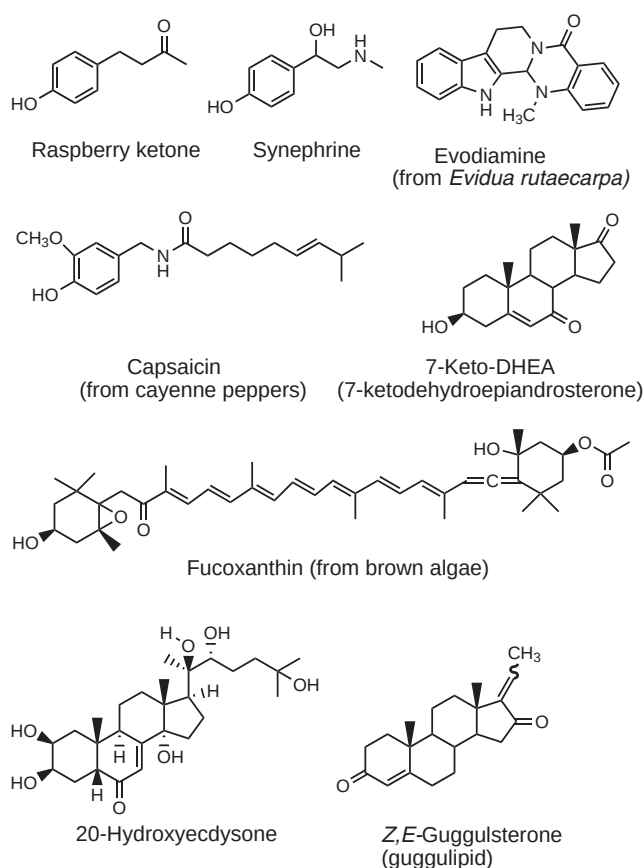


FIGURE 42.14 Partial list of ingredients in LipoFuze.

black pepper extract, green tea, capsaicin, synephrine hydrochloride, 7-oxo DHEA, cinnamon, phenylethylamine, ginger, and 20-hydroxyecdysone (Fig. 42.14). Classified as dietary supplements, the products are not tightly controlled by the FDA, and thus it is up to the consumer to be aware of their effectiveness and safety.

MICRONUTRIENTS

Vitamins

Vitamins are organic chemicals found in microorganisms, plants, and animals, but not man, that are essential to man and higher animals for the maintenance of growth, physiologic function, metabolism, and reproduction. The role of vitamins in the body is to act as catalysts for the biochemical reactions of the body, and although they are not consumed in these reactions, vitamins do have a normal turn over and elimination and therefore must be continuously resupplied to the body. The reader is referred to a biochemistry textbook for a discussion of the molecular mechanisms involving vitamins, as well as earlier chapters in this text (see Chapter 26 for vitamin K and Chapter 30 for vitamin D). The RDIs for the vitamins are given in Table 42.3. It is not our intention to review all of the vitamins in depth, but rather to describe nutritional properties of a select few vitamins.

The vitamins are divided into fat-soluble and water-soluble vitamins based on their physical-chemical properties. The fat-soluble vitamins include vitamins A, D, E, and K, whereas all of the remaining vitamins are considered water soluble (the B complex vitamins: thiamine [vitamin B₁], riboflavin [vitamin B₂], niacin [vitamin B₃], pantothenic acid [vitamin B₅], pyridoxal/pyridoxine [vitamin B₆], cobalamin [vitamin B₁₂], folic acid, and biotin; and vitamin C). Diseases associated with the vitamins most commonly occur due to deficiencies of the vitamins (hypovitaminosis) (Table 42.8), but can also occur due to excessive quantities of a vitamin (hypervitaminosis). Because of the lack of water solubility, the fat-soluble vitamins can build up in the body and therefore are commonly associated with hypervitaminosis, but the water-soluble vitamins can also cause hypervitaminosis. Vitamin deficiencies can be classified as either primary hypovitaminosis or secondary hypovitaminosis. Primary hypovitaminosis is the result of inadequate diet because of a lack of foods containing the vitamins or an inadequate diet due to poor use of foods. The latter may occur if the individual is dieting and not consuming a balanced diet or possibly because of food faddism where a variety of foods are not being used. Secondary hypovitaminosis may be the result of poor health or chronic disease states. Examples might be hyperthyroidism, chronic diarrhea, liver disease, or diseases of the GI tract. There are times in the life of an individual when demands for vitamins increase, such as during pregnancy and lactation in the female when the RDIs for all nutrients increase. Unless an increase in vitamin consumption occurs, the female will suffer hypovitaminosis. The use of drugs can also result in hypovitaminosis. An alcoholic may experience both primary and secondary deficiencies. While consuming alcohol, an alcoholic may not be eating a balanced diet and thus is at risk for a primary deficiency. Alcohol also damages both the intestinal lining and the liver, and nutrients cannot be absorbed through the damaged mucosal membrane or properly used by a damaged liver.

Vitamin A (Retinol, Retinal, Carotenes)

Vitamin A and its precursors, the carotenes, are fat soluble and found in high concentrations in liver and fish oils, yellow and green leafy vegetables, eggs, and whole-milk products. β -Carotene is also found in high levels in carrots and is highly lipophilic and readily absorbed. The carotenes, α - and β -carotene, are oxidatively converted into all-*trans*-retinal, which in turn can be converted into all-*trans*-retinol (Fig. 42.15), and thus the carotenes can be considered provitamin A's. *Trans*-retinol is generally considered the active form of vitamin A, whereas β -carotene has approximately half the activity of retinol, with α -carotene being a far less efficient source of vitamin A activity. Retinol is commonly found as a lipid-soluble ester with either palmitic acid or acetic acid. The carotinoids, a combination of

TABLE 42.8 Vitamins, Outward Disease(s), and Common Sources and Roles

Vitamin	Specific Disease(s)	Source	Role
Fat soluble			
Vitamin A	Xerophthalmia Nyctalopia	Yellow and green leafy vegetables, carrots, yellow fruits, fish oils, milk, fats and egg yolk, apricots, peaches, and sweet potatoes	Necessary for normal eye function, integrity of the epithelium, synthesis of adrenocortical steroids
Vitamin D	Rickets Osteomalacia	Egg yolks, dairy products, fish, and UV light	Calcium absorption and bone formation
Vitamin E	Spinocerebellar ataxia Skeletal myopathy	Cereals, nuts, unsaturated oils, leafy green and yellow vegetables, milk, muscle meats, and butter	Health of sensory neurons leading to peripheral neuropathy if absent
Vitamin K	Pigmented retinopathy Hypoprothrombinemia	Synthesized in the intestinal tract by bacteria and found in leafy green vegetables	Involved in the blood-clotting process
Water soluble			
Vitamin C (ascorbic acid)	Scurvy	Citrus fruits, cabbage, peppers, berries, melons, and salad greens	Prevents scurvy, necessary for proper bone and teeth formation
Vitamin B ₁ (thiamine)	Beriberi Wernicke encephalopathy	Yeast, beans, brown rice, lean pork, whole-wheat bread, liver, and outer layers of seeds of plants or unrefined cereal grains	Prevents beriberi (leg edema, paralysis, muscle atrophy)
Vitamin B ₂ (riboflavin)	Anemias	Liver, kidney, milk, yeast, heart, anaerobic fermenting bacteria, and many vegetables	Deficiency leads to lip lesions, seborrheic dermatitis
Vitamin B ₃ (niacin)	Pellagra	Liver, kidney, lean meat, soybeans, wheat germ, broccoli, and avocados	Prevents pellagra (dermatitis, diarrhea, dementia)
Vitamin B ₅ (pantothenic acid)	Rare, no specific disease	Muscle meats, outer layer of whole grains, broccoli, and avocados	Deficiencies are rare, but may appear as numbness, paresthesia, and muscle cramps
Vitamin B ₆ (pyridoxal)	Homocystinuria	Wheat germ, milk, yeast, liver	Prevents dermatitis sicca in adults and convulsions in infants, essential to biochemical reactions
Vitamin B ₁₂ (cobalamin)	Pernicious anemia	Liver, kidney, muscle meats	Prevents pernicious anemia
Folic acid	Megaloblastic anemia NTDs	Leafy vegetables, beans, peas, liver, kidney, baker's yeast, grains, various nuts, and sunflower seeds	Important for prevention of NTDs, necessary to DNA and RNA synthesis and other biochemical reactions
Biotin	Rare, no specific disease	Tomatoes, kidney, egg yolk, peanuts, chocolates, yeast, and liver	Deficiency causes myodermatitis, lassitude, gastrointestinal symptoms, and hyperesthesias, including muscle pain

NTD, neural tube defect; UV, ultraviolet.

all sources of vitamin A, have used various measures to define their biologic activity. Since 2001, the retinol activity equivalent (RAE) has been used, which defines 1 mcg of RAE as equal to 1 mcg of all-*trans*-retinol and equal to 2 mcg of β -carotene, which is equal to 12 mcg of food-based β -carotene and 24 mcg of other food-based all-*trans* provitamin A carotenoids. Presently, the recommendation is that men should receive a dose of 900 mcg daily, whereas women should receive 700 mcg/day. Of course, the female dose should be increased

during pregnancy and lactation (770 and 1,300 mcg, respectively) (43).

PHYSIOLOGIC ACTION OF VITAMIN A

Vision One commonly thinks of vitamin A and its role in vision. A simplistic outline of the process of vision is shown in Figure 42.16. To initiate the process, *trans*-retinol is isomerized to 11-*cis*-retinol, which in turn is oxidized to 11-*cis*-retinal, the aldehyde that forms a Schiff base with the protein opsin to form rhodopsin, which is the functional

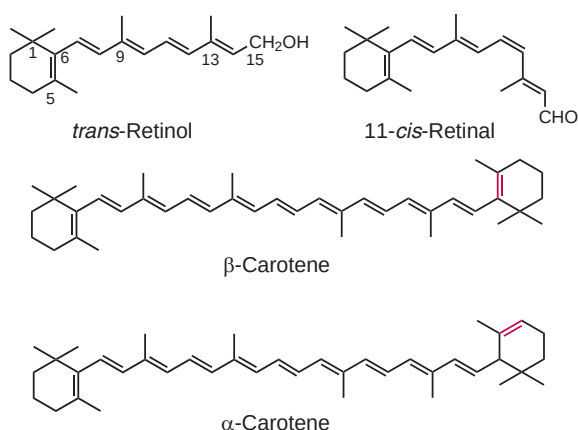


FIGURE 42.15 Naturally occurring carotenoids.

form of vitamin A. The absorption of a photon of light by the rhodopsin results in sodium conductance at a G protein-coupled receptor, leading to signaling in the visual center in the cortex via the optic nerve and release of opsin and *trans*-retinal, which is recycled through conversion back to *trans*-retinol. A deficiency of vitamin A (i.e., hypovitaminosis A) in the rods of the retina will initially lead to night blindness, nyctalopia, which is a reversible condition. Vitamin A deficiency in a child has much more severe consequences and is a leading cause of blindness (xerophthalmia) in many third-world nations.

Other Effects of Vitamin A Animal studies suggest additional roles for vitamin A in the development of epithelial tissue, mucus-producing tissue, reproduction, and bone growth. Whether these activities relate to human conditions is not known, and the evidence appears to point to retinoic acid as being important for these actions. Carotenoids have also been suggested to function as

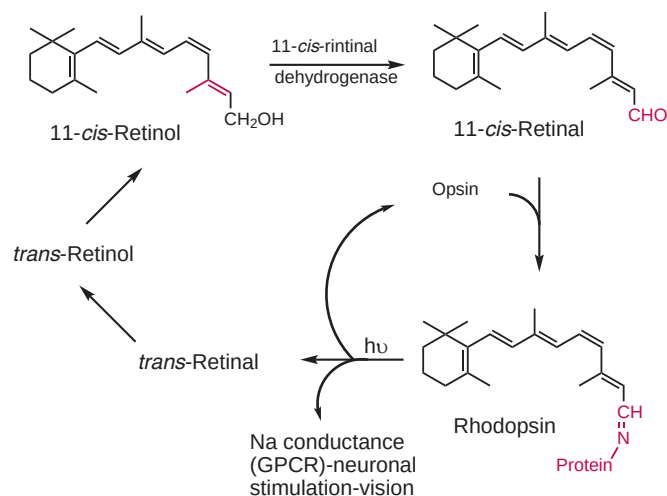
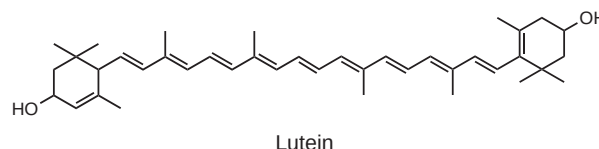


FIGURE 42.16 Role of vitamin A leading to visual signaling in the cortex.

antioxidants and therefore have been proposed as valuable agents in the prevention of heart disease and cancer. The data supporting this notion are weak at best. A specific carotenoid, lutein, as well as β-carotene in combination with vitamin E, vitamin C, and zinc have been claimed to show benefit in the treatment of age-related macular degeneration.



Once again, the scientific evidence supporting these claims is weak.

Absorption The absorbable form of vitamin A is usually considered to be retinol. When vitamin A is administered as a retinol ester, hydrolysis is required in the intestine to release the free alcohol. If the source of vitamin A is β-carotene, oxidative cleavage is required to produce two molecules of retinal, which in turn are reduced to all-*trans*-retinol. Studies have shown that β-carotene can be absorbed as such, but quite inefficiently. Oxidative cleavage of β-carotene does not always occur symmetrically, in which case the yield of vitamin A is less than two molecules per molecule of β-carotene. Approximately 70% to 90% of absorption of vitamin A occurs from the small intestine, with much of the vitamin A being transported to the liver for storage.

Hypervitaminosis A Lay press publications claiming unsubstantiated benefits from vitamin A have resulted in consumers ingesting vitamin A in doses beyond normal daily requirements (doses as high as 25,000 IU or 7,500 mcg). Because of the lipid nature of vitamin A, large quantities can be retained and stored in the body, leading to hypervitaminosis A. The signs and symptoms of hypervitaminosis A include headache, fatigue, insomnia, cracked bleeding lips, cirrhosis-like damage of the liver, birth defects, and yellow skin coloration. Adverse events associated with vitamin A will not normally be seen unless the vitamin is consumed in large quantities over an extended period of time, and these effects are reversible upon discontinuance of the vitamin. The upper safe limit for vitamin A consumption is considered to be 3,000 mcg. The relationship between vitamin A and birth defects appears to be primarily associated with the intake of 13-*cis*-retinoic acid (Accutane), which is a known teratogen if consumed during the first trimester of pregnancy. There does not appear to be evidence that retinol, retinal, or β-carotene are sources of retinoic acid or 13-*cis*-retinoic acid, but cautious intake of vitamin A should be advised.

Vitamin D (Sunshine Vitamin, Calcitriol, Calciferols)

Vitamin D is an interesting fat-soluble vitamin because the vitamin can be synthesized in the body via a series of

reactions including an ultraviolet-stimulated ring cleavage (ultraviolet B radiation) and hydroxylation reactions that occur in the liver and kidney (Fig. 42.17). Since the vitamin can be and is synthesized in the body, it should be considered a hormone rather than a vitamin. However, the absence of sufficient sunlight will lead to vitamin D-associated diseases, and therefore, additional quantities of vitamin D (actually its precursors) are required, and thus the vitamin nature of the calcitriol. Because the amount of sunlight that an individual receives can vary considerably, the actual daily needs for an individual can also vary considerably, and the recommended intake for vitamin D cannot be accurately defined. It is generally accepted that 5 mcg (200 IU) is sufficient for those under the age of 50 years, with increased quantities for those between 51 and 70 years (400 IU) and those over 71 years (600 IU). However, clinicians are recommending up to 2,000 mcg/day, especially for individuals living in the northern latitudes. A scan of the Internet shows that the consumer can purchase vitamin D₃ (cholecalciferol) in doses as high as 5,000 IU/dose.

PHYSIOLOGIC ACTION OF VITAMIN D The major role of vitamin D is to regulate the serum levels of ionic calcium and phosphate, which in turn affects bone ossification and neuromuscular function. 7-Dehydrocholesterol is found in most tissues, but that which is in the epidermis can be cleaved (between the C9 and C10 position) by the energy of absorbed ultraviolet light that

is unfiltered by window glass. Following isomerization, vitamin D₃ is formed and circulates in the blood (Fig. 42.17). In the liver, vitamin D₃ is hydroxylated to 25-hydroxycholecalciferol, the primary circulating form of vitamin D. Final activation of vitamin D occurs in the kidney where hydroxylation occurs at the 1-position to give the active 1,25-dihydroxycholecalciferol (calcitriol). Calcitriol is transported through the serum into the target tissue of the intestine and bone where it governs the transcription of mRNA, leading to synthesis of proteins, which increase the absorption of calcium from the gut or resorption of calcium from the bones in conjunction with parathyroid hormone (see Chapter 30). The termination of the action of calcitriol occurs through further hydroxylation, leading to any of a number of tri- and tetrahydroxycholecalciferols (24-hydroxylation is shown in Fig. 42.17).

Hypovitaminosis D, resulting from insufficient vitamin D serum levels, leads to poor absorption of calcium and phosphate from the intestine, which can result in mobilization of calcium from bones (resorption) in order to meet serum needs for the ion. In an infant, this can lead to rickets, whereas in an adult, osteomalacia will result. To prevent these conditions, it has become common practice to fortify milk and other food products with vitamin D as well as to recommend that women who breastfeed their infant should supplement their diet with vitamin D. There are indications that the vitamin D content of breast milk from many lactating women may

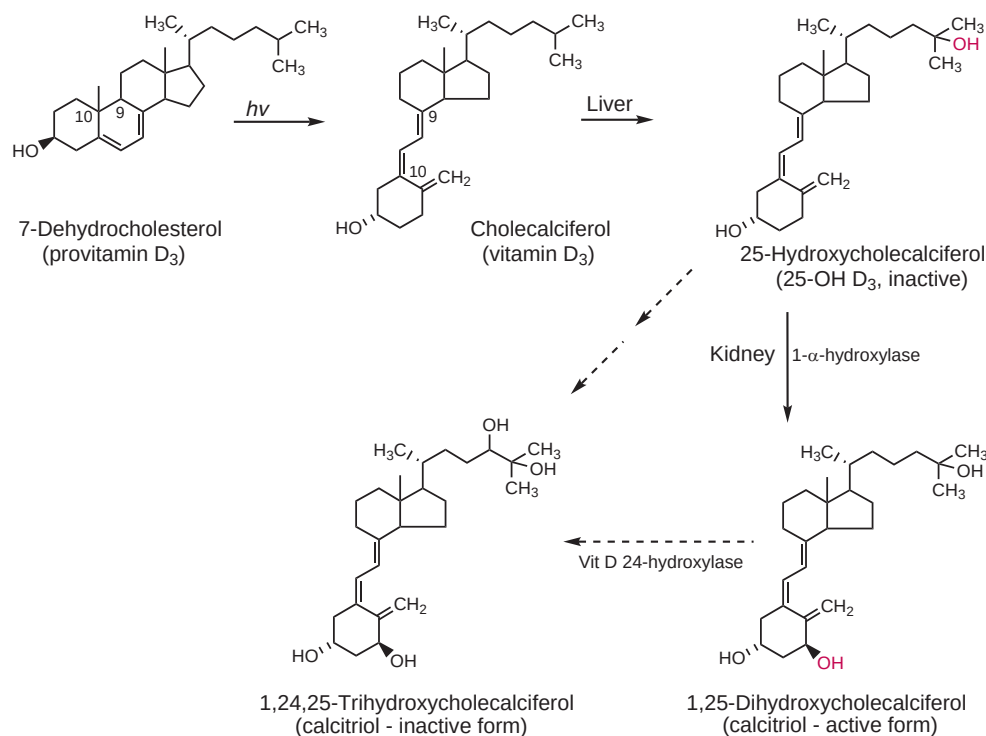
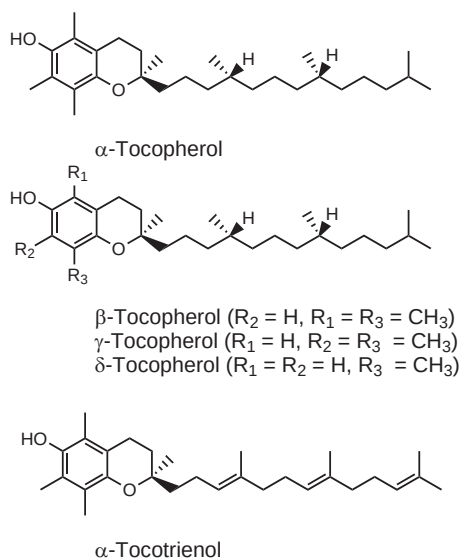


FIGURE 42.17 Bioactivation and inactivation of vitamin D.

be low in vitamin D. Hypovitaminosis D is common in the elderly due to the inability of the aged skin to form vitamin D₃ and because window glass filters out the beneficial ultraviolet light wavelengths. Hypovitaminosis D has also been found to be a factor in individuals with autoimmune diseases, such as diabetes, multiple sclerosis, irritable bowel syndrome, and rheumatoid arthritis, but the significance of the deficiency in these diseases has not been determined.

Hypervitaminosis D, resulting from consumption of high doses of vitamin D supplements, appears to be far less of a problem than earlier literature suggested, although doses of 10,000 IU/day or greater are considered problematic. High serum calcium levels increase the likelihood for precipitation of calcium salts in soft tissue such as kidney and heart. This could lead to a decrease in renal function and kidney stones, although the more common symptoms of hypervitaminosis D include constipation, bone pains and stiffness, and confusion. Treatment of hypervitaminosis D consists of removal of the source of vitamin D and reduction in exposure to the sun along with hydration and diuresis.

Vitamin E (Tocopherols)



Vitamin E consists of a group of fat-soluble compounds composed of α -, β -, γ -, δ -tocopherol and α -, β -, γ -, δ -tocotrienol. α -Tocopherol and γ -tocopherol represent the two major tocopherols, whereas the tocotrienols are normally found in low concentrations in nature and are the unsaturated analogs of the respective tocopherols. Since the tocopherols have three chiral centers, a total of eight stereoisomers are possible for each of the tocopherols. The tocotrienols have only a single asymmetric center, resulting in two isomers for each of these products. Both the tocopherols and tocotrienols make up what is normally called vitamin E. The interest in vitamin E arose from studies of infertile male rats, which were shown

to be deficient in a substance with vitamin E biologic activity. The substance was isolated and given the name of tocopherol, which is derived from the Greek words meaning “bring forth childbirth.” Vitamin E is known as the antisterility vitamin, although there is no known correlation between low levels of vitamin E in male humans and infertility. The commercially available vitamin E may be a single isomer of one of the tocopherols or a mixture of tocopherols, although it is generally considered that α -tocopherol or R,R,R - α -tocopherol is the more active form of vitamin E. The tocotrienols may also be present in commercial vitamin E, but far less is known about the effects of these materials. The tocopherols may come in the form of an ester of the primary alcohol with the acetate ester being a common form of the vitamin. The quantization of vitamin E is based on a direct analytical measure or an indirect biologic rat assay method.

PHYSIOLOGIC ACTION OF VITAMIN E The search for a human function for vitamin E has not been an easy task. Vitamin E is ubiquitous in nature, and therefore, human deficiency in this vitamin is uncommon. The source of vitamin E include cereals, nuts, unsaturated oils, leafy green and yellow vegetables, milk, muscle meats, and butter (Table 42.8), although the distribution of the tocopherols and tocotrienols in nature is not uniform. Tocopherols are found in oils, seeds, leaves, and other green part of higher plants, whereas tocotrienols are limited to bran and germ fractions of certain seeds and cereals. In either case, nearly every diet will have a source of vitamin E. A reference to generic vitamin E would usually include products containing both tocopherols and tocotrienols. The difficulty of identifying a role for vitamin E has been that the lay press contains numerous articles claiming favorable actions for vitamin E, including improving wound healing without scarring, slowing down the ageing process, protecting against various cancers, preventing heart disease, and improving the performance of athletes. Many of these claims are not substantiated by scientific studies, although a common thread linking these claims is that vitamin E has the chemical property of being a fat-soluble antioxidant. Located in cell membranes, vitamin E could act by virtue of its high reactivity toward reactive oxygen species (ROS), thus protecting cellular components from oxidation (antioxidant). A diagram representing the action of an antioxidant such as vitamin E is shown in Figure 42.18. The tocopherols are thought to protect PUFA from ROS-catalyzed reactions that result in lipid

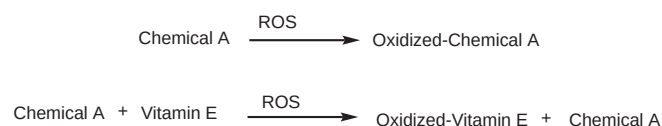
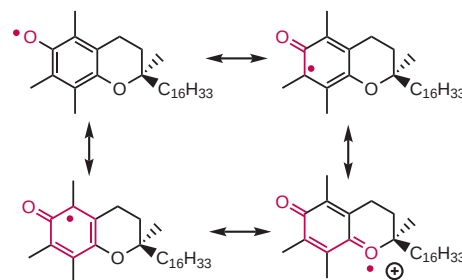


FIGURE 42.18 Vitamin E acting as an antioxidant to protect Chemical A.

peroxide radicals, which through a series of chain reactions damage PUFA.

MECHANISM OF ANTIOXIDANT ACTIVITY Vitamin E and especially α -tocopherol are substituted nicely to preferentially scavenge a lipid peroxyl radical. The mechanism and products of this reaction are shown in Figure 42.19. ROS can readily oxidize PUFA via abstraction of a hydrogen from the fatty acid, leaving behind a dienyl radical (44). This dienyl radical combines with oxygen to give a mixture of *cis/trans* lipid peroxyl radicals, which if left unchecked will lead to severe cellular damage through a series of hydrogen abstraction chain reactions with biomolecules (i.e., PUFA, DNA, RNA, etc.). In the presence of vitamin E, the lipid free radical abstracts a hydrogen from the phenolic group of α -tocopherol to produce a relatively stable tocopherol radical, thus effectively chain-breaking the auto-oxidation of the PUFA. α -Tocopherol is very efficient in acting as an antioxidant by virtue of the stability of the α -tocopherol radical, which results from the chemical substituents attached to the chroman ring system. Stabilization of the α -tocopherol radical is increased by the electron-donating *ortho* methyl groups, with only α -tocopherol having methyls in both *ortho* positions. Additionally, resonance stabilization by the *para* oxygen in the chroman

ring improved the stability of the α -tocopherol radical (45,46).



Resonance stabilization of α -tocopherol radical

The quenching of the α -tocopherol radical leads to formation of α -tocopherolquinone and the epoxytocopherones (Fig. 42.19).

ABSORPTION AND METABOLISM The absorption of vitamin E, both the tocopherols and the tocotrienols, occurs readily provided that two conditions are met. First, if vitamin E is administered as an ester, the ester must initially be hydrolyzed. This is easily done by esterases present in the intestine. Second, absorption is dependent on the presence of bile acids in the GI tract. Bile acids favor micelle

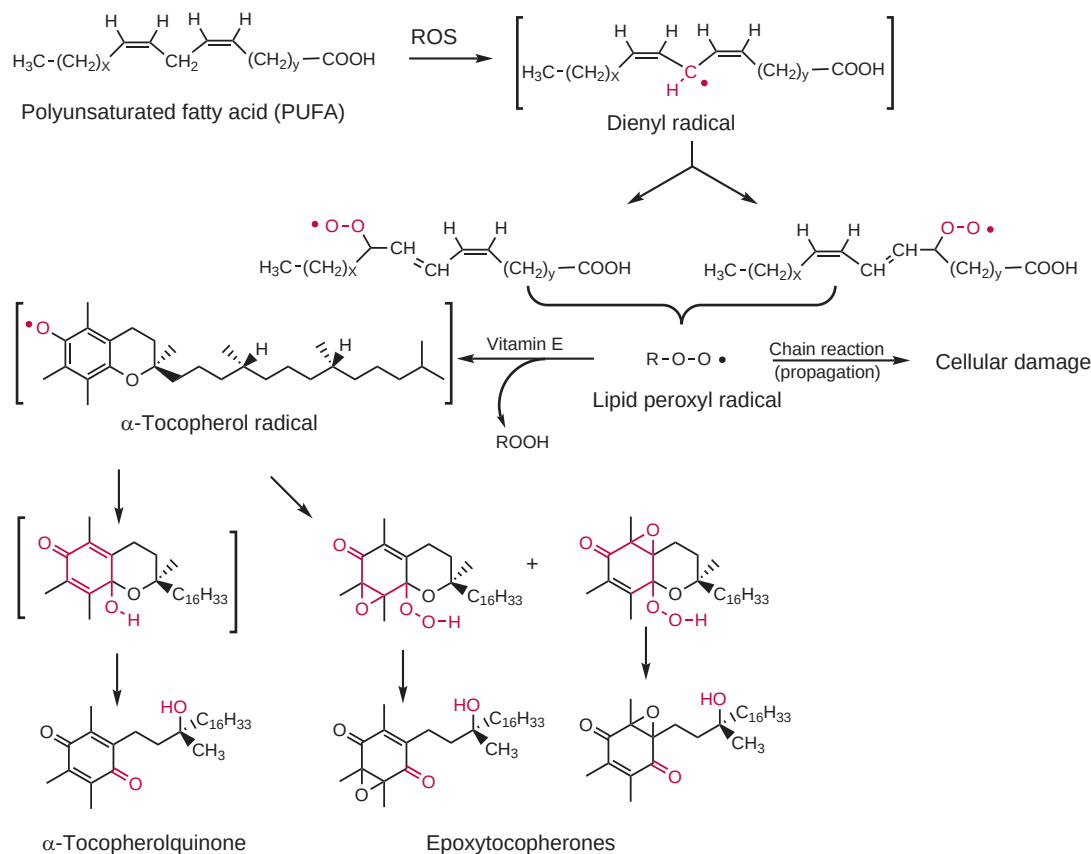
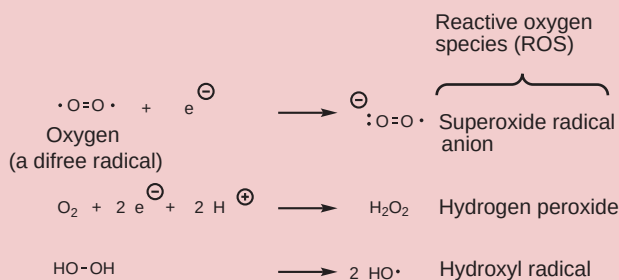


FIGURE 42.19 The action of vitamin E in decreasing cellular damage via reaction with PUFA free radicals.

REACTIVE OXYGEN SPECIES

A variety of chemicals found in the human body possess the chemical property of being an ROS. These include molecular oxygen, superoxide, hydrogen peroxide, and the hydroxy radical. A characteristic of an ROS is that it is highly reactive due to the presence of or ability to produce a single unshared valence shell electron, which is known as a free radical.



Free radicals such as molecular oxygen, superoxide radical anion, or hydroxy radical are very reactive and can attack cellular components leading to oxidation by abstracting an electron from the cellular component. DNA, RNA, protein, and PUFA are all prone to oxidative reactions catalyzed by the ROS. Generally, the body has sufficient enzymes, such as catalase, superoxide dismutase, and ubiquinone (coenzyme Q_{10}), along with natural antioxidants such as glutathione, vitamin E, vitamin C, vitamin A, and carotenes, to protect the cells against damage from ROS. Under oxidative stress, cellular damage may occur, and some theories suggest that chronic damage to key components of the cell may account for various cancers, cardiovascular disease, and even the ageing process.

formation, which is essential for absorption of these fat-soluble materials.

A general characteristic of the fat-soluble vitamins is their high lipophilicity and the potential for hypervitaminosis to occur; however in the case of vitamin E, hypervitaminosis is quite uncommon. This may be due to the fact that both the tocopherols and the tocotrienols are oxidized to more water-soluble chemicals that can be readily excreted from the body. The common metabolism consists of ω -oxidation of the lipophilic side chain of vitamin E followed by progressive shortening of the side chain through β -oxidation and the elimination of acetic and propionic acids (Fig. 42.20). The final products of these oxidations are the hydroxychromans as shown in Figure 42.20 (47–49). The ω -oxidation and apparently the β -oxidations are catalyzed by CYP4F2, and these series of reactions also occur with the tocotrienols, although the rates of these reactions and the amount of each of the metabolites may differ. It should be noted that whether the side chain of the vitamin E is saturated or unsaturated similar metabolic products are formed (how or when the alkene is reduced is not

presently known). Thus, the plasma level of the particular vitamin E derivative may differ depending on the tocopherols or tocotrienols that are administered, which in turn may affect the antioxidant benefit of vitamin E.

Vitamin K (Menadiones)

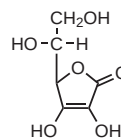
Vitamin K, the fourth of the fat-soluble vitamins, derives its name from the German word *koagulation* and is similar to vitamin E in that there is no single chemical entity that possesses the biologic activity attributed to vitamin K. The major activity associated with vitamin K is catalyzing a posttranslation carboxylation of several proteins that are involved in blood coagulation (see Chapter 26). The basic nucleus of all vitamin K compounds is the 2-methyl-1,4-naphthoquinone to which various chain lengths of the 5-carbon isoprenoid unit are attached at the 3-position (Fig. 42.21). Green plants produce phyloquinone or vitamin K_1 , which represents a major dietary source of the vitamin. Bacteria, some of which live in the intestine, produce various chain length methaquinones, where n varies from 4 to 9, that are abbreviated MK- n , and these compounds are referred to as vitamin K_2 . Animals, including man, but not bacteria, can synthesize MK-4 from phyloquinone as well as from menadione, which is a synthetic form of vitamin K.

SOURCES AND PROPERTIES OF VITAMIN K As previously indicated, vitamin K is found in leafy green vegetables such as collards, spinach, and salad greens, which have the highest concentrations (300 to 400 mcg/100 g), and from fats and oils such as soybean and canola oil (50,51). The RDA is rather low, being 1 mcg/kg body weight/day, while the DRI varies considerably from approximately 2 mcg/day for infants to 90 to 120 mcg/day for females and males between the ages of 19 and 70 years. One might expect that vitamin K deficiencies would be uncommon since various foods and intestinal bacterial flora are sources of this vitamin, and although this is generally true, the bioavailability of phyloquinone is in the range of 15% to 20%, and turnover of the vitamin occurs rapidly. The absorption of dietary vitamin K occurs via the lymphatic system, whereas the metabolism of vitamin K is not well characterized.

As discussed in Chapter 26, vitamin K is essential for the normal functioning of the blood clotting cascade. Additionally, vitamin K is thought to have a role in bone mineralization, although the exact role remains unclear.

Vitamin K toxicity has often been cited, and in fact, menadione is considered to be potentially toxic, a toxicity associated with interference with the functioning of glutathione. However, phyloquinone and the menaquinones have not been shown to produce toxic effects even at high doses.

Vitamin C (Ascorbic Acid)



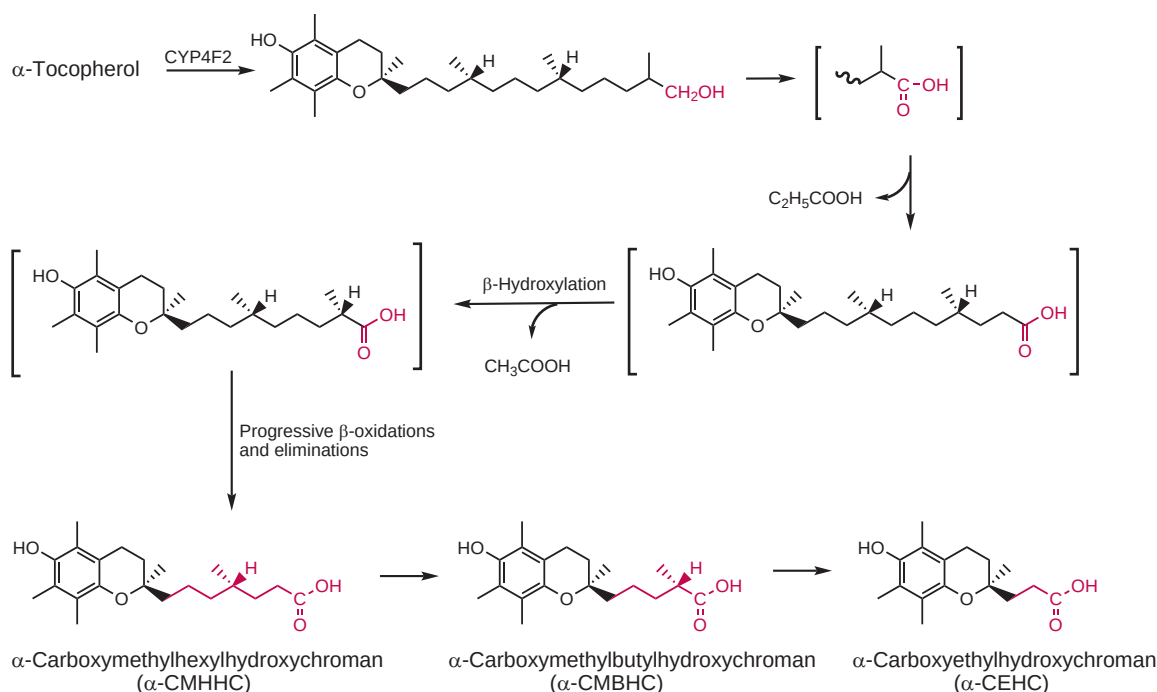


FIGURE 42.20 Metabolism of α -tocopherol.

Vitamin C, or ascorbic acid, is a water-soluble vitamin. The deficiency of vitamin C, known as scurvy, has a long history dating back to 3000 BC. Scurvy is a condition characterized by subcutaneous hemorrhage due to blood vessel rupture, commonly seen in the gingival, skin, and GI mucosa. With severe scurvy, loss of teeth, bone damage, internal bleeding, and infections occur, ultimately leading to death. Although the condition was described in the literature, it was not until the

1800s that its relationship to a chemical present in fruit juices became well known. It was another 100 years before Albert Szent-Gyorgyi identified and named the chemical in fruit juices that could prevent scurvy as ascorbic acid. Vitamin C is synthesized in plants from glucose and fructose, and most mammals can also synthesize this substance with the exception of humans, nonhuman primates, including guinea pigs, and several other animal species. Water-soluble vitamin C is often considered to be a vitamin that is not stored in the body and thus excessive levels will not build up in the body. Neither of these ideas is entirely correct. Vitamin C is stored in quantities sufficient to prevent scurvy for approximately 30 to 40 days, whereas large doses of vitamin C do have the potential for adverse effects (see below).

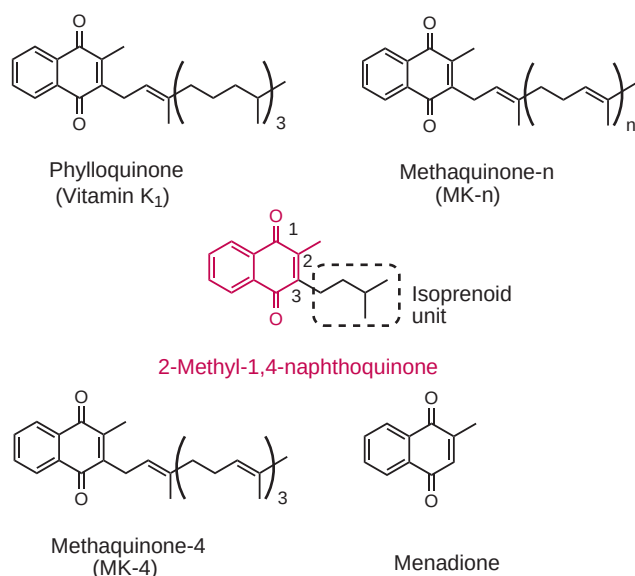


FIGURE 42.21 Structures of various vitamin K analogs.

BIOLOGIC ACTION Vitamin C is a water-soluble antioxidant by virtue of its ability to give up two electrons and in the process become oxidized (Fig. 42.22). The driving force for this activity resides in the fact that the ascorbyl free radical is stable and, with the loss of the second electron, ascorbic acid becomes dehydroascorbic acid, which following hydration is recycled back to ascorbic acid by reduction via glutathione. The relationship between the antioxidant property of vitamin C and scurvy is not known with certainty, but it is known that vitamin C is a cofactor in the hydroxylation of the amino acids proline and lysine, which are essential components of collagen. The hydroxylated amino acids have a role in stabilizing the helix structure of collagen,

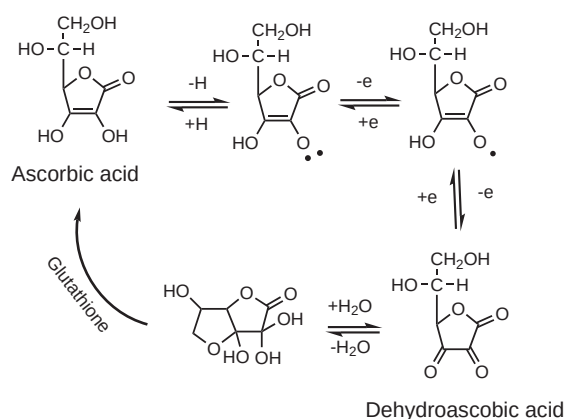


FIGURE 42.22 Antioxidant property of ascorbic acid.

and therefore, deficiency of vitamin C may impair the proper function of collagen. Collagen is part of the structural make up of connective tissue; thus vitamin C deficiency leads to gum deterioration and bleeding, adverse effects on bone integrity leading to teeth loss, and skin discoloration.

The antioxidant property of vitamin C has led to many suggestions as to additional biologic activity for vitamin C, which includes benefits in prevention of cardiovascular diseases. The anthrogenic properties of LDL are thought to relate to the oxidation of LDL to a peroxy radical, which then plaques onto the walls of blood vessels. Vitamin C may inhibit the initial oxidation. Increased consumption of vitamin C has been reported to correspond to a reduction in ischemic heart disease and a reduction in hypertension. Increased plasma levels of vitamin C, associated with increased consumption of fruits and vegetables (common sources of vitamin C), have been linked with a decrease in various cancers including oral cavity, esophagus, stomach, colon, and lung cancers. Suppression of ROS or radical intermediates serves as the basis for these claims. Doses in the gram quantity are often recommended for these actions, but clinical data are normally absent or complicated by other factors such as the benefits derived from simply increasing the intake of fruits and vegetables in the diet. Linus Pauling championed the concept that vitamin C was beneficial in treating some terminally ill cancer patients and could prevent or promote the recovery from the common cold. Data did not support the use of doses of up to 10 g/day for terminally ill cancer patients. As to the benefit of vitamin C in treatment of the common cold, it is felt that vitamin C will not prevent the common cold but may hasten the recovery from the cold. There is some evidence showing that vitamin C may increase the body's production of interferons, which in turn possess antiviral activity, thus supporting the claim for benefit in the common cold and potentially the treatment of viral-associated cancers.

HYPERVITAMINOSIS C The RDA for vitamin C is in the range of 60 to 75 mg/day, with higher amounts recommended for women during pregnancy and lactation and for smokers. Vitamin C-containing products are available with daily doses in the range of the RDAs all the way up to products that contain 1,000 mg per tablet or capsule. With such potent products available to the consumer, it is not uncommon for the patient to be consuming excessive quantities of vitamin C. High doses of vitamin C have been associated with increased oxalate excretion, with the potential for oxalate-based kidney stones, and will lower urine pH, affecting drug excretion. High doses, in the range of 3 g/day, may cause diarrhea and bloating, while even normal doses of vitamin C increase iron absorption, which in some individuals may cause health issues. Vitamin C prevents intestinal oxidation of ferrous ion to ferric ion, which is poorly absorbed from the GI tract.

Although vitamin C is an essential component in our diet, its role in good health and the biochemistry of the body is still not clear, and additional benefits can be expected to be elucidated in the future.

Sodium Chloride (Table Salt, NaCl)

The human body is composed of both organic and inorganic compounds. The inorganic nutrients generally are considered under the title of micronutrients and, just as with the organic micronutrients, are too numerous to cover in this chapter. The essential minerals include calcium, chromium, copper, iodine, iron, magnesium, manganese, molybdenum, phosphorus, selenium, and zinc, along with the essential electrolytes of sodium, potassium, and chloride. This discussion will be limited to sodium and chloride ions.

DIETARY ROLE Sodium ion, most commonly consumed in the form of sodium chloride, is essential for various roles in the body. What is not known is just how much sodium chloride represents the minimum amount needed for good health. It is estimated that Americans consume approximately 10 to 12 g/day, which qualifies it as the second most common food additive (~15 lb/year). There is no RDA set for sodium chloride, although it is thought that an intake of less than 1,000 mg/day would meet the needs for a normal adult. The adequate intake and upper limit for sodium chloride are given as approximately 3.8 and 9.1 g/day, respectively, which would result in 1.5 to 3.6 g/day of sodium and 2.3 to 5.5 g/day of chloride (39.6% of sodium chloride is sodium and 60.4% of sodium chloride is chloride). The 2005 dietary guideline for sodium chloride consumption was established at 5.8 g/day (~1 teaspoon), or 2.3 g/day of sodium.

The human appetite for sodium chloride has both a physiologic and an educated basis. Taste receptors in the oral cavity can detect a salty taste presumably as a means of encouraging and assuring the intake of sodium

chloride. But the human also has an educated appetite for salt. Commercially processed food commonly has added sodium, usually in the form of sodium chloride, as a preservative but also to improve the taste. It is estimated that as much as 65% to 85% of the daily consumed sodium chloride is present as “hidden salt” in processed food. In the case of baby food, sodium chloride may be added to meet the need to satisfy the parent who tastes the food rather than to meet a nutritional need of the child. As a result, the child grows up with a conditioned belief that a specific food must have a certain degree of saltiness, whether natural or “man enhanced.” Many adults salt their food prior to tasting it because again they have a belief that the food must meet an expected level of saltiness.

The concern about sodium intake is not a new concern as indicated by the White House Conference on Food, Nutrition, and Health, which issued a report in which the role of sodium in hypertension was emphasized. Since this report, additional reports and studies have appeared, all of which suggest that Americans need to reduce their intake of sodium in the form of sodium chloride and that high sodium intake is associated with elevated blood pressure, heart disease, stroke, congestive heart failure, and renal disease. In the 1980s, the FDA made an attempt to sensitize the public to this growing public health issue, which at best resulted in a public awareness of sodium intake from processed foods and resulted in some manufacturers producing low-salt products, but in general attempts to actually reduce salt intake have failed. These efforts by the FDA were directed primarily at consumer education, and it was estimated that only about 30% of the consumers actually attempted to reduce sodium intake. In 2009, the Institute of Medicine Committee on Strategies to Reduce Sodium Intake convened and issued their report in 2010 (52). This comprehensive report indicates that although dietary sources of sodium are plentiful, a major contributor to sodium intake is processed and restaurant foods, whereas sodium added to food from the salt shaker amounts to only 5% of the sodium consumed. The report reached the conclusion that standards for sodium intake need to be set by the FDA and that a progressive decrease in sodium content of processed food and restaurant food needs to occur but that this reduction needs to be done in a way acceptable to the consumer. It is also noted that sodium does appear in the GRAS list, giving the FDA authority to set standards, but that these standards need to be done within the confines of GRAS and that sodium should not be removed from GRAS. A total of five recommendations were made (39), and basically the GRAS status of salt needs to be changed

over time and in cooperation with the food industry. Throughout the changes in sodium content in foods, efforts need to be made to educate the public to the need for lowering sodium intake so as to lead to successful behavioral changes in the taste expectations of the public.

DIETARY SUPPLEMENTS

Two legislative acts that were passed by Congress in the 1990s have impacted nutrition in the United States. These are the NLEA of 1990 and the DSHEA of 1994. The NLEA allows companies, through petition to the FDA, to label their product with health claims linking food or food components to a reduced risk of a specific disease. The petitioning company is responsible for presenting evidence to support its claims. Among the claims that have appeared since this legislation went into effect are the following:

- Low sodium and a reduced risk of hypertension
- Calcium for the treatment of osteoporosis
- Fiber containing grains and a reduction in the risk of cancer
- Fruits, vegetables, and grains and the prevention of coronary heart disease
- Folic acid and the prevention of neural tube defects
- Soy protein and the prevention of coronary heart disease
- Whole-grain foods and the prevention of specific cancers
- Dietary lipids as the cause of cancer
- Dietary sugar alcohols and protection of dental caries
- Soluble fibers and the prevention of heart disease

The second act, DSHEA, allows companies to introduce dietary supplements without FDA approval provided “there is a history of use or other evidence to establishing safety.” This legislation allows “structure-function” claims but not health claims without supportive evidence. The manufacturer must show safety of ingredients. A dietary supplement is a product that contains dietary ingredients, is taken by mouth, and is intended to supplement one’s diet. Such products include vitamins, minerals, herbs, enzymes, amino acids, and metabolites.

As a result of these acts, various classes of nutrients have appeared on the market including nutraceuticals, functional foods, and the official use of foods titled as “organic foods.” Nutraceuticals are defined as dietary supplements, foods, or medical foods that possess health benefits and that are safe for human consumption in such quantities and such frequency as required to realize beneficial properties. A functional food is a product that is conveyed in a conventional food format and has health benefits. Through these acts, various natural products have appeared as dietary supplements including S-adenosylmethionine (SAM-e), plant stanols, soy protein in “milk” products, and a number of herbal products being sold as dietary supplements.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

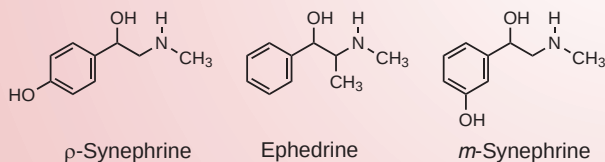
Beverly Lukawski, PharmD, and Jennifer Campbell, PharmD

JL was diagnosed with exercise-induced syncope. After noting no previous medical or family history of cardiac conditions, the pharmacist recommended that she stop taking Kwik Loss and discussed the benefits a low-calorie, high-fiber diet and exercise.

Chemical Analysis

Victoria Roche and S. William Zito

Bitter orange contains an ephedrine-like alkaloid known as *p*-synephrine. Unlike ephedrine, *p*-synephrine has a phenolic OH group located *para* to the ethanolamine side chain, which prevents its entry into the CNS. Because it cannot reach α -adrenoceptors in the brain, it cannot stimulate them to induce an anorexic effect. It also cannot act indirectly to increase central levels of the endogenous α -receptor agonist norepinephrine. Therefore, this compound would have no value in a product designed to induce weight loss.

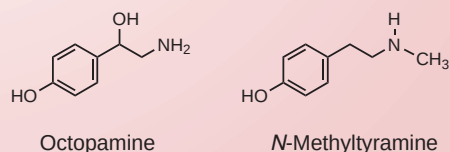


However, *p*-synephrine does have all the structural features required for direct stimulation of β -adrenergic receptors in the periphery. These features include a potentially cationic amine (ion-ion anchoring to Asp), benzylic OH (H donor in a hydrogen bond to Asn), and an N-CH_3 substituent (hydrophobic interaction with several residues within a spacious β -receptor lipophilic cleft). Although it is missing the catechol nucleus

found on the most potent direct-acting adrenoceptor stimulators, the *p*-hydroxyl group is known to be most important for high β -receptor affinity. Stimulation of the β_1 receptors of the heart can induce tachycardia and an increased force of myocardial contraction. The enhanced cardiac output can elevate blood pressure, albeit not as significantly as α -receptor stimulation. It is the β_1 -receptor stimulation of *p*-synephrine that is likely responsible for JL's rapid heart rate and (perhaps with some weak, residual α -receptor stimulation) her moderate, and hopefully transient, hypertension.

In contrast, a *m*-phenol selectively directs phenylethanolamines to α -adrenoceptors. *m*-Synephrine is commercially available as phenylephrine (Neo-Synephrine). Its vasoconstrictive properties make it useful as a nasal decongestant when administered locally (nasal spray) or orally, and it restores blood pressure in hypotensive crises when administered intravenously. *m*-Synephrine is not found naturally in bitter orange.

The other phenolic alkaloids associated with bitter orange are octopamine and *N*-methyltyramine. While extracts of bitter orange contain *N*-methyltyramine concentrations that are approximately 6% of *p*-synephrine's, the nutraceutical dosage form has concentrations of octopamine and *N*-methyltyramine that are 0.8% and 3.1% of the *p*-synephrine concentration, respectively (53). It is clear from their structures that both octopamine and *N*-methyltyramine would have activities similar to those exhibited by *p*-synephrine, and to the extent they are present, they would have contributed to JL's clinical presentation.



CASE STUDY

Victoria Roche and S. William Zito

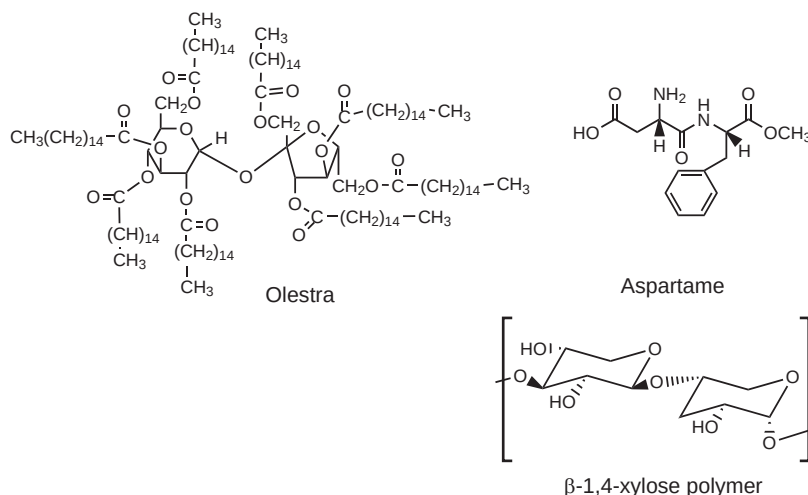
BW is a 58-year-old woman who has been approximately 30 pounds overweight for as long as you, her neighborhood pharmacist, have known her. She sees you regularly because you are a Certified Diabetes Educator and BW, a type 2 diabetic, is taking a glipizide/metformin combination product in an attempt to stay off insulin for as long as possible. She does pretty well on this therapy, although she occasionally experiences drug-induced loose stools.

BW has always had good intentions of losing weight and following your wise counsel for modifying her diet and increasing her level of physical activity. She even bought a membership to the local gym. However, she is self-conscious about being seen in exercise wear and always finds excuses to not go. In addition, she has an affinity for salty snacks (especially Kettle chips) that she knows are really bad for her, but she says she cannot help eating them when she feels hungry. Unfortunately, she generally enjoys a sugared soda when munching on those chips.

Today you are offering BW your condolences on the death of her father, who recently died from an aggressive form of

colon cancer. Knowing the genetic implications of this cancer (and many others), BW says she is now more determined than ever to regain and maintain good health and to reach her target weight by exercising and eating right. She asks your advice about a natural way to “jump start” her weight loss initiative so she can achieve some initial success in the first few months of her new lifestyle. Specifically, she asks about snack foods that contain olestra, switching to diet soda (which contains aspartame), or taking Metamucil fiber supplement, which contains a hemicellulose comprised of β -1,4-xylose units substituted with arabinose and additional xylose sugars. Which of those would you recommend to help BW on her way to a sustainably healthy lifestyle?

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.



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APPENDIX A. pK_a and CLogP Values for Some Drugs and pK_a Values for Miscellaneous Organic Acids and Bases

DAVID A. WILLIAMS

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Acebutolol		9.2	1.77	Amoxapine		7.6	1.89
Acenocoumarol	4.7		3.24	Amoxicillin	2.4	9.6	-1.99
Acetaminophen	9.7		0.8	Amphetamine		10.0	1.76
Acetanilide		0.5	1.14	Amphotericin B	5.5	10.0	2.30
Acetazolamide	7.4, 9.1		-0.26	Ampicillin	2.5	7.2	-1.28
α -Acetylmethadol		8.6	3.03	Amprenavir	1.76, 11.54		2.68
Acetylsalicylic acid	3.5		1.19	Anileridine		3.7, 7.5	2.96
Acyclovir	9.3	2.3	-1.47	Antazoline		2.5, 10.1	3.58
Adriamycin		8.2	0.10	Antipyrine		1.5	0.28
Ajmaline		8.2	1.81	Apomorphine	8.9	7.0	1.70
Albuterol	10.3	9.3	0.69	Apraclonidine		9.22	0.87
Alclofenac	4.3		2.48	Aprobarbital	8.0		1.28
Alfentanil		6.5	2.16	Ariprazole		7.6	3.76
Alfuzosin		8.13	1.26	Ascorbic acid	4.2, 11.6		-1.64
Allobarbital	7.8		0.94	Atomoxetine		9.23	3.36
Allopurinol	9.4		-0.14	Atenolol		9.6	0.33
Alphaprodine		8.7	2.49	Atropine		9.8	1.83
Alprenolol		9.7	2.89	Azatadine		9.3	2.69
Altretamine		10.3	2.52	Azathioprine	8.0		0.02
Amantadine		9.0	2.44	Azlocillin	2.8		0.83
Amiloride		8.7	0.93	Aztrenam	0.7, 2.9	3.9	0.49
Aminoacrine		10.0	2.74	Bacampicillin		6.8	3.52
p-Aminobenzoic acid	4.9	2.5	0.78	Baclofen	5.4	9.5	-0.96
Aminocaproic acid	4.4	10.8	-2.70	Barbital	7.9		0.70
Aminopterin	5.5		-0.9	Bendroflumethiazide	8.5		1.95
Aminopyrine		5.0	0.8	Benzocaine		2.5	1.86
p-Aminosalicylic acid	3.6	1.8	1.24	Benzphetamine		6.6	3.84
Aminothiadiazole		3.2	0.38	Benzquinamide		5.9	0.46
Amiodarone		6.6	7.81	Benztropine		10.0	4.03
Amitriptyline		9.4	4.41	Betaprodine		8.7	2.49
Amlodipine		8.7	3.01	Bethanidine		10.6	1.73
Amobarbital	7.8		1.58	Bitolterol			4.16

(Continued)

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases (Continued)

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Brimonidine		7.78	1.20	Cephacetrile			-1.60
Bosentan	5.46		4.36	Cephalexin	3.2		0.35
Bromazepam	11.0	2.9	1.69	L-Cephaloglycin	4.6	7.1	-0.38
Bromocriptine	15	6.6	-	Cephaloridine	3.4		-2.10
Bromodiphenhydramine		4.9	4.03	Cephalothin	2.5		-0.41
Brompheniramine		3.6, 9.8	3.24	Cephapirin			-0.41
Brucine		8.2, 2.5	1.41	Cephadrine			0.48
Bufuralol		8.9	3.54	Cetirizine	2.9	8.3	1.62
Bumetanide	5.2, 10.0		2.88	Cevimeline		8.5	2.44
Bunolol		9.3	1.96	Chloral hydrate	10.0		0.99
Bupivacaine		8.1	3.31	Chlorambucil	5.8		2.61
Buprenorphine		8.42	2.83	Chlorcyclizine		2.1, 8.2	3.24
Bupropion		7.0	2.32	Chlordiazepoxide		4.8	2.49
Butabarbital	7.9		1.67	Chlorhexidine		10.8	4.57
Butacaine		9.0	4.62	Chloroquin		8.1, 9.9	4.41
Butaclamol		7.2	3.81	Chlorothiazide	6.8, 9.5		-0.24
Butamben		5.4	2.85	Chlorpheniramine		9.0	2.97
Butorphanol		8.6	3.54	Chlorphentermine		9.6	2.78
Butylated hydroxytoluene	7.5		5.17	Chlorpromazine		9.3	5.35
Butylparaben	8.5		3.57	Chlorpropamide	4.9		2.30
Caffeine	>14.0	0.6	-0.63	Chlorprothixene		8.8, 7.6	5.31
Camptothecin		10.8	0.95	Chlortetracycline	3.3, 7.4	9.3	1.32
Captopril	3.7, 9.8		1.99	Chlorthalidone	9.4	1	0.70
Carbenicillin	2.7		1.13	Chlorzoxazone	8.3		2.25/1.82 lactam
Carbenoxolone	6.7, 7.1		6.63	Cimetidine		6.8	0.56
Carbidopa	7.8		-1.75	Cinchonine		4.3, 8.4	3.69
Carbinoxamine		8.1	2.69	Ciprofloxacin	6.0	8.8	1.63
Carisoprodol		4.2	2.10	Citalopram		9.59	3.47
Carpindolol		8.8	2.93	Clindamycin		7.5	1.75
Carvedilol		7.8	4.11	Clofibrate	3.5		3.88
Carteolol		9.74	1.34	Clonazepam	10.5	1.5	2.41
Cefaclor	1.5	7.2	0.14	Clonidine		8.3	1.52
Cefamandole	2.7		0.25	Clopendithiol		6.7, 7.6	3.91
Cefazolin	2.1		-0.70	Clopidogrel		4.55	2.50
Cefoperazone	2.6		-1.11	Clotrimazole		4.7	4.92
Cefotaxime	3.4		-0.51	Cloxacillin	2.8		2.83
Cefoxitin	2.2		-0.02	Clozapine		3.7, 7.4	3.94
Ceftazidime	1.8, 2.7	4.1	-2.50	Cocaine		8.7	2.28
Ceftizoxime	2.7	2.1	-0.65	Codeine		8.2	1.39
Ceftriaxone	3.2, 4.1	3.2	-0.54	Colchicine		1.9	1.07
Cefuroxime			0.26	Cromolyn	1.1, 1.9		2.00
Celecoxib	11.2		2.59	Cyclacillin	2.7	7.5	1.21

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases (Continued)

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Cyclamate	1.9		1.03	Diethazine		9.1	5.55
Cyclazocine		9.4	3.52	Diethylcarbamazepine		7.7	1.66
Cyclizine		8.0, 2.5	2.47	Diflunisal	3.0		3.65
Cyclobarbital	8.6		1.50	Dihydroergocryptine		6.9	6.37
Cyclobenzaprine		8.5	6.19	Dihydroergocristine		6.9	6.55
Cyclopentamine		11.5	2.46	Dihydroergotamine		6.9	5.69
Cyclopentolate		7.9	0.99	Dihydrostreptomycin		7.8	-1.49
Cycloserine		4.5, 7.4	-2.99	Dilevalol		9.5	3.09
Cyclothiazide	9.1, 10.5		1.73	Diltiazem		7.7	4.73
Cyproheptadine		8.9	4.92	Dimethadione	6.1		0.77
Cytarabine		4.3	-2.17	Dimethisoquin		6.3	4.04
Dacarbazine		4.4	-0.24	Dinoprost	4.9		2.59
Dantrolene	7.5		1.07	Dinoprostone	4.6		2.97
Dapsone		1.3, 2.5	0.99	Diperodon		8.4	4.65
Darifenicin		9.20	3.78	Diphenhydramine		9.1	3.27
Daunorubicin		8.4	0.54	Diphenoxin			3.97
Debrisoquin		11.9	0.75	Diphenoxylate		7.1	4.51
Dehydrocholic acid	5.12		1.89	Diphenylpyraline		8.9	3.43
Demeclocycline	3.3, 7.2	9.4	0.94	Dipipanone		8.5	5.10
Demoxepam		4.5, 10.6	0.73	Dipyridamole		6.4	3.35
Deprenyl		7.4	2.68	Disopyramide	10.2	8.4	2.33
Deserpidine		6.7	4.95	Donepezil		9.1	3.91
Desipramine		10.4	3.97	Dobutamine		9.5	2.53
Desloratadine		4.2, 9.7	3.50	Dopamine	10.6	8.9	0.05
Desvenlafaxine				Dozazosin		6.93	2.85
Dextroamphetamine		9.9	1.76	Doxepin		9.0	3.85
Dextrobrompheniramine		9.3	3.24	Doxorubicin		8.2, 10.2	0.24
Dextrochlorpheniramine		9.2	2.97	Doxycycline	3.4, 7.7	9.5	1.78
Dextrofenfluramine		9.1	3.55	Doxylamine		4.4, 9.2	2.34
Dextroindoprofen	4.6		2.82	Droperidol		7.6	3.10
Dextromethorphan		8.3	3.89	Duloxetine		9.34	4.81
Dextromoramide		7.0	2.53	Emetine		8.2, 7.4	3.82
Diacetylmorphine (heroin)		7.8	1.58	Enalapril	3.0	5.5	3.25
Diatrizoic acid	3.4		0.49	Enalaprilat	2.3, 3.4	8.0	3.63
Diazepam		3.4	2.82	Entacapone	4.5, 10.72		3.02
Diazoxide	8.5		1.09	Ephedrine		9.6	1.43
Dibenzepin		8.3	3.26	Epinephrine	8.9	10.0	-0.54
Dibucaine		8.9	4.40	Ergometrine		7.3	2.48
Dichlorphenamide	7.4, 8.6		0.34	Ergonovine		6.8	2.48
Diclofenac	4.5		4.55	Ergotamine		6.4	7.37
Dicloxacillin	2.8		3.10	Erythromycin		8.8	2.48
Dicoumarol	4.4, 8.0		2.05	Esmolol		9.5	1.92
Dicyclomine		9.0	4.64	Estrone	10.8		3.62

(Continued)

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases (Continued)

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Ethacrynic acid	3.50		2.84	Guanethidine		8.3, 11.9	1.10
Ethambutol		6.3, 9.5	-0.29	Guanabenz		8.1	2.66
Ethoheptazine		8.5	2.41	Guanoxan		12.3	0.83
Ethopropazine		9.6	4.77	Haloperidol		8.3	3.76
Ethosuximide	9.5		0.25	Hexobarbital	8.2		1.46
Ethyl biscoumacetate	7.5		1.61	Hexylcaine		9.1	3.65
Ethylmorphine		8.2	1.90	Hexylresorcinol	9.5		3.77
Ethylnorepinephrine		8.4	-0.40	Homatropine		9.7	1.30
Etidocaine		7.9	3.57	Hycanthone		3.4	3.81
Etomidate		4.2	3.05	Hydralazine		0.5, 7.1	1.00
Eugenol	9.8		2.40	Hydrochlorothiazide	7.0, 9.2		-0.02
Fenclofenac	4.5		4.59	Hydrocodone		8.9	2.57
Fenfluramine		9.1	3.55	Hydrocortisone	-		1.76
Fenoterol	10.0	8.6	1.33	Hydroflumethiazide	8.9, 10.7		-0.07
Fenpropfen	4.5		3.72	Hydromorphone		8.2	2.13
Fentanyl		8.4	3.68	Hydroxyamphetamine		9.3	1.14
fesoterodine		9.28	5.08	Hydroxyzine		2.0, 7.1	2.32
Fexofenadine	4.25	9.53	3.73	Hyoscyamine		9.7	1.79
Finasteride		12.4	3.83	Ibuprofen	5.2		3.50
Flurbiprofen	4.3		3.66	Imipramine		9.5	4.35
Flucloxacillin	2.7		2.89	Indacaterol			3.88
Flufenamic acid	3.9		5.22	Indapamide	8.8		1.96
Flumizole	10.7		4.26	Indomethacin	4.5		4.25
Flunitrazepam		1.8	2.13	Indoprofen	5.8		2.82
Fluoxetine		8.7	3.93	Indoramin		7.7	2.87
Flupenthixol		7.8	3.67	Iocetamic acid	4.1 or 4.3		4.57
Fluphenazine enanthate		3.5, 8.2	7.29	Iodipamide	3.5		5.10
Fluphenazine		3.9, 8.1	3.92	Iodoquinol	8.0		4.10
Flurazepam	8.2	1.9	4.84	Iopanoic acid	4.8		4.65
Flutamide	13.12		3.52	Iprindole	8.2		5.02
Formoterol	7.9	9.2	2.03	Iproniazid		8.1	0.41
Furosemide	3.9		2.30	Ipronidazole		2.7	0.96
Fusidic acid	5.4		5.76	Irbesartan	4.9		5.25
Fluvoxamine		9.39	3.71	Isocarboxazid		10.4	1.63
Gabapentin	3.7	10.7	1.08	Isoniazid		2.0, 3.5, 10.8	-0.77
Galantamine		8.2	-0.05	Isoproterenol	10.1, 12.1	8.6	0.32
Gentamicin		8.2	-3.55	Isoxsuprine	9.8	8.0	2.80
Glibenclamide	5.3		3.08	Kanamycin		7.2	-
Glipizide	5.9		1.88	Ketamine		7.5	3.01
Glutethimide	9.2		1.75	Ketobemidone		8.7	0.80
Glyburide	5.3		3.08	Ketoconazole		2.9, 6.5	4.04
Glycylamine		5.5	2.81	Ketoprofen	4.8		2.91
Guanfacine		7.1	1.33	Labetalol	8.7	7.4	2.72

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases (Continued)

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Lansoprazole		3.9	2.58	Methenamine		4.8	-0.17
Leucovorin	3.1, 8.1, 10.4		-1.43	Methicillin	3.0		1.02
Levallorphan	4.5	6.9	3.85	Methohexital	8.3		2.01
Levobunolol		9.2	1.96	Methotrexate	3.8, 4.8	5.6	-0.45
Levodopa	2.3, 9.7, 13.4	8.7	-1.15	Methotrimeprazine		9.2	4.94
Levonordefrin	9.8	8.6	-0.91	Methoxamine		9.2	0.61
Levorphanol		9.2	3.26, 3.26	Methoxyphenamine		10.1	2.12
Lidocaine		7.8	2.20	Methyclothiazide	9.4		1.42
Lincomycin		7.5	0.72	Methyl nicotinate		3.1	0.79
Liothyronine	8.4		3.91	Methyl paraben	8.4		1.88
Lisinopril	1.7, 3.3, 11.1	7.0	3.47	Methyl salicylate	9.9		2.52
Loperamide		8.6	4.15	Methyldopa	2.3, 10.4, 12.6	9.2	-0.74
Loratadine		5.0, 9.8	3.90	Methylergonovine		6.6	2.99
Lorazepam	11.5	1.3	2.35	Methylphenidate		8.8	2.31
Losartan	4.9, 4.7		3.46	Methylthiouracil	8.2		0.14
Loxapine		6.6	2.41	Methypylon	12.0		0.65
Lysergide		7.5	2.71	Methysergide		6.62	1.81
Maprotiline		10.2	4.36	Metoclopramide		0.6, 9.3	2.16
Mazindol		8.6	2.36	Metolazone	9.7		3.16
Mecamylamine		11.2	2.98	Metopon		8.1	2.41
Meclizine		3.1, 6.2	5.28	Metoprolol		9.88	1.63
Meclofenamic acid	4.0		5.44	Metronidazole		2.6	-0.14
Medazepam		6.2	3.89	Metyrosine	2.7, 10.1		00.01
Mefenamic acid	4.2		4.83	Mexiletine		9.1	2.12
Memantine		10.8	3.0	Mezlocillin	2.7		0.33
Mepazine		9.3	5.04	Miconazole		6.7	4.97
Meperidine		8.7	2.19	Midazolam		6.2	3.80
Mephentermine		10.4	2.61	Minocycline	2.8, 5.0, 7.8	9.5	2.12
Mephobarbital	7.7		1.23	Minoxidil		4.6	1.62
Mepindolol		8.9	2.30	Mitomycin		10.9	-0.30
Mepivacaine		7.7	1.78	Molindone		6.9	-1.32
Mercaptopurine	7.8	11.0	-0.04	Montelukast	5.7		5.81
Mesalamine	2.7	5.8	0.74	Morphine	9.9	8.0	0.87
Metaproterenol	11.8	8.8	0.28	Morpholine		8.33	-0.79
Metaraminol		8.6	0.25	Moxalactam	2.5, 7.7, 10.2		-2.56
Methacycline	3.5, 7.6	9.5	1.35	Nabilone	13.5		7.25
Methadone		8.3	3.93	Nadolol		9.67	0.56
Methamphetamine		10.0	2.20	Nafcillin	2.7		2.60
Methapyrilene		3.7, 8.9	2.78	Nalbuphine	10.0	8.7	1.21
Methaqualone		2.5	2.50	Nalidixic acid	6.0		0.03
Metharbital	8.2		1.50	Nalorphine		7.8	1.59
Methazolamide	7.3		0.13	Naloxone		7.9	1.78
Methdilazine		7.5	4.64	Naltrexone		8.13	2.05

(Continued)

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases (Continued)

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Naphazoline		10.9	2.99	Pentazocine	10.0	8.5	4.15
Naproxen	4.2		2.88	Pentobarbital	8.1		2.18
Nebivolol		7.1	4.08	Pentoxiphylline		0.3	0.29, 0.29
Nelfinavir	7.53	9.58	7.28	Pergolide		7.8	3.90
Niacin/nicotinic acid	2.0	4.8	0.22	Perphenazine		3.7, 7.8	3.94
Nicotinamide		0.5, 3.4	-0.37	Perhexilene			6.46
Nicotine		3.1, 8.0	0.57	Phenacetin		2.2	1.66
Nikethamide		3.5	0.92	Phencyclidine		8.5	4.25
Nitrazepam	10.8	3.2	2.35	Phendimetrazine		7.6	1.64
Nitrofurantoin	7.1		-0.47	Phenethicillin	2.8		2.29
Norcodeine		5.7	0.52	Phenindamine		8.3	3.81
Nordefrin	9.8	8.5	-0.91	Phenindione	4.1		3.13
Norepinephrine	9.8, 12.0	8.6	-1.26	Pheniramine		4.2, 9.3	2.20
Norfenephine		8.7	-0.60	Phenmetrazine		8.5	1.74
Normorphine		9.8	0.00	Phenobarbital	7.4		0.53
Nortriptyline		9.7	3.97	Phenolphthalein	9.7		1.91
Noscapine		6.2	2.38	Phenothiazine		2.5	4.15
Novobiocin	4.3, 9.1		3.74	Phenoxybenzamine		4.4	3.69
Nystatin	8.9	5.1	2.46	Phentermine		10.1	2.20
Octopamine	9.5	8.9	-0.59	Phentolamine		7.7	4.08
Olanzapine		5.0, 7.4	3.08	Phenylbutazone	4.5		3.38
Omeprazole		4.0	2.36	Phenylephrine	10.1	8.8	0.12
Ondansetron		7.7	1.55	Phenylpropanolamine		9.4	0.36
Orphenadrine		8.4	3.33	Phenylsalicylate	2.98, 8.80		2.75
Oxacillin	2.7		2.55	Phenyltoloxamine		9.1	3.46
Oxazepam	11.6	1.8	2.22	Phenylramidol		5.9	2.32
Oxprenolol		9.5	2.15	Phenytol	8.3		2.47
Oxybutynin		7.0	5.05	Physostigmine		2.0, 8.2	1.27
Oxycodone		8.9	1.59	Pilocarpine		1.6, 7.1	-0.24
Oxymorphone	9.3	8.5	1.15	Pimozide		7.3, 8.6	5.76
Oxyphenbutazone	4.7		3.28	Pindolol		8.8	1.68
Oxypurinol	7.7		0.18	Piperazine		5.6, 9.8	-1.50
Oxytetracycline	3.3, 7.3	9.1	0.48	Pipradrol		9.7	3.61
Pamaquine		1.3, 3.5, 10.0	4.38	Pirbuterol		3.0, 7.0, 10.3	0.04
Papaverine		6.4	2.95	Piroxicam	4.6		0.59
Pantoprazole	8.19	3.8	1.48	Pivampicillin		7.0	3.88
Pargyline		6.9	1.68	Polythiazide	9.8		2.05
Pemoline		10.5	0.42	Practolol		9.5	0.59
Penbutolol		9.3	4.02	Pramipexole		10.8	2.35
Penicillamine	1.8, 10.5	7.9	0.85	Pramoxine		6.2	2.59
Penicillin G	2.8		1.92	Prasugrel		5.4	4.31
Penicillin V	2.7		1.94	Prazepam		2.9	3.70
Pentamidine		11.4	2.85	Prazosin		6.5	2.14

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases (Continued)

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Pregabalin	4.2	10.6	1.09	Serotonin	9.8	4.9, 9.1	0.54
Prenalterol	10.0	9.5	1.27	Sertraline		9.5	5.08
Prilocaine		7.9	2.03	Silodosin			2.52
Probenecid	3.4		2.51	Solifenacin		6.44	3.70
Procainamide		9.2	1.32	Sotalol	8.5	9.8	0.24
Procaine		8.8	2.26	Spiperone		8.3, 9.1	3.25
Procarbazine		6.8	0.06	Streptozocin		1.3	-1.23
Prochlorperazine		3.7, 8.1	4.65	Strychnine		2.3, 8.0	1.93
Promazine		9.4	4.69	Sufentanil		8.0	3.95
Promethazine		9.1	4.89	Sulfacetamide	5.4	1.8	-0.96
Proparacaine		3.2	3.46	Sulfadiazine	6.5	2.0	0.12
Propoxycaine		8.6	2.70	Sulfaguanidine	12.1	2.8	-1.22
Propoxyphene		6.3	4.10	Sulfamerazine	7.1	2.3	0.11
Propranolol		9.5	2.90	Sulfamethazine	7.4	2.4	0.30
Propylhexedrine		10.4	2.98	Sulfamethoxazole	5.6		0.66
Propylthiouracil	7.8		1.15	Sulfasalazine	2.4, 9.7, 11.8		3.05
Pseudoephedrine		9.5	1.43	Sulfathiazole	7.1	2.4	0.05
Pyrazinazine		8.9	4.15	Sulfinpyrazone	2.8		1.89
Pyrazinamide		0.5	-0.17	Sulfisoxazole	5.0		1.01
Pyridoxine	8.96	5.0	-0.83	Sulindac	4.5		2.55
Pyrilamine		4.0, 8.9	2.67	Sulpiride		9.1	0.78
Pyrimethamine		7.3	2.75	Sulthiame	10.0		0.56
Pyrrobutamine		8.8	4.57	p-Synephrine	10.2	9.3	0.13
Quinacrine		8.2, 10.2	5.59	Talbutal	7.8		1.79
Quinidine		4.2, 7.9	2.82	Tamsulosin	10.23	8.37	2.14
Quinine		4.2, 8.5	2.82	Tamoxifen		8.9	5.13
Ranitidine		2.3, 8.2	-0.07	Tapentadol		9.34	3.02
Resperidone	3.11	8.24	3.04	Terazosin		7.1	0.80
Reserpine		6.6	3.65	Temazepam		1.6	2.19
Rifampin	1.7	7.9	2.05	Terbutaline	10.1, 11.2	8.8	0.70
Rimoterol	10.3	8.7	0.74	Tetracaine		8.4	3.75
Ritodrine		9.0	2.04	Tetracycline	3.3, 7.7	9.7	0.62
Rivastigmine		8.85	2.05	Tetrahydrocannabinol (THC)		10.6	6.84
Rolitetracycline	7.4		1.66	Tetrahydrozoline		10.5	2.85
Ropinirole		9.5	2.49	Theobromine	10.1	0.1	-1.06
Rotoxamine		8.1	2.69	Theophylline	8.6	3.5	-0.02
Saccharin	1.6		0.91	Thiazbendazole		4.7	2.47
Salicylamide	8.2		1.35	Thiamylal	7.5		2.81
Salicylic acid	3.0, 13.4		2.01	Thioguanine	8.2		-0.28
Salmeterol	9.26	9.99	3.71	Thiopental	7.5		2.63
Salsalate	3.5, 9.8		3.29	Thiopropazate		3.2, 7.2	4.76
Scopolamine		7.6	0.69	Thioridazine		9.5	5.90
Secobarbital	7.9, 12.6		2.30	Thiothixene		7.7, 7.9	3.72

(Continued)

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases (Continued)

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Thiouracil	7.5		0.85	Vidarabine		3.5, 12.5	-1.23
Thonzylamine		2.2, 9.0	1.86	Vigabatrin	4.02	9.72	-0.30
l-Thyroxine	2.2, 6.7	10.1	4.72	Viloxazine		8.1	1.24
tiagabine	3.3	9.4	4.03	Vinblastine		5.4, 7.4	5.92
Ticarcillin	2.6, 3.4		1.24	Vincristinej		5.0, 7.4	5.75
Ticrynafen	2.7		3.05	Vindesine		6.0, 7.7	4.94
Timolol		8.8	1.28	Voriconazole	2.72, 11.54		1.21
Timoprazole		3.1, 8.8	1.22	Warfarin	5.1		3.13
Tiotidine		6.8	0.35	Xylometazoline		10.2	4.59
Tobramycin		6.7, 8.3, 9.9	-4.22	Zimeldine		3.8, 8.74	3.07
Tocainide		7.5	0.81	Zonisamide		10.2	0.72
Tolamolol		7.9	2.03	Miscellaneous organic acids and bases			
Tolazamide	3.1	5.7	1.54	Acetic acid	4.8		
Tolazoline		10.3	1.80	Allylamine			
Tolbutamide	5.4		2.36	6-Aminopenicillanic acid	2.3		
tolcapone	10.63		2.12	Ammonia		9.3	
Tolmetin	3.5		2.68	Aniline		4.63	
Toloteridine		9.87	5.23	Benzoic acid	4.2		
Topiramate	8.67		2.16	Benzyl alcohol	18.0		
Tramadol		9.41	2.32	Benzylamine		9.33	
Tramazoline		10.7	2.78	Butyric acid	4.8		
Tranlycypromine		8.2	1.47	Carbonic acid	6.4, 10.4		
Trazodone		6.7	2.76	Citric acid	3.1, 4.8, 5.4		
Triamterene		6.2	1.16	Diethanolamine		8.9	
Trichlormethiazide	8.6		0.13	Diethylamine		11.09	
Trifluoperazine		3.9, 8.1	4.62	Dimethylamine		10.73	
Triflupromazine		9.2	5.16	Ethanol	15.6		
Trimeprazine		9.0	5.04	Ethanolamine		9.5	
Trimethobenzamide		8.3	1.26	Ethylamine		10.7	
Trimethoprim		6.6	0.91	Ethylenediamine		7.56, 10.71	
Trimipramine		8.0	4.71	Fumaric acid	3.0, 4.4		
Tripelennamine		4.2, 8.7	2.78	Gluconic acid	3.6		
Triprolidine		6.5	3.25	Glucuronic acid	3.2		
Troleandomycin		6.6	3.46	Guanidine		13.6	
Tropicamide		5.3	0.64	Imidazole		7.0, 14.9	
Tuaminoheptane		10.5	2.43	Isopropylamine		10.63	
Tubocurarine		8.1, 9.1	0.77	Lactic acid	3.9		
Tyramine	10.9	9.3	0.78	Maleic acid	1.9		
Valdenafil	6.13, 2.02	9.11	3.64	Mandelic acid	3.4		
Valproic acid	4.8		2.58	Methylamine		10.66	
Valsartan	6.0		4.02	Monochloroacetic acid	2.9		
Venlafaxine		9.4	2.47	N-propylamine		10.6	
Verapamil		8.9	4.02	Nitromethane	11.0		

TABLE A.1 pK_a Values for Some Drugs and Miscellaneous Organic Acids and Bases (Continued)

Drugs	pK_a Values		ClogP	Drugs	pK_a Values		ClogP
	HA	HB ⁺			HA	HB ⁺	
Phenol	9.9			Trichloroacetic acid	0.9		
Phthalic acid	2.9			Triethanolamine		7.8	
Resorcinol	9.2, 11.3			Triethylamine		10.75	
Sorbic acid	4.8			Trimethylamine		9.8	
Succinimide	9.6			Tropic acid	4.1		
Tartaric acid	3.0, 4.4			Tropine		10.8	
p-Toluidine		5.3		Uric acid	5.4		

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APPENDIX B. pH Values for Tissue Fluids

DAVID A. WILLIAMS

TABLE B.1 pH Values for Tissue Fluids

Fluid	pH
Aqueous humor	7.2
Blood, arterial	7.4
Blood, venous	7.4
Blood, maternal umbilical	7.3
Cerebrospinal fluid	7.4
Colon ^a	
Fasting	5–8
Fed	5–8
Duodenum ^a	
Fasting	4.4–6.6
Fed	5.2–6.2
Feces ^b	7.1 (4.6–8.8)
Ileum ^a	
Fasting	6.8–8.6
Fed	6.8–8.0
Intestine, microsurface	5.3
Lacrimal fluid (tears)	7.4
Milk, breast	7.0
Muscle, skeletal ^c	6.0
Nasal secretions	6.0
Prostatic fluid	6.5
Saliva	6.4
Semen	7.2
Stomach ^a	
Fasting	1.4–2.1
Fed	3–7
Sweat	5.4
Urine	5.8 (5.5–7.0)
Vaginal secretions, premenopause	4.5
Vaginal secretions, postmenopause	7.0

^aDressman JB, Amidon GL, Reppas C, et al. Dissolution testing as a prognostic tool for oral drug absorption. Pharm Res 1998;15:11–22.

^bValue for normal, soft, formed stools. Hard stools tend to be more alkaline, whereas watery, unformed stools are acidic.

^cStudies conducted intracellularly on the rat.

Drug Index

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